Alexander Grothendieck

Les Dérivateurs

Plain reading text

English translation

Edition by M. Künzer, J. Malgoire et G. Maltsiniotis

12 November 2026

Credits. Author: Alexander Grothendieck, *Les Dérivateurs* (1990–1991). The edition transcribed is the electronic edition by M. Künzer, J. Malgoire and G. Maltsiniotis (Universités de Montpellier II et Paris VII). The deciphering, edition and editors' notes are theirs. AI-transcribed by Claude Opus 5.5 (Anthropic). Companion commentary and AI-produced corrections by GPT-6 Astra (OpenAI, ChatGPT Pro mode), checked and integrated by Claude Opus 5.5 (Anthropic). Eight commentary blocks in II-a, III-V and XIIb-a, seven paragraphs of other blocks, nineteen rows of the claims register (II-a, III-V, VII-c, VIII-X, XIIb-a) and three bibliography entries are AI-produced by Claude Fable 5.2 (Anthropic) from GPT-6 Astra's answers that it reviewed; each relevant row names the answer and reviewer. Macro-definition corrections and changes in the checked texts absent from their tables were compared against the edition's pages by Claude Opus 5.5 (`transcription/DECISIONS.md`). AI-translated: translated by GPT-6.1 Sol (OpenAI, effort max). Glossary by Claude Fable 5.2 and Claude Opus 5.5; prompt, checks, fixes, completeness comparison and the published English book by Claude Opus 5.5 (Anthropic). Raw research answers AI-produced by GPT-6 Astra; reviews by Claude Fable 5.2. This reader build, apparatus and audit are by GPT-6 Astra (OpenAI, ChatGPT Pro mode).

https://webusers.imj-prg.fr/~georges.maltsiniotis/groth/Derivateurs.html

Rights

The rights in the original work belong to its author or the author's representatives or heirs. Where we hold rights, in the AI-produced transcription, translation, corrections and typesetting, they are dedicated to the public domain under CC0 1.0. This edition is published for educational, non-profit purposes, and we claim no rights we do not hold. If you hold rights in this work and object to its publication here, please open a GitHub issue and we will take it down.

Preface to the reading text. This volume gives the corrected text without our apparatus and without the original editors' notes, including their reports of words or passages struck out. The editors' afterword to Chapter XIV is omitted. The reference sections in Chapter XIII remain credited to the editors. Two quotations from Grothendieck's marginal annotations, transmitted in note 58 of Chapter XVIII, are retained in that note with the surrounding editorial commentary removed. All other words and the author's footnotes are retained. The neutral sign [...] marks places where the editors report illegible words or missing text. Edition and manuscript page numbers appear only in the margin; their order and arguments are retained. The apparatus volume and accompanying records provide the correction decisions and their sources.

Contents

The edition's table of contents 4

Chapter I 8

Chapter II 64

Chapter III 113

Chapter IV 119

Chapter V 125

Chapter VI 150

Chapter VII 248

Chapter VIII 327

Chapter IX 333

Chapter X 345

Chapter XI 348

Chapter XII, 1–4 353

Chapter XII, 5 384

Chapter XII, 6 441

Chapter XII, appendix 491

Chapter XIII 502

Chapter XIV 550

Chapter XV 555

Chapter XVI 618

Chapter XVII 664

Chapter XVIII 713

Chapter XIX 881

The Derivators

p. 1

Alexandre Grothendieck

I. Generalities on (pre)derivators. [1]-[126]

1. Preliminaries on the inverse image functors for presheaves on index categories, and their multiple adjoints ..... [1]
2. Heuristic preliminaries on derivators ..... [4]

Homotopically proper, homotopically smooth morphisms ..... [53]
Notions of $\mathbb{D}$-equivalence and of $\mathbb{D}$-contractibility ..... [61]
$\mathbb{D}$-fibrations ..... [76]

II. Cofinality (on the right and on the left) (preliminary to cohomological cofinality). [1]-[99]

1. Cohomological cofinality ..... [1]
2. The locally filtering categories are the locally totally 0-connected categories . [11]
3. Return to cofinality ..... [37]
4. Biaspheric functors ..... [49]
5. Formal duality ..... [69]
6. Transporters, kernels, stabilizers relative to W ..... [83]
7. Cofiltering and locally cofiltering morphisms ..... [90]

III. External Hom in derivators. [1]-[12]

1. Relation of derivators with the category Hot ..... [1]

IV. Substantial diagrams. [73]-[83]

Construction of substantial diagrams for a derivator ..... [73]

V. Categories of paths and localization. [1]-[52]

1. The category of paths ..... [1]
2. Calculation of a category of fractions $X\Sigma^{-1}$ ..... [5]
3. Passage to categories of diagrams ..... [28]

VI. HOT. [1]-[204]

The derivator HOT ..... [1]
New version of the theory of HOT_W ..... [171]

p. 2

VII. Categories of paths (2). [1]-[171]

1. Intervals and their variants ..... [1]
2. Full subcategories of an interval ..... [10]
3. Paths. Category of paths of a given type τ ..... [13]
4. Category of paths $\underline{\mathrm{Ch}}(X)$ ..... [24]
5. Properties of morphisms of intervals ..... [31]
6. Composition in $\underline{\mathrm{Ty}}$ ..... [48]
7. Explicitation of the categories of fractions $X\Sigma^{-1}$ in terms of the $\underline{\mathrm{Ch}}(X; a, b)$. [59]
8. The homotopy lemma for morphisms of intervals ..... [83]
9. Homotopy theorem for categories of paths ..... [94]
10. Comparison question for Cat-fibered categories satisfying the "homotopy theorem" (filtering base) ..... [99]
11. Quick return to the $(\mathcal{M}, \Sigma)$, from the viewpoint of derivators ..... [129]
12. Return to the "categories of paths" in Cat, as a category of models for Hot . [137]

13. Where we finally prove that $\underline{\mathrm{Ch}}(X;-,-)$ is as good as (or even better than) $\underline{\mathrm{Ch}}_\infty(X;-,-)$ ..... [157]

VIII. Relative homotopy 1-types: their integration ... [1]-[9]

1. Integration of relative homotopy 1-types over a category ..... [1]
2. Questions of base change ..... [6]

IX. Return to the categories $\mathcal{M}W^{-1}$, comparison with Quillen. [1]-[25]

1. Functoriality of $\mathrm{Ch}_\infty^\Sigma(a,b)$ in a, b in $\mathcal{M}\Sigma^{-1}$ ..... [1]
2. Comparison with Quillen ..... [14]
3. Attempt at constructing the functor Ω ..... [24]

X. Comparison of Cat with $\Delta^\wedge$. [1]-[5]

1. Reminder of the notions of $\mathbb{D}$-asphericity, and link with the "basic localizer $W_{\mathbb{D}}$" [1]

XI. Hot-fibrations, proper functors and smooth functors etc. [127]-[136]

(in Cat).

1. Hot-fibrations in Cat ..... [127]

XII. Characterization of W_∞. W-proper, W-smooth functors etc. Amalgamated sums and W-cocartesian squares in Cat. [1]-[275]

I Characterization of W_∞ ..... [1]
II Perplexities about the axioms and the terminology ..... [18]
III Remarkable classes of functors related to W ..... [22]
IV On axiom W(4) and its variants ..... [53]
V Amalgamated sums and W-cocartesian squares in Cat ..... [59]

1. Generalities on $U \overset{i}{\hookrightarrow} X \overset{j}{\hookleftarrow} T$ and amalgamated sums ..... [59]
   1.1. Gluing data between an open and a closed ..... [59]
   1.2. Relation with the gluing of topoi ..... [61]
   1.3. Arrows in Cat, with source or target a glued object (X,U,T) ..... [65] p. 3
   1.4. Construction of amalgamated sums from an open where closed immersion ..... [66]
2. Amalgamated sums and W-equivalences ..... [75]
   2.1. Cocartesian square axiom and cube axiom (or "Five Lemma") .... [75]
   2.2. Application to cocartesian squares deduced from an open or closed immersion ..... [80]
   2.3. Totally W-aspheric objects of Cat, and totally W-aspheric morphisms [87]
   2.4. Application to open Dwyerian immersions ..... [107]
   2.5. Complement: criterion for total asphericity of a W-proper category over another ..... [115]
3. Amalgamated sums and integration ..... [121]
   3.1. Canonical reduction to the case of a diagram of closed immersions [121]
   3.2. Comparison of the integral with an ordinary amalgamated sum. W-homotopically cocartesian squares ..... [128]
   3.3. Proper integration versus smooth integration. Statement of equivalence between the two points of view. Comments on the case of arbitrary W-homotopic limits ..... [133]
   3.4. Criteria for homotopically cocartesian squares ..... [146]
   3.5. Applications to amalgamated sums in a category $X^\wedge$ (with $X \in \mathrm{Ob}\,\mathrm{Cat}$) [158]

VI Criterion for W_∞-asphericity of the $\Omega^n(X,x)$ . . . . . [171]
1. Cogitations on W_n . . . . . [171]
2. Criterion for W_n-asphericity . . . . . [187]
3. On the philosophy of homotopy fibers . . . . . [191]
4. Axiomatization of "W-homotopy fibers" . . . . . [207]
5. Axiomatics of non-locally constant relative W-homotopy types $\mathrm{Ho}^{\mathcal{L}}_W(S)$ [218]
6. W-homotopy cofiber in Cat . . . . . [227]
7. Return to the axiomatics of the W-homotopy fiber . . . . . [240]
8. Appendix: Return to axioms W(7), W(7 bis) (pages 75, 76) . . . . . [265]

XIII. Categories of models (1). [1]-[98]
1. Quillen couples (Φ, Ψ) . . . . . [1]
2. Weak Quillen categories, and construction thereof (Conjectural theorem-scholie) [7]
3. Quillen categories (second version) . . . . . [27]
4. The variances u^*, u_*, $u_!$ for $(\mathcal{M}(I), W(I))$ (I variable) . . . . . [44]
5. Quillenizable categories of models . . . . . [50]
6. Quillenizable categories of models, and functors $u_!$, u_* . . . . . [57]
7. Cardinal filtrations and accessible subsets of a big category . . . . . [77]

p. 4

XIV. h-cartesian and h-cocartesian squares. [1]-[8]
1. Axioms for quadrangulated structures (an aborted attempt) . . . . . [1]
2. Examples . . . . . [7]

XV. Factorization theorems. (Models (2)). [1]-[137]
1. Fundamental construction: the transfinite factorization sequence of an arrow f [1]
2. Variation of the factorizations $f = p_\alpha i_\alpha$ with the generating subset C_0 . . . . . [12]
3. Bounding the cardinals . . . . . [16]
4. Complements on Quillen structures . . . . . [28]
5. Preliminary technical results on accessible subsets of $\mathrm{Ob}\,\mathcal{M}$, stable under filtering direct limits . . . . . [43]
6. Aborted attempt at a generation criterion for L-accessible subsets of $\mathrm{Fl}(\mathcal{M})$. [73]
7. Construction of closed Quillen triples with $C = \mathrm{Mon} \subset \mathrm{Cof}_W$ . . . . . [84]
8. Application to the categories of models $(\widehat{A}, W_A)$ . . . . . [100]
9. Revision of the notations, for the axioms of a basic localizer . . . . . [116]
10. Application to the derivator HOT_W . . . . . [134]

XVI. Basic localizers in Cat. [1]-[124]
1. Axioms Loc (1) to Loc (3) (and variants): basic localizers . . . . . [1]
2. Axioms Loc (4) and Loc (5) (of direct sums, of connectedness) . . . . . [8]
3. Axiom Loc (6) of the cocartesian square . . . . . [16′]
4. Axiom Loc (7) of W-fibrations . . . . . [27]
5. Limit axiom Loc (8), and axiom Loc (9) of accessibility . . . . . [71]
6. The condition $\mathrm{Loc}\,(5, \mathbb{D}_{W_\omega})$ implies $W \subset \mathrm{W}_\infty$ . . . . . [83]
7. Comments on the axioms . . . . . [97]

XVII. Categories with fibrations and with cofibrations. (Models (3)). [1]-[118]
1. Preliminaries . . . . . [1]
2. Description of $C_I \subset \mathrm{Fl}(\mathcal{M}(I^\circ))$, for I an ordered set . . . . . [25]
3. Categories of models with cofibrations . . . . . [39]
4. Special categories with cofibrations . . . . . [71]
5. Relation of special categories with cofibrations to special Quillen categories . . [84]

6. Application to the construction of derivators (associated with special Quillen triples) ..... [98]

XVIII. Accessible categories and sets. [50]-[492]

4. Paratopoi and the operation $\mathcal{M} \boxtimes \mathcal{N}$ ..... [50]
   4.1. Pseudo-topoi. Product problem ..... [50]
   4.2. Muddles over $\mathcal{M} \boxtimes \mathcal{N}$ ..... [57]
   4.3. The operation $\mathcal{M} \boxtimes \mathcal{N}$ and the accessibility axiom ..... [64]
   4.4. Study of $\mathrm{Ind}_\pi(C)$ (for C a small category stable under $\varinjlim$ of cardinality $\leq \pi$) [79]
   4.5. Canonical cardinal filtration of a $\mathcal{U}$-paratopos category ..... [106] p. 5
   4.6. Characterization of paratopos categories (proof of 4.3.6) ..... [139]
   4.7. $\mathrm{Ind}_\pi C$ (C arbitrary) and generalized cardinal filtrations ..... [161]
   4.8. $\mathcal{U}$-accessible categories ..... [187]
   4.9. The indicial representation theorem for $\mathrm{Ind}_\pi(C)$ ..... [217]
   4.10. The indicial representation theorem: proof ..... [243]
   4.11. Application to the existence of limits in a category of the form $\mathrm{Ind}_{\pi_0}(C)$ [267]
   4.12. Accessible functors between accessible categories ..... [273]
   4.13. Accessible sets of objects of an accessible category ..... [281]
   4.14. Accessibility criteria for big categories ..... [370]
   4.15. Accessible category over another ..... [425]
   4.16. Examples and questions ..... [464]
   4.17. Return to paratopoi, and to the operations $\mathcal{M} \boxtimes \mathcal{N}$, $\underline{\mathrm{Hom}}_!(\mathcal{M}, \mathcal{N})$ ..... [492]

Appendix. [1]-[19]

Ultraproducts of fields, and accessible categories ..... [1]
Application to an example of a non-accessible subcategory ..... [11]

XIX. Models (4). [1]-[50]

1. Reminders on Quillen couples (special Quillen couples) ..... [1]
2. Reminders on closed Quillen triples and special Quillen triples ..... [13]
3. Case of categories with cofibrations, of categories with fibrations ..... [33]

THE DERIVATORS

p. –

ALEXANDRE GROTHENDIECK

Chapter I

Generalities on (pre)derivators

p. –
p. 1
[page 1]

Derivators

1. Preliminaries on the inverse image functors of presheaves on index categories, and their multiple adjoints.

A "category of coefficients",
I "index category",
$A(I) = \underline{\mathsf{Hom}}(I^\circ, A)$ presheaves on I with values in A.

If $f : I \to J$, we get:

$$f^* : A(J) \to A(I) \quad ,$$

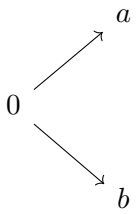
and we are interested in the existence of the successive adjoint functors on the left and on the right. Provisional notation:

$$f^? \; f_! \; f^* \; f_* \; f^! \quad .$$

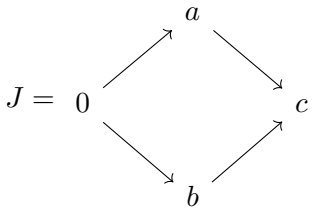
Here are the main facts:

a) Subject to the existence of the relevant $\varprojlim$ (resp. $\varinjlim$), f_* (resp. $f_!$) exists. For example if there exist $\varprojlim$ (resp. $\varinjlim$) that are *finite*, then f_* (resp. $f_!$) exists if I, J are finite categories.

b) Even if A is an abelian category, with $\varprojlim$ and $\varinjlim$ (for example Ab), the *biadjoints* do not always exist. For example, in the particular case where $J = e$ is the one-point category, and I the ordered set



(in the case of $f^!$), or the opposite ordered set (in the case of $f^?$). Even if f is an open immersion, $f^!$ does not necessarily exist (for example it does not exist if I is as above, and



),

and if f is a closed immersion, $f^?$ does not necessarily exist.

c) If f is a *closed immersion*, and A is *pointed* (*i.e.* admits an object both initial and final), then $f^!$ exists. If f is an open immersion, $f^?$ exists. (Always subject to the p. 2

existence in A of the relevant limits). If f is an immersion, f_* and $f_!$ fully faithful *i.e.* f^* a localization functor.

d) Let us assume only the existence of the final object e in A (plus if need be certain finite $\varprojlim$). If

[page 2]

$$i : Y \hookrightarrow S$$

is a closed immersion, with complement U

$$Y \overset{i}{\hookrightarrow} S \overset{j}{\hookleftarrow} U$$

so that j is an open immersion, then $A(Y)$ is identified by the functor i_* with the full subcategory of $A(S)$ consisting of the $F \in \mathsf{Ob}\ A(S)$ such that $F|U \simeq e_U$ (final presheaf). Similarly, if the initial object $\emptyset$ exists in A, and the relevant $\varinjlim$, then $A(U)$ is identified by $j_!$ with the full subcategory of $A(S)$ consisting of the F such that $F|Y = \emptyset_Y$ (initial presheaf, taking the value $\emptyset$ everywhere).

If A is even *pointed* ($e = \emptyset$), then we have, for every $F \in \mathsf{Ob}\ A(S)$, a square both cartesian and cocartesian, functorial in F

$$\begin{array}{ccc} & F & \\ {\scriptstyle \text{adj}}\nearrow & & \searrow{\scriptstyle \text{adj}} \\ j_!j^*F & & i_*i^*F \\ \searrow & & \nearrow \\ & e & \end{array} \quad . \tag{1}$$

In the case A abelian, this diagram takes the form of an exact sequence

$$0 \longrightarrow j_!j^*F \longrightarrow F \longrightarrow i_*i^*F \longrightarrow 0 \quad .$$

e) Same conditions for i and j. We assume that A has the relevant $\varprojlim$, including final object. We then have a cartesian square functorial in $F \in \mathsf{Ob}\ A(S)$

$$\begin{array}{ccc} F & \longleftarrow & i_*i^!F \\ {\scriptstyle \text{adj}}\downarrow & & \downarrow \\ j_*j^*F & \longleftarrow & e \end{array} \tag{2}$$

i.e. $i^!F$ is identified by the fully faithful functor i_* with the *kernel* of $F \to j_*j^*F$. In the abelian case, this takes the form of an exact sequence

$$0 \longrightarrow i_*i^!F \longrightarrow F \longrightarrow j_*j^*F \quad .$$

I dispense with writing down the dual cocartesian square, when A has the relevant $\varinjlim$, including initial object. It seems that it has not been of use in practice. The classically useful adjoint and biadjoint functors are included below (where the "not useful" ones are in brackets)

p. 3

[page 3]

$$[i_!]\ i^*\ i_*\ i^!$$
$$[j^?]\ j_!\ j^*\ j_*$$

i.e. the functors $i_!$ and $j^?$ have not been of use classically.

f) Case of the categories $\mathbf{\Delta}_i$ (standard simplices, $i \in \mathbb{N}$), and of the functors between them (semi-simplicial maps). It would seem that for such an f, if A is pointed, there

are *bi-adjoints* both on the left and on the right. Moreover, for $\mathbf{\Delta}_i$ and $\mathbf{\Delta}_j$ fixed when we look at all the $f : \mathbf{\Delta}_i \to \mathbf{\Delta}_j$ and all the $g : \mathbf{\Delta}_j \to \mathbf{\Delta}_i$, that the total sequences of multi-adjoint functors for each of the corresponding f^*, g^* are all the same . That looks true at least if $j = i+1$ (and I have checked it for $i = 0$, $i = 1$), we should find a "sequence of adjunctions" of length $2i+7 = (i+1)+(i+2)+2+2$. To be precise, the sequence has four "extreme" terms (the first two, the last two) and the $(i+1)+(i+2)$ intermediate terms corresponding to the functors of the form f^* or g^*, where $f : \mathbf{\Delta}_{i+1} \to \mathbf{\Delta}_i$ is one of the $(i+1)$ degeneracy maps, and $g : \mathbf{\Delta}_i \to \mathbf{\Delta}_{i+1}$ one of the $i+2$ face maps .

p. 4
[page 4]

2. Heuristic preliminaries on derivators.

We take a full subcategory Dia (as in "diagrams") of $\mathcal{C}at$. It can be all of $\mathcal{C}at$, or smaller. But we will most often assume that Dia contains at least the "ordered categories" (corresponding to ordered sets, *i.e.* such that *a*) every morphism is a mono and *b*) every iso is an identity). By abuse of language, we identify an ordered set with the corresponding category (and a category with the topos it defines?). We regard Dia as a 2-*category*.[1]

A *derivator* $\mathbb{D}$ (on Dia, as the "*domain*" of the derivator) is a contravariant 2-functor from Dia to $\mathcal{C}at$, satisfying certain conditions. Thus

$$\mathbb{D} : \mathsf{Dia}^\circ \longrightarrow \mathcal{C}at \quad .$$

If $I \in \mathsf{Ob}_0\mathsf{Dia}$ is a category (or a "type of diagram") in the domain of $\mathbb{D}$, we must therefore have a category $\mathbb{D}(I)$. The objects of $\mathbb{D}(I)$ will be called the *coefficient-objects* (or *coefficients*) of type $\mathbb{D}$, over I. For

$$f : I \longrightarrow J$$

a 1-arrow of Dia, we must have a functor

$$f^*_{\mathbb{D}} \text{ or } f^* : \mathbb{D}(J) \longrightarrow \mathbb{D}(I) \quad ,$$

called the *"inverse image functor"*. If F is a coefficient of type $\mathbb{D}$ on J its "*inverse image*" is therefore a coefficient of type $\mathbb{D}$ on I. We assume (to simplify!) that the 2-functor $\mathbb{D}$ is *strict*, hence that we have strict transitivity

$$(gf)^* = f^*g^* \ , \qquad \text{if} \qquad I \xrightarrow{f} J \xrightarrow{g} K \qquad \text{in } \mathsf{Dia} \ .$$

Moreover, if we have two functors $f, g : I \rightrightarrows J$, and a morphism between them [page 5]

$$I \ \underset{g}{\overset{f}{\underset{\downarrow u}{\rightrightarrows}}} \ J \quad ,$$

we take a morphism $u^*_{\mathbb{D}}$ or u^* between the functors g^* and f^*, whence

$$\mathbb{D}(I) \ \underset{g^*}{\overset{f^*}{\underset{\uparrow u^*}{\leftleftarrows}}} \ \mathbb{D}(J) \quad .$$

p. 5

[1] N.B. We need stability conditions for Dia, at least stability under $I \times J$ and $I \amalg J$, in fact, considerably more ...

We must also have transitivity for a composition of morphisms of functors:

$$I \;\begin{array}{c} \xrightarrow{f} \\ \downarrow u \\ \xrightarrow[g]{} \\ \downarrow v \\ \xrightarrow[h]{} \end{array}\; J$$

must give

$$(vu)^* = u^* v^*$$

thus

$$\mathbb{D}(I) \;\begin{array}{c} \xleftarrow{f^*} \\ \uparrow u^* \\ \xleftarrow[g^*]{} \\ \uparrow v^* \\ \xleftarrow[h^*]{} \end{array}\; \mathbb{D}(J) \quad .$$

Moreover, if f is an identity, f^* too

$$(\mathsf{id}_I)^* = \mathsf{id}_{\mathbb{D}(I)} \quad ,$$

and likewise if u is an identity

$$(\mathsf{id}_f)^* = \mathsf{id}_{f^*} \quad .$$

I indicate right away the most important axiom on a derivator (beyond the datum of a strict 2-contra functor): [page 6]

Der$_1$: If $f : I \to J$ in Dia, then $f^*_{\mathbb{D}} = f^* : \mathbb{D}(J) \to \mathbb{D}(I)$ has a right adjoint and a left adjoint.

These functors will be denoted (at least provisionally) $f^{\mathbb{D}}_*$ or f_* , and $f^{\mathbb{D}}_!$ or $f_!$, that is

$$f_!\,,\ f_* : \mathbb{D}(I) \rightrightarrows \mathbb{D}(J) \quad ,$$

so that we find a sequence of *three* adjoint functors between $\mathbb{D}(I)$ and $\mathbb{D}(J)$

$$f_!\,,\ f^*,\ f_* \quad .$$

N.B. For most of the developments we will not need the existence of both the $f_!$ and the f_* . A *left derivator* requires only the f_* (right adjoints of the f^*), and the existence of $f_!$ only when f is an *open immersion*. Dually, a *right derivator* requires only the $f_!$ (left adjoints of the f^*), and f_* only if f is a *closed immersion*. It is probably the left derivators that are the most important.

Der$_2$: If $f : I \to J$ is in Dia, if it is a *closed* immersion, the right biadjoint $f^!_{\mathbb{D}}$ or $f^!$ of f^* exists. If it is an open immersion, the left biadjoint $f^?$ exists.[2] In both cases, we want f^* to be a localization functor, which is equivalent to saying that f_* is fully faithful, or that $f_!$ is fully faithful.[3] [page 7] p. 6

N.B. In the case of a left derivator, we assume only the existence of $f^!$ if f is a closed immersion. Thus to summarize, for such a derivator, we assume the existence of

$$\begin{cases} f_* \text{ for any } f : I \to J \text{ in } \mathsf{Dia} \\ f_! \text{ if } f \text{ is an open immersion} \\ f^! \text{ if } f \text{ is a closed immersion (pointed case only).} \end{cases}$$

[2] The biadjoints should be required only for a pointed derivator. HOT does not satisfy this.

[3] We must in addition characterize the image of f_* if f is a closed immersion (the F such that $g^*F \xrightarrow{\sim} e_{\mathbb{D}(U)}$) and that of $f_!$ if it is an open immersion (the F such that $g^*F \simeq \emptyset_{\mathbb{D}(Y)}$), where g is the complementary immersion.

Dually, for a right derivator, we assume the existence of

$$\begin{cases} f_! \text{ for any } f : I \to J \text{ in } \mathsf{Dia} \\ f_* \text{ if } f \text{ is a closed immersion} \\ f^? \text{ if } f \text{ is an open immersion (pointed case only).} \end{cases}$$

I would like to formulate clever axioms, for the case of a pair

$$Y \overset{i}{\hookrightarrow} I \overset{j}{\hookleftarrow} U$$

of an open immersion j and a closed immersion i in Dia, complementary to one another. First a formulation if we assume $\mathbb{D}$ "*pointed*", *i.e.* the categories $\mathbb{D}(I)$ pointed, and the functors f^* compatible with the pointings. We want to "interpret" the square (1) on page 2:

[page 8]

$$\begin{array}{ccccc} & & F & & \\ & \overset{\text{adj}}{\nearrow} & & \overset{\text{adj}}{\searrow} & \\ j_!j^*F & & & & i_*i^*F \\ & \underset{}{\searrow} & & \underset{}{\nearrow} & \\ & & e & & \end{array} \quad , \tag{1}$$

Der$'_2$: For all $I, J \in \mathsf{Dia}$, we have

$$\mathbb{D}(I \amalg J) \xrightarrow[(\alpha^*,\beta^*)]{\sim} \mathbb{D}(I) \times \mathbb{D}(J) \quad ,$$

where α, β are the canonical injections of I and J into $I \amalg J$.

square both cartesian and cocartesian. We are thus looking for a square in $\mathbb{D}(I)$, but better than that. Let $J = \mathbf{\Delta}_1 \times \mathbf{\Delta}_1$ be the standard square p. 7

$$\begin{array}{ccccc} & & (1,0) & & \\ & \swarrow & & \nwarrow & \\ (1,1) & & & & (0,0) \\ & \nwarrow & & \swarrow & \\ & & (0,1) & & \end{array}$$

and let us consider the category $I \times J$. We have

$$\begin{aligned} J &\longrightarrow \underline{\mathsf{Hom}}(I, I \times J) \\ j &\longmapsto (i \longmapsto (i, j)) \end{aligned}$$

On the other hand, by $\mathbb{D}$

$$\underline{\mathsf{Hom}}(I, I \times J)^\circ \longrightarrow \underline{\mathsf{Hom}}(\mathbb{D}(I \times J), \mathbb{D}(I))$$

whence a composite

$$J^\circ \longrightarrow \underline{\mathsf{Hom}}(\mathbb{D}(I \times J), \mathbb{D}(I))$$

or again

$$\mathbb{D}(I \times J) \longrightarrow \underline{\mathsf{Hom}}(J^\circ, \mathbb{D}(I)) \quad .$$

[page 9]

This holds for every $J \in \mathsf{Ob}\,\mathsf{Dia}$. We can therefore say that the datum of a "coefficient" Φ on a product category $I \times J$ gives rise to a diagram of type J° in the category $\mathbb{D}(I)$ of coefficients on I. The philosophy of derivators consists in large part, conversely, whenever we have a diagram in a category of coefficients $\mathbb{D}(I)$ (on a diagram I), say diagram of type K: $K \to \mathbb{D}(I)$, in asking ourselves whether it does not come "canonically" from an object of $\mathbb{D}(I \times K^\circ)$ *i.e.* from a coefficient on $I \times K^\circ$.

Thus for the square p. 8. We would like it to come from $\mathbb{D}(I \times J)$, where $J = \mathbf{\Delta}_1 \times \mathbf{\Delta}_1$, and this functorially, so we want a "canonical" functor

$$\mathbb{D}(I) \longrightarrow \mathbb{D}(I \times \mathbf{\Delta}_1 \times \mathbf{\Delta}_1)$$

which on composing with

$$\mathbb{D}(I \times \mathbf{\Delta}_1 \times \mathbf{\Delta}_1) \longrightarrow \underline{\mathsf{Hom}}((\mathbf{\Delta}_1 \times \mathbf{\Delta}_1)^\circ, \mathbb{D}(I))$$

gives the functor described by square (1). Moreover, we must express in the present context the "bicartesian" condition on square (1), in the context of page 7. Analogous question for square (2) on page 2: p. 8

$$\begin{array}{ccc} & F & \\ \nearrow & & \searrow \\ i_*i^!F & & j_*j^*F \\ \searrow & & \nearrow \\ & e & \end{array} \quad , \tag{2}$$

where we must in addition interpret the cartesian condition. We are in fact going to define a coefficient on the following open subcategory W of $\mathbf{\Delta}_1 \times \mathbf{\Delta}_1$

$$\begin{array}{ccc} (1,0) & & \\ & \nwarrow & \\ & & (0,0) \\ & \swarrow & \\ (0,1) & & \end{array} \quad ,$$

more precisely on $W \times I$, and if α [page 10]

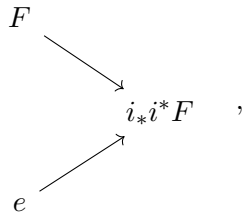
$$\alpha : W \times I \longrightarrow \mathbf{\Delta}_1 \times \mathbf{\Delta}_1 \times I$$

is the canonical inclusion, we will define the "cartesian" square as the image under α_* of this coefficient. This thus brings us back to the problem of first "globalizing" the partial diagram

$$\begin{array}{ccc} F & & \\ & \searrow & \\ & & j_*j^*F \\ & \nearrow & \\ e & & \end{array} \quad .$$

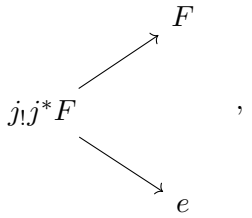
And likewise for square (1) except that we are at a loss if we choose to look at it mainly as cartesian (which suggests proceeding as just now) or as cocartesian (which p. 9

suggests proceeding in the dual way). But the two coefficients one will find by these two procedures, products either of



,

or of



,

should be *canonically* isomorphic.

If $\mathbb{D}$ is a derivator, the category [page 11]

$$\mathcal{A} = A_{\mathbb{D}} = \mathbb{D}(e)$$

plays an important role. It is called the *base category,* or *the fundamental category*, of the derivator, these objects are the *fundamental coefficients.* For arbitrary I in Dia, we thus have for $F \in \mathbb{D}(I)$, coefficient on I

$$I^\circ \longrightarrow \mathcal{A} \quad , \qquad i \longmapsto F_i$$

a diagram of type I° in $\mathcal{A}$, functorially, *i.e.*

$$\mathbb{D}(I) \longrightarrow \underline{\mathsf{Hom}}(I^\circ,\ \underset{\mathbb{D}(e)}{\mathcal{A}}\ )$$

(*cf.* page 8).

Der$_3$: For every $I \in \mathsf{Dia}$, this functor is *conservative,*[4] *i.e.* for

$$F \longrightarrow G$$

in $\mathbb{D}(I)$ to be isomorphism, it is necessary and sufficient that for every $i \in I$, p. 10

$$F_i \longrightarrow G_i$$

be one.

Corollary. *The canonical functor (for I, $J \in \mathsf{Dia}$)*

$$\mathbb{D}(I \times J) \xrightarrow{\theta} \underline{\mathsf{Hom}}(J^\circ, \mathbb{D}(I))$$

is conservative.

Indeed, if F is in $\mathbb{D}(I \times J)$, we have for $(i, j) \in I \times J$

$$F_{i,j} \simeq \theta(F)(j)_i \quad .$$

Intuitively, one regards a derivator $\mathbb{D}$ as "being" the fundamental category $\mathcal{A}$, with an [page 12]

[4] N.B. But this functor has no desire to be *faithful.*

additional "structure", wit the datum of the "extensions" of $\mathcal{A}$ (as category of point coefficients) to the diagrams $I \in \mathsf{Dia}$, contravariantly in I. A coefficient F in $\mathbb{D}(I) \overset{\text{def}}{=} \mathcal{A}_I$ is regarded as "being" the corresponding diagram $(F_i)_{i\in I}$ of fundamental objects, with an additional datum of "globalization". The F_i, or "fibers" of F at the $i \in I$, are the local characteristics, of the "global on I" coefficient F.

All known derivators are obtained in the following standard way. One starts from a "category of models"

$$\mathcal{M} \quad , \quad \Sigma$$

equipped with a set of arrows, called "equivalences", and satisfying the relevant axioms for passing to a category of fractions

$$\mathcal{A} = \mathcal{M}\Sigma^{-1}$$

which will be the "fundamental category". One defines for every I in Dia

$$\begin{aligned} \mathcal{M}_I &= \underline{\mathsf{Hom}}(I^\circ, \mathcal{M}) \\ \Sigma_I &= \{u \in \mathsf{Fl}(\mathcal{M}_I) \mid \forall i \in I, u_i \in \Sigma\} \\ \mathcal{A}_I &= \mathcal{M}_I\Sigma_I^{-1} \end{aligned}$$

It is clear that $\mathcal{A}_I \overset{\text{def}}{=} \mathbb{D}_{\mathcal{M},\Sigma}(I)$ depends on I in a contravariant functorial way, and even in a 2-functorial way. [page 13]

Of course, to really have a derivator, one must check the axioms. The strongest is the existence of the $f_!$, f_*. It is proved by Thomason when $(\mathcal{M}, \Sigma)$ comes from a "model category" in Quillen's sense. It is not clear to me that then $(\mathcal{M}_I, \Sigma_I)$ also comes from a model category (in which for example the fibrations and cofibrations would be defined from these data in $\mathcal{M}$, as earlier for the equivalences). p. 11

When one takes for $(\mathcal{M}, \Sigma)$ a category of models for Hot (for example the semi-simplicial sets $\mathcal{M} = \underline{\mathsf{Hom}}(\mathbf{\Delta}^\circ, \mathcal{E}ns)$, whence $\mathcal{M}_I \simeq \underline{\mathsf{Hom}}(\mathbf{\Delta}^\circ \times I^\circ, \mathcal{E}ns) \simeq \underline{\mathsf{Hom}}(\mathbf{\Delta}^\circ, \underbrace{\underline{\mathsf{Hom}}(I^\circ, \mathcal{E}ns)}_{\widehat{I}})$, category of semi-simplicial objects in the topos $\widehat{I}$), one finds a description of a derivator (nonadditive), which will play a particularly important role. I denote it by

$$\mathsf{HOT} \quad ,$$

its category of fundamental coefficients is Hot, the standard homotopy category.

The notion of "morphism of Derivators" is clear, it is a morphism between two strict 2-functors [page 14]

$$\mathsf{Dia}^\circ \longrightarrow \mathcal{C}at$$

Thus

$$u : \mathbb{D} \longrightarrow \mathbb{D}'$$

is defined by giving, for every $I \in \mathsf{Dia}$,

$$u_I : \mathbb{D}(I) \longrightarrow \mathbb{D}'(I)$$

with commutativity (strict?) for $f : I \to J$

$$\begin{array}{ccc} \mathbb{D}(J) & \xrightarrow{f^*_{\mathbb{D}}} & \mathbb{D}(I) \\ {\scriptstyle u_J}\downarrow & & \downarrow{\scriptstyle u_I} \\ \mathbb{D}'(J) & \xrightarrow{f^*_{\mathbb{D}'}} & \mathbb{D}'(I) \end{array} \quad . \tag{1}$$

We have

$$\begin{array}{ccc} \underline{\mathsf{Hom}}(I,J)^\circ & \xrightarrow{\mathbb{D}_{I,J}} & \underline{\mathsf{Hom}}(\mathbb{D}(J),\mathbb{D}(I)) \\ {\scriptstyle \mathbb{D}'_{I,J}}\downarrow & & \downarrow{\scriptstyle \phi\mapsto u_I\circ\phi} \\ \underline{\mathsf{Hom}}(\mathbb{D}'(J),\mathbb{D}'(I)) & \xrightarrow[\psi\mapsto\psi\circ u_J]{} & \underline{\mathsf{Hom}}(\mathbb{D}(J),\mathbb{D}'(I)) \end{array} \tag{2}$$

and the commutativity of (1) means the commutativity of (2) *for the objects*. But of course we want it for the arrows too, *i.e.* we want the commutativity of diagram (2).[5]

Probably we want a little more from morphisms of derivators - we want commutation with the f_* and (?) with the $f_!$ (this time only up to canonical isomorphism) ...[6]
For example, consider the derivator p. 12
[page 15]

$$\mathsf{HOTAB}$$

whose fundamental category is Hotab, the full subcategory of the derived category $\mathsf{Der}(\mathsf{Ab})$ of Ab, consisting of the complexes with vanishing cohomology in degree > 0. One can describe HOTAB by the category of models $\underline{\mathsf{Hom}}(\mathbf{\Delta}^\circ, \mathsf{Ab})$ of semi-simplicial abelian groups, with as equivalences the quasi-isomorphisms in the usual sense. The canonical "abelianization functor"

$$\mathsf{Hot} \longrightarrow \mathsf{Hotab}$$

extends to a morphism of derivators

$$\mathsf{HOT} \longrightarrow \mathsf{HOTAB} \quad .$$

The question of its compatibility with the f_* does not seem tautological. It would imply, for example, that if

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[j]{} & Y' \end{array}$$

is a cartesian diagram in $\mathcal{M} = \underline{\mathsf{Hom}}(\mathbf{\Delta}^\circ, \mathcal{E}ns)$ with f a fibration (of Thom in the setting of topological spaces, with f for example a locally trivial fibration) then the diagram in Hotab

$$\begin{array}{ccc} \mathbb{R}\Gamma_!(X) & \longleftarrow & \mathbb{R}\Gamma_!(X') \\ \downarrow & & \downarrow \\ \mathbb{R}\Gamma_!(Y) & \longleftarrow & \mathbb{R}\Gamma_!(Y') \end{array}$$

is hot-cartesian[7]. And likewise for a hot-amalgamated sum (hot-cocartesian square in Hot).
If this is true, and not known to boot, it would be scandalous all the same!

Proceeding as for the base change morphisms for the Rf_*, we find in any case a [page 16]
canonical morphism, for $f : I \to J$

$$u_J f_*^{\mathbb{D}} \longrightarrow f_*^{\mathbb{D}'} u_I$$

and likewise

p. 13

[5] Give the example of a morphism of derivators defined by a functor of categories of models $(\mathcal{M}, \Sigma) \to (\mathcal{M}', \Sigma')$.

[6] Commutation with f_* (resp. $f_!$) *left exact* (resp. *right exact*) morphism. The *exact* morphisms are those that are both left exact and right exact.

[7] This is false already if Y is the one-point space : trivially the $\mathbb{R}\Gamma_!$ does not transform products into sums! This functor is surely "left exact" (as a morphism of derivators).

$$f_!^{\mathbb{D}'} u_I \longrightarrow u_J f_!^{\mathbb{D}} \quad ,$$

and the question is whether these are isomorphisms. Analogous questions for the $f^!$, for f a closed immersion (or for $f^?$, for f an open immersion). Since the notion of a morphism of derivators ought all the same to "fit" the well-known notion of a (triangulated) morphism of triangulated categories[8], hence be exact in a suitable sense, this does show that one must require a morphism of derivators to commute with the f_* and with the $f_!$.[9] I should ask Thomason whether he has looked, for a morphism

$$\mathcal{M} \longrightarrow \mathcal{M}'$$

between categories of models, which is say *left exact*, and "compatible" with the equivalences, fibrations (and cofibrations (???)).

NOTATION : DER for the 2-category of *derivators*.

Duality. If $\mathbb{D}$ is a derivator, we are going to define a *dual* derivator $\mathbb{D}^\circ$ by [page 17]

$$\mathbb{D}^\circ(I) \stackrel{\text{def}}{=} \mathbb{D}(I^\circ)^\circ \quad .$$

The coefficients on I of type $\mathbb{D}^\circ$ are the coefficients on the category I° opposite to I of type $\mathbb{D}$, and as a category it is the category *opposite* to that of the coefficients of type $\mathbb{D}$ on I°. To define the 2-functorial law, we note that for I, J in Dia we have

$$\begin{array}{ccc} \underline{\mathsf{Hom}}(I,J)^\circ & \xrightarrow{\mathbb{D}_{I,J}} & \underline{\mathsf{Hom}}(\mathbb{D}(J),\mathbb{D}(I)) \\ \wr\downarrow \text{duality} & & \wr\downarrow \text{duality} \\ \underline{\mathsf{Hom}}(I^\circ,J^\circ) & \xrightarrow{\theta} & \underline{\mathsf{Hom}}(\mathbb{D}(J)^\circ,\mathbb{D}(I)^\circ)^\circ \end{array} \quad ,$$

where θ is the unique functor making the diagram commutative. Then $\theta^\circ$is a functor

$$\underline{\mathsf{Hom}}(I^\circ,J^\circ)^\circ \longrightarrow \underline{\mathsf{Hom}}(\mathbb{D}(J)^\circ,\mathbb{D}(I)^\circ) \quad ,$$

that is by definition $\mathbb{D}^\circ_{I,J}$.

If $\mathbb{D}$ is defined by $(\mathcal{M},\Sigma)$, then $\mathbb{D}^\circ$ by $(\mathcal{M}^\circ,\Sigma^\circ)$, since

$$(\mathcal{M}^\circ)_I \stackrel{\text{def}}{=} \underline{\mathsf{Hom}}(I^\circ,\mathcal{M}^\circ) \simeq \underline{\mathsf{Hom}}(I,\mathcal{M})^\circ \simeq (\mathcal{M}_{I^\circ})^\circ$$

and ${\Sigma^\circ}_I$ is deduced from Σ_{I°, whence

$$(\mathcal{M}^\circ)_I({\Sigma^\circ}_I)^{-1} \simeq (\mathcal{M}_{I^\circ}\Sigma_{I^\circ}^{-1})^\circ = \mathbb{D}(I^\circ)^\circ \quad . \qquad \text{p. 14}$$

Philosophy. Each time we have a functor

$$\mathcal{A} \xrightarrow{u_0} \mathcal{A}' \qquad [\text{page 18}]$$

between two categories which can be naturally interpreted as the fundamental categories of two derivators $\mathbb{D}$, $\mathbb{D}'$, the question arises of interpreting u_0 as coming from a morphism of derivators

$$\mathbb{D} \xrightarrow{u} \mathbb{D}' \quad .$$

The latter is, in practice, always canonically determined, up to canonical isomorphism. (But I nevertheless doubt that the restriction functor to the fundamental categories

$$\underline{\mathsf{Hom}}(\mathbb{D},\mathbb{D}') \longrightarrow \underline{\mathsf{Hom}}(\mathcal{A},\mathcal{A}')$$

[8] No! The case of triangulated categories is "misleading", because for these left and right exactness amount to the same thing.

[9] N.B. One can conceive of a functor between derivators being left exact or right exact or both.

is fully faithful …)

We are going to apply this philosophy to the functors f^*, $f_!$, f_* (and possibly $f^!$, $f^?$), for $f : I \to J$, thus to the functors between $\mathbb{D}(I)$ and $\mathbb{D}(J)$. This therefore leads us to regard $\mathbb{D}(I)$ and $\mathbb{D}(J)$ as fundamental categories of suitable derivators.

Induced derivator by a derivator $\mathbb{D}$, on an index category $I \in \mathsf{Dia}$. It is a derivator $\mathbb{D}_I$ defined by

$$\mathbb{D}_I(J) = \mathbb{D}(I \times J) \quad .$$

The functor associated with $\mathbb{D}_I$, for two categories J, $J' \in \mathsf{Dia}$

[page 19]

$$\underline{\mathsf{Hom}}(J, J')^\circ \xrightarrow{(\mathbb{D}_I)_{J,J'}} \underline{\mathsf{Hom}}(\mathbb{D}_I(J'), \mathbb{D}_I(J)) = \underline{\mathsf{Hom}}(\mathbb{D}(I \times J'), \mathbb{D}(I \times J))$$

is the composite

$$\underline{\mathsf{Hom}}(J, J')^\circ \longrightarrow \underline{\mathsf{Hom}}(I \times J, I \times J')^\circ \xrightarrow{\mathbb{D}_{I\times J, I\times J'}} \underline{\mathsf{Hom}}(\mathbb{D}(I \times J'), \mathbb{D}(I \times J)) \quad ,$$

where the first arrow is the functor

$$f \longmapsto \mathsf{id}_I \times f \quad .$$

In other words, $\mathbb{D}_I$ is the composite

$$\begin{array}{l} \mathsf{Dia}^\circ \longrightarrow \mathsf{Dia}^\circ \xrightarrow{\mathbb{D}} \mathcal{C}at \\ J \longmapsto I \times J \qquad . \end{array}$$

We have

$$f^*_{\mathbb{D}_I} = (\mathsf{id} \times f)^*_{\mathbb{D}} \quad .$$

If the f_* (resp. the $f_!$) exist for $\mathbb{D}$, they exist for $\mathbb{D}_I$. (Likewise for every kind of multiadjoints.) Since $J \mapsto I \times J$ takes closed immersion (resp. open) to likewise, axiom $\mathbf{Der}_2$ (more precisely, each of these ingredients, *cf.* marginal annotations p. 7) is stable under passage from $\mathbb{D}$ to $\mathbb{D}_I$. Likewise for $\mathbf{Der}_3$. (For each new axiom we bring out, we will have to check stability.)

Let

p. 15

[page 20]

$$f : I \longrightarrow I' \quad .$$

Consider between $\mathbb{D}(I)$, $\mathbb{D}(I')$ the functors

$$\mathbb{D}(I) \;\begin{array}{c} \xrightarrow{f_!} \\ \xleftarrow{f^*} \\ \xrightarrow[f_*]{} \end{array}\; \mathbb{D}(I')$$

(subject to the existence of the f_* resp. $f_!$). Can we interpret them in terms of morphisms of derivators

$$\mathbb{D}_I \;\begin{array}{c} \xrightarrow{\overline{f_!}} \\ \xleftarrow{\overline{f^*}} \\ \xrightarrow[\overline{f_*}]{} \end{array}\; \mathbb{D}_{I'} \qquad ?$$

So we must define, for J in Dia, corresponding functors between $\mathbb{D}_I(J)$ and $\mathbb{D}_{I'}(J)$, *i.e.* between $\mathbb{D}(I \times J)$ and $\mathbb{D}(I' \times J)$. Now we do have

$$f \times \mathsf{id}_J : I \times J \longrightarrow I' \times J$$

and we will set (subject to the existence of the last two)

$$\begin{cases} \overline{f^*}(J) = (f \times \mathsf{id}_J)^*_{\mathbb{D}} \\ \overline{f_!}(J) = (f \times \mathsf{id}_J)^{\mathbb{D}}_! \\ \overline{f_*}(J) = (f \times \mathsf{id}_J)^{\mathbb{D}}_* \quad . \end{cases}$$

In other words, we can regard [page 21]

$$\mathsf{Dia} \ni I \longmapsto \mathbb{D}_I \quad , \qquad \mathbb{D} \in \mathsf{DER} \quad ,$$

as functorial in I in three ways (one contravariant, the other two covariant), according as we use the f^*, the $f_!$ (if they exist) or the f_* (if they exist). We will denote by

$$f^{\mathbb{D}}_! \quad , \qquad f^*_{\mathbb{D}} \quad , \qquad f^{\mathbb{D}}_*$$

indifferently the functors between $\mathbb{D}(I)$ and $\mathbb{D}(I')$ or (more finely) between $\mathbb{D}_I$ and $\mathbb{D}_{I'}$, from which the preceding ones are deduced by passing to the corresponding fundamental categories. Likewise for $f^!$ or for $f^?$ subject to existence. For example, if we assume that the $f^!$ exist for $\mathbb{D}$, for every f a closed immersion, then for such an f we find a morphism of derivators

$$\mathbb{D}_I \xleftarrow{f^!_{\mathbb{D}}} \mathbb{D}_{I'} \quad .$$

We must see whether all these morphisms of derivators are exact, *i.e.* whether for

$$g : J \longrightarrow J'$$

in Dia, they commute with $g_!$, g_*.

To make this question explicit, we are led to look at the diagram of categories p. 16 [page 22 blank] [page 23]

$$\begin{array}{ccc} I \times J & \xrightarrow{G=\mathsf{id}_I \times g} & I \times J' \\ \downarrow{\scriptstyle F=f\times \mathsf{id}_J} & & \downarrow{\scriptstyle F'=f\times \mathsf{id}_{J'}} \\ I' \times J & \xrightarrow{G'=\mathsf{id}_{I'} \times g} & I' \times J' \end{array} \quad ,$$

which is moreover a cartesian square in $\mathcal{C}at$. The commutativity of $f^*_{\mathbb{D}}$ with the g_* is then made explicit by the commutativity in

$$\begin{array}{ccc} \mathbb{D}(I \times J) & \xrightarrow{G_*} & \mathbb{D}(I \times J') \\ \uparrow{\scriptstyle F^*} & \nwarrow^{\text{iso?}} & \uparrow{\scriptstyle F'^*} \\ \mathbb{D}(I' \times J) & \xrightarrow{G'_*} & \mathbb{D}(I' \times J') \end{array} \quad .$$

This commutativity will be ensured by later axioms on derivators. (cf. further below on page 59). Thus, $f^*_{\mathbb{D}}$ is right exact , and dually, it must also be left exact . So $f^*_D : \mathbb{D}_{I'} \to \mathbb{D}_I$ is *exact.* Consider $f^{\mathbb{D}}_*$, this time it is the commutativity of

$$\begin{array}{ccc} \mathbb{D}(I \times J) & \xrightarrow{G_*} & \mathbb{D}(I \times J') \\ \downarrow{\scriptstyle F_*} & & \downarrow{\scriptstyle F'_*} \\ \mathbb{D}(I' \times J) & \xrightarrow{G'_*} & \mathbb{D}(I' \times J') \end{array} \quad , \qquad \text{[page 24]}$$

it is trivial (up to canonical isomorphism). Thus, we find that $f_*^{\mathbb{D}} : \mathbb{D}_I \to \mathbb{D}_{I'}$ is *left exact*, and dually $f_!^{\mathbb{D}}$ is *right exact.* But is $f_*^{\mathbb{D}}$ also *right* exact? This therefore means commutativity of

$$\begin{array}{ccc} \mathbb{D}(I \times J) & \xrightarrow{G_!} & \mathbb{D}(I \times J') \\ \downarrow{\scriptstyle F_*} & \nearrow & \downarrow{\scriptstyle F'_*} \\ \mathbb{D}(I' \times J) & \xrightarrow{G'_!} & \mathbb{D}(I' \times J') \end{array} .$$

We do manage to define p. 17

$$G'_! F_* \xrightarrow{?} F'_* G_!$$

or what comes to the same thing by adjunction

$$F_* \xrightarrow{?} G'^* F'_* G_!$$

using the base change morphism

$$G'^* F'_* \xrightarrow{\alpha}_{\sim} F_* G^*$$

which is *isomorphism* (we have seen this), whence

$$\begin{array}{ccc} & & G'^* F'_* G_! \\ & \overset{?}{\nearrow} & \downarrow{\scriptstyle \wr\, \alpha * G_!} \\ F_* & & \\ & \underset{F_* * \beta}{\searrow} & \\ & & F_* G^* G_! \end{array}$$

where $\beta : \mathsf{id} \to G^* G_!$ is the adjunction morphism, whence the desired morphism. (We can also look for $F'^*(G'_! F_*) \to G_!$, and use the commutation $G_! F^* \xrightarrow{\sim} F'^* G'_!$, *i.e.* the *right* exactness of $f_{\mathbb{D}}^*$.)

But is this morphism an isomorphism? I think that in the additive case it may be reasonable, but in the nonadditive case there is no reason for it to be true. We see in any case that if for every $f : I \to I'$ in Dia, $f_*^{\mathbb{D}} : \mathbb{D}_I \to \mathbb{D}_{I'}$ is (not only left exact but also) right exact, then for every $g : J \to J'$ in Dia, $g_!^{\mathbb{D}}$ is (not only right exact but even) left exact. [page 25]

Let us take the derivator of presheaves of sets on variable I's, its fundamental category is $\mathcal{E}ns$. It is a nonpointed derivator. Consider $I \to I'$ a nonsurjective closed immersion, $f_* : \mathbb{D}(I) \to \mathbb{D}(I')$. If $f_*^{\mathbb{D}}$ were right exact f_* would have to take initial object of $\mathbb{D}(I)$ to initial object of $\mathbb{D}(I')$, unless I am mistaken (*cf.* below). But this is not so, since for every $F \in \mathsf{Im}(f_*^{\mathbb{D}} : \mathbb{D}(I) \to \mathbb{D}(I'))$ the restriction of F to $I' \setminus I$ is the final sheaf $e_U = (i \mapsto e)$, and not the initial sheaf $\emptyset_U = (i \mapsto \emptyset)$.

We are going to prove this:

Proposition. *If $\mathbb{D}$ is a left (right) derivator, then finite products (finite sums) exist in the $\mathbb{D}(I)$. Every morphism $\varphi : \mathbb{D} \to \mathbb{D}'$ of derivators which is left (right) exact commutes with the said products (sums), each I giving $\varphi(I) : \mathbb{D}(I) \to \mathbb{D}'(I)$. In particular, it takes final (initial) object to likewise.*

Saying that finite products indexed by the finite set I exist in the category $\mathcal{A} = \mathbb{D}(e)$ (say) is equivalent to saying that the diagonal functor p. 18 [page 26]

$$\mathcal{A} \xrightarrow{\Delta} \mathcal{A}^I$$

admits a right adjoint. Now if

$$f : I \to e$$

is the projection of I (regarded as a discrete ordered set) onto the one-point ordered set e, the functor Δ is identified with $f^* : \mathbb{D}(e) \to \mathbb{D}(I)$ (thanks to axiom $\mathbf{Der}_2$ on the behavior of the derivator for a discrete sum of index categories). So the existence of the product amounts to that of the right adjoint f_* of f^*, OK for left derivator. And a left exact homomorphism of left derivators, since it commutes with the f_*, commutes with finite products. Likewise for finite sums, and right derivators.

The preceding example uses the fact that the derivator under consideration is not pointed. But even for a pointed derivator $\mathbb{D}$, a morphism $f_*^{\mathbb{D}} : \mathbb{D}_I \to \mathbb{D}_{I'}$ is not necessarily right exact. If we take the derivator associated with an arbitrary category $\mathcal{M}$ (with $\Sigma = \emptyset$),

$$\mathbb{D}_{\mathcal{M}}(I) = \underline{\mathsf{Hom}}(I^\circ, \mathcal{M}) \overset{\text{def}}{=} \mathcal{M}(I)$$

then the induced derivator

$$(\mathbb{D}_{\mathcal{M}})_I = \{J \mapsto \mathbb{D}_{\mathcal{M}}(I \times J) = \underline{\mathsf{Hom}}((I \times J)^\circ, \mathcal{M}) \simeq \underline{\mathsf{Hom}}(J^\circ, \underline{\mathsf{Hom}}(I^\circ, \mathcal{M}))\}$$

is identified with [page 27]

$$\mathbb{D}_{\underline{\mathsf{Hom}}(I^\circ, \mathcal{M})} = \mathbb{D}_{\mathcal{M}(I)}$$

i.e.

$$(\mathbb{D}_{\mathcal{M}})_I = \mathbb{D}_{\mathcal{M}(I)}$$

N.B. Here $\mathcal{M}$ and $\mathbb{D}_{\mathcal{M}}$ determine each other and the functor

$$\mathcal{M} \longmapsto \mathbb{D}_{\mathcal{M}}$$

$$\mathcal{C}at \longrightarrow \mathsf{DER}$$

is 2-faithful, *i.e.* if $\mathcal{M}$, $\mathcal{M}'$ are two categories

$$\underline{\mathsf{Hom}}(\mathcal{M}, \mathcal{M}') \xrightarrow{\simeq} \mathsf{Hom}(\mathbb{D}_{\mathcal{M}}, \mathbb{D}_{\mathcal{M}'})$$

is an equivalence of categories. On the other hand, we check at once that $u : \mathcal{M} \to \mathcal{M}'$ is left exact (or right exact) if and only if $\mathbb{D}_u : \mathbb{D}_{\mathcal{M}} \to \mathbb{D}_{\mathcal{M}'}$ is so.

Thus, to see whether a morphism of derivators of the form

$$f_*^{\mathbb{D}} : \quad \begin{array}{ccc} (\mathbb{D}_{\mathcal{M}})_I & \longrightarrow & (\mathbb{D}_{\mathcal{M}})_{I'} \\ \| & & \| \\ \mathbb{D}_{\mathcal{M}(I)} & & \mathbb{D}_{\mathcal{M}(I')} \end{array}$$

is *right* exact, it comes to the same thing to say

that the functor for the associated fundamental categories

p. 19
[page 28]

$$\mathcal{M}(I) \xrightarrow{f_*^{\mathcal{M}}} \mathcal{M}(I')$$

is right exact. Now here it is a direct image functor of ordinary presheaves, and this functor has no desire at all to be *right* exact. (Even if $\mathcal{M}$ is *abelian*, for that matter!) Let us take for example for $f : I \hookrightarrow I'$ an open immersion with $I' = \mathbf{\Delta}_1 \times \mathbf{\Delta}_1$, $I = I' - \{c\}$, c a final object of I'

$$I' = \begin{array}{ccc} c & \longleftarrow & b \\ \uparrow & & \uparrow \\ a & \longleftarrow & 0 \end{array} \qquad I = \begin{array}{ccc} & & b \\ & & \uparrow \\ a & \longleftarrow & 0 \end{array}$$

The functor f_* is the one which associates to a diagram

$$F = \left\{ \begin{array}{ccc} & & F_b \\ & & \downarrow \\ F_a & \longrightarrow & F_0 \end{array} \right\}$$

the completed cartesian square

$$\overline{F} = \left\{ \begin{array}{ccc} F_a \times_{F_0} F_b & \longrightarrow & F_b \\ \downarrow & & \downarrow \beta \\ F_a & \xrightarrow{\alpha} & F_0 \end{array} \right\}$$

Now if $F \mapsto \overline{F}$ were right exact, $F \mapsto F_a \times_{F_0} F_b$ would be so too. Now $F_a \times_{F_0} F_b = K(F) = \mathsf{Ker}(F_a \times F_b \xrightarrow{\alpha pr_1 - \beta pr_2} F_0)$, as an additive functor in F, is not right exact, for example it does not necessarily transform epimorphism into epimorphism. Let us take for example

$$F' = \left\{ \begin{array}{ccc} & & F_b \\ & & \downarrow \\ F_a & \longrightarrow & 0 \end{array} \right\}$$

and the obvious homomorphism $F \to F'$ which is the identity on F_a and on F_b. It is an epimorphism and the corresponding homomorphism $\overline{F} \to \overline{F'}$ is the inclusion [page 29]

$$F_a \times_{F_0} F_b \longrightarrow F_a \times F_b$$

which is an epimorphism only if $\alpha = \beta = 0$.

Perplexity : One should be able to define a notion of an (additive) "*triangulated*" derivator (which would correspond to the intuition of triangulated categories; it is a derivator satisfying fairly strong axioms), so that for morphisms between these, left exactness and right exactness are *equivalent.* p. 20

"Base change theorem" for derivators. Let $\mathbb{D}$ be a left derivator (existence of f_*) and let us consider

$$f : I \longrightarrow J \qquad \text{in } \mathsf{Dia} \quad ,$$

let $j_0 \in \mathsf{Ob}\ J$, we propose to "calculate" the functor

$$F \mapsto f_*(F)_{j_0} = \alpha_{j_0}^*(f_*(F)) \quad ,$$

where

$$\alpha_{j_0} : e \longrightarrow J$$

is the inclusion of the one-point category e in J, with value j_0.

Let us consider the diagram [page 30]

$$\begin{array}{ccc} I/j_0 & \xrightarrow{q} & I \\ {\scriptstyle f'}\downarrow & \overset{\gamma}{\swarrow} & \downarrow{\scriptstyle f} \\ e & \xrightarrow[p=\alpha_{j_0}]{} & J \end{array} \quad , \tag{1}$$

where I/j_0 is the category $I \times_J J/j_0$ of pairs

$$\{(i,\varphi) \mid i \in \mathsf{Ob}\ I,\ \varphi : f(i) \to j_0\}$$

and q is the functor $(i,\varphi) \mapsto i$. This diagram is not commutative, but there is a canonical functor

$$fq \xrightarrow{\gamma} pf' \quad ,$$

where fq is the functor $(i,\varphi) \mapsto f(i)$ and pf' the constant functor $(i,\varphi) \mapsto j_0$ and where $\gamma(i,\varphi) : f(i) \to j_0$ is defined by $\gamma(i,\varphi) = \varphi$.

More generally, let us consider a diagram in Dia

$$\begin{array}{ccc} I' & \xrightarrow{q} & I \\ {\scriptstyle f'}\downarrow & \swarrow^{\gamma} & \downarrow{\scriptstyle f} \\ J' & \xrightarrow[p]{} & J \end{array} \quad , \tag{2}$$

with $\gamma : fq \to pf'$. We then define a *base change morphism for direct images* relative to this diagram p. 21

$$p^* f_* \xrightarrow{c_\gamma} f'_* q^*$$

in two ways, which give the same result (general story of adjoint functors to be made explicit once and for all, in terms of a diagram of functors [page 31]

$$\begin{array}{ccc} X' & \xleftarrow{q^*} & X \\ {\scriptstyle f'^*}\uparrow & \nearrow & \uparrow{\scriptstyle f^*} \\ Y' & \xleftarrow[p^*]{} & Y \end{array}$$

assuming that the right adjoint functors f'_*, f_* of the two vertical functors exist). An interesting case is the one where

$$pf' = fq \quad , \qquad \gamma = \mathsf{id}_{fq}$$

and where we put $c_\gamma = c$.

Here

$$p^* f_* \xrightarrow{?} f'_* q^*$$

amounts to

$$\begin{array}{lll} f_* \xrightarrow{?} p_* f'_* q^* \simeq & & (pf')_* q^* \\ & & \uparrow{\scriptstyle \gamma_* * q^*} \\ & (fq)_* q^* \simeq (f_* q_*) q^* = f_*(q_* q^*) & . \end{array}$$

Now we have

$$\mathsf{id} \xrightarrow{\alpha} q_* q^* \qquad \text{(adjunction)} \quad ,$$

whence

$$f_* \xrightarrow{f_* * \alpha} f_*(q_* q^*)$$

and composing with $\gamma * q^*$, we find what we want. And instead of this we can look for

$$\begin{array}{ccc} f'^*(p^* f_*) & \xrightarrow{\quad ? \quad} & q^* \\ \| & & \\ (pf')^* f_* & & \\ \downarrow{\scriptstyle \gamma^* * f_*} & & \\ (fq)^* f_* \simeq q^*(f^* f_*) & \xrightarrow{q^* * \gamma'} & q^* \end{array} \quad ,$$

where $\gamma' : f^* f_* \to \mathsf{id}$ is the adjunction.

Der$_4$: The base change morphism for direct images f_* associated with a diagram (1) is always an isomorphism (left derivators). Dual condition for right derivators and base change for direct images $f_!$, for the diagram

p. 22 [page 32]

$$\begin{array}{ccc} j_0\backslash I & \longrightarrow & I \\ \downarrow & \overset{\gamma}{\nearrow} & \downarrow f \\ e & \xrightarrow[p=\alpha_{j_0}]{} & J \end{array} \qquad (1')$$

associated with $f : I \to J$ in Dia, and $j_0 \in \mathsf{Ob}\ J$.

Der$_5$[10] : Let $f : I \to J$ be a *strict hot-equivalence, i.e.* $\forall j_0 \in \mathsf{Ob}\ J$, I/j_0 is contractible. Then f^* is fully faithful, *i.e.* the adjunction morphism

$$\mathsf{id}_{\mathbb{D}(J)} \longrightarrow f_* f^*$$

is an isomorphism ($\mathbb{D}$ a left derivator). Dually, if f is a co-strict equivalence, *i.e.* $f^\circ : I^\circ \to J^\circ$ is a strict equivalence, *i.e.* $\forall j_0 \in \mathsf{Ob}\ J$, $j_0\backslash I$ is contractible, then (if $\mathbb{D}$ is a right derivator) the functor f^* is fully faithful, *i.e.* the adjunction morphism

$$f_! f^* \longrightarrow \mathsf{id}_{\mathbb{D}(J)}$$

is an isomorphism.

N.B. Thus the functors f_* (resp. $f_!$) are localization functors.

N.B. Pages 33 to 40 have been deleted, because botched.

I am going to take up again the terminology and the axioms relating to derivators.

A *prederivator* is a 2-functor [page 41]

$$\mathsf{Dia}^\circ \xrightarrow{\mathbb{D}} \mathcal{C}at \quad ,$$

where Dia is a full 2-subcategory of $\mathcal{C}at$.[11] We assume at least that Dia contains the *finite ordered sets,* that Dia is stable under finite $\varprojlim$, under finite sums. That with every category C and every strictly full open (resp. closed) subcategory it contain the complementary subcategory.[12]

We define the prederivator $\mathbb{D}_{\mathcal{M},\Sigma}$ associated with a system

p. 23 [page 42]

$$(\mathcal{M}, \Sigma) \qquad \mathcal{M} \text{ Category}, \quad \Sigma \subset \mathsf{Fl}(\mathcal{M}) \quad .$$

If $\Sigma = \emptyset$, we write $\mathbb{D}_{\mathcal{M}}$.

Prederivators form a 2- category PREDER, and

$$(\mathcal{M}, \Sigma) \longmapsto \mathbb{D}_{\mathcal{M},\Sigma}$$

is a 2-functor

$$\mathsf{PREMOD}\left(\begin{array}{c}\text{“Category of categories}\\ \text{of models” (weak)}\end{array}\right) \longrightarrow \mathsf{PREDER} \quad .$$

For $\mathbb{D}$ given, we define the induced derivator ($I \in \mathsf{Dia}$)

$$\mathbb{D}_I = \left(J \longmapsto \mathbb{D}(I \times J)\right) \quad ,$$

[10] Totally botched, becomes reasonable only for constant coefficients on F.

[11] Dia *domain* of the derivator. The objects are the *index categories* for $\mathbb{D}$. The $\mathbb{D}(I)$ are the categories of coefficients for $\mathbb{D}$, the objects of $\mathbb{D}(I)$ the “coefficients of type $\mathbb{D}$ on I”.

[12] Notation f^* : inverse image of coefficients.

depending on I contravariantly, whence a 2-functor

$$\mathsf{Dia}^\circ \longrightarrow \mathsf{PREDER}$$
$$I \longmapsto \mathbb{D}_I$$

or again a bi-2-functor

$$\mathsf{PREDER} \times \mathsf{Dia}^\circ \longrightarrow \mathsf{PREDER}$$
$$(\mathbb{D}, I) \longmapsto \mathbb{D}_I \quad .$$

If $\mathbb{D}$ is defined by $(\mathcal{M}, \Sigma)$, then $\mathbb{D}_I$ is defined by $(\mathcal{M}(I), \Sigma(I))$:

$$(\mathbb{D}_{\mathcal{M},\Sigma})_I \simeq \mathbb{D}_{\mathcal{M}(I),\Sigma(I)} \quad ,$$

where $\mathcal{M}(I) = \underline{\mathsf{Hom}}(I^\circ, \mathcal{M})$ (presheaves on I with values in $\mathcal{M}$), and $\Sigma(I) \subset \mathsf{Fl}\ \mathcal{M}(I)$ is deduced from Σ by looking fiber by fiber (now assuming Σ saturated). [page 43]

For any axioms we shall single out for certain prederivators, we shall have to check that they are stable under passage to the induced prederivators $\mathbb{D}_I$, and also how they behave under duality (passage from $\mathbb{D}$ to $\mathbb{D}^\circ$).

Definition of the dual prederivator $\mathbb{D}^\circ$

$$\mathbb{D}^\circ(I) \overset{\text{def}}{=} \mathbb{D}(I^\circ)^\circ \qquad\qquad \textit{etc.}$$

Der 1 (First localization axiom) : *a*) If I, $J \in \mathsf{Dia}$, then

$$\mathbb{D}(I \amalg J) \longrightarrow \mathbb{D}(I) \times \mathbb{D}(J)$$

is an equivalence of categories.

b) $\mathbb{D}(\emptyset)$ is the final category.

Stable under passage to the induced prederivators, and passage to the dual.

Corollary. *Let* $(I_\alpha)_{\alpha\in A}$ be *finite family of categories in* Dia, $I = \coprod_{\alpha\in A} I_\alpha$. *Then*

$$\mathbb{D}(I) \longrightarrow \prod_{\alpha\in A} \mathbb{D}(I_\alpha)$$

is an equivalence of categories.

Der 2 (Second localization axiom) : For every $I \in \mathsf{Dia}$, the canonical functor

p. 24

$$\mathbb{D}(I) \longrightarrow \underline{\mathsf{Hom}}(I^\circ, \mathcal{A}_\mathbb{D}) \quad ,$$

$\mathcal{A}_\mathbb{D} = \mathbb{D}(e)$ fundamental category for the derivator $\mathbb{D}$, is conservative.
Stable under induction, and under duality.

A prederivator is called a *left derivator* if it satisfies **Der 1** and **Der 2**, and in addition the following axioms (of which here is the first). [page 44]

Der 3 (g) (Existence of direct images) : For every $f : I \to J$ in Dia, the functor $f^* : \mathbb{D}(J) \to \mathbb{D}(I)$ admits a *right adjoint* f_* .
Stable under induction.

We call f_* the direct image functor (cohomological) for f. The object $f_*(F) \in \mathbb{D}(J)$ (for $F \in \mathbb{D}(I)$) is also called the object of "*relative cohomology*" of I over J (via f), with coefficients in F. We have canonical isomorphisms

$$(gf)_* \simeq g_* f_* \quad .$$

Corollary. *The categories* $\mathbb{D}(I)$ *have finite products, the functors* f^* *and* f_* *commute with them.*

Uses **Der 1** for $I = \emptyset$ and $I = e \amalg e$, and **Der 3** for $I \xrightarrow{f} e$.

We have a dual axiom :

Der 3 (d) : existence of left adjoints $f_!$ to the f^*, the functor $f_!$ is called "homological direct image" via f, and we have a notion of *relative homology* of I over J (for f), with coefficients in F.

In what follows, the indication (g) or (d) next to the designation **Der** n of an axiom, means that it applies to left derivators, respectively to right derivators. Likewise for statements, definitions *etc.*

[page 45]

Der 4$_0$ (g)[13] : Let $U \overset{j}{\hookrightarrow} X \overset{i}{\hookleftarrow} Y$ be in Dia, open and closed immersions complementary to one another. Then the adjunction morphism in $\mathbb{D}(Y)$

$$i^* i_*(F_Y) \longrightarrow F_Y$$

is an isomorphism, hence

$$i_* : \mathbb{D}(Y) \longrightarrow \mathbb{D}(X)$$

is fully faithful, and $i^* : \mathbb{D}(X) \to \mathbb{D}(Y)$ is a localization functor. The essential image of i_* consists of the objects F of $\mathbb{D}(X)$ such that $j^*(F)$ is the final object of $\mathbb{D}(U)$.

Der 4$_0$ (d) : With the preceding notation for i, j, the adjunction morphism in $\mathbb{D}(U)$ p. 25

$$F_U \longrightarrow j^* j_! F_U$$

is an isomorphism, - $j_!$ is fully faithful, j^* a localization functor. The essential image of $j_!$ is the full subcategory of $\mathbb{D}(X)$ consisting of the F such that $i^*(F)$ is an initial object of $\mathbb{D}(Y)$.

There are the two squares of page 8 and 9, but as for interpreting the cartesian or cocartesian condition it is premature here.

[page 46]

Base change axioms.

Der 4 (g) (Fibers of a direct image) : Let $f : X \to Y$ be in Dia, $y \in Y$, let us consider the known diagram

$$\begin{array}{ccc} X/y & \xrightarrow{q} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ e & \xrightarrow[p=\alpha_y]{} & Y \end{array}$$

and the corresponding base change morphism

$$f_*(F)_y \overset{\text{def}}{=} p^* f_*(F) \longrightarrow f'_*(q^*(F)) \left(\overset{\text{def}}{=} \Gamma_{\mathbb{D}}(X/y, q^*(F))\right) \quad .$$

This is always an isomorphism.

[page 47]

Der 5 (g) (Cofinality axiom for cohomology)[14] : Let $\alpha : X' \to X$ in Dia be "cohomologically cofinal" *i.e.* the $x \backslash X'$ are contractible ($x \in X$) *i.e.* $\alpha^\circ : X'^\circ \to X^\circ$ is a hot-equivalence "locally over X°". Then $\forall F \in \mathbb{D}(X)$, the homomorphism

$$\Gamma_{\mathbb{D}}(X, F) \longrightarrow \Gamma_{\mathbb{D}}(X', \alpha^* F)$$

is an isomorphism (in $\mathcal{A} = \mathbb{D}(e)$).

[13] Unnecessary as an axiom, will become a theorem (follows from the other axioms).

[14] It is a consequence of weaker statements.

N.B. If $f : X \to e$, we set $\Gamma_{\mathbb{D}}(X, F) = f_*(F)$ ("absolute" cohomology of X, with coefficients in F).
Same thing[15] if α is "homologically cofinal", i.e. the X'/x are contractible .
Remark. We should be able to strengthen this axiom considerably, by requiring that p. 26

$$\alpha_!(\varepsilon_{X'}) \xrightarrow{\sim} \varepsilon_X \qquad \text{in} \quad \mathsf{HOT}(X) \quad ,$$

where $\varepsilon_{X'}$, ε_X the final objects of $\mathsf{HOT}(X')$, $\mathsf{HOT}(X)$, and where $\alpha_!(\varepsilon_{X'}) \to \varepsilon_X$ is the adjunction homomorphism, taking into account that $\varepsilon_{X'} \simeq \alpha^*(\varepsilon_X)$. We say that X' is *homotopically trivial over* X. (And if the dual condition is satisfied in $\mathsf{HOT}(X^\circ)$, we say that X' is cohomotopically trivial over X.) This condition is probably not self-dual. It is satisfied both if α is cohomologically cofinal, and if it is homologically cofinal (costrict or strict hot-equivalence).
N.B. For $f : X \to e$, saying that f is cohomologically cofinal, homologically cofinal, or that X is contractible, is equivalent.
Corollary 1 (g) (Asphericity theorem for constant coefficients in cohomology). *Let $f : X \to e$ be in* Dia*, X contractible, $M \in \mathbb{D}(e)$. Then the adjunction homomorphism*

$$M \xrightarrow{\sim} f_* f^*(M) \quad (\overset{\text{def}}{=} \Gamma(X, f^*(M)))$$

is an isomorphism.
Corollary 2 (g)[16] *Let $f : X \to Y$ be strict hot-equivalence (i.e. $\forall y \in Y$, X/y contractible). Then for $F \in \mathsf{Ob}\,\mathbb{D}(Y)$*

$$F \overset{\text{adj}}{\xrightarrow{\sim}} f_* f^* F$$

isomorphism.

[page 48]

We apply **Der 2**, **Der 4** (g), to reduce to looking, for $y_0 \in Y$, at the homomorphism α of the following (commutative!) diagram :

$$\begin{array}{ccc} \Gamma(\{y_0\}, F_{y_0}) = F_{y_0} & \xrightarrow{\quad\alpha\quad} & \Gamma_{\mathbb{D}}(X/y_0, f^*(F)) \\ \varphi\wr \uparrow & \nearrow_{\alpha'} & \\ \Gamma(Y/y_0, F) & & \end{array} \quad ,$$

where α' is deduced from the functor induced by f

$$X/y_0 \longrightarrow Y/y_0$$

and where φ is deduced from the functor

$$\{y_0\} \longrightarrow Y/y_0 \qquad \text{(final object)}\,.$$

Thus φ is an isomorphism by **Der 5** (g)[17] (N.B. If $z \in Z$, whence $\{z\} \to Z$, this functor is cohomologically cofinal if and only if z is a final object, homologically final if and only if z is an initial object). Thus the conclusion of the corollary is equivalent to saying that p. 27

[15] False!
[16] False, except for constant coefficients.
[17] *cf.* corollary 3, page 50, which we should move up here.

α' is an isomorphism. But $X/y_0 \to Y/y_0$ is also homologically cofinal, for if $u : y_1 \to y_0$ is in Y/y_0, we have a canonical isomorphism

$$(X/y_0)/(y_1, u) \simeq X/y_1$$

which is contractible. So by the second part of **Der 5** (g),[18] we win.

[page 49]

Corollary 4 (g) (base change criterion for cohomological direct images). *Let*

$$\begin{array}{ccc} X' & \xrightarrow{q} & X \\ {\scriptstyle f'}\downarrow & \overset{\gamma}{\swarrow} & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[p]{} & Y \end{array}$$

be in Dia (*cf.* page 30), *whence the base change homomorphism*

$$p^* f_*(F) \longrightarrow f'_* q^*(F) \qquad (F \in \mathsf{Ob}\, \mathbb{D}(X)) \quad .$$

For this to be an isomorphism for all F, it is enough that for every $y' \in \mathsf{Ob}\, Y'$, the functor induced by (p. 9)

$$X'/y' \longrightarrow X/p(y')$$

be a strict hot-equivalence (or costrict);[19] *this also means (in the strict case) that for*

$$y'_0 \in \mathsf{Ob}\, Y' , \quad x_0 \in \mathsf{Ob}\, X , \quad u_0 : f(x_0) \longrightarrow p(y'_0)$$

the category $K(y'_0, x_0, u_0)$ of objects of the form

$$(x', v', u) \mid x' \in \mathsf{Ob}\, X', \quad v' : f'(x') \to y'_0, \quad u : q(x') \overset{\leftarrow}{\longrightarrow} x_0$$

making the square in Y commutative

$$\begin{array}{ccc} f(q(x')) & \xrightarrow{\gamma(x')} & p(f'(x')) \\ {\scriptstyle f(u)}\downarrow & \text{case of costrict} \leftarrow \text{hot-equivalence} & \downarrow{\scriptstyle p(v')} \\ f(x_0) & \xrightarrow[u_0]{} & p(y'_0) \end{array} \qquad (*)$$

is contractible.

N.B. In the case of the costrict hot-equivalence, it is the same description of p. 28

$$K = (x_0, u_0) \backslash (X'/y'_0) \quad ,$$

except that the arrow u is in the opposite direction $x_0 \to q(x')$ and likewise in the commutative diagram $(*)$.

We apply **Der 2**, **Der 4** (g) and corollary 1 .

[page 50]

Corollary 5 (g). *Let $f : X \to Y$ be fully faithful in* Dia, *and $F \in \mathbb{D}(X)$. Then the adjunction homomorphism*

$$f^* f_*(F) \longrightarrow F$$

[18] (which is false)

[19] False for "strict", works only for "costrict".

is an isomorphism.

Indeed, it is enough to check it fiber by fiber (**Der 2**). Let $x \in X$, $y = f(x) \in Y$, then

$$f^* f_*(F)_x \simeq f_*(F)_y \simeq \Gamma_{\mathbb{D}}(X/y, F)$$

now because of full faithfulness $X/y \simeq X/x$.

Corollary 3 (move up to page 48).[20] *Let $Z \in \mathsf{Dia}$, suppose that Z has a final object e_Z. Let $p : e \to Z$ be the corresponding functor. Then the canonical homomorphism (for F in $\mathbb{D}(Z)$)*

$$\begin{array}{ccc} Z & & \\ {\scriptstyle p}\uparrow\downarrow{\scriptstyle f} & & f_*(F) \longrightarrow p^*(F) \\ e & & \end{array}$$

is an isomorphism.[21]

This homomorphism is special case of this one, for

$$\begin{array}{ccccc} Z' & & \xrightarrow{\quad p \quad} & & Z \\ & {\scriptstyle f'}\searrow & & \swarrow{\scriptstyle f} & \\ & & S & & \end{array}$$

(taking $Z' = S = e$, $f' = \mathsf{id}_e$),

$$f_*(F) \longrightarrow f'_*(p^*(F)) \quad , \qquad (\, F \in \mathsf{Ob}\, \mathbb{D}(Z)\,) \quad ,$$

and it is a special case[22] of the morphisms of base change for

p. 29 [page 51]

$$\begin{array}{ccc} Z' & \xrightarrow{p} & Z \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S & \xrightarrow{\mathsf{id}} & S \end{array} \quad .$$

Remark : We recover the fact that $i^* i_*(F_Y) \xrightarrow{\sim} F_Y$ is isomorphism, for i a closed immersion. Moreover, with the notation of **Der 4** (g), $j^*(i_*(F_Y))$ is computed by the base change theorem (corollary 2 above), which applies in every commutative situation

$$\begin{array}{ccc} X' & \xrightarrow{q} & X \\ {\scriptstyle f'}\downarrow & \text{cart} & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{p} & Y \end{array}$$

when p is an open immersion (and $\gamma = \mathsf{id}$) (then $X'/y' \to X/p(y')$ is even an isomorphism!), and we find, returning to the notation of **Der 4** (g),

$$\begin{array}{ccc} \emptyset & \xrightarrow{q} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle i} \\ U & \xrightarrow[j]{} & X \end{array}$$

[20] Will follow from the stronger **Der 6** (g) below.

[21] Axiom of the index category with a final object (arbitrary coefficient).

[22] already considered in cor. 2, p. 47

i.e. $j^*(i_*(F_Y)) \simeq f'_*(\varphi)$, where $\varphi = q^*(F_Y)$ is the unique element of the category (equivalent to a one-point category, by **Der 1**) $\mathbb{D}(\emptyset)$. On the other hand, f'_* takes final object to final object, whence $j^*i_*(F_Y)$ is a final object.

Conversely, let F be in $\mathbb{D}(X)$, such that $j^*(F) = e_{\mathbb{D}(U)}$. Consider

$$F \xrightarrow{u} i_*i^*F \qquad \text{(adjunction)} ;$$

I say that this is an isomorphism. Indeed, by the axiom of localization **Der 2**, it suffices to prove that $i^*(u)$ and $j^*(u)$ are isomorphisms. For $i^*(u)$ this follows from the fact that i^* is a localization functor, and for j^* from the fact that $j^*(F)$ and $j^*(i_*i^*(F))$ are final objects in $\mathbb{D}(U)$. [page 52]

p. 30
[page 53]

Homotopically proper, homotopically smooth morphisms.

Let $f : X \to Y$ be morphism of categories, consider $y_0 \in \mathsf{Ob}\, Y$, and the canonical functor

$$X_{y_0} \xrightarrow{\varphi} X/y_0 \qquad\qquad \varphi(x) = (x, \mathsf{id}_{f(x)}) \tag{1}$$

(X_{y_0} fiber of X at y_0). Let $(x_0, u_0) \in \mathsf{Ob}\, X/y_0$ hence $x_0 \in \mathsf{Ob}\, X$, $u_0 : f(x_0) \to y_0$. Consider the category coinduced by φ over (x_0, u_0)

$$(x_0, u_0)\backslash(X_{y_0}) \quad ,$$

to test whether φ is a costrict hot-equivalence *i.e.* homologically cofinal. We are going to interpret it via the functor induced by f

$$x_0\backslash X \xrightarrow{\psi} f(x_0)\backslash Y$$

going from the category of specializations of x_0 in X to that of specializations of $f(x_0)$ in Y. We then find a canonical isomorphism between the category coinduced $(x_0, u_0)\backslash X_{y_0}$ by X_{y_0} over the object (x_0, u_0) of X/y_0 and the fiber category $(x_0\backslash X)_{u_0}$ of $x_0\backslash X$ over $f(x_0)\backslash Y$ at u_0. Thus

Proposition. *For the functors $\varphi : X_{y_0} \to X/y_0$ ($y_0 \in \mathsf{Ob}\, Y$) to be costrict hot-equivalences, it is necessary and sufficient that the fibers of the functors $x_0\backslash X \to f(x_0)\backslash Y$ induced by f be contractible.*

We then say that f is *homotopically proper*. The dual notion is expressed, either by the condition that the functors induced by f [page 54]

$$X_{y_0} \longrightarrow y_0\backslash X$$

be strict hot-equivalences *i.e.* that they be cohomologically cofinal, or that the fibers of the morphisms induced on the "localized categories" (categories of generizations)

$$X/x_0 \longrightarrow Y/f(x_0)$$

be contractible. (N.B. We use here that a category Z is contractible if and only if Z° is.) We then say that f is *homotopically smooth.*

Proposition. *The notion of proper (resp. smooth) functor is stable under base change, under composition. Isomorphisms, and more generally, closed immersions are proper. The image of a proper morphism is a closed subcategory.*[23]

N.B. The functor $X \to e$ is always proper and smooth, for the categories $x_0\backslash X$ and X/x_0 are contractible (having an initial object or a final object). Thus every projection functor $X \times Y \xrightarrow{pr_2} Y$ is both proper and smooth. p. 31 [page 55]

PROOF. Stability under base change :

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & \text{cart} & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y' \end{array} \quad ,$$

whence for $x' \in \mathsf{Ob}\, X'$

$$\begin{array}{ccc} X/x & \longleftarrow & X'/x' \\ \downarrow & & \downarrow \\ Y/y & \longleftarrow & Y'/y' \end{array} \quad ,$$

where x, y', y are the images of x' in X, Y', Y. We check that this diagram is *cartesian*, hence the fibers of $X'/x' \longrightarrow Y'/y'$ are isomorphic to fibers of $X/x \longrightarrow Y/y$, hence contractible if these are.

In a case $X \xrightarrow{f} Y$ we check that for a bilocalized functor of f

$$X/x \xrightarrow{\varphi = f/x} Y/y \quad ,$$

every "bilocalized"

$$(X/x)/\xi \longrightarrow (Y/y)/\eta \qquad\qquad (\xi : x' \to x\,, \quad \eta : y' \to y)$$

of φ is again identified with a bilocalized of f

$$X/x' \longrightarrow Y/y' \quad .$$

Thus if f is smooth, the bilocalized f/x ($x \in \mathsf{Ob}\, X$) are so too. Let us note on the other hand.

Proposition.[24] *Every proper (or dually smooth) morphism with contractible fibers is a hot-equivalence.*

This comes from the fact that for such a proper functor, if F is a presheaf, or complex of [page 56]
abelian presheaves on X, the formation of the $Rf_*(F)$ commutes with passage to fibers, and likewise for $R^1 f_*(F)$ if F is a presheaf of groups (not even abelian).

Corollary 1. *If $f : X \to Y$ is cohomologically proper with contractible fibers, then Y is contractible if and only if X is.*

Thanks to this, we find that the composite of two proper (smooth) functors is proper (smooth).

N.B. Contrary to what happens in topology for the notion of properness, if gf is proper p. 32
and g proper, f is not necessarily so, *i.e.* an S-morphism of S-categories X, Y proper over S is not necessarily proper. Thus a section of X/S, for X proper over S, is not necessarily proper, *i.e.* is not necessarily a closed immersion. This is already false if $S = e$ (in which case $f : X \to e$ is always proper).

Corollary 2. *A proper or smooth morphism, with contractible fibers is both a strict and a costrict hot-equivalence.*

Indeed, we apply corollary 1 to $X/y \to Y/y$ or $y\backslash X \to y\backslash Y$.

[23] Beware, an equivalence of categories is not proper! If $X' \hookrightarrow X$ an inclusion of a subcategory of X, it is proper only if it is a closed immersion.

[24] *cf.* improvement in cor. 2, next page.

Theorem (g) *Consider a base change situation in* Dia

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & \text{cart} & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y' \end{array} \quad , \qquad p^* f_*(F) \xrightarrow{c} f'_* q^*(F)$$

($F \in \mathbb{D}(X)$), $\mathbb{D}$ a left exact derivator. If f is (homotopically) proper or p homotopically smooth, then c is an isomorphism.

The question is whether for every $y' \in Y'$, putting $y = p(y')$ and considering the functor [page 57] induced by q

$$X/y \longleftarrow X'/y'$$

the corresponding morphism

$$\Gamma_{\mathbb{D}}(X/y, F) \longrightarrow \Gamma_{\mathbb{D}}(X'/y', q^* F)$$

is isomorphism. In the case f proper, we consider

$$\begin{array}{ccc} X_y & \xleftarrow{\sim} & X'_{y'} \\ \sim \text{ costrict}\downarrow & & \downarrow\sim \text{ costrict} \\ X/y & \longleftarrow & X'/y' \end{array}$$

whence the conclusion, since by **Der 5** (g), in

$$\begin{array}{ccc} \Gamma(X_y) & \longrightarrow & \Gamma(X'_{y'}) \\ \uparrow & & \uparrow \\ \Gamma(X/y) & \longrightarrow & \Gamma(X'/y') \end{array}$$

the vertical arrows and the upper horizontal one are isomorphisms, hence also the lower horizontal one.

In the case p smooth, let us consider on the Y, Y' side the base change square [p. 33]

$$\begin{array}{ccc} Y & \longleftarrow & Y' \\ \uparrow & & \uparrow \\ Y/y & \longleftarrow & Y'/y' \end{array}$$

whence the adjoining diagram [page 58]

$$\begin{array}{ccccc} X & & \longleftarrow & & X' \\ & \nwarrow & & \swarrow & \\ & & X/y & \xleftarrow{\kappa} & X'/y' \\ & & \downarrow & & \downarrow \\ Y & \longleftarrow & & & Y' \\ & \nwarrow & & \swarrow & \\ & & Y/y & \xleftarrow[\pi]{} & Y'/y' \end{array} \quad ,$$

where the lateral (vertical) squares are cartesian. We note that π is smooth and has contractible fibers (for p is smooth) hence κ also. Thus κ is a costrict hot-equivalence, hence $\Gamma(X/y) \xrightarrow{\sim} \Gamma(X'/y')$, we win again.
N.B. Unless I am mistaken, we have a converse of the theorem, wit : If $f : X \to Y$ is such that the base change theorem for $f_*(F)$ holds for *every* derivator (it probably suffices to take HOTAB and HOT) and for every $Y' \to Y$, then f is homotopically proper. On the other hand, if $g : Y' \to Y$ is such that the base change theorem for g holds for every $X \to Y$ and every $F \in \mathbb{D}(X)$ ($\mathbb{D}$ any left exact derivator), then g is homotopically smooth.
On the homological side the smooth functors f are exactly those for which $f_!$ commutes with every base change. The proper functors g are exactly those for which base change by g commutes with $f_!$, for every $f : X \to Y$ and every $\mathbb{D}$ a right exact derivator.
Thus, the proper functors $f : X \to Y$ are characterized in *four* remarkable ways [page 59]

a) f_* commutes with every base change ($\forall$ left exact derivator $\mathbb{D}$);

b) the base change f^* commutes with every $g_!$ ($g : Y' \to Y$, $\mathbb{D}$ a right exact derivator);

c) $\forall y \in \mathsf{Ob}\, Y$, $X_y \to X/y$ is a costrict hot-equivalence *i.e.* is a homologically cofinal functor;

d) $\forall x \in \mathsf{Ob}\, X$ the functor induced on the colocalized

$$x\backslash X \longrightarrow f(x)\backslash Y$$

has contractible fibers. p. 34

Another standard situation where base change works (but where the functors involved have no reason to be proper or smooth).

Let

$$f : I \longrightarrow I' \quad , \qquad g : J \longrightarrow J'$$

be in Dia, whence the diagram (compare page 23)

$$\begin{array}{ccc} X' = I \times J & \xrightarrow{G'} & I \times J' = X \\ {\scriptstyle F'}\downarrow & & \downarrow{\scriptstyle F} \\ Y' = I' \times J & \xrightarrow{G} & I' \times J' = Y \end{array} \quad .$$

This diagram is such that for every left derivator [page 60]

we have base change

$$G^* F_* \simeq F'_* G'^* \quad ,$$

the reason here being that the induced functors

$$X'/y' \longrightarrow X/y$$

are costrict hot-equivalences, *i.e.* that the categories K (page 49) are contractible. Indeed, if $x_0 = (i_0, j'_0) \in \mathsf{Ob}X$, $y'_0 = (i'_0, j_0) \in \mathsf{Ob}Y'$, $u_0 = (u'_0, v'_0) : F(x_0) \to G(y'_0)$, where $u'_0 : f(i_0) \to i'_0$, $v'_0 : j'_0 \to g(j_0)$ are given, one checks that the category K sought is simply isomorphic to $(i_0, u'_0)\backslash(I/i'_0) \times (j'_0\backslash J)/(j_0, v'_0)$ which is contractible, being a product of a category having an initial object and a category having a final object . Whew!

I notice a very big blunder. I thought I remembered that for $f : X \to Y$ which is a strict hot-equivalence, one had for every $F \in \mathbb{D}(Y)$, $F \xrightarrow{\sim} f_* f^* F$ (for example for

$\mathbb{D} = \mathsf{Der}(Ab)$, or $\mathbb{D} = \mathsf{Hot}$), *i.e.* f^* fully faithful, whence also that $\Gamma_{\mathbb{D}}(Y, F) \to \Gamma_{\mathbb{D}}(X, f^*F)$ is isomorphism. But this is true only if F is a *constant* coefficient on Y, *i.e.* of the form $q^*(M)$ ($q : Y \to e$, $M \in \mathbb{D}(e) = \mathcal{A}_{\mathbb{D}}$). This is false, for general F, already in the case of the immersion $f : \{\varphi\} \to Y$, where φ is an initial object. Then, f is always a strict hot-equivalence, but the homomorphism $\Gamma(Y, F) \to \Gamma(X, f^*F)$ is in general not an isomorphism.

p. 35
[page 61]

Notions of $\mathbb{D}$-equivalence and of $\mathbb{D}$-contractibility.

So we are going to take up again the questions of acyclicity and base change.
Definition 1. Let $\mathbb{D}$ be a prederivator, and let

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & e & \end{array}$$

be in Dia. We say that f is a $\mathbb{D}$-*equivalence* if the functor

$$F \longmapsto f^*F \quad , \qquad \mathbb{D}(Y) \longrightarrow \mathbb{D}(X)$$

induces a fully faithful functor on the subcategory of $\mathbb{D}(Y)$ formed by the *constant* $\mathbb{D}$-coefficients, *i.e.* if for all $M, N \in \mathsf{Ob}\mathcal{A} = \mathbb{D}(e)$ ("absolute coefficients"), we have

$$\mathsf{Hom}(q^*M, q^*N) \xrightarrow{\sim} \mathsf{Hom}(p^*M, p^*N) \quad . \tag{1}$$

Suppose that p_* and q_* exist. We shall put, for a sheaf F on X

$$H^*_{\mathbb{D}}(X, F) \overset{\text{def}}{=} p_*(F) \qquad (\text{and likewise } \ H^{\mathbb{D}}_*(X, F) = p_!(F) \ ).$$

N.B. And if $F = p^*M$, $M \in \mathcal{A}$, we write simply $H^*_{\mathbb{D}}(X, M)$, $H^{\mathbb{D}}_*(X, M)$.
The homomorphism (1) is interpreted by adjunction as

$$\begin{array}{ccc} \mathsf{Hom}(M, q_*q^*N) & \longrightarrow & \mathsf{Hom}(M, p_*p^*N) \\ \| & & \| \\ \mathsf{Hom}(M, H^*_{\mathbb{D}}(Y, q^*N)) & \longrightarrow & \mathsf{Hom}(M, H^*_{\mathbb{D}}(X, p^*N)) \end{array}$$

deduced from the homomorphism

$$q_*q^*N \longrightarrow p_*p^*N \quad , \qquad i.e. \quad H^*_{\mathbb{D}}(Y, N) \longrightarrow H^*_{\mathbb{D}}(X, N) \quad . \tag{2}$$

Thus to say that (1) is an isomorphism for all M, N means that (2) is an isomorphism for every N. Likewise, if $p_!$, $q_!$ exist this is equivalent to saying that

$$p_!p^*N \longrightarrow q_!q^*N \quad , \qquad i.e. \quad H^{\mathbb{D}}_*(X, N) \longrightarrow H^{\mathbb{D}}_*(Y, N) \tag{2'}$$

is an isomorphism in $\mathcal{A}$ for every $N \in \mathcal{A}$.

p. 36
[page 62]

In summary:
Proposition 1. *If p_* and q_* exist, i.e. if $H^*_{\mathbb{D}}(X, \bullet)$ and $H^*_{\mathbb{D}}(Y, \bullet)$ exist (it even suffices that this exist for constant coefficients), then f is a $\mathbb{D}$-equivalence if and only if f induces an isomorphism for the $\mathbb{D}$-cohomologies with constant coefficients of X and Y. If $p_!$ and*

$q_!$ exist, i.e. if $H^{\mathbb{D}}_*(X,\bullet)$ and $H^{\mathbb{D}}_*(Y,\bullet)$ exist, f is a $\mathbb{D}$-equivalence if and only if f induces an isomorphism for the $\mathbb{D}$-homologies with constant coefficients of X and Y.

Remark 1. For f to be $\mathbb{D}$-equivalence, it *suffices* that for every F *constant* on Y the adjunction morphism $F \to f_* f^* F$ be isomorphism.

Remark 2. If f, $f' : X \to Y$ are in the same connected component of $\underline{\mathsf{Hom}}(X,Y)$, then f is a $\mathbb{D}$-equivalence $\Leftrightarrow$ f' a $\mathbb{D}$-equivalence.

Proposition 2. *Isomorphisms are $\mathbb{D}$-equivalences. If $X \xrightarrow{f} Y \xrightarrow{g} Z$, if two among f, g, gf are $\mathbb{D}$-equivalences, the third too.*

Definition 2. We say that X in Dia is $\mathbb{D}$-*aspheric* (or $\mathbb{D}$-*contractible*, or $\mathbb{D}$-*trivial*) if $X \xrightarrow{p} e$ is a $\mathbb{D}$-equivalence, *i.e.* if p^* is fully faithful.

If p_* exists, this therefore means that for $M \in \mathsf{Ob}(\mathcal{A})$

$$M \overset{\text{adj}}{\xrightarrow{\sim}} p_* p^* M \quad , \qquad i.e. \quad M \xrightarrow{\sim} H^*_{\mathbb{D}}(X, M) \quad ,$$

and if $p_!$ exists, this means that for every M

$$p_! p^* M \overset{\text{adj}}{\xrightarrow{\sim}} M \quad , \qquad i.e. \quad H^{\mathbb{D}}_*(X, M) \xrightarrow{\sim} M \quad .$$

Corollary. *If Y* (resp. X) *is contractible, then $f : X \to Y$ is a $\mathbb{D}$-equivalence if and only if X* (resp. Y) *is contractible.*

Example. Let us take $\mathbb{D} = \mathsf{HOT}$. (We can take Dia = category of small categories). Then, for $M \in \mathcal{A} = \mathsf{Hot}$ and X a small category, we have

$$H^*_{\mathsf{HOT}}(X, M) \simeq \mathsf{Hom}_{\mathcal{A}}(\mathsf{hot}(X), M) \quad ,$$

where

$$\mathsf{hot} : \mathcal{C}at \longrightarrow \mathcal{A} = \mathsf{Hot}$$

is the canonical functor which associates to every X its "homotopy type". We have

$$H^{\mathsf{HOT}}_*(X, M) = \mathsf{hot}(X) \times M \quad .$$

[page 63]

We deduce from this that $f : X \to Y$ is a HOT-equivalence (in the sense of definition 1) if and only if it is a hot-equivalence (or "weak equivalence") *i.e.* if and only if

$$\mathsf{hot}(f) : \mathsf{hot}(X) \xrightarrow{\sim} \mathsf{hot}(Y)$$

is an isomorphism. p. 37

Proposition 2. *Let $f : X \to Y$. Equivalent conditions (if $\mathbb{D}$* left *derivator*).

a) *f is a $\mathbb{D}$-equivalence, and remains so after every base change $Y' \to Y$ which is a local isomorphism (for example for $Y/F \to Y$, $F \in \mathsf{Ob}\widehat{Y}$).*

b) *For every $y \in \mathsf{Ob}Y$, the induced functor $X/y \to Y/y$ is a $\mathbb{D}$-equivalence.*

c) *For every $y \in \mathsf{Ob}Y$, X/y is $\mathbb{D}$-contractible.*[25]

N.B. We say that $g : Z \to Y$ is a *local isomorphism* if for every $z \in \mathsf{Ob}Z$, put $y = g(z)$, the induced morphism on the categories localized at z and y

$$Z/z \xrightarrow{\sim} Y/y$$

[25] Other equivalent conditions:

d) *For every constant sheaf M_Y on Y, we have $M_Y \xrightarrow{\sim} f_* f^* M_Y$.*

d′) *As in* d) *with locally constant sheaf.*

is an isomorphism. For example every open immersion, every induced functor $Y/y \to Y$ (localization functor) are local isomorphisms.

From this a) $\Rightarrow$ b) follows trivially.

b) $\Leftrightarrow$ c). Follows from the corollary on the preceding page, if one knows that Y/y is $\mathbb{D}$-contractible[26]. This will follow from **Der 5** below.

[page 64]

c) $\Rightarrow$ a) "Passage from local to global". Since hypothesis c) is stable under base change by open immersion, it suffices to prove that c) implies that f is a $\mathbb{D}$-equivalence, *i.e.* that for every $F = M_Y$ ($M \in \mathsf{Ob}\mathcal{A}$), constant coefficient on Y, the adjunction homomorphism

$$M_Y \longrightarrow f_* f^* M_Y = f_* M_X$$

is isomorphism. It suffices to prove that this is an isomorphism fiber by fiber (**Der 2**), hence that for every $y \in Y$,

$$M_y \longrightarrow f_*(M_X)_y$$

is an isomorphism. By **Der 4** (g), we have

$$f_*(M_X)_y \xrightarrow{\sim} \Gamma(X/y, M) \quad ,$$

and the hypothesis that X/y is $\mathbb{D}$-contractible gives what we want.

Proposition 2′. *Dually, if $\mathbb{D}$ is a right derivator , equivalent conditions on $f : X \to Y$:*

a) *f a $\mathbb{D}$-equivalence, remains so under every base change $g : Y' \to Y$ which is a "colocal isomorphism" (i.e. for every $y' \in \mathsf{Ob}Y'$, the induced functor on the "colocalized" categories $y'\backslash Y' \xrightarrow{\sim} y\backslash Y$ (where $y = g(y')$)).*

b) *For every $y \in Y$, the induced functor $y\backslash X \to y\backslash Y$ is a $\mathbb{D}$-equivalence.*

c) *For every $y \in Y$, $y\backslash X$ is $\mathbb{D}$-contractible.*

To prove it, one must use that the $y\backslash Y$ are $\mathbb{D}$-contractible (cf. cor. 1 of **Der 5** (d)).

p. 38

[page 65]

Definition 3. Let $\mathbb{D}$ be a prederivator (perhaps even a left or right derivator). We say that f is a *strict* $\mathbb{D}$-equivalence if f satisfies condition a) of proposition 2. We say that it is a *costrict* $\mathbb{D}$-equivalence if f satisfies that of proposition 2′; we also sometimes say that f is $\mathbb{D}$-cofinal. (The dual condition, *i.e.* strict $\mathbb{D}$-equivalence, would be called co-$\mathbb{D}$-cofinal!)[27]

Proposition 3. *Let $X \xrightarrow{f} Y \xrightarrow{g} Z$. If f, g are strict $\mathbb{D}$-equivalences, then gf too. If gf and f are strict $\mathbb{D}$-equivalences, then g too. If $\mathbb{D}$ left or right derivator, every equivalence is a strict $\mathbb{D}$-equivalence.* (For it has a right adjoint.)

We use the fact that the notion of local isomorphism is stable under base change.

[page 66]

Proposition 4. *Let $\mathbb{D}$ be a right derivator, $f : X \to Y$ a functor in* Dia*, such that for $y \in Y$, $y\backslash X$ is $\mathbb{D}$-contractible (hence f a costrict $\mathbb{D}$-equivalence, subject to $\mathbb{D}$ being a right derivator …). Then for every F in $\mathbb{D}(Y)$ (not necessarily with constant coefficients) we have*

$$H^*_{\mathbb{D}}(Y, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X, f^*(F))$$

is an isomorphism. Conversely, if the condition is true for every F, then the $y\backslash X$ are $\mathbb{D}$-contractible.

Proof. It amounts to the same thing to say that for every $M \in \mathsf{Ob}\mathcal{A} = \mathbb{D}(e)$, we have

$$\mathsf{Hom}(M, H^*_{\mathbb{D}}(Y, F)) \xrightarrow{\text{iso}} \mathsf{Hom}(M, H^*_{\mathbb{D}}(X, f^*(F))) \ .$$

[26] Cf. cor. 1 of **Der 5** (g)

[27] Better terminology: f is $\mathbb{D}$-*aspheric* (strict $\mathbb{D}$-equivalence), $\mathbb{D}$-*coaspheric* (co-strict $\mathbb{D}$-equivalence).

But by adjunction, this arrow is interpreted as

$$\mathsf{Hom}(M_Y, F) \longrightarrow \mathsf{Hom}(M_X, f^*F) \qquad (*)$$

On the other hand, if $f_!$ exists, we have

$$\mathsf{Hom}(M_X, f^*F) = \mathsf{Hom}(f_!(M_X), F) \ ,$$

and the homomorphism $(*)$ is deduced from the adjunction homomorphism

$$f_!(M_X) = f_!(f^*(M_Y)) \longrightarrow M_Y \ , \qquad (**)$$

by taking the Hom into F. Thus the condition is equivalent to saying that this homomorphism is isomorphism, whatever the constant coefficient M_Y on Y. To see whether this is an isomorphism, we look fiber by fiber (**Der 2**), now the fibers are interpreted as $H^{\mathbb{D}}_*(y\backslash X, M_X)$ (by **Der 4** (d)) and as M.
Thus the arrow under consideration $(**)$ is interpreted as

p. 39
[page 67]

$$H^{\mathbb{D}}_*(y\backslash X, M_X) \longrightarrow M$$

and this is the adjunction homomorphism $p_!(p^*(M)) \to M$.

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & e & \end{array}$$

Thus it is an isomorphism for every M if and only if $y\backslash X$ is $\mathbb{D}$-contractible.
We are going to state what we have proved.
Theorem 1. *Let*

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & e & \end{array}$$

be in Dia. *We assume that* $\mathbb{D}$ *is a prederivator satisfying* **Der 2**, *that* p_*, q_* *and* $f_!$ *exist, that* $f_!$ *satisfies* **Der 4** (d) *for the calculation of fibers. Equivalent conditions:*

a) f *is a costrict* $\mathbb{D}$*-equivalence, i.e.* (prop. $2'$) *for every* $y \in \mathsf{Ob}\, Y$, $y\backslash X$ *is* $\mathbb{D}$*-contractible;*

b) *for every* $F \in \mathsf{Ob}\, \mathbb{D}(Y)$, *the following homomorphism is isomorphism*

$$H^*_{\mathbb{D}}(Y, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X, f^*(F)) \ ;$$

c) *for every* $M \in \mathsf{Ob}(\mathcal{A} = \mathbb{D}(e))$, *the adjunction homomorphism is isomorphism*

$$f_!(M_X) = f_!(f^*(M_Y)) \xrightarrow{\sim} M_Y \ .$$

I am going to renumber the axioms on the prederivators.
Der 1, **Der 2** as before (p. 45).
Der 3[28] : Let X be in Dia, $p : X \to e$. Suppose X contractible.[29] Then

$$p^* : M \longmapsto M_X \quad , \qquad p^* : \mathcal{A} = \mathbb{D}(e) \longrightarrow \mathbb{D}(X)$$

is fully faithful.

[28] Axiom of aspheric indexing categories, *cf.* variants **Der 3** a) b) c) page 97, and especially pages 111-112.

p. 40

[29] N.B. It suffices to assume that X has a final object. But this is not a self-dual axiom.

[page 68]

Der 4 (g) : existence of the f_*.
Der 4 (d) : existence of the $f_!$.
Der 5 (g) Calculation of the fibers of $f_*(F)$.
Der 5 (d) Calculation of the fibers of $f_!(F)$.
The old axiom **Der 5** (g) follows from **Der 2**, **Der 3**, **Der 4** (g,d), **Der 5** (d), thanks to the preceding theorem.

An interesting aspect of the preceding theorem, if we view it as stating the equivalence of a) and b), is that this statement does not appeal to homological notions, but that its proof does appeal to them, in a relatively strong form (calculation of the fibers of $f_!(F)$). For the base change theorems for relative $\mathbb{D}$-cohomology, *i.e.* for the $f_*^{\mathbb{D}}$, we also constantly need the calculation of the fibers of $f_*(F)$. Thus it seems that we cannot dispense with any of the preceding axioms, including **Der 4** (d) and **Der 5** (d), even if we only want to concern ourselves with the formalism of the f^*, f_* (cohomological aspect). In what follows, we shall therefore assume these axioms satisfied.

Theorem 2. *Let $f : X \to Y$ be in* Dia. *Equivalent conditions:*

a) *For every $y_0 \in \mathsf{Ob}\, Y$, the canonical functor*

$$X_{y_0} \xrightarrow{\varphi} X/y_0 \quad (\simeq X \times_Y Y/y_0) \ ,$$

where X_{y_0} denotes the fiber at y_0, is a co-strict $\mathbb{D}$-equivalence.

b) *For every $x_0 \in \mathsf{Ob}\, X$, the induced functor on the categories colocalized at x_0 and at $y_0 = f(x_0)$*

$$x_0\backslash X \longrightarrow y_0\backslash Y$$

has $\mathbb{D}$-contractible fibers.

c) *The formation of the f_*F (F in $\mathbb{D}(X)$) commutes with every base change $Y' \to Y$,* [page 69]
and this remains true after every base change – i.e. if we have $Y'' \to Y' \to Y$, whence

$$\begin{array}{ccccc} X'' & \longrightarrow & X' & \longrightarrow & X \\ {\scriptstyle f''}\downarrow & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ Y'' & \xrightarrow[p']{} & Y' & \xrightarrow[p]{} & Y \end{array}$$

the "base change theorem" is true for f'_* and p'^*.

c′) *The formation of the $f_*(F)$ ($F \in \mathsf{Ob}\, \mathbb{D}(X)$) commutes with passage to the fibers and this remains true after every base change $p : Y' \to Y$.*

c″) *As in* c′) *with $Y' \to Y$ a localization morphism $Y/y \to Y$.*

d) *If $Y'' \xrightarrow{p} Y' \to Y$, with p coaspheric, then $q : X'' = X \times_Y Y'' \to X' = X \times_Y Y'$ is coaspheric.*

d′) *As in* d) *with $Y'' = e$* (where $e \longrightarrow Y'$ selects a final object of Y'). p. 41

d″) *As in* d) *with $Y'' = \{1\}$, $Y' = \mathbf{\Delta}_1$.*

.

Proof. a) ⇔ b) cf. proposition page 53. Let us show that these conditions imply c). Since they are stable under base change (*cf.* immediate proof page 55) therefore by **Der 2** we are reduced to proving that $H^*_{\mathbb{D}}(X/y, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X_y, F)$ ($F \in \mathsf{Ob}\, \mathbb{D}(X)$), which follows from theorem 1.[30] (Cf. proof of the theorem page 56, which works with more general hypotheses.)

We have c) ⇔ c′) ⇒ c″) (the equivalence c) ⇔ c′) by **Der 2**).

Let us prove c″) ⇒ a).

[page 70]

[30] Thus we use **Der 5** (d).

Consider the diagram with cartesian squares

$$\begin{array}{ccccc} X_{y_0} & \xrightarrow{q} & X/y_0 & \longrightarrow & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle f_0} & & \downarrow{\scriptstyle f} \\ \{y_0\} & \longrightarrow & Y/y_0 & \longrightarrow & Y \end{array} \quad .$$

Hypothesis c″) proves that for every F in $\mathbb{D}(X/y_0)$, we have $H^*_{\mathbb{D}}(X/y_0, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X_{y_0}, F)$. By theorem 1, this implies that q is a co-strict $\mathbb{D}$-equivalence.

Corollary 1. Dual statement of the theorem, for the functors $X_{y_0} \to y_0\backslash X$, $X/x_0 \to Y/y_0$, and base change for the $f_!(F)$ …

Definition 4. Under the equivalent conditions of theorem 2, we say that f is $\mathbb{D}$-*proper.* The dual condition is called f $\mathbb{D}$-*smooth.*

Corollary 2. *Let $f : X \to Y$ be in* Dia. *If f is $\mathbb{D}$-proper or $\mathbb{D}$-smooth, and has $\mathbb{D}$-contractible fibers, then f is a strict and costrict $\mathbb{D}$-equivalence. Better, f is a* universal *$\mathbb{D}$-equivalence i.e. is a $\mathbb{D}$-equivalence and remains so after every base change.*[31]

It suffices to prove that f is a $\mathbb{D}$-equivalence. Suppose for example f $\mathbb{D}$-proper. It suffices to prove that if Y is $\mathbb{D}$-contractible, then X is so (if need be, applying it to the Y/y_0 after base change, to show that the X/y_0 are contractible). For this, it suffices to prove that f is a $\mathbb{D}$-equivalence, and we know that this is the case if for every F *constant* on Y, the homomorphism $F \xrightarrow{\text{adj}} f_*f^*F$ is isomorphism. In fact, this is true for *every* F. p. 42 [page 71]

Corollary 3. *If $f : X \to Y$ is $\mathbb{D}$-proper or $\mathbb{D}$-smooth, with $\mathbb{D}$-contractible fibers, then $f^* : \mathbb{D}(Y) \to \mathbb{D}(X)$ is fully faithful, i.e. the adjunction homomorphisms*

$$F \longrightarrow f_*f^*F \quad , \qquad f_!f^*F \longrightarrow F$$

in $\mathbb{D}(Y)$ are isomorphisms.

It suffices to see this for f proper, and this follows from the calculation of the fibers of f_*f^*F …

Proposition 5. *The notion of a $\mathbb{D}$-proper functor* (resp. $\mathbb{D}$-*smooth*) *is stable under base change,* (trivial), *and under composition* (use corollary 2). *Every closed immersion is $\mathbb{D}$-proper, every open immersion $\mathbb{D}$-smooth. More generally, every local isomorphism is $\mathbb{D}$-smooth, every local co-isomorphism is $\mathbb{D}$-proper.*

Proposition 6. *Let X, Y be over S, $f : X \to Y$ an S-morphism. Suppose X, Y proper over S, or X and Y smooth over S, and that for every $s \in \mathsf{Ob}\, S$, $f_s : X_s \to Y_s$ is a $\mathbb{D}$-equivalence, then f also.*[32]

Theorem 3. *Let* [page 72]

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{p} & Y' \end{array}$$

[31] Generalizes: **Corollary 2′.** *Let*

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

with X, Y, $\mathbb{D}$-proper over S. If f induces a $\mathbb{D}$-equivalence on the fibers it is a $\mathbb{D}$-equivalence. Likewise in the smooth case.

This is proposition 6 below.

[32] Proof. We shall even prove that on the subcategory of coefficients on Y which come from $\mathbb{D}(S)$, $f^* : \mathbb{D}(Y) \to \mathbb{D}(X)$ induces a fully faithful functor. If p and q are proper, I interpret it as $q_*q^*F \xrightarrow{\sim} p_*p^*F$ for every F, and this follows from the calculation of p_*, q_* by passage to the fibers (proper), and from the hypothesis on the f_s. OK.

be cartesian in Dia. *Then the base change morphism*

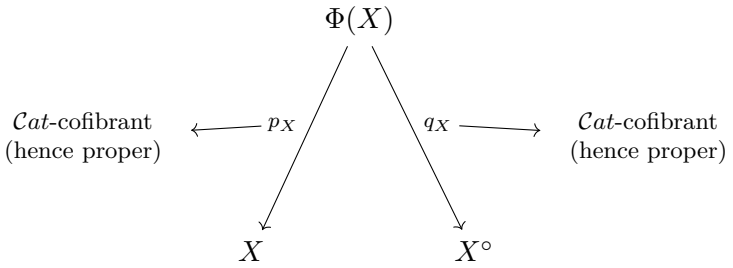
$$p^* f_* \longrightarrow f'_* q^*$$

is an isomorphism if f is $\mathbb{D}$-proper, or p $\mathbb{D}$-smooth.
The $\mathbb{D}$-proper case already proved. The $\mathbb{D}$-smooth case: the proof p. 56 is OK.

p. 43

Comparison of X and X°.

Let X be category, we define a category $\Phi(X)$ and functors



which will be hot-equivalences, and $\mathbb{D}$-equivalences if $\mathbb{D}$ is a derivator whose domain contains X and $\Phi(X)$. We set

$$\mathsf{Ob}\,\Phi(X) = \mathsf{Fl}(X) \quad ,$$

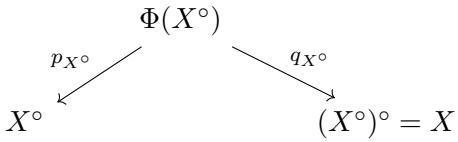
so the same objects as $\mathcal{F}\ell(X) = \underline{\mathsf{Hom}}(\mathbf{\Delta}_1, X)$, but the morphisms are different:

[page 73]

$$\begin{array}{ccc} Y_1 & \xrightarrow{u} & Y_0 \\ {\scriptstyle v_1}\uparrow & & \downarrow{\scriptstyle v_0} \\ Y'_1 & \xrightarrow[u']{} & Y'_0 \end{array}$$

a morphism from u to u' is a pair $v_0 : Y_0 \to Y'_0$, $v_1 : Y'_1 \to Y_1$ such that $v_0 u v_1 = u'$. Thus source and target $s(u)$ and $b(u)$ are the one contravariant, the other covariant in u. We take for q the source functor, for p the target functor.

If we apply this to X°, we find



We have $\mathsf{Ob}\,\Phi(X^\circ) = \mathsf{Fl}(X^\circ) \simeq \mathsf{Fl}(X)$

$$\begin{array}{ccc} Y_1 & \xrightarrow{u} & Y_0 \\ {\scriptstyle v_0}\uparrow & & \downarrow{\scriptstyle v_1} \\ Y'_1 & \xrightarrow[u']{} & Y'_0 \end{array} \qquad \begin{array}{l} \text{in } X \\ \\ \text{in } X \end{array}$$

and the same homomorphisms in the sense of $\Phi(X)$ and in the sense of $\Phi(X^\circ)$. Thus

$$\begin{cases} \Phi(X^\circ) = \Phi(X) \\ p_{X^\circ} = q_X \\ q_{X^\circ} = p_X \quad , \end{cases}$$

p. 44

the passage from X to X° amounts to interchanging p_X and q_X. The fiber of $\Phi(X)$ at $x \in \mathsf{Ob}\, X$ is $\simeq (X/x)^\circ$, that at $y^\circ \in \mathsf{Ob}\, X^\circ$ is $\simeq y\backslash X$

$$\begin{cases} p_X^{-1}(\{x\}) \simeq (X/x)^\circ \\ q_X^{-1}(\{y^\circ\}) \simeq y\backslash X \end{cases}$$

N.B. : p_X is W-proper with W-aspheric fibers, hence in W^u , q_X is W-proper with W-aspheric fibers, hence in W^u.

[page 74]

One checks

$$\begin{cases} \Phi(X)/x \simeq \Phi(X/x) & x \in \mathsf{Ob}\, X \\ \Phi(X)/x^\circ \simeq \Phi(x\backslash X) & x^\circ \in \mathsf{Ob}\, X^\circ \end{cases}$$

I say that p_X, q_X are $\mathbb{D}$-*strict equivalences* and *costrict.* By the preceding formulas, this means that the $\Phi(X/x)$, $\Phi(x\backslash X)$ are $\mathbb{D}$-contractible. Thus that if Z has a final object, $\Phi(Z)$ is $\mathbb{D}$-contractible. But let us consider

$$\begin{array}{ccc} \Phi(Z) & & \\ & \searrow^{q_Z} & \\ & & Z^\circ \end{array} \quad ;$$

it suffices to prove that this is a strict $\mathbb{D}$-equivalence (since Z°, having an initial object, is $\mathbb{D}$-contractible), hence that the $\Phi(Z)/z_0^{\mathrm{op}}$ are $\mathbb{D}$-contractible. We have by the preceding formulas

$$\Phi(Z)/z_0^{\mathrm{op}} \simeq \Phi(z_0\backslash Z) = \Phi(T) \quad ,$$

where this time T has an initial object $\emptyset$ and a final object e. Let us then consider the object

$$\emptyset \longrightarrow e$$

of $\Phi(T)$, it is a final object, since for every $x \to y$ in $\Phi(T)$, there is a unique arrow

$$\begin{array}{ccc} x & \longrightarrow & y \\ \uparrow & & \downarrow \\ \emptyset & \longrightarrow & e \end{array}$$

To summarize

p. 45

[page 75]

Proposition 1 : *p_X and q_X are strict* hot*-equivalences,* i.e. Hot*-aspheric (hence also strict $\mathbb{D}$-equivalences if $\Phi(X)$ is in the domain of $\mathbb{D}$) with fibers $(X/x)^\circ$ and $y\backslash X$ respectively (hence with contractible fibers).*

Corollary 1. *For X to be $\mathbb{D}$-contractible, it is necessary and sufficient that X° be so.*

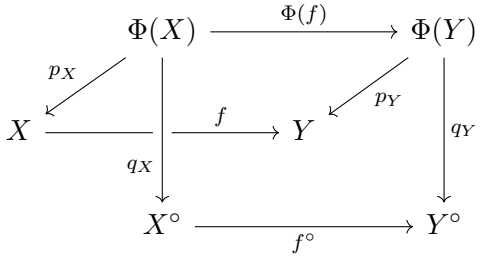
More generally :

Corollary 2. *Let $f : X \to Y$. For f to be a $\mathbb{D}$-equivalence, it is necessary and sufficient that f° be so.*

We shall use that the diagram

$$\begin{array}{ccccc} & & \Phi(X) & & \\ & \swarrow^{p_X} & & \searrow^{q_X} & \\ X & & & & X^\circ \end{array}$$

is functorial in X. We therefore have



the two squares commutative. Since p_X and p_Y are $\mathbb{D}$-equivalences, f is a $\mathbb{D}$-equivalence if and only if $\Phi(f)$ is, and likewise for f°. These arguments assume that if X is in the domain of $\mathbb{D}$, $\Phi(X)$ is too. This is the case if Dia consists of finite ordered categories.

p. 46

[page 76]

$\mathbb{D}$-fibrations

Proposition 1. *Let $f : X \to Y$ be in* Dia*, equivalent conditions :*

a) *For every $y \in Y$, the inclusion $X_y \to X/y$ is a $\mathbb{D}$-equivalence, and for every arrow $y_1 \to y_0$ of Y, $X/y_1 \to X/y_0$ is a $\mathbb{D}$-equivalence.*

b) *For every* constant *coefficient F on Y, f_*f^*F is locally constant,*[33] *and "is computed fiber by fiber"* i.e. *the canonical homomorphism*

$$f_*f^*(F)_y \xrightarrow{\sim} H^*_{\mathbb{D}}(X_y, F|X_y)$$

.

These conditions are equivalent, and imply the conditions of (b) for every F *locally* constant on Y.

The first condition in (a) means that for every $y \in Y$, and every $M \in \mathsf{Ob}\,\mathcal{A}$,

$$H^*_{\mathbb{D}}(X/y, M) \longrightarrow H^*_{\mathbb{D}}(X_y, M)$$

is an isomorphism. Now this arrow is interpreted as

$$f_*(M_X)_y \longrightarrow H^*_{\mathbb{D}}(X_y, M)$$

is an isomorphism, *i.e.* that $f_*(M_X)$ is computed fiber by fiber. Since $M_X \simeq f^*(M_Y)$, this is the second condition (b). The second condition in (a), for $y_1 \to y_0$ means that for M in $\mathcal{A}$,

$$H^*_{\mathbb{D}}(X/y_0, M) \longrightarrow H^*_{\mathbb{D}}(X/y_1, M)$$

is isomorphism, now this arrow is interpreted as

$$f_*(M_X)_{y_0} \longrightarrow f_*(M_X)_{y_1} \quad ,$$

that this is always an isomorphism, this is the first condition (b). That (b) also holds for F locally constant on Y is immediate, since this is a condition of a local nature with respect to Y.

The dual proposition calls on the functors

$$y_0\backslash X \longrightarrow y_1\backslash X \qquad \text{and} \qquad X_y \longrightarrow y\backslash X \quad ,$$

[page 77]

and the sheaves $f_!f^*M$, for M constant, or locally constant on Y.

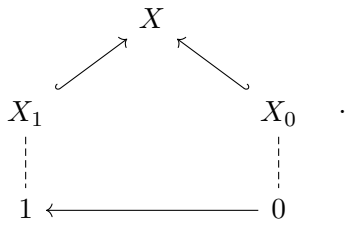
[33] By this we mean that for every $u : y \to y'$ in Y, $G_{y'} \to G_y$ is isomorphism. Without additional axioms on $\mathbb{D}$, it is not clear that this implies that the "restrictions" of G to the Y/y or $y\backslash Y$, are isomorphic to constant coefficients. See however axiom **Der 3** a′), page 97, which ensures that it comes to the same thing.

p. 47

Definition 1. If the equivalent conditions of proposition 1 are satisfied, we say that f is a $\mathbb{D}$-*cohomological fibration* (or $\mathbb{D}$-*left fibration*). If the dual conditions are satisfied, we say that it is a $\mathbb{D}$-*homological fibration* (or $\mathbb{D}$-*right fibration*).

N.B. Cohomological $\mathbb{D}$-fibrations are stable under any base change which is a local isomorphism. Homological $\mathbb{D}$-fibrations under any base change which is a local coisomorphism. But these are not notions stable under *every* base change[34]. But the notion of a cohomological $\mathbb{D}$-fibration is stable under $\mathbb{D}$-smooth base change, and that of a homological $\mathbb{D}$-fibration under $\mathbb{D}$-proper base change.

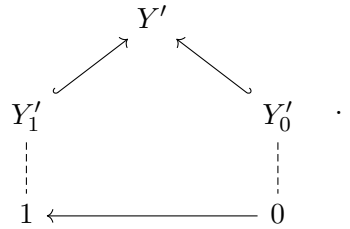
I am going to try to construct an example where base change gets botched, by setting $Y = \mathbf{\Delta}_1$. Giving X over $\mathbf{\Delta}_1$ amounts to giving an open subcategory X_0, and its closed complement X_1



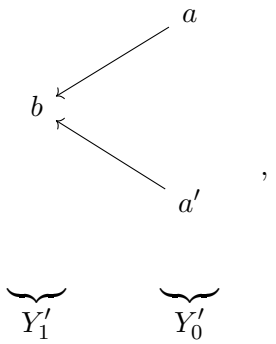
To say that $X \to Y = \mathbf{\Delta}_1$ is a cohomological $\mathbb{D}$-fibration simply means that the restrictions [page 78]

$$X_0 \xrightarrow{j} X \quad , \qquad X_1 \xrightarrow{i} X$$

are $\mathbb{D}$-equivalences. Let us take a base change $Y' \to Y = \mathbf{\Delta}_1$, *i.e.* a Y' with Y'_0, Y'_1



We see that the condition Y' smooth over $\mathbf{\Delta}_1$ means that for every $y'_1 \in \mathsf{Ob}\, Y'_1$, the category Y'_0/y'_1 of objects y_0 of Y'_0 over y'_1 (*i.e.* equipped with $y_0 \to y'_1$) is contractible. We see that if $Y'_1 \neq \emptyset$ then $Y'_0 \neq \emptyset$ – but an example with $Y'_0 = \emptyset$, or $Y'_1 = \emptyset$, is not interesting (for a projection $I \times J \to I$ is always a cohomological $\mathbb{D}$-fibration). Let us take for example for Y' p. 48



an amalgamated sum of two copies of $Y = \mathbf{\Delta}_1$, hence $X' = X \times_Y Y'$ can be viewed as an amalgamated sum $X \amalg_{X_1} X$. But that seems *not* to be a counterexample. We would need $X_1 \to X'$ *not* to be a $\mathbb{D}$-equivalence.

Proposition 2. *Let* $f : X \to Y$ *be in* Dia. *Equivalent conditions :* [page 79]

[34]? give example!

a) *For every* $Y'' \xrightarrow{p} Y' \to Y$*, whence*

$$\begin{array}{ccccc} X'' & \xrightarrow{q} & X' & \longrightarrow & X \\ \downarrow & \text{cart} & \downarrow & \text{cart} & \downarrow f \\ Y'' & \xrightarrow[p]{} & Y' & \longrightarrow & Y \end{array} \quad ,$$

if p *is a* $\mathbb{D}$*-equivalence, so is* q*.*[35]

b) f *is a "universal" cohomological* $\mathbb{D}$*-fibration,* i.e. *is a cohomological fibration, and remains so after any base change.*

b′) *Condition dual to* (b) *:* f *is a universal homological* $\mathbb{D}$*-fibration.*[36]

a) ⇒ *b*) : Let us assume *a*), let $Y' \to Y$ be a base change, whence

$$\begin{array}{ccc} X' & \longrightarrow & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \longrightarrow & Y \end{array} \quad .$$

To prove that f' is a cohomological $\mathbb{D}$-fibration, we look over Y' at the functors of the form $\{y'\} \to Y'/y'$ ($y' \in \mathsf{Ob}\, Y'$) and $Y'/y'_1 \to Y'/y'_0$ (if $u' : y'_1 \to y'_0$ an arrow of Y'), p. 49 and we see that the arrows in $\mathcal{C}at/Y'$ deduced from these by the base change $X' \xrightarrow{f'} Y'$ remain $\mathbb{D}$-equivalences

$$\begin{array}{ccccccc} X'_1 & \xrightarrow{q} & X'_0 & \longrightarrow & X' & \longrightarrow & X \\ {\scriptstyle f'_1}\downarrow & \text{cart} & \downarrow{\scriptstyle f'_0} & \text{cart} & \downarrow{\scriptstyle f'} & \text{cart} & \downarrow{\scriptstyle f} \\ Y'_1 & \xrightarrow[p]{} & Y'_0 & \longrightarrow & Y' & \longrightarrow & Y \end{array} \quad . \qquad \texttt{[page 80]}$$

Since p is a $\mathbb{D}$-equivalence, this indeed follows from a).

b) ⇒ *a*): As condition *b*) is obviously stable under base change, we are reduced to proving that if f satisfies *b*), and if we have

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & \text{cart} & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y' \end{array} \qquad (*)$$

if p is a $\mathbb{D}$-equivalence, then q too, *i.e.*

$$\begin{array}{ccc} H^*_{\mathbb{D}}(X, M) & \xrightarrow{\sim} & H^*_{\mathbb{D}}(X', M) \\ \wr\downarrow & & \wr\downarrow \\ H^*_{\mathbb{D}}(Y, f_*(M_X)) & & H^*_{\mathbb{D}}(Y', f'_*(M_{X'})) \end{array}$$

for every M in $\mathcal{A} = \mathbb{D}(e)$. But $F = f_*(M_X)$ is locally constant and is computed "fiber by fiber", likewise for $F' = f'_*(M_{X'})$.

[35] One must assume Y', Y'' Hot-aspheric, cf. page 96, and assume axiom Der 3 a') of page 97 to be satisfied.

[36] b) ⇒ a) botched, *cf.* marginal annotation,[37] *cf.* counterexample page 89. But it works for the derivators $\mathbb{D}_i$ ($i \in \mathbb{N}$) associated with the "basic localizers" W_i, *i.e.* with the theory of homotopy i-types. [page 81]

[37] The implication b) ⇒ a) not proved, only in a weaker form :

a′) if Y'', Y' are $\mathbb{D}$-aspheric and 1-connected, then $X'' \to X'$ is a $\mathbb{D}$-equivalence. (It even suffices to assume it for Y', Y'' having final objects (to imply b) or having initial objects (to imply b′).)

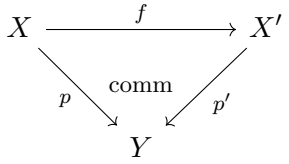
The equivalence $a) \Leftrightarrow b')$ follows from $a) \Leftrightarrow b)$ by duality (condition a) being self-dual), or by repeating the same argument for $a) \Rightarrow b')$, and for $b') \Rightarrow a)$ too (replacing cohomology by homology).

Definition 2. If the equivalent conditions of proposition 2 are satisfied, we say that f is a $\mathbb{D}$-*fibration.*

N.B. This is a notion *both* local and "colocal" on Y (whereas left fibration is local, right fibration colocal).

Proposition 3. a) *The notion of* $\mathbb{D}$*-fibration is stable under base change and under composition.*[38] p. 50

b) *Let*



with p, p' $\mathbb{D}$*-fibrations (it suffices that* p, p' *be cohomological* $\mathbb{D}$*-fibrations, or that they be homological* $\mathbb{D}$*-fibrations, or that* p, p' *be proper, or* p, p' *smooth). If* f *induces* $\mathbb{D}$*-equivalences in the fibers, then* f *is a* $\mathbb{D}$*-equivalence.*[39]

Proof. $a)$ is immediate under condition $a)$ (prop. 2) of the notion of $\mathbb{D}$-fibration. For $b)$, suppose that f induces $\mathbb{D}$-equivalences on the fibers. We want to prove that

$$H^*_{\mathbb{D}}(X', M) \longrightarrow H^*_{\mathbb{D}}(X, M)$$

is an isomorphism, for every M in $\mathcal{A}$, but this arrow is identified with

$$H^*_{\mathbb{D}}(Y, p'_*(M_{X'})) \longrightarrow H^*_{\mathbb{D}}(Y, p_*(M_X))$$

deduced from

$$p'_*(M_{X'}) \longrightarrow p_*(M_X) \qquad (\text{NB } \ M_X \simeq f^* M_{X'})$$

and it suffices to prove that this arrow is an isomorphism. For this, we look fiber by fiber, and we use the fact that "direct images are computed fiber by fiber". [page 82]

Question. Let $f : X \to Y$ be a $\mathbb{D}$-fibration, and let $F \in \mathbb{D}(Y)$. Then is $f_* f^* F$ computed by passing to the fibers? Same question for $f_! f^* F$. For the first property, concerning f_*, we show that it works if F is of the form $p_*(M_{Y'})$, where $p : Y' \to Y$ is $\mathbb{D}$-*proper,* and $M \in \mathsf{Ob}\,\mathcal{A}$. On the other hand, the full subcategory C of $\mathbb{D}(Y)$ consisting of the F for which it works is (unless I am mistaken) "stable under $\mathbb{D}$-hot-$\varprojlim$" in the following sense: if we have a category I with final object e forming a closed subcategory (*i.e.* $e \to x \Rightarrow x = e$), and if we have $\mathcal{F} \in \mathbb{D}(Y \times I)$, whence $I^\circ \to \mathbb{D}(Y)$, $i \mapsto \mathcal{F}_i$ (skeletal diagram), if the $\mathcal{F}_i$ for $i \neq e$ are in C, and if $\mathcal{F}$ is in the essential image of j_*, where $j : Y \times (I \setminus \{e\}) \hookrightarrow Y \times I$ is the inclusion, then $\mathcal{F}_e$ is also in C.

We may however consider that if we want $f : X \to Y$ to satisfy all the cohomological and homological properties that attach to the notion of a "fibration" ("locally trivial"), we should put among the conditions that f is *smooth*, and *proper*, so that base changes by f commute with the functors g_*, $g_!$.

Let us point out: p. 51

Proposition 4. *Let* $f : X \to Y$ *be* proper. *For* f *to be a* $\mathbb{D}$*-fibration, it is necessary and sufficient that for every* $y_1 \to y_0$ *in* Y*, the induced functor* $X/y_1 \to X/y_0$ *be a* $\mathbb{D}$*-equivalence* (i.e. *that* f *be a* cohomological $\mathbb{D}$*-fibration*). *Dually, if* f *is* $\mathbb{D}$*-smooth, for it to* [page 83]

[38] Stability under composition *is* not clear.

[39] Can be made considerably more precise in the case where p, p' are $\mathbb{D}$-fibrations, *cf.* prop. 6, page 84.

be a $\mathbb{D}$-fibration, it is necessary and sufficient that for every $y_1 \to y_0$ in Y, $y_0\backslash X \to y_1\backslash X$ be a $\mathbb{D}$-equivalence.

One could call *perfect* $\mathbb{D}$-fibrations the functors both proper and smooth that are cohomological $\mathbb{D}$-fibrations (or what amounts to the same thing homological $\mathbb{D}$-fibrations).

Corollary. *Suppose f a $\mathcal{C}at$-cofibration (hence proper). For f to be a $\mathbb{D}$-fibration, it is necessary and sufficient that the functors $X_{y_0} \to X_{y_1}$ (for $y_0 \to y_1$ in Y) be $\mathbb{D}$-equivalences. Dually, for $\mathcal{C}at$-fibration (hence smooth).*

Proposition 5. *Let $f : X \to Y$ be in* Dia. *Equivalent conditions:*

a) *f is a $\mathbb{D}$-equivalence, and remains so under every base change;*

b) *f is a $\mathbb{D}$-fibration with $\mathbb{D}$-contractible fibers.*

Proof. *a)* $\Rightarrow$ *b)*: That *a)* implies that f is a $\mathbb{D}$-fibration is immediate from its definition (condition *a)* of proposition 2), and that the fibers are contractible is tautological.

b) $\Rightarrow$ *a)*: It suffices to show that *b)* implies that f is a $\mathbb{D}$-equivalence. But this is a special case of proposition 3 *b)* (taking $X' = Y$ there).

When these conditions are satisfied, we expect that the functor [page 84]

$$f^* : \mathbb{D}(Y) \longrightarrow \mathbb{D}(X)$$

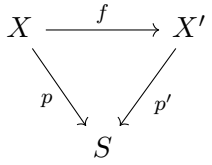
is fully faithful, *i.e.* that the adjunction morphisms

$$F \longrightarrow f_* f^* F \quad , \qquad f_! f^* F \longrightarrow F$$

in $\mathbb{D}(Y)$ are isomorphisms. (This is true if f is $\mathbb{D}$-proper or $\mathbb{D}$-smooth, cf. cor. 3, p.71.) This again amounts to saying that the $f_* f^* F$ are computed by passing to the fibers, or again likewise for the $f_! f^* F$. This brings us back to the question on page 82.

Definition 3. We then say that f is a *trivial* $\mathbb{D}$-fibration.

Proposition 6 (makes prop. 3 page 81 more precise). *Let us consider a commutative triangle in* Dia



.

a) *Suppose that p and p' are left $\mathbb{D}$-fibrations, or that p and p' are right $\mathbb{D}$-fibrations.* p. 52
Let S_0 be a subset of S that meets every connected component of S. For f to be a $\mathbb{D}$-equivalence, it suffices *that for every $s_0 \in S_0$, f induce a $\mathbb{D}$-equivalence $X_{s_0} \to X'_{s_0}$.*

b) *This condition is also necessary*[40] *if p and p' are $\mathbb{D}$-fibrations. So in this case, f is a $\mathbb{D}$-equivalence if and only if f induces $\mathbb{D}$-equivalences fiber by fiber.*

Proof. a) It suffices, by virtue of proposition 3, to show that the hypothesis implies [page 85]
that for *every* $s \in S$, f induces a $\mathbb{D}$-equivalence $f_s : X_s \to X'_s$. This follows at once from the lemma:

Lemma. *If p and p' are both either left $\mathbb{D}$-fibrations or right $\mathbb{D}$-fibrations, and if $s_1 \to s_0$ is an arrow in S, then f_{s_1} is a $\mathbb{D}$-equivalence if and only if f_{s_0} is. (Thus the set $S' \subset \mathsf{Ob}\, S$ of $s \in S$ such that f_s is a $\mathbb{D}$-equivalence is a union of connected components of S.)*

[40] Not proved![41] Even if $X' = S$, it is not clear, even if $\mathbb{D} = \mathsf{HOTAB}$, f proper and smooth. But OK if S is completely $\mathbb{D}$-aspheric.

[41] False, *cf.* counterexample, p. 87.

Suppose for example that p, p' are left $\mathbb{D}$-fibrations. Let us consider, for $s \in S$, the diagram

$$\begin{array}{ccc} X_s & \xrightarrow{f_s} & X'_s \\ \downarrow & & \downarrow \\ X/s & \xrightarrow[f/s]{} & X'/s \end{array} \quad .$$

The vertical arrows are $\mathbb{D}$-equivalences by the hypothesis on p, p', so f_s is a $\mathbb{D}$-equivalence if and only if f/s is. Let us now consider

$$\begin{array}{ccc} X/s_1 & \xrightarrow{f/s_1} & X'/s_1 \\ \downarrow & & \downarrow \\ X/s_0 & \xrightarrow[f/s_0]{} & X'/s_0 \end{array} \quad .$$

Here again, the hypothesis on p, p' implies that the vertical arrows are $\mathbb{D}$-equivalences, so f_{s_1} is a $\mathbb{D}$-equivalence if and only if f_{s_0} is.

If $X' = S$, the statement (not proved) says that if $f : X \to S$ is a $\mathbb{D}$-fibration and a $\mathbb{D}$-equivalence, then the fibers are $\mathbb{D}$-contractible. This is not certain even for a projection $X = T \times S \to S$. If $\eta = hot_0(S)$, $\xi = hot_0(T)$, the question reduces to this one: If for every M in $\mathcal{A}$, we have: the homomorphism [page 86]

$$\alpha_M : \xi \times_{\mathbb{D}} M \longrightarrow M$$

satisfies p. 53

$$\eta \times_{\mathbb{D}} \alpha_M : \eta \times_{\mathbb{D}} (\xi \times_{\mathbb{D}} M) \xrightarrow{\sim} \eta \times_{\mathbb{D}} M$$

is an isomorphism, then α_M is itself an isomorphism.

Here is a *counterexample*[42] which I first give in the context of topological spaces, instead of $\mathcal{C}at$. I am going to construct a locally trivial and nontrivial fibration [page 87]

$$\begin{array}{ccc} X & \supset & S^1 \\ {\scriptstyle f}\downarrow & & \\ S & = & S^1 \end{array} \quad ,$$

with fiber S^1, such that f is a $\mathbb{D}$-equivalence for every $\mathbb{D} = \mathsf{Der}(k\text{-}\mathsf{Mod})$, k being a ring in which 2 is invertible. This means that

$$H_*(X, k) \longrightarrow H_*(S, k)$$

is an isomorphism modulo 2-torsion (kernel and cokernel are annihilated by a power of 2 - in fact by 2 here), or again $H_*(X, k) \xrightarrow{\sim} H_*(S, k)$ if 2 invertible in k. But the fiber S^1 is not $\mathbb{D}$-contractible (unless $k = 0$!).[43]

The fibration is given up to isomorphism, by fixing a point $s \in S$, and giving oneself an automorphism u of the fiber X_s ("monodromy"). (The isomorphism class of the

[42] See an even simpler counterexample on pages 93, 94. (But it has the drawback of not having connected fibers.)

[43] Much simpler example still: we take $X = S^1$, and $X \to S$ the two-sheeted covering. If we take the n-sheeted covering, it is also OK if k a ring in which n is invertible.

bundle depends only on the isotopy class of u). I take for u the symmetry (reversing orientation) $z \mapsto z^{-1}$ of the standard circle $\{z \in \mathbb{C} \mid |z| = 1\}$. This fibration "is realized" by ordered sets $X \to S$, S^1 being represented (on the base, as on the fibers) by

$$S \;=\; \begin{array}{ccccc} & & a & & \\ & \nearrow & & \nwarrow & \\ 0 & & & & 0' \\ & \searrow & & \swarrow & \\ & & b & & \end{array} \qquad = \; S' \amalg_R S'' \quad ,$$

where R is the discrete category having two elements a, b. We define X as $X' \amalg_Y X''$, where $X' = X|S'$, $X'' = X|S''$, $Y = X|R$.[44]

We put

$$X' = S' \times \bar{S} \qquad\qquad X'' = S'' \times \bar{S}$$

p. 54
[page 88]

($\bar{S}$ an exact copy of S, cf. below) and, to glue X' and X'', we must give ourselves an isomorphism between $X'|R \simeq R \times \bar{S} = \bar{S} \amalg \bar{S}$ and $X''|R \simeq R \times \bar{S} = \bar{S} \amalg \bar{S}$ which is an R-functor, *i.e.* which comes from two isomorphisms

$$\bar{S} \xrightarrow{u_a} \bar{S}$$
$$\bar{S} \xrightarrow{u_b} \bar{S} \qquad .$$

We take

$u_a = \mathsf{id}_{\bar{S}}$

$u_b =$ reflection about the axis $\bar{0}, \bar{0}'$ *i.e.*

u_b leaves fixed $\bar{0}, \bar{0}'$ and interchanges $\bar{a}$ and $\bar{b}$.

(It is clearer if we disidentify the base and the fiber, taking fiber

$$\bar{S} \;=\; \begin{array}{ccccc} & & \bar{a} & & \\ & \nearrow & & \nwarrow & \\ \bar{0} & & & & \bar{0}' \\ & \searrow & & \swarrow & \\ & & \bar{b} & & \end{array} \qquad).$$

To see that X is a $\mathbb{D}$-fibration over S (whatever $\mathbb{D}$ is) let us prove that it is *proper*[45] over S - we are reduced to seeing that $X' \to S'$ and $X'' \to S''$ are proper, which is trivial. To see that it is a $\mathbb{D}$-fibration, we are likewise reduced to the case of $X' \to S'$ and $X'' \to S''$, where this is also trivial. We have used the:

Lemma. *Let $f : X \to S$, $(S_i)_{i \in I}$ a family of closed subcategories covering S. For every i, let $X_i = X \times_S S_i \xrightarrow{f_i} S_i$. For f to be proper, it is necessary and sufficient that the f_i be so. For f to be a proper $\mathbb{D}$-fibration it is necessary and sufficient that the f_i be so.*

Here is the little computation which shows that $f : X \to S$ is a $\mathbb{D}$-equivalence. We do the computation in cohomology.[46] We must prove that

[page 89]

$$H^*(S, k) \longrightarrow H^*(X, k) \qquad (*)$$

[44] N.B. S', S'', $R = S' \cap S''$ closed in S, X', X'', $Y = X' \cap X''$ closed in X.

[45] N.B. $f : X \to S$ is also *smooth*, for it is a trivial bundle (a product) over two opens $S - \{a\}$, $S - \{b\}$ which cover S. Better: f is both $\mathcal{C}at$-fibrant and $\mathcal{C}at$-cofibrant (for this condition uses only the properties of the $X \times_S \mathbf{\Delta}_2 \to \mathbf{\Delta}_2$, for $\mathbf{\Delta}_2 \to S$).

[46] But in the case of the 2-sheeted covering of S, there is not even any computation to do!

is an isomorphism (and likewise for any constant coefficient M on S). Now p. 55

$$H^*(X,k) \overset{\text{def}}{=} \mathbb{R}\Gamma_X(k_X) = \mathbb{R}\Gamma_S(\mathbb{R}f_*(k_X)) \quad ,$$

and $(*)$ is induced by

$$i : k_S \longrightarrow \mathbb{R}f_*k_X \quad .$$

Now we have

$$\mathbb{R}f_*k_X \simeq \underbrace{k_S}_{=\mathsf{Im}(i)} \quad \oplus \quad \underbrace{k_S[1]}_{k_S \text{ in degree } 1} \quad \underline{\text{tordu}}$$

namely "twisted" by the monodromy action on $S = S^1$, which is the symmetry $\lambda \mapsto -\lambda$. We have

$$\mathbb{R}\Gamma_S(k_S[1]^{\text{tordu}}) = H^1(\underbrace{\pi_1(S)}_{=\mathbb{Z}}, k_S^{\text{tordu}}) = k_{(\mathbb{Z})} = k/(u-\mathsf{id})k = k/2k = 0 \ ,$$

hence the result.

Let us note that if we look at

$$X' = X \times_S X \overset{f'}{\longrightarrow} S \quad ,$$

we have by relative Künneth

$$\mathbb{R}f'_*(k_{X'}) \simeq \mathbb{R}f_*(k_X) \otimes^{\mathbb{L}} \mathbb{R}f_*(k_X) \simeq k_S \oplus k_S[1]^t \oplus k_S[1]^t \oplus \underbrace{k_S[1]^t \otimes k_S[1]^t}_{\simeq k_S[2] \text{ untwisted}} \ ,$$

hence

$$H^*_{\mathbb{D}}(X',k) \simeq \mathbb{R}\Gamma_{X'}(k_{X'}) \simeq \mathbb{R}\Gamma_S(\mathbb{R}f'_*(k_{X'})) \simeq \mathbb{R}\Gamma_S(k_S) \oplus \mathbb{R}\Gamma_S(k_S)[2]$$

and

$$H^*_{\mathbb{D}}(S,k) \longrightarrow H^*_{\mathbb{D}}(X',k)$$

is *not* an isomorphism.

Thus $f : X \to S$ which is a $\mathbb{D}$ fibration (proper and smooth) does satisfy the equivalent conditions $b)$, $b')$ of proposition 2 (page 79), but not condition $a)$, since the base change [page 90]

$$\begin{array}{ccc} X' = X \times_S X & \xrightarrow{q=pr_1} & X \\ {\scriptstyle f'=pr_2}\downarrow & & \downarrow{\scriptstyle f} \\ S' = X & \xrightarrow[p=f]{} & S \end{array}$$

is a $\mathbb{D}$-equivalence $S' \to S$, whereas $X' \to X$ is not.

Here is another counterexample, on the same principle but a little simpler. Let G be a finite group (for example $G = \mathbb{Z}/2\mathbb{Z}$), let us make it act on an object Z of $\mathcal{C}at$, let $S = [G] = B_G$ be the category reduced to one object a, with $G = \mathsf{End}(a)$. Let E_G be the category (isomorphic to $(B_G)/a$)

$$\mathsf{Ob}\ E_G = G \quad , \qquad \mathsf{Hom}_{E_G}(g,g') = \{h \in G \mid g = g'h\} \quad ,$$

obvious composition - then E_G is a trivial groupoid (1-connected). The group G acts freely on it on the left by translations, and $B_G = E_G/G$.

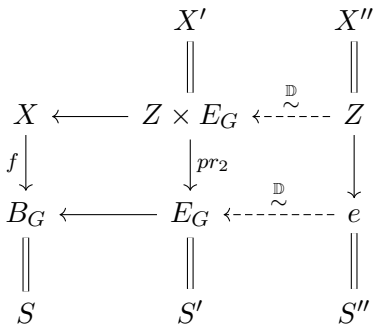
If G acts on Z on the left, we can then use it to twist Z, by

p. 56

$$X = E_G \times_G Z = (E_G \times Z)/\text{“diagonal” actions of G} \quad .$$

[page 91]

So we have a cartesian diagram



Since the base change $S' = E_G = (B_G)/a \to B_G = S$ transforms X over B_G into a product (which is a proper and smooth $\mathbb{D}$-bundle over S'), we see that X over B_G is a proper and smooth $\mathbb{D}$-bundle (for this is a local condition on B_G).

We now suppose that G is finite of order $n > 1$, and let k be a ring in which n is invertible, $\mathbb{D} = \mathsf{Der}(k - \mathsf{Mod})$. Then $E_G \to B_G$ is a $\mathbb{D}$-equivalence, or, what amounts to the same thing, $e \to B_G$ is.[47] Yet, after base change either by $S' = E_G$, or by $S'' = e$, $X' \to X$ (or what amounts to the same thing, $X'' \to X$) has hardly any reason to be one too. We have in cohomology

$$\mathbb{R}\Gamma_X(k_X) \simeq \mathbb{R}\Gamma_{BG}(f_*(k_X)) \simeq \mathbb{R}\Gamma_G(G, H^*(Z,k))$$

therefore (n being invertible in k): [page 92]

$$H^*(X,k) = \mathbb{R}\Gamma_X(k_X) \simeq H^*(Z,k)^G \longrightarrow H^*(Z,k) \quad ,$$

this is not in general an isomorphism. We can take for example $G = \mathbb{Z}/2\mathbb{Z}$, $Z \sim S^1$ as before, and G acting by reflection of S^1 as previously.

This example is particularly convincing, for it is an example *with S, S' $\mathbb{D}$-aspheric,* and $S' \to S$ a local isomorphism (in fact, a principal covering of S). It is also an example of a smooth and proper $\mathbb{D}$-bundle $X \to S$, over an *aspheric* base S, and such that the inclusion of a fiber in X

$$X_s \longrightarrow X$$

is *not* a $\mathbb{D}$-equivalence.

The only complaint we can make about this example, in comparison with the preceding one, is that the categories X, S are not ordered sets. And S cannot be a finite ordered set, for the classifying space of a finite group has cohomology in all dimensions (unless I am mistaken).

But we can take inspiration from this example to modify it so that it carries over to ordered sets. It suffices to realize, for given G: p. 57 [page 93]

a) A *free* action of G on an object E of $\mathcal{C}at$, such that $H^*(E,k)^G \simeq k$, *i.e.* $B = E/G$ is $\mathbb{D}$-aspheric.

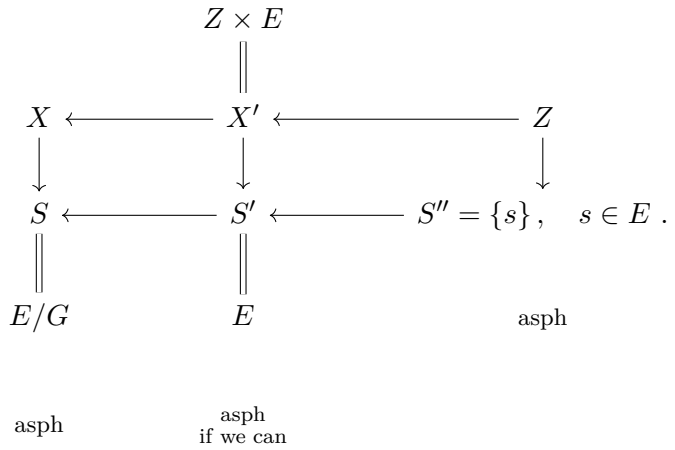
b) An action of G on an object Z, such that the inclusion

$$H^*(Z,k)^G \longrightarrow H^*(Z,k)$$

is not isomorphism.

[47] In fact, e, E_G, B_G are $\mathbb{D}$-aspheric.

c) Choose for E, Z ordered sets, and make sure that $B = E/G$ is also represented by an ordered category. Unless I am mistaken, the same will hold for $X = E \times Z/G$. We will have in any case



This gives another example with X a proper and smooth $\mathbb{D}$-bundle, over a $\mathbb{D}$-aspheric base S, such that the inclusion of a fiber X_s in X is not a $\mathbb{D}$-equivalence. We can even manage to have $H^*(Z,k)^G \simeq k$ which implies that X too is aspheric (but the fibers of [page 94] $X \to S$ are not $\mathbb{D}$-aspheric). This therefore gives another counterexample to b) of prop. 6 (p. 84).

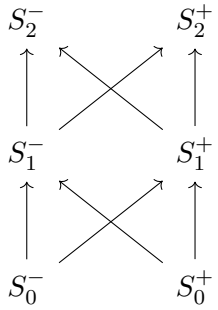
Here is the most economical way, I think, to realize the situation I have spoken of. Topologically, I take

$G = \{\pm 1\}$	
$E = S^2$	Euclidean sphere, on which $\{\pm 1\}$
	acts by antipodism $x \mapsto -x$
$B = S^2/\{\pm 1\}$	real projective plane
$Z = S^1$	where G acts as before by the reflection .

So this is an example very close to the one at the beginning (p. 87), where we could also consider the base ($B = S^1$) as the quotient of an E ($= S^1$) by $G = \{\pm 1\}$ acting freely (by the antipodism $z \mapsto -z$ in $\mathbb{C} \simeq \mathbb{R}^2$). But in this example, the base was not $\mathbb{D}$-aspheric. p. 58

The most economical way of realizing the sphere S^2 by an ordered set is probably by its "globular" decomposition into three pairs of cells of dimension 0, 1, 2, giving the ordered set

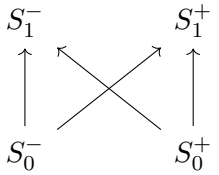
I =



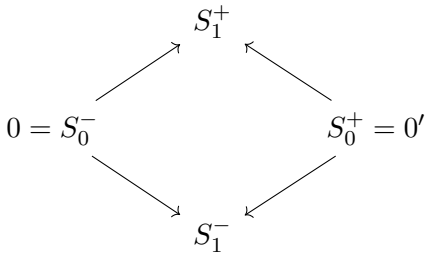
and the antipodism exchanges S_i^- and S_i^+, but the quotient is not ordered, it is the category (important for the theory of 2-categories): [page 95]

$$\begin{array}{c} S_2 \\ {\scriptstyle p_1^-}\uparrow\uparrow{\scriptstyle p_1^+} \\ S_1 \\ {\scriptstyle p_0^-}\uparrow\uparrow{\scriptstyle p_0^+} \\ S_0 \end{array} \qquad \text{with} \qquad \begin{cases} p_1^- p_0^- = p_1^+ p_0^+ \\ p_1^- p_0^+ = p_1^+ p_0^- \end{cases}$$

But let us take the barycentric subdivision of the preceding cellular decomposition of S whose vertices correspond to the elements of $\mathsf{Ob}\ I$, the simplices to the totally ordered subsets of $\mathsf{Ob}\ I$. We find the octahedron, having 6 vertices, 12 edges, 8 faces, and G acts freely on the set of simplices, hence the quotient: 3 vertices, 6 edges, 4 faces, a cellular decomposition of the projective plane. (Its decomposition by three lines.) We present S^1 by the ordered set (as at the beginning)



i.e.



and G acts on it by a reflection exchanging S_1^- and S_1^+. So the ordered set X will have $4 \times 13 = 52$ vertices and that is modest …

Finally, it seems that nothing remains of proposition 2 (p.79) introducing the notion of $\mathbb{D}$-fibration – not even the equivalence of the two conditions b) and b') (universal left $\mathbb{D}$-fibration, universal right $\mathbb{D}$-fibration). One thinks of their relation to a condition a) modified as follows (for $f : X \to Y$) : p. 59 [page 96]

a) For every $Y'' \to Y' \to Y$, with Y' and Y'' Hot-aspheric (and not only $\mathbb{D}$-aspheric) $X'' = X \times_Y Y'' \to X' = X \times_Y Y'$ is a $\mathbb{D}$-equivalence. (When one assumes that Y'' and Y' have final elements, this is the condition that f be a universal left $\mathbb{D}$-fibration; when one assumes that Y'' and Y' have initial elements, this is the condition universal right $\mathbb{D}$-fibration.)[48]

We obviously have $a) \Rightarrow b)$ and $a) \Rightarrow b')$ and we would like these to be equivalences. Here is an interesting axiom[49] on $\mathbb{D}$, which seems to hold in all known cases, which ensures the desired equivalence. At the same time I am going to reformulate **Der 3** in three parts (*cf.* p. 67).

Der 3 a) If $Z \in \mathsf{Dia}$ is Hot-aspheric, then $p : Z \to e$ is a $\mathbb{D}$-equivalence, *i.e.* for example $p^* : \mathcal{A} = \mathbb{D}(e) \to \mathbb{D}(Z)$ is fully faithful. [page 97]

b) Let $f : Z \to Z'$ be an arrow in Dia. If f is a Hot-equivalence, it is a $\mathbb{D}$-equivalence (*i.e.* $f^* : \mathbb{D}(Z') \to \mathbb{D}(Z)$ induces an equivalence between the subcategories of Z, Z' consisting of the constant objects).

a') (Makes a considerably more precise). If $Z \in \mathsf{Dia}$ is Hot-aspheric, then we have a), and the essential image of $\mathbb{D}(e) = \mathcal{A}$ in $\mathbb{D}(Z)$ consists of the locally constant objects of Z, *i.e.* of the objects such that the functor

$$Z^\circ \longrightarrow \mathcal{A} \qquad z \longmapsto F_z$$

[48] The reasoning must be made explicit. It suffices to test even on $Y'' \to Y'$, where Y'' is one-point, and where Y' has a final (resp. initial) object.

[49] *cf.* **Der 3** a') on the following page.

is locally constant (*i.e.* sends every arrow to an isomorphism) hence isomorphic to a constant functor (since Z is Hot-aspheric, hence 1-connected).

It must be clearly noted that in a'), it is not enough to assume that Z is $\mathbb{D}$-aspheric, instead of Hot-aspheric. As soon as $\mathbb{D}$-equivalence is strictly weaker than Hot, it seems that a') always becomes false, as a condition on $\mathbb{D}$, if one assumes there only that Z is $\mathbb{D}$-aspheric. For example, even if $\mathbb{D} = \mathbb{D}_i$ (theory of relative homotopy i-types), the category of locally constant objects on Z is expressed in terms of the homotopy $(i+1)$-type, but [page 98] not in terms of its homotopy i-type alone.

It does seem that it is axiom **Der** 3 (a') which ensures the existence of a good notion ("bilateral") of $\mathbb{D}$-fibration, which is stable under base change, and which reduces either to "universal left $\mathbb{D}$-fibration" or to "universal right $\mathbb{D}$-fibration" and thus which is in p. 60 addition a verifiable condition in practice. In the case where f is proper or smooth, the criterion is particularly simple, since it is limited to the condition that the $f_*f^*a_Y$, or the $f_!f^*a_Y$ (for $a \in \mathcal{A}$), be locally constant objects on Y (hence constant if Y is Hot-aspheric, by virtue of **Der** 3 (a')).

I am also going to introduce a strengthened variant of **Der** 3 b) :

Der 3 b') Let $Z' \to Z$ be a Hot-equivalence in Dia. Then $f^* : \mathbb{D}(Z) \to \mathbb{D}(Z')$ induces an equivalence between the categories of locally constant coefficients on Z and on Z'.[50]

In certain questions it suffices to know that the said functor is fully faithful. This [page 99] implies that

$$H^*_{\mathbb{D}}(Z, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(Z', f^*F)$$

when F is locally constant on Z[51] (but it is not clear that this condition always implies full faithfulness[52]) and similarly for homology

$$H^{\mathbb{D}}_*(Z', f^*F) \longrightarrow H^{\mathbb{D}}_*(Z, F) \quad .$$

Under this condition, we have the :

Proposition : *We assume* **Der** 3 b′) *holds. Let $f : X \to Y$ in* Dia. *For f to be a $\mathbb{D}$-fibration, it is necessary and sufficient that for $Y'' \to Y'$ over Y which is a* Hot*-equivalence, $X'' = X \times_Y Y'' \to X' = X \times_Y Y'$ be a $\mathbb{D}$-equivalence.*

To tell the truth, one can state something reasonable, without assuming axiom **Der** 3 b'). One must introduce the :

Definition : Let $f : Z' \to Z$ in Dia. We say that f is a 1-*complete* $\mathbb{D}$-equivalence if the functor $f^* : \mathbb{D}(Z) \to \mathbb{D}(Z')$ is fully faithful on the full subcategory of $\mathbb{D}(Z)$ consisting of the locally constant coefficients. We say that f is a 2-*complete* $\mathbb{D}$-equivalence if it induces an equivalence between the categories of locally constant coefficients.

We say that Z is *completely* $\mathbb{D}$*-aspheric*, if $Z \to e$ is a 2-complete $\mathbb{D}$-equivalence. [page 100]

Thus, axiom **Der** 3 a') can be reformulated, by saying that if Z is Hot-aspheric, then p. 61 Z is completely $\mathbb{D}$-aspheric. And likewise, **Der** 3 b') by saying that if $f : X \to Y$ in Dia is a Hot-equivalence, then f is a 2-complete $\mathbb{D}$-equivalence. We are going to give a weakened variant, of **Der** 3 a'), which suffices to give a reasonable notion of $\mathbb{D}$-fibration :

Der 3 a'') Every Z in Dia which has either a final object, or an initial object, is completely $\mathbb{D}$-aspheric.

---

[50] **Der** 3 b'_0) : the functor is faithful
b'_1) : the functor is fully faithful
b'_2) : the functor is an equivalence

[51] One may say that $f : Z' \to Z$ is a $\mathbb{D}$-equivalence 0-*complete on the left,* and 0-*complete on the right* in the homological case.

[52] But this is OK if one has internal Hom in $\mathbb{D}(Z)_{\text{loc.const.}}$ satisfying

$$\mathsf{Hom}(F, G) \simeq \mathsf{Hom}(e_Z, \underline{\mathsf{Hom}}(F, G)) \quad ,$$

where e is the final element of $\mathcal{A}$, the Hom being compatible with inverse images.

Definition. We say that $f : X \to Y$ in Dia is a $\mathbb{D}$-fibration, if for

$$Y'' \longrightarrow Y' \longrightarrow Y$$

with $Y'' \to Y'$ a 1-complete $\mathbb{D}$-equivalence (for example Y' and Y'' completely $\mathbb{D}$-aspheric), $X'' = X \times_Y Y'' \longrightarrow X' = X \times_Y Y'$ is a $\mathbb{D}$-equivalence.

This is equivalent to each of conditions b), b') of prop. 2 (page 79), *i.e.* that f be universally a left $\mathbb{D}$-fibration, or universally a right $\mathbb{D}$-fibration (thanks to **Der 3** b'').

[page 101]

Definition 1. $f : X \to Y$ in Dia is called a *cohomological $\mathbb{D}$-equivalence* (understood : *complete*) if for every $F \in \mathsf{Ob}\,\mathbb{D}(Y)$, F locally constant, we have

$$H^*_{\mathbb{D}}(Y, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X, f^*(F)) \quad .$$

Dual notion *homological $\mathbb{D}$-equivalence*

$$H^{\mathbb{D}}_*(X, f^*(F)) \xrightarrow{\sim} H^{\mathbb{D}}_*(Y, F) \quad .$$

If Y is completely $\mathbb{D}$-aspheric, then the locally constant coefficients on Y are constant, and f is a cohomological (resp. homological) $\mathbb{D}$-equivalence if and only if it is a $\mathbb{D}$-equivalence *i.e.* if X is $\mathbb{D}$-aspheric.[53]

Proposition 1 (Sorital). *Let $X \xrightarrow{f} Y \xrightarrow{g} Z$. If f and g are cohomological $\mathbb{D}$-equivalences, then gf too. If gf and f are cohomological $\mathbb{D}$-equivalences, then g too. If gf is a cohomological $\mathbb{D}$-equivalence, and if g is a 2-complete $\mathbb{D}$-equivalence* (hence every F locally constant on Y comes from a G locally constant on Z), *then f is a cohomological $\mathbb{D}$-equivalence.*

Definition 2. 1°) $f : X \to Y$ in Dia is called a *complete cohomological $\mathbb{D}$-fibration,* I do mean complete, if

a) For every $y \in \mathsf{Ob}\,Y$, $X_y \to X/y$ is a cohomological $\mathbb{D}$-equivalence (complete).

b) For every $y' \to y$ in Y, $X/y' \to X/y$ is a cohomological $\mathbb{D}$-equivalence (complete).

c) These conditions remain satisfied after every base change $Y' \to Y$.

p. 62

2°) Dual notion of complete homological fibration.

[page 102]

Example. If f is a Hot-fibration and is Hot-proper, the arrows of *a*) and *b*) are Hot-equivalences, hence by virtue of **Der 3** b'_1) (which must be admitted), they are cohomological $\mathbb{D}$-equivalences. This means that *a*) and *b*) hold, and *c*) too, for the notion of a Hot-proper arrow is stable under base change.

Proposition 2. *Let $f : X \to Y$ be a complete cohomological $\mathbb{D}$-fibration, $F \in \mathsf{Ob}\,\mathbb{D}(X)$ locally constant on X. Then $f_*(F)$ is locally constant, its formation commutes with every base change, and in particular is computed fiber by fiber.*[54]

Corollary 1. *Let there be a cartesian diagram*

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{p} & Y' \end{array} \quad .$$

We assume f a cohomological $\mathbb{D}$-fibration and p a cohomological $\mathbb{D}$-equivalence. Then q is a cohomological $\mathbb{D}$-equivalence.

[page 103]

Indeed, we have for F locally constant on X

$$H^*_{\mathbb{D}}(X, F) \simeq H^*_{\mathbb{D}}(Y, f_*F) \xrightarrow{\sim} H^*_{\mathbb{D}}(Y', p^*f_*F) \simeq H^*_{\mathbb{D}}(Y', f'_*q^*F) \simeq H^*_{\mathbb{D}}(X', q^*F)$$

[53] Axiom **Der** 3 b'_1) implies that every Hot-equivalence is a cohomological $\mathbb{D}$-equivalence.

[54] There is a converse *cf.* prop. 4.

qed.

Proposition 3. *The notion of complete cohomological $\mathbb{D}$-fibration is stable under base change* (trivial) *and under composition.*

For stability under composition, it suffices to note the following corollary of proposition 2.

Corollary 2 (of prop. 2). *Let $f : X \to Y$. For f to be a complete cohomological $\mathbb{D}$-fibration, it is necessary and sufficient that for every $Y'' \xrightarrow{p} Y' \to Y$ over Y, with p a cohomological $\mathbb{D}$-equivalence, the corresponding arrow $X'' = X \times_Y Y'' \longrightarrow X' = X \times_Y Y'$ also be a cohomological $\mathbb{D}$-equivalence.*

This is necessary by corollary 1, taking into account that $f' : X' = X \times_Y Y' \to Y'$ is also a cohomological $\mathbb{D}$-fibration. It is sufficient, for it suffices to apply it to the cases $\{y\} \to Y/\{y\}$, $Y/y \to Y/y'$, which are cohomological $\mathbb{D}$-equivalences by virtue of **Der** 3 b'_1) (or only **Der** 3 a')). [page 104]

Proposition 4. *We assume that* **Der** 3 a′) *is satisfied* (*i.e.* every Z in Dia which is Hot-aspheric, is completely $\mathbb{D}$-aspheric). *Let $f : X \to Y$ in* Dia. *The following conditions are equivalent* p. 63

1°) *f is a complete cohomological $\mathbb{D}$-fibration* (conditions *a*), *b*), *c*) of definition 2, page 101).

2°) *For every $F \in \mathsf{Ob}\,\mathbb{D}(X)$, locally constant, $f_*(F)$ is locally constant, is computed fiber by fiber; and that remains true after every base change $Y' \to Y$ (with F' locally constant on $X' = X \times_Y Y'$). (Then the formation of $f_*(F)$ commutes with every base change).*

3°) *For every $Y'' \xrightarrow{p} Y'$ over Y, if p is a cohomological $\mathbb{D}$-equivalence, then $X'' = X \times_Y Y'' \longrightarrow X' = X \times_Y Y'$ too.*

3′) *As in 3, assuming that Y'' is the one-point category, and that Y' has a final element : $X'' \to X'$ must be a cohomological $\mathbb{D}$-equivalence.*

Definition 3. We say that $f : X \to Y$ in Dia is a *complete $\mathbb{D}$-fibration* if it is a complete cohomological $\mathbb{D}$-fibration and a complete homological $\mathbb{D}$-fibration. [page 105]

This is stable under base change, under composition.

Definition 3 is equivalent to this : for every $p : Y'' \to Y'$ over Y, with Y'' and Y' Hot-aspheric, the corresponding morphism $X'' \to X'$ is at the same time a cohomological $\mathbb{D}$-equivalence, and a homological $\mathbb{D}$-equivalence. (And it even suffices to assume it for $Y'' = e$ and for Y' with a final object or an initial object.) One can think of strengthening this condition still further by requiring (without looking to the left or to the right) that $X'' \to X'$ be a 1-complete $\mathbb{D}$-equivalence. But it is not clear that this strengthened notion is stable under composition – unless one strengthens it still further, by requiring that if $Y'' \to Y'$ is a 1-complete $\mathbb{D}$-equivalence, then so is $X'' \to X'$. The trouble with this strengthened definition is that for lack of a cohomological or homological description of the notion of 1-complete $\mathbb{D}$-equivalence, I do not feel capable of giving interesting (not totally trivial) examples of $\mathbb{D}$-fibrations in this very strong sense. Except for the trivial Hot-fibrations (*i.e.* with Hot-aspheric fibers) – but that is no fun! [page 106]

But when one has associative and unital products $*$ and corresponding internal $\underline{\mathsf{Hom}}$ in the $\mathbb{D}(X)_{\text{loc. const.}}$, compatible with inverse images, so one will have on X if F and G are locally constant in $\mathbb{D}(X)$, and if ε_X denotes the unit for $*$ in $\mathbb{D}(X)$,

$$\mathsf{Hom}(F,G) \simeq \mathsf{Hom}(\varepsilon_X * F, G) \simeq \mathsf{Hom}(\varepsilon_X, \underline{\mathsf{Hom}}(F,G)) \simeq \mathsf{Hom}_{\mathcal{A}}(\varepsilon_e, p_*\,\underline{\mathsf{Hom}}(F,G))\ ,$$

for $\varepsilon_X = p^*(\varepsilon_e)$, where $p : X \to e$, so in this case an arrow $f : X \to Y$ in Dia is a 1-complete $\mathbb{D}$-equivalence if (and only if) it is a cohomological $\mathbb{D}$-equivalence. So in this case, if f is a complete cohomological $\mathbb{D}$-fibration, it is also a complete homological $\mathbb{D}$-fibration, hence also a complete $\mathbb{D}$-fibration (definition 3). (And even in the so-called

strengthened sense of just now : if $Y'' \to Y'$ over Y is a 1-complete $\mathbb{D}$-equivalence ($\Leftrightarrow$ a cohomological $\mathbb{D}$-equivalence!) then $X'' \to X'$ too).

The question arises of finding f which are cohomological $\mathbb{D}$-fibrations, or homological ones. I do not know whether I shall be able to find any, which are not already cohomological HOT-fibrations, or homological ones (this now amounts to the same thing[55]), for in practice I hardly know how to say that an arrow $Z' \to Z$ in $\mathcal{C}at$ is a complete cohomological $\mathbb{D}$-equivalence, except when this arrow is a HOT-equivalence (and assuming axiom **Der** 3 b'_1), or its weakened cohomological variant). p. 64 [page 107]

I shall therefore look for the HOT-fibrations, *i.e.* the $f : X \to Y$ such that for every $Y'' \to Y' \to Y$, with Y'' and Y' aspheric (understood: for HOT), $X'' \to X'$ is a Hot-equivalence. (It even suffices to suppose Y'' reduced to a point, Y' with a final object ...)

Suppose for example f proper (*i.e.* HOT-proper), so condition a) of definition 2, p. 101 is satisfied and likewise after every base change. Does it suffice to check in addition condition b), *i.e.* that

$$X/y \longrightarrow X/y' \quad , \qquad \text{for } y \to y' \text{ in } Y \ ,$$

are Hot-equivalences? (I mean staying over Y, without also supposing we have the same thing after base change). But does this remain true after every base change? [page 108]

The situation seems better here for smooth f (instead of proper), interpreting conditions a) and b) in homological form, as meaning that $f_!(e_X)$ is a locally constant object of $\mathsf{HOT}(Y)$, which is computed fiber by fiber. Since smooth f implies that $f_!$ commutes with every base change, this hypothesis on f (that f be a cohomological HOT-fibration, perhaps not complete) is stable under base change, hence implies that it is a complete cohomological HOT-fibration. But this argument uses the general properties of derivators for HOT, which I have not yet established - especially not in the whole domain $\mathcal{C}at$! So it is healthy for me to be wary. I prefer to feel out the difficulty while staying at first within the framework of the cohomological formalism (noncommutative), of which I have some mastery. I therefore suppose f proper again, and that the

$$X_y \longrightarrow X/y \quad , \qquad X/y \longrightarrow X/y'$$

are HOT-equivalences, I want to prove the same thing after every base change, at least that in the situation [page 109]

$$\begin{array}{ccccc} X & \xleftarrow{q} & X' & \xleftarrow{q'} & X'' \\ {\scriptstyle f}\downarrow & & {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f''} \\ Y & \xleftarrow[p]{} & Y' & \xleftarrow[p']{} & Y'' \end{array}$$

if Y', Y'' are aspheric, then q' is a Hot-equivalence. If I want to establish this by a cohomological criterion, I must start by taking on X' a locally constant sheaf of *sets* F, and prove that $\Gamma(X', F) \xrightarrow{\sim} \Gamma(X'', q'^* F)$. Since this sheaf does not come from X, I am at a loss how to apply the hypothesis on f, which concerns a priori constant sheaves of sets on X, or on the X/y. But we should have this: p. 65

Lemma. *Let $f : X'' \to X'$. Then f is a* HOT*-equivalence* i.e.

$$H^*_{\mathsf{HOT}}(X', a_{X'}) \xrightarrow{\sim} H^*_{\mathsf{HOT}}(X'', a_{X''})$$

[55] Cohomological HOT-equivalences = homological HOT-equivalences = HOT-equivalences plain and simple.

is isomorphism for every $a \in \mathsf{Hot} = \mathsf{HOT}(e)$*, if and only if* f *is a* Hot*-equivalence,*[56] *and then* f *is even a* 2*-complete* HOT*-equivalence* (I need only: complete cohomological and homological HOT-equivalence) i.e. *for every* $\xi \in \mathbb{D}(X')$ *locally constant on* X'*, we have*[57]

$$H^*_{\mathsf{HOT}}(X', \xi) \xrightarrow{\sim} H^*_{\mathsf{HOT}}(X'', f^*(\xi)) \quad .$$

[page 110]

The first assertion should follow from the formulas

$$H^*_{\mathsf{HOT}}(Z, a_Z) \simeq \underline{\mathsf{Hom}}_{\mathsf{Hot}}(\mathsf{hot}(Z), a) \qquad (*)$$

$$\text{(and also} \qquad H^{\mathsf{HOT}}_*(Z, a_Z) \simeq \mathsf{hot}(Z) \times a \qquad \text{)}$$

hence if

$$\mathsf{hot}(X'') \xrightarrow{\sim} \mathsf{hot}(X') \qquad (**)$$

is isomorphism, then

$$H^*_{\mathsf{HOT}}(X', a_{X'}) \longrightarrow H^*_{\mathsf{HOT}}(X'', a_{X''})$$

is isomorphism, since it is deduced from $(*)$ by taking $\underline{\mathsf{Hom}}_{\mathsf{Hot}}(\,.\,, a)$. Conversely, if $(*)$ is always isomorphism, expressing the two sides […] of $\underline{\mathsf{Hom}}_{\mathsf{Hot}}(\,.\,, a)$ and taking the π_0, we find that

$$\mathsf{Hom}_{\mathsf{Hot}}(\mathsf{hot}(X'), a) \xrightarrow{\sim} \mathsf{Hom}_{\mathsf{Hot}}(\mathsf{hot}(X''), a)$$

is an isomorphism for every a, hence $(**)$ is an isomorphism.

But the fact that this implies that f is a complete cohomological HOT-equivalence, *i.e.* that the relation $(*)$ remains true for a $\xi \in \mathbb{D}(X')$ locally constant on X', seems to me a delicate property of the derivator HOT, to which I shall have to return.

So the lemma is far from established, for want of having so much as constructed the derivator HOT, without even speaking of establishing its delicate special properties! But I have no reason to doubt the lemma, it is part of the litany of basic intuitions that I would like to put on a solid footing.

The lemma implies that a $\mathbb{D} =$ HOT-fibration (in the sense of definition 2 page 81, revised and corrected on page 96-97) is in fact already a complete cohomological and homological HOT-fibration. Now we have seen that if f is proper, f is a $\mathbb{D}$-fibration if and only if the preceding conditions over Y (without base change) are satisfied. And dually for smooth f.

p. 66

[page 111]

11.11.90

I feel the need to review the notions of $\mathbb{D}$-fibrations of all kinds (p. 76 ff). First the various useful variants of axiom **Der** 3 on derivators:

Der 3 *a*)[58] If $Z \in$ Dia is Hot-aspheric, it is $\mathbb{D}$-aspheric *i.e.* $\mathcal{A} = \mathbb{D}(e) \to \mathbb{D}(Z)$ is fully faithful.

a') As *a*), with in addition the characterization of the essential image of $\mathcal{A}$ in $\mathbb{D}(Z)$, as consisting of the locally constant $\mathbb{D}$-coefficients.

Der 3 *b*) If $f : Z' \to Z$ in Dia is a Hot-equivalence, then it is a $\mathbb{D}$-equivalence, *i.e.* the functor $f^* : \mathbb{D}(Z) \to \mathbb{D}(Z')$ induces an equivalence between the subcategories consisting of the constant coefficients (or again a fully faithful functor $\mathbb{D}(Z)_{\text{const.}} \longrightarrow \mathbb{D}(Z'))$.

b'_1) If f is as above, then f is a 1-complete $\mathbb{D}$-equivalence, *i.e.* the functor $f^* : \mathbb{D}(Z) \to \mathbb{D}(Z')$ induces a fully faithful functor between the full subcategories consisting

[page 112]

[56] Note here […] the distinction between HOT-equivalence and Hot-equivalence.

[57] Is this also true for the derivators $\mathbb{D}_i$?

[58] *cf.* also the series **Der 3** *c*. . . , page 124 ff.

of the locally constant objects.

b'_2) As b'_1), but with 2-complete $\mathbb{D}$-equivalence instead of 1-complete, *i.e.* f^* induces an equivalence between the subcategories consisting of the locally constant coefficients.

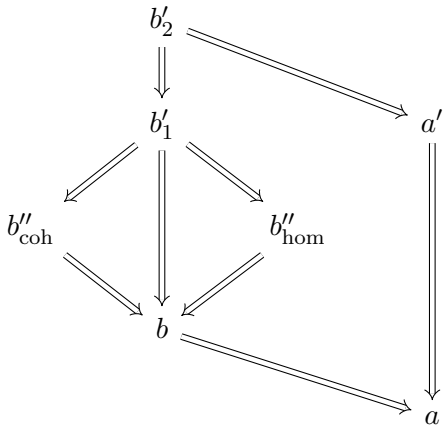
b''_{coh}) If $f : Z' \to Z$ in Dia is a Hot-equivalence, then it is a complete cohomological $\mathbb{D}$-equivalence, *i.e.* for every $F \in \mathsf{Ob}\ \mathbb{D}(Z)$ which is locally constant, we have

$$H^*_{\mathbb{D}}(Z, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(Z', f^*F) \quad .$$

b''_{hom}) If $f : Z' \to Z$ in Dia is a Hot-equivalence, it is a complete homological $\mathbb{D}$-equivalence, *i.e.* for every $F \in \mathsf{Ob}\ \mathbb{D}(Z)$ which is locally constant, we have[59]

$$H^{\mathbb{D}}_*(Z', f^*F) \xrightarrow{\sim} H^{\mathbb{D}}_*(Z, F) \quad .$$

We have the diagram of tautological implications for the variants of **Der** 3: p. 67



.

In all cases known to me, derivators satisfy the strongest condition **Der** 3 b'_2.

I also recall the: [page 113]

Definition 1: Let $f : X \to Y$ be in Dia. We say that f is a *cohomological* $\mathbb{D}$-*equivalence* (understood: "*complete*") if for every $\mathbb{D}$-coefficient F locally constant on Y,

$$H^*_{\mathbb{D}}(Y, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X, f^*F) \quad ;$$

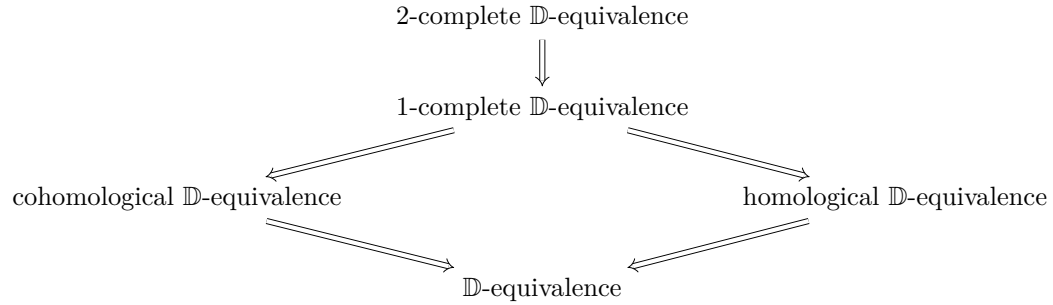
we say that f is a *homological* $\mathbb{D}$-*equivalence* ("*complete*") if for every F as above,

$$H^{\mathbb{D}}_*(X, f^*F) \xrightarrow{\sim} H^{\mathbb{D}}_*(Y, F) \quad .$$

We say that f is a 1-*complete* $\mathbb{D}$-*equivalence* (resp. 2-*complete*) if the functor

$$f^* : \mathbb{D}(Y)_{\text{loc. const.}} \longrightarrow \mathbb{D}(X)_{\text{loc. const.}}$$

is fully faithful (resp. an equivalence).



.

[59] *cf.* special ± variant for HOT pages 124-125.

If in the subcategories $\mathbb{D}(X)_{\text{loc. const.}}$ there is an operation $*$ (associative, unital, commutative) and associated internal $\underline{\mathsf{Hom}}$, preserved by inverse images, then cohomological $\mathbb{D}$-equivalences are already 1-complete and are therefore homological $\mathbb{D}$-equivalences, therefore in this particular case

p. 68

[page 114 blank]
[page 115]

$$\begin{array}{ccc}
\text{2-complete } \mathbb{D}\text{-equivalence} & & \\
\Downarrow & & \\
\text{1-complete } \mathbb{D}\text{-equivalence} & \Longleftrightarrow & \text{cohomological } \mathbb{D}\text{-equivalence} \\
 & & \Downarrow \\
 & & \text{homological } \mathbb{D}\text{-equivalence} \\
 & & \Downarrow \\
 & & \mathbb{D}\text{-equivalence}
\end{array} \quad .$$

In $\mathbb{D} = \mathsf{HOT}$, all these notions should be equivalent, and equivalent to the apparently still weaker condition, that $f : X \to Y$ be a Hot-equivalence. But I know only HOT as a derivator where the $\mathbb{D}$-equivalence of an arrow $f : X \to Y$ in Dia already implies cohomological $\mathbb{D}$-equivalence (complete), or homological $\mathbb{D}$-equivalence (complete), or even 1-complete $\mathbb{D}$-equivalence. (I shall come back to this presently.)

Proposition 1. *Let $f : X \to Y$ be in Dia. Equivalent conditions:*

a) *For every $y \in \mathsf{Ob}\, Y$, the inclusion $X_y \to X/y$ is a $\mathbb{D}$-equivalence, and for every arrow $y' \to y$ in Y, $X/y' \to X/y$ is a $\mathbb{D}$-equivalence.*

b) *For every constant coefficient F on Y, $f_* f^* F$ is locally constant, and is computed "fiber by fiber".*

(This is prop. 1, p. 76).

Proposition 1′. Dual.

[page 116]

Definition 2. Under the equivalent conditions of proposition 1, we say that f is a *weak cohomological $\mathbb{D}$-fibration.* Under the dual conditions, that f is a *weak homological $\mathbb{D}$-fibration.*

This is what in definition 1, p. 77 I called "cohomological $\mathbb{D}$-fibration" or "left $\mathbb{D}$-fibration", and dually. Now the notion seems to me too weak to deserve this name, which I reserve for the "complete" case.

Proposition 2. *Weak cohomological $\mathbb{D}$-fibrations are stable under smooth base changes, weak homological $\mathbb{D}$-fibrations stable under proper base changes.*

I should construct an example where f is a weak cohomological $\mathbb{D}$-fibration, and is no longer so after base change.

Proposition 3. *Let*

p. 69

$$\begin{array}{ccc}
X & \xrightarrow{\quad f \quad} & X' \\
 & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle p'} & \\
 & S &
\end{array}$$

be in Dia, with p and p' weak cohomological $\mathbb{D}$-fibrations (resp. *weak homological $\mathbb{D}$-fibrations*), *let $S_0 \subset S$ be such that $\pi_0(S_0) \to \pi_0(S)$ is surjective. For f to be a $\mathbb{D}$-equivalence, it suffices that the $f_s : X_s \to X'_s$, for $s \in S_0$, be so.*

Proposition 4. *If $f : X \to Y$ in Dia is $\mathbb{D}$-proper* (resp. $\mathbb{D}$-*smooth*), *then f is a weak cohomological* (resp. *homological*) *$\mathbb{D}$-fibration if and only if for every F constant on Y, $f_* f^* F$* (resp. *$f_! f^* F$*) *is locally constant on Y. And if this is so, for every $Y' \to Y$ in Dia, $f' : X' = X \times_Y Y' \to Y'$ is again a weak cohomological* (resp. *homological*) *$\mathbb{D}$-fibration.*

Immediate from criterion *b*) of proposition 1.

[page 117]

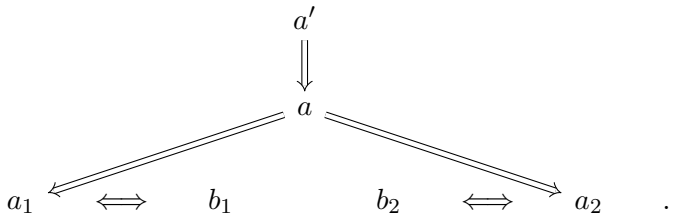
Proposition 5. *Let $f : X \to Y$ be in* Dia*. Consider the following conditions:*
a) *For every $Y'' \to Y'$ over Y in* Dia*, if Y'' and Y' are* Hot*-aspheric, then*

$$X'' = X \times_Y Y'' \longrightarrow X' = X \times_Y Y'$$

is a $\mathbb{D}$-equivalence.

$$\begin{array}{ccccc} X & \longleftarrow & X' & \longleftarrow & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & Y' & \xleftarrow[p]{} & Y'' \end{array}$$

a′) *As in* a), *but assuming $p : Y'' \to Y'$ is a* Hot*-equivalence (instead of assuming Y', Y''* Hot*-aspheric).*
a_1) *As in* a), *but assuming (instead of asphericity) that Y' has a final object, and Y'' is the unit category.*
a_2) *As in* a), *but assuming that Y' has an initial object, and Y'' is the unit category.*
b_1) *f is a weak cohomological $\mathbb{D}$-fibration, and remains so after any base change.*
b_2) *f is a weak homological $\mathbb{D}$-fibration, and remains so after any base change.*
We then have the diagram of tautological implications p. 70



When $\mathbb{D}$ satisfies axiom **Der** 3 a′ (p. 111), *then all these conditions, except* a′), *are equivalent. When $\mathbb{D}$ satisfies axiom* **Der** 3 b'_1 *(or only* **Der** 3 b''_{coh} or b''_{hom}*), then all the conditions (including* a′*) are equivalent.*

To simplify, we shall assume henceforth that axiom **Der** 3 a') is satisfied.

Definition 3. Under the equivalent conditions a, a_1, b_1, a_2, b_2 of proposition 5, we say that f is a $\mathbb{D}$*-fibration*.

Proposition 6. *Suppose that $f : X \to Y$ in* Dia *is $\mathbb{D}$-proper* (resp. *$\mathbb{D}$-smooth*). *Equivalent conditions:* [page 118]
a) *f is a $\mathbb{D}$-fibration.*
b) *f is a weak cohomological* (resp. *homological*) *$\mathbb{D}$-fibration.*
c) *For every $y' \to y$ in Y, $X/y' \to X/y$* (resp. *$y\backslash X \to y'\backslash X$*) *is a $\mathbb{D}$-equivalence.*
d) *For every F constant on Y, $f_* f^* F$* (resp. *$f_! f^* F$*) *is locally constant on Y.*
d′) *For every F locally constant on Y, $f_* f^* F$* (resp. *$f_! f^* F$*) *is locally constant on Y.*
This follows formally from Prop. 4 and Prop. 5.

Corollary. *Suppose $f : X \to Y$ is $\mathcal{C}at$-fibrant* (resp. *$\mathcal{C}at$-cofibrant*). *For f to be a $\mathbb{D}$-fibration, it is necessary and sufficient that for every $u : y \to y'$ in Y, $u^* : X_{y'} \to X_y$* (resp. *$u_* : X_y \to X_{y'}$*) *be a $\mathbb{D}$-equivalence.*

It is immediate that the notion of $\mathbb{D}$-fibration is stable under base change, but it is doubtful that it is so under composition.[60] (Here again, I have omitted to make a

[60] N.B. If the $\mathbb{D}$-equivalences are already Hot-equivalences, then the notion coincides with that of Hot-fibration, and it is stable under composition. Also, if the $\mathbb{D}$-equivalences are in the sense of cohomological, or homological, $\mathbb{D}$-equivalences, then the notion coincides with that of cohomological (resp. homological) $\mathbb{D}$-fibrations below, and is also stable under composition.

counterexample, with derivators $\mathbb{D} = \mathsf{Der}(k\text{-}\mathsf{Mod})$, say.) Another shortcoming of the notion: it is not clear, if f is a $\mathbb{D}$-fibration and F a $\mathbb{D}$-coefficient on Y, whether

$$f_* f^* F \qquad \text{and} \qquad f_! f^* F$$

are computed "fiber by fiber", as one would expect in the case of a "bundle". p. 71

On the other hand, here is a nice result:

Proposition 7. *Let $f : X \to Y$ be in* Dia*. Equivalent conditions:* [page 119]

a) *f is a "universal" $\mathbb{D}$-equivalence,* i.e. *for every $Y' \to Y$ in* Dia*, $f' : X' = X \times_Y Y' \to Y'$ is a $\mathbb{D}$-equivalence.*

b) *f is a $\mathbb{D}$-fibration with $\mathbb{D}$-aspheric fibers.*

Definition 4. Under the equivalent conditions of proposition 7, we say that f is a *trivial* $\mathbb{D}$-fibration.

But beware, f can be a $\mathbb{D}$-fibration and a $\mathbb{D}$-equivalence, without the $\mathbb{D}$-fibration being trivial *i.e.* without the fibers being $\mathbb{D}$-aspheric, cf. p. 87 and following. And also (should go above) even if $\mathbb{D}$ satisfies **Der** 3 b'_2 (the strongest of the variants of **Der** 3), thus when <u>all</u> the conditions of proposition 5, including a'), are equivalent, these (*i.e.* that f be a $\mathbb{D}$-fibration) nevertheless do not imply, in general, that in condition a') one can restrict oneself to $p : Y'' \to Y'$ a $\mathbb{D}$-equivalence (instead of a Hot-equivalence). In other words, if we have a cartesian diagram in Dia

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y' \end{array} \quad ,$$

with f a $\mathbb{D}$-fibration, and p a $\mathbb{D}$-equivalence, it does not necessarily follow that q is a $\mathbb{D}$-equivalence. (But this will nevertheless be the case, "of course", if $\mathbb{D} = \mathsf{HOT}$.)

On the other hand, we have: [page 120]

Proposition 8. *Let $f : X \to Y$ be in* Dia*. Equivalent conditions:*

a) *f is a $\mathbb{D}$-fibration.*

b) *For every $p : Y'' \to Y'$ over Y, if p is a* 1*-complete $\mathbb{D}$-equivalence* (def. 1, p. 113)*, then $q : X'' \to X'$ is a $\mathbb{D}$-equivalence.*

$$\begin{array}{ccccc} X & \longleftarrow & X' & \xleftarrow{q} & X'' \\ {\scriptstyle f}\downarrow & & {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f''} \\ Y & \longleftarrow & Y' & \xleftarrow[p]{} & Y'' \end{array}$$

Corollary. *Suppose that in* Dia*, every $\mathbb{D}$-equivalence is* 1*-complete.* (N.B. I know only the derivator HOT to satisfy this condition, which seems almost characteristic of HOT.) *Then $f : X \to Y$ is a $\mathbb{D}$-fibration if and only if for every $p : Y'' \to Y'$ over Y, if p is a $\mathbb{D}$-equivalence, then $q : X'' \to X'$ too. In this exceptional case among derivators, the notion of $\mathbb{D}$-fibration is stable under composition* (which is tautological from the criterion I have just given.)

We are going to strengthen the notion of $\mathbb{D}$-fibration, to make it more flexible, and in particular stable under composition. p. 72

Proposition 9. *Let $f : X \to Y$ be in* Dia*. Equivalent conditions:*

a) *For every $y \in Y$, $X_y \to X/y$ is a <u>cohomological</u> $\mathbb{D}$-<u>equivalence</u>* (def. 1, p. 113)*, and likewise for $X/y \to X/y'$ for every arrow $y \to y'$ in Y. And these conditions on f remain valid after any base change* (N.B. it suffices that the first of these conditions remain so).

b) *for every $F \in \mathsf{Ob}\ \mathbb{D}(X)$ <u>locally</u> <u>constant</u>, $f_* F$ is locally constant on Y and is computed "fiber by fiber" – and likewise after any base change.* [page 121]

c) *For every $p : Y'' \to Y'$ over Y, if p is a cohomological $\mathbb{D}$-equivalence, then the same holds for $q : X'' \to X'$.*

c$'$) *As in* c, *but assuming as a hypothesis on p that Y'' is the final category, and Y' has a final object.*

Proof. In the order ($\pm$ tautological)

$$\mathrm{a} \Longrightarrow \mathrm{b} \Longrightarrow \mathrm{c} \Longrightarrow \mathrm{c}' \Longrightarrow \mathrm{a} \quad .$$

Definition 5. Under the equivalent conditions of proposition 9, we say that f is a *cohomological $\mathbb{D}$-fibration*. Under the dual conditions, we say that f is a *homological $\mathbb{D}$-fibration*. We say that f is a *strong $\mathbb{D}$-fibration*, if it is both a cohomological $\mathbb{D}$-fibration and a homological $\mathbb{D}$-fibration.

These notions strengthen the notion of $\mathbb{D}$-fibration. They have the further advantage of being stable under composition, not only under base change.

The notion of strong $\mathbb{D}$-fibration was chosen to cover the two dual notions of cohomological and homological $\mathbb{D}$-fibration (for lack of a better notion) while remaining stable under composition. One could require that for every [page 122]

$$Y'' \longrightarrow Y'$$

over Y, with Y', Y'' Hot-aspheric, $q : X'' \to X'$ be a 1-complete $\mathbb{D}$-equivalence (or even 2-complete), call these the 1-*complete* (resp. 2-*complete*) $\mathbb{D}$-*fibrations*. This strengthens in a rather elegant way the notion of strong $\mathbb{D}$-fibration, but has the great fault of not being stable under composition (as far as I know!), except in the particular case where we have nice internal $\underline{\mathsf{Hom}}$ (cf. bottom of page 113), in which case the notion of cohomological $\mathbb{D}$-equivalence already implies that of 1-complete $\mathbb{D}$-equivalence, thus also p. 73 that of homological $\mathbb{D}$-equivalence. So in this case [page 123]

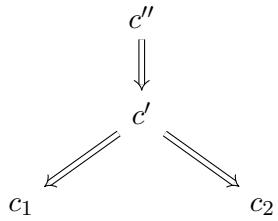
$$\begin{array}{ccc} \text{cohomological } \mathbb{D}\text{-fibration} & \Longleftrightarrow & \text{1-complete } \mathbb{D}\text{-fibration} \\ \Downarrow & & \\ \text{homological } \mathbb{D}\text{-fibration} & , & \end{array}$$

and there is no need to look further than cohomological $\mathbb{D}$-fibrations.

One could think, it is true, of defining (in the general case) a stronger notion, by requiring that for every $p : Y'' \to Y'$ over Y which is a 1-complete $\mathbb{D}$-equivalence, $q : X'' \to X'$ also be a 1-complete $\mathbb{D}$-equivalence. This time, the notion is stable under composition, but it nevertheless seems unusable to me (outside the case we have just mentioned, where it coincides with the notion of cohomological $\mathbb{D}$-fibration, or again of strong $\mathbb{D}$-fibration), for, not being in a position to describe the 1-complete $\mathbb{D}$-equivalences, I am also unable to give nontrivial examples of "strictly 1-complete" $\mathbb{D}$-fibrations.

To check that an $f : X \to Y$ in Dia is a cohomological $\mathbb{D}$-fibration, one must know [page 124] how to check that arrows $q : Z' \to Z$ in Dia are cohomological $\mathbb{D}$-equivalences, hence that $H^*_{\mathbb{D}}(Z, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(Z', q^*(F))$ is isomorphism for every $F \in \mathsf{Ob}\ \mathbb{D}(Z)$ which is locally constant on Z. Now I know how to check a statement of this nature (outside the case where I know in advance that Z is completely $\mathbb{D}$-aspheric, *i.e.* that the locally constant coefficients on Z are in fact constant) only if q is already a Hot-equivalence. That is to say that I know how to exhibit cohomological $\mathbb{D}$-fibrations, only when they are already cohomological HOT-fibrations. I want, in what follows, to give usable criteria for HOT-fibrations. But we are going to axiomatize these developments, keeping a general $\mathbb{D}$, on which I am going to make an assumption so draconian that practically, I know only the derivator HOT which satisfies it. This is c_1 below. [page 125]

Der 3 c_1) Every $\mathbb{D}$-equivalence is a cohomological $\mathbb{D}$-equivalence.
c_2) Every $\mathbb{D}$-equivalence is a homological $\mathbb{D}$-equivalence.
c') Every $\mathbb{D}$-equivalence is 1-complete.
c'') Every $\mathbb{D}$-equivalence is 2-complete.
N.B. We have



tautologically. I know only HOT as satisfying the two weakest conditions c_1, c_2 together. And in fact, subject to writing the theory of the derivator HOT, it must be true that it satisfies even the strongest form c''. (And the HOT-equivalences must be simply the Hot-equivalences.) (Cf. the "lemma" p. 109.) p. 74

The following proposition is a tautology.

Proposition 10. *Suppose that $\mathbb{D}$ satisfies condition* **Der** 3 c_1 (p. 124), *for example $\mathbb{D} = HOT$. Then the notion of $\mathbb{D}$-fibration coincides with that of cohomological $\mathbb{D}$-fibration. It is therefore stable under composition. For $f : X \to Y$ to be a $\mathbb{D}$-fibration, it is necessary and sufficient that for $p : Y'' \to Y'$ over Y, if p is a $\mathbb{D}$-equivalence, the same be true of $q : X'' \to X'$.*

Corollary 1. *If $f : X \to Y$ in* Dia *is $\mathbb{D}$-proper, then f is a cohomological $\mathbb{D}$-fibration if and only if for every arrow $y' \to y$ in Y, $X/y' \to X/y$ is a $\mathbb{D}$-equivalence* (i.e. *if and only if f is a weak cohomological $\mathbb{D}$-fibration*).

Apply criterion c) of prop. 6.

Corollary 2. *Suppose that $\mathbb{D}$ satisfies* **Der** 3 (c_1, c_2) (for example $\mathbb{D} = \mathsf{HOT}$). *Then for an arrow $f : X \to Y$ in* Dia, *it amounts to the same thing for f to be a $\mathbb{D}$-fibration, a cohomological $\mathbb{D}$-fibration, or a homological $\mathbb{D}$-fibration. If f is proper* (resp. *smooth*), *it amounts to the same thing to say that f is a cohomological $\mathbb{D}$-fibration* (resp. *homological*), *weak.* [page 126]

We therefore see that in $\mathcal{C}at$, we must have a single reasonable notion of HOT-fibration, which will coincide with that of cohomological or homological HOT-fibration. But for this notion to be available as of now, the derivator HOT would have to have already been constructed on the domain $\mathsf{Dia} = \mathcal{C}at$ in its entirety. Now this has not yet been done (and there could therefore have been no question of proving the properties of HOT just mentioned, say the strongest **Der** 3 c'').

Nevertheless, I would like to develop in a direct ("elementary") way the notion of Hot-fibration, without referring to the theory of the derivator HOT, and to the preceding theoretical developments.

p. –

THE DERIVATORS

ALEXANDRE GROTHENDIECK

Chapter II
Cofinality (on the right and on the left)
(preliminary to cohomological cofinality)

p. –
p. 1
[page 1]

1 Cohomological cofinality

Let

$$\alpha : C' \longrightarrow C$$

be a functor of (small) categories, $\mathcal{M}$ a category having the relevant $\varprojlim$, F a presheaf (thus contravariant in i) on C with values in $\mathcal{M}$. We have a canonical homomorphism in $\mathcal{M}$

$$\varprojlim_{C} F \longrightarrow \varprojlim_{C'} \underbrace{\alpha^*(F)}_{F\circ\alpha} \tag{1}$$

and we want conditions on α for this always to be an isomorphism ([1]). We are at once reduced to the case where $\mathcal{M} = (\mathrm{Ens})$. Let us say that a category I is *weakly left filtering* if $I \neq \emptyset$ and if $\forall\, i', i'' \in \mathrm{Ob}\, I$, $\exists\, i \in \mathrm{Ob}\, I$ and arrows

$$\begin{array}{ccc} & & i' \\ & \nearrow & \\ i & & \\ & \searrow & \\ & & i''. \end{array}$$

This also means that $\mathrm{Ob}\, I$ as an ordered set is left filtering.

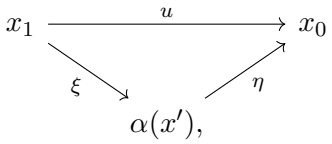
Proposition. *For* (1) *always to be an isomorphism* (α *"anodyne for* $\varprojlim$ *"), it suffices that for every* $x_0 \in \mathrm{Ob}\, C$*, the category* $x_0\backslash C'$ *of pairs* (x', u)*, with* $x' \in \mathrm{Ob}\, C'$*,* $u : x_0 \longrightarrow \alpha(x')$*, be weakly left filtering* ([2], [3]).

This does not imply the stronger conclusion

$$F \underbrace{\xrightarrow{\sim}}_{\text{adj.}} \alpha_*\alpha^* F$$

[1] One then says that α is *cofinal,* cf. SGA 4, I 8.1, or 3.3, p. 45.
[2] Cf. improvement in the remarks p. 5.
[3] Cf. 3.2, p. 43-44, for necessary and sufficient conditions for cofinality.

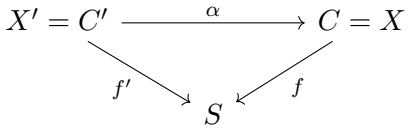
that the canonical adjunction morphism is an isomorphism. For this to be so, it suffices that $\forall\, x_0 \in C$, the functor $\alpha/x_0 : C'/x_0 \longrightarrow C/x_0$ induced by α be "anodyne for the $\varprojlim$ ", which is ensured when for every arrow $u : x_1 \longrightarrow x_0$ in C, the category $C'_{\vee u}$ formed of triples (x', ξ, η) with $x' \in \operatorname{Ob} C'$ and $\xi : x_1 \longrightarrow \alpha(x')$, $\eta : \alpha(x') \longrightarrow x_0$ rendering commutative the diagram [page 2]



p. 2

is weakly left filtering.

More generally, let us consider



and let us ask ourselves the question when the functorial homomorphism, with respect to a variable presheaf F on X,

$$f_*(F) \longrightarrow f'_*(\alpha^*(F))$$

is always an isomorphism. (The preceding case is the one where $S = X$, $f = \mathrm{id}_X$.) We find that it suffices that $\forall\, s_0 \in \operatorname{Ob} S$, the functor

$$\alpha/s_0 : X'/s_0 \longrightarrow X/s_0$$

be "anodyne for the $\varprojlim$ ", and for this, it suffices that for object (x_0, u_0) of X/s_0 (where

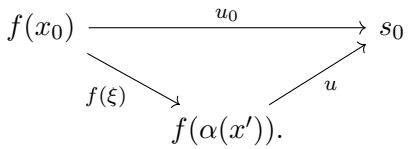
$$f(x_0) \xrightarrow{u_0} s_0)$$

the category $(x_0, u_0)\backslash(X'/s_0)$ be weakly left filtering.

This category is interpreted as that of triples (x', ξ, u) with [page 3]

$$\begin{cases} x' \in \operatorname{Ob} C' \\ u : f'(x') = f(\alpha(x')) \longrightarrow s_0 \\ \xi : x_0 \longrightarrow \alpha(x') \end{cases}$$

so as to render commutative the diagram



I would like to bring out variants of these statements in the cohomological setting, the functors $\varprojlim$ being replaced by $R\Gamma$'s (of abelian presheaves e.g. - but one also thinks of the "noncommutative cases", defined by derivators in particular.)

If I is a *finite ordered* category, the left filtering condition is equivalent to the existence of an initial object, which implies that I is contractible. In the preceding statements, it might be that the right philosophy is to replace "weakly left filtering" everywhere by "contractible". If e.g. C is the final category, then replace the condition that C' be left [page 4]

p. 3

filtering, by acyclicity. This is in any case in accord with the spirit of the axiom Der_5 (case where J is the final category). Indeed, in the case where $C = e$, the condition that α be anodyne for the $\varprojlim$ is visibly equivalent to saying that the functor $\alpha^* : \mathcal{M} \longrightarrow \underline{\mathrm{Hom}}(C'^o, \mathcal{M})$, associating to every $F \in \mathcal{M}$ the constant presheaf with value F on C', gives an adjunction morphism

$$\varprojlim_e F = F \longrightarrow \alpha_*(\alpha^*(F)) = \varprojlim_{C'} \alpha^* F$$

which is an isomorphism. And the axiom Der_5 ensures that it must be so if C' is contractible.

The filtering condition in the proposition is moreover a *sufficient* condition for anodyneness, but by no means necessary. If e.g. C is the final category, then the condition of anodyneness for $C' \longrightarrow C$ is simply that C' be connected and nonempty, i.e. $\pi_0(C') = \{0\}$. (Condition which visibly ignores the higher π_i.)

Let $\boxed{\alpha : X' \longrightarrow X}$ be a functor of small categories. The condition suggested by the proposition, that the $x_0\backslash X'$ be contractible, means that α is a co-strict hot-equivalence, i.e. that $\alpha^o : X'^o \longrightarrow X^o$ is a hot-equivalence "locally on X^o", hence also globally. If α itself were a strict hot-equivalence, instead of α^o, then we would have by Der_5, that for [page 5] $F \in \mathbf{D}(X)$, $F \xrightarrow{\sim} \alpha_*\alpha^* F$ ($\mathbf{D}$ a left derivator), which is much stronger than $\Gamma_{\mathbf{D}}(X, F) \xrightarrow[\sim]{\alpha^*} \Gamma_{\mathbf{D}}(X', \alpha^* F)$. The dual condition implies (if $\mathbf{D}$ right derivator) $\alpha_!\alpha^* F \xrightarrow{\sim} F$. But is it reasonable to expect that it also implies the condition "on the cohomology side, instead of homology",

$$\Gamma_{\mathbf{D}}(X, F) \xrightarrow{\sim} \Gamma_{\mathbf{D}}(X', \alpha^*(F)) \quad ?$$

The first case, which should carry conviction, is that of the derivator DerAb, i.e. of the derived categories of abelian sheaves:

Question. *Let $\alpha : X' \longrightarrow X$ be a co-strict* hot-*equivalence, F an abelian presheaf on X. Do we then have $H^i(X, F) \xrightarrow[\sim]{\alpha^*} H^i(X', \alpha^*(F))$ for every $i \in \mathbf{Z}$?*

If X admits an initial object $\emptyset$, then $\emptyset\backslash X' \simeq X'$, so X' contractible, so it's OK for F constant.

Remark. I realize that in proposition 1, instead of the condition that the $x\backslash C'$ be weakly left filtering, it would suffice that they be nonempty and connected, i.e.

0-connected ([4]). The fact that the $x\backslash C'$ nonempty implies that $\varprojlim_C \longrightarrow \varprojlim_{C'}$ is *injective.* For surjectivity, if $\xi' \in \varprojlim_{C'}$ (set-theoretic case), we define $\xi \in \varprojlim_C$ by the $\xi(x) \in F(x)$ with the help of [page 6]

$$x \xrightarrow{u} \alpha(x')$$

by taking $u^*(\xi'(x'))$ (where $\xi'(x') \in (\alpha^*(F))(x') = F(\alpha(x'))$), the essential point is to show that the $u^*(\xi'(x'))$ depend only on x, not on the choice of $(x', u) \in x\backslash C'$. Now for (x', u) variable in $x\backslash C'$, the map

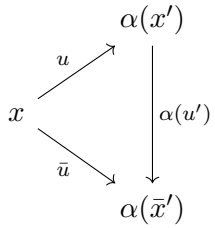
$$\begin{array}{rcl} \varphi : \mathrm{Ob}(x\backslash C') & \longrightarrow & F(x) \\ (x', u) & \longmapsto & u^*(\xi'(x')) \end{array}$$

is such that if there exists an arrow

$$(x', u) \xrightarrow{u'} (\bar{x}', \bar{u}) \qquad \text{where} \qquad u' : x' \longrightarrow \bar{x}'$$

[4] This also means that α is W_0-coaspheric. For a treatment ab ovo, cf. prop. 3.2, page 43.

(thus the diagram



is commutative), then $\varphi(x', u) = \varphi(\bar{x}', \bar{u})$. Thus φ is constant on the connected components of $x\backslash C'$, thus constant if $x\backslash C'$ is 0-connected, whence $\xi(x)$. We then prove at once that for variable x,

$$\xi \overset{\text{def}}{=} (\xi(x))_{x \in \operatorname{Ob} C} \in \varprojlim F,$$

and that $\alpha^*(\xi) = \xi'$, q.e.d.

We can say that for $x_0 \in \operatorname{Ob} C$ fixed, if we look at the *constant* presheaf on $x_0\backslash C'$ with value $F(x_0)$, say Φ_{x_0}, the map

$$(x', u) \longmapsto u^*(\xi'(x')) : x_0\backslash C' \longrightarrow F(x_0)$$

can be regarded as a section of this presheaf on $x_0\backslash C'$, defined by $\xi' \in \varprojlim_{C'} \alpha^* F$, so we have

$$\varprojlim_{C'} \alpha^* F \xrightarrow{c_{x_0}} \Gamma(x_0\backslash C', \Phi_{x_0}) \qquad (\xleftarrow{\sim} F(x_0) \text{ if } x_0\backslash C' \text{ 0-connected}).$$

If we have $\boxed{x_0 \xrightarrow{v} x_1}$ in C, we deduce from this

$$x_1\backslash C' \xrightarrow{V} x_0\backslash C'$$

(the $x\backslash C'$ are the fibers of a *fibered category* $C\backslash C'$ over C ...) and we have $F(x_1) \longrightarrow F(x_0)$, whence $\boxed{\Phi_{x_1} \longrightarrow V^*\Phi_{x_0}}$ for the corresponding constant sheaves on $x_1\backslash C'$ resp. $x_0\backslash C'$ and we have commutativity in

p. 5
[page 7]

$$\begin{array}{ccc}
\varprojlim_{C'} \alpha^* F & \xrightarrow{c_{x_0}} & \Gamma(x_0\backslash C', F(x_0)) \simeq \operatorname{Hom}(\pi_0(x_0\backslash C'), F(x_0)) \\
{\scriptstyle c_{x_1}}\downarrow & & \downarrow{\scriptstyle V^*} \\
\operatorname{Hom}(\pi_0(x_1\backslash C'), F(x_1)) \simeq \Gamma(x_1\backslash C', F(x_1)) & \xrightarrow[(F(x_1)\to F(x_0))]{v^*} & \Gamma(x_1\backslash C', F(x_0)) \simeq \operatorname{Hom}(\pi_0(x_1\backslash C'), F(x_0))
\end{array}$$

$$\begin{array}{ccc}
\pi_0(x_1\backslash C') & \xrightarrow{\pi_0(V)} & \pi_0(x_0\backslash C') \\
\vdots & & \vdots \\
F(x_1) & \xrightarrow{F(v)} & F(x_0)
\end{array}$$

commutative.

We consider the presheaf of sets

$$x \longmapsto \pi_0(x\backslash C') \qquad \text{on } C,$$

let us call it $\pi_0(C'/C)$, we will have a canonical morphism

$$\boxed{\varprojlim_{C'} \alpha^*(F) \xrightarrow{\sim} \operatorname{Hom}_{\substack{\text{presheaves} \\ \text{of sets on } C}} (\pi_0(C'/C), F),}$$

probably always isomorphism.

If we work in the derived category of presheaves of k-modules (k a ring), the preceding formula suggests conjecturally

$$\boxed{\mathbf{R}\Gamma(C', \alpha^* F) \simeq \mathbf{R}\underbrace{\mathrm{Hom}}_{\text{internal Hom}}(\underbrace{\mathbf{L}\alpha_!(k_{C'})}_{\substack{\text{relative}\\ \text{homology of}\\ C' \text{ over } C}}, F).}$$

The canonical augmentation

[page 8]

$$\mathbf{L}\alpha_!(k_{C'}) \longrightarrow k_C$$

gives back, by taking the $\mathbf{R}$Hom into F,

$$\mathbf{R}\Gamma(C, F) \longrightarrow \mathbf{R}\,\mathrm{Hom}(\mathbf{L}\alpha_!(k_{C'}), F) \xleftarrow{\sim} \mathbf{R}\Gamma(C', \alpha^* F)$$

the canonical homomorphism of "contravariance" of cohomology.

If on the other hand we work with the nonadditive derivator HOT, thus F a "relative homotopy type" over C, we must have

$$\boxed{\mathbf{R}\Gamma_{\mathrm{HOT}}(C', \alpha^* F) \xrightarrow{\sim} \mathbf{R}\Gamma(C\,,\ \underbrace{\mathrm{Hom}}_{\substack{\text{internal Hom}\\ \text{of the category of}\\ \text{relative homotopy}\\ \text{types over } C}}\ (\ \alpha_!\ (\ \underbrace{e_{C'}}_{\substack{\text{final object}\\ \text{of HOT}(C'),\\ \text{final relative}\\ \text{homotopy type}\\ \text{over } C'}}\ ),\ F)).}$$

In fact, these formulas are quasi-tautological, if we admit the existence of the functors $\alpha_!$ as *left* adjoints to the α^*. In the abelian case, say, the isomorphy of the H^0 of the two sides is the adjunction formula

$$\mathrm{Hom}_{C'}(k_{C'}, \alpha^* F) \simeq \mathrm{Hom}_C(\alpha_!(k_{C'}), F),$$

p. 6

[page 9]

whence the isomorphy for the H^i of the two sides, by shifting. Likewise in the case of HOT, the canonical isomorphy of the π_0 of the two sides is the adjunction formula. To have the finer isomorphism in $\mathbf{D}(e)$, the fundamental category of the derivator considered, in terms of internal Hom, we must admit the existence of unit objects (the k_X in one case, the e_X in the other), compatible with inverse images, in such a way $\mathrm{Hom\,int}(\varepsilon_X, \underbrace{F}_{\in \mathbf{D}_X}) \simeq F$. We will have

$$\Gamma(C', \alpha^* F) \overset{\mathrm{def}}{=} f'_*(\alpha^*(F)) \simeq f_*(\alpha_* \alpha^*(F)),$$

and the question is therefore to give an interpretation of $\alpha_* \alpha^* F$ as

$$\alpha_* \alpha^* F \simeq \mathrm{Hom\,int}(\alpha_!(\varepsilon_{C'}), F) \qquad \text{in } \mathbf{D}(C).$$

For this, we are going to write

$$\alpha^*(F) \simeq \mathrm{Hom\,int}_{C'}(\varepsilon_{C'}, \alpha^* F),$$

and the formula must follow from an "internal adjunction formula"

$$? \qquad \boxed{\alpha_*(\mathrm{Hom\,int}_{C'}(G', \alpha^* F)) \simeq \mathrm{Hom\,int}_C(\alpha_! G', F).}$$

(*"Internal" adjunction formula,* for $\alpha_!$ and α^*, valid in principle for *every* $\alpha : C' \longrightarrow C$ in Diag, every G' in $\mathbf{D}(C')$ and every F in $\mathbf{D}(C)$.)

How to prove this formula? By taking the product operation $\otimes$ associated with the operation Hom int, we are reduced (by taking the associated contravariant functors) to a formula

[page 10]

$$\alpha_!(\alpha^* G \otimes G') \xrightarrow{\sim} G \otimes \alpha_! G'.$$

We find by adjunction a canonical homomorphism, we must see whether it is always isomorphism.

p. 7

[page 11]

2 Locally filtering categories are the locally totally 0-connected categories

Let $\mathcal{T}$ be a topos. We say that $\mathcal{T}$ is *irreducible* if the ordered set $\mathrm{Ouv}(\mathcal{T})$ of subobjects of $e_{\mathcal{T}}$ has the following property: a) $\mathrm{Ouv}(\mathcal{T})$ is not reduced to one element, i.e. $\emptyset_{\mathcal{T}} \longrightarrow e_{\mathcal{T}}$ is not an isomorphism ($\mathcal{T}$ is not identical to its empty subtopos). b) If U', U'' are two "nonempty" objects, i.e. noninitial, of $\mathrm{Ouv}(\mathcal{T})$, their intersection is "nonempty". It amounts to the same for b) to say that if $\mathcal{T}'$, $\mathcal{T}''$ are two closed subtopoi distinct from $\mathcal{T}$, then their supremum $\mathcal{T}' \cup \mathcal{T}''$ is distinct from $\mathcal{T}$, or again that every open of $\mathcal{T}$ is connected. We say that $\mathcal{T}$ is *locally irreducible* if a) $\mathcal{T}$ is locally connected (i.e. each sheaf on $\mathcal{T}$ is a sum of 0-connected sheaves), and if b) for every 0-connected sheaf U on $\mathcal{T}$, the induced topos $\mathcal{T}/U$ is irreducible (i.e. the ordered set $\mathrm{Ouv}(U)$ of subobjects of U has property b) above (a) simply means that $U \not\simeq \emptyset_{\mathcal{T}}$, i.e. that U is not an initial object in $\mathcal{T}^\wedge$; or more simply): every subsheaf of a connected sheaf is connected ([5]) ([6]).

When $\mathcal{T}$ is of the form $\mathrm{Top}(T)$, where T is a topological space, then $\mathcal{T}$ irreducible means that T is irreducible (for then $\mathrm{Ouv}(\mathcal{T}) \simeq \mathrm{Ouv}(T)$). This already implies trivially that T is locally connected (every open is a disjoint sum of connected opens), thus $\mathrm{Top}(T)$ is locally connected. Moreover, this also implies that $\mathrm{Top}(T)$ is locally irreducible ([7]). (**NB** The 0-connected objects of $\mathrm{Top}(T)^\wedge$, i.e. the 0-connected sheaves of sets on T, are those whose "generic fiber" has exactly one point …) Thus $\mathcal{T} = \mathrm{Top}(T)$ is locally irreducible if and only if T locally connected, and the connected components of T are irreducible, i.e. if and only if $T = \coprod_i T_i$, the T_i irreducible. [page 12]

If $\mathcal{T}$ is of the form $\mathrm{Top}(X)$, X a category, we know that $\mathcal{T}$ is locally connected (and even locally ∞-connected). We see on the other hand that $\mathcal{T}$ is irreducible if and only if $\mathrm{Esp}(X)$ (the associated topological space, whose underlying set is $\mathrm{Ob}\, X$) is irreducible ([8]); and we then say that X is irreducible ([9]). As the connected components of X and those of $\mathrm{Esp}(X)$ or $\mathrm{Top}(X) = \mathcal{T}$ correspond to one another, we see therefore that the following conditions are equivalent: a) the topos $\mathrm{Top}(X)$ is locally irreducible, b) $\mathrm{Esp}(X)$ is locally irreducible, c) the connected components of X are irreducible, d) the X/x irreducible. We then say that X is locally irreducible ([10]). p. 8

This therefore implies that for every X' étale over X, with X' 0-connected, X' is irreducible.

State equivalent conditions.

a) $\mathrm{Top}(X)$ is locally irreducible.

b) The X/x irreducible.

c) Every 0-connected object of $X^\wedge$ is irreducible.

---

[5] **NB** Irreducible does not imply locally irreducible, any more than connected does not imply locally connected etc.

[6] $\mathcal{T}$ is locally irreducible if and only if it has a set of generators (U_α), with the U_α irreducible (hence 0-connected, hence $\mathcal{T}$ is locally connected).

[7] make the proof explicit.

[8] for the opens of $\mathrm{Top}(X)$, those of $\mathrm{Esp}(X)$ and those of X correspond to one another.

[9] This also means that for x, y in X, $\exists\, z$ with $z \nearrow x$, $z \searrow y$, i.e. $\mathrm{Ord}(X)$ is left filtering.

[10] False, we have (a $\iff$ d) $\implies$ (b $\iff$ c) X irreducible does *not* imply that the X/x are so, example $X = (0 \underset{\beta}{\overset{\alpha}{\rightrightarrows}} 1)$ $(\alpha \neq \beta)$, we have $X/1 = 0' \searrow 1 \nearrow 0''$, which is *not* irreducible, nor locally irreducible since it is connected. Yet it is étale over X.

d) The connected components of X are weakly cofiltering (i.e. weakly left filtering).

e) Condition PS 1_g of SGA 4, I 2.

If X is category, X irreducible also means that $X \neq \emptyset$ and $\forall x, y \in \mathrm{Ob}\, X$, exists s with $s \nearrow x$, $s \searrow y$. (This also means that X is *weakly filtering* ([11]) on the left, or weakly cofiltering.) X locally irreducible means that for x, y *belonging to the same connected component,* exists s with $s \nearrow x$, $s \searrow y$, i.e. $X/[x] \cap X/[y] \neq \emptyset$ (where $[x]$ is regarded as an element of the ordered set $\mathrm{Ob}\, X$). It again amounts to the same to say that the induced categories X/x are irreducible ([12]). [page 13]

Example. A category *having an initial object,* or merely a "smallest object", is irreducible. Conversely, if the *ordered* set $\mathrm{Ord}(X)$ associated with X (quotient of $\mathrm{Ob}(X)$ by the equivalence relation associated with its preorder) is finite, e.g. X finite, or only locally finite, e.g. X locally finite (i.e. the categories X/x finite), then if X is irreducible, $\mathrm{Ob}\, X$ has a smallest element, i.e. X has a smallest object. This is true more generally still, if $\mathrm{Ord}(X)$ is artinian, i.e. every decreasing sequence in $\mathrm{Ord}(X)$ is stationary.

But the interesting condition is the topos-theoretic condition of *local* irreducibility, which is equivalent to: I left pseudo-filtering, i.e. the connected components of I filtering, i.e. conditions PS 1, PS 2

PS 1 Every diagram $x \searrow$, $y \nearrow$ s can be completed to $z \nearrow x$, $z \searrow y$, $x \searrow s$, $y \nearrow s$ *commutative.*

PS 2 For every double arrow $x \rightrightarrows y$, exists an equalizer $z \longrightarrow x \rightrightarrows y$.

In summary, we have the p. 9 [page 14]

Proposition 2.1. *Let X be an object of* Cat. *Equivalent conditions:*

a) The associated topos $\mathrm{Top}(X)$ *is locally irreducible, i.e. (being locally connected) for every object F of $X^\wedge$ and every subobject F' of F, if F is connected, so is F'.*

a′) For every X' étale over X, if X' is connected, every open of X' is connected.

a″) Every open of a localized category X/a ($a \in \mathrm{Ob}\, X$) is connected (i.e. X/a is an irreducible category, i.e. the topos $\mathrm{Top}(X/a)$ *is).*

b) For every object a of X, $\mathrm{Esp}(X/a)$ *is an irreducible topological space.*

c) The categories X/a are weakly left filtering (or weakly cofiltering), i.e. for two objects ξ, η of such a category, exists an object ζ and two arrows $\zeta \nearrow \xi$, $\zeta \searrow \eta$ (i.e. the ordered set Ord(X/a) associated with X/a is left filtering)

[11] It is better to call it *pseudo-filtering,* and let us reserve weakly filtering for: locally weakly filtering + connected.

[12] false, cf. marginal annotation above.

c′) For every diagram in solid lines in X, $\begin{array}{ccccc} & & x & & \\ & \nearrow & & \searrow & \\ z & & & & s \\ & \searrow & & \nearrow & \\ & & y & & \end{array}$ *, we can complete it to a* commutative *square as indicated. (Condition* PS 1 *for locally filtering categories.)*

These conditions imply that every connected component X_α of X is irreducible, or what amounts to the same thing, that $\mathrm{Esp}(X_\alpha)$ is irreducible,
but are not implied by the latter. [page 15]

Counterexample, the 0-connected category $0 \underset{\beta}{\overset{\alpha}{\rightrightarrows}} 1$, with $\alpha \neq \beta$. We have $X/1 \simeq$ $\begin{array}{ccc} 0 & & \\ & \searrow & \\ & & 1 \\ & \nearrow & \\ 0' & & \end{array}$, which is *not* irreducible, since $\Phi\backslash\{1\} \simeq \varepsilon$ is a disconnected open .

I would like a "topossic" condition on X, equivalent to X "left pseudo-filtering", i.e. the connected components filtering, i.e. PS 1^{opp} (condition c′) above) + PS 2^{opp} (double arrow condition), and which moreover implies that every 0-connected object of $X^\wedge$ (i.e. every object X' of Cat étale over X and 0-connected) is W_∞-aspheric. And I am thinking of the notion of "total asphericity" of Pursuing Stacks.

Definition 2.2. ([13]) Let X be a category.

a) We say that X is *weakly filtering* ([14]) if $X \neq \emptyset$, and if for $a, b \in \mathrm{Ob}\, X$, $\exists\, c$ and arrows c $\begin{array}{ccc} & a & \\ \nearrow & & \\ & & \\ \searrow & & \\ & b & \end{array}$, i.e. $X/[a] \cap X/[b] \neq \emptyset$. (**NB** X is weakly left filtering ([15]) if and only if $\mathrm{Ord}(X)$ is, i.e. is left filtering. **ReNB** X weakly filtering $\Longrightarrow$ X 0-connected.)

p. 10

b) We say that X is *locally weakly left filtering* or more simply left prefiltering if the categories X/x ($x \in \mathrm{Ob}\, X$) are weakly left filtering, i.e. if X satisfies the condition: $\forall$ diagrams $\begin{array}{ccc} & a & \\ & \searrow & \\ x & & \\ & \nearrow & \\ & b & \end{array}$, $\exists\, c$ in X and arrows from c to a, b making the square *commutative*

$$\begin{array}{ccccc} & & a & & \\ & \nearrow & & \searrow & \\ c & & & & x \\ & \searrow & & \nearrow & \\ & & b & & \end{array}$$

([16]). (**NB** This condition implies that $\mathrm{Ord}(X)$ is locally weakly filtering, but is not equivalent to the latter. Example, $X = 0 \underset{\beta}{\overset{\alpha}{\rightrightarrows}} 1$, $X/1 =$ $\begin{array}{ccc} & & 0' \\ & \searrow & \\ 1 & & \\ & \nearrow & \\ & & 0'' \end{array}$ is not weakly left filtering.)

NB If X locally weakly left filtering, every X' étale over X likewise. [page 16]

[13] Cf. summary table pages 97, 98 and especially 99.

[14] rather say *pseudo-filtering*, reserve "weakly filtering" for "locally weakly filtering" (or *irreducible*) + connected, condition PS $1'_{\mathrm{g}}$.

[15] pseudo-filtering.

[16] Condition PS 1_{g}.

NB ([17]) X locally weakly left filtering $\iff$ the connected components are weakly left filtering, or again if X is 0-connected, X locally weakly left filtering $\iff$ X is weakly left filtering ([18]). We have indeed $\Longleftarrow$ tautologically ([19]), and $\Longrightarrow$ by showing that if PS 1 satisfied, then $PS(a,b)$ defined by $X/[a] \cap X/[b] \neq \emptyset$ is an equivalence relation, implied by the relation $C(a,b)$: $\{\mathrm{Hom}(a,b) \neq \emptyset$ *or* $\mathrm{Hom}(b,a) \neq \emptyset\}$, hence the equivalence relation (homotopy) generated by $C(a,b)$ is finer than $PS(a,b)$.

c) We say that X is *left filtering*, or *cofiltering,* if it is weakly left filtering ([20]) (hence *nonempty*), and satisfies (in addition to condition PS $1'_{\mathrm{g}}$ of a)) the following condition PS 2_{g}:

For every double arrow $a \underset{v}{\overset{u}{\rightrightarrows}} b$, exists arrow $c \xrightarrow{w} a$ such that $uw = vw$ ([21]).

It amounts to the same to say that X satisfies the three conditions a′) X is 0-connected (stronger than $X \neq \emptyset$), b′) PS 1, i.e. X locally weakly left filtering and c′) PS 2 as before.

d) We say that X is *locally left filtering* if the connected components are left filtering, i.e. if X satisfies conditions PS 1 and PS 2 ("pseudo-cofiltering" in the terminology of SGA 4). **NB** If X is locally left filtering, the same holds for every X' étale over X, for this is obvious for PS 1 which concerns the X/x, and for PS 2 this follows from the fact that $X' \longrightarrow X$ is faithful. So if X is locally cofiltering, the X/a are filtering, but the converse is false.

We have the dual notions: *weakly right filtering* (or simply weakly filtering), and *right filtering* or simply filtering. (Dual notions of a), c) above.) For the dual notion of b), namely that the $x\backslash X$ be weakly filtering, we can no longer call it "locally weakly filtering", but it would be *co*locally weakly filtering. p. 11

Example. $X = \left(0 \xrightarrow{\gamma} 1 \underset{\beta'}{\overset{\alpha'}{\rightrightarrows}} 2 \underset{\beta}{\overset{\alpha}{\rightrightarrows}} 3\right)$ with $(\alpha\alpha' = \beta\beta') \neq (\alpha\beta' = \beta\alpha')$.

Only the notion a) (weakly left filtering ([22])) depends only on $\mathrm{Ord}\, X$ - but of course if X is *pre*ordered, all three depend only on $\mathrm{Ord}\, X$. Let us note that saying it depends only on $\mathrm{Ord}\, X$, or only on $\mathrm{Esp}\,\mathrm{sep} X$, amounts to the same thing, since the two determine each other. Condition a) means that $\mathrm{Esp}\, X$, or also $\mathrm{Top}(X)$, is *irreducible.* We shall say that X is *irreducible* (instead of pseudo-filtering). The other three notions cannot be expressed in terms of $\mathrm{Esp}\, X$ or $\mathrm{Ord}\, X$ alone, but they can be expressed in terms of the $\mathrm{Topos}\, X$. Condition b) (X locally weakly filtering) means that $\mathrm{Top}(X)$ is *locally irreducible,* and we shall also say that X is *locally irreducible* (instead of locally weakly filtering). This means that for every X' étale over X and 0-connected, X' is weakly left filtering, i.e. is irreducible. [page 17]

It remains to express the filtering condition itself in terms of $\mathrm{Top}\, X$. I recall in this connection:

---

[17] **False.**

[18] pseudo-filtering.

[19] **false,** thus $0 \underset{\beta}{\overset{\alpha}{\rightrightarrows}} 1$ is 0-connected and weakly left filtering, but not locally weakly left filtering because

$$\begin{array}{ccc} 0 & \overset{\alpha}{\searrow} & \\ & & 1 \\ 0 & \underset{\beta}{\nearrow} & \end{array}$$

cannot be completed to a *commutative* square.

[20] pseudo-filtering.

[21] i.e. three conditions a) $X \neq \emptyset$ b) usual PS 1′ c) PS 2 (double arrow).

[22] pseudo-filtering.

Proposition 2.3. *The following conditions on a topos $\mathcal{X}$ are equivalent (when S (as a "site") denotes a generating subset of* $\mathrm{Fais}(\mathcal{X}) \simeq S^{\sim}$, consisting of 0-connected objects ([23])*.)*

a) If U, V are two 0-connected objects of $\mathrm{Ob}\,\mathcal{X}$*, $U \times V$ is 0-connected.*

b) If U is a 0-connected object of $\mathcal{X}^{\wedge}$, then $U \longrightarrow e$ is "W_0-aspheric", i.e. for every V, the projection $V \times U \longrightarrow V$ induces $\pi_0(U \times V) \xrightarrow{\sim} \pi_0(V)$.

c) If U, V in $\mathcal{X}^{\wedge}$, then

$$\pi_0(U \times V) \xrightarrow{\sim} \pi_0(U) \times \pi_0(V).$$

d) As in a), with U, V in S. (But we assume the objects of S to be 0-connected.)

([24]).

Moreover, these conditions imply that $\mathcal{X}$ is connected, hence 0-connected if $\mathcal{X} \neq \emptyset$. This moreover implies that $\mathcal{X}$ is W_∞-aspheric, cf. Pursuing Stacks §35 (Corollary p. 52) ([25]). [page 18]

It is essentially Proposition 1 of Pursuing Stacks §65, cf. loc. cit. p. 170-171 ([26]). Cf. also Pursuing Stacks, §35, pages 50-55.

Definition 2.4. We say that $\mathcal{X}$ is *totally 0-connected* if it is locally connected, nonempty and satisfies the equivalent conditions of 2.3. (NB this is a special case of the notion "totally W-aspheric" of loc. cit. Thus we have the notion of totally n-connected.) p. 12

Let us note that for an object X of Cat, saying that $\mathrm{Top}(X)$ is totally 0-connected amounts to saying that $X \neq \emptyset$, and that for $x, y \in \mathrm{Ob}\,X$, $x \times y$ in $X^{\wedge}$ is 0-connected, i.e. $X/x \times y$ is 0-connected ([27]). It is a very strong condition. We shall also say that X is *totally 0-connected.* Let us note here

Proposition 2.5. *Let $\mathcal{X}$ be a topos. If $\mathcal{X}$ is totally 0-connected, then $\mathcal{X}$ is irreducible. (In fact the conclusion holds, tautologically, assuming only that the product of two 0-connected objects of $\mathcal{X}^{\wedge}$ is nonempty, and it even suffices to assume this for subobjects of the final object $e_{\mathcal{X}}$.) In particular, if $\mathcal{X} = Top(X)$, where X is in* Cat*, then X is weakly left filtering.*

But of course, this does *not* imply that $\mathcal{X}$ is locally irreducible.

For example the category $\mathbf{\Delta}$, or any small category stable under binary products, is totally 0-connected, but $\mathbf{\Delta}$ is not locally irreducible (the induced categories $\mathbf{\Delta}/\mathbf{\Delta}^n$, for $n > 0$, are visibly *not* irreducible). (But let us recall that if $\mathcal{X} = \mathrm{Top}(T)$, T a topological space, then T irreducible implies that $\mathcal{X}$ is also *locally* irreducible. So T totally 0-connected implies $\mathrm{Top}(T)$ globally *and* locally irreducible, i.e. for every T' étale over T, and 0-connected, T' is irreducible.) [page 19]

Theorem 2.6. *Let $\mathcal{X}$ be a locally connected topos, $S \subseteq \mathcal{X}^{\wedge}$ a small generating subset consisting of 0-connected objects. Equivalent conditions (*except *c, c′, which seem* ([28]) *weaker, cf. proof).)*

a) Every induced topos $\mathcal{X}/U$ over U 0-connected in $\mathcal{X}^{\wedge}$ is totally 0-connected.

a′) Every fiber product $U \times_W V$ of 0-connected objects in $\mathcal{X}^{\wedge}$ is 0-connected.

b) $\mathcal{X}$ is locally irreducible, and its connected components are totally 0-connected.

---

[23] thus $\mathcal{X}$ is locally 0-connected.

[24] The proof should be made explicit together with that of 2.9 below (p. 25).

[25] **False,** loc. cit. says nothing of the kind.

[26] But there I restrict myself to topoi of the form $\mathrm{Top}(A)$, with A in Cat.

[27] This also means that $\mathrm{diag}_X : X \longrightarrow X \times X$ is co-cofinal, i.e. $X^o \longrightarrow X^o \times X^o$ is cofinal, and implies that the same holds for $X \longrightarrow X^E$ for every *finite* set E.

[28] **are.**

c) As in a), with U in S, i.e. the $\mathcal{X}/U$ ($U \in \mathrm{Ob}\, S$) are totally 0-connected.

c′) As in a′), with U, V, W in S.

d) S is locally left filtering, i.e. S satisfies the two conditions

PS* 1 *Every diagram* $\begin{array}{ccc} a & & \\ & \searrow & \\ & & x \\ & \nearrow & \\ b & & \end{array}$ *in S fits into* $\begin{array}{ccccc} & a & & & \\ c & \cdots\nearrow & & \searrow & x \\ & \cdots\searrow & & \nearrow & \\ & b & & & \end{array}$ *which is commutative.*

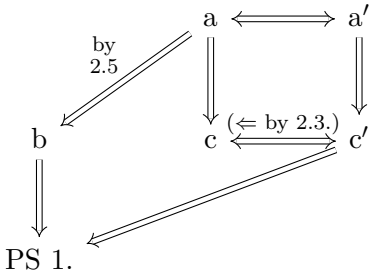
PS* 2 *Every diagram $a \rightrightarrows b$ fits into a commutative diagram $c \longrightarrow a \rightrightarrows b$.*

([29]).

These conditions imply that the 0-connected objects of $\mathcal{X}^\wedge$ (in particular the elements of S) are ∞-connected. p. 13 [page 20]

Corollary 2.7 (of the equivalence b $\iff$ d). *Let X be an object of* Cat. *For X to be left filtering it is necessary and sufficient that $Top(X)$ be totally 0-connected and locally (and therefore also globally) irreducible (or again that X be totally 0-connected and that for every $x \in \mathrm{Ob}\, X$, the localized category X/x be irreducible (it is even totally 0-connected), or again that X be nonempty and that for two objects a, b in X, $a \times b \in X^\wedge$ be 0-connected, and for* $\begin{array}{ccc} a & & \\ & \searrow & \\ & & x \\ & \nearrow & \\ b & & \end{array}$ *in X, $a \times_x b$ (in $X^\wedge$) be 0-connected)* ([31]).

Proof of 2.6. We have the diagram of ± tautological implications



Let us nevertheless show the two implications c′) $\Longrightarrow$ PS 1 and b $\Longrightarrow$ PS 1.
This is contained in the [page 21]

Corollary 2.8. *Let $\mathcal{X}$, S be as in 2.6. Equivalent conditions:*

a) $\mathcal{X}$ is locally irreducible, i.e. the induced topoi $\mathcal{X}/U$, where U is 0-connected, are irreducible.

b) Every fiber product $U \times_W V$ of 0-connected objects U, V, W is "nonempty".

b′) Every fiber product $U \times_W V$, with U, V "nonempty", W 0-connected, is nonempty.

c) The $\mathcal{X}/x$ with $x \in S$ are irreducible.

[29] NB We have equivalence of (a), a′), b), d),) which imply c), c′), which are equivalent. But it seems ([30]) that c), c′) are strictly weaker. Cf. however the variants e), e′) in 2.12 (p. 28, 29).

[30] It turns out. It is indeed true, c) and c′) do *not* imply the other conditions, cf. marginal annotation p. 28 (case of $X = B_G$).

[31] Example of X totally 0-connected, but not cofiltering, i.e. not locally irreducible? The category $\mathbf{\Delta}$ of standard simplices is totally 0-connected, but is not cofiltering because $\mathbf{\Delta}^0 = e \overset{\partial_0}{\underset{\partial_1}{\rightrightarrows}} \mathbf{\Delta}^1$ cannot be equalized! So $\mathbf{\Delta}^o = I$ is such that $I \longrightarrow I^E$ is cofinal for every finite set E, but I not filtering.

d) As in b), with U, V, W in S.

e) S is locally weakly left filtering, i.e. satisfies PS 1.*

Here the $\pm$ obvious implications are

p. 14

$$\begin{array}{ccccc} \mathrm{b}' & \Longrightarrow & \mathrm{b} & \Longrightarrow & \mathrm{a} \\ & & \Downarrow & & \Downarrow \\ & & \mathrm{d} & \Longleftrightarrow \mathrm{e} \Longleftrightarrow & \mathrm{c}. \end{array}$$

Let us prove the implications on the bottom line. Let us consider d). If $\begin{smallmatrix} & a & \searrow \\ & & & x \\ & b & \nearrow \end{smallmatrix}$ in S, to say that the fiber product in $\mathcal{X}$ is "nonempty" means, since S is a system of generators and in particular *nonempty,* that there exists c in S and $c \longrightarrow a \times_x b$, or what amounts to the same, $\begin{smallmatrix} & a & \\ c & \cdots & \\ & b & \end{smallmatrix}$, making the diagram $\begin{smallmatrix} & a & \searrow & \\ c & \cdots & & x \\ & b & \nearrow & \end{smallmatrix}$ commutative. Thus d) is equivalent to PS 1, i.e. to e) ($\mathcal{X}$ locally weakly left filtering). Let us consider condition c): the $\mathcal{X}/a = (S/a)^{\sim}$ are irreducible. This means that for two *nonempty* subobjects $U, V \subseteq a$ in $\mathcal{X}^{\wedge}$, we have $U \cap V \neq \emptyset$, or else that for two sieves

$\underline{U}$, $\underline{V}$ of S/x, if $\underline{U}$, $\underline{V}$ nonempty (which means that the subobjects of x in $\mathcal{X}^{\wedge}$ which they define are "nonempty", or that the objects of S are "nonempty" in $\mathcal{X}^{\wedge}$), then $\underline{U} \cap \underline{V}$ is nonempty. We may suppose that $\underline{U}$, $\underline{V}$ is each generated by a single arrow $a \xrightarrow{u} x$, $b \xrightarrow{v} x$. The intersection sieve consists of the objects $c \longrightarrow x$ of S/x such that $c \longrightarrow x$ factors through a $\begin{smallmatrix} & a & \\ c & \cdots & \\ & b & \end{smallmatrix}$, so to say that this sieve is not empty means that PS 1 is satisfied for the two given arrows u, v. Thus c) is equivalent to condition PS 1 e.

[page 22]

Thus we have established

$$\mathrm{b}' \Longrightarrow \mathrm{b} \Longrightarrow \mathrm{a} \Longrightarrow (\mathrm{c} \Longleftrightarrow \mathrm{d} \Longleftrightarrow \mathrm{e}),$$

it remains to prove that $\mathrm{c} \Longrightarrow \mathrm{b}'$. So let $\begin{smallmatrix} & U & \searrow & \\ & & W & \\ & V & \swarrow & \end{smallmatrix}$ be a diagram of "nonempty" objects, with W 0-connected, we must prove that the fiber product is nonempty. Considering $a \longrightarrow U$ and $b \longrightarrow V$ with $a, b \in S$, we are reduced to the case where $U = a$, $V = b$ in S. Let us prove that if $a \times_W b$ is "empty", then W is disconnected. For this, consider the subcategory C of S/W consisting of the $x \longrightarrow W$ such that $a \times_W x$ is "nonempty". Denoting by $w(a)$, $w(x)$ the image of a, x in W, this means that $w(a) \cap w(x) \neq \emptyset$.

[page 23]

Note that any quotient in $\mathcal{X}^{\wedge}$ of an irreducible object is irreducible, so $w(a)$, $w(x)$ are irreducible objects. This shows that for x in C and x' in S/W, we have x' in C, i.e. $w(a) \cap w(x') \neq \emptyset$ if and only if $w(x) \cap w(x') \neq \emptyset$. (Direct lemma: in an object W of a topos $\mathcal{X}$, if we have irreducible subobjects W_1, W_2, W_3, if one of the two meets the other two, these meet.) This shows that C is a sieve, since $x' \longrightarrow x$ implies $w(x') \subseteq w(x)$, so $w(x')$ meets $w(x)$, hence also $w(a)$. But then the cosieve $C' = (S/W) \backslash C$ is also a sieve, for if c is in C', i.e. $w(c)$ does not meet $w(x)$, a fortiori not any x' over x, for which we will have $w(x') \subseteq w(x)$. Thus, S/W is decomposed as a sum of two opens C, C', nonempty since one contains a, the other b. Consequently, W is decomposed as a sum of two nonempty subobjects, so W is not 0-connected.

p. 15

We have proved 2.8, hence also (returning to 2.6) the implications toward PS 1 in the diagram on page 20. It remains to prove

$$\begin{cases} \mathrm{b} \Longrightarrow \text{PS 2}, \ \mathrm{c}' \Longrightarrow \text{PS 2} \\ \text{PS 1} + \text{PS 2 (i.e. d)} \Longrightarrow \mathrm{a}' \end{cases}$$

([32]).

[page 24]

We therefore place ourselves under the hypothesis where PS 1 is satisfied (since b), c′) and d) imply PS 1), thus where we are under the equivalent conditions of 2.8 - $\mathcal{X}$ is locally irreducible. Let us then consider condition PS 2, and prove that it is implied both by b), and by c′), and that it implies a′). It means that for a double arrow $a \overset{u}{\underset{v}{\rightrightarrows}} b$ in S, the cokernel in $S^\wedge$, or, what amounts to the same, in $\mathcal{X}^\wedge = S^\sim \subseteq S^\wedge$, is "nonempty". We can write it in the form of a fiber product

$$\begin{array}{ccc} a & \xrightarrow{(u,v)} & b \times b \\ \uparrow & & \uparrow{\scriptstyle \mathrm{diag}} \\ \mathrm{Ker}(u,v) & \longrightarrow & b. \end{array}$$

Here by hypothesis a and b are 0-connected objects of $\mathcal{X}^\wedge$ (those of S being so), b) holds (($\mathcal{X}$ locally irreducible and) its connected components totally 0-connected), then $b \times b$ is 0-connected as a product of two 0-connected objects over the *same connected component,* which is totally 0-connected. Since $\mathcal{X}$ is locally irreducible, it follows that the fiber product is nonempty (2.8), hence b) $\Longrightarrow$ d). Let us prove d) $\Longrightarrow$ a), i.e. if

[page 25]

S is locally cofiltering, then the induced topoi $\mathcal{X}/W$, where W is 0-connected, are totally 0-connected. This induced topos is described by the site S/W, i.e. S/W is a system of generators. Now if S is locally filtering, S/W (étale over S) is so too. So we may suppose $W = e$, and to prove that if S is locally cofiltering and $\mathcal{X}$ 0-connected, then $\mathcal{X}$ is totally 0-connected. Note that $\mathcal{X}$ 0-connected is equivalent to S 0-connected - one should make this explicit all the same:

Lemma 2.9. *Let $\mathcal{X}$ be a topos, $S \subseteq \mathcal{X}^\wedge$ a full generating subcategory consisting of objects* that are 0-connected. *Then $\mathcal{X}$ is* 0*-connected if and only if S is - more generally, the ordered set of both open and closed subobjects of S in* Cat *(or else of the open subobjects = sieves of S that are also closed, i.e. cosieves) is isomorphic to that of the direct summands of $\mathcal{X}$.*

p. 16

I used it tacitly in the proof of 2.8, for the induced topos $\mathcal{X}/W$. I do not spell out the proof.

Thus (returning to the proof of d) $\Longrightarrow$ a) in 2.6.) S being 0-connected and locally cofiltering,

[page 26]

is in fact cofiltering. So for two objects a, b of S, $\exists$ c with $c \longrightarrow a$, $c \longrightarrow b$ (condition PS 1′, stronger than PS 1). To prove that $\mathcal{X}$ is totally 0-connected, we are reduced to proving by 2.3 that the $a \times b$ in $\mathcal{X}^\wedge$ are 0-connected. By 2.9, it amounts to the same to prove that they are 0-connected in $S^\wedge$. So we are reduced to proving the

Lemma 2.10. *Let S be a cofiltering category. If a, b are two objects of S, $a \times b$ is an object that is* 0*-connected in $S^\wedge$.*

We see at once that it suffices to prove (and in fact it amounts to the same, taking into account that $a \times b$ is locally irreducible) that if we have two arrows

$$\begin{array}{ccc} x & \longrightarrow & a \times b \\ & & \uparrow \\ & & y, \end{array}$$

[32] NB in the end we do not prove c′ $\Longrightarrow$ PS 2.

then $x \times_{a\times b} y \neq \emptyset$, i.e. $\exists\ z$ in S making commutative

$$\begin{array}{ccc} x & \longrightarrow & a \times b \\ \uparrow & & \uparrow \\ z & \longrightarrow & y. \end{array}$$

Now by PS 1′ there does indeed exist a z mapping to x and to y, but the two composite arrows

$$z \rightrightarrows a \times b,$$

i.e. each of the double arrows

$$z \rightrightarrows a \qquad (*)$$

$$z \rightrightarrows b \qquad (*')$$

may not be equal. But by PS 2,

we can find [page 27]

$$z' \longrightarrow z$$

equalizing the double arrow $(*)$, so we may suppose that this one is already trivial. p. 17 Applying PS 2 once again, by a $z'' \longrightarrow z'$, we similarly equalize the double arrow $(*')$, and we win. We have therefore proved, finally,

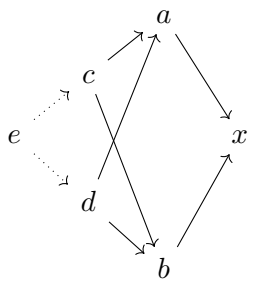
$$\begin{array}{ccccccc} \mathrm{a} & \Longleftarrow & \mathrm{a}' & \Longrightarrow & \mathrm{b} & \Longleftarrow & \mathrm{d}\ (=\mathrm{PS}\ 1 + \mathrm{PS}\ 2) \\ \Downarrow & & & & & & \\ \mathrm{c} & \Longleftarrow & \mathrm{c}', & & & & \end{array}$$

but not the reverse implication. In other words, if the induced topoi $\mathcal{X}/a$, for a in S, are totally 0-connected, I have not proved that the same holds for the $\mathcal{X}/W$, W a 0-connected object *whatsoever* of $\mathcal{X}^\wedge$. Thus, we are at the following point

Perplexity 2.11. *Let X be in* Cat. *If X is locally left filtering, then the fiber products $a \times_x b$ of objects in X are objects that are* 0*-connected in $X^\wedge$, and this condition in turn implies PS 1, i.e. that X is locally irreducible. But in the presence of this hypothesis PS 1, or X locally irreducible, the condition on the fiber products $a \times_x b$ is equivalent to the following: for every diagram solid lines* [page 28]

$$\begin{array}{ccc} c & \longrightarrow & a \times_x b \\ \vdots\uparrow & & \uparrow \\ e & \cdots\cdots\rightarrow & d, \end{array} \qquad \textit{with } x,\ a,\ b,\ c,\ d \textit{ in } X,$$

we can complete it to a diagram that is commutative, *with e in X. In other words, every commutative diagram in solid lines below can be completed by adding to it an "initial object" e*



([33]).

This condition is thus implied by X locally cofiltering, and from what precedes it is also satisfied more generally if and only if ([34]) the X/x are cofiltering ([35]). But is the converse true - does this condition of total 0-connectedness of the X/x imply that X is totally 0-connected? But I have not constructed an example where the X/x are cofiltering p. 18
without X being locally cofiltering (or what amounts to the same, of X 0-connected with the X/x cofiltering without X thereby being cofiltering)([36]) ; nor an example where the X/x are totally 0-connected, without the X/x being cofiltering. But let us note the

Corollary 2.12: *The equivalent conditions a), a′), b), d) (excluding c), c′), apparently weaker) of theorem 2.6*

are also equivalent to the following [page 29]

e) $\forall$ *x in S, the induced topos* $\mathcal{X}/x$ *and the induced topos* $\mathcal{X}/x \times x$ *is irreducible* ([37]),

or else to the following

e′) $\forall$ *x in S,* $S^\wedge/x$ *and* $S^\wedge/x{\times}x$ *are irreducible, i.e.* S/x *and* $S/x{\times}x$ *are irreducible* ([38]).

This is the apparently weakest "topological" form that I can find for the condition S locally cofiltering, i.e. $\mathcal{X}$ (or Top(S), at will) locally totally 0-connected.

Definition 2.13. We say that a topos $\mathcal{X}$ is *locally totally* 0*-connected* if it satisfies the equivalent conditions a, a′, b, d of theorem 2.6, i.e. if $\mathcal{X}$ is locally connected and if in addition every 0-connected induced topos is totally 0-connected - the fiber product of two 0-connected objects over a third is 0-connected. It amounts to the same to say that every (or that one) full generating subcategory S of $\mathcal{X}^\wedge$ consisting of 0-connected objects is locally cofiltering (still assuming $\mathcal{X}$ locally connected, or what amounts to the same, the existence of an S consisting of 0-connected objects).

We have not yet finished the proof of 2.6 - it remains to prove that if $\mathcal{X}$ (assumed locally totally 0-connected) is 0-connected, it is ∞-connected - or else that the connected components of $\mathcal{X}$, assumed locally totally 0-connected, are ∞-connected, i.e. W_∞-aspheric.

Corollary 2.14. *Let us note all the same that if* $\mathcal{X}$ *is of the form Top*(T)*,* T *a topological space, then* $\mathcal{X}$ *is locally totally* 0*-connected if and only if it is locally irreducible, i.e. if and only if* T *is locally irreducible.* [page 30]

We are reduced to proving that this last condition implies condition d) of 2.6, and we p. 19
can take for S the ordered set of nonempty connected open sets of T, this then becomes obvious.

---

[33] It suffices to require for e the commutativity of the two squares (e c b d) and (e d a c).

[34] if and only if?

[35] It means that X satisfies PS 1, and the condition

PS′ 2 Every commutative diagram

$$a \rightrightarrows b \longrightarrow x$$

fits into a commutative diagram

$$y \longrightarrow a \rightrightarrows b \longrightarrow x.$$

[36] But there is an obvious *counterexample*, if X is a 0-connected groupoid, the X/x are $\approx e$, hence cofiltering, but X is cofiltering (i.e. [...]) only if X is 1-connected. E.g. if G is a group $\neq 1$, then $X = B_G$ is such that the X/x are cofiltering, but X not cofiltering.

[37] NB $\mathcal{X}/x$ unnecessary.

[38] S/x unnecessary, for $x \xrightarrow{\text{diag}} x \times x$ is a nonempty open immersion, so if $x \times x$ is irreducible, x is too. But it amounts to the same to say that the S/x are irreducible (condition PS* 1, cf. 2.8, p. 21) *and* that the $x \xrightarrow{\text{diag}} x \times x$ are dominant (which indeed implies that $x \times x$ is irreducible). This immediately gives back the form PS* 2 of the cofiltering condition.

That said, we must remember that the cohomological theory of irreducible topological spaces is by no means trivial - as SERRE's FAC shows us! On the other hand, it was SERRE who had pointed out to me that for coefficients that are *constant,* the cohomology of an irreducible space is trivial.

In the present case, let us already prove, if $\mathcal{X}$ is 0-connected and locally totally 0-connected, that every covering of the final object of $\mathcal{X}$ is trivial. It suffices to see that if U is 0-connected, $U \longrightarrow e$ is an isomorphism. But let us consider $U \hookrightarrow U \times U$, we know that it is a direct summand, and as $U \times U$ is connected, it is an isomorphism. So $U \longrightarrow e$ monomorphism, hence a direct summand, and as $U \neq \emptyset$, we have $U \longrightarrow e$ an isomorphism.

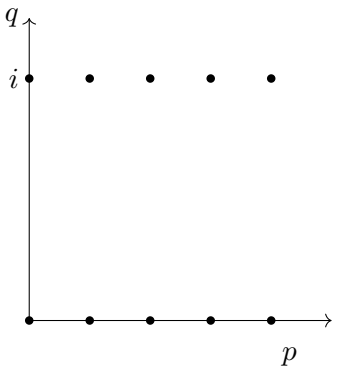
It remains to prove that for every constant abelian sheaf F on $\mathcal{X}$, we have $H^i(\mathcal{X}, F) =$ [page 31]
0 for $i > 0$. We proceed by induction on i, it is already done for $i = 1$. Suppose this has been proved for the $j < i$ (i fixed) for every topos that is *locally totally* 0*-connected.* Let $\xi \in H^i(\mathcal{X}, F)$, let $U \longrightarrow e$ be epimorphism such that $\xi | U = 0$, and consider the spectral sequence

$$E_2^{pq} = H^p(C(U^*, \mathcal{H}^q(F))) \Longrightarrow H^*(\mathcal{X}, F).$$

We have by the totally 0-connected hypothesis

$$\pi_0(U^q) \simeq \pi_0(U)^q, \qquad (*)$$

and the connected components of U^q are totally 0-aspheric,



so by the induction hypothesis

$$E_1^{pq} = 0 \qquad \text{if } 0 < q < i,$$

so we find an exact sequence

$$0 \longrightarrow H^i(C(U^*, F)) \longrightarrow H^i(\mathcal{X}, F) \longrightarrow H^0(\underbrace{H^i(U^*, F)}_{\substack{\text{complex} \\ \text{co-semi-simplicial} \\ \text{for variable Čech degree} \\ \text{at fixed } i}}) = \mathrm{Ker}(H^i(U, F) \rightrightarrows H^i(U^2, F)).$$

By the hypothesis on U, the image of ξ in the third term is zero, so ξ belongs to p. 20
the first term. But for $(*)$, the complex $C(U^*, F)$ is the *trivial* complex of the powers of $I = \pi_0(U)$, with coefficients in M (if $F = M_{\mathcal{X}}$). Its cohomology in degrees > 0 is zero, [page 32]
so $\xi = 0$, q.e.d.

Proposition 2.15. ([39]) *Let X be an object of* Cat*. Equivalent conditions:*

[39] This is already in SGA 4, I 8.

a) X is filtering, i.e. X is 0-connected and X^o satisfies the equivalent conditions of theorem 2.6 ($X^{o,\wedge}$ is locally totally 0-connected).

b) The final object of $X^\wedge$ is ind-representable.

PROOF. We know that we have

$$e_{X^\wedge} \simeq \varinjlim_{x\in X} x, \tag{2.15.1}$$

direct limit taken in $X^\wedge$. Indeed, this also means that for every object a in X, we have

$$\varinjlim_{x\in X} \underbrace{x(a)}_{\mathrm{Hom}_X(a,x)} \xrightarrow{\sim} e \qquad \text{(one-point set)}$$

But let us consider the object $X' = a\backslash X \longrightarrow X$ coétale over X, which is therefore Cat-cofibered over X, we know that we have

$$\pi_0(X') \simeq \varinjlim_{x\in X} \pi_0(X'_x),$$

where $X'_x = \mathrm{Hom}(a,x)$, so we find

$$\varinjlim_{x\in X} \mathrm{Hom}(a,x) = \pi_0(a\backslash X) = e$$

since $a\backslash X$ is 0-connected.

Formula (2.15.1) shows that if X is filtering, $e_{X^\wedge}$ is a filtering direct limit of repre- [page 33]
sentable functors. We know moreover that we can reindex by a filtering *ordered* category (cf. SGA 4, I 8.1.6 (this is due to P. DELIGNE!)).

We have proved a $\Longrightarrow$ b, let us prove b $\Longrightarrow$ a. So suppose

$$e_{X^\wedge} = \varinjlim_{i\in I} x_i,$$

where I is a (small) filtering ordered set, and $I \longrightarrow X$ a functor. Let us prove PS $1'$ (right-hand form), i.e. for a, b in X, exists $\begin{matrix} a & \searrow & \\ & & x \\ b & \nearrow & \end{matrix}$. Indeed, since $\varinjlim_I \mathrm{Hom}(a,x_i) = e$, it follows that $\mathrm{Hom}(a,x_i) \neq \emptyset$ for i large, likewise $\mathrm{Hom}(b,x_i) \neq \emptyset$, so it suffices to take $\begin{matrix} a & \searrow & \\ & & x_i \\ b & \nearrow & \end{matrix}$. Let us prove PS 2_d, i.e. that for double arrow $a \overset{u}{\underset{v}{\rightrightarrows}} b$, exists $b \longrightarrow x$ which equalizes.

Let ξ_{i_0} be in $\mathrm{Hom}(b, x_{i_0})$, its two images in $\mathrm{Hom}(a, x_{i_0})$ are not necessarily equal, but p. 21
become so in the direct limit, reduced to e, so become equal by composing with $x_{i_0} \longrightarrow x_i$ [page 34]
for i large enough, so the composite $b \longrightarrow x_{i_0} \longrightarrow x_i$

$$a \rightrightarrows b \longrightarrow x_{i_0} \longrightarrow x_i$$

equalizes the double arrow.

Corollary 2.16. *Let W be a basic localizer satisfying the limit axiom* Loc (8)*. If X in* Cat *is a filtering or cofiltering category, then X is W-aspheric.*

More generally and pleasantly:

Corollary 2.17. *Let W be as in 2.16. Then every filtering category is* totally W-aspheric ([40])*, i.e. for every arrow $f : X \longrightarrow Y$ in* Cat*, the object $f_!(e_X)$ of $Y^\wedge$ (also defined as* [page 35]

$y \longmapsto \pi_0(y\backslash X)$, and occurring in the canonical factorization of f as ip, with i étale and the cofibers of p 0-connected) is W-aspheric. In fact, $Y/f_!(e_{X^\wedge})$ is even a filtering category.

Indeed, the functor $f_!$ commutes with direct limits, so $f_!e_{X^\wedge}$ is a filtering direct limit of the $f_!(x_i) = f(x_i)$. So $Y/f_!(e_{X^\wedge})$ is a filtering direct limit of the $Y/f(x_i)$. So it is even a filtering category, a fortiori it is W-aspheric.

So finally, we find nontrivial examples of totally W-aspheric categories, i.e. examples of categories without a final object.

Corollary 2.18. ([41]) *Let X be in* Cat*, F in $X^\wedge$. For F to be ind-representable it is necessary and sufficient that X/F be filtering.*

Indeed, by (2.15), X/F is filtering if and only if the final object of $(X/F)^\wedge$, corresponding to F in $X^\wedge/F$, comes from an ind-object of X/F, but this also means that F is the direct limit in $X^\wedge$ of an ind-object of X, i.e. that F is ind-representable.

[page 36]

Question. *If X in* Cat *is totally W_∞-aspheric, i.e. the $f_!(e_{X^\wedge})$ are W_∞-aspheric for every arrow $f : X \longrightarrow Y$ in* Cat*, or again for every such arrow, with the $y\backslash X$ 0-connected ($y \in \mathrm{Ob}\, Y$), Y is W_∞-aspheric - does this imply that X is filtering?* That wouldn't surprise me - and even assuming only the $f_!(e_{X^\wedge})$ 1-connected ([42]) (they are 0-connected, as soon as X is). But I do not want to linger at present on this question.

The answer to the preceding question is negative, as we see from the:

p. 22

Proposition 2.19. ([43]) *Let X be in* Cat *such that $diag_X : X \longrightarrow X \times X$ is cofinal* ([44])*, i.e. such that X^o is totally* 0*-connected. Then X is totally W_∞-aspheric (hence also totally W-aspheric for every basic localizer W such that $W \supseteq W_\infty$).*

Indeed, let $f : X \longrightarrow Y$ be a cofinal functor, we must prove that Y is W_∞-aspheric. Now we know that totally 0-connected implies W_∞-aspheric ([45]) (cf. proposition 2.3, p. 17-18), so it suffices to see that Y^o is totally 0-connected, i.e. $Y \xrightarrow{d_Y} Y \times Y$ cofinal. Now we have the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{d_X} & X \times X \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f\times f} \\ Y & \xrightarrow{d_Y} & Y \times Y, \end{array}$$

where by hypothesis f, hence also $f \times f$, and d_X are cofinal. So $\varphi = (f \times f) \circ d_X$ is cofinal, and since $\varphi = d_Y f$ and f is also cofinal, we conclude that d_Y is, q.e.d.

Corollary. *$\mathbf{\Delta}$ being totally* 0*-connected, $\mathbf{\Delta}^o$ is totally W_∞-aspheric, yet it is not filtering.*

Question. *X totally W_∞-aspheric $\Longrightarrow$ X^o totally* 0*-connected? Or at least implies PS 1′?*

[page 36bis]

Counterexample 2.20 to the false proposition 2.19.

Let X be an object of Cat, let us adjoin to it the final category e as a closed subobject, i.e. let us consider an object X' of Cat in which X is an open subobject, e the complementary closed subobject:

$$X \overset{i}{\hookrightarrow} X' \overset{j}{\hookleftarrow} e.$$

[40] cf. XII, §V for this notion, which I have struggled a great deal to bring out and to understand!

[41] already in SGA 4, I 8.

[42] They always are (cf. XII, p. 101), it is 2-*connected* that should be read here!

[43] **False.**

[44] cf. definition 3.3, p. 45.

[45] This is **false.**

We know (cf. XII) that such an operation is determined, up to isomorphism, by the datum of an object

$$H \in \mathrm{Ob}\, X^\wedge$$

by putting

$$H(x) = \mathrm{Hom}_{X'}(x, e),$$

the category X/e being identified (up to isomorphism) with X/H. It is therefore 0-connected if and only if the object H of $X^\wedge$ is 0-connected. This also means that the morphism i is co-cofinal, i.e. that $i^o : X^o \longrightarrow X'^o$ is cofinal. Suppose now X W_∞-aspheric and totally 0-connected, i.e. the current Loc (6), cf. XVI. Proposition 2.19 asserts that, under these conditions, X' (which is itself also totally 0-connected) is even W-aspheric. It would therefore follow that $i \in W_\infty$. But by XII, th. 4, cor. 4 (p. 149), it would follow that X/e is W_∞-aspheric, i.e. H W_∞-aspheric. Thus, if X is totally 0-aspheric, every 0-connected object of $X^\wedge$ would be ∞-connected. But this is manifestly false, e.g. if we take for X a test category for W_∞, e.g. $X = \mathbf{\Delta}$. p. 23 [page 36ter]

This makes me suspect that the conjecture

$$X \text{ totally } W_\infty\text{-aspheric} \Longrightarrow X \text{ filtering}$$

might be valid.

p. 24 [page 37]

3 Return to cofinality

It is instructive to relate questions of cofinality (for an arrow $u : I' \longrightarrow I$ in Cat) to the derivator HOT_{W_0}, where $W_0 \subseteq \mathrm{Fl}\,\mathrm{Cat}$ is the coarsest of the localizers in Cat satisfying the connectedness axiom Loc 5, i.e.

$$W_0 = \{f \in \mathrm{Fl}\,\mathrm{Cat} \mid \pi_0(f) \text{ bijective}\}. \tag{3.1}$$

One then finds, for I in Cat, a canonical equivalence, 2-functorial in I,

$$\mathrm{HOT}_{W_0}(I) \xrightarrow{\sim} I^\wedge \overset{\mathrm{def}}{=} \underline{\mathrm{Hom}}(I^o, (\mathrm{Ens})), \tag{3.2}$$

in other words, one has an equivalence

$$\mathrm{HOT}_{W_0} \simeq \mathbf{D}_{(\mathrm{Ens})}. \tag{3.3}$$

When HOT_{W_0} is defined by localization starting from the category $\mathrm{Cat}(I) \overset{\mathrm{def}}{=} \underline{\mathrm{Hom}}(I^o, \mathrm{Cat})$, the arrow (3.2) is the one deduced, by localization, from the arrow

$$\begin{array}{rcl} \pi_0^I : \underline{\mathrm{Hom}}(I^o, \mathrm{Cat}) & \longrightarrow & \underline{\mathrm{Hom}}(I^o, (\mathrm{Ens})) \\ X & \longmapsto & \pi_0 \circ X, \end{array} \tag{3.4}$$

where

$$\pi_0 : \mathrm{Cat} \longrightarrow (\mathrm{Ens}) \tag{3.5}$$

is the "connected components" functor. If one describes $\mathrm{HOT}_{W_0}(I)$ as a localized category of Cat/I, (3.2) is deduced from the arrow

$$\left\{ \begin{array}{rcl} \mathrm{Cat}/I & \longrightarrow & I^\wedge = \underline{\mathrm{Hom}}(I^o, (\mathrm{Ens})) \\ X & \longmapsto & \underline{\pi_0}(X/I) = (i \longmapsto \pi_0(i\backslash X)). \end{array} \right. \tag{3.6}$$

[page 38]

Thus, the familiar functors, for an arrow

$$u : I' \longrightarrow I$$

in Cat

$$\left\{ \begin{array}{ccc} & \xrightarrow{u_!} & \\ I'^{\wedge} & \xleftarrow{u^*} & I^{\wedge}, \\ & \xrightarrow{u_*} & \end{array} \right. \tag{3.7}$$

are interpreted as the functors of the same name for the derivator HOT_{W_0}, or more directly, for the derivator (which one might mistakenly regard as hoax-like) $\mathbf{D}_{(\mathrm{Ens})}$. p. 25

This gives a meaning to the cohort of cohomological notions associated with a derivator, or more particularly, with a basic localizer W on Cat (giving rise to the derivator HOT_W). We shall designate them by the name of W_0-*fourbi* (gear) (instead of HOT_{W_0}-*fourbi*, or $\mathbf{D}_{(\mathrm{Ens})}$-*fourbi*), in particular for an arrow $u : I' \longrightarrow I$ of Cat such as the

notion

$$\left\{\begin{array}{lll}
W_0\text{-equivalence} & \big(\iff & \pi_0(u) \text{ bijective.}\big) \\
W_0\text{-aspheric} & \big(\iff & \text{the } I'/i \text{ for } i \in \mathrm{Ob}I \text{ are} \\
 & & \text{0-connected.}\big) \\
W_0\text{-coaspheric} & \big(\iff & \text{the } i\backslash I' \text{ for } i \in \mathrm{Ob}I \text{ are} \\
 & & \text{0-connected.}\big) \\
W_0\text{-proper } ([46]) & \big(\iff & \text{the inclusions } I'_i \longrightarrow I'/i \ (i \in \mathrm{Ob}I) \\
 & & \text{are } W_0\text{-coaspheric, i.e. the fibers} \\
 & & \text{of the colocalized } i'\backslash I' \longrightarrow i\backslash I \\
 & & (i = u(i')) \\
 & & \text{are 0-connected.}\big) \\
 & & \text{[page 39]} \\
W_0\text{-smooth } ([47]) & \big(\iff & \text{the inclusions } I'_i \longrightarrow i\backslash I' \\
 & & \text{are } W_0\text{-aspheric, i.e. the fibers} \\
 & & \text{of the localized arrows } I'/i' \longrightarrow I/i \\
 & & (i = u(i')) \text{ are 0-connected.}\big) \\
W_0\text{-perfect} & \big(\iff & W_0\text{-proper and } W_0\text{-smooth.}\big) \\
\text{universal } W_0\text{-equivalence } ([48]) & \big(\iff & \pi_0 \text{ bijective and remains so under any} \\
 & & \text{base change. These are also} \\
 & & \text{the } W_0\text{-fibrations (or weak } W_0\text{-fibrations)} \\
 & & \text{with } W_0\text{-aspheric fibers,} \\
 & & \text{i.e. 0-connected.}\big) \\
W_0\text{-fibration} & \big(\iff & u^* : \mathrm{Cat}/I \longrightarrow \mathrm{Cat}/I' \text{ transforms} \\
 & & W_0\text{-equivalences into } W_0\text{-equivalences.}\big) \\
\text{weak } W_0\text{-fibration } ([49]) & \big(\iff & \text{the base change} \\
 & & u^* : \mathrm{Cat}/I \longrightarrow \mathrm{Cat}/I' \text{ transforms} \\
 & & \text{homotopisms } J_1 \longrightarrow J_2 \text{ into} \\
 & & W_0\text{-equivalences - and it suffices to} \\
 & & \text{require it for the inclusion of a} \\
 & & \text{final object (or at one's choice initial) and for} \\
 & & \text{the inclusion of a point in } \mathbf{\Delta}^1.\big)
\end{array}\right. \tag{3.8}$$

p. 26

strict implication (and how)

$$\text{universal } W_0\text{-equivalence} \Longrightarrow W_0\text{-fibration}$$

[46] Example: the Cat-cofibrations.

[47] Example: the Cat-fibrations.

[48] Example: the W_0-smooth or W_0-proper arrows, with 0-connected fibers.

[49] Example: the W_0-smooth arrows such that $\pi_0(I'/I) = (i \longmapsto \pi_0(i\backslash I')) = u_!(e_{I'^\wedge})$ is locally constant on I, i.e. $\underline{\pi_0}(I'/I)$ is an étale *covering* of I, or if u^{opp} satisfies this condition.

strict implication (and how!) ([50])

$$W_0\text{-fibration} \Longrightarrow \text{weak } W_0\text{-fibration}.$$

Let us point out, as an example of the use of these notions, that for a cartesian diagram in Cat

$$\begin{array}{ccc} I' & \xleftarrow{p'} & J' \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle v} \\ I & \xleftarrow{p} & J, \end{array} \tag{3.9}$$

the base change arrow for u_*, v_* for presheaves is isomorphism if u is W_0-proper, or p W_0-smooth, and the base change arrow for $u_!$, $v_!$ is isomorphism if u is W_0-smooth or p W_0 proper ([51]). I point out in this connection that these statements, which a priori are formulated for presheaves with values in (Ens), remain valid for presheaves with values in any category $\mathcal{M}$ (subject to the existence of the cohomological or homological direct image operations under consideration). This can be seen e.g. by noting that [page 40]

$$W_0\text{-equivalence} \implies \mathbf{D}_{\mathcal{M}}\text{-equivalence} \tag{3.10}$$

for every $\mathcal{U}$-category $\mathcal{M}$, from which it follows that every ([52]) "W_0-notion" implies the corresponding $\mathbf{D}_{\mathcal{M}}$-notion, and thus allows one to apply to $\mathbf{D}_{\mathcal{M}}$ the results concerning it. (There is of course a direct, less learned argument, consisting in reducing to the case where $\mathcal{M}$ is small, and then using the fully faithful embedding $\mathcal{M} \hookrightarrow \mathcal{M}^\wedge$ resp. $\mathcal{M} \hookrightarrow ((\mathcal{M}^o)^\wedge)^o$.)

I recall for the record a (third?) characterization of the arrows $u : I' \longrightarrow I$ which are W_0-proper (resp. W_0-smooth): they are those for which the base change functor $u^* : \mathrm{Cat}/I \longrightarrow \mathrm{Cat}/I'$ transforms W_0-coaspheric (resp. W_0-aspheric) arrows into likewise. This gains in interest, from the fact that the notion of a W_0-coaspheric (resp. W_0-aspheric) functor is interpreted as that of a cofinal (resp. co-cofinal) functor, cf. below.

The notion of W_0-fibration appears extremely strong - much stronger in certain respects than that (say) of W_∞-fibration or of W_ω-fibration. It is equivalent to this (as one sees at once): for every connected component I'_α of I', denoting by I_α the corresponding connected component of I, and for every J_α 0-connected and over I_α, the fiber product $I'_\alpha \times_{I_\alpha} J_\alpha$ is again 0-connected. (That looks like those stories of total 0-connectedness of a topos!) But this also means that $I'_\alpha \longrightarrow I_\alpha$ is a universal W_0-equivalence. This implies that its fibers are 0-connected, and the converse is true if $u : I' \longrightarrow I$ is W_0-proper or W_0-smooth. Thus, the essential condition for a W_0-fibration is that the fibers of the induced arrows $I'_\alpha \longrightarrow I_\alpha$ be 0-connected. In fact, one has this p. 27 [page 41]

Proposition 3.1. *Let $u : I' \longrightarrow I$ be an arrow in* Cat.

a) For it to be a universal W_0-equivalence, it is necessary and sufficient that for every $\mathbf{\Delta}^1 \longrightarrow I$, i.e. every arrow α of I, the category $I' \times_I \mathbf{\Delta}^1 \overset{\text{def}}{=} I'|\alpha$ be 0-connected, or again that the fibers of u be 0-connected, and that for every nonidentity *arrow α of I, $I'|\alpha$ be 0-connected.*

b) For $u : I' \longrightarrow I$ to be a W_0-fibration, it is necessary and sufficient that for every connected component $I'_{\alpha'}$ of I', putting $\alpha = u(\alpha')$, the induced arrow $I'_{\alpha'} \longrightarrow I_\alpha$ be a universal W_0-equivalence (cf. criterion a) for this notion).

[page 42]

[50] But if the fibers are 0-connected, the two notions of W_0-fibrations are equivalent and mean that one has a universal W_0-equivalence …

[51] Unless I am mistaken, this gives *two characterizations* of W_0-proper morphisms, or of W_0-smooth morphisms.

[52] Not quite correct for *every* "W_0-notion", one must except the two notions of strong, and weak W_0-fibration.

c) *For it to be a weak W_0-fibration, it is* necessary *that $\forall\, i \in I$, the inclusion $I'_i \hookrightarrow i\backslash I'$ be a W_0-equivalence, i.e. $\pi_0(I'_i) \xrightarrow{\sim} \pi_0(i\backslash I')$ be bijective,* and *that $i \longmapsto \pi_0(i\backslash I')$ be a locally constant presheaf on I (i.e. $\underline{\pi_0}(I'/I)$ be an étale* covering *of I). This condition is also sufficient, if u is W_0-smooth. Moreover (for the record) u is a weak W_0-fibration if and only if u^{opp} is.*

The greatest difference between the notion of W_0-fibration, and that of weak W_0-fibration, seems to me that W_0-perfect morphisms (i.e. W_0-smooth and W_0-proper) are necessarily weak W_0-fibrations, but are not necessarily W_0-fibrations. The W-perfect morphisms, for a basic localizer $W \subseteq W_0$, are weak W-fibrations. The most striking example in this respect is probably that of étale coverings, which are W_ω-perfect (hence weak W_ω-fibrations and a fortiori weak W_0-fibrations), but not W_0-fibrations; for in the [page 43]
case where I', I are 0-connected, the fibers would have to be 0-connected (cf. b) above), i.e. u would have to be an isomorphism!

A nice property of W_0-smooth (resp. W_0-proper) morphisms, is that they are *open* (resp. closed) maps for the underlying topological spaces, which indeed means that the local fibers, i.e. of the arrows localized above $I'/i' \longrightarrow I/i$ (resp. colocal, i.e. the arrows colocalized below $i'\backslash I' \longrightarrow i\backslash I$), are surjective, i.e. with nonempty fibers (which is implied by the condition that these be 0-connected).

In all this I have still hardly spoken of W_0-aspheric resp. coaspheric arrows, which p. 28
nevertheless are here a main motivation for introducing W_0!

Proposition 3.2. *Equivalent conditions on the arrow $u : I' \longrightarrow I$ in* Cat:

a) *For every $\mathcal{U}$-category $\mathcal{M}$ and every functor $F : I \longrightarrow \mathcal{M}$, the canonical arrow (provided that small direct limits are representable in $\mathcal{M}$),*

$$\varinjlim_{I'} u^*(F) \longrightarrow \varinjlim_{I} F \tag{3.2.1}$$

is isomorphism. [page 44]

a′) *For every $\mathcal{U}$-category $\mathcal{M}$ in which small inverse limits are representable and every $F \in \mathcal{M}(I) = \underline{\mathrm{Hom}}(I^o, \mathcal{M})$, i.e. every presheaf on I with values in $\mathcal{M}$, the canonical arrow*

$$\varprojlim_{I} F \longrightarrow \varprojlim_{I'} u^*(F) \tag{3.2.2}$$

is isomorphism.

a″) ([53]) *As in a′, with $\mathcal{M} = (\mathrm{Ens})$, i.e. $F \in \mathrm{Ob} I^\wedge$.* ([54]).

b) *The arrow u is W_0-coaspheric, i.e. for every $i \in I$, $i\backslash I'$ is 0-connected.*

[53] This is a condition on morphisms of topoi, namely that if F in $\mathcal{X}^\wedge$, one has

$$H^0(\mathcal{X}, F) \xrightarrow{\sim} H^0(\mathcal{X}', u^*F).$$

There is similar notion for the H^i $(i \leq n)$, to be brought out. Notion of W_0-coaspheric morphisms of topoi! It is in the nature of a relative 0-connectedness condition, dual to the W_0-aspheric notion, which simply means that for U 0-connected in $\mathcal{X}^\wedge$, $u^*(U)$ is 0-connected. Likewise, one probably has notions ("dual" to the preceding ones) of W_n-coasphericity (well understood in principle if $\mathcal{X}$ is locally n-aspheric) (The W_n-coasphericity, being that mentioned above.)

[54] **NB** One can introduce

a‴) As in a), but with $\mathcal{M} = (\mathrm{Ens})$.

a^{IV}) As in a), with the canonical functor $I \longrightarrow I^\wedge$.

The equivalence of a, a′, a″ is ± tautological in terms of the definitions (cf. SGA 4, I, 8…). The form a″ reads, putting $\mathbf{D} = \mathrm{HOT}_{W_0}$, that $\forall\, F \in \mathbf{D}(I)$, one wants

$$\underbrace{H^\bullet_{\mathbf{D}}(I,F)}_{\overset{\text{def}}{=} p_{I,*}(F)} \longrightarrow \underbrace{H^\bullet_{\mathbf{D}}(I',u^*F)}_{\overset{\text{def}}{=} p_{I',*}(u^*F)} \tag{3.2.3}$$

($p_I : I \longrightarrow e$ and $p_{I'} : I' \longrightarrow e$ the canonical functors) to be isomorphism. One knows that this is equivalent to u **D**-coaspheric. The proof is an amusing game between $u_!$, u^*, u_* all at once. One must express that the Hom from an object M of $\mathbf{D}(e)$ (= (Ens) in this case) into the arrow (3.2.3) is bijective. Here, it

is more natural to take $M = e$ (one-point set) and to interpret the arrow (3.2.3) as p. 29

[page 45]

$$\mathrm{Hom}(e_I, F) \longrightarrow \mathrm{Hom}(e_{I'}, u^*F) \tag{3.2.4}$$

and by adjunction of $u_!$, u^* the second member is interpreted as $\mathrm{Hom}(u_!(e_{I'}), F)$, and the arrow is deduced from

$$u_! e_{I'} = u_! u^*(e_I) \xrightarrow{\text{adj.}} e_I \tag{3.2.5}$$

by taking Hom into F. Thus to say that (3.2.4) is isomorphism for every F means that (3.2.5) is isomorphism. Passing to the values of the two members at i, we find $\pi_0(i\backslash I') \xrightarrow{\sim} 1$, i.e. that $i\backslash I'$ is 0-connected, q.e.d.

Definition 3.3.

a) When the equivalent conditions of 3.2 are satisfied, thus when the $i\backslash I'$ ($i \in \mathrm{Ob}\, I$) are 0-connected, we say that $u : I' \longrightarrow I$ is a *cofinal* functor. We say that u is *co-cofinal* if $u^o : I'^o \longrightarrow I^o$ is cofinal, i.e. if the localized categories I'/i of I' over I ($i \in \mathrm{Ob}\, I$) are 0-connected.

b) If I' is a *subcategory* of I, we say that I' *is cofinal in* I if the inclusion functor is cofinal, *co-cofinal* if this functor is co-cofinal.

Corollary 3.4. ([55])

a) For u to be cofinal, it is necessary *that the colocalized categories $i\backslash I'$ be nonempty (condition F 1); F 0 is also required. Thus we must have*

F 0) $\pi_0(u)$ is bijective.

[page 46]

F 1) For every $i \in \mathrm{Ob}\, I$, exists $i' \in I'$ and an arrow $i \longrightarrow u(i')$ in I.

([56]).

b) If I' is weakly filtering, for u to be cofinal, it is sufficient *that it satisfy F 1, as well as the condition:*

F 2) For every double arrow $i \overset{\beta}{\underset{\gamma}{\rightrightarrows}} u(i')$ (with $i \in \mathrm{Ob}\, I$, $i' \in \mathrm{Ob}\, I'$), exists an arrow $i' \xrightarrow{\alpha'} j'$ in I', in such a way that $u(\alpha')$ equalizes (β, γ), i.e. $u(\alpha')\beta = u(\alpha')\gamma$.

and these conditions F 1, F 2 imply that the $i\backslash I'$ are weakly filtering, and I as well ([57]).

[55] **NB** For a complementary cofinality statement, cf. XVIII prop. 4.9.6.3 (p. 224-225, 228-242). A statement covering both would be needed.

p. 30

[56] **NB** F 1 already implies surjectivity of π_0, hence its bijectivity if I' is 0-connected - in this case F 0 follows from F 1.

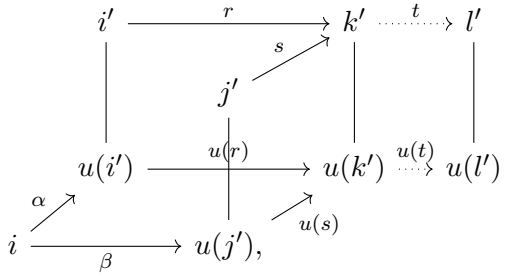
[57] If I' satisfies right PS 1, then for the cofinality of u, it suffices that we have F 0, F 1, F 2, and then the $i\backslash I'$ are weakly filtering, and I is prefiltering just like I'.

c) If I' is filtering, then u is cofinal if and only if *it satisfies the two conditions F1, F2, or again if and only if the $i\backslash I'$ are* filtering *(and not merely* 0*-connected). And this implies that I is also filtering* ([58]).

d) If I is right prefiltering (i.e. satisfying right PS 1), and u fully faithful, *for u to be cofinal, it is* necessary *and it is* sufficient *that u satisfy condition F 1 above. And then I' is also prefiltering, and the $i\backslash I'$ are weakly filtering. If moreover I is filtering, the $i\backslash I'$ and I' are as well. (Compare SGA 4, I 8.1.3.)*

PROOF. a) F 1 is obvious, since 0-connected $\Longrightarrow$ nonempty. F 0 because W_0-coaspheric $\Longrightarrow$ W_0-equivalence, i.e. $\pi_0(u)$ bijective. [page 47]

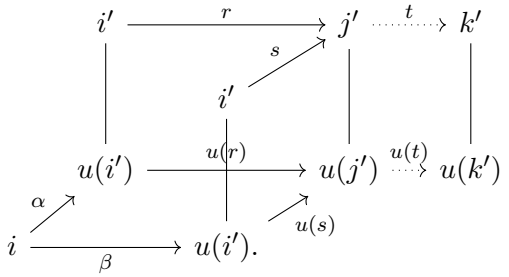
To *prove* b), it suffices to prove that F 1, F 2 (if I' is weakly filtering) imply that the $i\backslash I'$ are weakly filtering (a fortiori 0-connected) and moreover that I is as well. We already know that they are nonempty by F 1, remains to see that if (i', α), (j', β) are in $i\backslash I'$, then they are "dominated" by a third object of $i\backslash I'$. Since I' is weakly filtering, we find $\begin{matrix} i' & \overset{r}{\searrow} & \\ & & k' \\ j' & \underset{s}{\nearrow} & \end{matrix}$, whence $u(r)$, $u(s)$ in I (cf. diagram)



applying F 2 to the double arrow $u(r)\alpha, u(s)\beta : i \rightrightarrows u(k')$, we find $k' \xrightarrow{t} l'$ (dotted on diagram) such that $(u(t)u(r))\alpha = (u(t)u(s))\beta$, say γ, which shows that the object $(l', \gamma : i \longrightarrow u(l'))$ dominates (i', α) and (j', β).

The proof that if I' is weakly filtering and u satisfies F 1, then I is weakly filtering, is immediate.

Let us prove c). It remains to prove that if I' is *filtering* (not merely weakly filtering), then if u cofinal, then F 2 is satisfied, and moreover the $i\backslash I'$ and I are filtering (not merely weakly filtering). But let us note at once that $i\backslash I' \longrightarrow I'$ is coétale, hence I' filtering, hence locally filtering implies that $i\backslash I'$ is also locally filtering; since by hypothesis the $i\backslash I'$ are 0-connected, they are therefore filtering. Condition F 2 comes out easily now:



p. 31
[page 48]

Since $i\backslash I'$ is filtering, hence weakly filtering (PS 1), we can find $i' \overset{r}{\underset{s}{\rightrightarrows}} j'$ such that

$$u(r)\alpha = u(s)\beta,$$

[58] If I' is locally filtering, u is cofinal if and only if it satisfies F 0, F 1, F 2, i.e. if and only if the $i\backslash I'$ are filtering, and then I is locally filtering.

and since I' is filtering, we can find $t : j' \longrightarrow k'$ which equalizes the double arrow (r, s):

$$tr = ts (\overset{\text{def}}{=} \tau),$$

then

$$\underbrace{u(t)u(r)}_{u(tr)}\alpha = \underbrace{u(t)u(s)}_{u(ts)}\beta \qquad \text{i.e.} \qquad u(\tau)\alpha = u(\tau)\beta,$$

i.e. we have verified F 2.

Remains to prove that I is filtering (if I' is and $u : I' \longrightarrow I$ is cofinal). We already know that I is weakly filtering by b), remains to verify the double arrow condition

$$i \underset{\beta}{\overset{\alpha}{\rightrightarrows}} j$$

(PS 2). Let i' be in I' and $j \xrightarrow{\gamma} u(i')$, and consider

$$i \underset{\gamma\beta}{\overset{\gamma\alpha}{\rightrightarrows}} u(i'),$$

by hypothesis F 2 $\exists r : i' \longrightarrow j'$ in I' such that $u(r)(\gamma\alpha) = u(r)(\gamma\beta)$, hence $u(r)\gamma$ equalizes (α, β), q.e.d. [page 49]

The verification of d) by AQT .

p. 32

4 Biaspheric functors

Let again

$$u : I' \longrightarrow I$$

be an arrow of Cat, let $\mathcal{M}$ be a category, whence

$$u^* : \mathcal{M}(I) \longrightarrow \mathcal{M}(I'); \tag{4.1}$$

we seek conditions on u, $\mathcal{M}$ for this functor to be fully faithful. If the left adjoint functor $u_!$ (resp. right adjoint u_*) exists, we know that it amounts to the same to say that the adjunction morphism

$$u_! u^* F \longrightarrow F \tag{4.2}$$

resp.

$$u_* u^* F \longleftarrow F \tag{4.2 bis}$$

is isomorphism for every F in $\mathcal{M}(I)$. Let us then say that u is $\mathcal{M}$-connected, or $\mathcal{M}$-biaspheric, this terminology implying that this is a property which refers to the derivator $\mathbf{D}_{\mathcal{M}} = (S \longmapsto \mathcal{M}(S))$ on Cat defined by $\mathcal{M}$. Let us note that

> If u is $\mathcal{M}$-connected, then it is also $\mathcal{M}(S)$-connected, for every S in Cat, since $\mathcal{M}(S)(I) \simeq \mathcal{M}(I)(S) \overset{\text{def}}{=} \underline{\text{Hom}}(S^o, \mathcal{M}(I))$, (4.3)

and

> If $\mathcal{M}' \hookrightarrow \mathcal{M}$ is fully faithful, then $\mathcal{M}$-connected implies $\mathcal{M}'$-connected, (4.4)

[page 50]

finally, conversely,

> for u to be $\mathcal{M}$-connected, it is necessary and sufficient that for every *small* full subcategory $\mathcal{M}'$ of $\mathcal{M}$, u be $\mathcal{M}'$-connected. (4.5)

Since every small $\mathcal{M}'$ embeds in $\mathcal{M}'^{\wedge} = \mathrm{Ens}(\mathcal{M}')$, it follows from (4.3), (4.4), (4.5) that

$$\text{If } u \text{ is Ens-connected, it is } \mathcal{M}\text{-connected for every category } \mathcal{M}, \tag{4.6}$$

for this is the case if $\mathcal{M}$ is small by (4.3), hence for every $\mathcal{M}$ by (4.5).

Let us say that a functor u in Cat is $\mathcal{M}$-*cofinal* or $\mathcal{M}$-coaspheric if the functorial homomorphism

$$\varprojlim_{I} F \longrightarrow \varprojlim_{I'} u^*(F) \tag{4.7}$$

(arrow in $\underline{\mathrm{Hom}}(\mathcal{M}(I), \mathcal{M})$ if the $\varprojlim$ of types I, I' exist in $\mathcal{M}$, otherwise arrow in $\underline{\mathrm{Hom}}(\mathcal{M}(I), \mathcal{M}^{\wedge})$) is an isomorphism. Reducing to the case of small categories $\mathcal{M}$, each embedded in $\mathcal{M}^{\wedge}$, we see by criterion (4.2 bis) that

$$u \text{ is } \mathcal{M}\text{-connected if for every } i \in \mathrm{Ob}\, I, \tag{4.8}$$

the induced functor $I'/i \longrightarrow I/i$ is $\mathcal{M}$-cofinal. The converse is not clear, for u is $\mathcal{M}$-connected if and only if for every $i \in \mathrm{Ob}\, I$ and every $\mathcal{M}$-presheaf F_i on I/i *induced by a presheaf* F *on* I, the homomorphism $\varprojlim_{I/i} F_i \longrightarrow \varprojlim_{I'/i}(u/i)^*(F/i)$ is isomorphism. We can therefore assert straightaway that this implies likewise for *every* F_i on I/i, only if we know that $\alpha_i^* : \mathcal{M}(I) \longrightarrow \mathcal{M}(I/i)$ is essentially surjective. This is so for every i when I is *ordered,* for then $I/i \longrightarrow I$ is fully faithful, hence $\mathcal{M}(I/i) \longrightarrow \mathcal{M}(I)$ fully faithful, hence $\mathcal{M}(I) \longrightarrow \mathcal{M}(I/i)$ a localization functor, in particular essentially surjective.

p. 33
[page 51]

But when $\mathcal{M} = (\mathrm{Ens})$, this perplexity disappears. We find the

[page 52]

Proposition 4.1. *Let* $u : I' \longrightarrow I$ *be an arrow in* Cat. *Equivalent conditions:*

a) For every category $\mathcal{M}$, $u^*_{\mathcal{M}} : \mathcal{M}(I) \longrightarrow \mathcal{M}(I')$ *is fully faithful (i.e.* u *is* $\mathcal{M}$*-connected).*

a′) For every category $\mathcal{M}$ *in which direct limits of type* I, I' *exist, the adjunction arrow for* F *in* $\mathcal{M}(I)$

$$u_! u^* F \xrightarrow{\sim} F$$

is isomorphism.

a″) Dual of a′), with adjunction arrow

$$u_* u^* F \xleftarrow{\sim} F.$$

b) b′) b″) As in a) a′) a″), with $\mathcal{M} = (\mathrm{Ens})$.

c) As in b′), with F *representable, i.e. of the form* $j \longmapsto \mathrm{Hom}_I(j, i)$.

d) For every i *in* I, $I'/i \longrightarrow I/i$ *is cofinal.*

d′) For every i *in* I, $i\backslash I' \longrightarrow i\backslash I$ *is co-cofinal.*

d″) For every arrow $i_0 \xrightarrow{\alpha} i_1$ *in* I, *the category* $i_0\backslash\underset{\alpha}{I'}/i_1 \overset{\text{def}}{=} I' \times_I \underbrace{i_0\backslash\underset{\alpha}{I}/i_1}_{\substack{\text{biinduced category} \\ (i_0,\alpha)\backslash(I/i_1) \simeq (i_0\backslash I)/(i_1,\alpha)}}$

is

0*-connected.*

[page 53]

PROOF. We already know, from the preliminaries, the equivalence of a, a′, a″, b, b′, b″, let A be this condition - we shall say that u is W_0-*connected* or W_0-*biaspheric,* if these equivalent conditions are satisfied. We have moreover the self-duality

$$u \text{ is } W_0\text{-connected if and only if } u^o \text{ is,} \tag{4.8}$$

for we obviously have

$$u \text{ is } \mathcal{M}\text{-connected if and only if } u^o \text{ is } \mathcal{M}^o\text{-connected.} \tag{4.9}$$

This shows the equivalence of d, d′, which can also be seen from the equivalence of each of these two conditions with d″, by the cofinality criterion of the preceding paragraph. Let D be this condition on u. By (4.8) we have D $\Longrightarrow$ A, hence p. 34

$$\mathrm{D} \Longrightarrow \mathrm{A} \Longrightarrow \mathrm{c},$$

it remains to prove that c $\Longrightarrow$ D, more precisely c $\Longrightarrow$ d″ and c′ $\Longrightarrow$ d″ . Let us make explicit the adjunction morphism (for i in I)

$$u_! \underbrace{u^* i} \longrightarrow i,$$

i.e. for j in I

$$\underbrace{(u_! u^* i)(j)}_{\pi_0(j\backslash(I'/u^*(i)))} \longrightarrow i(j) = \mathrm{Hom}(j, i).$$

Now $I'/u^*(i) \simeq I'/i$, so we have the arrow [page 54]

$$(u_! u^*(i))(j) \simeq \pi_0(j\backslash(I'/i)) \longrightarrow \mathrm{Hom}_I(j, i). \tag{4.10}$$

Now we see at once that for $\alpha : j \longrightarrow i$ in $\mathrm{Hom}(j, i)$, the fiber of (4.10) at α identifies with $\pi_0(j\backslash_{\alpha} I'/i\ )$, more precisely, $j\backslash_{\alpha} I'/i$ is the open and closed subcategory of $J = j\backslash(I'/i)$ corresponding to the part of $\pi_0(J)$ which lies over α in (4.10). This completes the proof of 4.1.

NB Another way to prove that c $\Longrightarrow$ A is to note that the two sides in the adjunction morphism $u_! u^* F \longrightarrow F$ in $I^\wedge$ are functors $I^\wedge \longrightarrow I^\wedge$ which commute with arbitrary $\varinjlim$. Thus to check that it is isomorphism, it suffices to check it for F in I, since I generates $I^\wedge$ by $\varinjlim$. On the other hand, an argument of this type does not seem to prove that to check that we have an *isomorphism* $F \longrightarrow u_* u^* F$, it suffices to do so for representable F, i.e. to check b″ for representable F.

It would remain to see whether c′ $\Longrightarrow$ d″ . [page 55]

Let us make explicit the adjunction morphism

$$i \longrightarrow u_* u^* i$$

(for $i \in \mathrm{Ob}\, I$) by its values on j

$$\underbrace{i(j)}_{\mathrm{Hom}(j,i)} \longrightarrow \Gamma(I'/j, u^* i). \tag{$*$}$$

So let

$$u_j : I'/j \longrightarrow I,$$

we have p. 35

$$\Gamma(I'/j, \underbrace{u^*(i)|I'/j)}_{u_j^*(i)} \simeq \mathrm{Hom}(e_{I'/j}, u^*(i))$$

$$\overset{\mathrm{adj}}{\simeq} \mathrm{Hom}(u_{j,!}e_{I'/j}, i),$$

and the arrow $(*)$, functorial in i, is deduced from an arrow

$$u_{j,!}(e_{I'/j}) \longrightarrow j. \qquad (**)$$

We see at once, as above, that $(**)$ is isomorphism if and only if for every $\alpha : k \longrightarrow j$, the category $k\backslash\underset{\alpha}{I'}/j$ is 0-connected, and if we require this for every j, k, α, this is condition d″. But it is not clear that for $(**)$ to be an isomorphism, it suffices that the Hom of the two sides with values in every i in I (and not every F in $I^\wedge$) be isomorphism. The difficulty comes from the fact that $u_{j,!}(e_{I'/j})$ is not necessarily representable, and that an arbitrary F in $I^\wedge$ is not necessarily $\varprojlim$ of representable functors. So it is doubtful whether c′ suffices to imply that u is W_0-connected. [page 56]

Corollary 4.2. *For u to be W_0-connected, it* suffices *that it be a universal W_0-equivalence, i.e. (cf. 3.1) that the fibers be 0-connected, and that for every nonidentity arrow in I, defining $\mathbf{\Delta}^1 \longrightarrow I$, $I' \times_I \mathbf{\Delta}^1$ be 0-connected.*

4.3. This condition of universal W_0-equivalence is however *strictly* stronger than W_0-connectedness, as we see from the example where u is an *equivalence of categories* not surjective on objects (e.g. the inclusion of a vertex in a 1-connected groupoid having more than one object). More generally, every localization functor is W_0-connected, whereas its fibers are not necessarily connected, and a fortiori it is not a universal W_0-equivalence.

However, if every endomorphism p of an i in I, such that $p^2 = p$ (projector) is the identity, and if $I' \longrightarrow I$ is "transportable", e.g. if in I every isomorphism is an identity, then u W_0-connected implies that the fibers of u are 0-connected, as we see by expressing d″ for α an identity arrow. The conditions envisaged on I are satisfied if I is ordered. But even if I' and I are both ordered, u can be W_0-connected, without being a universal W_0-equivalence. Here is an example with $I = \mathbf{\Delta}^2$: [page 57]

$$I' = \left\{ \begin{array}{ccc} & & 1' \longrightarrow 2 \\ & & \downarrow \\ 0 & \longrightarrow & 1 \end{array} \right\} \qquad \downarrow$$

$$I = \left\{ 0 \longrightarrow 1 \longrightarrow 2 \right\}.$$

The bilocalized categories of I are the three vertices $\{0\}$, $\{1\}$, $\{2\}$, and the three segments $[0,1]$, $[1,2]$ and $[0,2] = I$, the inverse images are respectively p. 36

$$\bullet\,, \quad \downarrow\,, \quad \bullet\,, \quad \longrightarrow\!\downarrow\,, \quad \downarrow\!\longrightarrow\,, \quad I'\,,$$

they are all 0-connected, i.e. W_0-aspheric, and even contractible hence W_ω-aspheric (so u is even W_ω-connected, i.e. W_ω-biaspheric). But it is not a universal W_0-equivalence, since the inverse image of I' under $\mathbf{\Delta}^1 \overset{\partial}{\to} \mathbf{\Delta}^2$, $\partial(0) = 0$, $\partial(1) = 2$, is the disconnected category $\varepsilon = \{\bullet \quad \bullet\}$. [page 58]

I had hoped for a moment that the W_0-connected functors $I' \overset{u}{\to} I$ would be those which factor as $I' \overset{u_0}{\longrightarrow} I'' \overset{v}{\to} I$, with u_0 an equivalence, and v a universal W_0-equivalence. But the preceding example defeats this, with u even W_ω-connected. Indeed, I'' would

be derived from a preordered set, and v would factor through the corresponding ordered set, namely I', hence would be of the form

$$I'' \longrightarrow I' \xrightarrow{u} I,$$

where $I'' \longrightarrow I'$ is a passage to the quotient, and the inverse image of $\mathbf{\Delta}^1 \xrightarrow{\partial} I$ would again be disconnected (a π_0 of cardinality 2). So it is really hopeless.

However

Corollary 4.3. *Suppose that u is W_0-smooth, or W_0-proper, or a weak W_0-fibration. Then u is W_0-connected if and only if it is a universal W_0-equivalence, i.e. if and only if a) its fibers are 0-connected and b) $I' \times_I \mathbf{\Delta}^1$ 0-connected for every $\mathbf{\Delta}^1 \longrightarrow I$ (corresponding to a nonidentity arrow).*

In all three cases we know that u is a universal W_0-equivalence if and only if its fibers are W_0-connected. Now W_0-smoothness implies that $\forall i \in \mathrm{Ob}\, I$, the inclusion [page 59]

$$I'_i \longrightarrow i\backslash I'$$

is a W_0-equivalence, on the other hand $i\backslash I' \longrightarrow i\backslash I$ being W_0-aspheric, hence a W_0-equivalence by hypothesis, $i\backslash I'$ is 0-connected, hence I'_i also. The case u W_0-proper is dual. In the case u a weak W_0-fibration, we again know that $I'_i \longrightarrow i\backslash I'$ is a W_0-equivalence, and we conclude as in the smooth case (or, as one chooses, via $I'_i \longrightarrow I'/i$).

But it occurs to me that the W_0-connected functors might be exactly the localization functors? It amounts to the same thing to say that a *conservative* functor $u : I' \longrightarrow I$ (i.e. $u(\alpha)$ isomorphism $\Longrightarrow \alpha$ isomorphism) which is W_0-connected, is an equivalence. But this is false, as we see by taking $I = e$, I' a 0-connected and not 1-connected groupoid: $I' \longrightarrow e$ is W_0-connected (and even a universal W_0-equivalence), but it is not an equivalence (and not a localization functor).

I would like to generalize the notion of a W_0-biaspheric arrow to the case of an arbitrary derivator $\mathbf{D}$. p. 37 [page 60]

Proposition 4.4. *Let $\mathbf{D}$ be a derivator on* Diag, *and $u : I' \longrightarrow I$ an arrow in* Diag. *Suppose that u_* and $u_!$ exist, and are computed in the standard way, for every arrow u in* Diag. *(In fact, it suffices that this be the case for u and its localized arrows $u/i : I'/i \longrightarrow I/i$, or u and its colocalized arrows $i\backslash u : i\backslash I' \longrightarrow i\backslash I$.) Consider the conditions:*

a) $u^* : \mathbf{D}(I) \longrightarrow \mathbf{D}(I')$ *fully faithful.*

a′) $\forall$ F *in* $\mathbf{D}(I)$, *the arrow*

$$u_! u^* F \longrightarrow F$$

is an isomorphism.

a″) $\forall$ F *in* $\mathbf{D}(I)$, *the arrow*

$$u_* u^* F \longleftarrow F$$

is an isomorphism.

b) $\forall$ i *in* I, *the arrow* $i\backslash I' \longrightarrow i\backslash I$ *is* $\mathbf{D}$*-aspheric.*

b′) $\forall$ i *in* I, *the arrow* $I'/i \longrightarrow I/i$ *is* $\mathbf{D}$*-coaspheric.*

b″) $\forall$ $\alpha : i_0 \longrightarrow i_1$ *in* I, *the category* $i_0\backslash\!\underset{\alpha}{I'}/i_1$ *is* $\mathbf{D}$*-aspheric.*

[page 61]

We then have the implications

$$(b \iff b' \iff b'') \implies (a \iff a' \iff a'').$$

We have $a \implies b$ (i.e. all the legitimate conditions envisaged are equivalent) in each of the following cases.

1° *I is preordered.*

2° *$\forall\ i$ in I, $\alpha_i^* : \mathbf{D}(I) \longrightarrow \mathbf{D}(I/i)$ is essentially surjective.*

3° *$\forall\ i$ in I, considering $\alpha_i : I/i \longrightarrow I$, the functor $\alpha_{i,!} : \mathbf{D}(I/i) \longrightarrow \mathbf{D}(I)$ is conservative.*

4° $\mathcal{D} = \mathrm{HOT}_{W_0} \approx \mathbf{D}_{\mathrm{Ens}}$.

Proof. The equivalence of a, a′, a″ is clear, say A. Likewise the equivalence of b, b′, b″, say B.

Let us prove

$$B \implies A$$

assuming e.g. the existence of u_*. We tacitly assume, along with the existence of $u_* : \mathbf{D}(I') \longrightarrow \mathbf{D}(I)$, the method of computing the fibers of $u_*(-)$. Thus A, via a″, is equivalent to

$$H^\bullet_{\mathbf{D}}(I'/i, u^*F) \xleftarrow{iso} H^\bullet_{\mathbf{D}}(I/i, F) \simeq F(i)$$

for every F in $\mathbf{D}(I)$, every i in I. p. 38

Let

$$\alpha_i : I/i \longrightarrow I, \qquad F_i = \alpha_i^*(F),$$

we must therefore express that for every F_i coming from an F, we have

$$H^\bullet_{\mathbf{D}}(I/i, F_i) \xrightarrow{\text{iso}} H^\bullet_{\mathbf{D}}(I'/i, (u/i)^*(F_i)). \tag{$*$}$$

Now subject to the existence of $(u/i)_!$, to say that $(*)$ is isomorphism for *every* F_i in $\mathbf{D}(I/i)$, means that

$$u/i : I'/i \longrightarrow I/i$$

is $\mathbf{D}$-coaspheric, whence b′ $\implies$ a″. It is clear that the reverse implication holds if α_i^* is essentially surjective, this is case 2°. Now 2° is satisfied if I is preordered, whence case 1°. Case 4° has already been treated (4.1). We must still treat case 3°, which moreover includes case 2°, hence case 1° (and also case 4°, cf. below). [page 62]

We must express the isomorphy of the arrow $(*)$ for F_i of the form $\alpha_i^*(F)$. Let ξ be object of $\mathcal{A} = \mathcal{D}(e)$, we must express the isomorphy of $\mathrm{Hom}(\xi, -)$ in the two sides of $(*)$, i.e. of

$$\begin{array}{ccc} \mathrm{Hom}(\xi_{I/i}, F_i) & \longrightarrow & \mathrm{Hom}(\xi_{I'/i}, (u/i)^*(F_i)) \\ & \overset{\text{adj.}}{\simeq} & \mathrm{Hom}((u/i)_!(\xi_{I'/i}), F_i), \end{array} \tag{$**$}$$

arrow which is deduced from

$$(u/i)_!(u/i)^*(\xi_{I/i}) \xrightarrow{\text{adj.}} \xi_{I/i} \tag{$***$}$$

by taking the Hom into F_i. The condition u/i coaspheric (i.e. b′) means that $(***)$ is an isomorphism for every ξ in $\mathcal{A}$. Does it *suffice* for this that its Hom into an F_i be bijective, when we restrict F_i in $\mathbf{D}(I/i)$ to be in the image of $\mathbf{D}(I)$ under α_i^*? Now this arrow $(**)$, for $F_i = \alpha_i^* F$, is made explicit by the adjunction formula relative to the functor α_i, as [page 63]

$$\mathrm{Hom}(\alpha_{i,!}(\xi_{I/i}), F) \longrightarrow \mathrm{Hom}(\alpha_{i,!}(\ldots), F).$$

To say that it is an isomorphism for every F, thus means *exactly* that the transform of the arrow $(***)$ under $\alpha_!$ is an isomorphism. To conclude from this that $(***)$ itself is isomorphism, the natural condition to impose is therefore that $\alpha_{i,!} : \mathbf{D}(I/i) \longrightarrow \mathbf{D}(I)$ be conservative. This is case 3°. This completes the proof of 4.4.

4.5. Let us show that condition 3° on I is not always satisfied, in the case of $\mathbf{D} = (\mathrm{HOT}_{W_0})^{\mathrm{opp}} \approx (\mathbf{D}_{\mathrm{Ens}})^{\mathrm{opp}}$, although in this case, it is nevertheless true that for every I' over

I satisfying condition a$''$ (self-dual), condition b$'$ (also self-dual) is also satisfied (by case 4° above). I say therefore that there exist I and i in I, such that p. 39

$$\beta_{i,*} : (i\backslash I)^\wedge \longrightarrow I^\wedge$$

is not conservative. (**NB** On the other hand $\alpha_{i,!} : (I/i)^\wedge \longrightarrow I^\wedge$ is always conservative, which gives back case 4° as a special case of 3°). As $\beta_i : i\backslash I \longrightarrow I$ is W_0-proper (and even W_ω-proper!), $\beta_{i,*}$ "is computed fiber by fiber", so we have [page 64]

$$\beta_{i,*}(G)(j) \simeq \prod_{\alpha \in \mathrm{Hom}(i,j)} G(\alpha),$$

the fiber in question at j being discrete. Let us take

$$I = 0 \underset{\beta}{\overset{\alpha}{\rightrightarrows}} 1, \qquad \text{with } \alpha \neq \beta,$$

then $0\backslash I$ is $\Psi = \begin{array}{ccc} & \overset{\alpha'}{\nearrow} & 1 \\ 0 & & \\ & \underset{\beta'}{\searrow} & 1' \end{array}$, and $\Psi^\wedge$ is the category of diagrams of sets $G = \begin{array}{ccc} & \overset{\lambda}{\swarrow} & G_1 \\ G_0 & & \\ & \underset{\mu}{\nwarrow} & G_{1'} \end{array}$.

The functor β_* deduced from $\beta : i\backslash I \longrightarrow I$ transforms this diagram into

$$G_1 \times G_{1'} \underset{\mu\,\mathrm{pr}_2}{\overset{\lambda\,\mathrm{pr}_1}{\rightrightarrows}} G_0.$$

. If we restrict ourselves to the diagrams G for which $G_{1'} = \emptyset$, β_* on these gives

$$\emptyset \rightrightarrows G_0,$$

so the restriction of β_* to this subcategory (equivalent to $(\mathbf{\Delta}^1)^\wedge$) is not faithful, nor conservative, taking account only of the component G_0, and not of G_1.

Definition 4.6. *We say that $u : I' \longrightarrow I$ in* Diag *is* **D**-biaspheric *if it satisfies the equivalent conditions b, b$'$, b$''$ of 4.4,* **D**-connected *if it satisfies condition a, or again (subject to the existence of $u_!$ resp. of u_*) the equivalent conditions a$'$, a$''$.* [page 63bis]

Thus, **D**-biaspheric implies **D**-connected, and the converse is true when I satisfies (with respect to **D**) condition 3° of 4.4 (thus also, if it satisfies one of the stronger conditions 2°, 1°), or the condition dual to this one (since the notions **D**-biaspheric, **D**-connected are self-dual). In this regard, let us note the

Corollary 4.7. *Suppose that $\mathcal{A} = \mathbf{D}(e)$ satisfies the following condition: the functor*

$$\begin{cases} \mathcal{A} \times \mathcal{A} \longrightarrow \mathcal{A} \\ (\xi \ , \ \eta) \longrightarrow \xi \amalg \eta \end{cases}$$

is conservative (resp. faithful). Then for every étale arrow $u : J \longrightarrow I$ in Diag *(and in particular for $I/i \longrightarrow I$), the functor $u_! : \mathbf{D}(J) \longrightarrow \mathbf{D}(I)$ is conservative (resp. faithful)* p. 40 [page 64bis]

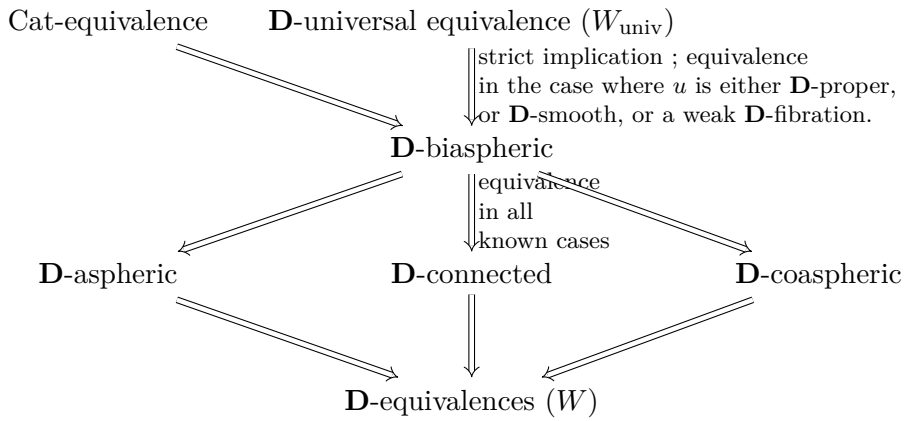
([4]). *If this condition, or the dual condition, involving the products $\xi \times \eta$ (ξ, η in $\mathcal{A}$), is satisfied, then for every arrow $u : I' \longrightarrow I$ in* Diag, *u is* **D***-connected if and only if u is* **D***-biaspheric.*

Proof. The hypothesis of conservativity, or of faithfulness carries over to sums indexed by an arbitrary set (subject to existence), as one sees by writing $\coprod_{\lambda} \xi_\lambda = \xi_{\lambda_0} \amalg \coprod_{\lambda \neq \lambda_0} \xi_\lambda$. The assertion made about $u_!$ then follows from the computation of $u_!$ fiber by fiber (which is valid since u is étale hence smooth), taking into account that the fibers are discrete.

I know of no example of a pair $(\mathbf{D}, u)$ of a derivator $\mathbf{D}$ and an arrow u in Diag, for which u is $\mathbf{D}$-connected without being $\mathbf{D}$-biaspheric. We have given conditions on $\mathbf{D}$, which ensure equivalence of the two notions. If $\mathbf{D}$ is of the form $\mathbf{D}_{\mathcal{M}}$, and if $\mathcal{M}$ satisfies the condition stated in 4.7 or its dual, then by 4.7 it is OK. So a counterexample with a $\mathbf{D}_{\mathcal{M}}$ would have to take this into account. I am not going to persist in trying to clear up how things stand.

Corollary 4.8. *An arrow u of* Diag *which is a universal* **D***-equivalence is* **D***-biaspheric, and* a fortiori **D***-connected. Conversely, if u is either* **D***-smooth, or* **D***-proper, or a weak* **D***-fibration, and if u is* **D***-biaspheric, it is a universal* **D***-equivalence.* [page 65]

The proof is that of the special case 4.3. We thus have a diagram of implications:



([5]). Recall that for a given $u : I' \longrightarrow I$

a) u is $\mathbf{D}$-aspheric if and only if $\forall\, F$ *constant* in $\mathbf{D}(I)$

$$F \xrightarrow[\sim]{\text{adj.}} u_* u^* F \qquad \text{is isomorphism.} \qquad (*)$$

p. 41

b) u is $\mathbf{D}$-coaspheric if and only if $\forall\, F$ *constant* in $\mathbf{D}(I)$

$$F \xleftarrow[\sim]{\text{adj.}} u_! u^* F \qquad \text{is isomorphism.} \qquad (**)$$

c) u is $\mathbf{D}$-connected if and only if for *every* F in $\mathbf{D}(I)$ (not necessarily constant, nor locally constant), $(*)$ is isomorphism, or what amounts to the same thing, for every F $(**)$ is isomorphism.

[4] **NB** This condition is satisfied if $\forall\, \xi, \eta \in \mathcal{A}$ we have $\mathrm{Hom}(\xi, \eta) \neq \emptyset$, (for it reduces to this: if u_1, u_2 are two arrows in Ens between sets *that are nonempty,* and if $u_1 \times u_2$ is isomorphism, u_1, u_2 are isomorphisms). This applies in particular if $\mathcal{A}$ admits a zero object, e.g. if $\mathcal{A}$ is additive. [page 66]

[5] **NB** These seven classes of morphisms in Cat are stable under composition including the notion of $\mathbf{D}$-connectedness, all these notions depend only on the set $W = W_{\mathbf{D}}$ of $\mathbf{D}$-equivalences.

Recall that the class W_{asph} W-aspheric arrows and that, Liss_W, of W-smooth arrows, determine each other, thus

$$\begin{cases} (f : X \longrightarrow Y) \in W_{\text{asph}} & \iff & \forall\, (Y' \xrightarrow{u} Y) \in \text{Liss}_W, \\ & & X' = X \times_Y Y' \longrightarrow Y' \text{ is } \in W. \\ (f : Y' \longrightarrow Y) \in \text{Liss}_W & \iff & \text{The functor } X \longmapsto X' = X \times_Y Y' \\ & & \text{from Cat}/Y \text{ to Cat}/Y' \text{ transforms} \\ & & \text{arrows in } W_{\text{asph}} \text{ in} \\ & & \text{arrows in } W_{\text{asph}}. \end{cases} \tag{4.10}$$

To every subset A of Fl(Cat) stable under isomorphisms, one can associate a subset $\mathbf{L}(A)$ of Fl(Cat), by the relation

$$\begin{array}{rcll} (u : Y' \longrightarrow Y) \in \mathbf{L}(A) & \iff & u^* : \text{Cat}/Y \longrightarrow \text{Cat}/Y' & \\ & & \text{transforms arrows} & (4.11) \\ & & \in A \text{ into arrows } \in A. & \end{array}$$

We see that $\mathbf{L}(A)$ contains the isomorphisms, is stable under composition, and is also stable under base change (as is the case for Liss_W). Conversely, to a subset L of Fl(Cat), stable under isomorphisms, we associate a subset $\mathbf{A}_W(L)$ by

$$\begin{array}{rcll} (f : X \longrightarrow Y) \in \mathbf{A}_W(L) & \iff & \forall\, (Y' \longrightarrow Y) \text{ in } L, & \\ & & X' = X \times_Y Y' \longrightarrow Y' & (4.12) \\ & & \text{is in } W. & \end{array}$$

If L is stable under base change, then $\mathbf{A}_W(L)$ is stable under composition. It contains the isomorphisms (since W contains them), is contained in W if L contains the identities. But no reason for it to be stable under arbitrary base change (and in fact, W_{asph} is not). [page 67]

(^{6}). We have, for a given $A \subseteq W \subseteq \text{Fl(Cat)}$, stable under isomorphisms,

$$A \subseteq \mathbf{A}_W(\mathbf{L}(A)) \overset{\text{def}}{=} A^\natural \subseteq W, \qquad \begin{array}{l} \text{subset stable under composition,} \\ \text{contains isomorphisms,} \end{array} \tag{4.13}$$

and for every subset L of Fl(Cat) which is stable under base change, and under composition p. 42

$$L \subseteq \mathbf{L}(\mathbf{A}_W(L)) \overset{\text{def}}{=} L^\flat, \qquad \begin{array}{l} \text{subset of Fl Cat stable} \\ \text{under composition and under base} \\ \text{change, containing the isomorphisms.} \end{array} \tag{4.14}$$

We have

$$\begin{cases} A^{\natural\natural} = A^\natural, & \text{since } \mathbf{A}_W(L)^\natural = \mathbf{A}_W(L) \\ L^{\flat\flat} = L^\flat, & \text{since } \mathbf{L}(A)^\flat = \mathbf{L}(A). \end{cases}$$

Examples:

$1^o)$ $L = \text{Ét}$, $\mathbf{A}_W(L) = W_{\text{asph}}$, $\mathbf{L}(\mathbf{A}_W(L)) = L^\flat = \text{Liss}_W$.

$2^o)$ $L = \text{Coét}$, $\mathbf{A}_W(L) = W_{\text{coas}}$, $\mathbf{L}(\mathbf{A}_W(L)) = L^\flat = \text{Prop}_W$.

$3^o)$ $A = W$, $\mathbf{L}(A) = \text{Fib}_W$, $A^\natural = W$.

$4^o)$ $L = \text{Fl Cat}$, $\mathbf{A}_W(L) = W_{\text{univ}}$, $L^\flat = L$.

[page 68 blank]

[6] Botched , cf. §6. p. 43

[page 69]

5 Formal duality

Let $\mathcal{P}$, $\mathcal{Q}$ be two ordered sets. I am trying to isolate a notion of "duality" (formal) linking $\mathcal{P}$ and $\mathcal{Q}$. This will be a relation between $\mathcal{P}$ and $\mathcal{Q}$

$$\mathcal{R} \subseteq \mathcal{P} \times \mathcal{Q} \tag{5.1}$$

having the following properties

(D) For every subset A of $\mathcal{P}$, the set

$$A^{\cup} \overset{\text{def}}{=} \{q \in \mathcal{Q} \mid (p,q) \in \mathcal{R} \text{ for every } p \in A, \text{ i.e. } A \times \{q\} \subseteq \mathcal{R}\}$$

is an open subset of $\mathcal{Q}$ having a greatest element $\beta(A) : A^{\cup} = \mathcal{Q}/\beta(A)$. Dually, for every subset B of $\mathcal{Q}$, the set

$$B^{\cap} \overset{\text{def}}{=} \{p \in \mathcal{P} \mid (p,q) \in \mathcal{R} \text{ for every } q \in B, \text{ i.e. } \{p\} \times B \subseteq \mathcal{R}\}$$

is an open subset of $\mathcal{P}$ having a greatest element $\alpha(B)$, i.e. $B^{\cap} = \mathcal{P}/\alpha(B)$.

Most often, $\mathcal{P}$ and $\mathcal{Q}$ will be stable under arbitrary suprema (hence also under arbitrary infima). In this case, (D) is equivalent, on the side of the $A^{\cup}$ (for $A \in \mathfrak{P}(\mathcal{P})$) to this:

(D_0) For every $p \in \mathcal{P}$, $\mathcal{R}(p) = \{q \in \mathcal{Q} \mid (p,q) \in \mathcal{R}\}$ is an open *filtering* subset of $\mathcal{Q}$, or what amounts to the same thing, a nonempty subset ($\Longleftrightarrow$ containing the least element $\emptyset_{\mathcal{Q}}$ of $\mathcal{Q}$) stable under suprema of two elements. Moreover, $\mathcal{R}(p)$ is stable under filtering suprema, or what amounts to the same thing (being itself filtering), it contains a greatest element. It amounts to the same thing to say that $\mathcal{R}(p)$ is an *open subset* of $\mathcal{Q}$ stable under arbitrary suprema (which implies that it contains $\emptyset_{\mathcal{Q}}$). Dually, for every $q \in \mathcal{Q}$, ${}^{s}\mathcal{R}(q) = \{p \in \mathcal{P} \mid (p,q) \in \mathcal{R}\}$ is an *open* subset of $\mathcal{P}$, stable under arbitrary suprema, i.e. an open subset having a greatest element. [page 70]

In other words, when one postulates condition (D) for A resp. B reduced to one element, then it remains satisfied for every subset. This comes from the formulas

$$\begin{cases} (\bigcup_i A_i)^{\cup} &= \bigcap_i A_i^{\cup} \\ (\bigcup_j B_j)^{\cap} &= \bigcap_j B_j^{\cap}, \end{cases} \tag{5.2}$$

and from the fact that the intersection of open subsets of the form $\mathcal{P}/\alpha_i$ in $\mathcal{P}$ is of the form $\mathcal{P}/\alpha$, where $\alpha = \inf_i \alpha_i$, and likewise in $\mathcal{Q}$. Thus we find

$$\begin{cases} \beta(\bigcup_i A_i) &= \inf_i \beta(A_i) \\ \alpha(\bigcup_j B_j) &= \inf_j \alpha(B_j). \end{cases} \tag{5.3}$$

We shall put, for a subset A of $\mathcal{P}$ resp. B of $\mathcal{Q}$ p. 44

$$A^{\natural} = (A^{\cup})^{\cap}, \qquad B^{\flat} = (B^{\cap})^{\cup} \tag{5.4}$$

so that we shall have

$$A \subseteq A^{\natural}, \qquad B \subseteq B^{\flat} \tag{5.5}$$

We shall say that A is $\mathcal{R}$-*closed* if $A = A^{\natural}$ (or again if A is of the form $B'^{\cap}$ for $B' \subseteq \mathcal{Q}$, cf. 5.6), that B is $\mathcal{R}$-*closed* if $B = B^{\flat}$. We shall have for every $A \subseteq \mathcal{P}$, $B \subseteq \mathcal{Q}$ [71]

$$\begin{cases} (A^{\cup})^{\flat} &= A^{\cup}, \text{ i.e. } A^{\cup} \text{ is } \mathcal{R}\text{-closed}, \\ (B^{\cap})^{\natural} &= B^{\cap}, \text{ i.e. } B^{\cap} \text{ is } \mathcal{R}\text{-closed}. \end{cases} \tag{5.6}$$

In particular

$$(A^\natural)^\natural = A^\natural, \qquad (B^\flat)^\flat = B^\flat, \tag{5.7}$$

i.e. the $\mathcal{R}$-closure $A^\natural$ of $A \subseteq \mathcal{P}$ is $\mathcal{R}$-closed, and likewise the $\mathcal{R}$-closure $B^\flat$ of $B \subseteq \mathcal{Q}$.

In fact, I realize that I am mixing two different kinds of facts. The relations (5.2), and (5.4) to (5.7), do not depend on the datum of order relations on $\mathcal{P}$, $\mathcal{Q}$, nor on axioms connecting these with the given relation $\mathcal{R}$, but only on this relation, which need not be subject to any condition. Let us note that

$$\emptyset^\cup = \mathcal{Q}, \qquad \emptyset^\cap = \mathcal{P}, \tag{5.8}$$

whereas

$$\emptyset^\natural = \mathcal{Q}^\cap, \qquad \emptyset^\flat = \mathcal{P}^\cup \tag{5.9}$$

are respectively the smallest $\mathcal{R}$-closed subset of $\mathcal{P}$ and of $\mathcal{Q}$.

The map

$$A \longmapsto A^\natural \tag{5.10}$$

is an *increasing* (and *idempotent*) map from $\mathfrak{P}(\mathcal{P})$ to itself, thus having the property [72]

$$(\bigcup_i A_i)^\natural = (\bigcup_i A_i^\natural)^\natural.$$

For every $A \subseteq \mathcal{P}$, $A^\natural$ is the smallest $\mathcal{R}$-closed subset of $\mathcal{P}$ containing A, and likewise p. 45
for every $B \subseteq \mathcal{Q}$, $B^\flat$ is the smallest $\mathcal{R}$-closed subset of $\mathcal{Q}$ containing B. We have moreover (this should even have been said before saying that $A \longmapsto A^\natural$ is increasing)

$$\begin{cases} A \subseteq A' & \Longrightarrow \quad A'^\cup \subseteq A^\cup \\ B \subseteq B' & \Longrightarrow \quad B'^\cap \subseteq B^\cap, \end{cases} \tag{5.11}$$

i.e. the maps

$$\mathfrak{P}(\mathcal{P}) \underset{\alpha}{\overset{\beta}{\rightleftarrows}} \mathfrak{P}(\mathcal{Q}) \qquad \begin{matrix} A \longmapsto A^\cup \\ B \longmapsto B^\cap \end{matrix} \tag{5.12}$$

are decreasing (which implies that the two composites $\alpha\beta = (A \longmapsto A^\natural)$ and $\beta\alpha = (B \longmapsto B^\flat)$ are increasing). Their restrictions to the subsets $\mathfrak{P}_\mathcal{R}(\mathcal{P})$ and $\mathfrak{P}_{^s\mathcal{R}}(\mathcal{Q})$ of the $\mathcal{R}$-closed subsets of $\mathcal{P}$ resp. of $\mathcal{Q}$, are, by definition of these as

$$\begin{aligned} \mathfrak{P}_\mathcal{R}(\mathcal{P}) &= \{A \subseteq \mathcal{P} \mid A^\natural \ (\overset{\text{def}}{=} \alpha\beta(A)) = A\} \\ \mathfrak{P}_{^s\mathcal{R}}(\mathcal{Q}) &= \{B \subseteq \mathcal{Q} \mid B^\flat \ (\overset{\text{def}}{=} \beta\alpha(B)) = B\}, \end{aligned}$$

inverse to one another. We thus find

Proposition 5.1. *Let $\mathcal{P}$, $\mathcal{Q}$ be two sets, $\mathcal{R} \subseteq \mathcal{P} \times \mathcal{Q}$ a relation. Then the maps α, β of (5.12) induce bijections inverse to one another, between the set $Cl_\mathcal{R}(\mathcal{P})$ of the $\mathcal{R}$-closed subsets of $\mathcal{P}$, and the set $Cl_{^s\mathcal{R}}(\mathcal{Q})$ of the $^s\mathcal{R}$-closed subsets (we shall say $\mathcal{R}$-closed by abuse of language) of $\mathcal{Q}$. These bijections are decreasing, hence antiisomorphisms for the natural order relations (of inclusion).* [73]

Corollary 5.2. *Let $(A_i)_i$ be a family of $\mathcal{R}$-closed subsets of $\mathcal{P}$. Then $\bigcap A_i$ is a $\mathcal{R}$-closed subset of $\mathcal{P}$, and*

$$(\bigcap_i A_i)^\cup = \underbrace{(\bigcup_i A_i^\cup)^\flat}_{\substack{\text{sup of the} \\ \text{subsets } \mathcal{R}\text{-closed} \\ A_i^\cup \text{ of } \mathcal{Q}}} . \tag{5.2.1}$$

But the relation $\mathcal{R}$ "extends" to a relation $\bar{\mathcal{R}}$ between

$$\bar{\mathcal{P}} = \mathfrak{P}(\mathcal{P}), \qquad \bar{\mathcal{Q}} = \mathfrak{P}(\mathcal{Q}) \tag{5.13}$$

by the prescription

$$\underbrace{\bar{\mathcal{R}}}_{\subseteq \bar{\mathcal{P}} \times \bar{\mathcal{Q}}} = \{(A, B) \in \bar{\mathcal{P}} \times \bar{\mathcal{Q}} \mid B \subseteq A^{\cup}, \text{ i.e. } A \subseteq B^{\cap}, \text{ i.e. } A \times B \subseteq \mathcal{R}\}. \tag{5.14}$$

The sets $\bar{\mathcal{P}}$, $\bar{\mathcal{Q}}$ are stable under arbitrary sups (union of subsets of $\mathcal{P}$ resp. $\mathcal{Q}$), and the relation $\bar{\mathcal{R}}$ on these ordered sets satisfies condition (D_0) above (p. 69), in an obvious way: for fixed A, the B such that $(A, B) \in \bar{\mathcal{R}}$ are those which are contained in $A^{\cup}$, so the set $\bar{\mathcal{R}}(A)$ is nothing other than $\bar{\mathcal{Q}}/A^{\cup}$, and dually ${}^s\bar{\mathcal{R}}(B)$ is nothing other than $\bar{\mathcal{P}}/B^{\cap}$. Thus p. 46 [74]

$$\beta(A) = A^{\cup}, \qquad \alpha(B) = B^{\cap}. \tag{5.14}$$

If I thought of introducing axiom (D_0), for a pair of arbitrary ordered sets, it is because of the cases where these are *not* of the form $\mathfrak{P}(\mathcal{P})$, $\mathfrak{P}(\mathcal{Q})$, and (even if they were) where the basic relation is not deduced as above from an arbitrary relation between sets $\mathcal{P}$, $\mathcal{Q}$.

I therefore return to the axiomatics of the beginning, with $\mathcal{P}$, $\mathcal{Q}$ ordered, and $\mathcal{R}$ satisfying (D). In all the cases I can think of, $\mathcal{P}$ and $\mathcal{Q}$ are stable under arbitrary sups, so (D) is equivalent to (D_0). Let us note that the map

$$\mathcal{P} \longrightarrow \mathfrak{P}(\mathcal{P}), \qquad p \longmapsto \mathcal{P}/p$$

is injective, and likewise for $\mathcal{Q} \longrightarrow \mathfrak{P}(\mathcal{Q})$. Thus, knowledge of $A \longmapsto A^{\cup}$ is equivalent to that of

$$\begin{array}{rcl} A & \longmapsto & \beta(A) \\ \beta : \mathfrak{P}(\mathcal{P}) & \longrightarrow & \mathcal{Q}, \end{array}$$

and likewise that of $B \longmapsto B^{\cap}$ is equivalent to that of

$$\alpha : \mathfrak{P}(\mathcal{Q}) \longrightarrow \mathcal{P}.$$

In fact, it suffices to give oneself [75]

$$\begin{array}{rcccl} \beta_0 : & \mathcal{P} & \longrightarrow & \mathcal{Q} & \text{maps} \\ \alpha_0 : & \mathcal{Q} & \longrightarrow & \mathcal{P} & \textit{decreasing} \end{array} \tag{5.15}$$

defined by

$$\left\{ \begin{array}{rcl} \beta_0(p) & = & \beta(\{p\}) \\ \alpha_0(q) & = & \alpha(\{q\}), \end{array} \right.$$

since we deduce β, α from them by

$$\left\{ \begin{array}{rcl} \beta(A) & = & \inf_{p \in A} \beta_0(p) \\ \alpha(B) & = & \inf_{q \in B} \alpha_0(q). \end{array} \right. \tag{5.16}$$

The relation $\mathcal{R}$ is deduced from the datum of β_0, or at one's choice of α_0, by

$$(p, q) \in \mathcal{R} \iff q \leq \beta_0(p) \iff p \leq \alpha_0(q). \tag{5.17}$$

This therefore implies that α_0, β_0 determine one another, by the formulas

$$\left\{ \begin{array}{rcl} \alpha_0(q) & = & \sup \underbrace{\{p \in \mathcal{P} \mid q \leq \beta_0(p)\}}_{\{q\}^{\cap}} \\ \beta_0(p) & = & \sup \underbrace{\{q \in \mathcal{Q} \mid p \leq \alpha_0(q)\}}_{\{p\}^{\cup}}. \end{array} \right. \tag{5.18}$$

p. 47

It remains to ask which maps $\beta_0 : \mathcal{P} \longrightarrow \mathcal{Q}$, or $\alpha_0 : \mathcal{Q} \longrightarrow \mathcal{P}$, correspond to a *"dualizing relation"* $\mathcal{R}$ on $\mathcal{P} \times \mathcal{Q}$. Here are the conditions on $\beta_0 : \mathcal{P} \longrightarrow \mathcal{Q}$: [76]

ⓐ β_0 is *decreasing*,

which implies that the relation

$$\underbrace{\mathcal{R}_{\beta_0}}_{\subseteq \mathcal{P}\times\mathcal{Q}} = \{(p,q) \in \mathcal{P}\times\mathcal{Q} \mid q \leq \beta_0(p)\}$$

satisfies condition (D_0) on the left side, and we still have to check it on the right side, i.e. look, for fixed q, at the set

$$\{q\}^{\cap} = \{p \in \mathcal{P} \mid q \leq \beta_0(p)\}.$$

As β_0 is decreasing, we see that $p \in \{q\}^{\cap}$, and $p' \leq p \Longrightarrow p' \in \{q\}^{\cap}$, i.e. $\{q\}^{\cap}$ is *open.* We must still check (assuming the existence of arbitrary sups in $\mathcal{P}$) that this set is stable under arbitrary sups - it will follow that it is of the form $\mathcal{P}/\alpha(q)$. This is equivalent to the condition

ⓑ β_0 transforms arbitrary sups in $\mathcal{P}$ into infs in $\mathcal{Q}$.

If we regard β_0 as a contravariant functor $\mathcal{P}^o \longrightarrow \mathcal{Q}$, this means that it commutes with inverse limits, or again that it has a left adjoint $\mathcal{Q} \longrightarrow \mathcal{P}^o$, which commutes with direct limits, and corresponds to the decreasing map $\alpha_0 : \mathcal{Q} \longrightarrow \mathcal{P}$, transforming sups into infs. [77]

Here it is convenient to consider $\natural$ and $\flat$ as operations on $\mathcal{P}$ resp. $\mathcal{Q}$, by

$$p^{\natural} = \alpha_0\beta_0(p), \qquad q^{\flat} = \beta_0\alpha_0(q). \tag{5.19}$$

The relation between these endomorphisms, and the operation of the same name on $\mathfrak{P}(\mathcal{P})$, $\mathfrak{P}(\mathcal{Q})$, is given by

$$\begin{cases} p^{\natural} &= \text{Greatest element of } \{p\}^{\natural} \\ A^{\natural} &= \{p \in \mathcal{P} \mid p \leq (\sup_{a\in A} a)^{\natural}\} = \{p \in \mathcal{P} \mid p \leq (\sup_{a\in A} a^{\natural})^{\natural}\}, \end{cases} \tag{5.20}$$

and symmetrically on the side of $\mathcal{Q}$. p. 48

We say that p is $\mathcal{R}$-*closed,* or β_0-closed, or α_0-closed, if $p = p^{\natural}$, and likewise $q \in \mathcal{Q}$ is said to be $\mathcal{R}$-*closed* etc. if $q = q^{\flat}$. The maps

$$\begin{cases} p &\longmapsto & p^{\natural} &: & \mathcal{P} &\longrightarrow & \mathcal{P} \\ q &\longmapsto & q^{\flat} &: & \mathcal{Q} &\longrightarrow & \mathcal{Q} \end{cases} \tag{5.21}$$

are *increasing* and *idempotent.* We can characterize $p^{\natural}$ as the smallest closed element of $\mathcal{P}$ lying above p, and likewise for $q^{\flat}$. We then find, by (5.19), that α_0 and β_0 induce bijections inverse to one another, between the set of closed elements of $\mathcal{P}$, and the set of [78] closed elements of $\mathcal{Q}$

$$\mathcal{P}^{\natural} \underset{\bar{\alpha}_0}{\overset{\bar{\beta}_0}{\underset{\sim}{\rightleftarrows}}} \mathcal{Q}^{\flat}. \tag{5.22}$$

But in practice, the maps α_0, β_0 are not obtained directly, but we have at our disposal the relation $\mathcal{R}$, satisfying condition (D_0), from which we deduce α_0, β_0. In all the cases I am thinking of here, either $\mathcal{P}$, $\mathcal{Q}$ are of the form $\mathfrak{P}(P_0)$, $\mathfrak{P}(Q_0)$, and $\mathcal{R}$ is deduced from a relation $R_0 \subseteq P_0 \times Q_0$ as above (by far the most frequent, it seems to me), or else $\mathcal{P}$ and $\mathcal{Q}$ appear as suitable subsets of sets $\mathfrak{P}(P_0)$, $\mathfrak{P}(Q_0)$, equipped with the induced order, and the relation $\mathcal{R}$ on $\mathcal{P}\times\mathcal{Q}$ is obtained in a more complex way.

Examples.

1) Relations of *orthogonality* between subsets of two modules over a ring, equipped with a duality pairing. In the case of vector of finite dimension and of a true duality, the closed subsets are the vector subspaces, in the infinite case, they are the vector subspaces closed for the "weak topology" defined by the pairing.

2) Relations of *orthogonality* between subsets of two locally compact commutative topological groups dual to one another, in the sense of Pontrjagin. [79]

   The closed subsets are the closed subgroups.

3) For two vector spaces "in duality" over a complete valued field, such as $\mathbf{R}$ or $\mathbf{C}$, the relation

$$|\langle x, x' \rangle| \leq 1.$$

   In the real or complex case, the closed subsets are the nonempty convex circled weakly closed subsets.

4) For real vector spaces, the relation of semi-polarity

$$\langle x, x' \rangle \geq -1.$$

   The closed subsets are the convex weakly closed subsets containing the origin.

p. 49

5) Let K be a field (for simplicity), L a Galois extension with group G. Relation between L and G

$$\gamma \cdot x = x, \qquad \text{i.e. } x \in L^{\gamma}.$$

   The closed subsets of L are the subextensions, the closed subsets of G the subgroups of G closed for the Krull topology.

There are various analogues in inseparable cases (by dint of p-Lie algebras for example), and descent statements of all kinds.

All these examples are of the 1st kind, the ordered sets that occur are of the form [page 80] $\mathfrak{P}(P_0)$, $\mathfrak{P}(Q_0)$, and the relation between them is deduced from a relation R_0 between P_0 and Q_0. Here now are more complex examples.

6) Let $\mathcal{M}$ be a category ("big", most often …). We take

$$\mathcal{P} = \mathcal{Q} = \mathfrak{P}(\mathrm{Fl}(\mathcal{M})).$$

   The relation $\mathcal{R} \subseteq \mathcal{P} \times \mathcal{Q}$ is Quillen's relation, which I write, for $\Phi, \Psi \in \mathfrak{P}$

$$\Psi \subseteq \Phi_*,$$

   or again equivalently

$$\Phi \subseteq \Psi^*,$$

   (Ψ has the LLP with respect to Φ, Φ the RLP with respect to Ψ). Thus the maps α_0, β_0 are given by

$$\beta_0(\Phi) = \Phi_*, \qquad \alpha_0(\Psi) = \Psi^*.$$

   Here it would occur to no one to introduce $\mathfrak{P}(\mathcal{P})$, $\mathfrak{P}(\mathcal{Q})$ - it is already big enough as it is! The notation for $\Phi^{\natural}$ is here $\tilde{\Phi}$, for $\Psi^{\flat}$ it is $\bar{\Psi}$. The closed subsets are those I call Q-closed on the left and on the right.

   The relation $\mathcal{R}$, which I write

$$\Phi \longleftrightarrow \Psi,$$ [page 81]

is not deduced from a relation between arrows of $\mathcal{M}$, i.e. from a subset of $\mathrm{Fl}(\mathcal{M}) \times \mathrm{Fl}(\mathcal{M})$, but is of a more complex nature. To describe it, one must introduce the set $\mathcal{D}$ of commutative square diagrams (solid lines) of the form

$$\begin{array}{ccc} A & \longrightarrow & X \\ \downarrow{\scriptstyle i} & \overset{f}{\nearrow} & \downarrow{\scriptstyle p} \\ B & \longrightarrow & Y \end{array} \qquad (*)$$

(which can be interpreted as $\mathrm{Fl}(\underline{\mathrm{Fl}}(\mathcal{M}))$, the set of arrows in $\underline{\mathrm{Fl}}(\mathcal{M})$). We have a subset $\mathcal{D}_0$ of $\mathcal{D}$, consisting of the diagrams for which exists f with two commutative triangles (it is thus the image of $\underbrace{\mathrm{Fl}_3(\mathcal{M})}_{\mathrm{Hom}(\mathbf{\Delta}^3,\mathcal{M})}$ in $\mathrm{Hom}(\mathbf{\Delta}^1 \times \mathbf{\Delta}^1, \mathcal{M})$, transpose of $\mathbf{\Delta}^1 \times \mathbf{\Delta}^1 \longrightarrow \mathbf{\Delta}^3$ given by $(0,0) \longmapsto 0$, $(0,1) \longmapsto 1$, $(1,0) \longmapsto 2$, $(1,1) \longmapsto 3$), and a map

p. 50

$$\mathcal{D}_0 \subseteq \mathcal{D} \xrightarrow{\varphi} \mathrm{Fl}(\mathcal{M}) \times \mathrm{Fl}(\mathcal{M}),$$

associating to $(*)$ the pair (i, j) (source and target of $(*)$, considered as an arrow in $\underline{\mathrm{Fl}}(\mathcal{M})$ from i to p). The relation $\mathcal{R}$ is then deduced from φ, $\mathcal{D}_0$ by the condition

[page 82]

$$\Phi \longleftrightarrow \Psi \qquad \Longleftrightarrow \qquad \varphi^{-1}(\Psi \times \Phi) \subseteq \mathcal{D}_0.$$

But to tell the truth, I realize that it is after all deduced from a relation between elements of $\mathrm{Fl}(\mathcal{M})$, and this relation between $i : A \longrightarrow B$ and $p : X \longrightarrow Y$ can be written as follows:

($**$) for all arrows $A \longrightarrow X$, $B \longrightarrow Y$ making the diagram $(*)$ commutative, the dotted arrow f in $(*)$ exists.

This makes me think that if we have a relation $\mathcal{R}$ between $\mathfrak{P}(P)$ and $\mathfrak{P}(Q)$, satisfying (D_0), hence described by a decreasing map

$$\mathfrak{P}(P) \longrightarrow \mathfrak{P}(Q)$$

transforming sup into inf, the datum of such a map is equivalent to that of an arbitrary map

$$P \longrightarrow \mathfrak{P}(Q),$$

or what amounts to the same thing, of a relation $\mathcal{R}$ on $P \times Q$. Thus this structure must be regarded as "of the first kind". The situation changes from the moment when, to describe the relation $\mathcal{R}$ with the desired properties, we find ourselves compelled to replace $\mathfrak{P}(P)$, $\mathfrak{P}(Q)$ by suitable subsets - when we look only at certain subsets of P, Q. This is what we are going to do in the next §. The case I am going to treat there is the only one of its kind that I have encountered - a case of "formal duality" which comes in naturally, and which is not "of the 1st kind".

[page 83]

p. 51

6 Transporters, kernels, stabilizers relative to W

Let $\mathcal{M}$ be a category, equipped with a subset

$$W \subseteq \mathrm{Fl}(\mathcal{M}). \qquad (6.1)$$

I am thinking of the case of a localizer, but will introduce an assumption on W only as needed. For simplicity, I assume $\mathcal{M}$ stable under fiber products.

I set

$$\left\{\begin{array}{rcll} ({}^{62})\mathcal{P} & = & \mathfrak{P}(W) = \{A \in \mathfrak{P}(\mathrm{Fl}(\mathcal{M})) \mid A \subseteq W\} & \\ \mathcal{Q} & = & \{L \in \mathfrak{P}(\mathrm{Fl}(\mathcal{M})) \mid & \begin{array}{l} L \textit{ stable} \text{ under base} \\ \text{change (by which I also mean,} \\ \text{stable under isomorphism)} \\ \text{and under composition. }\}. \end{array} \end{array}\right. \tag{6.2}$$

I introduce a relation

$$\mathcal{R} \subseteq \mathcal{P} \times \mathcal{Q}$$

in the following way. For $A \in \mathcal{P}$, $L \in \mathcal{Q}$, we set

$$(A, L) \in \mathcal{R} \iff \begin{array}{l} \text{For every cartesian diagram} \\ \begin{array}{ccc} X & \longleftarrow & X' \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'} \\ Y & \underset{p}{\longleftarrow} & Y' \end{array} \\ \text{with } f \in A,\ p \in L, \text{ we have } f' \in W \\ \text{(i.e. the base changes of arrows of } A \text{ by } p \in L \text{ belong to } W). \end{array} \tag{6.3}$$

When I order $\mathcal{P}, \mathcal{Q} \subseteq \mathfrak{P}(\mathrm{Fl}(\mathcal{M}))$ by the order relation induced by inclusion of subsets of $\mathrm{Fl}(\mathcal{M})$, we find ordered subsets of $\mathfrak{P}(\mathrm{Fl}(\mathcal{M}))$, stable under arbitrary unions, and also under arbitrary intersections, with the sole proviso that in the case of $\mathcal{P}$, the family of indices must be $\neq \emptyset$ (since then we find as intersection the whole of $\mathrm{Fl}(\mathcal{M})$, which is not in general contained in W). [page 84]

We are going to verify condition (D_0) on page 69, for $\mathcal{R}$ and the order relations. There are two checks to make.

1°) Let us give ourselves $A \in \mathcal{P}$, i.e. $A \subseteq W$, and consider the set $\mathcal{R}(A)$ of $L \in \mathcal{Q}$ such that we have $(A, L) \in \mathcal{R}$. It is clear that this is an open set (and likewise, that ${}^s\mathcal{R}(L)$ is an open set of $\mathcal{P}$ for every L in $\mathcal{Q}$) - i.e. $L \in \mathcal{R}(A)$, $L' \subseteq L$ implies $L' \in \mathcal{R}(A)$. p. 52
It remains to find a greatest element of $\mathcal{R}(A)$. Let us note that if $L \in \mathcal{R}(A)$, then for every $p \in L$, $p : Y' \longrightarrow Y$, the base change

$$p^* : \mathcal{M}/Y \longrightarrow \mathcal{M}/Y' : X \longmapsto X \times_Y Y'$$

must transform arrows $f \in A$ into arrows $g \in W$, as follows from the fact that L is stable under base change.

$$\begin{array}{ccc} X_1 & \xleftarrow{q_1} & X_1' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ X_2 & \xleftarrow{q_2} & X_2' \\ \downarrow & & \downarrow \\ Y & \xleftarrow{p} & Y'. \end{array}$$

Thus $L \subseteq \mathbf{L}_W(A)$, where we set [page 85]

$$\mathbf{L}_W(A) = \{(p : Y' \longrightarrow Y) \in \mathrm{Fl}(\mathcal{M}) \quad | \quad p^* : \mathcal{M}/Y \longrightarrow \mathcal{M}/Y' \text{ transforms arrows of } A \text{ into arrows } \textit{of } W.\}$$

[62] cf. correction p. 86.

([63]).

$$\begin{array}{ccc} X_1 & \longleftarrow & X_1' \\ {\scriptstyle f}\downarrow & \text{cart.} & \downarrow{\scriptstyle f'} \\ X_2 & \longleftarrow & X_2' \\ \downarrow & \text{cart.} & \downarrow \\ Y & \xleftarrow{p} & Y' \end{array}$$

$$f \in A \implies f' \in W.$$

It is clear on the other hand that this subset of $\mathrm{Fl}(\mathcal{M})$ is stable under base change, hence $\mathbf{L}_W(A) \in \mathcal{Q}$, and that $(A, \mathbf{L}_W(A)) \in \mathcal{R}$. Thus $\mathbf{L}_W(A)$ is a greatest element of $\mathcal{R}(A)$, so we find

$$\mathcal{R} = \{(A, L) \in \mathcal{P} \times \mathcal{Q} \mid L \subseteq \mathbf{L}_W(A)\}. \tag{6.5}$$

p. 53

$$\begin{array}{ccc} X_1 & \xleftarrow{q_1} & X_1' \\ {\scriptstyle f}\downarrow & \text{cart.} & \downarrow{\scriptstyle f'} \\ X_2 & \xleftarrow{q_2} & X_2' \\ \downarrow & \text{cart.} & \downarrow \\ Y_1 & \xleftarrow{p_1} & Y_1' \\ \downarrow & \text{cart.} & \downarrow \\ Y & \xleftarrow{p} & Y'. \end{array}$$

Conversely, let us start with $L \in \mathcal{Q}$, and show that the open set ${}^s\mathcal{R}(L)$ of $\mathcal{P}$ has a greatest element. For this, let us set

$$\mathbf{A}_W(L) = \{(f : X \longrightarrow Y) \in W \subseteq \mathrm{Fl}(\mathcal{M}) \quad | \quad \forall \quad \begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & \textit{cart.} & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y', \end{array} \text{ with } p \in L, \text{ we have } f' \in W\}. \tag{6.6}$$

(NB When L contains the identities, as is the case in particular if it is of the form $\mathbf{L}_W(A)$ - then it contains the isomorphisms, at least if A contains them - then $\mathbf{A}_W(L)$ is the set of $f \in \mathrm{Fl}(\mathcal{M})$ satisfying the condition stated after |, without having to specify $f \in W$.) We have $\mathbf{A}_W(L) \in \mathcal{P}$, and we verify that $(\mathbf{A}_W(L), L) \in \mathcal{R}$ (using the stability of L under composition), finally it is clear that $(A, L) \in \mathcal{R}$ implies $A \subseteq \mathbf{A}_W(L)$ (using $A \subseteq W$), so $\mathbf{A}_W(L)$ is a greatest element of ${}^s\mathcal{R}(L)$.

Let us note that if W *is stable under composition* (which we are going to assume from now on), the same is true of $\mathbf{A}_W(L)$ (which again uses the fact that L is stable under base change). Moreover, if W *contains the isomorphisms* (which we shall also assume), $\mathbf{A}_W(L)$ contains them too. As for $\mathbf{L}_W(A)$, it contains the isomorphisms *provided that* A is stable under isomorphisms; when A is stable under composition and contains the isomorphisms, this condition will automatically be satisfied. We insist on this condition,

[page 86]

[63] NB say it rather in terms of cartesian diagrams, since we have made no assumption of stability of W under isomorphisms.

that is why on the side of A, in the definition of $\mathcal{P}$, I am going to introduce this additional condition that A be *stable under isomorphisms.* So I set, correcting (6.2),

$$\begin{cases} \mathcal{P} &= \{A \in \mathfrak{P}(\mathrm{Fl}(\mathcal{M})) \quad | \quad \mathrm{Is}(\mathcal{M}) \subseteq A \subseteq W, \quad A \text{ stable under composition}\} \\ \mathcal{Q} &= \{L \in \mathfrak{P}(\mathrm{Fl}(\mathcal{M})) \quad | \quad \mathrm{Is}(\mathcal{M}) \subseteq L, \quad L \text{ stable under base change}\}. \end{cases} \tag{6.7}$$

We are of course interested above all in the closed elements of $\mathcal{P}$, $\mathcal{Q}$, i.e. the subsets A, L of $\mathrm{Fl}(\mathcal{M})$, containing the isomorphisms, stable under composition (and L also stable under base change, $A \subseteq W$) such that we have p. 54

$$A = \mathbf{A}_W(L), \qquad L = \mathbf{L}_W(A) \tag{6.8}$$

.

[page 87]

I shall keep the notations $\natural$, $\flat$ of the preceding paragraph. What are the two extreme pairs?

Starting from $A_0 = \mathrm{Is}(\mathcal{M})$ (smallest of the elements of $\mathcal{P}$), we find $\mathbf{L}_W(A_0) = \mathrm{Fl}(\mathcal{M})$ (largest element of $\mathcal{Q}$), and $A_0^\natural = \mathbf{A}_W(\mathrm{Fl}(\mathcal{M})) = W_{\mathrm{univ}}$, the set of arrows which are in W and remain so under any base change:

$$A_0^\natural = W_{\mathrm{univ}}, \qquad \beta_0(A_0) \overset{\mathrm{def}}{=} \mathbf{L}_W(A_0) = \mathrm{Fl}(\mathcal{M}), \qquad \text{where } A_0 = \mathrm{Is}(\mathcal{M}) \text{ smallest element of } \mathcal{P}. \tag{6.9}$$

Conversely, let us start from $L_0 = \mathrm{Is}(\mathcal{M})$, the smallest element of $\mathcal{Q}$. We have $\mathbf{A}_W(L_0) = W$ (W being stable under composition and containing the isomorphisms), and $L_0^\flat = \mathbf{L}_W(W) = \mathrm{Fib}_W$, the W-fibrations:

$$L_0^\flat = \mathrm{Fib}_W, \qquad \alpha_0(L_0) = W, \qquad \text{where } L_0 = \mathrm{Is}(\mathcal{M}), \text{ smallest element of } \mathcal{Q}. \tag{6.10}$$

So the left $\mathcal{R}$-closed subsets of $\mathrm{Fl}(\mathcal{M})$ wander between W_{univ} and W, the right $\mathcal{R}$-closed subsets between Fib_W and $\mathrm{Fl}(\mathcal{M})$.

Proposition 6.1. *Let $A \in \mathcal{P}$, $L \in \mathcal{Q}$ be such that $A = \alpha_0(L)$, $L = \beta_0(A)$ (i.e. $A = \mathbf{A}_W(L)$, $L = \mathbf{L}_W(A)$). For L to be stable under composition, it is necessary and sufficient that A be stable under base change by the arrows $f \in L$. (More generally, if $L \in \mathcal{Q}$ is any element of $\mathcal{Q}$, and if we assume in addition that L is stable under composition, then $A = \mathbf{A}_W(L)$ is stable under base change by L.)* [page 88]

Terminology: for $A \in \mathcal{P}$, i.e. A a subset of $\mathrm{Fl}(\mathcal{M})$ contained in W containing the isomorphisms and stable under composition, $\mathbf{L}_W(A)$ is called the *transporter of A into W*. (There is a dual notion of *cotransporter.*) It is an element of $\mathcal{Q}$, thus a subset of $\mathrm{Fl}(\mathcal{M})$ containing the isomorphisms and stable under base change. On the other hand, for L in $\mathcal{Q}$, the arrows $f \in \mathbf{A}_W(L)$ are called *W-L-aspheric,* or the *W-L-equivalences,* or the *W-equivalences that are L-universal,* and $\mathbf{A}_W(L)$ is called the *L-kernel of W*. We could denote it more simply by W_L (when this notation does not conflict with $W_S \subseteq \mathrm{Fl}(\mathcal{M}/S)$ for $S \in \mathrm{Ob}\,\mathcal{M}$). It is a subset of W which (like W itself) contains the isomorphisms and is stable under composition.

Finally, there is also reason to introduce the *stabilizer* $\mathrm{Stab}(A)$, or transporter of A into A, which does not depend on the choice of W, but only on the subset A of $\mathrm{Fl}(\mathcal{M})$, containing the isomorphisms and stable under composition. We could also denote it by Fib_A, and call its elements *A-fibrations* (A now playing the role of W). We have $\mathrm{Stab}(A) \subseteq \mathbf{L}_W(A)$, and it would seem that in practice, it is these stabilizers, more than the transporters $\mathbf{L}_W(A)$, that are important (when the two do not coincide). Thus in Cat, if W is a basic localizer, the W-smooth (resp. W-proper) morphisms Liss_W (Prop_W), which are the stabilizers of the sets W_{as} (W_{coas}) of the sets of W-aspheric (resp. p. 55 [page 89]

W-coaspheric) morphisms, playing an essential role, whereas I do not know how to characterize in a meaningful way the arrows of the transporters $\mathbf{L}_W(W_{\mathrm{as}})$, $\mathbf{L}_W(W_{\mathrm{coas}})$, which contain, in addition to the W-smooth (resp. W-proper) morphisms the W-fibrations Fib_W. I do not know how to characterize a class of morphisms which looks so motley! Likewise, there is reason to introduce and characterize the stabilizer of W_{bias} (set of W-biaspheric arrows), as the "W-neat" morphisms Net_W, by a description similar to that of the Liss_W, Prop_W, using the bilocalized categories $\backslash\underset{u}{X}/$, $\backslash\underset{v}{Y}/$, instead of the localized categories X/x, Y/y or the colocalized categories $x\backslash X$, $y\backslash Y$. Finally there is reason to determine the stabilizer of the set Equ of Cat-equivalences (which does not depend on the choice of a W - but we have $\mathrm{Equ} \in \mathcal{P}(W)$ for every basic W).

p. 56

[page 90]

7 Cofiltering and locally cofiltering morphisms

But I do not want to dwell now on these developments, which do not seem urgent to me, and would like to return to the questions connected with the notion of filtering or cofiltering categories.

Definition 7.1. Let $f : X \longrightarrow Y$ be an arrow in Cat. We say that f is *cofiltering* (resp. *locally cofiltering*, resp. *weakly cofiltering* ([64]), resp. *locally weakly cofiltering* or again *precofiltering*) if for every y in Y, the colocalized category $y\backslash X$ is cofiltering (resp. …). (Dually, we say that f is *filtering*, resp. …, if the categories X/y are filtering, resp. …, i.e. if $f^o : X^o \longrightarrow Y^o$ is cofiltering resp. ….)

Remark 7.2. Notions stable under coétale base change. Contains the isomorphisms and even the Cat-equivalences. The *étale* morphisms are locally cofiltering.

Proposition 7.3. *Let $f : X \longrightarrow Y$ be a cofiltering morphism (resp. locally cofiltering, resp. weakly cofiltering, resp. precofiltering), and suppose that Y is cofiltering (resp. locally cofiltering, resp. weakly cofiltering, resp. precofiltering). Then X itself is cofiltering (resp. …).*

Proof. It is a matter, in the four cases considered, of checking for X the condition $\underbrace{\mathrm{PS}_g1' \,/\, \mathrm{PS}_g1}_{\text{last two cases}} \,/\, \mathrm{PS}_g1' + \mathrm{PS}_g2 \,/\, \mathrm{PS}_g1 + \mathrm{PS}_g2$. But for each of the three conditions PS_g1, PS_g1', PS_g2 separately, we see immediately that if it is satisfied for Y and for the $y\backslash X$, it is satisfied for X, whence the proposition.

[page 91]

Corollary 7.3. *The four notions considered in (7.1) are stable under composition.*

Proof immediate, using remark (7.2).

$$\begin{array}{ccc} X & \longleftarrow & z\backslash X \\ \downarrow & & \downarrow \\ Y & \longleftarrow & z\backslash Y \\ \downarrow & & \downarrow \\ Z & \longleftarrow & z\backslash Z. \end{array}$$

Remark 7.4.

p. 57

1) Note that the notion of a cofiltering or weakly cofiltering morphism is *not* stable under arbitrary base change, nor even under an open immersion. Example: $e \longrightarrow$

[64] I have changed the terminology: now weakly cofiltering = 0-connected + locally weakly cofiltering, *stable under passage* to X' over X, X' étale and ∞-connected. Review the terminology in the weakly cofiltering case, which is a treacherous notion, for lack of being stable under passage an X' étale over X.

$\mathbf{\Delta}^1$, mapping e to 1, is cofiltering,

$$\begin{array}{ccc} & & e \\ & & \downarrow \\ 0 & \longrightarrow & 1 \end{array}$$

but its fiber at 0 is empty, hence is neither cofiltering nor weakly cofiltering. I have not seen an example showing that the notion of a *locally* cofiltering resp. precofiltering arrow is not stable under base change, and would not want to dwell on it, having no need to know the answer.

2) For X to be cofiltering (resp. …), it is necessary and sufficient that $X \longrightarrow e$ be so. Thus the absolute notion is reduced to a relative notion.

3) If $f : X \longrightarrow Y$ is cofiltering, it is W-coaspheric (thus $\in W$) for every W satisfying the direct limit axiom Loc (8). In particular, it is W_0-coaspheric, i.e. a cofiltering functor (and even merely weakly cofiltering) is *cofinal.* [page 92]

The main reason for my interest in cofiltering arrows is contained in the

Proposition 7.4. *Let $f : X \longrightarrow Y$ in* Cat *be locally cofiltering (resp. locally weakly cofiltering). Then:*

a) For every X' étale over X, the composite $f' : X' \longrightarrow X \longrightarrow Y$ is locally cofiltering (resp. …), and for every factorization of f' as $X' \xrightarrow{g'} Y' \longrightarrow Y$ with $Y' \longrightarrow Y$ étale, g' is locally cofiltering (resp. …).

b) $\forall$ x in X, the morphism on the localized categories

$$f/x : X/x \longrightarrow Y/y \qquad (y = f(x))$$

is cofiltering (resp. weakly cofiltering).

c) If x, y are as in b), and U is an open subset of X/x, denoting by V the open subset of Y/y generated by the image of $(f/x)|_U$ (i.e. the open subset consisting of the z in Y/y such that $z \backslash U$ is nonempty), the induced arrow $g : U \longrightarrow V$ cofiltering (resp. weakly cofiltering), a fortiori cofinal (and even W-coaspheric, for every W satisfying the limit axiom Loc (8)) ([65]).

[page 93]

Thus for every cosheaf *G on V, with values in a category $\mathcal{M}$ with $\varinjlim$, we have isomorphism*

$$\varinjlim_U g^*(G) \xrightarrow{\sim} \varinjlim_V G,$$

and for every presheaf F on V, with values in a category $\mathcal{M}$ with $\varprojlim$, we have isomorphism p. 58

$$\varprojlim_U g^*(F) \xleftarrow{\sim} \varprojlim_V F.$$

Proof. I shall first spell out the

Lemma 7.5. *Let $h : X' \longrightarrow X$ be an étale morphism in* Cat. *Then h is locally cofiltering. More precisely, for every x in X, we have*

$$\pi_0(x \backslash X') \xrightarrow{\sim} \mathrm{Ob}\, X'_x,$$

[65] It suffices here to be under the resp'ed hypothesis (*f weakly* locally cofiltering) except for the NB here .

every connected component of $x\backslash X'$ contains one and only one element of the discrete category $X'_x \subseteq x\backslash X'$. If x' is in X'_x (i.e. x' an object of X' such that $h(x') = x$), it is an initial *object of the connected component which contains it. (Thus this component is cofiltering.)*

This proves (and beyond) the first assertion of 7.4, in view of 7.3 (stability under composition), or because an étale category (here $y\backslash X'$) over a locally cofiltering (resp. locally weakly cofiltering) category (here $y\backslash X$) is likewise. [page 94]

$$\begin{array}{ccc} y\backslash X' & \longrightarrow & X' \\ \downarrow{\scriptstyle \text{ét.}} & \text{cart.} & \downarrow{\scriptstyle \text{ét.}} \\ y\backslash X & \longrightarrow & X \\ \downarrow & \text{cart.} & \downarrow \\ y\backslash Y & \longrightarrow & Y. \end{array}$$

For the second assertion in a), we can now assume $X' = X$. We must prove that for every y' in Y', $y'\backslash X \overset{\text{def}}{=} X \times_{Y'} y'\backslash Y'$ is locally cofiltering (resp. locally weakly cofiltering). Let y in Y be the image of y', and consider the diagram of three cartesian squares opposite.

$$\begin{array}{ccccc} y'\backslash X & \longrightarrow & y\backslash X & \longrightarrow & X \\ \downarrow & \textit{cart.} & \downarrow & \textit{cart.} & \downarrow \\ y'\backslash Y' & \longrightarrow & y\backslash Y' & \longrightarrow & Y' \\ & & \downarrow & \textit{cart.} & \downarrow \\ & & y\backslash Y & \longrightarrow & Y \end{array}$$

Lemma 7.5 implies that $y'\backslash Y' \longrightarrow y\backslash Y'$ is the inclusion into $y\backslash Y'$ of a connected component of $y\backslash Y'$, so it is an open immersion. So $y'\backslash X \longrightarrow y\backslash X$ is an immersion

open. Since $y\backslash X$ is locally cofiltering (resp. locally weakly cofiltering) by hypothesis, the same is therefore true of $y'\backslash X$. This proves a). Applying a) to the situation $X/x \longrightarrow Y/y$ in b), we see that the colocalized categories $z\backslash(X/x)$ of X/x over Y/y ($z \in \mathrm{Ob}\,(Y/y)$) are locally cofiltering (resp. weakly). But since X/x has a final object over the final object of Y/y, it follows that $z\backslash(X/x)$ has a final object, hence is 0-*connected.* p. 59 !

Thus these categories, being locally cofiltering (resp. weakly), are in fact cofiltering (resp. weakly cofiltering). [page 95]

Let us prove c). Note that for z in V, $z\backslash U$ is identified with the inverse image, under $z\backslash(X/x) \longrightarrow X/x$, of the open subset U of X/x, it is therefore an *open subset* of $z\backslash(X/x)$, nonempty by construction of V. But a nonempty open subset of a cofiltering (resp. weakly cofiltering) category is again cofiltering (resp. weakly cofiltering). This means that $g : U \longrightarrow V$ is cofiltering (resp. weakly cofiltering), a fortiori cofinal (and W-coaspheric, for every W satisfying Loc (8)). This completes the proof.

Remark 7.5.

1) The same argument as in c) shows that if $f : X \longrightarrow Y$ is cofiltering (resp. weakly cofiltering), then for every open subset U in X, if we denote by V the open subset of Y generated by $f(U)$, i.e. the open subset consisting of the y in Y such that $y\backslash U \neq \emptyset$, then the induced arrow $g : U \longrightarrow V$ is cofiltering (resp. weakly cofiltering), and

a fortiori g is cofinal (and more generally, if f is cofiltering, g is W-coaspheric for every W satisfying condition Loc (8)).

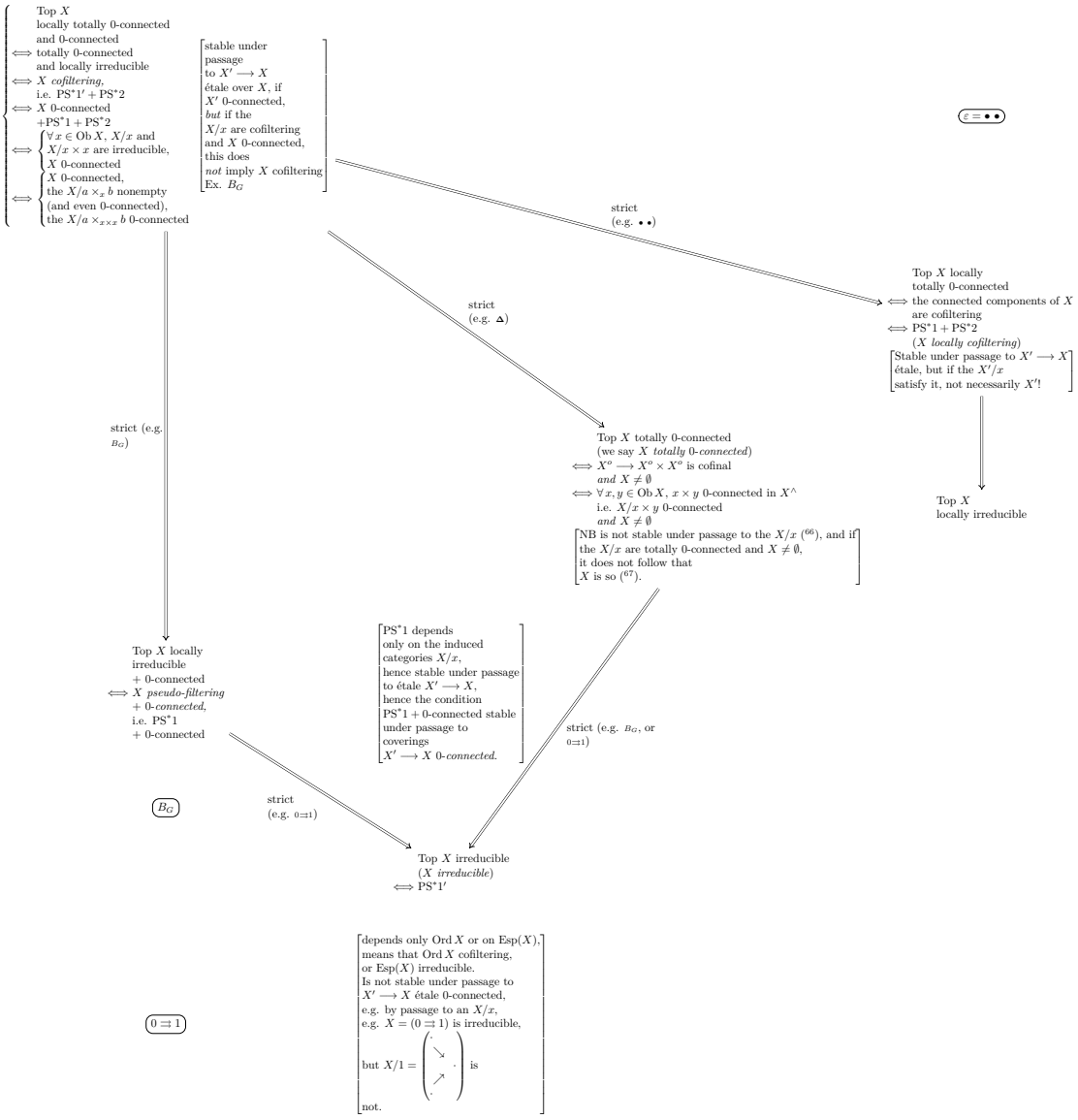
[page 96]

2) An interesting special case of 7.4 is the one where we assume that the $y\backslash X$ ($y \in \mathrm{Ob}\, Y$) are direct sums of categories each having an *initial object.* Then we see that this condition is stable under the operations of 7.4 a) , and consequently, the colocalized categories $z\backslash(X/x)$ of the morphisms localized upstairs $X/x \longrightarrow Y/y$ are categories with an initial object. This then implies that the same is true of the colocalized categories $z\backslash U$ ($z \in \mathrm{Ob}\, V$) in $g : U \longrightarrow V$. It follows that g is W-coaspheric for *every* basic localizer W (without having to worry about the axiom on limits).

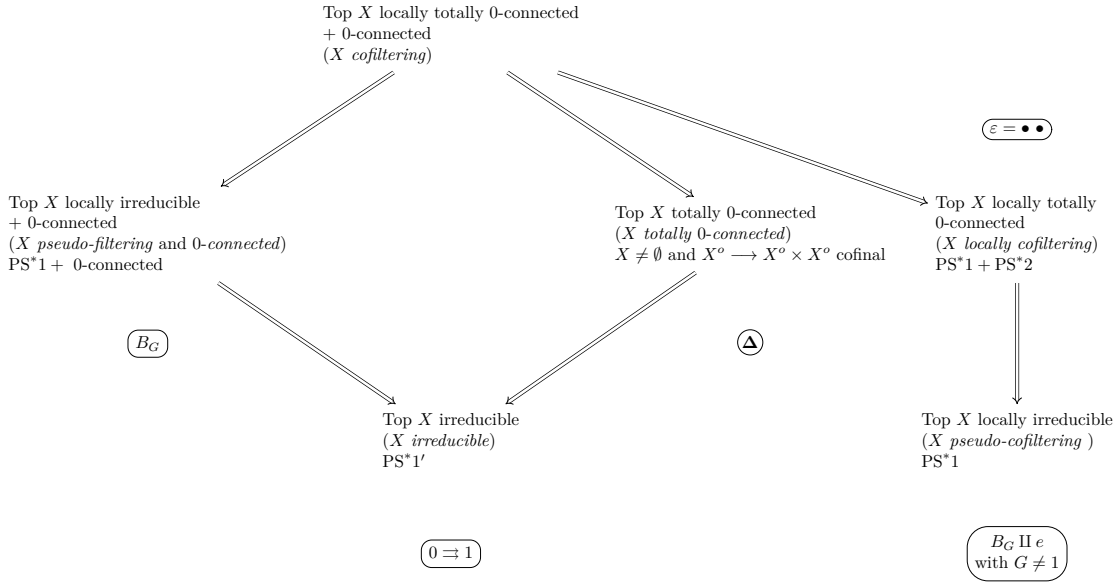
The condition considered on $f : X \longrightarrow Y$ is satisfied in particular in the following two cases (which seem to me by far the most important): a) f is étale and b) X is discrete.

p. 60

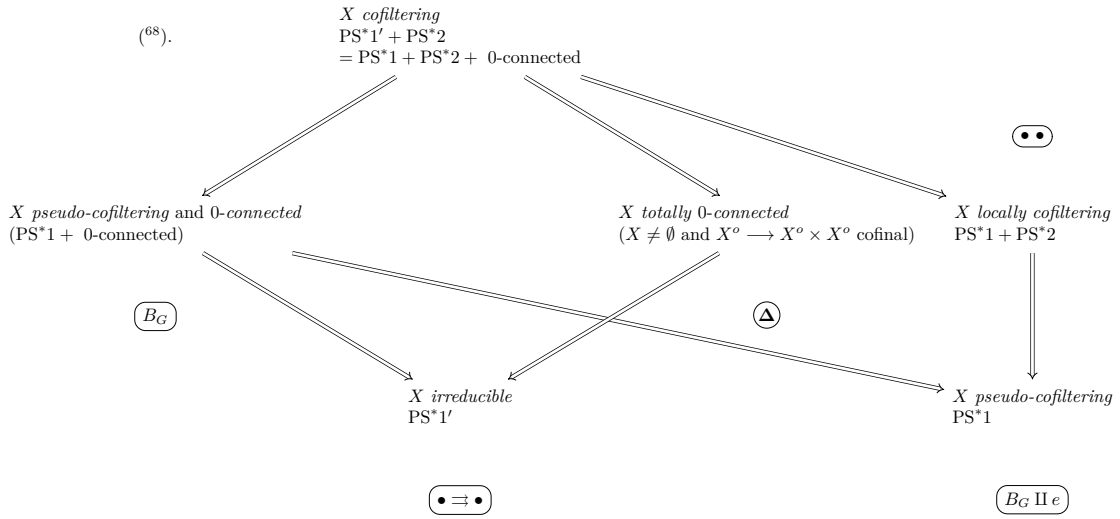
[page 97]



p. 61
[page 98]



([68]).



p. 62

$\bullet \rightrightarrows \bullet$	is neither pseudo-cofiltering, nor totally 0-connected.
B_G	is neither locally cofiltering, nor totally 0-connected.
$\boldsymbol{\Delta}$	is not pseudo-filtering.
$B_G \amalg e$	is not locally cofiltering, for B_G is not, nor belongs to one of the 4 classes of connected objects.

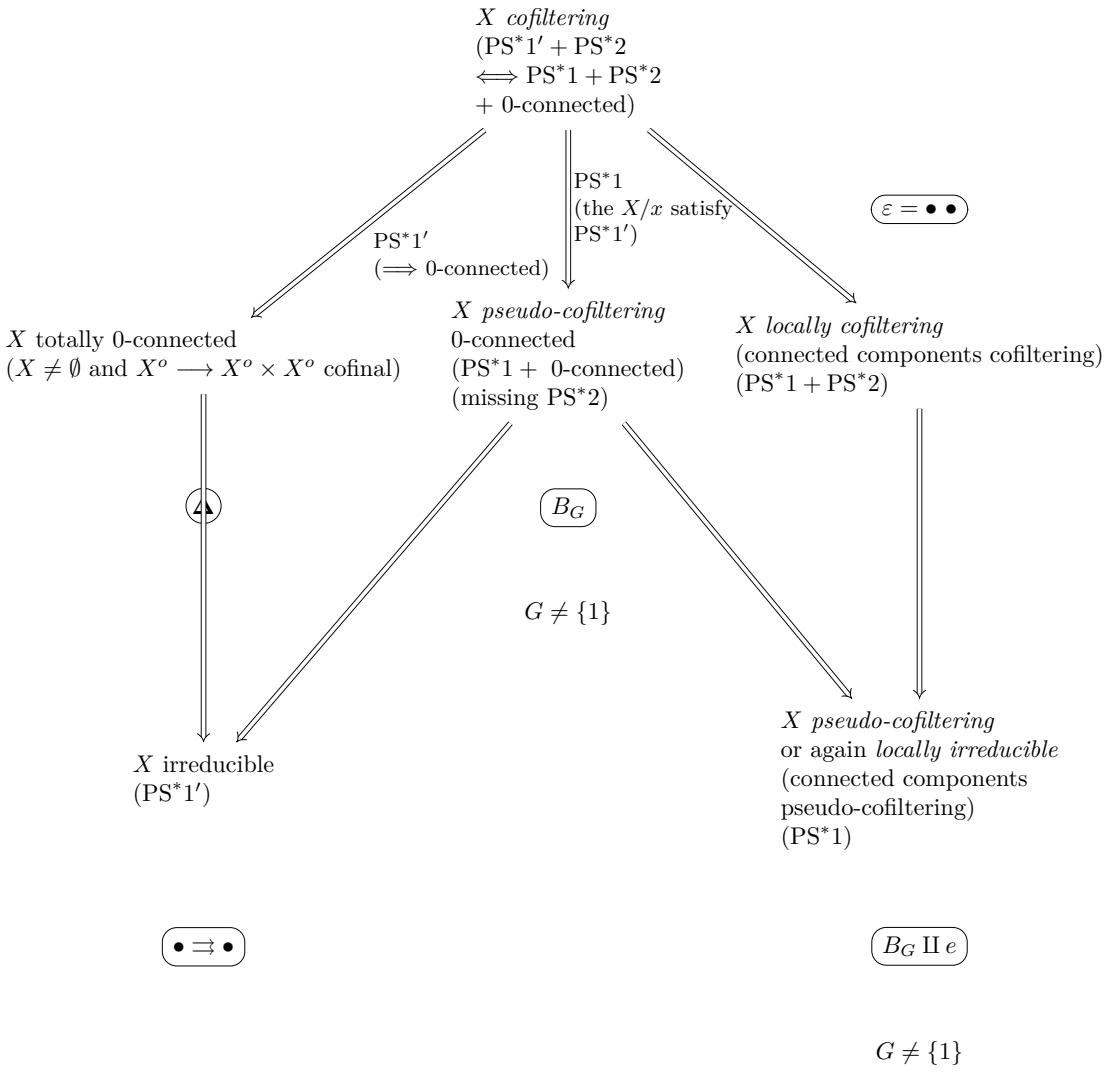
The same exclusion from the four connected classes for ex-$\varepsilon = \bullet\,\bullet$

[page 99]

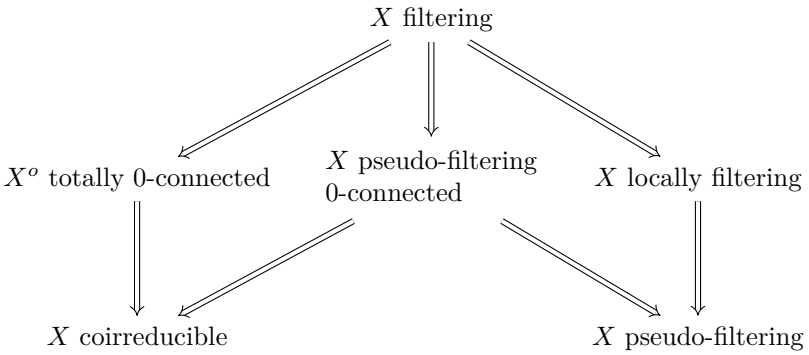
[66] **Ex.** $\boldsymbol{\Delta}$.

[67] **Ex.** B_G.

[68] **NB** The five typical categories B_G ($G \neq 1$), $\boldsymbol{\Delta}$, $0 \rightrightarrows 1$, $\varepsilon = \bullet\,\bullet$, $B_G \amalg e$ ($G \neq 1$) are such that they belong to the class C of objects of Cat under consideration, but to none of the five other classes except those which are linked to C by $C \Longrightarrow C'$ on the diagram of implications (or by a composite implication).



and dually we have



so we have a direct terminology except for "X^o totally 0-connected" - I don't dare say "X totally co-connected"!! p. 63

THE DERIVATORS

p. –

ALEXANDRE GROTHENDIECK

Chapter III

External Hom in derivators

p. –
p. 1

1 Relation of derivators with the category Hot

[page 1]

Let $\mathbf{D}$ be a derivator, $\mathcal{A}$ its fundamental category,

$$\mathcal{A} = \mathbf{D}(e).$$

I propose to define, if it can be done, functors

$$\left\{\begin{array}{rcccccl} \mathrm{Hot} & \times & \mathcal{A} & \xrightarrow{\pi_{\mathbf{D}}} & \mathcal{A} & (\xi, M) \longmapsto & \xi \times M \text{ or } \xi \otimes M \quad \text{(depending on the case)} \\ \mathrm{Hot}^o & \times & \mathcal{A} & \xrightarrow{\varpi_{\mathbf{D}}} & \mathcal{A} & (\xi, M) \longmapsto & \hom(\xi, M). \end{array}\right. \tag{1}$$

These functors must be interchanged by duality: if one takes the functors π^o, ϖ^o on the opposite categories, one finds

$$\left\{\begin{array}{rcccc} \mathrm{Hot} & \times & \mathcal{A}^o & \xrightarrow{\varpi^o_{\mathbf{D}}} & \mathcal{A}^o \\ \mathrm{Hot}^o & \times & \mathcal{A}^o & \xrightarrow{\pi^o_{\mathbf{D}}} & \mathcal{A}^o, \end{array}\right. \tag{1'}$$

and one must have

$$\left\{\begin{array}{rcl} \varpi^o_{\mathbf{D}} & = & \pi_{\mathbf{D}^o} \\ \pi^o_{\mathbf{D}} & = & \varpi_{\mathbf{D}^o}. \end{array}\right.$$

To tell the truth, considering (1) as functors

$$\begin{array}{rcl} \mathrm{Hot} & \xrightarrow{\pi_{\mathbf{D}}} & \underline{\mathrm{Hom}}(\mathcal{A}, \mathcal{A}) \\ \mathrm{Hot}^o & \xrightarrow{\varpi_{\mathbf{D}}} & \underline{\mathrm{Hom}}(\mathcal{A}, \mathcal{A}), \end{array}$$

one expects these functors to extend to

$$\left\{\begin{array}{rcl} \mathrm{Hot} & \xrightarrow{\pi_{\mathbf{D}}} & \underline{\mathrm{Hom}}(\mathbf{D}, \mathbf{D}) \\ \mathrm{Hot}^o & \xrightarrow{\varpi_{\mathbf{D}}} & \underline{\mathrm{Hom}}(\mathbf{D}, \mathbf{D}) \end{array}\right. \tag{2}$$

. This means, essentially, that for a given $\xi \in \mathrm{Ob}\,\mathrm{Hot}$, the operations $\xi \times_{\mathbf{D}} M$ and $\hom_{\mathbf{D}}(\xi, M)$ are defined not only in $\mathcal{A} = \mathbf{D}(e)$, but in all the $\mathbf{D}(I)$, and this in such a way as to commute with the functors f^*:

If [page 2]

$$f : I' \longrightarrow I,$$

then for $M \in \mathrm{Ob}\,\mathbf{D}(I)$, one must have canonical isomorphisms p. 2

$$\left\{\begin{array}{rcl} f^*(\xi \times_{\mathbf{D}} M) & \simeq & \xi \times_{\mathbf{D}} f^*(M) \\ f^*\hom_{\mathbf{D}}(\xi, M) & \simeq & \hom_{\mathbf{D}}(\xi, f^*(M)), \end{array}\right. \tag{3}$$

with the usual compatibilities for a composite fg, relative to $I'' \xrightarrow{g} I' \xrightarrow{f} I$.

But I prefer to work in $\mathcal{A}$, which should imply the similar constructions in all the $\mathbf{D}(I)$, by applying what has been said to the derivator $\mathbf{D}_I$ if need be.

The fundamental algebraic intuition is this, denoting by

$$X \longmapsto \mathrm{hot}(X) : (\mathrm{Cat}) \longrightarrow (\mathrm{Hot})$$

the localization functor from (Cat) to (Hot), obtained by inverting the arrows in (Cat) which are "weak equivalences". Suppose X is in Diag, let

$$p : X \longrightarrow e$$

be the functor with values in the one-point category. We want

$$\left\{ \begin{array}{rcccc} \mathrm{hot}(X) \times_{\mathbf{D}} M & \simeq & p_!(p^*(M)) & = & H^{\mathbf{D}}_{\bullet}(X, M) \\ \mathrm{hom}_{\mathbf{D}}(\mathrm{hot}(X), M) & \simeq & p_*(p^*(M)) & = & H^{\bullet}_{\mathbf{D}}(X, M) \end{array} \right. \tag{4}$$

(canonical isomorphisms).

This would give a principle of definition for the operations sought. A difficulty comes from the fact that in general, Diag is not equal to (Cat), and that the homotopy types coming from Diag do not give (even up to isomorphism) *all* homotopy types. Intuitively, one finds only "small enough" homotopy types. But one can localize Diag with respect to weak equivalences (it is better to say "quasi-isomorphisms"), one then finds a category $(\mathrm{Hot})_0$, and a canonical functor [page 3]

$$\mathrm{Hot}_0 \longrightarrow \mathrm{Hot},$$

if we are lucky, this functor will be fully faithful, and we shall try to describe the functors (1) or (2) by replacing Hot in them by the full subcategory of "small" homotopy types. One can also forget (temporarily!) Hot, and work with Hot_0. This is what we are going to do for the moment.

In order to be able to define the functors

$$\left\{ \begin{array}{ccccccc} \mathrm{Hot}_0 & \times & \mathcal{A} & \longrightarrow & \mathcal{A} & (\xi, F) & \longmapsto & \xi \times_{\mathbf{D}} F \\ \mathrm{Hot}_0^o & \times & \mathcal{A} & \longrightarrow & \mathcal{A} & (\xi, F) & \longmapsto & \mathrm{hom}_{\mathbf{D}}(\xi, F), \end{array} \right. \tag{5}$$

I note that for M fixed, X variable in Diag, the last members in (4)

p. 3

$$H^{\mathbf{D}}_{\bullet}(X, M), \qquad H^{\bullet}_{\mathbf{D}}(X, M) \tag{6}$$ [page 4]

are indeed, the one covariant, the other contravariant in X. Thus there is, up to unique isomorphism, a single pair of bifunctors (5) satisfying (bifunctorially) the isomorphisms (4), if and only if the expressions (6), considered as functorial in X, transform quasi-isomorphisms into isomorphisms. This is therefore equivalent to the validity of the

Theorem (?) ([1]). *Let $f : X' \longrightarrow X$ be in Diag. If f is a quasi-isomorphism, it is a* $\mathbf{D}$*-equivalence.*

When X is the one-point category, this is nothing other than axiom Der 3 (p. 67). I do not know whether it is a consequence of axioms Der 1 to Der 5 (d). If this is not the case, one would have to add it to the axioms, I think. I have not even proved that an equivalence of categories in Diag (in the sense of categories!) is a $\mathbf{D}$-equivalence!

Let us note the *adjunction formula*

$$\mathrm{Hom}(\xi \times_{\mathbf{D}} M, N) \xrightarrow{\sim} \mathrm{Hom}(M, \mathrm{hom}_{\mathbf{D}}(\xi, N)). \tag{7}$$

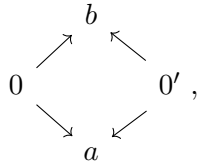
If ξ comes from $X \in \mathrm{Diag}$, then this reads

$$\mathrm{Hom}(p_! p^* M, N) \xrightarrow{\sim} \mathrm{Hom}(M, p_* p^* N), \tag{8}$$ [page 5]

now by adjunction both sides are canonically isomorphic to

$$\mathrm{Hom}_{\mathbf{D}(X)}(p^*M, p^*N).$$

Example. Taking for ξ the homotopy type of the standard circle (represented e.g. by the ordered set



where we agree to take 0 as origin, and to orient from 0 toward a), the two functors in question are (unless I am mistaken) the functors suspension Σ and "cosuspension" or "loop space" functor Ω. Subject to checking that ΩF does indeed coincide with what I have described elsewhere, concerning substantial cartesian squares in $\mathcal{A}$.

In Hot_0 we have a product operation, in fact this must be the categorical product in Hot_0. But on the models it is given by

$$\mathrm{hot}_0(X) \times \mathrm{hot}_0(Y) \;\simeq\; \mathrm{hot}_0(X \times Y) \tag{8}$$

. This being so, we have the canonical isomorphisms, for $\xi, \eta \in \mathrm{Ob}\,\mathrm{Hot}_0$, p. 4

$$\begin{cases} \xi \times_{\mathbf{D}} (\eta \times_{\mathbf{D}} M) \;\simeq\; (\xi \times \eta) \times_{\mathbf{D}} M \\ \mathrm{hom}_{\mathbf{D}}(\xi, \mathrm{hom}_{\mathbf{D}}(\eta, M)) \;\simeq\; \mathrm{hom}_{\mathbf{D}}(\xi \times \eta, M). \end{cases} \tag{9}$$

If ξ, η are represented by X, Y, the first formula e.g. is established by computing the second member as [page 6]

$$\varphi_!(M_{X\times Y}) \simeq p_!(\quad \underbrace{q'_!(M_{X\times Y})}_{\substack{\simeq\, q_!(M_Y)_X \\ \text{since } q \text{ is } \mathbf{D}\text{-smooth}}} \quad) \simeq \xi \times_{\mathbf{D}} (\eta \times_{\mathbf{D}} M),$$

O.K.

$$\begin{array}{ccc} X & \xleftarrow{q'} & X \times_e Y \\ {\scriptstyle p}\downarrow & \overset{\varphi}{\swarrow} & \downarrow{\scriptstyle p'} \\ e & \xleftarrow[q]{} & Y \end{array}$$

I also feel like proving:

Proposition 1. *Let $\xi \in \mathrm{Ob}\,\mathrm{Hot}_0$. Then $F \longmapsto \xi \times_{\mathbf{D}} F$, as a morphism of derivators $\mathbf{D} \longrightarrow \mathbf{D}$, is* right exact, *and $F \longmapsto \mathrm{hom}_{\mathbf{D}}(\xi, F)$ is left exact.*

We may assume that ξ is defined by X in Diag. Let $f : I \longrightarrow J$, let F be in $\mathbf{D}(I)$, we have to prove

$$f_*(\mathrm{hom}_{\mathbf{D}}(\xi, F)) \;\simeq\; \mathrm{hom}_{\mathbf{D}}(\xi, f_*(F)).$$

So let us consider

$$\begin{array}{ccc} X \times I & \xrightarrow{f'} & X \times J \\ {\scriptstyle p=(x,i)\longmapsto i}\downarrow & & \downarrow{\scriptstyle q=(x,j)\longmapsto j} \\ I & \xrightarrow{\quad f \quad} & J. \end{array}$$

We have

$$\mathrm{hom}_{\mathbf{D}}(\xi, F) = p_*(p^*F),$$

[1] New axiom?

so

$$f_*(\hom_{\mathbf{D}}(\xi, F)) = \underbrace{f_*p_*}_{(fp)_*}(p^*(F)) = (qf')_*p^*(F) \simeq q_*(f'_*p^*(F)).$$

But

$$f'_*p^*(F) \ \simeq \ q^*f_*(F),$$

the base change formula for the f_*, for the base change q, which is smooth, since obtained from $X \longrightarrow e$, which is smooth, by base change $J \longrightarrow e$. We find therefore

p. 5
[page 7]

$$f_*(\hom_{\mathbf{D}}(\xi, F)) \simeq q_*q^*(f_*(F)) \overset{\text{def}}{=} \hom_{\mathbf{D}}(\xi, f_*(F)),$$

q.e.d. The statement for $F \longmapsto \xi \times_{\mathbf{D}} F$ is dual to the one we have just proved.

Let us take up again, for $\xi \in \mathrm{Hot}_0$, the adjunction formula

$$\mathrm{Hom}(\xi \times_{\mathbf{D}} M, N) \simeq \mathrm{Hom}(M, \hom_{\mathbf{D}}(\xi, N)) \overset{\text{def}}{=} \mathrm{Hom}_\xi(M, N). \qquad (*)$$

If ξ is defined by X in Diag, the two members of $(*)$ are identified with

$$\mathrm{Hom}_{\mathbf{D}(X)}(p^*M, p^*N),$$

where $p : X \longrightarrow e$ is the structural functor.

Let us fix M, N, and consider the expression $(*)$ as a contrafunctor

$$(\mathrm{Hot}_0)^o \ \longrightarrow \ \mathrm{Ens}.$$

It would seem that in all known cases, this functor is "representable" by an object, if not of Hot_0 itself, at least an object in Hot. If we cannot define it as an object H of Hot_0, the fact that we have a canonical isomorphism

$$\mathrm{Hom}_\xi(M, N) \ \simeq \mathrm{Hom}_{\mathrm{Hot}}(\xi, H), \qquad (10)$$

functorial in $\xi \in \mathrm{Ob}\,\mathrm{Hot}_0$, does not necessarily define up to unique isomorphism. In fact, when

[page 8]

$$\mathbf{D} = \mathbf{D}_{\mathcal{M},\Sigma}$$

is defined by a category of models $(\mathcal{M}, \Sigma)$, then there is a natural definition of an operation $\hom\mathrm{ext}$ ("external" Hom) with values in Hot, namely a functor

$$\begin{cases} \mathcal{A}^o \times \mathcal{A} & \longrightarrow & \mathrm{Hot} \\ (M, N) & \longmapsto & \hom\mathrm{ext}(M, N), \end{cases} \qquad (11)$$

where $\mathcal{A} = \mathcal{M}\Sigma^{-1}$ is the fundamental category of $\mathbf{D} = \mathbf{D}_{\mathcal{M},\Sigma}$. It is true that I have not checked, if e.g. $(\mathcal{M}, \Sigma)$ underlies a Quillen model category, so that (by THOMASON) $\mathbf{D}$ is indeed a derivator, that this notion of $\hom\mathrm{ext}$ does indeed satisfy a formula of the form (10). Here I am going to posit it , if need be, these additional data: specification of (11), and of a functorial isomorphism

$$\underbrace{\mathrm{Hom}_\xi(M, N)}_{\substack{= \mathrm{Hom}(\xi \times_{\mathbf{D}} M, N) \\ = \mathrm{Hom}(M, \hom_{\mathbf{D}}(\xi, N))}} \quad \simeq \ \mathrm{Hom}_{\mathrm{Hot}}(\xi, \hom\mathrm{ext}(M, N)). \qquad (12)$$

If now η is a second element of Hot_0, we feel like writing

p. 6
[page 9]

$$\begin{cases} \hom\mathrm{ext}(\eta \times_{\mathbf{D}} M, N) \ \simeq \ \hom\mathrm{ext}(M, \hom_{\mathbf{D}}(\eta, N)) \\ \qquad\qquad\qquad\qquad \simeq \ \hom_{\mathrm{HOT}}(\eta, \hom\mathrm{ext}(M, N)), \end{cases} \qquad (13)$$

where in the second member, $\hom_{\mathrm{HOT}}$ is the operation (1) in the case where $\mathbf{D} = \mathrm{HOT}$, and which is also defined in terms of the cartesian product $\xi \times \eta$ in Hot, by

$$\mathrm{Hom}_{\mathrm{Hot}}(\xi, \hom_{\mathrm{HOT}}(\eta, \zeta)) \simeq \mathrm{Hom}_{\mathrm{Hot}}(\xi \times \eta, \zeta),$$

in other words, for η fixed, the functor

$$\zeta \longmapsto \hom_{\mathrm{HOT}}(\eta, \zeta)$$

is defined as the functor right adjoint to the functor

$$\xi \longmapsto \xi \times \eta.$$

To verify formula (13) heuristically (a verification valid "in due form" if the three objects under consideration are in fact defined as objects of Hot_0 ...), we take $\mathrm{Hom}_{\mathrm{Hot}}(\xi, -)$ with values in the three members, for ξ given in Hot_0. For the first, we find

$$\begin{aligned} \mathrm{Hom}_{\xi}(\eta \times_{\mathbf{D}} M, N) &= \mathrm{Hom}(\xi \times_{\mathbf{D}} (\eta \times_{\mathbf{D}} M), N) \\ &= \mathrm{Hom}((\xi \times \eta) \times_{\mathbf{D}} M, N). \end{aligned}$$

For the second

$$\begin{aligned} \mathrm{Hom}_{\xi}(M, \hom_{\mathbf{D}}(\eta, N)) &= \mathrm{Hom}_{\mathcal{A}}(\xi \times_{\mathbf{D}} M, \hom_{\mathbf{D}}(\eta, N)) \\ &\simeq \mathrm{Hom}_{\mathcal{A}}(\eta \times_{\mathbf{D}} (\xi \times_{\mathbf{D}} M), N) \\ &\simeq \mathrm{Hom}_{\mathcal{A}}((\eta \times \xi) \times_{\mathbf{D}} M, N), \end{aligned}$$

and finally for the third

[page 10]

$$\begin{aligned} \mathrm{Hom}_{\mathrm{Hot}}(\xi \times \eta, \hom \mathrm{ext}(M, N)) &\simeq \mathrm{Hom}_{\xi \times \eta}(M, N) \\ &\simeq \mathrm{Hom}((\xi \times \eta) \times_{\mathbf{D}} M, N), \end{aligned}$$

so always the same thing! No doubt about it, this must work.

If we have to give ourselves the hom ext (11) as an additional datum, with in addition a trifunctorial isomorphism (12), we must also give ourselves the trifunctorial isomorphisms (13), *"compatible"* with the isomorphism (12) in the sense of the preceding calculation.

For example, taking $\eta =$ homotopy type of the circle S^1, formulas (13) specialize to

p. 7

$$\begin{aligned} \hom \mathrm{ext}(\Sigma M, N) &\simeq \hom \mathrm{ext}(M, \Omega N) \\ &\simeq \Omega \hom \mathrm{ext}(M, N) \\ &\overset{\mathrm{def}}{=} \hom_{\mathrm{HOT}}(S^1, \hom \mathrm{ext}(M, N)). \end{aligned} \tag{14}$$

But this formula in Hot is probably not quite the right one, for the right formalism of Ω, Σ is carried out in the categories that are *pointed,* and in particular in the pointed homotopy category $\mathrm{Hot}_\bullet$.

I am going to spell out an example, taking the derivator

$$\mathrm{Dér}(A)$$

"derived" from an abelian category A. Thus $\mathbf{D}(e) =$ derived category of A. To avoid incidental difficulties here, I am going to assume that A is the category of k-modules, k an arbitrary ring.

[page 11]

To every $\xi \in \mathrm{Ob}\,\mathrm{Hot}$ is associated an "abelianized"

$$\xi_{\mathrm{ab}} \overset{\mathrm{def}}{=} H_\bullet(\xi) \in \underbrace{D(\mathrm{Ab})}_{\text{or } \mathbf{D}(\mathbf{Z}\text{-Mod})}, \qquad \text{derived category of abelian groups,}$$

an object which has cohomology only in degree ≤ 0. That said, if M is in $D(k\text{-Mod})$, we shall have

$$\begin{cases} \xi \times_{\mathbf{D}} M \ \simeq \ \xi_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} M \\ \hom_{\mathbf{D}}(\xi, M) \ \simeq \ \mathbf{R}\mathrm{Hom}_{\mathbf{Z}}(\xi_{\mathrm{ab}}, M), \end{cases} \tag{15}$$

a formula which is in cahoots with the Künneth formula for homology

$$(\xi \times \eta)_{\mathrm{ab}} \simeq \xi_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} \eta_{\mathrm{ab}}, \tag{16}$$

which again gives a familiar look to formulas (9), which reduce to

$$\begin{cases} \xi_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} (\eta_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} M) \ \simeq \ (\xi_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} \eta_{\mathrm{ab}}) \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} M \\ \mathbf{R}\mathrm{Hom}_{\mathbf{Z}}(\xi_{\mathrm{ab}}, \mathbf{R}\mathrm{Hom}_{\mathbf{Z}}(\eta_{\mathrm{ab}}, M)) \ \simeq \ \mathbf{R}\mathrm{Hom}_{\mathbf{Z}}(\xi_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} \eta_{\mathrm{ab}}, M). \end{cases} \tag{^{2}}$$

On the other hand, does a $\hom\mathrm{ext}(M, N)$ indeed exist, if M, N in $D(k\text{-Mod})$? We think of $\mathbf{R}\mathrm{Hom}_k(M, N)$, but to make a homotopy type out of it, we have to "truncate in degrees > 0". We could therefore take

$$\hom\mathrm{ext}(M, N) \ \simeq \ \tau_{\leq 0}(\mathbf{R}\mathrm{Hom}_k(M, N)).$$

Is the fundamental formula (10) then valid? The second member is indeed p. 8 [page 12]

$$\mathrm{Hom}_{\mathrm{Hot}}(\xi, H) \ \simeq \ \mathrm{Hom}_{\mathrm{Hotab}}(\xi_{\mathrm{ab}}, H) \qquad (^3)$$

for H is an abelian object of Hot (cf. the carpet on abelianization …), from there it is in the bag:

$$\mathrm{Hom}_{\underbrace{\mathrm{Hotab}}_{\substack{\approx\, D(\mathbf{Z}\text{-Mod})^{\leq 0},\\ \textit{full}\text{ subcategory}\\ \text{of } D(\mathbf{Z}\text{-Mod})}}}(\xi_{\mathrm{ab}}, \tau_{\leq 0}\mathbf{R}\mathrm{Hom}_k(M, N)) \ \simeq \ \mathrm{Hom}_{D(\mathbf{Z}\text{-Mod})}(\xi_{\mathrm{ab}}, \mathbf{R}\mathrm{Hom}_k(M, N))$$

$$\simeq \ \mathrm{Hom}_{D(k\text{-Mod})}(\xi_{\mathrm{ab}} \times M, N),$$

O.K.

On the other hand, the bifunctor

$$(M, N) \ \longmapsto \ \hom\mathrm{ext}(M, N) \ = \ \tau_{\leq 0}\mathbf{R}\mathrm{Hom}_k(M, N)$$

factors canonically through Hotab and a fortiori through $\mathrm{Hot}_{\bullet}$. The "right" $\hom\mathrm{ext}(M, N)$ would here rather be the one in Hotab, or even in the category $D^-(\mathbf{Z}\text{-Mod})$, obtained from that one "by making the translation functor invertible". It is evident, moreover, that here the $\xi_{\mathrm{ab}} \times_{\mathbf{D}} M$, where $\xi = S^1$, is not "the right" suspension, which must be the shift of degrees – ξ_{ab} should have cohomology only in degree -1, and not also in degree 0.

[2] Also spell out the adjunction formula (7) (p. 4).

[3] ? ? is this true ? ? ?

THE DERIVATORS

p.

ALEXANDRE GROTHENDIECK

Chapter IV

Substantial diagrams

p.

p. 1

Construction of substantial diagrams for a derivator.

[page 73]

Let $\mathbf{D}$ be a prederivator, I in Diag, let us consider
$\varphi_I : \mathbf{D}(I) \longrightarrow \underline{\mathrm{Hom}}(I^o, \mathcal{A})$, where $\mathcal{A} = \mathbf{D}(e)$ is the fundamental category of coefficients.

The objects of $\underline{\mathrm{Hom}}(I^o, \mathcal{A})$ are interpreted as *diagrams of type* I^o in $\mathcal{A}$. This language is especially convenient if I is ordered. Let us denote, for $F \in \mathrm{Ob}\,\mathbf{D}(I)$, by $(F_i)_{i\in \mathrm{Ob}\, I}$ the associated I^o-diagram in $\mathcal{A}$. By axiom Der 2, the functor

$$\varphi : F \longmapsto (F_i)_{i\in \mathrm{Ob}\, I}$$

is conservative. But it has no desire to be faithful, still less fully faithful. When one knows the diagram $(F_i)_i$, one is far from knowing F, not even up to nonunique isomorphism, I presume. It is by no means clear that if F, F' in $\mathbf{D}(I)$ are such that the associated diagrams are isomorphic, they are themselves isomorphic.

To insist on this point, the objects of $\mathbf{D}(I)$ will be called the "*substantial diagrams* of type I^o in $\mathcal{A}$" (or $\mathbf{D}$-substantial, if there is a risk of confusion). To support intuition and simplify the explanations, an object F of $\mathbf{D}(I)$ ($\mathbf{D}$-coefficients on I) will be represented graphically by the corresponding "weak" or "skeletal" diagram – one regards it as a sort of [page 74]
(pale) "skeleton" of the substantial diagram from which it comes. E.g. if $I = \mathbf{\Delta}_1 : 0 \longrightarrow 1$, one denotes a substantial diagram of type $(\mathbf{\Delta}_1)^o$ by the same graphical notation, as the ordinary diagram:

$$F_0 \longleftarrow F_1.$$

If

$$I = \begin{array}{ccc} a & \longrightarrow & c \\ \uparrow & & \uparrow \\ 0 & \longrightarrow & b, \end{array}$$

the substantial diagrams of type I^o are written

$$\begin{array}{ccc} F_a & \longleftarrow & F_c \\ \downarrow & & \downarrow \\ F_0 & \longleftarrow & F_b. \end{array}$$

It will of course be necessary to specify each time whether this graphical notation p. 2
designates a skeletal diagram, or a substantial one. Often one starts from a skeleton, and tries to "flesh it out" with a little flesh, to make it into a substantial diagram – if possible, in a canonical way. This is a point of derivatic philosophy essential for practice. One will often proceed step by step, by "substantifying" subdiagrams corresponding to a strictly increasing sequence $I_0 \subset I_1 \subset \cdots \subset I_n = I$ of I, as best suits the particular situation.

[page 75]

Here is a typical example, assuming $\mathbf{D}$ a *pointed* derivator (the $\mathbf{D}(I)$ are pointed, and the f^* take zero objects to zero objects). It is a matter of defining the "cosuspension" or "loop space" functor in $\mathcal{A}$,

$$\Omega : \mathcal{A} \longrightarrow \mathcal{A}.$$

It is understood that for M in $\mathcal{A}$, $\Omega(M)$ must fit into a commutative square

$$\begin{array}{ccc} e & \longleftarrow & \Omega M \\ \downarrow & & \downarrow \\ M & \longleftarrow & e, \end{array} \tag{C}$$

where e denotes the unit elements. This square however should be substantial, more precisely, be a substantial "**D**-*cartesian*" square. It being understood that an object of $\mathbf{D}(I)$ (substantial square in $\mathcal{A}$) is called **D**-cartesian, if it is in the essential image of $\mathbf{D}(I_1)$ under i_{1*}, where

$$i_1 : I_1 \hookrightarrow I,$$

I_1 being the open subcategory of I defined by

$$I_1 = \begin{array}{ccc} a & & \\ \uparrow & & \\ 0 & \longrightarrow & b. \end{array}$$

The restriction of the substantial square sought (C) to the subcategory I_1 is therefore [page 76]
a substantial diagram with skeleton

$$\begin{array}{ccc} e & & \\ \downarrow & & \\ M & \longleftarrow & e, \end{array} \tag{Sq}$$

from which it is obtained (up to canonical isomorphism) by applying i_* to the latter, p. 3
so as to complete it to $\begin{array}{ccc} e & \leftarrow & \Omega M \\ \downarrow & & \vdots \\ M & \longleftarrow & e \end{array}$. It therefore remains to describe a canonical substantiation of the skeleton (Sq), in terms of M. This leads us to look at the open subcategory

$$I_0 = \{0\} : \quad \begin{array}{ccc} \cdot & \cdots\cdots\rightarrow & \cdot \\ \uparrow & & \uparrow \\ 0 & \cdots\cdots\rightarrow & \cdot \end{array}$$

of I_1 (hence also of I) reduced to the initial object, and the inclusion

$$I_0 \overset{i_0}{\hookrightarrow} I_1,$$

and to substantify the skeleton (Sq) by $(i_0)_!(M)$.

One can say a posteriori that, if one knew the substantial square, there is, up to unique isomorphism, only one way of obtaining it as an $i_{1*}i_{0!}(?)$, with ? in $\mathbf{D}(I_0) \simeq \mathcal{A}$, for $i_{0!}$ and i_{1*} being fully faithful, the same holds for their composite.

In fact, one sees that the functor [page 77]

$$i_{1*}i_{0!} : \ \mathcal{A} \simeq \mathbf{D}(I_0) \longrightarrow \mathbf{D}(I)$$

induces an equivalence of $\mathcal{A}$ with the full subcategory of $\mathbf{D}(I)$ formed by the $\mathbf{D}$-cartesian squares whose fibers at a and b are zero. (The obvious quasi-inverse functor is i^*, where $i : I_0 \hookrightarrow I$.) This therefore means that in this case, the datum of the skeleton of a very particular nature ($F_a \simeq F_b \simeq 0_{\mathcal{A}}$), plus the additional $\mathbf{D}$-cartesian constraint, on the square sought, determines the latter up to unique isomorphism in terms of its skeleton: there is a substantiation of the latter not only "canonical", but even unique up to unique isomorphism, because of the $\mathbf{D}$-cartesian constraint.

I would like to begin by looking at the still simpler case where $I = \mathbf{\Delta}_1 = (0 \longrightarrow 1)$, so the skeletons envisaged are simply the arrows

$$F_1 \longrightarrow F_0 \qquad \text{in } \mathcal{A}.$$

The philosophy obliges us, for the most fundamental arrows in the theory of derivators $\mathbf{D}$ (arrows in $\mathcal{A}$, or by extension in a $\mathbf{D}(I) \simeq \mathcal{A}_{\mathbf{D}_I}$, fundamental category of the induced derivator $\mathbf{D}_I$) to ask ourselves whether there is not

a "canonical", "natural", "willed by God" way of "fleshing it out" into a substantial diagram. The most obvious arrows that present themselves are relative to a functor

p. 4
[page 78]

$$f : I \longrightarrow J$$

in Diag, and to the two adjunction morphisms for the pair (f^*, f_*), namely

$$\begin{aligned} G &\longrightarrow f_* f^* G \quad \text{in } D(J) \\ f^* f_* F &\longrightarrow F \qquad \text{in } D(I). \quad ([1]) \end{aligned}$$

I am going to try to flesh out these two arrows, and am going to generalize by considering the situation below

$$\begin{array}{ccc} I & & F \\ {\scriptstyle f}\downarrow & & \uparrow{\scriptstyle u} \\ J & & G, \end{array}$$

where

$$\mathrm{Hom}^f(G, F) \overset{\text{def}}{=} \mathrm{Hom}(f^*G, F) \overset{\text{adj.}}{\simeq} \mathrm{Hom}(G, f_*F).$$

Thus the same u is interpreted by *two* skeletal arrows

$$\begin{cases} f^*(G) \longrightarrow F & \text{in } \mathbf{D}(I) \\ \quad G \longrightarrow f_*(F) & \text{in } \mathbf{D}(J). \end{cases} \tag{1}$$

To flesh them out is to interpret them (if it can be done) in terms of two objects (canonical if possible) in $\mathbf{D}(I \times \mathbf{\Delta}_1)$ and $\mathbf{D}(J \times \mathbf{\Delta}_1)$ respectively.

I am going to be guided by the special case of a trivial derivator

$$\mathbf{D} = \mathbf{D}_{\mathcal{M}},$$

where $\mathcal{M}$ is a "naked" category of models ($\Sigma = \emptyset$).

For fixed f, we are going to construct an auxiliary category $K(f)$, in such a way that for every category $\mathcal{M}$, the systems (G, F, u) consisting of an $\mathcal{M}$-presheaf G on J, another F on I, and an f-homomorphism $G \longrightarrow F$, are identified (functorially) with the $\mathcal{M}$-presheaves on $K(f)$. That said, we are going to try to interpret the skeletal arrows

[page 79]
p. 5

[1] Of course, dual problem for $(f_!, f^*)$.

(1), but no matter here – all the **D**-skeletons are now fleshed out in terms of the geometry of $K(f)$.

$$\begin{cases} \operatorname{Ob} K(f) = \operatorname{Ob} I \amalg \operatorname{Ob} J \\ I,\ J \text{ being identified with full subcategories of } K(f), \\ \text{there are no arrows from a } j \in J \text{ to an } i \in I. \\ \text{(Thus } I \text{ is open in } K(f),\ J \text{ its closed complement.)} \\ \text{For } i \in \operatorname{Ob} I,\ j \in \operatorname{Ob} J, \text{ we set } \operatorname{Hom}_{K(f)}(i,j) = \operatorname{Hom}_J(f(i),j). \end{cases}$$

The composition of arrows in $K(f)$ is obvious. We are going to define a canonical retraction

$$\bar{f} : K(f) \longrightarrow J$$

of $K(f)$ onto its closed subcategory J. Thus $\bar{f}$ is known on $\operatorname{Ob} J$, and on $\operatorname{Ob} I$ it will be f:

$$\begin{cases} \bar{f}(j) = j & \text{if } j \in \operatorname{Ob} J \\ \bar{f}(i) = f(i) & \text{if } i \in \operatorname{Ob} I. \end{cases}$$

It remains to define, for $x, y \in \operatorname{Ob} K(f)$, [page 80]

$$\operatorname{Hom}_{K(f)}(x,y) \longrightarrow \operatorname{Hom}_J(\bar{f}(x), \bar{f}(y)).$$

If $x, y \in \operatorname{Ob} J$, it is the identity, if $x, y \in I$, it is f. If $x \in I$, $y \in J$, we must define an arrow

$$\operatorname{Hom}(f(x), y) \longrightarrow \operatorname{Hom}(f(x), y),$$

and we again take the identity. We have

$$\begin{array}{ccccc} J & \xhookrightarrow{j} & K(f) & \xhookleftarrow{i} & I \\ & \searrow & \downarrow{\scriptstyle \bar{f}} & & \\ & & J & & \end{array}$$

$$\begin{cases} \bar{f} \circ j = \mathrm{id}_J \\ j \circ \bar{f} = \begin{cases} j & \text{on } J \\ jf & \text{on } I, \end{cases} \end{cases} \qquad \text{i.e.} \quad \boxed{\begin{array}{ll} j\bar{f}j = j & \text{on } J \\ j\bar{f}i = jf & \text{on } I \end{array}},$$

and we have

$$\alpha :\ \mathrm{id}_K \longrightarrow j\bar{f} \qquad \text{i.e. functorial homomorphism}$$
$$x \xrightarrow{\alpha(x)} j(\bar{f}(x)), \quad x \in \operatorname{Ob} K,$$

defined as follows: p. 6

$$\begin{cases} \text{If } x \in J,\ \alpha(x) = \mathrm{id}_x \ :\ x \xrightarrow{\sim} j(\bar{f}(x)) = j(x) = x \\ \text{If } x \in I,\ \alpha(x) : \underbrace{x \longrightarrow f(x)}_{\text{in } K(f)},\ \text{or}\ \underbrace{f(x) \longrightarrow f(x)}_{\text{in } J},\ \alpha(x) = \mathrm{id}_{f(x)}. \end{cases}$$

We must check functoriality for x varying in K …

Let Φ be an $\mathcal{M}$-presheaf on K. It defines

$$\begin{aligned} F = i^*(\Phi) &\quad \in \operatorname{Ob} \mathbf{D}_{\mathcal{M}}(I) \\ G = j^*(\Phi) &\quad \in \operatorname{Ob} \mathbf{D}_{\mathcal{M}}(J). \end{aligned}$$

On the other hand $f^*(G)$ is interpreted as $(j\bar{f})^*(\Phi)|I$, since

$$\begin{aligned} f^*(G)(i) &= G(\underbrace{f(i)}_{=\bar{f}(i)\in J}) \\ &= j^*(\Phi)(\bar{f}(i)) \\ &= \Phi(j\bar{f}(i)) \\ &= (j\bar{f})^*(\Phi)(i). \end{aligned}$$

Thus [page 81]

$$f^*(G) \simeq (j\bar{f})^*(\Phi)|I.$$

Now the arrow

$$\alpha : \mathrm{id}_K \longrightarrow j\bar{f}$$

gives

$$\underbrace{\mathrm{id}_K(\Phi)}_{=\Phi} \xleftarrow{\Phi(\alpha)} (j\bar{f})^*(\Phi), \qquad \text{functorial in } \Phi,$$

whence, taking the restrictions to I,

$$i^*(\Phi) \overset{\mathrm{def}}{=} F \xleftarrow{u} f^*G.$$

In fact, without too much twisting and turning, we should be able to show that for every category $\mathcal{M}$, the category $\underline{\mathrm{Hom}}(K^o, \mathcal{M})$ of $\mathcal{M}$-presheaves on K is isomorphic to that of triples (F, G, u), with $F \in \mathrm{Ob}\,\underline{\mathrm{Hom}}(I^o, \mathcal{M})$, $G \in \underline{\mathrm{Hom}}(J^o, \mathcal{M})$, and $u : f^*(G) \longrightarrow F$. Thus, K appears as a solution of a universal problem

$$\underline{\mathrm{Hom}}(K^o, \mathcal{M}) \longrightarrow \{F,\ G,\ u \text{ as above}\}(\mathcal{M}).$$

But to specify the arrow $\longrightarrow$ here in a simple conceptual way, it is good to give ourselves K with the additional structures $J \underset{\bar{f}}{\overset{j}{\rightleftarrows}} K \xleftarrow{i} I$, $\bar{f}j = \mathrm{id}_J$, $\mathrm{id}_K \xrightarrow{\alpha} j\bar{f}$. And this system $\{K,\ i,\ j,\ \bar{f},\ \alpha\}$ is (unless I am mistaken) defined up to canonical isomorphism by the condition of representing the functor $\mathcal{M} \longmapsto \{F,\ G,\ u\}(\mathcal{M})$ … p. 7

In particular, the system defined by $G \in \mathrm{Ob}\,\mathbf{D}_{\mathcal{M}}(J)$ [page 82]

$$(f^*(G),\ G,\ \mathrm{id} : f^*(G) \longrightarrow f^*(G))$$

is defined unless I am mistaken by $\bar{f}^*(G)$ on $K(f)$. We propose to find a similar interpretation for

$$(F,\ f_*(F),\ f^*f_*F \longrightarrow F)$$

($F \in \mathrm{Ob}\,\mathbf{D}(I)$). The natural candidate here would be $\Phi = i_*(F)$. Let us grant this point (i.e. especially the verification of the fact that $j^*(i_*(F)) \simeq f_*(F)$). All this shows that if $\mathbf{D}$ is once again an arbitrary derivator, we can, by the preceding formulas, find a canonical object Φ of $\mathbf{D}(K)$, in terms either of a $\mathbf{D}$-coefficient F on I (taking $\Phi = i_*(F)$), or of a $\mathbf{D}$-coefficient G on J (taking $\Phi = \bar{f}^*(G)$). On the other hand, we can hope to define canonically

$$\begin{aligned} \mathbf{D}(K) &\longrightarrow \mathbf{D}(I \times \mathbf{\Delta}_1) \\ \mathbf{D}(K) &\longrightarrow \mathbf{D}(J \times \mathbf{\Delta}_1) \end{aligned}$$

whose composites with the "skeleton" functors

$$\begin{aligned} \mathbf{D}(I \times \mathbf{\Delta}_1) &\longrightarrow \underline{\mathrm{Hom}}(\mathbf{\Delta}_1^o, \mathbf{D}(I)) \\ \mathbf{D}(J \times \mathbf{\Delta}_1) &\longrightarrow \underline{\mathrm{Hom}}(\mathbf{\Delta}_1^o, \mathbf{D}(J)) \end{aligned}$$

[page 83]

give back (for $\Phi \in \mathrm{Ob}\,\mathbf{D}(K)$) the skeletal arrows

$$\begin{cases} f^*(G) \xrightarrow{u} F \\ G \xrightarrow{v} f_*F \end{cases}$$

associated with (F, G, u). To be precise, $f^*(G) \longrightarrow F$ is defined in terms of $\mathbf{D}$, and $\Phi \in \mathrm{Ob}\,\mathbf{D}(K)$, as above, and v is deduced from it by adjunction.

The natural idea here, would be to reduce the problem of fleshing out the skeletal adjunction arrows, to the case of open or closed immersions. Note that the open subcategories of K correspond precisely to the functors $K \longrightarrow \mathbf{\Delta}_1$, by taking the inverse images of the generic open. There is therefore a unique

$$\varphi : \ K(f) \longrightarrow \mathbf{\Delta}_1$$

with $I = \varphi^{-1}(\{0\})$. Thus

p.

THE DERIVATORS

ALEXANDRE GROTHENDIECK

Chapter V

Categories of paths and localization

p.
p. 1
[page 1]

1 The category of paths

Let Φ be the following category:

$$\begin{cases} \mathrm{Ob}\,\Phi &= \mathbf{Z} \\ \mathrm{Fl}\,\Phi &= \mathrm{Ob}\,\Phi \sqcup \mathbf{Z} \qquad (\mathrm{Ob}\,\Phi \leftrightarrow \text{identities}) \end{cases}$$

We denote by $\{\alpha\}$, or simply α, the object corresponding to $\alpha \in \mathbf{Z}$. We denote by u_α the arrow corresponding to the integer $\alpha \in \mathbf{Z}$.

$$\begin{aligned} s(u_\alpha) &= \alpha \qquad \text{if } \alpha \text{ even}, \quad \alpha+1 \quad \text{if } \alpha \text{ odd} \\ b(u_\alpha) &= \alpha+1 \qquad \text{if } \alpha \text{ even}, \quad \alpha \qquad \text{if } \alpha \text{ odd} \end{aligned}$$

No compositions to define, for if two arrows are composable, one of them is the identity. In fact, Φ is an *ordered* category, $\{\alpha\} < \{\beta\}$ if and only if α even, $\beta = \alpha + 1$ or $\beta = \alpha - 1$.

We denote, for $n \in \mathbf{N}$, by

$$\Phi_n \subseteq \Phi, \quad \mathrm{Ob}\,\Phi_n = \{\alpha \in \mathrm{Ob}\,\Phi \mid -n \leq \alpha \leq +n\},$$

the full subcategory of Φ whose objects are the $\{\alpha\}$ with $-n \leq \alpha \leq +n$. Thus

$$\Phi_1 = \{-1 \longleftarrow 0 \longrightarrow 1\},$$

the dual is

$$\Psi_1 = \Phi_1^o = \{-1 \longrightarrow 0 \longleftarrow 1\}.$$

We can regard Φ as an amalgamated sum of an infinite ordered sequence (on both sides) of copies of Φ_1, or also of a similar sequence of copies of Ψ_1.

$$\cdots \quad \overbrace{\underbrace{\longleftarrow -4 \longrightarrow -3 \longleftarrow -2 \longrightarrow -1 \longleftarrow 0 \longrightarrow 1 \longleftarrow 2 \longrightarrow 3 \longleftarrow 4 \longrightarrow}_{\Psi_1 \quad \Psi_1 \quad \Psi_1 \quad \Psi_1}}^{\Phi_1 \quad \Phi_1 \quad \Phi_1 \quad \Phi_1 \quad \Phi_1} \quad \cdots$$

The Φ_n, for n odd, are likewise sums of copies of Φ_1 put end to end, and the Φ_n, for n even > 0, sums of copies of Ψ_1.

The Φ_n form a direct system with fully faithful transition functors, of which Φ is the direct limit p. 2

$$\Phi = \varinjlim_n \Phi_n.$$

But for what follows we are going to make the Φ_n into an *inverse* system, by defining for every $n \leq m$ a *retraction* $\varphi_{n,m}$ of Φ_m onto Φ_n, by

[page 2]

$$\varphi_{n,m} : \Phi_m \longrightarrow \Phi_n \subset \Phi_m$$

$$\varphi_{n,m}(\{i\}) = \begin{cases} \{i\} & \text{if } i \in \operatorname{Ob}\Phi_n, \text{ i.e } -n \leq i \leq n \\ \{n\} & \text{if } n \leq i \leq m \\ \{-n\} & \text{if } -m \leq i \leq -n \end{cases}$$

We denote the inverse system of ordered sets thus obtained by

$$\mathbf{\Phi} = (\Phi_n)_{n \in \mathbf{N}} \in \operatorname{Pro}(\mathrm{Cat}).$$

Let X be a category, then for $n \in \mathbf{N}$

$$\underline{\mathrm{Ch}}_n(X) \overset{\text{def}}{=} \underline{\mathrm{Hom}}(\Phi_n, X)$$

is the category of diagrams in X of the form

$$A_{-n} \cdots \longleftarrow A_{-2} \xrightarrow{u_{-2}} A_{-1} \xleftarrow{u_{-1}} A_0 \xrightarrow{u_0} A_1 \xleftarrow{u_1} A_2 \xrightarrow{u_2} A_3 \longleftarrow \cdots A_n \qquad (*)$$

with $2n$ arrows end to end u_i $(-n \leq i \leq n-1)$, two consecutive arrows always going in opposite directions,

$$\text{either } \cdot \longrightarrow \cdot \longleftarrow \cdot, \quad \text{or } \cdot \longleftarrow \cdot \longrightarrow \cdot.$$

The canonical functor deduced from $\varphi_{n,m} : \Phi_m \longrightarrow \Phi_n$

$$\varphi_X^{m,n} : \underline{\mathrm{Ch}}_n(X) \longrightarrow \underline{\mathrm{Ch}}_m(X), \qquad n \leq m,$$

is the "degeneracy functor", consisting in completing a diagram $(*)$, with $m-n$ terms A_{-n} with identities on the left, and with $m-n$ terms with identities on the right, so for n even we find p. 3

$$\underbrace{A_{-n} \cdots \xrightarrow{\mathrm{id}} A_{-n}}_{m-n \text{ terms}} \xleftarrow{\mathrm{id}} A_{-n} \longrightarrow A_{-(n-1)} \cdots\cdots A_{n-1} \longleftarrow A_n \xrightarrow{\mathrm{id}} \underbrace{A_n \xleftarrow{\mathrm{id}} A_n \longrightarrow \cdots A_n}_{m-n \text{ terms}}.$$

This is a fully faithful functor (which corresponds to the fact that $\varphi_{n,m}$ is surjective on objects and on arrows). [page 3]

We set

$$\underbrace{\underline{\mathrm{Ch}}_\infty(X)}_{\text{or simply } \underline{\mathrm{Ch}}(X)} = \underline{\mathrm{Hom}}(\mathbf{\Phi}, X) \overset{\text{def}}{=} \varinjlim_n \underline{\mathrm{Ch}}_n(X)$$

([1]) ([2]). We have a fully faithful inclusion

$$\underline{\mathrm{Ch}}_\infty(X) \hookrightarrow \underline{\mathrm{Hom}}(\Phi, X),$$

where the second category is that of infinite diagrams (in both directions)

$$\cdots \longleftarrow A_{-2} \xrightarrow{u_{-2}} A_{-1} \xleftarrow{u_{-1}} A_0 \xrightarrow{u_0} A_1 \xleftarrow{u_1} A_2 \xrightarrow{u_2} \cdots$$

and where $\underline{\mathrm{Ch}}_\infty(X)$ is identified with the full subcategory formed by the objects for which the arrows u_i are identities, for $|i|$ sufficiently large.

We shall call $\underline{\mathrm{Ch}}_\infty(X)$ the *category of paths* in X, its objects will be the paths in X. Those which come from $\underline{\mathrm{Ch}}_n(X)$ will be called *standard paths of length* $2n$. (The term "standard" applies here to our notational conventions. Thus we consider only paths of even length, and for a given length, we choose one of the two reasonable kinds, for example for paths of length 2 we have taken only

$$A_{-1} \longleftarrow A_0 \longrightarrow A_1$$

and not

$$A_{-1} \longrightarrow A_0 \longleftarrow A_1.)$$

I realize that my notations u_i are not convenient for duality, I am going to change them, to u_i for the arrows between objects $\{i\}$ with $i \geq 0$, and v_i the arrow symmetric to u_i with respect to the origin. (The group of automorphisms of the Φ_n, for $n > 0$, is reduced to the group with two elements, the only nontrivial automorphism being the symmetry $\{i\} \longrightarrow \{-i\}$). So our paths will henceforth be denoted [page 4]

$$A_{-2} \xrightarrow{v_1} A_{-1} \xleftarrow{v_0} A_0 \xrightarrow{u_0} A_1 \xleftarrow{u_1} A_2 \longrightarrow \cdots . \qquad (*)$$

(This is not exciting either …) p. 4

Let

$$\varepsilon = \{-1, +1\}$$

be the ordered set, *discrete,* reduced to the two elements -1, $+1$, and let us define functors (inclusions and fully faithful for $n > 0$)

$$\alpha_n : \varepsilon \longrightarrow \Phi_n \quad \begin{matrix} \alpha_n(-1) = -n \\ \alpha_n(1) = n, \end{matrix}$$

these functors are compatible with the $\varphi_{n,m}$, whence

$$\alpha_\infty : \varepsilon \longrightarrow \mathbf{\Phi} \in \mathrm{ProCat},$$

whence for every X a functor

$$\kappa : \underline{\mathrm{Hom}}(\mathbf{\Phi}, X) \longrightarrow \underline{\mathrm{Hom}}(\varepsilon, X) \simeq X \times X$$

i.e.

[page 5]

[1] The set of objects of this category will be denoted by $\mathrm{Ch}_\infty(X)$ or $\mathrm{Ch}(X)$, likewise we define the $\mathrm{Ch}_n(X)$.

[2] in this formula: $\mathbf{\Phi}$ procategory

$$\underline{\mathrm{Ch}}(X) \longrightarrow X \times X$$
$$c \longrightarrow (\sigma(c), \beta(c)).$$

For every path c in X i.e. a diagram $(*)$ (satisfying the condition of degeneracy at infinity), we have a source or origin of the path, and a target or an *endpoint*

$$\begin{cases} \sigma(c) = A_i & \text{for } i \leq -\mathrm{long}(c) \\ \beta(c) = A_i & \text{for } i \geq \mathrm{long}(c) \end{cases}$$

. The fiber of the preceding functor at x, y will be denoted

$$\underline{\mathrm{Ch}}(X; x, y) \qquad \text{or simply } \underline{\mathrm{Ch}}(x, y),$$

category of paths from x to y.

p. 5

2 Calculation of a category of fractions $X\Sigma^{-1}$

Before continuing the formalism of paths (composition etc.), I am going to test the usefulness of the notion of category of paths, for the "calculation" of the $X\Sigma^{-1}$, where

$$\Sigma \subset \mathrm{Fl}(X), \qquad \Sigma \text{ contains the identities}$$

is a subset, otherwise *arbitrary*, of $\mathrm{Fl}(X)$. We know that $\mathrm{Ob}\, X\Sigma^{-1} = \mathrm{Ob}\, X$, and everything comes down to defining the

$$\mathrm{Hom}_{X\Sigma^{-1}}(x, y), \qquad x, y \in \mathrm{Ob}\, X,$$

and for $x, y, z \in \mathrm{Ob}\, X$ the composition

[page 6]

$$\mathrm{Hom}_\Sigma(x, y) \times \mathrm{Hom}_\Sigma(y, z) \xrightarrow{(u,v)\mapsto vu} \mathrm{Hom}_\Sigma(x, z),$$

(where we write Hom_Σ instead of $\mathrm{Hom}_{X\Sigma^{-1}}$, to lighten the notation).

Using Σ, we define a full subcategory

$$\underline{\mathrm{Ch}}_\Sigma(X) \hookrightarrow \underline{\mathrm{Ch}}(X),$$

([3]) by the condition on $c \in \mathrm{Ch}(X)$: For i even, i.e. $i - 1 \xleftarrow{u_i} i$ (u_i is called "retrograde"), $c(u_i) \in \Sigma$, i.e. the retrograde "segments" in the path c must be in Σ. We similarly define the full subcategories

$$\underline{\mathrm{Ch}}_\Sigma(X; x, y) = \{c \in \underline{\mathrm{Ch}}(X; x, y) \mid c \in \underline{\mathrm{Ch}}_\Sigma(X)\},$$

or again the fiber at (x, y) of the source-target functor for $\underline{\mathrm{Ch}}_\Sigma(X)$

$$\underline{\mathrm{Ch}}_\Sigma(X) \longrightarrow X \times X.$$

We define a canonical map

$$\mathrm{Ch}_\Sigma(X, x, y) \longrightarrow \mathrm{Hom}_\Sigma(x, y),$$

by associating to every path from x to y the Σ-homomorphism from x to y obtained by *inverting* (in $X\Sigma^{-1}$) the retrograde segments (which is possible, since they are in Σ), so that we find in $X\Sigma^{-1}$ a "direct path"

[page 7]

$$x = A_{-n} \longrightarrow \cdots \longrightarrow A_{-1} \longrightarrow A_0 \longrightarrow A_1 \longrightarrow A_2 \cdots A_n = A_{n+1} = \cdots = y.$$

p. 6

and we compose all these morphisms.

In fact, we find at once that we thus define a *functor*

$$\underline{\mathrm{Ch}}_\Sigma(X) \longrightarrow \underline{\mathrm{Fl}}(X\Sigma^{-1}),$$

from the fact that if we have a diagram

$$\begin{array}{ccc} A_{i-1} & \xleftarrow{u_i} & A_i \\ \downarrow{\scriptstyle f_{i-1}} & & \downarrow{\scriptstyle f_i} \\ A'_{i-1} & \xleftarrow{u'_i} & A'_i \end{array} \qquad (i \text{ even})$$

with u_i, u'_i in Σ, this diagram, as a diagram in $X\Sigma^{-1}$, is commutative with horizontal arrows *isomorphisms,* and therefore remains commutative on inverting the said arrows. Whence an induced functor

$$\underline{\mathrm{Ch}}_\Sigma(X; x, y) \longrightarrow \underline{\mathrm{Fl}}(\underbrace{X\Sigma^{-1}}_{X'}; x, y).$$

But for any category X', and objects x, y in X', the category $\underline{\mathrm{Fl}}(X', x, y)$ is a category, *discrete,* having as its set of objects $\mathrm{Hom}_{X'}(x, y)$. Thus the preceding functor in fact defines a canonical map [page 8]

$$\boxed{\pi_0\underline{\mathrm{Ch}}_\Sigma(X; x, y) \longrightarrow \mathrm{Hom}_\Sigma(x, y).}$$

I say that this map is always bijective. To see this, we are going to *construct* "from scratch" the category $X\Sigma^{-1}$, taking as objects those of X, and *defining*

$$\boxed{\mathrm{Hom}_\Sigma(x, y) = \pi_0\underline{\mathrm{Ch}}_\Sigma(X; x, y).}$$

We must therefore define the composition

$$\mathrm{Hom}_\Sigma(x, y) \times \mathrm{Hom}_\Sigma(y, z) \longrightarrow \mathrm{Hom}_\Sigma(x, z).$$

For this, we are going to define functors "composition of paths"

$$\begin{aligned} \underline{\mathrm{Ch}}_m(X; x, y) \times \underline{\mathrm{Ch}}_n(X; y, z) &\longrightarrow \underline{\mathrm{Ch}}_{m+n}(X; x, z) \\ (c, c') &\longmapsto c' \circ c, \end{aligned}$$

but restricting ourselves for simplicity to the case where m, n are *even,* so that by "putting end to end" c and c' (in this order), we do indeed find a path of type $n + m$.

This operation is "transposed" from the pair of functors [page 9]

$$\begin{aligned} \Phi_m &\xrightarrow{\alpha} \Phi_{m+n} \quad i \longmapsto (i - n) \\ \Phi_n &\xrightarrow{\beta} \Phi_{m+n} \quad i \longmapsto (i + m), \end{aligned}$$

where we will have p. 7

$$\alpha(\{m\}) = \beta(\{-n\}) = m - n,$$

and we will have

$$\Phi_{m+n} \simeq \Phi_m \sqcup_e \Phi_n,$$

[3]The set of its objects is denoted $\mathrm{Ch}_\Sigma(X)$. Its elements are the Σ-paths .

i.e. Φ_{m+n} is the "amalgamated sum" in (Cat), obtained by "gluing" Φ_m and Φ_n, by identification of the "target element" of Φ_m with the "source element" of Φ_n. The composition of paths functor induces

$$\underline{\mathrm{Ch}}_{m,\Sigma}(x,y) \times \underline{\mathrm{Ch}}_{n,\Sigma}(y,z) \longrightarrow \underline{\mathrm{Ch}}_{m+n,\Sigma}(x,z),$$

whence on passing to the π_0

$$\mathrm{Hom}_{\Sigma,m}(x,y) \times \mathrm{Hom}_{\Sigma,n}(y,z) \longrightarrow \mathrm{Hom}_{\Sigma,m+n}(x,z),$$

where we denote, for $a, b \in \mathrm{Ob}\, X$, by

$$\mathrm{Hom}_{\Sigma,n}(a,b) \overset{\mathrm{def}}{=} \pi_0(\underline{\mathrm{Ch}}_{n,\Sigma}(a,b)).$$

To deduce from this a map on the $\varinjlim$ [page 10]

$$\underbrace{\mathrm{Hom}_\Sigma(x,y)}_{=\varinjlim \mathrm{Hom}_{\Sigma,m}(x,y)} \times \underbrace{\mathrm{Hom}_\Sigma(y,z)}_{=\varinjlim \mathrm{Hom}_{\Sigma,n}(y,z)} \longrightarrow \underbrace{\mathrm{Hom}_\Sigma(x,z)}_{=\varinjlim \mathrm{Hom}_{\Sigma,n+m}(x,z)}$$

we must prove, for $n \leq n'$, $m \leq m'$, commutativity in

$$\begin{array}{ccc} \mathrm{Hom}_{\Sigma,m}(x,y) \times \mathrm{Hom}_{\Sigma,n}(y,z) & \longrightarrow & \mathrm{Hom}_{\Sigma,m+n}(x,z) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{\Sigma,m'}(x,y) \times \mathrm{Hom}_{\Sigma,n'}(y,z) & \longrightarrow & \mathrm{Hom}_{\Sigma,m'+n'}(x,z) \end{array}$$

In other words, that for $c : x \longrightarrow y$ and $c' : y \longrightarrow z$ two composable Σ-paths of length $2m$ and $2n$ respectively, the composite $c'c : x \longrightarrow z$ does not change, "up to homotopy" in $\underline{\mathrm{Ch}}_\Sigma(x,z)$, when we replace c, c' by paths $\bar{c}$, $\bar{c}'$ of length $2m'$, $2n'$ deduced from c and c' by "degeneracy at the ends".

More precisely,

$$\bar{c}'\bar{c} \sim \overline{c'c},$$

the composite path is homotopic to the one deduced from $c'c$ by "degeneracy at the ends".

I have just looked at $n = m = 1$, $n' = m' = 2$, and found that there is no natural arrow between $\bar{c}'\bar{c}$ and $\overline{c'c}$ - the homotopy sought must be defined with the help of a "path" in $\underline{\mathrm{Ch}}_{\Sigma,m'+n'}(X;x,z)$. In the general case, we are in fact going to find a homotopy [p. 8] [page 11]

$$h = h_{(n,m),(n',m')}$$

"making commutative" the diagram of functors

$$\begin{array}{ccc} \mathcal{A} = \underline{\mathrm{Ch}}_{\Sigma,m}(X;x,y) \times \underline{\mathrm{Ch}}_{\Sigma,n}(X;y,z) & \xrightarrow{*_{m,n}} & \underline{\mathrm{Ch}}_{\Sigma,m+n}(X;x,z) \\ \downarrow{\scriptstyle i_{m,m'} \times j_{n,n'}} & \overset{h}{\swarrow} & \downarrow{\scriptstyle k_{m+n,m'+n'}} \\ \underline{\mathrm{Ch}}_{\Sigma,m'}(X;x,y) \times \underline{\mathrm{Ch}}_{\Sigma,n'}(X;y,z) & \xrightarrow{*_{m',n'}} & \underline{\mathrm{Ch}}_{\Sigma,m'+n'}(X;x,z) = \mathcal{B}, \end{array}$$

where the $i_{\alpha,\beta}$, $j_{\alpha,\beta}$, $k_{\alpha,\beta}$ are the transition functors

$$\underline{\mathrm{Ch}}_{\Sigma,\alpha}(X;a,b) \longrightarrow \underline{\mathrm{Ch}}_{\Sigma,\beta}(X;a,b)$$

($\alpha \leq \beta$) for the (x,y), (y,z) and (x,z) (in place of (a,b)) respectively. The homotopy h will be a path in the category $\underline{\mathrm{Hom}}(\mathcal{A},\mathcal{B})$, going from

$$k_{m+n,m'+n'} \circ *_{m,n} \longrightarrow *_{m',n'} \circ (i_{m,m'} \times j_{n,n'}).$$ [page 12]

The existence of such a homotopy means neither more nor less than that the square diagram above is commutative, as a diagram in the category (Cat hot) of "categories up to homotopy," whose objects are the (small) categories, and where we define

$$\mathrm{Hom}_{\mathrm{Cat\,hot}}(\mathcal{A},\mathcal{B}) \overset{\mathrm{def}}{=} \underbrace{\pi_0(\underline{\mathrm{Hom}}(\mathcal{A},\mathcal{B}))}_{\substack{\text{homotopy classes}\\ \text{of functors from } \mathcal{A}\\ \text{to } \mathcal{B}}} .$$

It is this conclusion that will be useful to us, and not the explicit description of the path h. This existence granted (cf. the proof below), we see therefore that if we regard, for $a, b \in \mathrm{Ob}\, X$, the system of categories $\underline{\mathrm{Ch}}_{\Sigma,r}(X;a,b)$ ($r \in \mathbf{N}$) as an *ind-object of* Cat hot, which we denote by

$$\underline{\mathrm{Ch}}^{\mathrm{ind}}_{\Sigma}(X;a,b) \in \mathrm{Ob}\,\mathrm{Ind}(\mathrm{Cat\,hot}),$$

we will have a canonical morphism in Ind(Cat hot)

$$\underline{\mathrm{Ch}}^{\mathrm{ind}}_{\Sigma}(X;x,y) \times \underline{\mathrm{Ch}}^{\mathrm{ind}}_{\Sigma}(X;y,z) \overset{*}{\longrightarrow} \underline{\mathrm{Ch}}^{\mathrm{ind}}_{\Sigma}(X;x,z),$$

called *"ind-morphism of composition of paths."* We would have liked to deduce it from an honest *functor* "composition of paths"

$$\underline{\mathrm{Ch}}_{\Sigma}(X;x,y) \times \underline{\mathrm{Ch}}_{\Sigma}(X;y,z) \overset{?}{\longrightarrow} \underline{\mathrm{Ch}}_{\Sigma}(X;x,z),$$
$$\text{which we would write } (c,c') \longmapsto c' \circ c \text{ or } c' \cdot c,$$

but I doubt that such a functor exists,

which on passing to ind-objects gives back the preceding morphism, or, what amounts to the same thing, such that for every $m, n \in \mathbf{N}$ (of the same parity), there exists an $N \geq m+n$ large enough that we have a diagram

p. 9
[page 13]

$$\begin{array}{ccc}
\underline{\mathrm{Ch}}_{\Sigma,m}(X;x,y) \times \underline{\mathrm{Ch}}_{\Sigma,n}(X;y,z) & \xrightarrow{*_{m,n}} & \underline{\mathrm{Ch}}_{\Sigma,m+n}(X;x,z) \\
\big\downarrow & \searrow & \big\downarrow \\
 & & \underline{\mathrm{Ch}}_{\Sigma,N}(X;x,z) \\
 & & \big\downarrow \\
\underline{\mathrm{Ch}}_{\Sigma}(X;x,y) \times \underline{\mathrm{Ch}}_{\Sigma}(X;y,z) & \xrightarrow{\quad ? \quad} & \underline{\mathrm{Ch}}_{\Sigma}(X;x,z)
\end{array}$$

and where the upper triangle is *commutative in* Cat hot. (One must suppose that for N large enough, the composite $\downarrow\!\!\xrightarrow{\ ?\ }$ factors through $\underline{\mathrm{Ch}}_{\Sigma,N}(X;x,z)$ …) To see that the composition functors $*_{m,n}$ define a homomorphism of the ind-objects in (Cat hot), it suffices to prove it in the particular cases $(m',n') = (m+2,n)$ ou $(m',n') = (m,n+2)$. We are going to draw the figure assuming m, n *odd.*

I realize here that I have stupidly botched it, the composition of two paths of lengths m, n *odd is*

not a path of length $m+n$, i.e. a diagram of type Φ_{m+n}, but a diagram of *dual* type $\Phi^o_{m+n} = \Psi_{m+n}$. For example, for $m = n = 1$

[page 14]

$$\underbrace{A_0 \longleftarrow A_1 \longrightarrow A_2}_{c} = \underbrace{B_0 \longleftarrow B_1 \longrightarrow B_2}_{c'}$$

is of type Φ^o_2! So as not to complicate life too much, we are going to restrict ourselves in the preceding developments to m, n *even,* hence paths of length at least 4. It is a little

ridiculous! We are reduced to proving commutativity up to homotopy in the case where we leave c' fixed and where we send $\underline{\mathrm{Ch}}_{\Sigma,m}(X;x,y)$ into $\underline{\mathrm{Ch}}_{\Sigma,m+2}(X;x,y)$, or conversely c fixed and $\underline{\mathrm{Ch}}_{\Sigma,n}(X;y,z)$ into $\underline{\mathrm{Ch}}_{\Sigma,n+2}(X;y,z)$. Let us take the first case, and write a path c without negative indices (this is not a very happy notation!)

$$A_0 \longrightarrow A_1 \longleftarrow A_2 \cdots \longleftarrow A_{2m}$$

(NB because of the condition m even, all our paths now begin with an arrow from A_0 to A_1, never in the opposite direction).

I see that making the homotopy explicit, while offering no difficulty, is very tedious to explain. I am going to try to give a conceptual proof, without tedious "calculation".

The two composite functors p. 10 [page 15]

$$f = k \circ *_{m,n} \ , \quad g = *_{m',n'} \circ (i \times j)$$

in the square on page 11, which we have to prove are homotopic in $\underline{\mathrm{Hom}}(\mathcal{A},\mathcal{B})$, can be viewed as obtained in the following way, in terms of two functors

$$\varphi, \psi : \Phi_{m'+n'} \rightrightarrows \Phi_{m+n}, \tag{1}$$

whence two "transposed" functors

$$\underline{\mathrm{Hom}}(\Phi_{m+n}, X) \underset{G}{\overset{F}{\rightrightarrows}} \underline{\mathrm{Hom}}(\Phi_{m'+n'}, X) \tag{2}$$

which functors "induce" f and g. To be precise, denoting by a, b the source and target objects of Φ_{m+n}, by c the object which corresponds to the target of the path c of length m , to the source of the path c' of length n which is to be composed, finally by a_0, b_0 the source and target elements of $\Phi_{m'+n'}$, the functors under consideration (1) send a_0 to a, b_0 to b, hence define a commutative diagram

$$\begin{array}{ccc} \Phi_{m'+n'} & \overset{\varphi}{\underset{\psi}{\rightrightarrows}} & \Phi_{m+n} \\ {\scriptstyle\alpha'}\uparrow & \nearrow_{\alpha} & \\ \varepsilon = \{a_0, b_0\}. & & \\ \text{discrete category} & & \end{array} \tag{3}$$

This said, the target category $\mathcal{B}$ for f, g, namely $\underline{\mathrm{Ch}}_{\Sigma,m'+n'}(X;x,z)$, is interpreted as the full subcategory of $\underline{\mathrm{Hom}}(\Phi_{m'+n'}, X)$ consisting

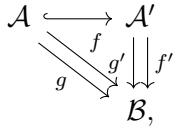
of the functors $c' : \Phi_{m'+n'} \longrightarrow X$ whose composite with α' is the functor [page 16]

$$\beta : \varepsilon \longrightarrow X \qquad \begin{array}{l} a_0 \longrightarrow x \\ b_0 \longrightarrow z, \end{array} \tag{$*$}$$

and likewise the source category $\mathcal{A}$ is interpreted as the full subcategory of $\underline{\mathrm{Hom}}(\Phi_{m+n}, X)$ consisting of the $c : \Phi_{m+n} \longrightarrow X$ whose restriction to $\{a,b,c\}$ is the functor

$$a \longrightarrow x, \ b \longrightarrow z, \ c \longrightarrow y.$$

It is contained in the full subcategory $\mathcal{A}'$ of functors whose restriction to $\{a,b\}$ is $a \longrightarrow x$, $b \longrightarrow z$, i.e. whose composite with α is the functor $\beta : \varepsilon \longrightarrow X$ already considered above. Thus the functors $f, g : \mathcal{A} \rightrightarrows \mathcal{B}$ factor through $\mathcal{A}'$ p. 11



functors deduced from φ, ψ above, giving rise to the commutative triangles (3), and from the functor $\beta : \varepsilon \longrightarrow X$, in the said way. To prove that f and g are homotopic, it suffices to prove it for f', g'. And for this, it suffices to show that φ and ψ are homotopic as objects of the category $\underline{\mathrm{Hom}}(\Phi_{m'+n'}, \Phi_{m+n}; \alpha')$ consisting of the functors whose restriction to ε is α .

[page 17]

Moreover, the two assertions (on f', g', for arbitrary X, x, z, and on φ, ψ, α') are in fact equivalent as one sees by taking $X = \Phi_{m+n}$, $x = a$, $z = b$. This will be contained in the more general lemma. ([4])

Lemma. *Let k and l be two integers. Then the full subcategory of $\underline{\mathrm{Hom}}_{cr}(\Phi_k, \Phi_l)$ consisting of the functors $\Phi_k \longrightarrow \Phi_l$ sending source to source, target to target, is connected.* (NB It is nonempty if and only if $k \geq l$.)

We are going to describe these functors completely (= increasing maps) $\Phi_k \xrightarrow{\varphi} \Phi_l$, identifying $\mathrm{Fl}(\Phi_k)$ with the interval of integers $[1, 2k]$, and associating to φ the subset

$$E_\varphi \subset \mathrm{Fl}(\Phi_k) = [1, 2k] \qquad\qquad E_\varphi = \{u \in \mathrm{Fl}(\Phi_k) \mid \varphi(u) \text{ an identity}\}$$

consisting of the $u \in \mathrm{Fl}(\Phi_k)$ which are "contracted" by φ to an identity. If E is a set of integers, we introduce on it the notion of connected components (by the equivalence relation in E generated by the successor relation $y = x + 1$). One sees at once that the connected components of E_φ which contain neither the first nor the last arrow of Φ_k are even, and that

$$\underbrace{\mathrm{cardFl}(\Phi_k)}_{2k} = \mathrm{card}\,(E_\varphi) + \underbrace{\mathrm{cardFl}(\Phi_l)}_{2l},$$

i.e.

[page 18]

$$\mathrm{card}\,(E_\varphi) = 2(k - l).$$

It is immediate that a functor φ is surjective on objects and on arrows, hence identifies p. 12
Φ_l with a quotient category of Φ_k - and the equivalence relation in Φ_k defined by φ is the one obtained by identifying all the objects that are source or target of arrows of one and the same connected component I of E_φ

$$\cdots \longleftarrow \underbrace{\longrightarrow \longleftarrow \longrightarrow \longleftarrow}_{I} \longrightarrow \cdots$$

and the arrows in this component with the corresponding identity. The conditions stated on E (independently of a φ which would give rise to it) ensure that the quotient category is isomorphic (with *preservation* of source and target) either to Φ_l, or to the dual $\Psi_l = \Phi_l^o$, according to the direction of the first arrow of $\bar{\Phi}_k$ (quotient of Φ_k). This "initial direction" is the same as that for Φ_k if the first arrow of Φ_k does not belong to E, or else if it belongs to it and its connected component in E has even cardinality. In the opposite case ($u \in E$ and its connected component in E has odd cardinality), the initial direction of $\bar{\Phi}_k$ is opposite to that of Φ_k.

One sees therefore that E must satisfy yet another condition, for the corresponding [page 19]
$\bar{\Phi}_k$ to be isomorphic to Φ_l, i.e. for E to come from a functor φ. I summarize all the conditions

[4] In the considerations that follow, we in fact tacitly take the functors under consideration to be *increasing* on the vertices with respect to the order of their numbering. Thus one should write $\underline{\mathrm{Hom}}_{cr}$ instead of $\underline{\mathrm{Hom}}$, in the statements that follow, including in the connectedness lemma opposite.

a) card$E = 2(k - l)$.

b) The connected components of E which contain no initial or final arrow have even cardinality.

c) If Φ_k and Φ_l have the same initial direction (k and l of the same parity), then the possible component of E containing the initial arrow must itself also have even cardinality (which, because of a) and b) , is equivalent to the same condition for the possible connected component containing the final arrow of Φ_k). If Φ_k and Φ_l have different initial directions, the initial arrow of Φ_k must belong to a connected component of *odd* cardinality in E (which is equivalent to likewise for the final arrow).

On the other hand, the functor φ is uniquely determined by the knowledge of E, for there is a *unique* isomorphism $\bar{\Phi}_k \xrightarrow{\sim} \Phi_l$ compatible with sources and targets, because of the

Lemma (immediate) : *An automorphism of Φ_l which is the identity on the source object (or on the target object) is the identity everywhere.*

The functors φ, ψ from earlier corresponded to the choice, one of $E_\varphi = \{1, 2, 2k - 1, 2k\}$, the other of $E_\psi = \{1, 2, i, i + 1\}$, 2 connected components in one case, two in the other. [page 20]

In the general case, one should make explicit the order relation on the set $\underline{\mathrm{Hom}}^!(\Phi_k, \Phi_l)$ (! means : compatible with source and target), in terms of the representative sets E_φ, E_ψ. I am not going to make this order relation explicit, but only a property of it which will suffice for us. Let φ_E be the functor defined by a subset $E \subset [1, 2k]$. First of all : p. 13

Lemma 1 (immediate) : *Let $E, E' \subset [1, 2k]$ be admissible, and let $j \in [1, 2k]$ be such that $E \cap [1, j] = E' \cap [1, j]$. Then φ_E and $\varphi_{E'}$ agree on the target of u_j (j an arrow of Φ_k) and on all the objects of Φ_k preceding it (in the sense of the numbering of the objects, and not of the order relation of the category Φ_k of course.) Dually, if $E \cap [j, 2k] = E' \cap [j, 2k]$, then φ_E and $\varphi_{E'}$ agree on the source of u_j, and on all the objects of Φ_k following it.*

Let us now consider a connected subset I of E (not necessarily a connected component), $I = [r, s] \subset [1, 2k]$ (r the smallest element of I, s its largest element) and suppose that

a) card$I$ is even (i.e. $s - r$ is odd)

b) $r \neq 1$, and $r - 1 \notin E$ (i.e. $r - 1 \in [1, 2k] \backslash E$)

and let

$$E' = (E \backslash \{s\}) \cup \{r - 1\} = E \backslash I \cup \overbrace{[r - 1, s - 1]}^{I'}$$

obtained from I by "translating" one notch to the left.

It is immediate that if E is admissible, E' is too (and conversely). This said, we have the [page 21]

Lemma 2. *E and E' being as above, there exists a homomorphism of functors between φ_E and $\varphi_{E'}$* (I am not saying in which direction the arrow goes!).

Corollary. Dual statement, when E' is obtained from E by translating to the right by one notch a connected interval of even length $I = [r, s]$ of E, $s \neq 2k$ and $s + 1 \notin E$.

Indeed, in this case E is obtained from E' by an operation of the type made explicit in lemma 2.

The connectedness of the category $\underline{\mathrm{Hom}}^!(\Phi_k, \Phi_l)$ is now clear. If k and l have the same parity, by pushing the connected components of E to the left, we write E in the "normal" form

$$E_0 = [1, 2(k-l)].$$

If k and l are of opposite parity, we cannot push the whole connected component of E which contains the last element $2k$, for it has odd cardinality. One must leave $2k$ in E and push the rest, and we reduce E to the normal form

$$E_0 = [1, 2(k-l)-1] \cup \{2k\}.$$

q.e.d.!

What precedes justifies our reasoning, which gave by composition of paths well-defined composition maps p. 14

$$\underbrace{\mathrm{Hom}_\Sigma(x,y)}_{=\pi_0\underline{\mathrm{Ch}}_\Sigma(x,y)} \times \underbrace{\mathrm{Hom}_\Sigma(y,z)}_{=\pi_0\underline{\mathrm{Ch}}_\Sigma(y,z)} \longrightarrow \underbrace{\mathrm{Hom}_\Sigma(x,z)}_{=\pi_0\underline{\mathrm{Ch}}_\Sigma(x,z)}.$$

I say that this composition is associative. This follows from the fact that if we have three paths (of *even* indices!) [page 22]

$$x \xrightarrow{c} y \xrightarrow{c'} z \xrightarrow{c''} t,$$

we have (without having to carry out homotopies …) strict associativity

$$c''(c'c) = (c''c')c.$$

I say that there are identities. We define

$$1_x \in \mathrm{Hom}_\Sigma(x,x)$$

by the path of index 0 (which is even!), i.e. by the diagram reduced to x, with no arrow (or if we want to interpret the diagram as a functor $e \longrightarrow X$, there is the implicit identity arrow). Our arguments did not exclude the case $n = 0$ or $m = 0$, and it is tautological that for a path

$$x \xrightarrow{c} y$$

we have

$$c \cdot 1_x = 1_y \cdot c = c.$$

So we have indeed defined a category, which I am going to call X_Σ. I am going to define a canonical functor

$$\theta : X \longrightarrow X_\Sigma$$

which will be the identity on objects, we must define it on arrows

$$\mathrm{Hom}_X(x,y) \longrightarrow \mathrm{Hom}_{X_\Sigma}(x,y) = \pi_0\underline{\mathrm{Ch}}_\Sigma(x,y)$$

. Unfortunately, an arrow

$$x \xrightarrow{u} y$$

cannot be interpreted directly as a path in our sense, we have to pay for our evasions! [page 23] But we are going to associate to such an arrow the path

$$x \longrightarrow y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{\mathrm{id}} y,$$

an element of $\underline{\mathrm{Ch}}_{\Sigma,2}(x,y)$, which defines the desired map. We must check compatibility p. 15

with identities and under composition. To the arrow $1_x : x \longrightarrow x$ is associated the element of $\underline{\mathrm{Ch}}_2(X; x, x)$

$$x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x,$$

we must prove that in $\pi_0(\underline{\mathrm{Ch}}_{\Sigma,2}(x, x))$ it becomes equal to the element deduced from $1_x \in \underline{\mathrm{Ch}}_{\Sigma,0}(X; x, x)$, by the transition functor. But this element is precisely none other than

$$x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x,$$

it is the same (without even having to pass to π_0)!

For compatibility with composition

$$x \xrightarrow{u} y \xrightarrow{v} z,$$

we have on the one hand

$$x \xrightarrow{vu} z \xleftarrow{\mathrm{id}} z \xrightarrow{\mathrm{id}} z \xleftarrow{\mathrm{id}} z,$$

on the other hand

$$x \xrightarrow{u} y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{\mathrm{id}} y \xrightarrow{v} z \xleftarrow{\mathrm{id}} z \xrightarrow{\mathrm{id}} z \xleftarrow{\mathrm{id}} z.$$

We complete the first path by degeneracy at both ends, to compare it with the second as an element of $\underline{\mathrm{Ch}}_4(x, z)$,

$$x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x \xrightarrow{vu} z \xleftarrow{\mathrm{id}} z \xrightarrow{\mathrm{id}} z \xleftarrow{\mathrm{id}} z \xrightarrow{\mathrm{id}} z \xleftarrow{\mathrm{id}} z.$$

[page 24]

We must prove that the two are homotopic. We thus have two functors

$$f, g : \underbrace{\overbrace{\underline{\mathrm{Fl}}(X; x, y)}^{\text{discrete category}} \times \overbrace{\underline{\mathrm{Fl}}(X; y, z)}^{\text{discrete category}}}_{\mathcal{A}} \underset{g}{\overset{f}{\rightrightarrows}} \underbrace{\underline{\mathrm{Ch}}_{4,\Sigma}(X; x, z)}_{\mathcal{B}}, \qquad (*)$$

we want to prove that they are homotopic. For this, we regard them (as a moment ago) as "transposes" of the two functors

$$\mathbf{\Delta}_1 \sqcup_{\mathbf{\Delta}_0} \mathbf{\Delta}_1 \simeq \mathbf{\Delta}_2 \underset{\psi}{\overset{\varphi}{\leftleftarrows}} \Phi_4 \qquad \alpha \swarrow \qquad \uparrow \alpha' \qquad \varepsilon,$$

since the categories $\mathcal{A}$, $\mathcal{B}$ of the two sides of (*) can be interpreted as full subcategories of $\underline{\mathrm{Hom}}(\mathbf{\Delta}_2, X)$ and $\underline{\mathrm{Hom}}(\Phi_4, X)$ respectively; namely the full subcategories consisting of the functors whose composite with α resp. α' is given by x and by z, and moreover (for $\mathcal{A}$) whose value at the middle object of $\mathbf{\Delta}_2$ is y. We can also replace $\mathcal{A}$ by the larger full subcategory $\mathcal{A}'$ of $\underline{\mathrm{Hom}}(\mathbf{\Delta}_2, X)$, obtained by ignoring this last condition, and it suffices to prove that the two "bigger" functors

p. 16

$$\mathcal{A}' \underset{g'}{\overset{f'}{\rightrightarrows}} \mathcal{B}'$$

are homotopic. And for this, it suffices to see again that φ and ψ are homotopic in the category of functors $\Phi_4 \longrightarrow \mathbf{\Delta}_2$ which are compatible with the source, and the target. We thus see that our homotopy lemma of a moment ago was not general enough. We would have to define the *generalized interval categories,* of arbitrary length n (n = number of

[page 25]

arrows end to end), a *standard* generalized interval category of length n has as its set of objects the interval of integers

$$[0, n],$$

and for every pair of consecutive integers $i-1$, i among the objects ($1 \leq i \leq n$), there is exactly one arrow connecting $i-1$, i . We assign to the interval $[i, i+1]$ the sign $+$ if $i \longrightarrow i+1$ (arrow in the direct direction), the sign $-$ if $i \longleftarrow i+1$ (arrow in the retrograde direction) - thus the standard categories are defined, for fixed n, by a system of signs $+$ or $-$, which we assign to the n consecutive "segments" $A_i := [i-1, i]$ $(1 \leq i \leq n)$ of the interval of integers $[0, n]$. There are thus exactly 2^n of them.

We expect the following statement to be true:

Lemma. *Let I, I' be two generalized intervals, equipped with source objects and target object (so that I, I' are* canonically *isomorphic to the corresponding standard generalized intervals). Let $\underline{\mathrm{Hom}}^!(I, I')$ be the category of functors from I to I', compatible with source and target. If this category is $\neq \emptyset$, it is 0-connected* (and, I presume, even contractible?). [page 26]

I am going to assume provisionally that this is so, at least in the particular case of Φ_4 (generalized interval of length 8) and $\mathbf{\Delta}_2$. This therefore gives a functor

$$\theta : X \longrightarrow X_\Sigma.$$

We have won, for the universal property of this functor is practically obvious. But I forgot to check that θ transforms elements of Σ into invertible arrows. So let

$$x \xrightarrow{u} y$$

with $u \in \Sigma$, we are going to define a path in the reverse direction

$$y \xrightarrow{c} x$$

which will define in X_Σ an arrow inverse to u. The natural idea is to take the "path"

$$y \xleftarrow{u} x$$

from y to x, but this is not a path in the sense of our definitions, we are going to complete it to p. 17

$$c : y \xrightarrow{\mathrm{id}} y \xleftarrow{u} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x.$$

I say that c does the job. We must prove two homotopy relations, of $c \cdot \bar{u}$ and of $\bar{u} \cdot c$ with the identity paths at x of length 8 (index 4, i.e. in $\underline{\mathrm{Ch}}_{\Sigma,4}(X; x, x)$ resp. $\underline{\mathrm{Ch}}_{\Sigma,4}(X; y, y)$), where $\bar{u}$ is the path associated to u [page 27]

$$x \xrightarrow{u} y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{\mathrm{id}} y.$$

We thus have to look at

$$x \xrightarrow{u} y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{u} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x$$

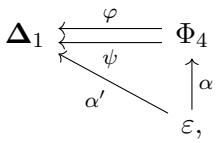
and see that it is homotopic to the trivial path

$$x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} \cdots x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x$$

(identities everywhere), and likewise for

$$y \xrightarrow{\mathrm{id}} y \xleftarrow{u} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x \xrightarrow{u} y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{\mathrm{id}} y$$

and the trivial path at y. This time, we again have two functors



compatible with source objects and target objects, and we must prove that they are homotopic in $\underline{\mathrm{Hom}}^!(\Phi_4, \mathbf{\Delta}_1)$. This must therefore follow from the same general lemma. We have won - by a simple interpretation of $X\Sigma^{-1}$, and in particular of the $\mathrm{Hom}_{X\Sigma^{-1}}(x,y)$ and of their composition, in terms of the geometry of path spaces:

$$\begin{cases} \mathrm{Hom}_{X\Sigma^{-1}}(x,y) \simeq \pi_0(\underline{\mathrm{Ch}}_\Sigma(X;x,y)) \\ \text{Composition of homomorphisms in } X\Sigma^{-1} \text{ corresponds to} \\ \text{the composition of } \Sigma\text{-paths in } X \end{cases}$$

p. 18

[page 28]

3 Passage to categories of diagrams

We should have denoted the category X by $\mathcal{M}$, for later on it will play the role of a "category of models". On the other hand, we shall denote by X the "index categories" on which we want to construct the "categories of diagrams" of type X

$$\mathcal{M}(X) = \underline{\mathrm{Hom}}(X, \mathcal{M}).$$

We thus take

$$\underbrace{\Sigma}_{\text{quasi-isomorphisms}} \subseteq \mathrm{Fl}(\mathcal{M}), \qquad \Sigma \supset \text{set of identity arrows},$$

and for every category X we define

$$\Sigma_X \subset \mathrm{Fl}\,\mathcal{M}(X)$$

as follows: if $F, G \in \mathrm{Ob}\,\mathcal{M}(X)$ $(F, G : X \rightrightarrows \mathcal{M})$, an arrow $u : F \longrightarrow G$ (homomorphism of functors) is in Σ_X if and only if $\forall x \in X$, $u(x) : F(x) \longrightarrow G(x)$ is in Σ ("criterion for quasi-isomorphy fiber by fiber", for the homomorphism between the copresheaves F, G under consideration).

Let

$$a, b \in \mathrm{Ob}\,\mathcal{M}$$

and consider the corresponding constant functors on X

$$a_X, b_X \in \mathrm{Ob}\,\mathcal{M}(X).$$

We propose to apply the theory of §2 to the "calculation" of [page 29]

$$\mathrm{Hom}_{\Sigma_X}(a_X, b_X),$$

we thus have a canonical bijection

$$\boxed{\mathrm{Hom}_{\Sigma_X}(a_X, b_X) \simeq \pi_0 \underline{\mathrm{Ch}}_{\Sigma_X}(\mathcal{M}(X); a_X, b_X).}$$

p. 19

We propose to make explicit the category of paths on the right-hand side. We have, leaving aside the data of Σ, a, b in $\mathcal{M}$, the expression for the category of *all* paths in $\mathcal{M}(X)$

$$\begin{aligned}
\underline{\mathrm{Ch}}(\mathcal{M}(X)) &\simeq \underline{\mathrm{Hom}}\Big(\underbrace{\mathbf{\Phi}}_{\substack{\text{pro-category}\\(\Phi_n)_{n\in\mathbf{N}}}}, \underbrace{\mathcal{M}(X)}_{\underline{\mathrm{Hom}}(X,M)}\Big)\\
&\overset{\text{def}}{=} \varinjlim_n \underline{\mathrm{Hom}}(\Phi_n, \mathcal{M}(X))\\
&\simeq \varinjlim_n \underline{\mathrm{Hom}}(\Phi_n \times X, \mathcal{M})\\
&\simeq \varinjlim_n \underline{\mathrm{Hom}}(X, \underline{\mathrm{Hom}}(\Phi_n, \mathcal{M}))\\
&\overset{\text{def}}{=} \varinjlim_n \underline{\mathrm{Hom}}(X, \underline{\mathrm{Ch}}_n(\mathcal{M}))\\
&\simeq \underline{\mathrm{Hom}}^!(X, \underline{\mathrm{Ch}}(\mathcal{M})),
\end{aligned}$$

where the sign ! means that we take the full subcategory of functors

$$X \longrightarrow \underline{\mathrm{Ch}}(\mathcal{M}),$$

which factor through one of the full subcategories $\underline{\mathrm{Ch}}_n(\mathcal{M})$. We can also say that if we consider the direct system of categories $\underline{\mathrm{Ch}}_n(\mathcal{M})$ as an ind-object of Cat, which we are going to denote [page 30]

$$\underline{\underline{\mathrm{Ch}}}(\mathcal{M}) \qquad\qquad \text{ind-object of Cat,}$$

we have

$$\underline{\mathrm{Ch}}(\mathcal{M}(X)) \xrightarrow{\sim} \underline{\mathrm{Hom}}(X, \underline{\underline{\mathrm{Ch}}}(\mathcal{M})).$$

If we turn to Σ, Σ_X, we find at once that a path in $\mathcal{M}(X)$ is a Σ_X-path, if and only if the corresponding functor $X \longrightarrow \underline{\mathrm{Ch}}(\mathcal{M})$ factors through $\underline{\mathrm{Ch}}_\Sigma(\mathcal{M})$. Thus we find

$$\underline{\mathrm{Ch}}_{\Sigma_X}(\mathcal{M}(X)) \simeq \underline{\mathrm{Hom}}^!(X, \underline{\mathrm{Ch}}_\Sigma(\mathcal{M})),$$

where the sign ! has the same meaning as above. If we introduce the ind-object $\underline{\underline{\mathrm{Ch}}}_\Sigma(\mathcal{M})$ defined by the $\underline{\mathrm{Ch}}_{\Sigma,n}(\mathcal{M})$, we thus also find

$$\underline{\mathrm{Ch}}_{\Sigma_X}(\mathcal{M}(X)) \xrightarrow{\sim} \underline{\mathrm{Hom}}(X, \underline{\underline{\mathrm{Ch}}}_\Sigma(\mathcal{M})).$$

Let c be a path in $\mathcal{M}(X)$, and [page 31]

$$X \longrightarrow \underline{\mathrm{Ch}}(\mathcal{M})$$

the functor it defines, let us compose it with the source-target functors

$$\underline{\mathrm{Ch}}(\mathcal{M}) \longrightarrow \mathcal{M} \times \mathcal{M},$$

we find a functor

$$X \longrightarrow \mathcal{M} \times \mathcal{M},$$

i.e. an object (F_0, F_1) of $\mathcal{M}(X) \times \mathcal{M}(X)$. I say that F_0 and F_1 are respectively the source and the target of the path c. A little more precisely, we have commutativity (strict) in the diagram of functors

p. 20

$$\begin{array}{ccc}
\underline{\mathrm{Ch}}_{\Sigma_X}(\mathcal{M}(X)) & \xrightarrow{\sim} & \underline{\mathrm{Hom}}^!(X, \underline{\mathrm{Ch}}_\Sigma(\mathcal{M}))\\
\Big\downarrow{\scriptstyle \sigma_{\mathcal{M}(X)}, \beta_{\mathcal{M}(X)}} & & \Big\downarrow{\scriptstyle \text{deduced from } \sigma_{\mathcal{M}}, \beta_{\mathcal{M}}}\\
\mathcal{M}(X) \times \mathcal{M}(X) & \xrightarrow{\sim} & \underline{\mathrm{Hom}}(X, \mathcal{M} \times \mathcal{M}),
\end{array}$$

where the functors σ are the source functors, the functors β the target functors.

If we give ourselves arbitrary objects F_0, F_1 in $\mathcal{M}(X)$, hence a functor

$$X \longrightarrow \mathcal{M} \times \mathcal{M},$$

$\underline{\mathrm{Ch}}_{\Sigma_X}(F_0, F_1)$ thus identifies with the category of "admissible" functors (i.e. factoring through a $\underline{\mathrm{Ch}}_{\Sigma,n}(\mathcal{M})$) $X \xrightarrow{C} \underline{\mathrm{Ch}}_{\Sigma}(\mathcal{M})$ which *lift* the functor (F_0, F_1) above, i.e. which give rise to a commutative triangle (strict) [page 32 blank] [page 33]

$$\begin{array}{ccc} & & \underline{\mathrm{Ch}}_{\Sigma}(\mathcal{M}) \\ & \overset{C}{\nearrow} & \big\downarrow {\scriptstyle (\sigma_{\mathcal{M}}, \beta_{\mathcal{M}})} \\ X & \xrightarrow[(F_0,F_1)]{} & \mathcal{M} \times \mathcal{M}. \end{array}$$

But if the functors F_0, F_1 are constant, with value a, b as before, i.e. (F_0, F_1) constant with value (a, b), then the category of functors C in question is tautologically isomorphic to that of functors from X into the fiber of $(\sigma_{\mathcal{M}}, \beta_{\mathcal{M}})$ at (a, b), whence the fundamental isomorphism of categories

$$\boxed{\underline{\mathrm{Ch}}_{\Sigma_X}(\mathcal{M}(X); a_X, b_X) \simeq \underline{\mathrm{Hom}}^!(X, \underline{\mathrm{Ch}}_{\Sigma}(\mathcal{M}; a, b)).}$$

We are going to make explicit the essential content of this formula in a

Scholie. Let $\mathcal{M}$ be a category with a set Σ of localization arrows, $a, b \in \mathrm{Ob}\,\mathcal{M}$. Then we can find an object $\underline{\underline{K}} = K(a, b)$, $K = (K_n)$, of $\mathrm{Ind}\,\mathrm{Cat}$ (indexed by $\mathbf{N}$ but this fact matters little), and a canonical isomorphism for every category X (isomorphism functorial in X)

$$\begin{aligned} \mathrm{Hom}_{\Sigma_X}(a_X, b_X) &\simeq \pi_0(\underline{\mathrm{Hom}}(X, \underline{\underline{K}})) \\ &\overset{\mathrm{def}}{=} \pi_0(\varinjlim \underline{\mathrm{Hom}}(X, K_n)) \\ &\simeq \varinjlim \pi_0(\underline{\mathrm{Hom}}(X, K_n)). \end{aligned}$$

I am going to dwell on this result, without worrying for the moment about the particular way in which the ind-category $\underline{\underline{K}}$ was obtained in terms of $\mathcal{M}$, Σ, a, b. p. 21 [page 34]

Since the $\pi_0\underline{\mathrm{Hom}}(X, K_n) = \mathrm{Hom}_{\mathrm{Cat\,hot}}(X, K_n)$ depend only on X and on K_n as objects of $\mathrm{Cat\,hot}$, I am going to interpret $\underline{\underline{K}}$ as an ind-object of $\mathrm{Cat\,hot}$, and write the fundamental formula in the form

$$\mathrm{Hom}_{\Sigma_X}(a_X, b_X) \simeq \mathrm{Hom}_{\mathrm{Ind\,Cat\,hot}}(X, \underline{\underline{K}}).$$

Thus, the statement of the scholie implies two things:

a) The contrafunctor $X \longmapsto \mathrm{Hom}_{\Sigma_X}(a_X, b_X)$ on Cat factors canonically through $\mathrm{Cat\,hot}$, via a canonical functor

$$\mathrm{Cat\,hot}^o \xrightarrow{H = H^{\mathcal{M},\Sigma}_{a,b}} \mathrm{Ens}\ ;$$

b) This latter contrafunctor on $\mathrm{Cat\,hot}$ is *ind-representable* by an ind-object $\underline{\underline{K}}$ of $\mathrm{Cat\,hot}$.

It is this ind-object of $\mathrm{Cat\,hot}$ which is defined up to unique isomorphism by $\mathcal{M}$, Σ, a, b, independently of any computational artifice for making the $\mathrm{Hom}_{\Sigma_X}(a_X, b_X)$ explicit.

Let us also note the following conclusion, by no means obvious a priori: [page 35]

Let us first note (this being implicit in the scholie) that every arrow in Cat

$$X \xrightarrow{f} Y$$

defines

$$\mathcal{M}(Y) \xrightarrow{\mathcal{M}(f)} \mathcal{M}(X) \qquad \text{mapping } \Sigma_Y \text{ into } \Sigma_X,$$

whence

$$\mathcal{M}(Y)\Sigma_Y^{-1} \overset{\text{def}}{=} \mathbf{D}_{\mathcal{M},\Sigma}(Y) \longrightarrow \mathcal{M}(X)\Sigma_X^{-1} = \mathbf{D}_{\mathcal{M},\Sigma}(X),$$

whence, for every $a, b \in \operatorname{Ob}\mathcal{M}$, since $\mathcal{M}(f)$ transforms a_Y, b_Y into a_X, b_X:

$$\boxed{\operatorname{Hom}_{\Sigma_Y}(a_Y, b_Y) \longrightarrow \operatorname{Hom}_{\Sigma_X}(a_X, b_X).}$$

This being so, we have:

Corollary: *If f is a homotopism, then for any $a, b \in \mathcal{M}$, the preceding arrow is an isomorphism.*

The natural question is to know whether the same conclusion is still valid, assuming only that f is a "*quasi-isomorphism*" (or "weak equivalence"), here that f defines an isomorphism in the category of fractions Hot of Cat. (Not to be confused with Cat hot, distinctly finer). If this were so, the contrafunctor

p. 22 [page 36]

$$X \longmapsto \operatorname{Hom}_{\Sigma_X}(a_X, b_X)$$

would factor not only through Cat hot, but even through the coarser category of fractions Hot. And then the question would arise whether this hypothetical contrafunctor

$$\text{Hot}^o \longrightarrow \text{Ens}$$

is still ind-representable; and preferably ([5]) ind-representable by the ind-object image of $\underline{\underline{K}}$ in Ind Hot under

$$\operatorname{Ind}(\text{Cat hot}) \longrightarrow \operatorname{Ind}(\text{Hot}),$$

deduced from the localization functor Cat hot $\longrightarrow$ Hot. (For this ind-object is a ready-made candidate.)

I am going to give a few categorical reminders, generalizing the situation of the canonical localization functor [page 37]

$$\text{Cat hot} \xrightarrow{\varphi} \text{Hot},$$

which we know admits a right adjoint

$$\text{Cat hot} \xleftarrow{\psi} \text{Hot}, \qquad \operatorname{Hom}_{\text{Hot}}(\varphi X, Y) \simeq \operatorname{Hom}_{\text{Cat hot}}(X, \psi Y),$$

where ψ is the "Kan envelope" or "Kan representative" functor. We are therefore going to consider the more general situation

$$C \underset{\varphi}{\overset{\psi}{\leftrightarrows}} C' \qquad \operatorname{Hom}_{C'}(\varphi X, X') \simeq \operatorname{Hom}_{C}(X, \psi X')$$

i.e.

$$\varphi, \psi \text{ pair of adjoint functors.}$$

We assume moreover

$$\begin{cases} \varphi \text{ a localization functor, i.e. } \psi \text{ fully faithful,} \\ \text{i.e. } \varphi\psi \xrightarrow[\sim]{\text{adj}} \text{id}_{C'} \quad . \end{cases}$$

p. 23

Let $W = W_\varphi \subset \mathrm{Fl}\, C$, defined by φ: $W_\varphi = \{u \in \mathrm{Fl}\, C \mid \varphi(u) \text{ isomorphism}\}$. Here is the proposition I had forgotten, and which I had trouble reconstructing, alas!

Proposition 1. *Let $K \in ObC$. The following conditions on K are equivalent:*

(i) *The functor*

$$X \longmapsto K(X) \overset{\text{def}}{=} \mathrm{Hom}_C(X, K) \qquad C^o \longrightarrow \mathrm{Ens}$$

transforms arrows of W ("quasi-isomorphisms") into bijections.

(i′) *Same condition for the adjunction arrows*

$$X \longrightarrow \psi\varphi X \qquad (\forall X \in \mathrm{Ob}\ \mathrm{C})$$

(i″) *Same condition for the adjunction arrow* ([6])

$$K \longrightarrow \psi\varphi K.$$

(ii) *K belongs to the essential image of ψ*

(ii′) *$K \longrightarrow \psi\varphi K$ (adjunction) is an isomorphism.*

(ii″) *The homomorphism functorial in X*

$$\mathrm{Hom}_C(X, K) \longrightarrow \mathrm{Hom}_{C'}(\varphi X, \varphi K)$$

is an isomorphism.

Proof. We first have as quasi-tautologies (ii′)$\Longrightarrow$ (ii)$\Longrightarrow$ (ii″)$\Longrightarrow$ (ii′), i.e. the equivalence of the last three conditions. Indeed, (ii′)$\Longrightarrow$ (ii) tautological, (ii)$\Longrightarrow$ (ii″) since if $K = \psi(K')$ we have by adjunction [page 38]

$$\mathrm{Hom}_C(X, K) = \mathrm{Hom}_C(X, \psi(K')) \overset{\text{adj}}{\simeq} \mathrm{Hom}_{C'}(\varphi X, K'),$$

now $\varphi(K) = \varphi\psi(K') \xrightarrow{\sim} K'$ (adjunction, and ψ fully faithful).

$$\begin{array}{ccc} \mathrm{Hom}_C(X,K) & \xrightarrow{\quad\sim\quad} & \mathrm{Hom}_{C'}(\varphi X, K') \\ & {\scriptstyle \varphi_{X,K}} \searrow \quad \nearrow {\scriptstyle \sim \atop \text{adj}} & \\ & \mathrm{Hom}_{C'}(\varphi X, \underbrace{\varphi K}_{=\varphi\psi(K')}) & \end{array}$$

which is completed to

which proves that $\varphi_{X,K}$ is an isomorphism.

(ii″) $\Longrightarrow$ (ii′). For the arrow

p. 24

$$\begin{array}{ccc} \mathrm{Hom}_C(X,K) & \xrightarrow{\alpha_X} & \mathrm{Hom}_{C'}(\varphi X, \varphi K) \\ & \searrow & \wr \,\big|\, \text{adj} \\ & & \mathrm{Hom}_C(X, \psi\varphi K) \end{array}$$

[5] this will be automatic if we have ind-representability as a functor on Hot, cf. below.

[6] N.B. It even suffices that the element id_K of $F_K(K)$ come from $F_K(\psi\varphi K)$, i.e.
(i‴) There exists a retraction of $\psi\varphi K$ onto K (NB This will even be an isomorphism).

is deduced from the adjunction arrow

$$K \longrightarrow \psi\varphi K$$

by taking $\mathrm{Hom}_C(X, -)$, so α_X is an isomorphism for every X if and only if this adjunction arrow is an isomorphism. So

$$(\mathrm{ii}) \Longleftrightarrow (\mathrm{ii}') \Longleftrightarrow (\mathrm{ii}'').$$

On the other hand,

$$(\mathrm{ii}'') \Longrightarrow (\mathrm{i}) \Longrightarrow (\mathrm{i}') \Longrightarrow (\mathrm{i}'')$$

are tautological, bearing in mind for $(\mathrm{i}) \Longrightarrow (\mathrm{i}')$ that the adjunction arrows

$$X \overset{u}{\to} \psi\varphi X \qquad\qquad \text{in } C$$

are indeed in W, for we have a commutative triangle

$$\begin{array}{ccc} \varphi(X) & \xrightarrow{\varphi(u)} & \varphi(\psi\varphi X) = (\varphi\psi)(\varphi X) \\ & \searrow^{\mathrm{id}_{\varphi(X)}} & \Big\downarrow \wr \text{ adj } \varphi\psi \longrightarrow 1, \text{ which is an isomorphism since } \psi \text{ is fully faithful} \\ & & \varphi(X), \end{array}$$

so $\varphi(u)$ is indeed an isomorphism.

It remains to prove $(\mathrm{i}'') \Longrightarrow (\mathrm{ii}')$, i.e. we suppose
that the functor $X \longrightarrow \mathrm{Hom}(X, K)$ transforms the particular adjunction morphism [page 39]

$$K \overset{i}{\to} \psi\varphi K$$

into

$$\begin{array}{rcl} u & \longrightarrow & u \circ i \\ \mathrm{Hom}(\psi\varphi(K), K) & \longrightarrow & \mathrm{Hom}(K, K), \end{array}$$

which is a bijection. So id_K is in the image, let p be an element of the first member which gives rise to it

p. 25

$$K \underset{p}{\overset{i}{\rightleftarrows}} \psi \underbrace{\varphi(K)}_{K'} \qquad\qquad \text{so } p \circ i = \mathrm{id}_K$$

Let us consider

$$ip = q : \psi(K') \longrightarrow \psi(K') \qquad\qquad (\text{NB } q^2 = q),$$

since ψ is fully faithful, there exists a unique

$$q_0 : K' \longrightarrow K' \qquad\qquad (K' \overset{\text{def}}{=} \varphi(K))$$

such that we have

$$q = \psi(q_0).$$

The relation $pi = \mathrm{id}_K$ gives, on applying φ,

$$\varphi(p)\varphi(i) = \mathrm{id},$$

so since $\varphi(i)$ is an isomorphism (it is

$$\begin{array}{ccc} \varphi(K) & \xrightarrow{\sim} & \varphi(\psi\varphi)(K) = (\varphi\psi)(\varphi K) \\ & \overset{\sim}{\searrow}{}^{\mathrm{id}} & \wr\downarrow{\scriptstyle \mathrm{adj}} \\ & & \varphi(K), \end{array}$$

making the triangle opposite commutative), $\varphi(p)$ is one. So $\varphi(q) = \varphi(i)\varphi(p)$ is one, since $\varphi(i)$ and $\varphi(p)$ are. But

$$\varphi(q) = \varphi\psi(q_0) \quad \underbrace{\overset{\sim}{\longrightarrow}}_{\mathrm{adj}\ \varphi\psi\ \overset{\sim}{\to}\ \mathrm{id}} \quad q_0,$$

so q_0 is an isomorphism, hence also $q = \psi q_0$. Since $q^2 = q$, this implies $q = \mathrm{id}$. So i and p are isomorphisms inverse to one another, which establishes (ii′) and completes the proof.

Corollary: *Let us consider functors* $C^o \xrightarrow{H}$ Ens, *and* $C'^o \xrightarrow{H'}$ Ens, *such that* $H \simeq H' \circ \varphi$. *For* H *to be representable, it is necessary and sufficient that* H' *be so, and in this case, the object* K *of* C
which represents H *satisfies the equivalent conditions of the proposition. If* K, K' *represent them, we have canonical isomorphisms* $K' \simeq \varphi K$, $K \simeq \psi K'$. [page 40]

If H' is representable by $K' \in \mathrm{Ob}C'$, we will therefore have

$$H' \circ \varphi(X) \simeq \mathrm{Hom}_{C'}(\varphi X, K') \overset{\mathrm{adj}}{\simeq} \mathrm{Hom}_C(X, \psi(K'))$$

so H is representable by $K = \psi(K')$, satisfying condition (ii) of the proposition. Conversely, if H is representable by an object K, then K satisfies condition (i), so by (ii″) we have p. 26

$$\mathrm{Hom}_C(X, K) \xrightarrow{\sim} \mathrm{Hom}_{C'}(\varphi X, \varphi K),$$

so the functor $H' : C'^o \longrightarrow$ Ens which factors H is given explicitly as $X' \longmapsto \mathrm{Hom}_{C'}(X', \varphi K)$, hence is representable by φK.

Let us pass to ind-objects. The functors φ, ψ pass to ind-objects, whence functors

$$\mathrm{Ind}\, C \underset{\Phi=\mathrm{Ind}\,\varphi}{\overset{\Psi=\mathrm{Ind}\,\psi}{\rightleftarrows}} \mathrm{Ind}\, C'.$$

We will again have the adjunction formula for $X = (X_i)_{i\in I}$ in $\mathrm{Ind}\, C$, $X' = (X'_{i'})_{i'\in I'}$ in $\mathrm{Ind}\, C'$, for

$$\begin{aligned} \mathrm{Hom}(X, \Psi X') &= \varprojlim_i \varinjlim_{i'} \mathrm{Hom}(X_i, \psi(X'_{i'})) \\ &= \varprojlim_i \varinjlim_{i'} \mathrm{Hom}(\varphi X_i, X'_{i'}) \\ &= \mathrm{Hom}(\Phi X, X'). \end{aligned}$$

We prove in the same way that Ψ is fully faithful, for if $X' = (X'_i)_{i\in I}$, $Y' = (Y'_j)_{j\in J}$ are objects of $\mathrm{Ind}\, C'$, we will have [page 41]

$$\begin{array}{ccccc} \mathrm{Hom}(X', Y') & \simeq \varprojlim_i \varinjlim_j & & \mathrm{Hom}_{C'}(X'_i, Y'_j) \\ \downarrow{\scriptstyle \Psi_{X',Y'}} & & & \downarrow{\scriptstyle \psi_{X'_i,Y'_j}} \\ \mathrm{Hom}(\Psi X', \Psi Y') & \simeq \varprojlim_i \varinjlim_j & & \mathrm{Hom}_C(\psi(X'_i), \psi(Y'_j)), \end{array}$$

so $\Psi_{X',Y'}$ is an isomorphism if the $\psi_{X'_i,Y'_j}$ are.

So proposition 1 can be applied to the situation Φ, Ψ between ind-objects. I am going to spell it out in the most convenient form:

Proposition 2. *Let $K = (K_i)_{i\in I}$ be an ind-object of C, and let H be the functor*

$$\begin{cases} H : X \longmapsto \underbrace{\mathrm{Hom}_C(X,K)}_{K(X)} \overset{\text{def}}{=} \varinjlim_i \mathrm{Hom}_C(X,K_i) \\ C^o \longrightarrow \mathrm{Ens} \end{cases}$$

which it ind-represents. Equivalent conditions

(i) *The functor H factors (up to isomorphism) through C' via φ, i.e. transforms the $u \in W_\varphi$ into isomorphisms.* p. 27

(i′) *Same condition for the adjunction arrows $X \longrightarrow \psi\varphi(X)$ (which are in W).*

(i″) *Same condition for the adjunction arrow $K \longrightarrow \psi\varphi(K)$.*

(ii) *K belongs to the essential image of* $\mathrm{Ind}\, C'$ *under Ψ.*

(ii′) *$K \longrightarrow \Psi\Phi(K)$ is an* isomorphism *in* $\mathrm{Ind}\, C$.

(ii″) *The functorial homomorphism*

$$K(X) = \mathrm{Hom}_{\mathrm{Ind}(C)}(X,K) \longrightarrow \mathrm{Hom}_{\mathrm{Ind}(C')}(\Phi(X),\Phi(K))$$

is an isomorphism.

Proof. The equivalence of (ii), (ii′), (ii″) is proposition 1 applied to Φ and Ψ, except that we must note that (ii″) obviously remains valid if we replace X in it by an arbitrary ind-object $(X_j)_{j\in J}$. We have

[page 42]

$$(\text{ii}'') \Longrightarrow (\text{i}) \Longrightarrow (\text{i}')$$

tautologically, it remains to prove that (i′) implies (i″) and (i″) $\Longrightarrow$ (ii′). We cannot apply proposition 1 directly, for hypothesis (i′) here concerns only the objects of $\mathrm{Ind}\, C$ which come from C, and it is not obvious a priori that it passes to arbitrary objects. But it is in the case of a homomorphism of ind-objects (X_i), (Y_i) on the same index set, given by an ordinary (term-by-term) homomorphism of direct systems on I, $X_i \longrightarrow Y_i$: if the $X_i \longrightarrow Y_i$ are in W, hypothesis (i) implies that $H(Y) \longrightarrow H(X)$ is an isomorphism, and likewise for (i′) for $\underbrace{X}_{=(X_i)_i} \longrightarrow \underbrace{\Psi\Phi(X)}_{=(\psi\varphi(X_i))_i}$, where the adjunction homomorphism is given term by term by the adjunction homomorphisms $X_i \longrightarrow \psi\varphi(X_i)$. So hypothesis (i′) here implies hypothesis (i′) of proposition 1 for Φ and Ψ, hence also (ii′) (by proposition 1), and we win.

Let us return to the situation where $C = \mathrm{Cat}\,\mathrm{hot}$, $C' = \mathrm{Hot}$, φ the functor localization. If $K \in \mathrm{ObCat}\,\mathrm{hot}$, then $\psi\varphi(K)$ is called *the Thomason-Kan envelope* of K - if K is a category, its envelope $\bar{K}$ and the arrow $K \longrightarrow \bar{K}$ are only defined in the category $\mathrm{Cat}\,\mathrm{hot}$, i.e. $\bar{K}$ is only defined "up to homotopism".

But we can always *choose* an object $\bar{K}$ of Cat, and an arrow

[page 43]

$$i : K \longrightarrow \bar{K} \qquad \textit{in } \mathrm{Cat},$$

not only in $\mathrm{Cat}\,\mathrm{hot}$, in such a way moreover that i is a "Thomason *cofibration*" (and in p. 28

particular an open immersion?). It follows that if we have a direct system

$$K = (K_n)_{n\in\mathbf{N}}$$

indexed by $\mathbf{N}$, and if we choose for each n a Thomason-Kan envelope

$$K_n \overset{\alpha_n}{\hookrightarrow} \bar{K}_n$$

in this way, we can find transition functors

$$\bar{K}_n \overset{\bar{i}_n}{\longrightarrow} \bar{K}_{n+1}$$

such that the squares

$$\begin{array}{ccc} K_n & \xhookrightarrow{\alpha_n} & \bar{K}_n \\ {\scriptstyle i_n}\downarrow & & \downarrow{\scriptstyle \bar{i}_n} \\ K_{n+1} & \xhookrightarrow{\alpha_{n+1}} & \bar{K}_{n+1} \end{array}$$

are commutative, so that we thus have a homomorphism of *direct systems of* Cat

$$K \longrightarrow \bar{K}$$

and a fortiori a homomorphism of ind-objects of Cat. This being so, conditions (i), (ii′) of proposition 2 show that the functor ind-represented by K *in* $C = \mathrm{Cat\,hot}$ transforms ordinary
quasi-isomorphisms (weak equivalences) into bijections, *if and only if* the preceding ho- [page 44]
momorphism is an isomorphism of ind-objects of $\mathrm{Ind\,Cat\,hot}$.

So that will be the natural condition to impose on the ind-categories of paths

$$\underline{\underline{\mathrm{Ch}}}_{\Sigma}(\mathcal{M}; a, b) \qquad\qquad (a, b \in \mathrm{ob}\,\mathcal{M})$$

coming from a "category of models" $\mathcal{M}$, Σ.

Of course, having at our disposal an explicit direct system of categories (not only of objects of Cat hot)

$$K_n = \underline{\underline{\mathrm{Ch}}}_{\Sigma,n}(\mathcal{M}; a, b),$$

it is tempting to pass to the direct limit

$$K_\infty = \varinjlim_n \underline{\underline{\mathrm{Ch}}}_{\Sigma,n}(\mathcal{M}; a, b) \overset{\mathrm{def}}{=} \underline{\underline{\mathrm{Ch}}}_{\Sigma}(\mathcal{M}; a, b),$$

and this all the more so since the transition functors are monomorphisms and fully faithful, so that the K_n are identified with a sequence of full subcategories of
K_∞, whose union is K_∞. But on the other hand I do not know how to give any p. 29
convincing interpretation of K_∞, like that of the ind-object of Cat hot defined by $\underline{\underline{K}} = (K_n)_{n\in\mathbf{N}}$, which ind-represents the remarkable functor

$$H = H_{\mathcal{M};x,y} : \mathrm{Cat\,hot}^o \longrightarrow \mathrm{Ens}$$

factoring the functor [page 45]

$$X \longmapsto H_{\mathcal{M};x,y}(X) \overset{\mathrm{def}}{=} \mathrm{Hom}_{\mathcal{M}(X)\Sigma_X^{-1}}(x_X, y_X),$$

by the formula, then

$$H(X) = \varinjlim_n \underbrace{\pi_0 \underline{\mathrm{Hom}}(X, K_n)}_{\mathrm{Hom}_{\mathrm{Cat\,hot}}(X,K_n)}.$$

I do not know how to give an interpretation of this type either of the category K_∞ itself, or of its image in Cat hot, or of its image in Hot. It is true that if I restrict the functor H to the full subcategory of (Cat) consisting of *finite* categories (Ob X and Fl X finite), then we will have

$$\underline{\mathrm{Hom}}^!(X, K_\infty) = \underline{\mathrm{Hom}}(X, K_\infty),$$

and the preceding isomorphism is interpreted as

$$\mathrm{Hom}_{\mathcal{M}(X)\Sigma_X^{-1}}(a_X, b_X) \overset{\mathrm{def}}{=} H(X) \simeq \pi_0 \underline{\mathrm{Hom}}(X, K_\infty) \overset{\mathrm{def}}{=} \mathrm{Hom}_{\mathrm{Cat\,hot}}(X, K_\infty).$$

This proves at least that knowledge of K_∞ is useful. But this formula, with X restricted to categories that are *finite,* will not define K_∞ as an object of Cat hot, up to unique isomorphism in this category - *except* in the case where K_∞ is itself homotopic to a finite category. (I do not quite see whether this happens very often … But one cannot found a general theory of derivators and "derivable" categories of models on an assumption of this sort.) [page 46]

But suppose that the functor H under consideration even factors through Hot, i.e. that it transforms quasi-isomorphisms into bijections. We have seen that this is equivalent to the condition that it can be represented by a direct system over **N** made up of Thomason-Kan categories. It must moreover be easy, if one wants to, to arrange for the transition functors to be monomorphisms and fully faithful, so as to have a nice $\varinjlim$. It will of course be a Thomason-Kan category. Is this category determined by the functor $H : \mathrm{Cat\,hot}^o \longrightarrow$ Ens up to unique isomorphism in Cat hot? Of course, there is no question of its representing the functor H, except the latter is really and truly representable, i.e. if the ind-object $\underline{\underline{K}}$ of Cat hot is essentially constant (which must practically never be the case). Let us already note that if u is a homomorphism of ind-objects *in* Cat, or, what now amounts to the same thing, a homomorphism in Cat p. 30 [page 47]

$$K_\infty \xrightarrow{u_\infty} K'_\infty$$

such that $\forall n$, $u_\infty(K_n) \subseteq K'_{n'}$ for a suitable $n' = \tau(n)$, and if we assume that the homomorphism of ind-objects induces an isomorphism for the ind-representable functors *on* Cat hot (for example, if $\underline{\underline{K}}$ and $\underline{\underline{K}}'$ ind-represent the same functor H, and the homomorphism of ind-objects is the one corresponding to id_H), then u_∞ is necessarily an isomorphism *in* Hot, hence also (unless I am mistaken) in Cat hot since K_∞, K'_∞ are Thomason-Kan categories. But what is not evident is the existence of a homomorphism u_∞ as above, and if it exists, that it is unique (or at least canonical?) as an arrow of Cat hot.

For the existence of a u_∞, we are reduced to proving that if we have a diagram with solid arrows

$$\begin{array}{ccc} K_n & \xrightarrow[\text{cofib.}]{i_n} & K_{n+1} \\ \downarrow{\scriptstyle u_n} & & \searrow \\ K'_{\tau(n)} & \xrightarrow[\text{transition}]{} & K'_N, \end{array}$$

and if an arrow u_{n+1} is given "in dotted lines" *in* Cat hot, so as to make the square commutative, then we can realize it by an arrow *in* Cat that makes the diagram commute for real (in Cat). We are reduced to the same problem in a situation [page 48]

$$X \underset{i}{\overset{p}{\leftrightarrows}} Y,$$

where X is a Thomason category, i a Thomason cofibration, p a retraction from Y to X *in* Cat hot, we want to realize it by a genuine retraction .

Admitting that this is the case, hence that every homomorphism of the ind-objects $\underline{\underline{K}}$, $\underline{\underline{K}}'$ *in* Cat hot can be represented by a homomorphism between the ind-objects *in* Cat, hence also by a functor $K_\infty \longrightarrow K'_\infty$, there remains the question whether the corresponding homomorphism in Cat hot is unique. So if we have two functors f, g which induce the same homomorphism on the ind-represented functors on Cat hot, whether f and g are homotopic. So

Question. Two functors $f, g : K_\infty \longrightarrow K'_\infty$ such that for every $n \in \mathbf{N}$, f_n, g_n obtained from them by restriction to K_n are homotopic (and even homotopic as functors from K_n to a suitable K'_N), are they homotopic? ([8])

I presume not - that it can happen that the minimum required length $l(n)$ for a path in $\underline{\mathrm{Hom}}(K_n, K'_{\tau(n)})$ joining f_n and g_n, tends to infinity if $n \longrightarrow \infty$. Does the hypothesis that the K'_n are Thomason categories change anything here (are two homotopic objects in $\underline{\mathrm{Hom}}(X, K')$, if K' is Thomason, joined by an arrow?) p. 31 [page 49]

Of course, one can claim that the categories $\underline{\mathrm{Ch}}_{\Sigma,n}(\mathcal{M}; a, b)$ have been defined "canonically", and their direct limit $\underline{\mathrm{Ch}}_{\Sigma}(\mathcal{M}; a, b)$ too. But the definition was rather arbitrary and outlandish - I am convinced that ten mathematicians, proposing to develop this same idea of categories of paths, with the same aim as me, would arrive at ten different definitions! Yet it is not ruled out that we will end up finding a transitive system of isomorphisms in Cat hot, between the categories of "infinite" paths (i.e. of unspecified length) $K_\infty = K_\infty(\mathcal{M}; a, b)$ that we will have constructed, with the main aim (let us say) of representing the functor *H on finite categories.* And that the isomorphisms obtained will, of course, be homomorphisms in Ind(Cat hot). I have the impression that I have not yet thoroughly axiomatized the very notion of paths, which after all has a fairly obvious and important geometric meaning in its own right. So by what is to be understood by "*a* theory of paths in a category". Such a theory $\mathcal{T}$ would thus associate to $\mathcal{M}$, Σ, a, b a direct system of categories $\mathrm{Ch}^{\mathcal{T}}_{n,\Sigma}(\mathcal{M}; a, b)$. And the natural geometric constraints to be imposed on such theories, also with a view to their use in making explicit [page 50]

$$\mathrm{Hom}_{\mathcal{M}(X)\Sigma_X^{-1}}(a_X, b_X) \xrightarrow{\sim} \pi_0 \underline{\mathrm{Hom}}(X, \underline{\underline{\mathrm{Ch}}}^{\mathcal{T}}(\mathcal{M}; a, b))$$

would give a transitive system of isomorphisms not only between the ind-objects of Cat hot, but perhaps also between the $\mathrm{Ch}^{\mathcal{T}}_{\infty,\Sigma}(\mathcal{M}; a, b)$, for different theories. If we have such a statement at our disposal, the category $\mathrm{Ch}_{\infty,\Sigma}(\mathcal{M}; a, b)$, or more exactly its images in Cat hot and Hot, would take on a convincing meaning. [page 51]

On the other hand there is another point of view, which perhaps could "canonify" the categories $\underline{\mathrm{Ch}}_\infty$ as an object of Cat hot and especially of Hot (one would pass to Cat hot by applying the functor ψ, Thomason-Kan envelope …). Indeed, we are interested in constructing, in terms of $\mathcal{M}$, Σ, a "derivator" (or a prederivator) $\mathbf{D}_{\mathcal{M},\Sigma}$ especially on the "domain" of finite categories. Extending it to a larger domain is perhaps important in a "second wind", but can be considered a luxury in the first movement of founding a theory of derivators. This said, we can ask ourselves questions of the following type.

Let

$$\mathrm{Hot}_0 \subset \mathrm{Hot}$$

be the full subcategory of Hot, defined by finite categories ([9]) - let us call the homotopy types "of finite type" (or of finite *presentation*?). We have an obvious functor p. 32 [page 52]

$$\begin{aligned} \mathrm{Hot} &\longrightarrow \mathrm{Hot}_0^\wedge = \underline{\mathrm{Hom}}(\mathrm{Hot}_0^o, (\mathrm{Ens}))\,, \\ X &\longmapsto \tilde{X} = (\xi \longmapsto \mathrm{Hom}_{\mathrm{Hot}}(\xi, X)). \end{aligned}$$

[8] K, K' Thomason categories, the K_n, K'_n too.

[9] *NB* Hot_0 is equivalent to a small category.

This functor is *conservative* , since a homomorphism $X \longrightarrow X'$ in Hot is an isomorphism if it induces isomorphisms for the functors $\tilde{X} \longrightarrow \tilde{X}'$, and even just for their values on the spheres S^n $(n \geq 0)$. (Unless I am mistaken. Are there any blunders, because the criterion for weak equivalence in terms of the π_i does not apply directly, in the absence of base points. That should not be very serious …). But is it perhaps even *fully faithful* ? So that Hot would be realized as a full subcategory of $\mathrm{Hot}_0^{\wedge}$, containing the representable functors?

p. [no number]

THE DERIVATORS

ALEXANDRE GROTHENDIECK

Chapter VI
HOT

p. [no number]
p. 1
[page 1]

The derivator HOT

14.11.90

(The first sheets, from two weeks ago, have been ditched, overtaken by VII)

I define Hot starting from the category of models

$$\mathcal{M} = \mathrm{Cat},\ W \subseteq \mathrm{Fl}(\mathrm{Cat}),$$

W being in principle the Hot-equivalences. But as far as possible I am going to axiomatize more, assuming given a more general set W, called "basic localizer" (basic localizer), e.g. $W = W_{\mathbf{D}}$, where $\mathbf{D}$ is an already given derivator (but assuming that its domain is all of (Cat)). We are going to note as we go along which properties of W we will need. For purposes of later reference, I am going to recall here those I isolated in Pursuing Stacks.
L1 Saturation of W

a) *The identities in* W.

b) *If two among* gf, f, g *in* W, *the third also,* $X \xrightarrow{f} Y \xrightarrow{g} Z$.

c) *If* gf *and* fg *are in* W, *then* f *and* g *too,* $X \underset{g}{\overset{f}{\rightleftarrows}} Y$.

L2 $\mathbf{\Delta}^1$ *is universally* W*-aspheric, i.e.* $\mathbf{\Delta}^1 \times X \longrightarrow X$ *is in* W *for every* X *in* Cat. This implies

L2 bis $\mathbf{\Delta}^1$ *is W-aspheric.* (This was L3′ in Pursuing Stacks.)
L3 *If X has a final object, X is W-aspheric.* ([1])
I note here strengthened variants.
L3 bis *If X is* Hot-*aspheric, it is W-aspheric.*
L3 ter *If X is* Hot-*aspheric, it is universally W-aspheric, i.e. $X \times Y \xrightarrow{\mathrm{pr}_2} Y$ is in W for every Y in* Cat. ([2])
L3 quater *Every* Hot-*equivalence is in W.* p. 2
L3 quater $\Longrightarrow$ L3 ter $\Longrightarrow$ L3 bis $\Longrightarrow$ L3.
L3 ter $\Longrightarrow$ L2 $\Longrightarrow$ L2 bis.

[page 2]

L4 *If $f : X \longrightarrow Y$ is W-aspheric (i.e. the $X/y \longrightarrow Y/y$ in W, i.e. (if we have L3) if the X/y are W-aspheric), then f too.* (Localization axiom.) ([3])

L5 (Fibration axiom). *If*

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

with X and Y fibered over S and f cartesian, if for every $s \in S$ $f_s : X_s \longrightarrow Y_s$ is in W, f too.

In Pursuing Stacks I always assumed $\boxed{\text{L1, L3, L4}}$ (implying L2) - it was these conditions that defined the notion of "localizer" (basic) in Cat. ([4]) On the other hand, $\boxed{\text{L1, L2 bis, L5}}$ imply L1,2,3,4,5.
We are going to assume here too that we have at least L1, L3, L4 (i.e. W a localizer). Applying the construction of I, p. 72-**75,** we see that $f : X \longrightarrow Y$ is in W if and only if $f^o : X^o \longrightarrow Y^o$ is. So L1, L3, L4 imply L3′, L4′ (axioms dual to L3, L4).
There were supplementary axioms

La $f \in W \Longrightarrow \pi_0(f)$ *bijective.*

Lb *The canonical localization functor*

$$\mathrm{Cat} \longrightarrow (\mathrm{Cat})W^{-1} \overset{\mathrm{def}}{=} \mathrm{Hot}_W$$

commutes with finite sums.

Lc *Likewise for finite products.*

One should all the same isolate concrete conditions on W for the validity of Lb, Lc. The p. 3
conditions L3 quater and La are written respectively

$$W_\infty \subseteq W, \quad W \subseteq W_0.$$

[page 3]

[1] There is an L3′ dual to L3.
[2] Suffices to imply L2, L3.
[3] there is an L4′ dual to L4.
[4] If W a localizer, then contractible categories are W-aspheric, homotopisms are in W, homotopic maps are equal in $(\mathrm{Cat})W^{-1} \overset{\mathrm{def}}{=} \mathrm{Hot}_W$.

Their conjunction is the very reasonable axiom

$$W_\infty \subseteq W \subseteq W_0.$$

Of course, axiom L5 now looks a bit cramped, i.e. feeble. We are going to see very soon what we will need!
We thus want to define a derivator

$$\mathrm{HOT}_W$$

starting from the category of models

$$(\mathrm{Cat}, W \subseteq \mathrm{Fl}(\mathrm{Cat})).$$

So to define

$$\mathrm{HOT}_W(S) \qquad\qquad (S \in \mathrm{Ob}\,\mathrm{Cat}),$$

the general recipe is to take the category

$$\mathcal{M}(S) = \mathrm{Cat}(S) = \underline{\mathrm{Hom}}(S^o, (\mathrm{Cat})),$$

to introduce in it the set of arrows

$$\begin{cases} W_S \subseteq \mathrm{Fl}(\mathcal{M}(S)) \\ u \in W_S \iff u_s \in W \quad \forall\, s \in \mathrm{Ob}\, S. \end{cases}$$

We must set

$$\mathrm{HOT}_W(S) \overset{\mathrm{def}}{=} \underline{\mathrm{Hom}}(S^o, \mathrm{Cat}) W_S^{-1}.$$

But the category $\underline{\mathrm{Hom}}(S^o, \mathrm{Cat})$ is interpreted, up to canonical isomorphism, as that

$$\underline{\mathrm{Fibsc}}\, S$$

of categories X Cat-fibered over S, equipped with a "splitting", i.e. with a choice of base change functors $u^* : X_t \longrightarrow X_s$ (for $t \overset{u}{\to} s$ in S) , with moreover *strict transitivity*

$$(vu)^* = u^* v^* \qquad\qquad \text{if } \; t \overset{u}{\to} s \overset{v}{\to} r \;\text{ in } S$$

and not only transitivity up to canonical isomorphism. Thus one can also write [page 4]

$$\mathrm{HOT}_W(S) \overset{\sim}{\longrightarrow} (\underline{\mathrm{Fibsc}}\, S) W_S^{-1}, \qquad (*)$$

p. 4

where this time W_S is interpreted as

$$W_S \subseteq \mathrm{Fl}(\underline{\mathrm{Fibsc}}\, S),$$

namely

$$W_S = \{u \in \mathrm{Fl}(\underline{\mathrm{Fibsc}}\, S) \mid u_s \in W \text{ for all } s \in \mathrm{Ob}\, S\}$$

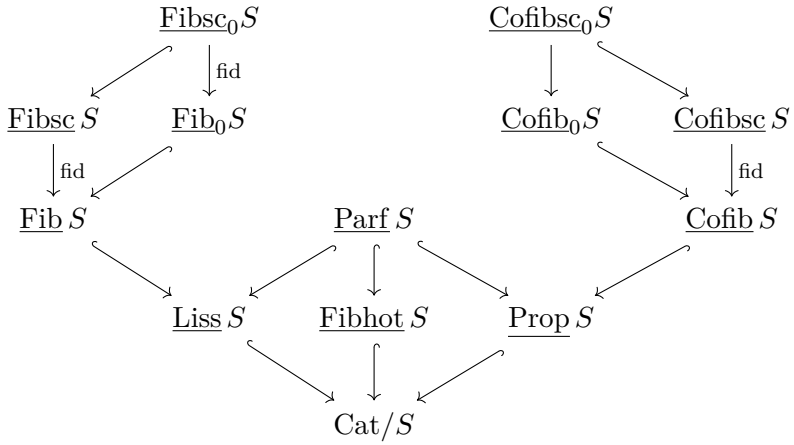
(strictly cartesian functors, i.e. commuting with the splittings, which induce W-equivalences between the fiber categories).

The arrow in $(*)$ is indeed an isomorphism of categories, not just an equivalence. But we are going to define other categories, more convenient in certain respects, which will be *equivalent* (we hope), but certainly not isomorphic, to $\mathrm{HOT}_W(S)$. (We shall deal later with the variance of $\mathrm{HOT}_W(S)$ in S.)

The categories $\underline{\mathrm{Hom}}(S^o, \mathrm{Cat})$, $\underline{\mathrm{Fibsc}}\, S$, as well as the variants we are now going to introduce, are in fact 2-categories. But to describe $\mathrm{HOT}_W(S)$, only the 1-category structures are used. One has the impression that this is quite crude, that one loses something [page 5]

in doing so. But what is lost will be recovered, and superabundantly, in the structure of the internal $\underline{\mathrm{Hom}}$ in the $\mathrm{HOT}(S)$, if one "remembers" that these internal $\underline{\mathrm{Hom}}$, giving rise to "relative homotopy types" $\underline{\mathrm{Hom}}_{\mathrm{HOT}_W}(S;\xi,\eta)$ (if ξ, η belong to them), can be interpreted in terms of ∞-stacks in groupoids over S, with all the richness of structure that belongs to them. But this is another story…

In fact, I see a diagram of 13 categories of "categories over S", to be examined now



where $\underline{\mathrm{Fib}}$, $\underline{\mathrm{Cofib}}$ denote the Cat-fibered, Cat-cofibered categories over S, $\underline{\mathrm{Liss}}$ and $\underline{\mathrm{Prop}}$ the categories smooth resp. proper over S, $\underline{\mathrm{Parf}}$ those that are both smooth and proper (perfect fibrations), finally $\underline{\mathrm{Fibhot}}$ the Hot-fibered categories over S. All the arrows are strictly fully faithful functors, except for the two (rather four) upper vertical arrows, which are faithful and no more. I have just added another 4 of them, categories: $\underline{\mathrm{Fibsc}}_0$, $\underline{\mathrm{Fib}}_0$, $\underline{\mathrm{Cofibsc}}_0$, $\underline{\mathrm{Cofib}}_0$ are the full subcategories of the corresponding categories without subscript, consisting of the fibered resp. cofibered categories X over S, for which the base change resp. cobase change functors

p. 5
[page 6]

$$u^* : X_t \longrightarrow X_s, \qquad \text{resp.} \qquad u_* : X_s \longrightarrow X_t,$$

for $s \xrightarrow{u} t$ in S, are Hot-equivalences. In terms of functors

$$S^o \xrightarrow{F} \mathrm{Cat} \qquad \text{resp.} \qquad S \xrightarrow{F'} \mathrm{Cat},$$

this means that this functor transforms arbitrary arrows in S into elements of W in Cat, i.e. that we have a commutative diagram

$$\begin{array}{ccc} S^o & \xrightarrow{\qquad F \qquad} & \mathrm{Cat} \\ {\scriptstyle \mathrm{can}}\downarrow & & \downarrow{\scriptstyle \mathrm{can}} \\ \Pi(S^o) \simeq (\Pi S)^o = S^o(\mathrm{Fl}\, S^o)^{-1} & \xrightarrow[\bar{F}]{} & \mathrm{Hot}_W = (\mathrm{Cat})W^{-1} \end{array}$$

(and likewise for F'), i.e. that we have a "local system" $\bar{F}$ on S, with values in Hot_W, defined using F. I recall that one also finds such a local system in the case of a Hot-fibered category over S (and even a local system with values in Hot itself, instead of Hot_W - we tacitly use here that $W_\infty \subseteq W$, i.e. L3 quater ([5])). And the same holds a fortiori for the objects of $\underline{\mathrm{Parf}}\, S$. For $\underline{\mathrm{Liss}}\, S$ resp. $\underline{\mathrm{Prop}}\, S$, one simply finds a functor

[page 7]

$$S^o \xrightarrow{\bar{F}} \mathrm{Hot} \qquad \text{resp.} \qquad S \xrightarrow{\bar{F}'} \mathrm{Hot},$$

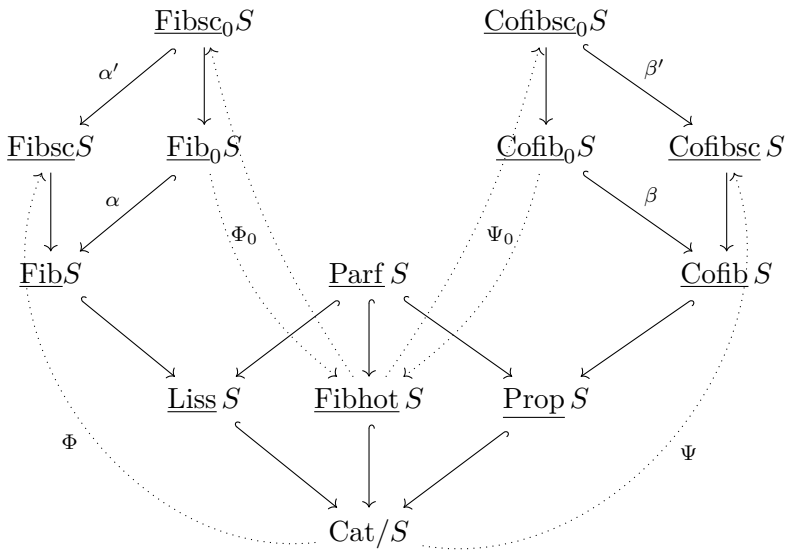
[5] We shall assume it true throughout what follows.

whence, by composing with Hot $\longrightarrow \mathrm{Hot}_W$, functors with values in Hot_W.

In each of these 14 categories, we introduce a set of arrows, which will serve to localize, and we shall always denote it by the same notation W_S. In all these categories except only Cat/S, W_S denotes W-equivalence "fiber by fiber". In Cat/S on the other hand, it will be just W-equivalence. All the arrows of the diagram are compatible with the localizing sets of arrows.

p. 6

[page 8]



I am almost sure of the

Theorem. ([6]) *All the preceding functors induce equivalences between the localized categories.*

As there are 16 functors, it looks like an exercise in patience. The general principle is that one manages to define functors in the opposite direction, compatible with the W_S, in such a way that the composites are linked to the identity functors by arrows which are in W_S, so that by passage to the localizations these functors induce (equivalences) quasi-inverses of the functors deduced from those of the diagram. All these functors in the opposite direction arise essentially by composition, from three very familiar functors. Two of these are indicated by dotted lines on the diagram, as Φ, Ψ.

The third is a canonical functor

[page 9]

$$C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Parf}}\, S,$$

which we are also going to make explicit.

p. 7

The functor Φ (resp. Ψ) applied to $\underline{\mathrm{Fib}}_0 S$ (resp. $\underline{\mathrm{Cofib}}_0 S$), induces a functor

$$\underline{\mathrm{Fib}}_0 S \longrightarrow \underline{\mathrm{Fibsc}}_0 S \qquad \text{resp.} \qquad \underline{\mathrm{Cofib}}_0 S \longrightarrow \underline{\mathrm{Cofibsc}}_0 S \qquad (*)$$

(which will have to be checked). This provides all the "inverse" arrows we need, with the sole exception of α, β, α', β'. But if we have proved, by the indicated method, that all the arrows of the diagram induce equivalences on the localized categories, to prove that the same is still true for α, β, α', β', it suffices to prove it for the two functors

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fib}}\, S, \qquad \underline{\mathrm{Cofibsc}}_0 S \longrightarrow \underline{\mathrm{Cofib}}\, S,$$

[6] This is totally botched, cf. comments page 32 ff., solution of the perplexities pages 41 ff. See especially the recapitulatory Theorem-Scholie p. 70-81.

so it remains to define the functors in the opposite direction

$$\underline{\mathrm{Fib}}\, S \longrightarrow \underline{\mathrm{Fibsc}_0} S, \qquad \underline{\mathrm{Cofib}}\, S \longrightarrow \underline{\mathrm{Cofibsc}_0} S,$$

and as the sources are full subcategories of Cat/S, it will suffice to define

$$\Phi_0 : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fibsc}_0} S, \qquad \Psi_0 : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Cofibsc}_0} S.$$

For this, we shall set

$$\Phi_0 = \Phi r C, \qquad \Psi_0 = \Psi r C, \qquad (**)$$

where $C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Parf}}\, S$, Φ and Ψ are the 3 "retrograde" functors inverted above, and $r : \underline{\mathrm{Parf}}(S) \hookrightarrow \mathrm{Cat}/S$ is the inclusion. We must therefore check that Φ_0, Ψ_0 defined by $(**)$, which a priori have as target categories $\underline{\mathrm{Fibsc}}\, S$ and $\underline{\mathrm{Cofibsc}}\, S$, take their value in the subcategories with subscript 0. [page 10]

I am going to give a single equivalence statement, which will cover (almost) all 16 cases at once:

Corollary. *Let us consider the diagram*

$$\begin{array}{ccccc} & & \underline{\mathrm{Parf}}\, S & & \\ & & \downarrow{\scriptstyle r} & & \\ \underline{\mathrm{Fibsc}_0} S & \overset{p}{\searrow} & & \overset{q}{\swarrow} & \underline{\mathrm{Cofibsc}_0} S \\ & & \mathrm{Cat}/S & & \end{array}$$

and let $\mathcal{F}$ be a full subcategory of Cat/S, such that one of the functors p, q, r factors through $\mathcal{F}$. Let $W_{\mathcal{F}}$ be induced on $\mathcal{F}$ by W_S in Cat/S. Then $\mathcal{F} \hookrightarrow \mathrm{Cat}/S$ induces an equivalence on the localized categories.

Proof. I say that it suffices to prove, for the three functors p, q, r, and for the functors in the opposite direction [page 11]

p. 8

$$\begin{array}{ccccc} & & \underline{\mathrm{Parf}}\, S & & \\ & & \uparrow{\scriptstyle C} & & \\ \underline{\mathrm{Fibsc}_0} S & \overset{\Phi_0}{\nwarrow} & & \overset{\Psi_0}{\nearrow} & \underline{\mathrm{Cofibsc}_0} S \\ & & \mathrm{Cat}/S & & \end{array}$$

that the six composites

$$p\Phi_0, \Phi_0 p, q\Psi_0, \Psi_0 q, rC, Cr$$

are linked to the identity functors (in the relevant category) by an arrow which is in the corresponding localizer. For each of these three pairs of functors (p, Φ_0), (q, Ψ_0), (r, C), this will already prove that on passing to the localizations, they induce equivalences quasi-inverse to one another. But we can do better with the help of the following formal lemma.

Lemma. *Let (C, W), (C', W') be two categories with localizers $W \subseteq \mathrm{Fl}\, C$, $W' \subseteq \mathrm{Fl}\, C'$. Let*

$$p : C \longrightarrow C'$$

be a functor compatible with the localizers, and suppose there exists a functor $\varphi : C' \longrightarrow C$ likewise compatible, and such that we have arrows (of unspecified direction)

$$\varphi p \xrightarrow{u} id_C, \qquad p\varphi \xrightarrow{v} id_{C'}, \qquad$$ [page 12]

which are in W resp. W' "argument by argument".

Then p and φ induce equivalences quasi-inverse to one another on passing to the localized categories (they have no merit). Moreover, for any full subcategory C'_0 of C', such that p factors (as $p_0 : C \longrightarrow C'_0$) through C'_0, if we denote by W'_0 the localizer induced in C'_0 by C', then $p_0 : C \longrightarrow C'_0$ and $\varphi_0 = \varphi|C'_0 : C' \longrightarrow C$ also induce equivalences quasi-inverse to one another on the localizations.

Indeed, $\varphi_0 p_0 = (\varphi i)p_0 = \varphi(ip_0) = \varphi p \xrightarrow{u} \mathrm{id}_C$ (here $i : C'_0 \hookrightarrow C'$ denotes the inclusion), and $F \overset{\text{def}}{=} p_0\varphi_0 = p_0\varphi i$ satisfies

$$iF = \underbrace{ip_0}_{p}\,\varphi i = (p\varphi)i.$$

To define $F \xrightarrow{v_0} \mathrm{id}_{C'_0}$, which is to be in W'_0 argument by argument, it suffices to define [page 13]

$$\underbrace{iF}_{=(p\varphi)i} \xrightarrow{\bar{v}_0} i,$$

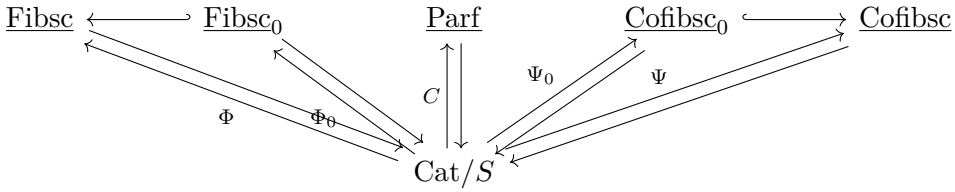
which is to be in W' argument by argument, since W'_0 induced by W' and since $F \longmapsto iF$

$$\underline{\mathrm{Hom}}(C'_0, C'_0) \longrightarrow \underline{\mathrm{Hom}}(C'_0, C')$$

is fully faithful (i being so). Now it suffices to take p. 9

$$\bar{v}_0 = v * i : (p\varphi) \circ i \longrightarrow \mathrm{id}_{C'} \circ i.$$

To sum up, we have to show that the composite of the pairs of quasi-inverse functors works in the following five cases



Besides, by duality, we are reduced to the three cases of $\underline{\text{Fibsc}}$, $\underline{\text{Fibsc}}_0$, $\underline{\text{Parf}}$. (Perhaps the case $\underline{\text{Fibsc}}_0$, with Φ_0 defined in terms of Φ and C, will reduce to the two cases surrounding it, of $\underline{\text{Fibsc}}$ and $\underline{\text{Parf}}$.)

I see that I have to adjust my aim, first of all because I forgot that the functors Φ, Ψ [page 14] I had in mind are *not* compatible with the localizers - an arrow $f : X \longrightarrow Y$ of Cat/S such that the corresponding arrow in Cat is in W, is not going to induce an arrow in $\underline{\text{Fibsc}}\,S$ or $\underline{\text{Cofibsc}}\,S$ which is in W on each fiber. This is not serious, for we can replace the functors by $\Phi' = \alpha'\Phi_0$, $\Psi' = \beta'\Psi_0$, which, unless I am mistaken, should work. But in view of the preceding Lemma, there is a perplexity, for I seem to have proved that if $\mathcal{F}$ is a full subcategory of Cat/S of which I assume only that it contains the essential image of one of the three categories $\underline{\text{Fib}}_0 S$, $\underline{\text{Cofib}}_0 S$, or $\underline{\text{Parf}}\,S$, then the inclusion into Cat/S induces an equivalence of the localized categories, taking in $\mathcal{F}$ as localizer the one

$$\begin{array}{ccc} f : X & \longrightarrow & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

induced by Cat/S, i.e. is in $W_{\mathcal{F}}$ if and only if as an arrow of Cat, f is in W. Now this does indeed coincide (via a nontrivial theorem to which I shall have to return in the setting of a general basic localizer W) with the set of localization arrows "fiber by fiber", in the case of $\mathcal{F} = \underline{\text{Parf}}\,S$, $\underline{\text{Fibhot}}\,S$, $\underline{\text{Fib}}_0 S$, $\underline{\text{Cofib}}_0 S$. [page 15]

On the other hand, this is not at all the case on taking $\mathcal{F} = \underline{\mathrm{Fib}}\, S, \underline{\mathrm{Liss}}\, S$ or $\underline{\mathrm{Cofib}}\, S$. Having chosen in these categories as localizers the notion of W-equivalences "fiber by fiber", I thus seem to be saying that I find for these four categories the *same* localized ones (up to equivalence) as if I took the much coarser localizer inherited by Cat/S. This seems hard to believe, and I shall have to be very careful in proving the key statements.

To see that the solid-line arrows of the big diagram p. 6 are all compatible with the localiz(at)ers , hence also the arrows composed by means of these, I already have to use this (for the case of the arrows $\underline{\mathrm{Liss}} \longrightarrow \mathrm{Cat}/S$, $\underline{\mathrm{Prop}} \longrightarrow \mathrm{Cat}/S$):

① If X, Y smooth or proper over S, and $f : X \longrightarrow Y$ an S-functor, if $f_s : X_s \longrightarrow Y_s$ is in W for every $s \in S$, then $f \in W$. ([7]). p. 10

This considerably strengthens condition L5 on W, where we assumed that f is a *cartesian* functor between categories *fibered* over S. (To sort out, for the various properties we need for W, to what standard properties we can reduce them. Everything will be OK if $W = W_\infty$.)

I) **Definition of Φ, Ψ.** [page 16]

We are going to copy the canonical construction of $\underline{\mathrm{Ch}}(X) \longrightarrow X \times X$ (or $\underline{\mathrm{Ch}}_\infty(X) \longrightarrow X \times X$) factoring $X \xrightarrow{\mathrm{diag}} X \times X$, and the corresponding canonical factorization of an $f : X' \longrightarrow X$ as $X' \xrightarrow{i} \bar{X}' \xrightarrow{\bar{f}} Y$. We shall simply replace $\underline{\mathrm{Ch}}(X)$ by the subcategory $\underline{\mathrm{Ch}}_+(X)$, which is none other than the category of arrows of X

$$\underline{\mathrm{Fl}}(X) = \underline{\mathrm{Hom}}(\mathbf{\Delta}_1, X),$$

equipped with the functors

$$X \xrightarrow{i_X} \underline{\mathrm{Fl}}(X) \xrightarrow{p_X = (\sigma_X, \beta_X)} X \times X,$$

where i_X is the arrow $x \longmapsto \mathrm{id}_x$, and σ_X, β_X are the source and target functors. The fiber of p_X at $(a, b) \in \mathrm{Ob}(X \times X)$ is the discrete category defined by the set $\mathrm{Hom}(a, b)$, that depends a great deal on (a, b) and not only on its connected component in $X \times X$ - so p_X is nothing like a Hot-fibration. It is in general neither proper nor smooth nor anything, on the other hand the two arrows composed with pr_1 and pr_2

$$\begin{array}{ccc} & \underline{\mathrm{Fl}}(X) & \\ {\scriptstyle \sigma_X}\swarrow & & \searrow{\scriptstyle \beta_X} \\ X & & X \end{array}$$

are nice, we have in fact

Lemma. *We have* [page 17]

$$\sigma_X^{-1}(\{x\}) \simeq x\backslash X, \qquad \beta_X^{-1}(\{x\}) \simeq X/x.$$

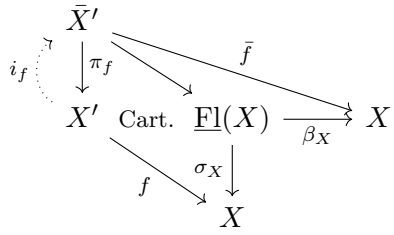
The functor σ_X is Cat*-fibrant with contractible fibers (hence aspheric, hence W-aspheric), the functor β_X cofibrant with contractible fibers (hence aspheric, hence W-aspheric). They are thus also* Hot*-equivalences, a fortiori W-equivalences* ([8]).

Corollary 1. *The functor $i_X : X \longrightarrow \underline{\mathrm{Fl}}(X)$ is a* Hot*-equivalence* (since its composite with the Hot-equivalence σ_X is the identity). p. 11

[7] OK if we assume L3 quater and L4.

[8] In fact, they are even homotopisms, cf. marginal remark below (more general - we recover the case of σ_X by taking $X' = X$.)

Here is the general construction of the factorization of $f : X' \longrightarrow X$, corresponding to the preceding choice of a "category of paths" $\mathcal{C}(X) = \underline{\mathrm{Fl}}(X)$



where i_f is deduced from the section i of $\underline{\mathrm{Fl}}(X)$ over X ([9]). Thus, $\bar{X}'$ is the category of pairs

$$\{(x', u) \mid x' \in \mathrm{Ob}\, X', \quad \underbrace{u}_{\in \mathrm{Fl}\,(X)} : f(x') \longrightarrow y\},$$

the functor $\bar{f}$ is defined by

$$\bar{f}(x', u) = \mathrm{but}(u) = y$$

and i_f by

$$i_f(x') = (x', \mathrm{id}_{f(x')} : f(x') \xrightarrow{\mathrm{id}} f(x')),$$

so we do indeed have

$$f(x') = \bar{f} i_f(x').$$

Corollary 2. *The functor* [page 18]

$$\bar{f} : \bar{X}' \longrightarrow X$$

has as fibers the categories X'/x, it is cofibrant, the base change functors are the functors $X'/x \longrightarrow X'/y$ associated with the arrows $x \longrightarrow y$ in X.

In fact, we see that this category is not only cofibered, but canonically *split.*

([10]).

Applying this to $f : X \longrightarrow S$, we find a construction

$$\mathrm{Cat}/S \xrightarrow{\Psi} \underline{\mathrm{Cofibsc}}\, S,$$

it is well and truly functorial in X varying in Cat/S. If

$$\psi : X \longrightarrow Y$$

in Cat/S, then ψ induces p. 12

$$\psi/s : X/s \longrightarrow Y/s\ ,$$

for $s \in S$, and this in a way compatible with the "cobase change" functors associated with the $u : s \longrightarrow t$ in S [page 19]

$$\begin{array}{ccc} X/s & \xrightarrow{u_*^X} & X/t \\ {\scriptstyle \psi/s}\downarrow & & \downarrow{\scriptstyle \psi/t} \\ Y/s & \xrightarrow{u_*^Y} & Y/t \end{array} \quad .$$

[9] **NB** $\pi_f i_f = \mathrm{id}$, and $i_f \pi_f \longrightarrow \mathrm{id}$, hence i_f, π_f are homotopisms inverse to each other in $\mathrm{Cat}\,\mathrm{hot}$. This makes more precise cor. 3, page 18.

[10] **Corollary 3.** *The arrow $i_f : X' \longrightarrow \bar{X}'$ is an X-morphism (if X' over X by f, $\bar{X}'$ by $\bar{f}$), i.e. $f = \bar{f} \circ i_f$, and it is a* Hot*-equivalence* (for $\pi_f : \bar{X}' \longrightarrow X'$ is Cat-fibrant, with contractible fibers, hence a Hot-equivalence).

This therefore defines a morphism of split categories over S

$$\Psi(X) \longrightarrow \Psi(Y),$$

that's the one!

We have a morphism functorial in X

$$X \xrightarrow{i_X} \Psi(X)$$

in Cat/X, and it is a Hot-equivalence, hence $\in W$. On the fibers, the morphism i_X induces

$$X_s \xrightarrow{i_{X,s}} X/s,$$

the canonical inclusion.

[page 20]

Corollary 1. *If $X \xrightarrow{f} Y$ in Cat/S is in W, then $\Psi(X) \xrightarrow{\Psi(f)} \Psi(Y)$ too.* But we shall beware of thinking that it is in W_S in the sense of $\underline{\mathrm{Cofib}}\, S$ or $\underline{\mathrm{Cofibsc}}\, S$.

For on the fibers of $\Psi(X)$, $\Psi(Y)$ we find the arrows

$$f/s : X/s \longrightarrow Y/s$$

induced by f, which have no reason to be in W. Their being so means that $f : X \longrightarrow Y$ is in W "locally over S" ([11]), which (for the W that interest us) is much stronger than being in W. (If $Y = S$, this means in fact that $p : X \longrightarrow S$ is W-aspheric, much stronger than $p \in W$.)

So to prove that

$$\underline{\mathrm{Cofib}}\, S \longrightarrow \mathrm{Cat}/S \qquad \text{or} \qquad \underline{\mathrm{Cofibsc}}\, S \longrightarrow \mathrm{Cat}/S$$

induce equivalences for the localized categories, the functor Ψ directly will not be usable. p. 13

However:

Corollary 2. *Suppose X, Y in $\underline{\mathrm{Prop}}\, S$ or in $\underline{\mathrm{Fibhot}}\, S$, and consider*

$$\Psi(f) : \Psi(X) \longrightarrow \Psi(Y),$$

for this to be a W-equivalence in $\underline{\mathrm{Cofibsc}}\, S$, i.e. a W-equivalence fiber by fiber, it is necessary and sufficient that f be a W-equivalence fiber by fiber (and this must imply ([12]) that f is a W-equivalence).

[page 21]

Indeed, in the commutative diagram

$$\begin{array}{ccc} X_s & \xrightarrow{f_s} & Y_s \\ \downarrow & & \downarrow \\ X/s & \xrightarrow{f/s} & Y/s \end{array}$$

the vertical arrows are in W, for X, Y proper over S (or X, Y in $\underline{\mathrm{Fibhot}}\, S$), so f_s is in W if and only if f/s is.

Φ is defined dually to Ψ, which amounts to interchanging the roles of σ_X, β_X in the diagram page 17.

[11] cf. (1) page 15.

[12] cf. (1) page 15.

([13])

II) Definition of $C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Parf}}\, S$

This time, it is the factorization of the arrows $p : X \longrightarrow S$ in Cat, as

$$X \xrightarrow{j_{X/S}} \bar{X} \xrightarrow{\bar{p}} S,$$

where i_X is a Hot-equivalence (hence in W), and $\bar{p}$ perfect. This is constructed formally as before, using $\underline{\mathrm{Ch}}_\infty(S)$ instead of $\underline{\mathrm{Fl}}(S)$, and the factorization

$$S \xrightarrow{j_S} \underline{\mathrm{Ch}}_\infty(S) \xrightarrow{p_S=(\sigma_S,\beta_S)} S \times S$$

of $\mathrm{diag}_S : S \longrightarrow S \times S$. The situation is prettier, for p_S is already "perfect", which formally implies that the $\bar{p}$ deduced from it, to factor $p : X \longrightarrow S$, is itself perfect.

Here again, the construction is functorial, we get the functor

p. 14

$$C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Parf}}\, S$$

[page 22]

which must be composed with the functor in the opposite direction

$$r : \mathrm{Cat}/S \longleftarrow \underline{\mathrm{Parf}}\, S.$$

First of all, let us check that C, just like r, is compatible with the localizers. Let $f : X \longrightarrow Y$ be in Cat/S, suppose f is in W, let us prove $C(f)$ is in W_S. We have commutativity in

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle j_X}\downarrow & & \downarrow{\scriptstyle j_Y} \\ CX & \xrightarrow{Cf} & CY \end{array}$$

and j_X, j_Y are in W, so f is in W if and only if Cf is. But in the case $W = W_\infty$ we know that this is equivalent to Cf being in W_∞ fiber by fiber. Admitting ① p. 15 for W, it remains to make sure of a converse in $\underline{\mathrm{Parf}}$, i.e.

② Let $\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$ be in $\underline{\mathrm{Parf}}\, S$ (or merely in $\underline{\mathrm{Fibhot}}\, S$). If f is in W, then so are the $f_s : X_s \longrightarrow Y_s$.

Let us try to prove it, by copying the proof in the case $C = C_\infty$ (XI, th. 2). We can suppose S connected ([14]), using La (i.e. $W \subseteq W_0$) and admitting the

Lemma. *Let $f : X \longrightarrow Y$ be in* Cat. *Then $f \in W$ if and only if $\pi_0(f)$ is bijective, and for every connected component X_i of X, if Y_i is the corresponding connected component of Y, $f_i : X_i \longrightarrow Y_i$ induced by f is in W.*

[page 23]

Let us then use the existence of

$$S' \longrightarrow S$$

a Hot-fibration with S' Hot-aspheric ([15]). We can even suppose that $S' \longrightarrow S$ is perfect. Suppose we know that base change by such perfect fibrations transforms W-equivalences into likewise. (That must be true for the W_i, but we have seen that it was false for k-homological equivalence, cf. I p. 87 ff.) Then $X' \xrightarrow{f'} Y'$ over S' is a W-equivalence.

[13] **Corollary 3.** *If $X \in Ob\, \underline{\mathrm{Fibhot}}\, S$, then $\Psi(X)$ is in $\underline{\mathrm{Fibsc}}_0 S$.*

This means in fact that the $X/s \longrightarrow X/t$ are in W, now the hypothesis on X implies that it is in W_∞. OK.

p. 15

[14] serves no purpose

[15] **NB** It suffices to have a Hot-fibration $S' \longrightarrow S_0 \hookrightarrow S$, where S_0 is the connected component of S containing the element $s \in \mathrm{Ob}\, S$ under consideration.

Now for $s' \in S'$, since $\{s'\} \longrightarrow S'$ is a Hot-equivalence (S' Hot-aspheric), and X', Y' being Hot-fibered over S, it follows that $X'_{s'} \longrightarrow X'$, $Y'_{s'} \longrightarrow Y'$ are Hot-equivalences, hence also $X'_{s'} \xrightarrow{f'_{s'}} Y'_{s'}$, as we see from the commutative square

$$\begin{array}{ccc} X'_{s'} & \xrightarrow{f'_{s'}} & Y'_{s'} \\ \wr\downarrow & & \wr\downarrow \\ X' & \xrightarrow{f'} & Y'. \end{array}$$

Since $S' \longrightarrow S$ is surjective, this shows that the $f_s : X_s \longrightarrow Y_s$ are in W, qed. [page 24]

Thus, (2) will be a consequence of (2.) Let $p : Y' \longrightarrow Y$ be a Hot-fibration (or even a perfect functor, if that can help …). Let $f : X \longrightarrow Y$ be an S-arrow (in Cat/S), then $f' : X' \longrightarrow Y'$ by base change. If $f \in W$, $f' \in W'$

$$\begin{array}{ccc} X' & \xrightarrow{q} & X \\ {\scriptstyle f'}\downarrow & \text{cart.} & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{p} & Y \end{array} \qquad f \in W \text{ and } p \text{ perfect} \Longrightarrow f' \in W$$

In any case, we absolutely need (2) to be able to conclude that C is compatible with the localizers.

Let us consider $rC : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S$, we have an arrow

$$\begin{cases} \mathrm{id}_{\mathrm{Cat}/S} \xrightarrow{j} rC \\ j_X : X \longrightarrow CX \quad \text{the canonical arrow } i_{X/S} \ \text{(p. 21)}, \end{cases}$$

and j_X is always in W.

Let us consider $Cr : \underline{\mathrm{Parf}}\, S \longrightarrow \underline{\mathrm{Parf}}\, S$, i.e. let us apply the construction C to an X perfect over S. Then we again have

$$\begin{cases} \mathrm{id}_{\underline{\mathrm{Parf}}\, S} \xrightarrow{j'} Cr \\ j'_X = j_{r(X)} : X = r(X) \xrightarrow{j_X} CX. \end{cases}$$

This morphism is in W, hence also a W-equivalence fiber by fiber, since source and target are perfect, hence Hot-fibered over S.

This already proves that the functors r and C induce equivalences quasi-inverse to one another on the localized categories. By the lemma p. 11, this shows that for every full subcategory $\mathcal{F}$ of Cat/S containing $\underline{\mathrm{Parf}}\, S$, the inclusion of $\mathcal{F}$ in Cat/S induces an equivalence on the localized categories, taking as localizer in $\mathcal{F}$ the "coarse" localizer induced by Cat/S. This applies in particular to the full subcategories [page 25] p. 16

$$\underline{\mathrm{Liss}}\, S, \quad \underline{\mathrm{Fibhot}}\, S, \quad \underline{\mathrm{Prop}}\, S,$$

but taking in $\underline{\mathrm{Liss}}\, S$ and $\underline{\mathrm{Prop}}\, S$ not the fine "fiber by fiber" localizer, as I had wrongly announced, but the coarse localizer induced by Cat/S. (For $\underline{\mathrm{Fibhot}}\, S$, there is no risk of ambiguity). I ought to convince myself that if I take the fine W-equivalence (in the case $W = W_\infty$, say), I do *not* find the "good" localized categories, i.e. the functor of localized categories, to the localized category of Cat/S, is not an equivalence. This in fact supposes that in $\underline{\mathrm{Fib}}\, S$ and $\underline{\mathrm{Cofib}}\, S$ as well, it is the *coarse* equivalence that should [page 26]

be taken. (In $\underline{\mathrm{Fib}}_0 S$ and $\underline{\mathrm{Cofib}}_0 S$, or $\underline{\mathrm{Fibsc}}_0 S$ and $\underline{\mathrm{Cofibsc}}_0 S$, the two equivalences are identical, since these four categories are contained in $\underline{\mathrm{Fibhot}}\, S$.) The sequel will show us whether this is indeed so. On the other hand, in $\underline{\mathrm{Fibsc}}\, S$ and $\underline{\mathrm{Cofibsc}}\, S$, there is no doubt, according to the general principles of constructing a derivator from an $(\mathcal{M}, W)$, that we must take the fine equivalence, fiber by fiber. In this connection, we must be aware that the vertical functors of the diagram p. 8

$$\underline{\mathrm{Fibsc}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S, \quad \underline{\mathrm{Cofibsc}}\, S \longrightarrow \underline{\mathrm{Cofib}}\, S,$$

though essentially surjective and faithful, are not fully faithful. They would be, and even equivalences of categories, if in $\underline{\mathrm{Fib}}\, S$ resp. $\underline{\mathrm{Cofib}}\, S$ I confined myself to the S-functors that are *cartesian* resp. *cocartesian*. So we can interpret, up to equivalence, $\underline{\mathrm{Fibsc}}\, S$ and $\underline{\mathrm{Cofibsc}}\, S$ as $\underline{\mathrm{Fib}}\, S$ and $\underline{\mathrm{Cofib}}\, S$, where we would have restricted the morphisms to just those S-functors that are *cartesian* resp. *cocartesian.* It would thus seem that by restricting the arrows in this way, but by restricting the localizer in a still more draconian way, we find equivalent localized categories.

That's rather strange - we'll see if it's really true! [page 27]

III) Construction of Φ_0, Ψ_0 : $\mathrm{Cat}/S \to \underline{\mathrm{Fibsc}}_0 S$, $\mathrm{Cat}/S \to \underline{\mathrm{Cofibsc}}_0 S$.

We must have

$$\Phi_0(X) = \Phi r C(X) \qquad \Psi_0(X) = \Psi r C(X),$$

where

$$r : \underline{\mathrm{Parf}}\, S \longrightarrow \mathrm{Cat}/S$$

is the canonical inclusion. As functors with values in $\underline{\mathrm{Fibsc}}$ resp. $\underline{\mathrm{Cofibsc}}$, this already defines Φ_0, Ψ_0 by

$$\Phi_0 = \Phi r C \qquad \Psi_0 = \Psi r C,$$

but we must see that Φ_0, Ψ_0 take their values in the full subcategories with index 0. p. 17
This is therefore equivalent to the following Lemma, applied to $Y = rC(X)$:

Lemma. *Let Y be perfect over S (or merely a* Hot*-fibered object over S). Then $\Phi(Y)$ is in $\underline{\mathrm{Fibsc}}_0 S$, $\Psi(Y)$ in $\underline{\mathrm{Cofibsc}}_0 S$.*

We must prove (for $\Psi(Y)$) that if $s \longrightarrow t$ is an arrow in S, then

$$\Psi(Y)_s \longrightarrow \Psi(Y)_t$$

is a W-equivalence. Now this arrow identifies with

$$Y/s \longrightarrow Y/t$$

and we know well that it is a Hot-equivalence, by definition of Hot-fibered objects, hence it is a W-equivalence. [page 28]

We thus have the functors

$$\underline{\mathrm{Fibsc}}_0 S \underset{\Phi_0}{\overset{p}{\rightleftarrows}} \mathrm{Cat}/S,$$

we must see that they satisfy the general conditions of the lemma p. 11 (and it will be the same for q, Ψ_0). We already know that p is compatible with the localizers. Let us prove it for Ψ_0. Let

$$f : X \longrightarrow Y$$

in Cat/S be a W-equivalence, then $rC(f)$ is one too. It suffices to prove this:

Let

$$q : X' \longrightarrow Y'$$

be a morphism in $\underline{\mathrm{Fibhot}}\, S$ (for example a morphism of $\underline{\mathrm{Parf}}\, S$), which is a W-equivalence, hence W-equivalences on the fibers. Then $\Psi(q) : \Psi(X') \longrightarrow \Psi(Y')$ is a W-equivalence fiber by fiber, i.e.

$$X'/s \longrightarrow Y'/s$$

a W-equivalence for every $s \in S$ (was seen in corollary 2, page 20).

Let us consider [page 29]

$$p\Phi_0 : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S,$$

this thus transforms X into $\Phi C(X)$, regarded as an object of Cat/S and nothing more. We have a functorial morphism

$$X \xrightarrow{j_X} C(X) \qquad\qquad j : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C,$$

hence a functorial morphism p. 18

$$\Phi(X) \xrightarrow{\Phi(j_X)} \Phi C(X) \qquad\qquad \Phi * j : \Phi \longrightarrow \Phi_0$$

and also a functorial morphism

$$X \xrightarrow{i_X} \Phi(X),$$

hence on composing

$$X \xrightarrow{i_X} \Phi(X) \xrightarrow{\Phi(j_X)} \Phi C(X), \qquad \alpha_X$$

i.e.

$$\alpha : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow p\Phi_0.$$

For X fixed in Cat/S, we know that j_X is in W, hence so is $\Phi(j_X)$ (though not fiber by fiber), cf. cor. 1 page 20. We also know that i_X is in W, hence so is the composite α_X.

Let us consider

$$\Phi_0 p : \underline{\mathrm{Fibsc}}_0\, S \longrightarrow \underline{\mathrm{Fibsc}}_0\, S \quad .$$

We start with X in $\underline{\mathrm{Fibsc}}_0\, S$, we forget the splitting and the very fact that it is Cat-fibrant (we have applied p), we apply Φ_0 to it, i.e. we regard ΦCX as an object of $\underline{\mathrm{Fibsc}}\, S$ (and even of $\underline{\mathrm{Fibsc}}_0\, S$). We want again to define [page 30]

$$X \xrightarrow{\beta_X} \Phi_0 p(X) = \Phi C(X) \qquad \textbf{in } \underline{\mathrm{Fibsc}}_0,$$

functorial in X. Now we have as before

$$X \xrightarrow{j_X} C(X),$$

hence on applying Φ

$$\Phi(X) \xrightarrow{\Phi(j_X)} \Phi C(X),$$

but regarded this time as an arrow in $\underline{\mathrm{Fibsc}}_0\, S$ (N.B. we do have $\Phi(X)$ in $\underline{\mathrm{Fibsc}}_0\, S$, for X, coming from $\underline{\mathrm{Fibsc}}_0\, S$, is in $\underline{\mathrm{Fibhot}}\, S$, and we apply the remark "Cor. 3" page 21.) On the other hand, we have an arrow in Cat/S

$$X \xrightarrow{i_X} \Phi(X),$$

is it an arrow in $\underline{\mathrm{Fibsc}}_0(S)$, i.e. is it cartesian? We see at once that no, that we have only, for an arrow $u : s \longrightarrow t$ in S, a commutation homomorphism (not an isomorphism) in

p. 19

$$\begin{array}{ccc} X_s & \xleftarrow{u_X^*} & X_t \\ \downarrow{\scriptstyle i_s} & \Downarrow \alpha & \downarrow{\scriptstyle i_t} \\ s\backslash X & \xleftarrow{u\backslash \mathrm{id}_X = u^*_{\Phi(X)}} & t\backslash X \end{array}$$

So I have to pack up! One can regard i_X at best as an arrow of $\underline{\mathrm{Fib}}_0\, S$, and taking the composites one finds an arrow in $\underline{\mathrm{Fib}}_0\, S$: [page 31]

$$X \xrightarrow{i_X} \Phi(X) \xrightarrow{\Phi(j_X)} \Phi C(X),$$

with $\beta_X (= \alpha_X)$ the arrow $X \longrightarrow \Phi C(X)$,

i.e. (now regarding Φ_0 as a functor with values in $\underline{\mathrm{Fib}}_0\, S$) a homomorphism of functors

$$\beta : \mathrm{id}_{\underline{\mathrm{Fib}}_0 S} \longrightarrow \Phi_0 p_0,$$

where $p_0 : \underline{\mathrm{Fib}}_0 S \hookrightarrow \mathrm{Cat}/S$ is the canonical inclusion. This β is in W argument by argument, and likewise for α page 29, interpreted this time as

$$\mathrm{id}_{\mathrm{Cat}/S} \longrightarrow p_0 \Phi_0.$$

What we have proved in the end is this:

Packing-up theorem. *Let $\mathcal{F}$ be a full subcategory of* Cat/S *which contains one of the three full subcategories* $\underline{\mathrm{Fib}}_0 S$, $\underline{\mathrm{Parf}}\, S$, $\underline{\mathrm{Cofib}}_0 S$, *and equipping $\mathcal{F}$ with the "coarse" localizer W (induced by that of* Cat/S*). Then the inclusion*

$$\mathcal{F} \longrightarrow \mathrm{Cat}/S$$

induces an equivalence on the localizers.

On the other hand, the localized categories of the four categories of split fibrations [page 32]

$$\underline{\mathrm{Fibsc}}\, S, \quad \underline{\mathrm{Cofibsc}}\, S, \quad \underline{\mathrm{Fibsc}}_0 S, \quad \underline{\mathrm{Cofibsc}}_0 S$$

escape our arguments. *And for good reason!* It is completely open, over S connected say (e.g. $S = \mathbf{\Delta}^1$), whether for a relative homotopy type ξ over S, the induced types ξ_s ($s \in S$) form a local system on S, with values in Hot_W, as would be the case if the theorem announced at the beginning were true. On the contrary, *every* arrow of Hot_W must come from an object of $\mathrm{Hot}_W(\mathbf{\Delta}^1)$. The localizations of the 9 other categories of the diagram p. 8, and of all those discussed in the theorem we have just proved, must correspond "visibly" to the full subcategories of $\mathrm{HOT}_W(S)$ formed by the locally constant HOT-coefficients on S.

Also, I realize that it was absurd to expect $\underline{\mathrm{Fibsc}}\, S$ and $\underline{\mathrm{Cofibsc}}\, S$ to give canonically equivalent localized categories. For the elements ξ of the first of the two categories each p. 20 [page 33]

give rise to a contravariant functor on S

$$\bar{\xi} : S^o \longrightarrow \mathrm{Hot}_W,$$

those η of the second (which is none other than $\underline{\mathrm{Fibsc}}(S^o)$) each give rise to a *covariant* functor

$$\bar{\eta} : S \longrightarrow \mathrm{Hot}_W,$$

which is not the same! It is only "the same" if these are local coefficients, which means exactly that ξ is in $\underline{\mathrm{Fibsc}}_0 S$ resp. η in $\underline{\mathrm{Cofibsc}}_0 S$. So one does indeed expect (1°) that the arrows

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fib}}_0 S, \quad \underline{\mathrm{Cofibsc}}_0 S \longrightarrow \underline{\mathrm{Cofib}}_0 S$$

induce equivalences on the localizations (something I was unable to prove a little while ago) ([16]). On the other hand, it is worse than unreasonable to expect the same to hold for the functors

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fibsc}}\, S, \quad \underline{\mathrm{Cofibsc}}_0 S \longrightarrow \underline{\mathrm{Cofibsc}}\, S,$$

that should give by localization the fully faithful inclusion functors, of the categories of locally constant objects into $\mathrm{HOT}_W(S)$ resp. into $\mathrm{HOT}_W(S^o)$. And likewise for the [page 34]
functors induced on the localizations by

$$(*) \qquad \underline{\mathrm{Fibsc}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S, \quad \underline{\mathrm{Cofibsc}}\, S \longrightarrow \underline{\mathrm{Cofib}}\, S.$$

Since the localizations of the target categories of these two functors look very much as though they identify with the preceding subcategories of locally constant objects, it would therefore seem that: (2°) The preceding functors define retractions of the category $\mathrm{HOT}_W(S)$ resp. $\mathrm{HOT}_W(S^o)$ onto the subcategory of locally constant objects on S, the localized functors of the inclusion functors

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fibsc}}\, S, \quad \underline{\mathrm{Cofibsc}}_0 S \longrightarrow \underline{\mathrm{Cofibsc}}\, S$$

being fully faithful and having as essential images the subcategories in question.

This retraction seems to me a wholly mysterious thing. This is tied to the fact that for the very special derivator $\mathbf{D} = \mathrm{HOT}$, when one has a $\xi \in \mathbf{D}(S)$, one deduces from it (up to HOT/S-equivalence) a type of diagram X over S, which one can then regard (via [page 35]
the construction C) as defining a relative homotopy type *locally constant* over S.

So to elucidate the situation, i.e. the relations between the 13 localized categories coming from the diagram on page 8, it remains to see whether the hypothetical assertions I have just made are indeed valid, namely p. 21

(1°) The functor deduced from

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fib}}_0 S$$

by passing to the localizations is an equivalence of categories (and in this question one can replace $\underline{\mathrm{Fib}}_0 S$ by any other full subcategory of Cat/S containing it - it amounts to the same thing), and

(2°) The functor deduced from

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fibsc}}\, S$$

by localization is fully faithful. (That its image is exactly made up of the locally constant objects of the first localization, interpreted as $\mathrm{HOT}_W(S)$, is tautological.)

[page 36]

There is no longer any need to worry about the side of cofibered categories, since all one will be able to say about it comes down to what I have just spelled out, for X^o instead of X.

In the direction of (1°), our preceding arguments by dint of Φ_0, p already give us a useful piece of information. But it will be more convenient now to drop $\underline{\mathrm{Fib}}_0 S$ or rather, to replace it by $\underline{\mathrm{Fib}}\, S$ (which is legitimate, we said so in (1°)), so that we now have a commutative diagram (solid lines)

$$\begin{array}{ccc} \underline{\mathrm{Fibsc}}\, S & \xleftarrow{\quad i \quad} & \underline{\mathrm{Fibsc}}_0 S \\ & {\scriptstyle j}\searrow \quad \swarrow{\scriptstyle k} & \uparrow \\ & \underline{\mathrm{Fib}}\, S, & {\scriptstyle \Phi_0 = \psi} \end{array}$$

where i is a (strictly) fully faithful inclusion, j (hence also k) faithful. Finally, we have a canonical functor ψ $(= \Phi_0)$ (dotted). The localizers on the top line are the fine localizers (for $\underline{\mathrm{Fibsc}}_0 S$ this is the same as coarse W-equivalence), on $\underline{\mathrm{Fib}}\, S$ on the other hand we take coarse W-equivalence. (We know that the localization is the same, up to equivalence, as if one restricts to the full subcategory $\underline{\mathrm{Fib}}_0 S$, which is quite remarkable).

All four functors i, j, k, Φ_0 are compatible with the localizers, hence pass to the localizations [page 37] p. 22

$$\begin{array}{ccc} \mathrm{HOT}_W(S) & \xleftarrow{\quad \bar{\imath} \quad} & \mathrm{HOT}^?_W(S) \\ & {\scriptstyle \bar{\jmath}}\searrow \quad \swarrow{\scriptstyle \bar{k}} & \uparrow \\ & \mathrm{HOT}^{\mathrm{lc}}_W(S), & {\scriptstyle \bar{\psi}} \end{array}$$

where the superscript ? in the notation means that I do not know how to interpret this category, unless I prove that $\bar{k}$ is an equivalence, and the superscript lc in the bottom category should suggest the intuition that in any case we are dealing with a sort of "relative homotopy types" (in a modified sense) *locally constant* on S. The problem is to understand this diagram. The natural conjecture is

$1^o)$ $\bar{k}$ is an equivalence. (Which implies that $\bar{\psi}$ is the quasi-inverse equivalence, that $\bar{k}$, $\bar{\psi}$ essentially allow one to identify $\mathrm{HOT}^?_W$ and $\mathrm{HOT}^{\mathrm{lc}}_W$, so that $\bar{\imath}$ and $\bar{\jmath}$ become functors between HOT_W and $\mathrm{HOT}^{\mathrm{lc}}_W$.)

$2^o)$ $\bar{\imath}$ is fully faithful (NB the essential image is well understood in any case.)

In the direction of $1^o)$, I know for now that one has

$$\bar{k}\bar{\psi} \simeq \mathrm{id}_{\mathrm{HOT}^{\mathrm{lc}}_W(S)}, \qquad \text{[page 38]}$$

so $\bar{k}$ is essentially surjective (on objects), and induces *surjections* on the $\mathrm{Hom}(-,-)$ between objects in the image of $\bar{\psi}$, while $\bar{\psi}$ is injective on the isomorphism classes of objects, and on the $\mathrm{Hom}(-,-)$ (i.e. *faithful*).

If the relations between $\mathrm{HOT}^?_W$ and $\mathrm{HOT}^{\mathrm{lc}}_W$ turn out to be impossible to make head or tail of (e.g. $\bar{\psi}$ not even fully faithful, which implies that $\bar{k}$ is not even faithful), then the category $\mathrm{HOT}^?_W$ deserves to be dumped in the wastebasket, and we will keep only

$$\begin{array}{c} \mathrm{HOT}_W(S) \\ {\scriptstyle \bar{\jmath}}\downarrow \quad \uparrow{\scriptstyle \bar{\psi}'} \\ \mathrm{HOT}^{\mathrm{lc}}_W(S), \end{array}$$

[16] The quasi-inverses being well and truly given by Φ_0, Ψ_0 - for by what else then ???

where we define $\bar{\psi}'$ as

$$\bar{\psi}' = \bar{\imath}\bar{\psi} = \overline{i\psi}.$$

And the question (more modest) now becomes:

3°) Is $\bar{\psi}'$ fully faithful? (The *essential* image is the same as that of $\underline{\mathrm{Fib}}_0$ ($\approx \underline{\mathrm{Fib}}$), it is made up, unless I am mistaken, of the locally constant objects - to be checked if need be …)

It is hard to imagine that $\bar{k}$, or what amounts to the same thing $\bar{\psi}$, could be an equivalence, without there being at least a "W_S-homotopy" p. 23 [page 39]

$$\mathrm{id}_{\underline{\mathrm{Fibsc}}_0 S} \leftrightarrow \leftrightarrow^h \leftrightarrow \leftrightarrow \psi k$$

in the category of endofunctors of $\underline{\mathrm{Fibsc}}_0 S$. To tell the truth, I now realize that the description of $\bar{\psi}$ starting from $\psi : X \longmapsto \Phi_0 X = \Phi C X$ is needlessly complicated, for we had made the detour through $C(X)$ solely to be sure of landing in $\underline{\mathrm{Fibsc}}_0 S$, not just in $\underline{\mathrm{Fibsc}}\, S$. But if we drop $\underline{\mathrm{Fibsc}}_0 S$, we don't give a damn, and the functor $\Phi : X \longmapsto \Phi(X)$ is much more natural. And if we do not drop it, there is no harm in replacing the source $\underline{\mathrm{Fib}}\, S$ by $\underline{\mathrm{Fib}}_0 S$ (which has the same localization, up to equivalence), and on $\underline{\mathrm{Fib}}_0 S$ again there is no need to burden ourselves with CX, for the pleasure of applying Φ to it, since $\Phi(X)$ itself is already in $\underline{\mathrm{Fibsc}}_0 S$.

So, the question is this: If X varies [page 40]

$$\text{in } \underset{\nearrow\uparrow\nwarrow}{\underline{\mathrm{Fibsc}}_0 S}, \quad \text{is there a homotopy in } \underset{\nearrow\uparrow\nwarrow}{\underline{\mathrm{Fibsc}}_0 S}, \quad \text{"natural", i.e. functorial in } X,$$

$$X \xrightarrow{\quad c \quad} \Phi(X) = \text{ Split fibered category with fibers the } s\backslash X$$

But a path which is made up of arrows

$$X = X_0 \text{ —— } X_1 \text{ —— } X_2 \cdots \text{ —— } X_n = \Phi(X)$$

which are all cartesian (and moreover, of course, in W!). This therefore means that we would have split fibered categories over S, X_1, …, X_{n-1}, depending *functorially* on X, which connect

$$X = X_0 \qquad \text{to} \qquad \Phi(X) = X_n$$

i.e.

$$X_s \qquad \text{to} \qquad s\backslash X \qquad ???$$

I see nothing on the horizon.

p. 24

15.11.90 [page 41]

I think I finally understood, last night, the exact situation. There is reason, in Cat/S and in the other categories of the diagram p. 8 (all mapping to Cat/S) to consider four sorts of "W-equivalences".

a) The *"coarse" W-equivalence* : $f : X \longrightarrow Y$ (with $X \to S$, $Y \to S$) in Cat/S is a coarse W-equivalence if as an arrow of Cat, it is a W-equivalence. ([17]).

[17] stable under base change both smooth and proper, more generally if $S' \longrightarrow S$ is a Hot-fibration.

b) The *"fiberwise" W-equivalence* (over S) : $\forall\ s \in S$, $f_s : X_s \longrightarrow Y_s$ is a W-equivalence. ([18]).

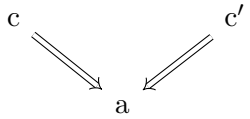
c) The *W-equivalence "local over S"* : $\forall\ s \in S$, $f/s : X/s \longrightarrow Y/s$ is a W-equivalence. Means that it is a W-equivalence, and remains so under any base change $S' \longrightarrow S$ which is a local isomorphism. ([19]).

c′) The *W-equivalence "colocal over S"* : $\forall\ s \in S$, $s\backslash f : s\backslash X \longrightarrow s\backslash Y$ is a W-equivalence. Means that it is a W-equivalence, and remains so under any base change $S' \longrightarrow S$ which is a local co-isomorphism. ([20]).

I am going to summarize in a recapitulative statement the relations between these four notions of W-equivalence, thinking of the case $W = W_\infty$. Once this is done, I shall concern myself with isolating the minimal conditions on a general W which ensure that it is true for W.

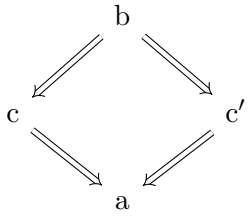
Proposition 1. *Let* $f : X \longrightarrow Y$ (with $X \to S$, $Y \to S$) *be in* Cat$/S$.

1) *Local W-equivalence over S, and colocal W-equivalence over S, imply coarse W-equivalence :*



p. 25

2) *If X, Y are proper over S, or X, Y smooth over S, or X, Y* Hot-*fibered over S, then fiberwise W-equivalence implies that it is a* universal *W-equivalence over S (W-equivalence, and remains so under* any *base change $S' \longrightarrow S$), hence in these three cases, we have*



([21]).

But we can be more precise about each of these three cases :

(2 fib) *If X, Y are* Hot-*fibered over S, then coarse W-equivalence of f implies that it is a fiberwise W-equivalence, hence the four notions of W-equivalence are equivalent for f.*

$$\text{a} \iff \text{c} \iff \text{c}' \iff \text{b}$$

([22])

[page 42]

[18] stable under any base change $S' \longrightarrow S$

[19] should be stable under *smooth* base change $S' \longrightarrow S$

[20] should be stable under *proper* base change over S

[21] It suffices X, Y W-proper over S, or W-smooth over S, or W-fibered over S.

[22] a $\Longrightarrow$ b only assuming L5, see below p. 43.

(2 smooth) *If X, Y are smooth over S, then fiberwise W-equivalence over S is equivalent to colocal W-equivalence over S and implies* universal *W-equivalence over S, hence in this case*

$$\begin{array}{ccc} \mathrm{b} & \Longleftrightarrow & \mathrm{c}' \\ \Downarrow & & \\ \mathrm{c} & & \\ \Downarrow & & \\ \mathrm{a} & & \end{array}$$

([23]). p. 26

(2 prop) *If X, Y are proper over S, then fiberwise W-equivalence over S is equivalent to local W-equivalence over S :*

$$\begin{array}{ccc} \mathrm{b} & \Longleftrightarrow & \mathrm{c} \\ \Downarrow & & \\ \mathrm{c}' & & \\ \Downarrow & & \\ \mathrm{a} & & \end{array}$$

([24]).

Proof 1) is none other than L4 ([25]) and its dual L4′ for a basic localizer W, and we have seen that if we assume L1, L3, then L4 is equivalent to L4′, and in the notion of basic localizer we shall always assume that this is satisfied. Also, we assume that L3 is true even in the stronger form L3 quater (p. 1) : $W_\infty \subseteq W$, i.e. every Hot-equivalence is a W-equivalence.

(2 smooth) If X, Y smooth over S, then the $X_s \longrightarrow s\backslash X$, $Y_s \longrightarrow s\backslash Y$ are W_∞-equivalences, hence W-equivalences (it would even suffice that X, Y be W-smooth/S), hence the commutative diagram

$$\begin{array}{ccc} X_s & \xrightarrow{f_s} & Y_s \\ {\scriptstyle i_s^X}\downarrow & & \downarrow{\scriptstyle i_s^Y} \\ s\backslash X & \xrightarrow{s\backslash f} & s\backslash Y \end{array}$$

shows us that f_s is a W-equivalence if and only if $s\backslash f$ is, whence b $\Longleftrightarrow$ c′. To see that fiberwise W-equivalence is even a *universal* W-equivalence over S, we are reduced to showing that it is a coarse W-equivalence (since smoothness stable under base change over S), and we win since c′ $\Longrightarrow$ a by 1).

The case (2 proper) is dual.

(2 fib) If X, Y are Hot-fibered over S, then the inclusions p. 27

[page 43]

$$X_s \longrightarrow s\backslash X, \qquad X_s \longrightarrow X/s,$$

and likewise for Y, are W_∞-equivalences, hence W-equivalences (it would even suffice that X, Y be W-fibrations over S ([26]), i.e. by definition that the preceding functors be

[23] it even suffices X, Y W-smooth over S.
[24] It even suffices X, Y W-proper over S.
[25] No, there is confusion, L4 is weaker, cf. p. 44 the axiom L4 bis.
[26] compare def. 3, p. 117, in (I).

W-equivalences, and that the same condition be satisfied after any base change over S), the argument in the smooth resp. proper case already shows that

$$\begin{array}{ccccc} \mathrm{c} & \Longleftarrow & \mathrm{b} & \Longleftrightarrow & \mathrm{c}' \\ & & \Downarrow & & \\ & & \mathrm{a}, & & \end{array}$$

and that fiberwise W-equivalence (or local W-equivalence over S, resp. colocal over S) already implies *universal* W-equivalence over S. ([27])

The question a $\Longrightarrow$ b is on the other hand distinctly more delicate, and we have seen that this is by no means the case for every basic localizer - probably not for the one associated to the derivator Der(**Z**-Mod), in any case not for the one associated to Der(k-Mod), when 2 is invertible in k. (It would even suffice that $\exists\ n \in \mathbf{N}$, $n \geq 2$, such that n invertible in k - the typical counterexample is $X = S^1 \xrightarrow{n} S^1 = Y = S$, the étale covering of order n of the circle.)

The implication a $\Longrightarrow$ b is thus a very strong condition on W, I am going to denote it L5 (the old L5, cf. page 2, not deserving to survive, for it is a botched compromise).

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

L5 *Let* *be in* Cat, *with* X, Y Hot-*fibered over* S. *If* f *is a* W-*equivalence ("coarse" relative to* S*), then it is a fiberwise* W-*equivalence* ([28]).

We have seen that up to an ε ("lemma" p. 23), L5 is a consequence of

L6 *Let* $f : X \longrightarrow Y$ *be a* W-*equivalence,* $Y' \longrightarrow Y$ *a perfect morphism (i.e. smooth and proper), then* $f' : X' = X \times_Y Y' \longrightarrow Y'$ *is a* W-*equivalence.*

Let us note the quite natural variant [page 44]

L6 bis *As in L6, but assuming only* $Y' \longrightarrow Y$ *a* Hot-*fibration.*

There is a chance that in practice, when L6 (or only L5) is satisfied, L6 bis is too. p. 28

We are going to introduce an axiom

L7 a) *Let* $(f_i : X_i \longrightarrow Y_i)_{i \in I}$ *be a family of arrows in* Cat, *let* $f : \coprod X_i \longrightarrow \coprod Y_i$ *be the corresponding arrow on the sum categories. For* $f \in W$, *it is necessary and sufficient that* $f_i \in W$ *for every* $i \in I$.

b) *Let* $f : X \longrightarrow Y$ *be in* W. *If* Y *is* 0-*connected,* X *too.*

This axiom is equivalent to the conjunction of La ($f \in W \Longrightarrow \pi_0 f$ bijective) and the validity of the "Lemma" page 23, which I put here again as a reminder :

Lemma. *The preceding axiom L7 on the basic localizer* W *is equivalent to the following condition :* $f : X \longrightarrow Y$ *is in* W *if and only if* $\pi_0(f)$ *is bijective, and* f *induces* W-*equivalences* $f_i : X_i \longrightarrow Y_i$ *on the connected components.*

I must come back once more to 1) of the proposition, it is not true that "it is" the condition L4 on W (which I assume, I said), it is L4 in the particular case where $Y = S$. On reflection, I see it as a strengthened variant of L4, which does not seem likely to follow from L4 :

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

L4 bis *Let* *be in* Cat, *let us suppose that* f *is a local* W-*equivalence over* S *(i.e.* $f/s : X/s \longrightarrow Y/s$ *a* W-*equivalence* $\forall\ s \in S$*), then* f *is a* W-*equivalence* ([29]).

[page 45]

[27] valid assuming only X, Y W-fibered over S.

[28] In XII (p. 12) we see that if W satisfies L1, L3 quater, L4 bis (cf. next page), L5, then $\boxed{W = W_\infty}$. So it is perfectly illusory, if we assume L5 to go on dragging a W along.

[29] equivalent to the dual axiom, in terms of the $s\backslash f$

In what follows, by a "basic localizer", we shall mean a set

$$W \subseteq \mathrm{Fl}\,\mathrm{Cat}$$

satisfying (at least) the axioms

L1, L3 quater, L4 bis, L7,

i.e.

a) *Saturation* of W (conditions a) b) c) p. 1)

b) $W_\infty \subseteq W$ $[\subseteq W_0]$ ([30]) ([31]).

c) If $f : X \longrightarrow Y$ over S is locally over S a W-equivalence, then $f \in W$ (L4 bis)

d) For $f : X \longrightarrow Y$ to be in W, it is necessary and sufficient that $\pi_0(f)$ be bijective, and that the $f_i : X_i \longrightarrow Y_i$ induced on the connected components be in W. p. 29

These axioms are satisfied by all the W that I know, and that one would feel like regarding as localizers. On the other hand, L5, L6, L6 bis (p. 43, 44) are axioms of a much more special nature. I feel like saying that they distinguish localizers of a strictly *"homotopical"* nature from those of a *"homological"* (or "cohomological") nature. I should make sure that they are satisfied for the W_i (i-homotopy equivalence) ([32]).

The fact that L4 bis, i.e. c) above, implies the dual axiom, follows from the fact that L4 bis implies this

Proposition 2. *Let $f : X \longrightarrow Y$ be in* Cat*, then $f \in W$ if and only if $f^o : X^o \longrightarrow Y^o$ is in W.*

Seen by the argument of I, page 75 (cor. 2) - uses only L4, not even L1, L3.

I am going to try to summarize what I think I have understood:

[page 46]

1) W-equivalence "by fibers" is of interest (for constructing localized categories) only in the case where one knows that it already implies, either local W-equivalence, or colocal W-equivalence (or even universal W-equivalence) over S, hence when one assumes X, Y to be at the very least both either W-smooth, or W-proper, or W-fibered over S.

2) One could call local W-equivalence over S *"left fine W-equivalence"* over S, and colocal W-equivalence over S *"right fine W-equivalence"* over S. They coincide in $\underline{\mathrm{Fibhot}}\,S$, a fortiori in $\underline{\mathrm{Parf}}\,S$ and in the four categories (cf. diagram p. 8) $\underline{\mathrm{Fib}}_0 S$, $\underline{\mathrm{Fibsc}}_0 S$, $\underline{\mathrm{Cofib}}_0 S$, $\underline{\mathrm{Cofibsc}}_0 S$ ([33]). On the other hand, they do not coincide on the *bordering* categories of the diagram, in particular on $\underline{\mathrm{Liss}}\,S$, $\underline{\mathrm{Prop}}\,S$, $\underline{\mathrm{Fib}}\,S$, $\underline{\mathrm{Cofib}}\,S$, $\underline{\mathrm{Fibsc}}\,S$, $\underline{\mathrm{Cofibsc}}\,S$. In this regard, it is the *fine W-equivalence* **on the right** (in terms of the $s\backslash X$) that is of interest on the (alas) left side of the diagram, i.e. for $\underline{\mathrm{Liss}}\,S$, $\underline{\mathrm{Fib}}\,S$, $\underline{\mathrm{Fibsc}}\,S$, whereas it is the *fine W-equivalence* **on the left** (in terms of the X/s) that is relevant on the right side, for $\underline{\mathrm{Prop}}\,S$, $\underline{\mathrm{Cofib}}\,S$, $\underline{\mathrm{Cofibsc}}\,S$. One will never have to look simultaneously at the two fine equivalences on one and the same category of the diagram, with the sole exception of Cat/S as a whole.

---

[30] I am no longer sure whether it is really useful to impose $W_\infty \subseteq W$. Take instead the final object axiom, namely b′). If we want a) b′) c) + L5, then we shall have $W \subseteq W_\infty$. **NB** a) b) c) + L5 implies $W = W_\infty$, cf. annotation page 43. [page 47]

[31] cf. d)

[32] This is *false*, cf. annotation above.

[33] One could call them the six *inner* categories of the diagram.

3) Coarse W-equivalence, for its part, is "good everywhere" - but that depends for whom. For the 6 categories that are above $\underline{\mathrm{Fibhot}}\, S$, i.e.

$$\underline{\mathrm{Fibhot}}\, S, \quad \underline{\mathrm{Parf}}\, S, \quad \underline{\mathrm{Fib}}_0 S, \quad \underline{\mathrm{Fibsc}}_0 S, \quad \underline{\mathrm{Cofib}}_0 S, \quad \underline{\mathrm{Cofibsc}}_0 S,$$

it coincides with all the other equivalences (assuming axiom L5), for the three upper left bordering categories p. 30

$$\underline{\mathrm{Liss}}\, S, \quad \underline{\mathrm{Fib}}\, S, \quad \underline{\mathrm{Fibsc}}\, S$$

of the diagram, fiberwise W-equivalence coincides with right fine W-equivalence, and dually on

$$\underline{\mathrm{Prop}}\, S, \quad \underline{\mathrm{Cofib}}\, S, \quad \underline{\mathrm{Cofibsc}}\, S$$

fiberwise W-equivalence coincides with left fine W-equivalence. As for Cat/S as a whole, it is the only category of the diagram where there are *three* different interesting notions of W-equivalence, namely: the coarse one, and the two fine ones (left, right). On the other six bordering categories of the diagram, there are in principle two: the coarse one, and a specified one of the two fine ones (forbidden to look at the other - at one's own risk and peril …): the right one on the left side, the left one on the right side. (NB I ought to redo the diagram, really, putting the fibrations on the right, not on the left.) As for the six inner categories of the diagram, in them there is only a single notion of W-equivalence left (since on them the four notions coincide).

(**NB** there is no page 48)

I now summarize what I think I have understood about the corresponding localized categories. There are, up to equivalence, only *three* different ones all told, when one takes the "admissible" localizers in the categories, as I have just said (that being in short the objective interest of the prescriptions!) To be precise: [page 48 (blank)] [page 49]

1°) The *coarse W-equivalence* gives "the same" localized category for the 13 categories of the diagram, i.e. all the transition arrows of the diagram induce equivalences for this coarse W-equivalence. The localized category will be denoted $\boxed{\mathrm{HOT}^{\mathrm{lc}}_W(S)}$, where the exponent lc means: (relative W-homotopy types over S) *"locally constant."* (I have only a small doubt left for $\underline{\mathrm{Fibsc}}\, S$ and $\underline{\mathrm{Cofibsc(S)}}$ ([34]), whether coarse equivalence there is really and truly reasonable. We will see when doing the proof.)

2°) The *right fine W-equivalence* gives the same result for all the categories of the diagram where it is considered as admitted, *and where it does not coincide with the coarse W-equivalence* already considered, hence on the four left bordering categories

$$\mathrm{Cat}/S, \quad \underline{\mathrm{Liss}}\, S, \quad \underline{\mathrm{Fib}}\, S, \quad \underline{\mathrm{Fibsc}}\, S$$

([35]). The localized category is denoted $\boxed{\mathrm{HOT}_W(S)}$.

$2'^o$) Dually, the *left fine W-equivalence* gives the same result on all the categories of the diagram where it is admitted, and where it does not coincide with the coarse one, i.e. on the 4 categories [page 50] p. 31

$$\mathrm{Cat}/S, \quad \underline{\mathrm{Prop}}\, S, \quad \underline{\mathrm{Cofib}}\, S, \quad \underline{\mathrm{Cofibsc}}\, S.$$

It is appropriate to denote the common category of fractions by the symbol

$$\mathrm{HOT}_W(S^o) \qquad (\text{or also } \mathrm{HOT}^o_W(S)),$$

since it is isomorphic to the one denoted by this symbol.

[34] This doubt is unfounded, everything works perfectly!

[35] **NB** On the last three this equivalence is W-equivalence by fibers.

If one wants unambiguous notations, designating well-defined categories, and not just "up to equivalence", I will choose

$$\mathrm{HOT}_W(S) \overset{\text{def}}{=} \underline{\mathrm{Fibsc}}(S)(W_S^{\mathrm{d}})^{-1}$$

whence

$$\mathrm{HOT}_W(S^o) \underbrace{\simeq}_{\text{isomorphism of categories}} \underline{\mathrm{Cofibsc}}(S)(W_S^{\mathrm{g}})^{-1},$$

where W_S^{d}, W_S^{g} denote the sets of arrows corresponding to right resp. left fine W-equivalence. For $\mathrm{HOT}_W^{\mathrm{lc}}(S)$, I prefer a definition as symmetric as possible in S, S^o, and moreover a category as small as possible, which leads me to choose

$$\mathrm{HOT}_W^{\mathrm{lc}}(S) \overset{\text{def}}{=} (\underline{\mathrm{Parf}}\, S)W_S^{-1},$$

where we have denoted by W_S the W-equivalences in $\underline{\mathrm{Parf}}\, S$. (No need this time to specify which ones - all four coincide in $\underline{\mathrm{Parf}}\, S$!)

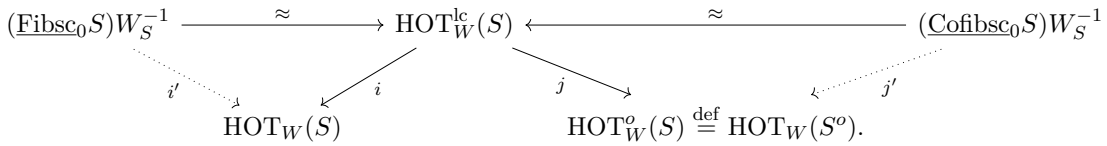
Let us note the isomorphism [page 51]

$$\underline{\mathrm{Parf}}(S^o) \simeq \underline{\mathrm{Parf}}\, S$$

(coming from the fact that $f : X \longrightarrow Y$ in Cat is a perfect fibration if and only if f^o is). This isomorphism $X \longmapsto X^o$ is compatible with the localizers (since $f \in W \iff f^o \in W$), whence

$$\mathrm{HOT}_W^{\mathrm{lc}}(S^o) \simeq \mathrm{HOT}_W^{\mathrm{lc}}(S)$$

(isomorphism of categories). We will have two canonical functors between the three categories of relative W-homotopy types over S



These arrows i, j are suggested by the dotted arrows i', j', coming from the inclusions p. 32

$$\underline{\mathrm{Fibsc}_0}S \hookrightarrow \underline{\mathrm{Fibsc}}\, S, \quad \underline{\mathrm{Cofibsc}_0}S \hookrightarrow \underline{\mathrm{Cofibsc}}\, S,$$

which would define i, j with the help of a choice of quasi-inverse functors for the horizontal arrows. But i, j would be defined only up to (unique) isomorphism. It will be prettier to choose the quasi-inverses in a canonical way, with the help of the arrows

$$\underline{\mathrm{Fibsc}_0}S \xleftarrow{\Phi_0} \underline{\mathrm{Parf}}\, S \xrightarrow{\Psi_0} \underline{\mathrm{Cofibsc}_0}S$$

(where $\Phi_0 = \Phi|\underline{\mathrm{Parf}}\, S$, $\Psi_0 = \Psi|\underline{\mathrm{Parf}}\, S$) of the diagram p. 8. The main question which remains unelucidated, is whether these functors i, j really are *fully faithful,* as I certainly expect. [page 52]

I now want to prove the preceding assertions about the localized categories.

① The localized category for coarse W-equivalence. It will suffice, up to an ε, to establish the conditions of the lemma p. 11, in the following three cases. $\underline{\mathrm{Parf}}\, S \xrightarrow{r} \mathrm{Cat}/S$ (that will give the case of $\underline{\mathrm{Fibhot}}\, S$, $\underline{\mathrm{Liss}}\, S$ and $\underline{\mathrm{Prop}}\, S$ over



Cat/S) and $\underline{\mathrm{Fibsc}}_0 S \xrightarrow{p} \mathrm{Cat}/S$, $\underline{\mathrm{Cofibsc}}_0 S \xrightarrow{q} \mathrm{Cat}/S$, which will give respectively the cases of

$$\underline{\mathrm{Fib}}_0 S,\ \underline{\mathrm{Fib}}\, S,\ \underline{\mathrm{Liss}}\, S \qquad \text{and} \qquad \underline{\mathrm{Cofib}}_0 S,\ \underline{\mathrm{Cofib}}\, S,\ \underline{\mathrm{Prop}}\, S \qquad \text{over } S.$$

But the trouble for p and q, is that the functors in the reverse direction that I had constructed (and again denoted Φ_0, Ψ_0 (but they were essentially $\Phi \circ r \circ C$ and $\Psi \circ r \circ c$, not to be confused with the functors Φ_0, Ψ_0 at the bottom of the preceding page, and also marked in the diagram p. 8), let us rather denote them Φ', Ψ', work well on the Cat/S side, i.e. $p\Phi'$, $q\Psi'$ receive the functor $\mathrm{id}_{\mathrm{Cat}/S}$ by a W-arrow, but not on the side of the target categories - one cannot manage to find a natural arrow between $\Phi' p$ resp. $\Psi' q$, and the identity functors of $\underline{\mathrm{Fibsc}}_0 S$ resp. $\underline{\mathrm{Cofibsc}}_0 S$. I do not see, indeed, that there are any - the difficulty, by this route, seems irremediable. On the other hand, if one regards Φ', Ψ' as functors with values in $\underline{\mathrm{Fib}}_0 S$, $\underline{\mathrm{Cofib}}_0 S$ respectively (thus in the categories where one does not require the arrows to be cartesian resp. cocartesian), one again finds W-morphisms [page 53]

$$\mathrm{id} \longrightarrow \Phi' p, \qquad \mathrm{id} \longrightarrow \Psi' q,$$

which at least already settles the case of all the categories of the diagrams p. 8 that are full subcategories of Cat/S ([36]). It remains only to treat the case of the four categories of "split" structures p. 33

$$\underline{\mathrm{Fibsc}}\, S, \quad \underline{\mathrm{Fibsc}}_0 S \qquad \text{and} \qquad \underline{\mathrm{Cofibsc}}\, S, \quad \underline{\mathrm{Cofibsc}}_0 S.$$

Of course it suffices to do it on one side, say on the fibered side, by duality ($S \longmapsto S^o$) this will establish the symmetric result. So we are going to look a little more closely at the square of functors

$$\begin{array}{ccc} \underline{\mathrm{Fibsc}}\, S & \xleftarrow{\alpha'} & \underline{\mathrm{Fibsc}}_0 S \\ {\scriptstyle\varphi}\big\downarrow{\scriptstyle p} & & {\scriptstyle p_0}\big\downarrow{\scriptstyle\varphi_0} \\ \underline{\mathrm{Fib}}\, S & \xleftarrow{\alpha} & \underline{\mathrm{Fib}}_0 S \end{array}$$

([37], [38]). The functor

$$\Phi : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fibsc}}\, S$$

induces functors in the reverse direction to p, p_0 on the full subcategories $\underline{\mathrm{Fib}}\, S$, $\underline{\mathrm{Fib}}_0 S$ of Cat/S, functors marked by dotted lines as φ, φ_0.

It is only necessary to check that if X/S is in $\underline{\mathrm{Fib}}_0 S$, then $\varphi(X) = \Phi(X)$ is in the full subcategory $\underline{\mathrm{Fibsc}}_0 S$ of $\underline{\mathrm{Fibsc}}\, S$. But we have seen that this is the case, provided only that X/S is in $\underline{\mathrm{Fibhot}}\, S$, for the base change morphisms in $\Phi(X)$, associated to the arrows $u : s \longrightarrow t$ in S [page 54]

$$u^*_{\Phi_X} : t\backslash X \longrightarrow s\backslash X$$

are W_∞-equivalences if X is Hot-fibered over S. On the other hand, for arbitrary X in Cat/S, one has an arrow, functorial in X,

$$X \longrightarrow \Phi(X),$$

[36] But it is idiotic to take such complicated functors, because one wants to reduce everything to Cat/S. It is moreover simple to compare $\underline{\mathrm{Fibhot}}\, S$ directly with $\underline{\mathrm{Fibsc}}_0 S$ and with $\underline{\mathrm{Cofibsc}}_0 S$. Φ_0 and Ψ_0 (which are in fact defined on $\underline{\mathrm{Fibhot}}$, not just on $\underline{\mathrm{Parf}}$), and one finds as below $\mathrm{id} \xrightarrow{W} \pi_0\Phi_0$ and $\Phi_0\pi_0 \xrightarrow{W} \mathrm{id}$ (likewise for Ψ_0 and $\underline{\mathrm{Cofibsc}}_0 S$), where $\pi_0 : \underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fibhot}}\, S$.

[37] NB It is a matter of proving only that p, p_0 induce equivalences for the localized categories (for coarse W-equivalence). We already know that this is so for α, and it will follow for α'.

[38] needless to look at this square, cf. marginal annotations above (top of the page)

i.e. $\mathrm{id}_{\mathrm{Cat}/S} \longrightarrow \pi\Phi$ where $\pi : \underline{\mathrm{Fibsc}}\, S \longrightarrow \mathrm{Cat}/S$ is the canonical functor, which is a W_∞-equivalence (even a homotopism), hence a W-equivalence. A fortiori the functors $p\varphi$, $p_0\varphi_0$ give rise ($\underline{\mathrm{Fib}}\, S$ and $\underline{\mathrm{Fib}}_0 S$ being full subcategories of Cat/S) to arrows

$$\mathrm{id} \longrightarrow p\varphi, \qquad \mathrm{id} \longrightarrow p_0\varphi_0.$$

It remains to find an arrow (and coarse W-equivalence) p. 34

$$\varphi p \longrightarrow \mathrm{id},$$

which will automatically induce [page 55]

$$\varphi_0 p_0 \longrightarrow \mathrm{id}$$

since $\underline{\mathrm{Fibsc}}_0 S$ is full subcategory of $\underline{\mathrm{Fibsc}}\, S$. In fact, we are going to define $\varphi p \longrightarrow \mathrm{id}$, hence

$$\boxed{\gamma_X : \Phi(X) \longrightarrow X}$$

when X is split fibered over S - an arrow that is *cartesian,* hence (if one likes) compatible with the splittings of $\Phi(X)$, and of X. In fact, we will have such an arrow, functorial in X, for X varying in $\underline{\mathrm{Fib}}\, S$ (thus without restricting ourselves to cartesian S-morphisms between fibered categories over S), but the arrow $\longrightarrow$ being of course *cartesian.* (This makes sense, without choosing a splitting on X, and without there necessarily being one.) The definition of this arrow γ makes essential use of the hypothesis that X be Cat-fibrant over S. Of course, composed with the obvious inclusion arrow

$$X \xrightarrow{i_X} \Phi(X)$$

we are going to have

$$\gamma_X i_X = \mathrm{id}_X$$

([39]). We define γ fiber by fiber,

$$\gamma_{X,s} : \Phi(X)_s \overset{\mathrm{def}}{=} s\backslash X \longrightarrow X_s,$$

thus by a *retraction* of $s\backslash X$ onto the full subcategory X_s. An object of $s\backslash X$ is a pair (x, u), $x \in \mathrm{Ob}\, X$, $u : s \longrightarrow p_X(x)$ ($p_X : X \longrightarrow S$ the structural morphism of X). We put

$$\gamma(x, u) = u_X^*(x) \in \mathrm{Ob}\, X_s,$$

which thus supposes that we have chosen inverse image functors $u_X^* : X_t \longrightarrow X_s$, for the [page 56]
Cat-fibered category X/S. (But up to unique isomorphism, γ does not depend on these choices.) I dispense with the tedious verification that this rule is indeed "functorial in s", i.e. does indeed define a cartesian arrow

$$\gamma_X : \Phi(X) \longrightarrow X,$$

and with verifying that the latter is functorial in X.

Perhaps it is more elegant to show that $\pi : \underline{\mathrm{Fibsc}}\, S \longrightarrow \mathrm{Cat}/S$ and $\Phi : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fibsc}}\, S$ are adjoint to one another: p. 35

$$\boxed{\mathrm{Homcart}_S(\Phi(\underbrace{X}_{\in \mathrm{Cat}/S}), \underbrace{Y}_{\in \underline{\mathrm{Fibsc}}}) \simeq \mathrm{Hom}_S(X, \pi Y),}$$

[39] NB Since we know that $i_X \in W$, it will indeed follow that $\gamma_X \in W$ as we want.

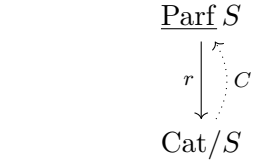
from which there will follow adjunction arrows

$$\Phi\pi \longrightarrow \mathrm{id}_{\underline{\mathrm{Fibsc}}}, \qquad \pi\Phi \longleftarrow \mathrm{id}_{\underline{\mathrm{Fib}}},$$

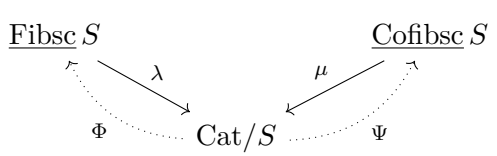
and we will have won. But to prove this adjunction formula, I have the impression that one has to use $\varphi p \longrightarrow \mathrm{id}$ … There are tedious verifications …

Finally, I now see that we have five cases to deal with, and three if we play on the symmetries:

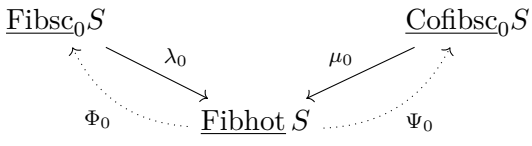
[page 57]



$$\begin{cases} rC & \longleftarrow & \mathrm{id} \\ Cr & \longleftarrow & \mathrm{id} \end{cases}$$



$$\begin{cases} \lambda\Phi & \longleftarrow & \mathrm{id} \\ \Phi\lambda & \longrightarrow & \mathrm{id} \end{cases} \qquad \begin{cases} \mu\Psi & \longleftarrow & \mathrm{id} \\ \Psi\mu & \longrightarrow & \mathrm{id} \end{cases}$$



$$\begin{cases} \lambda_0\Phi_0 & \longleftarrow & \mathrm{id} \\ \Phi_0\lambda_0 & \longrightarrow & \mathrm{id} \end{cases} \qquad \begin{cases} \mu_0\Psi_0 & \longleftarrow & \mathrm{id} \\ \Psi_0\mu_0 & \longrightarrow & \mathrm{id} \end{cases}$$

([40])

NB The last four homomorphisms of functors, relating to the third diagram (with the subscripts 0) are simply induced by those relating to the diagram preceding it. Thus the key verifications relate to the first two diagrams, and by symmetry, in the end we only have to look at the *two* pairs of functors (r, C), (λ, Φ), and for each of these pairs, exhibit the two functorial homomorphisms, of the two composites with the identity functors.

p. 36
[page 58]

② The localized categories for *right fine W-equivalence.* There is no longer any reason, as we have said, to look at the inner categories of the diagram p. 8, where the said equivalence coincides with coarse W-equivalence, for which we already know the localized categories. It remains to look at the categories

$$\mathrm{Cat}/S, \quad \underline{\mathrm{Liss}}\, S, \quad \underline{\mathrm{Fib}}\, S, \quad \underline{\mathrm{Fibsc}}\, S.$$

The lemma on page 11 tells us that it suffices to show that in the situation

$$\underline{\mathrm{Fibsc}}\, S \xrightarrow{\lambda} \mathrm{Cat}/S,$$

there is a functor in the opposite direction, with the specified properties. It will follow that any full subcategory $\mathcal{F}$ of Cat/S which contains $\underline{\mathrm{Fib}}\, S$, when we equip it with the localizer of right fine W-equivalence, gives a localized category equivalent, by the functor induced by the inclusion, to the corresponding localized category of Cat/S, i.e. essentially to $\mathrm{HOT}_W(S)$.

As functor in the opposite direction, it is natural to take Φ again. There are three things to verify:

[40] **NB** All the marked functors are compatible with coarse W-equivalence, and of the ten functorial homomorphisms are coarse W-equivalences.

$$\begin{array}{ccc} X & \xrightarrow{\ f\ } & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

a) Φ compatible with the localizers, i.e. if over S is a right fine W-equivalence, i.e. the $s\backslash X \longrightarrow s\backslash Y$ are W-equivalences, then so is $\Phi(f)$. But the said equivalence coincides in $\underline{\text{Fibsc}}\, S$ (as also in $\underline{\text{Fib}}\, S$ and $\underline{\text{Liss}}\, S$) with W-equivalence on fibers, and on the other hand $\Phi(f)_s$ is none other than the preceding arrow $s\backslash X \longrightarrow s\backslash Y$. Thus it is obvious that [page 59]

$$f \in \underbrace{W_S^{\mathrm{d}}}_{W\text{-equivalence on the right}} \iff \Phi(f) \in \underbrace{W_S^{\mathrm{f}}}_{W\text{-equivalence on fibers}} .$$

b) That $X \xrightarrow{i_X} \lambda\Phi(X)$ is in W_S^{d}, for every X in Cat/S. This thus means that i_X induces

$$s\backslash X \longrightarrow s\backslash \Phi(X)$$

which is a W-equivalence. This would be O.K. if it were clear that $X \longmapsto \Phi(X)$ commutes with every proper base change over S (hence in particular with $s\backslash S \longrightarrow S$), since we know that $X \longrightarrow \Phi(X)$ is a W-equivalence. This is probably asking too much, but at least that it commutes with base change which is a local co-isomorphism (such as the $s\backslash S \longrightarrow S$).

For such a base change $S' \xrightarrow{p} S$, verifying that the fiber of $\Phi(X')$ at s' is isomorphic to that of $\Phi(X)$ at $s = p(s')$, I will be content with that. Now this fiber is $s'\backslash X'$, obtained from X' over S' by the base change $s'\backslash S' \longrightarrow S'$, hence obtained from X by $s'\backslash S' \longrightarrow S$. Now $s'\backslash S' \longrightarrow s\backslash S$ is an isomorphism by the hypothesis on p (local co-isomorphism), hence the base change $s'\backslash S' \longrightarrow S$ is identified with $s\backslash S \longrightarrow S$, and the inverse image is indeed isomorphic to $s\backslash X$, OK. p. 37 [page 60]

c) That likewise

$$\Phi\lambda \longrightarrow \mathrm{id}$$

is not only in W_S (coarse), but that it is a W-equivalence on fibers. So we start with X fibered over S. We look at

$$\Phi(X) \longrightarrow X,$$

to see that it is a W-equivalence on fibers, it suffices to see that this is so for $X \longrightarrow \Phi(X)$,

$$X_s \longrightarrow s\backslash X$$

is a W-equivalence for every $s \in S$. But this is already true as soon as X is smooth over S.

Since the case of left fine W-equivalence reduces to the present case by $S \longmapsto S^o$, we have thus proved all that was expected, with the sole exception of the full faithfulness of the functors i, j on page 51.

But in the prospective summary of the situation for the localizers, p. 49-51, I forgot other functors besides i, j, of a nature if possible even more trivial. They are those which one can describe starting from the identity functor, on one of the categories of the diagram where two, or even three, different notions of W-equivalence are in use. These are, I recall [page 61]

1°) The four categories

$$(*) \qquad \underline{\text{Fibsc}}\, S \longrightarrow \underline{\text{Fib}}\, S \longrightarrow \underline{\text{Liss}}\, S \longrightarrow \mathrm{Cat}/S, \qquad W_S^{\mathrm{d}} \subseteq W$$

with right fine W-equivalence on the one hand, coarse on the other, and dually

$$(*') \qquad \underline{\text{Cofibsc}}\, S \longrightarrow \underline{\text{Cofib}}\, S \longrightarrow \underline{\text{Prop}}\, S \longrightarrow \mathrm{Cat}/S, \qquad W_S^{\mathrm{g}} \subseteq W$$

for left fine W-equivalence on the one hand, coarse on the other, and moreover

2°) Cat/S, which was included in the two preceding sequences, and which is the only category of the diagram giving rise to three different notions of W-equivalence

$$W_S^{\mathrm{d}} \hookrightarrow W_S \hookleftarrow W_S^{\mathrm{g}} \quad .$$

In each of the four cases (C, equipped with two localizers $\Sigma \subseteq \Sigma'$) of the sequence ($*$), the functor p. 38

$$C\Sigma^{-1} \longrightarrow C\Sigma'^{-1}$$

is essentially "the same", because of the comparison theorems. It suffices to look at this functor in a single case, e.g. one of the two extremes $\underline{\mathrm{Fibsc}}\, S$ or Cat/S. Same remark for the sequence ($*'$). So finally there remain two remarkable functors to examine, which do [page 62] not at all look as though they come from the functors of the diagram p. 8 or from some composite (that only gives canonical equivalences or else the functors i, j on page 51, when we go from the inner categories of the diagram to the bordering categories). The prettiest thing is probably to look at them starting from Cat/S, since it is only there that we have both at once

$$\begin{array}{ccc} (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1} \approx \mathrm{HOT}_W(S) & & \\ & \overset{\rho}{\searrow} & \\ & & (\mathrm{Cat}/S)W_S^{-1} \approx \mathrm{HOT}^{\mathrm{lc}}_W(S). \\ & \underset{\sigma}{\nearrow} & \\ (\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1} \approx \mathrm{HOT}^{o}_W(S) & & \end{array}$$

On the other hand, the functors (which we feel like calling "inclusion functors") i, j on page 51, in the opposite direction to ρ, σ, are those which come, by localization at W_S^{d} or W_S^{g} as the case may be, from one of the functors of the diagram page 8 (or from a composite of such functors), going from one of the six inner categories (where all the W-equivalences considered coincide) to one of the seven bordering categories, except when the latter is Cat/S, is already determined with which W-equivalence, W_S^{d} or W_S^{g}, we are going to work . The determi-nation that is perhaps the prettiest is by [page 63]

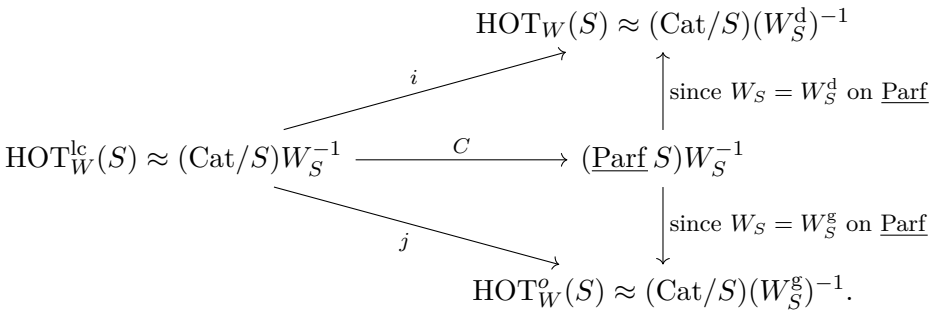
$$(*) \qquad \underline{\mathrm{Parf}}\, S \longrightarrow \mathrm{Cat}/S$$

which gives the functor i when we work with W_S^{d}, the functor j when we work with W_S^{g}. We then see at once that if we compose with the relevant functor, either ρ or σ, defined by the inclusion $W_S^{\mathrm{d}} \subseteq W_S$ resp. $W_S^{\mathrm{g}} \subseteq W_S$ (as subsets of $\mathrm{Fl}(\mathrm{Cat}/S)$), we find the functor

$$(**) \qquad (\underline{\mathrm{Parf}}\, S)W_S^{-1} \xrightarrow{\approx} (\mathrm{Cat}/S)W_S^{-1}$$

which is the *canonical equivalence* between these two categories. Thus the functor corresponding to i, j page 51, expressed in terms of Cat/S as a functor in the opposite direction to ρ, σ (p. 62), is the composite of the equivalence quasi-inverse to ($**$), which is made explicit by the functor C, and of the functor on the category localized at W_S^{d} or W_S^{g} obtained from ($*$), i.e.

p. 39



We see from this that we have canonical isomorphisms [page 64]

$$\rho i \simeq \mathrm{id}, \qquad \sigma j \simeq \mathrm{id}$$

in the category of endofunctors of $(\mathrm{Cat}/S)W_S^{-1}$ ($\approx \mathrm{HOT}_W^{\mathrm{lc}}(S)$), an isomorphism which is made explicit here by the fact that ρi, σj are deduced from the same functor C by passage to the localized categories W_S^{-1} (so in fact $\rho i = \sigma j$!), and that we have an arrow functorial in X

$$X \xrightarrow{i_X} C(X),$$

which is in W for every X. Thus, ρ and σ are, up to isomorphism, *retractions* of the categories $\mathrm{HOT}_W(S)$, $\mathrm{HOT}_W^{o}(S)$ onto the category $\mathrm{HOT}_W^{\mathrm{lc}}(S)$. Moreover, these retractions are what are called localization functors (as we see from the diagram on page 62). (We say that $C \xrightarrow{F} C'$ is a *localization functor* if, denoting by $\Sigma \subseteq \mathrm{Fl}\, C$ the set of arrows of C transformed into isomorphisms by F, the functor

$$C\Sigma^{-1} \xrightarrow{\bar{F}} C'$$

which factors F is an equivalence of categories.)

We know, moreover, that if a functor f has an adjoint g (left or right), then f is a functor localization if and only if the adjoint g is fully faithful. To prove that i, j are fully faithful, it would therefore be tempting to prove that they are adjoint to the functors ρ, σ. [page 65]

To get our bearings, I shall set

$$\begin{cases} \mathcal{A} = \mathrm{Cat}/S, \quad \Sigma_0 = W_S^{\mathrm{d}}, \quad \Sigma = W_S \qquad (\subseteq \mathrm{Fl}(\mathcal{A})) \\ C : \mathcal{A} \longrightarrow \mathcal{A} \quad \text{the familiar functor, which has the virtue} \\ \qquad\qquad C(\Sigma) \subseteq \Sigma_0 \quad . \end{cases}$$

We thus have the situation

$$(*) \qquad (\mathcal{A}, \Sigma_0) \underset{C}{\overset{\rho_0 = \mathrm{id}_{\mathcal{A}}}{\rightleftarrows}} (\mathcal{A}, \Sigma), \qquad \text{p. 40}$$

inducing by passage to the localized categories

$$\mathcal{A}\Sigma_0^{-1} \underset{\bar{C}}{\overset{\rho = \bar{\rho}_0}{\rightleftarrows}} \mathcal{A}\Sigma^{-1},$$

we want to check that ρ and C are adjoint to one another, C *surely on the right,* which means that we must have functorial homomorphisms

$$\begin{cases} \mathrm{id}_{\mathcal{A}\Sigma^{-1}} & \rightleftarrows & \rho\,\bar{C} & (= \overline{\rho_0 C}) \\ \mathrm{id}_{\mathcal{A}\Sigma_0^{-1}} & \rightleftarrows & \bar{C}\rho & (= \overline{C\rho_0}) \end{cases}$$

([41]) satisfying two compatibility relations to which we shall return. We may hope to find these functorial homomorphisms starting from

$$\begin{cases} \mathrm{id}_{\mathcal{A}} & \xrightarrow{i} & \rho_0 C & = & C \\ \mathrm{id}_{\mathcal{A}} & \xleftarrow{p} & C\rho_0 & = & C \end{cases}$$

and in our case we do indeed have [page 66]

$$i : \mathrm{id}_{\mathcal{A}} \longrightarrow C \qquad \text{i.e.} \quad i_X : X \longrightarrow C(X)$$

functorial in $X \in \mathrm{Ob}\,\mathcal{A}$ (and moreover i_X is in Σ). There is however no natural arrow in the opposite direction p_X

$$C(X) \xrightarrow{?} X.$$

I had thought of the projection $\pi_X : C(X) \longrightarrow X$, deduced from the construction $C(X) \simeq X \times_S (\underline{\mathrm{Ch}}_\infty(S), \sigma)$, but it is unfortunately not an S-morphism. As we are looking for an arrow in $\mathcal{A}\Sigma_0^{-1}$, we could make do with having a functorial arrow in the opposite direction, provided it is in Σ_0, hence can be inverted when we work in $\mathcal{A}\Sigma_0^{-1}$. The obvious candidate is $i_X : X \longrightarrow C(X)$ which we have just spoken of. The trouble is that i_X is only in Σ, not in Σ_0 ([42]). Yet, when X is itself of the form $C(Y)$ (more generally, if it is Hot-fibered over S), then the fact that $X \longrightarrow C(X)$ is in Σ already implies that it is in Σ_0. Let us therefore note this

$$\begin{cases} i : \mathrm{id}_{\mathcal{A}} \longrightarrow C \text{ such that} & 1^o) & i_X : X \longrightarrow C(X) \text{ in } \Sigma \\ & & \text{for every } X \in \mathrm{Ob}\,\mathcal{A} \\ & 2^o) & i_X \text{ in } \Sigma_0 \text{ if } X \text{ of the} \\ & & \text{form } C(Y) \end{cases}$$

The idea comes to me, after some shitty attempts , that $\bar{C}$ might be the right adjoint, and not the left , i.e. that the arrows sought might go in the opposite direction p. 41 [page 67]

$$\begin{cases} \mathrm{id}_{\mathcal{A}\Sigma^{-1}} & \xleftarrow{p} & \rho\bar{C} & (= \overline{\rho_0 C}) \\ \mathrm{id}_{\mathcal{A}\Sigma_0^{-1}} & \xrightarrow{i} & \bar{C}\rho & (= \overline{C\rho_0}) \end{cases}$$

(so I reverse the notations for p and i), thus we are led if possible to look for arrows

$$\begin{aligned} \mathrm{id}_{\mathcal{A}} & \xleftarrow{p} & \rho_0 C & = & C \\ \mathrm{id}_{\mathcal{A}} & \xrightarrow{i} & C\rho_0 & = & C, \end{aligned}$$

thus arrows functorial in X

$$X \xrightarrow{i_X} C(X) \xrightarrow{p_X} X.$$

We do indeed have a functorial i_X - always the same one! For p_X, we have no more than before, but recalling that it suffices to find it in $\mathcal{A}\Sigma^{-1}$ (and no longer $\mathcal{A}\Sigma_0^{-1}$!), it suffices to invert the arrow i_X, which is indeed an *isomorphism* in $\mathcal{A}\Sigma^{-1}$ (but not in $\mathcal{A}\Sigma_0^{-1}$). We shall therefore set

$$p_X = i_X^{-1} : C(X) \longrightarrow X \qquad \text{in } \mathcal{A}\Sigma^{-1}.$$

[41] but if C is on the *right,* the arrows here must go in the opposite direction and this is indeed what I must resign myself to on p. 67.

[42] If this were not so, the relative homotopy type over S defined by X/S would be locally constant (since this is so for the one defined by $C(X)$), which is not always the case.

We have to prove just two compatibilities, for the functors $\bar{C}$ and ρ between $\mathcal{A}\Sigma_0^{-1}$ and $\mathcal{A}\Sigma^{-1}$ (and I shall write C instead of $\bar{C}$): the two composites

$$\begin{cases} \rho \xrightarrow{\rho * i} \rho C \rho \overset{\mathrm{id}}{\underset{p*\rho}{\rightrightarrows}} \rho & \text{(relation for functors with values in } \mathcal{A}\Sigma^{-1}) \\ C \xrightarrow{i*C} C\rho C \overset{\mathrm{id}}{\underset{C*p}{\rightrightarrows}} C & \text{(relation for functors with values in } \mathcal{A}\Sigma_0^{-1}) \end{cases}$$

must be the identity id_ρ resp. id_C.

The first relation is checked tautologically, it amounts to saying that in $\mathcal{A}\Sigma^{-1}$, we have an identity composite of the [page 68]

$$X \xrightarrow{i_X} C(X) \xrightarrow{i_X^{-1}} X \qquad !$$

The second means that for every X, we have an identity composite in $\mathcal{A}\Sigma_0^{-1}$

$$C(X) \xrightarrow{i_{C(X)}} C(C(X)) \xrightarrow{C(p_X)} C(X).$$

Now $p_X : C(X) \longrightarrow X$ had been defined in $\mathcal{A}\Sigma^{-1}$ as the inverse i_X^{-1} of i_X, on the other hand the functor C on $\mathcal{A}$ defines by hypothesis $\mathcal{A}\Sigma^{-1} \longrightarrow \mathcal{A}\Sigma_0^{-1}$, more precisely $C(\Sigma) \subseteq \Sigma_0$ thus $C(i_X)$ is invertible in $\mathcal{A}\Sigma_0^{-1}$, and $C(p_X) = C(i_X)^{-1}$. So the compatibility p. 42 amounts to the relation

$$C(i_X)^{-1} i_{C(X)} = \mathrm{id} \qquad \textit{in} \quad \mathcal{A}\Sigma_0^{-1},$$

i.e.

$$\boxed{i_{C(X)} = C(i_X)} : C(X) \longrightarrow C^2(X).$$

In other words, we must prove the:

Lemma. *Let X be in* Cat/S*, let us consider the two arrows*

$$i_{C(X)},\ C(i_X) : C(X) \rightrightarrows C(C(X)).$$

These arrows are equal, if not in the absolute sense, at least in $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$.

Here $f_X : X \longrightarrow S$ is the standard projection and

$$C(X) = \{(x, c) \mid x \in \mathrm{Ob}\, X,\ c : f_X(x) \underbrace{\longrightarrow}_{\text{path in } S} t\}$$

$$C^2(X) = \{(x, c, c') \mid x \in \mathrm{Ob}\, X,\ \underbrace{f_X(x) \xrightarrow{c} t \xrightarrow{c'} t'}_{c,\ c' \text{ paths in } S}\},$$

the projections to S are given by

$$\begin{array}{rcrcl} (x, c) & \longmapsto & t & = & \text{target } c \\ (x, c, c') & \longmapsto & t' & = & \text{target } c' \end{array}$$

We find [page 69]

$$i_{C(X)}(x, c) = (x, c, 1_t) : \quad \begin{array}{c} x \\ \downarrow \\ f_X(x) \end{array} \xrightarrow{c} t \underset{\text{empty path}}{\xrightarrow{1_t}} t$$

$$C(i_X)(x,c) = (x, 1_{f_X(x)}, c) : \quad \begin{array}{c} x \\ \downarrow \\ f_X(x) \xrightarrow[\text{empty path}]{1_{f_X(x)}} f_X(x) \xrightarrow{c} t \end{array}$$

But we have a choice between two theories of paths, to construct $C(X)$: either with the $\underline{\mathrm{Ch}}_\infty(S)$, or with the $\underline{\mathrm{Ch}}(S)$. "It gives the same results", but depending on the situation one is more convenient than the other. I shall work with the $\underline{\mathrm{Ch}}(S)$, in which case the 1_S are left and right units, and we have a composition law for paths. This law allows us to define

$$\begin{array}{ccc} C^2(X) & \xrightarrow{\gamma} & C(X) \\ (x,c,c') & \longmapsto & (x, c' \circ c), \end{array}$$

and the composites of $i_{C(X)}$ and of $C(i_X)$ with this arrow are $\mathrm{id}_{C(X)}$ in both cases p. 43

$$C(X) \rightrightarrows C^2(X) \xrightarrow{\gamma} C(X). \qquad \gamma \circ \text{(each)} = \mathrm{id} \text{ (id)}$$

So to show that $i_{C(X)} = C(i_X)$ in $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$, it suffices to prove that γ is in W_S^{d}. But we know that the functor C transforms W_S into W_S^{d}, so $C(i_X)$ is in W_S^{d}, and since $\gamma \circ C(i_X) = \mathrm{id}$, γ is in W_S^{d}, OK!

I feel like grouping what has been proved into a summary statement. [page 70]

Theorem-Scholie. We consider the diagram of categories on page 8 ([43]).

Ⅰ When we equip all 13 categories of the diagram with the localizer W_S (coarse W-equivalence, i.e. W-equivalence in Cat, nothing more), *all* the functors of the diagram induce equivalences on the localized categories ([44]). Let us denote by

$$\mathrm{HOT}_W^{\mathrm{lc}}(S)$$

"the" common localized category.

Ⅱ Let us equip the four left bordering categories

$$(*) \qquad \mathrm{Cat}/S \longleftarrow \underline{\mathrm{Liss}}\, S \longleftarrow \underline{\mathrm{Fib}}\, S \longleftarrow \underline{\mathrm{Fibsc}}\, S$$

with the localizer W_S^{d} of *right* fine W-equivalence over S ($\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$ is in W_S^{d} if and only if the $s\backslash X \longrightarrow s\backslash Y$ are , for every $s \in \mathrm{Ob}\, S$). The functors of the diagram, i.e. those of $(*)$, between these categories, hence also their composites, induce equivalences between the localized categories ([45]). Let us denote by

$$\mathrm{HOT}_W(S)$$

"the" common localized category.

Dual statement for

$$(*') \qquad \mathrm{Cat}/S \longleftarrow \underline{\mathrm{Prop}}\, S \longleftarrow \underline{\mathrm{Cofib}}\, S \longleftarrow \underline{\mathrm{Cofibsc}}\, S,$$

---

[43] We assume for W the conditions a) b) c) of p. 45, in addition L5 (p. 43). Cf. comments p. 81.

[44] Moreover, if $\mathcal{F}$ is a full subcategory of Cat/S which contains one of the two subcategories $\underline{\mathrm{Fib}}_0 S$, $\underline{\mathrm{Cofib}}_0 S$ (which includes the case of the 8 categories of the diagrams, other than Cat/S itself), the inclusion $(\mathcal{F}, W_S|\mathcal{F})$ into $(\mathrm{Cat}/S, W_S)$ induces an equivalence for the localized categories. p. 44

[45] More generally, this works for every full subcategory of Cat/S which contains $\underline{\mathrm{Fib}}\, S$ (and dually, with W_S^{g} instead of W_S^{d}, if $\mathcal{F}$ contains $\underline{\mathrm{Cofib}}\, S$).

and the localizers W_S^{g} in these categories ($f \in W_S^{\mathrm{g}}$ if and only if the $f/s : X/s \longrightarrow Y/s$ are in W). We find a "same localized category", denoted

$$\mathrm{HOT}_W^o(S),$$

and we have tautologically [page 71]

$$\mathrm{HOT}_W^o(S) \approx \mathrm{HOT}_W(S^o).$$

(III) On the six inner categories of the diagram, the three notions of W-equivalence, in terms of W_S, W_S^{d}, W_S^{g} coincide, and coincide with fiberwise W-equivalence W_S^{f}. On the three categories of $(*)$ other than Cat/S, the localizers W_S^{f}, W_S^{d} coincide; on the three categories of $(*')$ other than Cat/S, the localizers W_S^{f}, W_S^{g} coincide. But in general ([46]), on these six categories, these localizers W_S^{f} are not equal to W_S, more precisely give nonequivalent localized categories, they are therefore each equipped with *two* essentially distinct localizers, W_S^{f} and W_S. The category Cat/S is the only category in the diagram on which the localizers W_S, W_S^{d} and W_S^{g} are all three essentially distinct. It is the only one that allows one to obtain the three localized categories simultaneously, as

$$\begin{array}{ccccc} & & \mathrm{HOT}_W^{\mathrm{lc}}(S) \simeq (\mathrm{Cat}/S)W_S^{-1} & & \\ & \overset{\rho}{\nearrow} & & \overset{\sigma}{\nwarrow} & \\ \mathrm{HOT}_W(S) \simeq (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1} & & & & \mathrm{HOT}_W^o(S) \simeq (\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1} \end{array}$$

Here, ρ and σ are localization functors (as is evident from the definition).

Any functor or composite functor in the diagram on page 8, going from an inner category of the diagram to a bordering category, induces a functor on the localized categories for W_S^{d} (when the target category is one of the four on the left, recalled in $(*)$, p. 70), or for W_S^{g} (when these categories are among the four on the right-hand side, recalled in $(*')$), which localizers also coincide with W_S^{f} except in the sole case of the category Cat/S. All the functors thus obtained reduce, up to equivalence, to two, according as the target localized category is $\mathrm{HOT}_W(S)$ (case of localization W_S^{d}), or is $\mathrm{HOT}_W^o(S)$ (case of localization W_S^{g}). We thus find functors (in the direction opposite to ρ and to σ) [page 72] p. 45

$$\begin{array}{ccc} & \mathrm{HOT}_W^{\mathrm{lc}}(S) & \\ \overset{i}{\swarrow} & & \overset{j}{\searrow} \\ \mathrm{HOT}_W(S) & & \mathrm{HOT}_W^o(S). \end{array}$$

The functors ρ, σ are interchanged by duality $S \longmapsto S^o$, and likewise for i, j.

(IV) The functors ρ, i are adjoint to one another in that order, i.e. we have for two objects $\xi \in \mathrm{Ob}\,\mathrm{HOT}_W(S)$, $\eta \in \mathrm{Ob}\,\mathrm{HOT}_W^{\mathrm{lc}}(S)$

$$\mathrm{Hom}(\rho(\xi), \eta) \simeq \mathrm{Hom}(\xi, i(\eta))$$

(isomorphism of bifunctors). And dually the functors σ, j are adjoint to one another in that order: for $\xi \in \mathrm{Ob}\,\mathrm{HOT}_W^o(S)$, $\eta \in \mathrm{Ob}\,\mathrm{HOT}_W^{\mathrm{lc}}(S)$

$$\mathrm{Hom}(\xi, j(\eta)) \simeq \mathrm{Hom}(\sigma(\xi), \eta).$$

It follows in particular that *the functors i, j are fully faithful.* [page 73]

[46] probably in all cases where S is not equivalent to a discrete category.

Ⓥ We have obvious functors

$$\begin{array}{lccc} (*) & \mathrm{HOT}_W(S) & \longrightarrow & \underline{\mathrm{Hom}}(S^o, \mathrm{Hot}_W) \\ (*') & \mathrm{HOT}^o_W(S) & \longrightarrow & \underline{\mathrm{Hom}}(S, \mathrm{Hot}_W). \end{array}$$

The first is interpreted on the four categories $(*)$ (bordering on the left) on page 70, which serve indifferently to describe $\mathrm{HOT}_W(S)$, by

$$s \longmapsto \mathrm{hot}_W(X_s)$$

in the case of the three categories distinct from Cat/S, and by

$$s \longmapsto \mathrm{hot}_W(s\backslash X)$$

in the case of Cat/S. The functorial dependence on s is obvious in the case of Cat/S (given by $s \longmapsto \mathrm{hot}_W(s\backslash X)$), and in the case of $\underline{\mathrm{Fibsc}}\, S$, $\underline{\mathrm{Fib}}\, S$, where we have, for $u : s \longrightarrow t$ in S, the base change

$$u^* : X_t \longrightarrow X_s.$$

In the case of $\underline{\mathrm{Liss}}\, S$, the functorial dependence is deduced from that of $\mathrm{hot}_W(s\backslash X)$, thanks to the fact that the inclusion

$$X_s \longrightarrow s\backslash X$$

is then a W-equivalence.

We have a dual interpretation of the map $(*)'$, [...] p. 46 [page 74]

$$s \longmapsto \mathrm{hot}_W(X_s) \qquad \text{or} \qquad s \longmapsto \mathrm{hot}_W(X/s),$$

with the functorial law in s deduced from the functor $t\backslash X \longrightarrow s\backslash X$ (for $u : s \longrightarrow t$) in the case of Cat/S, from the cobase change functor in the cases of $\underline{\mathrm{Cofib}}\, S$ and $\underline{\mathrm{Cofibsc}}\, S$, and from the variant for Cat/S and from the W-equivalence

$$X_s \longrightarrow X/s$$

in the case of $\underline{\mathrm{Prop}}\, S$.

We then find this: for an object ξ of $\mathrm{HOT}_W(S)$ (resp. $\mathrm{HOT}^o_W(S)$) to be in the essential image of $\mathrm{HOT}^{\mathrm{lc}}_W(S)$, it is necessary and sufficient that the associated functor $S^o \longrightarrow \mathrm{Hot}_W$ (resp. $S \longrightarrow \mathrm{Hot}_W$) be a local system ([47]), i.e. transform every arrow of S into an isomorphism, i.e. factor through $(\Pi S)^o \longrightarrow \mathrm{Hot}_W$ resp. $\Pi S \longrightarrow \mathrm{Hot}_W$ ([48]). Thus for an X in Cat/S, this means either that the functors

$$t\backslash X \longrightarrow s\backslash X$$

associated to $u : s \longrightarrow t$ in S are all W-equivalences (case of $\mathrm{HOT}_W(S)$), or that the functors

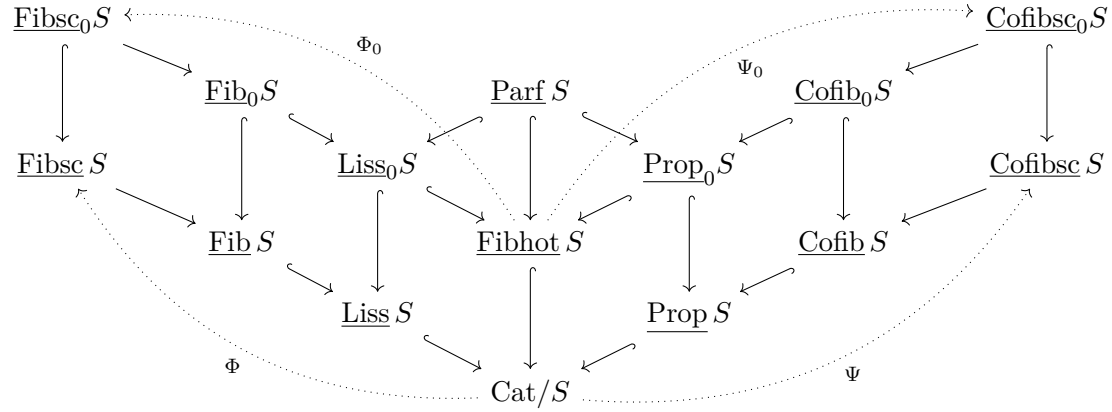
$$X/s \longrightarrow X/t$$

are (case of $\mathrm{HOT}^o_W(S)$). (NB. these conditions are by no means equivalent. But if X smooth S, the first implies the second, and the converse is true if X is proper over S, and [page 75]

[47] the verification will still have to be written

[48] where ΠS is the fundamental groupoid of S

in this case, the condition is equivalent to X being in $\underline{\mathrm{Fibhot}}$. Finally, if X is already in $\underline{\mathrm{Fibhot}}\, S$, then the two conditions are automatically satisfied ([49]).



([50]). Moreover, if two elements $\xi \in \mathrm{Ob}\,\mathrm{HOT}_W(S)$, $\xi' \in \mathrm{Ob}\,\mathrm{HOT}^{o}_W(S)$ come from the same $\eta \in \mathrm{Ob}\,\mathrm{HOT}^{\mathrm{lc}}_W(S)$, then the two corresponding functors p. 47

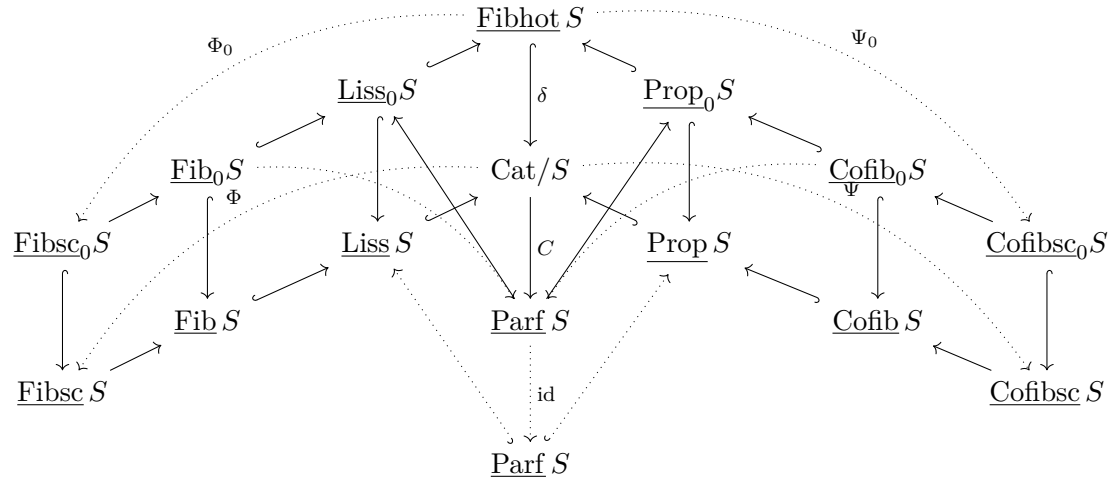
$$(\Pi S)^o \xrightarrow{\varphi} \mathrm{Hot}_W \xleftarrow{\varphi'} \Pi S$$

are *associated,* i.e. each is isomorphic to "the inverse" of the other.

$$\left\{ \begin{array}{lcl} \varphi^o(s) & = & \varphi(s) \\ \varphi^o(u) & = & \varphi(u)^{-1}. \end{array} \right.$$

(VI) NB In the new version of the diagram, where I have added the categories $\underline{\mathrm{Liss}}_0 S$ and $\underline{\mathrm{Prop}}_0 S$ to make the drawing more transparent, the categories called "bordering" become the "ground floor", and the "inner" categories, the "first upper floor" or "upper floor". All 7 squares of the diagram are cartesian. All the arrows are fully faithful inclusions, except the four outer horizontal arrows from the categories of Cat-fibered or Cat-cofibered categories that are *split*, with morphisms exclusively cartesian or cocartesian and even compatible with the splittings, to the corresponding categories of nonsplit structures.

[page 76]



[49] We should also have introduced in the diagram $\underline{\mathrm{Liss}}_0 S$, $\underline{\mathrm{Prop}}_0 S$!

[50] The diagram of *fifteen* remarkable categories.

I have drawn in green dotted lines the five functors Φ, Ψ, Φ_0, Ψ_0, C which by passage to the localized categories give quasi-inverses for all the equivalences of categories stated in Ⓘ Ⓘ of the present Scholie. To be precise, the functor

$$\Phi : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fibsc}}\, S$$

associates to every X in Cat/S the split category over S

$$s \longmapsto s\backslash X$$

over S. This functor, and its restrictions to $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Fib}}\, S$ and, more generally, to any p. 48
full subcategory $\mathcal{F}$ of Cat/S containing $\underline{\mathrm{Fib}}\, S$, is compatible with the localizers W_S on the one hand (on $\mathcal{F}$ and $\underline{\mathrm{Fibsc}}\, S$), W_S^{d} on the other hand, and therefore passes to the corresponding localized categories, to provide the desired quasi-inverse of

$$(\underline{\mathrm{Fibsc}}\, S)\Sigma^{-1} \longrightarrow \mathcal{F}\Sigma^{-1} \qquad (\Sigma = W_S \text{ or } W_S^{\mathrm{d}})$$

We can also, for any intermediate full subcategory $\mathcal{F}'$

$$\underline{\mathrm{Fib}}\, S \subseteq \mathcal{F}' \subseteq \mathcal{F} \subseteq \mathrm{Cat}/S,$$

regard Φ as a functor $\mathcal{F} \longrightarrow \mathcal{F}'$ (more precisely, take the composite

$$\mathcal{F} \xrightarrow{\Phi|\mathcal{F}} \underline{\mathrm{Fibsc}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S \hookrightarrow \mathcal{F}'\ ),$$

say $\Phi_{\mathcal{F}',\mathcal{F}}$, and from what has just been said it follows that the two functors

$$\mathcal{F}' \underset{\Phi_{\mathcal{F}',\mathcal{F}}}{\overset{}{\hookrightarrow}} \mathcal{F}$$

establish equivalences quasi-inverse to one another [page 77]

$$\mathcal{F}'\Sigma^{-1} \underset{\Phi_{\mathcal{F}',\mathcal{F}}}{\overset{\approx}{\longrightarrow}} \mathcal{F}\Sigma^{-1} \qquad (\Sigma = W_S \text{ or } W_S^{\mathrm{d}})$$

Dual statement for the functor

$$\Psi : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Cofibsc}}\, S,$$

associating to X in Cat/S the split cofibered category over S

$$s \longmapsto X/s.$$

It gives pairs of quasi-inverse functors

$$(\underline{\mathrm{Cofibsc}}\, S)\Sigma^{-1} \underset{\Psi_{\mathcal{F}}}{\overset{}{\longrightarrow}} \mathcal{F}\Sigma^{-1} \qquad (\Sigma = W_S \text{ or } W_S^{\mathrm{g}})$$

for any full subcategory $\mathcal{F}$ of Cat/S such that

$$\underline{\mathrm{Cofib}}\, S \subseteq \mathcal{F} \subseteq \mathrm{Cat}/S,$$

and pairs of quasi-inverse functors p. 49

$$\mathcal{F}'\Sigma^{-1} \underset{\Psi_{\mathcal{F}',\mathcal{F}}}{\overset{\approx}{\rightleftarrows}} \mathcal{F}\Sigma^{-1} \qquad (\Sigma = W_S \text{ or } W_S^{\mathrm{g}})$$

for two full subcategories $\mathcal{F}'$, $\mathcal{F}$

$$\underline{\mathrm{Cofib}}\, S \subseteq \mathcal{F}' \subseteq \mathcal{F} \subseteq \mathrm{Cat}/S.$$

The functors Φ, Ψ send the full subcategory $\underline{\mathrm{Fibhot}}\, S$ of "the upper floor", into the subcategories $\underline{\mathrm{Fibsc}}_0 S$, $\underline{\mathrm{Cofibsc}}_0 S$, respectively, hence induce on the upper floor functors Φ_0, Ψ_0

$$\underline{\mathrm{Fibsc}}_0 S \xleftarrow{\Phi_0} \underline{\mathrm{Fibhot}}\, S \xrightarrow{\Psi_0} \underline{\mathrm{Cofibsc}}_0 S,$$

playing on the upper floor exactly the same role as Φ, Ψ on the ground floor. The only difference is that on the upper floor we have $W_S = W_S^{\mathrm{d}} = W_S^{\mathrm{g}}$ $(= W_S^{\mathrm{f}})$, so there is only one sequence of localized categories to examine. We thus find [page 78]

$$(\underline{\mathrm{Fibsc}}_0 S)W_S^{-1} \underset{\Phi_{0,\mathcal{F}}}{\overset{\approx}{\rightleftarrows}} \mathcal{F}W_S^{-1}$$

$$\mathcal{F}'W_S^{-1} \underset{\Phi_{0,\mathcal{F}',\mathcal{F}}}{\overset{\approx}{\rightleftarrows}} \mathcal{F}W_S^{-1}$$

for $\mathcal{F}$ (or $\mathcal{F}$, $\mathcal{F}'$) full subcategory(ies) of $\underline{\mathrm{Fibhot}}\, S$ such that

$$\underline{\mathrm{Fib}}_0 S \subseteq \mathcal{F} \subseteq \underline{\mathrm{Fibhot}}\, S \qquad \text{resp.} \qquad \underline{\mathrm{Fib}}_0 S \subseteq \mathcal{F}' \subseteq \mathcal{F} \subseteq \underline{\mathrm{Fibhot}}\, S,$$

and likewise on the right-hand side for the full subcategories $\mathcal{F}$ or $\mathcal{F}$, $\mathcal{F}'$ of $\underline{\mathrm{Fibhot}}\, S$ satisfying

$$\underline{\mathrm{Cofib}}_0 S \subseteq \mathcal{F} \subseteq \underline{\mathrm{Fibhot}}\, S \qquad \text{resp.} \qquad \underline{\mathrm{Cofib}}_0 S \subseteq \mathcal{F}' \subseteq \mathcal{F} \subseteq \underline{\mathrm{Fibhot}}\, S,$$

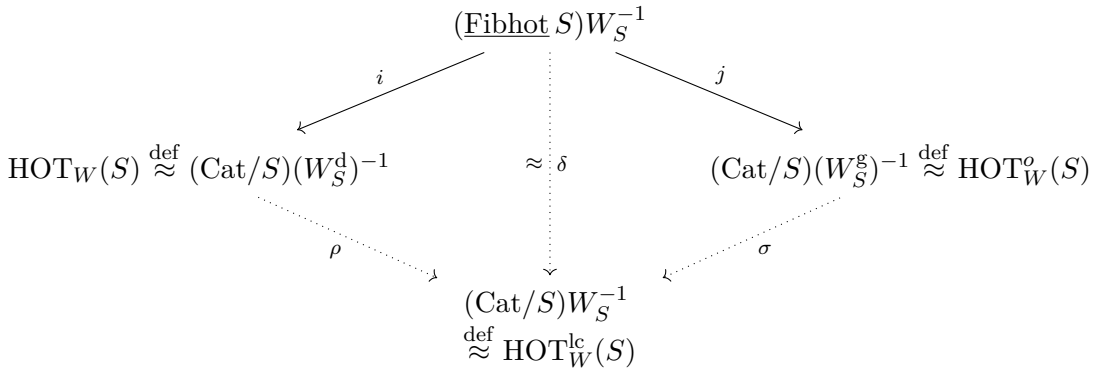
NB. The category $\underline{\mathrm{Parf}}\, S$ contains both $\underline{\mathrm{Fib}}_0 S$ and $\underline{\mathrm{Cofib}}_0 S$, and is therefore among the admissible categories just like $\mathcal{F}$, or $\mathcal{F}'$.

Thus, the four functors Φ, Ψ, Φ_0, Ψ_0 give all the quasi-inverses one can wish for on the ground floor on the one hand (Φ and Ψ), on the upper floor on the other (Φ_0, Ψ_0). They therefore reduce all the relevant localized categories to the following four

$$(\mathrm{Cat}/S)W_S^{-1},\ \ (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1},\ \ (\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1} \qquad \text{(on the ground floor)}$$
$$(\underline{\mathrm{Fibhot}}\, S)W_S^{-1} \qquad \text{(on the upper floor).}$$

(VII) It remains to compare the localized category on the upper floor with the localized categories on the ground floor. The vertical arrows descending from the upper floor to the ground floor, are compatible with W_S^{d} on the left-hand side of the diagram, with W_S^{g} on the right-hand side, so they induce essentially *two* different functors i, j on p. 50 [page 79]

$(\underline{\text{Fibhot}}\, S)W_S^{-1}$, into $(\text{Cat}/S)(W_S^{\text{d}})^{-1}$ and $(\text{Cat}/S)(W_S^{\text{g}})^{-1}$:



,

so as to make the square completed by the dotted lines commute, representing the localization functors ρ, σ. We have of course

$$(*) \qquad \rho i = \sigma j : (\underline{\text{Fibhot}}\, S)W_S^{-1} \longrightarrow (\text{Cat}/S)W_S^{-1}$$
induced by the inclusion
$$\delta : \underline{\text{Fibhot}}\, S \hookrightarrow \text{Cat}/S$$

We have stated in Ⓘ that this arrow is an equivalence, so that ρ and σ can be interpreted, up to isomorphism of functors, as retractions of $\text{HOT}_W(S)$ and of $\text{HOT}^{\text{o}}_W(S)$ onto a common subcategory, interpreted either as $(\underline{\text{Fibhot}}\, S)W_S^{-1}$, or as $(\text{Cat}/S)W_S^{-1}$. It remains to describe the quasi-inverse functor of $(*)$, which will at the same time give the quasi-inverses of all the vertical functors (from the upper floor to the ground floor) in the diagram p. 76, more precisely for their localized versions, for W_S.

[page 80]

This is what is obtained with the help of the "ascending" functor (from the ground floor to the upper floor) C, which one can regard as a functor $\text{Cat}/S \longrightarrow \underline{\text{Fibhot}}\, S$, but which in fact takes its values in $\underline{\text{Parf}}\, S$. (This is a detail of no great importance, it seems to me.)

$$\text{Cat}/S \xrightarrow{C} \underline{\text{Parf}}\, S \hookrightarrow \underline{\text{Fibhot}}\, S.$$

This functor is compatible with the localizers W_S on both sides, and establishes equivalences (quasi-inverses of the equivalences

$$(\underline{\text{Parf}}\, S)W_S^{-1} \xrightarrow{\sim} (\text{Cat}/S)W_S^{-1} \qquad \text{and} \qquad (\underline{\text{Fibhot}}\, S)W_S^{-1} \xrightarrow{\sim} (\text{Cat}/S)W_S^{-1}),$$
$$(\text{Cat}/S)W_S^{-1} \xrightarrow{\sim} (\underline{\text{Parf}}\, S)W_S^{-1} \left(\xrightarrow{\sim} (\underline{\text{Fibhot}}\, S)W_S^{-1}\right).$$

So, when the three HOT categories for S are interpreted in terms of localized categories of Cat/S, it is the functor C (more precisely the composite $\text{Cat}/S \xrightarrow{C} \underline{\text{Parf}}\, S \hookrightarrow \text{Cat}/S$) that provides both the inclusion functor (!) i, and the inclusion functor j into $\text{HOT}_W(S)$ and $\text{HOT}^{\text{o}}_W(S)$ respectively. The functor in question

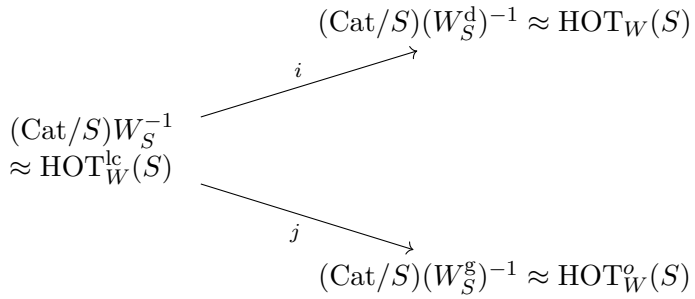
p. 51

$$C' : \text{Cat}/S \longrightarrow \text{Cat}/S$$

does indeed satisfy

$$C'(W_S) \subseteq W_S^{\text{d}} \cap W_S^{\text{g}}$$

and defines the two functors



This allows us to make explicit the adjunction formulas of (IV), when we interpret ξ, η there as coming from X, Y in Cat/S:

[page 81]

$$\boxed{\mathrm{Hom}_{W_S^{\mathrm{d}}}(X, CY) \simeq \mathrm{Hom}_{W_S^{\mathrm{g}}}(X, CY) \simeq \mathrm{Hom}_{W_S}(X, Y),}$$

where $\mathrm{Hom}_{W_S^{\mathrm{d}}}$, $\mathrm{Hom}_{W_S^{\mathrm{g}}}$ and Hom_{W_S} denote respectively the Hom in the localized categories for W_S^{d}, W_S^{g} and W_S, i.e. in $\mathrm{HOT}_W(S)$, $\mathrm{HOT}^o_W(S)$ and $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ respectively ([51]).

Comment. All the constructions and statements from (I) to (VI), with the exception of the fact that the descending functors induce equivalences for the *coarse* localized categories, and of the adjunction formulas of (IV), are of a virtually sorital nature, and require no conditions on the basic localizer $W \subseteq \mathrm{Fl}\,\mathrm{Cat}$ other than L1, L3 quater, L4 bis, summarized as a) b) c) on page 45. (We do not even have to use d).) It is true that in the absence of condition L5 (page 43), we do not know whether on the upper floor the two relevant localizers, W_S and W_S^{f} (equal to W_S^{d} on the left-hand side, to W_S^{g} on the right-hand side) are equal. Thus a priori we find on the upper floor *two* essentially distinct localized categories, and not just one, namely $(\underline{\mathrm{Fibhot}}\,S)\Sigma^{-1}$ with $\Sigma = W_S$ or $\Sigma = W_S^{\mathrm{f}}$ $(= W_S^{\mathrm{d}} = W_S^{\mathrm{g}})$.

On the other hand, the fact that the descent functors induce equivalences between the coarse localized categories is deeper, it requires (it seems to me) that we know an "ascending" functor C, say from Cat/S to $\underline{\mathrm{Fibhot}}\,S$, compatible with W_S, and equipped with

p. 52

$$X \xrightarrow{i_X} C(X),$$

which is in W_S for arbitrary X. When we know that Cat is a Quillen category, even without having the properties of the functors $\underline{\mathrm{Ch}}_\infty$ or $\underline{\mathrm{Ch}}$ ([52]), we should probably be able to manage. But the adjunction formula seems to me to remain unexpected and deep, even admitting the results of Quillen and Thomason ([53]). It seems linked here to a rather special property of the functor C, namely the lemma p. 68

[page 82]

$$\begin{array}{c} i_{C(X)} \\ = \\ \uparrow \end{array} C(i_X) : C(X) \longrightarrow C(C(X)),$$

$$\text{in } \underbrace{(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}}_{\mathrm{HOT}_W(S)} \text{ and in } \underbrace{(\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1}}_{\mathrm{HOT}^o_W(S)}$$

[51] This formula suggests that if $X \in \mathrm{Ob}\,\mathrm{Cat}/S$, $Y \in \mathrm{Ob}\,\underline{\mathrm{Fibhot}}\,S$, then

$$\mathrm{Hom}_{W_S^{\mathrm{d}}}(X, Y) \simeq \mathrm{Hom}_{W_S^{\mathrm{g}}}(X, Y) \simeq \mathrm{Hom}_{W_S}(X, Y),$$

but this follows from the adjunction formula and from $Y \longrightarrow C(Y)$ in $W_S^{\mathrm{d}} \cap W_S^{\mathrm{g}}$ for Y in $\underline{\mathrm{Fibhot}}\,S$.

[52] nor even being sure of the existence of a *functorial* factorization of $X \xrightarrow{f} S$ into $X \xrightarrow{i_X} C(X) \xrightarrow{\bar{f}} S$ with $i_X \in W$ and $\bar{f}$ perfect.

[53] See however theorem-scholie 2, p. 108-113. It is after all not as deep as it seemed to me, but virtually sorital!

which does not look purely formal. It would be interesting to examine to what extent this formula still makes sense in a more or less arbitrary "category of models" $(\mathcal{M}, W)$, or coming from a Quillen triple $(W, \mathrm{Cof}, \mathrm{Fib})$ in $\mathcal{M}$. It is clear that the factorization $X \xrightarrow{f} S$ into $X \xrightarrow{i_X} C(X) \xrightarrow{\bar{f}} S$ ($i_X \in W$, $\bar{f} \in \mathrm{Parf}$), playing the role of "fibrations" in the Quillen context, has an intrinsic meaning independent of the choice of a particular construction, something we can make perfectly precise. It is the description of the localizers W_S^{d}, W_S^{g} in $\mathcal{M}/S$ (for $S \in \mathrm{Ob}\,\mathcal{M}$) that is not understood in a general way, outside the case $\mathcal{M} = \mathrm{Cat}$. In other words, I do not understand how to construct categories $\mathrm{HOT}_{\mathcal{M},W}(S)$, $\mathrm{HOT}^o_{\mathcal{M},W}(S)$, even if $(\mathcal{M}, W)$ comes from a Quillen triple, I see only the description of a category $\mathrm{HOT}^{\mathrm{lc}}_{\mathcal{M},W}(S) = (\mathcal{M}/S)(W_S^{\mathrm{coarse}})^{-1}$, and a localization functor $\mathcal{M}/S \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_{\mathcal{M},W}(S)$. We have to manage to describe in $\mathcal{M}/S$ W-equivalences (*"relative to S"*) finer than the coarse one given by W on the source X and nothing more. In Cat I manage it by using either the family of local isomorphisms (to describe W_S^{g}), or that of colocal isomorphisms (to describe W_S^{d}) - in any case, there is a notion of "localization" (either on S, or on S^o …), with all the richness of the intuitions linked to such a notion. Perhaps, [page 83]
without knowing it I have always treated Cat not as a "bare" category, but as a site, or rather (since there is the involution $X \longmapsto X^o$ which cannot be evaded), as a "bi-site" p. 53
- we have *two* ways of localizing there, either by the X/x (Zariski topology!) or by the $x\backslash X$ (co-Zariski topology). There is also the smooth topology (probably very close to the intuition of étale cohomology), and the proper topology - and we expect the smooth one to give essentially the same results as the Zariski one (since the étale topology is equal to it - there is no need to make a fuss about it …); and the proper topology the same results (e.g. the same cohomology?) as the co-Zariski topology.

It does seem that in all the categories of models used up to now to describe Hot as a localized category, we have notions of localization: (Top), $\mathbf{\Delta}^\wedge$ (or $A^\wedge$, with A an arbitrary test category), (Cat); and in Cat we even have *two* such notions. On the other hand I am thinking of ∞-groupoids, it is perhaps the only category of models already so close to Hot itself, that we discern no notion of localization in it, and that the very idea of wanting to localize in it (except by "localizing" Hot itself) seems almost preposterous. [page 84]

This quick digression now convinces me that the said compatibility for the construction of $C(X)$ and $i_X : X \longrightarrow C(X)$, can in no case be purely formal, a mere matter of Quillen categories say. That will at best give the compatibility in $(\mathcal{M}/S)W_S^{-1}$, which is ridiculous! But if we take as localizer Σ in $\mathcal{M}/S$ the isomorphisms and nothing more, the compatibility relation will become false every time! As a slightly more reasonable set of arrows Σ, which is still very small, we can think e.g. of *universal* W-equivalence over S (distinctly finer still than W_S^{d}, W_S^{g} in the special case of Cat). Would we still have compatibility for this set of arrows? I am also a little perplexed by the fact (not well understood) that the adjunction formulas for the Zariski and co-Zariski topologies go in opposite directions. (Or might I have made a mistake in dualizing in a hurry?) But to tell the truth, this compatibility by itself, even when we use a site structure on $\mathcal{M}$, is not enough to write down an adjunction formula. We have two localized categories,

$$\mathrm{HOT}^{\mathrm{lc}}_{\mathcal{M}} = (\mathcal{M}/S)W_S^{-1}, \qquad \mathrm{HOT}_W(S) = (\mathcal{M}/S)(W_S^{\mathrm{f}})^{-1},$$

where W_S^{f} denotes a "fine" W-equivalence, corresponding to the notion of localization, more precisely to the "admissible" base changes we have chosen, and we have a localization functor [page 85]

$$\mathrm{Hot}_W(S) \longrightarrow \mathrm{Hot}^{\mathrm{lc}}_W(S),$$

but no functor in the reverse direction. But yes, but yes, working with the similar category $\underline{\mathrm{Fibhot}}\,S$, this must still make sense at least when we have a good notion of "fibration".

I see a little work of conceptual roughing-out to do, with perhaps an adjunction formula at the end of it, but I am not going to launch into it now.

p. 54

16.11.90

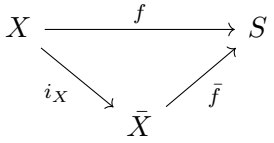
I am going to try all the same to do some "roughing-out" work, for the diagram of the 15 categories looks a bit bushy, after all! I give myself

$$(\mathcal{M}, W) \qquad W \subseteq \mathrm{Fl}\,\mathcal{M},$$

an $S \in \mathrm{Ob}\,\mathcal{M}$, whence $\mathcal{M}/S$, with $W_S \subseteq \mathrm{Fl}(\mathcal{M}/S)$ induced by W. We are going to put

$$\mathrm{Ho}^{\mathrm{lc}}(S) = (\mathcal{M}/S)W_S^{-1},$$

this will play the role of relatively constant homotopy types. I suppose on the other hand that we have singled out a class $F \subseteq \mathrm{Fl}(\mathcal{M})$ of arrows, playing the role of "fibrations" (these will be the Hot-fibrant morphisms in Cat, giving rise to the subcategory $\underline{\mathrm{Fibhot}}\,S$ of Cat/S). I foresee that we will have to suppose, besides $\mathcal{M}$ being stable under finite $\varprojlim$, that F is stable under arbitrary base change, and stable under composition, that it contains the isomorphisms. And above all that we have a factorization theorem [page 86]



$$i_X \in W,\ \bar{f} \in F$$

for every arrow f in $\mathcal{M}$. With Quillen one even requires that i_X have the left lifting property with respect to F, but that should not be essential, since I did not have to use this fact in Cat. When I work in the category $\mathcal{M}/S$, where f is understood for an object X of the said category, I will write (by abuse of language) $C(X)$ for $\bar{X}$, although it is not said that we have a canonical choice of $C(X)$, still less that $C(X)$ depends functorially on X.

Let us denote by

$$\underline{\mathrm{Fib}}\,S \subseteq \mathcal{M}/S \qquad (*)$$

the subcategory of $\mathcal{M}/S$ formed by the X whose structural morphism $f_X : X \longrightarrow S$ is in F. Let us denote (by abuse of notation) by W_S the localizer induced on $\underline{\mathrm{Fib}}\,S$ by W_S, then the inclusion $(*)$ defines

$$(\underline{\mathrm{Fib}}\,S)W_S^{-1} \xrightarrow{\delta} (\mathcal{M}/S)W_S^{-1} \overset{\mathrm{def}}{=} \mathrm{Ho}^{\mathrm{lc}}_W(S).$$

A first question is to see that this is an equivalence. We see that it is essentially surjective thanks to the existence of the $C(X)$, i_X. That is not much! We are going to try to define an arrow in the opposite direction [page 87]

$$\bar{C} : (\mathcal{M}/S)W_S^{-1} \longrightarrow (\underline{\mathrm{Fib}}\,S)W_S^{-1},$$

i.e. a *functor* p. 55

$$\gamma : \mathcal{M}/S \longrightarrow (\underline{\mathrm{Fib}}\,S)W_S^{-1}$$

which transforms $W_S \subseteq \mathrm{Fl}(\mathcal{M}/S)$ into isomorphisms. We will choose, for X in $\mathcal{M}/S$, a $C(X)$ in $\underline{\mathrm{Fib}}\,S$ and an

$$i_X : X \longrightarrow C(X)\ , \qquad i_X \in W_S,$$

and we will put

$$\gamma(X) = C(X).$$

The difficulty is to show that this does not depend on the choices made, up to canonical isomorphism, and that it is functorial in X - and the second implies the first.

To see that it works, clearly we have to suppose something more. I see two hypotheses which, either of them, could make it work: one is that we already give ourselves C as a functor and i_X functorial in X (this was the case in Cat just now). The other is that the i_X have, with respect to F, the lifting property in the Quillen style. [page 88]

I begin with the hypothesis that C is already a functor. The diagram

$$\begin{array}{ccc} X & \xrightarrow{i_X} & CX \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle C(u)} \\ X' & \xrightarrow{i_{X'}} & CX', \end{array}$$

where $i_X,\ i_{X'} \in W$, shows that C transforms W_S into W_S, hence into isomorphisms of $\mathrm{Ho}^{\mathrm{lc}}_W(S)$. So we have got the sought-for functor γ. In fact we have functors

$$\underline{\mathrm{Fib}}\, S \underset{C}{\overset{\overset{\text{incl.}}{D}}{\hookrightarrow}} \mathcal{M}/S$$

compatible with the localizers W_S, inducing δ and γ, and we have

$$\mathrm{id}_{\mathcal{M}/S} \longrightarrow DC, \qquad \text{in } W_S \text{ argument by argument,}$$

for we have

$$X \xrightarrow{i_X} DC(X) = C(X)$$

with $i_X \in W_S$. Likewise, we have

$$\mathrm{id}_{\underline{\mathrm{Fib}}\, S} \longrightarrow CD \qquad \text{in } W_S \text{ argument by argument,}$$

with the same p. 56

$$X \xrightarrow{i_X} CX \qquad (\text{this time } X \text{ in } \underline{\mathrm{Fib}}\, S).$$

So we have without tears the two functors δ, γ as quasi-inverses of each other. And the lemma on page 11 even shows that for *every* full subcategory $\mathcal{F}$ of $\mathcal{M}/S$, such that [page 89]

$$\underline{\mathrm{Fib}}\, S \subseteq \mathcal{F} \subseteq \mathcal{M}/S,$$

the inclusions induce equivalences on the W_S-localized categories

$$(\underline{\mathrm{Fib}}\, S)W_S^{-1} \xrightarrow{\sim} \mathcal{F}W_S^{-1} \xrightarrow{\sim} \underbrace{(\mathcal{M}/S)W_S^{-1}}_{\overset{\mathrm{def}}{=}\mathrm{Ho}^{\mathrm{lc}}_W(S)}.$$

I am now going to treat the case of Quillen type, to convince myself that the existence of the functor C, and the particular choice one can make of it, is irrelevant. (This last point is moreover obvious a posteriori, when we know that δ is an equivalence, for the quasi-inverse γ will then be unique up to unique isomorphism, and the choice of a functor C, and of a functorial homomorphism $i : \mathrm{id} \longrightarrow C$, is only one way among others of making the quasi-inverse explicit.) [page 90]

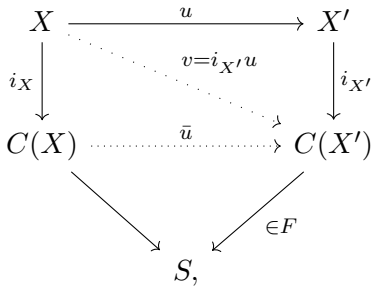
We therefore place ourselves under the Quillenian hypothesis on the i_X. We choose $C(X)$, i_X for each X in $\mathcal{M}/S$, and we put as before

$$\gamma(X) = C(X) \qquad \text{as an object of } (\mathcal{M}/S)W_S^{-1}.$$

Let us show that this is functorial in X. Let

$$u : X \longrightarrow X'$$

be in $\mathcal{M}/S$, let us consider the harmless (commutative) diagram



let us complete it with the coarsely dotted arrow, $v = i_{X'}u$ (making the upper triangle commutative). ([54]). The hypothesis on i_X implies that the finely dotted arrow $\bar{u}$ exists, so as to have two more commutative triangles. But this means exactly that there is an *S-morphism* $\bar{u}$, i.e. an arrow in $\mathcal{M}/S$, making the square commutative

p. 57

$$\begin{array}{ccc} X & \xrightarrow{u} & X' \\ {\scriptstyle i_X}\downarrow & & \downarrow{\scriptstyle i_{X'}} \\ C(X) & \xrightarrow{\bar{u}} & C(X'), \end{array}$$

and we must take

$$\gamma(u) = \text{ image of } \bar{u} \text{ in } \mathrm{Fl}((\underline{\mathrm{Fib}}\, S)W_S^{-1}).$$

[page 91]

We have to prove that this arrow does not depend on the choice of $\bar{u}$. We must isolate a Lemma, where we denote by Cof_0 the set of arrows of $\mathcal{M}$ which have the lifting property with respect to the elements of $\mathrm{Fib} = F$. (Cof_0 plays the role of trivial cofibrations.)

Lemma. *Let there be a solid-line diagram in* $\mathcal{M}/S$

$$\begin{array}{ccc} X & \xrightarrow{u} & X' \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle i'} \\ C & \overset{v}{\cdots\!\!\rightarrow} & C' \end{array}$$

with $i \in Cof_0$, *and* $C' \in Ob\,\underline{\mathrm{Fib}}\, S$ *i.e.* $(C' \longrightarrow S) \in Fib$. *Then* v *exists, so as to make the square commutative. Moreover, if* $i \in W$ *and* $C \in \mathrm{Ob}\,\underline{\mathrm{Fib}}\, S$, *the class of* v *as an arrow of* $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$ *is independent of the choice made.*

It remains to prove this independence, which will follow at once from the corollary:

[54] Lemma: let

$$\begin{array}{ccc} X & \xrightarrow{u} & X' \\ {\scriptstyle i}\downarrow & & \downarrow \\ C & \cdots\!\!\rightarrow & C' \end{array}$$

[page 92]

be in $\mathcal{M}/S$ with $i \in \mathrm{Cof}_0$, $(C' \longrightarrow S) \in F = \mathrm{Fib}$, then the dotted arrow exists in $\mathcal{M}/S$, so as to make the square commutative.

Corollary. *Let there be a diagram in* $\mathcal{M}/S$

$$X \xrightarrow{\ i\ } C \underset{v_2}{\overset{v_1}{\rightrightarrows}} C',$$

with C *and* C' *in* $\underline{\mathrm{Fib}}\, S$ *(i.e.* $C \longrightarrow S$ *and* $C' \longrightarrow S$ *in Fib), and* $i \in Cof_0$ *and* $i \in W$*. If* $v_1 i = v_2 i$*, then* v_1 *and* v_2 *define the same arrow of* $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$*.*

Proof. Let p. 58

$$v = (v_1, v_2) : C \longrightarrow C' \times_S C',$$

which by hypothesis makes the diagram commutative

$$\begin{array}{ccc} X & \xrightarrow{\ w\ } & C' \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle \mathrm{diag}_{C'}} \\ C & \xrightarrow{\ v\ } & C' \times_S C', \end{array}$$

where $w = v_1 i = v_2 i$. Let us factor $\mathrm{diag}_{C'}$ as

$$C' \xrightarrow{\ i'\ } \bar{C}' \xrightarrow{\ \delta'\ } C' \times_S C', \qquad \text{with } \delta' \in \mathrm{Fib},$$

and $i' \in W$. We find a commutative square

$$\begin{array}{ccccc} X & \xrightarrow{\ w'=i'w\ } & \bar{C}' & & \\ {\scriptstyle i}\downarrow & \overset{\bar{v}}{\nearrow} & \downarrow{\scriptstyle \delta'} & & \\ C & \xrightarrow{\ v\ } & C' \times_S C' & \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} & C' \end{array}$$

with $i \in \mathrm{Cof}_0$, $\delta' \in \mathrm{Fib}$, whence the existence of the dotted arrow, making the two triangles commutative. We want to prove $\mathrm{pr}_1 \circ v = \mathrm{pr}_2 \circ v$ in $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$, as $v = \delta'\bar{v}$ it suffices to prove that we have $\mathrm{pr}_1\delta' = \mathrm{pr}_2\delta'$ in $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$.

We only have to check first (to justify the argument) that this assertion makes sense, i.e. that $\bar{C}'$ is itself also in $\underline{\mathrm{Fib}}\, S$. But the hypotheses of stability of $F = \mathrm{Fib}$ under base change and composition ensure that 1°) $C' \times_S C'$ is in $\underline{\mathrm{Fib}}\, S$ (which is stable under products), and consequently, 2°) that $\bar{C}'$ is also, since $\delta' \in \mathrm{Fib}$. As C' is also in $\underline{\mathrm{Fib}}\, S$ and $C' \xrightarrow{\ i'\ } \bar{C}'$ is in W_S, to check $\mathrm{pr}_1\delta' = \mathrm{pr}_2\delta'$ in the localized category of $\underline{\mathrm{Fib}}\, S$, it suffices to see that $(\mathrm{pr}_1\delta')i' = (\mathrm{pr}_2\delta')i'$, now since $\delta' i' = \mathrm{diag}_{C'}$, we have the equality even in $\underline{\mathrm{Fib}}\, S$ (without localizing). [page 93]

We have therefore defined $\gamma(X)$, $\gamma(u)$ for objects and arrows of $\mathcal{M}/S$. Transitivity for $\gamma(u)$ is clear, and also $\gamma(\mathrm{id}) = \mathrm{id}$. So we have got the functor

$$\gamma : \mathcal{M}/S \longrightarrow (\underline{\mathrm{Fib}}\, S)W_S^{-1}.$$

Let now $\mathcal{F}$ be a full subcategory of $\mathcal{M}/S$, with

$$\underline{\mathrm{Fib}}\, S \subseteq \mathcal{F} \subseteq \mathcal{M}/S,$$

let us consider γ as a functor with values in $\mathcal{F}W_S^{-1}$. (I forgot to check that γ transforms W_S into W_S, but this is tautological.) Thus we have functors p. 59

$$\mathcal{F}W_S^{-1} \underset{\gamma}{\overset{\delta}{\rightleftarrows}} (\mathcal{M}/S)W_S^{-1}, \qquad \text{[page 94]}$$

we must see that they are quasi-inverse to one another.

On the $\mathcal{M}/S$ side: we must compare X and $\delta\gamma(X) = C(X)$, this is done by

$$i_X : X \longrightarrow C(X)$$

which is in W_S, hence the isomorphism in the localized category

$$X \simeq \delta\gamma(X) \qquad \text{in } (\mathcal{M}/S)W_S^{-1}.$$

We must check that this is functorial in X.

On the $\mathcal{F}$ side, compare X and $\gamma\delta(X)$, i.e. X and $C(X)$. Same story, we choose $i_X : X \simeq C(X) = \gamma\delta(X)$. Functoriality to be checked.

It is now a matter of introducing a finer localized category of $\mathcal{M}/S$. For this, we suppose given another class of morphisms, I call it $\mathcal{L}$ (as in "localization")

$$\mathcal{L} \subseteq \mathrm{Fl}(\mathcal{M}),$$

e.g. in the case of $\mathcal{M} = \mathrm{Cat}$, $\mathcal{L}$ will consist of the local isomorphisms ([55]) (if it is a matter of defining W_S^{g}), or the colocal isomorphisms ([56]) (if it is a matter of defining W_S^{d}). We shall assume that $\mathcal{L}$ contains the isomorphisms, and $\mathcal{L}$ stable under composition ([57]), but not under base change. We then define

$$W_S^{\mathcal{L}} \subseteq \mathrm{Fl}(\mathcal{M}/S)$$

by

$$f \in W_S^{\mathcal{L}} \overset{\text{def}}{\Longleftrightarrow} (f \times_S S') \in W, \quad \forall (S' \longrightarrow S) \in \mathcal{L}.$$

Since $\mathcal{L}$ contains id_S, we shall have [page 95]

$$W_S^{\mathcal{L}} \subseteq W_S,$$

hence a canonical functor

$$\underbrace{(\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1}}_{\overset{\text{def}}{=}\mathrm{Ho}_W^{\mathcal{L}}(S)} \xrightarrow{\rho} \underbrace{(\mathcal{M}/S)(W_S)^{-1}}_{\overset{\text{def}}{=}\mathrm{Ho}_W^{\mathrm{lc}}(S)}$$

which is a localization functor. I would like to define a functor in the reverse direction. For this, it is natural to assume (by analogy with the situation in Cat)

Hypothesis 1. *On* $\underline{\mathrm{Fib}}\, S$, $W_S = W_S^{\mathcal{L}}$, *i.e. if* $f : X \longrightarrow Y$ *is in* W *and* X, Y *in* $\underline{\mathrm{Fib}}\, S$, *then for every base change* $S' \longrightarrow S$ *which is in* $\mathcal{L}$, $f' : X_{S'} \longrightarrow Y_{S'}$ *is still in* W. p. 60

In the Quillen context, this hypothesis has a definite tendency to be satisfied whatever $\mathcal{L}$ may be, i.e. for *every* base change $S' \longrightarrow S$ without restriction. In fact it suffices that the Quillen structure be "proper", more precisely (and less restrictively), that $a \in W$ remain so after base change by a morphism $\in \mathrm{Fib}$. Let us single this out as a Lemma.

Lemma. *The hypothesis is satisfied if we assume that* $F = Fib$ *and* W *are related by the additional condition (besides the factorization* i_X*): for every cartesian square* [page 96]

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{p} & S', \end{array}$$

[55] Or also, *smooth* morphisms. That should give the same result.

[56] Or also proper morphisms.

[57] Perhaps this is not needed.

if f is in Fib and p in W, then $q \in W$.

PROOF. Let

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & S & \end{array}$$

with $p, q \in \mathrm{Fib}$, $f \in W$, let us show that for *every* base change $g : S' \longrightarrow S$, $f_{S'} : X_{S'} \longrightarrow Y_{S'}$ is still in W. We factor g as

$$S' \xrightarrow{\ i\ } \bar{S}' \xrightarrow{\ \bar{g}\ } S, \qquad\qquad i \in W,\ \bar{g} \in \mathrm{Fib},$$

and we are reduced to proving that the assertion is true in the two special cases where $g \in \mathrm{Fib}$, or $g \in W$. The case $g \in \mathrm{Fib}$ follows at once from the hypothesis of the Lemma (applied to the cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ X & \xrightarrow{f} & Y, \end{array}$$

where $p \in \mathrm{Fib}$.) p. 61

In the case where $S' \longrightarrow S$ is in W, it follows from the hypothesis of the Lemma applied to

$$\begin{array}{ccc} Y & \xleftarrow{p} & Y' \\ \downarrow{\scriptstyle \in \mathrm{Fib}} & & \downarrow \\ S & \longleftarrow & S' \end{array}$$

that $p : Y' \longrightarrow Y$ is $\in W$, and likewise for $q : X' \longrightarrow X$, and $f \in W$ again implies $f' \in W$. [page 97]

Let us return to the hypothesis p. 95. We shall therefore have a functor

$$(\underline{\mathrm{Fib}}\, S, W_S = W_S^{\mathcal{L}}) \longrightarrow (\mathcal{M}/S, W_S^{\mathcal{L}}),$$

hence for the localized categories a horizontal functor

$$\begin{array}{ccc} (\underline{\mathrm{Fib}}\, S)W_S^{-1} & \xrightarrow{\quad i_0 \quad} & (\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1} \overset{\mathrm{def}}{=} \mathrm{Ho}_W^{\mathcal{L}}(S) \\ \downarrow{\scriptstyle \approx} & \swarrow & \\ \mathrm{Ho}_W^{\mathrm{lc}}(S) \overset{\mathrm{def}}{=} (\mathcal{M}/S)W_S^{-1} & & \end{array}$$

which factors the vertical functor induced by the inclusion $(\underline{\mathrm{Fib}}\, S, W_S) \longrightarrow (\mathcal{M}/S, W_S)$ - a functor which we have seen (in the two cases considered) is an equivalence. By inverting the latter and composing it with i_0, we thus find a functor

$$\underbrace{\mathrm{Ho}_W^{\mathrm{lc}}(S)}_{\overset{\mathrm{def}}{=}(\mathcal{M}/S)W_S^{-1}} \xrightarrow{\ i\ } \underbrace{\mathrm{Ho}_W^{\mathcal{L}}(S)}_{\overset{\mathrm{def}}{=}(\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1}},$$

such that

$$\rho i \simeq \mathrm{id},$$

hence a *retraction* of $\mathrm{Ho}_W^{\mathcal{L}}(S)$ onto $\mathrm{Ho}_W^{\mathrm{lc}}(S)$. We expect i to be fully faithful, and more precisely still, that ρ and i be adjoint, i being the right adjoint of ρ: [page 98] p. 62

$$\mathrm{Ho}_W^{\mathrm{lc}}(S) \underset{i}{\overset{\rho}{\leftrightarrows}} \mathrm{Ho}_W^{\mathcal{L}}(S),$$

i.e. we seek

$$i\rho \quad \underbrace{\xleftarrow{i}}_{not\ \mathrm{iso}} \quad \mathrm{id} \quad \text{in } \mathrm{Ho}_W^{\mathcal{L}} \quad (\text{localization by } W_S^{\mathcal{L}})$$

$$\rho i \quad \underbrace{\xrightarrow{p}}_{\mathrm{iso}} \quad \mathrm{id} \quad \text{in } \mathrm{Ho}_W^{\mathrm{lc}} \quad (\text{localization by } W_S),$$

so for X in $\mathcal{M}/S$, we seek

$$\begin{aligned} i_X : \quad X \quad &\longrightarrow \quad C(X) \quad \text{at least in } (\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1} \\ p_X : \quad X \quad &\longleftarrow \quad C(X) \quad \text{at least in } (\mathcal{M}/S)(W_S)^{-1}. \end{aligned}$$

We shall choose for i_X the "canonical" arrow from X to $C(X)$, defined in $\mathcal{M}/S$ itself, without having to localize. But in the "Quillen case" (when C is not given as a functor $\mathcal{M}/S \longrightarrow \underline{\mathrm{Fib}}\, S$), we must see that this i_X is functorial in X, when we look at it in the localized category $(\mathcal{M}/S)W_S^{-1}$. But this is contained in the construction of C as a *functor*

$$\mathcal{M}/S \longrightarrow (\underline{\mathrm{Fib}}\, S)W_S^{-1}.$$

We shall note that i_X is in W_S and no more, not in $W_S^{\mathcal{L}}$, hence does not in general define an isomorphism of $\mathrm{Ho}_W^{\mathcal{L}}$.

We define p_X with the help of i_X, as the inverse of i_X *in* $(\mathcal{M}/S)W_S^{-1}$, which exists since i_X is in W_S, hence isomorphism in $(\mathcal{M}/S)W_S^{-1} = \mathrm{Ho}_W^{\mathrm{lc}}$, and p is therefore isomorphism. We thus recover $\rho i \simeq \mathrm{id}$, hence that ρ is a *retraction* of $\mathrm{Ho}_W^{\mathcal{L}}$ onto $\mathrm{Ho}_W^{\mathrm{lc}}$. [page 99]

To prove the adjunction, proceeding as in the case of Cat, we are reduced to proving that for every X,

$$\begin{cases} C(i_X) = i_{C(X)} : C(X) \rightrightarrows C(C(X)) \\ \textit{in } \mathrm{Ho}_W^{\mathcal{L}} \text{ (localization by } W_S^{\mathcal{L}}). \end{cases}$$

I had managed this in Cat, using a morphism

$$C(C(X)) \xrightarrow{c} C(X)$$

(essentially, a path composition morphism) such that $c \in W_S^{\mathcal{L}}$, and that $c \circ C(i_X) = c \circ i_{C(X)}$. That had looked very delicate, for it was with great difficulty that I had managed to construct a reasonable functor $\underline{\mathrm{Ch}}$ ("paths"), where we could really and truly compose paths! For the moment I give up axiomatizing the properties of the functor $\underline{\mathrm{Ch}}$ that I need to make the machine work (I should come back to this shortly). I am going to begin instead with the "Quillen case", with $i_X \in \mathrm{Cof}_0$.

We take up again the description of the "functor" C (with values in $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$, and then without quotation marks), we find that $C(i_X)$, where p. 63 [page 100]

$$X \xrightarrow{u=i_X} X' = C(X)$$

is defined simply (in $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$) by "the" arrow $C(u)$ making the diagram commute (in $\mathcal{M}/S$)

$$\begin{array}{ccc} X & \xrightarrow{u\ (=i_X)} & X'\ (=C(X)) \\ {\scriptstyle i_X}\downarrow & & \downarrow{\scriptstyle i_{X'}\ (=i_{C(X)})} \\ C(X) & \xrightarrow[C(u)\ (=C(i_X))]{} & C(X')\ (=C(C(X))), \end{array}$$

so that in $\mathcal{M}/S$ we have the diagram

$$X \xrightarrow{\overbrace{i_X}^{\in \mathrm{Cof}_0 \cap W}} \overbrace{C(X)}^{\in \mathrm{Ob}\,\underline{\mathrm{Fib}}\,S} \underset{C(i_X)}{\overset{i_{C(X)}}{\rightrightarrows}} \overbrace{C(C(X))}^{\in \mathrm{Ob}\,\underline{\mathrm{Fib}}\,S},$$

commutative, i.e. $i_{C(X)} i_X = C(i_X) i_X$. The corollary on page 92 tells us that the arrows $i_{C(X)}$ and $C(i_X)$ are equal in $(\underline{\mathrm{Fib}}\,S)W_S^{-1}$. But by hypothesis (p. 95), in $\underline{\mathrm{Fib}}\,S$ we have $W_S = W_S^{\mathcal{L}}$, so the arrows are equal in $(\underline{\mathrm{Fib}}\,S)(W_S^{\mathcal{L}})^{-1}$, and a fortiori in $(\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1}$, and we win.

Let us see what we can do in the case where we have a *functor* C at our disposal. We assume of course that this functor in $\mathcal{M}/S$ is not tied to the particular S in $\mathcal{M}$ that [page 101] we have chosen as base, that its construction comes in Cartan-Serre fashion from a functor

$$X \longmapsto \mathrm{Ch}(X) : \mathcal{M} \longrightarrow \mathcal{M},$$

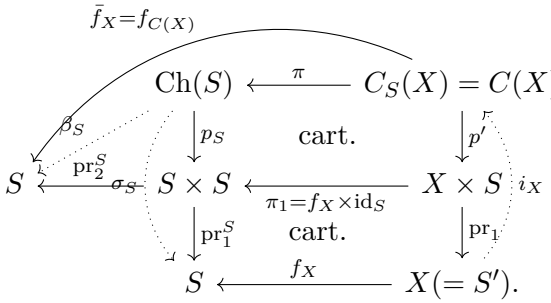
("path object") and functorial morphisms

$$X \xrightarrow{\alpha_X} \mathrm{Ch}(X) \xrightarrow{p_X} X \times X, \qquad \boxed{p_X \alpha_X = \mathrm{diag}_X}$$

([58]). I presume we shall have to assume above all the following

Hypothesis 2. $\alpha_X \in W$, $p_X \in \mathrm{Fib}$.

For an X over S, we shall construct $C_S(X)$ or $C(X)$ as follows p. 64



It is the inverse image of $(\mathrm{Ch}(S), \mathrm{pr}_1^S p_S = \sigma_S)$ under $X \xrightarrow{f_X} S$, but we have made the factorization of σ_S explicit in the diagram to reinterpret the structural arrow $\bar{f}_X = f_{C(X)}$, defined as

$$f_{C(X)} = \beta_S \pi (= \mathrm{pr}_2^S \underbrace{p_S \pi}_{= \pi_1 p'})$$

by the formula

$$f_{C(X)} = \underbrace{(\mathrm{pr}_2^S \pi_1)}_{\text{projection } \mathrm{pr}_2 \text{ from } X \times S \text{ to } S} p'.$$

Now [page 102]

$p' \in \mathrm{Fib}$, since obtained from $p_S \in \mathrm{Fib}$ by base change π_1,

$(\mathrm{pr}_2 : X \times S \longrightarrow S) \in \mathrm{Fib}$ provided that $X \longrightarrow \underbrace{e}_{\text{final object of } \mathcal{M}}$ is in Fib (same reason),

so if $X \in \underline{\mathrm{Fib}}\,e$, we have $f_{C(X)} \in \mathrm{Fib}$. Since this is an essential property that we want to have for *every* X in $\mathcal{M}/S$, we are going to make the additional hypothesis

[58] **NB** We put $\mathrm{pr}_1 \circ p_X = \sigma_X$, $\mathrm{pr}_2 \circ p_X = \beta_X$ ("source" and "target" arrows for the object X).

Hypothesis 3. $\forall\, X \in Ob\,\mathcal{M},\ X \longrightarrow e$ *is in Fib.*

As for

$$i_X : X \longrightarrow C(X),$$

it is obtained from the section α_S of $(\mathrm{Ch}(S), \sigma_S)$ over S by base change $X \longrightarrow S$. Let us now note that

$$\mathrm{Ch}(S) \underset{\beta_S}{\overset{\sigma_S}{\rightrightarrows}} S \qquad\qquad \text{are in } W,$$

since their composites with $S \xrightarrow{\alpha_S} \mathrm{Ch}(S)$ are the identity, and $\alpha_S \in W$. On the other hand, $\sigma_S = \mathrm{pr}_1^S \circ p_S$ is in Fib, as a composite of two elements in Fib (NB $S \longrightarrow e$ in Fib by Hypothesis 3). Thus σ_S, β_S are in $\mathrm{Fib} \cap W$. We shall need

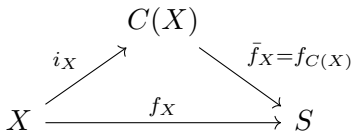
Hypothesis 4. *The set of arrows* $\mathrm{Fib} \cap W$ *is stable under every base change.*

This therefore implies that

p. 65
[page 103]

$$C(X) \xrightarrow{\mathrm{pr}_1 p'} X$$

is $\in W$ (being obtained from $\sigma_S : \mathrm{Ch}(S) \longrightarrow S$ by base change), so every section $i_X : X \longrightarrow C(X)$ of $C(X)$ over X is in W . We thus do indeed have a factorization, functorial in X



with

$$i_X \in W, \qquad\qquad f_{C(X)} \in \mathrm{Fib}.$$

This is all we had used to establish

$$(\underline{\mathrm{Fib}}\, S)W_S^{-1} \xrightarrow{\sim} (\mathcal{M}/S)W_S^{-1}.$$

But it is now a matter, under the hypothesis where we introduce an $\mathcal{L}$, of proving that for every X in $\mathcal{M}/S$, the two arrows

$$C(X) \underset{i_{C(X)}}{\overset{C(i_X)}{\rightrightarrows}} C(C(X))$$

are equal in $(\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1}$. Since by hypothesis we have $W_S^{\mathcal{L}} = W_S$ in $\underline{\mathrm{Fib}}\, S$ (Hypothesis 1), and $C(X),\ C(C(X)) \in \mathrm{Ob}\,\underline{\mathrm{Fib}}\, S$, it suffices to prove that the two arrows are equal *in* $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$. This statement is independent of the datum of $\mathcal{L}$, and we feel like proving it.

[page 104]

Perhaps there is a way to make a geometric language explicit, to find our way around. Let us identify the objects X of $\mathcal{M}$ with the functors $\mathcal{M}^o \longrightarrow (\mathrm{Ens})$ that they define,

$$X \longmapsto (T \longmapsto X(T) = \mathrm{Hom}_{\mathcal{M}}(T, X))\,.$$

Thus, for every T

$$\mathrm{Ch}(X)(T)$$

is a set, called *the set of paths in* $X(T)$. It maps (by p_X) functorially (in T) to $(X \times X)(T) = X(T) \times X(T)$, i.e. every path c in $X(T)$ has an origin $\sigma(c)$, and a target $\beta(c)$. Moreover, by α_X, $X(T)$ maps (functorially in T) to the set of paths of $X(T)$: for every element x in $X(T)$, there is therefore a well-defined path (denoted 1_x or α_x) with source and target x, which we call the trivial, or empty, path. This said,

p. 66
[page 105]

for X in $\mathcal{M}/S$, $C(X)$ represents the following contrafunctor in $\mathcal{M}$: for every T in $\mathcal{M}$,

$$C(X)(T) = \left\{(x,c) \;\middle|\; x \in X(T),\ c : f_X(x) \xrightarrow{c} s \ \text{ a path in } S(T), \text{ with source } f_X(x)\right\},$$

where $f_X : X \longrightarrow S$ is the structural morphism.

The "calculations" done, two (?) days ago show that we likewise have an interpretation

$$C(C(X))(T) = \left\{(x,c,c') \;\middle|\; x \in X(T),\ f_X(x) \xrightarrow{c} s \xrightarrow{c'} t,\ c \text{ a path with source } f_X(x),\ c' \text{ a path with source equal to } \beta(c)\right\}$$

and the maps $C(i_X)$ and $i_{C(X)}$ are identified respectively with

$$\begin{array}{lcll} (x,c) & \longmapsto & (x, 1_{f(x)}, c) & f(x) \xrightarrow{1_f(x)} f(x) \xrightarrow{c} s \\ (x,c) & \longmapsto & (x, c, 1_{\beta(c)}) & f(x) \longrightarrow s \xrightarrow{1_s} s \end{array}$$

. To compare them, we have an irresistible urge to write

$$c \circ 1_{f(x)} = 1_s \circ c = c,$$

but do we have the right? We would have to be able to compose the paths, and that is very, very difficult!

But we can always axiomatize. We therefore suppose that for X in $\mathcal{M}$, $\mathrm{Ch}(X)$ is not just an amorphous structure of (so-called) "paths" in X, just the datum of the source and target of the paths and of the trivial paths (these structures on $\mathrm{Ch}(X)$ depending functorially on X), but that we have a composition of paths when they are "composable", i.e. an arrow from a fiber product [page 106]

$$(c,c') \longmapsto c' \circ c : (\mathrm{Ch}(X), \beta_X) \times_X (\mathrm{Ch}(X), \sigma_X) \xrightarrow{\pi_X} \mathrm{Ch}(X),$$

i.e. we can compose paths in $X(T)$

$$x \xrightarrow{c} y \xrightarrow{c'} z$$

if and only if target c = source c', and moreover we of course require that source$(c' \circ c)$ = source c, target$(c' \circ c)$ = target c', which is expressed by a commutative diagram

$$\begin{array}{ccc} (\mathrm{Ch}(X), \beta_X) \times_X (\mathrm{Ch}(X), \sigma_X) & \longrightarrow & \mathrm{Ch}(X) \\ & \searrow \quad \swarrow_{p_X} & \\ & X \times X. & \end{array}$$

The natural thought is that with this composition and these identities, the $X(T)$, $\mathrm{Ch}(X)(T)$ become the objects and the arrows of a *category* - i.e. that $(X, \mathrm{Ch}(X), \alpha_X, p_X, \pi_X)$ define a structure of "category in $\mathcal{M}$". But for what we want to do now, we probably do not need everything, i.e. the relevant axiom of associativity of composition and the one on units, but only that the 1_x be two-sided units for composition. p. 67 [page 107]

Supposing this additional structure on the functor Ch given, and returning to the consideration of the associated functors

$$C : \mathcal{M}/S \longrightarrow \underline{\mathrm{Fib}}\, S,$$

we find a canonical morphism

$$C(C(X)) \xrightarrow{\gamma_X} C(X)$$

by

$$(x, c, c') \longmapsto (x, c' \circ c),$$

and the axiom of two-sided units assures us at once that

$$(*) \qquad \gamma_X \circ i_{C(X)} = \gamma_X \circ C(i_X) = \mathrm{id}_{C(X)}.$$

Since $i_{C(X)} \in W$, this already proves that $\gamma_X \in W$, so $\gamma_X \in W_S$. So γ_X becomes an isomorphism in $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$, and the relation $(*)$ does indeed imply $i_{C(X)} = C(i_X)$ in the localized category, q.e.d.

I am going to summarize again what I have worked out.

Theorem-Scholie 2. Let $\mathcal{M}$ be a category, equipped with two sets of arrows [page 108]

$$W,\ \mathrm{Fib} \subseteq \mathrm{Fl}(\mathcal{M}).$$

We make the following hypotheses.

1° W contains the isomorphisms, if f, g are composable, if two of f, g, gf are in W, the third also.

2° $\mathcal{M}$ is stable under finite $\varprojlim$, Fib contains the isomorphisms, is stable under composition and under base change ([59]).

For S in $\mathcal{M}$, $\boxed{\underline{\mathrm{Fib}}\, S}$ denotes the full subcategory of $\mathcal{M}/S$ formed by the $X \longrightarrow S$ which are in Fib. We distinguish two different versions of an axiom of factorization of an arbitrary $f : X \longrightarrow S$ in $\mathcal{M}$ as $X \xrightarrow{i} \bar{X} \xrightarrow{\bar{f}} S$, with $i \in W$, $\bar{f} \in \mathrm{Fib}$. p. 68

A) We are given, for every S in $\mathcal{M}$, a functor

$$\mathcal{M}/S \longrightarrow \mathcal{M}/S \qquad X \longmapsto C(X)\ \ (= \bar{X})$$

and a morphism of functors in X

$$X \xrightarrow{i_X} C(X)$$

with

$$i_X \in W, \qquad C(X) \in \mathrm{Ob}\, \underline{\mathrm{Fib}}\, S.$$

B) Denoting by

$$\mathrm{Cof}_0 \subseteq \mathrm{Fl}\, \mathcal{M}$$

the set of arrows of $\mathcal{M}$ which have the left lifting property with respect to Fib, we suppose "in the style of Quillen", that every arrow $f : X \longrightarrow S$ factors as

$$X \xrightarrow{i_X} C(X) \xrightarrow{\bar{f}} S,$$

with

$$i \in \mathrm{Cofib}_0 \cap W, \qquad \bar{f} \in \mathrm{Fib}.$$

. [page 109]

Let us take S in $\mathcal{M}$, consider

$$W_S \subseteq \mathrm{Fl}(\mathcal{M}/S)$$

$$f : X \longrightarrow Y$$
$$\searrow \quad \swarrow$$
$$S$$

the set of arrows induced by W (is in W_S if and only if it is on forgetting S), and let us denote by W_S (by abuse of notation) the set of arrows induced by W_S on $\underline{\mathrm{Fib}}\, S$, or more generally on any full subcategory $\mathcal{F}$ of $\mathcal{M}/S$, with

$$\underline{\mathrm{Fib}}\, S \subseteq \mathcal{F} \subseteq \mathcal{M}/S.$$

Ⓘ The functor

$$(*) \qquad (\underline{\mathrm{Fib}}\, S)W_S^{-1} \xrightarrow{\sim} (\mathcal{M}/S)W_S^{-1}$$

induced by the inclusion is an equivalence of categories, more generally for every $\mathcal{F}$ as above

$$(\underline{\mathrm{Fib}}\, S)W_S^{-1} \longrightarrow \mathcal{F}W_S^{-1} \longrightarrow (\mathcal{M}/S)W_S^{-1}$$

are equivalences of categories. A quasi-inverse functor is obtained (in both cases A) and B)) with the help of the functor

$$\gamma : \mathcal{M}/S \longrightarrow (\underline{\mathrm{Fib}}\, S)W_S^{-1}$$

which is given on objects by

$$\gamma(X) = C(X),$$

p. 69

and which on arrows is defined in the obvious way in case A) (functoriality of $C(X)$ in X), and in the case of B) is constructed ad hoc, as the only functorial law for which the homomorphisms

$$i_X : X \longrightarrow C(X)$$

(interpreted as arrows in $(\mathcal{M}/S)W_S^{-1}$) are themselves functorial in X.

We denote by

[page 110]

$$\mathrm{Ho}_W^{\mathrm{lc}}(S) \qquad \text{(lc as in "locally constant")}$$

the category of fractions $(\mathcal{M}/S)W_S^{-1}$, equivalent to $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$.

Ⓘ Let us take a localizer W_S^0 in $\mathcal{M}/S$

$$W_S^0 \subseteq W_S \subseteq \mathrm{Fl}(\mathcal{M}/S),$$

with the formal hypothesis similar to that on W (cf. 1°). We assume that

$3^o)$ $W_S^0 \cap \mathrm{Fl}(\underline{\mathrm{Fib}}\, S) = W_S|\underline{\mathrm{Fib}}\, S.$

(Cf. pages 94-96 for a general way of obtaining such localizers. One must clearly see that to define relative "homotopy types" over S the localizer W_S is much too coarse, morally it gives only a category of locally constant relative homotopy types, and this is precisely what is shown by the equivalence of Ⓘ , where $\underline{\mathrm{Fib}}\, S$ is visualized as made up of the "locally trivial fibrations" over S.) We then have a second localized category

$$\mathrm{Ho}_{W_S^0}(S) \overset{\mathrm{def}}{=} (\mathcal{M}/S)(W_S^0)^{-1},$$

and two functors

$$\mathrm{Ho}_{W_S^0}(S) \underset{i}{\overset{\rho}{\rightleftarrows}} \mathrm{Ho}_W^{\mathrm{lc}}(S), \qquad (*)$$

[page 111]

where ρ is the obvious (strict) localization functor, due to the inclusion $W_S^0 \subseteq W_S$, and where i is essentially the functor deduced from the inclusion

$$(\underline{\mathrm{Fib}}\, S, \underbrace{W_S|\underline{\mathrm{Fib}}\, S}_{=W_S^0|\underline{\mathrm{Fib}}\, S}) \longrightarrow (\mathcal{M}/S, W_S^0)$$

by localization *by means of* W_S^0, hence the functor

$$(\underline{\mathrm{Fib}}\, S)(W_S^0)^{-1} \longrightarrow (\mathcal{M}/S)(W_S^0)^{-1}, \qquad (**)$$

taking into account the equivalence $(*)$ of Ⓘ , i.e. i is the composite of the quasi-inverse of $(*)$, with the preceding functor $(**)$. Explicitly, it is defined by p. 70

$$i(X) = \text{class of C(X) in } (\mathcal{M}/S)(W_S^0)^{-1},$$

taking into account the fact that $X \longmapsto C(X)$ transforms W_S into W_S^0, thanks to condition $3^o)$ on W_S^0.

This being so, *under hypothesis* B ("Quillen"), we have: in the pair of functors (ρ, i) of $(*)$, page 110, *i is right adjoint to ρ,* i.e. we have for $\xi \in \mathrm{Ob}\,\mathrm{Ho}_{W_S^0}(S)$, $\eta \in \mathrm{Ob}\,\mathrm{Ho}_W^{\mathrm{lc}}(S)$

$$\mathrm{Hom}_{W_S^0}(\xi, i(\eta)) \simeq \mathrm{Hom}_{W_S}(\rho(\xi), \eta),$$

which is made explicit, for X, Y in $\mathcal{M}/S$, by

$$\boxed{\mathrm{Hom}_{W_S^0}(X, C(Y)) \simeq \mathrm{Hom}_{W_S}(X, Y)}$$

([60]). It follows that *the functor i is fully faithful.*

The same adjunction formula, and the full faithfulness of i, hold in case A, when one strengthens hypothesis A (relative to a particular S) to a more restrictive and more comprehensive hypothesis, as follows. [page 112]

A′) The functor $C : \mathcal{M}/S \longrightarrow \mathcal{M}/S$ and the functorial morphism $i_X : X \longrightarrow C(X)$ are constructed "in the manner of Cartan-Serre" from a functor defined on the whole of $\mathcal{M}$

$$\mathcal{M} \xrightarrow{\mathrm{Ch}} \mathcal{M},$$

of the "paths" type, i.e. equipped in addition with the following data

a)
$$X \xrightarrow{\alpha_X} \mathrm{Ch}(X) \xrightarrow{p_X=(\sigma_X,\beta_X)} X \times X$$

(structure of source and target of paths, and of trivial paths), and

b) A composition of paths, without requiring here the associativity of composition but only that trivial paths be two-sided units for composition. Moreover, we assume that for every X

$$\begin{cases} \alpha_X \in W,\ p_X \in \mathrm{Fib} \\ (X \longrightarrow e) \in \mathrm{Fib}. \end{cases}$$

[59] These stability hypotheses on Fib are also used in I under hypothesis B, pp. 92–93, and in II, in case A), taken up again in toto in the form A′ below (p. 112). p. 71 [page 113]

[60] **NB** The first member is independent of the choice of W_S^0! (Provided only that the hypothesis $W_S^0 \cap \mathrm{Fl}\,\underline{\mathrm{Fib}}\, S = W_S \cap \mathrm{Fl}\,\underline{\mathrm{Fib}}\, S$, in addition to $W_S^0 \subseteq W_S$, is satisfied.)

Corollary. Let X be in $\mathcal{M}/S$, Y in $\underline{\mathrm{Fib}}\, S$, then

$$\mathrm{Hom}_{W_S^0}(X,Y) \xrightarrow{\sim} \mathrm{Hom}_{W_S}(X,Y).$$

PROOF. By the adjunction formula (p. 111), the second member is $\mathrm{Hom}_{W_S^0}(X, CY)$. But as Y and $C(Y)$ are in $\underline{\mathrm{Fib}}\, S$, where W_S and W_S^0 coincide, $Y \xrightarrow{i_Y} C(Y)$ is not only in W_S, but in W_S^0, hence $\mathrm{Hom}_{W_S^0}(X,Y) \xrightarrow{\sim} \mathrm{Hom}_{W_S^0}(X, CY)$, whence the result.

Let us return to Cat, Cat/S. The preceding digression on $(\mathcal{M}, W, \mathrm{Fib})$ has made it possible to better situate what concerns the functor C, and the relations between the localized categories of

$$\mathrm{Cat}/S \ \ (\text{for } W_S,\ W_S^{\mathrm{d}},\ W_S^{\mathrm{g}}) \qquad \text{and} \qquad \underline{\mathrm{Parf}}\, S \ \ (\text{for } W_S^{\mathrm{f}} = W_S = W_S^{\mathrm{d}} = W_S^{\mathrm{g}})$$

as well as for the intermediate categories

$$\underline{\mathrm{Parf}}\, S \subseteq \mathcal{F} \subseteq \mathrm{Cat}/S$$

full in Cat/S (in the diagram p. 76, there are five others, namely $\underline{\mathrm{Fibhot}}\, S$, $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Liss}}_0 S$, $\underline{\mathrm{Prop}}\, S$, $\underline{\mathrm{Prop}}_0\, S$). As for the relations between the localized categories of these seven categories, which are at the center of the diagram, the statements of theorem-scholie 1 (p. 70-81) give nothing more than the content of theorem-scholie 2 (p. 108-113). To establish them, we therefore have the choice between the two hypotheses A) or B) of the said theorem. Either use an explicit functor C_S, with the help of a functor $X \longmapsto \underline{\mathrm{Ch}}(X)$ ("paths in X") on Cat, hence the structures of source, target and composition for paths, [page 114] and trivial paths (acting as two-sided units for composition). The construction of such a functor C does not look trivial and seems not to be known - I struggled to get there! (With, in addition, the associativity of composition …). Or else use, in the manner of QUILLEN, the existence of a class of morphisms called Q-fibrant, with good factorization properties. This is not obvious in Cat either, and it is established by THOMASON, by a somewhat complicated route. Perhaps I shall be led to come back to this - but only if necessary!

On the other hand, the assertions concerning the categories $\underline{\mathrm{Fib}}\, S$, $\underline{\mathrm{Fibsc}}\, S$, or $\underline{\mathrm{Cofib}}\, S$, $\underline{\mathrm{Cofibsc}}\, S$, and their variants with indices 0 (thus the eight wing-categories of the diagram), and their relations to Cat/S on the one hand, to $\underline{\mathrm{Fibhot}}\, S$ on the other (for the categories with index 0), cannot be contained in a general carpet on categories of models $(\mathcal{M}, W)$, it seems to me, in view of the very special nature of the four categories designated above, of their very particular role in the categorical formalism. Now at least the "split" categories $\underline{\mathrm{Fibsc}}\, S$, $\underline{\mathrm{Cofibsc}}\, S$, and consequently also their variants with index 0, play an essential role for the theory of the $\mathrm{HOT}(S)$, since it is they that intervene dually when we apply the general recipe for constructing a derivator from a category of [page 115] models $(\mathcal{M}, W)$. So I see no simplification of substance to be made to theorem-scholie 1. At most one could prune the diagram of categories under consideration, by dropping p. 72 $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Liss}}_0 S$, $\underline{\mathrm{Prop}}\, S$, $\underline{\mathrm{Prop}}_0 S$, which do not play a really useful role in it. As for the stated results, what concerns them is covered by what is said for the full subcategories $\mathcal{F}$ of Cat/S which contain (say) $\underline{\mathrm{Parf}}\, S$ (or at one's choice those which contain $\underline{\mathrm{Fib}}\, S$ or $\underline{\mathrm{Cofib}}\, S$).

The description of categories of the type $\mathrm{HOT}(S)$, for fixed S, now seems to me perfectly well understood. It is time to let S vary, and to understand the contravariant character in S. We thus take

$$S' \xrightarrow{f} S,$$

and we propose to describe the inverse image functors for the three types of categories $\mathrm{HOT}(S)$. But we shall now restrict ourselves to the $\mathrm{HOT}_W(S)$ and $\mathrm{HOT}_W^{\mathrm{lc}}(S)$, since

everything concerning $\mathrm{HOT}^o_W(S) = \mathrm{HOT}_W(S^o)$ will follow from this by the duality $S \longmapsto S^o$. So it is a matter of describing [page 116]

$$f^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S') \qquad (*)$$

and the induced functor

$$f^* : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S').$$

When we describe the categories $\mathrm{HOT}(S)$, in accordance with the general recipe, by means of functors

$$F : S^o \longrightarrow \mathrm{Cat}$$

(presheaves of categories on S), or, what amounts to the same thing, of split categories, the description of the arrow $(*)$ is obvious: it comes down to the law

$$\left|\begin{array}{rcl} F & \longmapsto & F \circ f^o \\ \underline{\mathrm{Hom}}(S^o, \mathrm{Cat}) & \longrightarrow & \underline{\mathrm{Hom}}(S'^o, \mathrm{Cat}), \end{array}\right.$$

by passing to the localized categories for W-equivalence argument by argument. In the language of fibered categories, it is well known that this trivial operation on presheaves of categories corresponds to the *inverse image* operation on categories over S - which operation passes to splittings. This therefore gives us

$$f^*(\mathrm{hot}_{W,S}(X)) = \mathrm{hot}_{W,S'}(\underbrace{X \times_S S'}_{X'}),$$

for every X split fibered over S. Now we know that the splitting of X plays no role, in the end, in the description of the class

$$\mathrm{hot}_{W,S}(X) \in \mathrm{HOT}_W(S)$$

associated to X. We can forget the splitting, and even (to study the $\mathrm{Hom}_{\mathrm{Hot}_W(S)}(X,Y)$) work with S-functors *of any kind* between categories fibered over S - no need even to assume them cartesian, or even compatible with splittings. This seems to me a very important thing for the formalism of the $\mathrm{HOT}_W(S)$, and very special to the case of the category of models Cat. This is somewhat tacit in the theorem-scholie - but my initial doubts concerning the category Fibsc itself show clearly that I felt there was something there that was not obvious, nor even (almost) plausible. And yet the existence of the functor Φ shows that it is indeed so. p. 73 [page 117]

We can therefore interpret f^* in terms of $\underline{\mathrm{Fib}}\, S$, $\underline{\mathrm{Fib}}\, S'$, by the base change functor

$$f^* : \underline{\mathrm{Fib}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S', \qquad X \longmapsto X \times_S S'/S',$$

which functor obviously preserves the fiberwise W-equivalence W^{f}_S, $W^{\mathrm{f}}_{S'}$, which is the relevant one here

$$f^*(W^{\mathrm{f}}_S) \subseteq W^{\mathrm{f}}_{S'},$$

so that f^* does indeed induce

$$(\underline{\mathrm{Fib}}\, S)(W^{\mathrm{f}}_S)^{-1} \longrightarrow (\underline{\mathrm{Fib}}\, S')(W^{\mathrm{f}}_S)^{-1}$$

, i.e. the functor $(*)$ [page 118]

$$f^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S').$$

On the other hand, $\mathrm{HOT}_W(S)$ can also be described as a localized category of Cat/S as a whole, with respect to the localizer W_S^{d}, which on $\underline{\mathrm{Fib}}\,S$, and even on $\underline{\mathrm{Liss}}\,S$, coincides with W_S^{f}. But here we run up against the difficulty that the inverse image functor

$$f^* : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S', \qquad X \longmapsto X \times_S S'/S',$$

is *not* compatible in general (if S is not isomorphic to the one-point category, for arbitrary S') with the localizers W_S^{d}, $W_{S'}^{\mathrm{d}}$. We see this e.g. by taking $S' = \{s\}$, and recalling that W_S^{d} does *not* imply, on the whole of Cat/S, fiberwise W-equivalence.

These troubles do not occur on $\underline{\mathrm{Liss}}\,S$, where W_S^{d} and W_S^{f} coincide. We can therefore describe f^* on the HOT also with the help of

$$f^* : \underline{\mathrm{Liss}}\,S \longrightarrow \underline{\mathrm{Liss}}\,S', \qquad X \longmapsto X \times_S S'/S',$$

by passing to the localized categories

$$f^* : (\underline{\mathrm{Liss}}\,S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Liss}}\,S')(W_{S'}^{\mathrm{f}})^{-1}.$$

In other words:

Proposition 1. *If X in Cat/S is smooth over S, then we have a canonical isomorphism*

$$f^*(\mathrm{hot}_{W,S}(X)) \simeq \mathrm{hot}_{W,S'}(X'), \qquad \textit{where } X' = X \times_S S'/S'.$$

A similar remark applies to $\underline{\mathrm{Fibhot}}\,S$, where $W_S^{\mathrm{d}} = W_S^{\mathrm{f}}$ as well. The base change functor induces

$$\underline{\mathrm{Fibhot}}\,S \longrightarrow \underline{\mathrm{Fibhot}}\,S', \qquad X \longmapsto X \times_S S'/S', \qquad \text{p. 74 [page 119]}$$

compatible with the relevant relative localizers (all those considered so far coincide on these categories, under the hypotheses made on W), whence

$$f^* : (\underline{\mathrm{Fibhot}}\,S)W_S^{-1} \longrightarrow (\underline{\mathrm{Fibhot}}\,S')W_{S'}^{-1}.$$

The localized categories this time are not the HOT_W, but the $\mathrm{HOT}_W^{\mathrm{lc}}$, so we find a nice description of

$$f^* : \mathrm{HOT}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{HOT}_W^{\mathrm{lc}}(S'),$$

still more advantageous than in terms of $\underline{\mathrm{Liss}}_0 S$, $\underline{\mathrm{Liss}}_0 S'$, since applying to somewhat larger categories (NB $\underline{\mathrm{Liss}}_0 \subseteq \underline{\mathrm{Fibhot}}$). Let us therefore note the

Corollary 1. *The formula of proposition 1 (p. 118) remains valid if X is assumed* Hot-*fibrant over S (instead of smooth over S).* (One then finds relative homotopy types *locally constant* over S, S'.)

We thus have a description of f^* which applies to all the categories on the left-hand side of the diagram 76, with the sole exception of Cat/S itself. If X is in Cat/S, without being either smooth or Hot-fibered over S, how to find the inverse image in $\mathrm{HOT}_W(S')$? We can use Φ to improve the situation! We know that X defines the same class in $\mathrm{HOT}_W(S)$ as $\Phi(X)$, while $C(X)$ represents the image of this class under $i\rho$. These two objects are, the one smooth (and even split fibered), the other Hot-fibered (and even perfect) over S. Thus [page 120]

Corollary 2. *If X in Cat/S is arbitrary, then*

$$f^*(\mathrm{hot}_{W,S}(X)) \simeq \mathrm{hot}_{W,S'}(\Phi(X) \times_S S'), \qquad f^* i\rho(\mathrm{hot}_{W,S}(X)) \simeq \mathrm{hot}_{W,S'}(C(X) \times_S S').$$

One may expect, on the other hand, that for certain particular base changes, the formula of proposition 1 is valid without restriction on X in Cat/S. It must come to $\pm$ the same thing that the inverse image functor

$$X \longmapsto X \times_S S'$$

be compatible with the localizers W_S^{d}, $W_{S'}^{\mathrm{d}}$. The condition of W-equivalence "colocal over S" must be viewed as a condition of a "homological" nature, for the derivator HOT. It is associated, for an arbitrary derivator $\mathbf{D}$, with the condition (for $f : X \longrightarrow Y$ in Cat/S) that the $s\backslash X \longrightarrow s\backslash Y$ are $\mathbf{D}$-equivalences, which is written here elegantly in terms of relative homology, by the condition that $\forall\ a \in \mathcal{A} = \mathbf{D}(e)$, the arrow induced by f p. 75 [page 121]

$$\underbrace{H_\bullet^{\mathbf{D}}(X/S, a_X)}_{\stackrel{\mathrm{def}}{=}(p_X)_!(a_X)} \longrightarrow \underbrace{H_\bullet^{\mathbf{D}}(Y/S, a_Y)}_{\stackrel{\mathrm{def}}{=}(p_Y)_!(a_Y)}$$

be isomorphism. (Use the "calculation" of the relative homology sheaves $(p_X)_!(a_X)$, $(p_Y)_!(a_Y)$, by the characterization of the fibers in Der 4 bis). Now, the $\mathbf{D}$-*proper* base changes $S' \longrightarrow S$ commute with the $p_!$, so that we find the

Lemma. *The notion of* $\mathbf{D}$*-colocal equivalence over* S *in* Cat/S *is stable under every base change* $S' \longrightarrow S$ *that is* $\mathbf{D}$*-proper. (Whatever the derivator* $\mathbf{D}$*.)*

It will be recalled that the $s\backslash S \longrightarrow S$ are precisely proper base changes! One can say that $\mathbf{D}$-colocal equivalence can be described by requiring that $f : X \longrightarrow Y$ be transformed into W-equivalences by certain types of base change $S' \longrightarrow S$, choosing at will one of the four types

a) The base changes $s\backslash S \longrightarrow S$ (which gives back the definition we had adopted).

b) The base changes that are local co-isomorphisms.

c) Proper base changes.

d) $\mathbf{D}$-proper base changes.

We cannot yet apply these results to the derivator HOT, for lack of having yet constructed it, and verified, the relevant axioms. We can simply state the [page 122]

Conjecture (sure!) : *Let* $p : S' \longrightarrow S$ *be a proper arrow (understood:* HOT*-proper) in* Cat*. Then the base change functor*

$$\mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S', \qquad X \longmapsto X \times_S S'/S'$$

is compatible with the localizers W_S^{d}, $W_{S'}^{\mathrm{d}}$*. (For a smooth functor* $S' \longrightarrow S$*, base change would be compatible with* W_S^{g}, $W_{S'}^{\mathrm{g}}$*.)* ([61]).

But for every base change $S' \longrightarrow S$ (proper or not) which has the preceding property, hence which induces a functor

$$\underbrace{(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}}_{\approx \mathrm{HOT}_W(S)} \longrightarrow \underbrace{(\mathrm{Cat}/S')(W_{S'}^{\mathrm{d}})^{-1}}_{\approx \mathrm{HOT}_W(S')},$$

it is immediate, using the comparison theorem (e.g. with $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Liss}}\, S'$) over S and over S', that this functor is none other (up to canonical isomorphism) than the inverse image functor f^* for relative homotopy types. So let us state the p. 76

Corollary 3. *The formula of proposition 1 remains valid without any hypothesis whatever on* X *in* Cat/S*, provided that the base change arrow* $f : S' \longrightarrow S$ *is such that the functor* $X \longmapsto X \times_S S'$ *from* Cat/S *to* Cat/S' *takes colocal* W*-equivalences over* S *to colocal* W*-equivalences over* S'*.* (Which should be the case if (and only if?) f is W-proper, hence also if f is ("absolutely") proper ([62]).) [page 123]

[61] NB This "conjecture" is proved without a lemma on page 168-169, without needing anything that is not already known here. Cf. theorem 4, case b) p. 162.

[62] cf. I, pages 55-59.

So far we have worked only with the localizers W_S^{d}, $W_{S'}^{\mathrm{d}}$, to the exclusion of W_S, $W_{S'}$. (Except on the categories contained in $\underline{\mathrm{Fibhot}}\, S$, where the two coincide.) When one makes no draconian hypothesis on $p : S' \longrightarrow S$, of the Hot-fibration type, one has no chance of $X \longmapsto X \times_S S'$ taking coarse W-equivalences into likewise, hence no chance of catching the inverse image of the "coarse" relative homotopy type of X, by taking that of $X' = X \times_S S'$. This is tied to the fact that for a general $S' \longrightarrow S$, the localization functors

$$\mathrm{HOT}_W(S) \xrightarrow{\rho_S} \mathrm{HOT}_W^{\mathrm{lc}}(S), \qquad \mathrm{HOT}_W(S') \xrightarrow{\rho_{S'}} \mathrm{HOT}_W^{\mathrm{lc}}(S')$$

have no desire to commute with the inverse image functor f^* on the HOT_W. We may hazard the

Proposition 2. *Equivalent conditions on the functor $S' \longrightarrow S$:*

a) The diagram of functors

$$\begin{array}{ccc} \mathrm{HOT}_W(S) & \xrightarrow{\rho_S} & \mathrm{HOT}_W^{\mathrm{lc}}(S) \\ \downarrow{\scriptstyle f^*} & & \downarrow{\scriptstyle f^*_{\mathrm{lc}}} \\ \mathrm{HOT}_W(S') & \xrightarrow{\rho_{S'}} & \mathrm{HOT}_W^{\mathrm{lc}}(S') \end{array}$$

commutes (up to isomorphism of composite functors).

[page 124]

b) The base change functor

$$\underline{\mathrm{Liss}}\, S \longrightarrow \underline{\mathrm{Liss}}\, S', \qquad X \longmapsto X \times_S S'/S'$$

takes W_S into $W_{S'}$.

p. 77

b′) Analogous statement for $\underline{\mathrm{Fib}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S'$.

b″) Analogous statement for $\underline{\mathrm{Fibsc}}\, S \longrightarrow \underline{\mathrm{Fibsc}}\, S'$.

([63], [64]).

Proof. If $\mathcal{F}$ is one of the three categories $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Fib}}\, S$ or $\underline{\mathrm{Fibsc}}\, S$, the functor ρ_S can be described as the canonical localization functor

$$\mathcal{F}(W_S^{\mathrm{d}})^{-1} \longrightarrow \mathcal{F}W_S^{-1}$$

(that would work for any full subcategory $\mathcal{F}$ of Cat/S which contains $\underline{\mathrm{Fib}}\, S$). Now if b), b′) or b″) is satisfied, we find a commutative diagram of functors induced by base change

$$\begin{array}{ccc} \mathcal{F}(W_S^{\mathrm{d}})^{-1} & \xrightarrow{\rho_S} & \mathcal{F}(W_S)^{-1} \\ {\scriptstyle f^*_{W_S^{\mathrm{d}}}}\downarrow & & \downarrow{\scriptstyle f^*_{W_S}} \\ \mathcal{F}'(W_{S'}^{\mathrm{d}})^{-1} & \xrightarrow{\rho_{S'}} & \mathcal{F}'(W_{S'})^{-1}, \end{array}$$

with the proviso, it is true, in the case $\mathcal{F} = \mathrm{Cat}/S$, that the inverse image functor is also compatible with W_S^{d}, $W_{S'}^{\mathrm{d}}$ (which is automatic if $\mathcal{F} = \underline{\mathrm{Fib}}\, S$ or $\underline{\mathrm{Fibsc}}\, S$). We know that $f^*_{W_S^{\mathrm{d}}}$ describes the functor f^* on relative homotopy types, and to prove a) as a consequence of b) (strengthened), b′ or b″, it remains to make sure that the functor $f^*_{W_S}$

[page 125]

[63] Not proved, only that b, b′, b″ imply a). For the reverse implication, one would need to know that every arrow of $\mathcal{F}'$ which becomes isomorphism in $\mathcal{F}'W_{S'}^{-1}$, is in $W_{S'}$ (strong saturation of $W_{S'}$ in $\mathcal{F}'$), something I do not know. But I believe this must be true.

[64] **NB** If we assume p proper, we should also be able to take $\mathcal{F} = \mathrm{Cat}/S$.

likewise describes the functor induced by the preceding one on locally constant relative homotopy types. Now we have the canonical inclusions $\mathcal{F}_0 \subseteq \mathcal{F}$, $\mathcal{F}'_0 \subseteq \mathcal{F}'$, and on $\mathcal{F}_0$, $\mathcal{F}'_0$ $W_S = W_S^{\mathrm{d}} = W_S^{\mathrm{f}}$, and the categories are OK for describing $\mathrm{HOT}_W^{\mathrm{lc}}(S)$ and $\mathrm{HOT}_W^{\mathrm{lc}}(S')$, and the functor f^* induced between them by inverse images ([65]) (**NB** The base change functor $X \longmapsto X \times_S S'$ is compatible with the localizers on these categories, for it is trivial in the form W_S^{f}.) We then have, by inverse images $X \longmapsto X \times_S S'$, a commutative diagram

$$\begin{array}{ccc} \mathcal{F}_0(W_S)^{-1} & \xrightarrow{\approx} & \mathcal{F}(W_S)^{-1} \\ {\scriptstyle (f^*_{W_S})_0}\downarrow & & \downarrow{\scriptstyle f^*_{W_S}} \\ \mathcal{F}'_0(W_{S'})^{-1} & \xrightarrow{\approx} & \mathcal{F}'(W_{S'})^{-1}, \end{array}$$

where the horizontal arrows are *equivalences* of categories, by theorem-scholie 1 ①. p. 78 This proves that $f^*_{W_S}$ too is good for calculating the inverse images of relative homotopy types.

We have proved that b, b′, b″ each imply a), let us prove that a) implies them. Now a) says exactly that the inverse images f^* of the elements of $\mathrm{HOT}_W^{\mathrm{lc}}(S)$ are, up to isomorphism, obtained by the inverse image functor on the whole of $\mathcal{F}$, interpreting $\mathrm{HOT}_W^{\mathrm{lc}}(S)$ as $\mathcal{F}W_S^{-1}$. But this does indeed imply that $\mathcal{F} \longrightarrow \mathcal{F}'$ must be compatible with the localizers. But I'm getting carried away - one only finds that if $u : X \longrightarrow Y$ in $\mathcal{F}$ is in W_S, then the arrow $u' : X' \longrightarrow Y'$ gives an isomorphism in $\mathcal{F}'(W_{S'})^{-1}$. I do not know, on the other hand, whether this implies that u' is in $W_{S'}$. [page 126]

Corollary 1. *Suppose that $p : S' \longrightarrow S$ is a* Hot*-fibration. Then the inverse image functor $X \longmapsto X \times_S S'$ from* Cat$/S$ *to Cat$/S'$ maps W_S into $W_{S'}$, hence the functor $p^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$ commutes with the localization functors ρ_S, $\rho_{S'}$ in these categories.* (**NB** One must make the appropriate hypothesis on W, namely L6 bis page 44. It is OK for W_∞.)

The fact that $X \longmapsto X \times_S S'$ is compatible with W_S, $W_{S'}$ is a tautology, in view of hypothesis L6 bis. It does not seem that it is also compatible with W_S^{d}, $W_{S'}^{\mathrm{d}}$ (I should make a counterexample, with of course S' not proper over S). So we are under the conditions of b), which implies a).

I do not feel capable of exhibiting a functor other than a Hot-fibration (or at least a W-fibration), for which the inverse image functor on HOT_W is compatible with the localization functors ρ_S, $\rho_{S'}$. Of course, if X is arbitrary in Cat$/S$, nothing prevents us from looking at its localized object for W_S, i.e. $\rho_S(\mathrm{hot}_{W,S}(X))$, and asking how to find its inverse image. If $p : S' \longrightarrow S$ is a Hot-fibration, then it suffices to take $\mathrm{hot}_{W,S'}(X')$, $X' = X \times_S S'$. In any other case (I mean, without a hypothesis on X/S, more precisely without assuming that X is itself Hot-fibered over S), this formula is hopelessly botched. We must, to calculate $f^*\rho_S(\mathrm{hot}_{W,S}(X))$, "calculate" $\rho_S(\mathrm{hot}_{W,S}(X))$ by "going up to the upper floor", as $\mathrm{hot}_{W,S}(C(X))$. We then find the inverse image $\mathrm{hot}_{W,S'}(C(X) \times_S S')$ (by corollary 1, p. 119). [page 127]

Now that the contravariant law of $\mathrm{HOT}_W(S)$ in $S \in \mathrm{Ob}\,\mathrm{Cat}$ seems well understood, it is a matter of seeing that we do indeed have a derivator HOT. I do not expect the $\mathrm{HOT}_W^{\mathrm{lc}}(S)$ to define a derivator as well, but I would have to make sure of this expressly. (It is for the existence of $f_!$, f_* that I must look.) [page 128]

To tell the truth, one would have to, to do things properly, look also at the 2-*functor* structure of

$$S \longmapsto \mathrm{HOT}_W(S) \qquad \mathrm{Cat}^o \longrightarrow \mathrm{Cat},$$

p. 79

[65] NB $(\mathrm{Cat}/S)_0 = \underline{\mathrm{Fibhot}}\, S$, for the other cases, it is clear.

so for S, S' given in Cat, define

$$\underline{\mathrm{Hom}}(S', S)^o \longrightarrow \underline{\mathrm{Hom}}(\mathrm{HOT}_W(S), \mathrm{HOT}_W(S')).$$

This comes down above all, for

$$f, g : S' \rightrightarrows S$$

two arrows from S' to S, and a homomorphism of functors

$$u : f \longrightarrow g,$$

to describing a homomorphism of functors

$$u^* : g^* \longrightarrow f^*$$

between $\mathrm{HOT}_W(S)$ and $\mathrm{HOT}_W(S')$. This is moreover already done, in the general setting of $(\mathcal{M}, W)$, when one describes $\mathrm{HOT}_W(S)$ and $\mathrm{HOT}_W(S')$ by localization of $\mathrm{Hom}(S^o, \mathrm{Cat})$ and $\mathrm{Hom}(S'^o, \mathrm{Cat})$. These categories are interpreted respectively as $\underline{\mathrm{Fibsc}}\, S$ and $\underline{\mathrm{Fibsc}}\, S'$, hence as categories of categories over S resp. S', so one would like to interpret u^* in terms of such categories, and while one is at it, for categories X over S as general as possible. We shall then expect to have to make assumptions, either on X, or on f and on g, which ensure that the images of $\mathrm{hot}_{W,S}(X)$ under f^* and g^* are none other than $\mathrm{hot}_{W,S}(f^*X)$, $\mathrm{hot}_{W,S}(g^*X)$ respectively, where we put [page 129]

$$f^*(X) = X \times_S (S', f), \qquad g^*(X) = X \times_S (S', g).$$

So we shall probably have to place ourselves under one of the following assumptions:

a) X is smooth over S.

b) X is Hot-fibered over S.

c) f and g are proper (and subject to the validity of the conjecture on page 122).

One then expects u^* to be associated with a homomorphism

$$u^* : g^*(X) \longrightarrow f^*(X) \qquad \textit{in } \mathrm{Cat}/S'$$

defined in terms of u. Is there a way of actually defining it, without assuming X at least fibered over S? Let us take $S' = e$, so f, g corresponding to objects $s, t \in \mathrm{Ob}\, S$, and u to an arrow

$$u : s \longrightarrow t,$$

the inverse images of X/S under f, g are the fibers X_s, X_t, and we seek

$$u^* : X_t \longrightarrow X_s.$$

We have no chance of finding one canonically, if we do not assume X fibered over S! So it is not a), b) or c) that can serve us, but only the hypothesis $X \in \mathrm{Ob}\, \underline{\mathrm{Fib}}\, S$. Thus one can hope to find a description of u^* between $\mathrm{HOT}_W(S)$ and $\mathrm{HOT}_W(S')$ in terms of operations on the categories over S, S', as just suggested, only by restricting oneself to "models" X which are in $\underline{\mathrm{Fib}}\, S$, at least. And I do not doubt indeed that this works on $\underline{\mathrm{Fib}}\, S$ no less well than directly on $\underline{\mathrm{Fibsc}}\, S$, in terms of $\underline{\mathrm{Hom}}(S^o, \mathrm{Cat})$. p. 80 [page 130]

For the essential axioms for derivators, I refer to I p. 67-68 and p. 111-112. We have seen that Der 1, Der 2 ([66]) are satisfied for every prederivator defined by a pair

[66] Botched for Der 2, I do not know whether it is true even assuming W *strongly* saturated, i.e. such that $\mathrm{hot}_W(f)$ isomorphism $\Longrightarrow f \in W$.

$(\mathcal{M}, W \subseteq \mathrm{Fl}\,\mathcal{M})$, so it is OK here for HOT_W. It follows that it is OK also for $\mathrm{HOT}^{\mathrm{lc}}_W$, in view of the characterization of the $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ as subcategories of $\mathrm{HOT}_W(S)$. For Der 1, does this work even for arbitrary sums of categories, i.e. do we have

$$\mathrm{HOT}_W(\coprod_i S_i) \xrightarrow{\sim} \prod_i \mathrm{HOT}_W(S_i)\,?$$

We have in any case

$$\underline{\mathrm{Fib}}(\overbrace{\coprod_i S_i}^{S}) \xrightarrow{\sim} \prod_i \underline{\mathrm{Fib}}(S_i),$$

and the localizer W^{f}_S in $\underline{\mathrm{Fib}}(S)$ is indeed identified with the product of the localizers $W^{\mathrm{f}}_{S_i}$ [page 131]
in the $\underline{\mathrm{Fib}}\,S_i$. But in the case of a perhaps infinite family of categories $\mathcal{F}_i$, equipped with localizers W_i, if we put

$$\mathcal{F} = \prod \mathcal{F}_i, \qquad W = \prod W_i \subseteq \mathrm{Fl}(\mathcal{F}),$$

is the canonical functor

$$\mathcal{F}W^{-1} \longrightarrow \prod_i (\mathcal{F}_i W_i^{-1})$$

always an equivalence? Even in the case I finite one needs (it seems to me) a rather careful argument, either by the universal property by induction on $\mathrm{card}\,I$ (one is reduced to the case of two functors), or by dint of categories of paths, noting that

$$\underline{\mathrm{Ch}}^W_\infty(\mathcal{F}; (s_i), (t_i)) \xrightarrow{\sim} \prod_i \underline{\mathrm{Ch}}^{W_i}_\infty(\mathcal{F}_i; s_i, t_i), \qquad (*)$$

and passing to the π_0, which gives

$$\mathrm{Hom}_W((s_i), (t_i)) \xrightarrow{\sim} \prod_i \mathrm{Hom}_{W_i}(s_i, t_i).$$

But formula $(*)$, visibly, only works for I finite. In the general case, one finds the subcategory $\prod_i^!$ of $\prod_i$ formed by the systems $(c_i)_{i\in I}$ of paths which are of bounded length.

So the question of the validity of Der 1 without a finiteness assumption on the indexing set I remains open for the moment. (I suspect that it is true, but one would need, to p. 81
prove it, to have in hand a description of the arrows in $\mathrm{HOT}_W(S)$ by paths of bounded [page 132]
length, say of length two or three. This is an interesting question, but one that I reserve for later, if need forces me to it ...)

More important is the question of the validity of Der 3 and of its variants (I, p. 111-112). The minimum one asks of a derivator $\mathbf{D}$ is that if $S' \longrightarrow S$ in Diag is a Hot-equivalence, it should be a $\mathbf{D}$-equivalence, i.e. (in the present case) that the functor

$$f^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$$

be fully faithful on the subcategory of *constant* objects in $\mathrm{HOT}_W(S)$. This condition becomes more tangible when one already has either the cohomology functors $H^\bullet_{\mathbf{D}}(S, -)$, $H^\bullet_{\mathbf{D}}(S', -)$, or the homology $H^{\mathbf{D}}_\bullet(S, -)$, $H^{\mathbf{D}}_\bullet(S', -)$. In terms of the latter, it means that

$$p_! p^*(a) \xrightarrow{\sim} p'_! p'^*(a) \qquad \text{isomorphism},$$

for every a in $\mathcal{A} = \mathbf{D}(e)$. (Where $p : S \longrightarrow e$, $p' : S' \longrightarrow e$ are the structural morphisms [page 133]

to the one-point category.) We have seen the general heuristic formula

$$p_! \underbrace{p^*(a)}_{a_S} = \mathrm{hot}(S) \times a,$$

where $\mathrm{hot}\, S$ is the class of $S \in \mathrm{Ob}\,\mathrm{Cat}$ in $\mathrm{Hot} = (\mathrm{Cat})W_\infty^{-1}$, and where the product $\times$ is an external operation of Hot on $\mathcal{A} = \mathbf{D}(e)$

$$\mathrm{Hot} \times \mathcal{A} \longrightarrow \mathcal{A}\ .$$

In the present case, $\mathcal{A} = \mathrm{Hot}$, and the operation

$$\mathrm{Hot} \times \mathrm{Hot} \longrightarrow \mathrm{Hot}$$

should be none other than the cartesian product functor, which is deduced from the similar functor in Cat. When one has this formalism, the desired result for an $f : S' \longrightarrow S$ simply reduces to saying that if f is in W, then so is $f \times \mathrm{id}_X$ for every X in Cat (taking X such that $\mathrm{hot}(X) \simeq a$). Now here this is a well-known fact, resulting from the axioms imposed on W.

So I have the impression that I have no chance of proving Der 3a′ (p. 111), nor even Der 3a) (the case where S = e), without having a mastery of the formalism of the $p_!$. And I have good hope that this mastery will be obtained at little cost, whereas the question of the p_* seems much more delicate. p. 82 [page 134]

I am therefore going to turn first to the axiom Der 4′ of existence of

$$f_! : \mathrm{HOT}_W(S') \longrightarrow \mathrm{HOT}_W(S),$$

as a left adjoint to f^*. I am going to try to construct ad-hoc a functor $f_!$, then the two adjunction arrows

$$\begin{array}{ccc} \mathrm{id} & \longrightarrow & f^* f_! \\ \mathrm{id} & \longleftarrow & f_! f^*, \end{array}$$

and finally, to check the two relevant compatibilities.

The "obvious" geometric idea is, for $X' \in \mathrm{Cat}/S'$, representing $\xi' \in \mathrm{Ob}\,\mathrm{HOT}_W(S')$, to define $f_!(\xi')$ by

$$f_!(\mathrm{hot}_{S',W}(X')) = \mathrm{hot}_{S,W}(X'),$$

where on the right-hand side, X' is regarded as an object of Cat/S, thanks to $f : S' \longrightarrow S$ and to the functor

$$f_! : \mathrm{Cat}/S' \longrightarrow \mathrm{Cat}/S$$

which it induces. The only problem is to see whether this functor is compatible with the localizers $W^{\mathrm{d}}_{S'}$, W^{d}_S, and if not, whether it is so at least on the subcategory $\underline{\mathrm{Fib}}\, S'$ or $\underline{\mathrm{Fibsc}}\, S'$, which suffices to compute $\mathrm{HOT}_W(S')$ by localization. We must therefore see whether an $X'_1 \longrightarrow X'_2$ in Cat/S', which is in $W^{\mathrm{d}}_{S'}$, i.e. which is in W and remains so under every base change of the form $s'\backslash S' \longrightarrow S'$, more generally for every local co-isomorphism, is also in W^{d}_S, i.e. whether base changes of the form $s\backslash S \longrightarrow S$ turn it into W-equivalences (coarse). Now local co-isomorphisms are stable under *every base change* (cf. I, p. 55 in the dual case of local isomorphisms), so we win! [page 135]

$$\begin{array}{ccc} \overline{S'} = s\backslash S' & \longrightarrow & S' \\ \downarrow & & \downarrow \\ \overline{S} = s\backslash S & \longrightarrow & S \end{array} \quad .$$

I am going to check it again all the same. If

$$\begin{array}{ccc} \overline{S'} & \longrightarrow & S' \\ \downarrow & & \downarrow \\ \overline{S} & \longrightarrow & S \end{array}$$

is a cartesian square in Cat, we said in loc. cit., essentially, that for $\overline{s'}$ in $\overline{S'}$, having images s', $\overline{s}$, s in S', $\overline{S}$, S respectively, the square of the colocalized categories p. 83

$$\begin{array}{ccc} \overline{s'}\backslash\overline{S'} & \longrightarrow & s'\backslash S' \\ \downarrow & & \downarrow \\ \overline{s}\backslash\overline{S} & \longrightarrow & s\backslash S \end{array}$$

is again cartesian. I have just rechecked it again this very instant, for the objects at least. To say that $\overline{S} \longrightarrow S$ is a local co-isomorphism, means that the $\overline{s}\backslash\overline{S} \longrightarrow s\backslash S$ are isomorphisms. But the preceding cartesian square shows that then the $\overline{s'}\backslash\overline{S'} \longrightarrow s'\backslash S'$ are also isomorphisms, we win! [page 136]

Theorem 1. *For every arrow $f : S' \longrightarrow S$ in* Cat*, the induced functor*

$$f_! : \mathrm{Cat}/S' \longrightarrow \mathrm{Cat}/S$$

is compatible with the localizers $(W^{\mathrm{d}}_{S'}, W^{\mathrm{d}}_{S})$, hence dually with the localizers $(W^{\mathrm{g}}_{S'}, W^{\mathrm{g}}_{S})$, and trivially with the localizers $(W_{S'}, W_S)$. It therefore induces a commutative diagram of functors on the localized categories

$$\begin{array}{ccccc} \mathrm{HOT}_W(S') & \xrightarrow{\rho_{S'}} & \mathrm{HOT}^{\mathrm{lc}}_W(S') & \xleftarrow{\sigma_{S'}} & \mathrm{HOT}^{o}_W(S') \\ \downarrow{\scriptstyle f_!} & & \downarrow{\scriptstyle f^{\mathrm{lc}}_!} & & \downarrow{\scriptstyle f^{o}_!} \\ \mathrm{HOT}_W(S) & \xrightarrow{\rho_{S}} & \mathrm{HOT}^{\mathrm{lc}}_W(S) & \xleftarrow{\sigma_{S}} & \mathrm{HOT}^{o}_W(S). \end{array}$$

Definition 1. *We have thus defined the vertical functors $f_!$, $f^{\mathrm{lc}}_!$, $f^{o}_! = (f^o)_!$, called homological direct image functors, or integration functors for f (and relative homotopy types).*

We have obtained at the same time, for free, the compatibility of these functors with the functors localization $\rho_{S'}$, $\sigma_{S'}$ for $\mathrm{HOT}(S')$, and ρ_S, σ_S for $\mathrm{HOT}(S)$. On the other hand, one will not expect them to commute with the fully faithful inclusion functors of the $\mathrm{HOT}^{\mathrm{lc}}_W$ into HOT_W. Indeed, if one does not assume f in Fibhot ([67]), it is quite immoral to expect that $f_!$ in $\mathrm{HOT}_W(S')$ should transform locally constant types into likewise. The situation is therefore exactly the reverse of that for the functor f^*, which was compatible with the inclusion functors i_S, $i_{S'}$ and j_S, $j_{S'}$ - this was so trivial that there was no question even of saying it, let alone making a proposition of it! The situation is [page 137]

even better still for the functor $f_!$ between the HOT, since the description of this functor can be given directly on the whole of Cat/S', without having to restrict oneself to more or less clever subcategories, where the induced functor would be compatible with the localizers (Liss S and Fibhot S in the case of f^*). p. 84 [page 138]

We have immediately the transitivity of $f_!$, $f^{\mathrm{lc}}_!$, $f^{o}_!$, i.e.

$$(gf)_! = g_! f_!, \qquad (gf)^{\mathrm{lc}}_! = g^{\mathrm{lc}}_! f^{\mathrm{lc}}_!, \qquad (gf)^{o}_! = g^{o}_! f^{o}_!.$$

[67] This case must be looked at carefully later.

But we are not going to move S, S', f for the moment, for we must now, before anything else, try to establish the adjunction of $f_!$ and f^* (in that order).

For the moment, let us deal with HOT_W only - the case of $\mathrm{HOT}^{\mathrm{o}}_W$ will follow from this by applying what we shall know to f^o. And we shall deal with $\mathrm{HOT}^{\mathrm{lc}}_W$ later.

1) *The homomorphism*

$$\alpha : \mathrm{id}_{\mathrm{HOT}_W(S')} \longrightarrow f^* f_!.$$

If X' is in Cat/S', $f_!(\mathrm{hot}_{S',W}(X'))$ is $\mathrm{hot}_{S,W}(X')$, but we cannot find its inverse image f^* over S' by taking the categorical inverse image. For even if we make draconian hypotheses on X' as an object of Cat/S' (e.g. X' fibered over S'), these will be reduced to nothing when we have composed with f, on which we do not want to make any hypothesis here. We must therefore first apply the functor Φ_S to X' over S, then take its inverse image over S': [page 139]

$$f^* f_!(\mathrm{hot}_{S',W}(X')) = \mathrm{hot}_{S',W}(\Phi_S(X'/S) \times_S S').$$

We must therefore find an arrow

$$\alpha_{X'} : X' \longrightarrow \Phi_S(X'/S) \times_S S'$$

in the localized $(\mathrm{Cat}/S')(W^{\mathrm{d}}_{S'})^{-1}$, or even an arrow in Cat/S' itself. But by the adjunction properties of the functors $f_!$ and f^* at the *categorical level,* such an arrow amounts to an arrow

$$f_!(X') \longrightarrow \Phi_S(X'/S)$$

in Cat/S, i.e. to an arrow which makes the square commute

$$\begin{array}{ccc} X' & \longrightarrow & \Phi_S(X'/S) \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S. \end{array}$$

Now we have such a canonical arrow, namely $i_{X'/S}$. It is functorial in X' over S (without even having to worry whether X' over S comes from an X' over S', i.e. whether the structural arrow $X' \longrightarrow S$ factors through $S' \longrightarrow S$). p. 85

2) *The homomorphism*

$$\beta : \mathrm{id}_{\mathrm{HOT}_W(S)} \longleftarrow f_! f^*.$$

[page 140]

We can describe $\mathrm{HOT}_W(S)$ by $(\underline{\mathrm{Fib}}\, S)(W^{\mathrm{d}}_S)^{-1}$ or $(\underline{\mathrm{Liss}}\, S)(W^{\mathrm{d}}_S)^{-1}$ for example, so that f^* has a simple description by

$$f^*(\mathrm{hot}_{S,W}(X)) = \mathrm{hot}_{S',W}(X \times_S S')$$

(for X smooth over S). We then have

$$f_! f^*(\mathrm{hot}_{S,W}(X)) = \mathrm{hot}_{S,W}(X \times_S S'),$$

but watch out, $X \times_S S'$ is not a priori in $\underline{\mathrm{Liss}}\, S$, only in Cat/S and nothing more. But that does not prevent it from giving what is needed, by localization $(W^{\mathrm{d}}_S)^{-1}$ in Cat/S. To define β_X, it therefore suffices to find an arrow

$$\beta_X : X \longleftarrow X \times_S S'$$

in Cat/S, and we do see pr_1^X ([68]). We thus have

$$\begin{cases} \alpha : \mathrm{id} & \longrightarrow & f^* f_! \\ \beta : \mathrm{id} & \longleftarrow & f_! f^* \end{cases}$$

[68] when $\xi \in \mathrm{HOT}_W(S)$ is defined by X in Cat/S not necessarily smooth, we must find $\Phi_S(X) \times_S S' \longrightarrow X$ in $(\mathrm{Cat}/S)(W^{\mathrm{d}}_S)^{-1}$, we take $(i_X)^{-1}\mathrm{pr}_1^{\Phi_S(X)}$ (cf. page 143).

3) *The compatibility*

$$f_! \xrightarrow{f_!*\alpha} f_! f^* f_! \xrightarrow{\beta * f_!} f_!.$$
$$\text{id}$$

We must therefore take X' over S', and make the arrows explicit starting from the object

$$\xi' = \text{hot}_{S',W}(X').$$

We have

$$f_!(\xi') = \text{hot}_{S,W}(X'/S),$$

p. 86
[page 141]

and to calculate $f^* f_!(\xi')$, we must as before take $\Phi_S(X')$ and we find

$$f^* f_!(\xi') = \text{hot}_{S',W}(\Phi_S(X') \times_S S').$$

Finally,

$$f_! f^* f_!(\xi') = \text{hot}_{S,W}(\Phi_S(X') \times_S S').$$

We must make the arrows explicit

$$\underbrace{f_!(\xi')} \xrightarrow{f_!(\alpha_{\xi'})} \underbrace{f_! f^* f_!(\xi')} \xrightarrow{\beta_{f_!(\xi')}} \underbrace{f_!(\xi')}.$$

$$X'/S \xrightarrow[\text{deduced from } i_{X'}]{\alpha_{X'}} \Phi_S(X') \times_S S' \xrightarrow{\beta_{X'}} X'/S \qquad (*)$$

$$\text{over } S$$

$\alpha_{X'}$ is deduced from $i_{X'} : X' \longrightarrow \Phi_S(X')$ as the unique S'-morphism making the diagram commute

$$\begin{array}{ccccc} & & i_{X'} & & \\ & \overset{\curvearrowright}{} & & \overset{\curvearrowright}{} & \\ X' & \xrightarrow{\alpha_{X'}} & \Phi_S(X') \times_S S' & \xrightarrow{\text{pr}_1} & \Phi_S(X') \\ \downarrow & \swarrow & & & \downarrow \\ S' & & \longrightarrow & & S. \end{array} \qquad (**)$$

To describe $\beta_{f_!(\xi')}$, where $f_!(\xi') = \text{hot}_{S,W}(X')$, we must give a general description of β_ξ that is valid for *every* representative X of ξ, not only in the case X smooth over S. To calculate $f^*(\xi)$, we must therefore take the $\text{hot}_{S',W}$ of $\Phi(X) \times_S S'$, then for $f_! f^*(\xi)$, we must take the $\text{hot}_{S,W}$ of the same $\Phi_S(X) \times_S S'$. We must define, if we can, an arrow

[page 142]

$$\Phi_S(X) \times_S S' \longrightarrow X \qquad \text{in Cat}/S,$$

or at least in $(\text{Cat}/S)(W_S^{\text{d}})^{-1}$. But we have

$$i_X : X \longrightarrow \Phi_S(X)$$

which is in W_S^{d}, and which by Cartesian product defines

$$i_X \times_S \text{id}_{S'} : X \times_S S' \longrightarrow \Phi_S(X) \times_S S'$$

(in other words we have taken the inverse image of i_X under $S' \longrightarrow S$). Is this arrow still in W_S^{d}? ([69]). If so, we have the diagram

p. 87

[69] No, it is not, for if $S' = \{s\}$, this would mean that $X'_s \longrightarrow s\backslash X$ is in W, which is false in general.

$$\begin{array}{ccc} \Phi_S(X) \times_S S' & \overset{\beta_X ?}{\dashrightarrow} & X \\ {\scriptstyle i_X \times_S \mathrm{id}_{S'}}\uparrow & \overset{\mathrm{pr}_1^X}{\nearrow} & \\ X \times_S S' & & \end{array} \quad = \pi_X$$

and we define the desired arrow β_X *in* $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$ by the commutativity of this triangle, thus as

$$\beta_X = \mathrm{pr}_1^X \circ \underbrace{\pi_X^{-1}}_{\text{inverse in } (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}} .$$

But that doesn't work - cf. annotation! We shall then do the reverse [page 143]

$$\begin{array}{ccc} \Phi_S(X) \times_S S' & \xrightarrow{\mathrm{pr}_1^{\Phi_S(X)}} & \Phi_S(X) \\ & \overset{\beta_X ?}{\searrow} & \uparrow {\scriptstyle i_X} \quad \text{in } W_S^{\mathrm{d}} \\ & & X \end{array}$$

([70]), we define β_X as $(i_X)^{-1}\mathrm{pr}_1^{\Phi_S(X)}$, where $(i_X)^{-1}$ is defined in $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$.

To prove the commutativity of $(*)$ (p. 141), since $i_{X'} : X' \longrightarrow \Phi_S(X')$ is in W_S^{d}, it suffices to prove, instead of $\beta_{X'}\alpha_{X'} = \mathrm{id}_{X'}$, the relation deduced by composing with $i_{X'} : X' \longrightarrow \Phi_S(X')$

$$\underbrace{i_{X'}\beta_{X'}}_{\mathrm{pr}_1^{\Phi_S(X')}} \alpha_{X'} = i_{X'},$$

i.e.

$$\mathrm{pr}_1^{\Phi_S(X')} \circ \alpha_{X'} = i_{X'},$$

which is nothing other than the defining relation of $\alpha_{X'}$ (cf. $(**)$ p. 141), we win!

4) *Compatibility*

$$f^* \xrightarrow{\alpha * f^*} f^* f_! f^* \xrightarrow{f^* * \beta} f^*. \qquad \text{(id)}$$

Let us therefore start with a $\xi \in \mathrm{ObHOT}_W(S)$, which we may suppose defined by $X \in \mathrm{Ob}\underline{\mathrm{Fib}}\, S$. Then [p. 88] [page 144]

$$\begin{aligned} f^*(\xi) &= \mathrm{hot}_{S',W}(X \times_S S'), \\ f_! f^*(\xi) &= \mathrm{hot}_{S,W}(X \times_S S'), \\ f^* f_! f^*(\xi) &= \mathrm{hot}_{S',W}(\Phi_S(X \times_S S') \times_S S'). \end{aligned}$$

We must now make explicit

$$\underbrace{f^*(\xi)}_{\mathrm{hot}_{S',W}(X')} \xrightarrow{\alpha_{f^*(\xi)}} \underbrace{f^* f_! f^*(\xi)}_{\mathrm{hot}_{S',W}(\Phi_S(X') \times_S S')} \xrightarrow{f^*(\beta_\xi)} \underbrace{f^*(\xi)}_{\mathrm{hot}_{S',W}(X')} ,$$

and check that the composite is the identity. We have set

$$X' = X \times_S S',$$

[70] Description of β_ξ for $\xi = \mathrm{hot}_{S,W}(X)$, with X arbitrary in Cat/S, to be made explicit in definition in due form.

and the diagram is therefore also

$$X' \xrightarrow{\alpha_{X'}} \Phi_S(X') \times_S S' \xrightarrow{f^*(\beta_\xi)} X', \qquad \mathrm{id}_{X'} : X' \longrightarrow X'$$

in $(\mathrm{Cat}/S')(W^{\mathrm{d}}_{S'})^{-1}$. To check commutativity, we can compose with

$$X \times_S S' = X' \xrightarrow{i_X \times_S \mathrm{id}_{S'}} \Phi_S(X) \times_S S' \qquad (\text{where } i_X : X \longrightarrow \Phi_S(X))$$

which is in $W^{\mathrm{d}}_{S'}$ ([71]), now we check, from the general construction of $\beta_\xi = \beta_X$ (preceding page), that we have commutativity in

$$\begin{array}{ccc} \Phi_S(X') \times_S S' & \xrightarrow{f^*(\beta_\xi)} & X' \\ & \searrow^{\Phi_S(\mathrm{pr}_1^X) \times_S S'} & \downarrow{\scriptstyle i_X \times_S S'} \\ & & \Phi_S(X) \times_S S'. \end{array}$$

We are reduced to the commutativity of

$$\begin{array}{ccc} X \times_S S' = X' & \xrightarrow{\alpha_{X'}} & \Phi_S(X') \times_S S' \\ & \overset{?}{\searrow}_{i_X \times_S S'} & \downarrow{\scriptstyle \Phi_S(\mathrm{pr}_1^X) \times_S S'} \\ & & \Phi_S(X) \times_S S' \end{array} \qquad (*)$$

[page 145]

in $(\mathrm{Cat}/S')(W^{\mathrm{d}}_{S'})^{-1}$. Let us try to prove that this is commutative in Cat/S', not just in the category localized by $W^{\mathrm{d}}_{S'}$. We can look at it as a diagram in Cat/S and prove its commutativity as such. This commutativity relation $vu = w$ then splits into two, according to the two factors $\Phi_S(X)$ and S' of the target object $\Phi_S(X) \times_S S'$. For the factor S', the relation simply means that $\alpha_{X'}$ is an S'-morphism, that's OK. For the factor $\Phi_S(X)$, we extend the diagram $(*)$ to the right by the square

p. 89

$$\begin{array}{ccccc} X' & \xrightarrow{\alpha_{X'}} & \Phi_S(X') \times_S S' & \xrightarrow{\mathrm{pr}_1^{\Phi_S(X')}} & \Phi_S(X') \\ & \searrow & \downarrow{\scriptstyle \Phi_S(\mathrm{pr}_1^X) \times_S S'} & & \downarrow{\scriptstyle \Phi_S(\mathrm{pr}_1^X)} \\ & & \Phi_S(X) \times_S S' & \xrightarrow{\mathrm{pr}_1^{\Phi_S(X)}} & \Phi_S(X), \end{array}$$

and taking into account that the composite of the first horizontal row is $i_{X'}$, by construction of $\alpha_{X'}$, we are reduced to the commutativity of

$$\begin{array}{ccc} X \times_S S' = X' & \xrightarrow{i_{X'}} & \Phi_S(X') \\ {\scriptstyle i_X \times_S S'}\downarrow & & \downarrow{\scriptstyle \Phi_S(\mathrm{pr}_1^X)} \\ \Phi_S(X) \times_S S' & \xrightarrow{\mathrm{pr}_1^{\Phi_S(X)}} & \Phi_S(X). \end{array} \qquad (*)$$

[page 146]

Now the morphism $i_X : X \longrightarrow \Phi_S(X)$ in Cat/S gives rise to the commutative square (as does any arrow $i : X \longrightarrow Y$)

$$\begin{array}{ccc} X \times_S S' & \xrightarrow{\mathrm{pr}_1^X} & X \\ {\scriptstyle i_X \times_S S'}\downarrow & & \downarrow{\scriptstyle i_X} \\ \Phi_S(X) \times_S S' & \xrightarrow{\mathrm{pr}_1^{\Phi_S(X)}} & \Phi_S(X), \end{array}$$

[71] **NB** Since X is smooth/S, and $\Phi_S(X)$ too, $X \longrightarrow \Phi_S(X)$ remains a colocal W-equivalence under any base change.

so that the lower composite of $(*)$ is none other than that of $(**)$, hence equal to the upper composite of the latter, and the commutativity of $(*)$ comes down to that of

$$\begin{array}{ccc} X \times_S S' = X' & \xrightarrow{i_{X'}} & \Phi_S(X') \\ {\scriptstyle \mathrm{pr}_1^X}\downarrow & & \downarrow{\scriptstyle \Phi_S(\mathrm{pr}_1^X)} \\ X & \xrightarrow{i_X} & \Phi_S(X), \end{array}$$

which is obvious by functoriality of $Z \xrightarrow{i_Z} \Phi_S(Z)$, applied to the arrow $\mathrm{pr}_1^X : X' \longrightarrow X$ of Cat/S.

We have thus proved that the functors

p. 90
[page 147]

$$\mathrm{HOT}_W(S') \underset{f^*}{\overset{f_!}{\rightleftarrows}} \mathrm{HOT}_W(S) \qquad (*)$$

are adjoint to each other, with adjunction morphisms

$$\begin{cases} \mathrm{id} & \xrightarrow{\alpha} & f^* f_! \\ \mathrm{id} & \xleftarrow{\beta} & f_! f^* \end{cases}$$

specified above. I presume that the same proof, working with the coarse localizers W_S, $W_{S'}$, instead of W_S^{d}, $W_{S'}^{\mathrm{d}}$, would give that the corresponding functors

$$\mathrm{HOT}_W^{\mathrm{lc}}(S') \underset{f_{\mathrm{lc}}^*}{\overset{f_!^{\mathrm{lc}}}{\rightleftarrows}} \mathrm{HOT}_W^{\mathrm{lc}}(S) \qquad (**)$$

are adjoint to each other, with essentially "the same" adjunction morphisms α, β. I do not feel like going over the preceding arguments again and prefer to *deduce* it from the adjunction in the situation $(*)$. This will give me the opportunity to better understand the essential formal features of the situation. I shall put for simplicity

$$\begin{array}{lcllcl} \mathrm{HOT}_W(S) & = & \mathcal{A}, & \mathrm{HOT}_W(S') & = & \mathcal{A}', \\ \mathrm{HOT}_W^{\mathrm{lc}}(S) & = & \mathcal{A}_0, & \mathrm{HOT}_W^{\mathrm{lc}}(S') & = & \mathcal{A}_0', \end{array}$$

[page 148]

and I use the pairs of adjoint functors

$$\mathcal{A} \underset{i}{\overset{\rho}{\rightleftarrows}} \mathcal{A}_0, \qquad \mathcal{A}' \underset{i'}{\overset{\rho'}{\rightleftarrows}} \mathcal{A}_0',$$

where ρ, ρ' are localization functors, i.e. i, i' fully faithful functors. I have a pair of adjoint functors

$$\mathcal{A}' \underset{f^*}{\overset{f_!}{\rightleftarrows}} \mathcal{A},$$

and I assume moreover

a) $f_!$ is compatible with the localizers of $\mathcal{A}'$, $\mathcal{A}$ defined by ρ', ρ respectively, hence defines a functor (well determined up to unique isomorphism)

$$f_!^0 : \mathcal{A}_0' \longrightarrow \mathcal{A}_0,$$

thus giving rise to a commutative diagram (up to functorial isomorphism)

$$\begin{array}{ccc} \mathcal{A}' & \xrightarrow{\rho'} & \mathcal{A}_0' \\ {\scriptstyle f_!}\downarrow & & \downarrow{\scriptstyle f_!^0} \\ \mathcal{A} & \xrightarrow{\rho} & \mathcal{A}_0. \end{array} \qquad (\mathrm{I})$$

p. 91

b) "Dually", $f^* : \mathcal{A} \longrightarrow \mathcal{A}'$ is compatible with the full subcategories, essential images of the fully faithful functors i, i', and then induces a functor, defined up to unique isomorphism: [page 149]

$$f_0^* : \mathcal{A}_0 \longrightarrow \mathcal{A}'_0,$$

giving rise to a square of functors, commutative up to canonical functorial isomorphism

$$\begin{array}{ccc} \mathcal{A}_0 & \xrightarrow{\ i\ } & \mathcal{A} \\ {\scriptstyle f_0^*}\downarrow & & \downarrow{\scriptstyle f^*} \\ \mathcal{A}'_0 & \xrightarrow{\ i'\ } & \mathcal{A}'. \end{array} \tag{II}$$

Under these conditions, the pair of functors

$$\mathcal{A}'_0 \underset{f_0^*}{\overset{f_!^0}{\rightleftarrows}} \mathcal{A}_0,$$

is a pair of adjoint functors. (NB when I speak of pairs of adjoint functors, the one written above the other is the left adjoint.)

Proof. Let

$$\eta_0 \in \operatorname{Ob} \mathcal{A}_0, \qquad\qquad \xi'_0 \in \operatorname{Ob} \mathcal{A}'_0,$$

we must define a bifunctorial bijection

$$\operatorname{Hom}_{\mathcal{A}_0}(f_!^0(\xi'_0), \eta_0) \simeq \operatorname{Hom}_{\mathcal{A}'_0}(\xi'_0, f_0^*(\eta_0))$$

For this, we are going to write ξ'_0 in the form

$$\xi'_0 = \rho'(\xi'), \qquad\qquad \xi' \in \operatorname{Ob} \mathcal{A}',$$

it will suffice to establish the bifunctorial bijection by replacing ξ'_0 by $\rho'(\xi')$, i.e.

$$\operatorname{Hom}_{\mathcal{A}_0}(f_!^0(\rho'(\xi')), \eta_0) \simeq \operatorname{Hom}_{\mathcal{A}'_0}(\rho'(\xi'), f_0^*(\eta_0))$$

(Indeed, the adjunction formula is interpreted as an isomorphism of bifunctors, or, what amounts to the same thing, of functors $\mathcal{A}'_0 \underset{\psi}{\overset{\varphi}{\rightrightarrows}} \mathcal{A}_0^\wedge$, now p. 92 [page 150]

$$\begin{array}{ccc} \underline{\operatorname{Hom}}(\mathcal{A}'_0, \mathcal{A}_0^\wedge) & \longrightarrow & \underline{\operatorname{Hom}}(\mathcal{A}', \mathcal{A}_0^\wedge) \\ \varphi & \longmapsto & \varphi \circ \rho' \end{array}$$

is fully faithful, since ρ' is a localization functor, so it amounts to the same thing to give oneself an isomorphism $\varphi \simeq \psi$, or $\varphi \circ \rho' \simeq \psi \circ \rho'$.) We must therefore transform the expression

$$\begin{array}{ccccc} & \operatorname{Hom}_{\mathcal{A}_0}(\underbrace{f_!^0(\rho'(\xi'))}_{\rho f_!(\xi')}, \eta_0) & \underbrace{\simeq}_{\text{commutativity of (I)}} & \operatorname{Hom}_{\mathcal{A}_0}(\rho f_!(\xi'), \eta_0) & \\ \underbrace{\simeq}_{\text{adjunction } \rho,\ i} & \operatorname{Hom}_{\mathcal{A}}(f_!(\xi'), i(\eta_0)) & \underbrace{\simeq}_{\text{adjunction } f_!,\ f^*} & \operatorname{Hom}_{\mathcal{A}'}(\xi', \underbrace{f^* i(\eta_0)}_{i'(f_0^*(\eta_0))}) & \\ \underbrace{\simeq}_{\text{commutativity of (II)}} & \operatorname{Hom}_{\mathcal{A}'}(\xi', i'(f_0^*(\eta_0))) & \underbrace{\simeq}_{\text{adjunction } \rho',\ i'} & \operatorname{Hom}_{\mathcal{A}'_0}(\underbrace{\rho'(\xi')}_{\xi'_0}, f_0^*(\eta_0)), & \end{array}$$

which is the announced bijection.

I summarize the essentials of the results obtained on $f_!$, $f_!^{\mathrm{lc}}$:

Theorem 2. *Let $f : S' \longrightarrow S$ be an arrow in* Cat*, let us consider the functors*

$$\mathrm{HOT}_W(S') \underset{f^*}{\overset{f_!}{\rightleftarrows}} \mathrm{HOT}_W(S)$$

$$\mathrm{HOT}^{\mathrm{lc}}_W(S') \underset{f^*_{\mathrm{lc}}}{\overset{f^{\mathrm{lc}}_!}{\rightleftarrows}} \mathrm{HOT}^{\mathrm{lc}}_W(S),$$

where the functors $f_!$, $f^{\mathrm{lc}}_!$ are described in theorem 1 *(p. 136). The adjunction morphisms* [page 151]

$$\begin{cases} \mathrm{id} & \longrightarrow & f^* f_! \\ \mathrm{id} & \longleftarrow & f_! f^* \end{cases}$$

are described on pp. 138 to 140.

We would like to make explicit the nice formula, obtained in passing

$$\boxed{f_! f^*(\xi) = \mathrm{hot}_{S,W}(S'/S) \times_S \xi}\ ,$$

for this we must make the right-hand side explicit. Let us note the

Proposition 4. *There exists a unique functor*

$$(\xi, \eta) \longmapsto \xi \times_S \eta : \mathrm{HOT}_W(S) \times \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S)$$

with the property that for X, Y varying in $\underline{\mathrm{Liss}}\, S$*, we have an isomorphism bifunctorial* p. 93
in X, Y

$$\mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y) \simeq \mathrm{hot}_{S,W}(X \times_S Y) \qquad (*)$$

([72]).

This is formal, using only that on $\underline{\mathrm{Liss}}\, S$, the bifunctor

$$(X, Y) \longrightarrow X \times_S Y$$

is compatible with the localizers $W^{\mathrm{d}}_S \times W^{\mathrm{d}}_S$ and W^{d}_S. To see this, we are reduced to seeing that if $X \xrightarrow{u} X'$ in $\underline{\mathrm{Liss}}\, S$ is in W^{d}_S, then the same holds for $X \times_S Y \xrightarrow{u \times_S \mathrm{id}_Y} X' \times_S Y$, for every Y in $\underline{\mathrm{Liss}}\, S$. In fact, we use only the smoothness of X, X', not that of Y, and [page 152]
this is immediate, for if $X \longrightarrow X'$ is in W^{d}_S, $X \times_S Y \longrightarrow X' \times_S Y$ is in W^{d}_Y, hence also in W^{d}_S (cf. theorem 1 applied to $Y \longrightarrow S$). This remark suggests at the same time:

Corollary 1. (False) *The bifunctorial formula $(*)$ remains valid for X in* $\underline{\mathrm{Liss}}\, S$*, Y arbitrary in* Cat/S*, and also in the reverse case.* ([73])

In the first case, let us consider

$$Y \xrightarrow{i_Y} \Phi(Y)$$

in W^{d}_S, hence an isomorphism in $(\mathrm{Cat}/S)(W^{\mathrm{d}}_S)^{-1}$, we shall therefore have

$$\begin{array}{ccc} \mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y) & \xrightarrow{\sim} & \mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(\Phi(Y)) \\ & \underbrace{\xrightarrow{\sim}}_{\text{since } X \text{ and } \Phi(Y) \text{ smooth over } S} & \mathrm{hot}_{S,W}(X \times_S \Phi(Y)), \end{array}$$

now I say that

$$X \times_S Y \xrightarrow{X \times_S i_Y} X \times_S \Phi(Y)$$

[page 153]

is again in W_S^{d}. It suffices to see that it is in W_X^{d} (cf. theorem 1), and for this, that the smooth base change $X \longrightarrow S$ induces $\mathrm{Cat}/S \longrightarrow \mathrm{Cat}/X$ compatible with the localizers W_S^{d}, W_X^{d}. But this is *false* (take $X = \{s\}$).

Let us show that the corollary is false. I am going to take $Y = \{s\}$, the one-point category at s, then $\Phi(Y)$ is the fibered category with discrete fibers defined by the representable functor F defined by s, so it is S/F and, in this case S/s:

$$\Phi(\{s\}) \simeq S/s \qquad \text{as a category over } S.$$

If e.g. s is the final element e of S, we find p. 94

$$\Phi(\underset{S}{e}) \simeq S,$$

hence

$$\mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(e) = \mathrm{hot}_{S,W}(X),$$

on the other hand

$$\mathrm{hot}_{S,W}(X \times_S e) = \mathrm{hot}_{S,W}(X_e).$$

If the two were isomorphic in $\mathrm{HOT}_W(S)$ by the canonical arrow [154]

$$X_e \longrightarrow X,$$

applying the functor $p_!$ (where $p : S \longrightarrow e$), it would follow that $X_e \longrightarrow X$ is in W. If for example $S = \mathbf{\Delta}^1$, this would imply that for every X smooth over S, $X_1 \longrightarrow X$ is in W, whereas normally smoothness implies that $X_0 \longrightarrow X$ is in W. So that would amount to saying that $\underline{\mathrm{Liss}}\, \mathbf{\Delta}^1 = \underline{\mathrm{Liss}}_0 \mathbf{\Delta}^1$, whence at once $\underline{\mathrm{Liss}}\, S = \underline{\mathrm{Liss}}_0 S$, for every S in Cat, we'd know about it! The most trivial counterexample is to take $X = \{0\} \hookrightarrow S = \mathbf{\Delta}^1$, the inclusion of the "generic" open, an open immersion, hence smooth, the fiber X_1 is empty!

Corollary 1 corrected. *If X is in* Cat/S*, not necessarily smooth over S, then the functor*

$$\mathrm{mult}_{X/S} : Y \longmapsto Y \times_S X : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S$$

induces a functor

$$\underline{\mathrm{Liss}}\, S \longrightarrow \mathrm{Cat}/S$$

compatible with the localizers W_S^{d}, W_S^{d}, hence induces a functor

$$(\underline{\mathrm{Liss}}\, S)(W_S^{\mathrm{d}})^{-1} \longrightarrow (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1},$$

or again (taking into account the equivalence induced by the inclusion

$$(\underline{\mathrm{Liss}}\, S)(W_S^{\mathrm{d}})^{-1} \xrightarrow{\sim} (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1} \quad \textit{)}, \qquad [155]$$

a functor, again denoted $\mathrm{multhot}_{X/S}$,

$$\mathrm{multhot}_{X/S} : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S).$$

This functor therefore satisfies the relation

$$\mathrm{multhot}_{X/S}(\mathrm{hot}_{S,W}(Y)) \quad \underbrace{\simeq}_{\text{canonical isomorphism}} \quad \mathrm{hot}_{S,W}(X \times_S Y)$$

for Y *smooth over* S, and for arbitrary Y in Cat/S,

[72] **NB** The multiplication $\times_S$ in $\mathrm{HOT}_W(S)$ is associative, commutative, unital *etc., etc.*, since this is the case in $\underline{\mathrm{Liss}}\, S$.

[73] This statement is surely *false!* Counterexample p. 153, cf. corollary 1 corrected on page 154-156. p. 95

$$\mathrm{multhot}_{X/S}(\mathrm{hot}_{S,W}(Y)) \quad \underbrace{\simeq}_{\text{canonical isomorphism}} \quad \mathrm{hot}_{S,W}(X \times_S \Phi_S(Y)).$$

When X is itself smooth over S, this functor is none other than multiplication by $\mathrm{hot}_{S,W}(X)$:

$$\mathrm{multhot}_{X/S} = \mathrm{mult}_{\mathrm{hot}_{S,W}(X)} \qquad \text{if } X \textit{ smooth over } S.$$

But its value for $\mathrm{hot}_{S,W}(Y)$, when Y is *not* smooth over S, must not be confused with $\mathrm{hot}_{S,W}(X \times_S Y)$.

Thus, if X and Y are in Cat/S, let us consider

$$X \times_S Y \qquad \text{and} \qquad \boxed{\mathrm{hot}_{S,W}(X \times_S Y).}$$

a) If neither X nor Y is assumed smooth over S, this element "has no natural interpretation in terms of operations on the categories $\mathrm{HOT}_W(S)$" ([74]).

[156]

b) If X and Y are smooth over S, this element is interpreted as

$$\mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y)$$

([75]).

c) If one of the two X, Y is smooth over S, without assuming anything about the other, we have an interpretation in terms of mult_X (if Y smooth), or mult_Y (if X smooth), in particular

$$\begin{aligned} \mathrm{hot}_{S,W}(X \times_S Y) &\simeq \mathrm{multhot}_X(\mathrm{hot}_{S,W}(Y)) \quad \text{if } Y \text{ smooth over } S \\ &\simeq \mathrm{multhot}_Y(\mathrm{hot}_{S,W}(X)) \quad \text{if } X \text{ smooth over } S. \end{aligned}$$

Corollary 2. *Let $f : S' \longrightarrow S$ be an arrow in* Cat*, whence $f_!$, f^* between $\mathrm{HOT}_W(S')$ and $\mathrm{HOT}_W(S)$. We have the relation*

$$f_! f^*(\xi) \simeq \mathrm{multhot}_{S'}(\xi),$$

hence if S' is smooth over S,

$$f_! f^*(\xi) \simeq \mathrm{hot}_{S,W}(S') \times_S \xi.$$

We are now in a position to verify the axioms Der 4′b, Der 5′ relating to direct images (I, p. 67, 68), and moreover Der 3b (I, p. 111). The axiom Der 4′ is none other than the existence of the left adjoints $f_!$ of the f^*. The axiom Der 5′ allows the "calculation" of the fibers of an object $f_!(\xi)$. It says that a certain base change arrow, relative to the diagram in Cat equipped with the natural transformation $pf' \Longrightarrow fq$

p. 96

[157]

$$\begin{array}{ccc} S' & \xleftarrow{q} & s\backslash S' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{p} & e \simeq \{s\} \end{array} \qquad (s \in \mathrm{Ob}\, S),$$

namely

$$f'_!(q^*(\xi')) \longrightarrow p^*(f_!(\xi'))$$

[74] Except in the case where one of the two is in Fibhot S, cf. annotation on the next page.

[75] But also if *one* of the two elements is in Fibhot S (and assuming the axiom L6 bis on W).

is an isomorphism (for $\xi' \in \mathbf{D}(S')$). So let

$$\xi' \in \mathrm{Ob}\,\mathrm{HOT}_W(S'), \qquad \xi' = \mathrm{hot}_{S',W}(X')$$

with X' in Cat/S'. Since

$$f_!(\xi') = \mathrm{hot}_{S,W}(X'),$$

and since X' has no reason to be smooth over S (if we do not assume S' itself smooth over S, in which case it might be useful to take X' smooth over S', hence also over S), to calculate $p^*(f_!(\xi'))$ there is no question of taking the fiber X'_s, but we must take $s\backslash X' = \Phi_S(X')_s$:

$$p^*(f_!(\xi')) \simeq \mathrm{hot}_W(s\backslash X').$$

$$\begin{array}{ccc} X' & \longleftarrow & s\backslash X' \\ \downarrow & & \downarrow \\ S' & \xleftarrow{q} & s\backslash S' \\ \downarrow & & \downarrow \\ S & \longleftarrow & s\backslash S \end{array}$$

Consider the base change diagram along $s\backslash S \longrightarrow S$ opposite , with cartesian squares. We first see that $q^*(\xi')$ is none other than the class in $\mathrm{HOT}_W(s\backslash S')$ of the inverse image of X' by the base change $s\backslash S' \longrightarrow S'$. To ensure this, it would suffice to take X' smooth over S' (which is permissible) - then the description of the inverse image is OK for every base change. But we have seen that in the case of a local co-isomorphism such as $s\backslash S' \longrightarrow S'$, there is not even any need for this artifice - the formula is OK without any smoothness restriction. Thus [158]

$$q^*(\xi') = \mathrm{hot}_{s\backslash S',W}(s\backslash X'),$$

whence p. 97

$$f'_!(q^*(\xi')) \simeq \mathrm{hot}_{e,W}(s\backslash X') = \mathrm{hot}_W(s\backslash X').$$

So we have indeed found

$$p^* f_! \simeq f'_! q^*.$$

We can now also prove Der 3b. We have seen that once we have the left adjoints $f_!$ of the f^*, this axiom is stated in *homological form* as follows: If

$$f : S' \longrightarrow S$$

is in W, we must prove that for every

$$\xi \in \mathrm{HOT}_W(e) \overset{\mathrm{def}}{=} \mathrm{Hot}_W,$$

the corresponding arrow

$$(*) \qquad p'_! p'^*(\xi) \longrightarrow p_! p^*(\xi)$$

is isomorphism in Hot_W, where $p : S \longrightarrow e$, $p' : S' \longrightarrow e$ are the structural arrows. Now since every object of Cat is smooth over e, the preceding corollary 2 allows us to identify this arrow $(*)$ with

$$\eta' \times \xi \longrightarrow \eta \times \xi$$

deduced from [159]

$$\mathrm{hot}_W(f): \quad \underbrace{\eta'}_{\stackrel{\mathrm{def}}{=}\mathrm{hot}_W(S')} \longrightarrow \underbrace{\eta}_{\stackrel{\mathrm{def}}{=}\mathrm{hot}_W(S)} .$$

Since the latter is isomorphism by hypothesis, we win.

So we have proved

Theorem 3. *The prederivator* HOT *satisfies the axioms* Der 1, Der 2 ([76]) *(for the record), and* Der 3b ([77]), Der 4′, Der 5′.

We also feel like proving the strengthened variants of Der 3, the strongest form being this one (Der $3b'_2$): if $f \in W$, then f^* induces an *equivalence* of categories

$$f_0^* : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S').$$

The full faithfulness of the functor f_0^* must follow from the natural variant Der $3b''_{\mathrm{coh}}$ of Der 3b, once we have developed the formalism of internal Hom in the $\mathrm{HOT}_W(S)$. But to give a meaning to Der $3b''_{\mathrm{coh}}$, we need the functors *cohomology,* i.e. p_* and p'_* (instead of $p_!$, $p'_!$) for HOT, so this is still premature for the moment. The construction of the f_* and that of internal Hom will moreover surely be done ± jointly.

But I realize just now that we find almost "by eye" the fact that f_0^* is indeed an *equivalence* of categories, and that $f_!^0$ is the quasi-inverse equivalence, by using the adjunction of the functors $f_!^0$, f_0^*, and recalling that to prove that they are equivalences quasi-inverse to one another, it amounts to the same thing to show that the adjunction morphisms

p. 98
[160]

$$\begin{array}{ccc} \mathrm{id} & \xrightarrow{\alpha_0} & f_0^* f_!^0 \\ \mathrm{id} & \xleftarrow{\beta_0} & f_!^0 f_0^* \end{array}$$

are isomorphisms. I am going to try to describe these adjunction arrows in the general case (without assuming $f \in W$), then to show that they are in W if f is.

$\xi = \mathrm{hot}^{\mathrm{lc}}_{S,W}(X)$ in $\mathrm{HOT}^{\mathrm{lc}}_W(S)$. We may assume $X \in \mathrm{Ob}\,\underline{\mathrm{Parf}}\,S$. Since X smooth over S, we have

$$f_0^*(\xi) = \mathrm{hot}^{\mathrm{lc}}_{S',W}(X \times_S S'), \qquad f_!^0 f_0^*(\xi) = \mathrm{hot}^{\mathrm{lc}}_{S,W}(X \times_S S').$$

So

$$\beta_{0,\xi} : f_!^0 f_0^*(\xi) \longrightarrow \xi$$

is an arrow

$$X \times_S S' \longrightarrow X \qquad \text{in } (\mathrm{Cat}/S)(W_S)^{-1}$$

and I presume that it is pr_1 ([78]). Is it in W if $f : S' \longrightarrow S$ is? Yes (for X is perfect over S), provided that W satisfies the axiom L6 (cf. page 43). [161]

2) Let

$$\xi' = \mathrm{hot}^{\mathrm{lc}}_{S',W}(X') \qquad \text{in } \mathrm{HOT}^{\mathrm{lc}}_W(S'),$$

we have

$$f_!^0(\xi') = \mathrm{hot}^{\mathrm{lc}}_{S,W}(X'),$$

[76] *not proved*, the "result" I had in mind is botched, cf. supplement proposition 5, page 164.

[77] And even the strongest variant Der $3b'_2$ of Der 3, cf. proof on the next page. **NB** For Der $3b'_2$, we used the axiom L6 on W - and this is essential. We see that the validity of Der $3b'_2$ implies L6, and even the stronger form L6 bis, and even if we assume only Der $3b'_1$, i.e. f_0^* fully faithful.

[78] **NB** We see that the axiom L6 is ± *necessary* for the result on f_0^*, $f_!^0$ to be valid.

but to make f_0^* of this element explicit, since X' has no reason to be Hot-fibered over S (in which case it would suffice to take $f^*(X') \overset{\text{def}}{=} X' \times_S S'$), we must first take $C_S(X')$, then apply f^*:

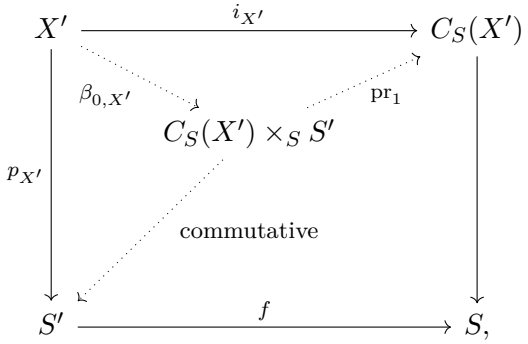
$$f_0^* f_!^0(X') = \mathrm{hot}^{\mathrm{lc}}_{S',W}(C_S(X') \times_S S').$$

We must therefore define

$$X' \xrightarrow{\alpha_{0,X'}} C_S(X') \times_S S' \qquad \text{in } (\mathrm{Cat}/S')W_{S'}^{-1}.$$

Now we find a canonical arrow in Cat/S', deduced from the S-morphism $i_{X'}$: $X' \longrightarrow C_S(X')$, i.e. from the commutative square in solid lines

p. 99



as the unique dotted arrow $\beta_{0,X'}$ which is an S'-morphism (left triangle commutative) and which makes the upper triangle commute.

If $f \in W$, then pr_1, obtained from f by base change $C_S(X') \longrightarrow S$, which is perfect, is in W by L6, we know that the same holds for $i_{X'}$, hence also for $\beta_{0,X'}$, we win.

To finish with the general properties of the functors $f_!$ for HOT_W, I would like to give the reasonable base change statements. This would be a

[page 162]

Theorem 4. *Let*

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{p} & S' \end{array}$$

be a cartesian diagram in Cat*, let us consider the base change arrow (in* $\underline{\mathrm{Hom}}(\mathrm{HOT}_W(X), \mathrm{HOT}_W(S'))$*):*

$$p^* f_! \longleftarrow f'_! q^*.$$

This is an isomorphism in each of the following two cases.

a) f is smooth.

b) p is proper.

Proof. *Case a).* We test with ξ in $\mathrm{HOT}_W(X)$, we may suppose ξ to be of the form

$$\xi = \mathrm{hot}_{X,W}(Z) \qquad \text{with } Z \in \mathrm{Ob}\,\underline{\mathrm{Liss}}\, X.$$

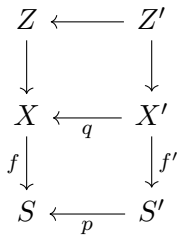
Let us note that Z is then also smooth over S, since f is smooth.

$$f_!(\xi) = \mathrm{hot}_{S,W}(Z), \qquad p^* f_!(\xi) = \mathrm{hot}_{S',W}(\underbrace{Z \times_S S'}_{Z',\ \text{cf. diagram}})$$

p. 100

$$q^*(\xi) = \mathrm{hot}_{X',W}(Z'), \qquad f'_! q^*(\xi) = \mathrm{hot}_{S',W}(Z'),$$

OK!



Case b). We have to admit that the base change functor by p: [page 163]

$$Z \longmapsto Z \times_S S' : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S'$$

is compatible with the localizers W_S^{d}, $W_{S'}^{\mathrm{d}}$ (cf. below) - that's, formally, the relevant property for p ([79]). It will follow that

$$p^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$$

is computed, for an element in $\mathrm{HOT}_W(S)$ given by any Z *whatsoever* in Cat/S (not necessarily smooth over S), by

$$p^*(\mathrm{hot}_{S,W}(Z)) = \mathrm{hot}_{S',W}(Z \times_S S').$$

Let us then start, as in a), from an element ξ of $\mathrm{HOT}_W(X)$ given by a Z in Cat/X (no need this time to suppose Z smooth), the preceding immediate calculation again establishes the commutation formula.

But it does not seem that the compatibility property of $X \longmapsto X \times_S S'$, with colocal equivalence over S and over S', follows in a $\pm$ obvious way from the condition that $p : S' \longrightarrow S$ be proper ([80]). I feel like deducing it from general *fourbi* on the formalism of derivators, namely from the base change theorem for the $f_!^{\mathbf{D}}$ for a $\mathbf{D}$-proper morphism.

In the dual form of a base change theorem for $f_*^{\mathbf{D}}$, by $\mathbf{D}$-smooth base change, this is done in I, p. 56 to 58, and we should be able to apply it here. Let us note that in [page 164] the proof of the theorem loc. cit., we use only L4 , i.e. the existence of the $f_*^{\mathbf{D}}$, to the exclusion of the $f_!^{\mathbf{D}}$ (L4′), plus L5 of course, so the dual form does not call on the existence of the $f_*^{\mathbf{D}}$, only of the $f_!^{\mathbf{D}}$. We thus have a good chance of being able to apply it to HOT_W (where I have not yet established the existence of the f_*^{HOT}).

Let us first note

Proposition 5. *Let $f : S' \longrightarrow S$ be an arrow in* Cat. *Then f is a HOT_W-equivalence* p. 101 if and only if $\mathrm{hot}_W(f)$ *is isomorphism. If W is strongly saturated, this therefore means that $f \in W$ (and in any case $f \in W \Longrightarrow f$ a HOT_W-equivalence)* ([81]).

This is contained in the proof of Der 3b earlier. I recall that f is a HOT-equivalence, by definition, if and only if for $a, b \in \mathrm{Hot}_W(= \mathrm{HOT}_W(e))$,

$$\mathrm{Hom}_{\mathrm{HOT}_W(S)}(a_S, b_S) \longrightarrow \mathrm{Hom}_{\mathrm{HOT}_W(S')}(a_{S'}, b_{S'}) \qquad (*)$$

is isomorphism, where a_S etc. denotes $p_S^*(a)$ etc. ($p_S : S \longrightarrow e$). The adjunction formulas for $p_{S!}$, p_S^* and for $p_{S'!}$, $p_{S'}^*$ allow us to interpret the arrow $(*)$ as

$$\mathrm{Hom}_{\mathrm{Hot}_W}(p_{S!}(a_S), b) \longrightarrow \mathrm{Hom}_{\mathrm{Hot}_W}(p_{S'!}(a_{S'}), b),$$

deduced by "transposition" $\mathrm{Hom}(-, b)$ from the arrow

[79] This must be *equivalent* to p proper. Compare with corollary 1, page 166.

[80] But yes, cf. proof page 168. The developments of the four pages that follow are unnecessary by this [page 165] proof.

[81] should be moved up toward theorem 3, p. 159.

$$p_{S'!}(a_{S'}) \longrightarrow p_{S!}(a_S)$$

induced by f - so we have to prove that this is an isomorphism (this is the form we had used earlier). Now this arrow is identified with

$$\eta' \times a \longrightarrow \eta \times a$$

induced by

$$\mathrm{hot}_W(f) : \underbrace{\eta'}_{=\mathrm{hot}_W(S')} \longrightarrow \underbrace{\eta}_{=\mathrm{hot}_W(S)} ,$$

and to say that this is isomorphism for any a in Hot_W amounts to saying that $\eta' \longrightarrow \eta$ is (since multiplication in Hot_W has a unit element …).

Now I remember that to have axiom Der 2 of derivators, I should also suppose W *strongly* saturated (i.e. equal to the localizer defined by the canonical functor $\mathrm{Cat} \longrightarrow (\mathrm{Cat})W^{-1} \overset{\mathrm{def}}{=} \mathrm{Hot}_W$). (This is what I assumed from the outset, starting from a general $(\mathcal{M}, W)$ to define a prederivator, to have axiom Der 2.)

Applying the dual of the theorem in I, p. 56, we thus find the base change theorem for $f_!$, for base change $f : S' \longrightarrow S$ *W-proper.*

But the proof surely uses Der 2 (fiberwise isomorphism criterion, in $\mathrm{HOT}_W(S)$), which I have not proved yet. It is incomplete for the moment. But I can already note the [page 166] ‖

Corollary 1 (of theorem 4, p. 162). *If the arrow $S' \overset{p}{\to} S$ in* Cat *gives rise (for every $f : X \longrightarrow S$ in Cat/S) to the base change formula for* p. 102

$$f_! : \mathrm{HOT}_W(X) \longrightarrow \mathrm{HOT}_W(S),$$

then the functor $p^* : X \longmapsto X \times_S S'$ from Cat/S to Cat/S' is compatible with the localizers W^{d}_S, $W^{\mathrm{d}}_{S'}$. (The converse implication has already been seen, cf. pages 161-163.)

This will follow from the

Lemma. *Let $u : X' \longrightarrow X$ be an arrow in Cat/S. For $u \in W^{\mathrm{d}}_S$, it is necessary and sufficient (if W strongly saturated) that for every $\xi \in \mathrm{Hot}_W$, the corresponding arrow*

$$u_* : p'_!(\xi_{X'}) \longrightarrow p_!(\xi_X)$$

be isomorphism in $\mathrm{HOT}_W(S)$.

This is sufficient, subject to strong saturation of W, for if u_* is isomorphism, passing to the fibers, we find by Der $5'$

$$H^{\mathrm{HOT}_W}_\bullet(s\backslash X', \xi) \longrightarrow H^{\mathrm{HOT}_W}_\bullet(s\backslash X, \xi),$$

if this is isomorphism, then $s\backslash X' \longrightarrow s\backslash X$ is by definition a HOT-equivalence, which implies that it is in W (proposition 5).

This is necessary, for if $\xi = \mathrm{hot}_W(T)$, hence $\xi_S = \mathrm{hot}_{S,W}(T_S)$ (where $T_S = T \times S/S$), and we therefore have [page 167]

$$p'_!(\xi_{X'}) = \mathrm{hot}_{S,W}(X' \times T), \qquad p_!(\xi_X) = \mathrm{hot}_{S,W}(X \times T),$$

and we have to prove that if $u : X' \longrightarrow X$ is in W^{d}_S, then $\underbrace{X' \times T}_{X' \times_S T_S} \longrightarrow \underbrace{X \times T}_{X \times_S T_S}$ also, which simply means that

$$s\backslash(X' \times T) \longrightarrow s\backslash(X \times T) \qquad (*)$$

is in W for every $s \in S$. But the base change $s\backslash S \longrightarrow S$ commutes with fiber products, so transforms $X \times T = X \times_S T_S$ into $(s\backslash X) \times_{s\backslash S} T_{(s\backslash S)} = s\backslash X \times T$, and likewise for X'. Thus the arrow is identified with

$$(s\backslash X') \times T \longrightarrow (s\backslash X) \times T,$$

obtained from

$$s\backslash X' \longrightarrow s\backslash X$$

by the functor $? \times T$. Since this functor transforms W into W, we win.
Finally, I realize that I had a stupid mental block, over the question of proper base change being compatible with colocal W-equivalence. Let

p. 103
[page 168]

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \quad \swarrow & \\ & S \xleftarrow{g} S', & \end{array}$$

with $f \in W_S^{\mathrm{d}}$ and g proper, we have to see that $f' : X' \longrightarrow Y'$ obtained from f by base change $S' \longrightarrow S$ is still in $W_{S'}^{\mathrm{d}}$. This means that every base change of the form $s'\backslash S' \longrightarrow S'$ transforms it into $\in W$. But let $s = g(s')$, we have the commutative square

$$\begin{array}{ccc} S & \longleftarrow & S' \\ \uparrow & & \uparrow \\ \underbrace{s\backslash S}_{=Z} & \longleftarrow & \underbrace{s'\backslash S'}_{=Z'}, \end{array}$$

and it suffices to see that for every arrow $\varphi : \mathcal{X} \longrightarrow \mathcal{Y}$ in Cat/Z which is in W, its inverse image by base change $Z' \longrightarrow Z$ is likewise. (Applying this to the case $\mathcal{X} = X \times_S (s\backslash S)$, $\mathcal{Y} = Y \times_S (s\backslash S)$, $\varphi = s\backslash f$.) Now this follows from the fact that, by the properness assumption, $Z' \longrightarrow Z$ is proper and has Hot-aspheric fibers, hence is a universal Hot-equivalence. It would even suffice for g to be W-proper, to imply that $Z' \longrightarrow Z$ is a universal W-equivalence, i.e. which remains so after every base change. Now we have the

[page 169]

Lemma. If $g : Z' \longrightarrow Z$ is a universal W-equivalence, then the functor

$$g^* : X \longmapsto X \times_Z Z' : \mathrm{Cat}/Z \longrightarrow \mathrm{Cat}/Z'$$

is compatible with the localizers W_Z, $W_{Z'}$, i.e. if $f : X \longrightarrow Y$ in Cat/Z is in W, likewise $f' : X' \longrightarrow Y'$ in Cat/Z'.

$$\begin{array}{ccccc} X & \xleftarrow{g_X} & X' & & \\ & \searrow^{f} & & \searrow^{f'} & \\ & & Y & \xleftarrow{g_Y} & Y' \\ & & \downarrow & & \downarrow \\ & & Z & \xleftarrow{g} & Z' \end{array}$$

This is obvious from the upper square of the diagram above, where g_X, g_Y are in W by the hypothesis on g, so f is in W if and only if f' is, q.e.d.

[page 170 missing]
p. 104
[page 171]

[...]

New version of the theory of HOT_W.

I realized that the hypothesis $W \supseteq W_\infty$ (L3 quater, p. 1), which I had ended up putting on W, is artificial, and that the hypothesis L5 (p. 43), or even L6, L6 bis (p. 43-44), is unnecessary for a large part of the theory of HOT_W. I have thus been led to review the theory (i.e. what I have done with it up to now) under the following hypotheses on W (cf. XII):

W(1) (*Saturation* of W): as before (L1 page 1)

W(2) (localization): this is the old L4 bis ([82]).

W(3) (final object): If X has a final object, X is W-aspheric (old L3).

These three axioms are sufficient for everything I had done in Pursuing Stacks, and in a large part of what has been done here on HOT. Sometimes however one needs the old L 5, which I restate here (compare XII, theorem [...]):

W(4b) Let X, Y be W-fibered over S, $f : X \longrightarrow Y$ an S-morphism. If $f \in W$, then $\forall s \in \mathrm{Ob}\, S$, $f_s : X_s \longrightarrow Y_s$ is in W.

For the notion of W-fibered (*weak*), I took a long time finally to find *the* right notion, right for a W satisfying only $\mathrm{W}(1,2,3)$ (and even only $\mathrm{W}(1,2\text{ bis},3)$, where W(2 bis) is the old L4). For this I refer to (XII, theorem 3, p. 51), where the notion is well pinned down by means of equivalent conditions. In fact, for the main applications to HOT_W, seen up to now, one needs only ([83]) a weakened variant of W(4b):

W(4b′) Like W(4b), but requiring in addition that X, Y be in $\underline{\mathrm{Parf}}_\omega\, S$.

[page 172]

For the notion of W-proper, W-smooth, W-perfect morphisms in Cat, I refer to XII, part III (p. 22-52), and the notations $\underline{\mathrm{Liss}}_W S$, $\underline{\mathrm{Prop}}_W S$, $\underline{\mathrm{Parf}}_W S$ (subcategories of Cat/S), and $\mathrm{Parf}_W \subseteq \mathrm{Fl(Cat)}$. In the statement of W(4b′) we have set

$$\underline{\mathrm{Parf}}_\omega S = \underline{\mathrm{Parf}}_{W_\omega} S, \qquad \text{where } W_\omega \subseteq \mathrm{Fl(Cat)} \text{ is the smallest subset of Fl(Cat) satisfying } \mathrm{W}(1,2\mathrm{bis},3).$$

I similarly write

$$W'_\omega \subseteq \mathrm{Fl(Cat)}, \qquad \text{smallest subset of Fl(Cat) satisfying } \mathrm{W}(1,2,3,4\mathrm{b}).$$

p. 105

The hypothesis $W \supseteq W_\infty$ which we had ended up imposing on W, and which appeared artificial in the course of the reflection in XII, is better replaced by

$$W \supseteq W'_\omega \qquad (\text{or by } W \supseteq \overline{W'_\omega},\ \text{where } \overline{W'_\omega} \text{ is the strong saturation of } W'_\omega)$$

([84]). But we do not impose it as a mandatory condition, on the same footing as $\mathrm{W}(1,2,3)$, any more than W(4b). The role I have had W_∞ play in the past weeks, seems to belong more naturally to W'_ω or $\overline{W'_\omega}$. This is so in particular also, it seems to me, in the statement of the conditions of type Der 3 on a derivator (I, p. 111-112), which ought to be reviewed in this light. But everything depends, whether (as is my impression) one can really do substantial homotopical algebra, with in particular the long exact sequences of fibration

[page 173]

and suspension, for HOT_W, with W's which are *strictly* contained in W_∞. Before even being sure, it is still too early for me to pronounce on the "right" formulation of Der 3,

which is the axiom relating the derivator $\mathbf{D}$ to W_∞, or to some other basic localizer.

I am now going to give a summary of the theory of HOT_W, reviewed from the viewpoint I have just stated: we assume for W *the conditions* $\mathrm{W}(1,2,3)$ *and no more,* unless explicitly mentioned.

① **Main localizers** on Cat/S and on the main categories of S-categories (cf. diagram p. 76).

They will all be induced by the three familiar localizers

$$W_S^{\mathrm{d}} \qquad W_S^{\mathrm{g}}$$
$$\searrow \quad \swarrow$$
$$W_S$$

on Cat/S. There is also reason to look at W_S^{f}, but never on the whole of Cat/S, but on other categories of the diagram, and always in cases where W_S^{f} coincides either with W_S^{d} or with W_S^{g}. But the W_S^{f} point of view is often more directly convenient, and moreover has the advantage of being symmetric.

ⓐ In $\underline{\mathrm{Liss}}_W S$, we have

p. 106

$$\begin{array}{ll} W_S^{\mathrm{f}} & = W_S^{\mathrm{d}} = W_S^{\mathrm{univ}} \\ \Downarrow & \\ W_S^{\mathrm{g}} & \\ \Downarrow & \\ W_S & \end{array}$$

Dually,

ⓑ in $\underline{\mathrm{Prop}}_W S$, we have

$$\begin{array}{ll} W_S^{\mathrm{f}} & = W_S^{\mathrm{g}} = W_S^{\mathrm{univ}} \\ \Downarrow & \\ W_S^{\mathrm{d}} & \\ \Downarrow & \\ W_S, & \end{array}$$

finally, [page 174]

[82] i.e. L5 of page 2. The axiom is taken up again (in a similar and equivalent form) at the bottom of page 43.

[83] This is ultimately an error, it seems now. I think that the axiom W(4b′) ultimately serves strictly no purpose (except in fact to keep false ideas alive.)

[84] $\overline{W'_\omega}$ is the smallest *strongly saturated* subset of Fl(Cat), satisfying W(4b), and it even satisfies condition W′(4a), the strongest variant of the axioms of type W(4).

ⓒ in $\underline{\mathrm{Fib}}_W S$, we have

$$\begin{array}{ccccccc} W_S^{\mathrm{f}} & \Longleftrightarrow & W_S^{\mathrm{d}} & \Longleftrightarrow & W_S^{\mathrm{g}} & \Longleftrightarrow & W_S^{\mathrm{univ}} \\ \Downarrow & & & & & & \\ W_S. & & & & & & \end{array}$$

This implication $W_S^{\mathrm{f}} \Longrightarrow W_S$ is strict if one does not impose on W axiom W(4b), whose usefulness here is exactly to ensure

> $W_S^{\mathrm{f}} \Longleftrightarrow W_S$ in $\underline{\mathrm{Fib}}_W S$ *if* W(4b) satisfied
> and then the five localizers considered above
> in $\underline{\mathrm{Fib}}_W S$ are all equal.

The relations between localizers in $\underline{\mathrm{Liss}}_W S$, $\underline{\mathrm{Prop}}_W S$, $\underline{\mathrm{Fib}}_W S$ carry over automatically to subcategories of these, more generally to all those which map by an inclusion functor into one of these. In the diagram p. 76, all the categories, with the sole exception of the whole of Cat/S, are in this case. p. 107

For the main results developed up to now on HOT_W, when axiom W(4b) is used, the weak form W(4b′) suffices. We have

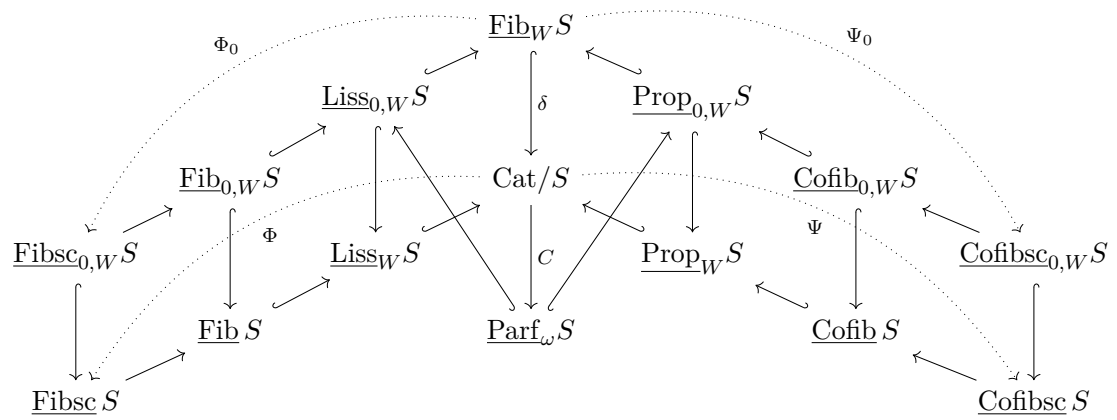
> $W_S^{\mathrm{f}} \Longleftrightarrow W_S$ in $\underline{\mathrm{Parf}}_\omega S$ *if* W(4b′) satisfied
> and then the five localizers W_S^{f}, W_S^{d}, W_S^{g}, W_S^{univ}, W_S
> are equal in $\underline{\mathrm{Parf}}_\omega S$.

② Main categories of categories over S.

These are, up to changes in lighting, those of the diagram p. 76. The categories $\underline{\mathrm{Fibsc}}\, S$, $\underline{\mathrm{Fib}}\, S$, $\underline{\mathrm{Cofibsc}}\, S$, $\underline{\mathrm{Cofib}}\, S$ and their variants with subscript 0, are unchanged - their role is to be as small

[page 175]

as possible to fulfill their function (of describing $\mathrm{HOT}_W(S)$, $\mathrm{HOT}^o_W(S)$, $\mathrm{HOT}^{\mathrm{lc}}_W(S)$, or $\mathrm{HOT}^{\mathrm{lc!}}_W(S)$ below). The category $\underline{\mathrm{Parf}}\, S$ of the diagram fulfilled a function of the same type, which is why we now replace it (bidding higher) by $\underline{\mathrm{Parf}}_\omega S$. The other categories of the diagram are, in addition to Cat/S, $\underline{\mathrm{Liss}}\, S$ and $\underline{\mathrm{Prop}}\, S$ and their variants with subscript 0, as well as $\underline{\mathrm{Fibhot}}\, S$. (All these categories, as well as $\underline{\mathrm{Parf}}\, S$, were thus constructed with reference to W_∞, not to W.) Their role is to be as large as possible to give rise to certain properties and relations. Now it turns out that here the reference to W_∞ was artificial, and one should put instead the categories $\underline{\mathrm{Liss}}_W S$, $\underline{\mathrm{Prop}}_W S$, $\underline{\mathrm{Fib}}_W S$, and in addition the variants $\underline{\mathrm{Liss}}_{0,W} S$, $\underline{\mathrm{Prop}}_{0,W} S$. Thus the diagram becomes



([85]) (all the squares cartesian, the functors denoted $\hookrightarrow$ are fully faithful),
where I have again marked in green the five *"inversion functors"*

p. 108
[page 176]

$$\Phi,\ \Psi,\ \Phi_0,\ \Psi_0,\ C,$$

which provide quasi-inverses of the functors on the localized categories induced by the *inclusion functors.*

③ **The four types of categories** HOT_W**.**

On each of the categories of the upper floor, there are *two* localizers to look at,

$$\begin{array}{ccc} W_S^{\mathrm{f}} & = W_S^{\mathrm{g}} = W_S^{\mathrm{d}} = W_S^{\mathrm{univ}} & \\ \Big\Downarrow & & \\ W_S, & & \end{array}$$

giving rise to two distinct localized categories, denoted

$$\mathrm{HOT}_W^{\mathrm{lc}}(S) \qquad \text{and} \qquad \mathrm{HOT}_W^{\mathrm{lc!}}(S),$$

it turns out that (thanks to Φ_0 and Ψ_0) these localized categories "are the same" for the 8 categories of the upper floor, with the exception of $\underline{\mathrm{Parf}}_\omega S$; more exactly, the inclusion functors (with the exception perhaps of those having as source $\underline{\mathrm{Parf}}_\omega S$) induce equivalences on the two types of localized categories, the quasi-inverse equivalences being obtained by Φ_0 (left wing) or Ψ_0 (right wing). When one localizes *for* W_S (to the exclusion of W_S^{f}), one also finds quasi-inverses by $C|\underline{\mathrm{Liss}}_{0,W}S$ resp. $C|\underline{\mathrm{Prop}}_{0,W}S$, for the two inclusions

$$\underline{\mathrm{Liss}}_{0,W}S \hookleftarrow \underline{\mathrm{Parf}}_\omega S \hookrightarrow \underline{\mathrm{Prop}}_{0,W}S$$

([86]). For reasons of symmetry, the prettiest seems to be to define

$$\begin{cases} \mathrm{HOT}_W^{\mathrm{lc}}(S) &= (\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1} \\ \mathrm{HOT}_W^{\mathrm{lc!}}(S) &= (\underline{\mathrm{Fib}}_W S)(W_S)^{-1}, \end{cases}$$

and we have a localization functor

$$\mathrm{HOT}_W^{\mathrm{lc}}(S) \xrightarrow{\lambda} \mathrm{HOT}_W^{\mathrm{lc!}}(S)$$

induced by the inclusion

$$W_S^{\mathrm{f}} \subseteq W_S.$$

For every full subcategory

$$\mathcal{F} \subseteq \underline{\mathrm{Fib}}_W S$$

p. 109
[page 177]

which contains one of the three subcategories

$$\underline{\mathrm{Fib}}_{0,W}S,\ \underline{\mathrm{Parf}}_\omega S,\ \underline{\mathrm{Prop}}_{0,W}S,$$

the inclusion of $\mathcal{F}$ into $\underline{\mathrm{Fib}}_W S$ defines equivalences

$$\mathcal{F}(W_S)^{-1} \approx \mathrm{HOT}_W^{\mathrm{lc!}}(S),$$

[85] NB the new notation would rather be $\underline{\mathrm{Fibf}}_W S$, for these are W-fibrations which I now call "weak" (and defined only in Cat, as opposed to W-fibrations, definable in any category of models).

[86] cf. below, what one can say for the localized categories for W_S^{f}.

and if $\mathcal{F}$ contains one of the two subcategories $\underline{\mathrm{Fib}}_{0,W}S$, $\underline{\mathrm{Prop}}_{0,W}S$ (to the exclusion of $\underline{\mathrm{Parf}}_{\omega}S$), one also finds

$$\mathcal{F}(W_S^{\mathrm{f}})^{-1} \approx \mathrm{HOT}_W^{\mathrm{lc}}(S).$$

The ground floor contributes two other variants of HOT, by localizing either by W_S^{d} (left wing), or by W_S^{g} (right wing). The inclusion functors define equivalences for the localized categories either for W_S^{d} (left wing), or for W_S^{g} (right wing). The prettiest (symmetry) seems to be to define these two localized categories by Cat/S

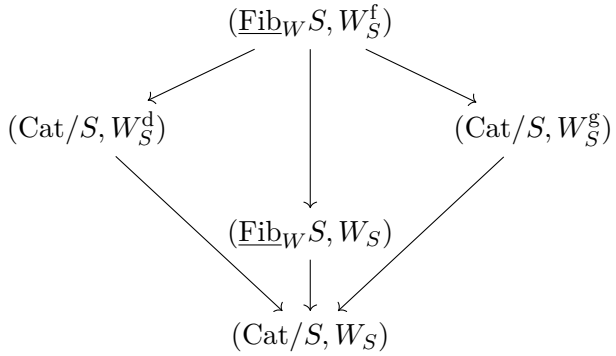
$$\mathrm{HOT}_W(S) = (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}, \qquad \mathrm{HOT}_W^{o}(S) = (\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1}.$$

The quasi-inverses for the equivalences of localized categories, induced by the inclusion functors, are given either by Φ (left wing), or by Ψ (right wing).

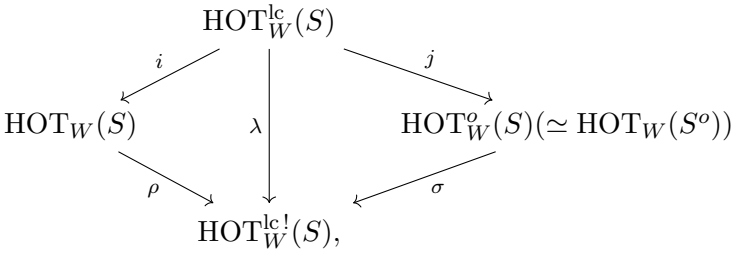
But Φ and Ψ supply quasi-inverses also when one localizes for W_S, the coarse W-equivalence instead of the fine - just as on the upper floor. One therefore finds on the ground floor a (unique) third localized category, which one can describe more prettily by

$$(\mathrm{Cat}/S)(W_S)^{-1},$$

but it is equivalent to $(\underline{\mathrm{Fib}}_W S)W_S^{-1}$ by the functor induced by the inclusion. (A quasi-inverse being supplied by C.) It is therefore $\mathrm{HOT}_W^{\mathrm{lc}!}(S)$. One thus finds, from the commutative diagram of categories with localizers [page 178]



(where the short vertical arrow induces an equivalence on the localized categories), the commutative diagram between the HOT_W categories linked to S: p. 110



$(*)$

where ρ, σ, λ are localization functors, and where i, j are felt to be inclusion functors. This diagram $(*)$ seems to exhaust the natural functors which link the four categories considered to one another.

The properties of the functors Φ, Ψ, C which have been used, to supply all the desired quasi-inverses of functors (between localized categories) induced by the inclusion functors of the diagram, are the following. [page 179]

ⓐ We have a functorial homomorphism in Cat/S

$$X \xrightarrow{i_X} \Phi(X), \qquad \underline{i_X \in W_S^{\mathrm{d}}}.$$

When X is in $\underline{\mathrm{Fib}}_W S$, $\Phi(X)$ is in $\underline{\mathrm{Fibsc}}_0 S$, so $i_X \in W_S^{\mathrm{f}}$ $(= W_S^{\mathrm{d}}$ on $\underline{\mathrm{Fib}}_W S)$. Thus Φ induces the functor Φ_0 of the diagram.

Dually, we have

$$X \xrightarrow{i_X} \Psi(X), \qquad \underline{i_X \in W_S^{\mathrm{g}}},$$

and when X is in $\underline{\mathrm{Fib}}_W S$, $\Psi(X)$ is in $\underline{\mathrm{Cofibsc}}_0 S$, hence the functor Ψ_0.

ⓑ We have a functorial homomorphism

$$\left|\begin{array}{l} X \xrightarrow{i_X} C(X), \qquad \underline{i_X \in W_S} \\ \text{and } C(X) \in \mathrm{Ob}\, \underline{\mathrm{Parf}}_\omega S. \end{array}\right.$$

||| But one must bear in mind that in the absence of axiom W(4b) on W, one can *not* assert $i_X \in W_S^{\mathrm{f}}$, even if X is in $\underline{\mathrm{Fib}}_W S$ (or even, in $\underline{\mathrm{Parf}}_\omega S$), the only case (at any rate) in which one is entitled to expect it. I note in this connection the diagram, *cartesian,* for $f : X \longrightarrow Y$ in Cat/S p. 111

$$\begin{array}{ccc} C(Y)_s & \xleftarrow{C(f)_s} & C(X)_s \\ {\scriptstyle \pi_Y^s}\downarrow & & \downarrow{\scriptstyle \pi_X^s} \\ Y & \xleftarrow{\quad f \quad} & X, \end{array} \qquad \text{vertical arrows in } \underline{\mathrm{Parf}}_\omega \subseteq \underline{\mathrm{Fib}}_W$$

which shows for example that one cannot expect that $f \in W_S \implies C(f) \in W_S^{\mathrm{f}}$ without [page 180] having axiom W(4a′) (or an axiom of comparable strength, such as W(4b)).

It is these negative facts (in the absence of W(4b)) that are the reason why I cannot say anything in particular about the functors induced by *fine localization* $(W_S^{\mathrm{f}})^{-1}$ of the two inclusion functors

$$\underline{\mathrm{Liss}}_0 S \longleftarrow \underline{\mathrm{Parf}}_\omega S \longrightarrow \underline{\mathrm{Prop}}_0 S,$$

or, what amounts to the same thing, about the one induced by the common composite of those with the inclusions into $\underline{\mathrm{Fib}}_W S$

$$\begin{array}{ccccc} & & \underline{\mathrm{Parf}}_\omega S & & \\ & \swarrow & \downarrow & \searrow & \\ \underline{\mathrm{Liss}}_0 S & & & & \underline{\mathrm{Prop}}_0 S \\ & \searrow & \downarrow & \swarrow & \\ & & \underline{\mathrm{Fib}}_W S, & & \end{array}$$

in other words, I can say nothing about the faithfulness properties of the functor

$$(\underline{\mathrm{Parf}}_\omega S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1},$$

and still less that it is an equivalence of categories. Thus, in the absence of W(4b), for the full subcategories $\mathcal{F}$ of $\underline{\mathrm{Fib}}_W S$ which contain $\underline{\mathrm{Parf}}_\omega S$, without containing either $\underline{\mathrm{Liss}}_0 S$ or $\underline{\mathrm{Prop}}_0 S$, it could a priori be that the localized categories

$$\mathcal{F}(W_S^{\mathrm{f}})^{-1}$$

are all different. I am thinking in particular of $\underline{\mathrm{Parf}}_W S$, as interesting in itself, after all, as $\underline{\mathrm{Parf}}_\omega S$. So in addition to the four localized categories of diagram $(*)$, p. 178, there is a whole infinite "teratological" series of localized categories which refuse to let [page 181] themselves be "closed up" by Φ_0 or Ψ_0, and for which even C is of no help, for lack of axiom W(4b). p. 112

④ Case where axiom W(4b) is satisfied.

Then the perplexities I have just spoken of disappear. We have at once, by virtue of W(4b):

$$|||\qquad \text{If } X \in \mathrm{Ob}\,\underline{\mathrm{Fib}}_W S, \text{ then } i_X : X \longrightarrow C(X) \text{ is in } W_S^{\mathrm{f}} \quad (\textit{assuming axiom } \mathrm{W(4b)})$$

and C supplies quasi-inverses for the "inclusion" functors between fine localized categories (for W_S^{f}) considered on the preceding page. It supplies one as well for the functor λ considered above ((∗) page 178)

$$\underbrace{(\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1}}_{\overset{\mathrm{def}}{=}\mathrm{HOT}_W^{\mathrm{lc}}(S)} \longrightarrow \underbrace{(\mathrm{Cat}/S)(W_S)^{-1}}_{\simeq\mathrm{HOT}_W^{\mathrm{lc}\,!}(S)},$$

which is therefore an equivalence. Another, more obvious, way of saying the same thing is that on each of the categories of the upper floor the two localizers W_S^{f} and W_S are now identical (since they are all contained in $\underline{\mathrm{Fib}}_W S$, where they are so precisely by axiom W(4b)), so the functor λ between these two localized categories (original definition of λ, in the case $\mathcal{F} = \underline{\mathrm{Fib}}_W S$, cf. page 176) is even the *identity functor,* the two localized categories being *equal:*

$$\mathrm{HOT}_W^{\mathrm{lc}}(S) = \mathrm{HOT}_W^{\mathrm{lc}\,!}(S) \qquad !$$

Thus the diagram (∗) p. 178 takes the form

$$\begin{array}{ccccc} & & \mathrm{HOT}_W^{\mathrm{lc}}(S) & & \\ & \overset{\rho}{\nearrow}\!\!\underset{i}{\swarrow} & & \overset{\sigma}{\nwarrow}\!\!\underset{j}{\searrow} & \\ \mathrm{HOT}_W(S) & & & & \mathrm{HOT}_W^{o}(S). \end{array} \qquad (*)$$

When one interprets not only the two lateral categories $\mathrm{HOT}_W(S)$, $\mathrm{HOT}_W^o(S)$, but also the one in the middle, as localized categories of Cat/S, for the localizers W_S^{d}, W_S^{g} and W_S respectively, the functors ρ, σ are those deduced from the two inclusions [page 182]

$$\begin{array}{ccc} W_S^{\mathrm{d}} & & W_S^{\mathrm{g}} \\ & \searrow \quad \swarrow & \\ & W_S & \end{array} \qquad \text{as subsets of } \mathrm{Fl}(\mathrm{Cat}/S),$$

whereas the functors i, j are deduced, by passage to the localized categories, from one and the same functor

$$\mathrm{Cat}/S \xrightarrow{C} \mathrm{Cat}/S,$$

this functor having the virtue of mapping W_S into $W_S^{\mathrm{d}} \cap W_S^{\mathrm{g}}$,

$$f \in W_S \overset{\text{assuming W(4b)}}{\Longrightarrow} C(f) \in W_S^{\mathrm{d}} \cap W_S^{\mathrm{g}} \qquad (\text{and even } \in W_S^{\mathrm{univ}}).$$

(And for good reason, since C factors through $\underline{\mathrm{Fib}}_W S$ (and even through $\underline{\mathrm{Parf}}_\omega S$), where W_S is identical to W_S^{d}, W_S^{g}, W_S^{univ}. That $f \in W_S$ implies $C(f) \in W$ is independent of axiom W(4b), and is seen from the diagram p. 113

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ {\scriptstyle i_X}\downarrow & & \downarrow{\scriptstyle i_Y} \\ C(X) & \xrightarrow{C(f)} & C(Y), \end{array}$$

where the arrows i_X, i_Y are in W_S.)

That said, it is true that the pairs

$$(\rho, i),\ (\sigma, j)$$

are pairs of adjoint functors, hence (ρ and σ being localization functors) that i and j are *fully faithful.* The adjunction formula can be read as

$$\boxed{\mathrm{Hom}_{W_S^{\mathrm{g}}}(X, CY) \simeq \mathrm{Hom}_{W_S^{\mathrm{d}}}(X, CY) \simeq \mathrm{Hom}_{W_S}(X, Y)} \qquad (*)$$

for X, Y in Cat/S with nothing more. A Hom_Σ indicates the Hom in $(\mathrm{Cat}/S)\Sigma^{-1}$. [page 183]

Corollary. *If X in Cat/S, Y in $\underline{\mathrm{Fib}}_W S$, then*

$$\begin{array}{ccc} \mathrm{Hom}_{W_S^{\mathrm{g}}}(X,Y) & & \mathrm{Hom}_{W_S^{\mathrm{d}}}(X,Y) \\ & {\scriptstyle\sim}\searrow \quad \swarrow{\scriptstyle\sim} & \\ & \mathrm{Hom}_{W_S}(X,Y) & \end{array}$$

are bijective.

Indeed, Y in $\underline{\mathrm{Fib}}_W(S)$ implies that $Y \xrightarrow{i_Y} C(Y)$ is in $W_S^{\mathrm{univ}} \subseteq W_S^{\mathrm{d}} \cap W_S^{\mathrm{g}}$, so

$$\mathrm{Hom}_{W_S^{\mathrm{g}}}(X,Y) \xrightarrow{\sim} \mathrm{Hom}_{W_S^{\mathrm{g}}}(X, CY) \underbrace{\xrightarrow{\sim}}_{\text{adjunction}} \mathrm{Hom}_{W_S}(X,Y),$$

and likewise for $\mathrm{Hom}_{W_S^{\mathrm{d}}}(X,Y)$.

⑤ Perplexities about the weakened variants of axiom W(4b).

What remains of all these beautiful things (only three HOT categories left, instead of $4+\infty$!) when one now assumes only W(4b′) instead of W(4b), i.e. the equality of W_S and W_S^{f} in

$\underline{\mathrm{Parf}}_\omega S$? Finally, it seems to me (contrary to what I had thought I saw) that almost nothing remains ([87]). We still do not then know, for X in $\underline{\mathrm{Fib}}_W S$, whether p. 114

$$i_X : X \longrightarrow C(X)$$

might not be in W_S^{f}, and not only in W_S; we only know it for $X \in \underline{\mathrm{Parf}}_\omega S$, but in this case we do not need it, for the point of the functor $C(X)$ and of the inclusion $i_X : X \longrightarrow C(X)$ is "to improve" X and to make it perfect, or even W-perfect. But if it already is, it is no longer worth the trouble. There will be, as before, 4 essentially distinct localized [page 184] categories HOT_W, $\mathrm{HOT}_W^{\mathrm{o}}$, $\mathrm{HOT}_W^{\mathrm{lc}}$, $\mathrm{HOT}_W^{\mathrm{lc}!}$, plus infinitely many others $\mathcal{F}(W_S^{\mathrm{f}})^{-1}$ for $\mathcal{F}$ between $\underline{\mathrm{Parf}}_\omega S$ and $\underline{\mathrm{Fib}}_W S$ (cf. page 180), with the sole benefit that we can now exclude the case $\mathcal{F} = \underline{\mathrm{Parf}}_\omega S$, for in this case $\mathcal{F}(W_S^{\mathrm{f}})^{-1} = \mathcal{F}(W_S)^{-1} = \mathrm{HOT}_W^{\mathrm{lc}!}(S)$ is one of the four canonical ones. The decisive thing is that we still have no way of proving that

$$\lambda : \mathrm{HOT}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{HOT}_W^{\mathrm{lc}!}(S)$$

is an equivalence, and I doubt that this is the case. We would have to look with $W = W_{\mathbf{D}}$, the $\mathbf{D}$-equivalence, where $\mathbf{D}$ is a derivator $\mathrm{Der}(k\text{-Mod})$, k a given ring (e.g. $k = \mathbf{Z}$, or k

[87] **NB** If W satisfies the axiom of limits, then W(4b′) implies W(4b), cf. the summary theorem in XVI 4.4.13 (p. 58-62), in particular 2°). Condition W(4b) is condition B of this statement, and W(4b′) condition D. (It is implied by W(4a) = C, which I did not suspect in the least when writing XII …)

a field $\mathbf{Z}/p\mathbf{Z}$ or $\mathbf{Q}$ …). I am not going to do it now, however. (It is true that the axiom W(4b′) is not satisfied in this case either …)

If in the statement of the axiom, we replaced $\underline{\mathrm{Parf}}_\omega S$ by $\underline{\mathrm{Parf}}_W S$, nothing essential would be changed - at most I would replace $\underline{\mathrm{Parf}}_\omega S$ by $\underline{\mathrm{Parf}}_W S$ in the brilliant diagram p. 175. Finally, I am now convinced that it has to be all or nothing: either *four* categories $\mathrm{HOT}(S)$ for W, being content with W(1, 2, 3), or else only three, i.e. no $\mathrm{HOT}^{\mathrm{lc}\,!}_W(S)$ - and for that, it does not seem that we can make do with less than W(4b).

[page 185]

It would suffice, it is true, to assume that $W_S = W_S^{\mathrm{f}}$ on any one of the seven categories of the upper floor other than $\underline{\mathrm{Parf}}_\omega S$. This would then imply that in the other six, we will have $\overline{W_S} = \overline{W_S^{\mathrm{f}}}$ (same strong saturations in $\mathrm{Fl}(\mathcal{F})$). One of the two weakest of the axioms in question would be:

> If $f : X \longrightarrow Y$ is a *cartesian* functor of Cat-fibered and W-fibered categories ([88]) over another S, if $f \in W$, then $f_s \in W$ for every $s \in S$,

(and the other is the dual version, for cocartesian functors of cofibered categories). Unless I am mistaken, this axiom must already imply that $\overline{W}$ satisfies the strongest version W(4a) of all (a fortiori W(4b)) of the axioms W(4). This axiom therefore now seems to me much more reasonable than W(4b′), and this all the more so as it refers directly to the description of $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ in terms of functors (or of "pseudofunctors", if we do not want to assume the categories split and the functors compatible with the splittings) $S^o \longrightarrow \mathrm{Cat}$. Besides, although it does not look it, this axiom *is* self-dual, since the dual version, for fixed S, is the direct version for S^o! This makes this axiom still more appetizing. So to formulate W(4a′), I now feel like dropping $\underline{\mathrm{Parf}}_\omega$, and doing the same. If we keep two versions W(4a′) (the old one) and W(4a″) (the new one), both have the virtue of implying W(4b). And W(4b′) (new version) must imply W(4a) (ultrastrong version), if not for W, at least for $\overline{W}$.

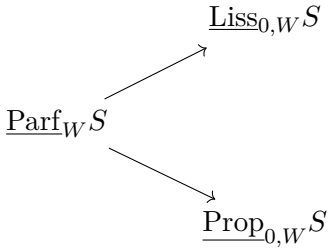
p. 115

My ideas were tenaciously muddled until this very day, for I had the idea that, thanks to the wondrous functor C, the category $\underline{\mathrm{Parf}}_\omega S$, like the seven other categories of the upper floor, could serve to define $\mathrm{HOT}^{\mathrm{lc}}_W(S)$. But this is not so, it seems, unless we assume the axiom W(4b), or the variant W(4b′) (new version). The old version of W(4b′) turns out to be perfectly unusable.

[page 186]

On this subject, I have still not given a clear answer, however, not a "yes" or a "no", to the question:

> **Question:** *do the inclusions*



> *induce, even in the absence of W(4b) (or of its weakened variants), equivalences for the fine localized categories (by W_S^{f})? Same question for $\underline{\mathrm{Parf}}_\omega S$ instead of $\underline{\mathrm{Parf}}_W S$ ([89]). Same question for*
>
> $$\underline{\mathrm{Bifib}}\, S \subseteq \underline{\mathrm{Parf}}_\omega S,$$

[88] "W-fibered" meaning here that for $u \in \mathrm{Fl}\, S$, the base change functors $u^*\ X_{s'} \longrightarrow X_s$, $Y_{s'} \longrightarrow Y_s$ are in W.

[89] OK in the presence of the axiom W(6bis) on the $\varinjlim$, cf. p. 192 ff.

the category of categories over S which are both Cat-*fibered and* Cat-*cofibered.*

I cannot tell, even granting (this time) the axiom W(4a), whether $\underline{\mathrm{Bifib}}\, S$ is big enough to be able to describe all the locally constant relative homotopy types over S, i.e. whether

$$(\underline{\mathrm{Bifib}}\, S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1}$$

is at least essentially surjective, if not an equivalence of categories. What is missing is a functor

$$\mathrm{Cat}/S \longrightarrow \underline{\mathrm{Bifib}}\, S \qquad \text{or at least} \qquad \underbrace{\underline{\mathrm{Fib}}_W S}_{\text{or } \underline{\mathrm{Fib}}_\omega S\,?} \longrightarrow \underline{\mathrm{Bifib}}\, S,$$

which could play the role of Φ and of Ψ *simultaneously.* I have not even checked that Φ does *not* transform objects of $\underline{\mathrm{Prop}}_W S$ or of $\underline{\mathrm{Prop}}_{0,W}(S)$ into W-proper objects (hence W-perfect, being in any case W-smooth and even W_ω-smooth) over S ([90]). If this were nevertheless true on $\underline{\mathrm{Prop}}_W S$, that would show that the essential image of $\underline{\mathrm{Prop}}_W S$ in $\mathrm{HOT}_W(S)$ $(= (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1})$ is contained in $\mathrm{HOT}_W^{\mathrm{lc}}(S)$. Taking $S = \mathbf{\Delta}^1$, this would mean that if in X we have a closed subcategory X_1 such that $X_1 \overset{i}{\hookrightarrow} X$ is W_ω-aspheric, then $1\backslash X \longrightarrow 0\backslash X$ is in W - but $0\backslash X \simeq X$, $1\backslash X \simeq X_1$, the arrow is $X_1 \longrightarrow X$, which is indeed in W being W-aspheric! This is strange and almost immoral ([91]). But by dint of scribbling I have just convinced myself that even if X is cofibered over S (hence also W-proper), $\Phi(X)$ is not W-proper (which would imply W-perfect), nor even a W-fibered one, which would mean that

p. 116 [page 187] [page 188]

$$s'\backslash X \longrightarrow s\backslash X \qquad \text{associated to } s \longrightarrow s' \text{ in } S$$

would always be in W. Setting $Z' = s'\backslash S$, $Z = s\backslash S$, this would mean that if we have a *local co-isomorphism*

$$i : \underbrace{Z'}_{\simeq z_0\backslash Z} \longrightarrow Z,$$

where Z', Z have initial elements, and if X is cofibered over Z, X' its inverse image over Z', then $X' \longrightarrow X$ is in W. In fact, if Z also has a *final* element e (as was the case just now for $\mathbf{\Delta}^1$), then $Z' \simeq z_0\backslash Z$ also has one at e', which maps to e, and we will have the commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\ j\ } & X \\ {\scriptstyle W}\uparrow & & \uparrow{\scriptstyle W} \\ X'_{e'} & \xrightarrow{\ \sim\ } & X_e, \end{array}$$

whence $j \in W$. So to make a counterexample, we must take Z without a final element.

But let us take Z ordered set, with an initial object 0, but without a final object, and let us take as z_0 a *maximal* element of Z, so that $Z' = z_0\backslash Z$ is isomorphic to $\{z_0\}$. We must give in Cat an example where the inclusion of X_{z_0} in X is not in W. We are going to take

[page 189]

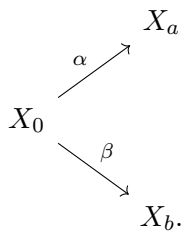
p. 117

$$Z = \quad \begin{array}{ccc} & & a \\ & \nearrow & \\ 0 & & \\ & \searrow & \\ & & b = z_0, \end{array}$$

[90] See the answer to this interesting question on page 192.

[91] We observe below that this is so whenever S has a final object.

so giving X cofibered cosplit over Z is equivalent to a diagram in Cat



I think that under strong enough conditions (e.g. if α, β are open immersions, or closed immersions - to be checked!), the category X is isomorphic in Hot to the amalgamated sum in Cat

$$X_a \coprod_{X_0} X_b,$$

and there no reason (unless we assume $\alpha \in W$, and even then ...) for $X_b \longrightarrow X_a \coprod_{X_0} X_b$ to be in W. Let us take $X_0 = X_b = e$, $\beta = \mathrm{id}_e$, so α is given by the datum of an element t of $\mathrm{Ob}(X_a = T)$. The amalgamated sum is T, which is not isomorphic to e in Hot_W, if T has not been taken W-aspheric. But I am going to try to show directly that the [page 190] inclusion

$$T = X_a \longrightarrow X$$

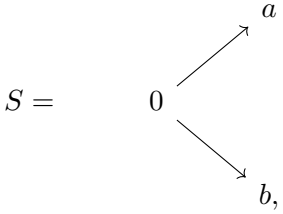
is $\in W$. For this I look at the open subcategory Z'' of Z, complementary to the full closed subcategory Z', and I note that $X'' = X|Z''$ being proper over Z'' and a being a final object of Z'', we have

$$X_a \longrightarrow X'' \qquad \text{in } W,$$

it remains to see that the same holds for $X'' \longrightarrow X$. Now X is obtained from X'' by adding a single element ξ (the one of X_b), which does not map to any element of X'', and which receives no element of X_a - on the other hand for the unique element ξ_0 of X_0 there is a unique arrow from it to ξ. We thus have a retraction of X onto X'', sending ξ to ξ_0 and the preceding arrow to id_{ξ_0}. Is this a homotopism? It suffices to that as an endomorphism φ of X, it is $\sim \mathrm{id}_X$. Now we do have $\mathrm{id}_X \longleftarrow \varphi$, whose value on X'' is the identity, and the value on ξ is the unique $\varphi(\xi) = \xi_0 \longrightarrow \xi$. This is a little brief, but it must be true. (In any case, X is the union of two opens X'', and of the open $X|\{0, b\} = X'''$, with $X''' \simeq \mathbf{\Delta}^1$, and $X'' \cap X''' \simeq e$, so at least for the cohomology with locally constant coefficients of X I am at ease by Mayer-Vietoris, and know full well that it is that of the piece X'', hence that of X_a [...] not at all that of e!)

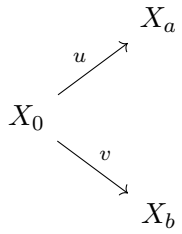
We have thus seen at least this: If X is cofibered hence W-proper over S, it by no means follows that $\Phi(X)$ is still W-proper over S ([92]), nor that it is a W-fibered one. But what happens if $X \in \underline{\mathrm{Prop}}_{0,W} S$, for example if it comes from $\underline{\mathrm{Cofibsc}}_0 S$? Then X is in $\underline{\mathrm{Fib}}_W S$, so we already know that $\Phi(X)$ is in $\underline{\mathrm{Fibsc}}_{0,W} S$, so it is not that way that we will make a counterexample. But let us again consider X cofibered over

p. 118
[page 191]

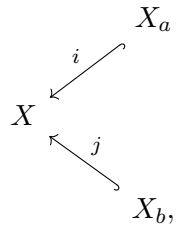


[92] see a more precise result on the next page!

given by the diagram



in Cat, assuming this time $u, v \in W$ ($X \in \underline{\mathrm{Cofibsc}}_{0,W}S$). Then $\Phi(X)$ is the split fibered category over S defined by the contravariant diagram on S



where i, j are the inclusions (which are closed immersions). To say that $\mathcal{X} = \Phi(X)$ is proper over S also means that its restrictions over the open subcategories $\{0, a\}$ and $\{0, b\}$ of S are so (for every arrow $\mathbf{\Delta}^1 \longrightarrow S$ factors through one or the other). This therefore also means that the inclusion of the closed fibers [page 192]

$$X_a \longrightarrow X, \qquad X_b \longrightarrow X$$

are coaspheric. But $X_a \longrightarrow X$ can be coaspheric only if $X_b = \emptyset$, for $x \in X_b \subseteq X$, $x \backslash X_a$ (as well as X_a/x) is empty, hence *not* W-aspheric (unless we assume $W = \mathrm{Fl}(\mathrm{Cat})$, a case which we exclude). So $\Phi(X)$ is proper over S *only* if X_a and $X_b = \emptyset$ (a result valid for *every* X over S, whether it is cofibered over S or W-proper, or not). If we want $u, v \in W$, this therefore implies that $X_0 = \emptyset$, so we have proved:

p. 119

||| Let $X \in \mathrm{Cat}/S$, $S = \begin{array}{ccc} & & a \\ & \nearrow & \\ 0 & & \\ & \searrow & \\ & & b \end{array}$. If $\Phi(X)$ is proper over S, then $X = \emptyset$ (if we assume $W \neq \mathrm{Fl}(\mathrm{Cat})$).

But along the way a bizarre idea appeared. Consider the infinite diagram, for X in Cat/S:

$$X \longrightarrow \Phi(X) \longrightarrow \Psi\Phi(X) \longrightarrow \Phi\Psi\Phi(X) \longrightarrow \cdots$$

If X is arbitrary, these arrows are alternately in W_S^{d} and W_S^{g}, so their composites are in W_S and no more. But if X is in $\underline{\mathrm{Fib}}_W S$, since this category is *stable* under Φ, Ψ, and since on $\underline{\mathrm{Fib}}_W S$ $W_S^{\mathrm{d}} = W_S^{\mathrm{g}} = W_S^{\mathrm{f}} = W_S^{\mathrm{univ}}$, we see that in this case the arrows are all in W_S^{f}. We now feel like setting

$$\Phi_\infty(X) = \varinjlim \text{ of all that}$$

and we would define symmetrically [page 193]

$$\Psi_\infty(X) = \varinjlim(X \longrightarrow \Psi(X) \longrightarrow \Phi\Psi(X) \longrightarrow \cdots).$$

For this to be usable, we must postulate the axiom W(6 bis) (compare XII, pages 14, 15). The direct limit of a sequence of morphisms $f_i : X_i \longrightarrow Y_i$, for a morphism of direct systems in Cat indexed by $\mathbf{N}$, is in W if the f_i are.

This axiom is close to W(6) of loc. cit., but is implied by it only with W(4b). It is stronger in the case of direct systems indexed by $\mathbf{N}$ (so also, more generally, of direct systems on filtering ordered sets having a countable cofinal set), but says nothing about direct systems on more general ordered index sets.

This axiom also implies that a sequential filtering direct limit of morphisms f_i is W-smooth, resp. W-proper, if the f_i are. We then find that $\Phi_\infty(X)$ is both W-smooth (as a direct limit of $\Phi(X)$, $\Phi\Psi\Phi(X)$, …, $\Phi(\Psi\Phi)^n(X)$, which are W-smooth over S) and W-proper (as a direct limit of $\Psi\Phi(X)$, …, $(\Psi\Phi)^n(X)$ …). On the other hand, that $X \longrightarrow \Phi_\infty(X)$ is in W_S^{f} - for W_S^{f} is stable under countable filtering direct limits in Cat/S. We have thus found a functor [page 194]

$$\Phi_\infty : \underline{\mathrm{Fib}}_W S \longrightarrow \underline{\mathrm{Parf}}_W S \subseteq \underline{\mathrm{Fib}}_W S$$

and a functorial morphism

$$i_X : X \longrightarrow \Phi_\infty(X), \qquad \boxed{i_X \in W_S^{\mathrm{f}}} \qquad \text{if } X \in \underline{\mathrm{Fib}}_W S.$$

This functor then allows us, in the absence of W(4b), to do what the functor C absolutely refused. It is true that C has the advantage of being defined on all of Cat/S, and of being reasonable and even very useful there, when we have W(4b). But Φ_∞ too p. 120 is defined on all of Cat/S, as is i_X

$$\Phi_\infty : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Parf}}_W S$$
$$i : \mathrm{id} \longrightarrow \Phi_\infty \qquad \text{i.e.} \begin{cases} i_X : X \longrightarrow \Phi_\infty(X) \\ i_X \in W_S, \end{cases}$$

with the sole handicap (inevitable) that if X is not in $\underline{\mathrm{Fib}}_W S$, we can say no better than $i_X \in W_S$, to the exclusion of $i_X \in W_S^{\mathrm{f}}$. But for $C(X)$ it was necessarily the same. So we get the impression that Φ_∞, take your pick, Ψ_∞, six of one, half a dozen of the other, plays in the presence of W(6 bis) the same role as C in the presence of W(4b). At most (in the presence of both axioms at once) the functor C has the aesthetic advantage of taking its values in $\underline{\mathrm{Parf}}_\omega S$, instead of merely $\underline{\mathrm{Parf}}_W S$ ([93]). But it is imperative to realize [page 195] now that Φ_∞ has an advantage of exactly the same order over C, by introducing

$$W''_\omega = \text{smallest subset } \Sigma \text{ of } \mathrm{Fl}(\mathrm{Cat}) \text{ satisfying } \mathrm{W}(1, 2 \text{ bis}, 3, 6 \text{ bis})$$

and note that

$$\Phi_\infty(X) \in \mathrm{Ob}\, \underline{\mathrm{Parf}}_{W''_\omega} S, \qquad i_X \in W''_\omega$$

- but here I notice that we have

$$W_\omega \subseteq W''_\omega,$$

so C is still a bit better in this respect …

When we have Φ_∞ with the properties we have just made explicit, we find that the inclusion

$$\underline{\mathrm{Parf}}_W S \longrightarrow \underline{\mathrm{Fib}}_W(S)$$

induces an equivalence not only for the categories localized for W_S, but also for the fine localized categories

$$(\underline{\mathrm{Parf}}_W S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Fib}}_W(S))(W_S^{\mathrm{f}})^{-1} \overset{\mathrm{def}}{=} \mathrm{HOT}_W^{\mathrm{lc}}(S)$$

[93] *In fact* $\Phi_\infty = C$, if C is defined simply via one-sided infinite paths. If we define C via *finite* paths of all types c, then it does indeed take its values in $\underline{\mathrm{Parf}}_\omega S$.

([94]). But we cannot for all that reduce the number of useful localized categories to three, in the absence of W(4b). At most, having a functor Φ_∞ which is simpler than C, in the absence of W(4b), because we know in any case that

$$i_X : X \longrightarrow \Phi_\infty(X)$$

is in W_S^{f} if X is in $\underline{\mathrm{Fib}}_W S$, can we hope to get by with the weakened variant of W(4b) obtained by restricting ourselves in the statement of W(4b) (p. 171) to the case where X, Y are in $\underline{\mathrm{Parf}}_{W''_\omega} S$. This time, that does indeed suffice to imply that the functor

p. 121
[page 196]

$$\lambda : (\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Fib}}_W S)(W_S)^{-1}$$

is an equivalence of categories, since we can "compute" it just as well using $\underline{\mathrm{Parf}}_{W''_\omega} S$ instead of $\underline{\mathrm{Fib}}_W S$, now the hypothesis W(4b(W$''_\omega$)) implies that on $\underline{\mathrm{Parf}}_{W''_\omega} S$ the two localizers W_S^{f}, W_S are *equal.*

We need a usable practical conclusion from all these perplexities and observations. Let us try

Proposition. *Suppose that we are under one of the following conditions.*

1°) *The axiom* W(4b) *is satisfied in the case of categories X, Y which are in* $\underline{\mathrm{Fibsc}}_{0,W} S$ ([95]) *(this condition on W is in fact self-dual).*

2°) *The axiom* W(4b) *is satisfied in the case of categories X, Y which are in* $\underline{\mathrm{Parf}}_{W''_\omega} S$*, where W''_ω is the smallest set $\Sigma \subseteq Fl(\mathrm{Cat})$ satisfying the conditions* W(1, 2 bis, 3, 6 bis). *(We can also make do with the condition, stronger but more understandable, that* W(4b) *is OK if X, Y are in* $\underline{\mathrm{Parf}}_W S$ ([96])*.) Moreover, we assume that W satisfies* W(6 bis).

Under these conditions, the corresponding functor λ

$$\mathrm{HOT}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{HOT}_W^{\mathrm{lc!}}(S)$$

is an equivalence of categories (cf. diagram p. 178), so (ρ, i), (σ, j) can be regarded as pairs of functors between two categories (cf. $(*)$ p. 181). These are pairs of adjoint functors, *so i, j are fully faithful, and we have the adjunction formula $(*)$, page 182.*

To prove the adjunction of (ρ, i) (or dually of (σ, j)), we again express the two categories at hand $\mathrm{HOT}_W(S)$, $\mathrm{HOT}_W^{\mathrm{lc}}(S)$ as $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$, $(\mathrm{Cat}/S)(W_S)^{-1}$, the functor ρ being the canonical localization going from the first to the second, and the functor i being given by an inversion functor

[page 197]

$$\gamma : (\mathrm{Cat}/S, W_S) \longrightarrow (\mathrm{Cat}/S, \overline{W_S^{\mathrm{d}}})$$

which is C in case 1°, Φ_∞ in case 2°. The adjunction morphisms

$$\rho i \longrightarrow \mathrm{id}, \qquad i\rho \longleftarrow \mathrm{id}$$

should then be described as in the case where we simply assume W(4b), and the compatibilities are seen in the same way: we are reduced to proving that two arrows

$$\gamma(X) \underset{\gamma(i_X)}{\overset{i_{\gamma(X)}}{\rightrightarrows}} \gamma(\gamma(X))$$

are equal in $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$. The verification in case 1°), where $\gamma = C$, should not be changed. In case 2°) where $\gamma = \Phi_\infty$, I have not done it. I will do it only under the pressure of need.

p. 122

[94] We can even replace $\underline{\mathrm{Parf}}_W S$ there by $\underline{\mathrm{Parf}}_{W''_\omega} S$, *or* by any other full subcategory of $\underline{\mathrm{Fib}}_W S$ containing $\underline{\mathrm{Parf}}_{W''_\omega} S$.

[95] but this is *equivalent* to W(4b)! Cf. XVI 4.9. loc. cit.

[96] But this also implies W(4b), cf. loc. cit.

⑥ Contravariant law of $\mathrm{HOT}_W(S)$, $\mathrm{HOT}^{\mathrm{lc}}_W(S)$.

There is for

$$f : S' \longrightarrow S$$

a homomorphism from the diagram

$$\begin{array}{ccc} & \mathrm{HOT}^{\mathrm{lc}}_W(S') & \\ \swarrow & & \searrow \\ \mathrm{HOT}_W(S') & & \mathrm{HOT}^{\mathrm{o}}_W(S') \end{array}$$

into the similar diagram for S , by functors f^* which are defined respectively on [page 198]

$$\begin{array}{rcccc} \underline{\mathrm{Liss}}_W S & \text{for} & \mathrm{HOT}_W(S) & \longrightarrow & \mathrm{HOT}_W(S') \\ \underline{\mathrm{Prop}}_W S & \text{for} & \mathrm{HOT}^{\mathrm{o}}_W(S) & \longrightarrow & \mathrm{HOT}^{\mathrm{o}}_W(S') \\ \underline{\mathrm{Fib}}_W S & \text{for} & \mathrm{HOT}^{\mathrm{lc}}_W(S) & \longrightarrow & \mathrm{HOT}^{\mathrm{lc}}_W(S'), \end{array}$$

through the functors

$$f^* : X \longmapsto X \times_S S' : \begin{cases} \underline{\mathrm{Liss}}_W S & \longrightarrow & \underline{\mathrm{Liss}}_W S' \\ \underline{\mathrm{Prop}}_W S & \longrightarrow & \underline{\mathrm{Prop}}_W S' \\ \underline{\mathrm{Fib}}_W S & \longrightarrow & \underline{\mathrm{Fib}}_W S', \end{cases}$$

which functor is compatible with the relevant localizers, which are W^{f}_S in all cases. On the other hand, in the absence of draconian hypotheses on W or on S, we cannot say that f^* is compatible with W_S, $W_{S'}$ on any of the categories of the diagram p. 175, allowing us to define $\mathrm{HOT}^{\mathrm{lc!}}_W(S)$, so as to give us a functor

$$\mathrm{HOT}^{\mathrm{lc!}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc!}}_W(S').$$

In the absence of axioms on W, the only case known to me is that where f is *a universal W-equivalence,* i.e. a W-fibration with W-aspheric fibers, in which case it is obvious that for *every* arrow $u : X \longrightarrow Y$ in Cat/S, $u \in W_S$ if and only if $(u' : X' \longrightarrow Y') \in W_{S'}$. We then find a homomorphism from the complete diagram $(*)$ p. 176 for S, into the corresponding diagram for S'.

If we assume W(4b), then $W_S = W^{\mathrm{f}}_S$ on $\underline{\mathrm{Fib}}_W S$, so on that category the functor f^* is compatible (no merit!) with the localizers W_S, $W_{S'}$, but this is not really new compared p. 123
with what was said above, for $\underline{\mathrm{Fib}}_W S$, $\underline{\mathrm{Fib}}_W S'$, and for the $\mathrm{HOT}^{\mathrm{lc}}_W$. [page 199]

But if we assume W(4a), and $f \in \underline{\mathrm{Fib}}_W$, then f^* on all of Cat/S is compatible with the localizers W_S, $W_{S'}$, and we find a computation of f^* on $\mathrm{HOT}^{\mathrm{lc!}}_W(S) \approx \mathrm{HOT}^{\mathrm{lc}}_W(S)$ by

$$f^*(\mathrm{hot}^{\mathrm{lc}}_{S,W}(X)) = \mathrm{hot}^{\mathrm{lc}}_{S',W}(X \times_S S').$$

So this is a case, exceptionally, where the functors f^* for the HOT_W are compatible not only with the inclusion functors i, j (of page 176), but also with the localization functors ρ, σ.

Omission: If $p : S' \longrightarrow S$ is W-proper, then $p^* : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S'$ is compatible with W^{d}_S, $W^{\mathrm{d}}_{S'}$, and $p^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$ can be "computed" by the formula

$$p^*(\mathrm{hot}_{S,W}(X)) = \mathrm{hot}_{S',W}(X \times_S S')$$

for *every* X in Cat/S.

⑦ Product in $\mathrm{HOT}_W(S)$.

The functor

$$(X,Y)\longmapsto X\times_S Y : \underline{\mathrm{Liss}}_W S\times \underline{\mathrm{Liss}}_W S\longrightarrow \underline{\mathrm{Liss}}_W S$$

is compatible with the localizers $W_S^{\mathrm{f}}\times W_S^{\mathrm{f}}$ and W_S^{f}, so passes to the localized categories as

$$\begin{cases}\mathrm{HOT}_W(S)\times\mathrm{HOT}_W(S) & \longrightarrow & \mathrm{HOT}_W(S)\\ (\xi,\eta) & \longmapsto & \xi\times_S\eta\end{cases}$$

([97]). This product is associative, unital, commutes with all the usual compatibilities, as follows from the fact that that's how it is in $\underline{\mathrm{Liss}}_S$. The unit element is the class of $S\xrightarrow{\mathrm{id}}S$. The subcategory $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ is stable under the product, which induces [page 200]

$$\mathrm{HOT}^{\mathrm{lc}}_W(S)\times\mathrm{HOT}^{\mathrm{lc}}_W(S)\longrightarrow\mathrm{HOT}^{\mathrm{lc}}_W(S),$$

itself induced by the product functor $(X,Y)\longrightarrow X\times_S Y$ in $\underline{\mathrm{Fib}}_W S$.

If X is arbitrary in Cat/S, the functor

$$\mathrm{mult}_{X/S} : Y\longmapsto X\times_S Y : \underline{\mathrm{Liss}}_W S\longrightarrow \mathrm{Cat}/S$$

is compatible with the localizers W_S^{d}, for this functor can be written as the composite of the corresponding functor *with values in* $\underline{\mathrm{Liss}}_W X$, which is compatible with the localizers W_S^{d}, W_X^{d}, and the "forget X" functor from $\underline{\mathrm{Liss}}_W X$ (even from Cat/X) to Cat/S, p. 124 which is compatible with W_X^{d} and W_S^{d}. We thus get a functor

$$\begin{cases}\mathrm{mult}_{X/S} : \mathrm{HOT}_W(S)\longrightarrow\mathrm{HOT}_W(S)\\ \mathrm{mult}_{X/S}(\mathrm{hot}_{S,W}(Y))\overset{\mathrm{def}}{=}\mathrm{hot}_{S,W}(X\times_S Y)\qquad \text{if } Y \textit{ smooth} \text{ over } S\end{cases}$$

([98]), since $\mathrm{HOT}_W(S)$ can be obtained as the localized category of both $\underline{\mathrm{Liss}}_W S$ and Cat/S, for W_S^{d}. When X is itself W-smooth over S, this functor is none other than

$$\begin{cases}\mathrm{mult}_\xi : \eta\longmapsto\xi\times_S\eta,\\ \text{where }\ \xi=\mathrm{hot}_{S,W}(X).\end{cases}$$

But this is not so if X is not smooth, i.e. we must distinguish in this case, even for Y W-smooth over S,

$$\mathrm{hot}_{S,W}(X\times_S Y)\qquad\text{and}\qquad \mathrm{hot}_{S,W}(X)\times_S\mathrm{hot}_{S,W}(Y).$$

The second term can be made explicit as $\mathrm{hot}_{S,W}(\Phi(X)\times_S Y)$ which receives $\mathrm{hot}_{S,W}(X\times_S Y)$, i.e. we have a canonical homomorphism [page 201]

$$\underbrace{\mathrm{hot}_{S,W}(X\times_S Y)}_{=\mathrm{mult}_{X/S}(Y)}\longrightarrow\underbrace{\mathrm{hot}_{S,W}(X)\times_S\mathrm{hot}_{S,W}(Y)}_{=\mathrm{hot}_{S,W}(\Phi(X)\times_S Y)},$$

induced by

$$X\times_S Y\longrightarrow\Phi(X)\times_S Y,$$

but this last arrow has no reason to be in W_S^{d}, and thus does not necessarily induce an isomorphism in $\mathrm{HOT}_W(S)$. (Counterexample page 154.)

[97] **NB** If X, Y are *arbitrary* in Cat/S, we have $\mathrm{hot}_{S,W}(X)\times_S\mathrm{hot}_{S,W}(Y)=\mathrm{hot}_{S,W}(\Phi_S(X)\times_S\Phi_S(Y))$. If X or Y is smooth over S, we can replace $\Phi(X)$ resp. $\Phi(Y)$ there by X resp. Y.

[98] This formula is also valid if Y is in $\underline{\mathrm{Fib}}_W S$, cf. $(*)$, p. 202.

We must look separately at the case *where X is in* $\underline{\mathrm{Fib}}_W S$, I feel like saying that in this case we have

$$\mathrm{hot}_{S,W}(X \times_S Y) \xrightarrow{\sim} \underbrace{\mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y)}_{=\mathrm{hot}_{S,W}(\Phi(X)\times_S \Phi(Y))} \qquad (*)$$

([99]) for *every* Y in Cat/S (not necessarily W-smooth), a fortiori that in this case

$$\begin{cases} \mathrm{mult}_{X/S} = \mathrm{mult}_\xi & \text{in } \mathrm{HOT}_W(S) \\ \text{where } \xi = \mathrm{hot}_{S,W}(X/S) \text{ (if } X \text{ in } \underline{\mathrm{Fib}}_W S). \end{cases} \qquad (**)$$

In the case where Y is smooth over S (which suffices to establish $(**)$), it suffices p. 125
to recall that $X \longrightarrow \Phi(X)$ is in W_S^{f} (follows from the fact that X is W-fibered over S ([100]), hence $X_s \longrightarrow s\backslash X$ is in W, $\forall\, s \in S$), hence $\underbrace{X \times_S Y}_{X_Y} \longrightarrow \underbrace{\Phi(X) \times_S Y}_{\Phi(X)_Y}$ is also in W_Y^{f}, hence (since X_Y, $\Phi(X)_Y$ are W-fibered over Y) in W_Y^{d}, hence also in W_S^{d}. In fact, even without assuming Y smooth, this shows that we have [page 202]

$$\mathrm{hot}_{S,W}(X \times_S Y) \xrightarrow{\sim} \mathrm{mult}_{Y/S}(\mathrm{hot}_{S,W}(X)) \qquad \text{if } X \text{ is in } \underline{\mathrm{Fib}}_W S. \qquad (*)$$

To have $(*)$ for arbitrary Y, we can by the foregoing assume X smooth (replacing it by $\Phi(X)$ if need be), hence X in $\underline{\mathrm{Liss}}_{0,W} S$. We must see that

$$X \times_S Y \longrightarrow X \times_S \Phi(Y)$$

is still in W_S^{d}. Now if X is W-*proper* over S, the base change $X \longrightarrow S$ is compatible with W_S^{d}, W_X^{d} (on all of Cat/S), and we win. (This must be false without an assumption of W-properness on X, *or* of W-smoothness on Y …)

When we have

$$f : S' \longrightarrow S,$$

then

$$f^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$$

is compatible with the product structures.

⑧ The functors $f_!$: adjunction theorem, properties of the derivator HOT_W.

Let

$$f : S' \longrightarrow S$$

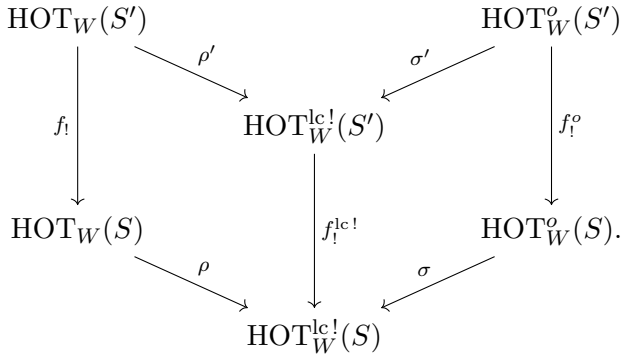
be in Cat, then the induced functor

$$f_! : \mathrm{Cat}/S' \longrightarrow \mathrm{Cat}/S$$

is compatible with the localizers $(W_{S'}^{\mathrm{d}}, W_S^{\mathrm{d}})$, or $(W_{S'}^{\mathrm{g}}, W_S^{\mathrm{g}})$, or $(W_{S'}, W_S)$. We thus get an induced functor for the localized categories, and commutative diagram [page 203]

[99] This formula is established only by assuming either Y W-smooth (in addition to X W-fibered over S), or X W-proper, e.g. X W-perfect.

[100] even W-presmooth over S suffices



We do not have a functor $f_!^{\mathrm{lc}}$ at the level of the $\mathrm{HOT}_W^{\mathrm{lc}}$, unless we make an assumption, such as W(4b), which ensures that $\mathrm{HOT}_W^{\mathrm{lc}\,!} \approx \mathrm{HOT}_W^{\mathrm{lc}}$. p. 126

The main theorem, that the functor $f_!$ is left adjoint to f^*, is valid *under the sole assumption* W(1, 2, 3) ([101]). Thus dually for $f_!^o$, f_o^*. But since we do not have a functor $f_{\mathrm{lc}\,!}^*$, the question of adjunction does not arise for general f, and without an assumption W(4b) on W. It only arises, apparently, for f a trivial W-fibration (cf. end of ⑥, p. 198), and in this case the adjunction should not present any difficulties. But as long as I do not have a very clear feeling about the meaning of the $\mathrm{HOT}_W^{\mathrm{lc}\,!}$, this is not a result to get much of a hard-on over .

The adjunction of $f_!$, f^* for HOT_W already allows us to show that the prederivator HOT_W satisfies the conditions Der 1, Der 4′, Der 5′ for derivators. Moreover, we have [page 204] what we can call the property Der 3b for HOT_W as a W-derivator, i.e.

> *If $f \in W$, then f is a* HOT_W*-equivalence* ([102]) (assuming only the axioms W(1, 2, 3) on W). More precisely, f is a HOT_W-equivalence if and only if $\mathrm{hot}_W(f)$ is isomorphism, i.e. $f \in \overline{W}$, the strong saturation of W.

If W *satisfies* W(4b), then we also have a pair of adjoint functors

$$\mathrm{HOT}_W^{\mathrm{lc}}(S') \underset{f_{\mathrm{lc}}^*}{\overset{f_!^{\mathrm{lc}}}{\rightleftarrows}} \mathrm{HOT}_W^{\mathrm{lc}}(S)$$

for every arrow $S' \longrightarrow S$ in Cat. We also see in this case that

> if $f : S' \longrightarrow S$ is in W, then f_{lc}^* and $f_!^{\mathrm{lc}}$ between $\mathrm{HOT}_W^{\mathrm{lc}}(S)$ and $\mathrm{HOT}_W^{\mathrm{lc}}(S')$ are equivalences quasi-inverse to one another (assuming W(4b)),

which is axiom Der $3b_2$ "W-derivators version". We cannot expect any better, on the Der 3 side!

Omission: Let $f : S' \longrightarrow S$, we then have $\boxed{f_! f^*} : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S)$ is equal to $\mathrm{mult}_{S'/S}$ (cf. p. 200). It is therefore equal to mult_ξ, where $\xi = \mathrm{hot}_{S,W}(S')$, if we assume either S' W-smooth over S, or S' W-fibered over S (cf. (∗∗), p. 201).

⑨ Base change theorem for $f_!$ in HOT_W.

If we have p. 127

[101] i.e. with these new notations Loc(1, 2, 3 bis) (cf. XVI).

[102] We prove this by interpreting the notion of a HOT_W-equivalence in terms of HOT_W*-homology.* This comes down in the end to this: If $f : S' \longrightarrow S$ is in W, then $\forall\, T$, $S' \times T \longrightarrow S \times T$ too!

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{g} & S', \end{array}$$

with f W-smooth *or* g W-proper [...]

THE DERIVATORS

p.

ALEXANDRE GROTHENDIECK

Chapter VII

Categories of paths (2)

Since the "simplified" formalism of paths that I wanted to work out in a hurry has turned out to be unbearably cumbersome, enough to make it virtually unusable, I am going to try to work out a flexible formalism, sticking to intuition, at the cost only of an additional conceptual effort.

p. p. 1 [page 1]

1 Intervals and their variants

We call a *standard interval category* a category defined by an ordered set, whose set of elements is an interval of integers $[0, n]$ ($n \in \mathbf{N}$, we do not exclude the case $n = 0$, i.e. of the one-point category $\{0\}$):

$$\mathrm{Ob}\, I = [0, n] \subseteq \mathbf{N},$$

and satisfying the following hypothesis:

a) for every pair of two consecutive vertices $i-1$, i of I ($1 \leq i \leq n$), there is an arrow joining them, thus either $i \longrightarrow i+1$, or $i \longleftarrow i+1$ (this arrow is denoted by $\underline{u}_i$,

$$(i-1) \xrightarrow{\underline{u}_i} i \qquad \text{or} \qquad (i-1) \xleftarrow{\underline{u}_i} i \ \ ).$$

b) The category I is generated by the set of its arrows $\underline{u}_i$ (called *transition arrows*).

The transition arrow is called *direct* if it goes from $i-1$ to i, *retrograde* in the opposite direction. If we assign to the interval $[i-1, i]$ the sign $+$ or $-$ according as $\underline{u}_i$ is direct or retrograde, we thus see that the standard intervals of given length n correspond one-to-one to the systems of signs $+$ or $-$ on the n transition intervals $[i-1, i]$ ($1 \leq i \leq n$), so there are 2^n of them. The vertex 0 is the *origin* or the *source* of the interval, the vertex n the *endpoint* or the *target* of the interval. [page 2]

A standard interval is *oriented,* in that we have the natural *total* order relation on the set of its vertices, namely that of the integers in $[0, n]$. When we speak of the order relation on I, that a vertex i *"precedes"* or *"follows"* another one j ($i \leq j$ or $i \geq j$ respectively), it is this order relation that we will mean. The one associated with the category structure will never be regarded as an order relation. There is also a natural total order relation on the set of transition arrows $\underline{u}_i$ ($1 \leq i \leq n$), given by the order relation of the integers i, $1 \leq i \leq n$, that index them. We will say that a transition arrow i *precedes* or *follows* another , or that it is the *immediate successor* or *immediate predecessor* of another (if $i = j+1$ or $i = j-1$).

If we have a set of integers

$$E \subseteq \mathbf{Z},$$

p. 2

we will consider in E the equivalence relation generated by the relation between two elements (i, j) of E: j is the immediate successor of i *in* $\mathbf{Z}$ (i.e. $j = i+1$). Two distinct elements i, j of E are congruent for this relation if and only if the interval of integers that they determine is $\subseteq E$, thus

if $i < j$, i and j are *"connected"* in E if and only if $[i, j] \subseteq E$. [page 3]

When we speak of the "connected components" of E, it will always be a matter of the classes under this equivalence relation.

We will identify the set of transition arrows $\mathrm{Fl\,tr}(I)$ of I with the interval of integers $[1, n]$, hence with a subset of $\mathbf{Z}$. If we have a subset $E \subseteq \mathrm{Fl\,tr}(I)$, we thus have in it the notion of transition arrows "connected" in E, and the notion of connectedness for E. If $i < j$, $\underline{u}_i$ and $\underline{u}_j$ are connected if and only if all the intermediate transition arrows between $\underline{u}_i$ and $\underline{u}_j$ are in E. And E is connected if and only if it is either empty, or formed of a sequence of transition arrows each of which is the immediate successor of the one preceding it,

$$E = \{\underline{u}_i, \underline{u}_{i+1}, \dots, \underline{u}_{i+l-1}\},$$

where l is the cardinal of E. Such an E is called a *chain* of transition arrows, and l is its length. A chain is called *composable* if the composite $\underline{u}_{i+l-1} \circ \cdots \circ \underline{u}_{i+1} \circ \underline{u}_i$ makes sense, i.e. if they are all in the same direction, either direct, or retrograde.

We denote by $\mathrm{Fl\,tr}^+(I)$, $\mathrm{Fl\,tr}^-(I)$ the set of direct, or retrograde, transition arrows, respectively. Their connected components are nothing other than the maximal composable chains. Thus, the maximal composable chains form a partition of the set of transition arrows, into mutually disjoint intervals of integers. The chains themselves form a totally ordered set

$$C_1,\ C_2,\ \dots\ C_r \qquad (1 \leq r \leq n).$$

The integer r is called the *"useful length"* of the interval I, it is thus the cardinal of the [page 4]
set of its maximal composable chains.

Two maximal composable chains are always of opposite sign, if one is direct the other is retrograde, and conversely. The first maximal composable chain C_1 (let us say simply *"structural chain"*) has the same sign as $\underline{u}_1$; say that the *initial direction* of I is direct, resp. retrograde, if this is so for $\underline{u}_1$, or what amounts to the same thing, for C_1. Likewise we define the *final direction* of I as that of $\underline{u}_n$, or again that of C_r. Initial and final directions are equal if the useful length r is odd, unequal if it is even. (**NB** It is only p. 3
for $n = 0$, i.e. when $I = \{0\}$ is the one-point category, that initial direction and final direction are not defined.)

The composable chains correspond to the full subcategories of I whose vertices are consecutive in the total order and that are isomorphic to a category $\mathbf{\Delta}_i$ (where i is the length of the chain). Let us note that $\mathbf{\Delta}_i$ has no automorphism other than the identity, so the maximal chains of given length $i \geq 1$ correspond exactly to the functors

$$\mathbf{\Delta}_i \hookrightarrow I$$

that are injective on the vertices and whose image is an entire structural interval. Such a functor is fully faithful, and a monomorphism. This functor defines a strictly increasing or strictly decreasing map on the vertices, according as the chain is direct or retrograde. [page 5]

Let us note that every standard simplex $\mathbf{\Delta}_i$, $i \geq 0$, can be viewed as an interval, of length i, of useful length 1 (if $i \geq 1$, 0 if $i = 0$). Its origin is 0, its endpoint is i. But let us call an arbitrary category an *interval category* if it is isomorphic to a standard interval. Let us note here

Proposition 1. *Let I be a standard interval of length $n \geq 1$.*

a) Let $i \in I$. For $i \in \{0, n\}$, i.e. for i to be one of the two "ends" *(or* "endpoints"*) (source or target) of I, it is necessary and sufficient that one of the two categories I/i or $i\backslash I$ be the one-point category and that the other be isomorphic to a $\mathbf{\Delta}_j$ ([1]).*

[1] where j is the length of the unique structural chain that "touches" i (i.e. such that i is in the full subcategory defined by the said chain).

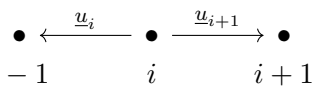
(Intrinsic characterization of the ends of an interval.) For i to be the source, it is necessary and sufficient that $i\backslash I$

b) Every automorphism $I \longrightarrow I$ that fixes the two endpoints of I is the identity.

Corollary. *The group of automorphisms of I has at most two elements. If it has two, the nontrivial automorphism interchanges the two endpoints. (We then say that the interval is* symmetric*.)*

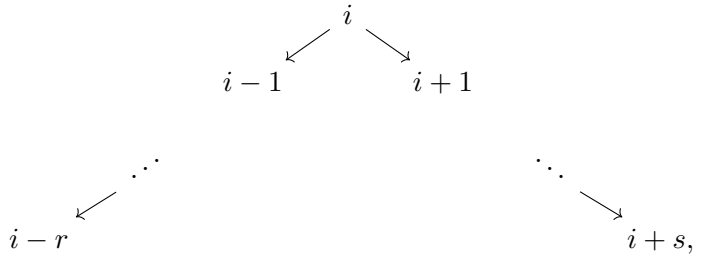
[page 6]

NB If i is not an end, and if nevertheless one of the two categories $i\backslash I$, I/i is one-point, say I/i, then in the immediate neighborhood of i, I has the form



($\underline{u}_i$ retrograde, $\underline{u}_{i+1}$ direct - if $i\backslash I$ were the one-point one, then $\underline{u}_i$ would be direct, $\underline{u}_{i+1}$ retrograde - in any case these two arrows have opposite directions, i.e. not composable). We thus find that $i\backslash I$ is a category of the form

p. 4



where r is the length of the structural chain containing $\underline{u}_i$, s that of the structural chain containing $\underline{u}_{i+1}$. This category is an amalgamated sum $\mathbf{\Delta}_r \amalg \mathbf{\Delta}_s$ (glued together along their origin), but is not isomorphic to a $\mathbf{\Delta}_j$, e.g. for lack of a final object.

This observation proves a). To prove b), it suffices to prove that the total order on I is determined intrinsically, by the choice of *the origin* s_0 of I (among its two endpoints). Suppose that we have already determined intrinsically the vertices s_0, s_1, ..., s_i ($i \geq 0$). How to determine s_{i+1}? (When $i < n$.) Let I' be the full subcategory of I formed by the $j \in I$ which are not among the s_j already constructed, except the last s_i (thus $s_i \in I'$, $\operatorname{card} I' \geq 2$). Then I' is isomorphic to a standard interval, with s_i corresponding to the origin. One of the categories $s_i\backslash I'$, I'/s_i is one-point, the other isomorphic to a category $\mathbf{\Delta}_j$, and moreover, i is an initial or final object of $\mathbf{\Delta}_j$. If it is the initial object 0, we take for s_{i+1} the immediate successor of i in $\mathbf{\Delta}_j$. If it is the final object j, we take for s_{i+1} the immediate predecessor $j-1$.

[page 7]

If we had started from the target n of I, instead of starting from the source, we would have found the opposite total order relation. Assigning the target element the order number 0, we find a renumbering of I, i becoming $n-i$, and the transition arrows which were direct for the initial order relation, become retrograde in the opposite order relation, and conversely. We thus see:

Proposition 2. *If I is a standard interval, there exists a unique pair (I', θ), where I' is a standard interval, and θ an isomorphism $I \xrightarrow{\sim} I'$ which takes the target of I into the source of I' (and the source of I into the target of I').*

This interval I' is called the *opposite* interval of I. We must see it, intrinsically, as the "same" category I, but with an opposite "orientation".

[page 8]

A clearer way to see this is the following. Let J be an interval category, i.e. isomorphic to a standard interval. Then there are exactly (if $I \not\simeq e$) *two* pairs (I, θ) of a standard

interval I and an isomorphism $\theta : I \xrightarrow{\sim} J$. These two isomorphisms induce on J two total orders, opposite to one another. Each pair is uniquely determined by the said total order, which we shall call an *orientation* of the interval J. Thus J has two orientations, and each determines an *oriented isomorphism* (i.e. respecting the orientations) of J with p. 5
a standard interval ([2]). These two intervals are the same if and only if there exists an automorphism of J (necessarily unique) which interchanges its two orientations, or what amounts to the same thing, which interchanges the two endpoints. (Thus the categories opposite to one another



are each symmetric intervals.) Choosing an orientation also amounts to choosing the origin among the two endpoints.

Let us note that the *opposite category* of an interval is an interval, and the opposite category of a standard interval is the standard interval deduced from it by changing all the signs on the transition intervals $[i-1, i]$ $(1 \leq i \leq n)$. We shall take care not to confuse the property $I \simeq I^o$, with the property of symmetry.

Thus the two categories (Ψ_1 and Φ_1) above are symmetric, but opposite to one [page 9]
another without being isomorphic. On the other hand, the $\mathbf{\Delta}_i$ are not symmetric, but each isomorphic to the opposite category.

If I is isomorphic to the opposite category I^o, and if θ is such an isomorphism (necessarily unique), then it is immediate that θ reverses the orientation (otherwise it would be the identity on the vertices, and the transition arrows would be both direct and indirect), hence $\theta(0) = n$, $\theta(1) = n-1$, and we want $\underline{u}_1$ and $\underline{u}_n$ to be of the same sign. But if I is symmetric, $\underline{u}_1$ and $\underline{u}_n$ are of opposite sign. Thus I cannot be both symmetric and isomorphic to the opposite interval .

For any category $\mathbf{\Delta}_i$, this being an interval, there are (if $i > 0$) *two orientations.* As oriented intervals, there are $\mathbf{\Delta}_i^+$ (with the ordinary orientation, given by the ordinary order on $\mathrm{Ob}\,\mathbf{\Delta}_i = [0, i]$), and $\mathbf{\Delta}_i^-$ with the opposite orientation.

p. 6
[page 10]

2 Full subcategories of an interval

Let I be an oriented interval category, of length $n \geq 1$. We shall identify $\mathrm{Ob}\, I$ with the set of integers $[0, n]$. For a chain $\underline{u}_i$, $\underline{u}_{i+1}$, …, $\underline{u}_{i+l-1}$ of transition arrows in I, we call the *domain* of the chain, the set of sources and targets of these elements. This is thus an interval of integers $[i-1, i+l-1]$, of length l equal to that of the chain, whose origin is determined by the first arrow $\underline{u}_i$ of the chain (it is its source, if $\underline{u}_i$ is direct, and its target if $\underline{u}_i$ is retrograde), and whose endpoint is determined by the last $\underline{u}_{i+l-1}$. The intervals of integers (viewed as vertices of I) corresponding to the structural chains are called the *structural intervals* or *critical intervals* of I. Unlike the structural chains, these intervals do not form a partition of I, since two structural intervals may have a common endpoint. If the structural chains are

$$C_1,\ C_2,\ \ldots\ ,\ C_r \qquad (r = \text{ useful length of } I)$$

of lengths

$$l_1,\ l_2,\ \ldots\ ,\ l_r,$$

[2] **NB** We shall speak of the *oppositely oriented* interval I^{opp} of an *oriented* interval I, which we must take care not to confuse with the opposite category I^o. Here I^{opp} has the same underlying category as I, but with an opposite orientation.

then the structural intervals are

$$\underbrace{[0, l_1]}_{I_1},\ \underbrace{[l_1, l_1 + l_2]}_{I_2},\ \dots\ ,\ \underbrace{[l_1 + \dots + l_{r-1}, l_1 + \dots + l_{r-1} + l_r]}_{I_r},$$

a sequence of intervals of integers of which two consecutive ones have a common endpoint, and such that if $i < j$, each element of I_i is $\leq$ each element of I_j. We call *critical vertex* of I the endpoints of the structural intervals.

Proposition 3. *Let $s \in I$, let $I' = I/s$ (full subcategory of I of the s' such that there exists $s' \longrightarrow s$), $I'' = I/s$, $I'_0 = (I/s)\backslash\{s\}$, $I''_0 = (s\backslash I)\backslash\{s\}$ ([3]).* For s to be critical, it is necessary and sufficient that $((I/s) \cup (s\backslash I))\backslash\{s\}$ be the sum category of I'_0 and I''_0.

Proposition 4. *Let i, j be in I, with $i < j$. Equivalent conditions:* [page 11]

a) There exists an arrow between i and j (either $i \longrightarrow j$, or $j \longrightarrow i$).

b) i and j belong to the same structural interval (necessarily unique).

c) The full subcategory of I defined by the set of objects $[i, j]$ is isomorphic to a $\mathbf{\Delta}_\alpha$ (we shall have $\alpha = j - i$).

We shall then say that i and j are *directly connected* in I. (And we shall use the same terminology in an arbitrary category: i and j are directly connected if and only if $\mathrm{Hom}_I(i, j)$ *or* $\mathrm{Hom}_I(j, i)$ is nonempty.) p. 7

Proposition 5. *Let $J \subseteq I$ be a subset of I, let us consider the full subcategory of I which it defines. Then:*

a) The connected components J_α of J are intervals and the order induced on a J_α by that of I is an orientation (called the induced orientation*) of J_α.*

b) Suppose $\mathrm{card}\, J > 1$. *Let i be the smallest element of J, j the largest element, thus $i < j$. For J itself to be an interval, it is necessary and sufficient that every critical vertex in $[i, j]$ be an element of J.*

([4]).

NB If $s \in [i, j]$ is a critical vertex of I which is not in J, then by proposition 3 it would disconnect J, i.e. J would be a direct sum of two nonempty subcategories J' and J'' ($i \in J'$, $j \in J''$), hence J would not be an interval.

Corollary. *The set of critical vertices of I is also the set of vertices of the smallest subinterval of I containing the endpoints. More generally, if $i < j$ are two vertices of I, the smallest subinterval of I containing i, j is formed by i, j and the critical vertices s such that $i < s < j$.* [page 12]

Definition. An interval is said to be *reduced* (or *irredundant*) if all its vertices are critical, or what amounts to the same thing, if its structural chains are of length 1.

A *reduced* oriented interval is known, up to isomorphism, when we know

a) its length n,

b) its initial direction + (direct) or − (retrograde).

[3] i.e. that we have: there are no arrows between an element of I'_0 and an element of I''_0.

[4] We call a *subinterval* of I a full subcategory which is connected, i.e. which is itself an interval. This fullness condition is part of the definition: a subcategory I' of I which is an interval is not necessarily full, cf. below on functors between intervals.

NB This direction is the same as the final direction if n is odd, it is opposite if n is even.

Thus, according to the two cases of initial direction, we find

$$0 \longrightarrow 1 \longleftarrow 2 \longrightarrow \cdots \quad \cdots\ n \qquad (\text{With} \quad \begin{array}{ll} n-1 \longrightarrow n & \text{if } n \text{ odd,} \\ n-1 \longleftarrow n & \text{if } n \text{ even.)} \end{array}$$

$$0 \longleftarrow 1 \longrightarrow 2 \longleftarrow \cdots \quad \cdots\ n \qquad (\text{With} \quad \begin{array}{ll} n-1 \longleftarrow n & \text{if } n \text{ odd,} \\ n-1 \longrightarrow n & \text{if } n \text{ even.)} \end{array}$$

Proposition 6. *A subinterval J, having i and j $(i < j)$ as smallest and largest vertex, is reduced if and only if every $s \in J$ distinct from i and from j is critical.* p. 8

Corollary. *The full subcategory of I formed by the critical vertices is a subinterval. We call it the reduced interval associated to I, denoted I_{red}.*

p. 9

[page 13]

3 Paths. Category of paths of a given type τ.

Let I be an oriented interval, and X a category. We call a *path of type I in X* any functor $I \longrightarrow X$, and the *category of paths of type I in X* the category

$$\underline{\mathrm{Ch}}_I(X) = \underline{\mathrm{Hom}}(I, X).$$

Since I is *canonically* isomorphic to a standard interval category I_0, we can identify the paths of type I, with those of type I_0. The type τ of a path of length n $(= \mathrm{card}(\mathrm{Ob}\, I) - 1)$ in X can therefore be described by a sequence of n signs $+$ or $-$

$$\tau_1,\ \tau_2,\ \ldots\ ,\ \tau_n \qquad (\tau_i \in \{-1, +1\} \text{ for } 1 \leq i \leq n).$$

Because of the fact that I is generated by its transition arrows, a functor $c : I \longrightarrow X$ is determined when we know

a) The $c(i)$ $(i \in I)$, i.e. a sequence of $n+1$ objects of X:

$$a_0,\ a_1,\ \ldots\ ,\ a_n\ ({}^5);$$

b) The $c(\underline{u}_i)$ (where the $\underline{u}_i$ are the transition arrows in I), i.e. for every $i \in [1, n]$ an arrow u_i in X between a_{i-1} and a_i, going from a_{i-1} to a_i if $\underline{u}_i$ is direct $(\tau_i = +1)$, and from a_i to a_{i-1} if $\underline{u}_i$ is retrograde $(\tau_i = -1)$,

and to any such datum a), b) there corresponds a functor. Thus we shall be able to identify the paths of type I in X with the diagrams of the specified type in X. And the morphisms between paths likewise correspond to the morphisms between diagrams a and a', an arrow $a \xrightarrow{\alpha} a'$ thus being a sequence of $n+1$ arrows

[page 14]

$$\alpha_i : a_i \longrightarrow a'_i, \qquad i \in [0, n],$$

satisfying the commutativity condition for the n squares

$$\begin{array}{ccc} a_{i-1} & \xrightarrow{u_i} & a_i \\ {\scriptstyle \alpha_{i-1}}\downarrow & & \downarrow{\scriptstyle \alpha_i} \\ a'_{i-1} & \xrightarrow{u'_i} & a'_i \end{array} \qquad (\text{case } \underline{u}_i \text{ direct})$$

p. 10

or

$$\begin{array}{ccc} a_{i-1} & \xleftarrow{u_i} & a_i \\ {\scriptstyle\alpha_{i-1}}\downarrow & & \downarrow{\scriptstyle\alpha_i} \\ a'_{i-1} & \xleftarrow{u'_i} & a'_i. \end{array} \qquad \text{(case } \underline{u}_i \text{ retrograde)}$$

If we have an arrow between any two vertices i, j in I, with $i < j$ and j not the immediate successor of i, then i, j belong to the same critical interval, and in I we shall have

$$i \xrightarrow{\underline{u}_{i+1}} i+1 \xrightarrow{\underline{u}_{i+2}} \cdots\ j-1 \xrightarrow{\underline{u}_j} j \qquad \text{a chain of length } j-i \geq 2$$

or

$$i \longleftarrow i+1 \longleftarrow \quad \cdots\ j-1 \longleftarrow j,$$

and the corresponding arrow between a_i and a_j is obtained by composing the corresponding arrows u_i.

Examples. [page 15]

1. $I = \mathbf{\Delta}_0 = e$, $\underline{\mathrm{Ch}}_I(X) \simeq X$. The paths of length 0 are identified with the objects a of X. The unique element id_e of $\mathrm{Fl}(I)$ gives id_a in X.

2. $I = \mathbf{\Delta}_1^+$, $\underline{\mathrm{Ch}}_I(X) \simeq \underline{\mathrm{Fl}}(X)$. The paths of type $\mathbf{\Delta}_1^+$ are those of length 1 and *direct,* i.e. given by an arrow

   $$a_0 \longrightarrow a_1$$

   whose source is also the source a_0 of the path. On the other hand, the paths of type $\mathbf{\Delta}_1^-$ are given by diagrams

   $$a_0 \longleftarrow a_1,$$

   i.e. by arrows whose target is regarded as the source of the path. Of course, the categories $\underline{\mathrm{Ch}}_{\mathbf{\Delta}_1^+}(X)$ and $\underline{\mathrm{Ch}}_{\mathbf{\Delta}_1^-}(X)$ are canonically isomorphic, but by an isomorphism which *exchanges source and target of the path.* We shall therefore take good care not to confuse the two preceding paths. [page 16]

3. $\mathbf{\Delta}_2^+$, $\mathbf{\Delta}_2^-$, corresponding to the diagrams

   $$a_0 \longrightarrow a_1 \longrightarrow a_2$$

   resp.

   $$a_0 \longleftarrow a_1 \longleftarrow a_2,$$

   we can make a similar remark as for $\mathbf{\Delta}_1^+$, $\mathbf{\Delta}_1^-$. p. 11

4. Φ_1, Ψ_1, these are respectively the intervals

   $$\Phi_1 :\ 0 \xleftarrow{\underline{u}_0} 1 \xrightarrow{\underline{u}_1} 2 \qquad \text{or} \qquad \begin{array}{ccc} & & 0 \\ & \nearrow & \\ 1 & & \\ & \searrow & \\ & & 2 \end{array},$$

[5] a_0 is called the *source* of the path, a_n its target.

and

$$\Psi_1 :\ 0 \xrightarrow{u_0} 1 \xleftarrow{u_1} 2 \qquad \text{or} \qquad \begin{array}{ccc} & & 0 \\ & \swarrow & \\ 1 & & \\ & \nwarrow & \\ & & 2 \end{array} .$$

They correspond respectively to the diagrams

$$a_0 \xleftarrow{u_0} a_1 \xrightarrow{u_1} a_2$$

and

$$a_0 \longrightarrow a_1 \longleftarrow a_2.$$

The categories Φ_1, Ψ_1 are opposite to one another, and are not isomorphic. So there is in general no isomorphism between $\underline{\mathrm{Ch}}_{\Phi_1}(X)$ and $\underline{\mathrm{Ch}}_{\Psi_1}(X)$.

Let I be an oriented interval, and let us consider the *opposite interval* I^{opp}, having the same underlying category but the opposite orientation. As the definition of the category of paths of type I does not involve the orientation of I, we thus have a canonical isomorphism

$$\underline{\mathrm{Ch}}_I(X) \xrightarrow{\sim} \underline{\mathrm{Ch}}_{I^{\mathrm{opp}}}(X).$$

Nevertheless we shall take care not to confuse a path of type I with the opposite path of type I^{opp}, the source of the one being the target of the other and conversely. This is to say that a path must be seen as *more* than a mere functor [page 17]

$$I \longrightarrow X,$$

we must in addition give ourselves the *orientation of* I as forming part of the path structure. So the reasonable formal definition, which fits the intuition of a path as going from a well-defined *source* toward an equally well-defined *target*, is that a path (defined in terms of an interval category I) is the datum of a pair (ω, c), where ω is an orientation of I, and c a functor $c : I \longrightarrow X$. When we look at the category of paths of a fixed "type", this *type* is (I, ω), and not I. It is when ω is fixed once and for all that this category can with impunity be identified with $\underline{\mathrm{Hom}}(I, X)$ without more.

We must view this category of paths as equipped with the essential structure

$$\begin{array}{c} \underline{\mathrm{Ch}}_{(I,\omega)}(X) \\ \big\downarrow{\scriptstyle (\sigma,\beta)} \\ X \times X \end{array} \qquad \begin{array}{l} c \longmapsto (\sigma(\varphi, v), \beta(\varphi, v)) \\ \text{where } \sigma(c) \text{ is the } \textit{source,} \\ \qquad \beta(c) \text{ the } \textit{target} \text{ of } c. \end{array}$$

This functor is identified with the composition functor p. 12 [page 18]

$$c \longmapsto c \circ \alpha$$

where α is the endpoint functor (an inclusion when the length is positive)

$$\begin{cases} \varepsilon \xrightarrow{\alpha} I \\ \varepsilon = \{0, 1\}, \quad \text{discrete category with two elements,} \\ \alpha \text{ given by} \quad \alpha(0) = \ \text{source of } I \\ \qquad\qquad\quad \alpha(1) = \ \text{target of } I. \end{cases}$$

If ω' is the opposite orientation of I, we have a commutative diagram

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{(I,\omega)}(X) & \xrightarrow[\sim]{\varphi} & \underline{\mathrm{Ch}}_{(I,\omega')}(X) \\ \downarrow{\scriptstyle (\sigma,\beta)} & & \downarrow{\scriptstyle (\sigma',\beta')} \\ X \times X & \xrightarrow{\ s\ } & X \times X, \end{array}$$

where s is the symmetry functor

$$s(x,y) = (y,x)$$

([6]). Thus the isomorphism φ does not respect the structural "source-target" arrows (σ, β).

There is however a difficulty when there exists an isomorphism of *oriented* intervals

$$\theta : (I,\omega) \xrightarrow{\sim} (I,\omega'),$$

i.e. when I is symmetric. This is not the case for the $\mathbf{\Delta}_i$ $(i \geq 1)$ (these categories have no automorphisms other than the identity), but it is the case for the categories Φ_1, Ψ_1 of example 4°, more generally for the reduced categories whose length is *even.* As every time we have an isomorphism θ of intervals that are *oriented,* we find an isomorphism between the categories of paths of type (I,ω) and $(I,\omega)^{\mathrm{opp}} = (I,\omega')$, *compatible* this time with the functors (σ,β): [page 19]

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{(I,\omega)}(X) & \xrightarrow[\sim]{\Psi_\theta} & \underline{\mathrm{Ch}}_{(I,\omega')}(X) \\ & {\scriptstyle (\sigma,\beta)} \searrow \quad \swarrow {\scriptstyle (\sigma',\beta')} & \\ & X \times X. & \end{array}$$

It is good to regard a *type* of paths as a noncategorical, but combinatorial, datum, a simple sequence τ of n signs τ_i $(1 \leq i \leq n)$ + or − (n = length of the paths under consideration). This combinatorial datum defines a standard oriented interval, I_τ. When we have an arbitrary oriented interval (I,ω), it defines a type τ, and we identify it canonically with the type I_τ. It is better to write $\underline{\mathrm{Ch}}_\tau(X)$, rather than $\underline{\mathrm{Ch}}_{(I,\omega)}(X)$, to make it quite clear that for given X, this is a totally "pinned down" category, without automorphisms tied to the particular nature of τ (to automorphisms of τ!). p. 13 [page 20]

We can make this point precise as follows. Let τ' be another type of intervals, and let us set out to determine "laws" which allow us to associate to any path of type τ in an arbitrary category X, a path of type τ' of this same category, and this in a "functorial" way, i.e. compatible with the horizontal maps

$$\begin{array}{ccc} \mathrm{Ch}_\tau(X) & \xrightarrow{\mathrm{Ch}_\tau(f)} & \mathrm{Ch}_\tau(X') \\ {\scriptstyle L_X}\downarrow & & \downarrow{\scriptstyle L_{X'}} \\ \mathrm{Ch}_{\tau'}(X) & \xrightarrow{\mathrm{Ch}_{\tau'}(f)} & \mathrm{Ch}_{\tau'}(X') \end{array}$$

associated with a functor $f : X \longrightarrow X'$.

For example, taking the type $\mathbf{\Delta}_2^+$ (i.e. $+1,+1$), corresponding to the paths [page 21]

$$a_0 \xrightarrow{u_1} a_1 \xrightarrow{u_2} a_2,$$

[6] The functor φ is the one which associates to every path c, of type τ, the *opposite path* c^{opp} or c^{-1}, of type τ^{opp}. If $\tau = \tau^{\mathrm{opp}}$, this is therefore a path *of the same type,* and φ is an *automorphism* of $\underline{\mathrm{Ch}}_\tau(X)$. This automorphism (functorial in X) is deduced from an automorphism θ of I, cf. below.

we can associate to such a path the path of type

$$a_0 \xrightarrow{u_2u_1} a_2,$$

or also the path of type $\mathbf{\Delta}_2^+$

$$a_0 \xrightarrow{u_2u_1} a_2 \xrightarrow{\mathrm{id}} a_2,$$

etc. Since the functor

$$\begin{cases} X & \longmapsto & \mathrm{Ch}_\tau(X) & (= \mathrm{Ob}\,\underline{\mathrm{Ch}}_\tau(X)) \\ \mathrm{Cat} & \longrightarrow & (\mathrm{Ens}) \end{cases}$$

is *represented* by the object I_τ of X , the sorites of representable functors tells us that such laws of formation of paths of type τ from paths of type τ' correspond bijectively to the arrows in Cat

$$I_{\tau'} \xrightarrow{\varphi} I_\tau\ ,$$

by simply taking the "law" corresponding to φ p. 14

$$c \longmapsto c \circ \varphi \qquad \text{for } c : I_\tau \longrightarrow X.$$

More precisely, the natural map

$$\underbrace{\mathrm{Hom}_{\mathrm{Cat}}(I_{\tau'}, I_\tau)}_{=\mathrm{Ob}\,\underline{\mathrm{Hom}}(I_{\tau'}, I_\tau)} \longrightarrow \mathrm{Hom}(\mathrm{Ch}_\tau, \mathrm{Ch}_{\tau'}),$$

where the second Hom is regarded as a Hom of objects in $\mathrm{Cat}^\vee = \underline{\mathrm{Hom}}(\mathrm{Cat}, \mathrm{Ens})$. Thus, in the two examples above, where $\tau = \mathbf{\Delta}_2^+$ and $\tau' = \mathbf{\Delta}_1^+$ or $\mathbf{\Delta}_2^+$, the functors in the reverse direction are [page 22]

$$\begin{array}{lll} \mathbf{\Delta}_2^+ \Big\{ & 0 \longrightarrow 1 \longrightarrow 2 \\ & \uparrow \qquad\qquad \uparrow \\ \mathbf{\Delta}_1^+ \Big\{ & 0 \longrightarrow\!\!\!\!\!\!\longrightarrow 1 \end{array}$$

and

$$\begin{array}{lll} \mathbf{\Delta}_2^+ \Big\{ & 0 \longrightarrow 1 \longrightarrow 2 \\ & \uparrow \qquad \nearrow \quad \uparrow \\ \mathbf{\Delta}_2^+ \Big\{ & 0 \longrightarrow 1 \longrightarrow 2\,. \end{array}$$

This is a general principle for reducing all the ± canonical "operations" on paths in arbitrary categories, and statements concerning them, to questions where X disappears and where we work in the interval categories themselves. One of the technical advantages is that these are very small categories, giving rise moreover to a sure topological intuition (which risks evaporating at times when working in categories of paths), and also that in such categories, hence also in the categories of functors between them, there is always at most *one single arrow* going from a given object to another. In other words, in such categories all diagrams are commutative.

Of course, we also see that if we have a law of formation L of paths of type τ' from paths of type τ (with the condition of functoriality with respect to the category X in which we take the paths), this law is necessarily *functorial with respect to paths:* if we have a homomorphism [page 23]

$$c \xrightarrow{v} c' \qquad \text{in } \underline{\mathrm{Ch}}_\tau(X),$$ p. 15

then the law L "extends" to such v to give

$$L(c) \xrightarrow{L(v)} L(c'),$$

with the usual relations

$$L(\mathrm{id}_c) = \mathrm{id}_{L(c)}, \qquad L(v'v) = L(v')L(v).$$

This follows from the description of L as an operation L_φ in terms of a well-determined functor

$$I_{\tau'} \xrightarrow{\varphi} I_\tau,$$

which in turn gives for every X a *functor*

$$\begin{cases} \underline{\mathrm{Ch}}_\tau(X) & \longrightarrow & \underline{\mathrm{Ch}}_{\tau'}(X) \\ c & \longmapsto & c \circ \varphi \end{cases}$$

more precise than simply a *map* $\mathrm{Ch}_\tau(X) \longrightarrow \mathrm{Ch}_{\tau'}(X)$ between the corresponding sets of objects.

We now have the choice between [page 24]

a) A more or less systematic study of functors between intervals. I prefer to wait for this until I need this and that, and to do only the strict minimum for the constructions I have in view.

b) A description of the composition of paths.

c) A formalism of categories of paths *of unspecified type.*

Since my aim is above all to describe such categories $\underline{\mathrm{Ch}}(X)$ (without a subscript τ!), with good properties, I will start with c), then look at b) as a crucial test for the usability of the notions obtained, and finally take up a) when I know what I will need of it.

p. 16

4 Category of paths $\underline{\mathrm{Ch}}(X)$

I want to "include" all the categories $\underline{\mathrm{Ch}}_\tau(X)$ of paths of all the types one wants, in one and the same category $\underline{\mathrm{Ch}}(X)$. Recall

$$\underline{\mathrm{Ch}}_\tau(X) = \underline{\mathrm{Hom}}(I_\tau, X),$$

where I_τ is the standard (oriented) interval of type τ.

The set of all types of intervals can be identified with the free monoid Ty on two generators Δ^+ and Δ^-, whose elements are arbitrary finite sequences (possibly empty) of objects alternately equal to Δ^+ and to Δ^-. If the first element is Δ^+, we have a type of path with direct initial direction, retrograde otherwise. The decomposition of I_τ, or of $\mathrm{Fl}\,(I_\tau)$, into critical intervals corresponds to the reduced expression of a word as [page 25]

$$(\Delta^+)^{l_1}(\Delta^-)^{l_2} \cdots (\Delta^?)^{l_r} \qquad (\Delta^? = \Delta^+ \text{ if } r \text{ odd}, \Delta^? = \Delta^- \text{ if } r \text{ even}),$$

or as

$$(\Delta^-)^{l_1}(\Delta^+)^{l_2} \cdots (\Delta^?)^{l_r} \qquad (\Delta^? = \Delta^- \text{ if } r \text{ odd}, \Delta^? = \Delta^+ \text{ if } r \text{ even}).$$

The l_i are the lengths of the successive critical intervals (or of the successive structural chains), their number is the reduced length of the interval.

We are going to regard Ty as the set of objects of a category $\underline{\mathrm{Ty}}$:

$$\begin{cases} \mathrm{Ob}\,\underline{\mathrm{Ty}} = \mathrm{Ty} = \text{ free monoid on the two generators } \Delta^+,\ \Delta^- \text{ above.} \\ \text{If } \tau, \tau' \in \mathrm{Ty},\ \mathrm{Hom}(\tau, \tau') = \mathrm{Hom}_{\mathrm{Cat}}(I_\tau, I_{\tau'}). \end{cases}$$

([7]). We will note that we have not only a set $\mathrm{Hom}_{\mathrm{Cat}}(I_\tau, I_{\tau'})$, but even a category $\underline{\mathrm{Hom}}(I_\tau, I_{\tau'})$, which here amounts to saying that $\mathrm{Hom}(I_\tau, I_{\tau'})$ (i.e. $\mathrm{Hom}(\tau, \tau')$) can be viewed as an *ordered* set. But for the moment we are going to ignore this order relation.

The composition of morphisms is the composition of functors between categories I_τ. [page 26]
Thus we have a *fully faithful* ([8]) canonical functor

$$\tau \longmapsto I_\tau : \underline{\mathrm{Ty}} \xrightarrow{\rho} (\mathrm{Cat}) \qquad ([9]),$$

by which the category I_τ could be identified with a full subcategory of (Cat) (that of p. 17
the I_τ). But I prefer not to identify $\tau \in \mathrm{Ty}$ (combinatorial datum) with I_τ (categorical datum).

Let X be a (small?) category. We have a contrafunctor

$$\begin{cases} S & \longmapsto & \underline{\mathrm{Hom}}(S, X) \\ (\mathrm{Cat})^o & \longrightarrow & (\mathrm{Cat}) \end{cases}$$

defined by X, whence by composing with ρ a contrafunctor

$$\underline{\mathrm{Ty}}^o \xrightarrow{\mathrm{CH}_X} \mathrm{Cat}.$$

This is the contrafunctor

$$\begin{cases} \tau \longmapsto \underline{\mathrm{Hom}}(I_\tau, X) \overset{\mathrm{def}}{=} \underline{\mathrm{Ch}}_\tau(X), \\ \text{i.e. } \mathrm{CH}_X(\tau) = \underline{\mathrm{Ch}}_\tau(X). \end{cases}$$

But a contrafunctor on a small category $\underline{\mathrm{Ty}}$ gives rise to a *fibered category* (and even split) over it.

I am going to denote this fibered category by $\underline{\mathrm{Ch}}(X)$. (Provisionally - if it turns out [page 27]
to be too big, I will restrict it …) Thus we have

$$\begin{array}{c} \underline{\mathrm{Ch}}(X) \\ \Big\downarrow{\scriptstyle \pi_X} \\ \underline{\mathrm{Ty}}\ . \end{array}$$

The fiber of $\underline{\mathrm{Ch}}(X)$ at $\tau \in \mathrm{Ob}\,\underline{\mathrm{Ty}}$ is the category $\underline{\mathrm{Ch}}_\tau(X)$ of paths of type τ in X. This therefore tells us what the arrows in $\underline{\mathrm{Ch}}(X)$ over the identity arrows in $\underline{\mathrm{Ty}}$ are. But what are the more general arrows? We give ourselves an arrow in $\underline{\mathrm{Ty}}$

$$\varphi : \tau' \longrightarrow \tau, \qquad \text{i.e. } I_{\tau'} \xrightarrow{\varphi} I_\tau,$$

[7] A somewhat botched definition, one must restrict oneself to homomorphisms from I_τ to $I_{\tau'}$ compatible with source and target - cf. p. 29.

[8] false, cf. marginal annotation on the preceding page - we do not take *all* the functors in Cat between I_τ and $I_{\tau'}$.

[9] *realization* functor of a (combinatorial) type of intervals.

what are the arrows of $\underline{\mathrm{Ch}}(X)$ over φ? By the general sorites of fibered categories, if

$$\begin{aligned} c \in \underline{\mathrm{Ch}}(X)_\tau &= \underline{\mathrm{Ch}}_\tau(X) \overset{\text{def}}{=} \underline{\mathrm{Hom}}(I_\tau, X) \\ c' \in \underline{\mathrm{Ch}}(X)_{\tau'} &= \underline{\mathrm{Ch}}_{\tau'}(X) \overset{\text{def}}{=} \underline{\mathrm{Hom}}(I_{\tau'}, X) \end{aligned}$$

we have

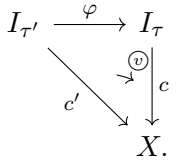
$$\boxed{\mathrm{Hom}_\varphi(c', c) \simeq \mathrm{Hom}(c', \underbrace{\varphi^*(c)}_{=c\circ\varphi}).}$$

Thus: the homomorphisms from the path c' of type τ' in X, to the path c of type τ, "over" the morphism φ from τ' to τ, i.e. the functor

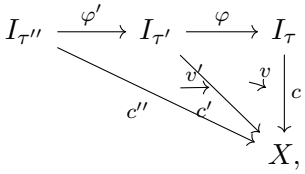
p. 18
[page 28]

$$\varphi : I_{\tau'} \longrightarrow I_\tau,$$

is a homomorphism v from c' to $c \circ \varphi$, i.e.



The composition of arrows is obvious:



we have

$$\begin{aligned} v : \quad c' &\longrightarrow c \circ \varphi, \quad \text{whence } c' \circ \varphi' \xrightarrow{v*\varphi'} \underbrace{(c \circ \varphi) \circ \varphi'}_{=c\circ(\varphi\varphi')} \\ v' : \quad c'' &\longrightarrow c' \circ \varphi' \end{aligned}$$

and

$$v'' = (v * \varphi') \circ v', \qquad \text{thus} \qquad c'' \xrightarrow{v'} c' \circ \varphi' \xrightarrow{v*\varphi'} c \circ (\varphi\varphi').$$

But do we have source and target functors

$$\underline{\mathrm{Ch}}(X) \underset{\beta}{\overset{\sigma}{\rightrightarrows}} X \qquad \begin{aligned} c &\longmapsto \mathrm{source}(c) \\ c &\longmapsto \mathrm{target}(c) \end{aligned} \qquad ?$$

It would therefore be necessary for a homomorphism v as above

$$v : c' \longrightarrow c \circ \varphi$$

to define

$$\begin{aligned} \sigma(c) : \quad \mathrm{source}(c') &\longrightarrow \mathrm{source}(c) \\ \beta(c) : \quad \mathrm{target}(c') &\longrightarrow \mathrm{target}(c). \end{aligned}$$

But since

$$\begin{cases} \sigma(c) = c(0_\tau), \ \beta(c) = c(1_\tau) \\ \text{where } 0_\tau, \ 1_\tau \text{ are the source and target objects of the standard oriented interval } I_\tau \ , \end{cases}$$

p. 19
[page 29]

and similarly for τ', v will indeed induce

$$\underbrace{\sigma(c')}_{=c'(0_{\tau'})} \longrightarrow c(\varphi(0_{\tau'})) \stackrel{?}{=} \underbrace{\sigma(c)}_{c(0_\tau)},$$

and likewise for $\beta(c')$ and $\beta(c)$, only if we impose

$$\boxed{\begin{cases} \varphi(0_{\tau'}) &= 0_\tau \\ \varphi(1_{\tau'}) &= 1_\tau, \end{cases}}$$

i.e. that the functor φ under consideration be *compatible with the source and the target.* This is what visibly had to be imposed in the definition of

$$\mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau', \tau) = \underline{\mathrm{Hom}}^{\textcircled{\scriptsize int}}(I_{\tau'}, I_\tau) \quad \text{-}$$

one must take the functors φ respecting the interval structures, i.e. making commutative the diagram

$$\begin{array}{ccc} I_{\tau'} & \longrightarrow & I_\tau \\ & {\scriptstyle i_{\tau'}} \nwarrow \quad \nearrow {\scriptstyle i_\tau} & \\ & \varepsilon = \{0,1\} & \end{array}$$

defining the orientation of $I_{\tau'}$ and of I_τ ([10]). This rectification made, I have good hope that the category $\underline{\mathrm{Ch}}(X)$ we have just constructed is "the right one". One could denote by $\underline{\mathrm{CH}}(X)$ the larger category, over the category $\underline{\mathrm{Typ}}$ having the same objects as $\underline{\mathrm{Ty}}$, but more morphisms: namely

$$\begin{array}{lcl} \mathrm{Hom}_{\underline{\mathrm{Typ}}}(\tau, \tau') & = & \mathrm{Hom}_{\mathrm{Cat}}(I_\tau, I_{\tau'}) \\ \mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau, \tau') & = & \mathrm{Hom}^{\mathrm{int}}_{\mathrm{Cat}}(I_\tau, I_{\tau'}) \\ & & \text{(homomorphisms respecting the interval structure).} \end{array}$$

[page 30]

We will have

$$\underline{\mathrm{Ch}}(X) \simeq \underline{\mathrm{CH}}(X) \times_{\underline{\mathrm{Typ}}} \underline{\mathrm{Ty}},$$

i.e. $\underline{\mathrm{Ch}}(X)$ is the *restriction* of the fibered category $\underline{\mathrm{CH}}(X)$ over $\underline{\mathrm{Typ}}$, to the (non-full) subcategory $\underline{\mathrm{Ty}}$ of $\underline{\mathrm{Typ}}$.

NB The fibered categories $\underline{\mathrm{CH}}(X)$, $\underline{\mathrm{Ch}}(X)$ over $\underline{\mathrm{Typ}}$, $\underline{\mathrm{Ty}}$ depend functorially on X, varying in (Cat), as it should be. As is only right, we are interested in them above all when X *is not* in (Cat), i.e. when X is a "big" category (such as (Cat) itself). I dispense with making explicit here the pertinent Universe questions. p. 20

Two essential tests, to ascertain whether the notion is good

a) The theory of the $\mathcal{M}\Sigma^{-1}$.

b) The composition of paths must work well.

We have the diagram of canonical functors

$$\begin{array}{ccc} \underline{\mathrm{Ty}} & \hookrightarrow & \varepsilon\backslash\mathrm{Cat} \\ \downarrow & & \downarrow \\ \underline{\mathrm{Typ}} & \hookrightarrow & \mathrm{Cat}, \end{array}$$

[page 31]

where the horizontal arrows are fully faithful.

[10] In fact, one must probably restrict the morphisms still further by requiring that they be *increasing* on objects - cf. page 31. p. 21

5 Properties of morphisms of intervals

I have more or less convinced myself that to lend itself to calculations and to simple arguments, one must restrict the morphisms still further in the definition of the category Ty. I will require that the morphisms $\tau \longrightarrow \tau'$ be functors

$$w : I_\tau \longrightarrow I_{\tau'}$$

which are compatible with the *oriented interval* structure of I_τ, $I_{\tau'}$, i.e. that they take source to source, target to target, but moreover that they be *increasing* on objects (for the order relation of progression of paths). This is a restriction which geometrically seems quite natural to me. (It is similar to the analogous restriction, in the pre-categorical definition of morphisms between $\mathbf{\Delta}_i$ …) It is this hypothesis that allows one to make a fairly simple structure theory of morphisms of intervals. But for the moment I am going to distinguish "morphisms" and "increasing morphisms".

[page 32]

Let

$$f : I_\tau \longrightarrow I_{\tau'}$$

be a morphism of oriented intervals. Consider the full subcategories, isomorphic to Δ_i^+ or Δ_i^- as oriented intervals, defined by the critical intervals of I_τ. They form a sequence Σ_τ of subcategories of I_τ

$$\Sigma_\tau = (\mathbf{\Delta}_{l_1}^{\varepsilon_1}(\tau), \mathbf{\Delta}_{l_2}^{\varepsilon_2}(\tau), \ldots, \mathbf{\Delta}_{l_r}^{\varepsilon_r}(\tau))$$

of which I_τ is the amalgamated sum (we will come back to this point further), r being the useful length of τ. We call them the *critical subintervals* of I_τ. Thus f decomposes into a sequence of morphisms

$$f_i : \mathbf{\Delta}_{l_i}^{\varepsilon_i} \longrightarrow I_{\tau'}, \qquad i \in [1, r],$$

such that

a) for $i \in [1, r-1]$, the target of f_i is equal to the source of f_{i+1}.

Moreover,

b) f_1 takes source to source, and f_r takes target to target.

Conversely, when we are given f_i satisfying these two conditions, they define a unique morphism of intervals f. (And if condition b) is not required, we find a description of the morphisms of categories $I_\tau \longrightarrow I_{\tau'}$.)

[page 33]

We are first going to look at an arbitrary functor

$$f : \mathbf{\Delta}_i^\varepsilon \longrightarrow I_{\tau'} \qquad (\varepsilon \in \{\pm 1\}) \quad .$$

p. 22

Suppose to fix ideas that $\varepsilon = +1$, i.e. that the first member is a direct critical interval … We find that f factors through a critical subinterval of $I_{\tau'}$, say $\mathbf{\Delta}_j^{\varepsilon'}$. It is therefore defined by an increasing map from $\mathbf{\Delta}_i^\varepsilon$ to $\mathbf{\Delta}_j^{\varepsilon'}$ (if ε, ε' of the same sign), a decreasing map in the opposite direction ([11]). But if we require that f be increasing, then it is necessary that *either* f be constant (in which case, if its value is an interior critical vertex of $I_{\tau'}$, there are exactly *two* critical intervals of $I_{\tau'}$ through which f factors) *or* f nonconstant

[11] **Lemma.** *If $I_{\tau'}$ is an interval of strictly positive length, every subset J of $I_{\tau'}$ such that for two elements of J, there is always an arrow from one to the other, is contained in a critical interval (unique if* $\operatorname{card} J \geq 2$*).*

and $\varepsilon = \varepsilon'$ (and in any case, if f nonconstant, then the critical interval through which f factors is unique). To sum up:

Proposition 7. *Let I_τ, $I_{\tau'}$ be two oriented intervals,*

$$f : I_\tau \longrightarrow I_{\tau'}$$

a functor. Then for every critical interval $J \subseteq I_\tau$ such that $f|J$ is nonconstant, there exists a unique critical interval $J' \subseteq I_{\tau'}$ such that $f|J$ factors through J', i.e. such that $f(J) \subseteq J'$. If f is increasing, then J and J' are of the same sign, i.e. both direct, or both retrograde. [page 34]

Thus, if we look at the sequence $s_0, s_1, \ldots, s_n$ of critical vertices of τ, then

$$t_0 = f(s_0),\ t_1 = f(s_1),\ \ldots\ ,\ t_n = f(s_n)$$

form a sequence of vertices of $I_{\tau'}$, such that two consecutive ones are joined by an arrow of $I_{\tau'}$. Thus the full subcategory of $I_{\tau'}$ defined by these t_i is connected, it is therefore a subinterval. If f is increasing, this subinterval has endpoints t_0, t_n and convex hull $[t_0, t_n]$. If f is a morphism of intervals, it contains all the critical vertices of $I_{\tau'}$. In all cases, taking t_i, t_j $(i < j)$ and applying the above to $f|J$, where J is the subinterval of I with origin s_i, endpoint s_j, we see that every critical vertex of $I_{\tau'}$ lying between t_i and t_j is among the t_k $(i \leq k \leq j)$. To sum up:

Corollary. *Let $s_i \leq s_j$ be two critical vertices of I_τ. Then every critical vertex of $I_{\tau'}$ lying between $f(s_i) = t_i$ and $f(s_j) = t_j$, is the image of a critical vertex s_k lying between s_i and s_j (i.e. $i \leq k \leq j$). In particular, if f is a morphism of intervals,*

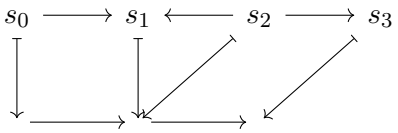
$$f(I_{\tau\,\mathrm{red}}) \supseteq I_{\tau'\,\mathrm{red}}$$

(where $I_{\tau\,\mathrm{red}}$ denotes the set of critical vertices of I_τ or also the reduced subinterval it defines).

One should be aware that even if f is increasing, one *does not have* in general $f(I_{\tau\,\mathrm{red}}) \subseteq I_{\tau'\,\mathrm{red}}$, i.e. f does not necessarily take critical vertices to critical vertices, Minimal example: [page 35] p. 23

Φ_3^+ :

$\boldsymbol{\Delta}_2^+$:



Φ_3^+ is reduced, the two critical vertices taken to the noncritical vertex 1.

Let us denote by $\mathrm{Cr}(\tau)$ the set of critical intervals of I_τ, in one-to-one correspondence (respecting the total order) with the set of transition arrows of $I_{\tau\,\mathrm{red}}$,

$$\mathrm{Cr}(\tau) \simeq \mathrm{Fl\,tr}\,(I_{\tau\,\mathrm{red}}).$$

Let, for a functor

$$f : I_\tau \longrightarrow I_{\tau'},$$

$\mathrm{Cr}(\tau, f)$ be the set of critical intervals of I_τ on which f is not constant. We then have a map

$$\mathrm{Cr}(\tau, f) \longrightarrow \mathrm{Cr}(\tau') \qquad (*)$$

induced by f via proposition 7, associating to each $J \in \mathrm{Cr}(\tau, f)$ the unique $J' \in \mathrm{Cr}(\tau')$ such that $f(J) \subseteq J'$. We have moreover

Corollary 2.

[page 36]

a) The image of $(*)$ *is a segment of the totally ordered set* $\mathrm{Cr}(\tau')$*, i.e. if two critical intervals* J'*,* J'' *in* $I_{\tau'}$ *are in the image, the same is true of all those lying between them.*

b) This image also contains the critical interval of $I_{\tau'}$ *(there is at most one) which contains* $f(0_\tau)$ *(resp.* $f(1_\tau)$*) and whose intersection with the segment* $[f(0_\tau), f(1_\tau)]$ *has cardinality* ≥ 2*, as well as all the critical intervals contained in the said segment.*

c) If f *is a morphism of oriented intervals, then the induced map*

$$\mathrm{Cr}(f) : \mathrm{Cr}(\tau, f) \longrightarrow \mathrm{Cr}(\tau')$$

is surjective. More precisely, let $J' \in \mathrm{Cr}(\tau')$*, and let* $J = f^{-1}(J')$*. Then* $\mathrm{Cr}(f)^{-1}(\{J'\})$ *consists of the critical intervals in* $\mathrm{Cr}(\tau, f)$ contained in J*. We have moreover,* if f is increasing,

$$\mathrm{long}(J') = \sum_{K \in \mathrm{Cr}(f)^{-1}(\{J'\})} \pi_{\mathrm{n}}(K, f),$$

where $\pi_{\mathrm{n}}(K, f)$ *is the "part" (numerical) of* $f|K$*, i.e. the length of the interval* $[f(0_K), f(1_K)]$*. (***NB** *If* K *is a critical interval of* I *such that* $K \notin Cr(\tau, f)$*, i.e.* $f|K$ *is constant, we will put* $\pi_{\mathrm{n}}(K, f) = 0$*. Thus the preceding formula also reads*

$$\mathrm{long}(J') = \sum_{K \in \mathrm{Cr}(I_\tau),\ K \subseteq J} \pi_{\mathrm{n}}(K, f).\ )$$

Corollary 3. *If* f *is an* increasing *morphism of oriented intervals, then for every* $J' \in \mathrm{Cr}(\tau')$*, we have* p. 24

$$(0 \leq)\ \mathrm{card}\, \mathrm{Cr}(f)^{-1}(\{J'\}) \leq \mathrm{long}(J'),$$

equality being obtained only if the parts of f *on the critical intervals of* I_τ *which lie over* J' *are all equal to* 0 *or* 1 *(hence if and only if on the* $J \in \mathrm{Cr}(f)^{-1}(\{J'\})$*, they are equal to* 1*).* [page 37]

Corollary 4. *Suppose moreover that* I' *is reduced. Then*

$$Cr(f) : Cr(\tau, f) \longrightarrow Cr(\tau')\ (\simeq \mathrm{Fl}\,\mathrm{tr}(\tau'))$$

is bijective.

Thus in this case, τ' reduced, we obtain a map in the opposite direction

$$\begin{array}{ccc} \mathrm{Tr}(\tau') \simeq \mathrm{Cr}(\tau') & \xrightarrow{\mathrm{Cr}^*(f)} & \mathrm{Cr}(\tau) \\ & {\scriptstyle \mathrm{Cr}(f)^{-1}} \searrow \quad \nearrow {\scriptstyle \text{inclusion}} & \\ & \mathrm{Cr}(\tau, f), & \end{array}$$

a map which transforms the critical interval J' $(= \{i-1, i\})$ into the *unique* critical interval J of I_τ such that

$$f(J) = J'.$$

But $J \simeq \mathbf{\Delta}^\varepsilon_i$, and we therefore have a *surjective* morphism [page 38]

$$f_0 : \mathbf{\Delta}^\varepsilon_i \longrightarrow J' \simeq \mathbf{\Delta}^\varepsilon_1.$$

We see right away that there then exists a *unique* transition arrow u of $\mathbf{\Delta}^\varepsilon_i$, such that $f(u) \neq \mathrm{id}$. Thus we find the following corollary, more precise than corollary 4:

Corollary 5. *Let $f : I_\tau \longrightarrow I_{\tau'}$ be an increasing morphism of oriented intervals, with $I_{\tau'}$ reduced, and let $\mathrm{Tr}(\tau, f)$ be the set of transition arrows $\underline{u}_i$ of I_τ such that $f(\underline{u}_i) \neq \mathrm{id}$, i.e. such that $\pi(u_i, f) = 1$. Then f induces a map*

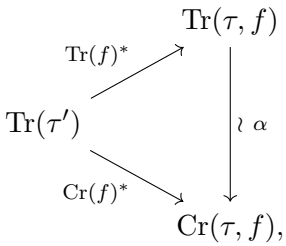
$$\mathrm{Tr}(\tau, f) \longrightarrow \mathrm{Tr}(\tau')$$

which is bijective, and (of course) increasing, hence an isomorphism of ordered sets.

Thus the map $\mathrm{Tr}(\tau') \longrightarrow \mathrm{Cr}(\tau, f)$ from earlier is made more precise by a bijective map

$$\mathrm{Tr}(f)^* : \mathrm{Tr}(\tau') \longrightarrow \mathrm{Tr}(\tau, f),$$

related to $\mathrm{Cr}(f)^*$ above by the commutative triangle



the map α associating to every $u_i \in \mathrm{Tr}(\tau, f)$ the unique critical interval which "contains" u_i (i.e. such that u_i is an arrow of the said interval, or again, is part of the associated "critical chain" …). p. 25 [page 39]

We have better still:

Proposition 8. *Let $\tau, \tau' \in \mathrm{Ty}$, τ' reduced. For every increasing morphism of oriented intervals*

$$f : I_\tau \longrightarrow I_{\tau'}$$

(we shall simply say that f is an arrow in $\underline{\mathrm{Ty}}$, and write $f \in \mathrm{Hom}_{\underline{\mathrm{Ty}}}(I_\tau, I_{\tau'})$), let us associate to it the subset $\mathrm{Tr}(\tau, f)$ of transition arrows of I_τ nondegenerate for f. These arrows form a totally ordered set equipped with signs $+$ or $-$ (direct or retrograde character of the arrows), and f induces a bijection of totally ordered sets

$$\underbrace{E_f}_{\subseteq \mathrm{Tr}(\tau)} = \mathrm{Tr}(\tau, f) \simeq \mathrm{Tr}(\tau'),$$

and even an isomorphism of totally ordered sets "equipped with signs". Thus, to every f we associate a subset E_f of $\mathrm{Tr}(\tau)$, and such that its "type" is equal to τ'. Let us denote by

$$\mathfrak{P}_{\tau'}(\mathrm{Tr}(\tau)) \subseteq \mathfrak{P}(\mathrm{Tr}(\tau))$$

the set of subsets E of $\mathrm{Tr}(\tau)$ whose type is τ'.

Then the map [page 40]

$$f \longmapsto \mathrm{Tr}(\tau, f) : \mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau, \tau') \longrightarrow \mathfrak{P}_{\tau'}(\mathrm{Tr}(\tau))$$

is bijective *: giving oneself a morphism $f : \tau \longrightarrow \tau'$ in $\underline{\mathrm{Ty}}$,* amounts to the same thing *as giving oneself a strictly increasing sequence of transition arrows*

$$\underline{u}_{i_1},\ \underline{u}_{i_2},\ \dots\ ,\underline{u}_{i_{n'}} \qquad (n' = \log \tau')$$

in I_τ ($i_1 < i_2 < \cdots < i_{n'}$), such that the corresponding sequence of signatures

$$(\varepsilon_{i_1}, \varepsilon_{i_2}, \dots, \varepsilon_{i_{n'}}) = \tau'$$

is equal to τ'.

Corollary 1. *For every τ, we have canonical bijections*

$$\begin{cases} \mathrm{Tr}^+(\tau) & \simeq & \mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau, \Delta^+) \\ \mathrm{Tr}^-(\tau) & \simeq & \mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau, \Delta^-), \end{cases}$$

functorial in τ *(varying in* $\underline{\mathrm{Ty}}$*).*

To give the corollary a meaning, we have to define p. 26

$$\tau \longmapsto \mathrm{Tr}^+(\tau), \qquad \tau \longmapsto \mathrm{Tr}^-(\tau)$$

as *functors* contravariant in τ, thus for a morphism

$$f : I_{\tau_2} \longrightarrow I_{\tau_1}$$

in $\underline{\mathrm{Ty}}$, define maps [page 41]

$$\begin{array}{rcl} \mathrm{Tr}^+(f)^* : \mathrm{Tr}^+(\tau_1) & \longrightarrow & \mathrm{Tr}^+(\tau_2) \\ \mathrm{Tr}^-(f)^* : \mathrm{Tr}^-(\tau_1) & \longrightarrow & \mathrm{Tr}^-(\tau_2), \end{array}$$

in other words, a map

$$\mathrm{Tr}(f)^* : \mathrm{Tr}(\tau_1) \longrightarrow \mathrm{Tr}(\tau_2)$$

"compatible with the signatures". In corollary 5 above, we defined such a map only for τ' reduced ([12]). For an arrow u of I_τ, let us call *set-theoretic part* of f on u, or of u for f, the set-theoretic part of the arrow $f(u)$ in $I_{\tau'}$, i.e. the segment $[f(i-1), f(i)] \subseteq I_{\tau'}$ (here $u = \underline{u}_i$). Let

$$\underline{u}_{i_1}, \ \ldots \ , \underline{u}_{i_N} \qquad \text{or } (v_1, v_2, \ldots, v_N)$$

be the increasing sequence of elements of $\mathrm{Tr}(\tau, f)$, we then see by induction on i that for $i \in [1, N-1]$, we have

$$\begin{array}{rcl} f(\mathrm{but}(v_i)) & = & f(\mathrm{source}(v_{i+1})) \ \ (= t_i), \\ f(\mathrm{source}(v_1)) & = & 0_{\tau'}, \\ f(\mathrm{but}(v_N)) & = & 1_{\tau'}. \end{array}$$

Let us consider the sequence

$$\{\overbrace{0_{\tau'}}^{=t_0}, t_1, t_2, \ldots, t_{N-1}, \overbrace{1_{\tau'}}^{=t_N}\},$$

it is defined by the corresponding subset of $I_{\tau'}$, which we shall call the set of *critical vertices of $I_{\tau'}$ for f*. It is a sequence containing $0_{\tau'}$, $1_{\tau'}$, and such that two consecutive elements are connected in $I_{\tau'}$ by an arrow, which is equivalent to saying that it contains the critical vertices (intrinsic) of $I_{\tau'}$. On the other hand, for every transition interval $[i-1, i]$ of $I_{\tau'}$ ($i \in [1, n']$), there exists one and only one transition interval $[t_{j-1}, t_j]$ for p. 27 this sequence ($j \in [1, N]$), such that we have $[i-1, i] \subseteq [t_{j-1}, t_j]$. This says that there [page 42] exists one and only one $v_j \in \mathrm{Tr}(\tau, f)$ such that the part of f on v_j (namely precisely $[t_{j-1}, t_j]$) contains (that of) $\underline{u}_i$. To sum up:

Corollary 2. *Let $f : \tau \longrightarrow \tau'$ be in $\underline{\mathrm{Ty}}$. Then for every $u \in \mathrm{Tr}(\tau') = \mathrm{Tr}(I_{\tau'})$, there exists one and only one $v \in \mathrm{Tr}(\tau)$ such that the part of f on v contains u. Moreover, the signatures of u and v are equal.*

[12] What follows does not involve the signatures - it is a sorites on increasing maps between finite totally ordered sets.

It is in this way that we therefore obtain a map

$$\mathrm{Tr}(f)^* : \mathrm{Tr}(\tau') \quad \longrightarrow \quad \mathrm{Tr}(\tau) \qquad \text{(factoring through } \mathrm{Tr}(\tau, f))$$

inducing

$$\begin{array}{rcl} \mathrm{Tr}^+(f)^* : \mathrm{Tr}^+(\tau') & \longrightarrow & \mathrm{Tr}^+(\tau) \\ \mathrm{Tr}^-(f)^* : \mathrm{Tr}^-(\tau') & \longrightarrow & \mathrm{Tr}^-(\tau). \end{array}$$

Transitivity for a composite

$$\tau \xrightarrow{f} \tau' \xrightarrow{g} \tau''$$

or

$$I_\tau \xrightarrow{f} I_{\tau'} \xrightarrow{g} I_{\tau''}$$

is immediate. If u, u', u'' are transition arrows in I_τ, $I_{\tau'}$, $I_{\tau''}$ respectively, such that

$$\begin{array}{rcll} \pi(g, u') & \supseteq & \pi(u''), & \text{i.e. } u' = \mathrm{Tr}(g)^*(u'') \\ \pi(f, u) & \supseteq & \pi(u'), & \text{i.e. } u = \mathrm{Tr}(f)^*(u'), \end{array}$$

then we have

$$\pi(gf, u) \supseteq \pi(u''), \qquad \text{i.e. } u = \mathrm{Tr}(gf)^*(u'').$$

If the arrows in question have as intrinsic part $[i-1, i]$, $[i'-1, i']$, $[i''-1, i'']$ respectively, [page 43] the hypothesis is

$$\begin{array}{rcl} g(i'-1) \leq i''-1, & & g(i') \geq i'' \\ f(i-1) \leq i'-1, & & f(i) \geq i' \end{array}$$

and since g is increasing, we have deduced

$$\begin{array}{rcll} gf(i-1) & \leq & g(i'-1) & (\leq i''-1) \\ gf(i) & \geq & g(i') & (\geq i''), \end{array}$$

q.e.d.

Thus, corollary 1 above acquires a precise meaning and it is proved at the same time, [p. 28] since the correspondences of corollary 1 are given by

$$\underbrace{f}_{\in \mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau, \Delta^\varepsilon)} \qquad \longmapsto \mathrm{Tr}(f)^*(\underline{u}),$$

where $\underline{u}$ is the unique transition arrow of Δ^ε.

We are going to use these sorites to give, for every $\tau \in \underline{\mathrm{Ty}}$, a simple combinatorial description of the categories $\underline{\mathrm{Ty}}/\tau$ and $\tau\backslash\underline{\mathrm{Ty}}$, or at least of their sets of objects.

1° Description of $\mathrm{Ob}\,(\tau\backslash\underline{\mathrm{Ty}})$**.**

To $f \in \mathrm{Ob}(\tau\backslash\underline{\mathrm{Ty}})$, i.e.

$$f : \tau \longrightarrow \tau' \qquad \text{i.e. } f : I_\tau \longrightarrow I_{\tau'}$$

in $\underline{\mathrm{Ty}}$, is associated $E = E_f = \mathrm{Tr}(\tau, f) \subseteq \mathrm{Tr}(\tau)$, the set of critical transition arrows (i.e. nondegenerate) for f. To each $u \in E$, let us associate the integer $\mathrm{long}(\pi(f, u))$, the *part* [page 44] of f on u. Then (τ', f) is uniquely determined by the pair

$$(E, \pi), \qquad E \in \mathfrak{P}(\mathrm{Tr}(\tau)), \qquad \underbrace{\pi}_{\text{weighting}} : E \longrightarrow \underbrace{\mathbf{N}^*}_{\substack{=\text{set of} \\ \text{integers} \geq 1}} .$$

Or if one prefers, by the map $\pi = \pi_f$

$$\mathrm{Tr}(\tau) \xrightarrow{\pi} \mathbf{N}, \qquad u \longmapsto \pi(u) = \pi(u, f) = \mathrm{long}(\pi_{\tau'}(f(u)))$$

(which gives back E as

$$E = \pi^{-1}(\mathbf{N}^*) \qquad).$$

This map is subject to no restriction,

$$\mathrm{Ob}\,(\tau\backslash\underline{\mathrm{Ty}}) \xrightarrow[\sim]{\text{bij.}} \mathrm{Hom}_{\mathrm{Ens}}(\mathrm{Tr}(\tau), \mathbf{N}).$$

We recover τ' from π as follows. We take $E = \pi^{-1}(\mathbf{N}^*)$ (transition arrows nondegenerate for π), E is totally ordered by the order induced by that of $\mathrm{Tr}(\tau)$, it is moreover signatured by the signaturation of $\mathrm{Tr}(\tau)$, let

$$(\varepsilon_1, n_1),\ \ldots\ ,\ (\varepsilon_r, n_r)$$

be the sequence of signatures and weights, of the sequence of elements of E ($\varepsilon_i \in \{\pm 1\}$, $n_i \in \mathbf{N}^*$ for $i \in [1, r]$). Then,

$$\tau' = \Delta_{\varepsilon_1}^{n_1} \Delta_{\varepsilon_2}^{n_2} \cdots \Delta_{\varepsilon_r}^{n_r}, \qquad r = \mathrm{card}\, E,$$

in the free monoid Ty, where we have denoted by Δ_+, Δ_- (instead of Δ^+, Δ^-) the two generators. The map

$$I_\tau \longrightarrow I_{\tau'}$$

is described uniquely by saying that it is *constant* on each of the connected components of the graph $E' = E'(f)$ whose vertices are all the vertices of $I_\tau = [0, n]$ and whose edges are the transition arrows degenerate for f, i.e. in $\mathrm{Tr}(\tau)\backslash E$. And on the α-th element of E ($\alpha \in [1, r = \mathrm{card}\, E]$), of index i_α in $\mathrm{Tr}(\tau)$, f induces the functor

p. 29 [page 45]

$$\mathbf{\Delta}_{\varepsilon_\alpha} \simeq \underbrace{J_{i_\alpha}}_{\substack{\text{transition}\\ \text{interval}\\ \text{of index } i_\alpha\\ \text{in } I_\tau}} \longrightarrow I_{\tau'}$$

given by *the unique* $\underline{\mathrm{Ty}}$*-arrow* in $I_{\tau'}$ from J_{i_α} to the α-th factor, of type $\Delta_{\varepsilon_\alpha}^{n_\alpha}$, of τ', given by $(*)$

$$\Delta_{\varepsilon_\alpha}^{n_\alpha} \simeq \text{type of } \mathbf{\Delta}_{n_\alpha}^{\varepsilon_\alpha} \quad \ldots$$

2° Determination of $\mathrm{Ob}\,(\underline{\mathrm{Ty}}/\tau')$.

If $(\tau, f) \in \mathrm{Ob}\,(\underline{\mathrm{Ty}}/\tau')$, thus

$$f : I_\tau \longrightarrow I_{\tau'},$$

we first of all associate with it the subset

$$\mathrm{Cr}(\tau', f) = E'_f = E' \subseteq I_{\tau'}$$

consisting of the vertices (called "critical for f", cf. page 41) of $I_{\tau'}$ which are source or target of an arrow $f(u)$, where $u \in \mathrm{Tr}(\tau, f)$, i.e. $f(u)$ is not degenerate ([13]). When $I_{\tau'}$ is a point, we set $E' = \{0\}$, so that in every case $E' = f(I_\tau)$. E' contains $f(I_{\tau\,\mathrm{red}})$ (i.e. the images under f of the critical vertices of I_τ), but can be larger. (E.g. if $\tau = \tau'$, $f = \mathrm{id}$, $E' = I_{\tau'}$, which is thus strictly larger than $I_{\tau'\,\mathrm{red}}$ if τ' is not reduced.) For every

[page 46]

[13] isolate a lemma f increasing unnecessary.

$i \in E'$, let us consider $f^{-1}(\{i\}) \subseteq I_\tau$. Since f increasing, it is an interval of the totally ordered set I_τ, so it is a subinterval of the interval category I_τ. As such, it has a signature $\tau_i \in \mathrm{Ty}$. We thus have a map

$$\eta_f \text{ or } \eta : E' \longrightarrow \mathrm{Ty}.$$

I say that $(\tau, f) \in \mathrm{Ob}\,(\underline{\mathrm{Ty}}/\tau')$ is uniquely determined by this pair

$$(E', \eta) \begin{cases} E' \in \mathfrak{P}(I_{\tau'}) = \mathfrak{P}([0, n']) \\ \eta : E' \longrightarrow \mathrm{Ty}\ . \end{cases}$$

Moreover, the latter is subject to the sole restriction that

$$E' \supseteq \mathrm{Cr}(\tau') = \text{set of critical vertices of } I_{\tau'}$$

(or what amounts to the same thing, that E' contains $0_{\tau'}$, $1_{\tau'}$ and that two consecutive elements of E' are joined by an arrow of $I_{\tau'}$): p. 30

$$\begin{cases} \mathrm{Ob}\,(\mathrm{Ty}/\tau') \simeq \text{set of pairs } (E', \eta), \text{ with} \\ \qquad\qquad E' \subseteq [0, \underbrace{n'}_{=\log \tau'}], \quad E' \supseteq \mathrm{Cr}(\tau'), \\ \qquad\qquad \eta : E' \longrightarrow \mathrm{Ty}\ . \end{cases}$$

Let

$$0 = t_0 < t_1 < \cdots < t_N = n'$$

be the sequence of elements of E', let for $\boxed{i \in [1, N]}$

$$\left\| \begin{array}{l} l_i = t_i - t_{i-1} \\ \varepsilon_i \quad \text{signature of } [t_{i-1}, t_i] \text{ in } I_{\tau'}\ , \end{array} \right.$$

and for $\boxed{i \in [0, N]}$, let

$$\| \quad \tau_i = \eta(t_i). \qquad \texttt{[page 47]}$$

Then the type τ is given by

$$\tau = \tau_0 \Delta_{\varepsilon_1} \tau_1 \Delta_{\varepsilon_2} \cdots \Delta_{\varepsilon_N} \tau_N, \qquad (*)$$

length $N + \sum_{i=0}^{N} \underbrace{\mathrm{long}(\tau_i)}_{\lambda_i \geq 0}$.

3° Description of $\mathrm{Fl}(\underline{\mathrm{Ty}})$.

If

$$f : \tau \longrightarrow \tau' \qquad \text{in } \underline{\mathrm{Ty}}\ ,$$

we associate with it as before an integer $\underline{N \geq 0}$ ($N = \mathrm{card}\,\mathrm{Cr}(\tau', f) - 1$, $N = 0$ if and only if $\tau' = 1$, type of the one-point interval), maps

$$\begin{cases} [1, N] \longrightarrow \{+1, -1\} \times \mathbf{N}^*, & i \longmapsto (\varepsilon_i, l_i) \\ [0, N] \longrightarrow \mathrm{Ty} & i \longmapsto \tau_i \end{cases}$$

([14]). The arrow f in $\underline{\mathrm{Ty}}$ is determined by these data. The type τ is given by the formula $(*)$ above, the type τ' by

$$\tau' = \Delta_{\varepsilon_1}^{l_1} \Delta_{\varepsilon_2}^{l_2} \cdots \Delta_{\varepsilon_N}^{l_N}, \qquad \text{length } n' = \textstyle\sum l_i, \qquad (*') \quad \text{p. 31}$$

and the increasing map from $I_\tau = [0, N + \sum \mathrm{long}(\tau_i)]$ to $I_{\tau'} = [0, \sum l_i]$ is given thus:

$$\begin{array}{lcl}
\text{constant on } [0, 0+\lambda_0] & \longmapsto & 0 \\
\text{constant on } [\lambda_0+1, \lambda_0+1+\lambda_1] & \longmapsto & l_1 \\
\text{constant on } [\lambda_0+\lambda_1+2, \lambda_0+\lambda_1+2+\lambda_2] & \longmapsto & l_1+l_2 \\
\dots & & \\
\text{constant on } [\lambda_0+\lambda_1+\dots+\lambda_i+(i+1), \lambda_0+\dots+\lambda_i+\lambda_{i+1}+(i+1)] & \longmapsto & \sum_{h=1}^{i+1} l_h.
\end{array}$$

p. 32
[page 48]

6 Composition in $\underline{\mathrm{Ty}}$

We have a composition law (everywhere defined) in $\mathrm{Ob}\,(\underline{\mathrm{Ty}}) = \mathrm{Ty}$, it is that of the free monoid on two generators Δ_+, Δ_-. This composition law is interpreted, at the categorical level (and no longer merely combinatorial) as an amalgamated sum

$$I_{\tau\tau'} = I_\tau \amalg_{(e, \underline{1}_\tau, \underline{0}_{\tau'})} I_{\tau'},$$

where e is the unit category, and where $\underline{1}_\tau$, $\underline{0}_{\tau'}$ now represent the morphisms $e \longrightarrow I_\tau$, $e \longrightarrow I_{\tau'}$, having as image the target element and the source element respectively of the oriented interval in question. We thus have a cocartesian square in (Cat)

$$\begin{array}{ccc}
 & e & \\
 {\scriptstyle \underline{1}_\tau}\swarrow & & \searrow{\scriptstyle \underline{0}_{\tau'}} \\
I_\tau & \text{cocart.} & I_{\tau'} \\
 {\scriptstyle i_{\tau,\tau'}}\searrow & & \swarrow{\scriptstyle j_{\tau,\tau'}} \\
 & I_{\tau\tau'}, &
\end{array}$$

where all the arrows are monic and fully faithful functors. That this diagram (obviously commutative) is indeed cocartesian in (Cat) amounts, by definition, to saying that for every X in Cat, the transposed diagram

$$\begin{array}{ccccc}
 & & X & & \\
 & {\scriptstyle \text{(target)}}\ {\scriptstyle \beta_\tau}\nearrow & & \nwarrow{\scriptstyle \sigma_{\tau'}}\ {\scriptstyle \text{(source)}} & \\
\underline{\mathrm{Ch}}_\tau(X) & & & & \underline{\mathrm{Ch}}_{\tau'}(X) \\
 & \nwarrow & & \nearrow & \\
 & & \underline{\mathrm{Ch}}_{\tau\tau'}(X), & &
\end{array}$$

which amounts to the following tautology that a path C in X of type $\tau\tau'$ "is nothing other" than a pair (c, c') of a path c of type τ and a path c' of type τ' "put end to end" (in this order), i.e. such that [page 49]

$$\mathrm{but}(c) = \mathrm{source}(c')$$

(and we then have

$$\begin{array}{rcl}
\mathrm{source}(C) & = & \mathrm{source}(c) \\
\mathrm{but}(C) & = & \mathrm{but}(c') \qquad\qquad).
\end{array}$$

Here arises the everlasting problem whether we are going to denote this path by $c{\cdot}c'$, p. 33
which at first sight seems to impose itself, or rather $c' \circ c$, regarding the path c from

[14] **NB** If $N = 0$, $[1, N] = \emptyset$, and there are no ε_i, l_i; the map $\underbrace{[0, N]}_{=\{0\}} \longrightarrow \mathrm{Ty}$ reduces to the datum of $\tau_0 \in \mathrm{Ty}$.

$x = \text{source}(c)$ to $y = \text{but}(c)$ as a sort of morphism from x to y (cf. below the formalism of categories of fractions), c' as a morphism from $y = \text{source}(c')$ to $z = \text{but}(c')$, and interpreting the composite C as the composite of these two morphisms c, c' (in this order from the point of view of the operations to be carried out), which is traditionally denoted $c' \circ c$ or $c' \cdot c$, and not $c \circ c'$ or $c \cdot c'$, alas! It is this notation that we are going to follow below.

But for the moment we are not going to concern ourselves with paths in arbitrary categories X, but return to the formalism of the τ and the I_τ themselves. The categorical interpretation of $\tau\tau'$ as an amalgamated sum assures us that this composition operation is functorial:

[page 50]

$$\tau_1 \xrightarrow{f} \tau_2, \quad \tau_1' \xrightarrow{f'} \tau_2' \qquad \text{gives } \tau_1\tau_1' \xrightarrow{f*f'} \tau_2\tau_2'$$

(we shall not write $f \cdot f'$ for the arrow denoted $f * f'$, because of the possible confusion with the composition of arrows in $\underline{\text{Ty}}$). We have transitivity in an obvious way: if moreover $\tau_2 \xrightarrow{g} \tau_3$, $\tau_2' \xrightarrow{g'} \tau_3'$,

$$(gf) * (g'f') = (g * g') \circ (f * f').$$

Thus we have a composition bifunctor

$$\underline{\text{Ty}} \times \underline{\text{Ty}} \longrightarrow \underline{\text{Ty}} \qquad (c, c') \longmapsto c \cdot c' \text{ or } c * c'.$$

(In fact, restricting here the morphisms of $\underline{\text{Typ}}$ to functors preserving the two endpoints, we also have a bifunctor

$$\underline{\text{Typ}} \times \underline{\text{Typ}} \longrightarrow \underline{\text{Typ}},$$

but we prefer now to forget the category $\underline{\text{Typ}}$, which gives a distinctly less simple formalism.)

This composition is *strictly* associative and *strictly unital* (the unit being of course the unit element 1 of the free monoid Ty, corresponding to the one-point category $\{0\}$), and not just "up to isomorphism". And with good reason, since in $\underline{\text{Ty}}$ the isomorphisms are identities. *Thus* $\underline{\text{Ty}}$ *is a monoid in the category* (Cat) of small categories!

This implies that not only $\text{Ob}\,\underline{\text{Ty}} = \text{Ty}$, but also $\text{Fl}(\underline{\text{Ty}})$, and more generally all the sets of the form $\text{Hom}_{\text{Cat}}(X, \underline{\text{Ty}})$ are monoids, and this functorially in X, for example the components $\text{Hom}_{\text{Cat}}(\mathbf{\Delta}_i, \underline{\text{Ty}})$ of the semi-simplicial set "nerve" of $\underline{\text{Ty}}$,

[page 49′ (the numbering is shifted)]

$$\text{Nerf}_i(\underline{\text{Ty}}) = \text{Hom}_{\text{Cat}}(\mathbf{\Delta}_i, \underline{\text{Ty}}),$$

are monoids, and the semi-simplicial maps between them are monoid homomorphisms: the nerve of $\underline{\text{Ty}}$ is a *semi-simplicial monoid.*

We are going to describe these monoids $\text{Nerf}_i(\underline{\text{Ty}})$, and observe that they are all monoids that are *free,* just like $\text{Nerf}_0(\underline{\text{Ty}}) = \text{Ob}\,\underline{\text{Ty}} = \text{Ty}$. The semi-simplicial maps between these monoids are therefore known when we know them on the *generators* of these free monoids (which generators are also the indecomposable elements, hence defined intrinsically in terms of the multiplicative structure).

p. 34

For $\text{Nerf}_0(\underline{\text{Ty}}) = \text{Ty}$, this is something we have seen, the generators are

$$\Delta_+, \Delta_-,$$

corresponding respectively to the oriented intervals

$$\mathbf{\Delta}_+ : \quad 0 \quad \longrightarrow \quad 1$$
$$\mathbf{\Delta}_- : \quad 0 \quad \longleftarrow \quad 1.$$

[page 50′]

For $\mathrm{Nerf}_1(\underline{\mathrm{Ty}}) = \mathrm{Fl}(\underline{\mathrm{Ty}})$, i.e. for the arrows, result 3°) of the preceding paragraph (determination of $\mathrm{Fl}(\underline{\mathrm{Ty}})$) is also interpreted by saying that $\mathrm{Fl}(\underline{\mathrm{Ty}})$ is the free monoid generated by an infinite system of generators

$$\mathfrak{f}_{\varepsilon,n}\ , \qquad (\varepsilon, n) \in \{+1, -1\} \times \mathbf{N},$$

indexed by the product set $\{+1, -1\} \times \mathbf{N}$. Here $\mathfrak{f}_{\varepsilon,n}$ denotes the unique arrow in $\underline{\mathrm{Ty}}$

$$\mathfrak{f}_{\varepsilon,l} : \Delta_\varepsilon \longrightarrow \Delta_\varepsilon^l$$

(where of course the exponent l denotes an exponentiation for the composition law in $\underline{\mathrm{Ty}}$

$$\Delta_\varepsilon^l = \underbrace{\Delta_\varepsilon \circ \Delta_\varepsilon \circ \ \cdots \ \circ \Delta_\varepsilon}_{l \text{ copies}},$$

and not the Cartesian power in Cat, or in $\underline{\mathrm{Ty}}$ (where it exists only if $l \in \{0, 1\}$)). In the description of $\mathrm{Fl}(\underline{\mathrm{Ty}})$ in the preceding paragraph, we wrote the "general term" of the free monoid by grouping together the groups of successive functors of the form $f_{+,0}$, $f_{-,0}$, i.e. corresponding to the two indecomposable degeneracy arrows (for $*$)

$$\begin{cases} f_{+,0} : & \Delta_+ & \longrightarrow & e \\ f_{-,0} : & \Delta_- & \longrightarrow & e \end{cases}$$

([15]). The free submonoid of $\mathrm{Fl}(\underline{\mathrm{Ty}})$ generated by these two elements is the monoid of arbitrary (pure) degeneracy arrows in $\underline{\mathrm{Ty}}$:

p. 35
[page 51]

$$f_\tau : I_\tau \longrightarrow e \qquad (\tau \in \mathrm{Ty}),$$

and we have

$$f_\tau * f_{\tau'} = f_{\tau\tau'},$$

i.e. the map

$$\tau \longmapsto f_\tau : \quad \underbrace{\mathrm{Ob}\,(\underline{\mathrm{Ty}})}_{= \mathrm{Ty} = \mathrm{Nerf}_0(\underline{\mathrm{Ty}})} \quad \longrightarrow \quad \underbrace{\mathrm{Fl}(\underline{\mathrm{Ty}})}_{\mathrm{Nerf}_1(\underline{\mathrm{Ty}})}$$

is a monoid homomorphism, taking Δ_ε ($\varepsilon \in \{+1, -1\}$) to $\mathfrak{f}_{\varepsilon,0}$. This is not the degeneracy map from Nerf_0 to Nerf_1, i.e. the map $\tau \longmapsto \mathrm{id}_\tau$. The latter is given by

$$\begin{array}{ccc} \mathrm{Ob}\,(\underline{\mathrm{Ty}}) & \longrightarrow & \mathrm{Fl}(\underline{\mathrm{Ty}}) \\ \tau & \longmapsto & \mathrm{id}_\tau \end{array} \quad : \quad \left| \begin{array}{lcll} & \Delta_\varepsilon & \longmapsto & \mathfrak{f}_{\varepsilon,1} \qquad (\varepsilon \in \{+1, -1\}) \\ \text{i.e.} & \Delta_+ & \longmapsto & \mathfrak{f}_{+,1} \\ & \Delta_- & \longmapsto & \mathfrak{f}_{-,1}\ . \end{array} \right.$$

As for the source and target homomorphisms

$$\mathrm{Fl}(\underline{\mathrm{Ty}}) \underset{\beta \text{ (target)}}{\overset{\sigma \text{ (source)}}{\rightrightarrows}} \mathrm{Ob}\,(\underline{\mathrm{Ty}}),$$

they are obvious on the generators

$$\begin{cases} \sigma(\mathfrak{f}_{\varepsilon,l}) &= \Delta_\varepsilon \\ \beta(\mathfrak{f}_{\varepsilon,l}) &= \Delta_\varepsilon^l. \end{cases}$$

I am now going to describe $\mathrm{Nerf}_2(\underline{\mathrm{Ty}})$, i.e. the systems

$$\tau \xrightarrow{f} \tau' \xrightarrow{g} \tau''.$$

[page 52]

Giving $f \in \mathrm{Fl}(\underline{\mathrm{Ty}})$ amounts to giving an ordered sequence of generators $f_{\varepsilon,l}$, hence a sequence

$$\mathfrak{f}_{\varepsilon_1,l_1},\ \mathfrak{f}_{\varepsilon_2,l_2},\ \ \cdots\ \ ,\mathfrak{f}_{\varepsilon_n,l_n}$$
$$((\varepsilon_i,l_i))_{1\leq i\leq n} \in [\{\pm 1\}\times \mathbf{N}]^n.$$

We then have

$$x = \Delta_{\varepsilon_1}\Delta_{\varepsilon_2}\cdots\Delta_{\varepsilon_n} \qquad \log x = n$$
$$y = \Delta_{\varepsilon_1}^{l_1}\Delta_{\varepsilon_2}^{l_2}\cdots\Delta_{\varepsilon_n}^{l_n} \qquad \log y = \textstyle\sum_{i\in[1,n]} l_i.$$

Giving $g : y \longrightarrow z$, i.e. an object of $y\backslash\underline{\mathrm{Ty}}$, amounts to giving a map from the set of the $m = \sum l_i$ transition arrows of y to $\mathbf{N}$. In fact, we have p. 36

$$\mathrm{Tr}(y) = \mathrm{Tr}(\Delta_{\varepsilon_1}^{l_1}) \amalg \mathrm{Tr}(\Delta_{\varepsilon_2}^{l_2}) \amalg\ \cdots\ \amalg \mathrm{Tr}(\Delta_{\varepsilon_n}^{l_n})$$

("sum set", i.e. disjoint union), hence

$$\mathrm{Hom}_{\mathrm{Ens}}(\mathrm{Tr}(y),\mathbf{N}) \simeq \prod_{i\in[1,n]} \mathrm{Hom}_{\mathrm{Ens}}(\underbrace{\mathrm{Tr}(\Delta_{\varepsilon_i}^{l_i})}_{[1,l_i]},\mathbf{N}),$$

hence giving g is equivalent to giving a system of n maps

$$\begin{cases} \pi_i : [1,l_i] \longrightarrow \mathbf{N} & \text{(weighting of } \Delta_{\varepsilon_i}^{l_i}) \\ \text{for } i \in [1,n]. \end{cases}$$

But for $i \in [1,n]$, let us consider the element of Nerf_2

$$\Delta_{\varepsilon_i} \xrightarrow{\mathfrak{f}_{\varepsilon_i,l_i}} \Delta_{\varepsilon_i}^{l_i} \xrightarrow{g_{\varepsilon_i,\pi_i}} \underbrace{\Delta_{\varepsilon_i}^{\pi_i(1)}\Delta_{\varepsilon_i}^{\pi_i(2)}\cdots\Delta_{\varepsilon_i}^{\pi_i(l_i)}}_{\Delta_{\varepsilon_i}^{\sum_j \pi_i(j)\ \overset{\mathrm{def}}{=}\ \log(\pi_i)}}, \tag{1}$$

where g_{ε_i,π_i} is the arrow defined in terms of π_i in the preceding paragraph. But this is also nothing other than [page 53]

$$g_{\varepsilon_i,\pi_i} = \mathfrak{f}_{\varepsilon_i,\pi_i(1)} * \mathfrak{f}_{\varepsilon_i,\pi_i(2)} * \cdots * \mathfrak{f}_{\varepsilon_i,\pi_i(l_i)}. \tag{2}$$

The element of Nerf_2 under consideration depends only on

$$(\varepsilon_i,l_i,\pi_i), \qquad \text{where } (\varepsilon_i,l_i) \in \{\pm 1\}\times\mathbf{N},\ \pi_i : [1,l_i] \longrightarrow \mathbf{N}.$$

(**NB** If $l_i = 0$, $[1,l_i] = [1,0] = \emptyset$ by convention, and π_i is the unique map $\emptyset \longrightarrow \mathbf{N}$, so the datum $(\varepsilon_i,l_i,\pi_i)$ then amounts to (ε_i,l_i).)

If

$$(\varepsilon,l,\pi)\ \text{ as above}, \qquad (\varepsilon,l) \in \{\pm 1\}\times\mathbf{N},\ \pi : [1,l] \longrightarrow \mathbf{N},$$

let us denote by

$$\mathfrak{f}_{\varepsilon,l,\pi}$$

the element of Nerf_2 given by

$$\Delta_\varepsilon \xrightarrow{\mathfrak{f}_{\varepsilon,l}} \Delta_\varepsilon^l \xrightarrow{\mathfrak{f}_{\varepsilon,\pi(1)}\ *\cdots*\ \mathfrak{f}_{\varepsilon,\pi(l)}} \Delta_\varepsilon^{\pi(1)}\Delta_\varepsilon^{\pi(2)}\cdots\Delta_\varepsilon^{\pi(l)}.$$

We can formulate the preceding construction by saying that every element of Nerf_2 can be decomposed uniquely into a product p. 37

$$(f,g) = \mathfrak{f}_{\varepsilon_1,l_1,\pi_1} * \mathfrak{f}_{\varepsilon_2,l_2,\pi_2} * \cdots * \mathfrak{f}_{\varepsilon_n,l_n,\pi_n}, \qquad \text{[page 54]}$$

which proves that $\mathrm{Nerf}_2(\underline{\mathrm{Ty}})$ is the free monoid on the preceding generators $\mathfrak{f}_{\varepsilon,l,\pi}$. (**NB** to the empty sequence corresponds

$$1 \xrightarrow{\mathrm{id}_1} 1 \xrightarrow{\mathrm{id}_1} 1 \qquad .)$$

If the object of Nerf_2 is

$$x \xrightarrow{f} y \xrightarrow{g} z,$$

the preceding construction tells us, in terms of the combinatorial datum, i.e. of the sequence of data

$$(\varepsilon_1, l_1, \pi_1),\ \ldots\ ,\ (\varepsilon_n, l_n, \pi_n),$$

what x, y, z are: each of them is the product in $\mathrm{Nerf}_0 = \mathrm{Ty}$ of the n factors x_i, y_i or z_i respectively, for $i \in [1,n]$, given in the definition of the $\mathfrak{f}_{\varepsilon_i,l_i,\pi_i}$, i.e. of

$$\Delta_{\varepsilon_i},\ \Delta_{\varepsilon_i}^{l_i},\ \text{ and } \Delta_{\varepsilon_i}^{\pi_i(1)}\Delta_{\varepsilon_i}^{\pi_i(2)}\cdots\Delta_{\varepsilon_i}^{\pi_i(l_i)} = \Delta_{\varepsilon_i}^{\mathrm{long}(\pi_i)}$$

(where $\mathrm{long}(\pi_i) = \sum_{j\in[1,l_i]} \pi_i(j)$). Thus

$$\begin{cases} x &= \prod_{i\in[1,n]}^{\mathrm{Ty}} \Delta_{\varepsilon_i} \\ y &= \prod_{i\in[1,n]}^{\mathrm{Ty}} \Delta_{\varepsilon_i}^{l_i} \\ z &= \prod_{i\in[1,n]}^{\mathrm{Ty}} \Delta_{\varepsilon_i}^{\mathrm{long}(\pi_i)} \end{cases}$$

([16]), and we similarly find f, g as products (in the sense $*$) of n factors f_i resp. g_i ($i \in [1,n]$) made explicit in formulas (1), (2) (p. 52, 53):

$$\begin{aligned} f_i &= \mathfrak{f}_{\varepsilon_i,l_i} : \quad \Delta_{\varepsilon_i} \longrightarrow \Delta_{\varepsilon_i}^{l_i} \\ g_i &= g_{\varepsilon_i,\pi_i} : \quad \Delta_{\varepsilon_i}^{l_i} \longrightarrow \Delta_{\varepsilon_i}^{\mathrm{long}(\pi_i)}, \end{aligned}$$

g_i itself being represented as a $*$-product of l_i indecomposable factors, so that f and g appear in the form of $*$-products of n, resp. of $\sum_{i\in[1,n]} l_i$ factors of the form $\mathfrak{f}_{\varepsilon,\nu}$. To finish getting the truncated semi-simplicial object $(\mathrm{Nerf}_i(\underline{\mathrm{Ty}}))_{0\le i\le 2}$ in hand, it remains to describe the two degeneracy maps $\mathrm{Nerf}_1 \longrightarrow \mathrm{Nerf}_2$ (the case where f or g is an identity arrow) and above all the *composition* map for arrows [page 55]

$$\mathrm{Nerf}_2(\underline{\mathrm{Ty}}) \xrightarrow{\gamma} \mathrm{Nerf}_1(\underline{\mathrm{Ty}}) = \mathrm{Fl}(\underline{\mathrm{Ty}}).$$

It suffices to give γ on the generators, and we have tautologically, by definition of the $\mathfrak{f}_{\varepsilon,l,\pi}$

$$\gamma(\mathfrak{f}_{\varepsilon,l,\pi}) = \mathfrak{f}_{\varepsilon,\mathrm{long}(\pi)}.$$

(**NB** There is at most one morphism from Δ_ε (from Δ_+ or from Δ_-) to an interval I_τ. It exists if and only if τ is of the form Δ_ε^l, $l \in \mathbf{N}$.)

The case $g = \mathrm{id}$ corresponds to the case where p. 38

$$\pi_i(j) = 1 \qquad \forall j \in [1,l_i],$$

the $\mathfrak{f}_{\varepsilon_i,l_i,\pi_i}$ are then determined in terms of the $\mathfrak{f}_{\varepsilon_i,l_i}$, and the degeneracy map is

$$\mathfrak{f}_{\varepsilon,l} \longmapsto \mathfrak{f}_{\varepsilon,l,1_l} \qquad (\text{where } 1_l \text{ is the constant function with value 1 on } [1,l]).$$

The case $f = \mathrm{id}$ is the one where the l_i are equal to 1. Then each $\pi_i : [1,1] \longrightarrow \mathbf{N}$ can be viewed as an element of $\mathbf{N}$, giving $(\varepsilon_i, l_i, \pi_i)$ amounts to giving the $(\varepsilon_i, n_i) \in \{\pm 1\} \times \mathbf{N}$, i.e. [page 56]

giving a sequence of n generators among the $\mathfrak{f}_{\varepsilon,l}$. The degeneracy map $\mathrm{Nerf}_1 \longrightarrow \mathrm{Nerf}_2$ is here given on the generators by

$$\mathfrak{f}_{\varepsilon,l} \longmapsto \mathfrak{f}_{\varepsilon,1,l}.$$

Fortunately, we no longer have to prove the associativity of composition.

NB The simplicity of these descriptions reinforces my feeling that the category $\underline{\mathrm{Ty}}$ now defined, with the *increasing morphisms of oriented intervals* $I_\tau \longrightarrow I_{\tau'}$ as arrows of $\underline{\mathrm{Ty}}$, is indeed "the right one". We shall see for a start whether it gives a nice description of the categories of fractions $X\Sigma^{-1}$, without contortions as before.

But we have promised to determine $\mathrm{Nerf}_r(\underline{\mathrm{Ty}})$, for every $r \in \mathbf{N}$. The general generator of Nerf_r is of the form (for $\varepsilon \in \{\pm 1\}$ fixed)

$$\underbrace{\Delta_\varepsilon}_{=\Delta_\varepsilon^{l_0} \text{ with } l_0=1} \xrightarrow{f_1} \Delta_\varepsilon^{l_1} \xrightarrow{f_2} \Delta_\varepsilon^{l_2} \longrightarrow \cdots \xrightarrow{f_r} \Delta_\varepsilon^{l_r},$$

where the $f_i : \Delta_\varepsilon^{l_{i-1}} \longrightarrow \Delta_\varepsilon^{l_i}$, $i \in [1,r]$, are given by maps

$$\pi_{i-1} : [1, l_{i-1}] \longrightarrow \mathbf{N} \qquad \text{such that} \qquad \underbrace{\mathrm{long}(\pi_{i-1})}_{\overset{\text{def}}{=} \Sigma_{j\in[1,l_{i-1}]}\pi_{i-1}(j)} = l_i.$$

(NB $\pi_0 : [1,1] = \{1\} \longrightarrow \mathbf{N}$ is identified with $\mathrm{long}(\pi_0) = \pi_0(1) = l_1$.)

To avoid making the definition of the domain of the π_i ($i \in [0, r-1]$) depend on the integers l_i, we can also consider the π_i as maps [page 57]

$$\mathbf{N}^* \xrightarrow{\pi} \mathbf{N} \qquad (\mathbf{N}^* = \{i \in \mathbf{N} \mid i \geq 1\})$$

with finite supports (i.e. such that $\pi(i) = 0$ for i large), and require precisely

$$\begin{aligned} &\mathrm{supp}\,\pi_0 \ \subseteq\ \{1\} \\ &\mathrm{supp}\,\pi_1 \ \subseteq\ [1, \mathrm{long}(\pi_0)] \\ &\qquad \cdots \\ &\boxed{\mathrm{supp}\,\pi_i \ \subseteq\ [1, \mathrm{long}(\pi_{i-1})]}, \qquad \text{for } \boxed{i \in [1, r-1]} \quad . \end{aligned} \tag{$**$}$$

On the other hand, setting $l_i = \mathrm{long}(\pi_{i-1})$ (for $i \in [1,r]$), p. 39

$$f_i : \Delta_\varepsilon^{l_{i-1}} \longrightarrow \Delta_\varepsilon^{l_i\ (=\mathrm{long}(\pi_{i-1}))}$$

is defined as

$$f_i = \mathfrak{f}_{\varepsilon,\pi_{i-1}(1)} * \mathfrak{f}_{\varepsilon,\pi_{i-1}(2)} * \cdots * \mathfrak{f}_{\varepsilon,\pi_{i-1}(l_{i-1})},$$

so that the element $(*)$ of $\mathrm{Nerf}_r(\underline{\mathrm{Ty}})$ is uniquely determined by the data of the $\varepsilon \in \{\pm 1\}$ and of

$$\pi_\bullet = (\pi_i)_{i\in[0,r-1]} \in (\mathrm{Hom}_{\mathrm{Ens}}(\mathbf{N}^*, \mathbf{N}))^r$$

satisfying the conditions $(**)$. Let $F_{\varepsilon,\pi_\bullet}$ be this element of $\mathrm{Nerf}_r(\underline{\mathrm{Ty}})$, and let N_r be the set of $\pi_\bullet$ we have just made explicit. Then

$$(F_{\varepsilon,\pi_\bullet})_{(\varepsilon,\pi_\bullet)\in\{\pm1\}\times N_r}$$

is a free system of generators of the monoid $\mathrm{Nerf}_r(\underline{\mathrm{Ty}})$.

If $l_i = 0$ (i.e. $\boxed{\pi_{i-1} = 0}$), for $i < r$, we must have $\mathrm{supp}(\pi_i) = \emptyset$, i.e. $\pi_i = 0$, whence $l_{i+1} = 0$, and likewise for the following l_j and π_j. Geometrically the hypothesis [page 58]

is expressed by the condition that the target of f_i is the element 1 of Ty (corresponding to the one-point interval), and since $\mathrm{Hom}(1,\tau)=\emptyset$ for $\tau\neq 1$, this means that the arrows following f_i must all be equal to the arrow $\mathrm{id}_1 : 1 \longrightarrow 1$.

Remark. In the combinatorial description of $\underline{\mathrm{Ty}}$ and of $\mathrm{Nerf}_\bullet(\underline{\mathrm{Ty}})$, the set of "signatures" $\{\pm 1\}$ plays a rather phony role. It can be replaced by any set Σ (called the "set of signatures"), and we then construct a monoidal category (or a categorical monoid) having as its set of objects the free monoid generated by Σ, interpreted as intervals of integers $[0,n]$, equipped with "signatures" in Σ for each transition interval, or what amounts to the same thing, Σ-weighted directed "graphs" of combinatorial type an interval $\mathbf{\Delta}_i$, which amounts to giving an (ordered) sequence of n elements $\varepsilon_1,\ \varepsilon_2,\ \ldots\ ,\ \varepsilon_n \in \Sigma$, i.e. a "word" $\varepsilon_1\varepsilon_2\ldots\varepsilon_n$, which we prefer to write $\Delta_{\varepsilon_1}\Delta_{\varepsilon_2}\cdots\Delta_{\varepsilon_n}$. We can define in the obvious way the *increasing* homomorphisms of such weighted directed graphs (understood: with the source element and the target element in the structure, to be respected by the morphisms), without needing a categorical interpretation. The functoriality of the operation of composition of weighted intervals, for this notion of morphism, is obvious. The choice [page 59] $\Sigma=\{\pm 1\}$ is to my knowledge the only useful one, because of its link with the notion of paths and the calculation of π_0 of an object X of Cat , etc.

p. 40

7 Explicit description of the categories of fractions $X\Sigma^{-1}$ in terms of the $\underline{\mathrm{Ch}}(X;a,b)$

If X is a category, we denote by $\underline{\mathrm{Ch}}(X)$ the category of paths in X (cf. §4), and for $a,b\in \mathrm{Ob}\,X$, by

$$\underline{\mathrm{Ch}}(X;a,b) \qquad \text{or simply } \underline{\mathrm{Ch}}(a,b)$$

that of paths from a to b, i.e. with source a, target b. The geometric intuition (which we will make explicit) is that such a path c defines a "direct" path from a to b (i.e. a path of type τ' "direct", i.e. all the transition arrows are direct, i.e. τ' of the form Δ_+^n), of the same length as the initial path c, obtained by inverting all the retrograde arrows in the path c. This operation is actually possible in X itself if and only if the said arrows are invertible. We then obtain a composable sequence of arrows, whose total composite (which we can call "the useful effect" of the path) is an arrow from a to b. In the general [page 60] case, paths from a to b can be seen as a sort of "generalized arrows" going from a to b, whose composition would be the obvious composition, and the identities the "empty paths" relative to the $a\in \mathrm{Ob}\,X$. We do moreover find in this way a category, having the same objects as X, but where the

$$\mathrm{Hom}_{\mathrm{Ch}}(a,b)=\mathrm{Ch}(a,b)=\mathrm{Ob}\,\underline{\mathrm{Ch}}(a,b)$$

are prohibitively large. This is expressed in particular by the fact that the only invertible arrows in X_{Ch} are the identities! We have no natural functor between X_{Ch} and X that is the identity on objects. For to a path $a \xrightarrow{c} b$ we cannot associate any X-morphisms $a\longrightarrow b$ (unless c of type Δ_+^n, or more generally if its retrograde arrows are invertible), and to an X-morphism $a\xrightarrow{u} b$ we can associate a path $c(u)$ of type Δ_+, but if $a\xrightarrow{u} b\xrightarrow{v} c$, $c(vu)$ is still of type Δ_+ whereas $c(v)c(u)$ is of type Δ_+^2, they are not equal. On the other hand, there is a natural arrow between these two elements of $\underline{\mathrm{Ch}}(a,c)$, an arrow over the

[15] **NB** The empty "word" of $\mathrm{Fl}(\underline{\mathrm{Ty}})$ corresponds to the arrow id_1, where 1 is the unit element of Ty, realizing the one-point "interval".

[16] careful, the products must be taken respecting the order of the factors given by the index notation in i.

unique homomorphism

$$\Delta_+ \xrightarrow{\mathfrak{f}_{+,2}} \Delta_+^2.$$

We then have [page 61]

$$\mathfrak{f}_{+,2}^*(\underbrace{a \xrightarrow{u} b \xrightarrow{v} c}_{c(v)c(u)}) = (\underbrace{a \xrightarrow{vu} c}_{c(vu)}),$$

whence a canonical arrow

$$c(vu) \longrightarrow c(v)c(u) \qquad \text{in } \underline{\mathrm{Ch}}(a,c).$$

This shows that $c(vu)$ and $c(v)c(u)$ are at least in the same connected component of $\underline{\mathrm{Ch}}(a,c)$, or (as we also say) that they are "homotopic". This suggests here modifying the definition of $\mathrm{Hom}_{\mathrm{Ch}}(a,b)$ to

$$\pi(a,b) = \pi_0(\underline{\mathrm{Ch}}(a,b)).$$

The path composition functor p. 41

$$\begin{array}{ccccc} \underline{\mathrm{Ch}}(a,b) & \times & \underline{\mathrm{Ch}}(b,c) & \longrightarrow & \underline{\mathrm{Ch}}(a,c) \\ (c & , & c') & \longmapsto & c' \circ c \end{array}$$

([17]) induces a map on the π_0, which gives a composition law

$$\pi(a,b) \times \pi(b,c) \longrightarrow \pi(a,c)$$

on arrows. It is immediate that it is associative (the composition of paths being so), and unital. It admits as identity arrow at x either the "empty loop at x", which is (directly) homotopic in $\underline{\mathrm{Ch}}(x,x)$ to the "degenerate" path $x \xrightarrow{\mathrm{id}} x$ of type Δ_+, or also to [page 62] the degenerate path of any type τ, say

$$x \xleftarrow{\mathrm{id}} x \xrightarrow{\mathrm{id}} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x,$$

such a path being none other than $p_\tau^*(x)$, where

$$p_\tau : \tau \longrightarrow 1$$

is the degeneracy morphism (the unique morphism in $\underline{\mathrm{Ty}}$ to 1, which is a final object), whence

$$p_\tau^* : \underline{\mathrm{Ch}}_1(x,x) = \{x\} \longrightarrow \underline{\mathrm{Ch}}_\tau(x,x) \subseteq \underline{\mathrm{Ch}}(x,x).$$

The category $\Pi(X)$ thus obtained must be called the *fundamental groupoid* of X. Indeed,

a) all the arrows of $\Pi(X)$ are invertible, i.e. $\Pi(X)$ is indeed a groupoid, and

b) the canonical functor

$$X \longrightarrow \Pi(X)$$

which is the identity on objects, and which is defined on arrows by $u \longmapsto c(u)$ as above, is *universal* (in Cat) for functors (i.e. Cat-morphisms) from X to *groupoids.* (More generally still, it is universal for functors $X \xrightarrow{F} Y$ such that F transforms every arrow of X into invertible arrow.)

p. 42
[page 63]

To prove that every arrow in $\Pi(X)$ is invertible, since every nonempty path is a composite of "elementary" paths of type Δ_+ or Δ_-, it suffices to prove that these paths are invertible. In fact, they correspond bijectively,

$$\underline{\mathrm{Ch}}_{\Delta_+}(a,b) \simeq \underline{\mathrm{Ch}}_{\Delta_-}(b,a),$$

by associating to the path from a to b

$$c(u) : a \xrightarrow{u} b \qquad \text{of type } \Delta_+$$

the path

$$c(u)^o : b \xleftarrow{u} a \qquad \text{of type } \Delta_-$$

from b to a. I say that

$$\begin{array}{rcl} c(u)^o c(u) & \sim & 1_a \\ c(u)c(u)^o & \sim & 1_b \end{array}$$

$$\left.\begin{array}{ccccc} a & \xrightarrow{u} & b & \xleftarrow{u} & a \\ \uparrow{\scriptstyle \mathrm{id}} & & \uparrow{\scriptstyle u} & & \uparrow{\scriptstyle \mathrm{id}} \\ a & \xrightarrow{\mathrm{id}} & a & \xleftarrow{\mathrm{id}} & a \end{array}\right\} \quad \text{we have a homomorphism in } \underline{\mathrm{Ch}}_{\Delta_+\Delta_-}(a,a) \text{ from } 1_a \text{ to } c(u)^o c(u).$$

$$\uparrow \quad a$$

$$\left.\begin{array}{ccccc} b & \xleftarrow{u} & a & \xrightarrow{u} & b \\ \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle \mathrm{id}} \\ b & \xleftarrow{\mathrm{id}} & b & \xrightarrow{\mathrm{id}} & b \end{array}\right\} \quad \text{likewise in } \underline{\mathrm{Ch}}_{\Delta_-\Delta_+}(b,b), \text{ from } 1_b \longleftarrow c(u)c(u)^o$$

Let us note that, more generally, for any path p. 43 [page 64]

$$c : a \longrightarrow b, \qquad \text{of type } \tau,$$

we define an *opposite path*

$$c^o : b \longrightarrow a \qquad \text{of type } \tau^o,$$

where

$$\tau \longmapsto \tau^o : \mathrm{Ty} \longrightarrow \mathrm{Ty}$$

is the anti-involution of the free monoid Ty which exchanges the two generators Δ_-, Δ_+, and which is therefore given on the general element by

$$(\Delta_{\varepsilon_1}\Delta_{\varepsilon_2}\cdots\Delta_{\varepsilon_n})^o = \Delta_{-\varepsilon_n}\Delta_{-\varepsilon_{n-1}}\cdots\Delta_{-\varepsilon_1}.$$

By induction on the length of c ([18]), using the preceding result, that $c(u)$, $c(u)^o$ are inverses of one another in $\Pi(X)$, we find

$$\left.\begin{array}{rcl} c^o c & = & \mathrm{id}_a \\ cc^o & = & \mathrm{id}_b \end{array}\right\} \text{in } \Pi(X).$$

[17] This functor is deduced from the functors

$$\underline{\mathrm{Ch}}_{\tau}(a,b) \times \underline{\mathrm{Ch}}_{\tau'}(b,c) \longrightarrow \underline{\mathrm{Ch}}_{\tau\tau'}(a,c)$$

which are obvious, cf. page 77.

[18] unnecessary, follows directly from formula $(*)$ following page, by decomposing c into composites of "elementary" paths of type Δ_+ or Δ_-.

Example. If

$$\begin{aligned} \tau &= \Delta_+\Delta_-\Delta_+^2\Delta_- \\ \tau^o &= \Delta_+\Delta_-^2\Delta_+\Delta_-, \end{aligned}$$

and if we have a path

$$c : a_0 \xrightarrow{u_1} a_1 \xleftarrow{u_2} a_2 \xrightarrow{u_3} a_3 \xrightarrow{u_4} a_4 \xleftarrow{u_5} a_5$$

of type τ, we deduce from it the path

$$c^o : a_5 \xrightarrow{u_5} a_4 \xleftarrow{u_4} a_3 \xleftarrow{u_3} a_2 \xrightarrow{u_2} a_1 \xleftarrow{u_1} a_0$$

of type τ^o. The operation $c \longmapsto c^o$ satisfies in fact

[page 65]

$$\begin{array}{ccccc} a & \xrightarrow{\gamma} & b & \xrightarrow{\gamma'} & c \\ c & \xrightarrow{\gamma'^o} & b & \xrightarrow{\gamma^o} & a \end{array}$$

the relation

$$(\gamma'\gamma)^o = \gamma^o\gamma'^o, \qquad (*)$$

which is in accord with the intuition that the operation $\gamma \longmapsto \gamma^o$ plays the role of passing to the inverse ("up to homotopy"), and we have the formula

p. 44

$$(\gamma'\gamma)^{-1} = \gamma^{-1}\gamma'^{-1}$$

for two invertible arrows γ, γ' composable (in the order γ', γ).

As for the universal property of

$$X \longrightarrow \Pi(X)$$

(which will allow us to write

$$\Pi(X) \simeq X\Sigma^{-1}, \qquad \text{with } \Sigma = \mathrm{Fl}(X) \ \ ),$$

we may as well establish it right away in the general setting where we give ourselves a subset

$$\Sigma \subseteq \mathrm{Fl}(X),$$

and we define, for $\tau \in \mathrm{Ty}$,

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(X) \subseteq \underline{\mathrm{Ch}}_\tau(X)$$

as the full subcategory of $\underline{\mathrm{Ch}}_\tau(X)$ consisting of the paths in X for which *all the retrograde arrows*

are in Σ . We shall note that for a path c and its opposite c^o both to be Σ-*paths,* it is necessary and sufficient that *all* the arrows of the path c be in Σ (we shall then say that c is a Σ-*strict path*), since the retrograde arrows of c^o are the direct arrows of c.

[page 66]

For the full subcategories $\underline{\mathrm{Ch}}_{\tau,\Sigma}(X)$ to be stable under the inverse image functors

$$f^* : \underline{\mathrm{Ch}}_{\tau'}(X) \longrightarrow \underline{\mathrm{Ch}}_\tau(X)$$

associated with the arrows

$$f : \tau \longrightarrow \tau'$$

in $\underline{\mathrm{Ty}}$, or simply for there to be stability for the two functors f^* associated with an $f : \Delta_- \longrightarrow 1$ and $\Delta_- \longrightarrow \Delta_-^2$, it is necessary and sufficient that Σ be stable under composition and contain all the identity arrows

$$\begin{cases} x \in \mathrm{Ob}\, X \implies \mathrm{id}_x \in \Sigma \\ u, v \in \Sigma, \quad u,\, v \text{ composable} \implies uv \in \Sigma. \end{cases}$$

Indeed, the arrows in a path $f^*(c') \in \underline{\mathrm{Ch}}_\tau(X)$ are either identity arrows (which correspond to the transition arrows of I_τ degenerate for f), or arrows among those of the path c' of type τ', or composites of a sequence of consecutive composable arrows of this path (which correspond to the transition arrows in $I_{\tau'}$ included in the part of a transition arrow of I_τ), with preservation of signatures.

We assume these conditions satisfied, so that the $\underline{\mathrm{Ch}}_{\tau,\Sigma}(X)$ form the fibers of a full fibered subcategory of $\underline{\mathrm{Ch}}(X)$ over $\underline{\mathrm{Ty}}$, which we shall denote by p. 45 [page 67]

$$\underline{\mathrm{Ch}}_\Sigma(X) \hookrightarrow \underline{\mathrm{Ch}}(X).$$

It is obvious that this subcategory is stable under the operation of composition of paths (but not, in general, as we have said, under passage from c to c^o (for this Σ would have to contain *all* the arrows of X, since for every $u : a \longrightarrow b$, which is thus in $\underline{\mathrm{Ch}}_{\Delta_+,\Sigma}(X)$, u^o given by $b \xleftarrow{u} a$ in $\underline{\mathrm{Ch}}_{\Delta_-}(X)$ would have to be a Σ-path, whence $u \in \Sigma$)).

We are going to define a canonical category

$$\Pi_\Sigma(X),$$

having the same objects as X, and a canonical functor

$$X \longrightarrow \Pi_\Sigma(X)$$

which is the identity on objects, and which transforms the $u \in \Sigma$ into invertible arrows. For this we copy the description of the fundamental groupoid (but $\Pi_\Sigma(X)$ will not in general be a groupoid), by putting

$$\mathrm{Hom}_\Sigma(x, y) = \pi_0(\underline{\mathrm{Ch}}_\Sigma(x, y)),$$

and defining the composition of "Σ-arrows" in terms of the composition of paths. [page 68] The argument which proved, in the case of $\Pi(X)$, that *every* arrow of X, viewed as a path c of type Δ_+, was invertible in $\Pi(X)$, having as inverse c^o of type Δ_-, is still valid, provided that c^o is indeed still a Σ-path, which requires that $u \in \Sigma$.

We are going to interpret it in terms of a canonical functor

$$\kappa : X \longrightarrow \Pi_\Sigma(X) \qquad ([19]) \qquad (*)$$

which is again described as in the case of the fundamental groupoid, by associating to every arrow u of X the homotopy class of the path $c(u)$ it defines. We thus see that this canonical functor transforms the $u \in \Sigma$ into invertible arrows.

When the arrows of Σ are already invertible in X, I say that the functor $(*)$ is an isomorphism, i.e. that for every $a, b \in \mathrm{Ob}\, X$,

$$\alpha : \mathrm{Hom}_X(a, b) \longrightarrow \pi_0 \underline{\mathrm{Ch}}_\Sigma(a, b), \qquad u \longmapsto \underbrace{\mathrm{ch}_\Sigma(u)}_{=\mathrm{ch}(u)}$$

is bijective. To see this, we are going to define

$$\beta : \pi_0(\underline{\mathrm{Ch}}_\Sigma(a, b)) \longrightarrow \mathrm{Hom}_X(a, b),$$

which will be inverse to α. We define

$\beta(c)$, for a Σ-arrow of type τ, the length n, by inverting all the retrograde arrows (which is possible, since they are in Σ), so as to obtain a path $c^\natural$ of type Δ_+^n, and by p. 46 [page 69]

[19] On arrows, κ is defined, for $u : a \longrightarrow b$, by $\kappa(u)$ = homotopy class of $\mathrm{ch}(u)$ in $\underline{\mathrm{Ch}}_\Sigma(a, b)$, where $\mathrm{ch}(u)$ is the path of type Δ_+ from a to b defined by u.

taking the composite (in X) of the arrows of this path $c^\natural$. It is then obvious that $\beta\alpha = \mathrm{id}$, we must prove that we also have $\alpha\beta = \mathrm{id}$, i.e. that in $\Pi_\Sigma(X)$, the arrows $u = \int c^\natural$ and c are equal, i.e. u and c are homotopic in $\underline{\mathrm{Ch}}_\Sigma(a,b)$. Let us consider the path $c^\natural$, of type Δ^n_+:

$$a = a_0 \xrightarrow{u_1} a_1 \xrightarrow{u_2} a_2 \longrightarrow \cdots \longrightarrow a_{n-1} \xrightarrow{u_n} a_n = b,$$

and let u_i ($i \in [1,n]$) be an arrow of this path, corresponding to a transition arrow $\underline{u}_i$ that is *retrograde* in τ, hence by construction

$$u_i = v_i^{-1} \qquad \text{in } X, \text{ with } v_i \in \Sigma$$

the arrow with index i in the initial path c of type τ. If $\underline{u}_i$, on the other hand, is direct, we will have

$$u_i = v_i.$$

(NB We denote by v_i the i-arrow in the path c.) Now we have

$$c = c_n * c_{n-1} * \cdots * c_1 \qquad \text{in } \Pi_\Sigma(X) \tag{$*$}$$

and the factor c_i is either the homotopy class of $\mathrm{ch}(v_i) = \mathrm{ch}(u_i)$ (the case where the transition arrow $\underline{u}_i$ of τ is direct), or that of the path *opposite* to the path $\mathrm{ch}(v_i) : a_i \xrightarrow{v_i} a_{i-1}$ of type Δ_+, hence the path $\mathrm{ch}(v_i)^o : a_{i-1} \xleftarrow{v_i} a_i$ of type Δ_-. I say that

$$\boxed{\mathrm{ch}(v_i)^o = \mathrm{ch}(u_i) \qquad \textit{in } \Pi_\Sigma(X),} \tag{1}$$

which proves that in $\Pi_\Sigma(X)$, c is equal to the composite of the $\mathrm{ch}(u_i)$ ($i \in [1,n]$), hence to $c^\natural$. We must prove moreover that

$$c^\natural = \mathrm{ch}(u) \qquad (u = u_n u_{n-1} \cdots u_1) \qquad \text{in } \Pi_\Sigma(X). \tag{2}$$

So there are two verifications to make (where I again assume Σ arbitrary): [page 70]

1°) Let $u \in \Sigma$, $u : a \longrightarrow b$, suppose Σ invertible in X. Then

$$\mathrm{ch}(u)^o = \mathrm{ch}(u^{-1}) \qquad \text{in } \Pi_\Sigma(X),$$

i.e. the two paths from b to

$$b \xleftarrow{u} a \quad \text{of type } \Delta_-$$
$$b \xrightarrow{u^{-1}} a \quad \text{of type } \Delta_+$$

are homotopic in $\underline{\mathrm{Ch}}_\Sigma(b,a)$. Now we already know, since $\kappa : X \longrightarrow \Pi_\Sigma(X)$ is a functor, that p. 47

$$\mathrm{hot}\,\mathrm{ch}(u^{-1}) = (\mathrm{hot}\,\mathrm{ch}(u))^{-1} \qquad \text{in } \Pi_\Sigma(X)$$

where $\mathrm{hot}(\bullet)$ denotes the homotopy class of a chain (here a class in $\underline{\mathrm{Ch}}_\Sigma(b,a)$ and in $\underline{\mathrm{Ch}}_\Sigma(a,b)$ respectively). So the statement to prove is that $\mathrm{hot}(\mathrm{ch}(u)^o)$ is equal to the *inverse* of $\mathrm{hot}(\mathrm{ch}(u))$ in $\Pi_\Sigma(X)$. This is thus what we proved for *every* $u \in \mathrm{Fl}(X)$, in the case of $\Pi(X)$. The argument must work, so long as it makes sense, i.e. that $\mathrm{ch}(u)^o$ really is a Σ-path, i.e. $u \in \Sigma$.

2°) Formula (2) means that for every path c of type Δ^n_+ (thus tautologically a Σ-path), going from a to b, its class in $\mathrm{Hom}_\Sigma(a,b)$ is none other than that of $\mathrm{ch}(u)$, where u is the composite $u_n u_{n-1} \cdots u_1$. But we have [page 71]

$$c = c_1 * c_2 * \cdots * c_n,$$

whence in $\Pi_\Sigma(X)$

$$\mathrm{hot}(c) = \mathrm{hot}(c_n)\mathrm{hot}(c_{n-1})\cdots\mathrm{hot}(c_1),$$

now $\mathrm{hot}(c_i) = \mathrm{hot}\,\mathrm{ch}(u_i) = \kappa(u_i)$, hence

$$\begin{aligned}\mathrm{hot}(c) &= \kappa(u_n)\kappa(u_{n-1})\cdots\kappa(u_1)\\ &= \kappa(\underbrace{u_n u_{n-1}\cdots u_1}_{u}),\end{aligned}$$

since κ is a functor, hence

$$\mathrm{hot}(c) = \mathrm{hot}\,\mathrm{ch}(u),$$

q.e.d.

Let us now note that if (X', Σ') is a second pair consisting of a category X' and an *"admissible"* set of arrows of X', and if

$$f : X \longrightarrow X'$$

is a functor such that

$$f(\Sigma) \subseteq \Sigma',$$

then there exists a *unique* functor

$$\bar{f} = \Pi_\Sigma(f) : \Pi_\Sigma(X) \longrightarrow \Pi_{\Sigma'}(X')$$

which "extends" f, i.e. such that the square

p. 48
[page 72]

$$\begin{array}{ccc} X & \xrightarrow{\kappa_X} & \Pi_\Sigma(X) \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \bar{f}=\Pi_\Sigma(f)} \\ X' & \xrightarrow{\kappa_{X'}} & \Pi_{\Sigma'}(X') \end{array}$$

is commutative. Uniqueness follows from the fact that on objects $\bar{f}$ must be given by f, and that on arrows the commutativity of the square implies that $\bar{f}$ is known on arrows of the form $\kappa_X(u)$, and thus also on the inverses of such arrows, when the arrows under consideration are invertible, hence in particular on the $\kappa_X(u)^{-1}$ for $u \in \Sigma$. Now every arrow of $\Pi_\Sigma(X)$, coming from a path, is a composite of a sequence of arrows of this form. For the existence of $\bar{f}$, we note that f induces functors

$$\underline{\mathrm{Ch}}_\tau(X; a, b) \longrightarrow \underline{\mathrm{Ch}}_\tau(X'; f(a), f(b))$$

which commute with the operations φ^* on paths associated with the functors

$$\varphi : \tau \longrightarrow \tau',$$

whence a homomorphism of fibered categories over $\underline{\mathrm{Ty}}$

$$f_{a,b} : \underline{\mathrm{Ch}}(X; a, b) \longrightarrow \underline{\mathrm{Ch}}(X'; f(a), f(b)).$$

Whence a map on the π_0, i.e.

[page 73]

$$\bar{f}_{a,b} : \mathrm{Hom}_\Sigma(a, b) \longrightarrow \mathrm{Hom}'_\Sigma(f(a), f(b)),$$

given by

$$\bar{f}_{a,b}(\mathrm{hot}(c)) = \mathrm{hot}(f_{a,b}(c)).$$

We obviously have, for two paths $\gamma : a \longrightarrow b$ and $\gamma' : b \longrightarrow c$,

$$f_{a,c}(\gamma'\gamma) = f_{b,c}(\gamma')f_{a,b}(\gamma),$$

which proves that $\bar{f}$ is compatible with composition. It is trivial that it is compatible with identities (for $f_{a,a}$ transforms the empty path at a into the empty path at $f(a)$), so we have a functor $\bar{f}$, and it really does make the square commutative.

We can now prove the universal property of $\Pi_\Sigma(X)$ and of

$$X \xrightarrow{\kappa} \Pi_\Sigma(X),$$

namely that every functor $f : X \longrightarrow X'$ such that $u \in \Sigma \Longrightarrow f(u)$ invertible, factors *uniquely* through $\Pi_\Sigma(X)$. Uniqueness is seen as before ([20]). For existence, we note that f maps Σ into the set Σ'

of the invertible arrows of X' (which set does indeed satisfy the two admissibility conditions). Thus by what precedes we have a commutative diagram p. 49
[page 74]

$$\begin{array}{ccc} X & \xrightarrow{\kappa} & \Pi_\Sigma(X) \\ {\scriptstyle f}\downarrow & \overset{\tilde{f}}{\nearrow} & \downarrow{\scriptstyle \bar{f}=\Pi_\Sigma(f)} \\ X' & \xrightarrow[\sim]{\kappa'} & \Pi_{\Sigma'}(X'), \end{array}$$

where $\bar{f}$ is uniquely determined. Since κ' is an isomorphism (as we saw earlier, since Σ' made up of the invertible arrows), the existence and uniqueness of $\bar{f}$ is equivalent to the announced conclusion,

q.e.d.

One more sorites on the homotopy classes of elements of a category X. In general, one sees that X and $\Pi_\Sigma(X) = X\Sigma^{-1}$ have the same connected components, for if a and b in X are joined by an arrow of $X\Sigma^{-1}$, i.e. by a Σ-path, then they are in the same connected component of X. But the converse is also true, as one sees e.g. by taking $\Sigma = \mathrm{Fl}(X)$, and by saying that in the *groupoid* $\Pi(X)$, where two vertices are in the same component if and only if $\exists\ x \longrightarrow y$ (a relation which is not only transitive and reflexive, but also symmetric in a groupoid). A path [page 75]

$$a \xrightarrow{c} b$$

will also be called a *homotopy* from a to b. We say that a and b are *homotopic* if there exists a homotopy c from a to b, in which case there also exists an "inverse" homotopy c^o from b to a. This also means that a and b are in the same connected component of X, i.e. define the same element of $\pi_0(X)$.

The homotopy will be called *simple* if c is of type Δ_+ or Δ_-, i.e. if it is given by an arrow $u : a \longrightarrow b$ or $v : b \longrightarrow a$. If one is looking for a homotopy between a and b, one will try to find it as "short" as possible. If one has a homotopy c of length l from a to b, then on replacing τ by τ_{red}, hence I_τ by $I_{\tau\,\mathrm{red}}$, the corresponding reduced subinterval, one finds a homotopy from a to b of type τ_{red}, whose length is equal to the *"useful length"* of the type τ of c (which justifies the terminology). Let us note that if [page 76]

$$\tau = \Delta_\varepsilon^{l_1}\Delta_{-\varepsilon}^{l_2}\Delta_\varepsilon^{l_3}\cdots\Delta_{(-1)^{r-1}\varepsilon}^{l_r},$$

then

$$\tau_{\mathrm{red}} = \Delta_\varepsilon\Delta_{-\varepsilon}\Delta_\varepsilon\cdots\Delta_{(-1)^{r-1}\varepsilon},$$

where r is the useful length of τ (the not useful length being $n = \sum_{i\in[1,r]} l_i$).

[20] This is unnecessary, cf. below. p. 50

When one starts with a homotopy between *functors*

$$X \longrightarrow Y,$$

it is understood that we are dealing with homotopies in the category $\underline{\mathrm{Hom}}(X,Y)$. (Or, depending on the context, in a specified subcategory of $\underline{\mathrm{Hom}}(X,Y)$.)

On page 61, I invoked the existence of a *functor* of *"composition of paths"* [page 77]

$$\underline{\mathrm{Ch}}_{\Sigma}(X;a,b) \times \underline{\mathrm{Ch}}_{\Sigma}(X;b,c) \;\longrightarrow\; \underline{\mathrm{Ch}}_{\Sigma}(X;a,c),$$

which on the fiber of the first member over $(\tau,\tau') \in \mathrm{Ob}(\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}})$, would be given by

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(X;a,b) \times \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X;b,c) \;\longrightarrow\; \underline{\mathrm{Ch}}_{\Sigma,\tau\tau'}(X;a,c), \qquad (*)$$

the composition of paths of specified types τ, τ'. That we have here a *functor,* not just a map on the sets of objects, is fairly obvious, and can be expressed for example with the help of the amalgation diagram

$$\begin{array}{ccccc} & & \{1_\tau = 0_{\tau'}\} & & \\ & \swarrow & & \searrow & \\ I_\tau & & & & I_{\tau'} \\ & \searrow & & \swarrow & \\ & & I_{\tau\tau'} & & \end{array}$$

by transforming it by $\underline{\mathrm{Hom}}(\cdot,X)$ (instead of transforming it by $\mathrm{Hom}_{\mathrm{Cat}}(\cdot,X)$ only), at least in the particular case where $\Sigma = \mathrm{Fl}(X)$ (which was the one on page 61). But the passage to arrows is fairly obvious - for two composites $c_1' \circ c_1$ and $c_2' \circ c_2$, from

$$\begin{array}{ccccc} a & \xrightarrow{c_1} & b & \xrightarrow{c_1'} & c \\ \| & & \| & & \| \\ a & \xrightarrow[c_2]{} & b & \xrightarrow[c_2']{} & c \end{array} \qquad \begin{array}{c} \\ w \\ \Downarrow \\ \\ \end{array} \qquad \begin{array}{c} \\ \\ \Downarrow w' \\ \\ \end{array}$$

sees that the homomorphisms w'' from the first to the second, in $\underline{\mathrm{Ch}}_{\Sigma}(X;a,c)$, which are the identity on the vertex of index n of the two paths (where $n = \log\tau$), corresponding to 1_τ and to $0_{\tau'}$, "are" exactly the pairs formed of a homomorphism $w : c_1 \longrightarrow c_2$ and $w' : c_1' \longrightarrow c_2'$ in $\underline{\mathrm{Ch}}_{\Sigma,\tau}(a,b)$ and $\underline{\mathrm{Ch}}_{\Sigma,\tau'}(b,c)$ respectively, and if w'', w, w' correspond in this way, we shall set [p. 51] [page 78]

$$w'' = w * w'.$$

There is an obvious transitivity to check to see that this defines a *functor* $(*)$. We can express the situation with the help of the diagram

$$\begin{array}{ccccc} & & \underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}} & \xrightarrow{\pi \text{ (composition)}} & \underline{\mathrm{Ty}} \\ & \swarrow^{p_1} & & \searrow_{p_2} & \\ \underline{\mathrm{Ty}} & & & & \underline{\mathrm{Ty}}, \end{array}$$

where p_1, p_2 are the two projections. We consider over $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$ the three fibered categories

$$p_1^*(\underline{\mathrm{CH}}_{\Sigma,\tau}(a,b)), \quad p_2^*(\underline{\mathrm{CH}}_{\Sigma,\tau'}(b,c)), \quad \pi^*(\underline{\mathrm{Ch}}_{\Sigma,\tau\tau'}(a,c))$$

, whose fibers at (τ, τ') are respectively

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(a,b), \quad \underline{\mathrm{Ch}}_{\Sigma,\tau'}(b,c), \quad \underline{\mathrm{Ch}}_{\Sigma,\tau\tau'}(a,c).$$

(We write $\underline{\mathrm{CH}}$ instead of $\underline{\mathrm{Ch}}$, to indicate that we regard the categories under consideration as (fibered) categories *over* ($\underline{\mathrm{Ty}}$) ([21]).) Over $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$, we have a (Cartesian) homomorphism of fibered categories

$$p_1^*(\underline{\mathrm{CH}}_\tau(a,b)) \times_{\underline{\mathrm{Ty}}\times\underline{\mathrm{Ty}}} p_2^*(\underline{\mathrm{CH}}_{\tau'}(b,c)) \longrightarrow \pi^*(\underline{\mathrm{CH}}_{\tau\tau'}(a,c)) \qquad (**)$$

, for it is OK on the fibers ((∗) page 77), and if [page 79]

$$(\varphi, \varphi') : (\tau_1, \tau_1') \longrightarrow (\tau, \tau') \qquad (\varphi : \tau_1 \longrightarrow \tau,\ \varphi' : \tau_1' \longrightarrow \tau')$$

is an arrow in $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$, we deduce from it a commutative diagram

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{\Sigma,\tau}(a,b) \times \underline{\mathrm{Ch}}_{\Sigma,\tau'}(b,c) & \xrightarrow{\text{comp.}} & \underline{\mathrm{Ch}}_{\Sigma,\tau\tau'}(a,c) \\ \downarrow{\scriptstyle \varphi^*_{a,b}\times(\varphi'_{b,c})^*} & & \downarrow{\scriptstyle (\varphi*\varphi')^*_{a,c}} \\ \underline{\mathrm{Ch}}_{\Sigma,\tau_1}(a,b) \times \underline{\mathrm{Ch}}_{\Sigma,\tau_1'}(b,c) & \xrightarrow{\text{comp.}} & \underline{\mathrm{Ch}}_{\Sigma,\tau_1\tau_1'}(a,c) \end{array}$$

, i.e. the composition of paths with fixed endpoints is compatible with the inverse images of paths under arrows between the types of paths. p. 52

But the Cartesian homomorphism (∗∗) of fibered categories over $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$ is not quite the same thing as an innocuous functor

$$\underline{\mathrm{Ch}}_\Sigma(a,b) \times \underline{\mathrm{Ch}}_\Sigma(b,c) \longrightarrow \underline{\mathrm{Ch}}_\Sigma(a,c). \qquad (*)$$

The formula (∗) defines in any case a map on the sets of objects, since the sets of objects of each of the three categories under consideration decompose as a sum of sets $\underline{\mathrm{Ch}}_{\Sigma,\tau}(\bullet,\bullet)$, where τ runs through $\mathrm{Ty} = \mathrm{Ob}\,\underline{\mathrm{Ty}}$. But does this extend in a natural way to morphisms?

But now I remember that this famous Cartesian product $p_1^*(\textit{fourbi}) \times p_2^*(\text{thingamajig})$ is none other than the plain old Cartesian product [page 80]

$$\underline{\mathrm{Ch}}_\Sigma(a,b) \times \underline{\mathrm{Ch}}_\Sigma(b,c),$$

sent to $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$ by the Cartesian product of the two projection functors

$$\begin{aligned} p_{a,b} : \underline{\mathrm{Ch}}_\Sigma(a,b) &\longrightarrow \underline{\mathrm{Ty}} \\ p_{b,c} : \underline{\mathrm{Ch}}_\Sigma(b,c) &\longrightarrow \underline{\mathrm{Ty}}, \end{aligned}$$

hence

$$\begin{array}{c} \underline{\mathrm{Ch}}_\Sigma(a,b) \times \underline{\mathrm{Ch}}_\Sigma(b,c) \\ \downarrow{\scriptstyle p_{a,b}\times p_{b,c}} \\ \underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}} \end{array}$$

which is well and truly fibrant (as a Cartesian product of two fibrant functors) and has the right fibers! So we do find, by the general sorites on fibrant functors (here over $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$) a functor of fibered categories over $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$ [page 81]

$$\underline{\mathrm{Ch}}_\Sigma(a,b) \times \underline{\mathrm{Ch}}_\Sigma(b,c) \longrightarrow \underbrace{\pi^*(\underline{\mathrm{CH}}_\Sigma(a,c))}_{=(\underline{\mathrm{Ty}}\times\underline{\mathrm{Ty}})\times_{\underline{\mathrm{Ty}}}\underline{\mathrm{Ch}}_\Sigma(a,c)\ ([22])} .$$

p. 53

Since on the other hand we have a functor from the last member (fiber product) to the factor $\underline{\mathrm{Ch}}_\Sigma(a,c)$, we win. (We would have found it again by the direct argument I was about to make, and which I crossed out - but I am happier this way!)

To do things properly, we would have to prove the *associativity* of the composition functors, not just at the level of objects, by arguments like this, but it comes down simply to a relation of the type

$$(w'' * w') * w = w'' * (w' * w)$$

for a system of three arrows in

$$\underline{\mathrm{Ch}}_\Sigma(a,b),\ \underline{\mathrm{Ch}}_\Sigma(b,c)\ \text{ and }\ \underline{\mathrm{Ch}}_\Sigma(c,d)$$

respectively. It can only be trivially true, but tedious to spell out. [page 82]

The cold sweats are over!

In summary, the formalism of the $\underline{\mathrm{Ch}}_\Sigma(X)$ and of the $\underline{\mathrm{Ch}}_\Sigma(X;a,b)$ developed up to now gives complete satisfaction on the following points:

a) Good formalism for the composition of paths (which was disgusting before).

b) Elegant treatment of the construction of categories of fractions $\mathcal{M}\Sigma^{-1}$. But I expect that it will be less pretty than before, when it comes to replacing an $(\mathcal{M},\Sigma)$ by $(\underbrace{\mathcal{M}(X)}_{=\underline{\mathrm{Hom}}(X^o,\mathcal{M})}, \Sigma_X)$'s ($X$ an index category) . It is to remedy this that I will have to develop comparison statements, between the category $\underline{\mathrm{Ch}}_\Sigma$ constructed here, and others whose type in Hot lends itself to computations in a more direct way, as in the original state of the question. Moreover, we will surely have to turn to Ind-categories, in order to be able to follow the variance of $\mathcal{M}(X)\Sigma_X^{-1}$ for variable X.

c) Good formalism for "homotopies" and for compositions of homotopies (which is another way of speaking of paths and of their composition).

p. 54
[page 83]

8 The homotopy lemma for morphisms of intervals

It is the following:

Lemma 1. *Let $\tau,\tau' \in \mathrm{Ty}$, and let $f,g : \tau \rightrightarrows \tau'$ be two morphisms in $\underline{\mathrm{Ty}}$. Then f, g are homotopic in the category $\underline{\mathrm{Hom}}\,\underline{\mathrm{int}}(I_\tau, I_{\tau'})$ of increasing morphisms of oriented intervals from I_τ to $I_{\tau'}$.*

This category could legitimately be denoted by $\underline{\mathrm{Hom}}_{\underline{\mathrm{Ty}}}(\tau,\tau')$, or $\underline{\mathrm{Hom}}(\tau,\tau')$. It is a subcategory of the category $\underline{\mathrm{Hom}}(I_\tau, I_{\tau'})$ of *all* functors from I_τ to $I_{\tau'}$, and a full subcategory of the category $\underline{\mathrm{Hom}}^\varepsilon(I_\tau, I_{\tau'})$ of functors f that are compatible with the datum of the sources and targets, i.e. that make the diagram commute

$$\begin{array}{ccc} & \varepsilon = \{0,1\} & \\ {\scriptstyle (\sigma_\tau,\beta_\tau)}\swarrow & & \searrow{\scriptstyle (\sigma_{\tau'},\beta_{\tau'})} \\ I_\tau & \xrightarrow{\quad f \quad} & I_{\tau'}. \end{array}$$

[21] But for p_1^*, p_2^* and π^* these are ordinary Cartesian products in Cat, there is no need to be afraid and to write $\underline{\mathrm{CH}}$'s …

[22] fiber product along $\pi : \underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}} \longrightarrow \underline{\mathrm{Ty}}$.

If f, g are in this category (not necessarily increasing morphisms, by the way), an arrow from f to g is thus the datum, for every $i \in I_\tau$, of an arrow $f(i) \xrightarrow{w_i} g(i)$ in $I_{\tau'}$, with the two conditions:

a) If $i \in \{0_\tau, 1_\tau\}$, $w_i = \mathrm{id}$, and

b) the usual commutative square for an arrow $i \longrightarrow j$ in I_τ. (It suffices to have this compatibility for the transition arrows $\underline{u}_i : i-1 \longrightarrow i$ if $\tau_i = +$, or $\underline{u}_i : i \longrightarrow i-1$ if $\tau_i = -$, in I_τ ($i \in [1, \log \tau]$).)

Another way of describing this category $\underline{\mathrm{Hom}}^\varepsilon$ is to say that it is the category [page 84]

$$\underline{\mathrm{Ch}}_\tau(I_{\tau'}; 0_{\tau'}, 1_{\tau'}) \qquad (\simeq \underline{\mathrm{Hom}}^\varepsilon(I_\tau, I_{\tau'})),$$

but $\underline{\mathrm{Hom}}_{\mathrm{Ty}}(\tau, \tau')$ is the full subcategory of this category consisting of the *increasing paths* in $I_{\tau'}$. The homotopy lemma also means that this category $\underline{\mathrm{Hom}}_{\mathrm{Ty}}(\tau, \tau')$ is always *connected.* It is not necessarily 0-connected, however, for it can be empty. Let us note here, for later, the

Proposition 1: *Let $\tau, \tau' \in \mathrm{Ty}$, let r, r' be their useful lengths. For $\mathrm{Hom}_{\mathrm{Ty}}(\tau, \tau')$ to be nonempty, it is necessary and sufficient that one or the other of the following two conditions be satisfied:*

a) $r \geq r' + 1$

b) $r = r'$, and (if $r \neq 0$) τ and τ' have the same initial direction (so that $\tau_{\mathrm{red}} = \tau'_{\mathrm{red}}$). p. 55

(Proof ± immediate.)

Let us note that if τ, τ' do not have the same initial direction, then every morphism $\tau \longrightarrow \tau'$ is necessarily degenerate on the first critical interval of I_τ. We will thus have $\tau = \Delta_\varepsilon^{l_0} \cdot \tau_0$ (where l_0 is the length of the first critical interval, and where τ_0 has initial direction opposite to that of τ), and an isomorphism

$$\underline{\mathrm{Hom}}(\tau, \tau') \simeq \underline{\mathrm{Hom}}(\tau_0, \tau'),$$

where this time τ_0, τ' have the same initial direction. We will therefore assume henceforth that *the initial direction of τ, τ' is the same.* The condition $\underline{\mathrm{Hom}}(\tau, \tau') \neq \emptyset$ is now equivalent to $r \geq r'$, which we can assume. I will then describe a canonical element F_0 of $\underline{\mathrm{Hom}}(\tau, \tau')$, as follows. Let [page 85]

$$I_1, I_2, \ldots, I_r$$

be the critical intervals of I_τ, and

$$I'_1, I'_2, \ldots, I'_{r'} \qquad (r' \leq r)$$

those of $I_{\tau'}$. The functor F_0 will come from morphisms of oriented intervals (such as

$$\left\{ \begin{array}{lcl} I_1 & \xrightarrow{f_1} & I'_1 \\ I_2 & \xrightarrow{f_2} & I'_2 \\ & \cdots & \\ I_{r'} & \xrightarrow{f_{r'}} & I'_{r'} \end{array} \right. \quad)$$

and will therefore necessarily be trivial on the critical intervals I_α for $r' < \alpha \leq r$ (there are some if and only if $r > r'$). Since I_τ, $I_{\tau'}$ have the same initial direction, it follows

that for $\alpha \in [1, r']$, I_α and I'_α have the same signature (namely $\varepsilon_\alpha = (-1)^{\alpha+1}\varepsilon_1$, where ε_1 is the common signature of I_1 and I'_1). Thus

$$I_\alpha = \Delta^{l_\alpha}_{\varepsilon_\alpha}, \qquad I'_\alpha = \Delta^{l'_\alpha}_{\varepsilon_\alpha},$$

and we define f_α as

$$f_\alpha = \mathfrak{f}_{\varepsilon_\alpha, l'_\alpha} * \mathrm{triv}_\alpha,$$

where

$$\mathfrak{f}_{\varepsilon_\alpha, l'_\alpha} : \Delta_{\varepsilon_\alpha} \longrightarrow \Delta^{l'_\alpha}_{\varepsilon_\alpha}$$

and where

$$\mathrm{triv}_\alpha : \Delta^{l_\alpha - 1}_{\varepsilon_\alpha} \longrightarrow 1.$$

In other words, for $\varepsilon = +1$ (e.g.) we have to define a morphism of the standard intervals $\underbrace{\mathbf{\Delta}^+_{l_\alpha}}_{=[0,l_\alpha]} \longrightarrow \underbrace{\mathbf{\Delta}^+_{l'_\alpha}}_{=[0,l'_\alpha]}$, and we set (with the numberings internal to $\mathbf{\Delta}_{l_\alpha}$, $\mathbf{\Delta}_{l'_\alpha}$)

$$f_\alpha(0) = 0, \qquad f_\alpha(i) = l'_\alpha \ \text{ for } i \in [1, l_\alpha].$$

The idea then, is to be in the frame of mind of a traveler who, bound to a travel plan in stages described by τ, would like to arrive as quickly as possible at $1_{\tau'}$, starting from $0_{\tau'}$. He will, from the very first stage $\Delta_\varepsilon \subseteq I_1$, go to the end of I'_1, regretting not being able to go farther (because of the travel plan). He has to remain inactive until he has reached the moment marked by the final endpoint of I_1 (if this is beyond that of his first transition arrow, i.e. if $l_1 > 1$), then he takes his second big step to the terminal endpoint of I'_2 etc. He can hope, at each moment of his journey, to be ahead of (or at least not behind) the other travelers, who might have followed different tactics for advancing. If this hope were justified, that would mean that F_0 would be a greatest element of $\underline{\mathrm{Hom}}(\tau, \tau')$ (**NB** all these categories $\underline{\mathrm{Hom}}$ between intervals are "ordered", i.e. there is at most one arrow between two objects), i.e. a final object. This would imply that $\underline{\mathrm{Hom}}(\tau, \tau')$ would even be contractible - something perhaps true, but surely not in so simplistic a way I fear. I have botched it, by the way: F_0 probably is indeed a "greatest" element, but for the order defined by the numbering of the vertices of $I_{\tau'}$ (so $F \leq G$ if and only if $F(i) \leq G(i)\ \forall i \in I_\tau$), and not at all for the order defined by the category structure of $\underline{\mathrm{Hom}}(\tau, \tau')$.

p. 56 [page 86]

The idea for proving the connectedness of $\underline{\mathrm{Hom}}(\tau, \tau')$ is to bring any $F \in \mathrm{Hom}(\tau, \tau')$, by a sequence of simple homotopies, to the normal form. Let us recall that F is given by an increasing sequence of transition arrows

[page 87]

$$\underline{u}_{i_1},\ \underline{u}_{i_2},\ \ldots\ ,\ \underline{u}_{i_s}, \qquad i_1 < i_2 < \cdots < i_s,$$

and a sequence of integers ≥ 1

$$\lambda_1,\ \lambda_2,\ \ldots\ ,\ \lambda_s$$

with the condition that

$$\Delta^{\lambda_1}_{\varepsilon_{i_1}} \Delta^{\lambda_2}_{\varepsilon_{i_2}} \cdots \Delta^{\lambda_s}_{\varepsilon_{i_s}} = \tau'. \qquad (*)$$

The $\underline{u}_{i_\alpha}$ are the transition arrows of I_τ *critical* for F, i.e. nondegenerate for F, and λ_α is the *part* of $\underline{u}_{i_\alpha}$ for F. The normalized ("hurried") functor F_0 is distinguished by the following conditions a) and b) . First of all:

a) The sequence of signs $\varepsilon_{i_1}, \varepsilon_{i_2}, \ldots, \varepsilon_{i_s}$ is "alternating", i.e. $\varepsilon_{i_\alpha} \neq \varepsilon_{i_{\alpha+1}}$ $(\alpha \in [1, s-1])$. (This expresses the fact that the hurried traveler does not do in several steps what he could do in a single one.)

Since the first of these signs is necessarily $\varepsilon'_1 = \varepsilon_1$, this means that the sequence of signs is the same as that of the ε_1, ε_2, ..., $\varepsilon_{r'}$ (apart from the length). But better still, that the expression $(*)$ for τ' is its unique reduced expression, hence that $s = r'$ (useful length of τ'), and that the λ_i are the lengths l'_i of the critical intervals of $I_{\tau'}$.

p. 57
[page 88]

Let then $\alpha \in [1, s = r']$, and let us consider the F-critical transition arrow $\underline{u}_{i_\alpha}$, let j_α be the index marking the critical interval of I_τ containing the part of $\underline{u}_{i_\alpha}$ in I_τ,

$$\underline{u}_{i_\alpha} \in \mathrm{Fl}(I_{j_\alpha}),$$

then F induces (increasing) morphisms of oriented intervals

$$F_\alpha = F|I_{j_\alpha} : I_{j_\alpha} \longrightarrow I'_\alpha$$

and is trivial on the other critical intervals (those not of the form I_{j_α}). The datum of the F_α thus determines F on the I_{j_α}, hence on *all* the I_j, hence everywhere (by the operation $*$ of composition by gluing). Moreover condition a) ensures that each F_α is of a rather special nature: a single critical transition arrow $\underline{u}_{i_\alpha}$. Moreover, for us to have $F = F_0$, it will be necessary:

b) $\forall \alpha \in [1, r']$, $\underline{u}_{i_\alpha}$ is the *first* transition arrow of the critical interval I_{j_α} that contains it.

And finally:

c) The sequence of critical intervals of I_τ

$$I_{j_1},\ I_{j_2},\ \ldots\ ,\ I_{j_{r'}} \qquad \text{(of alternating signs } \varepsilon_1,\ -\varepsilon_1,\ \varepsilon_1,\ \ldots)$$

associated with the F-critical arrows, is none other than that

$$I_1,\ I_2,\ \ldots\ ,\ I_{r'}$$

of the first r' critical intervals of I_τ, i.e.

$$(j_1, j_2, \ldots, j_{r'}) = (1, 2, \ldots, r').$$

[page 89]

This being seen, the natural idea for "deforming" F into F_0, is to find a sequence of homotopies that transform F so as to make it acquire successively the characteristics a), b), c) of the hurried traveler. If need be, to get to the goal faster, we will use the induction hypothesis on $n = \log \tau$, that the connectedness theorem is already proved when I is of length $\leq n-1$. (NB This theorem is trivial for $n \leq 1$, for then the categories $\underline{\mathrm{Hom}}(\tau, \tau')$ are all either empty, or one-point.)

[page 90]

1) We want to ensure a). For this, we use the induction hypothesis, to replace a group of [...] consecutive critical arrows of the same sign, by a single critical arrow. This method only breaks down if the critical arrows are all of the same sign ε and include $\underline{u}_1$ and $\underline{u}_n$. But the induction hypothesis applied to $I = [0, n-1] \subseteq I_\tau$ reduces us to the case where there are only two critical arrows $\underline{u}_1$ and $\underline{u}_n$. If the first critical interval is not reduced to Δ_ε, i.e. is of length at least 2, then the induction hypothesis applied to $I = [1, n] \subseteq I_\tau$ reduces us to the case where the two critical arrows are $\underline{u}_1$ and $\underline{u}_2$, and we are then reduced to the case where $\boxed{I = \Delta^2_\varepsilon,\ \tau' = \Delta^2_\varepsilon.}$ Otherwise, to the case where

p. 58
[page 91]

$\boxed{\tau = \Delta_\varepsilon \Delta^{l_2}_{-\varepsilon} \Delta_\varepsilon,\ \tau' = \Delta^2_\varepsilon,}$ and where the two critical arrows are again the first and the last, and $\tau' = \Delta^2_\varepsilon$. Those are the two irreducible cases, which will have to be treated separately.

These cases granted, we thus have a way to deform F so as to satisfy a). Next we want:

2) To ensure that the sequence of critical intervals I_{j_1}, I_{j_2}, ..., $I_{j_{r'}}$ which contains the critical arrows $\underline{u}_{i_1}$, ..., $\underline{u}_{i_{r'}}$, is the sequence I_1, ..., $I_{r'}$. This is again immediate by the induction hypothesis, the only irreducible case being that where $r' = 1$ (a single critical arrow), and where it is a matter of bringing this critical arrow down to $\underline{u}_1$. If I_1 is of length > 1, then we bring it down to $\underline{u}_2$ by the induction hypothesis, and the irreducible case is that where $\boxed{\tau = \Delta_\varepsilon^2,\ \tau' = \Delta_\varepsilon,}$ a critical arrow $\underline{u}_2$, bring it down to $\underline{u}_1$. If on the other hand I_1 of length 1, the irreducible case is $\boxed{\tau = \Delta_\varepsilon \Delta_{-\varepsilon}^{l_2} \Delta_\varepsilon,\ \tau' = \Delta_\varepsilon,}$ a critical arrow $\underline{u}_{l_2+2}$, bring it down to $\underline{u}_1$.

Subject to treating the two new irreducible cases, we can deform F into a functor that satisfies a) and c). It remains to normalize each of the $F_\alpha : I_\alpha \longrightarrow I'_\alpha$ ($\alpha \in [1, r']$). Thus we are reduced to $\Delta_\varepsilon^n \longrightarrow \Delta_\varepsilon$: a single critical arrow $\underline{u}_\alpha$, $\alpha \in [1, n]$, to be brought down to $\underline{u}_1$. If $n \geq 2$, by the induction hypothesis it comes down to $\underline{u}_2$, case assumed treated in 2°). So we win, subject to treating the 4 irreducible cases. [page 92]

A) Case $\tau = \Delta_\varepsilon^2$, $\tau' = \Delta_\varepsilon^2$ or Δ_ε.

The category $\underline{\mathrm{Hom}}(\Delta_\varepsilon^2, \Delta_\varepsilon)$ has two elements, the morphisms $\Delta_\varepsilon^2 \longrightarrow \Delta_\varepsilon$ correspond to the transition arrows of Δ_ε^2 ($= \boldsymbol{\Delta}_2^+$ or $\boldsymbol{\Delta}_2^-$). In the case $\varepsilon = +1$, these are the two maps from $\{0, 1, 2\}$ to $\{0, 1\}$, which map 0 to 0, 2 to 1, and 1 either to 1 (F_0), or to 0 (F_1).

$$0 \longrightarrow 1 \longrightarrow 2$$

$$0 \longrightarrow 1$$

We obviously have an arrow $F_1 \longrightarrow F_0$ (the hurried traveler arrives faster ...). Similar for $\varepsilon = -1$,

$$\underline{\mathrm{Hom}}(\Delta_\varepsilon^2, \Delta_\varepsilon) \simeq \boldsymbol{\Delta}^1, \qquad \text{0-connected category.}$$

The category $\underline{\mathrm{Hom}}(\Delta_\varepsilon^2, \Delta_\varepsilon^2)$ has three elements, identified with the set of nonempty subsets of the two-element set $\mathrm{Tr}(\tau)$ of transition arrows $\underline{u}_1$, $\underline{u}_2$ of τ: there are $F_{\underline{u}_1} = F_0$, $F_{\underline{u}_2}$, $F_{\underline{u}_1, \underline{u}_2}$. If $\varepsilon = +1$, these are the increasing maps (nothing more) from $\{0, 1, 2\}$ to $\{0, 1, 2\}$, mapping 0 to 0, 2 to 2, and 1 to whatever it likes.

Thus $1 \longmapsto 2$ corresponds to F_0 p. 59

$1 \longmapsto 1$ corresponds to F_1

$1 \longmapsto 0$ corresponds to F_2

and of course we have $F_2 \longrightarrow F_1 \longrightarrow F_0$,

$$\underline{\mathrm{Hom}}(\Delta_\varepsilon^2, \Delta_\varepsilon^2) \simeq \boldsymbol{\Delta}^2, \qquad \text{0-connected category.}$$

In these two cases, two elements of $\mathrm{Hom}(\tau, \tau')$ are joined by a homotopy which is *simple.*

B) Case $\tau = \Delta_\varepsilon \Delta_{-\varepsilon}^l \Delta_\varepsilon$, $\tau' = \Delta_\varepsilon^2$ or Δ_ε. [page 93]

Let us note that we have, for arbitrary $\tau, \tau' \in \underline{\mathrm{Ty}}$, an isomorphism of categories

$$\underline{\mathrm{Hom}}(\tau, \tau') \longrightarrow \underline{\mathrm{Hom}}(\tau^o, \tau'^o),$$

simply expressing that

$$\tau \longmapsto \tau^o$$

is a -functor from $\underline{\mathrm{Ty}}$ to $\underline{\mathrm{Ty}}$, and even an automorphism of this -category. Thus to say that $\underline{\mathrm{Hom}}(\tau, \tau')$ is connected amounts to saying that $\underline{\mathrm{Hom}}(\tau^o, \tau'^o)$ is. Thus we can suppose

above that $\varepsilon = +1$ (and the same in A!).

$$\tau : \quad \longrightarrow \underbrace{\longleftarrow \longleftarrow \cdots \longleftarrow}_{l} \longrightarrow$$
$$\tau' : \quad \longrightarrow \longrightarrow \quad \text{or} \quad \longrightarrow .$$

But all the morphisms $\tau \longrightarrow \tau'$ are trivial on the critical interval I_2 (retrograde), which implies that

$$\underline{\mathrm{Hom}}(\tau, \tau') \xleftarrow{\sim} \underline{\mathrm{Hom}}(\Delta^2_\varepsilon, \tau'),$$

via the canonical contraction functor

$$I_\tau \longrightarrow I_{\Delta^2_\varepsilon} = \mathbf{\Delta}^2.$$

Now in A we saw that $\underline{\mathrm{Hom}}(\Delta^2_\varepsilon, \tau')$ is connected (for τ' of the form Δ_ε or Δ^2_ε).
We've won!

p. 60
[page 94]

9 Homotopy theorem for categories of paths

Let X be a category, $\Sigma \subseteq \mathrm{Fl}(X)$ admissible set of arrows in X, $a, b \in \mathrm{Ob}\, X$. Let us consider for variable $\tau \in \mathrm{Ty}$, the categories

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(X), \qquad \underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b),$$

which form the fibers of the Cat-fibered categories

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{\Sigma}(X) & \xleftarrow{\text{incl.}} & \underline{\mathrm{Ch}}_{\Sigma}(X; a, b) \\ & \searrow \quad \swarrow & \\ & \underline{\mathrm{Ty}} & \end{array}$$

over the category $\underline{\mathrm{Ty}}$. Thus, for $\tau, \tau' \in \mathrm{Ty}$, and for any

$$\varphi : \tau \longrightarrow \tau',$$

we have a commutative square of homomorphisms

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X) & \xleftarrow{\text{incl.}} & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X; a, b) \\ {\scriptstyle \varphi^*_X}\downarrow & & \downarrow{\scriptstyle \varphi^*_{a,b}} \\ \underline{\mathrm{Ch}}_{\Sigma,\tau}(X) & \xleftarrow{\text{incl.}} & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b), \end{array}$$

with transitivity for variable φ (composition $\psi \circ \varphi$ of arrows in $\underline{\mathrm{Ty}}$).

In fact, for φ variable in $\underline{\mathrm{Hom}}(\tau, \tau')$, the inverse image functors φ^*_X, $\varphi^*_{a,b}$ depend *functorially* on φ; i.e. we have *functors*

$$\underline{\mathrm{Hom}}(\tau, \tau') \longrightarrow \begin{array}{l} \underline{\mathrm{Hom}}(\underline{\mathrm{Ch}}_{\Sigma,\tau'}(X), \underline{\mathrm{Ch}}_{\Sigma,\tau}(X)) \\ \underline{\mathrm{Hom}}(\underline{\mathrm{Ch}}_{\Sigma,\tau'}(X; a, b), \underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b)) \end{array} \qquad (*)$$

p. 61
[page 95]

This comes from the inclusion

$$\underline{\mathrm{Hom}}(\tau,\tau') \hookrightarrow \underline{\mathrm{Hom}}^{\varepsilon}(I_\tau, I_{\tau'})$$

and from the description of the categories $\underline{\mathrm{Ch}}_\tau$, $\underline{\mathrm{Ch}}_{\tau'}$ as categories of functors, such as

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(X) \simeq \underline{\mathrm{Hom}}_{\Sigma_\tau,\Sigma}(I_\tau, X),$$

where Σ_τ denotes the set of arrows in I_τ which are either invertible or *retrograde.* (This set of arrows is admissible, since stable under composition.) We denote by $\underline{\mathrm{Hom}}_{\Sigma_\tau,\Sigma}$ the full subcategory of $\underline{\mathrm{Hom}}$ formed by the functors which map Σ_τ into Σ. Now in general, if

$$(I,\Sigma_I),\quad (I',\Sigma_{I'}),\quad (X,\Sigma)$$

are three categories, equipped with admissible sets of arrows, considering the functors $\varphi : I \longrightarrow I'$ with $\varphi(\Sigma_I) \subseteq \Sigma_{I'}$, then the familiar functor

$$\underline{\mathrm{Hom}}(I,I') \longrightarrow \underline{\mathrm{Hom}}(\underline{\mathrm{Hom}}(I',X), \underline{\mathrm{Hom}}(I,X))$$

(transpose of the composition functor

$$\begin{array}{ccccc} \underline{\mathrm{Hom}}(I,I') & \times & \underline{\mathrm{Hom}}(I',X) & \longrightarrow & \underline{\mathrm{Hom}}(I,X) \\ (\varphi & , & F') & \longmapsto & F'\circ\varphi \quad), \end{array}$$

generalizes to a functor

$$\underline{\mathrm{Hom}}_{\Sigma_I,\Sigma_{I'}}(I,I') \longrightarrow \underline{\mathrm{Hom}}(\underline{\mathrm{Hom}}_{\Sigma_{I'},\Sigma}(I',X), \underline{\mathrm{Hom}}_{\Sigma_I,\Sigma}(I,X)), \qquad (*)$$

transpose of the composition functor

$$\underline{\mathrm{Hom}}_{\Sigma_I,\Sigma_{I'}}(I,I') \times \underline{\mathrm{Hom}}_{\Sigma_{I'},\Sigma}(I',X) \longrightarrow \underline{\mathrm{Hom}}_{\Sigma_I,\Sigma}(I,X)$$

induced by $(*)$.

Since every

$$\varphi : \tau \longrightarrow \tau' \qquad \texttt{[page 96]}$$

induces

$$\varphi : I_\tau \longrightarrow I_{\tau'},$$

transforming retrograde arrows into identity or retrograde arrows, i.e.

$$\varphi(\Sigma_\tau) \subseteq \Sigma_{\tau'},$$

we find the first of the two functors $(*)$ announced on page 94, which is associated by transposition with the functor γ

p. 62

$$\begin{array}{ccc} \underline{\mathrm{Hom}}(\tau,\tau') \times \underbrace{\underline{\mathrm{Ch}}_{\Sigma,\tau'}(X)}_{\simeq \underline{\mathrm{Hom}}_{\Sigma_{\tau'},\Sigma}(I_{\tau'},X)} & \xrightarrow{\quad\gamma\quad} & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X) \\ \Big\downarrow{\scriptstyle \text{can.}} & & \Big\| \\ \underline{\mathrm{Hom}}_{\Sigma_\tau,\Sigma_{\tau'}}(I_\tau, I_{\tau'}) \times \underline{\mathrm{Hom}}_{\Sigma_{\tau'},\Sigma}(I_{\tau'},X) & \xrightarrow{\quad\gamma_0\quad} & \underline{\mathrm{Hom}}_{\Sigma_\tau,\Sigma}(I_\tau,X) \end{array} \qquad (*)$$

deduced from the composition functor γ_0 in this diagram: $\gamma = \gamma_0 \circ \mathrm{can}$. On the other hand, this composition functor γ visibly induces a functor on the subcategories of paths with fixed source and target

$$\underline{\mathrm{Hom}}(\tau,\tau') \times \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X;a,b) \longrightarrow \underline{\mathrm{Ch}}_{\Sigma,\tau}(X;a,b), \qquad \texttt{[page 97]}$$

whence the second of the two arrows announced in (∗) p. 94.

This therefore shows that if

$$\varphi_1, \varphi_2 : \tau \rightrightarrows \tau'$$

are two morphisms from τ to τ' homotopic in $\underline{\mathrm{Hom}}(\tau, \tau')$, then the induced functors between categories of Σ-paths or between categories of Σ-paths with specified endpoints:

$$\begin{array}{rrcl} \varphi_{1\,X}^*, \varphi_{2\,X}^* : & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X) & \rightrightarrows & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X), \\ (\varphi_1)_{a,b}^*, (\varphi_2)_{a,b}^* : & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X; a, b) & \rightrightarrows & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b) \end{array}$$

are also homotopic.

Now the homotopy lemma assures us that the hypothesis $\varphi_1 \sim \varphi_2$ is *always* satisfied. Thus:

Homotopy theorem on the universal operations on paths. *Let X, Σ, a, b be as above, and $\tau, \tau' \in \mathrm{Ty}$ such that $\mathrm{Hom}(\tau, \tau') \neq \emptyset$. Then the functors φ_X^*, $\varphi_{X;a,b}^*$ induced by a $\varphi \in \mathrm{Hom}(\tau, \tau')$*

$$\begin{cases} \varphi_X^* : & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X) & \longrightarrow & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X) \\ \varphi_{X;a,b}^* : & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X; a, b) & \longrightarrow & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b) \end{cases}$$

do not depend on φ, up to homotopy.

We can make this statement more precise, in such a way that the statement for the $\varphi_{X;a,b}^*$ appears, as it should, as a corollary to the one on φ_X^*. For this, we are going to look at the categories of paths in X as categories *over* $X \times X$, by the "source and target" functors:

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{\Sigma,\tau}(X) & & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X) \\ & {\scriptstyle (\sigma_\tau, \beta_\tau)} \searrow \quad \swarrow {\scriptstyle (\sigma_{\tau'}, \beta_{\tau'})} & \\ & X \times X & \end{array}$$

Then the functor (∗) (p. 94) factors through the subcategory $\underline{\mathrm{Hom}}_{\mathrm{Cat}/X\times X}(\cdot, \cdot)$ of $\underline{\mathrm{Hom}}_{\mathrm{Cat}}(\cdot, \cdot)$ formed by the functors which commute with the projection functors to $X \times X$, and the homomorphisms between such functors which are effected by "vertical" arrows of the target category (i.e. by arrows over identity arrows of the base category $X \times X$): p. 63 [page 98]

$$\underline{\mathrm{Hom}}(\tau, \tau') \longrightarrow \underline{\mathrm{Hom}}_{\mathrm{Cat}/X\times X}(\underline{\mathrm{Ch}}_{\Sigma,\tau'}(X), \underline{\mathrm{Ch}}_{\Sigma,\tau}(X)).$$

Thus in the statement of the homotopy theorem, we can specify that the homotopy takes place in this subcategory, i.e. is effected by a sequence of simple homotopies which are morphisms of functors between categories over $X \times X$. This then implies the homotopy statement on the $\varphi_{X;a,b}^*$.

p. 64 [page 99]

10 Comparison question for Cat-fibered categories satisfying the "homotopy theorem" (filtering base)

We are now going to generalize to the case of a (Cat)-fibered category over another

$$\begin{array}{c} C \\ {\scriptstyle \pi}\downarrow \\ T, \end{array}$$

thinking mainly of the case $T = \underline{\mathrm{Ty}}$, $C = \underline{\mathrm{Ch}}_{\Sigma}(X; a, b)$. The essential hypothesis is the following:

(Homotopy hypothesis) If $t, t' \in \mathrm{Ob}\, T$ are such that $\mathrm{Hom}(t, t') \neq \emptyset$, then the functors $\varphi^* : C_{t'} \longrightarrow C_t$ associated to the $\varphi \in \mathrm{Hom}(t, t')$ are homotopic.

They therefore define one and the same arrow in Cat hot. I denote by T_{ord} or T_{o} the set $\mathrm{Ob}\,(T)$, with as order relation the one deduced from the category structure:

$$t \leq t' \qquad \Longleftrightarrow \qquad \mathrm{Hom}_T(t, t') \neq \emptyset,$$

so that we have a functor

$$T \longrightarrow T_{\mathrm{o}}$$

which is *universal* for functors from T into "ordered" categories (i.e. where always $\#\mathrm{Hom}(a, b) \leq 1$) or for functors F on T such that F is constant on every $\mathrm{Hom}(t, t')$. The homotopy hypothesis can be expressed by saying that the composite functor

$$T^o \longrightarrow \mathrm{Cat} \longrightarrow \mathrm{Cat\,hot}$$

$$t \longmapsto C_t$$

factors (obviously in a unique way) as a functor [page 100]

$$H_C : T_{\mathrm{o}}^o \longrightarrow \mathrm{Cat\,hot},$$

i.e. that we have a commutative diagram p. 65

$$\begin{array}{ccc} T^o & \xrightarrow{t \longmapsto C_t} & \mathrm{Cat} \\ \downarrow & & \downarrow \\ T_{\mathrm{o}}^o & \xrightarrow{H_C} & \mathrm{Cat\,hot}. \end{array}$$

We are going to assume in addition:

(Filtering hypothesis) The ordered set T_{o} associated to T is downward *filtering* (left filtering), i.e. if $t, t' \in \mathrm{Ob}\, T$, there exist $t'' \in \mathrm{Ob}\, T$ and arrows $t'' \longrightarrow t$, $t'' \longrightarrow t'$.

This hypothesis is indeed satisfied in $\underline{\mathrm{Ty}}$, since for $\tau, \tau' \in \mathrm{Ob}\,\underline{\mathrm{Ty}} = \mathrm{Ty}$, it suffices to take $\tau\tau'$, which maps to τ by $\mathrm{id}_\tau * p_{\tau'}$, to τ' by $p_\tau * \mathrm{id}_{\tau'}$, where $p_\tau : \tau \longrightarrow 1$ is the morphism from τ to the final object.

Thus, H_C defines an ind-object of Cat hot, which we are going to denote by $\underline{\underline{C}}$. If we assume in addition:

(Countability hypothesis, always satisfied if $\mathrm{card}\, T_{\mathrm{o}} \leq \aleph_0$) T_{o} has a *countable* subset cofinal on the left.

then this ind-object of Cat hot can be defined, in many ways, in terms of a direct system in Cat (not just in Cat hot) [page 101]

$$(C_n)_{n \in \mathbf{N}},$$

from which we recover the ind-object $\underline{\underline{C}}$ in Cat hot by passing to the corresponding direct system in Cat hot, then to the associated ind-object.

To construct the direct system C_n, it suffices to take an infinite chain

$$t_0 \longleftarrow t_1 \longleftarrow t_2 \longleftarrow \cdots \longleftarrow t_n \longleftarrow t_{n+1} \longleftarrow \cdots$$

of morphisms in T, such that the set of the t_i is cofinal on the left in T_o. We are going to assume in addition that C is *strictly fibered* over T (i.e. coming from a genuine functor $T^o \longrightarrow \mathrm{Cat}$, i.e. we can choose the inverse image functors

$$\varphi^* : C_{t'} \longrightarrow C_t$$

in a strictly transitive way, not just transitive up to isomorphism) - a case to which we can always reduce, moreover, by the trick of GIRAUD. This hypothesis is moreover verified in the case $\underline{\mathrm{Ch}}_\Sigma(X; a, b)$ over $\underline{\mathrm{Ty}}$. Thus, the

$$(C_{t_i})_{i\in\mathbf{N}}$$

form a direct system of categories indexed by $\mathbf{N}$, the transition morphism

$$C_n \longrightarrow C_{n'} \qquad \text{i.e. } C_{t_n} \longrightarrow C_{t_{n'}}$$

being the inverse image functor $\varphi^*_{n,n'}$, where

p. 66
[page 102]

$$\varphi_{n,n'} : t_{n'} \longrightarrow t_n$$

is the composite of the $t_{n'} \longrightarrow t_{n'-1} \longrightarrow \cdots \longrightarrow t_n$. (If we did not have strict transitivity, we would define $\varphi^*_{n,n'}$ as the composite of the inverse image functors along successive transition arrows $t_{n'} \longrightarrow t_{n'-1} \longrightarrow \cdots \longrightarrow t_{n+1} \longrightarrow t_n$, this would be $\varphi^*_{n,n'}$ only up to canonical isomorphism.)

Let

$$C_\infty = \varinjlim_n C_n = \varinjlim_n C_{t_n}$$

be the category $\underline{\varinjlim}$. (It will be especially nice if we have been able to choose the system $(t_n)_{n\in\mathbf{N}}$ in such a way that the transition functors

$$C_{t_{n-1}} \longrightarrow C_{t_n}$$

are fully faithful, and perhaps even monomorphisms, in such a way that the C_n identify with full subcategories of their direct limit C_∞.) I would like to understand the precise relations between the following three kinds of objects:

ⓐ C itself (as an object in Cat, not of Cat/T), and especially the objects of Cat hot and of Hot which correspond to it.

ⓑ C_∞, and the objects of Cat hot and of Hot which correspond to it. To what extent are these latter independent of the choice of the inverse system cofinal on the left $(t_n)_{n\in\mathbf{N}}$ in T? What is the role of C_∞? Is it canonically isomorphic to C in Cat hot, or in Hot? (In which case the question of its "role" would seem to me resolved!)

[page 103]

ⓒ The ind-object H_C in Cat hot, and the ind-object of Hot which it defines.

I note that the ind-object H_C, and also the categories C_∞ constructed as above, do not change when we replace T by a subcategory T' cofinal on the left, or also by a category of the form T/t_0 ($t_0 \in \mathrm{Ob}\, T$), which gives rise to the associated ordered set

$$(T/t_0)_o \longrightarrow T_o/t_0 = \{t \in T_o \mid t \leq t_0\}$$

. If the relation between the objects in a), and those in b) and in c), is very close, we would conclude from this that the "type" of C in Cat hot, or in Hot as the case may be, does not change on replacing C by $C \times_T T'$ or by $C/t_0 \simeq C \times_T T/t_0$.

p. 67
[page 104]

Better still, it would not change on replacing T by $(\mathbf{N})^o$, the (ordered) category opposite to $\mathbf{N}$ (thus having a unique final object), and the fibered category C over T by the category $C_{\underline{t}}$ induced by C over $(\mathbf{N})^o$, via the functor

$$\underline{t} : (\mathbf{N})^o \longrightarrow T$$

defined by the inverse system $\underline{t} = (t_n)_{n\in\mathbf{N}}$ in T. This would thus mean that to understand the homotopy type (in the strict sense, i.e. in Cat hot, or broad, i.e. in Hot) of C, we can "clean" it in a draconian way, the base T being viewed as a sort of somewhat muddy gangue, or at least plethoric.

ⓓ In any case, if T' is another left filtering category, and

$$\alpha : T' \longrightarrow T$$

a functor cofinal on the left, so that C over T, and $C' = C \times_T T'$ over T', define essentially the "same" object of Ind Cat hot, and the category C'_∞ for C' over T' is among the categories C_∞ for C over T - we may ask whether the canonical functor [page 105]

$$C' \longrightarrow C$$

is also always an isomorphism in Cat hot, or if not, at least in Hot.

A particular interest attaches to $\pi_0(C)$, and moreover to its variance for C over T varying in the following way. If S is in Cat, we look at the categories

$$\underline{\mathrm{Hom}}_{\mathrm{Cat}}(S, C_t) \qquad (t \in T),$$

we find a functor

$$t \longmapsto \underline{\mathrm{Hom}}_{\mathrm{Cat}}(S, C_t), \qquad T^o \longrightarrow (\mathrm{Cat}),$$

whence a fibered category over T, which I am going to denote by

$$\underline{\mathcal{H}om}_T(S_T, C)$$

(Unless I am mistaken, it is interpreted as the fibered category over T corresponding to the stack on $\mathrm{Top}(T) = T^\wedge$ of the morphism of T-stacks from S_T to C, where $S_T = S \times T$ (considered as a fibered category over T with essentially constant fiber S) to C. Here I identify fibered category over T, and stacks on the topos $T^\wedge$.) It is immediate that the homotopy property (p. 99) is stable under passage from C to $\underline{\mathcal{H}om}_T(S_T, C)$. We are interested in the variation of

$$\pi_0(\underline{\mathcal{H}om}_T(S_T, C))$$

as a contrafunctor in S, p. 68 [page 106]

$$H : S \longmapsto \pi_0\underline{\mathcal{H}om}_T(S_T, C) \qquad (\mathrm{Cat})^o \longrightarrow (\mathrm{Ens}),$$

and in a possible expression of this contrafunctor H in terms of the ind-object H_C of Cat hot, by the hypothetical formula

$$H(S) \simeq \mathrm{Hom}_{\mathrm{Ind}(\mathrm{Cat\,hot})}(\gamma(S), H_C),$$

where

$$\gamma : \mathrm{Cat} \longrightarrow \mathrm{Cat\,hot}$$

is the canonical functor. If we had such a canonical isomorphism functorial in S, this would give a description (up to unique isomorphism) of the ind-object H_C in terms of the preceding functor H on Cat.

A doubt comes to me here, in view of the example $T = \underline{\mathrm{Ty}}$, $C = \underline{\mathrm{Ch}}_\Sigma(X; a, b)$. The three types of objects a) b) c) have the same π_0, and it is certain that "it's the right one". On the other hand, for a variable parameter category S, it is also certain that the right functor in S is

$$S \longmapsto \pi_0(\underline{\mathrm{Hom}}_T(S_T, C)) \simeq \mathrm{Hom}_{\mathrm{Ind\,Cat\,hot}}(S, H_C),$$

also isomorphic, *if S is finite,* to

$$\mathrm{Hom}_{\mathrm{Cat\,hot}}(S, C_\infty),$$

which shows that, at least for arguments $S \in (\mathrm{Cat})$ which are *finite* categories, the objects C_∞ and H_C are "good", they allow us to describe an important functor which dominates [page 107] the situation under consideration. Under a very strong hypothesis, but one satisfied in the important cases, we can in these two formulas replace Cat hot by Hot.

By contrast, except for the case where S is reduced to a point and where $\mathrm{Hom}_{\mathrm{Cat\,hot}}$ or $\mathrm{Hom}_{\mathrm{Hot}}$ from S to C is none other than $\pi_0(C)$ ($\simeq \pi_0(C_\infty) \simeq \pi_0(H_C)$), thus indeed "good", I do not know whether

$$\underbrace{\mathrm{Hom}_{\mathrm{Cat\,hot}}(S, C)}_{=\pi_0\underline{\mathrm{Hom}}(S,C)} \qquad \text{or} \qquad \mathrm{Hom}_{\mathrm{Hot}}(S, C)$$

is related in a reasonable way to the preceding sets (determined by H_C, resp. by C_∞), e.g. whether they are isomorphic to it in a natural way, and whether they have a reasonable geometrico-topological meaning. I am beginning to suspect *not* - hence that we would have here a case where a relation of a very general type concerning a π_0, here thus

$$\pi_0(C) \simeq \pi_0(C_\infty) \simeq \pi_0(H_C),$$

does not extend to an isomorphism on the higher π_i, and to a property of Hot-equivalence.

I find no direct arrow, for that matter, between C and either of the objects C_∞, H_C; [p. 69] [page 108] all I see in this direction is

$$\begin{array}{ccc} C_{\mathbf{N}^o} & \xrightarrow[\text{functor}]{\text{localization}} & C_\infty \\ \downarrow & & \\ C, & & \end{array}$$

where the vertical arrow is the projection arrow of $C_{\mathbf{N}^o} = C \times_T \mathbf{N}^o$ (with $\mathbf{N}^o \xrightarrow{t} T$). This establishes a close topological relation between C (as an object of Cat hot or simply of Hot) and C_∞, only if the two preceding arrows are at least Hot-equivalences. And the typical (problematic) case where the question of the relation between C and C_∞ arises, is therefore the one where $T = \mathbf{N}^o$, i.e. that of an arbitrary direct system of (small) categories, indexed by $\mathbf{N}$, and interpreted as defined by a fibered category $C = C_{\mathbf{N}^o}$ over $\mathbf{N}^o$. In this particular case, one has at least a canonical arrow

$$C \longrightarrow C_\infty$$

about which one can ask oneself the question whether it is a Hot-equivalence. If not, this [page 109] will prove that in the general case, the Hot-type of C (and a fortiori its isomorphism class

in Cat hot) has no interesting geometric meaning. That is to say that C is interesting only *as a fibered category over* T, making it possible in particular to construct the ind-object H_T of Cat hot or of Hot, but not as a "bare category".

But I have just convinced myself that $C_{\mathbf{N}^o} \longrightarrow C_\infty$ is indeed a Hot-equivalence. So the arrow $C_{\mathbf{N}^o} \longrightarrow C$ defines an arrow *in* Hot

$$C_\infty \longrightarrow C$$

(by composing with the inverse in Hot of the preceding one). Is it an isomorphism? So for this question, one is reduced to the question for $C_{\mathbf{N}^o} \longrightarrow C$. But if one takes $C = T$, this already poses the same question for $\mathbf{N}^o \longrightarrow T$, hence the question whether T is aspheric, under the sole hypothesis that T_o is left filtering and countable at infinity (on the left). But of course this is false (e.g. if T has only one object, it is not aspheric for all that!).

This therefore convinces me that in the absence of an additional draconian hypothesis on T, the "bare" category C is of no particular interest, apart from its π_0 (which is also given by C_∞, and by H_C). It is not certain, to be sure, that $\underline{\mathrm{Ty}}$ does not satisfy these "draconian conditions". But since "bare" C does not fit into a congenial formalism (e.g. one has no geometric interpretation of $\mathrm{Hom}_{\mathrm{Cat\,hot}}(S, C)$, or of $\mathrm{Hom}_{\mathrm{Hot}}(S, C)$), I think that in what follows I can leave it out of account. Exit a)! p. 70 [page 110]

5.11.90

All the same, I spent the whole day yesterday trying to understand the relation of C and of its type in Hot, to that of C_∞. Indeed, since one finally has an arrow in Hot

$$C_\infty \longrightarrow C,$$

deduced from the diagram in (Hot)

$$\begin{array}{ccc} C_{\mathbf{N}^o} & \xrightarrow{\sim} & C_\infty \\ \downarrow & & \\ C, & & \end{array}$$

to prove that $C_\infty \longrightarrow C$ is an isomorphism in Hot seems the simplest, and perhaps the only, way of proving that the image of C_∞ in Hot does not depend, up to unique isomorphism, on the choices made to construct C_∞. Not to mention that in the motivating case, $C = \underline{\mathrm{Ch}_\Sigma}(X)$ or $\underline{\mathrm{Ch}_\Sigma}(X; a, b)$, $T = \underline{\mathrm{Ty}}$, it is the category of paths C that gives rise to an impeccable composition law for paths, functorial, associative and all, whereas there is nothing of the sort for C_∞. One does not find a composition law [page 111]

$$C_\infty \times C_\infty \longrightarrow C_\infty$$

in Cat, nor even in Cat hot or in Hot - or rather, if one does nevertheless manage to construct an arrow $C_\infty \times C_\infty \longrightarrow C_\infty$ in Cat, it is still laboriously by dint of $\pm$ arbitrary choices, notch by notch, and without knowing whether what one obtains has a uniqueness property, even if only in Hot; one finds a tangible geometric interpretation of this composition only in $\mathrm{Hot}_0^\wedge$ (where one has a genuine groupoid law), or in H_T as an object of Ind Cat hot or of Ind Hot. (So the direct limit in $\mathrm{Hot}_0^\wedge$ is none other than the element of $\mathrm{Hot}_0^\wedge$ defined by C_∞.) p. 71 [page 112]

My efforts to prove that

$$C_\infty \longrightarrow C$$

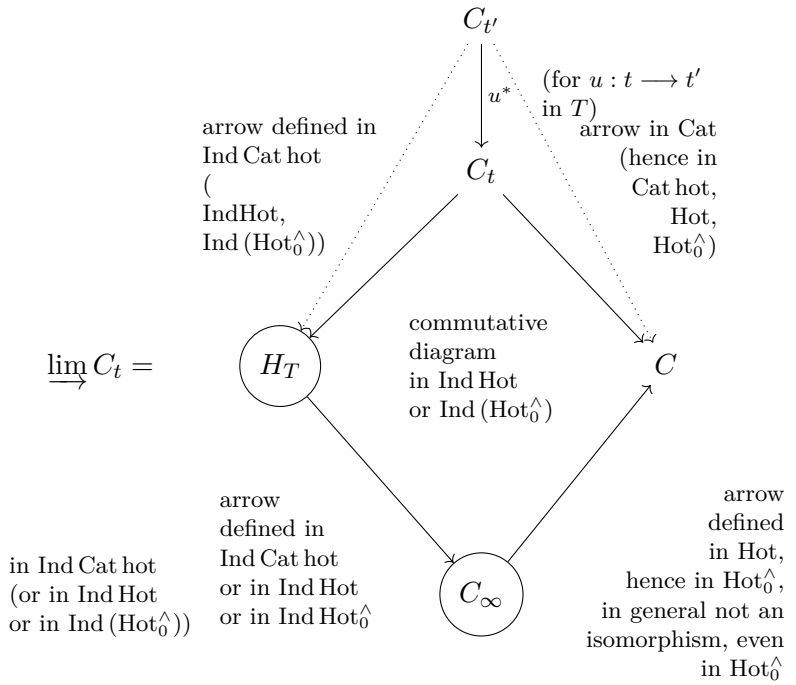
was an isomorphism in Hot, or even just a 1-isomorphism (i.e. an isomorphism not only for the π_0, which is always true, but also for the π_1 of all the components), have failed. I found that it was enough for the functor

$$T \longrightarrow T_\mathrm{o}$$

to be either smooth and coaspheric, or proper and aspheric - so everything is OK if $T = T_\mathrm{o}$, i.e. if T is ordered. But in the case $T = \underline{\mathrm{Ty}}$, the functor $T \longrightarrow T_\mathrm{o}$ is neither proper, nor smooth (not even 0-proper or 0-smooth).

Finally I have come to the conviction that in the case we are concerned with ($T = \underline{\mathrm{Ty}}$, $C = \underline{\mathrm{Ch}}_\Sigma(X; a, b)$), the arrow $C_\infty \longrightarrow C$ is in general *not* an isomorphism in Hot. Indeed, in the case $a = b$, if this , one would have a description of the Hot-group $\Omega(X, a) = \underline{\mathrm{Ch}}(X, a)$ ([23]) in terms of a category C equipped with an *associative* (and unital) composition law *in* Cat, giving back the group law of Ω on passing from (Cat) to Hot. Taking the nerve of C, one would find a semi-simplicial monoid which, in Hot, would give the group law of $\Omega(X, a)$. Now I remember ([24]) that it was explained to me that in general, the group law of $\Omega(X, a)$ could not be obtained by an *associative* law on a semi-simplicial set. Thus, the failure of my efforts seems to be due, not to my clumsiness, but to the nature of things. [page 113]

I am going to summarize the positive facts I have arrived at in the following diagram, which takes place in the general situation $C \longrightarrow T$ considered at the beginning of the present paragraph: [page 114] p. 72



$$= \varinjlim H_T \text{ in } \mathrm{Hot}_0^\wedge$$

[23] Here we take $\Sigma = \mathrm{Fl}(X)$.

[24] This is an error - it must be for *other* group objects of Hot than loop groups. I saw in Quillen that $\Omega(X, x)$ must be describable by a semi-simplicial group (when X is a semi-simplicial set, x a vertex). So the problem whether $C_\infty \simeq C$ in Hot remains completely open.

The important objects are H_T and C_∞, whereas C (over T) and the C_t take on an intermediate technical role, to construct H_T and C_∞. One can consider C_∞ as a category (object of (Cat)), but whose construction depends on many arbitrary choices, unlike H_T in $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$. It is the object of $\mathrm{Hot}_0^\wedge$ deduced from C_∞ that has a canonical meaning, and can be described as the direct limit of H_T as an object of $\mathrm{Ind}\,\mathrm{Hot}_0^\wedge$, i.e. as representing the same functor on Hot_0 as H_T.

In the geometric case $T = \underline{\mathrm{Ty}}$, and C of the form $\underline{\mathrm{Ch}}_\Sigma(X; a, b)$, giving rise to $C_\infty(a, b)$, one *does not have* a natural definition of a bifunctor of composition of paths [page 115]

$$C_\infty(a,b) \times C_\infty(b,c) \longrightarrow C_\infty(a,c).$$

One can define, to be sure, pairings

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(a,b) \times \underline{\mathrm{Ch}}_{\Sigma,\tau'}(b,c) \longrightarrow \underline{\mathrm{Ch}}_{\Sigma,\tau\tau'}(a,c),$$

or, with the notation relating to the datum of

$$\underline{t} : \mathbf{N}^o \longrightarrow \underline{\mathrm{Ty}},$$
$$\underline{\mathrm{Ch}}_n(a,b) \times \underline{\mathrm{Ch}}_m(b,c) \longrightarrow \underline{\mathrm{Ch}}_{n+m}(a,c),$$

but these pairings *are not* compatible with the transition morphisms of the three direct systems of categories $(\underline{\mathrm{Ch}}_n)_{n\in\mathbf{N}}$, they are so *only up to homotopy.* This is enough, however, to have perfect pairings p. 73

$$H_T(a,b) \times H_T(b,c) \longrightarrow H_T(a,c),$$

with associativity, and (if $\Sigma = \mathrm{Fl}(X)$ in its entirety) existence of *true* inverses, i.e. of arrows (in $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$) of "passage to the opposite path" [page 116]

$$H_T(a,b) \longrightarrow H_T(b,a), \qquad c \longmapsto c^o, \tag{$*$}$$

giving rise to the commutative diagram expressing that this is indeed an inverse for composition. Another way of saying it is that for every object Y of $\mathrm{Cat}\,\mathrm{hot}$ (or of $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$), the sets

$$\mathrm{Hom}_{\mathrm{Ind}(\mathrm{Cat}\,\mathrm{hot})}(Y, H_T(a,b)) \overset{\mathrm{def}}{=} H_T^Y(a,b)$$

form, for variable $a, b \in X$, the arrows of a *groupoid* ([25]) $\mathrm{Ch}_\Sigma^Y(X)$ having as set of vertices $\mathrm{Ob}\,X$, and $H_T^Y(a,b)$ being made up of the arrows from a to b. The map

$$H_T^Y(a,b) \longrightarrow H_T^Y(b,a), \qquad c \longmapsto c^o,$$

deduced from the functor $(*)$ above is none other than the map of passage to the inverse. This category is moreover none other than the one deduced from

$$X(Y)\Sigma_Y^{-1}, \qquad \text{where} \begin{cases} X(Y) = \underline{\mathrm{Hom}}_{\mathrm{Cat}}(Y, X) \\ \Sigma_Y \subseteq \mathrm{Fl}(X(Y)), \\ \text{defined in terms of } \Sigma \text{ "argument by argument",} \end{cases}$$

by means of the functor

$$X \longrightarrow X(Y), \qquad a \longmapsto a_Y,$$

associating to every $a \in X$ the *constant* functor on Y with value a, taking [page 117] p. 74

$$H_T^Y(a,b) \simeq \mathrm{Hom}_{X(Y)\Sigma_Y^{-1}}(a_Y, b_Y)\ .$$

In the same geometric case, I have convinced myself, moreover, that C_∞ was, as an object of Hot (not only of $\mathrm{Hot}_0^\wedge$) *unique up to* Hot-*isomorphism,* but (strange thing) perhaps not up to *unique* Hot-isomorphism. In general, returning to the

conditions at the beginning of the paragraph, C_∞ must be written $C_\infty(\underline{t})$, it depends on the choice of the cofinal functor

$$\underline{t} : \mathbf{N}^o \longrightarrow T.$$

When $\underline{t}$ varies in the category

$$\underline{\mathrm{Hom\ cof}}(\mathbf{N}^o, T),$$

the $C_\infty(\underline{t})$ vary contravariantly in $\underline{t}$ (thus define a fibered category over $\underline{\mathrm{Hom\ cof}}(\mathbf{N}^o, T)$). Moreover, the functor

$$C_\infty(\underline{t'}) \longrightarrow C_\infty(\underline{t})$$

associated to a morphism of inverse systems

$$\underline{t} \longrightarrow \underline{t'}$$

is an *isomorphism* in Hot. So the functor [page 118]

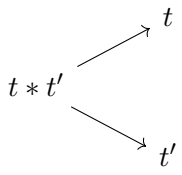
$$\underline{t} \longmapsto C_\infty(\underline{t}), \qquad \underline{\mathrm{Hom\ cof}}(\mathbf{N}^o, T) \longrightarrow (\mathrm{Hot})$$

factors through the fundamental groupoid

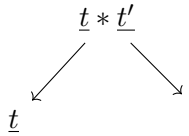
$$\Pi(\underline{\mathrm{Hom\ cof}}(\mathbf{N}^o, T)) \longrightarrow (\mathrm{Hot}).$$

When this groupoid, i.e. when the category $\underline{\mathrm{Hom\ cof}}$ itself is connected, we can therefore conclude that all the $C_\infty(\underline{t})$ are isomorphic to one another in Hot. To conclude even that there is a *transitive* system of isomorphisms between the $C_\infty(\underline{t})$, i.e. that $C_\infty(\underline{t})$ is "unique up to *unique* isomorphism", one would have to know in addition that the fundamental groupoid is equivalent to the one-point category (trivial fundamental group), or, as one says, that $\underline{\mathrm{Hom\ cof}}$ is *simply connected,* or 1-connected.

The connectedness of $\underline{\mathrm{Hom\ cof}}$ is assured when the category T has (as is the case for $\underline{\mathrm{Ty}}$) a law of composition $(t,t') \longmapsto t * t'$, with morphisms functorial in t, t'



(this last condition being automatic if the law of composition admits the final object of T, assumed to exist, as a two-sided unit). In this case, $\underline{t} = (t_n)_n$ and $\underline{t'} = (t'_n)_n$ are linked to one another by $\underline{t} * \underline{t'} = (t_n * t'_n)_n$, since we have a morphism of inverse systems p. 75 [page 119]



[25] If $\Sigma = \mathrm{Fl}(X)$. If Σ is smaller, instead of a groupoid, one finds a *category*, nothing more.

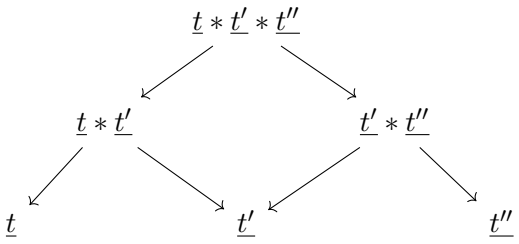
I do not know, for that matter, whether the arrow in Hot

$$C_\infty(\underline{t}) \longrightarrow C_\infty(\underline{t}')$$

associated to $\underline{t}' \longrightarrow \underline{t}$ in $\underline{\mathrm{Hom\,cof}}$ depends only on $\underline{t}$, $\underline{t}'$, and not on the choice of $\underline{t}' \longrightarrow \underline{t}$. What is clear is that its image in $\mathrm{Hot}_0^\wedge$ is unique, it is the identity when we identify $C_\infty(\underline{t})$ and $C_\infty(\underline{t}')$ with one and the same element of $\mathrm{Hot}_0^\wedge$ defined by $\varinjlim H_T$.

And yet, I realize just now that, under the hypothesis of the existence of the law of composition and of its two-sided unit, and assuming in addition that the law in question is associative up to isomorphism, *we have transitivity* in the isomorphisms (canonical!) between the $C_\infty(\underline{t})$ which we have just indicated!

(Without having for all that to compute the fundamental group of $\underline{\mathrm{Hom\,cof}}(\mathbf{N}^o, T)$!) [page 120]
If we have three elements $\underline{t}$, $\underline{t}'$, $\underline{t}''$ of $\underline{\mathrm{Hom\,cof}}$, it suffices to look at the diagram deduced from the following commutative diagram



by applying to it the contrafunctor $\underline{s} \longrightarrow C_\infty(\underline{s})$ from $\underline{\mathrm{Hom\,cof}}$ to Hot.

So in the case of the $C(a,b) = \underline{\mathrm{Ch}_\Sigma}(X; a, b)$, I no longer have any scruples about regarding the category $C_\infty(a,b)$ as a *canonical* object of Hot, linked to the canonical ind-object $H_T(a,b)$ in Hot by a canonical homomorphism

$$C_\infty(a,b) \longleftarrow H_T(a,b), \qquad (*)$$

which in $\mathrm{Hot}_0^\wedge$ gives an isomorphism

$$C_\infty(a,b) \simeq \varinjlim H_T(a,b), \qquad (**)$$

i.e. the arrow $(*)$ induces, for $\xi \in \mathrm{Ob\,Hot}$, a set-theoretic map

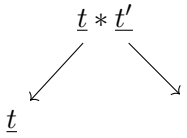
$$\mathrm{Hom}_{\mathrm{Hot}}(\xi, C_\infty(a,b)) \longleftarrow \underbrace{\mathrm{Hom}_{\mathrm{Ind\,Hot}}(\xi, H_T(a,b))}_{\varinjlim_n \mathrm{Hom}_{\mathrm{Hot}}(\xi, C_n(a,b))},$$

which is *bijective* if ξ is in Hot.

Of course, the arrows

$$C_\infty(\underline{t}') \longrightarrow C_\infty(\underline{t}) \qquad (*)$$

in Hot, associated to the arrows $\underline{t} \longrightarrow \underline{t}'$ in $\underline{\mathrm{Hom\,cof}}(\mathbf{N}^o, T)$, are in fact arrows in (Cat) itself - and even fully faithful functors, in the "geometric" case and for the two canonical arrows p. 76 [page 121]



They are isomorphic *in* Hot, because they define an isomorphism on the associated ind-objects of Cat hot, hence also for the associated ind-objects in Hot and in $\mathrm{Hot}_0^\wedge$, from which it follows that

$$\mathrm{Hom}_{\mathrm{Hot}}(Y, C_\infty(\underline{t}')) \longrightarrow \mathrm{Hom}_{\mathrm{Hot}}(Y, C_\infty(\underline{t}))$$

associated to $(*)$ is a bijection if Y *is finite* ([26]), this arrow then identifies with

$$\varinjlim_n \mathrm{Hom}_{\mathrm{Hot}}(Y, C_{t'_n}) \longrightarrow \varinjlim_n \mathrm{Hom}_{\mathrm{Hot}}(Y, C_{t_n}).$$

(For $Z \longmapsto \mathrm{Hom}_{\mathrm{Hot}}(Y, Z)$ commutes with filtering direct limits, if Y is a category which is *finite*. This comes from the fact that a filtering $\varinjlim$ of Thomason-Kan categories is likewise …) But I do not know, on the other hand, whether $(*)$ is isomorphism in Cat hot. Only that for a *finite* category Y, it induces

$$\mathrm{Hom}_{\mathrm{Cat\,hot}}(Y, C_\infty(\underline{t}')) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Cat\,hot}}(Y, C_\infty(\underline{t})),$$

the two sides then identifying with [page 122]

$$H_T(Y)_{\mathrm{Cat\,hot}} = \mathrm{Hom}_{\mathrm{Ind\,Cat\,hot}}(Y, H_T).$$

To sum up these insights and perplexities about the $C_\infty(\underline{t})$:

a) To every $\underline{t} = (t_n)_{n\in\mathbf{N}^o} \in \underline{\mathrm{Hom\,cof}}(\mathbf{N}^o, T)$ is associated a direct system, indexed by $\mathbf{N}$, of categories $C_n(\underline{t})$ $(= C_{t_n})$, and their direct limit

$$C_\infty(\underline{t}).$$

b) This direct system, and also its limit is contravariant functor in $\underline{t}$. If $\underline{t}, \underline{t}' \in \mathrm{Ob}\,\underline{\mathrm{Hom\,cof}}$, then we can compare $C_\infty(\underline{t})$, $C_\infty(\underline{t}')$ and the corresponding direct systems, thanks to p. 77

$$\begin{array}{ccc} & \underline{t} * \underline{t}' & \\ \swarrow & & \searrow \\ \underline{t} & & \underline{t}'. \end{array}$$

c) The object

$$(C_n(\underline{t}))_{n\in\mathbf{N}} \in \mathrm{Ob}\,\mathrm{Ind}\,\mathrm{Cat\,hot}$$

defined by the direct system $(C_n(\underline{t}))_{n\in\mathbf{N}}$ in Cat is (up to unique isomorphism) independent of the choice of $\underline{t}$, say H_C or H. The homomorphisms

$$C_\infty(\underline{t}') \xrightarrow{u^*} C_\infty(\underline{t}) \qquad \text{associated to } u : \underline{t} \longrightarrow \underline{t}' \qquad (*)$$

are compatible with this datum, i.e. we have commutativity in

$$\begin{array}{ccc} \varinjlim_{\mathrm{Ind\,Cat\,hot}} C_*(\underline{t}') & \longrightarrow & \varinjlim_{\mathrm{Ind\,Cat\,hot}} C_*(\underline{t}) \\ & \nwarrow^{\sim} \quad \nearrow_{\sim} & \\ & H & \end{array}$$

d) It follows that the homomorphisms $(*)$ are *isomorphisms* in Hot, hence all the $C_\infty(\underline{t})$ are isomorphic *in* Hot. The trick with the $t * t'$ (in the case where we have a law $t * t'$ which is associative and admits the final object as unit object) even gives *canonical* isomorphisms [page 123]

$$C_\infty(\underline{t}') \longrightarrow C_\infty(\underline{t}) \qquad \text{in Hot},$$

[26] Which suffices to ensure that $(*)$ is isomorphism in Hot: $\mathrm{Hot} \longrightarrow \mathrm{Hot}_0^\wedge$ is conservative!

and this is a transitive system of isomorphisms. So we have in Hot a canonical object

$$C_\infty \in \mathrm{Ob}\,\mathrm{Hot},$$

whose relation to $H \in \mathrm{Ob}\,\mathrm{Ind}\,\mathrm{Hot}$ is deduced from c), it is a canonical arrow

$$H \longrightarrow C_\infty \qquad \text{in } \mathrm{Ind}\,\mathrm{Hot}\ (^{27}).$$

This arrow induces an isomorphism for the corresponding objects of $\mathrm{Hot}_0^\wedge$; but viewing H not as an ind-object of $\mathrm{Hot}_0^\wedge$, but taking the functor H_0 induced on Hot_0 by the ind-representable functor H on Hot. This induced functor is "representable" by the object C_∞ of the larger category

$$H_0 \simeq C_\infty \qquad \text{in } \mathrm{Hot}_0^\wedge.$$

(If the canonical functor $\mathrm{Hot} \longrightarrow \mathrm{Hot}_0^\wedge$ is fully faithful, then this formula describes C_∞ as an object of Hot, up to unique isomorphism, in terms of H_0, hence also in terms of $H \in \mathrm{Ob}\,\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$.)

e) I do not know, on the other hand, whether the arrows

p. 78
[page 124]

$$C_\infty(\underline{t}') \longrightarrow C_\infty(\underline{t})$$

are isomorphisms in Cat hot, although they are defined in terms of homomorphisms of direct systems of categories, which are isomorphisms in Ind Cat hot. This is so, even for the arrows $C_\infty(\underline{t}), C_\infty(\underline{t}') \rightrightarrows C_\infty(\underline{t} * \underline{t}')$ used to compare $C_\infty(\underline{t})$ and $C_\infty(\underline{t}')$ canonically. Thus, I do not see how to define arrows in the reverse direction in Cat, which would be candidates to be inverse isomorphisms in Cat hot.

f) In the geometric case ($T = \underline{\mathrm{Ty}}$, situation described in terms of X, Σ, and $a, b \in \mathrm{Ob}\,X$, hence $C_\infty(\underline{t}; a, b)$, $C_\infty(\overline{a, b})$, $H(a, b)$ etc.), we *do not have* a path composition operation at the level of the *categories* $C_\infty(\underline{t}; a, b)$ and $C_\infty(\underline{t}; b, c)$, but only at the level of the objects $H(a, b)$, $H(b, c)$ of Ind Cat hot or of Ind Hot. Am I even sure I know how to define *in* Hot

$$C_\infty(a, b) \times C_\infty(b, c) \longrightarrow C_\infty(a, c) \ \ ?$$

Surely, there are no problems for the images in $\mathrm{Hot}_0^\wedge$, since there they are the same as those of $H(a, b)$, $H(b, c)$, $H(a, c)$ respectively, and we have perfect, associative pairings, with unit, and with inverses where there ought to be some. But not for the $C_\infty(a, b)$ themselves?

[page 125]

It is a bit vexing that with all this work it would seem that we haven't even been able to damn well define the analogue in Cat of the classical loop space $\Omega(X, x)$ as a group *in* Hot. (What does THOMASON do?) But here is a thinkable idea . To define *all the same*

$$C_\infty(a, b) \times C_\infty(b, c) \longrightarrow C_\infty(a, c)$$

in Hot, we are going to describe the two functors $C_\infty(a, b)$ and $C_\infty(b, c)$ by different inverse systems $\underline{t}$, $\underline{t}'$. We can make it explicit by starting from a τ_0 of useful length ≥ 2, and taking in both cases

$$t_n = t'_n = \tau_0^n,$$

[27] **NB** H is practically never essentially constant, so the arrow opposite not an isomorphism in Ind Hot.

but different transition morphisms

$$(t_{n+p} \longrightarrow t_n) = \pi_p * \mathrm{id}_{\tau_0^n}$$
$$(t'_{n+p} \longrightarrow t'_n) = \mathrm{id}_{\tau_0^n} * \pi_p,$$

where $\pi_p : \tau_0^p \longrightarrow 1$ is the total degeneracy. We will define $C_\infty(a,c)$ in terms of $\underline{t''} = \underline{t} * \underline{t'}$, hence

$$\begin{cases} t''_n = \tau_0^{2n} \\ (t''_{n+p} \longrightarrow t''_n) = \pi_p * \mathrm{id}_{\tau_0^{2n}} * \pi_p. \end{cases}$$

This time, the composition functors p. 79

$$C_n(\underline{t}; a, b) \times C_n(\underline{t'}; b, c) \longrightarrow C_n(\underline{t''}; a, c),$$

are compatible with the transition functors for varying n, and one finds [page 126]

$$C_\infty(\underline{t}; a, b) \times C_\infty(\underline{t'}; b, c) \longrightarrow C_\infty(\underline{t''}; a, c),$$

hence an arrow in Hot

$$\mathrm{comp}_{\tau_0} : C_\infty \times C_\infty \longrightarrow C_\infty.$$

We must see that it does not depend on the choice of τ_0, but if we have τ_0 and τ'_0, we compare them by $\tau_0 * \tau'_0 \begin{array}{l} \longrightarrow \tau_0 \\ \longrightarrow \tau'_0 \end{array}$, so it suffices to see that if there exists an arrow $\tau_0 \longrightarrow \tau'_0$, then $\mathrm{comp}_{\tau_0} = \mathrm{comp}_{\tau'_0}$. But such an arrow induces arrows $\underline{t}_{\tau_0} \longrightarrow \underline{t}_{\tau'_0}$, $\underline{t'}_{\tau_0} \longrightarrow \underline{t'}_{\tau'_0}$, $\underline{t''}_{\tau_0} \longrightarrow \underline{t''}_{\tau'_0}$, whence a diagram in Cat

$$\begin{array}{ccc} C_\infty(\underline{t}_{\tau'_0}; a, b) \times C_\infty(\underline{t'}_{\tau'_0}; b, c) & \xrightarrow{\mathrm{comp}_{\tau'_0}} & C_\infty(\underline{t''}_{\tau'_0}; a, c) \\ \downarrow & & \downarrow \\ C_\infty(\underline{t}_{\tau_0}; a, b) \times C_\infty(\underline{t'}_{\tau_0}; b, c) & \xrightarrow{\mathrm{comp}_{\tau_0}} & C_\infty(\underline{t''}_{\tau_0}; a, c), \end{array}$$

which one checks at once is commutative. Now this diagram can be rewritten in Hot as

$$\begin{array}{ccc} C_\infty(a, b) \times C_\infty(b, c) & \xrightarrow{\mathrm{comp}_{\tau'_0}} & C_\infty(a, c) \\ \downarrow \mathrm{id} & & \downarrow \mathrm{id} \\ C_\infty(a, b) \times C_\infty(b, c) & \xrightarrow{\mathrm{comp}_{\tau_0}} & C_\infty(a, c), \end{array}$$

whence $\mathrm{comp}_{\tau_0} = \mathrm{comp}_{\tau'_0}$.

I shrink from the real difficulty - that of proving the associativity of this composition law. It becomes associative in $\mathrm{Hot}_0^\wedge$ - but for two arrows in Hot to be equal (here then [page 127]

$$C_\infty(a, b) \times C_\infty(b, c) \times C_\infty(c, d) \rightrightarrows C_\infty(a, d) \quad),$$

is it enough that this be so for their images in $\mathrm{Hot}_0^\wedge$ - in other words, is the functor

$$\mathrm{Hot} \longrightarrow \mathrm{Hot}_0^\wedge,$$

(besides being conservative) at least faithful (if not fully faithful)?

There is also the fact that this composition law is compatible with that of the $H(a,b)$ in Ind Hot, for the canonical arrows p. 80

$$H(a, b) \longrightarrow C_\infty(a, b) \qquad \text{in Ind Hot},$$

if we would know that these are epimorphisms, and remain so under cartesian products, we would have won.

Of course, we will know that it is associative, once we know that the composition defined here is compatible with the composition of paths in the semi-simplicial framework. But how does one proceed in this framework, to compose paths of unbounded length? Is there not exactly the same difficulty here? Unless one gets out of it by reducing [page 128] at once, for the theory of Ω operations, to Kan complexes, and to paths of type Δ_+. But to compose them *functorially,* there still seems to me to be the same problem - one will find paths of type $\Delta_+\Delta_+ = \mathbf{\Delta}_2^+$, and not Δ_+, but one can then compose the direct homotopies (amounts to restricting to $\Delta_+ \simeq \{0,2\} \subseteq \mathbf{\Delta}_2^+$). Now I worked out the formalism of categories of paths without worrying about conditions of Kan type. I presume that in the semi-simplicial framework, the $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1, X)$ do *not* give the right path object, for X not Kan. (For otherwise, it would seem that for $X = X_0$ whose components are *finite,* the calculation of the π_i ought to be finite, which is nevertheless not the case, I have always been assured …)

I presume nevertheless, when one takes $\Sigma = \mathrm{Fl}(X)$, that the objects $\underline{\mathrm{Ch}}_\infty(X; a, b)$ and $\underline{\mathrm{Ch}}_\infty(\bar{X}; a, b)$ are isomorphic in Hot, when one embeds X in a Thomason-Kan envelope $\bar{X}$.

p. 81
[page 129]

11 Quick return to the $(\mathcal{M}, \Sigma)$, from the perspective of derivators

When one is interested in the formalism of the $\underline{\mathrm{Ch}}_\Sigma(X; a, b)$ with a view to a *derivator* $\mathbf{D}_{X,\Sigma}$ that one would like to define in terms of X, Σ, it would seem that the prederivator one does indeed obtain only deserves the name of derivator when one assumes that for every arrow

$$Y \longrightarrow Y'$$

in Cat which is a Hot-equivalence, the corresponding arrows

$$\mathrm{Hom}_{\mathrm{Ind\,Cat\,hot}}(Y', H(a,b)) \longrightarrow \mathrm{Hom}_{\mathrm{Ind\,Cat\,hot}}(Y, H(a,b)),$$

or again

$$\varinjlim_n \mathrm{Hom}_{\mathrm{Cat\,hot}}(Y', \underline{\mathrm{Ch}}_{n,\Sigma}(a,b)) \longrightarrow \varinjlim_n \mathrm{Hom}_{\mathrm{Cat\,hot}}(Y, \underline{\mathrm{Ch}}_{n,\Sigma}(a,b))$$

are bijections. I looked in V at what this means for the object $H(a,b) = (\underline{\mathrm{Ch}}_{n,\Sigma}(a,b))_n$ of Ind Cat hot. It is this condition alone that ensures, from the point of view of derivators, a convincing interpretation of $\underline{\mathrm{Ch}}_{\infty,\Sigma}(a,b)$,

$$\underline{\mathrm{Ch}}_{\infty,\Sigma}(a,b) \;=\; \varinjlim_n \underline{\mathrm{Ch}}_{n,\Sigma}(a,b)$$

as an object of Hot, or more exactly of $\mathrm{Hot}_0^\wedge$, for then one has for Y a *finite* category

$$\underbrace{\mathrm{Hom}_{\mathrm{IndCat\,hot}}(Y, H(a,b))}_{\simeq \mathrm{Hom}_{X(Y)\Sigma_Y^{-1}}(a_Y, b_Y)} \;\simeq\; \mathrm{Hom}_{\mathrm{Hot}}(Y, \underline{\mathrm{Ch}}_{\infty,\Sigma}(a,b)),$$

for I do want the type in Hot to be the right one, that it does correspond to the homotopy of path spaces. So one would like a morphism [page 130]

$$X \xrightarrow{f} X'$$

in (Cat), which is a Hot-equivalence, to define

$$\underline{\mathrm{Ch}}_\infty(X; a, b) \longrightarrow \underline{\mathrm{Ch}}_\infty(X'; f(a), f(b)),$$

which are themselves also Hot-equivalences. On the other hand, this is obviously compatible with the brilliant composition of paths, which I managed to define between these objects of Hot. So (subject to having checked what I have just said) the verification of associativity would reduce to the case where X is Thomason-Kan. But in this case, there is a chance that, just as in the semi-simplicial case, the $\underline{\mathrm{Ch}}_\infty(X; a, b)$ can be defined (as objects of (Hot)) as being simply the paths of type Δ_+ from a to b, which one then composes in the trivial way. But no, these categories are discrete!

I give up for the moment pursuing the question of associativity any further. Ask THOMASON about the properties of the functor Hot $\longrightarrow \mathrm{Hot}_0^\wedge$.

This means, very nearly, that one has an interpretation of $\underline{\mathrm{Ch}}_{\infty,\Sigma}(a, b)$, as an object of Hot, as an "external Hom" $\mathrm{Hom\,ext}(a, b)$ with values in Hot, defined on the category $\mathcal{M} = X$ using just the admissible set of arrows $\Sigma \subseteq \mathrm{Fl}(\mathcal{M})$, an operation satisfying the condition

p. 82
[page 131]

$$\boxed{\mathrm{Hom}_{\mathrm{Hot}}(Y, \mathrm{Hom\,ext}(a, b)) \;\simeq\; \mathrm{Hom}_{\mathcal{M}(Y)\Sigma_Y^{-1}}(a_Y, b_Y)} \qquad (*)$$

(which can also be interpreted, either as

$$\mathrm{Hom}_{\mathbf{D}(*)}(p_!(a_Y), b)$$

if $p_! : \underbrace{\mathcal{M}(Y)\Sigma_Y^{-1}}_{\mathbf{D}(Y)} \longrightarrow \mathcal{M}\Sigma^{-1}$ exists, or as

$$\mathrm{Hom}_{\mathbf{D}(*)}(a, p_*(b_Y))$$

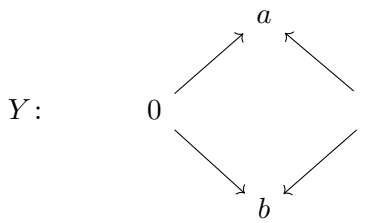
if $p_* : \underbrace{\mathcal{M}(Y)\Sigma_Y^{-1}}_{\mathbf{D}(Y)} \longrightarrow \underbrace{\mathcal{M}\Sigma^{-1}}_{=\mathbf{D}(*)}$ exists); with the one small nuisance that this formula is valid only for Y *finite.* If one wants to express the right-hand side for Y varying arbitrarily in Cat, an object of Hot ($\mathrm{Hom\,ext}(a, b)$) on the left-hand side does not suffice, but one needs the ind-object $H(a, b) = (\underline{\mathrm{Ch}}_n(a, b))_{n\in\mathbf{N}}$ in Hot. So if one insists on working in a "domain" Diag of the derivator $\mathbf{D} = \mathbf{D}_{\mathcal{M},\Sigma}$ as large as possible, while restricting oneself to finite categories, and one wants a formula $(*)$ to remain valid for every Y in the domain, one must "in all rigor" define the $\mathrm{Hom\,ext}(a, b)$ not as an object of Hot, but of Ind Hot. It would therefore then be described as

[page 132]

$$\mathrm{Hom\,ext}(a, b) = (\underline{\mathrm{Ch}}_{n,\Sigma}(a, b))_n \qquad \text{(ind-object in Hot)}\ ,$$

and the object $\underline{\mathrm{Ch}}_{\infty,\Sigma}$ in Hot is a sort of direct limit of it.

I wondered whether the preliminary condition, namely that the functors $H(a, b)$ on Cat turn Hot-equivalences $Y' \longrightarrow Y$ into bijections, was really a (strong) restriction. I convinced myself that this is indeed the case, already in the special case where $\Sigma = \mathrm{Fl}(X)$. It seems to me that when the connected components of $\Omega(X, a)$ are not contractible, the said condition is practically never satisfied. I thought e.g. of the case where X has the homotopy type (in Hot) of a 2-sphere, so that its $\Omega(X, a) = \underline{\mathrm{Ch}}_\infty(X, a, a)$ must have a $\pi_1 \simeq \mathbf{Z}$. Let us take Y having the type in Hot of a circle, say



,

p. 83
[page 133]

so the functors

$$f : Y \longrightarrow \underline{\mathrm{Ch}}_\infty(X; \bullet, \bullet) \stackrel{\mathrm{def}}{=} \Omega(X, \bullet)$$

correspond to *certain* "loops" in $\Omega(X, \bullet)$ (but loops with unspecified origin), they have a homotopy type in the topological sense ("number of turns" around the sphere X) given by an integer $n \in \mathbf{Z}$ (the "degree" of the loop λ). The obvious intuition is that the set of degrees δ which one can obtain in this way, with a given *finite* category Y which represents the circle $S^1 \in (\mathrm{Hot})$, is finite - $|\delta|$ surely bounded above by the number of these arrows (hence at most by 4 in the case under consideration, each arrow of Y making one go *at most* once around the pseudo-circle $\Omega(S^2, \bullet)$) ([28]). On the other hand, when one represents S^1 by arbitrarily large categories Y_n (say obtained from the intervals $\Phi_n = (\Delta_+\Delta_-)^n$ by identifying source and target), one must find arbitrarily high degrees. Now every Y_n maps to $Y(= Y_2)$ by an arrow

$$Y_n = Y' \longrightarrow Y = Y_2$$

which is of degree 1, hence which is a Hot-equivalence. But the corresponding map [page 134]

$$\pi_0\underline{\mathrm{Hom}}(Y, \Omega(X, \bullet)) \longrightarrow \pi_0\underline{\mathrm{Hom}}(Y', \Omega(X, \bullet))$$

is not bijective. This kind of argument seems to me bound to work whenever the set of homotopy groups $\pi_j(X, x)$, for $j \geq 2$ (hence the set of homotopy groups π_i of $\Omega(X, x)$, for $i \geq 1$) is *infinite,* taking for Y, Y' different representatives Y_n of the sphere S^i, say the Y_n obtained from Φ_n ($n \geq 2$) by taking $\Phi_n^i = \Phi_n \times \Phi_n \times \cdots \times \Phi_n$ (which is a sort of combinatorial i-cube), and identifying its "boundary" with a single point. I presume that the number of elements in $\pi_i(\Omega(X, x))$ that one can obtain from arrows $Y_n \longrightarrow \Omega(X, x)$ is limited by the size of Y_n, but one can surely obtain each of them, by taking n large enough. So if $\xi \in \pi_i(\Omega)$ escapes Y_{n_0} ($n_0 \geq 2$, say), one takes an n' large enough that ξ does not escape $Y_{n'}$, and one chooses

$$Y_{n'} \longrightarrow Y_n \ ,$$

which is a Hot-equivalence (e.g. with the aid of a retraction of $\Phi_{n'}$ onto Φ_n, compatible with sources and targets). [page 135]

This draws strong attention to the fact that if Y, C are two categories, the category $\underline{\mathrm{Hom}}(Y, C)$ *is not* a representative of $\underline{\mathrm{Hom}}_{\mathrm{Hot}}(\mathrm{hot}(Y), \mathrm{hot}(C))$, one has only a canonical arrow

$$\mathrm{hot}(\underline{\mathrm{Hom}}(Y, C)) \longrightarrow \underline{\mathrm{Hom}}_{\mathrm{Hot}}(\mathrm{hot}(Y), \mathrm{hot}(C)), \qquad (*)$$

i.e. p. 84

$$\underbrace{\mathrm{hot}(\underline{\mathrm{Hom}}(Y, C)) \times \mathrm{hot}(Y)}_{\mathrm{hot}(\underline{\mathrm{Hom}}(Y,C)\times Y)} \longrightarrow \mathrm{hot}(C),$$

which arrow is simply deduced from

$$\underline{\mathrm{Hom}}(Y, C) \times Y \longrightarrow C.$$

But the arrow $(*)$ has no desire to be isomorphism. It is because it is already not one in the special case $Y = \mathbf{\Delta}^1$, essentially, which forces us, in order to make a theory of paths in categories that is topologically reasonable, not to confine ourselves to $\mathbf{\Delta}^1$ as a type of paths, but to take composite intervals $\tau \in \mathrm{Ty}$.

On the other hand, $(*)$ is isomorphism when C is a Thomason-Kan category. It suffices to see that for every Z in (Cat), the Hom from $\mathrm{hot}(Z)$ into the two sides is [page 136]

[28] probably one finds at most only the degrees δ between -2 and $+2$.

isomorphic via the arrow deduced from $(*)$. Now if C is Thomason-Kan, so is $\underline{\mathrm{Hom}}(Y,C)$ (immediate from the definition of Thomason-Kan in terms of extensions of $A_0 \overset{i}{\hookrightarrow} A$ with $A_0 \to C$ …[29]), hence

$$\mathrm{Hom}_{\mathrm{Hot}}(\mathrm{hot}(Z),\mathrm{hot}(\underline{\mathrm{Hom}}(Y,C))) \overset{\sim}{\longleftarrow} \pi_0(\underbrace{\underline{\mathrm{Hom}}_{\mathrm{Cat}}(Z,\underline{\mathrm{Hom}}(Y,C))}_{\underline{\mathrm{Hom}}_{\mathrm{Cat}}(Z\times Y,C)}) \simeq \pi_0\underline{\mathrm{Hom}}(Z\times Y,C),$$

and

$$\mathrm{Hom}_{\mathrm{Hot}}(\mathrm{hot}(Z),\underline{\mathrm{Hom}}_{\mathrm{Hot}}(\mathrm{hot}(Y),\mathrm{hot}(C))) \simeq \mathrm{Hom}_{\mathrm{Hot}}(\underbrace{\mathrm{hot}(Z)\times\mathrm{hot}(Y)}_{\mathrm{hot}(Z\times Y)},\mathrm{hot}(C))$$

$$\underbrace{\overset{\sim}{\longleftarrow}}_{\text{since } C \text{ is Thomason-Kan}} \qquad \pi_0(\underline{\mathrm{Hom}}_{\mathrm{Cat}}(Z\times Y,C)),$$

O.K.

We thus see that the categories $\underline{\mathrm{Ch}}_{\infty,\Sigma}(X;a,b)$ have no tendency to be Thomason-Kan categories. Yet, its type in Hot must be "the right one", without first having to replace X by a Thomason-Kan envelope $\bar{X}$.

p. 85

12 Back to the "categories of paths" in Cat, as a category of models for Hot

[page 137]

I see exactly two interesting applications of the formalism of paths in categories. One, which above all has forced my hand, is to be able to construct Hom ext, *external* Hom with values in Hot, for every prederivator $\mathbf{D}$ defined by a pair $(\mathcal{M},\Sigma)$ (and I know of no prederivator that is not defined this way).

The other application is to have in (Cat) a *functorial construction,* in terms of an arbitrary object X or an arrow $X \longrightarrow Y$, of the analog of the formalism of Cartan-Serre on path spaces. In this setting, it is understood that one always takes $\Sigma_X = \mathrm{Fl}(X)$, the set of *all* arrows of X. To construct functorially in X in (Cat)

$$\begin{array}{c} \underline{\mathrm{Ch}}_\infty(X) \\ \big\downarrow \\ X\times X, \end{array}$$

one must have fixed an inverse system

$$\underline{t} = (t_n)_{n\in\mathbf{N}}$$

in $\underline{\mathrm{Ty}}$ which is cofinal in $\underline{\mathrm{Ty}}$, which simply means that among the t_n there are some of arbitrarily large useful length (and since

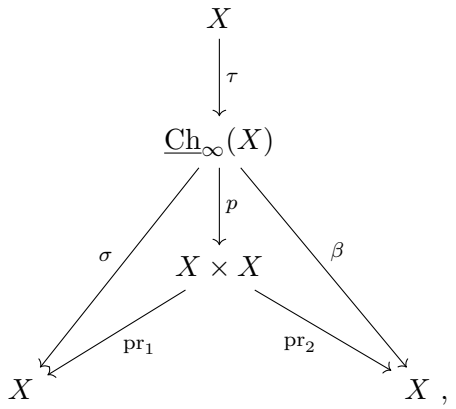
[page 138]

$$n \longmapsto \mathrm{long.ut.}(t_n)$$

is increasing, this implies that $\mathrm{long.ut.}(t_n)$ tends to $+\infty$ for $n \longrightarrow +\infty$). The main facts to check next (if it can be done) are the essential properties of the arrows in the diagram

[29] a bit hasty! One has to know that the cofibrant maps i of Thomason remain so after cartesian multiplication by Y …

p. 86

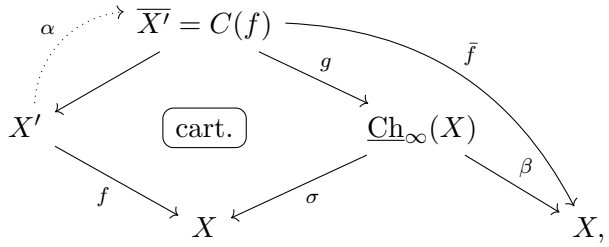


$$\left|\begin{array}{l}\sigma\tau = \beta\tau = \mathrm{id}_X \\ p\tau = \mathrm{diag}_X\end{array}\right.$$

and more generally, for an arrow

$$X' \xrightarrow{f} X$$

in Cat, of the arrows in the corresponding diagram



$$\bar{f}\alpha = (\beta g)\alpha = f,$$

and more particularly of βg. The arrow in dotted lines α is the section of $\overline{X'}$ over X' [page 139] deduced from the section τ of $\underline{\mathrm{Ch}}_\infty(X)$ over X which realizes the decomposition of f into

$$X' \xrightarrow{\alpha} \overline{X'} \xrightarrow{\bar{f}} X,$$

where $\bar{f}$ should be a *"fibrant"* arrow, α a Hot-equivalence and a *"cofibrant"* arrow in the sense of THOMASON. (But whether this is so, remains to be seen!) Here for the moment are the essential properties I would like to check.

1°) p is smooth and proper, and is a HOT-fibration.

2°) So the same holds for $\sigma = \mathrm{pr}_1 p$ and $\beta = \mathrm{pr}_2 p$. Moreover, these HOT-fibrations are *trivial,* i.e. with aspheric fibers.

p. 87

3°) More generally (as regards β), $\overline{X'} \xrightarrow{\bar{f}} X$ should still be proper and smooth and a HOT-fibration, but of course in general not trivial. (It should be trivial if and only if f is a Hot-equivalence.) ([30]).

A word on "duality". The statements about σ (resp. β) translate into statements about β (resp. σ) by passing from $\underline{t}$ to $\underline{t}^o = (t^o_n)$, where $\tau \longmapsto \tau^o$ is the functor in $\underline{\mathrm{Ty}}$ "passage to the opposite interval". We have in fact

an isomorphism functorial in X (passage to the opposite path) [page 140]

$$\boxed{\begin{array}{ccc} c & \longmapsto & c^o \\ \underline{\mathrm{Ch}}^{\underline{t}}_\infty(X) & \longrightarrow & \underline{\mathrm{Ch}}^{\underline{t}^o}_\infty(X)\end{array}} \qquad ([31])$$

[30] follows formally from 1° and 2°, cf. page 170.

which "exchanges source and target". Of course, we do well to take $\underline{t}$ such that we have $\underline{t} = \underline{t}^o$ (this is what I had done, by a sound instinct, right from the start in V), so that we have a canonical involutive functor

$$\begin{array}{ccc} c & \longmapsto & c^o \\ \underline{\mathrm{Ch}}_\infty(X) & \longrightarrow & \underline{\mathrm{Ch}}_\infty(X) \end{array}$$

(I again drop the superscript $\underline{t}$), which "exchanges source and target". So we find, up to canonical isomorphism, the "same" factorization of $f : X' \longrightarrow X$ into $\bar{f} \circ \alpha$, if we do as on page 138 to define $\overline{X'}$ and α, or if in this diagram we interchange the roles of σ and β.

Finally:

4°) If

$$c : a \longrightarrow b$$

is an arbitrary path in X, we would like to deduce from this an isomorphism in Hot (of "transport by c")

$$\overline{X'}_a \xrightarrow[\sim]{\bar{c}} \overline{X'}_b,$$

with of course transitivity for composition,

[page 141]

which will also give

$$\overline{c^{\mathrm{opp}}} = \bar{c}^{-1}$$

(assuming that the trivial paths $a \xrightarrow{\mathrm{id}} a$ give the identity $\overline{X'}_a \xrightarrow[\sim]{\mathrm{id}} \overline{X'}_a$!) ([32],[33]). Of course, $\bar{c}$ must depend only on the *homotopy class* of c (in $\underline{\mathrm{Ch}}_\tau(X; a, b)$, or even in $C = \underline{\mathrm{Ch}}(X; a, b)$, category of all paths in X, from a to b, of unspecified type). In other words, we must have a functor

p. 88

$$\begin{array}{ccc} \Pi(X) & \longrightarrow & \mathrm{Hot} \\ a & \longmapsto & \mathrm{hot}(\overline{X'}_a) \end{array}$$

from the fundamental groupoid of X to Hot. To tell the truth, this has nothing to do with the particular situation studied here - it must work whenever we have a HOT-fibration, here $\overline{X'} \longrightarrow X$, but also $\underline{\mathrm{Ch}}_\infty(X) \longrightarrow X \times X$ ([34]). And for a $\mathbf{D}$-fibration $X' \longrightarrow X$ ($\mathbf{D}$ an arbitrary derivator), and $a \in \mathbf{D}(e)$, we must likewise have two functors

$$\Pi(X) \rightrightarrows \mathbf{D}(e)$$

corresponding to

$$\begin{cases} x & \longmapsto & H^\bullet_{\mathbf{D}}(X'_x, \bullet) \\ x & \longmapsto & H^{\mathbf{D}}_\bullet(X'_x, \bullet). \end{cases}$$

[page 142]

The case we are concerned with here would be the one where $\mathbf{D} = \mathrm{HOT}$, where we take $H^{\mathbf{D}}_\bullet$, and $a =$ final object of $\mathbf{D}(e) = \mathrm{Hot}$.)

5°) If 1°, 2°, 3° are verified, we are morally sure that the construction of the $\underline{\mathrm{Ch}}_\infty(X)$ is indeed the exact equivalent of the formalism of Cartan-Serre. To be completely sure of it, i.e. to link this theory explicitly to the classical constructions, one would have to

[31] cf. page 143 for the passage from $\underline{t}$ to $\underline{t}^-$.

[32] Above all, not to confine oneself to the case where c would be one of the t_n!

[33] describe the effect on the π_0 all the same!

[34] Here too, one should describe the effect of the transport $\bar{c}$ on the π_0 of the fibers.

prove a *comparison statement* with semi-simplicial theory, which would essentially say this: If we denote, for X in Cat, by $\tilde{X}$ the associated object in $\mathbf{\Delta}^\wedge$ (the "nerve" of X), then the arrow

$$\underline{\mathrm{Ch}}_\infty(X)\tilde{} \longrightarrow \underbrace{(X\times X)\tilde{}}_{\simeq \tilde{X}\times\tilde{X}}$$

in $\mathbf{\Delta}^\wedge/(X\times X)\tilde{}$, is W-equivalent over $(X\times X)\tilde{}$ to the arrow

$$\underline{\mathrm{Ch}}(\tilde{X}) \longrightarrow \tilde{X}\times\tilde{X}$$

of the semi-simplicial theory of paths (assuming that there is such a classical arrow defined for *every* object X of $\mathbf{\Delta}^\wedge$). Or that we have at least canonical isomorphisms in Hot

$$\mathrm{hot}_{\mathrm{Cat}}(\underline{\mathrm{Ch}}_\infty(X;a,b)) \;\simeq\; \mathrm{hot}_{\mathbf{\Delta}^\wedge}(\underline{\mathrm{Ch}}(\tilde{X};\tilde{a},\tilde{b})).$$

In 1° and 3°, we want to prove that certain arrows (p and $\bar{f}$) are both *proper* and *smooth* - which is really ideal! I say that each time it suffices to prove one of the two, that the other follows from it (as a "special case") by duality. For this, let us denote by p. 89 [page 143]

$$\tau \longmapsto \tau^-$$

the automorphism (*not anti*-automorphism) of $\underline{\mathrm{Ty}}$ which consists in changing the sign of all the transition arrows - it is thus, at the level of objects, the unique automorphism of the free monoid Ty, which interchanges Δ_+ and Δ_-. To the inverse system $\underline{t}$ in $\underline{\mathrm{Ty}}$ corresponds the inverse system $\underline{t}^-$. That said, we have an isomorphism of categories

$$\boxed{\underline{\mathrm{Ch}}^{\underline{t}}_\infty(X^o) \;\simeq\; (\underline{\mathrm{Ch}}^{\underline{t}^-}_\infty(X))^o} \qquad (^{35})$$

and the commutative diagram

$$\begin{array}{ccc} \underline{\mathrm{Ch}}^{\underline{t}}_\infty(X^o) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}^{\underline{t}^-}_\infty(X)^o \\ \downarrow{\scriptstyle p^{\underline{t}}_{X^o}} & & \downarrow{\scriptstyle (p^{\underline{t}^-}_X)^o} \\ X^o\times X^o & = & (X\times X)^o. \end{array}$$

Thus, to say that $p^{\underline{t}}_X$ is smooth, i.e. that $(p^{\underline{t}}_X)^o$ is proper, is equivalent to saying that $p^{\underline{t}}_{X^o}$ is proper.

So we are left, on this subject, with only two facts to check: the smoothness of $p:\underline{\mathrm{Ch}}_\infty(X)\longrightarrow X\times X$, and that of $\bar{f}:\overline{X'}\longrightarrow X$. [page 144]

a) Smoothness of $p:\underline{\mathrm{Ch}}_\infty(X)\longrightarrow X\times X$. We take an arrow

$$w:(a,b)\longrightarrow(a',b'), \qquad \text{i.e.} \quad \begin{cases} u: & a & \longrightarrow & a' \\ v: & b & \longrightarrow & b', \end{cases}$$

in the base, an object c' over the target, i.e.

$$c'\in \underline{\mathrm{Ch}}_\infty(X;a',b'),$$

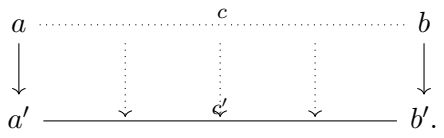
and we look at the category of arrows in $\underline{\mathrm{Ch}}_\infty(X)$, with target c', that lift w,

$$\begin{array}{ccc} \underset{\text{variable}}{c} & \xrightarrow{\varphi} & \underset{\text{fixed}}{c'} \\ \big| & & \big| \\ (a,b) & \xrightarrow{w} & (a',b'), \end{array}$$

p. 90

i.e. the category of pairs (c, φ), where

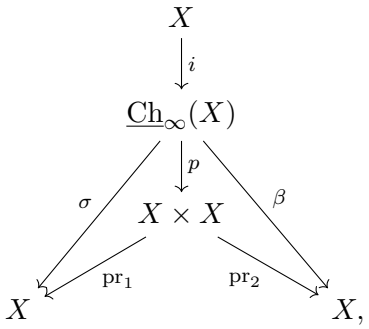
$$\begin{cases} c \in \underline{\mathrm{Ch}}_\infty(X; a, b) \\ \varphi : c \longrightarrow c' \ \ (\in \underline{\mathrm{Ch}}_\infty(X)) \\ \text{an arrow inducing} \ \ u \ \text{ on the sources} \\ \qquad\qquad\qquad\quad v \ \text{ on the targets,} \end{cases}$$



[page 145]

9.11.90

For the preceding question, concerning the arrows in the diagram



we must absolutely specify the description of $\underline{\mathrm{Ch}}_\infty(X)$ *in* Cat, so we must specify the inverse system $\underline{t} = (t_n)_{n \in \mathbf{N}}$. We are in fact going to take

$$t_n = (\underbrace{\Delta_+ \Delta_-}_{\tau_0})^{2n},$$

with the transition morphisms described elsewhere

$$\underbrace{t_{n+p}}_{=\tau_0^{2n+2p}} \longrightarrow \underbrace{t_n}_{\tau_0^{2n}},$$

being given by $\delta_p * \mathrm{id}_{\tau_0^{2n}} * \delta_p$, where $\delta_p : \tau_0^p \longrightarrow 1$ is the degeneracy of τ_0^p. Thus we can regard $\underline{\mathrm{Ch}}_\infty(X)$ as being the category of bi-infinite paths in X p. 91

$$\overbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}^{\underline{\mathrm{Ch}}_2}$$
$$\overbrace{\qquad\qquad\qquad\qquad\qquad}^{\underline{\mathrm{Ch}}_1}$$
$$\cdots \longrightarrow a_{-5} \longleftarrow a_{-4} \longrightarrow a_{-3} \overset{u_{-2}}{\longleftarrow} a_{-2} \overset{u_{-1}}{\longrightarrow} a_{-1} \overset{u_0}{\longleftarrow} \overbrace{a_0}^{\underline{\mathrm{Ch}}_0} \overset{u_1}{\longrightarrow} a_1 \overset{u_2}{\longleftarrow} a_2 \overset{u_3}{\longrightarrow} a_3 \longleftarrow a_4 \longrightarrow a_5 \longleftarrow \cdots,$$

with the u_i identities for $|i|$ large enough.

The (full) subcategory

$$\underline{\mathrm{Ch}}^W_{t_n}(X) \ = \ \underline{\mathrm{Ch}}^W_{(\Delta_+\Delta_-)^{2n}}$$

is identified with the one for which we have

> u_i an identity for $i \notin [-2n+1, 2n]$, i.e. the only "significant" u_i are the $4n$ arrows u_{-2n+1}, u_{-2n+2}, …, u_{2n}.

[page 146]

But it will be more convenient to regard $\underline{\mathrm{Ch}}_\infty(X)$ as the direct limit of a doubly indexed family of full subcategories

$$\underline{\mathrm{Ch}}_{n,m}(X), \qquad n,m \in \mathbf{Z},$$

defining this category as *empty* if $n > m$, and for $n \leq m$ as consisting of the objects

$$\left\{ \begin{array}{l} (a_i,u_i)_{i\in\mathbf{Z}} \in \underline{\mathrm{Ch}}^W_{n,m}(X) \iff u_i \text{ an identity for } i \notin [n+1,m] \\ \overset{\mathrm{id}}{\cdots} a_{n-1} \overset{\mathrm{id}}{\longrightarrow} \underbrace{a_n \overset{u_{n+1}}{\longrightarrow} a_{n+1} \cdots \longrightarrow a_{m-1} \overset{u_m}{\longrightarrow} a_m}_{\text{“useful part” of the path}} \overset{\mathrm{id}}{\longrightarrow} a_{m+1} \overset{\mathrm{id}}{\cdots}. \end{array} \right.$$

Thus for $n = m$, we find the full subcategory isomorphic to X, consisting of the “totally degenerate” infinite paths, i.e. such that *all* the u_i are identities. (This subcategory of $\underline{\mathrm{Ch}}_\infty$ therefore does not depend on the choice of $n = m$.)

$$\overset{\mathrm{id}}{\longrightarrow} \underbrace{a_{-1}}_{=a} \overset{\mathrm{id}}{\longleftarrow} \underbrace{a_0}_{=a} \overset{\mathrm{id}}{\longrightarrow} \underbrace{a_1}_{=a} \overset{\mathrm{id}}{\longleftarrow}$$

All the vertices of such a path are equal to one and the same object of X, and the transition arrows between them are identical.

The structure of $\underline{\mathrm{Ch}}^W_{n,m}(X) \simeq \underline{\mathrm{Ch}}^W_{\tau_{n,m}}(X)$ depends only on the length $m-n$ of $\tau_{n,m}$ and on the *parity* of n. If n is even, the initial direction of $\tau_{n,m}$ is positive, and

$$\tau_{n,m} = \underbrace{\Delta_+\Delta_-\cdots}_{m-n \text{ factors in alternation}} = \left\{ \begin{array}{ll} (\Delta_+\Delta_-)^{\frac{m-n}{2}} & \text{if } m \text{ even} \\ (\Delta_+\Delta_-)^{\frac{m-n-1}{2}}\Delta_+ & \text{if } m \text{ odd} \end{array} \right.$$

, whereas for n odd it is just the reverse

p. 92
[page 147]

$$\tau_{n,m} = \underbrace{\Delta_-\Delta_+\cdots}_{m-n \text{ alternating factors}} = \left\{ \begin{array}{ll} (\Delta_-\Delta_+)^{\frac{m-n}{2}} & \text{if } m \text{ odd} \\ (\Delta_-\Delta_+)^{\frac{m-n-1}{2}}\Delta_- & \text{if } m \text{ even} \end{array} \right.$$

. The initial direction agrees with the parity of n, i.e. equals $(-1)^n$, whereas the terminal direction is opposite to the parity of m, i.e. equals $(-1)^{m+1}$.

I note that we have $\tau_0 = (\tau_0)^o$, whence

$$\underline{t} = \underline{t}^o, \qquad \text{but } \underline{t} \neq \underline{t}^-,$$

and yet we have isomorphisms functorial in X

$$\underline{\mathrm{Ch}}_{\underline{t}}(X) \simeq \underline{\mathrm{Ch}}_{\underline{t}^-}(X)$$

respecting source and target, because of an isomorphism

$$\underline{t} \simeq \underline{t}^- \qquad \text{in } \mathrm{Pro}(\underline{\mathrm{Ty}}).$$

We can say e.g. that the *translation functor* T^p of amplitude p (for $p \in \mathbf{Z}$) is defined on $\underline{\mathrm{Ch}}^W_{\underline{t}}(X)$, by

$$\left\{ \begin{array}{l} T^p(((a_i),(u_i))_{i\in\mathbf{Z}}) = ((b_j),(v_j))_{j\in\mathbf{Z}} \\ \text{with} \left\{ \begin{array}{lcl} b_j & = & a_{j-p} \\ v_j & = & u_{j-p} \end{array} \right. \end{array} \right.$$

[35] cf. page 140 for the passage from $\underline{t}$ to $\underline{t}^o$.

and that

$$T^p : \left\{ \begin{array}{ccc} \underline{\mathrm{Ch}}_t(X) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}_{-t}(X) \\ \underline{\mathrm{Ch}}_{-t}(X) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}_t(X) \end{array} \right\} \text{ if } p \text{ odd}$$

$$T^p : \left\{ \begin{array}{ccc} \underline{\mathrm{Ch}}_t(X) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}_t(X) \\ \underline{\mathrm{Ch}}_{-t}(X) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}_{-t}(X) \end{array} \right\} \text{ if } p \text{ even.}$$

In other words, we can formally regard the sum category [page 148]

$$\underline{\mathrm{Ch}}_{\pm}(X) = \underline{\mathrm{Ch}}_{+}(X) \amalg \underline{\mathrm{Ch}}_{-}(X)$$

and the translation operation T as an automorphism which interchanges the two pieces, and T^p is nothing other than its p-th iterate, which interchanges the pieces $+$ and $-$ if p odd, which stabilizes them if p even.

There is also the involutive automorphism of passage to the opposite path, which p. 93
interchanges source and target, and which I define (in accordance with the canonical identity isomorphism

$$\underline{t} \xrightarrow{\sim} \underline{t}^o \quad)$$

as the symmetry S

$$\left\{ \begin{array}{l} S((a_i),(u_i)) = ((b_j),(u_j)) \text{ with} \\ \quad b_j = a_{-j} \\ \quad u_j = u_{-j+1}. \end{array} \right.$$

If I regard it as acting on $\underline{\mathrm{Ch}}^W_{\pm}(X)$, it is related to the functor T by

$$\left\{ \begin{array}{rcl} S^2 & = & \mathrm{id} \\ STS^{-1} & = & T^{-1}, \end{array} \right.$$

so that it is the semidirect group of $\mathbf{Z}/2\mathbf{Z}$ by $\mathbf{Z}$ (affine group of dimension 1 of the ring $\mathbf{Z}$) that acts on $\underline{\mathrm{Ch}}_{\pm}$.

We are embarrassed to find an automorphism T^2 of the situation $\underline{\mathrm{Ch}}_{\infty}(X) \longrightarrow X \times X$, [page 149]
without it being obvious that in Hot, this functor induces the identity! I see no way to connect a path c and T^2c by an arrow in $\underline{\mathrm{Ch}}_{\infty}$, nor even a "canonical" path. But I tell myself that this automorphism is described by an automorphism, not of the inverse system

$$(\tau_{n,m})_{n,m \in \mathbf{Z}^o \times \mathbf{Z},\ n \leq m}$$

(where $\mathbf{Z}^o$ denotes the ordered set *opposite* to $\mathbf{Z}$), but of the *pro-object* of $\underline{\mathrm{Ty}}$ which it defines (which is of course "the same" for all the inverse systems cofinal on the left in $\underline{\mathrm{Ty}}$). It already follows from this that T^2 induces the identity on the ind-object of $\mathrm{Cat}\,\mathrm{hot}$, $\underline{\underline{\mathrm{Ch}}}_{\infty}(X)$, and a fortiori on the corresponding ind-object of Hot, or of $\mathrm{Hot}_0^{\wedge}$, or finally for the object of $\mathrm{Hot}_0^{\wedge}$ itself. Likewise for the $\underline{\mathrm{Ch}}_{\infty}(X;a,b)$. It follows, returning to $\underline{\mathrm{Ch}}_{\infty}(X)$ itself and to the $\underline{\mathrm{Ch}}_{\infty}(X;a,b)$, that the automorphism induced by T^2 is the identity in $\mathrm{Hot}_0^{\wedge}$. I hope that $\mathrm{Hot} \longrightarrow \mathrm{Hot}_0^{\wedge}$ is at least faithful, so that this will prove [page 150]
that T^2 induces the identity in Hot as well.

I return to questions 1° to 4° of a few days ago.

For 1°), we must prove that

$$p : \underline{\mathrm{Ch}}_{\infty}(X) \longrightarrow X \times X$$

is smooth and proper and a Hot-fibration. By duality, p proper will follow from p smooth. But $\underline{\mathrm{Ch}}_{\infty}(X)$ is $\varinjlim$ of the $\underline{\mathrm{Ch}}_{n,m}(X)$, with $n \longrightarrow -\infty$, $m \longrightarrow +\infty$, and where we can p. 94

restrict ourselves to n and m *even.* Now we see that for n, m even, i.e. $\tau_{n,m}$ with forward initial arrow, with final arrow

retrograde, $\underline{\mathrm{Ch}}_{n,m}(X) \simeq \underline{\mathrm{Ch}}_{\tau_{n,m}}(X)$ is Cat-*fibered* over $X \times X$, hence smooth (i.e. HOT-smooth), and this passes to $\varinjlim$ ([36]).

Thus $\underline{\mathrm{Ch}}_\infty(X) \xrightarrow{p} X \times X$ is both proper and smooth ([37]). To say that it is a HOT-fibration, amounts e.g. to saying that for

$$(a,b) \xrightarrow{(u,v)} (a',b')$$

in $X \times X$, the corresponding arrow

$$\underline{\mathrm{Ch}}_\infty(X)/(a,b) \longrightarrow \underline{\mathrm{Ch}}_\infty(X)/(a',b')$$

is a Hot-equivalence. Now, since p is proper, the inclusions of the fibers at (a,b) and at (a',b') into the respective terms are Hot-equivalences: [page 151]

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_\infty(X)/(a,b) & \xrightarrow{\rho} & \underline{\mathrm{Ch}}_\infty(X)/(a',b') \\ \wr\uparrow & & \wr\uparrow \\ \underline{\mathrm{Ch}}_\infty(X;a,b) & \xrightarrow{\varphi} & \underline{\mathrm{Ch}}_\infty(X;a',b') \end{array} \qquad (*)$$

Thus we find a dotted arrow in Hot, which we have to prove is a Hot-equivalence. But by considering

$$u : a \longrightarrow a' \qquad v : b \longrightarrow b'$$

as *paths* in X, and

$$u^o : a' \longleftarrow a$$

as a retrograde path from a' to a, the general variance theory (in IX) does indeed give us an isomorphism in (Hot):

$$\underline{\mathrm{Ch}}_\infty(X;a,b) \longrightarrow \underline{\mathrm{Ch}}_\infty(X;a',b'),$$

we feel like saying that it is φ, i.e. that by inserting it in place of φ, the square above becomes commutative (which would prove what we want). I think that if we work with the ind-objects $\underline{\underline{\mathrm{Ch}}}_\infty(X)/(a,b)$, $\underline{\underline{\mathrm{Ch}}}_\infty(X;a,b)$ etc., for which we have pairings which can be spelled out nicely at the finite stages, and give rise to the φ, this should be easy (though a little tedious). This would not for all that prove the commutativity of $(*)$ in Hot, but only in $\mathrm{Hot}_0^\wedge$, which is perhaps not sufficient. But this suffices to prove that p. 95 the arrow ρ in $(*)$ becomes isomorphism in $\mathrm{Hot}_0^\wedge$, hence it is isomorphism. [page 152]

2°) We have to show that the HOT-fibrations

$$\underline{\mathrm{Ch}}_\infty(X) \underset{\beta}{\overset{\sigma}{\rightrightarrows}} X$$

are trivial, or what amounts to the same thing, that they have aspheric fibers. It suffices to see this for σ, hence that

$$\underline{\mathrm{Ch}}_\infty(X;a,?)$$

[36] **NB** $\underline{\mathrm{Ch}}_\infty(X)$ is neither fibered nor cofibered over $X \times X$ - for the inclusion functors between the $\underline{\mathrm{Ch}}_{n,m}$ are not cartesian resp. cocartesian.

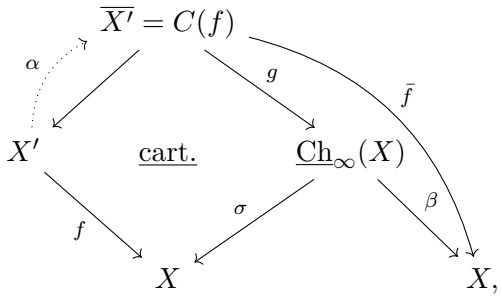
[37] 12.11. But that already implies that it is a fibration!

is aspheric. But this category is the direct limit of the $\underline{\mathrm{Ch}}_n(X; a, ?)$, i.e. of the

$$\underline{\mathrm{Hom}}^{\text{fixed source}}(\tau^n, X).$$

But since the interval I_{τ^n} retracts by deformation onto its origin, it is immediate that this $\underline{\mathrm{Hom}}$ category is even contractible.

3°) Let us consider



I would like to prove that $\bar{f}$ is a proper and smooth **D**-fibration ([38]). [page 153]

$\overline{X'}$ is the category of pairs

$$(x', c) \ \in \ \mathrm{Ob}\, X' \times \mathrm{Ch}_\infty(X)$$

such that

$$\sigma(c) = f(x')$$

and

$$\bar{f}(x', c) = \beta(c), \qquad \text{target of } c.$$

The fiber of $\overline{X'}$, $\bar{f}$ at $x_0 \in X$ is thus the category of pairs $(x', c) \in \mathrm{Ob}\, X' \times \mathrm{Ch}_\infty(X)$, such that

$$\sigma(c) = f(x'), \qquad \beta(c) = x_0.$$

We can also regard $\overline{X'}$ as the direct limit of the subcategories $\overline{X'}_{n,m}$, corresponding to the pairs (x', c) with $c \in \mathrm{Ch}_{n,m}(X)$. We find that for m even (terminal direction of $\tau_{n,m}$ retrograde) the $\overline{X'}_{n,m}$ are Cat-fibered over X, for m odd (terminal direction of $\tau_{n,m}$ direct) they are Cat-cofibered. So we conclude again as before, $\overline{X'}$ being (at will) a filtering direct limit of subcategories smooth over X, or proper over X. p. 96

Finally we have to prove that $\overline{X'} \xrightarrow{\bar{f}} X$ is a **D**-fibration ([39]), which amounts e.g. to saying that for an arrow [page 154]

$$a \xrightarrow{v} b$$

in X, the induced arrow

$$\overline{X'}/a \xrightarrow{\bar{v}} \overline{X'}/b$$

is a Hot-equivalence. We already know as before that we have a diagram with Hot-equivalences as vertical arrows

$$\begin{array}{ccc} \overline{X'}/a & \longrightarrow & \overline{X'}/b \\ \wr\uparrow & & \wr\uparrow \\ \overline{X'}_a & \dashrightarrow & \overline{X'}_b, \end{array}$$

[38] That follows formally from what has already been seen, cf. pages 170-171.

[39] This already follows from the fact that it is proper and smooth!

and it all comes down to showing that a certain arrow between $\overline{X'}_a$ and $\overline{X'}_b$ in Hot is isomorphism. I feel like defining such an arrow directly in Hot, or even in Cat. Now for odd m, the $\overline{X'}_{n,m}$ are Cat-cofibered over X. For $\overline{X'}_{-m,m}$ (m odd) we therefore find

$$(\overline{X'}_{-m,m})_a \xrightarrow{v_*} (\overline{X'}_{-m,m})_b,$$

the trouble is that they are not compatible with the transition morphisms for variable m. [page 155]
They surely are, however, *up to homotopy*, so we will find at least an arrow of ind-objects in Cat hot

$$\overline{\overline{X'}}_a \xrightarrow{v_*} \overline{\overline{X'}}_b,$$

which by passing to $\mathrm{Hot}_0^\wedge$ gives an arrow all the same

$$\overline{X'}_a \xrightarrow{v_*} \overline{X'}_b.$$

Working with even m, we likewise find functors in the opposite directions

$$(\overline{X'}_{-m,m})_a \xleftarrow{v^*} (\overline{X'}_{-m,m})_b,$$

which do not glue together for variable m, but which glue together up to homotopy, to give an arrow in Ind Cat hot p. 97

$$\overline{\overline{X'}}_a \xleftarrow{v^*} \overline{\overline{X'}}_b,$$

whence also

$$\overline{X'}_a \xleftarrow{v^*} \overline{X'}_b \qquad \text{in } \mathrm{Hot}_0^\wedge.$$

On the other hand, we should be able to show soritally, stage by stage, that v_*v^* and [page 156]
v^*v_* in Ind Cat hot are equal to the identity, hence v_* and v^* are isomorphisms inverse to each other. We must check on the other hand soritally that

$$\begin{array}{ccc} \overline{\overline{X'}}/a & \xrightarrow{\bar{v}} & \overline{\overline{X'}}/b \\ \uparrow & & \uparrow \\ \overline{\overline{X'}}_a & \xrightarrow{v_*} & \overline{\overline{X'}}_b, \end{array}$$

in Ind Cat hot is commutative. Passing to the corresponding diagram in $\mathrm{Hot}_0^\wedge$, by taking direct limits, we find a commutative square in $\mathrm{Hot}_0^\wedge$

$$\begin{array}{ccc} \overline{X'}/a & \xrightarrow{\bar{v}} & \overline{X'}/b \\ \uparrow\wr & & \uparrow\wr \\ \overline{X'}_a & \xrightarrow[\sim]{v_*} & \overline{X'}_b, \end{array}$$

whence it follows that $\bar{v}$ is isomorphism in $\mathrm{Hot}_0^\wedge$, hence also in Hot.

To check that the HOT-fibration $\bar{f}$ is trivial if and only if f is a HOT-equivalence, we would have to have developed in Cat the formalism of the exact sequence of loops - we are very close, evidently!

p. 98

13.11.90 [page 157]

13 Where we finally prove that $\underline{\mathrm{Ch}}(X;-,-)$ is as good as (or even better than) $\underline{\mathrm{Ch}}_\infty(X;-,-)$.

I believe after all that the categories $\underline{\mathrm{Ch}}(X)$, $\underline{\mathrm{Ch}}(X;a,b)$ (without the subscript ∞!) are "good", that the canonical arrows in Hot

$$\begin{cases} \underline{\mathrm{Ch}}_\infty(X) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}(X) \\ \underline{\mathrm{Ch}}_\infty(X;a,b) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}(X;a,b) \end{cases}$$

are isomorphisms. (The first is even isomorphism in $\mathrm{HOT}(X \times X)$, which implies the second relation, on passing to the fibers …) I believe that I am now (after XI) in a position to prove it. This will follow from the following two key results:

1) The canonical functor
$$X \longrightarrow \underline{\mathrm{Ch}}(X)$$
(associating to every $x \in \mathrm{Ob}\, X$ the empty path with source and target x) is a Hot-equivalence, and:

2) The source-target functor
$$\underline{\mathrm{Ch}}(X) \longrightarrow X \times X$$
is *smooth* and *proper* (hence a "perfect Hot-fibration").

This should give what we are looking for. Since the $X \times X$-arrow $\underline{\mathrm{Ch}}_\infty(X) \longrightarrow \underline{\mathrm{Ch}}(X)$ (and the deduced arrows $\underline{\mathrm{Ch}}_\infty(X;a,b) \longrightarrow \underline{\mathrm{Ch}}(X;a,b)$) was defined by way of the category

$$\mathcal{C}_{\mathbf{N}^o} = \mathbf{N}^o \times_{\underline{\mathrm{Ty}}} \mathcal{C}$$

(where $\mathcal{C} = \underline{\mathrm{Ch}}(X)$ considered as a fibered category over $\underline{\mathrm{Ty}}$, and $\mathbf{N}^o \longrightarrow \underline{\mathrm{Ty}}$ is the inverse system of the $\underbrace{(\Delta_+\Delta_-)^{2n}}_{t_n}\dots$)

with the help of the two canonical arrows [page 158]

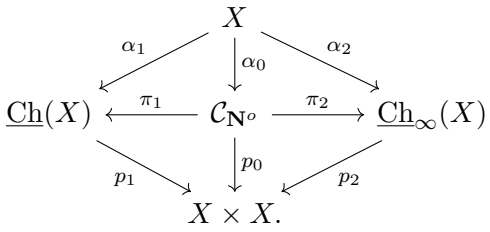
$$\begin{array}{ccc} \mathcal{C}_{\mathbf{N}^o} & \longrightarrow & \underline{\mathrm{Ch}}_\infty(X) \\ \downarrow & & \\ \underline{\mathrm{Ch}}(X), & & \end{array}$$

we will also have to prove the results corresponding to 1) 2) for $\mathcal{C}_{\mathbf{N}^o}$, i.e. p. 99

1′)
$$\begin{array}{ccc} X & \longrightarrow & \mathcal{C}_{\mathbf{N}^o} \\ \wr\downarrow & & \\ \mathcal{C}_{t_0} = \mathcal{C}_0 & & \end{array}$$
(the canonical inclusion of $\mathcal{C}_0$, fiber of $\mathcal{C}_{\mathbf{N}^o}$ (fibered over $\mathbf{N}^o$) at $0 \in \mathbf{N}^o$) is a Hot-equivalence, and:

2′) $\mathcal{C}_{\mathbf{N}^o} \longrightarrow X \times X$ is proper and smooth.

With these four technical results, which I am going to prove shortly (and $1'$ and $2'$ come as no surprise to me), we win. Let us indeed consider the diagram



By 1) and $1'$) (concerning α_1 and α_0) and the similar result already known for α_2, these three functors are Hot-equivalences. So the same holds for π_1, π_2. Since p_1, p_0, p_2 are Hot-fibrations (being proper and smooth), it follows that π_1 and π_2 induce Hot-equivalences on the fibers [page 159]

$$\underline{\mathrm{Ch}}(X;a,b) \xleftarrow{\pi_1(a,b)} \mathcal{C}_{\mathbf{N}^o}(a,b) \xrightarrow{\pi_2(a,b)} \underline{\mathrm{Ch}}_\infty(X;a,b).$$

I had checked this (without difficulty) already for π_2 and the $\pi_2(a,b)$, which had made it possible, by inverting π_2 in Hot, to define *in* Hot

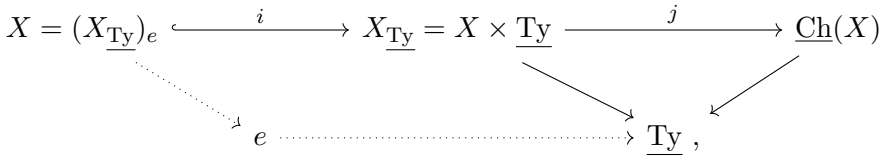
$$\begin{cases} \underline{\mathrm{Ch}}_\infty(X) & \longrightarrow & \underline{\mathrm{Ch}}(X) \\ \underline{\mathrm{Ch}}_\infty(X;a,b) & \longrightarrow & \underline{\mathrm{Ch}}(X;a,b), \end{cases}$$

and to say that these are isomorphisms means precisely that π_1 and the $\pi_1(a,b)$ are isomorphisms.

I do not think that $\mathcal{C}_{\mathbf{N}^o}$ is in itself of any other interest, than to make it possible to compare the $X \times X$-categories $\underline{\mathrm{Ch}}_\infty(X)$ and $\underline{\mathrm{Ch}}(X)$, with the help of the preceding diagram. On the other hand, $\underline{\mathrm{Ch}}_\infty(X)$ and $\underline{\mathrm{Ch}}(X)$ appear to me as the two essential ingredients of the formalism of paths in Cat.

I am going to prove 1, 2, $1'$, $2'$.

1) Let us consider the diagram [p. 100] [page 160]



where $X_{\underline{\mathrm{Ty}}}$ and $\underline{\mathrm{Ch}}(X)$ are considered as Cat-fibered categories over $\underline{\mathrm{Ty}}$, and the functor j is deduced from the canonical functors

$$X \longrightarrow \underline{\mathrm{Ch}}(X)_\tau \overset{\mathrm{def}}{=} \underline{\mathrm{Ch}}_\tau(X),$$

associating to $x \in \mathrm{Ob}\, X$ the path $p_\tau^*(x)$, where $p_\tau : \tau \longrightarrow e$ is the projection from τ to the unit element (and final element) $e = 1_{\underline{\mathrm{Ty}}}$ of $\underline{\mathrm{Ty}}$, and $\underline{\mathrm{Ch}}_e(X)$ is identified with X. We have to check the

Lemma. *For every $\tau \in \underline{\mathrm{Ty}}$, the preceding functor is a* Hot*-equivalence (and even a homotopism).*

But we know that for two categories $\mathcal{X}$, $\mathcal{C}$ smooth over another, T, every T-morphism from $\mathcal{X}$ to $\mathcal{C}$ that induces Hot-equivalences on the fibers, is a Hot-equivalence. Thus j is a Hot-equivalence. On the other hand, $\underline{\mathrm{Ty}}$, having a final object, is aspheric, hence $X \times \underline{\mathrm{Ty}} \xrightarrow{\mathrm{pr}_1} X$ is a Hot-equivalence, whence it follows at once that i is a Hot-equivalence. [page 161] So likewise ji, now $ji = \alpha_1$, q.e.d.

2) Let us prove e.g. the smoothness of $\underline{\mathrm{Ch}}(X) \longrightarrow X \times X$, properness will follow from it by applying this result to X^{opp}. So let

$$(a', b') \longrightarrow (a, b), \qquad \text{i.e. } a' \xrightarrow{u} a, \quad b' \xrightarrow{v} b,$$

be an arrow in $X \times X$, and c an object of $\underline{\mathrm{Ch}}(X)$ over (a, b):

$$\begin{array}{ccc} a' & \overset{c'}{\dashrightarrow} & b' \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle v} \\ a & \xrightarrow{c} & b. \end{array}$$

We look at the category $\mathcal{C}(c; u, v)$ of pairs (c', w) that lift $(c, (u, v))$, where therefore c' is a path (of unspecified type) from a' to b', and

$$w : c' \longrightarrow c$$

an arrow in $\underline{\mathrm{Ch}}(X)$ which on the source objects induces u, and on the target objects induces v (thus c' is a path from a' to b'). We have to prove that this category $\mathcal{C}(c; u, v)$ is aspheric. For this, we look at the composite path

p. 101 [page 162]

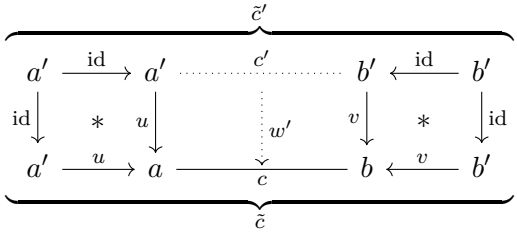
$$\tilde{c} = \mathrm{ch}(u)\ c\ \mathrm{ch}(v)^o,$$

where $\mathrm{ch}(u)$ is the (direct) path from a' to a defined by $u : a' \longrightarrow a$, and $\mathrm{ch}(v)^o$ therefore denotes the (retrograde) path from b to b' defined by $v : b' \longrightarrow b$. If c is of type τ, $\tilde{c}$ is therefore of type $\Delta_+ \tau \Delta_-$. We are going to compare $\mathcal{C}(c; u, v)$ and the category $\underline{\mathrm{Ch}}(X)/\tilde{c}$, with the help of two functors

$$\mathcal{C}(c; u, v) \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \underline{\mathrm{Ch}}(X; a', b')/\tilde{c}\ ,$$

which we will prove define arrows inverse to one another in Hot. Since $\underline{\mathrm{Ch}}(X)/\tilde{c}$ is contractible (having a final object), hence aspheric, the same holds for $\mathcal{C}(c; u, v)$, and we will win.

We define φ by $\boxed{\varphi(c', w') = (\tilde{c}', \tilde{w}'),}$ where (c' being of type τ') $\tilde{c}'$ is the path of type $\Delta_+ \tau' \Delta_-$ obtained from c' as inverse image under the degeneracy $\Delta_+ \tau' \Delta_- \longrightarrow \tau'$, i.e. we extend c' by a "Δ_+ step" $a' \xrightarrow{\mathrm{id}} a'$ on the left and by a "Δ_- step" $b' \xleftarrow{\mathrm{id}} b'$ on the right, cf. figure.



We can say that

[page 163]

$$\tilde{c}' = a'_{\Delta_+} \circ c' \circ b'_{\Delta_-},$$

where in general, if $x \in X$ and $\tau \in \mathrm{Ty}$, x_τ denotes the "constant path" of value x, type τ. On the other hand we have obvious morphisms

$$a'_{\Delta_+} \longrightarrow \mathrm{ch}(u), \qquad b'_{\Delta_-} \longrightarrow \mathrm{ch}(v)^{\mathrm{opp}},$$

as shown by the vertical arrows $(\mathrm{id}_{a'}, u)$ and $(v, \mathrm{id}_{b'})$ in the two squares marked $*$ in the diagram, arrows which lie one in $\underline{\mathrm{Ch}}_{\Delta_+}(X)$, the other in $\underline{\mathrm{Ch}}_{\Delta_-}(X)$. This said, we set

$$\tilde{w}' : \tilde{c}' \longrightarrow \tilde{c}, \qquad \tilde{w}' = \lambda(u) * w' * \mu(v),$$

where

$$\begin{cases} \lambda(u) : & a'_{\Delta_+} & \longrightarrow & \mathrm{ch}(u) \\ \mu(v) : & b'_{\Delta_-} & \longrightarrow & \mathrm{ch}(v)^{\mathrm{opp}} \end{cases}$$

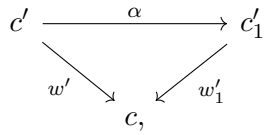
(cf. figure). The pair p. 102

$$(\tilde{c}', \tilde{w}') \in \mathrm{Ob}\, \underline{\mathrm{Ch}}(X)/\tilde{c}$$

thus defined depends functorially on (c', w') varying in $\mathcal{C}(c; u, v)$: If (c'_1, w'_1) is a second object of this category, an arrow from the first to the second is an arrow

$$c' \xrightarrow{\alpha} c'_1$$

in $\underline{\mathrm{Ch}}(X; a', b')$, such that we have commutativity in

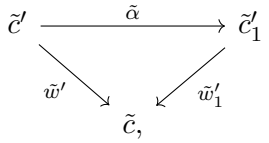


$$w' = w'_1 \alpha.$$

Such an arrow α defines

$$\tilde{\alpha} = a'_{\Delta_+} * \alpha * b'_{\Delta_-} : \tilde{c}' \longrightarrow \tilde{c}'_1$$

in $\underline{\mathrm{Ch}}(X; a', b')$, and we check at once that we have commutativity in [page 164]



i.e. $\tilde{w}' = \tilde{w}'_1 \tilde{\alpha}$,

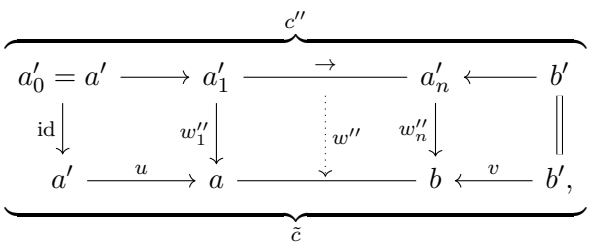
i.e. that $\tilde{\alpha}$ is indeed an arrow in $\underline{\mathrm{Ch}}(X; a', b')/\tilde{c}$. Likewise, that $\alpha \longmapsto \tilde{\alpha}$ defines a functorial dependence, i.e. $(\beta\alpha)\tilde{} = \tilde{\beta}\tilde{\alpha}$, is obvious from the definition of $\tilde{\alpha}$. So, we have indeed defined

$$\mathcal{C}(c; u, v) \xrightarrow{\varphi} \underline{\mathrm{Ch}}(X; a', b')/\tilde{c},$$

let us now define

$$\mathcal{C}(c; u, v) \xleftarrow{\psi} \underline{\mathrm{Ch}}(X; a', b')/\tilde{c}.$$

Let (c'', w'') be an object of $\underline{\mathrm{Ch}}(X; a', b')/\tilde{c}$,



thus p. 103

$$w'' : c'' \longrightarrow \tilde{c}, \qquad \text{arrow in } \underline{\mathrm{Ch}}(X; a', b'),$$

let us note that we have an obvious retraction arrow

$$\tilde{c} \xrightarrow{\rho_c} c \qquad \text{associated with } \Delta_+\tau\Delta_- \longrightarrow \tau \text{ (retraction)},$$

inducing u and v on the source and the target:

$$\begin{array}{ccccccc}
a' & \xrightarrow{u} & a & \xrightarrow{\quad c \quad} & b & \xleftarrow{v} & b' \\
\downarrow{\scriptstyle u} & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle v} \\
a & \xrightarrow{\mathrm{id}} & a & \xrightarrow{\quad c \quad} & b & \xleftarrow{\mathrm{id}} & b
\end{array}
\qquad \left.\begin{array}{l} \\ \\ \end{array}\right\} \tilde{c} \qquad \in \underline{\mathrm{Ch}}_{\Delta_+\tau\Delta_-}(a', b')$$

$$\left.\begin{array}{l} \\ \\ \end{array}\right\} \delta^*_\tau(c) \qquad \in \underline{\mathrm{Ch}}_{\Delta_+\tau\Delta_-}(a, b)$$

$$a \xrightarrow{\quad c \quad} b \qquad \left.\begin{array}{l} \\ \end{array}\right\} c \qquad \in \underline{\mathrm{Ch}}_{\tau}(a, b).$$

So we define an object [page 165]

$$(c', w') = \psi(c'', w'') \ \in \mathrm{Ob}\,\mathcal{C}(c; u, v)$$

by

$$c' = c'', \qquad w' = \rho_c w'',$$

and this is functorial in (c'', w'').

The composite $\varphi\psi$ being an endomorphism of the contractible category $\underline{\mathrm{Ch}}(X; a', b')/\tilde{c}$, is homotopic to the identity. We have to check that the same is true of $\psi\varphi$, an endomorphism of $\mathcal{C}(c; u, v)$. This functor associates to (c', w'), $(\tilde{c}', \overline{w'})$, where $\tilde{c}'$ is defined as above, $\tilde{c}' = a'_{\Delta_+} \circ c' \circ b'_{\Delta_-}$, and where $\overline{w'}$ is the composite

$$\underbrace{\tilde{c}'}_{=\delta'^*_{\tau'}(c')} \quad \underbrace{\xrightarrow{\rho_{c'}}}_{\substack{\text{canonical} \\ \text{arrow}}} \quad c' \xrightarrow{w'} c.$$

So we have

$$(\psi\varphi)(c', w') = (\tilde{c}', \overline{w'}) \xrightarrow{\rho_{c'}} (c', w'),$$

this is indeed a morphism in $\mathcal{C}(c; u, v)$, i.e. we have commutativity in

$$\begin{array}{ccc}
\tilde{c}' & \xrightarrow{\quad \rho_{c'} \quad} & c' \\
& {\scriptstyle \overline{w'}} \searrow \quad \swarrow {\scriptstyle w'} & \\
& c &
\end{array}
\qquad \text{(arrow in } \underline{\mathrm{Ch}}(X; a', b')\text{)}.$$

Thus, the functor $\psi\varphi$ is indeed homotopic to the identity functor. We have won!

Let us pass to $1'$, $2'$) for $\mathcal{C}_{\mathbf{N}^o}$. The assertion $1'$) is proved in exactly the same way as 1) for $\underline{\mathrm{Ch}}(X)$, by regarding $\mathcal{C}_{\mathbf{N}^o}$ as fibered over $\mathbf{N}^o$ (which has a final object 0), and using the diagram [p. 104] [page 166]

$$\begin{array}{ccccc}
X = (X_{\mathbf{N}^o})_e & \xrightarrow{\quad i' \quad} & X_{\mathbf{N}^o} = X \times \mathbf{N}^o & \xrightarrow{\qquad j' \qquad} & \mathcal{C}_{\mathbf{N}^o} \\
& & & \searrow \quad \swarrow & \\
& & & \mathbf{N}^o\,, &
\end{array}$$

where j' is (cartesian and) a Hot-equivalence on each fiber by the lemma, hence a Hot-equivalence globally, and i' is a Hot-equivalence since $\mathbf{N}^o$ is aspheric (having a final object). Thus $j'i'$ is a Hot-equivalence, q.e.d. [page 167]

It remains to prove that

$$\mathcal{C}_{\mathbf{N}^o} \longrightarrow X \times X$$

is proper and smooth. Here again, it is exactly the same argument as before, I presume. There is, however, a technical complication. We had taken, to define

$$\underline{t} : \mathbf{N}^o \longrightarrow \underline{\mathrm{Ty}}, \qquad \underline{t} = (t_n)_n,$$

the choice

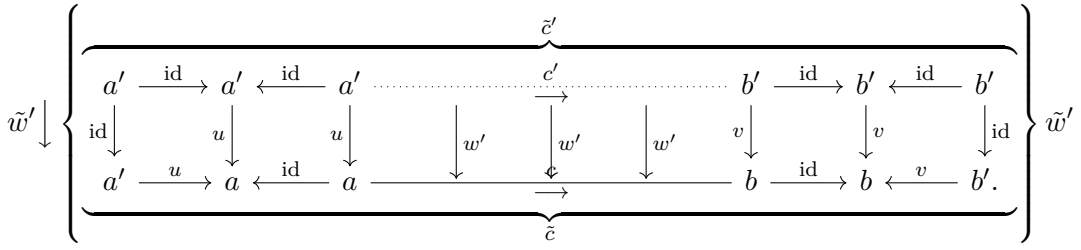
$$t_n = (\Delta_+\Delta_-)^{2n},$$

or if one prefers

$$t_n = (\Delta_+\Delta_-)^{n},$$

in any case the t_n have initial direction $+$, terminal $-$, and they are reduced types. If one takes $\Delta_+ t_n \Delta_-$, one finds a nonreduced type, which is not among the t_n. It would be of no use to change the definition of t_n to $t_n = (\Delta_-\Delta_+)^{\text{something}}$, for this time $\Delta_+ t_n \Delta_-$ will indeed be reduced, but will not have the right initial and terminal signs. Thus to define $\tilde{c}$ in terms of c, as an object of $\mathcal{C}_{\mathbf{N}^o}$, it will be necessary, if c is of type $\tau = t_n$, to take $\tilde{c}$ of type $\Delta_+\Delta_-\tau\Delta_+\Delta_- = t_{n+1}$ (taking $t_{n+1} = (\Delta_+\Delta_-)^{2n+2}$), and likewise to define $\tilde{c}'$ in terms of c'. The relevant definitions of the passage from (c', w') to $(\tilde{c}', \tilde{w}')$ are described in the figure:

[page 168]
p. 105



Thus

$$\begin{cases} \tilde{c} = \xi_u \circ c \circ \eta_v & \text{where } \left\{ \begin{array}{llllll} \xi_u : & a' & \xrightarrow{u} & a & \xleftarrow{\mathrm{id}} & a \\ \eta_v : & b & \xrightarrow{\mathrm{id}} & b & \xleftarrow{v} & b' \end{array} \right\} \text{ type } \Delta_+\Delta_- \\ \tilde{c}' = a'_{\Delta_+\Delta_-} \circ c' \circ b'_{\Delta_+\Delta_-} & \\ \tilde{w}' = \lambda(u) * w' * \mu(v) & \text{with } \left\{ \begin{array}{lll} \lambda(u) : & a'_{\Delta_+\Delta_-} & \longrightarrow \xi_u \\ \mu(v) : & b'_{\Delta_+\Delta_-} & \longrightarrow \eta_v \end{array} \right\} \text{ as in figure.} \end{cases}$$

One thus finds, mutatis mutandis,

$$\mathcal{C}_{\mathbf{N}^o}(c; u, v) \xrightarrow{\varphi} \mathcal{C}_{\mathbf{N}^o}(a', b')/\tilde{c}, \qquad \varphi(c', w') = (\tilde{c}', \tilde{w}'),$$

and one defines the functor

$$\mathcal{C}_{\mathbf{N}^o}(c; u, v) \xleftarrow{\psi} \mathcal{C}_{\mathbf{N}^o}(a', b')/\tilde{c}$$

as follows: to (c'', w'') in $\mathcal{C}_{\mathbf{N}^o}(a', b')/\tilde{c}$, thus a path c'' of type $t_n = (\Delta_+\Delta_-)^{2n}$ from a' to b' and a homomorphism $w'' : c'' \longrightarrow \tilde{c}$ with fixed endpoints, one associates (c', w') in $\mathcal{C}_{\mathbf{N}^o}(c; u, v)$, with

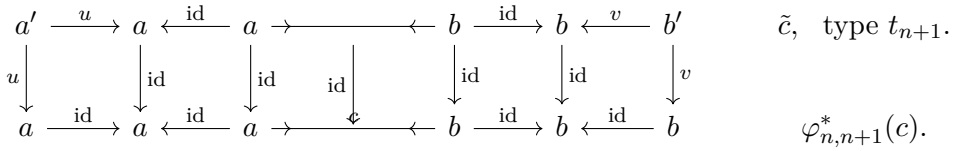
$$c' = c'', \qquad w' = \rho_c w'',$$

[page 169]

where

$$\rho_c : \tilde{c} \longrightarrow c$$

is defined in a similar way as before, by



One need only check that the endofunctor $\psi\varphi$ of $\mathcal{C}_{\mathbf{N}^o}(c; u, v)$ is homotopic to the identity. But this functor transforms (c', w') into $(\tilde{c}' = a'_{\Delta_+\Delta_-}c'b'_{\Delta_+\Delta_-}, \overline{w'})$, where $\overline{w'}$ is again the composite p. 106

$$\tilde{c}' \xrightarrow{\text{can.}} c' \xrightarrow{w'} c.$$
$$\overline{w'}$$

We therefore have

$$\psi\varphi \longrightarrow \mathrm{id},$$

$\psi\varphi$ is homotopic to the identity functor.

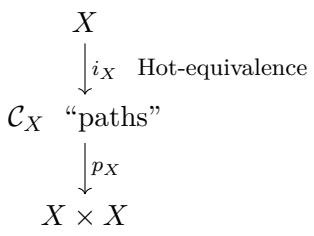
This proves the smoothness of $\mathcal{C}_{\mathbf{N}^o} \longrightarrow X \times X$. For properness, one must prove the smoothness in the case of X^o, and of the inverse system $\underline{t}^{\mathrm{opp}} \simeq (t_n^{\mathrm{opp}})_n$. But we have $\underline{t}^{\mathrm{opp}} = \underline{t}$, and it suffices to apply what we have already proved to the case of X^o, to see that $\mathcal{C}_{\mathbf{N}^o} \longrightarrow X \times X$ is proper.

Only the lemma remains to be proved. It must be essentially trivial, and I give up writing out a proof. [page 170]

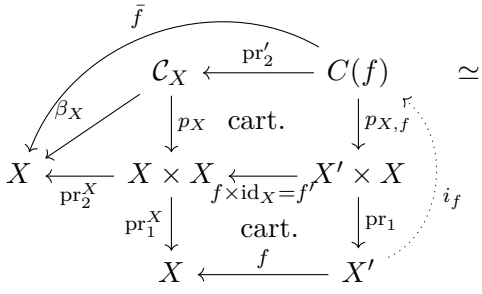
I am now going to show "formally" that the "canonical" factorization of an arrow in Cat $f : X' \longrightarrow X$ as

$$X' \xrightarrow{i_f} \overline{X'} \xrightarrow{\bar{f}} X, \qquad i_f \text{ Hot-equivalence},$$

with (we hoped) $\bar{f}$ proper and smooth, starting from a similar factorization for the diagonal arrow $X \xrightarrow{\mathrm{diag}_X} X \times X$ as



is a formal matter. (Cf. page 139 for the notations and problematics.) We look at the diagram



p. 107

([40]). Since p_X is proper and smooth, i.e. "perfect", its inverse image over $X' \times X$, namely $p_{X,f}$, is so too; on the other hand $\mathrm{pr}_2^X \circ f'$ is the projection pr_2 of $X' \times X$ onto X, hence it is also a perfect morphism (since $X' \longrightarrow e$ is perfect, hence the same holds for the composite [page 171]

$$(\mathrm{pr}_2^X \underbrace{f'}) \circ \underbrace{p_{X,f}} = \underbrace{\mathrm{pr}_2^X (p_X} \mathrm{pr}_2') = \beta_X \mathrm{pr}_2' \overset{\mathrm{def}}{=} \bar{f} \ ).$$

I recall that the section i_f of $C(f)$ over X' is defined from the section i_X of $(\mathcal{C}_X, \sigma_X)$ over X. It is a Hot-equivalence, for $\sigma_X = \mathrm{pr}_1^X p_X$ is perfect with aspheric fibers, hence universally a Hot-equivalence, hence $\mathrm{pr}_1 p_{X,f} : C(f) \longrightarrow X'$ is also a Hot-equivalence, and any section is therefore one too.

[40]NB $C(f)$ is defined as the inverse image over X' of the bundle C_X over X (by $\sigma_X = \mathrm{pr}_1^X \circ p_X$) by $f : X' \longrightarrow X$.

THE DERIVATORS

p.

ALEXANDRE GROTHENDIECK

Chapter VIII

Relative homotopy 1-types: their integration ...

p.

p. 1

1 Integration of relative homotopy 1-types over a category. [page 1]

Relative homotopy types over a (small) category S are defined by *stacks* (in categories) on $S^\wedge$, which correspond essentially to contrafunctors $S^o \longrightarrow (\mathrm{Cat})$, or again to categories X over S which are Cat-fibered over S. (In $(\mathrm{Cat}/S)_{\mathrm{fib}}$, one inverts the arrows which are Hot-equivalences on the fibers.) On the other hand, relative homotopy 1-types over S should be described by *stacks in groupoids* on S. If one has a relative homotopy type, defined by

$$X \longrightarrow S, \qquad \text{Cat-fibered category over } S,$$

or by

$$\begin{array}{ccc} S^o & \longrightarrow & \mathrm{Cat} \\ s & \longmapsto & X_s\ , \end{array}$$

the associated homotopy 1-type is defined by

$$s \longmapsto \Pi(X_s), \qquad S^o \longrightarrow (\mathrm{Grp}), \qquad \text{category of groupoids},$$

where $\Pi(X_s)$ is the fundamental groupoid of X_s. We denote the corresponding fibered category over S by $\boxed{\Pi(X/S)}$. It is true that if

$$X \longrightarrow X'$$

is a homomorphism of fibered categories over S which induces a Hot-equivalence on the fibers, it induces a fibered equivalence

$$\Pi(X/S) \longrightarrow \Pi(X'/S),$$

but not an isomorphism. There would thus be work to do to define a functor

$$\underbrace{\mathrm{Hot}(S)}_{\substack{\text{category of}\\ \text{relative homotopy}\\ \text{types over } S}} \xrightarrow{\xi \longmapsto \Pi(\xi/S)} \underbrace{\mathrm{Grp}(S)}_{\substack{\text{category of}\\ \text{groupoids}\\ \text{fibered over } S\\ \text{(or categories}\\ \text{of stacks in}\\ \text{groupoids}\\ \text{on } S^\wedge)}} .$$

But this is not my purpose for the moment. Rather, to give a transitivity formula in noncommutative homology, analogous to $R(gf)_* \simeq Rg_*Rf_*$ in cohomology.

If

p. 2

$$f : S \longrightarrow T$$

[page 2]

is a functor, we are going to define a functor

$$f_! : \underline{\mathrm{Grp}}(S) \longrightarrow \underline{\mathrm{Grp}}(T)$$

(direct image of relative homotopy 1-types). But let us first note that if

$$p : X \longrightarrow S$$

is any S-category *whatsoever* (not necessarily Cat-fibered), one nevertheless associates to it a relative homotopy type, via the fibered category $\tilde{X}^{(S)}$, or $\tilde{X}$, having as fibers the $s\backslash X$ (noting that $s' \longrightarrow s$ in S does indeed define $s\backslash X \longrightarrow s'\backslash X$) ([1]). We have a canonical S-functor

$$X \xrightarrow{i} \tilde{X}, \qquad x \longmapsto (x, p(x), \underbrace{p(x)}_{=\text{`}s\text{'}}, \xrightarrow{\mathrm{id}} p(x)).$$

This is an S-Hot-equivalence if X was already fibered. In fact, on each fiber it is a homotopism, for one then defines $\tilde{X}_{s_0} \xrightarrow{j} X_{s_0}$ in an obvious way, in such a way that

$$X_{s_0} \underset{j}{\overset{i}{\rightleftarrows}} \tilde{X}_{s_0}, \qquad ji \simeq \mathrm{id}, \qquad \underbrace{ij \xrightarrow{\text{simple homotopy}} \mathrm{id}}_{\substack{\text{NB this arrow}\\ \text{is not}\\ \text{isomorphism}\\ \text{in general.}}}$$

Let us then return to $f : S \longrightarrow T$, and consider a fibered groupoid over S,

$$G \longrightarrow S,$$

this is not a fibered category over T, but one can associate to it

$$t \longmapsto t\backslash G, \qquad T^o \longrightarrow (\mathrm{Cat}),$$

hence a fibered category over T, of which one takes the relative Π. But it is simpler to take directly

$$t \longmapsto \Pi(t\backslash G),$$

which defines a category fibered in groupoids over T.

Thus we do indeed find a functor (in fact, even a 2-functor of 2-categories) p. 3 [page 3]

$$f_! : \underline{\mathrm{Grp}}(S) \longrightarrow \underline{\mathrm{Grp}}(T).$$

If

$$u : G \longrightarrow G'$$

is a homomorphism of fibered groupoids over S, which induces Cat-equivalences on the fibers, then

$$f_!(u) : f_!(G) \longrightarrow f_!(G')$$

is also a Cat-equivalence on the fibers. (This is formal, from the fact that $f_!$ is a 2-functor, taking into account that if one has an equivalence on the fibers, then there is a quasi-inverse – this is a general story about fibered categories …)

Now let

$$p : X \longrightarrow S,$$

[1] Having $\tilde{X}$, one defines $\Pi(X/S) \overset{\mathrm{def}}{=} \Pi(\tilde{X}/S)$.

and let us take the case $G = \Pi(X/S)$. I say that we then have a T-equivalence

$$\boxed{f_!(\Pi(X/S)) \ \simeq \ \Pi(X/T),} \qquad (*)$$

in other words, to know $\Pi(X/T)$ (up to T-equivalence), it suffices to know $\Pi(X/S)$, and to apply to it the functor $f_!$ of integration of relative homotopy 1-types, for $f : S \longrightarrow T$.

Let us note that we trivially have

$$\Pi(X/S) \ = \ p_!(e_X),$$

where e_X denotes the trivial fibered X-groupoid (all fibers one-point). Thus the preceding formula is written as a transitivity formula

$$f_!(p_!(\Gamma)) \ \simeq \ (fp)_!(\Gamma),$$

where Γ is the trivial fibered groupoid over X. This suggests generalizing the formula $(*)$ to [page 4]

$$\boxed{(fp)_!(\Gamma) \ \simeq \ f_!(p_!(\Gamma))}$$

for *every* fibered groupoid over X, in a situation

$$X \xrightarrow{p} S \xrightarrow{f} T.$$

This will be formal, if we establish a 2-adjunction property for the functor $f_!$:

Theorem. *Let $f : S \longrightarrow T$ be an arrow in* (Cat)*, G a stack in groupoids over S, H a stack in groupoids over T. We then have a canonical equivalence of categories* p. 4

$$\boxed{\underline{\mathrm{Hom}}_{\underline{\mathrm{Grp}}(S)}(G, f^*H) \ \simeq \ \underline{\mathrm{Hom}}_{\underline{\mathrm{Grp}}(T)}(f_!G, H).}$$

The adjunction arrows

$$\begin{cases} f_!f^*H & \longrightarrow & H & \qquad & H & \in & \mathrm{Ob}\,\underline{\mathrm{Grp}}(T) \\ f^*f_!G & \longleftarrow & G & \qquad & G & \in & \mathrm{Ob}\,\underline{\mathrm{Grp}}(S) \end{cases}$$

should be definable without any problem. For the second, we have to define for $s \in S$

$$G_s \ \longrightarrow \ (f^*(f_!(G)))_s = (f_!(G))_{f(s)} \overset{\mathrm{def}}{=} \Pi(f(s)\backslash_T G),$$

now we have

$$G_s \longrightarrow f(s)\backslash_T G,$$

hence

$$\underbrace{\Pi(G_s)}_{\substack{= G_s \\ \text{since } G_s \\ \text{is a} \\ \text{groupoid}}} \longrightarrow \Pi(f(s)\backslash_T G),$$

O.K. This is functorial in s …

For the arrow $f_!f^*H \longrightarrow H$ over T, we have to find for every $t \in \mathrm{Ob}\,T$, and functorially in t, [page 5]

$$\underbrace{(f_!f^*H)_t}_{\overset{\mathrm{def}}{=}\Pi(t\backslash f^*(H))} \quad \longrightarrow \quad \underbrace{H_t}_{=\Pi(H_t)} \ ,$$

it would therefore suffice to define

$$t\backslash f^*(H) \ \longrightarrow \ H_t,$$

but we have the cartesian diagram

$$\begin{array}{ccc} f^*H = H \times_T S & \longrightarrow & H \\ \downarrow & & \downarrow \\ S & \xrightarrow{\quad f \quad} & T \ni t \end{array}$$

inducing

$$f^*H \longrightarrow H,$$

hence p. 5

$$t\backslash f^*H \longrightarrow t\backslash H \qquad \text{(instead of mapping into } H_t\text{)},$$

but on the other hand, since H is fibered over T, hence smooth over T, the inclusion

$$H_t \longrightarrow t\backslash H$$

is a Hot-equivalence. Or rather, let us say: $f^*H \longrightarrow H$ is a morphism of categories over T, inducing

$$\underbrace{\Pi(f^*H/T)}_{\overset{\text{def}}{=} f_!(f^*H)} \longrightarrow \Pi(H/T),$$

now the canonical T-functor [page 6]

$$H \longrightarrow \Pi(H/T) \qquad (= \text{category over } T \text{ defined by } t \longmapsto \Pi(t\backslash H))$$

is an equivalence, because H is a groupoid fibration over T. So we can find a T-quasi-inverse

$$\Pi(H/T) \longrightarrow H$$

and by composing we find

$$f_!(f^*H) \longrightarrow \Pi(H/T) \longrightarrow H$$

the arrow sought.

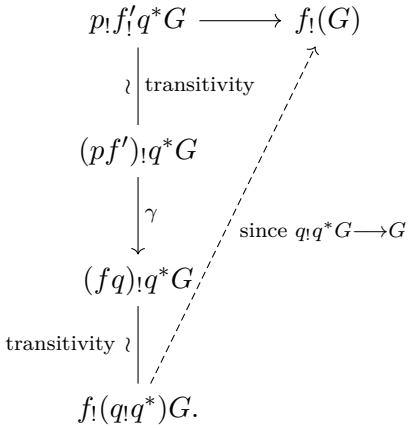
Having the adjunction 2-arrows, I stop there with the verifications! p. 6

2 Base change questions

$$\begin{array}{ccc} S & \xleftarrow{\quad q \quad} & S' \\ {\scriptstyle f}\downarrow & \swarrow_{\gamma} & \downarrow{\scriptstyle f'} \\ T & \xleftarrow[\quad p \quad]{} & T' \end{array} \qquad p^*f_!(G) \longleftarrow f'_!q^*(G) \qquad (*)$$

The arrow is defined, provided we have commutativity or a commutativity arrow $\gamma : pf' \longrightarrow fq$ (not necessarily isomorphism). By adjunction, it amounts to the same

thing to give



We wonder when the arrow is isomorphism.

The following theorem must be true, and follows from my general *fourbis*. [page 7]

Theorem. *If the square* $(*)$ *is* cartesian *in* (Cat)*,* γ *an isomorphism, and if we assume that* f *is* 1*-smooth, or* p 1*-proper, then the* T'*-functor of base change is an equivalence of* T'*-categories.*

The general "*fourbis*" would be in terms of an ad hoc derivator $\mathbf{D}$, that of relative homotopy 1-types. I will return to this later.

We ask the question when the 2-functor associated to $f : S \longrightarrow T$

$$f^* : \underline{\underline{\mathrm{Grp}}}(T) \longrightarrow \underline{\underline{\mathrm{Grp}}}(S)$$

is 0-faithful, 1-faithful, 2-faithful ([2]). If G, H are groupoid fibrations over T, we consider p. 7

$$\underline{\mathrm{Hom}}_{\underline{\underline{\mathrm{Grp}}}(T)}(G, H) \longrightarrow \underline{\mathrm{Hom}}_{\underline{\underline{\mathrm{Grp}}}(S)}(f^*G, f^*H),$$

and according as this functor is always faithful, or fully faithful, or an equivalence of categories, we say that f^* is 0-, 1-, or 2-faithful. In the 2-faithful case, this implies that if K is a groupoid fibration over S, if it is equivalent to an f^*G, then G is determined up to T-equivalence, itself determined up to unique isomorphism.

Let $i \in \mathbf{N}$. We say that f is i-aspheric if $\forall t \in T$, $Z = S/t$ is i-aspheric, i.e. [page 8]

$$\pi_0(Z), \ \ldots, \ \pi_i(Z) \qquad \text{trivial.} \qquad (*)$$

We also define (-1)-aspheric: the S/t are nonempty. Likewise for the notion of i-coaspheric.

In fact, for $i \in [-1, +\infty[\subseteq \mathbf{Z}$, we define a derivator $\mathbf{D}_i$ of relative homotopy i-types (cf. below). Hence the notions of a $\mathbf{D}_i$-aspheric object ($= i$-aspheric), of a $\mathbf{D}_i$-equivalence, of a $\mathbf{D}_i$-aspheric and $\mathbf{D}_i$-coaspheric functor, of a $\mathbf{D}_i$-smooth and $\mathbf{D}_i$-proper functor, of a $\mathbf{D}_i$-fibration and of a $\mathbf{D}_i$-fibration that is *trivial* (i.e. with $\mathbf{D}_i$-aspheric fibers). We should have the

Theorem 3. *Let* $f : S \longrightarrow T$ *be in* (Cat)*,* $i \in \{0, 1, 2\}$*. Suppose one of the following conditions is satisfied:*

a) f *is* $(i-1)$*-proper, and* $(i-1)$*-aspheric (is expressed by the condition that certain categories, such as the* S/t*, be* $(i-1)$*-aspheric).*

[2] The i-faithfulness of f^* ($i \in \{0, 1, 2\}$) is equivalent to that of the functors (obtained by taking $G = e_T$)

$$\underline{\Gamma}(H/T) \longrightarrow \underline{\Gamma}(f^*H/S).$$

b) f is $(i-1)$-smooth, and $(i-1)$-coaspheric.

Then:

1°) The inverse image functor f^*

$$f^* : \underline{\mathrm{Grp}}(T) \longrightarrow \underline{\mathrm{Grp}}(S)$$

is i-faithful.

*2°) In the case $i = 2$ (f^* 2-faithful), we also have this: if $G \in \mathrm{Ob}\,\underline{\mathrm{Grp}}(S)$ is a groupoid fibration over S, then G is equivalent to a groupoid fibration of the form $f^*(H)$ if and only if for every $t \in T$, the restriction G_t of G to S_t is equivalent to a constant groupoid fibration over S_t. (**NB** Condition a) or b) already implies that S_t is 1-connected, i.e. that its fundamental groupoid is equivalent to e.)*

p. 8
[page 9]

3°) Let Z be an object 1-connected in (Cat), G a groupoid fibration over Z. For G to be Z-equivalent to a constant groupoid over Z, it is necessary and sufficient that for every arrow u in Z, the morphism of groupoids u^* induced on the fibers of G be an equivalence of groupoids ([3]).

Corollary. *The necessary and sufficient condition of 2°) can also be written as follows: for every arrow u of S "vertical" over T, i.e. such that $f(u)$ is an identity arrow, the functor u^* induced by u on the fibers of G be an equivalence of groupoids.*

[3] We then say that G is *essentially locally constant* over Z.

THE DERIVATORS

p. [unnumbered]

ALEXANDRE GROTHENDIECK

Chapter IX
Return to the categories of fractions $\mathcal{M}W^{-1}$, comparison with Quillen

p. [unnumbered]
p. 1
[page 1]

1 Functoriality of $\mathrm{Ch}^{\Sigma}_{\infty}(a,b)$ in a, b in $\mathcal{M}\Sigma^{-1}$

We are given a category $\mathcal{M}$ with admissible set of arrows $\Sigma \subseteq \mathrm{Fl}(\mathcal{M})$, whence (taking e.g. $\underline{t} = (t_n)$, with $t_n = (\underbrace{\Delta_+\Delta_-}_{\tau_0})^{2n}$, and $\underbrace{t_{n+1}}_{=\tau_0^{2n+2}} \longrightarrow \underbrace{t_n}_{=\tau_0^{2n}}$ defined as $p * \mathrm{id}_{\tau_0^{2n}} * p$, where $p : \tau_0 \longrightarrow 1$ the degeneracy.) We then have a theory of the $\underline{\mathrm{Ch}}_{\Sigma,\infty}(\mathcal{M},a,b)$, which I denote simply $\mathrm{Ch}_\infty(a,b)$ as an object of Hot. Pairings

$$\mathrm{Ch}_\infty(a,b) \times \mathrm{Ch}_\infty(b,c) \longrightarrow \mathrm{Ch}_\infty(a,c) \qquad \text{in Hot}$$

with (we shall admit it) *associativity* in Hot. This therefore defines pairings on the $\pi_0(\mathrm{Ch}_\infty(a,b)) \overset{\text{def}}{=} \pi_0(a,b)$, which is none other than the composition of arrows in $\mathcal{M}\Sigma^{-1}$. On the other hand, we have for $a,b \in \mathrm{Ob}\,\mathcal{M}$

$$\mathrm{Hom}_{\mathcal{M}}(a,b) \longrightarrow \pi_0(\mathrm{Ch}_\infty(a,b)) = \pi_0(a,b),$$

compatible with compositions. But we can ignore it for the moment, and look at $\mathrm{Ob}\,\mathcal{M}$, or simply $\mathcal{M}_0$, with only the structure

$$\begin{cases} \mathcal{M}_0 \times \mathcal{M}_0 \longrightarrow \mathrm{Hot} & (a,b) \longmapsto \mathrm{Ch}_\infty(a,b) \\ \mathcal{M}_0 \times \mathcal{M}_0 \times \mathcal{M}_0 \longrightarrow \mathrm{Fl\,Hot} & (a,b,c) \longmapsto (\mathrm{Ch}_\infty(a,b) \times \mathrm{Ch}_\infty(b,c) \underbrace{\longrightarrow}_{\substack{\text{arrow}\\ \text{in Hot}}} \mathrm{Ch}_\infty(a,c)) \end{cases}$$

with condition associativity, and also a condition on units:

$$\mathrm{Ch}_\infty(a,a) \text{ is equipped with a } 1_a \in \pi_0\mathrm{Ch}_\infty(a,a) = \pi_0(a,a),$$

playing the role of a "two-sided identity", in the following sense. Let us note that the pairings

$$\begin{array}{ccccc} \mathrm{Ch}_\infty(a,b) & \times & \mathrm{Ch}_\infty(b,c) & \longrightarrow & \mathrm{Ch}_\infty(a,c) \\ P & \times & Q & \longrightarrow & R \end{array}$$

define

[page 2]

$$\begin{array}{ccc} P & \longrightarrow & \underline{\mathrm{Hom}}_{\mathrm{Hot}}(Q,R) \\ Q & \longrightarrow & \underline{\mathrm{Hom}}_{\mathrm{Hot}}(P,R), \end{array}$$

whence on passing to the π_0

$$\begin{array}{ccc} \pi_0(P) & \longrightarrow & \mathrm{Hom}_{\mathrm{Hot}}(Q,R) \\ \pi_0(Q) & \longrightarrow & \mathrm{Hom}_{\mathrm{Hot}}(P,R), \end{array}$$

in other words,

p. 2

$$\begin{aligned}\pi_0(a,b) &\longrightarrow \mathrm{Hom}_{\mathrm{Hot}}(\mathrm{Ch}_\infty(b,c),\mathrm{Ch}_\infty(a,c))\\ \pi_0(b,c) &\longrightarrow \mathrm{Hom}_{\mathrm{Hot}}(\mathrm{Ch}_\infty(a,b),\mathrm{Ch}_\infty(a,c)).\end{aligned}$$

We deduce from this in particular for $a,b\in\mathcal{M}_0$

$$\left\{\begin{array}{lll} 1) & \pi_0(a,a) \longrightarrow & \mathrm{Hom}_{\mathrm{Hot}}(\mathrm{Ch}_\infty(a,b),\mathrm{Ch}_\infty(a,b))\\ 2) & \pi_0(a,a) \longrightarrow & \mathrm{Hom}_{\mathrm{Hot}}(\mathrm{Ch}_\infty(b,a),\mathrm{Ch}_\infty(b,a)),\end{array}\right. \qquad (*)$$

and an object α_a of $\pi_0(a,a)$ is a two-sided unit if its two images are the identity homomorphisms in $\mathrm{Ch}_\infty(a,b)$ resp. $\mathrm{Ch}_\infty(b,a)$. Passing to the π_0 of these, this implies that α_a is a two-sided unit for the composition of the $\pi_0(x,y)$, relative to b. If we want this to be the case for *every* b, this means (taking $b=a$) that α_a is a two-sided unit object for the structure of associative composition (hence of a monoid) in $\pi_0(a,a)$, hence that $\pi_0(a,a)$ is a monoid *with unit,* having α_a as unit object. The element α_a is therefore uniquely determined. And we want α_a to exist for every $a\in\mathcal{M}_0$, so $\mathcal{M}_0$, equipped with [page 3]
the $\pi_0(a,b)$ and their composition, becomes a *category,* say $\mathcal{M}_\Pi$; and its identity arrows must play the role of a two-sided unit also for the "external Hom" $\mathrm{Ch}_\infty(a,b)$, with values in Hot.

We can say that for fixed $a=a_0$, the $\mathrm{Ch}_\infty(a_0,b)$ (b variable) form the values of a covariant *functor*

$$\mathcal{M}_\Pi \longrightarrow \mathrm{Hot} \qquad b\longmapsto \mathrm{Ch}_\infty(a_0,b),$$

by associating to

$$b\xrightarrow{u} b', \qquad u\in\pi_0(b,b'),$$

the corresponding arrow

$$\mathrm{Ch}_\infty(a_0,b)\longrightarrow \mathrm{Ch}_\infty(a_0,b')$$

$*_1$. That this is indeed a functorial law should follow formally from the associativity of the pairings on the Ch_∞. Likewise, we have a functorial law, for fixed b,

$$\mathcal{M}_\Pi^o \longrightarrow \mathrm{Hot}, \qquad a\longmapsto \mathrm{Ch}_\infty(a,b),$$

thanks to associativity. In fact, associativity will also imply that we have a *bifunctor*

$$\begin{aligned}\mathcal{M}_\Pi^o\times\mathcal{M}_\Pi &\longrightarrow \mathrm{Hot}\\ (a\,,\,b) &\longmapsto \mathrm{Ch}_\infty(a,b),\end{aligned}$$

for if p. 3
[page 4]

$$a'\xrightarrow{u} a, \qquad b\xrightarrow{v} b',$$

we have the diagram

$$\begin{array}{ccc}\mathrm{Ch}_\infty(a,b) & \xrightarrow[v]{\text{comp. with}} & \mathrm{Ch}_\infty(a,b')\\ \Big\downarrow{\scriptstyle \text{comp. with } u} & & \Big\downarrow{\scriptstyle \text{comp. with } u}\\ \mathrm{Ch}_\infty(a',b) & \xrightarrow[v]{\text{comp. with}} & \mathrm{Ch}_\infty(a',b'),\end{array}$$

which is commutative thanks to associativity for

$$\mathrm{Ch}_\infty(a',a),\ \mathrm{Ch}_\infty(a,b),\ \mathrm{Ch}_\infty(b,b').$$

Corollary of this bifunctorial dependence: *If $u : a' \longrightarrow a$, $v : b \longrightarrow b'$ are invertible in $\mathcal{M}_\Pi$, then the corresponding arrow*

$$\mathrm{Ch}_\infty(a,b) \longrightarrow \mathrm{Ch}_\infty(a',b') \qquad \textit{in } \mathrm{Hot}$$

is an isomorphism.

The same arguments should work if instead of the Hom ext with values in Hot, we took them in Ind Hot, or in Cat hot, or in Ind Cat hot – I am thinking of course of the $\mathrm{CH}_\infty(a,b)$ (in the theory of localization) in Ind Cat hot, which can be regarded as coming from a bifunctor (where $\mathcal{M}_\Pi = \mathcal{M}\Sigma^{-1}$)

$$\mathcal{M}_\Pi^o \times \mathcal{M}_\Pi \longrightarrow \mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}.$$

Here is the definitive axiomatization (?). We replace Hot by a category [page 5]

$\mathcal{H}$, category with product $*$ associative, commutative (?) ([1]), unital, and all the usual compatibilities as needed,

the unit object for $*$ will be denoted 1, and we shall write

$$\begin{cases} \pi_0(\xi) & \overset{\mathrm{def}}{=} & \mathrm{Hom}_{\mathcal{H}}(1,\xi) \\ \pi_0 : \mathcal{H} & \longrightarrow & \mathrm{Ens}. \end{cases}$$

If we have

$$P * Q \xrightarrow{u} R,$$

we deduce from this

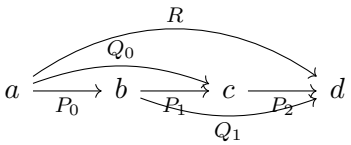
$$\begin{cases} \pi_0(P) & \xrightarrow{c_u^{\mathrm{l}}} & \mathrm{Hom}_{\mathcal{H}}(Q,R) \\ \pi_0(Q) & \xrightarrow{c_u^{\mathrm{r}}} & \mathrm{Hom}_{\mathcal{H}}(P,R) \end{cases}$$ p. 4

as follows:

$$\begin{array}{rrcl} \text{If } \alpha \in \pi_0(P), \text{ i.e. } 1 \xrightarrow{\alpha} P, \text{ then} & \alpha * \mathrm{id}_Q : Q \longrightarrow P * Q & & \\ \text{and} & c_u^{\mathrm{l}}(\alpha) & = & u \circ (\alpha * \mathrm{id}_Q) \\ \text{and dually} & c_u^{\mathrm{r}}(\beta) & = & u \circ (\mathrm{id}_P * \beta). \end{array}$$

If we have three objects P_0, P_1, P_2 in $\mathcal{H}$, pairings

$$\begin{cases} P_0 * P_1 & \xrightarrow{u_1} & Q_0 \\ P_1 * P_2 & \xrightarrow{u_2} & Q_1 \\ P_0 * Q_1 & \xrightarrow{u_3} & R \\ Q_0 * P_2 & \xrightarrow{u_4} & R \end{cases}$$



with an associativity relation, saying that

$$\begin{array}{ccc} P_0 * P_1 * P_2 & \longrightarrow & Q_0 * P_2 \\ \downarrow & & \downarrow \\ P_0 * Q_1 & \longrightarrow & R \end{array}$$ [page 6]

[1] We should not need commutativity.

is commutative, then we deduce from this two transitivity relations and one commutation relation, for the operations associated with the elements of $\pi_0(P_0)$, $\pi_0(P_1)$, $\pi_0(P_2)$, which we will not spell out, except for one:

$$\begin{array}{rrcl} \text{Let } \beta \in \pi_0(P_1),\ \gamma \in \pi_0(P_2), & & & \\ \text{whence by } \beta: & c_{u_1}(\beta): & P_0 \longrightarrow & Q_0, \\ \text{and by } \gamma: & c_{u_4}(\gamma): & Q_0 \longrightarrow & R, \\ \text{and by } \beta *_{u_2} \gamma \in \pi_0(Q_1): & c_{u_3}(\beta\,\gamma): & P_0 \longrightarrow & R, \end{array}$$

, well, we have p. 5

$$\boxed{c(\beta *_{u_2} \gamma) \;=\; c(\gamma)c(\beta).}$$

We still have to define $\beta *_{u_2} \gamma$, via the pairing $P_1 * P_2 \xrightarrow{u_2} Q_1$, we have in fact

$$\begin{array}{ccc} \pi_0(P_1) \times \pi_0(P_2) & \longrightarrow & \pi_0(P_1 * P_2) \\ (\beta\,,\,\gamma) & \longmapsto & \beta * \gamma, \end{array}$$

and we compose with

$$\pi_0(u_2) : \pi_0(P_1 * P_2) \longrightarrow \pi_0(Q_1)$$

.

We give ourselves a set $\mathcal{M}_0$, and [page 7]

$$\left\{ \begin{array}{rcll} \mathcal{M}_0 \times \mathcal{M}_0 & \xrightarrow{\text{Ch}} & \mathcal{H} \quad \text{denoted} & (a,b) \longmapsto \text{Ch}(a,b) \text{ or } \operatorname{Hom}\operatorname{ext}_{\mathcal{H}}(a,b) \\ \mathcal{M}_0 \times \mathcal{M}_0 \times \mathcal{M}_0 & \xrightarrow{\text{comp.}} & \operatorname{Fl}\mathcal{H} & (a,b,c) \longmapsto \big(\text{Ch}(a,b) * \text{Ch}(b,c) \longrightarrow \text{Ch}(a,c)\big) \end{array} \right.$$

with two axioms

a) Associativity for the compositions, for $(a,b,c,d) \in \mathcal{M}_0^4$

$$\begin{array}{ccc} \text{Ch}(a,b) * \text{Ch}(b,c) * \text{Ch}(c,d) & \longrightarrow & \text{Ch}(a,c) * \text{Ch}(c,d) \\ \downarrow & & \downarrow \\ \text{Ch}(a,b) * \text{Ch}(b,d) & \longrightarrow & \text{Ch}(a,d). \end{array}$$

b) Existence of two-sided unit objects

$$1_a \in \pi_0(\text{Ch}(a,a)) \qquad \text{for every } a \in \mathcal{M}_0$$

Then the $\pi_0(\text{Ch}(a,b))$ and their composition define on $\mathcal{M}_0$ as set of objects a *category* structure $\mathcal{M}_{0\,\Pi}$, having the 1_a as identities, and we find that

$$(a,b) \longmapsto \text{Ch}(a,b)$$

can be interpreted in terms of a canonical *bifunctor*

$$\mathcal{M}_{0\,\Pi}^o \times \mathcal{M}_{0\,\Pi} \longrightarrow \mathcal{H},$$

whose composite with

$$\mathcal{H} \xrightarrow{\pi_0} \text{Ens}$$

is none other than the bifunctor $(a,b) \longrightarrow \operatorname{Hom}_{\mathcal{M}_{0\,\Pi}}(a,b)$, p. 6 [page 8]

$$\operatorname{Hom}_{\mathcal{M}_{0\,\Pi}}(a,b) \;\simeq\; \pi_0(\text{Ch}(a,b)).$$

This relation is the *definition* of $\underline{\mathrm{Hom}}_{\mathcal{M}_{0\,\Pi}}$, but here the relation is understood as a *functorial* bijection in a, b.

We find in particular that if

$$u : a' \longrightarrow a, \qquad v : b \longrightarrow b'$$

are *invertible* arrows in $\mathcal{M}_{0\,\Pi}$, then

$$\mathrm{Ch}(a,b) \longrightarrow \mathrm{Ch}(a',b')$$

corresponding in $\mathcal{H}$ is *isomorphism.*

We will apply all this mainly to the two situations deduced from a situation $(\mathcal{M}, \Sigma)$, with

$$\begin{array}{llllll} 1^o) & \mathcal{H} & = & \mathrm{Hot}, & \mathrm{Ch}(a,b) & = & \mathrm{Ch}^{\Sigma}_{\infty}(\mathcal{M}; a, b) \\ 2^o) & \mathcal{H} & = & \mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}, & \mathrm{Ch}(a,b) & = & \mathrm{CH}^{\Sigma}_{\infty}(\mathcal{M}; a, b)\ (^2). \end{array}$$

In both cases, we thus have a functor on $(\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1})$, with values either in Hot, or in $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$. (**NB** The two category structures on $\mathcal{M}_0$ deduced from the these two functors are canonically isomorphic.)

$$\begin{array}{ccc} & \nearrow\ \mathrm{Hot} & (a,b) \longmapsto \mathrm{Ch}^{\Sigma}_{\infty}(a,b) \\ (\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1}) & & \\ & \searrow\ \mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot} & (a,b) \longmapsto \mathrm{CH}^{\Sigma}_{\infty}(a,b) \end{array}$$

As the canonical functors

$$\begin{array}{ccccccc} \mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot} & \xrightarrow{\mathrm{Ind}\,\varphi} & \mathrm{Ind}\,\mathrm{Hot} & \xrightarrow{\mathrm{Ind}\,\psi} & \mathrm{Ind}\,\mathrm{Hot}^{\wedge}_0 & \xrightarrow{\varinjlim} & \mathrm{Hot}^{\wedge}_0, \\ \uparrow & & \uparrow & & \uparrow & \nearrow_{\mathrm{id}} & \\ \mathrm{Cat}\,\mathrm{hot} & \xrightarrow{\varphi} & \mathrm{Hot} & \xrightarrow{\psi} & \mathrm{Hot}^{\wedge}_0 & & \end{array}$$

[page 9]

commuting with finite products, any theory of "external Hom" with values in one of these categories, defines one with values in each of those into which it maps. Besides, these functors are not all fully faithful (this is in any case false for φ and $\mathrm{Ind}\,\varphi$, and I do not know what the situation is for ψ and $\mathrm{Ind}\,\psi$ – the vertical arrows on the other hand are fully faithful (^{3}).) But they induce bijections for the π_0. So the category $\mathcal{M}_{0\,\Pi}$ deduced from a theory of external Hom with values in one of these categories, and the one deduced in another into which it maps, is the same up to canonical isomorphism. As far as this definition of $\mathcal{M}_{0\,\Pi}$ is concerned, nothing is changed then when we regard the CH_{∞} as objects of $\mathrm{Ind}\,\mathrm{Hot}$, or only of $\mathrm{Ind}\,\mathrm{Hot}^{\wedge}_0$, or even only of $\mathrm{Hot}^{\wedge}_0$, or when we regard the Ch_{∞} as objects in $\mathrm{Hot}^{\wedge}_0$. But of course, the datum of the composition pairings loses precision each time (except perhaps when passing from Hot to $\mathrm{Hot}^{\wedge}_0$, if ψ is fully faithful …). When we take the images of the $\mathrm{CH}^{\Sigma}_{\infty}(a,b)$ and $\mathrm{Ch}^{\Sigma}_{\infty}(a,b)$ in $\mathrm{Hot}^{\wedge}_0$ (the last category of the diagram), we find canonically isomorphic objects (functorially in a, b in $\mathcal{M}_{0\,\Pi} \simeq \mathcal{M}\Sigma^{-1}$). When we regard them as being respectively in $\mathrm{Ind}\,\mathrm{Hot}$ and in $\mathrm{Hot} \hookrightarrow \mathrm{Ind}\,\mathrm{Hot}$, we find a functorial homomorphism

p. 7

[page 10]

$$\mathrm{CH}^{\Sigma}_{\infty}(a,b) \longrightarrow \mathrm{Ch}^{\Sigma}_{\infty}(a,b) \qquad \text{in } \mathrm{Ind}\,\mathrm{Hot} \text{ or in } \mathrm{Ind}\,\mathrm{Hot}^{\wedge}_0 \text{ as we choose.} \qquad (*)$$

[2] $*$ in Hot and $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$ is the cartesian product, nothing more.

[3] and also, $\mathrm{Ind}\,\mathrm{Hot}^{\wedge}_0 \xrightarrow{\varinjlim} \mathrm{Hot}^{\wedge}_0$ is *not* fully faithful.

The $\varinjlim$ of $\mathrm{CH}^{\Sigma}_{\infty}(a,b)$ in Hot does not exist in general (unless the ind-object is essentially constant), but it exists in $\mathrm{Hot}^{\wedge}_0$, and the canonical isomorphism

$$\varinjlim_{\mathrm{Hot}^{\wedge}_0} \mathrm{CH}^{\Sigma}_{\infty}(a,b) \xrightarrow{\sim} \mathrm{Ch}^{\Sigma}_{\infty}(a,b) \qquad \text{in } \mathrm{Hot}^{\wedge}_0 \qquad (**)$$

is the arrow deduced from $(*)$ in $\mathrm{Ind}\,\mathrm{Hot}^{\wedge}_0$.

In the two cases CH and Ch (which we are going to denote by the common symbol $C_\infty(a,b)$), the functoriality of this object $C_\infty(a,b)$ (even in the weakest sense, as an object of $\mathrm{Hot}^{\wedge}_0$) with respect to a and b (even in the weakest sense, as a functor $\mathcal{M}^o\times\mathcal{M} \longrightarrow \mathrm{Hot}^{\wedge}_0$ and not even $(\mathcal{M}\Sigma^{-1})^o\times(\mathcal{M}\Sigma^{-1}) \longrightarrow \mathrm{Hot}^{\wedge}_0$) is not at all evident from the direct definition in terms of the explicit direct system in Cat [page 11]

$$C_\infty(a,b) \;=\; (\mathrm{Ch}_{\tau^{2n}_0}(a,b))_{n\in\mathbf{N}},$$

for an arrow $a' \longrightarrow a$ or $b \longrightarrow b'$ in $\mathcal{M}$, and even were it in Σ, in no way defines an arrow in Cat

$$\begin{array}{rcl} C_\infty(a,b) & \longrightarrow & C_\infty(a',b) \\ \text{or } C_\infty(a,b) & \longrightarrow & C_\infty(a,b'). \end{array}$$

This difficulty recalls the one that Quillen had to overcome, in Homotopical Algebra, to face the difference between "left homotopies" and "right homotopies". I recall how we can nevertheless define such arrows, as resulting from a (much richer) arrow of "composition" of paths between a' and a, and a and b, or between a and b, and b and b', that is in general then the pairing [p. 8] [page 12]

$$C_\infty(a,b)\times C_\infty(b,c) \longrightarrow C_\infty(a,c),$$

whose direct definition is no less problematic, and can in any case *not* be given as an arrow in Cat itself, or in Ind Cat. Interpreting e.g. $\mathrm{Ch}_\infty(a,b)$ in terms of *infinite* paths (on both sides) from a to b, in an obvious sense (but "essentially finite" on both sides), we can also define

$$\mathrm{Ch}^{-}_{\infty}(a,b), \qquad \mathrm{Ch}^{+}_{\infty}(a,b)$$

as the full subcategories formed respectively of paths which are totally degenerate in degrees ≥ 0 (i.e. $a_0 \longleftarrow a_1 \longrightarrow a_2 \longleftarrow a_3 \cdots$ are all identities), which amounts to paths infinite only to the left, i.e. stopping at a_0, and symmetrically for $\mathrm{Ch}^{+}_{\infty}(a,b)$. Same story for the CH^{+}, CH^{-}.

That said, we have this (returning to the ambivalent notion C_∞):

[page 13]

a) the fully faithful inclusions

$$C^{-}_{\infty}(a,b) \hookrightarrow C_\infty(a,b) \hookleftarrow C^{+}_{\infty}(a,b)$$

are isomorphisms in $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$ resp. in Hot, and

b) We really do have a composition of paths

$$C^{-}_{\infty}(a,b)\times C^{+}_{\infty}(b,c) \longrightarrow C_\infty(a,c).$$

The *associativity* in the CH case moreover offers no difficulty, because of the possibility of reducing to the

$$C_m(a,b)\times C_n(b,c)\times C_p(c,d) \rightrightarrows C_N(a,d)$$

for N large, and proving equality as arrows in $\mathrm{Cat}\,\mathrm{hot}$ – which will follow from the homotopy lemma on intervals in Cat. In the case of Ch_∞ on the other hand I have not found a proof. (Only for lack of having tried seriously to get down to it? I don't much feel like it, seeing that it might be that $\mathrm{Hot} \longrightarrow \mathrm{Hot}^{\wedge}_0$ is fully faithful, in which case the work becomes useless …) [p. 9] [page 14]

2 Comparison with Quillen

A certain number of questions arise about this functor $(\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1}) \longrightarrow \mathrm{Hot}$, and its relation to the classical constructions, in particular to the theory of Quillen. I will now write

$$\mathrm{Ch}^{\Sigma}_{\infty}(a,b) \;=\; \mathrm{Hom}\,\mathrm{ext}_{\mathrm{Hot}}(a,b) \quad \text{or simply} \quad \mathrm{Hom}\,\mathrm{ext}(a,b).$$

1°). Let $f, g : a \rightrightarrows b$ be in $\mathcal{M}$, then (if $\mathcal{M}$ is a "model category") Quillen defines a set

$$\pi_1(a,b;f,g) \qquad \text{or} \qquad \pi_1(f,g)$$

of homotopy classes from f to g, in such a way that $\mathrm{Hom}_{\mathcal{M}}(a,b)$ becomes the set of objects of a groupoid, whose arrows are the elements of $\pi_1(f,g)$, for a natural notion of composition. I suspect that one has

$$\pi_1(f,g) \;=\; \mathrm{Hom}_{\underbrace{\Pi(\underline{\mathrm{Ch}}^{\Sigma}_{\infty}(a,b))}_{\substack{\text{fundamental}\\ \text{groupoid}\\ \text{of } \underline{\mathrm{Ch}}^{\Sigma}_{\infty}(a,b)}}}(f,g),$$

when f and g are considered as "infinite paths" (by inserting each $f : a \longrightarrow b$ into

$$\cdots \underbrace{a_{-2}}_{=a} \xleftarrow{\mathrm{id}} \underbrace{a_{-1}}_{=a} \xrightarrow{f} \underbrace{a_0}_{=b} \xleftarrow{\mathrm{id}} \underbrace{a_1}_{=b} \longrightarrow \cdots \quad)$$

So the definition of the $\pi_1(f,g)$ (if that is indeed it) extends in a natural way to arbitrary infinite (essentially finite) paths. [page 15]

2°) Suppose that $\mathcal{M}$ admits an object both initial and final $0_{\mathcal{M}}$ ($\mathcal{M}$ "pointed"). I say that then all the categories $\underline{\mathrm{Ch}}^{\Sigma}_{\infty}(a,b)$ are pointed in a natural way: namely, by the infinite path from a to b defined (as above) starting from $f = 0_{a,b} : a \longrightarrow b$, the zero map from a to b. One then feels like viewing the $\underline{\mathrm{Ch}}^{\Sigma}_{\infty}(a,b)$ as defining elements in $\mathrm{Hot}_{\bullet}$ ("pointed Hot"). This is finer than saying that one gives oneself a neutral class

$$e \;\longrightarrow\; \mathrm{Ch}(a,b) \qquad \text{in Hot},$$

for the functor from $\mathrm{Hot}_{\bullet}$ to Hot (or to $e\backslash\mathrm{Hot}$) is not faithful. The problem is to define the arrows

$$\mathrm{Ch}(a,b) \times \mathrm{Ch}(b,c) \;\longrightarrow\; \mathrm{Ch}(a,c)$$

in $\mathrm{Hot}_{\bullet}$, and not just in Hot, and with associativity (and the units) in $\mathrm{Hot}_{\bullet}$ as well. If one applies the definition with $\mathrm{Ch}^{-}(a,b)$ and $\mathrm{Ch}^{+}(b,c)$, the composite of the zero elements $a \xrightarrow{0} b$ of $\mathrm{Ch}^{-}(a,b)$ and $b \xleftarrow{\mathrm{id}} b \xrightarrow{0} c$ of $\mathrm{Ch}^{+}(b,c)$ is the following element of $\mathrm{Ch}_{\infty}(a,c)$ p. 10 [page 16]

$$\begin{array}{ccccccccccccc}
 & & a_{-2} & \longleftarrow & a_{-1} & \longrightarrow & a_0 & \longleftarrow & a_1 & \longrightarrow & a_2 & & \cdots \\
 & & & & & & \| & & & & & & \\
\mathrm{id} & \cdots & a & \xleftarrow{\mathrm{id}} & a & \xrightarrow[0]{} & b & \xleftarrow{\mathrm{id}} & b & \xrightarrow[0]{} & c & \xleftarrow{\mathrm{id}} & \cdots_{\mathrm{id}}
\end{array}$$

this is *not* the zero element of $\mathrm{Ch}_{\infty}(a,c)$! But one has a canonical arrow

$$e^{-}_{a,b} \circ e^{+}_{b,c} \;\longrightarrow\; e_{a,c} \qquad \text{in } \mathrm{Ch}_{\infty}(a,c)$$

which is the identity on all the vertices except those of indices 0 and 1, where its components are zero. There is a sorites to develop, to the effect that a pair

$$(f : K_\bullet \longrightarrow L_\bullet, c)$$

of a functor between pointed categories, and of a path $c : f(e_{K_\bullet}) \longrightarrow e_{L_\bullet}$, defines in $\mathrm{Hot}_\bullet$ a canonical arrow

$$\mathrm{hot}_\bullet(K_\bullet) \longrightarrow \mathrm{hot}_\bullet(L_\bullet)$$

with transitivity if one has

$$(f' : L_\bullet \longrightarrow M_\bullet, c' : f'(e_{L_\bullet}) \longrightarrow e_{M_\bullet}),$$

whence [page 17]

$$f'' = f' \circ f : K_\bullet \longrightarrow M_\bullet, \qquad \text{and } \; c'' : f''(e_{K_\bullet}) \longrightarrow e_{M_\bullet},$$

defines thus

$$f'(f(e_{K_\bullet})) \xrightarrow{f'(c)} f'(e_{L_\bullet}) \xrightarrow{c'} e_{M_\bullet}, \quad c'' \text{ by definition}$$

? So this does indeed give us pairings in $\mathrm{Hot}_\bullet$ (and also in $\mathrm{Cat}\,\mathrm{hot}_\bullet$?). It is hard to believe that they could fail to be associative! Besides, $\mathrm{Hot}_\bullet$, with the product $\divideontimes$ with unit S^0 ([4]), satisfies the earlier axiomatics for $\mathcal{H}$, and $\mathrm{Hot}_\bullet \longrightarrow \mathrm{Hot}$ is equipped with the lax monoidal comparison from this product ([5]), and with π_0.

If all this works, one would thus have a theory of the $\mathrm{Hom}\,\mathrm{ext}_\bullet$ with values in $\mathrm{Hot}_\bullet$; and probably also for the CH_∞ as objects of $\mathrm{Ind}\,\mathrm{Hot}_\bullet$, and not only of $\mathrm{Ind}\,\mathrm{Hot}$. But in any case one cannot in general drop Cat hot, which remains important – so one would have to have objects $\mathrm{CH}_\infty(a, b)$ in $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}_\bullet$, to have a well-tailored setup. p. 11

The $\pi_0(C(a, b))$ will now be pointed sets, and this will be by the image of the zero map $a \xrightarrow{0} b$, i.e. the composite

$$a \longrightarrow 0 \longrightarrow b$$

which one must compose with the element deduced from the truncated path [page 18]

$$a \longrightarrow \underbrace{0}_{=a_0} \xleftarrow{\mathrm{id}} 0 \longrightarrow b \qquad \dots$$

One would have to show that 0 is also a neutral object of $\mathcal{M}\Sigma^{-1}$, i.e. that

$$\pi_0(a, 0) = \pi_0(0, a) = \{0\} \qquad \text{for every } a \in \mathrm{Ob}\,\mathcal{M}$$

and interpret the zero element in $\pi_0(a, b)$ as the zero map in $\mathcal{M}\Sigma^{-1}$. In general, one needs conditions which ensure (without $\mathcal{M}$ necessarily being pointed) that

$$\mathcal{M} \xrightarrow{\gamma} \mathcal{M}\Sigma^{-1}$$

commutes with finite sums, with finite products (so $\gamma(\varphi_{\mathcal{M}}) = \varphi_{\mathcal{M}\Sigma^{-1}}$, resp. $\gamma(e_{\mathcal{M}}) = e_{\mathcal{M}\Sigma^{-1}}$) ([6]).

Suppose finally that $\mathcal{M}$ is *additive,* and even k-linear (k a fixed ring), i.e. one has in addition a ring homomorphism

$$k \longrightarrow \mathrm{End}(\mathrm{id}_{\mathcal{M}}).$$

One must assume some compatibility of Σ with the additive structure, e.g. that the direct sum of two arrows in Σ is in Σ, and probably something stronger, of the "five [page 19]

lemma" kind for Σ. (But one does not necessarily assume $\mathcal{M}$ to be abelian, so one would have trouble formulating a five lemma?)

But perhaps here a subtler axiom is needed, e.g. (if $\underbrace{\Sigma}_{\text{suspension}}$ and Ω exist ([7])) that the set of adjunction morphisms $\Sigma\Omega \longrightarrow \mathrm{id}$, $\mathrm{id} \longrightarrow \Omega\Sigma$ be isomorphisms, which implies that Ω resp. Σ is fully faithful. In any case, the natural expectation here is that $\mathrm{Ch}_\infty^\Sigma(a,b)$ be not only in $\mathrm{Hot}_\bullet$, but even in the category $D^{\leq 0}(k\text{-Mod})$, a full subcategory of the derived category $D^-(k\text{-Mod})$ consisting of the K such that $H^i(K) = 0$ if $i > 0$. One can also denote it $D_{\geq 0}(k\text{-Mod})$, interpreting it in terms of complexes of "chains" rather than of p. 12
cochains. Here the pairings must correspond to the operation $* = \overset{\mathbf{L}}{\otimes}_k$ in $D_{\geq 0}(k\text{-Mod})$, admitting k as unit object, and

$$\pi_0(K_\bullet) \;\simeq\; H_0(K_\bullet).$$

So one must have

$$\mathrm{Hom\,ext}(a,b) \overset{\mathbf{L}}{\otimes}_k \mathrm{Hom\,ext}(b,c) \;\longrightarrow\; \mathrm{Hom\,ext}(a,c),$$

where this time Hom ext denotes the object of $D_{\geq 0}(k\text{-Mod})$, instead of just an object of Hot, or of $\mathrm{Hot}_\bullet$. One uses the functors

$$D_{\geq 0}(k\text{-Mod}) \longrightarrow \mathrm{Hot}_\bullet \longrightarrow \mathrm{Hot}$$

to make the link with the pointed theory in $\mathrm{Hot}_\bullet$, or general theory in Hot. If one [page 20]
denotes by

$$\gamma :\; D_{\geq 0}(k\text{-Mod}) \;\longrightarrow\; \mathrm{Hot}_\bullet$$

the first of these functors, it does not commute with the products $*$ ($\overset{\mathbf{L}}{\otimes}_k$ and $\times$ respectively), but one has, for $P_\bullet$, $Q_\bullet$ varying in $D_{\geq 0}(k\text{-Mod})$, a canonical functorial arrow

$$\gamma(P_\bullet \overset{\mathbf{L}}{\otimes}_k Q_\bullet) \xleftarrow{\;can.\;} \gamma(P_\bullet) \times \gamma(Q_\bullet)$$

"k-bilinear", which shows that giving pairings

$$P_\bullet * Q_\bullet \xrightarrow{c_k} R_\bullet,$$

whence

$$\gamma(P_\bullet * Q_\bullet) \xrightarrow{\gamma(c_k)} \gamma(R_\bullet), \qquad (*)$$

is in either case finer than giving oneself only

$$\gamma(P_\bullet) * \gamma(Q_\bullet) \xrightarrow{c_\bullet} \gamma(R_\bullet), \qquad (*')$$

seeing that a datum $(*)$ will imply a datum $(*')$ by

$$\begin{array}{ccc} & \gamma(P_\bullet * Q_\bullet) & \\ {\scriptstyle can.}\nearrow & & \searrow{\scriptstyle \gamma(c)} \\ \gamma(P_\bullet) * \gamma(Q_\bullet) & \xrightarrow{\qquad c_\bullet \qquad} & \gamma(R_\bullet). \end{array}$$

In view of this, one does indeed expect to have a *strong* theory of the pairings

[4] No, one must take the product $\xi_\bullet \divideontimes \eta_\bullet$ ($\xi_\bullet \times \eta_\bullet$, with $\xi_\bullet \vee \eta_\bullet$ contracted to e). p. 13

[5] No, but one gets by all the same …

[6] to do. [page 21]

[7] cf. below what this means.

$$\mathrm{Hom}\,\mathrm{ext}_k(a,b) \overset{\mathbf{L}}{\otimes}_k \mathrm{Hom}\,\mathrm{ext}_k(b,c) \;\longrightarrow\; \mathrm{Hom}\,\mathrm{ext}_k(a,c),$$

from which one will deduce the weaker pairings in $\mathrm{Hot}_\bullet$ (instead of $D_{\geq 0}(k\text{-Mod})$)

$$\mathrm{Hom}\,\mathrm{ext}_{\mathrm{Hot}_\bullet}(a,b) \times \mathrm{Hom}\,\mathrm{ext}_{\mathrm{Hot}_\bullet}(b,c) \;\longrightarrow\; \mathrm{Hom}\,\mathrm{ext}_{\mathrm{Hot}_\bullet}(a,c).$$

3°) Returning to the case $\mathcal{M}$ pointed and nothing more, Quillen defines, for a, b in $\mathcal{M}$,

$$\pi_1(a,b) \;=\; \pi_1(a,b\,;0_{a,b},0_{a,b})$$

as a bifunctor on $(\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1})$ (with the notation of 1°)). If what I put forward without proof in 1°) is true, it must follow that one has a canonical isomorphism

$$\pi_1(a,b) \;\simeq\; \pi_1(\mathrm{Ch}^{\Sigma}_{\infty}(a,b)),$$

where $\mathrm{Ch}^{\Sigma}_{\infty}(a,b)$ is viewed as an object of $\mathrm{Hot}_\bullet$, so its π_1 is defined. One has in any case, without any reference to cofibrations and fibrations, a bifunctor with values in Groups, composed in

$$(\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1}) \xrightarrow{(a,b)\longmapsto \mathrm{Ch}^{\Sigma}_{\infty}(a,b)} \mathrm{Hot}_\bullet \xrightarrow{\pi_1} (\mathrm{Groups}).$$

If for fixed a, variable b,

$$b \;\longmapsto\; \pi_1(a,b)$$

is representable, one denotes the object representing it by $\Sigma(a)$ (*suspension* – do not confuse this Σ with $\Sigma \subseteq \mathrm{Fl}(\mathcal{M})$, which it is better now to denote by W, as in "weak equivalence"). So

[page 22]

$$\mathrm{Hom}(\Sigma a,b) \;\simeq\; \pi_1(a,b),$$

and Σa depends functorially on a, subject to existence. If it always exists, one has

$$\Sigma :\; \mathcal{M}W^{-1} \;\longrightarrow\; \mathcal{M}W^{-1}.$$

One defines likewise, subject to existence,

$$\Omega :\; \mathcal{M}W^{-1} \;\longrightarrow\; \mathcal{M}W^{-1}$$

by

$$\mathrm{Hom}(a,\Omega b) \;\simeq\; \pi_1(a,b),$$

and if both exist, one has the adjunction formula

$$\mathrm{Hom}(\Sigma a,b) \;\simeq\; \mathrm{Hom}(a,\Omega b) \qquad (\simeq \pi_1 \mathrm{Hom}\,\mathrm{ext}_{\mathrm{Hot}_\bullet}(a,b)).$$

The existence of

p. 14

$$\Omega : \mathcal{M}W^{-1} \;\longrightarrow\; \mathcal{M}W^{-1}, \quad \text{or of}$$
$$\Sigma : \mathcal{M}W^{-1} \;\longrightarrow\; \mathcal{M}W^{-1},$$

or even of both at once, appears as an extremely strong hypothesis ⓐ on $(\mathcal{M},W)$. This condition *plus* the one ⓑ that the $\underline{\mathrm{Ch}}_\infty(a,b)$ be homotopic to Thomason-Kan categories, or (which comes to the same thing, more or less?) that $Y \longmapsto \mathrm{Hom}_{\mathcal{M}(Y)W_Y^{-1}}(a_Y,b_Y)$ transform Hot-equivalences of Cat into bijections (even if only for fixed Y …), seems the crucial test for knowing whether "$(\mathcal{M},\Sigma)$ lends itself to a homotopy theory "in $\mathcal{M}$" ". I doubt that

[page 23]

these conditions can be satisfied for finite $\mathcal{M}$, except in the trivial case where $\mathcal{M} = \emptyset$, where $\mathcal{M}\Sigma^{-1}$ is equivalent to the one-point category.

A third test ⓒ which seems important to me (but perhaps this follows from the rest), is the formulas in $\mathrm{Hot}_\bullet$

$$\begin{cases} \mathrm{Hom\,ext}_\bullet(\Sigma a, b) \simeq \mathrm{Hom\,ext}_\bullet(a, \Omega b) \\ \qquad\qquad\quad \simeq \Omega\mathrm{Hom\,ext}_\bullet(a, b), \end{cases} \tag{$*$}$$

where $\mathrm{Hom\,ext}_\bullet$ denotes the external Hom with values in $\mathrm{Hot}_\bullet$. One must view it separately (assuming the existence either of Ω, or of Σ) as two formulas "dual" to one another,

$$\begin{cases} \mathrm{Hom\,ext}_\bullet(a, \Omega b) \simeq \Omega\mathrm{Hom\,ext}_\bullet(a, b) \\ \mathrm{Hom\,ext}_\bullet(\Sigma a, b) \simeq \Omega\mathrm{Hom\,ext}_\bullet(a, b), \end{cases} \tag{$**$}$$

isomorphisms of *groups* in $\mathrm{Hot}_\bullet$. (**NB** Ωb is a group object of $\mathcal{M}W^{-1}$, for it represents a contrafunctor with values in Gr, and likewise Σa is a cogroup of $\mathcal{M}W^{-1}$. The functor $y \longmapsto \mathrm{Hom\,ext}(a, y)$ must commute with finite products, and thus transform group objects into likewise. And likewise $x \longmapsto \mathrm{Hom\,ext}(x, b)$ must transform finite sums into finite products, hence cogroups into groups ([8]).)

I have convinced myself that the isomorphisms $(**)$ have a chance of existing only if one has the prerequisite condition ⓑ.

3 Attempt at constructing the functor Ω

p. 15
[page 24]

There is probably little chance of proving anything about the functor

$$\Omega : \mathcal{M}W^{-1} \longrightarrow \mathcal{M}W^{-1},$$

without having in hand a more or less explicit *construction* of it. I am going to try, taking my inspiration from Quillen, to work out general conditions on $(\mathcal{M}, W)$ (more general than those of Quillen) which ensure the existence of Ω, and allow one to construct it. The geometric idea goes back to Serre-Cartan. The task is, for an object S of $\mathcal{M}$, to construct an object $\mathrm{Ch}(S)$ *in* $\mathcal{M}$, playing the role of a "path space", and which must fit into a factorization diagram for $\mathrm{diag}_S : S \longrightarrow S \times S$ (we assume that in $\mathcal{M}$ finite $\varprojlim$, hence finite products, exist)

$$\begin{array}{ccc} & & S \\ & \overset{\mathrm{diag}_S}{\searrow} & \downarrow{\scriptstyle i_S} \\ & & \mathrm{Ch}(S) \\ & & \downarrow{\scriptstyle p_S} \\ S \times S & \longleftarrow & e. \end{array}$$

The essential property is that p be a *"fibration"* (in a suitable sense, to be worked out).

As a subsidiary matter it may be useful (perhaps) that i be a "cofibration" ([9]). One defines $\Omega(S)$ as the fiber of $\mathrm{Ch}(S)$ over $S \times S$ (with respect to the unique morphism $e = e_{\mathcal{M}} \longrightarrow S \times S$, where $e_{\mathcal{M}}$ is the zero object of $\mathcal{M}$). [page 25]

$$\begin{array}{ccc} \mathrm{Ch}(S) & \longleftarrow & \Omega S \\ \downarrow & & \downarrow \\ S \times S & \longleftarrow & e. \end{array}$$

[8] The functor Ω must come from an endomorphism of the derivator $\mathbf{D} = \mathbf{D}_{\mathcal{M},W}$, and the latter must be left exact, and dually for Σ.

[9] and perhaps that $S \longrightarrow e$ also be a fibration?

This is an object of $\mathcal{M}$, which we shall regard mainly as an object of $\mathcal{M}W^{-1}$. We must make sure that we have a bijection functorial in T in $\mathcal{M}W^{-1}$ p. 16

$$\underbrace{\pi_1 \underline{\mathrm{Ch}}^W_\infty(T,S)}_{\stackrel{\mathrm{def}}{=} \pi_1(T,S)} \simeq \underbrace{\mathrm{Hom}_{\mathcal{M}W^{-1}}(T,\Omega S)}_{\stackrel{\mathrm{def}}{=} \pi_0 \underline{\mathrm{Ch}}^W_\infty(T,\Omega S)}.$$

For this, one must compare

$$\underline{\mathrm{Ch}}^W_\infty(T,\Omega S) \qquad \text{and} \qquad \underbrace{\Omega_{\mathrm{Hot}_\bullet}(\underline{\mathrm{Ch}}^W_\infty(T,S))}_{\stackrel{\text{“by definition” for } \Omega \text{ of } \mathrm{Hot}_\bullet}{=} \underline{\mathrm{Ch}}_\infty(\mathrm{Ch}^W_\infty(T,S);0,0)},$$

since it is a matter of constructing a bijection between their π_0.

THE DERIVATORS

p.

ALEXANDRE GROTHENDIECK

Chapter X
Comparison of Cat with $\mathbf{\Delta}^\wedge$

p.

p. 1

1 Recall of the notions of D-asphericity, and link with the "basic localizer $W_{\mathbf{D}}$" [page 1]

For a derivator **D**, the notions in question are the following (cf. I):

(i) *Object* of Cat that is **D**-*aspheric* (i.e. $Z \longrightarrow e$ is a **D**-equivalence). NB I prefer to say "**D**-aspheric" instead of "**D**-contractible", which sometimes leads to confusion.

(ii) **D**-equivalence for an arrow $f : X \longrightarrow Y$ in Cat.

(iii) *Functor* $f : X \longrightarrow Y$ that is **D**-*aspheric* (i.e. the X/y are **D**-aspheric) and **D**-*coaspheric* (the $y\backslash X$ are **D**-aspheric). This is what I had called "strict **D**-equivalence" (or "costrict"), or homologically **D**-cofinal functor (or cohomologically **D**-cofinal), but the terminology "aspheric" seems to me the simplest and the best. (Fished out of PS, Pursuing Stacks.)

(iv) *Functor* $f : X \longrightarrow Y$ that is **D**-*smooth,* **D**-*proper* (i.e. such that the fibers of the induced functors $X/x \longrightarrow Y/f(x)$, respectively $x\backslash X \longrightarrow f(x)\backslash Y$, are **D**-aspheric).

(v) **D**-*fibration* $f : X \longrightarrow Y$ (self-dual notion), i.e. such that for $Y'' \xrightarrow{u} Y'$ over Y, if Y'', Y' Hot-aspheric, then $u_X : X'' \longrightarrow X'$ in

$$\begin{array}{ccccc} X & \longleftarrow & X' & \longleftarrow & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & Y' & \longleftarrow & Y'' \end{array}$$

is a **D**-equivalence ([1]). The **D**-*fibration* is called *trivial* (or *aspheric*) if and only if its fibers are aspheric, or again if and only if f is aspheric, or if and only if f is coasperic. This also means that f is a *universal* **D**-equivalence, i.e. that for every $Y' \longrightarrow Y$, $X' = X \times_Y Y' \longrightarrow Y'$ is a **D**-equivalence. [page 2]

(vi) We also have weakened variants of the notion of **D**-fibrations, namely the "**D**-*left fibrations",* or the f that are "*cohomologically* **D**-*fibrant",* and likewise the "**D**-*right fibrations",* i.e. the f that are "*homologically* **D**-*fibrant*" ([2]). The "left" or cohomological notion for $f : X \longrightarrow Y$ means that the functors $X_y \hookrightarrow X/y$ ($y \in \mathrm{Ob}\, Y$) and $X/y \longrightarrow X/y'$ (for $(u : y \longrightarrow y') \in \mathrm{Fl}\, Y$) are **D**-equivalences. And dually with the $X_y \hookrightarrow y\backslash X$, $y'\backslash X \longrightarrow y\backslash X$ for the "right" or "homological" notion. The **D**-fibrations are none other than the universal left (cohomological) **D**-fibrations or the universal right (homological) **D**-fibrations. p. 2

[1] We shall also say that f is **D**-*fibrant.* (**NB** There is no notion of "**D**-cofibrant" to my knowledge.)

[2] New terminology: *weak* cohomological (resp. homological) fibrations, cf. I, pages 111-126 for a corrected revision of the various notions of "**D**-fibrations".

Finally, this whole string of notions is deduced from the single notion of **D**-equivalence (ii). If I denote by

$$W_{\mathbf{D}} \subseteq \mathrm{Fl}(\mathrm{Cat})$$

the set of these **D**-equivalences, one finds what in Pursuing Stacks I called a "localizer" (localizer) in Cat ([3]). In Pursuing Stacks it was such a subset

$$W \subseteq \mathrm{Fl}(\mathrm{Cat}),$$

and not the datum of a derivator, that was taken as the fundamental starting datum for [page 3] doing homotopical algebra of all kinds. I had moreover developed a string of notions of the same kind (i) to (vi), starting from this datum W. It could be that the statements for the notions I developed in I, in terms of a derivator, are valid more generally in terms of a localizer W, at least insofar as they still make sense. (So for the moment I set aside the strictly cohomological or homological statements, which appeal directly to **D**.) Although it could be that under conditions to be made explicit on W, which imply that (Cat, W) do indeed define a derivator $\mathbf{D}_{\mathrm{Cat},W} = \mathbf{D}$, one still manages to interpret W as a $W_{\mathbf{D}}$, so that the cohomological and homological statements still make sense.

I feel like first looking in Pursuing Stacks at the axioms I have singled out for localizers, and checking whether they are all satisfied for $W_{\mathbf{D}}$. These conditions L 1 to L 5 are stated [page 4] in Pursuing Stacks, §64 (page 164).

L 1 Saturation of W_D. O.K.

This follows formally from the fact that we have a functor

$$\mathrm{Cat} \xrightarrow{\varphi} C \qquad (= \mathrm{Cat}^o)$$

such that for $u \in \mathrm{Fl}(\mathrm{Cat})$, $u \in W_{\mathbf{D}} \iff \varphi(u)$ isomorphism. (Take for $\varphi(X)$ the subcategory of $\mathbf{D}(X)$ consisting of the **D**-coefficients *constant* on X).

L 2 "$\mathbf{\Delta}^1$ is W-aspheric over e", i.e. for every X in Cat, $X \times \mathbf{\Delta}^1 \longrightarrow X$ is in W.

This also means that $\mathbf{\Delta}^1 \longrightarrow e$ is a trivial **D**-fibration, which is indeed the case.

L 3 If $X \in \mathrm{Ob}\,\mathrm{Cat}$ has a final object, then $X \longrightarrow e$ is in W. p. 3

In fact, we require for derivators that this be true assuming only X aspheric (in the absolute sense), and one even concludes from this that $X \longrightarrow e$ is W-aspheric, i.e. $X \times Y \xrightarrow{\mathrm{pr}_2} Y$ is in W for every Y in Cat ([4]).

L 4 If $f : X \longrightarrow Y$ is such that $X/y \longrightarrow Y/y$ are in W, then f is in W.

This is OK.

L 5 If X, Y are Cat-fibered over S, and $X \longrightarrow Y$ is an S-morphism that is *cartesian,* [page 5] inducing on the fibers $X_s \longrightarrow Y_s$ which are in W, then f is in W.

For $W_{\mathbf{D}}$, this is proposition 6, page 71 of I.

These axioms L 1 to L 5 (equivalent to L 1, L 3′ ($\mathbf{\Delta}^1 \longrightarrow e$ is in W) and L 5) are in the nature of stability axioms, ensuring that W is large enough. There is another condition which limits W to not being too large:

L a) If $f \in W$, then $\pi_0(f)$ is bijective.

This is not satisfied for all the $W_{\mathbf{D}}$ – e.g. if the derivator **D** is *zero,* then $W_{\mathbf{D}} = \mathrm{Fl}\,\mathrm{Cat}$! But for the derivators useful in practice, this always seems to be satisfied.

L b), c) The functor

$$\mathrm{Cat} \longrightarrow \mathrm{Cat}\,W^{-1}$$

commutes with finite sums and finite products ([5]).

---

[3] But beware, I assume here that the domain Diag of **D** is the whole of Cat!

[4] In fact L 2 $\Longrightarrow$ L 3.

[5] This works if (Cat, W) comes from a Quillen triple (with Q-fibrations, Q-cofibrations, Q-weak equivalences).

To examine what this means in concrete terms on W – I have not checked that it is O.K. for the $W_{\mathbf{D}}$.

L d) If $f : X \longrightarrow Y$ in W, then $f^o : X^o \longrightarrow Y^o$ as well.

This is corollary 2, page 75 of I.

p. –

THE DERIVATORS

ALEXANDRE GROTHENDIECK

Chapter XI

Hot-fibrations, proper functors and smooth functors etc. (in Cat)

p. –
p. 1
[page 127]

1 Hot-fibrations in Cat

Theorem 1. *Let $f : X \longrightarrow Y$ be in* Cat. *Equivalent conditions:*

a) For every $Y'' \xrightarrow{p} Y'$ over Y, whence $X'' = X \times_Y Y'' \xrightarrow{q} X' = X \times_Y Y'$, if p is a Hot*-equivalence, so is q.*

a′) As in a), with $Y'' = e$, Y' having a final object.

a″) As in a), with $Y'' = e$, Y' having an initial object ([1]).

b′) For every $y \in \mathrm{Ob}\, Y$, $X_y \longrightarrow X/y$ is a Hot*-equivalence, for every $y \longrightarrow y'$ in Y, $X/y \longrightarrow X/y'$ is a* Hot*-equivalence; finally these conditions still hold for $f' : X' \longrightarrow Y'$, for every base change $Y' \longrightarrow Y$ in* Cat. *(In fact, it suffices to require this for the first of the two conditions, concerning $X_y \longrightarrow X/y$.)*

b″) Dual of b′), with $X_y \longrightarrow y\backslash X$ and $y'\backslash X \longrightarrow y\backslash X$.

c) For every locally constant sheaf of sets F on X, $f_*(F)$ is locally constant and is computed "fiber by fiber", likewise for $R^1 f_*(F)$ if F is a sheaf of groups, and for the $R^i f_*(F)$ if F is an abelian sheaf. Moreover, we want these conditions to remain satisfied after every base change. ([2])

Proof. We have the ± tautological implications

$$\mathrm{a} \implies \mathrm{a}' \implies \mathrm{b}' \implies \mathrm{c},$$

[1] keep a″) and b″), conditions dual to a′), b′) .

[2] *c′) (homological version). 1°) in terms of direct images of locally constant groupoids, and 2°) in terms of the $L_i f_!(L)$, for L a locally constant abelian sheaf.*

and it remains to prove that c $\implies$ a. Since a is self-dual, being equivalent to a′, b′, it is also equivalent to the conditions dual to these, a″ and b″ (cf. cor. 2, page 75). We are reduced to proving that if c) is satisfied, and if we have

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y' \end{array}$$

cartesian in Cat, if p is a Hot-equivalence, so is q. We use the Artin-Mazur criterion. p. 2
1°) If F is a locally constant sheaf of sets on X, we must prove that $H^0(X,F) \xrightarrow{\sim} H^0(X', q^*F)$.
But [page 128]

$$H^0(X,F) \simeq H^0(Y, \underbrace{f_*(F)}_{\substack{\text{locally constant}\\ \text{by hypothesis c)}}}) \quad \underbrace{\longrightarrow}_{\substack{\sim\\ \text{since } p\\ \text{Hot-equivalence}}} \quad H^0(Y', p^* f_*(F)),$$

but

$$p^* f_*(F) \xrightarrow{\sim} f'_*(q^*(F)),$$

since $f_*(F)$ and $f'_*(q^*F)$ are computed "fiber by fiber" by hypothesis. Thus

$$H^0(X,F) \simeq H^0(Y', f'_*(q^*(F))) \simeq H^0(X', q^*(F)),$$

q.e.d.

2°) If F is a locally constant sheaf of groups on X, we must prove

$$H^1(X,F) \xrightarrow{\sim} H^1(X', q^*F).$$

The fact that it is injective already follows from 1°), which implies that the functor

$$\begin{array}{ccc} F\text{-torsors on } X & \longrightarrow & q^*(F)\text{-torsors on } X' \\ P & \longmapsto & q^*P \end{array}$$

is fully faithful, by the well-known argument using the $\mathcal{H}om$ sheaves. We must see that this functor is essentially surjective.
We have a homomorphism of exact sequences of pointed sets [page 129]

$$\begin{array}{ccccccccc} H^1(Y, f_*F) & \xrightarrow{u} & H^1(X,F) & \xrightarrow{v} & \overbrace{H^0(Y, R^1 f_*(F))}^{\ni \eta} & \xrightarrow{w} & H^2(Y, f_*F) \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \delta} \\ H^1(Y', f'_*F') & \xrightarrow{u'} & \underbrace{H^1(X', F')}_{\ni \xi'} & \xrightarrow{v'} & \underbrace{H^0(Y', R^1 f'_*(F'))}_{\ni \eta'} & \xrightarrow{w'} & H^2(Y', f'_*F') \end{array}$$

([3]), where the sheaves f_*F, $R^1 f_*(F)$ are locally constant by hypothesis and are computed fiber by fiber, the sheaves $f'_*(F')$ and $R^1 f'_*(F')$ also, which implies that the latter are canonically isomorphic (by the base change arrows) to the inverse images of their counterparts $f_*(F)$, $R^1 f_*(F)$. Since p is a Hot-equivalence, this implies that the arrows p. 3
α, γ are isomorphisms. Thus if $\xi' \in H^1(X', F')$, its image η' in $H^0(Y', R^1 f'_*(F'))$ comes

[3] where we have set $F' = q^*(F)$.

from a well-determined $\eta \in H^0(Y, R^1 f_*(F))$. We must see that this η comes from a $\xi \in H^1(X, F)$. This (if I remember correctly) is possible if and only if a certain obstruction $w(\eta)$ in $H^2(Y, f_*(F))$ is in the neutral subset of H^2. **NB** The stack on $Y^\wedge$ of F-torsors on the $X^\wedge/U$ that lift η is an $\mathcal{F} = f_*F$-gerbe $\mathcal{G}_\eta$, and we must see that it is neutral. But there is something better to do:

[page 130]

We note that this gerbe $\mathcal{G}$ is locally constant (since it corresponds to a locally constant band $f_*(G)$) and that $\mathcal{G}'$ is its inverse image. But since p is a Hot-equivalence, it follows that the functor

$$\underline{\Gamma}(Y, \mathcal{G}) \longrightarrow \underline{\Gamma}(Y', \mathcal{G}')$$

is an equivalence of categories, so ξ' comes (up to isomorphism) from a unique ξ of $\underline{\Gamma}(Y, \mathcal{G})$, q.e.d.

3°) Let F be a locally constant abelian sheaf on X, we must see

$$H^i(X, F) \xrightarrow{\sim} H^i(X', F') \qquad (F' = q^*F), \tag{$*$}$$

for every $i \in \mathbf{N}$. But the two terms are the abutment of the Leray spectral sequences, with initial terms

$$E_2^{pq} \simeq H^p(Y, R^q f_*(F))$$
$$E_2'^{pq} \simeq H^p(Y', R^q f'_*(F')),$$

and $(*)$ is associated with a homomorphism of spectral sequences, which for the initial terms is

$$H^p(Y, R^q f_*(F)) \xrightarrow{\sim} H^p(Y', R^q f'_*(F')), \tag{$**$}$$

associated with

$$p^*(R^q f_*(F)) \longrightarrow R^q f'_*(F'). \tag{$***$}$$

But by hypothesis the $R^q f_*(F)$, $R^q f'_*(F')$ are locally constant and are computed fiber by fiber,

[page 131]

whence it follows that the base change homomorphisms $(***)$ are isomorphisms, and since we are dealing with locally constant sheaves and p is a Hot-equivalence, it follows that $(**)$ is isomorphism, hence also $(*)$ on the abutments, q.e.d.

Definition 1. Under the equivalent conditions of theorem 1, we say that f is a Hot-*fibration.*

Proposition 1. *This notion is stable under base change (trivial), under composition (formal from a)). For f to be a* Hot*-fibration, it is necessary and sufficient that f^o be one.*

p. 4

Proposition 2. *Let $f : X \longrightarrow S$ be a* Hot*-fibration. Then $s \longmapsto \mathrm{hot}(X_s)$ is a local system on S, with values in* Hot*. Thus the X_s for s in the same connected component of S are* Hot*-isomorphic to one another (but not canonically, if this component is not* 1*-connected).*

Theorem 2. *Let* $\begin{array}{ccc} X' & \xrightarrow{f} & X \\ & {\scriptstyle p'}\searrow \quad \swarrow{\scriptstyle p} & \\ & S & \end{array}$ *be a commutative diagram in* Cat*, with p, p'* Hot*-fibrations, and let $S_0 \subseteq S$ be a subset of S such that $S_0 \longrightarrow \pi_0(S)$ is surjective. For f to be a* Hot*-equivalence, it is necessary and sufficient that the $f_s : X'_s \longrightarrow X_s$ $(s \in S_0)$ be so.*

a) Sufficiency is proved as before, with the Artin-Mazur criterion.

[page 132]

Let us note that the hypothesis on S_0 ensures that if F, F' are two locally constant sheaves of sets on S, and $u : F \longrightarrow F'$ a homomorphism such that $u_s : F_s \xrightarrow{\sim} F'_s$ for $s \in S_0$, then u is isomorphism.

1°) Let F be a locally constant sheaf of sets on X, F' its inverse image on X', we must prove

$$H^0(X,F) \xrightarrow{\sim} H^0(X',F').$$

Now

$$H^0(X,F) \simeq H^0(S, \underbrace{p_*(F)}_{\substack{\text{locally}\\ \text{constant on } S}}) \longrightarrow H^0(S, p'_*(F')) \simeq H^0(X',F'),$$

we must prove that

$$p_*(F) \xrightarrow{\sim} p'_*(F') \qquad (*)$$

is isomorphism. Now by the hypothesis on p, p', the two sheaves are locally constant and are computed fiber by fiber, from which it follows that $(*)$ is an isomorphism for the fibers at $s \in S_0$, hence an isomorphism everywhere. OK.

2°) It already follows from this, if F is a sheaf of *groups,* that the functor

$$(F\text{-torsors on } X) \longrightarrow (F'\text{-torsors on } X')$$

is fully faithful, hence $H^1(X,F) \longrightarrow H^1(X',F')$ injective; it remains to prove that this functor is essentially surjective. Because of full faithfulness, this is *local on* S (**NB** We see directly that the hypothesis made on the $s \in S_0$ implies the same hypothesis for every $s \in S$, using the fact that p, p' are fibrations) ([4]). Thus we are reduced to the case where $S = S/s_0$ has a final object s_0.

We then have

p. 5
[page 133]

$$H^1(X,F) \simeq H^0(S, R^1p_*(F))$$
$$H^1(X',F') \simeq H^0(S, R^1p'_*(F')),$$

and we see again that

$$R^1p_*(F) \longrightarrow R^1p'_*(F')$$

is isomorphism (for it is computed fiber by fiber, and we use the hypothesis on the f_s), hence induces an isomorphism

$$H^1(X,F) \longrightarrow H^1(X',F').$$

3°) If F is even abelian, we must prove

$$H^i(X,F) \xrightarrow{\sim} H^i(X',F')$$

for every i, and this again follows from the Leray spectral sequence for (p,F) and for (p',F').

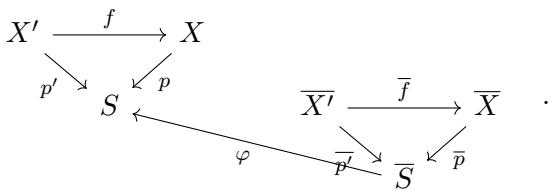
b) Conversely, suppose that f is a Hot-equivalence, and let us prove that the $f_s : X'_s \longrightarrow X_s$ are Hot-equivalences. I am going to use the

Lemma. *Let $S \in \mathrm{Ob}\,\mathrm{Cat}$, S 0-connected. Then there exists an $\overline{S} \in \mathrm{Ob}\,\mathrm{Cat}$, $\overline{S}$* Hot-*aspheric, and $\overline{S} \longrightarrow S$ a* Hot-*fibration.*

Let us see how this implies theorem 2 (converse part). We may suppose S 0-connected, and we make the base change by $\overline{S} \longrightarrow S$

as in the lemma:

[page 134]



[4]The hypotheses made are indeed local on S (at least if $S_0 = S$).

Then $\overline{p'}$ and $\overline{p}$ are Hot-fibrations, and since φ is a Hot-fibration, base change by φ transforms the Hot-equivalence f into a Hot-equivalence $\overline{f}$. Consider then $\overline{s} \in \mathrm{Ob}\,\overline{S}$, and the diagram

$$\begin{array}{ccc} \overline{X'} & \xrightarrow{\overline{f}} & \overline{X} \\ {\scriptstyle i'}\uparrow & & \uparrow{\scriptstyle i} \\ \overline{X'}_{\overline{s}} & \xrightarrow{\overline{f}_{\overline{s}}} & \overline{X}_{\overline{s}} \end{array} \quad .$$

Since $\{\overline{s}\} \longrightarrow \overline{S}$ is a Hot-equivalence, the arrow i deduced from it by the base change p. 6
$\overline{p} : \overline{X} \longrightarrow \overline{S}$ is one as well, since $\overline{p}$ is a Hot-fibration, and likewise for i'. Thus in the preceding commutative square, $\overline{f}$, i and i' are Hot-equivalences, hence so is $\overline{f}_{\overline{s}}$. But if $s = \varphi(\overline{s})$, $\overline{f}_{\overline{s}}$ is identified with $f_s : X'_s \longrightarrow X_s$. Besides, it is immediate that φ, being a Hot-fibration

with nonempty source, and connected target, is surjective. (**NB** all the fibers of one [page 135]
and the same connected component of S are Hot-equivalent, hence if one is nonempty, they all are.) This completes the proof, modulo the lemma.

It remains to prove it. Since $S \neq \emptyset$, I can choose an "origin"

$$s_0 \in S,$$

and look at the category

$$\underline{\mathrm{Ch}}_\infty(S; s_0, -)$$

of infinite paths in S, with origin s_0, fiber of $\underline{\mathrm{Ch}}_\infty(S)$ under the source functor

$$\sigma : \underline{\mathrm{Ch}}_\infty(S) \longrightarrow S.$$

We know that the fibers of σ are Hot-aspheric (VII), and the category constructed here is none other than (with the notations of VII, p. 152) than the category $\overline{S'}$ which enters into the canonical factorization of

$$f : \{s_0\} = S' \longrightarrow S$$

into

$$\{s_0\} = S' \underbrace{\longrightarrow}_{\text{Hot-equivalence}} \overline{S'} \xrightarrow{\overline{f}} S,$$

where here $\overline{f}$ is interpreted simply as the target functor (i.e. the functor induced on [page 136]
$\underline{\mathrm{Ch}}_\infty(S; s_0, -)$ by the target functor

$$\beta : \underline{\mathrm{Ch}}_\infty(S) \longrightarrow S \quad).$$

We proved in loc. cit. that $\overline{f}$ is a Hot-fibration ("proper" and "smooth"), but in a context which seemed to appeal tacitly to the theory of the derivator HOT. So to put the "topological" results of loc. cit. on a solid basis, we must develop the relevant notions of proper functor, smooth functor, without reference to a HOT theory. This is what I had in fact already done in 1983, in the days of Pursuing Stacks, but without a proper write-up on this subject. So I must now return to these notions, in a somewhat systematic way. Besides, I would have trouble constructing a single Hot-fibrant functor (except for the trivial examples, such as isomorphisms, or the products $Y \times Z \xrightarrow{\mathrm{pr}_1} Y$), if I did not have at my disposal a theory of smooth functors and proper functors ([5]).

[5] I am exaggerating a little all the same – after all it must be fairly obvious that the Thomason-Kan functors are Hot-fibrations (but there are still a few too few of them!).

THE DERIVATORS

p. –

ALEXANDRE GROTHENDIECK

Chapter XII

Characterization of W_∞. W-proper, W-smooth functors etc. Amalgamated sums and W-cocartesian squares in Cat**.**

19.11.90

p. –
p. 1
[page 1]

I Characterization of W_∞.

I would like to characterize W_∞ as a subset of Fl(Cat), by its properties as a localizer, and without appealing to knowledge of notions already using W_∞, such as properness etc. I would like at least some properties of a localizer W, familiar for W_∞, which are so strong that they imply $W \subseteq W_\infty$. Of course, in the proofs I allow myself to use everything I know about W_∞.

Of course, at least a form of saturation will be needed, perhaps a very weak one, I presume the following will suffice ([1]).

W(1) (Saturation axiom.) a) The identities are in W, b) if f, g composable and two among f, g, gf are in W, the third too, c) if $fg \in W$, $gf \in W$, then $f, g \in W$.

A particularly crucial property, which already distinguishes W_∞ from the set of homotopisms (a notion too strict in many respects), is the axiom I called L4 bis:

W(2) (Localization axiom.) Let $f : X \longrightarrow Y$ be an arrow in Cat/S, such that $\forall\ s \in \mathrm{Ob}\, S$, $f_s : X/s \longrightarrow Y/s$ is in W (we shall say that f is in W "locally over S"). Then $f \in W$.

In the case $Y = S$, this gives

Corollary: *If $f : X \longrightarrow Y$ is W-aspheric, i.e. $X/y \longrightarrow Y/y$ in W for every $y \in \mathrm{Ob}\, Y$, then $f \in W$* ([2]).

Proposition 1. *If W satisfies* W(1) *and* W(2)*, the following properties of W are equivalent:*

a) For every X in Cat*, $X \times \mathbf{\Delta}^1 \xrightarrow{\mathrm{pr}_1^X} X$ is in W (i.e. $\mathbf{\Delta}^1 \longrightarrow e$ is "universally in W").*

b) If X in Cat *has a final object, then X is W-aspheric, i.e. $X \longrightarrow e$ is in W.*

c) Every homotopism is in W.

Proof. b) $\Longrightarrow$ a) ([3]), for to see that $\mathrm{pr}_1 \in W$, applying the corollary above, we must prove that $\forall\ x \in X$ p. 2

$$(X \times \mathbf{\Delta}^1)/x \longrightarrow X/x \qquad (*)$$

[1] (no, I had to strengthen it in 2).

[2] One can consider the validity of the corollary as a weakened variant W(2 bis) of W(2) above. In proposition 1, one uses only W(2 bis), a weak form of W(2).

[3] uses W(1) only in weak form, without the saturation condition c). NB Only b) $\Longrightarrow$ c) uses W(2 bis), the implications a) $\Longrightarrow$ c) $\Longrightarrow$ b) are valid under the sole hypothesis W(1).

is in W. Now

$$(X \times \mathbf{\Delta}^1)/x \simeq X/x \times \mathbf{\Delta}^1,$$

$$\begin{array}{ccccc} \mathbf{\Delta}^1 & \longleftarrow & X \times \mathbf{\Delta}^1 & \longleftarrow & X/x \times \mathbf{\Delta}^1 \\ \downarrow & & \downarrow & & \downarrow \\ e & \longleftarrow & X & \longleftarrow & X/x \end{array}$$

and the two categories X/x, $X/x \times \mathbf{\Delta}^1$ are W-aspheric since they have a final object, so $(*)$ is in W by weak saturation W(1).

a) $\Longrightarrow$ c). I recall that

$$f : X \longrightarrow Y$$

is a homotopism if there exists

[page 2]

$$g : Y \longrightarrow X$$

such that

$$gf \sim \mathrm{id}_X, \quad fg \sim \mathrm{id}_Y, \qquad (*)$$

where the relation $(\sim)$ denotes homotopy in $\underline{\mathrm{Hom}}(X, X)$ and $\underline{\mathrm{Hom}}(Y, Y)$ respectively (defined by paths going from one element to the other). Let us prove the

Lemma. *Suppose that W satisfies* W(1) *and the* a) *of the proposition, i.e. $X \times \mathbf{\Delta}^1 \longrightarrow X$ in W for every X in* Cat. *Then if X, Y in* Cat *and f, g in $\underline{\mathrm{Hom}}(X, Y)$ are homotopic, we have*

$$\mathrm{hot}_W(f) = \mathrm{hot}_W(g).$$

Moreover, $f \in W$ if and only if $g \in W$.

It suffices to prove this when there exists a homomorphism $f \longrightarrow g$. This can be interpreted as the existence of an arrow

$$X \times \mathbf{\Delta}^1 \xrightarrow{F} Y$$

whose restrictions to $X \times \{0\}$ and $X \times \{1\}$ respectively are f, g. To prove that $\mathrm{hot}_W(f) = \mathrm{hot}_W(g)$, it suffices to prove that

$$i_0, i_1 : X \longrightarrow X \times \mathbf{\Delta}^1 \qquad x \longmapsto (x, 0),\ x \longmapsto (x, 1)$$

are equal in Hot_W, since

$$f = F \circ i_0, \quad g = F \circ i_1.$$

p. 3

Now this follows from the fact that they are sections of $X \times \mathbf{\Delta}^1$ over X, and that in Hot_W the projection $X \times \mathbf{\Delta}^1 \xrightarrow{p} X$ is isomorphism (hence $i_0 = i_1 = p^{-1}$ in Hot_W). Moreover, the diagram

$$\begin{array}{ccc} X & \xrightarrow{i_0} & X \times \mathbf{\Delta}^1 \\ & \searrow^{f} & \downarrow F \\ & & Y, \end{array}$$

where $i_0 \in W$, shows that $f \in W$ if and only if $F \in W$, and similarly $g \in W$ if and only if $F \in W$, hence $f \in W \Longleftrightarrow g \in W$.

Let us return to $(*)$, we see that in Hot_W $g \circ f = \mathrm{id}_X$, $f \circ g = \mathrm{id}_Y$, so f and g are Hot_W-isomorphisms inverse to one another. On the other hand, $gf \in W$, $fg \in W$, since $\mathrm{id}_X, \mathrm{id}_Y \in W$. So we win, if we have formulated W(1) strongly enough. We have thus proved a) $\Longrightarrow$ c).

c) $\Longrightarrow$ b) is trivial, since if X has a final object, $X \longrightarrow e$ is a homotopism.

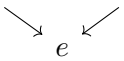
We must of course in what follows suppose that W satisfies the equivalent conditions of the proposition. We shall take e.g. the form

[page 3]

W(3) (Final element axiom.) If X has a final object, then X is W-aspheric.

Corollary 1. *For $f : X \longrightarrow Y$ to be W-aspheric, it is necessary and sufficient that the X/y be W-aspheric ($y \in \mathrm{Ob}\, Y$).*

For in order that $X/y \longrightarrow Y/y$ be in W, since Y/y is W-aspheric (final object),



it is necessary and sufficient that X/y be so as well.

Corollary 2. *If X in* Cat *has an initial object, it is W-aspheric.* (Indeed, $X \longrightarrow e$ is a homotopism, and we apply prop. 1.) ([4])

I shall put down for the record

Corollary 3. *Every homotopism in* Cat *is in W. If $f, g : X \rightrightarrows Y$ in* Cat *are homotopic, then they are equal in* Hot_W*, and f is in W if and only if g is.*

Proposition 2. Let $f : X \longrightarrow Y$ be in Cat. If $f \in W$, $f^\circ \in W$

We use the proof of I, p. 75, with the help of the category called $\Phi(X)$ ($\mathrm{Ob}\, \Phi(X) = \mathrm{Fl}(X)$, but with a somewhat peculiar notion of arrow in $\Phi(X)$). We find that $f \in W \iff \Phi(f) \in W$, and likewise $f^\circ \in W \iff \Phi(f) \in W$. p. 4

NB. In the proof of proposition 2, as in that of proposition 1 (for the implication b) $\Longrightarrow$ a)), we used the localization property W(2) only in the weak form of the corollary, i.e. in the form f W-aspheric $\Longrightarrow f \in W$.

Corollary (of prop. 2): *Let $W \subseteq \mathrm{Fl}(\mathrm{Cat})$ satisfy the axiom* W(1) (saturation). *Then the conjunction of the axioms* W(2), W(3) *is equivalent to that of the dual axioms* W(2′), W(3′) ([5]).

It suffices to prove that 1,2,3 $\Longrightarrow$ 2′,3′.

For 3′ , this is corollary 2 above. For 2′ , this follows at once from proposition 2. (NB This statement would remain valid, if we would have taken W(2) in the weak form only, cf. the NB on the previous page.) [page 4]

I still have to put in an axiom of the form L5, L6 (cf. VI, p. 43-44). But the statement of these axioms uses the notion of Hot-fibered objects, or of perfect functors, i.e. calls on knowledge of W_∞, which I would like to avoid (if I can). I must therefore introduce the notion of W-fibration, on the model of what I did in the context of derivators, in I (cf. p. 111-126). We introduce the notion of **D**-fibration on page 117 (prop. 5), by equivalent conditions, some of which make sense in terms of the datum of W alone. So let

$$f : X \longrightarrow Y$$

be in Cat and consider the conditions

a) If $Y'' \longrightarrow Y'$ in Cat/Y is in W, then so is $X'' \longrightarrow X'$ (base change by $X \longrightarrow Y$ takes W into W).

a′) As in a), but with Y'', Y' W-aspheric.

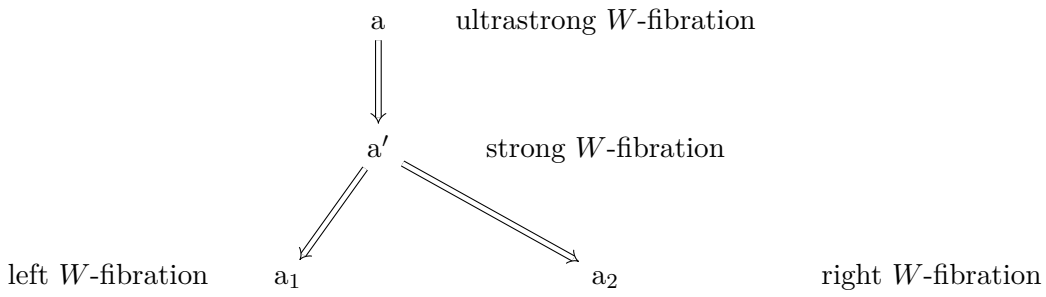
a_1) As in a), in the particular case where Y' has a final object and Y'' is the one-point category.

a_2) Dual of a_1), with an initial object of Y'.

[4] This is thus the statement dual to the axiom W(3) of the final object.

[5] Likewise if in this statement we replace W(2) by W(2 bis).

We thus have



and in the case $W = W_\infty$ all these conditions are equivalent. I have a mind take the strongest of these conditions, but I must give myself some means of finding W-fibrations in so strong a sense. For example I would like to know that if X is in Cat, then $\underline{\mathrm{Ch}}_\infty(X) \longrightarrow X \times X$ or $\underline{\mathrm{Ch}}(X) \longrightarrow X \times X$ is a W-fibration. In both cases, and (say) for W_∞, I prove it by proving that p is both smooth, and proper. So we would have to go over the notions of smoothness and properness again in the context of the datum of a W. p. 5 [page 5]

If I remember correctly, property a_1) was something fairly easily checked, it sufficed to suppose that the

$$X_y \longrightarrow X/y \qquad \text{and} \qquad X/y \longrightarrow X/y'$$

(for $y \in \mathrm{Ob}\, Y$ resp. $u : y \longrightarrow y'$ in $\mathrm{Fl}\, Y$) are in W, and that this remains true under every base change - which amounts to condition a_1) on f. We could therefore formulate a condition on W as follows:

W(4) (Axiom of W-fibrations ([6]).) Let $f : X \longrightarrow Y$ in Cat satisfy the W-fibration condition a_1) above (p. 4), then it even satisfies a), i.e. base change by f takes arrows in W to arrows in W ([7]).

Thanks to proposition 2, it follows that the dual condition a_2) on f implies the strong W-fibration condition, so the axiom W(4), apparently asymmetric, also means that the four fibration conditions on f are equivalent, it is therefore in fact symmetric.

NB I do not see, without additional axioms on W, how to pass from a_1 to a' alone. The difficulty is to show, by means of a_1 alone, that if Y is W-aspheric, the inclusion $X_y \longrightarrow X$ (where $y \in \mathrm{Ob}\, Y$) is in W. In the context of derivators **D**, this could be seen by **D**-cohomological means.

This very strong axiom suffices to imply that if X, Y are W-fibered over S, and if $f : X \longrightarrow Y$ is an S-morphism which is in W, then $f_s : X_s \longrightarrow Y_s$ is in W ($\forall\ s \in S$). Let us at least show the converse for now.

Proposition 3. *Let X, Y be in* Cat$/S$, *$f : X \longrightarrow Y$ an S-arrow. Suppose (X, Y W-fibered) that $\forall\ s \in S$, $f_s : X_s \longrightarrow Y_s$ is in W, then $f \in W$* ([8]). [page 6]

We have to use the localization axiom, hence look at the f/s induced by f p. 6

$$\begin{array}{ccc} X/s & \xrightarrow[f/s]{} & Y/s \\ {\scriptstyle i_s^X}\uparrow & & \uparrow{\scriptstyle i_s^Y} \\ X_s & \xrightarrow{f_s} & Y_s \end{array} .$$

[6] See sufficient weakened variant, in the NB p. 10. See also of variants pages 13, 14 below.

[7] i.e. a left W-fibration is an ultrastrong fibration.

[8] *Without* axiom W(4). This is the first time we need W(2) in its strong form, to the exclusion of W(2 bis).

Condition a_1 suffices to imply that i_s^X and i_s^Y are in W, so f/s is in W if and only if f_s is. We did not have to use axiom W(4), and it sufficed to suppose that X, Y are "W-cohomological" fibered objects (or left W-fibered) over S, or dually, right W-fibered objects.

Moreover:

Corollary ([9]). *Let $S_0 \subseteq \mathrm{Ob}\, S$ be a subset of S such that $S_0 \longrightarrow \pi_0(S)$ surjective. Then in proposition 3 it suffices to suppose $f_s \in W$ for $s \in S_0$.*

It suffices to see that if $u : s \longrightarrow s'$ is an arrow in S, then $f_s \in W \iff f_{s'} \in W$. We use the fact that $f_s \in W \iff f/s \in W$, and likewise for $f_{s'}$. So to see that $f/s \in W \iff f/s' \in W$. Now we have

$$\begin{array}{ccc} X/s & \xrightarrow{f/s} & Y/s \\ {\scriptstyle X/u}\downarrow & & \downarrow{\scriptstyle Y/u} \\ X/s' & \xrightarrow{f/s'} & Y/s' \end{array} ,$$

where X/u and Y/u are in W, whence the result.

But let us return to the converse: if $f \in W$, then do we have $f_s \in W$? There is an argument to this effect in VI, p. 23, which I am going to take up again. The idea is this. We consider the connected component S_0 of S containing s, and we construct a W-fibered object (in the *strong sense* a)) [page 7]

$$S' \longrightarrow S_0 \subseteq S$$

with S' *W-aspheric.* Since $X \longrightarrow Y$ over S is in W, the same will therefore hold for $X' \longrightarrow Y'$ over S'. Since the fibers of X', Y' at s' are identified with those of X, Y at the image of s' in S, we are reduced to the case where S is W-aspheric. But then we have

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle i_s^X}\uparrow & & \uparrow{\scriptstyle i_s^Y} \\ X_s & \xrightarrow{f_s} & Y_s \end{array} ,$$

where i_s^X, i_s^Y are in W, since obtained from $\{s\} \longrightarrow S$ (which is in W, S being W-aspheric) by the W-fibrant base changes $X \longrightarrow S$, $Y \longrightarrow S$. Hence $f \in W \implies f_s \in W$. p. 7

NB We used the fibration property for X, Y only in the sense a′) ([10]).

We must therefore prove, if we can, the

Lemma (?) ([11]). *Let S be a 0-connected object of* Cat*. Then there exists a W-aspheric object S' of* Cat*, and a W-fibration (in the* strong *sense of a), page 4) $S' \longrightarrow S$.*

The natural idea is to take, for fixed $s_0 \in \mathrm{Ob}\, S$,

$$S' = \underline{\mathrm{Ch}}(S; s_0, -) \qquad \text{or} \qquad S' = \underline{\mathrm{Ch}}_\infty(S; s_0, -),$$

and the arrow

$$\beta : c \longmapsto \text{target } c : S' \longrightarrow S.$$

We know that S' is W_∞-aspheric and that β is W_∞-fibrant (and even perfect). We must see that S' is also W-aspheric, and β W-fibrant. We must therefore go over the proof again in the case of W_∞, and prove that it transforms to W, given the axioms needed

[9] Without axiom W(4). [page 8]

[10] And even, in a′) we could require that $Y'' = e$, Y' W_ω-aspheric, where W_ω defined on page 14, $W_\omega \subseteq W \cap W_\infty \ldots$

[11] Is proved by means of axiom W(4) (+ some small later verifications).

on W. I have a mind to look first at the choice $S' = \underline{\mathrm{Ch}}(S; s_0, -)$, without passing to a direct limit which would force me to put a limit axiom on W. The delicate and essential homotopy properties of $\underline{\mathrm{Ch}}(S)$ are established in VII, p. 157-169, namely that in the diagram

$$S \xrightarrow{i_S} \underline{\mathrm{Ch}}(S) \xrightarrow{p_S} S \times S,$$

$i_S \in W_\infty$ and p_S is a W_∞-fibration (in fact, *proper* and *smooth* for W_∞) ([12]). The fact that $i_S \in W_\infty$ implies that the source and target functors

$$\underline{\mathrm{Ch}}(S) \underset{\beta}{\overset{\sigma}{\rightrightarrows}} S \qquad \sigma = \mathrm{pr}_1 \circ p_S,\ \beta = \mathrm{pr}_2 \circ p_S$$

are in W_∞ (since $\sigma i_S = \beta i_S = \mathrm{id}_S$). Since these are Hot-fibrations, it follows that the fibers $\underline{\mathrm{Ch}}(S; s, -)$ resp. $\underline{\mathrm{Ch}}(S; -, s)$ are Hot-aspheric. But even granting that σ, β are in W and are W-fibrations, I cannot conclude right away that the fibers are W-aspheric - there would be a vicious circle! So I need a direct proof.

Consider the functor

$$\begin{array}{c} \underline{\mathrm{Ch}}(S; s, -) \\ \Big\downarrow{\scriptstyle p} \\ \underline{Ty} \end{array},$$

it is a Cat-fibrant functor, hence W-smooth (I will have to develop the notion of W-smoothness ([13]) carefully, I will assume for the moment the familiar formal properties, and will justify them later). Because of W-smoothness, to prove $p \in W$, it suffices to prove that the fibers of p are W-aspheric ([14]). Now the fibers are the $Z_\tau = \underline{\mathrm{Ch}}_\tau(S; s, -)$, which are *contractible* categories (i.e. $Z_\tau \longrightarrow e$ is a homotopism), hence W-aspheric by W(3) (cf. cor. 3, page 3).

p. 8

[page 9]

On the other hand, if we grant that $\underline{\mathrm{Ch}}(S) \xrightarrow{p_S} S \times S$ is W-smooth (and W-proper), hence $\underline{\mathrm{Ch}}(S) \xrightarrow{\sigma_S} S$ also, having W-aspheric fibers as we have just seen, it follows that $\sigma_S \in W$, whence $S \xrightarrow{i_S} \underline{\mathrm{Ch}}(S)$ also in W (since $\sigma_S \circ i_S = \mathrm{id}_S$). So it remains only to prove that

$$\underline{\mathrm{Ch}}(S) \xrightarrow{p_S} S \times S \qquad (*)$$

is W-proper and W-smooth. It will follow that

$$\underline{\mathrm{Ch}}(S; s_0, -) \xrightarrow{\beta} S$$

is likewise W-proper and W-smooth, since obtained from p_S by the base change

$$S \longrightarrow S \times S \qquad s \longmapsto (s_0, s).$$

Next we must see that W-smooth and W-proper implies the W-fibration property *in its strongest form* a) (whereas I foresee that a priori we will only have it in the weak forms a_1, a_2 - not even in the form a′). So it is here that the axiom W(4) of W-fibrations will be essential ([15])!

---

[12] In the statement of the "essential properties" of $\underline{\mathrm{Ch}}(S)$, we can work with W_ω, the smallest subset of Fl(Cat) satisfying axioms W(1), W(2 bis), W(3) (cf. page 14), so $W_\omega \subseteq W \cap W_\infty$.

[13] $f : X \longrightarrow Y$ is W-smooth if the $X_y \longrightarrow y\backslash X$ are W-aspheric.

[14] Here we use W(2 bis), not $W(2)$.

[15] in the weakened form W(4a″) (p. 13).

Here is the argument for the smoothness of (∗), in loc. cit. p. 161 ff. We must prove that for $c \in \mathrm{Ob}\,\underline{\mathrm{Ch}}(S)$, and an arrow $(u, v) \in \mathrm{Fl}(S \times S)$, a certain category of "liftings of (u, v) with target c " , $\mathcal{C} = \mathcal{C}(c, u, v)$, is W-aspheric ([16]). For this, we compare it to a category $\mathcal{C}'$ with a final object, hence contractible and a fortiori W-aspheric, with the help of two functors

$$\mathcal{C} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \mathcal{C}',$$

and the homotopies $\varphi\psi \sim \mathrm{id}_{\mathcal{C}'}$ (no merit, $\mathcal{C}'$ being contractible) and $\psi\varphi \sim \mathrm{Id}_{\mathcal{C}}$ (we even find an arrow $\psi\varphi \longrightarrow \mathrm{id}_{\mathcal{C}}$). Thus φ is a homotopism, hence a W-equivalence (cor. 3, p. 3), hence $\mathcal{C}$ is W-aspheric, $\mathcal{C}'$ being so. The properness of p_S is equivalent to the smoothness of $p_{(S^\circ)}$, it is therefore proved at the same time. [page 10]

Thus, the lemma on page 7 is proved, using axiom W(4) which ensures that an arrow which is both W-smooth and W-proper satisfies the very strong condition a) of W-fibrations. p. 9

NB In fact, it would suffice, instead of axiom W(4) in its strong form, to assume that an $f : X \longrightarrow Y$ which is both "homotopically smooth" and "homotopically proper", i.e. the fibers of the $X/x \longrightarrow Y/y$ and of the $x\backslash X \longrightarrow y\backslash Y$ are contractible (hence W-aspheric for every W satisfying W(1,2,3)), is a strong W-fibration. Let us state all the same what we have proved.

Proposition 4. *If W satisfies* W(1,2,3) ([17]) (we always assume this in what follows) *plus* W(4) (in a somewhat weakened sufficient form, cf. preceding remark), *we have this: Let $f : X \longrightarrow Y$ be an arrow in* Cat/S, *with X, Y W-fibered over S. If $f \in W$, then $f_s : X_s \longrightarrow Y_s$ is in W for every $s \in S$.*

I wonder whether I know enough to be able to prove

$$W \subseteq W_\infty.$$

We must all the same eliminate joke cases, such as $W = \mathrm{Fl}(\mathrm{Cat})$, hence put in the axiom

W(5) (Axiom of connectedness ([18]).) If $f : X \longrightarrow Y$ is in W, $\pi_0(f) : \pi_0(X) \longrightarrow \pi_0(Y)$ bijective.

I then factor f as ([19]) [page 11]

$$X \overset{i}{\to} \bar{X} \overset{\bar{f}}{\to} Y,$$

with $\bar{f}$ a W_∞-fibration and a W-fibration ([20]), $i \in W \cap W_\infty$ - this works after all, by the construction in VII, p. 170-171 (of Cartan-Serre), starting from the properties of $Y \longrightarrow \underline{\mathrm{Ch}}(Y) \longrightarrow Y \times Y$ and nothing more. As $\bar{f}i = f \in W$ and $i \in W$, we deduce $\bar{f} \in W$. As $i \in W_\infty$, we conclude $f \in W_\infty \iff \bar{f} \in W_\infty$. Thus to prove that f is in W_∞, we are reduced to the case where f is a W-fibration and a W_∞-fibration. But then proposition 4 shows us that the fibers of f are W-aspheric ([21]), and it suffices to see that they are W_∞-aspheric (proposition 3 ([22]) applied to W_∞). Thus we are reduced to the case where $Y = e$, i.e. to proving that if X in Cat is W-aspheric, it is W_∞-aspheric.

[16] In fact, we prove that $\mathcal{C}(c, u, v)$ is *contractible*, hence it is W_ω-aspheric.

[17] For W(2), it suffices in the form W(2 bis).

[18] See on this subject the corollary on page 22.

[19] Once again, in these arguments it pays to work with W_ω.

[20] in fact, $\bar{f}$ is both W_∞-perfect (i.e. W_∞-smooth and proper) and W-perfect (in fact, W_ω-perfect).

[21] we have used here only this property of the fibers, as a consequence of W(4). (We use W(4) in its weakest form W(4b″), cf. pages 13, 14).

[22] we use proposition 3 only in the case $Y = S$, for which W(2 bis) suffices.

Let $x_0 \in \mathrm{Ob}\, X$, let us consider as before

$$E = \underline{\mathrm{Ch}}(X, x_0, -) \xrightarrow{p} X,$$

we know that E is $W \cap W_\infty$-aspheric, and that it is a fibration for W and for W_∞. As X is W-aspheric (like E), $p \in W$, hence again by proposition 4 ([23]), the fibers of p are W-aspheric, in particular (by W(5)) 0-connected. We have therefore proved the p. 10

Lemma. ([24]) *Let* (X, x_0) *be in* $\mathrm{Cat}_\bullet$. *If* X *is* W*-aspheric, then* $\Omega(X, x_0) \overset{\mathrm{def}}{=} \underline{\mathrm{Ch}}(X; x_0, x_0)$ *is too* (assuming axioms W(1,2,3,4) on W). *Thus (by iteration) the* $\Omega^n(X, x_0)$ *are* W*-aspheric* ($\forall\, n \in \mathbf{N}$). [page 12]

Corollary. *If moreover* W *satisfies the axiom* W(5) *of connectedness, then* X *is* W_∞*-aspheric.*

Indeed, "we know" that this means that all the $\Omega^n(X, x_0)$ are 0-connected. Now the connectedness axiom W(5) ensures that W-aspheric implies 0-connected.

But the "we know" is a tiny bit flimsy ([25]). To justify it, it would be necessary, either that I make the link with classical 1/2-simplicial theory, and prove that the Ω^n which I construct are "the right ones", i.e. that their nerve does indeed give back (up to isomorphism in Hot) the classical Ω^n (??). In particular, that the $\pi_0(\Omega^n(X, x_0))$ is indeed "everybody's" $\pi_n(X, x_0)$. Or else that I prove directly that the 0-connectedness of the $\Omega^n(X, x_0)$ is equivalent to the W_∞-asphericity of X, defined cohomologically in the manner of Artin-Mazur. So this is a piece of work that I shall absolutely have to do. But it is now no more than a simple piece of routine work, and I can state right now the

Theorem 1. ([26],[27],[28]) *Let* $W \subseteq \mathrm{Fl}(\mathrm{Cat})$ *satisfy the conditions* W(1,2,3,4,5) (axiom of saturation, of localization, of the final object, of fibration, of connectedness). *Then*

$$W \subseteq W_\infty.$$

Corollary. *We can characterize* W_∞ *as the largest subset of* $\mathrm{Fl}(\mathrm{Cat})$ *satisfying the conditions* W(1) *to* W(5) *above.* [page 13]

Of course, one would like a more interesting axiom 6, which joined to the preceding ones ensures that $W = W_\infty$ ([29]). Or do these properties already ensure that $W = W_\infty$? To see this, perhaps I would have to know more about the structure of W_∞-aspheric objects? If this were false, I am thinking of an axiom of "homotopy descent", which would imply that a W_∞-aspheric object is also W-aspheric.

I note right now that if some W_i ($i \in I$) satisfy axioms W(1) to W(5), the same is true of $\bigcap_i W_i$ ([30]). Thus there is a smallest set of arrows W_ω satisfying these conditions. If it is not equal to W_∞, i.e. if axioms W(1) to W(5) do not already characterize W_∞, it would be a really very interesting localizer. p. 11

I realize that I have used only a rather weak form of axiom W(4) - essentially, the validity of proposition 4. I am going to summarize different variants of the fibration axiom.

W(4)

[23] in fact, condition W(4b″) on W.

[24] Uses W(1, 2 bis, 3, 4b″) on W.

[25] This is proved quite simply in the corollary to "theorem 1", page 190.

[26] It even suffices to have W(4) in a weakened form, with "fibrant" functors in an "absolute" sense. Cf. the NB p. 10 - and above all, cf. pages 13, 14.

[27] NB Instead of W(5), it suffices to assume $W \neq \mathrm{Fl}(\mathrm{Cat})$, cf. th. 2, p. 20.

[28] The useful hypotheses on W are in the end W(1), W(2 bis), W(3), W(4b″), and $W \neq \mathrm{Fl}(\mathrm{Cat})$.

[29] Cf. axiom W(6) on the next page.

[30] This is true for any set of axioms in the nature of stability axioms.

a$_1$) If $f : X \longrightarrow Y$ in Cat is a left W-fibration, it is an ultrastrong W-fibration.

a$_2$) If $f : X \longrightarrow Y$ is a right W-fibration, it is an ultrastrong W-fibration.

a$'$) If $f : X \longrightarrow Y$ is W-perfect, i.e. both W-proper and W-smooth ([31]), then f is an ultrastrong W-fibration.

a$''$) If $f : X \longrightarrow Y$ is both homotopically proper and homotopically smooth (cf. NB page 10), f is an ultrastrong W-fibration. (It is better to say: If f is W_ω-perfect, it is an ultrastrong W-fibration.)

b) Let $f : X \longrightarrow Y$ be an arrow in Cat$/S$, with X and Y W-fibered over S. If $f \in W$, then $\forall\, s \in S$, $f_s : X_s \longrightarrow Y_s$ is W.

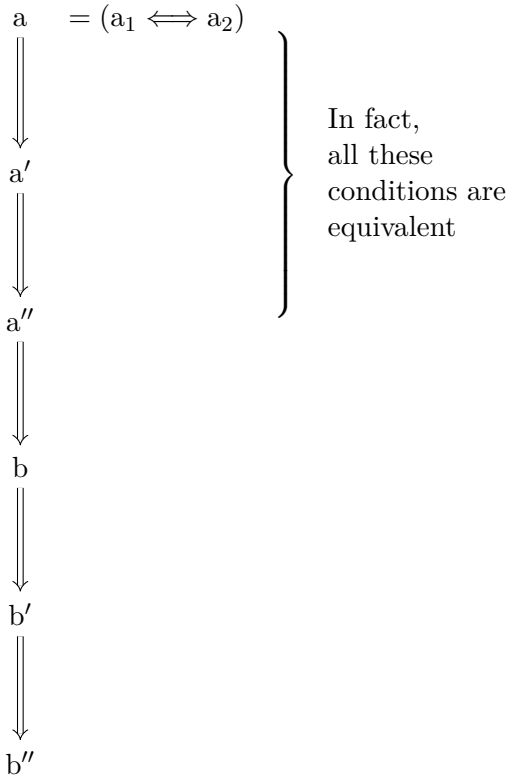
b$'$) Let $f : X \longrightarrow Y$ be an arrow in Cat$/S$ such that $X \longrightarrow S$ and $Y \longrightarrow S$ are W_ω-perfect. If $f \in W$, then $\forall\, s \in S$, $f_s : X_s \longrightarrow Y_s$ is in W.

[page 14]

b$''$) Let $W_\omega \subseteq W \cap W_\infty$ be the smallest subset of Fl(Cat) satisfying conditions W(1,2,3) ([32]) (axioms of saturation, of localization, of the final object). Let $f : X \longrightarrow Y$ be an arrow in Cat which is W_ω-perfect. If $f \in W$, then the fibers of f are W-aspheric .

Among these properties of W, we have the tautological implications (except a$''$) $\Longrightarrow$ b), which is essentially proposition 4)

p. 12



([33]).

Well, in the lemma on page 12, it sufficed to assume that W satisfies the weakest version b$''$) of the axiom of fibrations (seven versions!). Indeed, the fibrations $\bar{X} \longrightarrow Y$

[31] which implies that it is a left W-fibration *and* a right one.

[32] It is better to take W(2 bis) instead of W(2), to find W_ω as small as possible!

[33] I return to the various variants of W(4) below, p. 53-56.

and $E \longrightarrow X$ used in the proof of the lemma are W_ω-perfect, and it sufficed to know that if they are in W, then the fibers are W-aspheric.

But I do not think that even the strongest version W(4a) of the axiom of fibrations, joined to the three preceding axioms and to W(5) (connectedness), implies $W = W_\infty$. It would at least be necessary to add to it the axiom

W(6) (Axiom of filtering limits.) Let $X = \varinjlim_i X_i$ be a filtering direct limit in Cat. If the X_i are W-aspheric, X too.

This should be equivalent to the property, apparently stronger, in terms of arrows of W:

Proposition 5. *Suppose we have* W(6) (in addition to W(1) to W(4) ([34])), *and let* [page 15]

$$(f_i)_{i\in I} : (X_i)_{i\in I} \longrightarrow (Y_i)_{i\in I}$$

a homomorphism of direct systems in Cat*, indexed by a filtering ordered set, let* $X = \varinjlim X_i$, $Y = \varinjlim Y_i$, $f = \varinjlim f_i$, $f : X \longrightarrow Y$. *If* f_i *is in* W *for every* i*, then* $f \in W$. p. 13

We factor each f_i as

$$X_i \xrightarrow{\alpha_i} \bar{X}_i \xrightarrow{\bar{f}_i} Y_i, \qquad \alpha_i \in W_\omega,\ \bar{f}_i \in \operatorname{Parf} W_\omega$$

([35]) following the patented system. The construction being functorial, the $\bar{X}_i$ too form a direct system. Moreover, as the functor $S \longrightarrow \underline{\mathrm{Ch}}(S)$ which gives rise to the canonical factorization of morphisms, commutes with filtering $\varinjlim$ (immediate), it follows that, if we put

$$\bar{X} = \varinjlim_i \bar{X}_i$$

and look at the arrows

$$X \xrightarrow{\alpha} \bar{X} \xrightarrow{\bar{f}} Y, \qquad \alpha = \varinjlim \alpha_i,\ \bar{f} = \varinjlim \bar{f}_i,$$

with $\bar{f}\alpha = f$, this factorization of f is indeed its canonical factorization. Thus $\alpha \in W$, $\bar{f} \in \operatorname{Parf} W_\omega$. This having been seen, by virtue of prop. 3, $f \in W$, i.e. $\bar{f} \in W$, is true if the fibers of $\bar{f}$ are W-aspheric. But the fibers of $\bar{f}$ are interpreted as filtering direct limits of fibers of the $\bar{f}_i$, which are W-aspheric by condition W(4b″) on fibrations. Thus axiom W(6) gives what is needed.

Conjecture (bold? foolish?). *If* $W \subseteq \mathrm{Fl}(\mathrm{Cat})$ *satisfies the axioms* W(1) *to* W(6) *(with* W(4) *in its weakest form* W(4b″) ([36])*), then* $W = W_\infty$.

But it is quite possible that one must strengthen W(1) (saturation) a little, by requiring that every retract of an $f \in W$ be in W. [page 16]

Remark. Let us consider, for $n \in \mathbf{N}$, the set $W_n \subseteq \mathrm{Fl}(\mathrm{Cat})$ of "n-homotopy equivalences" in Cat:

$f \in W_n \iff \pi_0(f)$ bijective, and $\pi_i(f,x) : \pi_i(X,x) \longrightarrow \pi_i(Y,f(x))$ bijective for $0 < i \leq n$.

I thought that this satisfied all that one can ask of a localizer, without having taken the trouble to write it down carefully. But this cannot satisfy W(1) through W(4b″), since we do not have $W_n \subseteq W_\infty$!

[34] W(4) in the weakest form W(4b″) .

[35] W_ω introduced in condition W(4b″), page 14.

[36] and perhaps W(2) in the weaker form W(2 bis).

W(1) is trivial from the definition (and we have without tears even the *strong saturation* of W_n, i.e. if f in Cat is isomorphism in $\mathrm{Hot}_{W_n} = (\mathrm{Cat})W_n^{-1}$, then $f \in W_n$). I think I remember having checked W(2), by the same method as for W_∞, using a cohomological criterion in the manner of ARTIN-MAZUR, to express $f \in W_n$, and using the Leray spectral sequences for $X \longrightarrow S$ and $Y \longrightarrow S$. Condition W(3) is trivially satisfied since $W_n \supseteq W_\infty$ and it is for W_∞. So it gets botched where I

least expected: the fibration axiom. Let us take the weak form: Let $f : X \longrightarrow Y$ be a W_ω-fibration and $f \in W_i$, are the fibers X_s W_i-aspheric ([37])? We may suppose X and Y 0-connected, let $x \in X_s$, the homotopy exact sequence gives p. 14

$$\pi_{i+1}(X) \xrightarrow[\text{if } i\leq n-1]{\text{iso}} \pi_{i+1}(Y) \longrightarrow \pi_i(X_s) \longrightarrow \pi_i(X) \xrightarrow[\text{if } i\leq n]{\text{iso}} \pi_i(Y),$$

so

$$\pi_i(X_s) = 1 \qquad \text{if } 0 \leq i \leq n-1,$$

but for $i = n$ we find [page 17]

$$\pi_n(X_s) = \begin{cases} \mathrm{Coker}\,(\pi_{n+1}(X) \longrightarrow \pi_{n+1}(Y))\,, & n \geq 1, \\ \pi_1(Y)/\,\mathrm{im}(\pi_1(X) \longrightarrow \pi_1(Y))\,, & n = 0 \quad \text{(pointed set of cosets).} \end{cases},$$

there is no reason for this to be zero, as it should if $X_s \longrightarrow e$ were in W_n.

It is true that sometimes one does strengthen the condition for W_n, let us say one defines $W'_n \subseteq W_n$ by

$$f \in W'_n \iff \begin{cases} \text{a)} & \pi_0(f) \text{ bijective,} \\ \text{b)} & \pi_i(f,x) : \pi_i(X,x) \longrightarrow \pi_i(Y, f(x)) \text{ bijective for } 0 < i \leq n \text{ and } x \in X, \\ \text{c)} & \pi_{n+1}(f,x) : \pi_{n+1}(X,x) \longrightarrow \pi_{n+1}(Y, f(x)) \textit{ surjective,} \end{cases}$$

that is to say also

$$f \in W'_n \iff \begin{cases} \text{the "homotopy fibers" } Z \text{ of } X \longrightarrow Y \text{ (i.e. the fibers of } \bar{X} \xrightarrow{\bar{f}} Y) \\ \text{are } n\text{-connected, i.e. 0-connected and } \pi_i(Z_y, z) = 0 \text{ for } 1 \leq i \leq n. \end{cases}$$

Then X in Cat will be W'_n-aspheric if and only if it is W_n-aspheric, i.e. n-connected. So by form 2 of the definition of W'_n, axiom W(4b″) is satisfied. It must go wrong elsewhere! Perhaps saturation already.

If gf and g are in W'_i, the same is not necessarily true of f,

for if $\bar{g}\bar{f}$ (and hence $\bar{g}$) is surjective, the same is not necessarily true for $\bar{f}$! So W'_n does not deserve the name of "localizer".

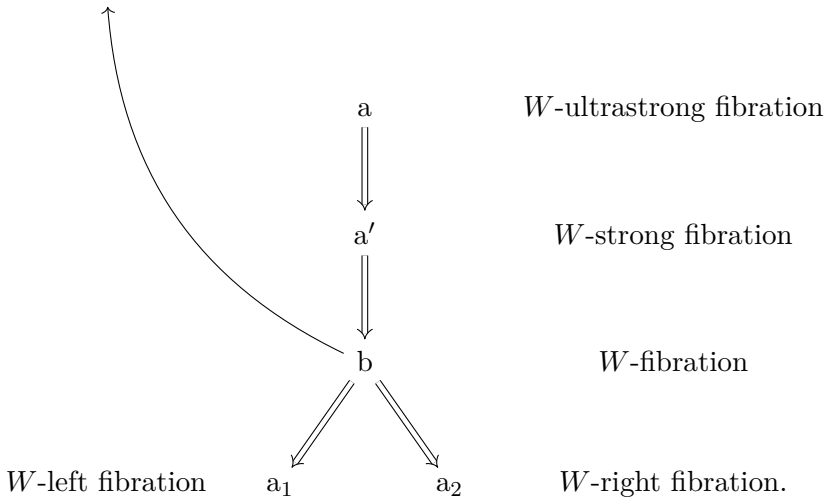
On the other hand, W_n seems excellent to me, if one does not ask the impossible of it - to satisfy a "fibration axiom" which does seem (more or less) to characterize W_∞ alone among the "good" localizers. In any case the connectedness axiom W(5) is satisfied, and the filtering limits axiom W(6) as well. Moreover W(1) (and even strong saturation), W(2), W(3) are satisfied by every $W_{\mathbf{D}}$ associated with a derivator - which confirms their importance once again. The axioms W(4) (the "draconian" fibration axiom, ± special to W_∞), W(5) (connectedness) and W(6) (limits) appear to me to be of a more special nature, I would not think of imposing any of them for the terminological use of "basic localizer". p. 15 [page 18] p. 16

[37] $\omega > \infty$, $W_\omega \subset W_\infty \subset W_n$

II Perplexities about the axioms and the terminology.

In I and most of the time, in working with derivators, I have assumed $\boxed{W_{\mathbf{D}} \supseteq W_\infty}$, a hypothesis which seemed essential to me (wrongly ([38]) or rightly). So we see that if $W_{\mathbf{D}}$ satisfies the fibration axiom W(4), even in its weakest form W(4b″), it is *equal* to W_∞. This does show that this axiom must be seen as practically a *characterization* of W_∞ (among the basic localizers which contain W_∞ – and these localizers seem to me the most important, and perhaps the only really useful ones).

The theory of **D**-fibrations in I, p. 111-126, was done in the spirit just mentioned, assuming $W_{\mathbf{D}} \supseteq W_\infty$. The notion of **D**-fibrations on which I had finally settled was consequently different from the notions considered here in a different spirit (page 4). It amounts to testing condition a) (almost always too strong, except for $W = W_\infty$), characterizing ultrastrong W-fibrations, by $Y'' \longrightarrow Y'$, where Y'', Y' are both W_∞-aspheric (which is more demanding than W-aspheric). It is only when Y'' and Y' are W_∞-aspheric (instead of W-aspheric), that one asks that $X'' \longrightarrow X'$ be in W. We thus find a less demanding notion than that of strong W-fibration. If we denote this condition here by b), we find (provided that $W_\infty \subseteq W$), with the notations of page 4 ([39]) [page 19]



a_1), a_2) were also called cohomological **D**-fibrations resp. homological **D**-fibrations ([40]). When **D** satisfies Der 3a′ (I, p. 111) on locally constant **D**-coefficients on the X that are W_∞-aspheric, then the notions b), a_1), a_2) are equivalent. (I presume this is the case for all the **D** and all the W that are really interesting.) On the other hand, I have the impression that the notion of strong W-fibration, or even ultrastrong, is hardly usable except in the case $W = W_\infty$, where it coincides with the preceding ones. That is to say that one can probably forget the terminology, and remember on the other hand that in the case $W = W_\infty$, the W-fibrations have properties of extreme strength, expressed by the two properties singled out in W(4a) and W(4b). p. 17

In the theory of HOT_W, influenced by the general reflection on derivators and axiom Der 3b, I have worked under the hypothesis $W \supseteq W_\infty$, but using a fibration axiom L5 = W(4b) which, it turns out, ultimately implies $W = W_\infty$. I have the impression that so far (properties of $f_!$ for HOT) I have not used $W \supseteq W_\infty$, so it may be useful, since "it costs no more" (and forces one to be more attentive to what one uses), to drop $W \supseteq W_\infty$ [page 20]

[38] *wrongly* it now turns out.

[39] Cf. the right notion of W-fibration in a context where one does not assume $W_\infty \subseteq W$, in th. 3 and def. 3, p. 51, 52.

[40] No, this is a still stronger notion, by dint of locally constant coefficients, cf. I, p. 121.

(implying $W = W_\infty$), and on the other hand to put on W axioms as strong (notably of fibration) as the properties of W_∞, "without depriving oneself of anything". So I shall assume W(1), W(2), W(3), W(4a) (the strongest form of the fibration axiom, cf. pages 13, 14) - that is all that has been useful up to now. If one adds W(5) (connectedness axiom) to this it implies $W \subseteq W_\infty$. But for the moment I do not think I have a need for it.

But meditating on the question, I believe I have found the

Theorem 2 (!) ([41]): *Let* $W \subseteq \mathrm{Fl(Cat)}$ *satisfy* W(1,2,3,4b″). *If* W *does not satisfy* W(5)*, then* $W = \mathrm{Fl(Cat)}$.

Proof. Let $f : X \longrightarrow Y$ be in Cat, such that $\pi_0(f)$ is not bijective. Let I be the discrete category having as set of objects the set $\pi_0(Y)$, so we have a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & I, & \end{array}$$

on the other hand a functor from a category into a discrete category is always W_ω-perfect, so p, q are W-fibrations. Since $f \in W$, it follows that $f_i \in W$ for all $i \in I$ ([42]). [page 21]

If $\pi_0(f)$ were not surjective, taking i which is not in the image, we would have $X_i = \emptyset \longrightarrow Y_i \neq \emptyset$, which would be in W. So let

$$f : X = \emptyset \longrightarrow Y$$

be in W. Then $\forall$ Z in Cat, $f \times \mathrm{id}_Z : \underbrace{X \times Z}_{=\emptyset} \longrightarrow Y \times Z$ is in W (since Z is W-fibered over e), so $\emptyset \longrightarrow Y \times Z$ is in W. So too the projection $Y \times Z \longrightarrow Y$. Since this p. 18

$$\begin{array}{ccc} Y \times Z & \longrightarrow & Y. \\ & \nwarrow \quad \nearrow & \\ & \emptyset & \end{array}$$

projection is a W-fibration, it follows by W(4b″) that its fibers are W-aspheric. Since $Y \neq \emptyset$, there is such a fiber, it is $\simeq Z$. So Z is W-aspheric. So we find that *every* category Z in Cat is W-aspheric. So every arrow $Z \longrightarrow Z'$ in Cat is in W, q.e.d.

Suppose now $\pi_0(f)$ surjective, but not injective. There would be a fiber of $\pi_0(X) \longrightarrow \pi_0(Y) = I$ which would be of cardinality > 1. Taking the corresponding f_i, we find $X_i \longrightarrow Y_i$ with Y_i 0-connected and $\mathrm{card}\,\pi_0(X_i) \geq 2$. So we are reduced to the case

$$f : X \longrightarrow Y \qquad \begin{cases} f \in W \\ Y \text{ 0-connected} \\ \mathrm{card}\,\pi_0(X) \geq 2. \end{cases}$$

Let us consider the canonical factorization

$$X \xrightarrow{i_f} \bar{X} \xrightarrow{\bar{f}} Y,$$

with $i_f \in W$, so $\bar{f} \in W$. So by W(4b″) the fibers of $\bar{f}$ are W-aspheric. But we see at once, on decomposing $\bar{X}$ according to the connected components X_i of X, that these fibers have a π_0 of cardinality $\geq \mathrm{card}\,\pi_0(X) \geq 2$. So we are reduced to the case where $Y = e$, i.e.

$$X \text{ } W\text{-aspheric}, \qquad \mathrm{card}\,\pi_0(X) \geq 2.$$

Let us consider $X \times X$, which is also W-aspheric, and the diagonal map

[41] Cf. for a stronger result XV 9.12, p. 133 bis. **NB** The result is more general, but not really stronger, since it does not give $W = \mathrm{Fl(Cat)}$. [page 22]

[42] We use W(4) in a form stronger than W(4b″)!

$$d : X \longrightarrow X \times X = Y,$$

which is therefore in W. But $\pi_0(d)$ is identified with the diagonal map $\pi_0(X) \longrightarrow \pi_0(X) \times \pi_0(X)$, which is not surjective since $\operatorname{card} \pi_0(X) \geq 2$. So for the first part, $W = \mathrm{Fl(Cat)}$.

So in more positive form:

Corollary. *Let* $W \subseteq \mathrm{Fl(Cat)}$ *satisfy* W(1,2,3,4b″)*, and be such that* $W \neq \mathrm{Fl(Cat)}$*. Then* W *satisfies the connectedness axiom* W(5)*, and consequently (th. 1, page 12)* $W \subseteq W_\infty$.

p. 19

III Remarkable classes of functors related to W.

Finally, I will call *"basic localizer"* a subset W of Fl(Cat) satisfying the following three conditions:

(i) Saturation W(1) (cf. reminder p. 1).

(ii) Final element axiom W(3) (page 3).

(iii) The weakened variant W(2 bis) of the localization axiom W(2) given in the corollary on page 1 (formerly it was L4′ in Pursuing Stacks).

When I need more from W, I will say so explicitly. These three axioms have sufficed for practically everything I have done with W in Pursuing Stacks, the only exception being a single use (if I recall), somewhere, of a weakened variant of W(2) (for the *cartesian* S-functors between *fibered* S-categories - if on every fiber it is in W, it is itself in W).

We therefore suppose henceforth given a basic localizer

$$W \subseteq \mathrm{Fl(Cat)}.$$

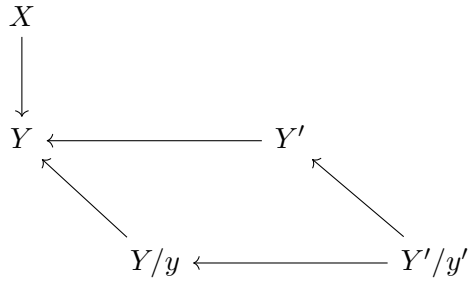
Proposition 1. *Let* $f : X \longrightarrow Y$ *be in* Cat*. Equivalent conditions:*

a) For every local isomorphism $Y' \longrightarrow Y$*,* $f' : X' \longrightarrow Y'$ *is* $\in W$*.*

b) Same condition, in the particular case $Y' = Y/y \longrightarrow Y$*, so* $\forall\ y \in Y$*,* $X/y \simeq X \times_Y (Y/y)$ *is* W*-aspheric.*

Definition 1. Under these equivalent conditions a), b) of proposition 1, we say that f is W*-aspheric.* We say f is W*-coaspheric* if f° is W-aspheric, i.e. if the $y\backslash X$ are W-aspheric (for $y \in \mathrm{Ob}\, Y$). [page 23]

Proof. It suffices to prove b) $\Longrightarrow$ a). By definition of a local isomorphism, we see that condition b) is stable under base change $Y' \longrightarrow Y$, with Y' a local isomorphism (since the $Y'/y' \xrightarrow{\sim} Y/y$, hence $X'/y' \xrightarrow{\sim} X/y$). It therefore suffices to see that if f satisfies b), then $f \in W$. But this is axiom W(2 bis).



p. 20

Remark. If f has a right adjoint g, then for every $y \in Y$, $X/y \simeq X/g(y)$, so X/y has a final object and is therefore W-aspheric. Thus if f has a right (left) adjoint, then f is W-aspheric (W-coaspheric).

But even if f has no right adjoint, there is still a generalized adjoint

$$g : Y \longrightarrow X^\wedge$$

and we have

$$X/y \simeq X/g(y) \qquad \text{(for the fully faithful inclusion } X \hookrightarrow X^\wedge\text{)},$$

so $\mathrm{Top}(X/y)$ is identified with the induced topos $\mathrm{Top}(X)/g(y)$. The condition f W-aspheric therefore means that the sheaves $g(y)$ in $\mathrm{Top}(X)$ are W-aspheric. For many other equivalent formulations, I refer to Pursuing Stacks. [page 24]

Proposition 2. *Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be in* Cat. *If f and g are W-aspheric (W-coaspheric), likewise gf. If gf and f are W-aspheric (W-coaspheric), g too.*

Recall. A morphism $p : S' \longrightarrow S$ in Cat is a *local isomorphism* if $\forall\ s' \in \mathrm{Ob}\, S'$, the induced morphism

$$p_{s'} : S'/s' \xrightarrow{\sim} S/s \qquad (s = p(s'))$$

is an isomorphism: p induces an isomorphism on the "localized categories" of S' and S at s' and $s = p(s')$ respectively. It is immediate that this notion is stable under composition, that if gf and g are local isomorphisms, then f is one too, that every isomorphism is a local isomorphism. Moreover, local isomorphisms are *stable under base change,* which follows from the fact (trivial and important) that if we have a cartesian square in Cat

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{p} & S', \end{array}$$

and if $x' \in \mathrm{Ob}\, X'$ has images x, s', s in X, S', S, then the diagram

$$\begin{array}{ccc} X/x & \xleftarrow{q/x'} & X'/x' \\ {\scriptstyle f/x}\downarrow & & \downarrow{\scriptstyle f'/x'} \\ S/s & \xleftarrow{p/s'} & S'/s' \end{array}$$

is also cartesian. (So if p/s' is isomorphism, q/x' too.)

Let us return to proposition 2. The stability of the notion of aspheric morphism under composition follows at once from form a) of the condition. Likewise for the condition that if gf and f are W-aspheric, g too - we must show that if $Z' \longrightarrow Z$ is local isomorphism, then g' is in W, now the hypothesis implies that $g'f'$ and f' are. p. 21 [page 25]

$$\begin{array}{ccc} X & \xleftarrow{r} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{q} & Y' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g'} \\ Z & \xleftarrow{p} & Z' \end{array}$$

Corollary 1. *The inverse image of an $f : X \longrightarrow Y$ which is W-aspheric (resp. W-coaspheric) under a local isomorphism (resp. under a local co-isomorphism) is W-aspheric (W-coaspheric).*

$$\begin{array}{ccccc} X & \longleftarrow & X' & \longleftarrow & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y & \underset{\text{local iso}}{\longleftarrow} & Y' & \underset{\text{local iso}}{\longleftarrow} & Y'' \end{array}$$

Follows formally from the definition by a) and from the fact that a composite of two local isomorphisms (resp. local co-isomorphisms) is likewise.

NB Of course, the notion of W-aspheric functor is not stable under arbitrary base change. For example, the inclusion $X = \{y_0\} \longrightarrow Y$ is W-aspheric if and only if the discrete categories $\mathrm{Hom}(y_0, y)$ are W-aspheric, i.e. (provided that we have W(5), the connectedness axiom) if and only if the $\mathrm{Hom}(y_0, y)$ have cardinality 1, i.e. y_0 *initial object* of Y. But if $Y \neq \{y_0\}$, taking y in Y, $y \neq y_0$, and setting $Y' = \{y\} \longrightarrow Y$, we have $X' = X \times_Y Y' = \emptyset$ and $\emptyset \longrightarrow e$ is not aspheric!

On the other hand, we expect the W-aspheric morphisms to be stable under W-*smooth* [page 26] base changes (which generalize local isomorphisms), and the W-coaspheric morphisms under W-proper base changes. Cf. below.

Corollary 2. *Let $f : X \longrightarrow Y$ in* Cat *be an* equivalence *of categories. Then f is W-aspheric and W-coaspheric.*

Indeed, f has a left adjoint and a right adjoint, and we use the remark on page 23.

Proposition 3. *Let $f : X \longrightarrow Y$ be in* Cat. *Equivalent conditions:*

p. 22

a) For every $y \in \mathrm{Ob}\, Y$, the inclusion functor

$$X_y \longrightarrow X/y \qquad (*)$$

is W-coaspheric.

b) For every $x_0 \in X$, the induced morphism for the colocalized categories

$$x_0\backslash X \longrightarrow y_0\backslash Y \qquad \textit{(where } y_0 = f(x_0)\textit{)}$$

has W-aspheric fibers.

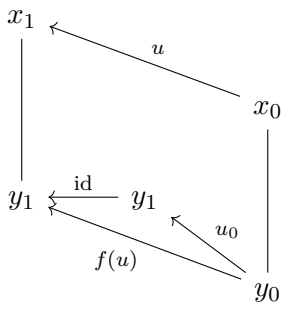
c) For every morphism $\mathbf{\Delta}^1 = Y' \longrightarrow Y$, let us consider $X' = X \times_Y Y'$ over $\mathbf{\Delta}^1$, the inclusion $X'_1 \longrightarrow X'$ of the closed fiber of X' (fiber over the closed point of Y') is W-coaspheric.

d) For every arrow $u_0 : y_0 \longrightarrow y_1$ in Y, and every x_0 in X_{y_0}, the category $X(x_0; u_0)$ formed of the arrows $u' : x_0 \longrightarrow x$ in X with source x_0, and which lift u_0 , is W-aspheric.

Proof. To spell out condition a), we must spell out the category $\xi_0\backslash(X_y)$, for an arbitrary object $\xi_0 = (x_0, u_0)$ of X/y, $x_0 \in \mathrm{Ob}\, X$, [page 27]

$$u_0 : y_0 \longrightarrow y_1 = y \quad \text{in } Y, \text{ with } y_0 = f(x_0).$$

This category $\xi_0\backslash X_{y_1}$ is then spelled out as the category of the x_1 in X_{y_1}, *equipped* with an arrow from ξ_0 to the image $\xi_1 = (x_1, \mathrm{id}_{f(x_1)} = \mathrm{id}_{y_1})$ of x_1 in X/y_1 in ξ_0 , i.e. with an arrow u from x_0 to x_1, such that $f(u)$ gives rise to the commutative triangle in Y, i.e. $f(u) = u_0$.



So this is the category denoted $X(x_0; u_0)$ in d). We see at once that it is also isomorphic to the fiber of $x_0\backslash X \longrightarrow y_0\backslash Y$ at the element u_0 of $y_0\backslash Y$, referred to in b). Hence the equivalence of a) b) d). On the other hand, giving u_0 in Y is equivalent to giving a functor $\mathbf{\Delta}^1 \longrightarrow Y$, as in c), and to express then that $\underbrace{X'_1}_{\simeq X_{y_1}} \longrightarrow X'$ is coaspheric, it suffices to express that the $x'\backslash X'_1$ ($x' \in \operatorname{Ob} X'$) are W-aspheric. Now this is trivial if $x' \in \operatorname{Ob} X'_1$ (initial object), so it suffices to verify it for x' in $X'\backslash X'_1 = X'_0 \simeq X_{y_0}$. Then x' is identified with an element x_0 of X_{y_0}, and we check at once that $x_0\backslash X'_1$ is isomorphic to the category $X(x_0; u_0)$ of d). This completes the proof.

Definition 2. Under the conditions of proposition 3, we say that f is *W-proper.* We say that f is *W-smooth* if f° is W-proper, i.e. if the inclusion morphisms $X_y \longrightarrow y\backslash X$ are W-aspheric etc. (conditions dual to those of proposition 3). p. 23 [page 28]

Remark. Condition a) of proposition 3 is satisfied in particular if the inclusion $(*)$ has a left adjoint (cf. remark on page 23). This means that the categories $X(x_0; u)$ have an initial object. This also means that the category X over Y is Cat-cofibered over Y in the weak sense, i.e. that there exists, for $u_0 : y_0 \longrightarrow y_1$ an arrow in Y, and x_0 in X_{y_0}, an object $u_{0*}(x_0)$ in X_{y_1} which represents the covariant functor $X_{y_1} \longrightarrow (\mathrm{Ens})$, $x_1 \longmapsto \operatorname{Hom}_u(x_0, x_1)$. Thus, if X is Cat-cofibered in the weak sense over Y, and a fortiori if it is Cat-cofibered, then $f : X \longrightarrow Y$ is W-proper. Dually, if f is weakly Cat-fibered, f is smooth.

Proposition 4. ([43]) *The notion of W-proper morphism is stable under composition, under base change. Local co-isomorphisms are proper (trivial from b)), a fortiori isomorphisms are proper, closed immersions are proper.*

Dual statement for smooth morphisms: stability under base change, under composition, local isomorphisms are smooth, in particular open immersions.

Stability under base change is immediate from form c). Stability under composition less trivial: [page 29]

$$\begin{array}{cccccc}
x & X & \longleftarrow & x\backslash X & \longleftarrow & \bar{X} \\
 & \downarrow{\scriptstyle f} & & \downarrow & & \downarrow \\
y & Y & \longleftarrow & y\backslash Y & \longleftarrow & \bar{Y} \\
 & \downarrow{\scriptstyle g} & & \downarrow & & \downarrow \\
z & Z & \longleftarrow & z\backslash Z & \longleftarrow & \{u\}
\end{array}$$

In the diagram with cartesian squares opposite (deduced from $x \in \operatorname{Ob} X$, whence $y = f(x)$, $z = g(y) = gf(x)$, and from $u \in \operatorname{Ob} z\backslash Z$), we must prove that $\bar{X}$ is W-

[43] Say also: If $X \xrightarrow{f} Y \xrightarrow{g} Z$, if gf and f W-smooth, then g is smooth at the points of $\operatorname{Im} f$, ditto for proper.

aspheric. We know that $\bar{Y}$ is (g proper) and that $\bar{X} \longrightarrow \bar{Y}$ has W-aspheric fibers. It therefore suffices to use proposition 5 below, and the immediate remark

Lemma. *If f is proper, the same holds for $x\backslash f : x\backslash X \longrightarrow y\backslash Y$, for every x in X, $y = f(x)$.*

Indeed, if u' is in $x\backslash X$ and v' in $y\backslash Y$, and if x', y' are their images in X, Y respectively, then p. 24

$$u'\backslash(x\backslash X) \simeq x'\backslash X, \qquad v'\backslash(y\backslash Y) \simeq y'\backslash Y,$$

and the arrow between the two is identified with the localization arrow

$$x'\backslash X \longrightarrow y'\backslash Y.$$

Proposition 5. *Let $f : X \longrightarrow Y$ be in* Cat*, suppose f W-proper or W-smooth, and with W-aspheric fibers, then $f \in W$ (and even f is* universally *in W, i.e. it is in W and remains so under every base change).*

As the hypotheses are stable under base change, it suffices to prove $f \in W$. Suppose e.g. X and Y proper over S, let us prove that f is W-*aspheric*, i.e. that the X/y ($y \in \mathrm{Ob}\, Y$) are W-aspheric. But $X_y \longrightarrow X/y$, being W-coaspheric by the properness hypothesis, is in W (by W(2 bis)), and since X_y is W-aspheric by hypothesis, X/y too. But by W(2 bis), f W-aspheric $\Longrightarrow f \in W$. The case f smooth is dual. [page 30]

Remark. Thus, if f is proper, if it has W-aspheric fibers it is "universally W-aspheric", i.e. is and remains W-aspheric under every base change. Conversely, if f is universally in W (not necessarily proper), then in particular its fibers are W-aspheric. But that of course doesn't imply that f is proper, or smooth ([44]).

Corollary. *If W satisfies* W(2) (not just W(2 bis))*, then for an S-morphism $X \longrightarrow Y$ of categories which are both W-proper over S, or both W-smooth, if $f_s : X_s \longrightarrow Y_s$ is in W for every $s \in S$, then $f \in W$.*

Suppose e.g. f proper, let us consider, for $s \in \mathrm{Ob}\, S$ [page 31]

$$\begin{array}{ccc} X_s & \xrightarrow{f_s} & Y_s \\ {\scriptstyle i_s^X}\downarrow & & \downarrow{\scriptstyle i_s^Y} \\ X/s & \xrightarrow{f/s} & Y/s, \end{array}$$

as X, Y proper, the vertical arrows are in W (much less than properness would suffice …), hence $f_s \in W \Longrightarrow f/s \in W$. Thus, "$f$ is in W locally over S", which implies $f \in W$ by W(2).

Proposition 6. *Let* p. 25

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow[p]{} & S', \end{array}$$

be cartesian in Cat*. If f W-aspheric, p W-smooth, then f' is W-aspheric.*

We must prove $X' \times_{S'} S'/s'$ is W-aspheric ($s' \in \mathrm{Ob}\, S'$). Let $s = p(s')$, we have

$$\begin{array}{ccc} S & \xleftarrow{p} & S' \\ & \nwarrow \quad \nearrow & \\ & S/s \xleftarrow[p_0]{} S'/s', & \end{array}$$

[44] But "universally in W" already implies "universally W-aspheric", since the notion of W-asphericity is defined by proposition 1 a).

with p_0 W-smooth (lemma p. 29) and with W-aspheric fibers, hence a universal W-equivalence (prop. 5). As the inverse image of X over S/s is W-aspheric (hypothesis for f), it

that in the cartesian square opposite [page 32]

$$(*) \qquad \begin{array}{ccc} X/s & \xleftarrow{W} & X'/s' \\ \downarrow & & \downarrow \\ S/s & \xleftarrow{W} & S'/s' \end{array}$$

the horizontal arrows are in W, and the first vertical arrow is so by the hypothesis on f, hence the second also, hence X'/s' W-aspheric.

Corollary (generalizes prop. 6, obtained by taking $Y = S$). *Let $p : S' \longrightarrow S$ be W-smooth morphism in* Cat. *Then the base change functor*

$$\mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S'$$

is compatible with the localizers W_S^{g}, $W_{S'}^{\mathrm{g}}$, i.e. transforms $f : X \longrightarrow Y$ which is in W locally *over S, into $f' : X' \longrightarrow Y'$ in W locally over S'.*

This time, we look at the diagram similar to $(*)$

$$\begin{array}{ccccc} X/s & \xleftarrow{W} & X'/s' & & \\ & \searrow^{f/s} & \downarrow & \searrow^{f'/s'} & \\ & Y/s & \xleftarrow{W} & Y'/s' & \\ \downarrow & \swarrow & \downarrow & \swarrow & \\ S/s & \xleftarrow{W} & S'/s', & & \end{array}$$

where the horizontal arrows are in W, hence $f/s \in W$ (hypothesis) implies $f'/s' \in W$.

Dually, if p is W-proper, the inverse image functor is compatible with the localizers p. 26
W_S^{d}, $W_{S'}^{\mathrm{d}}$. [page 33]

If $f : X \longrightarrow S$ is W-proper, we are going to define a canonical functor

$$\begin{array}{rccl} \mathrm{hot}_*^{X/S,W} : S & \longrightarrow & \mathrm{Hot}_W \overset{\mathrm{def}}{=} (\mathrm{Cat})W^{-1} \\ \text{given on objects by} \quad s & \longmapsto & \mathrm{hot}_W(X_s). \end{array}$$

We must define, for an arrow u in S, $u : s \longrightarrow s'$, an arrow

$$\mathrm{hot}_W(X_s) \longrightarrow \mathrm{hot}_W(X_{s'})$$

in Hot_W - arrow which we can denote by $\mathrm{hot}_*^{X/S,W}(u)$. We define it by the diagram

$$\begin{array}{ccc} X_s & \dashrightarrow & X_{s'} \\ \downarrow{\scriptstyle i_s^X} & & \downarrow{\scriptstyle i_{s'}^X} \\ X/s & \xrightarrow{X/u} & X/s', \end{array}$$

where the vertical arrows are W-coaspheric, hence in W. They are therefore isomorphisms in Hot_W, whence the unique dotted arrow in Hot_W, making the square commutative. From this definition it is immediate that

$$u \longmapsto \mathrm{hot}_W(u)$$ [page 34]

is compatible with composition, and units.

This functor is thus defined assuming only that the

$$X_y \longrightarrow X/y$$

are in W, which is distinctly weaker than proper. If we suppose that this hypothesis is satisfied "universally" (i.e. remains satisfied after every base change $S' \longrightarrow S$) ([45]), then we therefore have, for every S' in Cat/S, a functor

$$\mathrm{hot}_*^{X'/S',W} : S' \longrightarrow \mathrm{Hot}_W,$$

and we check at once that it is none other than the composite in

p. 27

$$(*) \qquad \begin{array}{ccc} S' & \xrightarrow{\quad p \quad} & S \\ & {\scriptstyle \mathrm{hot}_*^{X'/S',W}} \searrow \quad \swarrow {\scriptstyle \mathrm{hot}_*^{X/S,W}} & \\ & \mathrm{Hot}_W & \end{array}$$

. We have dual observations if X is smooth over S, more generally if the inclusions

$$X_s \longrightarrow s\backslash X$$

are in W. We then find a contrafunctor

$$\begin{cases} S^\circ & \longrightarrow & \mathrm{Hot}_W \\ s & \longmapsto & \mathrm{hot}_W(X_s) \\ u & \longmapsto & \mathrm{hot}^*_{X/S,W}(u) \text{ or } \mathrm{hot}^*_{X/S}(u) \end{cases}$$

and a compatibility similar to $(*)$, if the hypothesis is satisfied *universally* over S, e.g. f W-smooth. [page 35]

Proposition 7. *Suppose $f : X \longrightarrow S$ both W-proper and W-smooth, more generally that the morphisms*

$$X_s \longrightarrow X/s, \qquad X_s \longrightarrow s\backslash X \qquad (s \in S)$$

are in W, and that this hypothesis remains satisfied after every base change. Then the corresponding functors

$$\begin{array}{rcl} \lambda = \mathrm{hot}_*^{X/S,W} : & S & \longrightarrow \mathrm{Hot}_W \\ \lambda^\circ = \mathrm{hot}^*_{X/S,W} : & S^\circ & \longrightarrow \mathrm{Hot}_W \end{array}$$

transform every arrow u of S into isomorphism of Hot_W, and we have

$$\lambda^\circ(u) = \lambda(u)^{-1} \qquad \textit{in} \quad \mathrm{Hot}_W,$$

i.e. $\lambda(u)$ and $\lambda^\circ(u)$ are inverses of each other.

Proof. The datum of $u : s_0 \longrightarrow s_1$ is equivalent to that of

$$S' = \mathbf{\Delta}^1 \longrightarrow S,$$

whence

$$X' = X \times_S S' \qquad \text{over } S' = \mathbf{\Delta}^1,$$

with the two fibers

p. 28

$$X'_0 \simeq X_{s_0}, \qquad X'_1 \simeq X_{s_1}.$$

[page 36]

The hypothesis ensuring the existence of λ is expressed by the condition that

$$X'_1 \longrightarrow X'/1 \simeq X'$$

is in W, so that the diagram

$$X'_0 \xrightarrow{i_0} X' \xleftarrow[W]{i_1} X'_1$$

defines an arrow $\lambda(u) : X'_0 \longrightarrow X'_1$ in Hot_W, namely $\lambda(u) = i_1^{-1} i_0$. The hypothesis which ensures the existence of λ° is expressed by the symmetric condition $i_0 \in W$, which makes it possible to define the arrow $\lambda^\circ(u) : X'_1 \longrightarrow X'_0$ by $\lambda^\circ(u) = i_0^{-1} i_1$. When the two conditions are satisfied simultaneously, we therefore see that $\lambda(u)$, $\lambda^\circ(u)$ are inverses of each other.

Remarks. To define λ "universally", it was not indispensable to consider the X/s and the inclusions $X_s \longrightarrow X/s$, instead of that it sufficed to look at the base changes $S' = \mathbf{\Delta}^1 \xrightarrow{u} S$, and to make the hypothesis that for every such base change, the inclusion $X'_1 \xrightarrow{i_1} X' = X_{(\mathbf{\Delta}^1, u)}$ is in W. This hypothesis is automatically stable under arbitrary base change. (The W-properness of $f : X \longrightarrow S$ means that $X'_1 \longrightarrow X'$ is not only in W, but even W-coaspheric.) Dually, for the "universal" definition of λ°, it would suffice that for every base change $\mathbf{\Delta}^1 \longrightarrow S$ as above, the inclusion $X'_0 \xrightarrow{i_0} X'$ is in W.

The weakest reasonable notion of W-fibered, seems to me to be the one where we [page 37]
suppose both things at once. The hypotheses we have made up to now, excluding the strong form of W-properness or of W-smoothness, always amount to supposing that the *base change by* $f : X \longrightarrow S$ transforms certain arrows in Cat/S into W : To define λ, these arrows are

$$\text{either } \{s\} \longrightarrow S/s, \qquad \text{or } \{1\} \longrightarrow \mathbf{\Delta}^1 (\longrightarrow S),$$

to define λ°, they are

$$\text{either } \{s\} \longrightarrow s\backslash S, \qquad \text{or } \{0\} \longrightarrow \mathbf{\Delta}^1 (\longrightarrow S),$$

the "either" depending on the definition we have adopted. In these examples, it is always a matter of the inclusion of the final object of a category S', into the said category, or conversely of the initial object. This shows moreover (if we require universality), that the first hypothesis (ensuring λ) is *exactly* that (case of the inclusion of a final object s' - since for such an object, we have $S'/s' \simeq S'$), whereas the second hypothesis requires less, only for the case of $\mathbf{\Delta}^1$. And dually for λ° . But it should be noted that in practice, the most convenient way, and by far,

to verify such properties, is either by the W-properness of f, or by W-smoothness - [page 38]
which reduces to verifying that certain well-specified categories are W-aspheric, without having to worry about base changes and about verifying that the arrows constructed in terms of arbitrary base changes are in W. And the use of W-properness and W-smoothness is all the more congenial, as these notions are stable under composition,

whereas all the derived notions, as well as the notions of fibrations to boot, have an p. 29
unfortunate tendency *not* to be stable under composition.

We could call pseudo-W-proper resp. quasi-W-proper (a little better than pseudo) what we have just brought out - I hear snickering! And if we want both at once - λ and λ° associated as we said? I see the same principle of description with $S'' \longrightarrow S'$ over S, of requiring that $X'' \longrightarrow X'$ be transformed into W. With the following choices which look nice

[page 39]

a$_1$) Y' has a final object, $Y'' = e$. The definition thus comes down to the condition that the

$$X_s \longrightarrow X/s, \qquad X/s' \longrightarrow X/s$$

(for $s \in \mathrm{Ob}\, S$ where $u : s' \longrightarrow s$ in S) be in W, and that it stay that way after every base change $S' \longrightarrow S$.

a$_2$) Y' has an initial object, $Y'' = e$, this is the dual condition by dint of

$$X_s \longrightarrow s\backslash X, \qquad s\backslash X \longrightarrow s'\backslash X.$$

a$_0$) $Y' = \mathbf{\Delta}^1$, $Y'' = e$ ([46]).

b) Y' is contractible, Y'' too (or if preferred, Y' contractible, $Y'' = e$)

thus the condition comes down to this: for every *contractible* Y' over Y (notion independent of W, that is its main advantage) the inclusions of the fibers of $X' = X \times_Y Y'$ over Y' into X' are W.

I feel that even for general W, b) remains in the realm of the verifiable, whereas we go soft if we go to the extreme of generality in this direction:

b′) As b), but assuming only Y' W-aspheric ([47]).

Finally (condition a) of page 4)

c) If $Y'' \longrightarrow Y'$ is W without more, $X'' \longrightarrow X'$ too.

p. 30
[page 40]

The only one of these notions that has the slightest desire to be stable under composition is the strongest c). On the other hand, the notion of a W-perfect arrow (cf. def. 3) is spontaneously stable under composition:

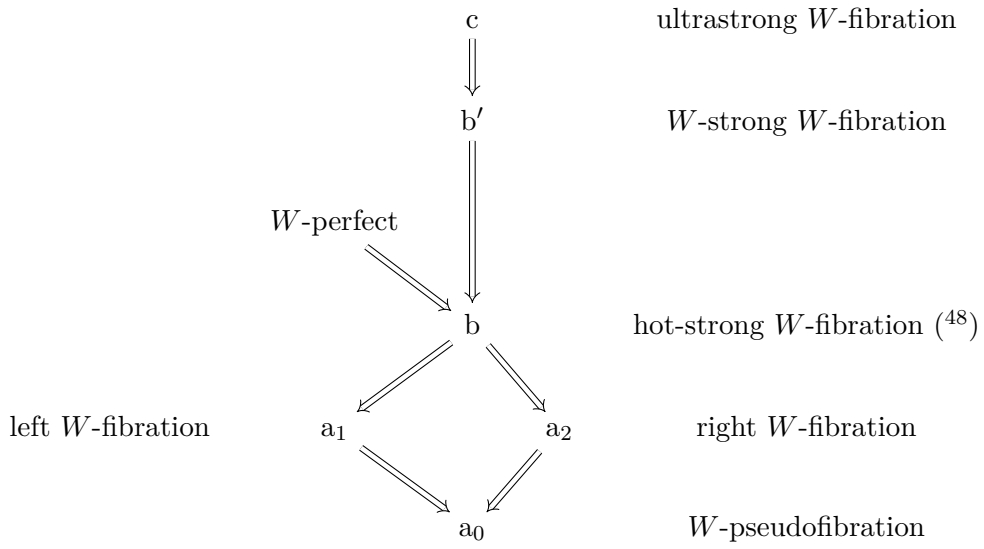
Definition 3. We say that $f : X \longrightarrow S$ is *W-perfect* if it is both W-proper and W-smooth.

In the case $W = W_\infty$, all the variants on the preceding page, with the sole exception of a$_0$), are equivalent. It would be interesting to bring out minimal conditions on W to ensure that this is so. (And the question arises whether in the case of W_∞, a$_0$) suffices to imply the rest - I hardly expect it …)

In the general case, we have the diagram of implications (and even so I have limited

[45] We shall say that f is *W-preproper*.
[46] NB a$_1$), a$_2$) are denoted the same way on p. 4, but a$_0$) was forgotten.
[47] This is essentially a′) of p. 4

myself)



.

Proposition 8. *Let $f : X \longrightarrow S$ be in* Cat. *The following conditions are equivalent.* [page 41]

a) f is a left W-fibration, i.e. condition a_1 *of page 39 is satisfied.*

b) f is W-preproper, and $\forall\ s_0 \longrightarrow s_1$ in S, $X/s_0 \longrightarrow X/s_1$ is in W.

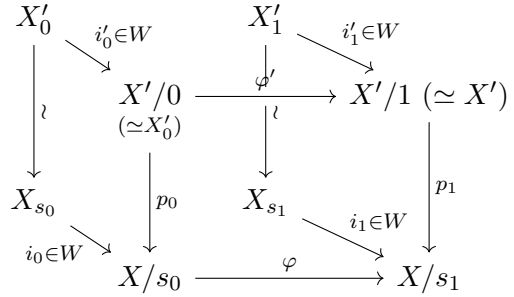
c) f is W-preproper, and a W-pseudofibration, i.e. condition a_0 of page 39 is satisfied.

([49]).

NB These conditions are satisfied if f is at the same time W-presmooth, cf. prop. 7. p. 31

Proof. We have a) $\Longrightarrow$ b) trivially (take $Y'' = Y/s$, $Y'' = Y/s'$).

b) $\Longleftrightarrow$ c) by the reasoning of proposition 7, for in terms of $X' = X_u = X \times_S (\mathbf{\Delta}^1, u)$, condition b) for a given $u : s_0 \longrightarrow s_1$ is interpreted in the commutative diagram



where i_0, i_1, i'_0, i'_1 are in W because of the hypothesis of W-preproperness (stable under base change).

Thus p_0, p_1 are W, thus $\varphi \in W$ if and only if $\varphi' \in W$, now $\varphi' \in W$ is condition a_0. [page 42]

[48] It would seem that this is, in the end, the "right notion" of W-fibration, *equivalent* to a_1 and a_2, which would be equivalent to each other. Cf. prop. 9, p. 48.

[49] **Corollary.** *When f is* Cat*-cofibrant, then these conditions are also equivalent to this: $\forall\ u : s_0 \longrightarrow s_1$ in S, $u_* : X_{s_0} \longrightarrow X_{s_1}$ is in W.*

Let us finally prove that b), c) imply a). Since c) (hence also b)) is stable under base change, it suffices to show this: If S has a final object e, then for every $s \in \mathrm{Ob}\, S$, the inclusion $X_s \xrightarrow{i} X$ is in W. Now we have the commutative diagram

$$\begin{array}{ccc} X_s & \xrightarrow{i_s \in W} & X/s \\ & \searrow^{i} & \downarrow{\scriptstyle u \in W \text{ by b)}} \\ & & X/e = X, \end{array}$$

which indeed shows $i = ui_s \in W$.

Dually, if f is W-presmooth, then condition a_0 (f a W-pseudofibration) is equivalent to a_2 (f a right W-fibration), and also means that the

$$s_1\backslash X \longrightarrow s_0\backslash X$$

are in W. But the notion of "W-fibration" that we land on (if we are not content with the paltry a_0!) is different in the two cases dual to each other. We would like in both cases to find "hot-strong" W-fibrations (condition b), at least by going all out (if necessary) - i.e. for f W-perfect. So it comes down to this.

Question: *Let $X \xrightarrow{f} S$, with f W-perfect, S contractible, $s \in \mathrm{Ob}\, S$. The inclusion $X_s \longrightarrow X$, is it in W?*

We are going to assume only that f is W-presmooth and a W-pseudofibration (cf. prop. 8). We are going to reduce to the case where X is even fibrant over S, by introducing $\Phi_S(X)$ and

p. 32
[page 43]

$$X \xrightarrow{i_X} \Phi_S(X).$$

The trouble is that I have done the theory of HOT_W up to now with the hypothesis (which turns out to be idiotic) that $W \supseteq W_\infty$ - I will therefore be obliged to review quickly under what more reasonable hypotheses on W the familiar results are valid. For example I would like $i_X \in W$ ([50]), so that we have

$$\begin{array}{ccc} X_s & \xrightarrow[\text{since } X \text{ } W\text{-smooth}]{i_s \in W} & \Phi_S(X)_s = s\backslash X \\ {\scriptstyle j_s^X}\downarrow & & \downarrow{\scriptstyle j_s^{\Phi_S(X)}} \\ X & \xrightarrow{i_X \in W} & \Phi_S(X), \end{array}$$

we therefore see that to prove that the inclusion j_s^X is W, it suffices to prove the same thing for $\Phi_S(X)$, which is fibrant (not just W-smooth). Moreover, the pseudofibration hypothesis on X surely passes to $\Phi_S(X)$ (I leave the two verifications for later ([51])). We are thus reduced to the case of

$$X \in \mathrm{Ob}\, \underline{\mathrm{Fib}}_0 S, \qquad S \text{ contractible},$$

to prove that the inclusion

$$X_s \longrightarrow X$$

is in W. I am going to prove this:

Lemma. *Let S be in* Cat, *$X \in \mathrm{Ob}\, \underline{\mathrm{Fib}}_0 S$, and let us consider, for an S' in* Cat,

[page 44]

$$\begin{array}{rcl} f & \longmapsto & f^*(X/S) = X \times_S (S', f) \\ \underline{\mathrm{Hom}}(S', S)^\circ & \longrightarrow & \underline{\mathrm{Fib}}_0 S'. \end{array}$$

[50] in fact, $i_X \in (W_\omega)_S^{\mathrm{d}} \subseteq W_S^{\mathrm{d}} \subseteq W_S$.
[51] O.K.

(This functor is defined thanks to the fact that X is fibrant over S, and can therefore be interpreted, up to S-equivalence, in terms of a functor $S^\circ \longrightarrow$ Cat … ([52])). The arrows of $\underline{\mathrm{Hom}}(S', S)$ are transformed into arrows $\in W^{\mathrm{d}}_{S'}$.

Indeed, in terms of functors

p. 33

$$S'^\circ \longrightarrow \mathrm{Cat},$$

if X is defined by

$$S^\circ \xrightarrow{\mathcal{X}} \mathrm{Cat},$$

then $f^*(X)$ is defined by $\mathcal{X} \circ f$, and the arrow

$$\mathcal{X} \circ g^{\mathrm{op}} \longrightarrow \mathcal{X} \circ f^{\mathrm{op}}$$

defined by $u : f \longrightarrow g$ is none other than $\mathcal{X} * u$. Its value at $s' \in \mathrm{Ob}\, S'$ is obtained by taking

$$u(s') : f(s') \longrightarrow g(s'),$$

whence by $\mathcal{X}$

$$\mathcal{X}(u(s')) : \underbrace{\mathcal{X}(g(s'))}_{(\mathcal{X}\circ g)(s')} \longrightarrow \underbrace{\mathcal{X}(f(s'))}_{(\mathcal{X}\circ f)(s')}$$

([53]). But the hypothesis $X \in \underline{\mathrm{Fib}}_0 S$ means that the $\mathcal{X}(v)$, $v \in \mathrm{Fl}(S)$, are all in W. Thus $\mathcal{X} * u$ is in W fiber by fiber, thus in $W^{\mathrm{d}}_{S'}$.

[page 45]

Corollary 1. *The functor of the lemma, composed with*

$$\underline{\mathrm{Fib}}_0 S' \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S') \overset{\mathrm{def}}{=} (\underline{\mathrm{Fib}}_0 S')(W^{\mathrm{d}}_{S'})^{-1}$$

factors through the fundamental groupoid of $\underline{\mathrm{Hom}}$ *:*

$$\begin{array}{ccc} \underline{\mathrm{Hom}}(S', S)^\circ & \longrightarrow & \underline{\mathrm{Fib}}_0 S' \\ \downarrow & & \downarrow \\ \prod(\underline{\mathrm{Hom}}(S', S))^\circ & \longrightarrow & \mathrm{HOT}^{\mathrm{lc}}_W(S'). \end{array}$$

If f, g are in the same connected component of $\underline{\mathrm{Hom}}(S', S)$, then f^*X and g^*X are isomorphic (noncanonically, in general) in $\mathrm{HOT}^{\mathrm{lc}}_W(S')$.

It is cleverer not to fix X and to consider

$$\underline{\mathrm{Hom}}(S', S) \longrightarrow \underline{\mathrm{Hom}}(\mathrm{HOT}^{\mathrm{lc}}_W(S), \mathrm{HOT}^{\mathrm{lc}}_W(S')) \qquad (*)$$

corresponding to

$$\begin{array}{ccccc} \underline{\mathrm{Hom}}(S', S) & \times & \mathrm{HOT}^{\mathrm{lc}}_W(S) & \longrightarrow & \mathrm{HOT}^{\mathrm{lc}}_W(S') \\ (f & , & \xi) & \longmapsto & f^*(\xi). \end{array}$$

The lemma will say that the arrows of $S' \longrightarrow S$ are transformed by $(*)$ into isomorphisms, whence

$$\prod \underline{\mathrm{Hom}}(S', S) \longrightarrow \underline{\mathrm{Hom}}(\mathrm{HOT}^{\mathrm{lc}}_W(S), \mathrm{HOT}^{\mathrm{lc}}_W(S')),$$

so if $f, g : S' \rightrightarrows S$ are two homotopic functors, then

p. 34

$$f^* \simeq g^* : \mathrm{HOT}^{\mathrm{lc}}_W(S) \rightrightarrows \mathrm{HOT}^{\mathrm{lc}}_W(S').$$

[page 46]

[52] **NB** $\Phi_S(X)$ was even already split, defined by the functor $s \longmapsto s\backslash X$.

[53] to bring out a separate statement.

The following follows at once

Corollary 2. *If f is a homotopism, $f^* : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S')$ is an equivalence of categories.*

Indeed, taking $g : S \longrightarrow S'$ such that

$$fg \sim \mathrm{id}_S, \qquad gf \sim \mathrm{id}_{S'},$$

we find that

$$g^*f^* \simeq (\mathrm{id}_S)^* = \mathrm{id}_{\mathrm{HOT}^{\mathrm{lc}}_W(S)}$$
$$f^*g^* \simeq (\mathrm{id}_{S'})^* = \mathrm{id}_{\mathrm{HOT}^{\mathrm{lc}}_W(S')},$$

so we find that (adjusting these functorial isomorphisms if need be) f^* and g^* are equivalences quasi-inverse to one another.

I have the impression ([54]) that all this should not use more than the standard axioms on W as a basic localizer, at most W(2) instead of the weak form W(2 bis).

Let us return to X smooth over contractible S. Corollary 2 implies that

$$\mathrm{Hot}_W \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S)$$
$$\xi \longmapsto p_S^*(\xi),$$

where $p_S : S \longrightarrow e$, is an equivalence. So in the category $\mathrm{HOT}^{\mathrm{lc}}_W(S)$, X is isomorphic to a Cat-fibered object of the form $Z \times S \xrightarrow{\mathrm{pr}_2} S$, with Z in Cat. Now for $X_1 = Z \times S \longrightarrow S$, it is immediate that $X_{1\,s} \longrightarrow X_1$ is in W, for a Cartesian product of arrows W (namely id_Z and $s \longrightarrow S$) is in W. (In fact, the inclusion is even a homotopism!) Let us now note that if X_0, X_1 are two objects in $\underline{\mathrm{Liss}}_0 S$, such that there exists an S-arrow $X_0 \longrightarrow X_1$ which is in W^{d}_S, then $X_{0\,s} \longrightarrow X_0$ is in W if and only if $X_{1\,s} \longrightarrow X_1$ is. Indeed, we have [page 47]

$$\begin{array}{ccc} X_{0\,s} & \xrightarrow{\in W} & X_{1\,s} \\ {\scriptstyle i_s^{X_0}}\downarrow & & \downarrow{\scriptstyle i_s^{X_1}} \\ X_0 & \xrightarrow{\in W} & X_1 \end{array}$$

here we probably use W(2), not only W(2 bis),

which proves the result.

It is true that the fact that $X = X_0$ and $X_1 = Z \times S$ are isomorphic in $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ does not imply, by itself, the existence of a chain of isomorphisms coming from arrows in W^{d}_S. But in fact, if in the statement of corollary 2 above we compare X in $\underline{\mathrm{Fib}}_0 S$ with $g^*f^*(X)$, thanks to a homotopy $\mathrm{id}_S \sim fg$, we do indeed find a sequence of functors $\varphi_i : S \longrightarrow S$, with $\varphi_0 = \mathrm{id}_S$, $\varphi_N = fg$, and φ_i and φ_{i+1} linked by an arrow in $\underline{\mathrm{Hom}}(S, S)$, so we have a sequence of objects $X_i = \varphi_i^*(X)$ of $\underline{\mathrm{Fib}}_0 S$ (with $X_0 = X$, $X_N = g^*f^*X$), and X_i and X_{i+1} linked by an arrow *in* $\underline{\mathrm{Fib}}_0 S$ (not only in $\mathrm{HOT}^{\mathrm{lc}}_W(S)$), which arrow is in $W^{\mathrm{d}}_S = W^{\mathrm{f}}_S$. So we win, subject to various routine checks ([55]). So modulo these, we will have [p. 35] [page 48]

Proposition 9 ([56],[57]). *Suppose that X in Cat/S is a left W-fibration (cf. prop. 8). Then it is even a W-fibration (in the sense of b), page 40, where it is called "W-fibration* hot-*strong"), in other words, if $S'' \longrightarrow S'$ is over S with S'', S' contractible, then $X'' \longrightarrow X'$ is in W ([58]). We probably have better: if $S'' \longrightarrow S'$ is a homotopism,*

[54] O.K.

[55] O.K.

[56] It's ok assuming that W is *strongly* saturated, i.e. $\mathrm{hot}_W(f)$ isomorphism $\Longrightarrow f \in W$.

[57] So left W-fibration $\Longleftrightarrow$ right W-fibration $\Longleftrightarrow$ W-fibration.

[58] we only find that $\mathrm{hot}_W(X'') \longrightarrow \mathrm{hot}_W(X')$ isomorphism.

then $X'' \longrightarrow X'$ is in W.

[page 49]

Let us prove (if we can) this nice complement. I point out that I had to make the hypothesis of W-presmoothness (or of W-preproperness) to be sure that the inverse image $X \times_S S'$, under $f' : S' \longrightarrow S$, really does coincide with f^* on the $\mathrm{HOT}_W^{\mathrm{lc}}$ - otherwise I have no chance. "We know" (everything has to be looked at again!) that it's O.K. in the three cases X W-smooth, X W-proper, X a W-fibration (but in what sense???). The check to be made (when working with the W_S^{d}) is this:

$$\begin{aligned} &X \xrightarrow{i_X} \Phi_S(X) = Y \text{ is in } W_S^{\mathrm{d}} \text{ (without merit), and} \\ &\text{remains so under any base change.} \end{aligned} \tag{$*$}$$

But for $X, Y \in \mathrm{Ob}\,\mathrm{Cat}/S$ which are W-presmooth, W_S^{d}-equivalence coincides with W_S^{f}, which is preserved by base change, as is the hypothesis on X, Y. So we can rest easy.

We are reduced to proving that if X is W-presmooth over S and a W-pseudofibration, and if $S' \longrightarrow S$ is a homotopism, then $X' \longrightarrow X$ is in W. By the earlier argument using $X \longrightarrow \Phi_S(X)$, we reduce to the case where X is split fibered.

p. 36

[page 50]

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \xrightarrow{f} & S \\ & {\scriptstyle p'}\searrow \quad \swarrow{\scriptstyle p} & \\ & e & \end{array}$$

The arrow $X' = f^*(X) \longrightarrow X$ is interpreted in Hot_W as

$$p'_!(f^*(\xi)) \longrightarrow p_!(\xi)\ , \qquad \text{where } \xi = \mathrm{hot}_{S,W}(X). \tag{$*$}$$

I say that this arrow is isomorphism, i.e. that for every $\eta \in \mathrm{Hot}_W$, the transposed arrow

$$\mathrm{Hom}_W(p_!(\xi), \eta) \longrightarrow \mathrm{Hom}_W(p'_!(f^*(\xi)), \eta) \tag{$**$}$$

is bijection. Granting the adjunction formula, the arrow $(**)$ is interpreted as

$$\mathrm{Hom}_W(\xi, \underbrace{p^*(\eta)}_{\eta_S}) \longrightarrow \mathrm{Hom}_W(f^*(\xi), \underbrace{p'^*(\eta)}_{f^*(\eta_S)}),$$

i.e.

$$\mathrm{Hom}_W(\xi, \eta_S) \longrightarrow \mathrm{Hom}_W(f^*(\xi), f^*(\eta_S)),$$

that this is isomorphism therefore follows from f^* fully faithful on the $\mathrm{HOT}_W^{\mathrm{lc}}$. Now we have seen that f^* is an equivalence (cor. 2, page 46).

I am going to take up propositions 9 and 8 again, a bit badly put together. [page 51]

Theorem 3. *We assume that W satisfies* W(1,2,3)*. Let $f : X \longrightarrow S$. Consider the following conditions.*

a) $\forall$ $S'' \longrightarrow S'$ over S, with S'', S' contractible*, $X'' \longrightarrow X'$ is in W.*

a') $\forall$ S' over S, with S' contractible, and $s' \in S'$, the inclusion

$$X_{s'} \hookrightarrow X' = X \times_S S'$$

is in W.

p. 37

a_1) *As in a′), but assuming that S' has a final object (instead of S' contractible).*

a_2) *As in a′), but assuming that S' has an initial object.*

b_1) *f is W-*preproper *(i.e. a′) is satisfied when s' is a final object of S', with no other condition on S' in* Cat$/S$*), and moreover for every arrow $s_0 \longrightarrow s_1$ in S, the induced arrow $X/s_0 \longrightarrow X/s_1$ is in W (i.e. a) is satisfied for $S/s_0 \longrightarrow S/s_1$).*

b_2) *f is W-*presmooth *(i.e. a′) is satisfied when s' is an initial object of S', with no other condition on S'), and moreover the arrows $s_1\backslash X \longrightarrow s_0\backslash X$ (for $s_0 \longrightarrow s_1$ in S) are in W, (i.e. a) satisfied for $s_1\backslash S \longrightarrow s_0\backslash S$).*

c_1) *f is W-preproper and a W-pseudofibration, i.e. condition a′) is satisfied in the following two cases: 1°) s' is a final object of S', 2°) $S' = \mathbf{\Delta}^1$.*

c_2) *f is W-presmooth and a W-pseudofibration, i.e. a′) in the two cases: 1°) s' is an initial object of S', 2°) $S' = \mathbf{\Delta}^1$.*

All these 8 conditions are equivalent. They imply the following:

d) Let $S'' \xrightarrow{u} S'$ be a homotopism in Cat$/S$*, and consider $u_X : X'' \longrightarrow X'$. Then* $\mathrm{hot}_W(u_X)$ *is isomorphism. (So if W is strongly saturated, $u_X \in W$. So under this condition on W, d) is equivalent to the preceding conditions.)*

Definition 4. We say that $f : X \longrightarrow Y$ in Cat is a *W-fibration*, if it satisfies the equivalent conditions a) to c_2) of theorem 3. [page 52]

Corollary 1. *If f is W-proper or W-smooth, for f to be a W-fibration, it is necessary and sufficient that f be a W-pseudofibration, i.e. that for $S' = \mathbf{\Delta}^1 \longrightarrow S$, the inclusions of the two fibers*

$$X'_0 \xrightarrow{i_0} X' \xleftarrow{i_1} X'_1$$

of $X' = X \times_S S'$ be in W.

Corollary 2. *Suppose f* Cat*-fibrant. Then f is a W-fibration if and only if for every arrow $u : s \longrightarrow s'$ in S, $u^* : X_{s'} \longrightarrow X_s$ is in W. Dual statement if f is* Cat*-cofibrant.*

Corollary 3. *Suppose f W-preproper and W-presmooth (for example f W-perfect). Then f is a W-fibration.*

Proposition 10. *Let $f : X \longrightarrow Y$ be an arrow in $\underline{\mathrm{Fib}}_W S$ ([59]) (full subcategory of* Cat$/S$ *formed by the W-fibered objects over S). If $\forall\ s \in S$, $f_s : X_s \longrightarrow Y_s$ is in W, then $f \in W$.*

Let us consider p. 38

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle i_X}\searrow \qquad \swarrow{\scriptstyle i_Y} & \\ & \Phi_S(X) \xrightarrow[\Phi_S(f)]{} \Phi_S(Y), & \end{array}$$

where $i_X, i_Y \in W$, so $\Phi_S(f) \in W \Rightarrow f \in W$. Now the fact that X W-fibered over S implies that $i_X \in W_S^{\mathrm{f}}$ (it even suffices that X be presmooth over S), likewise for Y, i_Y. So the hypothesis $f \in W_S^{\mathrm{f}}$ implies $\Phi_S(f) \in W_S^{\mathrm{f}}$, so $\Phi_S(f) \in W$ by the corollary to proposition 5 (page 30).

p. 39

[59] X and Y both preproper or both presmooth suffices.

[page 53]

IV On axiom W(4) and its variants.

([60]). I have spent days trying to prove that all these variants are equivalent. I am going to give a list of those, among these variants, which in retrospect still seem to me to be of some use. (With notations different from those on pages 13-14.)

W(4a) Every W-fibration (cf. th. 3, p. 51, def. 4, p. 52) is ultrastrong, which amounts to this: for

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{p} & Y' \end{array}$$

cartesian in Cat, if $f \in \underline{\mathrm{Fib}}_W$, $p \in W$, then $q \in W$.

W(4a′) As in a), assuming $f \in \mathrm{Parf}_\omega$, i.e. f W_ω-perfect (hence a W_ω-fibration), where $W_\omega \subseteq W$ is the smallest subset Σ of Fl(Cat) satisfying W(1, 2 bis, 3).

W(4b) Consider a commutative diagram $X \xrightarrow{f} Y$ in Cat, with $p, q \in \mathrm{Fib}_W$, $X \xrightarrow{p} S \xleftarrow{q} Y$, $f \in W$. Then $\forall s \in S$, $f_s : X_s \longrightarrow Y_s$.

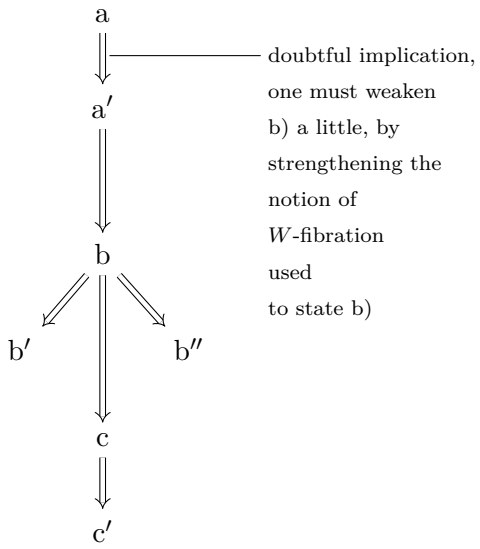
W(4b′) As in b), assuming X and Y split fibered over S, f compatible with splittings (hence a cartesian functor).

W(4b″) As in b), assuming p, q *W-perfect.*

W(4c) Let $f : X \longrightarrow Y$, with $f \in \mathrm{Fib}_W \cap W$, then the fibers of f are W-aspheric.

W(4c′) As in c), assuming $f \in \mathrm{Parf}_\omega \cap W$. (This is the old W(4b″), stripped of the W_∞-parasite.)

I have circled the notions that have been really useful. In practice, one can bet that it is the strongest notion, W(4a), which is the only important one. Here are the tautological implications (with the exception of W(4a′) $\Longrightarrow$ W(4b), which is proved in the proof of proposition 4 (pages 6-10) ([61])). [page 54]



p. 40

[60] Cf. also part VI (HOT_W), p. 186-196

[61] This proof uses a stronger notion of W-fibration than that of theorem 3, p. 51.

I have mainly sweated over showing that c′) $\Longrightarrow$ c), or c) $\Longrightarrow$ b), without getting there. The effort has allowed me to get rid of various false ideas, and to replace them with right ones, cf. below. There is all the same a positive result that stays afloat:

Proposition 1. *Condition b) (and probably even b′) and b″) ([62]), but I am not going to prove it here …) probably implies a) for $\bar{W}$, the strong saturation of W. In any case, if W is strongly saturated, conditions a), a′), b) (and surely also b′) and b″)) are equivalent.*

Proof of b) $\Longrightarrow$ a), when W is strongly saturated. In fact, assuming b) (and without even assuming W strongly saturated), we are going to prove that if we have

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{p} & Y' \end{array}$$

cartesian in Cat, with f a W-fibration, and $p \in W$, then $q \in \bar{W}$, i.e. $\mathrm{hot}_W(q)$ isomorphism. For this, we interpret the arrow q in Hot_W in the following way. Let [page 55]

$$\xi = \mathrm{hot}_{Y,W}(X/Y), \qquad \xi' = p^*(\xi),$$

which says that q is identified with the canonical arrow [p. 41]

$$H_\bullet^{\mathrm{HOT}_W}(Y', \xi') \longrightarrow H_\bullet^{\mathrm{HOT}_W}(Y, \xi) \qquad \text{in } \mathrm{Hot}_W = \mathrm{HOT}_W(e), \tag{$*$}$$

where we note moreover that $\xi \in \mathrm{HOT}_W(Y)$ is a coefficient *locally constant* on Y.

Let us first establish this interpretation of q. Note that since $X \in \underline{\mathrm{Fib}}_W Y$, the inverse image $\xi' = p^*(\xi)$ is none other than $\mathrm{hot}_{Y',W}(X' = X \times_Y Y')$. On the other hand, if $g : Y \longrightarrow e$, $g' : Y' \longrightarrow e$ are the structural arrows, the two sides of $(*)$ are interpreted resp. as $g'_!(\xi')$ and $g_!(\xi)$ (the $g'_!$ and $g_!$ in the sense of the prederivator HOT_W). Now the calculation of $g'_!$, $g_!$ gives us that the two sides are identified resp. with

$$\mathrm{hot}_W(X'), \qquad \mathrm{hot}_W(X),$$

and we admit that the arrow

$$\mathrm{hot}_W(X') \longrightarrow \mathrm{hot}_W(X)$$

is none other than $\mathrm{hot}_W(q)$. Besides, since X is W-fibered over Y, practically by definition, ξ is in $\mathrm{HOT}^{\mathrm{lc}}_W(Y)$. So it will suffice for us to prove the

Corollary. *Suppose that W satisfies W(4b), and let $p : S' \longrightarrow S$ be in W, $\xi \in \mathrm{HOT}^{\mathrm{lc}}_W(S)$, $\xi' = p^*(\xi)$. Then* [page 56]

$$H_\bullet(S', \xi') \xrightarrow{\sim} H_\bullet(S, \xi)$$

is an isomorphism in Hot_W.

Proof. It suffices to prove that for every $\eta \in \mathrm{Ob}\,\mathrm{Hot}_W$ we have a bijection for the $\mathrm{Hom}(-, \eta)$. By the adjunction formula on S, S', this map between the Hom in Hot_W is identified

$$\mathrm{Hom}_{\mathrm{HOT}_W(S)}(\xi, \eta_S) \longrightarrow \mathrm{Hom}_{\mathrm{HOT}_W(S')}(\xi', \eta_{S'}),$$

where $\eta_S = g^*(\eta)$, $\eta_{S'} = g'^*(\eta)$, hence $\eta_{S'} = p^*(\eta_S)$, and the arrow $(*)$ is the arrow on the Hom induced by the functor

$$p^*_{\mathrm{lc}} : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S').$$

[62] and assuming in the case of b″), that W satisfies the condition on direct limits W(6 bis), cf. prop. 5 of the "corollary" of the, page 14.

Now assuming axiom W(4b) (which we have not used so far), the said functor is an *equivalence of categories* if $p \in W$ (cf. VI, p. 204).

q.e.d.

For fear of forgetting them, I am now going to make a list of the little questions I have elucidated along the way while scribbling.

1°) If X, Y are W-smooth over S, and $f : X \longrightarrow Y$ an S-morphism such that $f_s : X_s \longrightarrow Y_s$ is W-smooth for every $s \in \mathrm{Ob}\, S$, it does *not* follow that f is W-smooth. But this is nevertheless true when X and Y are split fibered over S (which already implies their W-smoothness over S), and f compatible with the splittings and a fortiori cartesian. Dual statement for split cofibered categories and the strictly cartesian functors between them that are W-proper fiber by fiber: they are W-proper. p. 42 [page 57]

2°) I was interested, in a case of split fibered categories X, Y over S and a strictly cartesian functor $f : X \longrightarrow Y$, assumed W-perfect fiber by fiber, in concluding that f should also be W-perfect. (It was a case where in addition one knows that X, Y are also W-fibered over S, i.e. the base change functors $X_{s'} \longrightarrow X_s$, $Y_{s'} \longrightarrow Y_s$ are in W. But that was not much use …) The question that remained, in 1°), was that of the *W-properness* of f. But this is hopeless (even for $Y = S$), by virtue of the brilliant

Proposition 2. *Suppose that* $S = 0 \begin{array}{c} \nearrow a \\ \\ \searrow b \end{array}$ *, and let Z be a category over S. Then $\Phi(Z)$ is proper over S if and only if $Z = \emptyset$.*

Let us then take in 2°) above $X = \Phi(Z)$, $Y = S$, $f : \Phi(Z) \longrightarrow S$ the structural morphism (so the f_s are W-perfect), then f is proper only if $X = \emptyset$. Yet, if we take Z W-fibered over S, $\Phi(Z)$ is so too, and we have a counterexample to 2°).

But we have all the same convinced ourselves of the following result:

Proposition 3. *Let X, Y be split fibered categories over S, $f : X \longrightarrow Y$ a strictly cartesian S-functor, such that for every $s \in S$, $f_s : X_s \longrightarrow Y_s$ is* W-fibrant. *Then,* if W satisfies W(4a), *f is W-fibrant.* [page 58]

(I was hoping to get that f is W-fibrant "for free", to prove that W(4c) implies W(4b). But I had to assume W(4a) …)

V Amalgamated sums and W-cocartesian squares in Cat

p. 43
[page 59]

1 Generalities on $U \overset{i}{\hookrightarrow} X \overset{j}{\hookleftarrow} T$, and amalgamated sums.

1.1 Gluing data between an open and a closed.

Let X be in Cat, U an open in X, T the closed complement. (**NB** The opens do indeed correspond to the opens of $\mathrm{Ob}\, X$, topologized by its natural preorder relation $x \leq y \iff \mathrm{Hom}_X(x,y) \neq \emptyset$, likewise for the closed ones. The topologies associated to preorder relations are those where every intersection of opens is still open, i.e. every union of closed sets is still closed. Thus the opens of such a topology are the closed sets of a dual topology, corresponding to the opposite preorder relation. The preorder relation associated to the topology is

$$x \leq y \qquad \iff \qquad y \in \overline{\{x\}},$$

the topology associated to the preorder on I is the one for which the opens are the unions of sets of the form I/x (which form a basis of opens), or again the *"sieves"* ($x \in U$, $y \leq x \Longrightarrow y \in U$), the closed sets are the unions of sets $x\backslash I$, that is to say the cosieves. For every subset A of I, there is a smallest open containing A, which is also its smallest neighborhood, it is the union of the I/x for $x \in A$. And dually there is a largest closed set contained in A, whose complement is the smallest open containing $B = I\backslash A$, hence $\bigcup_{x\in B} I/x$. The closure of A is the union of the closures $x\backslash I$ of its points.)

The situation (X,U,T) is reconstructed, up to unique isomorphism, from the knowledge of [page 60]

1°) The two categories U, T.

2°) The bifunctor $\begin{cases} H : U^\circ \times T \longrightarrow \mathrm{Ens} \\ H(u,t) = \mathrm{Hom}_X(u,t). \end{cases}$

Moreover, these data 1°) 2°) are subject to no restriction. We have, more precisely

Proposition 1. *An isomorphism between the category $\mathcal{T}$ whose objects are the triples (X,U,T), with $X \in \mathrm{Ob}\,\mathrm{Cat}$, $U \in \mathrm{Ouv}(X)$, $T \in \mathrm{Ferm}(X)$, U and T complementary to one another, the morphisms from (X,U,T) to (X',U',T') being the functors $f : X \longrightarrow X'$ which induce $U \longrightarrow U'$, $T \longrightarrow T'$ (whence $U = f^{-1}(U')$, $T = f^{-1}(T')$), with the category $\mathcal{T}'$ of triples (U,T,H), of two categories U, T and of a bifunctor* [p. 44]

$$H : U^\circ \times T \longrightarrow \mathrm{Ens}$$

(or again a presheaf $U^\circ \longrightarrow T^\vee$ on U, hence an object of $\underline{\mathrm{Hom}}(U^\circ, T^\vee)$)),
a morphism from (U,T,H) to (U',T',H') being a triple

$$(f_U, f_T, u) \qquad \begin{cases} \left.\begin{array}{rcl} f_U : & U & \longrightarrow & U' \\ f_T : & T & \longrightarrow & T' \end{array}\right\} \text{functors} \\ u : H \longrightarrow H' \circ (f_U^\circ \times f_T), \\ \text{i.e. } H(u,t) \longrightarrow H'(f_U(u), f_T(t)) \\ \text{a functorial homomorphism} \\ \text{(arrow in } (U \times T^\circ)^\wedge \simeq \underline{\mathrm{Hom}}(U^\circ, T^\vee))), \end{cases}$$

composition of arrows obvious.

If $U \hookrightarrow X \hookleftarrow T$ is a triple, and

$$f_{U'} : U' \longrightarrow U, \ f_{T'} : T' \longrightarrow T$$ [page 61]

arrows in Cat, there is a privileged way to "glue" U' and T', in such a way that the object (X', U', T') maps into (X, U, T) inducing

$f_{U'}$, $f_{T'}$:

$$\begin{array}{ccccc} U' & \xhookrightarrow{i'} & X' & \xhookleftarrow{j'} & T' \\ \downarrow{\scriptstyle f_{U'}} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_{T'}} \\ U & \xhookrightarrow{i} & X & \xhookleftarrow{j} & T, \end{array}$$

it is obtained with the help of the bifunctors

$$\begin{cases} H'(u', t') \overset{\text{def}}{=} H(f_{U'}(u'), f_{T'}(t')) \\ \text{i.e. } H' = H \circ (f_{U'}^{\circ} \times f_{T'}). \end{cases}$$

This triple, i.e. this object of $\mathcal{T}$, represents the following functor on $\mathcal{T}$:

$$(X'', U'', T'') \longmapsto \mathrm{Hom}_{\mathcal{T}}((X'', U'', T''), (X, U, T)) \times_{\substack{\mathrm{Hom}_{\mathrm{Cat}}(U'', U) \\ \times \mathrm{Hom}_{\mathrm{Cat}}(T'', T)}} (\mathrm{Hom}_{\mathrm{Cat}}(U'', U') \times \mathrm{Hom}_{\mathrm{Cat}}(T'', T')),$$

i.e. for every object (X'', U'', T'') of $\mathcal{T}$ we have a canonical bijection, functorial in this object

$$\mathrm{Hom}_{\mathcal{T}}((X'', U'', T''), (X', U', T')) \simeq \mathrm{Hom}_{\mathcal{T}}((X'', U'', T''), (X, U, T)) \times_{\substack{\mathrm{Hom}_{\mathrm{Cat}}(U'', U) \\ \times \mathrm{Hom}_{\mathrm{Cat}}(T'', T)}} (\mathrm{Hom}_{\mathrm{Cat}}(U'', U') \times \mathrm{Hom}_{\mathrm{Cat}}(T'', T')). \qquad (*)$$

No doubt we even have p. 45

$$\underline{\mathrm{Hom}}_{\mathcal{T}}((X'', U'', T''), (X', U', T')) \simeq \underline{\mathrm{Hom}}_{\mathcal{T}}((X'', U'', T''), (X, U, T)) \times_{\substack{\underline{\mathrm{Hom}}_{\mathrm{Cat}}(U'', U) \\ \times \underline{\mathrm{Hom}}_{\mathrm{Cat}}(T'', T)}} (\underline{\mathrm{Hom}}_{\mathrm{Cat}}(U'', U') \times \underline{\mathrm{Hom}}_{\mathrm{Cat}}(T'', T')), \qquad (**)$$

just as functorial, from which $(*)$ is deduced by passing to the sets of objects. We say that the triple (X', U', T') is the triple *induced* by the triple (X, U, T), via the functors $f_{U'}$, $f_{T'}$.

NB If $f_{U'}$, $f_{T'}$ are fully faithful, then so is $f : X' \longrightarrow X$. We shall use this operation of privileged gluing mainly when $U' = U$, $f_{U'} = \mathrm{id}_U$, or conversely $T' = T$, $f_{T'} = \mathrm{id}_T$.

1.2 Relation with the gluing of topoi.

The situation of a triple

$$(\mathcal{X}, \mathcal{U}, \mathcal{T})$$

of a topos $\mathcal{X}$, of an open $\mathcal{U}$ of $\mathcal{X}$, and of a

complementary closed subtopos, is entirely similar (**NB** If $\mathcal{X} = \mathrm{Top}(X)$, with X in Cat, then $\mathrm{Ouv}(\mathcal{X}) \simeq \mathrm{Ouv}(X)$, since the opens correspond to the subsheaves of the terminal sheaf on $\mathcal{X}$, i.e. of the terminal presheaf on X. This bijection is compatible with passage to complements.) Topos theory says that the pairs $(\mathcal{X}, \mathcal{U}, \mathcal{T})$ are reconstructed, up to an equivalence defined up to unique isomorphism, with the help of a triple [page 62]

$$\begin{cases} \mathcal{U},\ \mathcal{T} \text{ two topoi} \\ \varphi : \mathrm{Fais}\,(\mathcal{U}) \longrightarrow \mathcal{T} \text{ a left exact functor ("accessible"),} \end{cases}$$

this functor φ being obtained, when we start from

$$\mathcal{U} \xrightarrow{i} \mathcal{X} \xleftarrow{j} \mathcal{T},$$

as

$$\varphi = j^* i_*.$$

Moreover, the triple $(\mathcal{U}, \mathcal{T}, \varphi)$ can be chosen arbitrarily, with the sole restriction (practically harmless, since always satisfied for all the natural functors we encounter) that φ be (besides being left exact) "accessible".

In the present case $\mathcal{X} = \mathrm{Top}(X)$, $\mathcal{U} = \mathrm{Top}(U)$, $\mathcal{T} = \mathrm{Top}(T)$, the functor $\varphi = j^* i_*$ is not only left exact, but it even commutes with arbitrary $\varprojlim$, since this is so (always) for i_*, and also (because the topos-theoretic j is induced by the category-theoretic $j : T \longrightarrow X$) for j^*. In fact, we find this : p. 46

Proposition 2 : *Let us consider a situation of gluing two topoi $\mathcal{U}$, $\mathcal{T}$ by $\varphi : \mathrm{Fais}\,\mathcal{U} \longrightarrow \mathrm{Fais}\,\mathcal{T}$, a left exact functor. Let us suppose that $\mathcal{U} = \mathrm{Top}(U)$, $\mathcal{T} = \mathrm{Top}(T)$, with U and T in* Cat*. For the glued topos $\mathcal{X}$ to be equivalent to a topos $\mathrm{Top}(X)$, with X in* Cat*, it is necessary and sufficient that the gluing functor φ commute with arbitrary $\varprojlim$, or what amounts to the same thing, that it have a left adjoint :* [page 63]

$$\psi : \underbrace{\mathrm{Fais}\,\mathcal{T}}_{=T^\wedge} \longrightarrow \underbrace{\mathrm{Fais}\,\mathcal{U}}_{=U^\wedge}.$$

We then expect that the gluing datum φ, or what amounts to the same thing the functor ψ having a right adjoint, i.e. commuting with arbitrary $\varinjlim$, amounts to that of a gluing datum H

$$H : U^\circ \times T \longrightarrow \mathrm{Ens}$$

for U and T. But we know well that the functors $\psi : T^\wedge \longrightarrow \mathcal{C}$ into a category $\mathcal{C}$ with arbitrary $\varinjlim$, which commute with the said $\varinjlim$, form a category equivalent (by the functor $\psi \longmapsto \psi | T (\subseteq T^\wedge)$) to that of functors

$$\psi_0 : T \longrightarrow \mathcal{C} \quad (= \underbrace{U^\wedge}_{\underline{\mathrm{Hom}}(U^\circ, \mathrm{Ens})} \text{ in this case}),$$

and giving such a functor ψ_0 does indeed amount to giving an H. Thus [page 64]

Corollary. *The category of topos-theoretic gluing data $\varphi : U^\wedge \longrightarrow T^\wedge$ between* Top(U), Top(T)*, such that the glued topos is a "*paradisiacal *topos", i.e. equivalent to a topos of the form $X^\wedge$ (with X in* Cat*), is equivalent to that of gluing data $H : U^\circ \times T \longrightarrow$* Ens *between U and T. The topos-theoretic datum φ associated to H is the functor right adjoint to $T^\wedge \longrightarrow U^\wedge$ deduced from $T \xrightarrow{H} U^\wedge$ by extension commuting with $\varinjlim$, or again it is given by*

$$\varphi(\underbrace{F}_{\in U^\wedge})(t) \simeq \mathrm{Hom}_{U^\wedge}(H(t), F),$$

where on the right-hand side H is interpreted as a functor $T \longrightarrow U^\wedge$. The topos glued by φ is canonically equivalent to Top(X)*, where X is the object of* Cat *obtained by gluing with the help of H.*

From the point of view of sheaf theory, this gives the interpretation of H, as expressing the functor

$$\varphi = j^* i_* : U^\wedge \longrightarrow T^\wedge$$

by the preceding formula. p. 47 [page 65]

1.3 Arrows in Cat, with source or target a glued object (X, U, T).

When (X, U, T) is given by a gluing datum

$$U,\ T,\ H : U^\circ \times T \longrightarrow \mathrm{Ens},$$

the arrows

$$f : X' \longrightarrow X$$

in Cat are immediately described as follows : giving f first defines

$$U' = f^{-1}(U), \qquad T' = f^{-1}(T),$$

a pair of an open of X' and the complementary closed set, so that this datum is equivalent to that of

1°) a pair (U', T') in X'

2°) a morphism (in $\mathcal{T}$) from (X', U', T') to (X, U, T), which by 1.1 is equivalent to a triple

$$\begin{cases} f_{U'} : & U' & \longrightarrow & U \\ f_{T'} : & T' & \longrightarrow & T \\ u : & H_{U',T'} & \longrightarrow & H \circ (f^\circ_{U'} \times f_{T'}), \end{cases}$$

where we have put

$$\begin{cases} H_{U',T'} : U'^\circ \times T' \longrightarrow \mathrm{Ens} \\ H_{U',T'}(u', t') = \mathrm{Hom}_{X'}(u', t'), \end{cases}$$

i.e. $H_{U',T'}$ is the gluing datum which corresponds to the triple (X', U', T') in $\mathcal{T}$.

We shall be more interested in the reverse situation, i.e. in the question of determining [page 66] the functors

$$\underbrace{X}_{=X_{U,T,H}} \longrightarrow X'$$

from X, obtained from (U, T, H), to an arbitrary object of Cat. We find tautologically

Proposition 3. $\mathrm{Hom}_{\mathrm{Cat}}(X, X')$ *is in canonical bijective correspondence, functorial in* X' *in* Cat*, with the set of triples*

p. 48

$$\begin{cases} \left.\begin{array}{l} f_U : U \longrightarrow X' \\ f_T : T \longrightarrow X' \end{array}\right\} \text{arrows in Cat} \\ u : H \longrightarrow H_{X'}(f^\circ_U \times f_T), \text{ arrow in } \underline{\mathrm{Hom}}(U^\circ \times T, \mathrm{Ens}), \end{cases}$$

where for every X' *in* Cat*, we have put*

$$H_{X'} = ((x, y) \longmapsto \mathrm{Hom}_{X'}(x, y)) : X'^\circ \times X' \longrightarrow \mathrm{Ens}.$$

Surely we will even have an isomorphism of categories

$$\underline{\mathrm{Hom}}(X, X') \simeq \text{category of triples } (f_U, f_T, u) \text{ as in } (*).$$

[page 67]

1.4 Construction of amalgamated sums from an open immersion or closed immersion.

For a construction of a sort of amalgamated sum, in terms of a triple $(X, U, T) + \ldots$, which would be symmetric in U, T, I feel like looking first at the "mixed" situation, of a diagram in solid lines

$$\begin{array}{ccccc} U & \xhookrightarrow{i} & X & \xhookleftarrow{j} & T \\ {\scriptstyle f_U}\downarrow & & \vdots{\scriptstyle f} & & \downarrow{\scriptstyle f_T} \\ U' & \xhookrightarrow{i'} & X' & \xhookleftarrow{j'} & T', \end{array} \qquad (*)$$

which we would like to complete "canonically", if it can be done, to a morphism of triples (i.e. to an arrow in $\mathcal{T}$). The natural idea is that the X' which fits into it would be the direct limit, in Cat, of the initial diagram. If $T' = T$, $f_T = \mathrm{id}_T$, this therefore means that X' identifies with the amalgamated sum $X \amalg_U U'$, and conversely if $U' = U$, $f_U = \mathrm{id}_U$.

Let us denote by X' the amalgamated sum , and make explicit, for any X'' in Cat, the set of arrows g from X' to X'', as a functor in X''. It is the set of triples

$$(g_X, g_{U'}, g_{T'}) \qquad \begin{cases} g_X : X \longrightarrow X'' \\ g_{U'} : U' \longrightarrow X'' \\ g_{T'} : T' \longrightarrow X'' \end{cases} \quad \text{with} \quad \begin{array}{l} g_X \circ i = g_{U'} f_U \\ g_X \circ j = g_{T'} f_T. \end{array} \qquad (*)$$

p. 49

$$\begin{array}{ccccc} U & \xhookrightarrow{i} & X & \xhookleftarrow{j} & T \\ {\scriptstyle f_U}\downarrow & & \downarrow{\scriptstyle g_X} & & \downarrow{\scriptstyle f_T} \\ U' & & & & T' \\ & {\scriptstyle g_{U'}}\searrow & \downarrow & \swarrow{\scriptstyle g_{T'}} & \\ & & X'' & & \end{array}$$

Note that for such a triple, g_X is partially known already in terms of $g_{U'}$, $g_{T'}$, by its restrictions $g_X \circ i$ and $g_X \circ j$ to U and T respectively, by the formulas

$$g_U = g_{U'} f_U, \qquad g_T = g_{T'} f_T.$$

Thus the datum of a triple $(*)$ is equivalent to that of a triple [page 68]

$$(g_{U'}, g_{T'}, g_X) \qquad \begin{cases} g_X \text{ an arrow } X \longrightarrow X'' \text{ corresponding to} \\ g_U = g_{U'} \circ f_U \text{ and } g_T = g_{T'} \circ f_T \text{ already given (via } g_{U'},\ g_{T'}), \end{cases}$$

so by virtue of 1.3. this corresponds to

$$(g_{U'}, g_{T'}, u) \qquad \begin{cases} g_{U'} : U' \longrightarrow X'' \\ g_{T'} : T' \longrightarrow X'' \\ u : H \longrightarrow \underbrace{H_{X''}\left((g_{U'} \circ f_U)^\circ \times (g_{T'} \circ f_T)\right)}_{(u,t) \longmapsto \mathrm{Hom}_{X''}(g_{U'} f_U(u), g_{T'} f_T(t))}. \end{cases} \qquad (*)$$

We are going to interpret the datum of u, in terms of a functor

$$F^* = (H' \longmapsto H' \circ F) : \underline{\mathrm{Hom}}(U'^\circ \times T', \mathrm{Ens}) \longrightarrow \underline{\mathrm{Hom}}(U^\circ \times T, \mathrm{Ens}) \qquad (**)$$

where

$$F : U^\circ \times T \longrightarrow U'^\circ \times T' \qquad \text{is given by } F = f_U^\circ \times f_T.$$

We have an object $H'' = H''_{g_{U'}, g_{T'}}$ of the source category of $(**)$ determined in terms of the pair $(g_{U'}, g_{T'})$, by

$$H''(u', t') = \mathrm{Hom}_{X''}(g_{U'}(u'), g_{T'}(t')),$$

and the target bifunctor of the arrow u in $(*)$ is interpreted as

$$F^*(H'') = H'' \circ F : (u, t) \longmapsto \underbrace{H''(f_U(u), f_T(t))}_{=\mathrm{Hom}(g_{U'} f_U(u), g_{T'} f_T(t))} .$$

Thus, u is interpreted as an arrow

$$u : H \longrightarrow F^*(H'').$$

Now the functor F^* has a left adjoint $F_!$, i.e. we have

$$\mathrm{Hom}(H, F^*(H'')) \simeq \mathrm{Hom}(F_!(H), H''). \qquad (***)$$

with

$$F_! : \underline{\mathrm{Hom}}(U^\circ \times T, \mathrm{Ens}) \longrightarrow \underline{\mathrm{Hom}}(U'^\circ \times T', \mathrm{Ens}).$$

p. 50
[page 69]

Put

$$H' = F_!(H) : U'^\circ \times T' \longrightarrow \mathrm{Ens},$$

it is a bifunctor well determined by the sole datum of the initial diagram $(*)$ p. 67, or what amounts to the same thing, in terms of

$$(U, T, H, f_U : U \longrightarrow U', f_T : T \longrightarrow T').$$

And we see that the datum of an arrow $X' \longrightarrow X''$, X'' a variable object of Cat, is equivalent to that of

$$g_{U'} : U' \longrightarrow X'', \ g_{T'} : T' \longrightarrow X'', \ u : H' \longrightarrow H'' = H_{X''}(g^\circ_{U'} \times g_{T'}),$$

that is to say, by virtue of 1.3., to that of an arrow

$$X'_0 \longrightarrow X'',$$

where X'_0 denotes the object of Cat obtained from U', T' by gluing by means of H'. This therefore shows that the direct limit sought is none other than X'_0. We have thus found the

Theorem 1. *The direct limit X' in* Cat *of the diagram $(*)$ page 67, defined in terms of a triple (X, U, T) in $\mathcal{T}$ and arrows $f_U : U \longrightarrow U'$, $f_T : T \longrightarrow T'$ in* Cat*, exists (without merit) and the canonical arrows*

$$U' \overset{i'}{\hookrightarrow} X' \overset{j'}{\hookleftarrow} T'$$

[page 70]

are open and closed immersions complementary to one another. The gluing datum which defines X' via U', T' is given by

$$H' = (\underbrace{f^\circ_U \times f_T}_{F})_!(H) : U'^\circ \times T' \longrightarrow \mathrm{Ens},$$

where H is the gluing datum which defines X, i.e. $H : U^\circ \times T \longrightarrow$ Ens is the bifunctor $(u, t) \longmapsto \mathrm{Hom}_X(u, t)$.

Thus we have, for u' in U' and t' in T'

$$\mathrm{Hom}_{X'}(u', t') = F_!(H)(u', t'),$$

p. 51

and it remains to make $F_!(H)$ explicit. Here $F = f_U^\circ \times f_T$, and H is a covariant functor on $U^\circ \times T$, we can interpret it as a presheaf on $U \times T^\circ$, and hence write

$$\underbrace{F_!(H)(u',t')}_{\mathrm{Hom}_{X'}(u',t')} = \varinjlim_{(u',t')\backslash U\times T^\circ} \underbrace{\mathrm{Hom}_X(u,t)}_{H(u,t)} \ . \qquad (*)$$

The category $(u',t')\backslash U \times T^\circ$ over which we take the $\varinjlim$ is made explicit as a product

$$(u',t')\backslash U \times T^\circ \simeq (u'\backslash U) \times (T/t')^\circ.$$

The maps

$$\mathrm{Hom}_X(u,t) \longrightarrow \mathrm{Hom}_{X'}(u',t') \qquad (= F_!(H)(u',t'))$$

deduced from this formula are none other than the following: u, t are given as objects of $u'\backslash U$, T/t', i.e. with arrows [page 71]

$$u' \xrightarrow{\alpha} f_U(u), \qquad f_T(t) \xrightarrow{\beta} t',$$

whence

$$\mathrm{Hom}_X(u,t) \xrightarrow{\text{via } f=f_X:X\to X'} \mathrm{Hom}_{X'}(\underbrace{f(u)}_{f_U(u)}, \underbrace{f(t)}_{f_T(t)}) \xrightarrow{\text{via } \alpha,\ \beta} \mathrm{Hom}_{X'}(u',t').$$

Let us note for the record that for every object Z of Cat, and every presheaf G on Z, we have the formula

$$\varinjlim_Z G \simeq \pi_0(Z/G),$$

where Z/G is interpreted in terms of the canonical immersion $Z \longrightarrow Z^\wedge$ ($G \in \mathrm{Ob}\, Z^\wedge$). It is therefore a fibered category over Z with discrete fibers, and we have in fact an isomorphism between the full subcategory of Cat/Z formed by such fibered categories with discrete fibers, with $Z^\wedge$. In this perfect dictionary between presheaves of sets and fibered categories with discrete fibers, the functor $\varinjlim$ on $Z^\wedge$ corresponds simply to the functor π_0 on the category of objects over Z in Cat, itself induced by the functor (independent of Z)

$$\mathrm{Cat} \xrightarrow{\pi_0} \mathrm{Ens}.$$

[page 72]

We can therefore rewrite the fundamental formula $(*)$ p. 70, in terms of the functor H defining the gluing X of U and of T, as

$$\boxed{\underbrace{\mathrm{Hom}_{X'}(u',t')}_{H'(u',t')} \simeq \pi_0(\underbrace{(u',t')\backslash U \times T^\circ/H}_{((u'\backslash U)\times(T/t')^\circ)/H}).} \qquad (*)$$

In what follows, we will probably only have to use the case where we have either $T' = T$, $f_T = \mathrm{id}_T$, or the reverse. We will work mainly in the first case, deducing what we will need for the dual case, by the principle of duality (automorphism $X \longmapsto X^\circ$ of the category Cat). In this case, in the fundamental formula $(*)$ the element t' of T' is written as an element t of T, and the formula is rewritten as p. 52

$$\boxed{\mathrm{Hom}_{X'}(u',t) \simeq \pi_0(\ \underbrace{u'\backslash U/t}_{\text{category of triples } (u,\alpha,\beta'),\ \begin{cases} u \in \mathrm{Ob}\, U \\ \alpha : u \longrightarrow t \\ \beta' : u' \longrightarrow f_U(u) \end{cases}}\ ).} \qquad (**)$$

This is the formula that I had found directly in this case, without yet bothering to vary T at the same time as U. I would like to make the link between the two formulas $(*)$ and $(**)$, by topological means, thus establish a canonical bijection, functorial etc.

$$\pi_0(u'\backslash U/t) \xrightarrow{\sim} \pi_0(((u'\backslash U) \times (T/t)^\circ)/H).$$

We surely have a functor [page 73]

$$u'\backslash U/t \longrightarrow ((u'\backslash U) \times (T/t)^\circ)/H$$

which associates to a triple

$$(u, \alpha, \beta') : u \in \mathrm{Ob}\, U,\ \alpha : u \longrightarrow t,\ \beta' : u' \longrightarrow f_U(u)$$

the triple $(\bar{u}, \bar{t}, \xi)$ $(\xi \in H(\bar{u}, \bar{t}))$ on the right-hand side, with $\bar{u}$ in $u'\backslash U$, $\bar{t}$ in T/t, defined as follows:

$$\bar{u} = (u, \beta'),\ \bar{t} = (t, \mathrm{id}_t), \underbrace{\xi}_{=\alpha} \in H(\bar{u}, \bar{t}) = \mathrm{Hom}(u, t).$$

In short

$$u'\backslash U/t = \text{fiber of } ((u'\backslash U) \times (T/t)^\circ)/H \text{ at the initial object } \bar{t} \text{ of } (T/t)^\circ$$

under the natural projection, composite arrow

$$\begin{array}{c} ((u'\backslash U) \times (T/t)^\circ)/H \\ \downarrow \\ (u'\backslash U) \times (T/t)^\circ \\ \downarrow \pi \\ (T/t)^\circ. \end{array}$$

We have to see that the inclusion of the fiber gives a bijection on the π_0, or what amounts to the same thing, that the inclusion of the fiber $u'\backslash U$ *of* π at the initial object (t, id_t) of $(T/t)^\circ$ gives a bijection on the $\varinjlim$ of the functor H. So the inclusion had better be aspheric, or at least co-cofinal.

But the inclusion of an initial object is always aspheric, and remains so under product p. 53 with any object (i.e. $u'\backslash U$) of Cat (and even under any smooth base change …), whence [page 74] the conclusion.

In what follows, we are mainly interested in the situation of amalgamated sums (where we change notation a little, now)

$$\begin{array}{ccccc} U & \xrightarrow{\ i\ } & X & \xleftarrow{\ j\ } & T \\ {\scriptstyle p}\downarrow & & {\scriptstyle q}\downarrow & & \vdots \\ U' & \xrightarrow{\ i'\ } & X' & \longleftarrow & T' \simeq T. \end{array} \qquad (*)$$

The closed subcategory T' complementary to U' in X' is then canonically isomorphic to the closed subcategory T complementary to U in X, the isomorphism being induced by q. Often we shall identify T' and T, which is legitimate on viewing X' as obtained from the gluing of U' and t , by the bifunctor $U'^\circ \times T \longrightarrow$ Ens made explicit in formula $(**)$ p. 72. We are mainly interested in isolating conditions which ensure the following two implications [page 75]

a) $i \in W \Longrightarrow i' \in W$ (and even $\Longleftrightarrow$).

b) $p \in W \Longrightarrow q \in W$ (and even $\Longleftrightarrow$).

Proposition 4. *Let $t \in \mathrm{Ob}\, T$. For $t \in \mathrm{Ob}\, \bar{U}$ to hold, it is necessary and sufficient that $q(t) \in \mathrm{Ob}\, \overline{U'}$. In other words, the "boundary" $\dot{T} = T \cap \bar{U}$ of T in X is "the same" as its boundary in X'.*

Proof. To say that $t \in \mathrm{Ob}\, \bar{U}$ also means

$$U/t \neq \emptyset$$

and implies, by continuity, that $q(t) \in \overline{q(U)} \subseteq \overline{U'}$. Conversely, if $q(t) \in \overline{U'}$, i.e. if there exists $u' \in U'$ such that

$$\mathrm{Hom}(u', q(t))(= \pi_0(u'\backslash U/t)) \neq \emptyset,$$

it follows that $u'\backslash U/t \neq \emptyset$, a fortiori U/t (which receives $u'\backslash U/t) \neq \emptyset$, q.e.d.

Proposition 5. *Let*

$$\begin{array}{ccc} Y & \xrightarrow{i} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ Y' & \xrightarrow{i'} & X' \end{array}$$

be a cocartesian square in Cat*. If i is an open immersion (resp. closed), then i' too, and* p. 54
the square is also cartesian. Ditto for p and q. If i and p are open immersions ([1]) (resp. …), then the situation is isomorphic to that of an object X' of Cat*, of two open (resp. closed) subcategories X, Y' of X' having as union X', as intersection Y. Conversely, in such a situation, the corresponding square is cocartesian, i.e. X is identified with the amalgamated sum*

$$X' \simeq X \amalg_Y Y'.$$

Thus we also have proposition 15, page 158, and its corollaries.

p. 55

2 Amalgamated sums and W-equivalences.

2.1 Cocartesian square axiom and cube axiom (or "Five Lemma").

We now suppose given a basic localizer

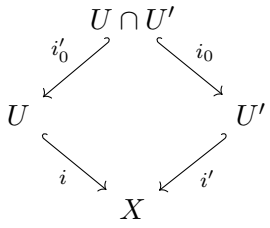
$$W \subseteq \mathrm{Fl}(\mathrm{Cat}),$$

so W satisfies W(1, 2 bis, 3). We do not assume for the moment for the other standard axioms W(4) (fibration axiom), W(5) (connectedness axiom), W(6) (direct limit axiom). But we are now going to introduce two additional stability axioms, of a different nature, concerning amalgamated sums.

W(7 bis) (Cocartesian square axiom.) Let X be an object of Cat, U, U' two open subcategories of X such that $U \cup U' = X$, let us consider the corresponding cocartesian diagram (cf. prop. 5 in the margin of the page):

[page 76]

[1] **NB** We use *"open immersion"*, *"closed immersion"* in the *strict sense*: it is a fully faithful *monomorphism* whose image is a subcategory that is *full,* either open or closed.



$(*)$

Then

$$i_0 \in W \qquad \Longleftrightarrow \qquad i \in W$$

(and symmetrically

$$i'_0 \in W \qquad \Longleftrightarrow \qquad i' \in W).$$

NB By duality, this axiom is equivalent to the similar axiom, with *closed subcategories* T, T' of X with union X.

Corollary. *Assume* W(7 bis). *Consider the following conditions on the preceding square.*

a) $U \cap U'$ *W-aspheric.*

b) U, U' *W-aspheric.*

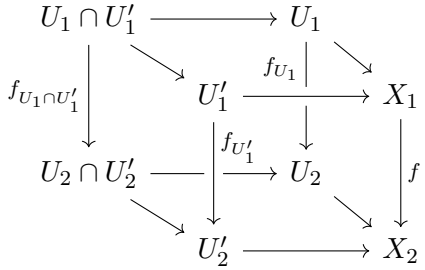
c) X *W-aspheric.*

Then (a and b) implies c, and (b and c) implies a.

NB Morally, (c and a) ought also to imply b, i.e. if $U \cap U'$ and X are W-aspheric, the same is true of U and U'. But it does not seem that the cocartesian square axiom allows one to prove this. On the other hand, the axiom which follows will imply the preceding corollary in its complete form.

p. 56

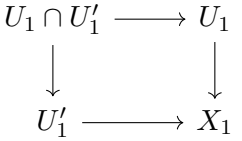
W(7) ("Cube axiom", or "Five Lemma".) Let there be a commutative cubical diagram



representing a morphism (vertical arrows) from a

[page 77]

cocartesian square



of the type considered in W(7 bis) (X_1 union of two open subcategories U_1, U'_1) into another (with subscripts 2). Consider the following conditions.

a) $f_{U_1 \cap U'_1} \in W$.

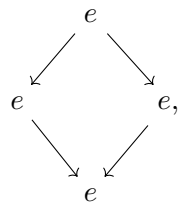
b) f_{U_1} and $f_{U'_1}$ are W.

c) $f \in W$.

Then any two of these conditions imply the third. In other words, when one of these conditions is satisfied, the other two are equivalent to each other.

NB This axiom is equivalent to the dual axiom, in terms of objects X_1, X_2 each represented as a union of two closed subcategories.

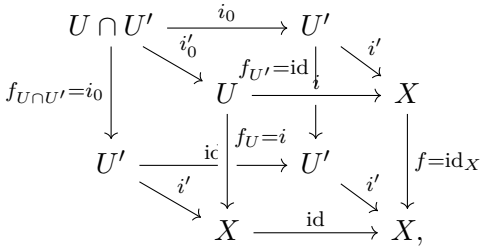
Corollary. *Assume* W(7) *satisfied. Then the corollary to* W(7 bis) *(an axiom which we do not have to use here), in its completed form (cf. the NB following it) is valid: any two of the conditions a), b), c) imply the third.*

It suffices to look at the unique morphism from the diagram (∗), p. 76, into the trivial diagram  , corresponding to the case where X is the final category e, and U, U' equal to X. Then the corollary is nothing other than W(7) applied to this particular case.

This argument makes one feel that W(7) is probably stronger than W(7 bis). Prompted by this feeling, I notice with pleasure that it is well founded:

Proposition 6. *The axiom* W(7) *(five lemma) implies* W(7 bis) *(cocartesian square axiom).* p. 57 [page 78]

We look at the following cubical diagram



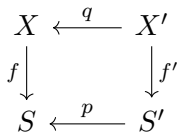
from the given cocartesian square into the cocartesian square obtained from X and its two open subcategories X, U', with intersection U'. Here the two vertical arrows f and $f_{U'}$ are isomorphisms, hence W. With the notation of W(7), we therefore have c) ($f \in W$), hence a) and b) are equivalent, i.e. $i_0 \in W$ and $i \in W$ are equivalent.

q.e.d.

Theorem 2. *The basic localizer W_∞ satisfies the axiom* W(7).

This is surely a consequence of the cohomological criterion of Artin-Mazur, taking into account the Mayer-Vietoris exact sequence for both of the two cocartesian squares, and the noncommutative variants of Mayer-Vietoris. I have not written out the verification in due form, and should probably repair this omission.

Remark. The dual version (in the sense of Quillen) of the cocartesian square axiom W(7 bis), i.e. the corresponding axiom for cartesian squares, is not valid for W_∞. The relevant statement would be the following: let [page 79]



be a cartesian square, with f a *W-fibration.* Then $f \in W \iff f' \in W$. If W satisfies W(4c) (which is the case for W_∞), then $f \in W$ does indeed imply that its fibers are W-aspheric, hence that it is a universal W-equivalence, i.e. $f' \in W\ \forall p : S' \longrightarrow S$. But for the reverse implication $f' \in W \Longrightarrow f \in W$, one must evidently assume that

$$\pi_0(p) : \pi_0(S') \longrightarrow \pi_0(S)$$

is *surjective.* (E.g. if $S' = \emptyset$, we have $X' = \emptyset$, hence $f' \in W$, without that implying $f \in W$!) The fact that in the cocartesian case of W(7 bis), $i_0 \in W \Longrightarrow i \in W$ must be regarded as "going without saying" - otherwise the square does not deserve, from the W-homotopic point of view, the name "cocartesian". On the other hand, the reverse implication $i \in W \Longrightarrow i_0 \in W$ appears to me as a very particular feature of the derivator HOT_W (when it has it), and notably of $\mathrm{HOT} = \mathrm{HOT}_{W_\infty}$. The beginning of the remark simply means that the dual derivator HOT° does not have this same property. I must refer in this connection to the two axioms for derivators corresponding to W(7 bis), W(7), which I will probably do shortly, so as to give a more precise meaning to the present comments.

p. 58

[page 80]

2.2 Application to cocartesian squares deduced from an open or closed immersion.

The axioms W(7), W(7 bis) (of the cube, and of the cocartesian square) concern cocartesian squares of a very particular nature. We are now going to see how to deduce from them statements of the same type for much more general cocartesian squares, including for squares which have nothing cocartesian about them (in Cat) - and which we will call "homotopically cocartesian" (relative to W), or simply W-cocartesian. (There will also be W-cartesian squares one day …) In the case where $W = W_\infty$, I will also call them Hot-cocartesian.

I will begin with the particular case of a cocartesian square associated with an open immersion (diagram $(*)$, p. 74):

$$(*) \qquad \begin{array}{ccccc} U & \xhookrightarrow{i} & X & \xhookleftarrow{j} & T \\ {\scriptstyle p}\downarrow & & {\scriptstyle q}\downarrow & & \vdots \\ U' & \xhookrightarrow{i'} & X' & \xhookleftarrow{j'} & T' \simeq T. \end{array}$$

[page 81]

The big trap here that we have to manage to exorcise is that it can happen that $i \in W$, without our having $i' \in W$, and also $p \in W$ without our having $q \in W$. I had convinced myself of this in 1983 while writing Pursuing Stacks (during the weeks of "scratchwork"), and don't feel like looking for a counterexample here, unless it presents itself on its own (which it probably will).

I am going to give an important case where these troubles disappear:

Proposition 6: *Consider the closures $\bar{U}$ and $\overline{U'}$ of U and U' in X, X' respectively, and suppose that the inclusions*

$$U \xrightarrow{i_0} \bar{U}, \qquad U' \xrightarrow{i'_0} \overline{U'}$$

are in W ([64]). *Then*

p. 59

$$\begin{cases} \text{ⓐ } i \in W & \iff & i' \in W & \text{if we have W(7 bis)} \\ \text{ⓑ } p \in W & \iff & q \in W & \text{if we have W(7).} \end{cases}$$

[64] cf. th. 3 p. 107 for general conditions ensuring that this hypothesis is satisfied.

PROOF. Let us first consider case ⓐ of i, i'. Let

$$k : \bar{U} \hookrightarrow X$$

be the inclusion. The triangle $U \xrightarrow[i]{i_0} \bar{U} \xrightarrow{k} X$, where $i_0 \in W$, shows that

$$i \in W \iff k \in W.$$

On the other hand, X is the union of the closed subcategories $\bar{U}$, T, so we have a cocartesian square in Cat

$$(*) \qquad \begin{array}{ccc} \bar{U} \cap T = \dot{T} & \xrightarrow{k_0} & T \\ {\scriptstyle j_0}\downarrow & & \downarrow{\scriptstyle j} \\ \bar{U} & \xrightarrow{k} & X. \end{array}$$

The axiom W(7 bis) (in its dual form (i.e. applied to the square formed by the opposite categories …)) implies that [page 82]

$$k \in W \iff k_0 \in W,$$

whence

$$i \in W \iff (k_0 : \dot{T} \longrightarrow T) \in W.$$

Likewise we find, applying this to (X', U') in place of (X, U)

$$i' \in W \iff (k'_0 : \dot{T}' \longrightarrow T') \in W.$$

Now we have $T \xrightarrow{\sim} T'$ (isomorphism), and by proposition 4 this isomorphism takes $\dot{T}$ to $\dot{T}'$. Thus

$$k_0 \in W \iff k'_0 \in W \qquad \text{(trivially)},$$

whence a).

Let us prove b). The commutative square induced by $(*)$, p. 80, p. 60

$$\begin{array}{ccc} U & \xrightarrow{i_0} & \bar{U} \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \bar{p}} \\ U' & \xrightarrow{i'_0} & \overline{U'}, \end{array}$$

where $i_0, i'_0 \in W$ by hypothesis, shows that

$$p \in W \iff \bar{p} \in W,$$

so everything comes down to proving that

$$\bar{p} \in W \overset{?}{\iff} q \in W.$$

But q induces a homomorphism from the square $(*)$ deduced from $(X, \bar{U}, T)$ (with $X = \bar{U} \cup T$), into the similar square for X'

$$\begin{array}{ccc} \overline{U'} \cap T' = \dot{T}' & \xrightarrow{k'_0} & T' \\ {\scriptstyle j'_0}\downarrow & & \downarrow{\scriptstyle j'} \\ \overline{U'} & \xrightarrow{k'} & X', \end{array}$$

[page 83]

the vertical arrows being denoted by $q_{\dot T}$, q_T, $\bar p$, q. We know that $q_{\dot T}$, q_T are isomorphisms, hence in W, so the "five lemma" W(7) assures us that $\bar p \in W \iff q \in W$, q.e.d.

Comment. This proposition leads us to look for conditions on an open subcategory U of an X, which ensure that the inclusion $i_0 : U \longrightarrow \bar U$ is W, and *that this property is preserved by every cobase change* $U \longrightarrow U'$. Moreover, we seek conditions which are of a local nature on $\bar U$, more precisely on $\bar U \cap T = \dot U = \dot T$. We can motivate this concern (a posteriori) by saying that we would like the property $i_0 \in W$ to be preserved, not only by an arbitrary cobase change $U \longrightarrow U'$, but also, in addition, by passing from X to any full subcategory X_1 containing U, thus giving rise to a situation

$$U \overset{i_Y}{\hookrightarrow} X_1, \qquad U \xrightarrow{(i_Y)_0} \bar U^{X_1} = \bar U \cap X_1$$

, similar to the starting situation. More generally still, we would like it to be preserved on passing from the triple (X, U, T) to an "induced" triple (X_1, U, T_1), by an arbitrary arrow $g : T_1 \longrightarrow T$ with target T in Cat, giving rise to the pair of arrows

$$U \overset{\mathrm{id}}{\longrightarrow} U, \ g : T_1 \longrightarrow T$$

(cf. pages 60, 61). The preceding case is the one where T_1 is a full subcategory of T, g the inclusion arrow. Let us note in this connection the sorital proposition p. 61 [page 84]

Proposition 7. *Equivalent conditions on the open subcategory U of an object X of* Cat (assuming W(7 bis)):

a) The inclusion $i_0 : U \longrightarrow \bar U$ is W-aspheric, i.e. for every $t \in \dot U = \bar U \backslash U$, the category U/t is W-aspheric.

b) The inclusion $i_0 : U \longrightarrow \bar U$ is in W, and this condition remains satisfied, when we replace (X, U) by (X_1, U), where (X_1, U) is deduced from (X, U) by means of an arrow $T_1 \longrightarrow T(= X \backslash U)$ in Cat*, as recalled above.*

b′) As in b), in the case where T_1 is the one-point category e, and where moreover its image is in $\dot U$.

Proof. It is obvious that if in b), b′) we replace the condition "$i_0 \in W$" by the stronger condition "i_0 is W-aspheric", the four conditions a) to b′) are equivalent. It remains to prove that the condition that the U/t be W-aspheric ($t \in \bar U$ - but this is trivial for $t \in U$, so it suffices to express it for $t \in \bar U \backslash U = \dot U$) is already implied by the weakest b′ and we guess that we have this:

Corollary. (Assuming W(7 bis).) *Let X be in* Cat*, U an open subcategory of X such that $X \backslash U = \overline{\{t\}}$ is the final category. Then the inclusion $U \overset{i}{\to} X$ is in W if and only if it is W-aspheric, i.e. if and only if U/t is W-aspheric (where t is the unique element of T).* [page 85]

We prove this corollary later ([65]). Proposition 7 will be of no use to us until then - it will have only a purely aesthetic role to play, to justify our deliberate intention of looking for conditions on $i : U \longrightarrow X$ (to ensure that $U \longrightarrow \bar U$ is in W and remains so, when we replace (X, U) by other pairs (X_1, U_1) which are deduced from it by certain operations …) which are of a *local nature* on $\bar U$ (in fact, at the points of $\dot U$), and which in any event imply the *W-asphericity* of $U \longrightarrow \bar U$. [page 86]

I had hoped that this asphericity condition (suggested by the notion "of a Dwyer immersion") would be stable under an arbitrary cobase change $U \longrightarrow U'$, which would

[65] cf. cor. 4 of proposition 15, page 149.

at the same time have ensured that after such an operation, the open immersion $i' : U' \longrightarrow X'$ obtained still satisfies the condition that $i'_0 : U' \longrightarrow \overline{U'}$ is in W. We will see that this expectation is unfounded - the condition of W-asphericity is *not* stable under cobase change, and it does *not* even suffice to ensure that after a cobase change, $i'_0 : U' \longrightarrow \overline{U'}$ is in W. A stronger notion will be needed, which is suggested by the p. 62
case where $T = \{t\}$ is the final category and contained in $\bar{U}$ (the case to which we tried to reduce ourselves in proposition 7). In this case, as we said (preceding corollary), the condition that $i : U \longrightarrow \bar{U} = X$ is in W, is equivalent to the W-asphericity of i, i.e. to the W-asphericity of U/t, and likewise for $i' : U' \longrightarrow \overline{U'} = X'$: it is in W if and only if U'/t' is W-aspheric, where t' is the unique element of $T'(\simeq T) = X' \backslash U'$. So we need [page 87]
conditions on U/t which ensure that not only U/t, but also all the similar categories U'/t , are W-aspheric. In the general case where T is arbitrary, it is this condition that we are going to impose on the U/t ($t \in \mathrm{Ob}\, T$), and it will turn out that this time, they are stable under every cobase change $U \xrightarrow{p} U'$, i.e. that the U'/t' ($t' \in \mathrm{Ob}\, T'$) will then satisfy the same condition. It is this condition that we are now going to reformulate.

2.3 Totally W-aspheric objects of Cat, and totally W-aspheric morphisms.

Let us take up again the situation

$$U \overset{i}{\hookrightarrow} X \overset{j}{\hookleftarrow} T, \qquad \text{with } T = \{t\}, \text{ final category, } t \in \bar{U}, \text{ i.e. } X = \bar{U},$$

and that of a cobase change $U \longrightarrow U'$, giving rise to a cocartesian square in Cat

$$(*) \qquad \begin{array}{ccccc} U & \overset{i}{\longrightarrow} & X & \overset{j}{\dashleftarrow} & T = \{t\} \\ {\scriptstyle p}\downarrow & & {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle \text{iso}} \\ U' & \overset{i'}{\longrightarrow} & X' & \overset{j'}{\dashleftarrow} & T' = \{t'\}. \end{array}$$

The gluing datum H between U and T, $H : \underbrace{U^\circ \times T}_{\simeq U^\circ} \longrightarrow \mathrm{Ens}$, can be interpreted as a presheaf on U, $H \in \mathrm{Ob}\, U^\wedge : U^\circ \longrightarrow \mathrm{Ens}$, and we have [page 88]

$$U/t \simeq U/H,$$

where U/H is interpreted in terms of the canonical embedding

$$U \hookrightarrow U^\wedge.$$

Likewise, the gluing between U' and T' will be given by a presheaf

$$H' \in \mathrm{Ob}\, U'^\wedge : U'^\circ \longrightarrow \mathrm{Ens},$$

and we will have p. 63

$$U'/t' \simeq U'/H'.$$

The relation between H' and H was made explicit in §1, as

$$H' = p_! H, \qquad \text{where } p_! : U^\wedge \longrightarrow U'^\wedge$$

is the left adjoint of $p^* : U'^\wedge \longrightarrow U^\wedge$, given by

$$p^* F' = F' \circ p.$$

[page 89]

We have

$$p_!(H)(u') \simeq \pi_0(u'\backslash U/H).$$

Definition 1. ⓐ Let U be an object of Cat, $H \in \mathrm{Ob}\, U^\wedge$ a presheaf on U. We say that H is *totally W-aspheric* if for every arrow $p : U \longrightarrow U'$ of Cat, $p_!(H)$ in $U'^\wedge$ is a W-aspheric object, i.e. $U'/p_!H$ is W-aspheric. ⓑ We say that an object V of Cat is *totally W-aspheric,* if the terminal functor e_V on V ($v \longmapsto \{\emptyset\}$) is totally W-aspheric, i.e. if for every arrow $f : V \longrightarrow V'$ in Cat, the presheaf

$$\pi_0(f) \overset{\text{def}}{=} f_!(e_V) \overset{\text{def}}{=} (v' \longmapsto \pi_0(v'\backslash V))$$

on V' is W-aspheric, i.e. if the category

$$\underline{\pi}_0(f) \quad \text{or} \quad \underline{\pi}_0(V/V') \overset{\text{def}}{=} V'/\pi_0(f)$$

is W-aspheric.

Let us note that the definition of the functors $p_!$ (or $f_!$) as adjoints, and their "calculation" in terms of the π_0 of certain categories, does not depend on the localizer W. Let us give a few details on this subject. We have an adjunction morphism in $V^\wedge$

$$f^* f_!(F) \longleftarrow F,$$

hence

$$V/F \longrightarrow V/f^* f_!(F) \simeq V \times_{V'} (V'/f_!(F)), \qquad (*)$$

the last isomorphism coming from the fact that we have commutativity, up to canonical isomorphism, in the diagram of functors

$$\begin{array}{ccc} V'^\wedge & \xrightarrow{f^*=(F' \longmapsto F' \circ f)} & V^\wedge \\ {\scriptstyle F' \longmapsto V'/F'}\downarrow & & \downarrow{\scriptstyle F \longmapsto V/F} \\ \mathrm{Cat}/V' & \xrightarrow{f^*_{\mathrm{Cat}}=(W' \longmapsto W' \times_{V'} V)} & \mathrm{Cat}/V, \end{array}$$

in other words, the operator f^* on presheaves is interpreted, in geometric terms, as the operation $W' \longmapsto W' \times_{V'} V$ of base change by $V \longrightarrow V'$, on the categories over V' which are fibered with discrete fibers (a property which is preserved by base change). p. 64 [page 90]

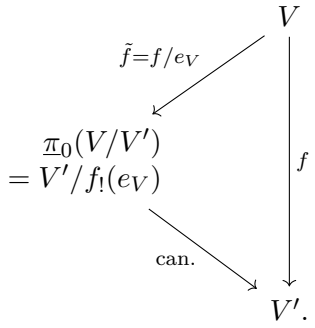
That said, giving an arrow $(*)$ is equivalent, by the universal property of base change, to giving an arrow (canonical in this case)

$$f/F : V/F \longrightarrow V'/f_!(F)$$

making the square commutative

$$\begin{array}{ccc} V/F & \xrightarrow[\text{or source functor}]{\text{forgetting } F} & V \\ {\scriptstyle f/F}\downarrow & & \downarrow{\scriptstyle f} \\ V'/(F' = f_!(F)) & \xrightarrow{\text{forgetting } F'} & V'. \end{array} \qquad (*)$$

In the case that interests us especially here, $F = e_V$, $V/F \longrightarrow V$ is an isomorphism, and we obtain a canonical factorization of functors [page 91]



(∗∗)

Let us make the V'-morphism $\tilde{f}$ explicit on the fibers,

$$\tilde{f}_{v'} : V_{v'} \longrightarrow \underline{\pi}_0(V/V')_{v'} \overset{\text{def}}{=} \pi_0(v'\backslash V), \qquad (***)$$

where in the last term, we have identified the set $\pi_0(v'\backslash V)$ with the discrete category it defines. To make this arrow $(***)$ explicit, let us note that we have a canonical inclusion

$$V_{v'} \longrightarrow v'\backslash V$$

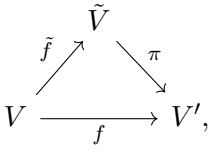
and for every category W, a canonical (adjunction) morphism $W \longrightarrow \pi_0(W)$, hence a composite

$$V_{v'} \longrightarrow v'\backslash V \longrightarrow \pi_0(v'\backslash V),$$

which is none other than $\tilde{f}_{v'}$ in $(***)$.

It should be easy to show that the factorization $(**)$ of f is characterized, up to unique isomorphism, as a factorization in Cat/V'

p. 65
[page 92]



by the following properties:

a) π is fibered with discrete fibers, i.e. a local isomorphism (i.e. $\tilde{V}$ is isomorphic in Cat/V' to a category V'/F', with F' in $V'^{\wedge}$).

b) $\tilde{f}_!(e_V) = e_{\tilde{V}}$, i.e. for every $\tilde{v} \in \mathrm{Ob}\,\tilde{V}$, the category $\tilde{v}\backslash V$ is 0-connected: the colocalizations of V over $\tilde{V}$ are 0-connected.

Let us note here, in passing, that for *every* morphism $\pi : W' \longrightarrow W$ in Cat which is fibered with discrete fibers, i.e. such that W' is isomorphic in Cat/W to a category of the form W/F ($F \in \mathrm{Ob}\,W^{\wedge}$), we have

$$\pi_!(e_{W'}) = F, \qquad \text{i.e. } W' \underbrace{\simeq}_{\text{in } \mathrm{Cat}/W} W/\pi_!(e_{W'}). \qquad (*)$$

We must check that the functor

$$w \longmapsto \pi_0(\underbrace{w\backslash(W/F)}_{\overset{\text{def}}{=} w\backslash W/F})$$

is canonically isomorphic to F. But the inclusion

$$\underbrace{W'_w}_{=F(w)\ \text{(discrete category)}} \longrightarrow w\backslash W'$$

[page 93]

does indeed define as before

$$F(w) = W'_w \longrightarrow \pi_0(w\backslash W'),$$

we must prove that this is bijective. Now, W' being W_∞-smooth over W, we know that the inclusion $W'_w \hookrightarrow w\backslash W'$ is in W_∞, a fortiori induces an isomorphism on the π_0.

This established, if we have a factorization of $f : V \longrightarrow V'$ as $f = \pi\tilde{f}$ having the properties a) b) of the preceding page, we already see that we have

$$f_!(e_V) \simeq \pi_!(\underbrace{\tilde{f}_!(e_V)}_{\simeq e_{\tilde{V}} \text{ by b)}}) \simeq \pi_!(e_{\tilde{V}}),$$

which already determines $\tilde{V}$ up to canonical isomorphism (thanks to a) as

$$\tilde{V} \simeq V'/f_!(e_V) \overset{\text{def}}{=} \underline{\pi}_0(V/V'),$$

and it should not be very hard to convince oneself that $\tilde{f}$ is none other than the canonical arrow of page 91. p. 66

Proposition 8. *For an object H of $U^\wedge$ to be totally W-aspheric, it suffices that the object U/H of* Cat *be totally W-aspheric.*

Let us indeed suppose this latter condition satisfied and prove that H is totally W-aspheric, i.e. that for every $p : U \longrightarrow U'$, $U'/p_!(H)$ is W-aspheric. [page 94]

$$\begin{array}{ccc} V \overset{\text{def}}{=} U/H & \xrightarrow{\alpha} & U \\ & \searrow^{f} & \downarrow{\scriptstyle p} \\ & & U' \end{array}$$

Let f be the composite functor in the diagram opposite , we have $H \simeq \alpha_!(e_V)$ (by $(*)$ p. 92), hence

$$p_!(H) \simeq p_!\alpha_!(e_V) \simeq (p\alpha)_!(e_V) \simeq f_!(e_V),$$

which is W-aspheric by the hypothesis on $V = U/H$.

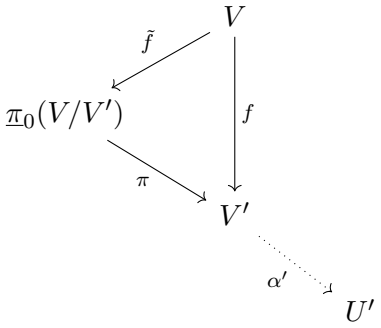
Does the converse of prop. 8 hold, i.e. if H is totally W-aspheric, is the same true of $V = U/H$? The difficulty comes from the fact that this means that for *every* arrow $f : V \longrightarrow V'$ in Cat with source V, we have $f_!(e_V)$ W-aspheric, whereas the hypothesis is equivalent to postulating it only for arrows that factor through U. Let us note that if we do not make this hypothesis on f, but suppose instead that f fits into a commutative square [page 95]

$$\begin{array}{ccc} V & \xrightarrow{\alpha} & U \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle p} \\ V' & \xrightarrow{\alpha'} & U', \end{array}$$

with α' also a local isomorphism, then we can conclude that $\underline{\pi}_0(f) = \underline{\pi}_0(V/V')$ is indeed aspheric. Indeed, we have a canonical isomorphism in Cat/U',

$$\underline{\pi}_0(\alpha' f) \simeq \underline{\pi}_0(f), \qquad \text{i.e. } \underline{\pi}_0(V/V') \simeq \underline{\pi}_0(V/U'),$$

from the fact e.g. that the canonical factorization of f p. 67



provides a factorization of the same type for $\alpha' f$, as

$$\alpha' f = (\alpha' \pi) \circ \tilde{f}.$$

That said, we can apply the hypothesis to p and we do indeed find the W-asphericity of $\underline{\pi}_0(f) \simeq \underline{\pi}_0(\alpha' f) = \underline{\pi}_0(p\alpha)$.

The natural idea, for a converse to prop. 8, would be to take for U' the amalgamated sum of U and V' over V, which will work provided we know that α' really is a local isomorphism just like α. This is indeed the case e.g. if $V \longrightarrow U$ is an open immersion, i.e. H a subpresheaf of the terminal presheaf. But I fear that this is not true in general - it looks a little too good to be true! This is indeed already false in the case [page 96]

$$\begin{array}{ccc} V = E_G & \xrightarrow{\alpha \text{ can}} & B_G = U \\ \Big\downarrow{\scriptstyle f} & & \\ V' = B_{G'}, & & \end{array}$$

in the case e.g. where $G = \{\pm 1\}$, the group with two elements, and where the element α of G', the image under f of one of the two nonidentity arrows of E_G is $\neq 1$. If H' is the normal subgroup generated by α in G', the amalgamated sum is identified with $B_{G'/H'}$, but $B_{G'} \longrightarrow B_{G'/H'}$ is *not* a local isomorphism (for $(B_{G'})/t_0 \longrightarrow (B_{G'/H'})/t'_0$ is not bijective on objects, being identified with $G' \xrightarrow{q} G'/H' \ \ldots$).

But on the other hand the condition of total W-asphericity on an object H of $U^\wedge$ already seems so demanding, that I can hardly see myself constructing an example where it is satisfied, without U/H itself being totally W-aspheric.

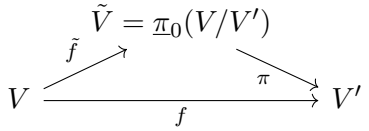
The existence of totally W-aspheric objects in Cat is not so obvious a priori. That there are a good bunch of them is guaranteed by the [page 97]

Proposition 9.

a) Every object V of Cat *having a final object is totally W-aspheric - hence every presheaf of sets H* representable *on a category U (representability meaning precisely that U/H has a final object) is totally W-aspheric.* p. 68

b) ([66]) *(Suppose* W(7 bis) *is satisfied.) Let V be in* Cat, *V', V'' two opens with union V. If V', V'', $V' \cap V''$ are totally W-aspheric, the same holds for V.*

Proof of a). Take V with a final object and $f : V \longrightarrow V'$, whence the factorization



[66]!! Probably *false* as it stands, cf. correction on the next page.

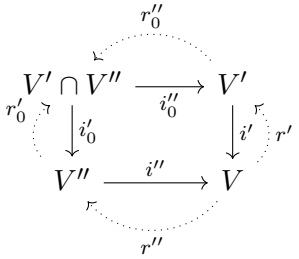
of f. One proves without difficulty ([67]) (a matter of cofinality) the following:

Corollary. *The image under $\tilde{f}$ of a final object of V is a final object of $\tilde{V} = \underline{\pi}_0(V/V')$.*

Since an object of Cat with a final object is W-aspheric by W(3), this establishes a).

Let us prove b). I feel like saying that if we have $f : V \longrightarrow W$ in Cat, then $\underline{\pi}_0(V/W)$ is a union of two opens isomorphic to $\underline{\pi}_0(V'/W)$, $\underline{\pi}_0(V''/W)$, whose intersection is isomorphic to $\underline{\pi}_0(V' \cap V''/W)$. This would mean neither more nor less than this: $f_!$ transforms [page 98] the monomorphisms $V' \longrightarrow V$, $V'' \longrightarrow V$ (interpreted as arrows in $V^\wedge$, with target e_V) into monomorphisms, and the fiber product $V' \cap V'' = V' \times_V V''$ into the corresponding fiber product in $W^\wedge$. But the functor $f_!$ commutes with arbitrary $\varinjlim$ (being a left adjoint), and I see no a priori reason why it should commute (say) with fiber products (which would also imply that it transforms monomorphisms into likewise). If I remember correctly, this is ultrafalse in general. So I now very much doubt that statement b) is true, without an additional hypothesis. To ensure monomorphy, one thinks of requiring the existence of *retractions* of V onto V', V'' and of V' or V'' onto $V' \cap V''$.

I have just checked this, and give the **Corollary 1** (correction of prop. 9, part b) p. 69 ([68]): *The statement is valid if one assumes that there exists a* retraction r'_0 *of V'' onto $V' \cap V''$, and similarly a retraction r''_0 of V' onto $V' \cap V''$.* ([69])



(**NB** We will see later that the existence of *one* of these retractions suffices to ensure the conclusion.)

Proof. Since V is the amalgamated sum of V' and V'', there exists a retraction r' of V onto V' (given by $\mathrm{id}_{V'}$ on V', by $i''_0 r'_0$ on V'') which induces r'_0 on V'', and similarly there exists a retraction r'' of V onto V'', inducing r''_0 on V'. The existence of these [page 99] retractions already ensures that for every functor $f : V \longrightarrow W$, the transform of the diagram

$$\begin{array}{ccc} V' \cap V'' & \longrightarrow & V' \\ \downarrow & & \downarrow \\ V'' & \longrightarrow & V \end{array}$$

[67] Here is a simple proof: For every V in Cat we have $e_V = \varinjlim_{v \text{ in } V} v$ (direct limit of the inclusion functor of V into $V^\wedge$), for $\forall\, x$ in V

$$\varinjlim_{v \text{ in } V} \mathrm{Hom}(x, v) \simeq \pi_0(x \backslash V) = e.$$

Since $f_!$ commutes with $\varinjlim$, we therefore have $f_!(e_V) = \varinjlim_{v \text{ in } V} f(v)$. If v_0 is a final object of V, this shows that $f_!(e_V) = f(v_0)$, q.e.d.

[68] NB There is a slightly more general statement, starting from a category U and a diagram $\begin{array}{ccc} V' \cap V'' & \longrightarrow & V' \\ \downarrow & & \downarrow \\ V'' & \longrightarrow & V \end{array}$ *in* $U^\wedge$: if V', V'', $V' \cap V''$ are totally W-aspheric, the same holds for V.

[69] Corrected corollary also false - the retractions are not arrows of $V^\wedge$, and do not pass into $p_!$.

(interpreted as a diagram in $V^\wedge$) consists of monomorphisms ([70])

$$\begin{array}{ccc} f_!(V' \cap V'') & \hookrightarrow & f_!(V') \\ \downarrow & & \downarrow \\ f_!(V'') & \hookrightarrow & f_!(V), \end{array}$$

and moreover that $f_!(V') \cap f_!(V'')$ (which contains $f_!(V' \cap V'')$) is in fact *equal to it.* It follows that we have a diagram both cartesian and cocartesian of open immersions in Cat

p. 70

$$\begin{array}{ccc} W/f_!(V' \cap V'') & \hookrightarrow & W/f_!(V') \\ \downarrow & & \downarrow \\ W/f_!(V'') & \hookrightarrow & W/\Big(f_!(V) \overset{\text{def}}{=} f_!(e_V)\Big), \end{array}$$

which is also identified with

$$\begin{array}{ccc} \underline{\pi}_0(V' \cap V''/W) & \hookrightarrow & \underline{\pi}_0(V'/W) \\ \downarrow & & \downarrow \\ \underline{\pi}_0(V''/W) & \hookrightarrow & \underline{\pi}_0(V/W). \end{array}$$

The hypothesis on V', V'', $V' \cap V''$ implies that the three vertices of the square other than $\underline{\pi}_0(V/W)$ are W-aspheric, hence by W(7 bis) the same holds for the latter, q.e.d. [page 100]

Corollary 2 ([71]). *Assuming* W(7 bis)*, the category* $\Psi = \begin{array}{ccc} & & a \\ & \nearrow & \\ 0 & & \\ & \searrow & \\ & & b \end{array}$ *(or* $0 \longleftarrow 1 \longrightarrow 2$*)*

is totally W-aspheric ([72]).

One applies corollary 1 to Ψ and to its two open subcategories, Ψ/a, Ψ/b (isomorphic to $\mathbf{\Delta}^1$), and to their intersection $\{0\} \simeq \mathbf{\Delta}^0 = e$. Since these three categories have a final object (unlike Ψ itself, which has only an initial object …), they are totally W-aspheric. The corollary applies, the existence of the retractions r'_0, r''_0 being obvious here.

It is especially part b) (corrected) of proposition 9 (ms.97–100) which ensures the existence of totally W-aspheric categories other than those having a final object ([73]). It is true that conversely one should still make sure that a W-aspheric object X of Cat is not *always* totally W-aspheric! The counterexample is not so obvious after all - I spent hours before finding one and getting a little clarity! It suffices, in short, to exhibit an example of an arrow in Cat [page 101]

$$f : X \longrightarrow Y$$

such that

p. 71

a) $\underline{\pi}_0(f) \overset{\text{def}}{=} \underline{\pi}_0(X/Y) \xrightarrow{\sim} Y$, i.e. the $\pi_0(y\backslash X)$ are 0-connected (for $y \in \mathrm{Ob}\, Y$),

[70] **false,** already in the case where $V = \Psi$, V', V'' its two canonical opens isomorphic to $\mathbf{\Delta}^1$, and $V' \cap V'' = e$.

[71] More generally, if I is a category *ordered and finite,* having an initial object, then

[72] **False,** Ψ is *not* totally W-aspheric.

[73] I now conjecture that these are the only ones! ([74])

[74] False, there are the filtering categories (if W satisfies the axiom of limits).

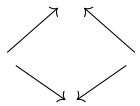
and such that moreover

b) X is W-aspheric, and Y not.

The difficulty is the following, when one looks for "stupid" counterexamples: it is that hypothesis a) already implies, when X is W_∞-aspheric (or only 1-connected) that Y is 1-connected: the functor

$$Y' \longmapsto X' = X \times_Y Y'$$

from the category of étale coverings Y' of Y to that of étale coverings X' of X is fully faithful (immediate verification), whence the stated result. So a counterexample must call upon the existence of a nonzero $\pi_i(Y, y)$, with $i \geq 2$. This implies that there is no [page 102] counterexample with categories of the form $Y = Z/F$ (deduced from a functor $q : X \longrightarrow Z$ by taking $F = q_!(e_X) = \pi_0(X/Z)$), where Z is a category such as $\mathbf{\Delta}^1$ (?), Φ, Ψ, or



(having the homotopy type of the circle) etc., for the connected components of such categories Y are $K(\pi, 1)$'s - the higher π_i ($i \geq 2$) are zero.

This difficulty seen, the counterexample becomes obvious. I start from a 1-connected object *whatsoever* Y of X , and I take $y_0 \in \mathrm{Ob}\, Y$,

$$X = \underline{\mathrm{Ch}}(Y; y_0, -) \xrightarrow{f} Y.$$

We know that X is W_∞-aspheric, and that f is W_∞-perfect, hence the inclusions

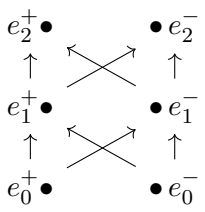
$$X_y \longrightarrow y\backslash X$$

are W_∞-equivalences. Hypothesis a) that the $y\backslash X$ be 0-connected is therefore equivalent to the same hypothesis on the fibers X_y, or again on a single fiber (f being perfect, hence [page 103] a W_∞-fibration, and Y 0-connected), namely

$$X_{y_0} \simeq \Omega(Y, y_0).$$

Its 0-connectedness is equivalent to the hypothesis of 1-connectedness of Y (so far we have used only that of 0-connectedness). So for a counterexample, it suffices to take Y 1-connected and not ∞-connected. The simplest case is the one where Y has the homotopy type of the sphere S^2, realized e.g. by the ordered category shown opposite with 6 vertices.

p. 72



([75]). I summarize the result obtained.

[75] It is in short this example (up to topological homotopy) that I realize (without at first noticing it in the Appendix p. 10-12, but with finite categories).

Proposition 10. *Let Y be a small category which is 1-connected ($y_0 \in \mathrm{Ob}\, Y$). Then $\underline{\mathrm{Ch}}(Y; y_0, -)$ is W_∞-aspheric, but is not totally W_∞-aspheric if Y itself is not ∞-connected, i.e. W_∞-aspheric.* ([76]).

One may ask whether Y W_∞-aspheric implies that $\underline{\mathrm{Ch}}(Y; y_0, -)$ is totally W_∞-aspheric. I very much doubt it!

Let us now return to the situation at the beginning of the section, diagram (∗), p. 87, [page 104]

$$\begin{array}{ccccc} U & \xhookrightarrow{i} & X & \overset{j}{\dashleftarrow\!\dashrightarrow} & T = \{t\} \\ {\scriptstyle p}\downarrow & & {\scriptstyle q}\downarrow & & \vdots\ \text{iso} \\ U' & \xhookrightarrow{i'} & X' & \overset{j'}{\dashleftarrow\!\dashrightarrow} & T' = \{t'\}. \end{array} \qquad (*)$$

We can now give an example where i is in W (and even W-aspheric) without i' being so:

$$i \in W, \qquad i' \notin W. \qquad (**)$$

One can even assume that t is a final object of X (which means that the object H of $U^\wedge$ is isomorphic to e_U, the final object), and t' a final object of X' (i.e. $p_!(e_U) = e_{U'}$, i.e. the $u'\backslash U$ are 0-aspheric). In this case, since X, X' are W-aspheric, the relations (∗∗) are equivalent to

$$U \text{ is } W\text{-aspheric, and } U' \text{ not}, \qquad (***)$$

which is precisely the situation we have just constructed (p. 101-103).

One would also like to find an example where in (∗) one has $p \in W$, $q \notin W$.

I will again take $(X, U, \{t\})$ such that t is a final object of X, but if we want $q \notin W$, we cannot suppose t' a final element of X'. Instead of this, I am going to suppose p. 73 [page 105]

a) U, U' (and also X, having final object t) W-aspheric, hence $i \in W$ (and even i W-aspheric), $p \in W$, but

b) X' not W-aspheric, hence $i', q \notin W$.

For this, let us note that

$$U'/t' \simeq U'/p_!(e_U) \overset{\mathrm{def}}{=} \underline{\pi}_0(U/U').$$

The corollary to proposition 7 (cf. p. 85), not yet proved, tells us that $i' \in W$ (i.e. here, X' W-aspheric) if and only if i' is W-aspheric, i.e. U'/t' is W-aspheric. We shall admit it here. Everything thus comes down to finding an arrow

$$p : U \longrightarrow U'$$

in Cat with the following properties.

a) U and U' are W-aspheric,

b) $\underline{\pi}_0(U/U')$ is not W-aspheric.

We shall take up the preceding example again

$$f : X \longrightarrow Y$$

(p. 101-103), taking

$$U = X, \qquad \underline{\pi}_0(U/U') = Y, \qquad \text{[page 106]}$$

so it all comes down, starting from the given example $X \xrightarrow{f} Y$, to constructing an arrow

$$\pi : Y \longrightarrow U' \qquad (*)$$

in Cat which satisfies the conditions

a) π is a local isomorphism,

b) U' is W-aspheric.

We shall then take

$$p = \pi f.$$

But from the way X has been constructed from a 1-connected and not ∞-connected Y, otherwise arbitrary, we are finally reduced to finding $(*)$ satisfying a), b), and moreover

p. 74

c) Y is 1-connected, but not ∞-connected.

(NB We now suppose $W = W_\infty$.) It amounts to the same thing to find

> U' an *aspheric* object of Cat, $F \in \mathrm{Ob}\, U'^{\wedge}$ such that U'/F is 1-connected and not ∞-connected.

One thinks for example of simply taking an open subcategory U'_0 of U', which is 1-connected and not ∞-connected. Let us start with such a category U'_0 - for example a homotopy realization in Cat, of the sphere S^2. We immerse it as an open in an aspheric U', by adding a final object to it. We have won.

[page 107]

Let us make a statement in due form.

Proposition 11. *One can find a situation* $(*)$ (p. 104) *with*

a) t a final object of X.

b) U, U' (and X) W_∞-aspheric, hence $i, p \in W_\infty$ (and even i W_∞-aspheric).

c) X' not W_∞-aspheric, hence $i', q \notin W_\infty$.

For this it suffices to start with a 1-connected object Y of Cat, to adjoin to it a final object u'_0, which gives U', to take $U = \underline{\mathrm{Ch}}(Y; y_0, -)$, $p : U \longrightarrow U'$ the composite $U \xrightarrow{\text{target}} Y \xrightarrow[\text{can.}]{\text{incl.}} U'$, and X deduced from U by adding a final object t to it. We then have $U'/t' \simeq Y$, whence

$$X' \ W_\infty\text{-aspheric (i.e. } i' \in W_\infty) \Longleftrightarrow Y \ W_\infty\text{-aspheric.}$$

We have the desired counterexample by taking Y (1-connected but) not ∞-connected.

---

[76] But a more striking example is $\Psi = \begin{array}{ccc} & & b \\ & \nearrow & \\ a & & \\ & \searrow & \\ & & c \end{array}$, which is W_∞-aspheric, but not totally!

2.4 Application to open Dwyerian immersions.

Let us give all the same the *positive* statement we were after.

Theorem 3. *Let (X, U, T) be a triple $\in \mathcal{T}$ (U open in X in* Cat*, $T = X\backslash U$), and for every arrow $p : U \longrightarrow U'$ an arrow of* Cat*, the cocartesian square in* Cat

$$\begin{array}{ccccc} U & \overset{i}{\hookrightarrow} & X & \dashleftarrow & T \\ {\scriptstyle p}\downarrow & & {\scriptstyle q}\downarrow & & \vdots\,{\scriptstyle \text{iso}} \\ U' & \overset{i'}{\hookrightarrow} & X' & \dashleftarrow & T'. \end{array}$$

Suppose that for every $t \in \dot{T} = \dot{U}$, the presheaf of sets p. 75 [page 108]

$$i^*(t) = (u \longmapsto \mathrm{Hom}(u, t)) \in U^\wedge$$

is totally W-aspheric (def. 1, p. 88) ([77]) - e.g. that the category U/t is totally W-aspheric (prop. 8, page 93). (NB the case where the U/t have a final object, cf. prop. 9 a), i.e. the one where the inclusion $U \longrightarrow \bar{U}$ admits a right adjoint, is the one envisaged by Dwyer, that of open Dwyer immersions ([78]).) Then for every $p : U \longrightarrow U'$, the open immersion

$$i'_0 : U' \longrightarrow \overline{U'}$$

is W-aspheric (and even, just like i_0, strictly W-aspheric), and a fortiori it is W. Consequently (prop. 6) we have the following.

ⓐ	$i \in W$	$\Longleftrightarrow$	$i' \in W$	*assuming* W(7 bis).
ⓑ	$p \in W$	$\Longleftrightarrow$	$q \in W$	*assuming* W(7). ([79])

Corollary 1. *Assume* W(7)*. If $i : U \hookrightarrow X$ is an open immersion such that $i_0 : U \hookrightarrow \bar{U}$ is totally W-aspheric, then $i : U \longrightarrow X$ is a W-cofibration ("ultrastrong"), by which I mean this : for every arrow $U' \xrightarrow{f} U''$ in $U\backslash$*Cat*,*

*such that $f \in W$, the arrow $g : X' \longrightarrow X''$ which is deduced from it by cobase change from the object X of $U\backslash$*Cat*, is also in W.* [page 109]

Proof of the corollary : we apply theorem 3 to the open immersion $U' \overset{i'}{\hookrightarrow} X'$ deduced from $U \overset{i}{\hookrightarrow} X$ by the cobase change $U \xrightarrow{p} U'$, noting in addition that the hypothesis of theorem 3 on i is preserved in passing from i to i'.

We must all the same introduce adequate terminology :

Definition 2 :

a) An arrow $f : X \longrightarrow Y$ is said to be *strictly W-aspheric* if for every $y \in \mathrm{Ob}\, Y$, $f^*(y) = (x \longmapsto \mathrm{Hom}_Y(f(x), y))$ is a totally W-aspheric object of $X^\wedge$ (cf. def. 1, page 88). It is said to be *totally W-aspheric* if for every $y \in \mathrm{Ob}\, Y$, X/y ($\simeq X/f^*(y)$) is a totally W-aspheric object of Cat.

p. 76

b) An open immersion $i : U \longrightarrow X$ is said to be *W-Dwyerian* (resp. *totally Dwyerian*) if the induced immersion $i_0 : U \longrightarrow \bar{U}$ is strictly (resp. totally) W-aspheric. A closed immersion $j : T \longrightarrow X$ is said to be *W-Dwyerian* (resp. *totally W-Dwyerian*)

[77] **NB** One says under these conditions that $i_0 : U \longrightarrow \bar{U}$ is *strictly W-aspheric* (cf. def. 2 next page).

[78] I know of no other case where th. 3 applies - for lack of knowing of a totally W-aspheric category which does not have a final object, nor of an object F of a category $U^\wedge$ which is totally W-aspheric without being representable, i.e. U/F has a final object. Now, I know that *filtering* categories are totally W-aspheric if W satisfies the axiom of direct limits.

[79] Proof page 112.

if and only if $j^\circ : T^\circ \longrightarrow X^\circ$ is an open W-Dwyerian immersion (resp totally W-Dwyerian).

c) An arrow $i : U \longrightarrow X$ of Cat (not necessarily an immersion)

is called a *W-cofibration,* if for every arrow $f : U' \longrightarrow U''$ in $U\backslash$Cat, the arrow $g : X' \longrightarrow X''$ which is deduced from it by cobase change by X (object of $U\backslash$Cat) is in W when f is. [page 110]

NB This notion of W-cofibration is stable under composition and under cobase change, and the isomorphisms of Cat are W-cofibrations. If f is a W-cofibration, f° is one and conversely.

(The corollary is stated by saying that an open or closed immersion which is W-Dwyerian is a W-cofibration.)

d) An arrow $i : U \longrightarrow X$ in Cat is called a *trivial W-cofibration*, or a *couniversal W-equivalence,* if for every U' in $U\backslash$Cat, i.e. every arrow $p : U \longrightarrow U'$ in Cat, the arrow deduced from i,

$$i' : U' \longrightarrow X' = U' \amalg_U X \quad ,$$

is in W. (NB It is immediate that this indeed implies that i is a W-cofibration.)

(This notion of trivial W-cofibration is also stable under composition and under cobase change, and the isomorphisms in Cat are such.)

Corollary 2. (Assuming W(7).) *Let $i : U \longrightarrow X$ be an open W-Dwyerian immersion. For i to be a trivial W-cofibration, it is necessary and sufficient that $i \in W$, or what amounts to the same thing* (cf. page 82), *that $\dot{T} \longrightarrow T$ be in W.* [page 111]

This is obviously necessary, and sufficiency follows from theorem 3 a).

Open questions.

a) I do not know whether an open immersion i which is a W-cofibration, or even a trivial W-cofibration, is necessarily W-Dwyerian ([80]). But by the corollary to proposition 7 (p. 85), I know that this is the case if we suppose T isomorphic to the final category e. At least this follows : For not only $i : U \longrightarrow X$, but also all the open immersions $i_1 : U \longrightarrow X_1$ which are deduced from it by the procedure of prop. 7 b) by means of arbitrary arrows $T_1 \longrightarrow T$ with target T in Cat, to be W-cofibrations, it is necessary and sufficient that i be Dwyerian.

b) Is there a good notion of a Dwyerian immersion, applicable to all immersions and including the case of open immersions and closed immersions? p. 77 [page 112]

Proof of theorem 3 (p. 107). It suffices to prove that if i is W-Dwyerian (i.e. i_0 strictly W-aspheric), then i' is so as well - for it will follow that i'_0 is W-aspheric and a fortiori $\in W$. We must therefore verify that for every $t' \in \dot{U}' = \dot{T}$, the element $i'^*(t') = (u' \longmapsto \mathrm{Hom}_{X'}(u', t'))$ of $U'^\wedge$ is totally W-aspheric. Now t' comes from a unique element t of $\dot{T}$, and we know that we have

$$i'^*(t') = p_!(i^*(t)) \quad , \qquad \text{where } \ p_! : U^\wedge \longrightarrow U'^\wedge.$$

By hypothesis, $F = i^*(t)$ is a totally W-aspheric element of $U^\wedge$, which means that for every arrow $p : U \longrightarrow U'$ in Cat with source U, $p_!(F)$ is a W-aspheric element of $U'^\wedge$. In fact, we have the immediate lemma (by transitivity $(p'p)_! \simeq p'_! p_!$).

[80] and perhaps even a Dwyer immersion?

Lemma. *Let U be in* Cat*, and F in $U^\wedge$. If F is totally W-aspheric (resp. if U/F is totally W-aspheric), then for every arrow $p : U \longrightarrow U'$ in* Cat *with source U, $F' = p_!(F)$ is a totally W-aspheric object of $U'^\wedge$ (resp.*

U'/F' is a totally W-aspheric object of Cat*.)* [page 113]

Corollary 3 (to theorem 3). *If $i : U \longrightarrow X$ is an open or closed W-Dwyerian immersion (resp. totally W-Dwyerian), then for every arrow $p : U \longrightarrow U'$ in* Cat*, $i' : U' \longrightarrow X' = U' \amalg_U X$ is so too.*

(NB The non-resp. case is essentially theorem 3. It does not depend on any additional axiom on W, but is of interest only in view of the equivalences ⓐ and ⓑ (p. 108) which result from it, and which are valid assuming W(7 bis) resp. W(7) only.)

Comment on motivation : open or closed W-Dwyerian immersions, or totally W-Dwyerian ones, are stable under cobase change (by the preceding corollary), but do not at all seem to be so under composition. In this respect, "Dwyer immersions" (where one supposes the existence of the right adjoint to $i_0 : U \longrightarrow \bar{U}$, in the case of an open immersion) behave better, being stable under composition. (**NB** It does not seem

clear either that strictly W-aspheric arrows, or totally W-aspheric ones, are stable under composition - sorry!) My motivation for wearing myself out over the W-Dwyerian notion is twofold. [page 114]

a) I would like to have conditions at once comprehensible, and as general as possible, on an immersion (open e.g.), for it to be a W-cofibration, a visibly important property (just like the dual notion of W-fibration).

p. 78

b) I would like, for later applications, to have at my disposal a class of open immersions having the following properties:

   1°) They are W-cofibrations, and

   2°) If $U \overset{i}{\hookrightarrow} X$ is in the class, and if $\varepsilon \overset{j}{\to} I$ is the "source-target" inclusion of the discrete category $\{0, 1\}$ into the interval

$$\Phi_n = 0 \longrightarrow 1 \longleftarrow 2 \longrightarrow \ldots 2n-1 \longleftarrow 2n$$

   ($j(0) = 0$, $j(1) = 2n$), $n \geq 2$, the inclusion α of $(U \times I) \cup (X \times \varepsilon)$ into $X \times I$ is in the class.

Now this last stability property is *not* satisfied by Dwyer immersions. (Take $i : \{0\} \longrightarrow \mathbf{\Delta}^1 = \{0, 1\}$, the inclusion α is not a Dwyer immersion.)

[page 115]

2.5 Complement: criterion for total asphericity of a W-proper category over another.

Let

$$f : X \longrightarrow I$$

be a proper category over an index category I. If I has a final object i_0, then X/i_0 is identified with X, and consequently the inclusion

$$X_{i_0} \longrightarrow X$$

of the fiber of X at i_0 is W-coaspheric , and a fortiori is in W. Thus X is aspheric if and only if X_{i_0} is. We want to extend this result to a criterion for *total* W-asphericity, more precisely to prove the

Proposition 11: *Let $f : X \longrightarrow I$ be W-proper, and let i_0 be a final object of I. If X_{i_0} is totally W-aspheric* (def. 1, page 88), *the same is true of X.* (Provided that W satisfies the connectedness axiom W(5).)

The proof will be given below, after some preliminaries of independent interest. First of all, let us recall that if X is W-proper over I, then it gives rise to a functor

$$\begin{array}{ccc} I & \longrightarrow & \mathrm{Hot}_W \\ i & \longmapsto & \mathrm{hot}_W(X_i)(\simeq \mathrm{hot}_W(X/i)); \end{array}$$

composing it with the functor

$$\pi_0 : \mathrm{Hot}_W \longrightarrow \mathrm{Ens}$$

(provided that the connectedness axiom W(5) (p. 10) is satisfied) we find a functor p. 79 [page 116]

$$i \longmapsto \underbrace{\pi_0(X_i)}_{\simeq \pi_0(X/i)} : I \longrightarrow \mathrm{Ens}.$$

This being so, we have the sorital result

Proposition 12: *Let $f : X \longrightarrow I$ be W-proper, and suppose that W satisfies the connectedness axiom* W(5) (otherwise, W-properness will be of no use to us …) ([81]). *We then have a canonical bijection*

$$\pi_0(X) \xleftarrow{\sim} \varinjlim_I \pi_0(X_i), \qquad (*)$$

where the arrow is induced by the canonical maps $\pi_0(X_i) \longrightarrow \pi_0(X)$.

Proof. Passing to $f^\circ : X^\circ \longrightarrow I^\circ$, we are reduced to proving the same result for X *W-smooth* over I, with the only difference that the functor

$$i \longmapsto \pi_0(X_i)$$

on I will be contravariant, i.e. a presheaf, which offers the advantage of the intuition attached to the formalism of presheaves. Let

$$g : I \longrightarrow e, \qquad \text{whence } gf = p : X \longrightarrow e,$$

it is immediate, by the standard calculation

$$p_!(F) = \pi_0(X/F) \qquad (F \in \mathrm{Ob}\, X^\wedge)$$

(where we have identified $e^\wedge$ with (Ens)), that we have

$$\pi_0(X) = p_!(e_X) = g_!(f_!(e_X)).$$

Now the presheaf

$$i \longmapsto \pi_0(X_i) \qquad (\simeq \pi_0(i\backslash X), \text{ since } X \text{ } W\text{-smooth over } I)$$

is none other than $f_!(e_X)$. On the other hand, for any presheaf G on I, we have

$$g_!(G) \simeq \pi_0(I/G) \simeq \varinjlim_{I^\circ} G(i), \qquad \text{[page 117]}$$

and applying this to $f_!(e_X)$, we find the formula $(*)$ of proposition 12.

Let us return to the situation with *f W-proper,* and consider a functor

$$p : X \longrightarrow Y, \qquad \text{p. 80}$$

we are interested in "calculating"

$$p_!(e_X) = (y \longmapsto \pi_0(y\backslash X)) \in \mathrm{Ob}\, Y^\wedge.$$

Let us consider for this the following diagram

$$\begin{array}{ccccc} y\backslash X_i & \longrightarrow & y\backslash X & \longrightarrow & y\backslash Y \\ \downarrow & & \downarrow & & \downarrow \\ X_i & \longrightarrow & X & \xrightarrow{p} & Y \\ \downarrow & & \downarrow f & & \\ \underset{\simeq e}{\{i\}} & \longrightarrow & I & & \end{array} \quad ,$$

where the three squares are cartesian. Since $y\backslash Y \longrightarrow Y$ is W-proper, the same is true of $y\backslash X \longrightarrow X$ by base change, hence of the composite $y\backslash X \longrightarrow X \longrightarrow I$ by transitivity. The diagram shows on the other hand that the fiber of $y\backslash X$ at $i \in \mathrm{Ob}\, I$ is identified with $y\backslash X_i$.

Proposition 12 therefore gives us the formula [page 118]

$$\pi_0(y\backslash X) = \varinjlim_I \pi_0(y\backslash X_i),$$

which formula cannot help being functorial in y, just as that of prop. 12 is in $\underline{\mathrm{Prop}}_W I$. Now for y variable in Y, the presheaf on Y

$$y \longmapsto \pi_0(y\backslash X_i)$$

is none other than $p_{i\,!}(e_{X_i})$, where

$$p_i : X_i \longrightarrow Y$$

is the composite $X_i \longrightarrow X \xrightarrow{p} Y$ (induced by p). We have thus proved

Corollary 1 (to prop. 12). *Let $f : X \longrightarrow I$ be W-proper ([82]), and $p : X \longrightarrow Y$ any functor. Let, for every $i \in \mathrm{Ob}\, I$, $p_i : X_i \longrightarrow Y$ be the functor induced on the fiber X_i of X at i. We then have a natural* functorial *law*

$$i \longmapsto p_{i\,!}(e_{X_i}) : I \longrightarrow Y^\wedge,$$

and we have the formula

$$p_!(e_X) \xleftarrow{\sim} \varinjlim_I p_{i\,!}(e_{X_i}), \qquad (*)$$

where the arrow is induced by the canonical morphisms

$$p_{i\,!}(e_{X_i}) \longrightarrow p_!(e_X)$$

induced by the inclusions $X_i \longrightarrow X$. p. 81

The functorial law of $p_{i\,!}(e_{X_i})$ in i has been made explicit in the course of the proof. [page 119]

Let us note that when X over I is cofibered and split, i.e. comes from a functor

$$i \longmapsto X_i, \quad I \longrightarrow \mathrm{Cat},$$

[81] It is better to say: Let f be W_0-proper, where W_0 is the localizer of 0-connectedness …

[82] say W_0-proper …

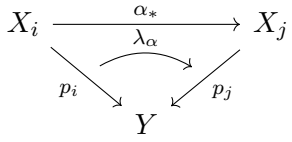
then the functorial law of $p_{i!}(e_{X_i})$ in i is made explicit in a particularly pleasant way: to an arrow

$$\alpha : i \longrightarrow j, \qquad \text{whence } \alpha_* : X_i \longrightarrow X_j,$$

is associated

$$p_!(\alpha) : p_{i!}(e_{X_i}) \longrightarrow p_{j!}(e_{X_j}),$$

noting that we have a diagram



with a commutativity arrow

$$\lambda_\alpha : p_i \longrightarrow p_j \alpha_*$$

(not isomorphism in general), whence

$$p_{i!}(e_{X_i}) \xrightarrow{(\lambda_\alpha)_!} p_{j!}\underbrace{\alpha_{*!}(e_{X_i})}_{\longrightarrow e_{X_j}} \longrightarrow p_{j!}(e_{X_j}),$$

no more doubt that's the one! ([83]).

Corollary 2: *Under the conditions of corollary 1, suppose that I has a final object i_0. Then* [page 120]

$$(p_{i_0})_!(e_{X_{i_0}}) \longrightarrow p_!(e_X)$$

is an isomorphism in $Y^\wedge$.

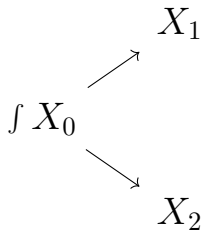
Whence at once proposition 11: if X_{i_0} is totally W-aspheric, then X is too, since for every $p : X \longrightarrow Y$, $p_!(e_X) \simeq p_{i_0!}(e_{X_{i_0}})$, and this latter element of $Y^\wedge$ is W-aspheric by the hypothesis on X_{i_0}.

Let us note the special case:

Corollary 3. *Let Z be a totally W-aspheric object in* Cat. *Then its product $Z \times I$ with any I in* Cat *having a final object is also totally W-aspheric.* p. 82

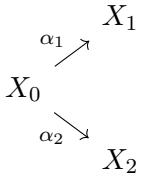
[page 121]

3 Amalgamated sums and integration



3.1 Canonical reduction to the case of a diagram of closed immersions

Let us consider a diagram



(∗)

[83] This arrow should be made explicit with care!

in Cat. We can regard it as a contrafunctor

$$\Phi^\circ \longrightarrow \mathrm{Cat},$$

where

$$\Phi = \begin{pmatrix} & & 1 \\ & \swarrow & \\ 0 & & \\ & \nwarrow & \\ & & 2 \end{pmatrix},$$

thus as defining a (split) fibered category $\mathcal{X}$ over Φ. It is sometimes convenient to represent this category $\mathcal{X}$ by the "integration sign"

$$\begin{array}{ccc} & \nearrow & X_1 \\ \int X_0 & & \\ & \searrow & \\ & & X_2, \end{array}$$

and regard it as a sort of "homotopy amalgamated sum" of X_1 and X_2 under X_0, or as a "homotopy direct limit" of the original diagram $(*)$.

By this one is to understand that (for a given $W \subseteq \mathrm{Fl}(\mathrm{Cat})$) the image $\mathrm{hot}_W(\mathcal{X})$ of $\mathcal{X}$ in Hot_W has a remarkable "homotopy meaning" in terms of $(*)$, more precisely, the diagram $(*)$ defines an object ξ of $\mathrm{HOT}_W(\Phi)$ (which is a category of fractions of the category of all diagrams of type $(*)$ …), and if [page 122]

$$\pi : \Phi \longrightarrow e$$

denotes the absolute projection of Φ, we have

$$\pi_!(\xi) = \mathrm{hot}_W(\mathcal{X}),$$

by the known computation of the functor $\pi_!$, left adjoint to the functor π^* p. 83

$$\mathrm{HOT}_W(e) \overset{\mathrm{def}}{=} \mathrm{Hot}_W \longrightarrow \mathrm{HOT}_W(\Phi)$$

([84]). Now for every object I in Cat, with absolute projection $p_I : I \longrightarrow e$, the functor

$$p_{I\,!} : \mathrm{HOT}_W(I) \longrightarrow \mathrm{HOT}_W(e) = \mathrm{Hot}_W$$

must be regarded as the "homotopic" equivalent of the direct limit functor $\varinjlim_I$ on diagrams of type I° in the category Hot_W - which direct limits *do not exist* in Hot_W (except for very special categories - the only ones I know are the discrete categories (which perhaps even have to be assumed finite as well)). [page 123]

Let us note that we have a diagram of canonical functors

$$\begin{array}{ccccc} & & X_0 & & \\ & \swarrow^{\alpha_1} & & \searrow^{\alpha_2} & \\ X_1 & & \text{not commuta-tive!} & & X_2 \\ & \searrow_{i_1} & & \swarrow_{i_2} & \\ & & \mathcal{X}, & & \end{array} \qquad (*)$$

[84] This assumes that we have W(2), not only W(2 bis).

which diagram has nothing commutative about it in Cat. *It is however commutative in* Hot_W (whatever the basic localizer W), and even in Cat hot, for if we denote by

$$i_0 : X_0 \longrightarrow \mathcal{X}$$

the canonical inclusion, we have canonical homomorphisms

$$i_1\alpha_1 \longrightarrow i_0 \longleftarrow i_2\alpha_2,$$

which shows that $i_1\alpha_1$ and $i_2\alpha_2$ are homotopic, hence define the same arrow in Hot_W.

Let us consider in Φ the two closed subsets isomorphic to $\mathbf{\Delta}^1$ [page 124]

$$\begin{aligned} F_1 = 1\backslash\Phi \quad &= \{0,1\} \\ F_2 = 2\backslash\Phi \quad &= \{0,2\}, \end{aligned}$$

we have

$$F_1 \cap F_2 = 0\backslash\Phi = \{0\}, \qquad F_1 \cup F_2 = \Phi.$$

Let p. 84

$$\begin{aligned} 1\backslash\mathcal{X} = \mathcal{X} \times_\Phi 1\backslash\Phi \quad &= \mathcal{X} \times_\Phi F_1 \\ 2\backslash\mathcal{X} = \mathcal{X} \times_\Phi 2\backslash\Phi \quad &= \mathcal{X} \times_\Phi F_2 \end{aligned}$$

be the inverse images of F_1, F_2 in $\mathcal{X}$, we therefore have

$$1\backslash\mathcal{X} \cap 2\backslash\mathcal{X} = X_0, \qquad 1\backslash\mathcal{X} \cup 2\backslash\mathcal{X} = \mathcal{X},$$

so the diagram

$$\begin{array}{ccccc} & & X_0 & & \\ & \overset{\alpha'_1}{\swarrow} & & \overset{\alpha'_2}{\searrow} & \\ 1\backslash\mathcal{X} & & & & 2\backslash\mathcal{X} \\ & \underset{i'_1}{\searrow} & & \underset{i'_2}{\swarrow} & \\ & & \mathcal{X}, & & \end{array} \qquad (**)$$

deduced from the similar diagram

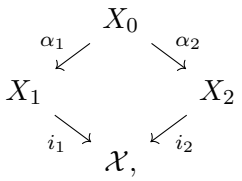
$$\begin{array}{ccccc} & & \{0\} & & \\ & \swarrow & & \searrow & \\ F_1 & & & & F_2 \\ & \searrow & & \swarrow & \\ & & \Phi, & & \end{array}$$

is (like the latter) a square of closed immersions both cartesian and cocartesian. On the other hand, we have a canonical homomorphism from the diagram $(*)$ to $(**)$, which is [page 125] the identity on the vertices X_0 and $\mathcal{X}$, and which on X_1, X_2 is given by the canonical inclusions

$$\underbrace{X_1}_{\mathcal{X}_1 \text{ fiber of } \mathcal{X} \text{ at } 1} \longrightarrow 1\backslash\mathcal{X}, \qquad \underbrace{X_2}_{\mathcal{X}_2 \text{ fiber of } \mathcal{X} \text{ at } 2} \longrightarrow 2\backslash\mathcal{X}.$$

These inclusions, as we know, are W-aspheric (they even have a right adjoint), a fortiori they are in W. Thus the noncommutative diagram $(*)$ is isomorphic, as a diagram in Hot_W, to the commutative cartesian and cocartesian diagram $(**)$ formed of closed immersions. Since $\mathcal{X}$ is indeed an amalgamated sum of its two closed subsets $1\backslash\mathcal{X}$, $2\backslash\mathcal{X}$ under their intersection X_0, this supports the intuition which sees $\mathcal{X}$ as a "homotopic amalgamated sum". Let us give at once the following application: p. 85

Proposition 13. *Let there be a diagram* $X_0 \xrightarrow{\alpha_1} X_1$, $X_0 \xrightarrow{\alpha_2} X_2$ *in* Cat, *whence the diagram* (∗), *p. 123,*



commutative in Hot_W *(but not in* Cat, *unless* $X_0 = \emptyset$*). Let us suppose that* W *satisfies* W(7 bis) (cocartesian square axiom, cf. p. 75). *Then we have* [page 126]

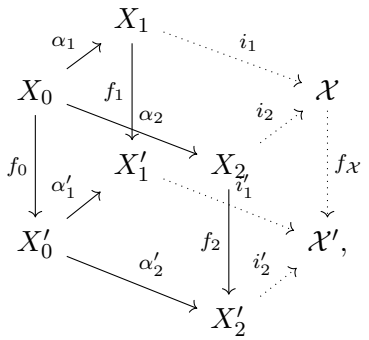
$$\begin{cases} \alpha_1 \in W \iff i_2 \in W \\ \alpha_2 \in W \iff i_1 \in W. \ ([85]) \end{cases}$$

Proof. Using the preceding homomorphism from (∗) to (∗∗), we see that each of the arrows α_1, α_2, i_1, i_2 is in W, if and only if the corresponding arrow α'_1, α'_2, i'_1 or i'_2 is. Thus the above equivalences are equivalent to

$$\begin{aligned} \alpha'_1 \in W &\iff i'_2 \in W \\ \alpha'_2 \in W &\iff i'_1 \in W, \end{aligned}$$

which are true by virtue of W(7 bis) (dual formulation) applied to $\mathcal{X}$ and to the two closed subsets $1\backslash\mathcal{X}$, $2\backslash\mathcal{X}$ of $\mathcal{X}$.

Corollary: *Let us consider* two *diagrams of type* Φ°*, and a homomorphism* $f = (f_0, f_1, f_2)$ *from one to the other (cf. the diagram in solid lines below):*



whence a homomorphism [page 127]

$$f_{\mathcal{X}} = \int f : \quad \mathcal{X} = \int \begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array} \quad \longrightarrow \quad \mathcal{X}' = \int \begin{array}{ccc} & & X'_1 \\ X'_0 & \nearrow & \\ & \searrow & \\ & & X'_2, \end{array}$$

fitting into a cubical diagram (cf. the preceding diagram completed by dotted lines). Let us consider the following conditions. p. 86

[85] **NB** For the implication $\Longrightarrow$ we need only the direct form of W(7 bis), i.e. (with the notations of XVI) of Loc(6′a).

a) $f_0 \in W$.

b) f_1 *and* f_2 *are* W.

c) $f_{\mathcal{X}} = \int(f_0, f_1, f_2) \in W$.

Let us suppose the axiom W(7) ("5 lemma", cf. page 76) *is satisfied. Then two among the conditions a), b), c) imply the third*

$$\begin{cases} a \,\&\, b & \Longrightarrow & c \\ b \,\&\, c & \Longrightarrow & a \\ c \,\&\, a & \Longrightarrow & b. \end{cases}$$

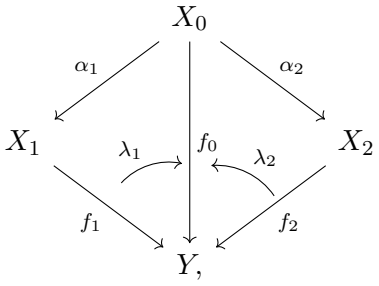
Indeed, we have a homomorphism from the cubical diagram above to the similar diagram, where X_1, X_2, X'_1, X'_2 are replaced respectively by the closed subsets $1\backslash\mathcal{X}$, $2\backslash\mathcal{X}$ of $\mathcal{X}$, and $1\backslash\mathcal{X}'$, $2\backslash\mathcal{X}'$ of $\mathcal{X}'$, the two squares, upper and lower, being inclusions of closed subsets of $\mathcal{X}$ resp. of $\mathcal{X}'$. Thus, conditions a), b), c) are equivalent respectively to the similar conditions for the new cubical diagram (a) and c) moreover not having changed, and b) giving [page 128]

b′) $f'_1 : 1\backslash\mathcal{X} \longrightarrow 1\backslash\mathcal{X}'$ and $f'_2 : 2\backslash\mathcal{X} \longrightarrow 2\backslash\mathcal{X}'$ are in W.

). Then the conclusion of the corollary is equivalent to W(7) (dual formulation) in the case of the new cubical diagram, q.e.d.

3.2 Comparison of $\int \begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array}$ with an ordinary amalgamated sum. W-homotopically cocartesian squares.

The integrated category $\mathcal{X} = \int \begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array}$ is a solution of the following universal problem, a 2-categorical variant of the one which defines the direct limit of diagrams. It is a matter, for Y a variable object in Cat, of considering the systems of three functors f_0, f_1, f_2 and of the two functorial homomorphisms λ_1, λ_2: p. 87



$(*)$

$$\begin{aligned} \lambda_1 : \quad f_1\alpha_1 &\longrightarrow f_0 \\ \lambda_2 : \quad f_2\alpha_2 &\longrightarrow f_0. \end{aligned}$$

[page 129]

To give oneself an arrow of Cat

$$\mathcal{X} \longrightarrow Y$$

"comes to the same thing" as giving oneself a system $(f_0, f_1, f_2, \lambda_1, \lambda_2)$ as above, and this "functorially in Y". This is the special case of the general determination of the functors from the "integrated" fibered category of a contrafunctor

$$I^\circ \longrightarrow \mathrm{Cat}, \qquad \begin{array}{ccc} i & \longmapsto & X_i \\ u & \longmapsto & u^* \end{array}$$

to a category Y, in terms of a system of functors

$$X_i \xrightarrow{f_i} Y,$$

and of functorial homomorphisms

$$f_i \circ u^* \xrightarrow{\lambda_u} f_j$$

for $u : i \longrightarrow j$, whence $u^* : X_j \longrightarrow X_i$, with transitivity conditions for composite arrows, which we need not take into account in the present case, seeing that there are no composite arrows to look at in $I = \Phi$ (except for composites with an identity arrow, which we can dispense with looking at here …).

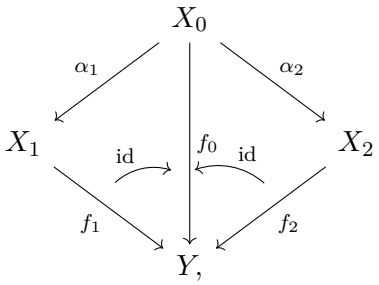
If in particular we have a *commutative* diagram in Cat [page 130] p. 88

$$Q \quad = \quad \begin{array}{ccccc} & & X_0 & & \\ & \overset{\alpha_1}{\swarrow} & & \overset{\alpha_2}{\searrow} & \\ X_1 & & & & X_2 \\ & \underset{f_1}{\searrow} & & \underset{f_2}{\swarrow} & \\ & & Y, & & \end{array}$$

we can complete it by

$$f_0 \overset{\text{def}}{=} f_1\alpha_1 = f_2\alpha_2$$

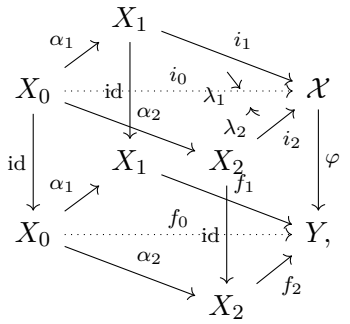
into a diagram



which shows that we have a datum of the type $(*)$ above, thus defining an arrow

$$\varphi_Q : \quad \mathcal{X} = \int \begin{array}{c} X_1 \\ \nearrow \\ X_0 \\ \searrow \\ X_2 \end{array} \qquad \longrightarrow \qquad Y$$

which gives rise to a cubical diagram [page 131]



commutative except for the top square, i.e. we have p. 89

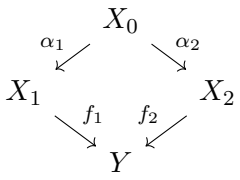
$$\varphi i_1 = f_1, \qquad \varphi i_2 = f_2, \qquad \text{and moreover} \quad \varphi i_0 = f_0,$$

φ moreover transforming the functorial homomorphisms

$$\lambda_1 : i_1\alpha_1 \longrightarrow i_0, \qquad \lambda_2 : i_2\alpha_2 \longrightarrow i_0$$

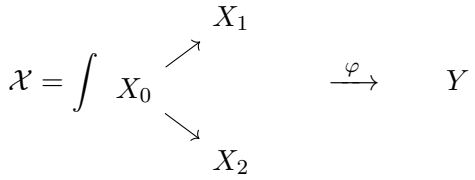
into the relevant identity functorial homomorphisms.

Definition 3. We say that the commutative diagram



$(*)$

is *W-homotopically cocartesian* ([86]) if the canonical homomorphism



it defines is in W. ([87])

We then have the: [page 132]

Corollary 1. *Suppose* W(7 bis). *If the square* $(*)$ *is W-homotopically cocartesian, then*

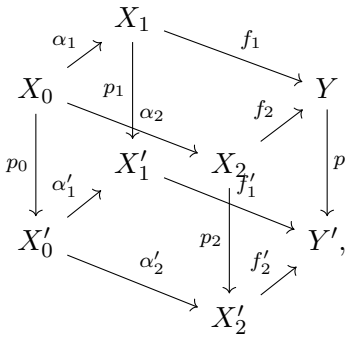
$$\begin{cases} \alpha_1 \in W & \Longleftrightarrow \quad f_2 \in W \\ \alpha_2 \in W & \Longleftrightarrow \quad f_1 \in W. \end{cases}$$

This follows at once from the cubical diagram on the preceding page, where all the vertical arrows are in W, and from the similar statement concerning the top face of the cube, given in proposition 13, p. 125.

Corollary 2. *Suppose* W(7). *Consider a cubical diagram* p. 90

[86] or simply W-cocartesian, or that it defines Y as a W-homotopic amalgamated sum of etc.

[87] Introduce φ in a proposition in due form, and give its essential properties: strictly cartesian arrows are transformed into *identity* arrows + lemmas 1 and 2 of pp. 152, 154.



where the two faces, bottom and top, of the cube are W-homotopically cocartesian squares. Consider the conditions

[page 133]

a) $p_0 \in W$.

b) p_1 *and* $p_2 \in W$.

c) $p \in W$.

Then two of these conditions imply the third.

Follows at once from definition 3, and from the similar result (cor. page 126) for the similar cubical diagram, with Y, Y' replaced by $\mathcal{X} = \int \begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & \\ & \searrow & \\ & & X_2 \end{array}$ and $\mathcal{X}' = \int \begin{array}{ccc} & & X'_1 \\ & \nearrow & \\ X'_0 & & \\ & \searrow & \\ & & X'_2 \end{array}$.

3.3 Proper integration versus smooth integration. Statement of equivalence between the two points of view. Comments on the case of arbitrary W-homotopic limits.

Definition 3, and the preceding corollaries, are of interest only insofar as we have workable criteria, allowing us to assert that a commutative square really is homotopically cocartesian. This is what we are going to try to do shortly. There is also a perplexity, concerning the privileged role played, in the theory developed here, by closed immersions, to the exclusion of open immersions. We have in short reduced, from the homotopical p. 91

[page 134]

point of view, arbitrary diagrams $\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & \\ & \searrow & \\ & & X_2 \end{array}$ in Cat, to the case of a diagram of closed immersions, where the amalgamated sum takes on a familiar look. But why not just as well reduce it to the case of open immersions? We would then consider the initial

diagram as defining a *cofibered* category over

$$\Phi^\circ = \Psi = \quad \begin{array}{ccc} & & 1 \\ & \nearrow & \\ 0 & & \\ & \searrow & \\ & & 2, \end{array}$$

say $\tilde{\mathcal{X}}$. We have in Ψ the two opens

$$\underbrace{\Psi/1}_{=\{0,1\}} = U_1, \qquad \underbrace{\Psi/2}_{=\{0,2\}} = U_2, \qquad U_1 \cap U_2 = \{0\},$$

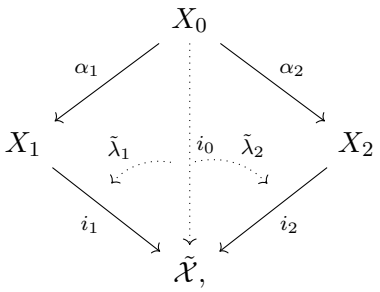
whence in $\tilde{\mathcal{X}}$ two inverse image opens

$$\tilde{\mathcal{X}}/1, \qquad \tilde{\mathcal{X}}/2, \qquad \text{with } \tilde{\mathcal{X}}/1 \cap \tilde{\mathcal{X}}/2 = X_0,$$

whence a commutative diagram of open inclusions

$$\begin{array}{ccccc} & & X_0 & & \\ & \swarrow^{\alpha_1'} & & \searrow^{\alpha_2'} & \\ \tilde{\mathcal{X}}/1 & & & & \tilde{\mathcal{X}}/2 \\ & \searrow_{i_1'} & & \swarrow_{i_2'} & \\ & & \tilde{\mathcal{X}}, & & \end{array} \qquad (*)$$

which receives the noncommutative diagram [page 135]



the arrows

$$X_1 \longrightarrow \tilde{\mathcal{X}}/1, \qquad X_2 \longrightarrow \tilde{\mathcal{X}}/2$$ p. 92

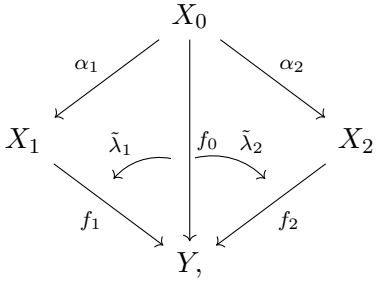
being in W (and even W-coaspheric - they even have a left adjoint).

So we find, as before

$$\begin{cases} \alpha_1 \in W & \Longleftrightarrow \quad i_2 \in W \\ \alpha_2 \in W & \Longleftrightarrow \quad i_1 \in W, \end{cases}$$

and likewise we can formulate a "five lemma" like the corollary on page 125 , but with $\tilde{\mathcal{X}}$, $\tilde{\mathcal{X}}'$ instead of $\mathcal{X}$, $\mathcal{X}'$.

Moreover, $\tilde{\mathcal{X}}$ is the solution of a universal problem in Cat, in terms of data



[page 136]

the only difference from the one formulated on page 128 being the opposite direction of the arrows λ_1, λ_2. For every commutative square

$$\begin{array}{ccccc} & & X_0 & & \\ & \overset{\alpha_1}{\swarrow} & & \overset{\alpha_2}{\searrow} & \\ X_1 & & & & X_2 \\ & \underset{f_1}{\searrow} & & \underset{f_2}{\swarrow} & \\ & & Y, & & \end{array} \qquad (*)$$

we again find a canonical arrow

$$\tilde{\varphi} : \tilde{\mathcal{X}} \longrightarrow Y.$$

It would be just as reasonable to say that the square $(*)$ is W-homotopically cocartesian, when $\tilde{\varphi}$ is W, as to adopt definition 3, page 131 - we would find exactly the same corollaries. Would the two definitions be equivalent??

I note that $\tilde{\mathcal{X}}$ being cofibered over $\Psi = \Phi^\circ$, $\tilde{\mathcal{X}}^\circ$ is fibered over $\Psi^\circ = \Phi$. More precisely, we will have p. 93

$$\tilde{\mathcal{X}}^\circ = \int \begin{array}{ccc} & \overset{\alpha_1^\circ}{\nearrow} & (X_1)^\circ \\ (X_0)^\circ & & \\ & \underset{\alpha_2^\circ}{\searrow} & (X_2)^\circ, \end{array}$$

and it is thus a matter of comparing the two integrals [page 137]

$$\mathcal{X} = \int \begin{array}{ccc} & \overset{\alpha_1}{\nearrow} & X_1 \\ X_0 & & \\ & \underset{\alpha_2}{\searrow} & X_2 \end{array} , \qquad \tilde{\mathcal{X}}^\circ = \int \begin{array}{ccc} & \overset{\alpha_1^\circ}{\nearrow} & (X_1)^\circ \\ (X_0)^\circ & & \\ & \underset{\alpha_2^\circ}{\searrow} & (X_2)^\circ \end{array} ,$$

in particular as objects of $\mathrm{HOT}_W(\Phi)$. For this, I take up again the construction of the functor

$$X \longmapsto \Phi(X) : \mathrm{Cat} \longrightarrow \mathrm{Cat}$$

from I, p. 72-75, but I prefer here to denote this functor by $\Sigma(X)$ (for Φ already denotes a small category $\begin{array}{ccc} & & 1 \\ & \swarrow & \\ 0 & & \\ & \nwarrow & \\ & & 2 \end{array}$). We will use the homomorphisms functorial in X:

$$\begin{array}{ccccc} & & \Sigma(X) & & \\ & \overset{p_X}{\swarrow} & & \overset{q_X}{\searrow} & \\ X & & & & X^\circ, \end{array}$$

which are W-aspheric, hence $\in W$. Moreover, $\Sigma(X^\circ) = \Sigma(X)$, $p_{X^\circ} = q_X$, $q_{X^\circ} = p_X$.

Thus, if

$$X_* \quad = \quad \begin{array}{ccc} & & X_1 \\ & \overset{\alpha_1}{\nearrow} & \\ X_0 & & \\ & \underset{\alpha_2}{\searrow} & \\ & & X_2 \end{array} \qquad \in \quad \mathrm{Hom}(\Phi^\circ, \mathrm{Cat})$$

$$X_*^\circ \quad = \quad \begin{array}{ccc} & & (X_1)^\circ \\ & \overset{\alpha_1^\circ}{\nearrow} & \\ (X_0)^\circ & & \\ & \underset{\alpha_2^\circ}{\searrow} & \\ & & (X_2)^\circ \end{array} \qquad \in \quad \mathrm{Hom}(\Phi^\circ, \mathrm{Cat}),$$

we find [page 138]

$$\Sigma(X_*) = \left(\begin{array}{ccc} & \overset{\Sigma(\alpha_1)}{\nearrow} & \Sigma(X_1) \\ \Sigma(X_0) & & \\ & \underset{\Sigma(\alpha_2)}{\searrow} & \\ & & \Sigma(X_2) \end{array} \right)$$

and the arrows p. 94

$$\begin{array}{ccc} & \Sigma(X_*) & \\ \overset{p_{X_*}}{\swarrow} & & \overset{q_{X_*}}{\searrow} \\ X_* & & X_*^\circ, \end{array}$$

in $\underline{\mathrm{Hom}}(\Phi^\circ, \mathrm{Cat})$. These arrows are in W term by term, so the homomorphisms in Cat

$$\begin{array}{ccc} & \int \Sigma(X_*) & \\ \overset{\int p_{X_*}}{\swarrow} & & \overset{\int q_{X_*}}{\searrow} \\ \mathcal{X} = \int X_* & & \int X_*^\circ = \tilde{\mathcal{X}}^\circ \end{array}$$

are in W.

Finally, the innocent commutative square

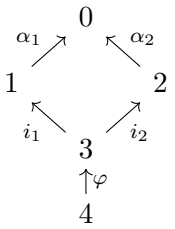
$$Q \quad = \quad \begin{array}{ccc} & X_0 & \\ \swarrow & & \searrow \\ X_1 & & X_2 \\ \searrow & & \swarrow \\ & Y & \end{array} \qquad (*)$$

(and the corresponding diagram

$$\widehat{Q} \quad = \quad \begin{array}{ccc} & X_0 & \\ \swarrow & & \searrow \\ X_1 & & X_2 \\ \searrow & & \swarrow \\ & \mathcal{X} = \int X_* & \\ & \downarrow{\scriptstyle \varphi_Q} & \\ & Y & \end{array} \qquad) \qquad (**)$$

gives rise, by transforming it by $X \longmapsto X^\circ$ and by $X \longmapsto \Sigma(X)$, and by taking into [page 139]

account the homomorphisms functorial in X $\begin{array}{ccc} & \Sigma(X) & \\ \swarrow & & \searrow \\ X & & X^\circ \end{array}$, to three diagrams of type $(**)$, i.e. modeled on the opposite category of

Γ = 

(i.e. the free diamond ([88]), to which we have added an initial object)

([89]), namely p. 95

$$\widehat{Q}, \qquad \widehat{\Sigma(Q)}, \qquad \widehat{Q^\circ},$$

and two homomorphisms of diagrams ($\underline{\mathrm{Hom}}(\Gamma^\circ, \mathrm{Cat})$)

$\begin{array}{ccc} & \widehat{\Sigma(Q)} & \\ \swarrow & & \searrow \\ \widehat{Q} & & \widehat{Q^\circ} \end{array}$ (deduced from $\begin{array}{ccc} & \Sigma(Q) & \\ \swarrow & & \searrow \\ Q & & Q^\circ \end{array}$), $(***)$

which are in W term by term (i.e. are in $W_\Gamma \subseteq \mathrm{Fl}(\underline{\mathrm{Hom}}(\Gamma^\circ, \mathrm{Cat}))$). Explicitly, at the vertices $i = 0, 1, 2$, resp. 4 of Γ, $(***)$ is none other than

$$D_i = \left(\begin{array}{ccc} & \Sigma(X_i) & \\ {\scriptstyle p_{X_i}} \swarrow & & \searrow {\scriptstyle q_{X_i}} \\ X_i & & X_i^\circ \end{array} \right) \quad \text{resp.} \quad \begin{array}{ccc} & \Sigma(Y) & \\ {\scriptstyle p_Y} \swarrow & & \searrow {\scriptstyle q_Y} \\ Y & & Y^\circ, \end{array}$$

at vertex 3 it is

$$\begin{array}{ccc} & \int \Sigma(X_*) & \\ {\scriptstyle \int p_{X_*}} \swarrow & & \searrow {\scriptstyle \int q_{X_*}} \\ \mathcal{X} = \int X_* & & \int X_*^\circ = \tilde{\mathcal{X}}^\circ, \end{array}$$

obtained by integration of the arrows

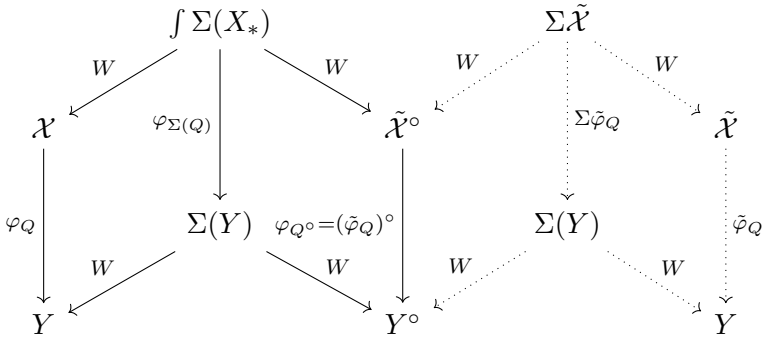
$$\begin{array}{ccc} & \Sigma(X_*) & \\ {\scriptstyle p_{X_*}} \swarrow & & \searrow {\scriptstyle q_{X_*}} \\ X_* & & X_*^\circ. \end{array}$$

Looking at the restriction of the two morphisms of diagrams $(***)$ to the parts of the three diagrams of type Γ° over the tail $\{3, 4\}$ of Γ we find the following diagram in solid [page 140]

[88] false, for the square is not commutative

[89] We have $\alpha_1 i_1 \neq \alpha_2 i_2$, but the relation $\alpha_1 i_1 \varphi = \alpha_2 i_2 \varphi$.

lines



which I have completed with a second panel in dotted lines. In this diagram of four adjoining squares, the horizontal sides of the squares represent arrows of Cat which are in W. We thus find: p. 96

$$\varphi_Q \in W \iff \varphi_{\Sigma Q} \in W \iff (\varphi_{Q^\circ} = \tilde{\varphi}_Q^\circ) \in W \iff \Sigma\tilde{\varphi}_Q \in W \iff \tilde{\varphi}_Q \in W.$$

We also find that the five vertical arrows are isomorphic, as arrows of Hot_W. To summarize:

Proposition 14. *Let us consider a diagram (commutative square)*

$$Q \quad = \quad \begin{array}{ccc} & X_0 & \\ \swarrow & & \searrow \\ X_1 & & X_2 \\ \searrow & & \swarrow \\ & Y & \end{array}$$

in Cat*, and the "opposite" diagram deduced by* $X \longmapsto X^\circ$

$$Q^\circ \quad = \quad \begin{array}{ccc} & X_0^\circ & \\ \swarrow & & \searrow \\ X_1^\circ & & X_2^\circ \\ \searrow & & \swarrow \\ & Y^\circ, & \end{array}$$

and the corresponding arrows [page 141]

$$\begin{cases} \varphi_Q : \ \mathcal{X}_Q \ = \ \left(\int_{X_0} \begin{array}{c} \nearrow X_1 \\ \\ \searrow X_2 \end{array} \right) \longrightarrow Y \\ \varphi_{Q^\circ} : \ \mathcal{X}_{Q^\circ} \ = \ \left(\int_{X_0^\circ} \begin{array}{c} \nearrow X_1^\circ \\ \\ \searrow X_2^\circ \end{array} \right) \longrightarrow Y^\circ. \end{cases}$$

Finally, let us also consider the canonical arrows $\tilde{\varphi}_Q$*,* $\tilde{\varphi}_{Q^\circ}$*, relative to the proper integral* $\int_{\mathrm{pr}}$ *(instead of the smooth integral* $\int = \int_{\mathrm{lis}}$*)* p. 97

$$\left\{\begin{array}{lcll} \tilde{\varphi}_Q : \quad \tilde{\mathcal{X}}_Q & = & \left(\int_{\mathrm{pr}} X_0 \begin{array}{c}\nearrow X_1 \\ \searrow X_2\end{array}\right) & \longrightarrow \quad Y \\ \\ \tilde{\varphi}_{Q^\circ} : \quad \tilde{\mathcal{X}}_{Q^\circ} & = & \left(\int_{\mathrm{pr}} X_0^\circ \begin{array}{c}\nearrow X_1^\circ \\ \searrow X_2^\circ\end{array}\right) & \longrightarrow \quad Y^\circ. \end{array}\right.$$

We then have the following:

ⓐ $\varphi_{Q^\circ} \simeq (\tilde{\varphi}_Q)^\circ$, $\varphi_Q \simeq (\tilde{\varphi}_{Q^\circ})^\circ$ *(from which it already follows that* $\left\{\begin{array}{lcl} \varphi_Q \in W & \iff & \tilde{\varphi}_{Q^\circ} \in W \\ \varphi_{Q^\circ} \in W & \iff & \tilde{\varphi}_Q \in W \end{array}\right.$ *(*[90]*)).*

ⓑ *The relations*

$$\varphi_Q \in W, \qquad \varphi_{Q^\circ} \in W, \qquad \tilde{\varphi}_Q \in W, \qquad \tilde{\varphi}_{Q^\circ} \in W$$

are equivalent, in other words: for the square to be W-homotopically cocartesian (smooth integral version), it is necessary and sufficient that the opposite square Q° be so, and these conditions are also equivalent to the corresponding conditions in terms of the proper integrals (instead of the smooth integrals).

ⓒ *The four arrows φ_Q, φ_{Q°, $\tilde{\varphi}_Q$, $\tilde{\varphi}_{Q^\circ}$ of* Cat *as arrows in* Hot_W*, are isomorphic in* $\underline{\mathrm{Fl}}(\mathrm{Hot}_W)$*. More precisely, as arrows of* Cat*, they define isomorphic objects of* $\mathrm{HOT}_W(\mathbf{\Delta}^1)$*.* [page 142]

ⓓ *More precisely still, the four diagrams in* Cat *of type* Γ°*, where*

$$\Gamma = \begin{array}{ccccccc} & & & & 1 & & \\ & & & \overset{\alpha_1}{\swarrow} & & \overset{i_1}{\nwarrow} & \\ 0 & & & & & & 3 \overset{\varphi}{\longleftarrow} 4, \\ & & & \underset{\alpha_2}{\nwarrow} & & \underset{i_2}{\swarrow} & \\ & & & & 2 & & \end{array}$$

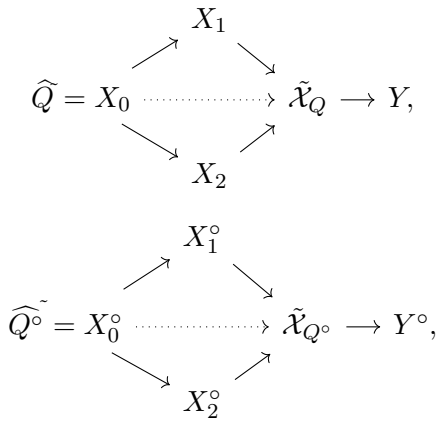
$\alpha_1 i_1 \varphi = \alpha_2 i_2 \varphi$*, into which the four arrows we have just considered fit, namely*

$$\widehat{Q} = \begin{array}{ccccc} & & X_1 & & \\ & \overset{\alpha_1}{\nearrow} & & \overset{i_1}{\searrow} & \\ X_0 & \cdots\cdots\cdots & & \cdots\cdots\cdots & \mathcal{X}_Q \overset{\varphi_Q}{\longrightarrow} Y, \\ & \underset{\alpha_2}{\searrow} & & \underset{i_2}{\nearrow} & \\ & & X_2 & & \end{array}$$

$$\widehat{Q^\circ} = \begin{array}{ccccc} & & X_1^\circ & & \\ & \overset{\alpha_1^\circ}{\nearrow} & & \overset{i_1'}{\searrow} & \\ X_0^\circ & \cdots\cdots\cdots & & \cdots\cdots\cdots & \mathcal{X}_{Q^\circ} \longrightarrow Y^\circ, \\ & \underset{\alpha_2^\circ}{\searrow} & & \underset{i_2'}{\nearrow} & \\ & & X_2^\circ & & \end{array}$$

p. 98

[90] and that the arrows φ_Q and $\tilde{\varphi}_{Q^\circ}$ on the one hand, φ_{Q° and $\tilde{\varphi}_Q$ on the other hand, are isomorphic as arrows in Hot_W.



(^{91}) *are isomorphic in* $\mathrm{Hot}_W(\Gamma)$. *More precisely still, we have the isomorphisms in* $\underline{\mathrm{Hom}}(\Gamma^\circ, \mathrm{Cat})$

$$(\widehat{\widetilde{Q}})^\circ \simeq \widehat{\widetilde{Q^\circ}},$$

whence, applying it to Q° *instead of* Q,

$$(\widehat{\widetilde{Q^\circ}}\,\tilde{})^\circ \simeq \widehat{Q},$$

which implies (via the canonical isomorphisms valid in $\mathrm{HOT}_W(S)$ *for every object* S *in* Cat *and every* $X_* \in \mathrm{Ob}\,\underline{\mathrm{Hom}}(S^\circ, \mathrm{Cat})$, $X_* \simeq X_*^\circ$ (^{92})*), that we have the canonical isomorphisms*

$$\left.\begin{array}{r} \widehat{Q} \simeq \widehat{\widetilde{Q^\circ}}\,\tilde{} \\ \widehat{\widetilde{Q^\circ}} \simeq \widehat{\widetilde{Q}} \end{array}\right\} \qquad \textit{in } \mathrm{HOT}_W(\Gamma),$$

but also directly

$$\widehat{Q} \simeq \widehat{\widetilde{Q^\circ}}, \qquad \widehat{\widetilde{Q}} \simeq \widehat{\widetilde{Q^\circ}}\,\tilde{} \qquad \textit{in } \mathrm{HOT}_W(\Gamma)$$

(and it remains to check that we thus have a transitive *system of canonical isomorphisms in* $\mathrm{HOT}_W(\Gamma)$*, between the four objects considered, which must be essentially sorital …).* [page 143]

Remarks.

1) What we have just said in the case of diagrams in Cat of type $\Phi^\circ = \Psi$, seems still to make sense for diagrams of arbitrary type

$$X_* : S^\circ \longrightarrow \mathrm{Cat}, \qquad s \longmapsto X_s$$

(S an object of Cat). We can integrate it in a smooth way, $\int X_* = \int_{\mathrm{lis}} X_*$, obtaining a fibered category (split) over S, or in a proper way, $\int_{\mathrm{pr}} X_*$, obtaining a split cofibered category over S°. We can also consider X_*°, the composite of X_* with the automorphism $Y \longmapsto Y^\circ$ in Cat, and integrate it in either a proper or a smooth way. In all these cases, we find a diagram of type $\widehat{S}^\circ$, where $\widehat{S}$ is obtained from S by adding an element which maps by a "canonical" (given) arrow to each of the others (*without* commutation relations!) A *direct system* of morphisms in Cat

p. 99

$$f \qquad X_s \xrightarrow{f_s} Y$$

[page 144]

then gives a morphism

$$\int_{\mathrm{lis}} X_* \xrightarrow{\varphi_Q} Y,$$

[91] **NB** We should still complete these (noncommutative) diagrams with the dotted arrows …

[92] via the functor Σ used above …

where Q denotes the diagram, of type $\widehat{S}^\circ$, made up of the X_s, Y, the $u^* : X_t \longrightarrow X_s$ (for $u : s \longrightarrow t$ in S) and the f_s. Likewise we will have

$$\int_{\text{prop}} X_* \xrightarrow{\tilde{\varphi}_Q} Y,$$

and if

$$\Gamma = (\widehat{S} + \text{initial object}),$$

we find diagrams $\widehat{Q}$, $\widetilde{\widehat{Q}}$ of type Γ° in Cat. Unless I am mistaken, the relations between $\widehat{Q}$, $\widehat{Q}$, $\widehat{Q^\circ}$, $\widehat{Q^\circ}$ are the same as those made explicit in proposition 14 above. I am not going to dwell on this with the requisite care, unless needed. We will say that a diagram Q defines Y as a $\varinjlim_I$ that is W-homotopic if $\varphi_Q \in W$. It amounts to the same thing to say that $\varphi_{Q^\circ} \in W$, or that $\tilde{\varphi}_Q \in W$ or that $\tilde{\varphi}_{Q^\circ} \in W$ … [page 145]

2) The "universal" case, for $\begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array}$ given, of a commutative square $Q =$

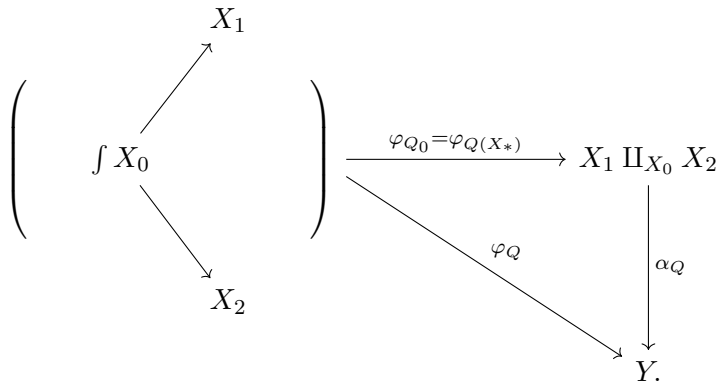
$\begin{array}{ccccc} & & X_1 & & \\ & \nearrow & & \searrow & \\ X_0 & & & & Y \\ & \searrow & & \nearrow & \\ & & X_2 & & \end{array}$ in Cat, giving rise to

$$\varphi_Q : \left(\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ \int X_0 & & \\ & \searrow & \\ & & X_2 \end{array} \right) \longrightarrow Y,$$

is that of $Y = X_1 \amalg_{X_0} X_2$, giving rise to p. 100

$$Q_0 = Q(X_*) = \begin{array}{ccccc} & & X_1 & & \\ & \nearrow & & \searrow & \\ X_0 & & & & X_1 \amalg_{X_0} X_2. \\ & \searrow & & \nearrow & \\ & & X_2 & & \end{array}$$

The general case reduces to this one, for we will have a commutative diagram



But there must be interesting cases where φ_{Q_0} is *not* in W, i.e. the Cat-cocartesian square $\begin{array}{ccc} & X_1 & \\ \nearrow & & \searrow \\ X_0 & & X_1 \amalg_{X_0} X_2 \\ \searrow & & \nearrow \\ & X_2 & \end{array}$ is not W-homotopically cocartesian, and where nevertheless we can find a commutative square Q completing X_*, in such a way that this square is W-homotopically cocartesian , i.e. that $\varphi_Q = \alpha_Q \varphi_{Q_0}$ is in W. This is all the more plausible since in Cat, amalgamated sums tend not to have good homotopy properties, even if one of the arrows α_1, α_2 is monomorphism (or even an open immersion), contrary to what happens in $\mathbf{\Delta}^\wedge$, and more generally in categories of the form $Z^\wedge$ (Z in Cat). [page 146]

3.4 Criteria for homotopically cocartesian squares.

The criteria I have obtained are all contained in the following statement:

Theorem 4 ([93],[94]). *Let us consider a commutative square diagram in* Cat

$$Q \quad = \quad \begin{array}{ccc} X_0 & \xrightarrow{\alpha_1} & X_1 \\ {\scriptstyle \alpha_2}\downarrow & \overset{f_0}{\searrow} & \downarrow{\scriptstyle f_1} \\ X_2 & \xrightarrow[f_2]{} & Y, \end{array} \qquad (*)$$

and consider the canonical morphism p. 101

$$\varphi_Q : \quad \mathcal{X} = \Big(\int \quad \begin{array}{ccc} & X_1 & \\ {\scriptstyle \alpha_1} \nearrow & & \\ X_0 & & \\ {\scriptstyle \alpha_2} \searrow & & \\ & X_2 & \end{array} \Big) \quad \longrightarrow \quad Y,$$

the integral being the integral which is smooth. *(There is a dual statement concerning* [page 147]

[93] **NB** In this theorem, and corollaries 1, 2, 3, we need only the *weak form* of W(7 bis), namely $i_0 \in W \Longrightarrow i \in W$ (cf. p. 79) (Loc(6′a) in the notation of XVI).

[94] **NB** All in all, the theorem (forward part) holds without any hypothesis W(7 bis) or weak W(7 bis) on W, and the same is true of its corollaries 1 to 3 (to the exclusion of corollaries 4, 5, which do use W(7 bis) in its full strength).

the proper integral of

$$\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & .) \\ & \searrow & \\ & & X_2 \end{array}$$

a) For φ_Q to be W-smooth, it is necessary and sufficient that f_1, f_2 and $f_0 = f_1\alpha_1 = f_2\alpha_2$ be so. (Assuming W(7 bis) ([95])).

b) The fiber of φ_Q at an element y of Y is isomorphic to

$$\int \quad \begin{array}{ccc} & & X_{1\,y} \\ & \nearrow & \\ X_{0\,y} & & \\ & \searrow & \\ & & X_{2\,y} \end{array} \qquad (\simeq \mathcal{X}_y). \tag{$**$}$$

Corollary 1. ([97]) *Suppose f_1, f_2 and $f_1\alpha_1 = f_2\alpha_2$ W-smooth, and the integrated categories $(**)$ W-aspheric for every $y \in \mathrm{Ob}\,Y$. (NB This is still the smooth integral, but it does not really matter, for we have seen (cf. the upper part of the diagram p. 140, which implies that $\mathcal{X}$ W-aspheric if and only if $\tilde{X}$ is) that one is W-aspheric if and only if the other is.) Then $\varphi_Q \in W$, i.e. the square $(*)$ is W-homotopically cocartesian . By duality, same conclusion if we suppose f_1, f_2, $f_1\alpha_1 = f_2\alpha_2$* proper *instead of smooth. (For Q W-homotopically cocartesian , if and only if Q° is, by prop. 14.)*

Indeed, a morphism in Cat which is W-smooth and has W-aspheric fibers is in W (and it is even a universal W-equivalence …). [page 148]

Corollary 2. *We assume that W satisfies* W(7 bis) (cocartesian square axiom) ([98]). *Suppose the square Q cocartesian in* Cat ([99]), *and that α_1 is an open immersion (resp. closed) and f_1 smooth (resp. proper). Then Q is W-homotopically cocartesian.* ([100]) p. 102

It suffices to establish the non-resp'ed result, the other will follow from it by duality (since Q is W-cocartesian if and only if Q° is). By hypothesis f_1 is W-smooth, on the other hand α_1 and f_2 are open immersions, hence W-smooth, hence $f_1\alpha_1$ is W-smooth too. (Besides, α_2 too, for the square is also cartesian, hence α_2 obtained from f_1 (which is W-smooth by hypothesis) by base change $X_2 \longrightarrow Y$.) Thus φ_Q is smooth, and it suffices to see that its fibers are W-aspheric. The calculation $(**)$ gives, for y in $Y \backslash X_2$,

$$\mathcal{X}_y \simeq \quad \int \quad \begin{array}{ccc} & & e \\ & \nearrow & \\ \emptyset & & \\ & \searrow & \\ & & \emptyset \end{array} \qquad \simeq e, \qquad \text{O.K.}$$

[95] For the forward part (f_0, f_1, f_2 smooth $\Longrightarrow$ φ_Q smooth), we need only W(7 bis) ([96]), weak form. It is only for the converse (which has never yet been of use to me) that we need W(7 bis), in its full strength.

[96] We do not even need W(7 bis) in weak form, cf. annotation on page 156. We use a restrictive hypothesis on W only in the converse (φ_Q smooth $\Longrightarrow$ f_0, f_1, f_2 smooth).

[97] Valid without any hypothesis on W (other than Loc(1,2,3 bis)).

[98] unnecessary.

[99] NB We do not really use that Q is cocartesian, rather that Q is *cartesian,* and f_2 (hence also α_1) an open immersion (resp. closed), and that for $y \in Y \backslash X_2$, $X_{1\,y}$ is W-aspheric. (In the case where Q is cocartesian, $X_{1\,y}$ is $\simeq e$.)

[100] Here we need W(7 bis), forward form, i.e. Loc(6′a) (notation of XVI).

And for y in X_2

$$\mathcal{X}_y \simeq \int \begin{array}{ccc} & & X_{1\,y} \\ & \overset{\sim}{\nearrow} & \\ X_{0\,y} & & \\ & \searrow & \\ & & e \end{array} .$$

Since $X_{0\,y} \longrightarrow X_{1\,y}$ is an isomorphism, hence is in W, it follows that $e \longrightarrow \mathcal{X}_y$ is too (prop. 13, p. 125), assuming W(7 bis), hence $\mathcal{X}_y$ is W-aspheric, O.K. ([101]). [page 149]

Corollary 3. (Assuming W(7 bis) ([102]).) *Let X be in* Cat*, U_1, U_2 two open subcategories of X such that $X = U_1 \cup U_2$, $U_0 = U_1 \cap U_2$ the open intersection. Then the square of open immersions*

$$\begin{array}{ccc} U_0 & \hookrightarrow & U_1 \\ \downarrow & & \downarrow \\ U_2 & \hookrightarrow & X \end{array}$$

is W-cocartesian. (Dual statement for two closed subcategories T_1, T_2 such that $T_1 \cup T_2 = X$, with $T_0 = T_1 \cap T_2$.)

We apply corollary 2. We know that the square is cocartesian in Cat ([103]), moreover $U_1 \hookrightarrow X$ is indeed W-smooth, being an open immersion.

p. 103

Corollary 4. (Assuming W(7 bis) ([104]).) *Let $U \overset{i}{\hookrightarrow} X$ be an open immersion such that $T = X \backslash U$ is isomorphic to the one-point category e. For $i \in W$ to hold, it is necessary and sufficient that i be W-aspheric, i.e. that U/t (where t is the unique object of T) be W-aspheric.*

It is of course enough to prove that

$$i \in W \Longrightarrow U/t \text{ } W\text{-aspheric.}$$

[page 150]

Consider the cartesian square

$$\begin{array}{ccc} U/t & \overset{i/t}{\hookrightarrow} & X/t \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ U & \overset{i}{\hookrightarrow} & X, \end{array}$$

we are going to prove that it is W-cocartesian. Assuming W(7 bis), it will follow that $i \in W$ implies $i/t \in W$, and since X/t is W-aspheric (having a final object), this means that U/t is W-aspheric, which is what we want to prove.

It is enough, by virtue of corollary 2, to prove that the square is cocartesian in Cat, since i/t is an open immersion (i being one), and $X/t \longrightarrow X$ is a local isomorphism, hence W-smooth. We see that $(X/t)\backslash(U/t) \simeq e$ (one-point category), moreover the unique element of $(X/t)\backslash(U/t)$ is the final element $t_0 : t \overset{\mathrm{id}_t}{\longrightarrow} t$ of X/t. Thus $H_{t_0} \simeq e_{U/t}$ (final presheaf), and $p_!(H_{t_0}) \simeq H'_t \overset{\text{def}}{=} (u \longmapsto \mathrm{Hom}_X(u,t))$, which characterizes the amalgamated sum in Cat …

Corollary 4 is none other than the unproved corollary of proposition 7 (p. 85).

[101] We need only the forward form of W(7 bis), i.e. (notation of XVI) Loc(6′a).
[102] unnecessary.
[103] We do not even need to know this, cf. marginal annotation to corollary 2.
[104] Here we need W(7 bis) in strong form, i.e. $i_0 \in W \iff i \in W$ (cf. annotation th. 4).

Here is a variant of corollary 4.

Corollary 5. (Assuming W(7 bis) ([105]).) *Let $i : U \hookrightarrow X$ be an open immersion, let $t_0 \in \mathrm{Ob}\, T$ be a* closed *point of $T = X\backslash U$, such that $\{t_0\}$ is isomorphic to the one-point category, i.e.* $\mathrm{Hom}_X(t_0, t_0) = \{\mathrm{id}\}$. *Suppose that for every $t \in T\backslash\{t_0\}$, U/t is W-aspheric. For $i \in W$ to hold, it is necessary and sufficient that i be aspheric, i.e. that U/t_0 also be W-aspheric.* [page 151]

We must prove that $i \in W$ implies that U/t_0 is W-aspheric. Consider the inclusion

$$i_0 : U \longrightarrow U' \overset{\text{def}}{=} X\backslash\{t_0\},$$

the assumption on the U/t, for $t \in T\backslash\{t_0\}$, means that i_0 is W-aspheric. Thus i_0 is in W, and since the composite $i : U \xrightarrow{i_0} U' \xrightarrow{i'} X$ is in W, this implies that $i' : U' \longrightarrow X$ is in W. We are now in the situation of corollary 4, which implies that i' is W-aspheric. Thus $i = i'i_0$ is W-aspheric, as a composite of two W-aspheric morphisms, q.e.d. p. 104

Proof of th. 4.

ⓐ Smoothness criterion for $\varphi_Q : \mathcal{X} = \int X_0 \longrightarrow Y$.

$$\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ \int X_0 & & \longrightarrow Y. \\ & \searrow & \\ & & X_2 \end{array}$$

Since $\{1\}$, $\{2\}$ are opens of $\Phi = \begin{array}{ccc} & & 1 \\ & \swarrow & \\ 0 & & \\ & \nwarrow & \\ & & 2 \end{array}$, $i_1 : X_1 \longrightarrow \mathcal{X}$ and $i_2 : X_2 \longrightarrow \mathcal{X}$ are open immersions. Since $\varphi_Q \circ i_1 = f_1$, $\varphi_Q \circ i_2 = f_2$, we already see that [page 152]

f_1 and f_2 smooth $\iff$ φ_Q smooth on the open union of $i_1(X_1)$ and $i_2(X_2)$, i.e. on $\mathcal{X}\backslash i_0(X_0)$.

It remains to express the smoothness of φ_Q at the points x_0 of $i_0(X_0)$. Let x_1, x_2, y be the images of x_0 in X_1, X_2, Y respectively. We must express that the fibers of

$$(\varphi_Q)/x_0 : \mathcal{X}/x_0 \longrightarrow Y/y \qquad (*)$$

are aspheric. For this we are going to make $\mathcal{X}/x_0$ and the arrow $(*)$ explicit in terms of the commutative square

$$(**) \qquad \begin{array}{ccc} X_0/x_0 & \xrightarrow{\alpha_1/x_0} & X_1/x_1 \\ {\scriptstyle \alpha_2/x_0}\downarrow & & \downarrow{\scriptstyle f_1/x_1} \\ X_2/x_2 & \xrightarrow{f_2/x_2} & Y/y, \end{array}$$

induced by Q. I say

Lemma 1. *Let there be a commutative square*

$$Q \quad = \quad \begin{array}{ccc} X_0 & \xrightarrow{\alpha_1} & X_1 \\ {\scriptstyle \alpha_2}\downarrow & & \downarrow{\scriptstyle f_1} \\ X_2 & \xrightarrow{f_2} & Y \end{array}$$

[105] strong form.

in Cat, $x_0 \in \mathrm{Ob}\, X_0$, x_1, x_2, y *its images in* X_1, X_2, Y *respectively,* Q/x_0 *the square* (∗∗) *above, induced by* Q *on the localized categories* X_0/x_0 *etc.,* $\mathcal{X} = \int X_0$  *. We then have a canonical isomorphism*

p. 105

[page 153]

$$\mathcal{X}/x_0 \quad \simeq \quad \int X_0/x_0 \quad \begin{array}{c} \nearrow X_1/x_1 \\ \\ \searrow X_2/x_2 \end{array} ,$$

and the arrow $(\varphi_Q)/x_0$*, the localized arrow of* $\varphi_Q : \mathcal{X} \longrightarrow Y$ *at* x_0*, is identified with the arrow* $\varphi_{(Q/x_0)}$ *:*

$$\varphi_{(Q/x_0)} : \quad \int X_0/x_0 \quad \begin{array}{c} \nearrow X_1/x_1 \\ \\ \searrow X_2/x_2 \end{array} \quad \longrightarrow \quad Y/y.$$

Proof of the Lemma. We know that

$$\mathcal{X}/x_0 \longrightarrow \mathcal{X}$$

is Cat-fibered (split) (with discrete fibers), and since

$$\mathcal{X} \longrightarrow \Phi$$

is also Cat-fibered (split), it follows that the same is true of the composite arrow

$$\mathcal{X}/x_0 \longrightarrow \Phi.$$

Thus to "calculate" $\mathcal{X}/x_0$, as a category over Φ, we must make explicit its three fibers, and the two base change functors which connect them. We find at once that these fibers are canonically isomorphic to

$$X_0/x_0, \qquad X_1/x_1, \qquad X_2/x_2,$$

and we shall admit (!) that the base change functors

$$X_0/x_0 \longrightarrow X_1/x_1, \qquad X_0/x_0 \longrightarrow X_2/x_2$$

[page 154]

(defined up to unique isomorphism) are none other than the localized arrows α_1/x_0 and α_2/x_0 of $\alpha_1 : X_0 \longrightarrow X_1$ and $\alpha_2 : X_0 \longrightarrow X_2$. This therefore shows that we have an isomorphism

$$\mathcal{X}/x_0 \quad \simeq \quad \int X_0/x_0 \quad \begin{array}{c} \nearrow X_1/x_1 \\ \\ \searrow X_2/x_2 \end{array} ,$$

p. 106

and we shall also admit that the arrow $(\varphi_Q)/x_0 : \mathcal{X}/x_0 \longrightarrow Y/y$ is identified with $\varphi_{(Q/x_0)}$.

To express the W-smoothness of φ_Q at $x_0 \in \operatorname{Ob} X_0 \subseteq \operatorname{Ob} \mathcal{X}$, we are thus reduced to expressing the W-asphericity of the fibers of

$$\varphi_{(Q/x_0)} : \quad \int X_0/x_0 \quad \begin{array}{c} \nearrow X_1/x_1 \\ \\ \searrow X_2/x_2 \end{array} \quad \longrightarrow \quad Y/y.$$

To make these fibers explicit, we need another auxiliary lemma

Lemma 2 ([106]). *Let there be a commutative square in* Cat

$$Q \quad = \quad \begin{pmatrix} X_0 & \xrightarrow{\alpha_1} & X_1 \\ {\scriptstyle \alpha_2}\downarrow & & \downarrow{\scriptstyle f_1} \\ X_2 & \xrightarrow{f_2} & Y \end{pmatrix},$$

whence

$$\varphi_Q : \quad \left(\int X_0 \quad \begin{array}{c} \nearrow X_1 \\ \\ \searrow X_2 \end{array} \right) \quad \longrightarrow \quad Y,$$

and let $y \in \operatorname{Ob} Y$. Then the fiber of φ_Q at y [page 155]

$$\mathcal{X}_y = \varphi_Q^{-1}(\underbrace{\{y\}}_{\text{one-point category}})$$

is given by (as a category over Φ)

$$\mathcal{X}_y \quad \simeq \quad \int X_{0\,y} \quad \begin{array}{c} \nearrow X_{1\,y} \\ \\ \searrow X_{2\,y} \end{array} .$$

Proof. It is immediate that we have

$$\mathcal{X}_y \cap \underbrace{\mathcal{X}_s}_{X_s} \simeq (X_s)_y \qquad \text{for } s \in \operatorname{Ob} \Phi = \{0,1,2\},$$

and (chance does things well) the base change functors p. 107

$$X_0 \longrightarrow X_1, \qquad X_0 \longrightarrow X_2$$

[106] NB This is b) of th. 4.

send the (nonfull) subcategory $X_{0\,y}$ of X_0 into the (nonfull) subcategories $X_{1\,y}$ and $X_{2\,y}$ of X_1 and X_2 respectively.

$$\begin{array}{ccccc} X_{s\,y} & \longrightarrow & \mathcal{X}_y & \longrightarrow & \{y\} \\ \downarrow & \text{cart.} & \downarrow & \text{cart.} & \downarrow \\ X_s & \longrightarrow & \mathcal{X} & \longrightarrow & Y \\ \downarrow & \text{cart.} & \downarrow & & \\ \{s\} & \longrightarrow & \Phi & & \end{array}$$

We thus have a fibered subcategory of $\mathcal{X}$, subject to checking that the strictly cartesian arrows

$$\alpha_1(x_0) \longrightarrow x_0, \qquad \alpha_2(x_0) \longrightarrow x_0$$

in $\mathcal{X}$, for x_0 in $X_{0\,y}$, are in $\mathcal{X}_y$, i.e. are over the identity arrow of y. Now that is the essential property of the functor $\varphi_Q : \mathcal{X} \longrightarrow Y$, which we forgot to make explicit in 3.2. It really seems to work (like clockwork!) [page 156]

Let us apply Lemma 2 to the square diagram Q/x_0 and to $\varphi_{(Q/x_0)}$ (page 154), we thus see that the fibers to be examined at the $u \in \mathrm{Ob}\, Y/y$ $(u : y' \longrightarrow y)$ can be written as the integrals

$$\mathcal{X}(x_0, u) \quad \stackrel{\text{def}}{=} \quad \int (X_0/x_0)_u \begin{array}{c} \nearrow (X_1/x_1)_u \\ \\ \searrow (X_2/x_2)_u \end{array} .$$

Let us now use the fact that f_1, f_2 are W-smooth (expressing that φ_Q is W-smooth in $\mathcal{X}\backslash X_0$), hence that $(X_1/x_1)_u$ and $(X_2/x_2)_u$ are W-aspheric. In the W-cocartesian square

$$\begin{array}{ccc} & (X_1/x_1)_u & \\ {\scriptstyle \alpha_1(x_0,u)} \nearrow & & \searrow {\scriptstyle i_1(x_0,u)} \\ (X_0/x_0)_u & & (\mathcal{X}/x_0)_u \simeq \mathcal{X}(x_0,u) \\ {\scriptstyle \alpha_2(x_0,u)} \searrow & & \nearrow {\scriptstyle i_2(x_0,u)} \\ & (X_2/x_2)_u & \end{array}$$

([107]), we must express that $(\mathcal{X}/x_0)_u$ is W-aspheric, knowing that $(X_1/x_1)_u$ and $(X_2/x_2)_u$ are. So this is equivalent to p. 108

$$i_1(x_0, u) \in W \qquad (\text{or } i_2(x_0, u) \in W),$$

hence also (assuming W(7 bis)) to [page 157]

$$\alpha_1(x_0, u) \in W, \qquad \text{i.e. } (X_0/x_0)_u \ W\text{-aspheric.}$$

[107] careful, this is not a commutative square. Say rather that $(\mathcal{X}/x_0)_u \simeq \int \begin{array}{c} \nearrow \\ \searrow \end{array}$ is smooth over Φ with W-aspheric fibers (if f_0, f_1, f_2 are smooth), hence W-aspheric since Φ is.

To say that this is true for every u in Y/y, means that $f_0 : X_0 \longrightarrow Y$ is smooth at x_0 .

So we have proved a), in the more precise form:

Corollary 6. *Let* $x \in \mathrm{Ob}\,\mathcal{X}$ *(where* $\mathcal{X} = \int X_0 \begin{array}{c} \nearrow X_1 \\ \\ \searrow X_2 \end{array}$ *.) We want to express the* W*-smoothness of*

$$\varphi_Q : \mathcal{X} \longrightarrow Y$$

at x*.*

a) If $x \in \mathrm{Ob}\,X_i$ *(*$i \in \{1, 2\}$*),* φ_Q *is* W*-smooth at* x *if and only if* $f_i : X_i \longrightarrow Y$ *is.*

b) Assume W(7 bis) ([108])*. Let* $x_0 \in \mathrm{Ob}\,X_0$*, and suppose* f_1*,* f_2 *smooth at* x_1*,* x_2*. Then* φ_Q *is* W*-smooth at* x_0 *if and only if* $f_0 : X_0 \longrightarrow Y$ *is.*

So (assuming W(7 bis)) *in all three cases* $x \in \mathrm{Ob}\,X_i$ *(*$i \in \{0, 1, 2\}$*), under the corresponding hypotheses of a) and b), we find that* φ_Q *is* W*-smooth at* x *if and only if* $f_i : X_i \longrightarrow Y$ *is* .

ⓑ As for b) of theorem 4, it is lemma 2 above. So everything is proved!

[page 158]

3.5 Applications to amalgamated sums in a category $X^\wedge$ (with $X \in \mathrm{Ob}\,\mathrm{Cat}$).

First a characterization of amalgamated sums involving an open immersion:

Proposition 15. *Let* p. 109

$$Q : \quad \begin{array}{ccc} U & \xrightarrow{i} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ U' & \xrightarrow{i'} & X' \end{array}$$

be a commutative square in Cat*. For this square to be cocartesian in* Cat*, with* i *(hence* i'*) an open immersion, it is necessary and sufficient that the following conditions be satisfied:*

a) Q *is cartesian.*

b) i' *is an open immersion (hence* i *too).*

c) If $T' = X'\backslash U'$*, and* $T = X \times_{X'} T' = X\backslash U$*,* $T \longrightarrow T'$ *is an isomorphism.*

d) For every $t \in T$*, if* t' *is its image in* T'*, the functor induced by* p

$$p/t : \underbrace{U/t}_{=V_t} \longrightarrow \underbrace{U'/t'}_{=V'_{t'}}$$

satisfies $(p/t)_!(e_{U/t}) \simeq e_{U'/t'}$*, or what amounts to the same, the "colocal fibers" of* p/t*,* $v'\backslash V_t$*, with* v' *in* $V'_{t'}$*, are* 0*-connected.*

[page 159]

Immediate proof - this should rather go back to §1.

Corollary 1. *Let* Q *be a square that is* cartesian (cf. page 158)*, with* i' *an* open immersion *(hence* i *too), and* q W-smooth *(hence* p *too). We assume that* W *satisfies the*

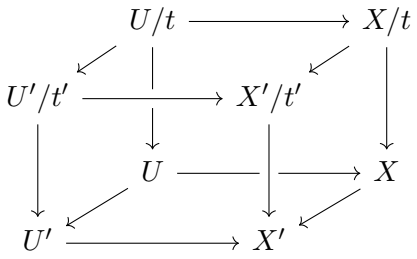
[108] Unnecessary for the forward direction, assuming .

axiom W(5) *of connectedness* ([109]). *Let* $T' = X'\backslash U'$, $T = X\backslash U$, *suppose that* $T \longrightarrow T'$ *is an isomorphism. Then* Q *is cocartesian.*

It suffices to prove condition d) of prop. 15. Let us consider the square

$$\begin{array}{ccc} U/t & \xrightarrow{i/t} & X/t \\ {\scriptstyle p/t}\downarrow & & \downarrow{\scriptstyle q/t} \\ U'/t' & \xrightarrow{i'/t'} & X'/t', \end{array} \qquad (*)$$

I say that it is *cartesian.* This follows from the sole assumption that Q is cartesian, without any smoothness assumption on q nor that i' is an open immersion, and can be seen e.g. by looking at the cubical diagram: p. 110



([110]). In this diagram, the front and back face of the cube is cartesian, by definition of U/t and of U'/t' respectively, and the base is by assumption. It follows that the top face [page 160] of the cube is also a cartesian square.

This seen, since q is smooth, the same is true of q/t in the cartesian square $(*)$, hence also of p/t by base change. On the other hand, the smoothness of q implies that q/t has W-aspheric fibers, hence by base change the same is true of p/t (which is thus a universal W-equivalence, i.e. a trivial W-fibration). Moreover, because of the smoothness of p/t, for v' in U'/t', the inclusion

$$(U/t)_{v'} \longrightarrow v'\backslash(U/t)$$

is in W, hence (by W(5)) defines a bijection on the π_0. Since $(U/t)_{v'}$ is W-aspheric by what precedes, it follows (by W(5)) that its π_0 has cardinality 1, hence the same is true of $v'\backslash(U/t)$, i.e. the latter is 0-connected. Now that is condition d) of proposition 15, which was to be established.

[page 161]

Corollary 2. *Let* Q *be a cartesian square* (p. 158), *with* i' *(hence also* i*) an open immersion, and* q *(hence also* p*) a* local isomorphism. *Let* $T = X\backslash U$, $T' = X'\backslash U'$ *(hence* $T \simeq X \times_{X'} T'$*), and suppose* $T \longrightarrow T'$ *an isomorphism. Then* Q *is cocartesian.*

This is a special case of corollary 1, but with no explicit reference to a localizer W. We can take $W = W_\infty$, or $W = W_0$, to reduce to corollary 1. But it is more reasonable to say that in the cartesian square $(*)$, the assumption that q is a local isomorphism implies that q/t is an isomorphism, hence p/t is too. It does indeed follow that the $v'\backslash(U/t)$ are 0-connected, this category being isomorphic to $v'\backslash(U'/t')$, hence has an initial object.

Now let X be an object of Cat, and let us consider a cocartesian diagram in $X^\wedge$:

$$\begin{array}{ccc} G & \xrightarrow{i} & F \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ G' & \xrightarrow{i'} & F', \end{array} \qquad (*)$$

where i is a monomorphism. We know that the cocartesian character of the square is p. 111

[page 162]

checked "argument by argument" by the cocartesian character of

$$\begin{array}{ccc} G(x) & \xrightarrow{i(x)} & F(x) \\ {\scriptstyle p(x)}\downarrow & & \downarrow{\scriptstyle q(x)} \\ G'(x) & \xrightarrow{i'(x)} & F'(x), \end{array} \qquad (*)$$

in Ens, $\forall\, x$ in X. Now such a cocartesian square in Ens, with $i(x)$ monomorphism, is also cartesian. This implies that the square Q in $X^\wedge$ is also cartesian.

Let us consider, on the other hand, the canonical functor

$$\varphi_X : F \longmapsto X/F : X^\wedge \longrightarrow \mathrm{Cat}.$$

It is better, moreover, to look at it as a functor

$$\varphi_X : X^\wedge \longrightarrow \mathrm{Cat}/X,$$

and I recall, for the record, that this functor is fully faithful, and that its essential image consists of the categories X' over X such that

$$X' \longrightarrow X$$

is a local isomorphism, or what amounts to the same, that X' is *fibered* over X with discrete fibers: [page 163]

$$\varphi_X : X^\wedge \xrightarrow{\sim} \underbrace{\underline{\mathrm{Fib\ disc}}\, X}_{\substack{\text{fibered categories} \\ \text{with discrete fibers}}} .$$

Let us note that every arrow $X' \longrightarrow X''$ in $\underline{\mathrm{Fib\ disc}}\, X$ is a local isomorphism (tautological on $X'/x' \longrightarrow X''/x''$), which shows that if we have a diagram

$$\begin{array}{ccc} & & X_1 \\ & \overset{\text{iso.}}{\searrow} & \\ X_2 & \underset{\text{iso.}}{\swarrow} & \\ & X/x & \end{array}$$

$$\begin{array}{ccc} X_1 & & X_2 \\ & \searrow \quad \swarrow & \\ & X_0 & \end{array}$$

in $\underline{\mathrm{Fib\ disc}}\, X$, the fiber product $X_3 = X_1 \times_{X_0} X_2$ in Cat is such that $X_3 \longrightarrow X_2$ is a local isomorphism, hence also $X_3 \longrightarrow X$. That is to say $\underline{\mathrm{Fib\ disc}}(X)$ is stable under fiber products. This implies that fiber products exist in $\underline{\mathrm{Fib\ disc}}\, X$, and that they are the fiber products in Cat, or in Cat/X.

Likewise for finite products in $\underline{\mathrm{Fib\ disc}}\, X$, they are the finite products in Cat/X. In short [p. 112] [page 164]

Lemma 1. *The full subcategory* $\underline{\mathrm{Fib\ disc}}\, X$ *of* Cat/X *is stable under finite* $\varprojlim$*, hence the* $\varprojlim$ *exist in* $\underline{\mathrm{Fib\ disc}}\, X$ *and the inclusion functor*

$$\underline{\mathrm{Fib\ disc}}\, X \longrightarrow \mathrm{Cat}/X$$

commutes with them, and in particular transforms cartesian squares of $\underline{\mathrm{Fib\ disc}}\, X$ *into cartesian squares of* Cat/X*, or what amounts to the same, of* Cat*. Moreover, by virtue of the equivalence*

$$\varphi_X : X^\wedge \xrightarrow{\sim} \underline{\mathrm{Fib\ disc}}\, X \hookrightarrow \mathrm{Cat}/X,$$

[page 165]

the functor φ_X from $X^\wedge$ to Cat$/X$ *is left exact, in particular it transforms cartesian squares of $X^\wedge$ into cartesian squares of* Cat$/X$*, or again of* Cat.

Let us also note that being left exact, the functor φ_X from $X^\wedge$ to Cat$/X$ transforms monomorphisms into monomorphisms. On the other hand:

Lemma 2. *The monomorphisms which are local isomorphisms are exactly the open immersions. So in* Fib disc X*, the monomorphisms are the open immersions, and the functor $\varphi_X : F \longmapsto X/F$ transforms monomorphisms into open immersions.*

Let us then return to the cocartesian diagram $(*)$ (p. 161), and to the corresponding diagram in Cat (and even in Fib disc X)

$$\begin{array}{ccccc} X/G & \xrightarrow{X/i} & X/F & \dashleftarrow & T \\ {\scriptstyle X/p}\downarrow & & \downarrow{\scriptstyle X/q} & & \vdots \\ X/G' & \xrightarrow{X/i'} & X/F' & \dashleftarrow & T'. \end{array}$$

Since $(*)$ is cartesian, the preceding square is cartesian, by what precedes. I want to prove that it is also cocartesian (in Cat$/X$, or in Cat, it amounts to the same), and for [page 166] this I apply corollary 2 of proposition 15 (p. 161), taking into account that X/q is a local isomorphism, and X/i' (being a local isomorphism monomorphism) an open immersion. It suffices to prove that, T' and T being the closed subcategories complementary to the opens X/G' of X/F' and to X/G of X/F, that $T \longrightarrow T'$ is isomorphism.

The verification is immediate: T is the category of arrows $x \longrightarrow G$ in $X^\wedge$ (with source x in X) which do not factor through F , T' that of arrows $x \longrightarrow G'$

which do not factor through F', the functor $T \longrightarrow T'$ being the functor of composition p. 113 of $x \longrightarrow G$ with $G \xrightarrow{q} G'$

$$\begin{array}{ccc} F & \hookrightarrow & G \\ \downarrow & & \downarrow \\ F' & \hookrightarrow & G' \end{array}$$

(NB The composite does not factor through F', for otherwise, because of the cartesian character of Q, $x \longrightarrow G$ would factor through F.) But for given x, the datum of $x \longrightarrow G$ not factoring through F is that of an element of $G(x)\backslash F(x)$, and because of the [page 167] cocartesian character of Q,

$$(G(x)\backslash F(x)) \longrightarrow (G'(x)\backslash F'(x))$$

is bijective, i.e. "it amounts to the same thing" to give $x \longrightarrow G$ not factoring through F, or $x \longrightarrow G'$ not factoring through F'.

Thus we have proved the

Theorem 5. *Let X be in* Cat*, and*

$$Q: \qquad \begin{array}{ccc} F & \xhookrightarrow{i} & G \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ F' & \xhookrightarrow{i'} & G' \end{array}$$

[109] So it suffices q W_0-smooth, where $W_0 = \{f \in \mathrm{Fl}\,\mathrm{Cat} \mid \pi_0(f) \text{ bijective}\}$.

[110] NB Here t is an arbitrary element of $\mathrm{Ob}\,X$, t' its image in X'.

a cocartesian square in $X^\wedge$, with i monomorphism. Then i' is monomorphism, the square Q is also cartesian, and the corresponding square in Cat$/X$ *or in* Cat

$$\begin{array}{ccc} X/F & \xrightarrow{X/i} & X/G \\ {\scriptstyle X/p}\downarrow & & \downarrow{\scriptstyle X/q} \\ X/F' & \xrightarrow{X/i'} & X/G' \end{array}$$

is both cartesian and cocartesian. Moreover, if W is a basic localizer in Cat*, satisfying* W(7 bis) (in addition to W(1,2,3))*, then the square X/Q in* Cat *is W-cocartesian.* [page 168]

This last statement follows from what preceded and from corollary 2 of theorem 4, taking into account that X/q is a local isomorphism, hence is W-smooth. p. 114

Corollary: ([111]) *We have* (assuming W(7 bis)) *the equivalences*

$$\begin{aligned} i \in W_X \quad &\Longleftrightarrow \quad i' \in W_X \\ p \in W_X \quad &\Longleftrightarrow \quad q \in W_X, \end{aligned}$$

where

$$W_X \subseteq Fl(X^\wedge)$$

is defined by (cf. Pursuing Stacks …)

$$u \in W_X \qquad \Longleftrightarrow \qquad X/u : X/F \longrightarrow X/G \quad \textit{is in } W.$$

Thus, in a category $X^\wedge$, amalgamated sums involving a monomorphism behave perfectly from the homotopy point of view, more precisely for the localizers W_X, provided only that W satisfies W(7 bis). In particular, the monomorphisms in $X^\wedge$ are "W_X-cofibrations", in a sense similar to that of definition 2 c) (pages 109, 110). [page 169]

[page 170 does not exist.]

[111] **NB** If we only want $i \in W_X \Longrightarrow i' \in W_X$, $p \in W_X \Longrightarrow q \in W_X$, it suffices to have W(7 bis) *weak*, i.e. $i_0 \in W \Longrightarrow i \in W$ in the statement of W(7 bis), p. 76 (Loc(6′a) with the notation of XVI).

VI Criterion for W_∞-asphericity of the $\Omega^n(X,x)$.

p. 115
[page 171]

1 Cogitations on W_n.

We are going to define, for $n \in \mathbf{N}$,

$$W_n \subseteq \mathrm{Fl}(\mathrm{Cat})$$

by the following conditions (where $f : X \longrightarrow Y$ is an arrow in Cat).

$f \in W_0 \iff$ a) $\pi_0(f)$ bijective.

$\iff$ b) For every presheaf of sets F *constant* on Y, $H^0(Y,F) \longrightarrow H^0(X,f^*F)$ is bijective.

$\iff$ c) The functor $F \longmapsto f^*F$ from presheaves of sets *constant* on Y to constant sheaves of sets on X is fully faithful.

NB If F_0 is a set, $F_{0\,Y}$ the constant sheaf of sets defined by F_0 on Y, we have

$$H^0(Y,F_{0\,Y}) \simeq \mathrm{Hom}_{\mathrm{Ens}}(\pi_0(Y),F_0) \qquad \text{(functorial in } Y\text{)}$$
$$\mathrm{Hom}_{Y^\wedge}(F_{0\,Y},G_{0\,Y}) \simeq \mathrm{Hom}_{\mathrm{Ens}}(\pi_0(Y),\mathrm{Hom}_{\mathrm{Ens}}(F_0,G_0)) \qquad \text{(functorial in } Y\text{)},$$

which implies that a) $\Longrightarrow$ c) $\overset{\text{tautologically}}{\underset{\text{taking } \mathrm{Hom}_{Y^\wedge}(e_Y,F)}{\Longrightarrow}}$ b) $\Longrightarrow$ a).

$f \in W_1 \iff$ a) $\pi_0(f)$ bijective, and for every $x \in \mathrm{Ob}\,X$, $\pi_1(f,x) : \pi_1(X,x) \longrightarrow \pi_1(Y,f(x))$ is bijective. (NB It suffices to check this for *one* x in each connected component of X.)

$\iff$ a′) $\Pi(X) \xrightarrow{\Pi f} \Pi(Y)$ is an equivalence of categories.

$\iff$ b) $H^0(Y,F) \longrightarrow H^0(X,f^*F)$ bijective for every sheaf of sets *locally constant* on Y, and $H^1(Y,F) \longrightarrow H^1(X,f^*F)$ bijective if in addition F is a sheaf of groups.

$\iff$ b′) As in b), but for H^1 we restrict ourselves to F *constant*, and require $H^1(f)$ *surjective* only.

$\iff$ c) The functor $F \longmapsto f^*F$ from the category of locally constant sheaves of sets on Y to that of locally constant sheaves on X is an equivalence of categories.

NB The group $\pi_1(X,x)$ is defined via the theory of the fundamental groupoid. We know that it is in canonical bijection with $\pi_0(\Omega(X,x))$, where $\Omega(X,x) = \underline{\mathrm{Ch}}(X;x,x)$ (which allows us to have a monoidal composition law on $\Omega(X,x)$, giving the group law on π_0), or at one's choice $\Omega(X,x) = \underline{\mathrm{Ch}}_\infty(X;x,x)$ (more convenient in certain situations, but not directly giving the composition law in $\pi_0(\Omega(X,x))$).

p. 116
[page 172]

I am going to set, for $n \in \mathbf{N}$,

$$\pi_n(X,x) = \pi_0(\Omega^n(X,x)),$$

Ω^n denoting the n-th iterate of the functor $(X,x) \longmapsto \Omega(X,x)$ from the category $\mathrm{Cat}_\bullet$ of pointed objects of Cat to itself. I prefer here to work with the $\underline{\mathrm{Ch}}$ theory instead of $\underline{\mathrm{Ch}}_\infty$. Sooner or later we will have to give a comparison statement …

Theorem 1 ([112]). *Let $f : X \longrightarrow Y$ be in* Cat*, and let $n \in \mathbf{N}$, $n \geq 2$, S a subset of* $\mathrm{Ob}\,X$ *such that the composite $S \longrightarrow \mathrm{Ob}\,X \longrightarrow \pi_0(X)$ is epimorphism. The following conditions are equivalent.*

[112] Probably *false*. We should have a) $\Longrightarrow$ b), but b) $\Longrightarrow$ a) probably false already for $n = 2$ (except in the case X, Y 1-connected). Cf. corrected statement .

a) $\pi_i(f,x)$ *an isomorphism for every* $x \in \mathrm{Ob}\, X$ *and every* $i \in [0,n]$.

a′) As in a), but with $x \in S$.

b) We have

$$\begin{array}{lll} H^0(Y,F) & \xrightarrow{\sim} & H^0(X,f^*F) \quad \textit{for } F \textit{ a sheaf of sets} \\ & & \quad \textit{locally constant on } Y. \\ H^1(Y,F) & \xrightarrow{\sim} & H^1(X,f^*F) \quad \textit{if in addition } F \textit{ a sheaf of groups.} \\ H^i(Y,F) & \xrightarrow{\sim} & H^i(X,f^*F) \quad \textit{if in addition } F \textit{ an abelian sheaf, } i \in [2,n]. \end{array}$$

NB The equivalence of a) and a′) should be trivial, but for this we need to have defined the $\pi_n(X,x)$, for x varying in $\mathrm{Ob}\, X$, as a local system on X, something I have not yet done. I am going to try to do it in a draft of air. First of all, the $\Omega(X,x)$ are the fibers of a perfect W_ω-fibration ([113]), fitting into the cartesian square [page 173]

$$\begin{array}{ccc} \underline{\mathrm{Ch}}(X) & \longleftarrow & \Omega(X,-) \\ {\scriptstyle p_X=(\sigma_X,\beta_X)}\downarrow & \overset{i_X}{\nearrow} & \downarrow \\ X \times X & \xleftarrow{\mathrm{diag}_X} & X. \end{array}$$

We then know that we have a functor p. 117

$$\begin{array}{l} \Pi X \longrightarrow \mathrm{Hot}_{W_\omega} \\ \quad x \longmapsto \mathrm{hot}_{W_\omega}(\Omega(X,x)). \end{array} \qquad (*)$$

Now we have, for $i \geq 1$,

$$\pi_i(X,x) = \pi_{i-1}(\underbrace{\Omega(X,x)}_{\text{pointed object}}).$$

This gives the desired local system of the π_i, if only we had a functor a little more precise than $(*)$, namely

$$\Pi X \longrightarrow \mathrm{Hot}_{\bullet\, W_\omega} \qquad (\textit{pointed } W_\omega\text{-homotopy category})$$

([114]). Now the arrow $i_X : X \longrightarrow \underline{\mathrm{Ch}}X$ lifting p_X gives a canonical section of $\Omega(X,-)$ over X - precisely the one which assigns to every $x \in X$ the marked point 1_x of $\Omega(X;x,x)$ ("empty loop at x"). The question is thus this: [page 174]

> We have a W-fibration $\Omega \longrightarrow X$, with a given section σ. Does this define a canonical functor
>
> $$X \longrightarrow \mathrm{Hot}_{\bullet\, W},$$
>
> turning every arrow into isomorphism?

The natural idea is to take

$$x \longmapsto \mathrm{hot}_{\bullet\, W}(\Omega/x)$$

by pointing Ω/x at the marked point $\sigma(x)$ of $\Omega_x \hookrightarrow \Omega/x$. The trouble is that the transition morphisms $\Omega/x \longrightarrow \Omega/x'$ (for $x \xrightarrow{u} x'$ in X) are not compatible with the marked points. But we have an arrow (namely $\sigma(u)$) going from the image $\sigma(x)/x'$ of the marked point

[113] **NB** W_ω is the smallest localizer of Cat (satisfying W(1, 2 bis, 3)). [page 175]

[114] But one must also check that the $\pi_n(X,x)$ $(n \geq 1)$ are functors on $\mathrm{Hot}_{\bullet\, W_\infty}$, i.e. transform W_∞-equivalences into likewise. This follows from W(4a) for W_∞, which implies that $(X,x) \mapsto \Omega(X,x)$ factors through $\mathrm{Hot}_{\bullet\, W_\infty}$ (as a functor with values in $\mathrm{Hot}_{\bullet\, W_\infty}$).

of Ω/x to the marked point $\sigma(x')/x'$ of Ω/x'. This should suffice to define the arrow

$$(\Omega/x, \sigma(x)/x) \longrightarrow (\Omega/x', \sigma(x')/x')$$

as an arrow in $\mathrm{Hot}_{\bullet W}$, with transitivity for a composite $x \xrightarrow{u} x' \xrightarrow{v} x''$ in X. For the moment I will confine myself to an indication, leaving detailed verifications for later. If need be, I will use right now that for $i \in \mathbf{N}$, $i \geq 1$, p. 118 [page 176]

$$\pi_i(f, x) : \pi_i(X, x) \longrightarrow \pi_i(Y, f(x))$$

can be interpreted as coming from a homomorphism of local systems

$$\underset{\sim i}{\pi}(X, -) \longrightarrow f^*(\underset{\sim i}{\pi}(Y, -))$$

on X. Condition a) of Theorem 1 is interpreted by saying that this homomorphism is an *isomorphism* for $i \in [1, n]$, and condition a′) then merely recalls that it suffices to test that this is so for *one* point of each connected component of X.

The statement of Theorem 1 retains a meaning for $n = 1$, and in this case was "recalled" previously (and is ± tautological). This will allow us to proceed by induction on n, assuming that the theorem is proved for $n - 1$. This means that we can suppose that conditions a) *and* b) are satisfied by f for $i \in [1, n-1]$, and confine ourselves to proving that, under these conditions, condition a) (or a′), for $i = n$, is equivalent to condition b) for $i = n$.

So we simply have to prove this: [page 177]

Lemma ([115]). *Suppose that $\pi_i(f, x)$ isomorphism for $i \in [0, n-1]$ and b) satisfied for $i \in [0, n-1]$, where $n \in \mathbf{N}$, $n \geq 2$, is fixed. Then the following conditions are equivalent:*

a) $\pi_n(f, x) : \pi_n(X, x) \longrightarrow \pi_n(Y, f(x))$ *isomorphism for every* $x \in \mathrm{Ob}\, X$.

b) $H^n(Y, F) \xrightarrow{\sim} H^n(X, f^*F)$ *isomorphism for every locally constant abelian sheaf on* Y.

Let us note that f induces an arrow in $\mathrm{Cat}_\bullet$,

$$\Omega(f, x) : \Omega(X, x) \longrightarrow \Omega(Y, f(x)),$$

whence

$$\begin{array}{ccc} \pi_i(\Omega(X, x)) & \longrightarrow & \pi_i(\Omega(Y, f(x))) \\ \wr| & & \wr| \\ \pi_{i+1}(X, x) & \longrightarrow & \pi_{i+1}(Y, f(x)), \end{array}$$

identifying with $\pi_{i+1}(f, x)$:

$$\pi_i(\Omega(f, x)) \simeq \pi_{i+1}(f, x).$$

Hypothesis a) of the lemma implies that $\pi_i(\Omega(X, x)) \xrightarrow{\sim} \pi_i(\Omega(Y, y))$ for $i \in [0, n-1]$. Here we are dealing with the π_i based at the unit points. We will probably have to prove that, because this is a morphism of monoids in Cat (satisfying certain conditions?) this remains true for the π_i at *every* point ξ of $\Omega(X, x)$. By the induction hypothesis, we will thus know that $\Omega(X, x) \longrightarrow \Omega(Y, y)$ induces an isomorphism on cohomology with locally constant coefficients, up to and including dimension $n - 1$. p. 119 [page 178]

Finally, I have ± convinced myself that the statement of Theorem 1 is false as it stands. Here is a statement that I hope is true, and that I am going to try to prove in a

[115] Probably false.

"utilitarian" spirit, using everything I know in homotopy and cohomology (and not just what I have developed "from scratch" in the modelizer Cat).

Theorem 1 corrected. *Let us consider, for a given integer $n \geq 2$, the following conditions on an arrow $f : X \longrightarrow Y$ in* Cat *(or a continuous map of topological spaces, or a morphism in $\mathbf{\Delta}^\wedge$ etc. etc.):*

a_n) *If Z is a "homotopy fiber" of f (NB there is essentially one per connected component of Y), then Z is n-connected, i.e. (e.g.) $\pi_i(Z) = 0$ for $i \in [0, n]$.* [page 179]

b_n) *$\pi_0(f)$ is bijective, and $\forall\ x \in X$, $\pi_i(f, x) : \pi_i(X, x) \longrightarrow \pi_i(Y, f(x))$ is isomorphism for $i \in [1, n]$. (**NB** It suffices to say this for* one *point x of each connected component of X.)*

c_n) *$H^i(Y, F) \longrightarrow H^i(X, f^*F)$ isomorphism for $0 \leq i \leq n$ and F a locally constant sheaf on Y (with F abelian if $i \geq 2$, F a sheaf of groups if $i = 1$, F a sheaf of sets if $i = 0$).*

We then have

$$a_n \implies b_n \implies ({}^{116})\ c_n \implies ({}^{117})\ a_{n-1}$$

(and I presume that each of these implications is strict).

Proof. $a_n \Longrightarrow b_n$ is an immediate consequence of the homotopy exact sequence for a Hot-fibration $\bar{X} \xrightarrow{\bar{f}} Y$ which factors $X \xrightarrow{f} Y$ as $f = \bar{f} \circ i$, $i \in W_\infty$. In fact, a_n *is equivalent* to b_n + the surjectivity of the $\pi_{n+1}(X, x) \longrightarrow \pi_{n+1}(Y, f(x))$, which of course does *not* follow from b_n, i.e. from the bijectivity of the $\pi_i(f, x)$ for $i \leq n$ (take some $K(\pi, n+1)$). Thus the implication is indeed strict.

$b_n \Longrightarrow c_n$. We can suppose that f is a Hot-fibration (and even Hot-perfect if we feel like it …). We already know that for [p. 120] [page 180]

$$H^i(Y, F) \longrightarrow H^i(X, f^*F) \qquad (*)$$

everything is O.K. if $i \in \{0, 1\}$. So we can suppose F abelian. If that gets us anywhere, we can suppose, by induction, (that the implication $b_{n-1} \Longrightarrow c_{n-1}$ is known, hence) that $(*)$ is isomorphism for $i < n$. Remains to see it for $i = n$. We are reduced of course, since $\pi_0(f)$ is bijective, to the case where X, Y are 0-connected, so there is essentially a single fiber Z. The homotopy exact sequence tells us that

$$\pi_i(Z, z) = 0 \qquad \text{for } i \leq n-1 \ (Z \text{ is } (n-1)\text{-connected})$$

and moreover

$$\pi_n(Z, z) \longrightarrow \pi_n(X, z) \qquad \text{is zero}$$

(since $\pi_n(X, z) \longrightarrow \pi_n(Y, y)$ isomorphism hence monomorphism), so

$$\pi_{n+1}(Y, y) \longrightarrow \pi_n(Z, z) \qquad \text{epimorphism.}$$

As Z is $(n-1)$-connected (hence also 1-connected), we have

$$H^i(Z, G) = 0\ , \qquad \text{if } 0 < i < n\ , \qquad (*)$$

for G locally constant (hence constant) on Z.

[116] if X, Y 1-connected, this is really proved, otherwise, there are verifications to be made (which look sorital …). [page 181]

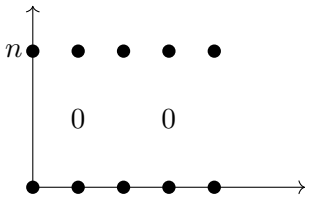
[117] **NB** Even if X, Y 1-connected, it is not clear that $c_n \Longrightarrow b_n$, for one must show that $\pi_n(Z) \longrightarrow \pi_n(X)$ is zero, and one only finds that the composite with $\pi_n(X) \longrightarrow H_n(X)$ is zero. I presume that there are counterexamples.

Let us write the Leray spectral sequence for $f : X \longrightarrow Y$ and $f^*F = F_X$:

$$H^*(X, f^*F) \Longleftarrow E_2^{pq} = H^p(Y, H^q(Z, f^*F)),$$

where "$H^q(Z)$" is to be viewed as a local system on Y. So by $(*)$ we have

$E_2^{pq} = 0$ if $0 < q < n$,



,

so we find the exact sequence

$$0 \longrightarrow H^n(Y, F) \longrightarrow H^n(X, F_X) \xrightarrow{\alpha} H^0(Y, \underbrace{H^n(Z, \overbrace{F_Z}^{M_Z})}_{\substack{= \mathrm{Hom}(\pi_n(Z,z),M) \\ \text{by Hurewicz thm.}}}) \longrightarrow H^{n+1}(Y, F).$$

Here I fix a point x of X, Z denotes the fiber of f at $y = f(x)$, and M that of F at y, $M = F_y$. The group $\pi_1(Y)$ acts on it, for that matter $\pi \overset{\text{def}}{=} \pi_1(X, x) \simeq \pi_1(Y, y)$ does too. The $H^0(Y, H^n(Z, F_Z))$ can be interpreted, unless I am mistaken, as p. 121

$$\mathrm{Hom}_\pi(\pi_n(Z, x), M),$$

taking into account the action of $\pi = \pi_1(X, x)$ on $\pi_n(Z, x)$ as well. (To be checked.) We have to interpret the arrow

$$H^n(X, F_X) \longrightarrow \mathrm{Hom}_\pi(\pi_n(Z, x), M),$$

via the composite map

$$\pi_n(Z, z) \longrightarrow \pi_n(X, x) \longrightarrow H_n(X, \mathbf{Z}). \qquad (*)$$

If F is *constant* on Y (e.g. if Y 1-connected), thus with value M, we have the canonical arrows

$$H^n(X, F_X) = H^n(X, M) \longrightarrow \mathrm{Hom}(H_n(X, \mathbf{Z}), M) \longrightarrow \mathrm{Hom}(\pi_n(Z, z), M)^{(118)} \qquad (*')$$

,

and surely (apart from any hypothesis of type b_n or other) the direct canonical map "of localization on Y", followed by …, [page 182]

$$H^n(X, F_X) \longrightarrow H^0(Y, H^n(Z, F_Z)) \longrightarrow \mathrm{Hom}_\pi(\pi_n(Z), M) \qquad (**)$$

is none other than the composite of the canonical maps above. In the present case where we suppose b_n, the second arrow in $(**)$ is isomorphism, so the composite "is none other" than the localization arrow itself. On the other hand $\pi_n(Z, z) \longrightarrow \pi_n(X, x)$ in $(*)$ is zero, so also $\pi_n(Z, z) \longrightarrow H_n(X, \mathbf{Z})$, so also the composite arrow $(*')$ deduced from it. Let us then return to the Leray exact sequence (previous page), we find that $H^n(Y, F) \xrightarrow{\sim} H^n(X, F_X)$.

[118] thanks to composition with $(*)$.

How to recover the case where F is *not* constant? In the case $n = 2$, this worked, by passing to the universal covers $\tilde{X}$, $\tilde{Y}$ of X, Y (with respect to the chosen points x, y), and writing the spectral sequences

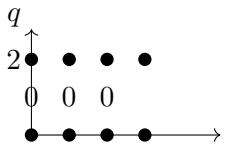
$$\begin{array}{ccccc} H^*(Y,F) & \Longleftarrow & E_2^{pq} & = & H^p(\pi, H^q(\tilde{Y}, F_{\tilde{Y}})) \\ \downarrow & & \downarrow & & \\ H^*(X,F_X) & \Longleftarrow & E_2'^{pq} & = & H^p(\pi, H^q(\tilde{X}, F_{\tilde{X}})). \end{array}$$

The hypothesis b_n is preserved in passing from $f : X \longrightarrow Y$ to $\tilde{f} = \tilde{f}_x : \tilde{X} \longrightarrow \tilde{Y}$, so we know that p. 122

$$E_2^{pq} \xrightarrow{\sim} E_2'^{pq} \qquad \text{isomorphism if } q \leq n. \qquad (*)$$ [page 183]

In the case $n = 2$, using that

$$E_2^{p1} = E_2'^{p1} = 0 \qquad \text{since } H^1(\tilde{X}) = H^1(\tilde{Y}) = 0,$$



,

we find two exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H^2(\pi, H^0(\tilde{Y})) & \longrightarrow & H^2(Y) & \longrightarrow & H^0(\pi, H^2(\tilde{Y})) & \longrightarrow & H^3(\pi, H^0(\tilde{Y})) \\ & & \downarrow \text{iso} & & \downarrow & & \downarrow \text{iso} & & \downarrow \text{iso} \\ 0 & \longrightarrow & H^2(\pi, H^0(\tilde{X})) & \longrightarrow & H^2(X) & \longrightarrow & H^0(\pi, H^2(\tilde{X})) & \longrightarrow & H^3(\pi, H^0(\tilde{X})), \end{array}$$

and a homomorphism from one to the other, which allows us to conclude by the 5 lemma that $H^2(Y,F) \longrightarrow H^2(X,F_X)$ is isomorphism. But for arbitrary n, the relation $(*)$ on the E_2 terms (for $q \leq n$) does not seem to me to imply the similar relation for the higher E_n, and from there for the E_∞ terms, which would allow us to conclude.

So I return to the exact sequence p. 181, with F a sheaf that is *twisted.* I feel like finding a canonical map

$$H^n(X,G) \longrightarrow \overbrace{\mathrm{Hom}_\pi(\pi_n(X,x), G_x)}^{\text{Homs } \textit{additive} \text{ of course, i.e. of } \mathbf{Z}[\pi]\text{-modules}} \qquad ,$$

where G is any locally constant abelian sheaf on X, and $\pi = \pi_1(X,x)$. If I interpret $\pi_n(X,x)$ in terms of homotopy classes of pointed maps [page 184]

$$(S^n, x_0) \xrightarrow{\varphi} (X,x)$$

I do have, for such a map φ,

$$\varphi^* : H^n(X,G) \longrightarrow H^n(S^n, \overbrace{\varphi^*G}^{\simeq (G_x)_{S^n}}) \simeq G_x$$

(taking into account that S^n is 1-connected, and that its n-cohomology with values in a constant sheaf M is M.) Surely the map

$$H^n(X,G) \longrightarrow G_x$$

thus defined depends only on the pointed homotopy class of φ, hence p. 123

$$\pi_n(X,x) \times H^n(X,G) \longrightarrow G_x,$$

and we must also prove a) that this is compatible with the actions of $\pi = \pi_1(X,x)$, and b) that it is additive in $\pi_n(X,x)$, to have the desired map

$$H^n(X,G) \longrightarrow \mathrm{Hom}_\pi(\pi_n(X,x), G_x).$$

If I admit this, it is only one step to admitting also that the arrow that really interests us (in the exact sequence p. 181) [page 185]

$$H^n(X,F_X) \longrightarrow \mathrm{Hom}_\pi(\pi_n(Z,z), \underbrace{M}_{=F_y})$$

is none other than the composite of the preceding arrow (previous page) for $G = F_X$, with

$$\mathrm{Hom}_\pi(\pi_n(X,x), M) \longrightarrow \mathrm{Hom}_\pi(\pi_n(Z,x), M)$$

deduced from

$$\pi_n(Z,x) \longrightarrow \pi_n(X,x).$$

Now this arrow is zero, by the hypothesis b_n. So the arrow α in the exact sequence page 181 is zero, so $H^n(Y,F) \longrightarrow H^n(X,F_X)$ is isomorphism, q.e.d. (but there is quite a bit of checking to do …).

$\mathrm{c}_n \Longrightarrow \mathrm{a}_{n-1}$. We can proceed by induction on n. (NB The statement makes sense even for $n = 1$, where it is proved, and even $\mathrm{c}_1 \Longrightarrow \mathrm{b}_1$, and $\mathrm{b}_1 \Longrightarrow \mathrm{a}_0$, more generally $\mathrm{b}_n \Longrightarrow \mathrm{a}_{n-1}$, immediate from homotopy exact sequence.) So we already know, in addition to the hypothesis c_n, that we have a_{n-2}, i.e. Z is $(n-2)$-connected. We have to prove that it is even $(n-1)$-connected, i.e. $\pi_{n-1}(Z) = 0$. The case $n = 2$ requires perhaps special attention, since we do not even know yet that Z is 1-connected. But we know that c_2 implies c_1, which is equivalent to b_1, so we have [page 186]

$$\pi_1(X,x) \longrightarrow \pi_1(Y,y) \qquad \text{isomorphism,}$$

so

$$\pi_1(Z,x) \longrightarrow \pi_1(X,x) \qquad \textit{zero,}$$

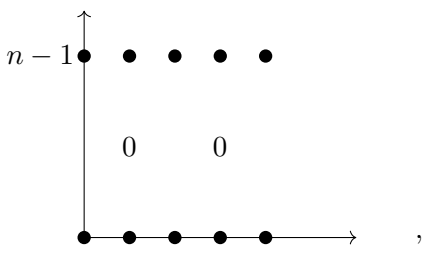
so

$$\pi_2(Y,y) \longrightarrow \pi_1(Z,x) \qquad \text{epimorphism,}$$

which implies at least that $\pi_1(Z,x)$ is *abelian.* We must view it as a π-module, and prove its vanishing as such (using c_2).

Let us pass to the general case. We have for F locally constant abelian on Y p. 124

$$H^*(X,F_X) \Longleftarrow \underbrace{E_2^{pq} = H^p(Y, H^q(Z,F_Z))}_{\substack{=0 \text{ if } 0<q<n-1 \\ \text{since } Z \text{ is } (n-2)\text{-connected}}},$$



,

[page 187]

Hence

$$0 \longrightarrow H^{n-1}(Y,F) \longrightarrow H^{n-1}(X,F_X) \longrightarrow \underbrace{H^0(Y,H^{n-1}(Z,F_Z))}_{\substack{\mathrm{Hom}_\pi(\pi_{n-1}(Z,x),F_x)\\ \text{by Hurewicz}}} \longrightarrow H^n(Y,F) \longrightarrow H^n(X,F_X).$$

The hypothesis c_n implies that we have

$$\mathrm{Hom}_\pi(\pi_{n-1}(Z,x), \underbrace{M}_{=F_x}) = 0$$

and this *whatever the π-module M,* since $\pi = \pi_1(X,x) \xrightarrow{\sim} \pi_1(Y,y)$. This implies (taking e.g. $M = \pi_{n-1}(Z,x)$!) that $\pi_{n-1}(Z,x) = 0$. (But keeping in mind, in the case $n = 2$, that we already know that $\pi_1(Z,x)$ is abelian).

So the theorem corrected on page 178 does look true. It is a bit sad, since in each of the implications

$$\mathrm{a}_n \implies \mathrm{b}_n \implies \mathrm{c}_n \implies \mathrm{a}_{n-1}$$

we lose a smidgen, and in the end we come out with the magnificent implication

$$\mathrm{a}_n \implies \mathrm{a}_{n-1}.$$

In any case, it is ± clear that the "right" localizer W_n, the one that is to define the "n-homotopy types" as objects of Hot_{W_n}, is defined by the condition b_n.

2 Criterion for W_n-asphericity.

I am going to try to do better when Y is reduced to a point. This time, I

want to be strict, and give (if I can) complete proofs in the framework in which I place myself at present, that of the modelizer Cat, and using only what I know how to prove there directly at present, without referring to semi-simplicial or topological theory. p. 125 [page 188]

Theorem 2 (?). *Let X be in* Cat, $n \in \mathbf{N}$ *an integer* ≥ 1. *Then the following conditions are equivalent:*

a_n) X is 0-connected, and if x is in X, $\pi_i(X,x) = 0$ for $i \in [1,n]$ (a condition which does not depend on the choice of x), i.e. the $\Omega^i(X,x)$ for $i \in [1,n]$ are 0-connected.

b_n) For every group G, $G \longrightarrow H^0(X,G_X)$ is an isomorphism, and $H^1(X,G_X) = 0$. Moreover, if G is abelian, we also have $H^i(X,G_X) = 0$, for $i \in [1,n]$.

Proof. For $n = 1$ this equivalence is ± tautological and was recalled (in a slightly different form) at the beginning (p. 171). We proceed by induction. We may therefore assume that $n \geq 2$, that X satisfies a_{n-1}, b_{n-1}, and prove that then a_n, b_n are equivalent, i.e. $\pi_n(X,x) = 0$ is equivalent to $H^n(X,M) = 0$ for every abelian group M.

We use the W_∞-fibration (perfect) [page 189]

$$\begin{array}{ccc} E = \underline{\mathrm{Ch}}(X;x,-) & & E\ W_\infty\text{-aspheric} \\ \downarrow & & \\ X & & \end{array}$$

with fiber Z at x equal to $\Omega(X,x)$.

Suppose a_n) for X, hence a_{n-1}) for $\Omega(X,x)$. By the induction hypothesis, we therefore have

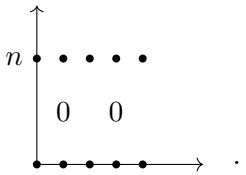
$$H^i(Z,M)=0 \qquad \text{for } i\in[1,n-1],$$

M an abelian group. So in the Leray spectral sequence,

$$H^*(E,M) \Longleftarrow E_2^{pq}=H^p(X,H^q(Z,M))$$

we have

$E_2^{pq}=0$ if $q\in[1,n-1]$,



.

We therefore have the exact sequence p. 126

$$0\longrightarrow H^n(X,M)\longrightarrow H^n(E,M)$$

and since E is W_∞-aspheric, we have $H^n(E,M)=0$, hence $H^n(X,M)=0$.

Conversely, suppose b_n), let us prove a_n). We already know that we have a_{n-1}), hence a_{n-2} for $\Omega(X,x)$, which implies

$$H^i(Z,M)=0 \qquad \text{for } i\in[1,n-2],$$

hence

$$E_2^{pq}=0 \qquad \text{if } q\in[1,n-2].$$

We therefore have the exact sequence

[page 190]

$$0\longrightarrow \underbrace{H^{n-1}(X,M)}_{=0}\longrightarrow \underbrace{H^{n-1}(E,M)}_{=0}\longrightarrow H^0(X,H^{n-1}(Z,M))\longrightarrow \underbrace{H^n(X,M)}_{=0}\longrightarrow H^n(E,M),$$

hence

$$H^0(X,\ \underbrace{H^{n-1}(Z,M)}_{\substack{\text{local system}\\ \textit{constant} \text{ on } X,\\ \text{since } X \text{ is 1-connected}}}\ )=0,$$

hence

$$H^{n-1}(Z,M)=0,$$

which, by the induction hypothesis, implies $\pi_{n-1}(Z,z)=0$, hence $\pi_n(X,x)=0$, q.e.d.

Corollary. *For X to be W_∞-aspheric, i.e. for us to have $G\xrightarrow{\sim} H^0(X,G)$, $H^1(X,G)=0$ every group , and $H^i(X,G)=0$ if $i\geq 2$ for G abelian, it is necessary and sufficient that* 1°) *X be 0-connected, and (x being a point of X)* 2°) *the $\Omega^i(X,x)$ 0-connected for $i\geq 1$.*

[page 191]

3 On the philosophy of homotopy fibers.

The criterion of asphericity (for W_∞) that I have just obtained, thanks to the functor $\underline{\mathrm{Ch}}(X)$ (or at one's choice, $\underline{\mathrm{Ch}}_\infty(X)$) and its derivatives $\Omega(X,x)$ and the iterates, removes all doubt (if any could still remain) that the "path spaces" that I have constructed (quite laboriously!) are indeed "the right ones". I would now like to get as precise an idea as

possible of their "canonicity", to what extent these constructions are independent of a choice (e.g. that between the theory $\underline{\mathrm{Ch}}$, and $\underline{\mathrm{Ch}}_\infty$).

Everything comes down, in the end, to questions of factorization of an arbitrary arrow

$$f : X \longrightarrow Y$$

of Cat as p. 127

$$X \xrightarrow{i_f} \bar{X} \xrightarrow{\bar{f}} Y,$$

with

$$i_f \in W, \qquad \bar{f} \in W\text{-fib}$$

(the fact that $\bar{f}$ can even be chosen W_ω-perfect, not seeming essential to me here).

This is indeed what is involved in constructing
an object "paths in X" [page 192]

$$\underline{\mathrm{Ch}}(X),$$

by factoring the diagonal map

$$X \xrightarrow{\mathrm{diag}_X} X \times X,$$

and also when we construct $\Omega(X, x)$ as the fiber at x of a W-fibration

$$E(X, x) \longrightarrow (X, x)$$

which factors the inclusion

$$\{x\} \longrightarrow X$$

(thus $E(X, x)$ being W-aspheric).

We can be more or less demanding, and consider, either for a "target" Y (or rather an S) fixed in Cat, a functorial factorization

$$C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fib}}_W S \hookrightarrow \mathrm{Cat}/S$$

of the structural morphism $X \longrightarrow S$ of an object, as

$$X \xrightarrow{i_X} C(X) \longrightarrow S,$$

i.e. we give ourselves the functor C, *plus*
the functorial morphism

$$i_C : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C$$ [page 193]

with

$$(i_X : X \longrightarrow C(X)) \in W \qquad \forall\ X \text{ in } \mathrm{Cat}/S,$$

or be still more demanding and vary the source and target of $f : X \longrightarrow Y (= S)$ at the same time, and ask for a functor

$$\begin{array}{rcl} C : \underline{\mathrm{Fl}}(\mathrm{Cat}) & \longrightarrow & \mathrm{Cat} \\ f & \longmapsto & C(f), \end{array}$$

with morphisms functorial in f

$$\underbrace{s(f)}_{=X} \xrightarrow{i_f} \underbrace{C(f)}_{=\bar{X}} \xrightarrow{p_f} \underbrace{b(f)}_{=Y}$$ p. 128

(i, p arrows in $\underline{\mathrm{Hom}}(\underline{\mathrm{Fl}}(\mathrm{Cat}), \mathrm{Cat})$), with

$$i_f \in W, \qquad p_f \in W\text{-fib}$$ [page 194]

for every arrow f in Cat.

In both points of view, in what sense do we have *uniqueness* of such a construction?

Let us first take the first point of view, factorization of arrows with fixed target S, hence question in Cat/S. I forgot an important property, but it follows from what has been said: the functor

$$X \longmapsto C(X): \qquad \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S$$

$$\underline{\mathrm{Fib}}_W S$$

transforms arrows W into likewise, as we see from the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{i_X} & C(X) \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle C(f)} \\ Y & \xrightarrow{i_Y} & C(Y), \end{array}$$

where i_X, i_Y are in W, hence

$$f \in W \qquad \Longleftrightarrow \qquad C(f) \in W.$$

Let

$$(C', i')$$

be a second functor, $\mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S$,

equipped with a morphism [page 195]

$$i' : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C'$$

which is in W argument by argument, and such that C' factors through $\underline{\mathrm{Fib}}_W S$. We can compare C and C' by means of $C \circ C'$, for

$$\begin{array}{ccccc} & & C \circ C' & & \\ & {\scriptstyle C*i'}\nearrow & & \nwarrow{\scriptstyle i*C'} & \\ C = C \circ \mathrm{id} & & & & C' = \mathrm{id} \circ C', \end{array}$$

i.e. we have, for every X in Cat/S, p. 129

$$\begin{array}{ccccc} & & C(C'(X)) & & \\ & {\scriptstyle C(i'_X)}\nearrow & & \nwarrow{\scriptstyle i_{C'(X)}} & \\ C(X) & & & & C'(X), \end{array}$$

the two arrows being in W (the one on the left because $C(W) \subseteq W$). These are arrows in $\underline{\mathrm{Fib}}_W S$, and if we assume that W satisfies W(4b), they induce W-equivalences in each fiber. This therefore already shows

that up to isomorphism (perhaps noncanonical), the composite functor defined by C, [page 196]

$$X \longmapsto C(X) \longmapsto (s \longmapsto \mathrm{hot}_W(C(X)_s))$$

$$\mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fib}}_W S \longrightarrow \underline{\mathrm{Hom}}(S^\circ, \mathrm{Hot}_W),$$

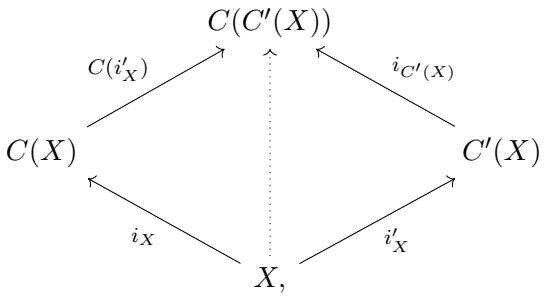
$$\varphi_C$$

does not depend on the particular choice of C, i.e. two such "homotopy fibers" functors (corresponding to two choices C, C') are always isomorphic. But instead of using $C \circ C'$ to define such an isomorphism, we could have used $C' \circ C$, or some other procedure. We would like to understand whether there is genuine uniqueness, an intrinsic characterization of these isomorphisms $\varphi_C \xrightarrow{\sim} \varphi_{C'}$, independently of one construction or another.

I presume that $C \circ C'$ is itself also equipped with

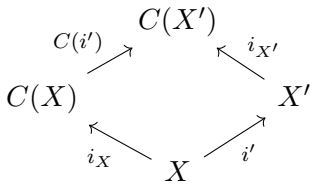
$$\mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C \circ C',$$

and even a priori in two possible ways [page 197]



either by $C(i'_X) \circ i_X$, or by $i_{C'(X)} \circ i'_X$, fortunately they are equal, since for every arrow $X \xrightarrow{i'} X'$ in Cat/S, the square

p. 130



is commutative (by functoriality of $i_X : X \longrightarrow C(X)$), and we apply this to the case of $X' = C'(X)$, $i' = i'_X$.

This remark encourages us to make a leap into abstraction, and to introduce the category [page 198]

$$\mathcal{C} = \mathcal{C}_S = \begin{cases} \text{category of pairs } (C, i), \text{ with} \\ C : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S \text{ factoring through } \underline{\mathrm{Fib}}_W S, \\ i : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C \text{ such that } i_X \in W \ \forall\ X \text{ in } \mathrm{Cat}/S, \end{cases} \qquad (*)$$

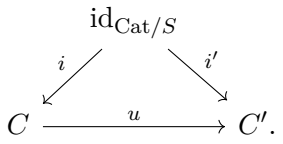
an arrow

$$(C, i) \longrightarrow (C', i')$$

being an arrow

$$C \xrightarrow{u} C' \qquad (\text{in } \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{Cat}/S))$$

making the square commutative



In other words, $\mathcal{C}$ is defined as a full subcategory of

$$\mathrm{id}_{\mathrm{Cat}/S} \backslash \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{Cat}/S), \qquad \text{[page 199]}$$

namely the one whose objects (C, i) satisfy the two conditions $(*)$ above (both involving the localizer W).

These conditions imply this

a) $\forall\ X$ in Cat/S, $(u_X : C(X) \longrightarrow C'(X)) \in W$, since

$$\begin{array}{ccc} & X & \\ {\scriptstyle i_X \in W}\swarrow & & \searrow{\scriptstyle i'_X \in W} \\ C(X) & \xrightarrow{u_X} & C'(X), \end{array}$$

and p. 131

b) if W satisfies W(4b), so that $u_X \in W_S^{\mathrm{f}}$, then this defines an *isomorphism* between $\varphi_C(X)$ and $\varphi_{C'}(X)$

$$\begin{array}{ccc} s & \longmapsto & \mathrm{hot}_W(C(X)_s) \\ & & \downarrow{\scriptstyle u_{X,s}} \\ s & \longmapsto & \mathrm{hot}_W(C'(X)_s). \end{array}$$

Better still, in this case, the composite functor

$$\mathcal{C} \times \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fib}}_W S \longrightarrow \mathrm{HOT}_W^{\mathrm{lc}}(S)$$

gives rise to

$$\mathcal{C} \longrightarrow \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{HOT}_W^{\mathrm{lc}}(S))$$

which transforms every arrow of $\mathcal{C}$ into *isomorphism,* hence factors through the fundamental groupoid of $\mathcal{C}$,

$$\Pi\mathcal{C} \longrightarrow \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{HOT}_W^{\mathrm{lc}}(S)).$$

As for the functor considered earlier [page 200]

$$\begin{array}{ccc} \mathcal{C} & \longrightarrow & \underbrace{\underline{\mathrm{Hom}}\,(\mathrm{Cat}/S, \underline{\mathrm{Hom}}(S^\circ, \mathrm{Hot}_W))}_{\simeq \underline{\mathrm{Hom}}((\mathrm{Cat}/S)\times S^\circ, \mathrm{Hot}_W)} \\ C & \longmapsto & ((X,s) \longmapsto \mathrm{hot}_W(C(X)_s)\,), \end{array}$$

it is deduced from the preceding one by composing with

$$\begin{array}{ccc} \mathrm{HOT}_W^{\mathrm{lc}}(S) & \longrightarrow & \underline{\mathrm{Hom}}(S^\circ, \mathrm{Hot}_W), \\ & \searrow \quad \nearrow & \\ & \underline{\mathrm{Hom}}(\Pi S^\circ, \mathrm{Hot}_W) & \end{array}$$

more exactly, with the functor deduced from it

$$\underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{HOT}_W^{\mathrm{lc}}(S)) \longrightarrow \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \underline{\mathrm{Hom}}(S^\circ, \mathrm{Hot}_W)).$$

We see emerging here a plausible principle of canonicity of isomorphisms, as regards the functors

$$C_W : \mathrm{Cat}/S \longrightarrow \mathrm{HOT}_W^{\mathrm{lc}}(S)$$

defined by the $C \in \mathrm{Ob}\,\mathcal{C}$, by the relation

p. 132

$$\Pi\mathcal{C} \approx \text{ one-point category, i.e. } \mathcal{C} \text{ is 1-connected.}$$

[page 201]

Moreover, the arguments used so far are of so formal a nature, that the specific aspects of the Modelizer Cat seem to play practically no role in them, or a very unobtrusive role.

The square on page 197 shows above all that the category $\mathcal{C}$ is filtering - (C, i), (C', i') both map into (C'', i''), with $C'' = C \circ C'$, and $i'' = (C * i') \circ i = (i * C') \circ i'$. There is even a law of composition in $\mathcal{C}$, by no means commutative or unital, that's for sure, but surely associative, and (if this composition is denoted by $*$) with functorial morphisms

$$\xi \xrightarrow{\alpha_{\xi,\xi'}} \xi * \xi' \xleftarrow{\beta_{\xi,\xi'}} \xi'$$

which ensure the upward filtering character of $\mathcal{C}$. These arrows seem somehow to replace the missing two-sided unit, and one expects them to have the same excellent properties as if one had such a unit, concerning the double composites $(\xi * \xi') * \xi'' \simeq \xi * (\xi' * \xi'')$,

[page 202]

$$(\xi * \xi') * \xi'' \xrightarrow{\sim} \xi * (\xi' * \xi'').$$

There are two ways of mapping a binary product

$$\xi * \xi', \qquad \xi' * \xi'', \qquad \xi * \xi''$$

into the ternary product, and well, they must in each of the three cases, be equal. And likewise there are, this granted, again two ways of mapping ξ into the ternary product, factoring either through $\xi * \xi'$, or through $\xi * \xi''$, and likewise for the other two functors ξ', ξ''; and there again, these two ways must be equal, in each of these three cases - so $3 + 3 = 6$ equalities in all to write down and to check - we've seen worse!

The computational philosophy should be this: If we have an n-fold product

$$\xi_1 * \xi_2 * \cdots * \xi_n$$

(where I dispense with putting in parentheses …), for every partial m-fold product ($m \in [1, n-1]$)

[page 203]

$$\xi_{i_1} * \xi_{i_2} * \cdots * \xi_{i_m} \qquad (i_1 < i_2 < \cdots < i_m),$$

using the associativity isomorphisms and the $\alpha_{\xi,\xi'}$, $\beta_{\xi,\xi'}$ left and right, there is nevertheless only one way of mapping it into the n-fold product. I really have the impression that this must be true. (Does this already imply that $\mathcal{C}$ is 1-connected?)

In fact, there is indeed a two-sided unit, but it is not in $\mathcal{C}$, but in the larger category

$$\mathrm{id}_{\mathrm{Cat}/S} \backslash \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{Cat}/S),$$

which itself, like any category

$$\mathrm{id}_{\mathcal{X}} \backslash \underline{\mathrm{Hom}}(\mathcal{X}, \mathcal{X}),$$

is a monoidal category, i.e. with an associative *and* unital product. The arrows $\alpha_{\xi,\xi'}$, $\beta_{\xi,\xi'}$ are the arrows that result from the presence of this unit (seeing that $\mathcal{C}$ is a full subcategory), and then it should become clear that we have the said compatibilities.

p. 133

But even a monoidal category, is it 1-connected? For example B_G, for G a group say, is indeed a monoidal category and even "groupal", a G-category (as strict as can be), and yet it is *not* 1-connected!

[page 204]

Of course, we can short-circuit the question of the 1-connectedness of $\mathcal{C}$, by noting that after all we already have a priori a privileged object in

$$\underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{HOT}^{\mathrm{lc}}_W(S)),$$

i.e. a canonical functor

$$\mathrm{Cat}/S \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S),$$

namely the composite functor

$$\mathrm{Cat}/S \underbrace{\longrightarrow}_{\substack{\text{localization}\\ \text{by } W_S}} (\mathrm{Cat}/S)W_S^{-1} \longrightarrow \underbrace{\mathrm{HOT}_W^{\mathrm{lc}}(S)}_{\stackrel{\text{def}}{=}(\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1}} \quad , \qquad (*)$$

where the second arrow is defined as the quasi-inverse of

$$(\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\mathrm{Cat}/S)W_S^{-1} \qquad (**)$$

deduced from the inclusion [page 205]

$$(\underline{\mathrm{Fib}}_W S, W_S^{\mathrm{f}}) \hookrightarrow (\mathrm{Cat}/S, W_S).$$

The fact that $(**)$ is an equivalence, is proved precisely by the fact of the existence of a functor $C : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S$ and of an $i : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C$, having the properties we know, i.e. from the fact that $\mathcal{C} \neq \emptyset$. Moreover, the datum of an object C of $\mathcal{C}$ gives a convenient way of making the quasi-inverse of the said functor explicit. But this quasi-inverse has an existence that is canonical and independent of the choice of a C, i.e. of a mode of calculation. The fact that the composite functor φ in $(*)$ can be "lifted" (up to isomorphism) to a functor

$$C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fib}}_W S$$

between $\varphi(X) = \mathrm{hot}_{W,S}(X)$ and $\mathrm{hot}_{W,S}(C(X))$ and that the isomorphism can be made explicit, it too, by an X-functorial arrow

$$i_X : X \longrightarrow C(X), \qquad i_X \in W,$$

in Cat/S itself (and not only in the localized $(\mathrm{Cat}/S)W_S^{-1}$), is a particular virtue of the [...], which should not for all that make us forget the intrinsic character of the arrows $(*)$, independently of any mode of calculation. p. 134 [page 206]

‖ The key to the situation, and to the questions of canonicity, seems to me to be in the fact that $(**)$ is an equivalence of categories - and one must admit that, however familiar and almost banal it has finally become to me, this is a quite remarkable fact and by no means obvious a priori. (But probably one can also get it by the general *fourbis* à la Quillen-Thomason, without fighting with a formalism of paths ...) The "canonical isomorphisms" that we can establish between a "theory C " and a "theory C' " (via two different constructions, e.g., of categories of paths, or via the Quillen-Thomason approach, without paths) must always come back to those

that we can define via the comparison of the "theory C " on the one hand, the "theory C' " on the other, with what we get directly from the composite arrow $(*)$ (p. 204). Finally, it seems to me that *the existence* of a theory of the "homotopy fibers" (or "W-homotopic") of an arrow $X \longrightarrow S$ in Cat, results from the equivalence $(**)$ (p. 204), and this theory could probably to a certain extent be developed from there, without a choice of a functor C or of a model category structure in the sense of Quillen. [page 207]

4 Axiomatization of "W-homotopic fibers".

Yesterday's reflection makes me return to the axiomatics in "categories of models"

$$(\mathcal{M}, W)$$

([119]) more general than Cat, in the spirit of VI (cf. theorem-scholie 2, p. 108-113). For the moment, we are going to suppose only $\mathcal{M}$ stable under fiber products (we do not need a final object). We introduce the set

$$\mathrm{Fib}_W \subseteq \mathrm{Fl}(\mathcal{M})$$

of *W-fibrations* ("ultrastrong" being understood!) by the condition [page 208]

$f : X \longrightarrow Y$ is in Fib_W if and only if the base change functor by X,

$$\begin{array}{rcl} Y' & \longmapsto & X' = X \times_Y Y' \\ \mathcal{M}/Y & \longrightarrow & \mathcal{M}/X \end{array}$$

p. 135

takes W_Y into W_X, where $W_X \subseteq \mathrm{Fl}(\mathcal{M}/X)$ defined by

$u \in W_X \iff \sigma(u) \in W$ ($\sigma : \mathcal{M}/X \longrightarrow \mathcal{M}$ the "source" functor, or "forgetting X").

([120]). Let us put, for every S in $\mathcal{M}$,

$$\mathrm{Ho}^!_W(S) = (\mathcal{M}/S)W_S^{-1},$$

which plays the role of a category of relative W-homotopy types *locally constant* over S. We would like to describe a contravariant law in S for $\mathrm{Ho}^!_W(S)$, i.e. for

$$f : S' \longrightarrow S$$

an arbitrary arrow in $\mathcal{M}$, define

$$f^*_W : \mathrm{Ho}^!_W(S) \overset{?}{\to} \mathrm{Ho}^!_W(S'). \qquad (*)$$

The difficulty is, of course, that the base change functor

$$f^* : \mathcal{M}/S \longrightarrow \mathcal{M}/S'$$

is *not* in general compatible with the localizers W_S, $W_{S'}$. The notion of W-fibration thus [page 209] serves in the first place to bring out the most obvious condition on an arrow $f : S' \longrightarrow S$ in $\mathcal{M}$, for the base change functor to be compatible with the localizers W_S, $W_{S'}$ (*that* is the definition), hence induces $(*)$ on the localized categories.

The *"problem of homotopy fibers"* is precisely to define $(*)$ for *every* arrow in $\mathcal{M}$, not only for W-fibrations, in such a way that the $\mathrm{Ho}^!_W(S)$ form, for S variable, the fibers of a remarkable fibered category over $\mathcal{M}$. When $\mathcal{M}$ admits a final object, so that we can regard the arrows

$$p : e \longrightarrow S \qquad (p \text{ as "point"})$$

in $\mathcal{M}$ as "points" of S, the "W-homotopy fiber" of an arbitrary object X/S of $\mathcal{M}/S$ at the said point will by definition be the element of

$$\mathrm{Ho}_W = \mathrm{Ho}_W(\mathcal{M}) \simeq \mathrm{Ho}^!_W(e)$$

defined by

$$p^*(\mathrm{ho}^!_{W/S}(X/S)),$$

p. 136

where [page 210]

$$\mathrm{ho}^!_{W/S} : \mathcal{M}/S \longrightarrow \mathrm{Ho}^!_W(S) \overset{\mathrm{def}}{=} (\mathcal{M}/S)(W_S)^{-1}$$

is the canonical localization functor.

For every S in $\mathcal{M}$, let

$$\underline{\mathrm{Fib}}_W S \subseteq \mathcal{M}/S$$

be the full subcategory formed by the objects X/S of $\mathcal{M}/S$ such that $X \longrightarrow S$ is W-fibrant. We are going to define a variant

$$\mathrm{Ho}^{\mathrm{lc}}_W(S)$$

of $\mathrm{Ho}^!_W(S)$, as

$$\mathrm{Ho}^{\mathrm{lc}}_W(S) \overset{\mathrm{def}}{=} (\underline{\mathrm{Fib}}_W S)(W^{\mathrm{u}}_S)^{-1},$$

where

> $W^{\mathrm{u}}_S \subseteq \mathrm{Fl}(\underline{\mathrm{Fib}}_W S)$ is the localizer formed by the arrows $u : X \longrightarrow Y$ which are "universally over S in W", i.e. such that for every arrow $S' \longrightarrow S$ of $\mathcal{M}$ with target S, the arrow deduced by base change,
>
> $$u' : X' = X \times_S S' \longrightarrow Y' = Y \times_S S'$$
>
> is in $W_{S'}$.

We have "done what was needed" so that $\mathrm{Ho}^{\mathrm{lc}}_W(S)$ is contravariant in S for *arbitrary* arrows $f : S' \longrightarrow S$ of $\mathcal{M}$, they define

$$f^{*\,\mathrm{lc}}_W : \mathrm{Ho}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{Ho}^{\mathrm{lc}}_W(S'), \qquad (**)$$

since the W-fibrations are stable under base change, and since consequently [page 211]
$f^* : \mathcal{M}/S \longrightarrow \mathcal{M}/S'$ induces

$$f^*_{\underline{\mathrm{Fib}}_W} : \underline{\mathrm{Fib}}_W S \longrightarrow \underline{\mathrm{Fib}}_W S'$$

which, by definition of W^{u}_S and $W^{\mathrm{u}}_{S'}$, is compatible with the localizers

$$f^*_{\underline{\mathrm{Fib}}_W} : (\underline{\mathrm{Fib}}_W S, W^{\mathrm{u}}_S) \longrightarrow (\underline{\mathrm{Fib}}_W S', W^{\mathrm{u}}_{S'}),$$

and thus passes to the categories of fractions as an arrow $(**)$.

We may wonder why not be more generous, and define $\mathrm{Ho}^{\mathrm{lc}}_W(S)$ as $(\mathcal{M}/S)(W^{\mathrm{u}}_S)^{-1}$, defining W^{u}_S in $\mathcal{M}/S$ in the same way as just now in $\underline{\mathrm{Fib}}_W S$ - which costs no more. A first answer is from experience the localizer W^{u}_S in the whole of $\mathcal{M}/S$ turns out to be practically unusable, on the one hand because membership in W^{u}_S (for an arrow between objects of $\mathcal{M}/S$ which are not very special, e.g. W-fibered over S) is unverifiable (except for isomorphisms and things like that …), and because on the other hand the category of fractions is prohibitively big and can serve (it seems) no purpose. Another answer is that p. 137
we want the category obtained to correspond well to the intuition of relative homotopy [page 212]
types *locally constant* over S - and we therefore want that for an arbitrary base change $f : S' \longrightarrow S$ by an

[119] W satisfies the axiom of weak saturation: isomorphisms in W, and if two among f, g, gf in W, the third also.

[120] **NB** Give here the proposition: Fib_W *stable under composition and under base change, and contains the isomorphisms, stable under direct factors.*

S' in $\mathcal{M}/S$, whence a composite functor

$$\mathcal{F}_S(W_S^{\mathrm{u}})^{-1} \xrightarrow{f_W^{*\,\mathrm{lc}}} \mathcal{F}_{S'}(W_{S'}^{\mathrm{u}})^{-1} \longrightarrow \mathrm{Ho}_W$$
$$\varphi_{S'}$$

([121]) deduced from the functors of categories with localizers

$$(\mathcal{F}_S, W_S^{\mathrm{u}}) \xrightarrow{f^*} (\mathcal{F}_{S'}, W_{S'}^{\mathrm{u}}) \xrightarrow{\text{forgetting } S'} (\mathcal{M}, W)$$

(composite functor $\varphi_{S'}$ which can be viewed as the functor *"homotopy fiber at the object S' over S"*, this object being visualized as a "point" of S "with values in S' " following the general functorial yoga …), the functor

$$\begin{aligned} S' &\longmapsto \varphi_{S'}(\xi) \qquad \text{for } \xi \text{ in } \mathcal{F}_S(W_S^{\mathrm{u}})^{-1} \\ \mathcal{M}/S &\longrightarrow \mathrm{Ho}_W \end{aligned}$$

take arrows in W_S to isomorphisms: the "W-homotopy fiber" at S'/S of an object of $\mathcal{F}_S(W_S^{\mathrm{u}})^{-1}$ (viewed as a W-homotopy type *locally constant* over S) must be "the same", if S' replaced by S'' with $S'' \longrightarrow S'$ given and in W_S. It is for this reason that it is reasonable to take $\mathcal{F}_S$ "small enough" (namely $\underline{\mathrm{Fib}}_W S$, which is exactly adjusted for this), rather than take the whole of $\mathcal{M}/S$. [page 213]

We can therefore say that the category $\underline{\mathrm{Fib}}_W S$, and the localizer W_S^{u} have been constructed ad-hoc, as simply as possible, to give rise to a localized $\mathrm{Ho}_W^{\mathrm{lc}}(S)$ which corresponds to the intuition of locally constant relative W-homotopy types in the following two ways:

1°) There is spontaneously a contravariant law in S, for arbitrary arrows $S' \longrightarrow S$ in $\mathcal{M}$ (we have defined W_S^{u} just for this purpose accordingly).

2°) that the notion of "W-homotopy fiber at S'/S" of an object ξ of $\mathrm{Ho}_W^{\mathrm{lc}}$ satisfy the preceding condition, i.e. an arrow $S'' \longrightarrow S'$ in $\mathcal{M}/S$ which is in W_S define an *isomorphism* (in Ho_W) of homotopy fibers

$$\underbrace{\xi_{S''}}_{\varphi_{S''}(\xi)} \longrightarrow \underbrace{\xi_{S'}}_{\varphi_{S'}(\xi)}$$

(and we have chosen $\mathcal{F}_S = \underline{\mathrm{Fib}}_W(S)$ just for that purpose).

We thus have an excellent notion of W-homotopy fibers for X W-fibered over S - this time excellent for *arbitrary* base changes - and there is no merit in that since we have defined everything on purpose for that! p. 138

To put it otherwise, we have defined *two* notions of relative W-homotopy types over S, one so as to give such a homotopy type $\mathrm{ho}_{W,S}^{!}(X)$ for *every* object of $\mathcal{M}/S$, but there is a priori a good notion of base change only for the base changes $S' \longrightarrow S$ which are W-fibered, the other, which corresponds more directly to the intuition of a locally constant W-homotopy type over S, associates such a type $\mathrm{ho}_{W,S}^{\mathrm{lc}}(X)$ only to the X of $\mathcal{M}/S$ which are W-fibered over S, on the other hand this notion admits a good notion of inverse image and of "W-homotopy fiber" for arbitrary base changes. [page 214]

A "good" theory of W-homotopy fibers must combine the two preceding notions into a single one, so that the advantages of the two add up, and the disadvantages cancel out! It is obtained with the help of a draconian axiom on the situation (whereas until now we [page 215]

have used nothing about $W \subseteq \mathrm{Fl}(\mathcal{M})$, except at most weak saturation, in addition to the existence of fiber products):

ⓐ (*Homotopy fiber axiom.*) For every object S in $\mathcal{M}$, the canonical functor

$$\mathrm{Ho}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{Ho}_W^{!}(S)$$

deduced from the inclusion

$$(\underline{\mathrm{Fib}}_W(S), W_S^{\mathrm{u}}) \longrightarrow (\mathcal{M}/S, W_S)$$

is an *equivalence of categories.*

This axiom implies that in $\underline{\mathrm{Fib}}_W S$, W_S^{u} and the localizer induced by W_S (which I still denote by W_S by abuse of notation) have the same strong saturation. In fact, it is natural to supplement the preceding axiom with the following one.

ⓑ (Adjunct axiom to the homotopy fiber axiom.) On $\underline{\mathrm{Fib}}_W S$, the localizers W_S^{u} and W_S coincide, i.e. we have this:

> Let X, Y be in $\underline{\mathrm{Fib}}_W S$, $X \xrightarrow{u} Y$ an S-morphism such that $u \in W_S$, then $u \in W_S^{\mathrm{u}}$ (i.e. the morphism $u' : X' \longrightarrow Y'$ deduced by any base change $S' \longrightarrow S$ is still in W).

I presume that a) will rarely be satisfied without b) being so (and conversely for that matter), and that the proof of a) and b) will be done practically in a single breath. [page 216]

If $\mathcal{M} = \mathrm{Cat}$, W a basic localizer, then the notion of W-fibration adapted here (read: "ultrastrong", in the specific terminology developed in Cat) is so strong, that it already practically implies b) (assuming W(4a′), it is true, cf. p. 53), and a) is hardly verifiable unless we have W(4a) ([122]) (loc. cit.). So here it is "the fibration axiom" on W, in its strong form W(4a), which is the keystone of a good theory of homotopy fibers. p. 139

Such a theory (returning to the general case) amounts to saying, in essence, that there are "enough" W-fibrations ("ultrastrong") to be able to describe *every* object of a $\mathcal{M}/S$, up to W_S-equivalence, as a W-fibrant object over S. In the cases known to me, we have moreover this: we have for every S in $\mathcal{M}$ a subcategory $\mathcal{F}_{0\,S}$ of $\underline{\mathrm{Fib}}_W S$ (I am thinking for example of $\underline{\mathrm{Parf}}_W S$, or even $\underline{\mathrm{Parf}}_{W_\omega} S$, in the case where $\mathcal{M} = \mathrm{Cat}$), such that for every full subcategory $\mathcal{F}$ of $\mathcal{M}/S$ such that [page 217]

$$\mathcal{F}_{0\,S} \subseteq \mathcal{F} \subseteq \mathcal{M}/S$$

(a fortiori, for a full $\mathcal{F}$ such that $\underline{\mathrm{Fib}}_W S \subseteq \mathcal{F} \subseteq \mathcal{M}/S$), the inclusions induce equivalences of categories

$$\mathcal{F}_{0\,S}(W_S)^{-1} \xrightarrow{\sim} \mathcal{F}(W_S)^{-1} \xrightarrow{\sim} \underbrace{(\mathcal{M}/S)(W_S)^{-1}}_{\overset{\mathrm{def}}{=}\mathrm{Ho}_W^{!}(S)}.$$

It does not seem possible to fix the choice of $\mathcal{F}_{0\,S}$ in "the best possible" way, once and for all in the axiomatic context adopted here. It is an additional element of structure in $\mathcal{M}$ (besides the localizer W), on the same footing as the datum of the classes of Q-fibrations and Q-cofibrations, in the manner of Quillen. I do not know whether the axioms a), b) above (homotopy fiber axiom) already imply the variant of ⓐ:

ⓒ For every full subcategory $\mathcal{F}$ of $\mathcal{M}/S$ such that [page 218]

$$\underline{\mathrm{Fib}}_W S \subseteq \mathcal{F} \subseteq \mathcal{M}/S$$

[121] $\mathcal{F}$ here denotes a fibered subcategory of $\mathcal{M}/\mathcal{M} = \underline{\mathrm{Fl}}(\mathcal{M})$ over $\mathcal{M}$, a possible substitute for the one formed by $\underline{\mathrm{Fib}}_W S$ or by the whole of $\mathcal{M}/S$.

[122] No, it seems to me that W(4b) should suffice …

the inclusions induce equivalences

$$\underbrace{(\underline{\mathrm{Fib}}_W S)(W_S)^{-1}}_{\substack{=\mathrm{Ho}^{\mathrm{lc}}_W(S)\\ \text{assuming a)}}} \xrightarrow{\sim} \mathcal{F}(W_S)^{-1} \xrightarrow{\sim} \underbrace{(\mathcal{M}/S)(W_S)^{-1}}_{=\mathrm{Ho}^{!}_W(S)}.$$

But I presume that in practice, when we can verify ⓐ, we will be able to verify ⓒ as well ([123]).

p. 140

5 Axiomatics of non-locally constant relative W-homotopy types $\mathrm{Ho}^{\mathcal{L}}_W(S)$.

We have said nothing up to this point about possible relative W-homotopy types (over an object S of $\mathcal{M}$) which would *not* be "locally constant". Yet they were discussed in loc. cit. (VI, theorem-scholie 2, part II of the theorem pages 109-113.) I take the opportunity to come back to this. We must introduce, in each $\mathcal{M}/S$, a second localizer

$$W^0_S \subseteq \mathrm{Fl}(\mathcal{M}/S), \qquad W^0_S \subseteq W_S$$

(finer than W_S), which will make it possible to define [page 219]

$$\mathrm{Ho}_W(S) \overset{\mathrm{def}}{=} (\mathcal{M}/S)(W^0_S)^{-1}.$$

That there is no *canonical* choice which imposes itself here, deduced from the structure $(\mathcal{M}, W)$ alone, is clear from the example of (Cat, W), in which case we find *two* symmetric choices, transformed into one another by the involution

$$X \longmapsto X^\circ : (\mathrm{Cat}, W) \xrightarrow{\sim} (\mathrm{Cat}, W),$$

namely W^{g}_S and W^{d}_S. The choice of one or the other, say (to fix ideas) of W^{g}_S, is associated with a notion of *local isomorphism* (or of "local co-isomorphism" in the case of W^{d}_S) which I would like to axiomatize. It does seem that for the main categories of models for Hot (and for its variants), we have in a fairly natural way such a notion of local isomorphism (without the encumbrance, most often, of an involutive automorphism providing a symmetric notion …).

We shall therefore suppose, as in loc. cit. p. 94-96, given a class

$$\mathcal{L} \subseteq \mathrm{Fl}(\mathcal{M})$$

of morphisms, playing the role of "arrows of localization" or "of local isomorphisms", [page 220] satisfying the following conditions.

(∗) $\mathcal{L}$ contains the isomorphisms, is stable under composition, and under base change.

We put, for S in $\mathcal{M}$,

$$\mathcal{L}_S \subseteq \mathcal{M}/S \qquad \text{full subcategory of the } S'/S \text{ such that } S' \longrightarrow S \text{ in } \mathcal{L}.$$

We then define for every S in $\mathcal{M}$

$$W^{\mathcal{L}}_S \subseteq \mathrm{Fl}(\mathcal{M}/S)$$
$$(X \xrightarrow{u} Y) \in W^{\mathcal{L}}_S \iff \forall\, S' \in \mathcal{L}_S,\ (u_{S'} : X \times_S S' \longrightarrow Y \times_S S') \text{ is in } W.$$

p. 141

This is therefore a variant of the localizer W_S^{u}, where now we take as base changes only the "$\mathcal{L}$-localizing" arrows, i.e. $\in \mathcal{L}$. We thus have

$$W_S^{\mathrm{u}} \subseteq W_S^{\mathcal{L}} \subseteq W_S$$

as subsets of $\mathrm{Fl}(\mathcal{M}/S)$, and likewise in any full subcategory $\mathcal{F}$ of $\mathcal{M}/S$. In the case of $\underline{\mathrm{Fib}}_W S$, by virtue of axiom (b) above ($W_S^{\mathrm{u}} = W_S$ in $\underline{\mathrm{Fib}}_W S$), the three localizers coincide.

We put

$$\mathrm{Ho}_W^{\mathcal{L}}(S) \text{ (or } \mathrm{Ho}_W(S)) \overset{\mathrm{def}}{=} (\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1},$$

and we gave in loc. cit. (part II of the theorem-scholie) an interesting result on this invariant, for fixed S. My aim at present is to see what happens for variable S. Visibly, [page 221] a notion of relative W-homotopy type over an arbitrary object S of $\mathcal{M}$ is satisfactory, only if it allows inverse images by *arbitrary* arrows

$$f : S' \longrightarrow S$$

in $\mathcal{M}$, which absolutely must define

$$f_W^{*\mathcal{L}} \text{ or } f_W^* : \mathrm{Ho}_W^{\mathcal{L}}(S) \longrightarrow \mathrm{Ho}_W^{\mathcal{L}}(S').$$

We are therefore faced with the same perplexity as just now - and it is *not,* this time, the introduction of $\underline{\mathrm{Fib}}_W(S)$ that is going to save the situation, for

$$(\underline{\mathrm{Fib}}_W S)(\underbrace{W_S^{\mathcal{L}}}_{\substack{=W_S \\ =W_S^{\mathrm{u}}}})^{-1} \longrightarrow (\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1}$$

has no desire, this time, to be an equivalence of categories, but much rather an inclusion (fully faithful) of the relative W-homotopy types constant over S, into arbitrary W-homotopy types. An axiom is visibly needed to replace the "homotopy fiber" axiom.

In the case of Cat, the class $\mathcal{L}$ of local isomorphisms (say) allows us to define, as [page 222] above, the localizer $W_S^{\mathcal{L}} = W_S^{\mathrm{g}}$ in the Cat/S, as consisting of the $u : X \longrightarrow Y$ over S which are in W and remain so under every base change $S' \longrightarrow S$ which is in $\mathcal{L}$. But there is reason to *saturate* $\mathcal{L}$, by considering the class $\mathcal{L}^\wedge$ of arrows $f : S' \longrightarrow S$ of Cat such that the base change

$$f^* : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S'$$

sends W_S^{g} into $W_{S'}^{\mathrm{g}}$. This $\mathcal{L}^\wedge$ is tautologically stable under composition, and contains the isomorphisms (better, contains $\mathcal{L}$ = set of local isomorphisms). It is not clear, from the p. 142 definition given here, that it is stable under base change - we are going to strengthen the definition accordingly, by requiring, for $f : S' \longrightarrow S$ to be in $\mathcal{L}^\wedge$, that for every T over S, let us consider $f_T : T' = S' \times_S T \longrightarrow T$, $f_T^* : \mathrm{Cat}/T \longrightarrow \mathrm{Cat}/T'$ be compatible with the localizers W_T^{g}, $W_{T'}^{\mathrm{g}}$. Then $\mathcal{L}^\wedge$ is (just like $\mathcal{L}$) stable under composition and under base change, and moreover it contains $\mathcal{L}$. Moreover, for every S, we have $W_S^{\mathcal{L}} = W_S^{\mathcal{L}^\wedge}$. This little sorites could of course be carried out for any modelizer $(\mathcal{M}, W)$. But the interesting thing here is that $\mathcal{L}^\wedge$ contains the set Liss_W of W-smooth arrows. (I do not know whether it is equal.) We could therefore have just as well started from the set of [page 223] arrows Liss_W, to define the localizers W_S^{g} as

$$W_S^{\mathrm{g}} = W_S^{\mathrm{Liss}_W} \qquad \text{instead of } W_S^{\mathrm{g}} = W_S^{\mathrm{Isoloc}},$$

[123] Cf. page 240 ff. We should here develop a little more the formalism of $\mathrm{Ho}_W^{\mathrm{lc}}(S)$ and homotopy fibers, under the hypothesis (a)(b) above.

which is equivalent.

On the other hand, in working with the localized categories $\mathrm{HOT}^{\circ}_{W}(S) = (\mathrm{Cat}/S)(W^{\mathrm{g}}_{S})^{-1}$, the role of $\underline{\mathrm{Fib}}_{W}S$ in the theory of the $\mathrm{HOT}^{\mathrm{lc}}_{W}(S)$ is played here by $\underline{\mathrm{Prop}}_{W}(S)$. We thus see that in the theory of non-locally constant relative W-homotopy types, HOT°_{W} mode, the sets of arrows Liss_{W} and Prop_{W} play a role that is in a sense complementary: the arrows $f : S' \longrightarrow S$ of Liss_{W} are (practically) those which allow us directly to define

$$f^{*\circ}_{W} : \mathrm{HOT}^{\circ}_{W}(S) \longrightarrow \mathrm{HOT}^{\circ}_{W}(S')$$

as induced by

$$f^{*} : (\mathrm{Cat}/S, W^{\mathrm{g}}_{S}) \longrightarrow (\mathrm{Cat}/S', W^{\mathrm{g}}_{S'}),$$

whereas the arrows of Prop_{W} allow us to define for every S in Cat the full subcategory

$$\underline{\mathrm{Prop}}_{W}S \subseteq \mathrm{Cat}/S,$$

giving rise to an equivalence of categories

$$\begin{cases} (\underline{\mathrm{Prop}}_{W}S)(W^{\mathrm{u}}_{S})^{-1} \xrightarrow{\sim} \underbrace{(\mathrm{Cat}/S)(W^{\mathrm{g}}_{S})^{-1}}_{=\mathrm{HOT}^{\circ}_{W}(S)} \\ \text{with} \\ W^{\mathrm{u}}_{S} = W^{\mathrm{g}}_{S} \text{ on } \underline{\mathrm{Prop}}_{W}S \end{cases}$$

(which visibly is the announced substitute for "the W-homotopy fiber axiom"). This equivalence, which allows us to describe $\mathrm{Hot}^{\circ}_{W}(S)$ in terms of $\underline{\mathrm{Prop}}_{W}S \subseteq \mathrm{Cat}/S$ and a localizer W^{u}_{S} *stable under* **every** *base change,* allows us on the other hand to define the expected contravariant law for $\mathrm{HOT}^{\circ}_{W}(S)$. [page 224]

On the other hand, when we interchange the role of Liss_{W} and Prop_{W}, we find the complementary theory of $\mathrm{HOT}_{W}(S)$, p. 143

$$\begin{cases} \mathrm{HOT}_{W}(S) \overset{\mathrm{def}}{=} (\mathrm{Cat}/S)(W^{\mathrm{d}}_{S})^{-1} \xleftarrow{\sim} (\underline{\mathrm{Liss}}_{W}S)(W^{\mathrm{u}}_{S})^{-1}, \\ \text{on } \underline{\mathrm{Liss}}_{W}S,\ W^{\mathrm{d}}_{S} = W^{\mathrm{u}}_{S}, \end{cases}$$

the description of $\mathrm{HOT}_{W}(S)$ in terms of $\underline{\mathrm{Liss}}_{W}S$ and W^{u}_{S} allows us to define its contravariant character in S, for *every* arrow $f : S' \longrightarrow S$ in Cat. Whereas if we want to be able to pass to the categories of fractions on Cat/S and Cat/S' in their entirety (via W^{d}_{S}, $W^{\mathrm{d}}_{S'}$), we must restrict ourselves to W-proper base changes.

So the perplexity comes to me whether this complementary, or even this duality, between two sets of arrows, the "smooth" and the "proper", might not be a general fact which turned up at least in the categories of models close to the description of HOT or of a Hot_{W}.

It may well be that in $\mathbf{\Delta}^{\wedge}$ we have a perfect duality between these two kinds of notions (I shall have to examine the matter), perhaps even in any category $\mathcal{A}^{\wedge}$ (but in $\mathbf{\Delta}^{\wedge}$ there is an involutive automorphism $K \longmapsto K^{\circ}$, corresponding to the $X \longmapsto X^{\circ}$ in Cat …). And after all, even in the context of topological spaces, the notions of properness on the one hand, of local asphericity on the other, for an arrow $X \longrightarrow Y$, are well defined, and might well fit together as do the corresponding notions in Cat, and (who knows) give rise to two complementary versions, or "dual" to one another, of a notion of relative homotopy type. [page 225]

Not being clear about this, I am a little undecided how to present an axiomatic description of the theory of the $\mathrm{Ho}_W^{\mathcal{C}}(S)$, for a general category of models $(\mathcal{M}, W)$. Should I give myself from the outset

$$\mathrm{Liss}, \mathrm{Prop} \subseteq \mathrm{Fl}(\mathcal{M}),$$

stable under composition, under base change, containing the isomorphisms, and such that (perhaps)

$$\mathrm{Liss} \cap \mathrm{Prop} \subseteq \mathrm{Fib}_W,$$

but above all they allow us to define, for S in $\mathcal{M}$ [page 226]

$$\begin{cases} W_S^{\mathrm{g}} \overset{\mathrm{def}}{=} W_S^{\mathrm{Liss}} \subseteq \mathrm{Fl}(\mathcal{M}/S) \\ W_S^{\mathrm{d}} \overset{\mathrm{def}}{=} W_S^{\mathrm{Prop}} \subseteq \mathrm{Fl}(\mathcal{M}/S) \end{cases}$$

with the following properties:

ⓐ If $f : S' \longrightarrow S$ is in Prop (resp. Liss), then $f^* : \mathcal{M}/S \longrightarrow \mathcal{M}/S'$ sends W_S^{d} into $W_{S'}^{\mathrm{d}}$ (resp. W_S^{g} into $W_{S'}^{\mathrm{g}}$),

which is a pure tautology, and does not yet concern the relations of Prop with Liss, but on the other hand: p. 144

ⓑ

$$\begin{cases} (\underline{\mathrm{Liss}}S)(W_S^{\mathrm{u}})^{-1} \xrightarrow{\sim} (\mathcal{M}/S)(W_S^{\mathrm{d}})^{-1} \text{ an equivalence} \\ \text{and } W_S^{\mathrm{u}} = W_S^{\mathrm{d}} \text{ on } \underline{\mathrm{Liss}}S, \end{cases}$$

and dually

$$\begin{cases} (\underline{\mathrm{Prop}}S)(W_S^{\mathrm{u}})^{-1} \xrightarrow{\sim} (\mathcal{M}/S)(W_S^{\mathrm{g}})^{-1} \text{ an equivalence} \\ \text{and } W_S^{\mathrm{u}} = W_S^{\mathrm{g}} \text{ on } \underline{\mathrm{Prop}}S. \end{cases}$$

Moreover, there is reason to state ⓒ two adjunction properties (dual to one another), for which I refer to VI, theorem-scholie 2, p. 108-113. Here the subsidiary question arises of simplifying the conditions of validity of the adjunction formula, somewhat disparate in loc. cit.

On the other hand, the question arises of developing the theory of the covariant laws $f_!$ and f_* for $\mathrm{Ho}_W^{\mathrm{lc}}(S)$ and for $\mathrm{Ho}_W^{\mathrm{Prop}}(S)$, $\mathrm{Ho}_W^{\mathrm{Liss}}(S)$.

I have good hope that the functor $f_!$ can be described, and its adjunction with f^* [page 227] established, at as little cost in this general context as in Cat. But I do not feel like dwelling on it at present.

6 W-homotopy cofiber in Cat.

On the other hand, I feel like examining here under what conditions on the basic localizer W, there exists in (Cat, W) a theory dual to that of the W-homotopy fiber.

The key question is, of course, that of having a good grip on W-cofibrations. Now so far I know *only* the open Dwyer immersions, the closed Dwyer immersions, and what follows from them by composition etc. The work of Thomason may motivate the hope that knowledge of those immersions, which seem here to play a role similar (dual) to that of W-perfect morphisms, will suffice to develop a good theory of "W-homotopy cofibers". p. 145

||| [page 228]

I assume throughout this section that W satisfies W(7 bis) (axiom of cocartesian squares), to have good properties of open Dwyer immersions ([124]), and more generally, for W-cofibrations.

So the question arises of the dual factorization of an arrow $f : X \longrightarrow Y$ in Cat into

$$X \xrightarrow[i_f]{W\text{-cofibration}} \tilde{Y} \xrightarrow[p_f]{\in W} Y.$$

The results of THOMASON ensure that this exists, with i an open Dwyer immersion, and p a W-fibration - at least in the case $W = W_\infty$. I am going to try to see it directly, by dually copying the construction of CARTAN-SERRE.

So it is first a matter of factoring

$$X \amalg X \xrightarrow{\text{codiag}_X} X,$$

and for this I choose a fixed interval

$$I \;=\; \Phi_n \;=\; (0 \longrightarrow 1 \longleftarrow 2 \longrightarrow \cdots \longrightarrow 2n-1 \longleftarrow 2n)$$

with

$$n \geq 2$$

so that the inclusion

$$\varepsilon = \text{discrete category } \{0,1\} \xrightarrow{\partial} I, \qquad \partial(0) = 0, \quad \partial(1) = 2n,$$

is an open Dwyer immersion. This property is stable under Cat-fibrant base change, in particular on taking a product

$$\underbrace{X \times \varepsilon}_{\simeq X \coprod X} \xrightarrow{i_X} X \times I,$$

on the other hand

$$X \times I \xrightarrow{\text{pr}_1} X$$

p. 146
[page 229]

is W_ω-perfect, whence the desired factorization

$$X \amalg X \xrightarrow{i_X} X \times I \xrightarrow{p_X = \text{pr}_1^X} X$$

---

[124]

a) These are W-cofibrations (follows from b) and c)).

b) If

$$\begin{array}{ccc} U & \xrightarrow{i} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ U' & \xrightarrow{i'} & X' \end{array}$$

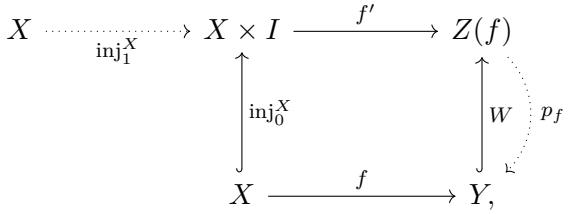
with i an open Dwyer immersion, then

$$\begin{aligned} i \in W &\iff i' \in W \\ p \in W &\iff q \in W, \end{aligned}$$

so I need W(7 bis) *and* even W(7) for the second implication.

c) Open Dwyer immersions stable under cobase change and under composition, contain the isomorphisms.

- nothing could be simpler! How to use it to construct the factorization of an arbitrary arrow $f : X \longrightarrow Y$, by dualizing the constructions relating to path spaces? We define $Z(f)$ by the cocartesian square in Cat



we have

$$Z(f) \xrightarrow{\alpha} Y \qquad \text{induced by } \mathrm{pr}_1 : X \times I \longrightarrow X,$$

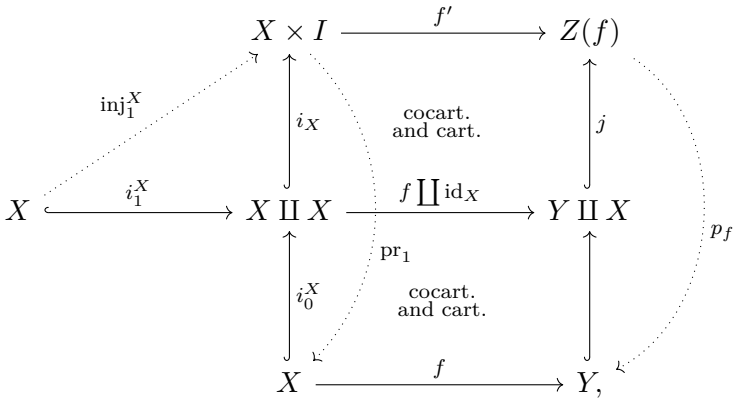
and the desired factorization of f is

$$X \xrightarrow{i_f = f' \circ \mathrm{inj}_1^X} Z(f) \xrightarrow{p_f} Y.$$

I say that

$$\begin{cases} i_f \text{ is an open Dwyer immersion, hence a } W\text{-cofibration.} \\ p_f \in W. \end{cases}$$

We see from the more complete diagram [page 230]



p. 147

$$i_f = f' \circ \mathrm{inj}_1^X = f' i_X i_1^X = j \underbrace{(f \amalg \mathrm{id}_X) i_1^X}_{\substack{\text{canonical inclusion} \\ \text{of } X \text{ into} \\ Y \coprod X}} ,$$

it is open Dwyer immersion, as a composite of $X \longrightarrow Y \amalg X$ and $j : Y \amalg X \longrightarrow Z(f)$ which are such (j obtained by cobase change from i_X, which is an open Dwyer immersion).

On the other hand, $X \xrightarrow{\mathrm{inj}_0^X} X \times I$ is open Dwyer immersion and in W, hence couniversally in W, hence $Y \longrightarrow Z(f)$ is also in W, hence also the retraction p_f.

On the other hand, the proposition dual to

$$\underline{\mathrm{Fib}}_W S \cap W_S = W_{S\,\mathrm{univ}}^{\mathrm{u}}$$

is valid for W-cofibrations: [page 231]

Proposition 16 ([125]). *On the full subcategory* $\underline{\mathrm{Cofib}}_W(S)$ *of* $S\backslash\mathrm{Cat}$*, the localizer* ${}_SW$ *is equal to* $({}_SW)^{\mathrm{univ}}$ *(couniversal* W*-equivalences), i.e. we have this: Let*

$$S \longrightarrow X, \qquad S \longrightarrow Y$$

be W*-cofibrations, and let* $f : X \longrightarrow Y$ *be a morphism of* $S\backslash\mathrm{Cat}$ *such that* $f \in W$*. Then for every cobase change* $S \longrightarrow S'$*, the corresponding arrow*

$$f' : X' = X \amalg_S S' \longrightarrow Y' = Y \amalg_S S'$$

is in W*.*

Proof. We factor $S \longrightarrow S'$ into

$$S \xrightarrow{i} S'_0 \xrightarrow{j} S',$$

with i an open Dwyer immersion (hence a W-cofibration), and $j \in W$. So we have the diagram with cocartesian squares

$$\begin{array}{ccccc}
S & \longrightarrow & X & \xrightarrow{f} & Y \\
{\scriptstyle i}\downarrow & & \downarrow & & \downarrow \\
S'_0 & \longrightarrow & X'_0 & \xrightarrow[W]{f'_0} & Y'_0 \\
{\scriptstyle \in W}\downarrow & & \downarrow{\scriptstyle W} & & \downarrow{\scriptstyle W} \\
S' & \longrightarrow & X' & \xrightarrow{f'} & Y'.
\end{array}$$

Since i is a W-cofibration and $f \in W$, we have $f'_0 \in W$. On the other hand, since $S \longrightarrow X$, hence $S'_0 \longrightarrow X'_0$ is a W-cofibration, $X'_0 \longrightarrow X'$ is in W, and for the same reason, $Y'_0 \longrightarrow Y'$ is too. So f' is too. p. 148 [page 232]

NB Here, this is a formal argument, valid (in dual form) in any $(\mathcal{M}, W)$ where we have a factorization statement for an arbitrary arrow $f : X \longrightarrow Y$ into $X \xrightarrow{i_f} \bar{X} \xrightarrow{\alpha} Y$, with $i_f \in W$ and α a W-fibration. Then, if X, Y are W-fibered over S, every S-arrow from X to Y that is in W, is in W_S^{u}. And dually, if every arrow $X \longrightarrow Y$ factors as $X \longrightarrow \tilde{Y} \longrightarrow Y$, with $X \longrightarrow \tilde{Y}$ a W-cofibration and $\tilde{Y} \longrightarrow Y$ in W.

A (small) worry comes to me here: have I at least proved that if

$$S \longrightarrow X$$

is a W-cofibration, then for every arrow $S \longrightarrow S'$ in Cat, the cocartesian square in Cat

$$\begin{array}{ccc}
S & \longrightarrow & X \\
\downarrow & & \downarrow \\
S' & \longrightarrow & X'
\end{array}$$

is also W-cocartesian ([126])? Otherwise, what good would that square be?! To prove it, let us again factor $S \longrightarrow S'$ into [page 233]

$$\underset{\substack{\text{open}\\ \text{Dwyer}\\ \text{immersion}}}{S \hookrightarrow} \quad S'_0 \underset{\in W}{\longrightarrow} S',$$

[125] Has nothing to do with Cat - a general thing when there are enough W-cofibrations.

[126] in the old sense, special to Cat (def. 3, p. 131). To compare it with the general theory, cf. prop. 20, p. 262.

whence the diagram of 2 cocartesian squares

$$\begin{array}{ccc} S & \xrightarrow{\mathrm{Cof}_W} & X \\ {\scriptstyle D_W}\big\downarrow & & \big\downarrow{\scriptstyle D_W} \\ S'_0 & \xrightarrow{\mathrm{Cof}_W} & X'_0 \\ {\scriptstyle \in W}\big\downarrow & & \big\downarrow{\scriptstyle \in W} \\ S' & \xrightarrow{\mathrm{Cof}_W} & X', \end{array}$$

where $X'_0 \longrightarrow X'$ is $\in W$ by the hypothesis on X under S. So we must verify: p. 149

a) A cocartesian square built on an arrow which is Dwyer immersion, is W-cocartesian.

b) A commutative square in which two opposite sides are in W, is W-cocartesian.

c) The composite of two W-cocartesian squares is W-cocartesian.

Let us provisionally admit b) and c), and prove a), for a cocartesian square

$$\begin{array}{ccc} U & \xhookrightarrow{i} & X \\ {\scriptstyle p}\big\downarrow & & \big\downarrow{\scriptstyle q} \\ U' & \xhookrightarrow{i'} & X', \end{array}$$

where i is an open Dwyer immersion. We proceed again as before, by factoring p into $U \overset{j}{\hookrightarrow} U'_0 \xrightarrow{p'} U'$, where j is open Dwyer immersion, $p' \in W$. So we have a diagram of two adjoining cocartesian squares [page 234]

$$\begin{array}{ccc} U & \xhookrightarrow{i \in D_W} & X \\ {\scriptstyle D_W}\big\downarrow & & \big\downarrow \\ U'_0 & \xhookrightarrow{D_W} & X'_0 \\ {\scriptstyle p' \in W}\big\downarrow & & \big\downarrow{\scriptstyle q' \in W} \\ U' & \hookrightarrow & X', \end{array}$$

where the first square is made up of open immersions. We then know that it is W-cocartesian. In the second, we have $p' \in W$, hence $q' \in W$ since i is a Dwyer immersion, hence a W-cofibration. So it is also W-cocartesian. We then conclude thanks to c) above.

Let us prove b), for $\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1, \\ {\scriptstyle i_2 \in W}\big\downarrow & & \big\downarrow{\scriptstyle f_1 \in W} \\ X_2 & \xrightarrow{f_2} & Y \end{array}$ by inserting it into $\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1, \\ {\scriptstyle i_2 \in W}\big\downarrow & & \big\downarrow{\scriptstyle j_1} \\ X_2 & \xrightarrow{j_2} & \mathcal{X} \end{array}$ with $f_1 : X_1 \to Y$, $\varphi : \mathcal{X} \to Y$, $f_2 : X_2 \to Y$,

where $\mathcal{X} = \int X_0 \begin{array}{c} \nearrow X_1 \\ \searrow X_2 \end{array}$.

Since $i_2 \in W$, we know that it follows that $j_1 \in W$, and since $f_1 \in W$, it follows that $\varphi \in W$, q.e.d. p. 150

It remains to prove c). I am going to deduce it from a

Proposition 17. *Let there be a commutative square*

$$Q \quad = \quad \begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1 \\ {\scriptstyle i_2}\downarrow & & \downarrow{\scriptstyle f_1} \\ X_2 & \xrightarrow{f_2} & Y \end{array}$$

in Cat*, and let $\mathcal{X}_Q$ be the fibered category over*

$$\tilde{\Phi} \quad (\simeq \mathbf{\Delta}^1 \times \mathbf{\Delta}^1) \quad = \quad \begin{array}{ccc} 0 & \longleftarrow & 1 \\ \uparrow & & \uparrow \\ 2 & \longleftarrow & 3 \end{array}$$

which integrates it. Let $\mathcal{X}$ be the inverse image of the closed subset

$$\begin{array}{ccc} 0 & \longleftarrow & 1 \\ \uparrow & & \\ 2 & & \end{array}$$

in $\mathcal{X}_Q$, which integrates the diagram [page 235]

$$\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & \\ & \searrow & \\ & & X_2 \end{array}$$

. For Q to be W-cocartesian, it is necessary and sufficient that the inclusion

$$i : \mathcal{X} \longrightarrow \mathcal{X}_Q$$

be in W.

Since $\mathcal{X}_Q$ is smooth over $\tilde{\Phi}$, and 3 is an initial object of $\tilde{\Phi}$, it follows that the inclusion of the fiber of $\mathcal{X}_Q$ at the vertex 3 of $\tilde{\Phi}$ is in W,

$$Y \xrightarrow{i_Y} \mathcal{X}_Q \qquad\qquad \in W.$$

Consider then the diagram

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{\varphi_Q} & Y \\ & {\scriptstyle \alpha}\swarrow & \\ {\scriptstyle i}\searrow & & \swarrow{\scriptstyle i_Y \in W} \\ & \mathcal{X}_Q. & \end{array}$$

This diagram is not commutative, but there is a commutativity arrow p. 151

$$\alpha : i_Y \varphi_Q \longrightarrow i.$$

We know that this implies that

$$i \in W \qquad\qquad \Longleftrightarrow \qquad\qquad (i_Y \varphi_Q) \in W$$

[page 236]

(cf. lemma p. 2), and since $i_Y \in W$,

$$(i_Y \varphi_Q) \in W \qquad \Longleftrightarrow \qquad \varphi_Q \in W,$$

whence

$$i \in W \qquad \Longleftrightarrow \qquad \varphi_Q \in W,$$

q.e.d.

We can now prove

Proposition 18 ([127]). (Assuming W(7 bis).) *Consider a commutative diagram*

$$\begin{array}{ccccc} X_0 & \longrightarrow & X_1 & \longrightarrow & X_2 \\ \downarrow & 1 & \downarrow & 2 & \downarrow \\ X'_0 & \longrightarrow & X'_1 & \longrightarrow & X'_2 \end{array} \qquad (*)$$

whence a composite square

$$\begin{array}{ccc} X_0 & \longrightarrow & X_2 \\ \downarrow & 3 & \downarrow \\ X'_0 & \longrightarrow & X'_2. \end{array}$$

Suppose that square 1 *is* W*-cocartesian. Then square* 2 *is* W*-cocartesian if and only if square* 3 *is.*

PROOF.

Let

p. 152

$$I \quad = \quad \left(\begin{array}{ccccc} 0 & \longleftarrow & 1 & \longleftarrow & 2 \\ \downarrow & & \downarrow & & \downarrow \\ 0' & \longleftarrow & 1' & \longleftarrow & 2' \end{array} \right)$$

and let $\mathcal{X}$ be the integral of the diagram $(*)$, a fibered category over I. We denote by $\mathcal{X}_{010'}$ the full subcategory of $\mathcal{X}$ which is the inverse image of the full subcategory

$$\underbrace{I_{010'}}_{=\{0,1,0'\}} \subseteq I,$$

and likewise for every other subset A of $\operatorname{Ob} I = \{0, 1, 2, 0', 1', 2'\}$, giving a full subcategory $\mathcal{X}_A$ of $\mathcal{X}$. We put

$$U \quad = \quad \begin{array}{ccccc} 0 & \longleftarrow & 1 & \longleftarrow & 2 \\ \downarrow & & & & \\ 0' & & & & \end{array} \quad , \qquad V \quad = \quad \begin{array}{ccccc} 0 & \longleftarrow & 1 & \longleftarrow & 2 \\ \downarrow & & \downarrow & & \\ 0' & \longleftarrow & 1' & & \end{array} \quad ,$$

whence

$$\mathcal{X}_U = \mathcal{X}_{0120'}, \qquad \mathcal{X}_V = \mathcal{X}_{0120'1'}.$$

The arrows between the $\mathcal{X}_A$ are the inclusions

deduced from inclusions $A \hookrightarrow A'$ in $\mathfrak{P}(\operatorname{Ob} I)$. Thus, the W-cocartesian character of squares 1, 2, 3 is expressed respectively by the condition that the inclusions

[page 237]

$$\begin{array}{ccc} \mathcal{X}_{010'} & \xrightarrow{\varphi_1} & \mathcal{X}_{010'1'} \\ \mathcal{X}_{121'} & \xrightarrow{\varphi_2} & \mathcal{X}_{121'2'} \\ \mathcal{X}_{020'} & \xrightarrow{\varphi_3} & \mathcal{X}_{020'2'} \end{array}$$

p. 153

be in W. Consider then the diagram of inclusion arrows

$$\begin{array}{ccc}
\mathcal{X}_{020'} & \xrightarrow{\varphi_3} & \mathcal{X}_{020'2'} \\
{\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\beta} \\
\mathcal{X}_U & \xrightarrow{\psi_3} & \mathcal{X} \\
{\scriptstyle\psi_1}\downarrow & & \downarrow{\scriptstyle\mathrm{id}} \\
\mathcal{X}_V & \xrightarrow{\psi_2} & \mathcal{X} \\
{\scriptstyle\alpha'}\uparrow & & \uparrow{\scriptstyle\beta'} \\
\mathcal{X}_{121'} & \xrightarrow{\varphi_2} & \mathcal{X}_{121'2'}.
\end{array}$$

I say that in this diagram we have

$$\alpha, \beta, \alpha', \beta' \in W \qquad (*)$$

and

$$\varphi_1 \in W \ \text{ (i.e. square 1 } W\text{-cocartesian)} \qquad \Longleftrightarrow \qquad \psi_1 \in W. \qquad (**)$$

The relations $(*)$ already imply that

$$\begin{aligned} \varphi_2 \in W \quad &\Longleftrightarrow \quad \psi_2 \in W \\ \varphi_3 \in W \quad &\Longleftrightarrow \quad \psi_3 \in W, \end{aligned}$$

and under the W-cocartesian hypothesis $(**)$, implying $\psi_1 \in W$,

we find [page 238]

$$\psi_2 \in W \qquad \Longleftrightarrow \qquad \psi_3 \in W,$$

whence

$$\varphi_2 \in W \qquad \Longleftrightarrow \qquad \varphi_3 \in W,$$

which is what was to be proved.

It remains to establish $(*)$ and $(**)$. The relations $(*)$ come from the observation that if $i : A \longrightarrow A'$ is an inclusion of full subcategories of I, such that i is W-*aspheric* (i.e. A/j W-aspheric for $j \in \mathrm{Ob}\, A' \backslash \mathrm{Ob}\, A$), then $\mathcal{X}_A \longrightarrow \mathcal{X}_{A'}$ is so too (smooth base change), hence is W. We must verify W-asphericity for the four inclusions

$$\begin{array}{rcll}
\{0,2,0'\} & \longrightarrow & U = \{0,1,2,0'\} & \boldsymbol{\Delta}^0 \\
\{0,2,0',2'\} & \longrightarrow & I & \boldsymbol{\Delta}^0/\boldsymbol{\Delta}^1 \\
\{1,2,1'\} & \longrightarrow & V = \{0,1,2,0',1'\} & \boldsymbol{\Delta}^1,\ \Delta^2 \\
\{1,2,1',2'\} & \longrightarrow & I & \boldsymbol{\Delta}^1,\ \boldsymbol{\Delta}^1 \times \boldsymbol{\Delta}^1.
\end{array}$$

(NB I have marked in the margin the categories we obtain as A/j, up to isomorphism, in the four cases considered - they all have an initial object, hence are W-aspheric.) p. 154

It remains to prove $(**)$. Let

$$U' = \{0,1,0',1'\}, \qquad \text{hence } V = U \cup U', \quad U \cap U' = \{0,1,0'\},$$

where U, U' are *closed subsets* of V which cover V. Thus we will likewise have

$$\mathcal{X}_V = \mathcal{X}_U \cup \mathcal{X}_{U'}, \qquad \mathcal{X}_U \cap \mathcal{X}_{U'} = \mathcal{X}_{010'},$$

[page 239]

[127] should be true in every category of models having enough cofibrations.

$$\begin{array}{ccc} \mathcal{X}_{U\cap U'} & \longrightarrow & \mathcal{X}_U \\ \downarrow{\scriptstyle \varphi_1} & & \downarrow{\scriptstyle \psi_1} \\ \mathcal{X}_{U'} & \longrightarrow & \mathcal{X}_V. \end{array}$$

Thus by W(7 bis) (dual case of closed immersions), we have

$$\varphi_1 \in W \qquad \Longleftrightarrow \qquad \psi_1 \in W,$$

q.e.d.

Let us note for the record the corollary which had motivated us.

Corollary. *Assume* W(7 bis) (as throughout this section). *Let there be a cocartesian square in* Cat

$$\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1 \\ {\scriptstyle i_2}\downarrow & & \downarrow \\ X_2 & \longrightarrow & X_3, \end{array}$$

with i_1 (or i_2) a W-cofibration (e.g. a Dwyer immersion). Then this square is W-cocartesian.

This digression (p. 232 ff.) is not necessary, for that matter, to formulate right now the conclusion of our cogitations on W-homotopic cofibers: *There is such a theory in* Cat, provided only that W satisfies W(7 bis). Indeed, everything follows from the fact that every arrow $f : X \longrightarrow Y$ in Cat factors as

$$X \xrightarrow{i_f} \tilde{Y} \xrightarrow{p_f} Y,$$

with i_f a W-cofibration (and even an open Dwyer immersion), and $p_f \in W$.

p. 155

[page 240]

7 Return to the axiomatics of the W-homotopic fiber.

I come back to the reflection in n° 4. In the end, for

$$(\mathcal{M}, W \subseteq \mathrm{Fl}(\mathcal{M}))$$

given with W weakly saturated, it appears that an excellent theory of W-homotopic fibers, satisfying the conditions ⓐⓑⓒ of n° 4 (pages 215-218), exists as soon as the following innocent-looking axiom is satisfied (in addition to the existence of the final object and fiber products ([128])).

Left factorization: ([129],[130]). Every arrow $f : X \longrightarrow Y$ of $\mathcal{M}$ factors as

$$X \xrightarrow{i_f} \bar{X} \xrightarrow{p_f} Y,$$

with

$$i_f \in W, \qquad p_f \in \mathrm{Fib}_W.$$

[128] and even that! it seems to me that it suffices to assume that the $X \longrightarrow Y$ which are W-fibrant are "squarable".

[129] Under these conditions, $\mathcal{M}$ equipped with (W, F_W) is a category with fibrations of K.S. Brown-Anderson, and "everything is O.K.", but one would have to check a) b) c) above (p. 215, 218), I am not sure I have done it.

[130] **NB** The left factorization axiom is sufficient for the existence of a good theory of homotopy fibers (it seems) only in the case where every X in $\mathcal{M}$ is W-fibrant, i.e. is in $\underline{\mathrm{Fib}}_W e$.

The dual axiom gives rise to a good theory of W-homotopic cofibers.

Right factorization: Every arrow $f : X \longrightarrow Y$ in $\mathcal{M}$ factors as

$$X \xrightarrow{j_f} \tilde{Y} \xrightarrow{q_f} Y,$$

with

$$q_f \in W, \qquad j_f \in \mathrm{Cofib}_W.$$

I would now like to develop somewhat the formalism of W-homotopic fibers, under the assumption (say) that W satisfies the left factorization axiom. [page 241]

Let us consider a cartesian diagram

$$\begin{array}{ccc} X_1 & \xleftarrow{q_2} & Y \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle q_1} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$$

in $\mathcal{M}$. If p_1 is in Fib_W, then Y over X_2 can be interpreted as the W-homotopic fiber of X_1/S at X_2/S. More precisely, this is so for the class $\mathrm{ho}^{\mathrm{lc}}_{W,X_2}(Y)$, p. 156

$$\mathrm{ho}^{\mathrm{lc}}_{W,X_2}(Y) \simeq p_2^*(\mathrm{ho}^{\mathrm{lc}}_{W,S}(X_1/S)). \qquad (*)$$

But as an object over X_1, Y can also be interpreted as a W-homotopic fiber of X_2/S at X_1/S:

$$\mathrm{ho}^{\mathrm{lc}}_{W,X_1}(Y/X_1) \simeq p_1^*(\mathrm{ho}^{\mathrm{lc}}_{W,S}(X_2/S)). \qquad (**)$$

To tell the truth, one would have to prove it. For this, let us consider a factorization

$$X_2 \xrightarrow{i_2} \bar{X}_2 \xrightarrow{\bar{p}_2} S \qquad i_2 \in W \ , \quad \bar{p}_2 \in \mathrm{Fib}_W,$$

so that we find a diagram of cartesian squares

$$\begin{array}{ccccc} X_1 & \xleftarrow{\bar{q}_2} & \bar{Y} & \xleftarrow[W]{j_2} & Y \\ \downarrow & & \downarrow & & \downarrow \\ S & \xleftarrow{\bar{p}_2} & \bar{X}_2 & \xleftarrow[W]{i_2} & X_2. \end{array}$$

By definition of p_1^*, we have [page 242]

$$p_1^*(\mathrm{ho}^{\mathrm{lc}}_{W,S}(X_2/S)) \simeq \mathrm{ho}^{\mathrm{lc}}_{W,X_1}(\bar{Y}/X_1),$$

on the other hand, since $X_1 \longrightarrow S$ is in Fib_W, and $i_2 \in W$, we have $j_2 \in W$, hence

$$\mathrm{ho}^{\mathrm{lc}}_{W,X_1}(Y/X_1) \xrightarrow{\sim} \mathrm{ho}^{\mathrm{lc}}_{W,X_1}(\bar{Y}/X_1),$$

whence formula $(**)$.

Thus, the object Y has two different homotopical interpretations, depending on whether one regards it as being over X_1, or over X_2. One would then like an interpretation which encompasses both at once. We are going to consider the diagram

$$\begin{array}{ccc} X_1 & & \\ \downarrow & & \\ S & \longleftarrow & X_2 \end{array}$$

of type Ψ°, or

$$\Psi \quad = \quad \begin{array}{ccc} & & 1 \\ & \nearrow & \uparrow \\ 0 & \longrightarrow & 2 \end{array} \qquad \text{or} \qquad \begin{array}{ccc} & \nearrow & 1 \\ 0 & & \\ & \searrow & 2 \end{array} \quad ,$$

from which Y is obtained by fiber product, by completing this diagram to a diagram of type $\tilde{\Psi}^\circ$, where p. 157

$$\tilde{\Psi} = \begin{array}{ccccc} & \nearrow & 1 & \searrow & \\ 0 & & & & 3 \\ & \searrow & 2 & \nearrow & \end{array} \qquad (\simeq \mathbf{\Delta}^1 \times \mathbf{\Delta}^1).$$

For every category I, let [page 243]

$$\mathrm{Ho}_{\mathcal{M},W}(I) \text{ or } \mathrm{Ho}_W(I) \overset{\text{def}}{=} \mathcal{M}(I)W(I)^{-1},$$

where

$$\begin{cases} \mathcal{M}(I) & = & \underline{\mathrm{Hom}}(I^\circ, \mathcal{M}) \\ W(I) & \subseteq & \mathrm{Fl}(\mathcal{M}(I)) \text{ consists of the } u : F \longrightarrow G \text{ such that} \\ & & u_i : F(i) \longrightarrow G(i) \text{ in } W \text{ for every } i \text{ in } \mathrm{Ob}\, I. \end{cases}$$

Thus, the diagram $\begin{array}{ccc} & & X_1 \\ & \swarrow & \\ S & & \\ & \nwarrow & \\ & & X_2 \end{array}$ defines an object ξ of $\mathcal{M}(\Psi)$, hence of $\mathrm{Ho}_W(\Psi)$, and

the diagram $\begin{array}{ccccc} & & X_1 & & \\ & \swarrow & & \nwarrow & \\ S & & & & Y \\ & \nwarrow & & \swarrow & \\ & & X_2 & & \end{array}$ can likewise be interpreted as an object $\tilde{\xi}$ of $\mathrm{Ho}_W(\tilde{\Psi})$, whose restriction to Ψ is ξ.

This being laid down, I would like to define a functor

$$i_* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi})$$

(where

$$i : \Psi \longrightarrow \tilde{\Psi}$$

is the inclusion, which is an open immersion), which must correspond, in the case of an object $\begin{array}{ccc} & & X_1 \\ & \swarrow & \\ S & & \\ & \nwarrow & \\ & & X_2 \end{array}$ of $\mathrm{Ho}_W(\Psi)$ with $p_1 \in \mathrm{Fib}_W$, to the fiber product operation ([131]).

p. 158

[131] **NB** This construction, which is done here rather laboriously, becomes immediate, and its main properties too, if one admits that the category with fibrations $(\mathcal{M}, W, \mathrm{Fib}_W)$ defines a *derivator* on the domain of finite ordered sets, which can indeed be proved quite simply (with no more trouble than I am taking here). [page 244]

The construction of this functor i_* should be essentially sorital, with what we have at our disposal. We shall first define, for S *fixed,* a canonical functor

$$\mathrm{Ho}_W^{\mathrm{lc}}(S) \times \mathrm{Ho}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{Ho}_{W,S}(\Phi), \qquad (*)$$

where the second member is defined, by analogy with the $\mathrm{Ho}_W(I)$, by

$$\mathrm{Ho}_{W,S}(I) \overset{\mathrm{def}}{=} ((\mathcal{M}/S)(I))(W_S(I))^{-1}$$

- i.e. it is the same definition, except that $(\mathcal{M}, W)$ is replaced by $(\mathcal{M}/S, W_S)$. As for $\Phi = \begin{array}{ccc} 1 & & \\ & \searrow & \\ & & 3 \\ & \nearrow & \\ 2 & & \end{array}$, we regard it as a closed subcategory of $\tilde{\Psi}$, which is obtained from it by adjoining an initial object 0 (just as $\tilde{\Psi}$ is obtained from Ψ by adjoining a final object 3). On the other hand, we have a tautological functor

$$\mathrm{Ho}_{W,S}(\Phi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi}), \qquad (*')$$

which on objects consists in completing a diagram $\begin{array}{ccc} & & X_1 \\ & \nwarrow & \\ & & Y \\ & \swarrow & \\ X_2 & & \end{array}$ of $\mathcal{M}/S$ to the square diagram $\begin{array}{ccccc} & & X_1 & & \\ & \swarrow & & \nwarrow & \\ S & & & & Y \\ & \nwarrow & & \swarrow & \\ & & X_2 & & \end{array}$ of $\mathcal{M}$.

The composite of $(*)$ and $(*')$ will therefore give a functor [page 245]

$$\mathrm{Ho}_W^{\mathrm{lc}}(S) \times \mathrm{Ho}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi}), \qquad (**)$$

already very close to the functor i_* to be constructed. To construct $(*)$, we interpret (up to equivalence of categories) the first member as

$$(\underline{\mathrm{Fib}}_W S \times \underline{\mathrm{Fib}}_W S)(W_S \times W_S)^{-1},$$

so we have to construct a functor

$$\underline{\mathrm{Fib}}_W S \times \underline{\mathrm{Fib}}_W S \longrightarrow \mathrm{Ho}_{W,S}(\Phi),$$

which transforms arrows in $W_S \times W_S$ into isomorphisms. We shall take the functor

$$(X_1, X_2) \longmapsto \begin{pmatrix} X_1 & \longleftarrow & X_1 \times_S X_2 \\ & & \downarrow \\ & & X_2 \end{pmatrix},$$

and we only have to check that if p. 159

$$X_1 \xrightarrow{\alpha_1} X_1', \qquad X_2 \xrightarrow{\alpha_2} X_2'$$

are two arrows in $\underline{\mathrm{Fib}}_W S$, whence

$$X_1 \times_S X_2 \xrightarrow{\alpha=\alpha_1\times_S\alpha_2} X_1' \times_S X_2',$$

then

$$\alpha_1 \in W,\ \alpha_2 \in W \qquad \Longrightarrow \qquad \alpha \in W,$$

which is immediate from the factorization

$$X_1 \times_S X_2 \underbrace{\longrightarrow}_{\substack{\in W \\ \text{since } X_2 \in \underline{\mathrm{Fib}}_W S}} X_1' \times_S X_2 \underbrace{\longrightarrow}_{\substack{\in W \\ \text{since } X_1' \in \underline{\mathrm{Fib}}_W S}} X_1' \times_S X_2'.$$

We have proved the

[page 246]

Proposition 18. *Let S be in $\mathcal{M}$. There exists, up to unique isomorphism, a functor and only one*

$$\mathrm{Ho}_W^{\mathrm{lc}}(S) \times \mathrm{Ho}_W^{\mathrm{lc}}(S) \xrightarrow{\pi} \mathrm{Ho}_{W,S}(\Phi)$$

giving rise to a square commutative up to isomorphism

$$\begin{array}{ccc} & X_1 & \\ & \nwarrow & \\ (X_1,X_2) \longmapsto & X_1\times_S X_2 & \\ & \swarrow & \\ & X_2 & \end{array}$$

$$\begin{array}{ccc} \underline{\mathrm{Fib}}_W S \times \underline{\mathrm{Fib}}_W S & \longrightarrow & (\mathcal{M}/S)(\Phi) \\ \downarrow & & \downarrow \\ \mathrm{Ho}_W^{\mathrm{lc}}(S) \times \mathrm{Ho}_W^{\mathrm{lc}}(S) & \longrightarrow & \mathrm{Ho}_{W,S}(\Phi) \\ (= (\mathcal{M}/S)(W_S)^{-1} \times (\mathcal{M}/S)(W_S)^{-1}) & & (= \mathcal{M}/S(\Phi)(W_S(\Phi))^{-1}). \end{array}$$

For X_1, X_2 in $\mathcal{M}/S$, the diagram

$$\pi(X_1, X_2) \overset{\mathrm{def}}{=} \pi(\mathrm{ho}_{W,S}^{\mathrm{lc}}(X_1), \mathrm{ho}_{W,S}^{\mathrm{lc}}(X_2))$$

is "computed" as follows, up to canonical isomorphism: one chooses

$$X_1 \xrightarrow{\alpha_1} \bar{X}_1, \qquad X_2 \xrightarrow{\alpha_2} \bar{X}_2 \qquad \textit{arrows in } \mathcal{M}/S$$

with

$$\alpha_1, \alpha_2 \in W_S, \qquad \bar{X}_1, \bar{X}_2 \in \mathrm{Ob}\,\underline{\mathrm{Fib}}_W S,$$

which defines

$$\pi(\alpha_1,\alpha_2) : \pi(X_1,X_2) \xrightarrow{\sim} \pi(\bar{X}_1,\bar{X}_2) = \mathrm{ho}_{S,W,\Phi}\left(\begin{array}{ccc} \bar{X}_1 & \longleftarrow & \bar{X}_1 \times_S \bar{X}_2 \\ & & \downarrow \\ & & \bar{X}_2 \end{array}\right).$$

Corollary. *Description of the functor*

$$(\xi_1,\xi_2) \longmapsto \xi_1 \times_S \xi_2 : \mathrm{Ho}_W^{\mathrm{lc}}(S) \times \mathrm{Ho}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{Ho}_W^{\mathrm{lc}}(S).$$

p. 160

[page 247]

This functor really is a cartesian product in $\mathrm{Ho}_W^{\mathrm{lc}}(S)$ ([132]), *where finite products exist* ([133]).

[132] but this is a special case of the cases of Ho_W (if $\mathcal{M}$ has binary products), replacing $\mathcal{M}$ by $\mathcal{M}/S$.

[133] Say also that the functors $f^* : \mathrm{Ho}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{Ho}_W^{\mathrm{lc}}(S')$ (for $f : S' \longrightarrow S$) commute with finite products.

To prove this last point, setting

$$\mathcal{H} = \mathrm{Ho}^{\mathrm{lc}}_W(S),$$

we must show that the functors

$$\mathcal{H} \underset{\pi}{\overset{\delta}{\rightleftarrows}} \mathcal{H} \times \mathcal{H}$$

are adjoint, with

$$\begin{array}{rcccccc} \delta & = & \mathrm{diag}_{\mathcal{H}} & : & \xi & \longmapsto & (\xi, \xi) \\ \pi & = & \text{“product”} & : & (\xi_1, \xi_2) & \longmapsto & \xi_1 \times_S \xi_2. \end{array}$$

We are going to define the adjunction homomorphisms

$$\begin{array}{rcccccc} \mathrm{id}_{\mathcal{H}} & \overset{\alpha}{\longrightarrow} & \pi\delta & \text{i.e.} & \xi & \overset{\alpha_\xi}{\longrightarrow} & \xi \times_S \xi \\ \mathrm{id}_{\mathcal{H}\times\mathcal{H}} & \overset{\beta}{\longleftarrow} & \delta\pi & \text{i.e.} & (\xi_1, \xi_2) & \overset{\beta_{\xi_1,\xi_2}}{\longleftarrow} & (\xi_1 \times_S \xi_2, \xi_1 \times_S \xi_2). \end{array}$$

For this, we describe $\mathcal{H}$ as

$$\mathcal{H} = (\underline{\mathrm{Fib}}_W S)(W_S)^{-1}$$

and we note that δ, π come from adjoint functors at the level of $\mathcal{H}_0 = \underline{\mathrm{Fib}}_W S$

$$\mathcal{H}_0 \underset{\pi_0}{\overset{\delta_0}{\rightleftarrows}} \mathcal{H}_0 \times \mathcal{H}_0 \ ,$$

which give rise to α_0, β_0

$$\begin{array}{rcl} \mathrm{id}_{\mathcal{H}_0} & \overset{\alpha_0}{\longrightarrow} & \pi_0\delta_0 \\ \mathrm{id}_{\mathcal{H}_0\times\mathcal{H}_0} & \overset{\beta_0}{\longleftarrow} & \delta_0\pi_0. \end{array}$$

We define α, β by passage to the categories of fractions for α_0, β_0. The 2 compatibilities, being satisfied for α_0, β_0, are also satisfied for α, β. Q.e.d. p. 161 [page 248]

I single out here the

Lemma. *Let* $\mathcal{X} \underset{g}{\overset{f}{\rightleftarrows}} \mathcal{Y}$ *be a pair of adjoint functors, and let* Σ, Σ' *be localizers in* $\mathcal{X}$, $\mathcal{Y}$ *such that* $f(\Sigma) \subseteq \Sigma'$, $g(\Sigma') \subseteq \Sigma$, *so that we have a pair of functors*

$$\mathcal{X}\Sigma^{-1} \underset{\bar{g}}{\overset{\bar{f}}{\rightleftarrows}} \mathcal{Y}\Sigma'^{-1} \ .$$

Then $\bar{f}$, $\bar{g}$ *are adjoint, the adjunction arrows being deduced from those for* f, g.

Corollary. *Let* $(\mathcal{M}, W)$ *be a category of models, suppose that in* $\mathcal{M}$ *binary products (resp. finite products) exist, and that for* $u_1 : X_1 \longrightarrow X_1'$ *and* $u_2 : X_2 \longrightarrow X_2'$, *if* $u_1, u_2 \in W$, *then* $u_1 \times u_2 : X_1 \times X_2 \longrightarrow X_1' \times X_2'$ *is in* W. *Then in* $\mathcal{M}W^{-1}$ *binary products (resp. finite products) exist, and the localization functor* $\mathcal{M} \longrightarrow \mathcal{M}W^{-1}$ *commutes with them.*

We apply the lemma to

$$\mathcal{M} \underset{\pi}{\overset{\delta}{\rightleftarrows}} \mathcal{M} \times \mathcal{M} \qquad \text{and } W,\ W \times W,$$

and for the final object,

$$\mathcal{M} \underset{\bullet}{\overset{e}{\rightleftarrows}} e \qquad \text{and } W,\ W_0 = \{\mathrm{id}_e\}.$$

[page 249]

I am now going to construct the functor

$$i_* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi}).$$

I must first prove the

Lemma 1. *Let $\mathcal{F}$ be the full subcategory of $\mathcal{M}(\Psi)$, category of diagrams* $X_1 \xleftarrow{p_1} S \xrightarrow{p_2} X_2$ *in S , consisting of those for which p_1, p_2 are in Fib_W. Let $W_{\mathcal{F}}$ be the localizer induced on $\mathcal{F}$ by $W(\Psi)$ (termwise W-equivalences). Then*

$$\mathcal{F}W_{\mathcal{F}}^{-1} \xrightarrow{\sim} \mathcal{M}(\Psi)W(\Psi)^{-1} \ (\overset{\text{def}}{=} \mathrm{Ho}_W(\Psi)).$$

We are going to prove this lemma below. Assuming it, to construct i_*, it suffices to construct p. 162

$$\mathcal{F} \longrightarrow \mathrm{Ho}_W(\tilde{\Psi})$$

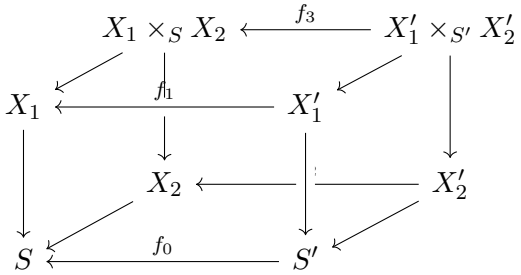
which maps $W_{\mathcal{F}}$ to isomorphisms. I shall take, of course,

$$\begin{array}{ccc} X_1 & & X_1 \\ \swarrow\ \nwarrow & \longmapsto & \swarrow\ \nwarrow \quad \swarrow\ \nwarrow \\ S \qquad\quad & & S \qquad\quad X_1 \times_S X_2 \\ \nwarrow\ \swarrow & & \nwarrow\ \swarrow \quad \nwarrow\ \swarrow \\ X_2 & & X_2 \end{array}$$

$$\mathcal{F} \quad \longrightarrow \quad \mathrm{Ho}_W(\tilde{\Psi}).$$

Let us check that it transforms arrows of $W_{\mathcal{F}}$ into isomorphisms.

[page 250]



We must prove that if $f_0, f_1, f_2 \in W$, then $f_3 \in W$. But let us consider the inverse image of the cartesian square Q under $f_0 : S' \longrightarrow S$, whence a cartesian square $Q_{S'}$ and a factorization

$$Q \xleftarrow{\tilde{f}} Q_{S'} \xleftarrow{g} Q', \qquad Q' \xrightarrow{f} Q$$

of $Q' \longrightarrow Q$. Since $f_0 \in W$, and since X_1, X_2, $X_1 \times_S X_2$ are in $\underline{\mathrm{Fib}}_W S$, the horizontal arrows realizing $Q_{S'} \longrightarrow Q$ are all in W. This, plus the fact that $f_1, f_2 \in W$, imply that in the morphism g, the arrows g_1, g_2 are in W. It follows that the arrow

$$g_3 : X_1' \times_{S'} X_2' \longrightarrow (X_1 \times_S X_2) \times_S S'$$

is as well (theory of the product in $\underline{\mathrm{Fib}}_W S'$ and $\mathrm{Ho}_W^{\mathrm{lc}}(S')$), hence $f_3 = \tilde{f}_3 g_3 \in W$, q.e.d.

[page 251]

Let us note that we have a canonical isomorphism

$$\alpha : i^* i_* \xrightarrow{\sim} \mathrm{id}_{\mathrm{Ho}_W(\Psi)}. \qquad (*)$$

I feel like proving that the functors

$$\mathrm{Ho}_W(\Psi) \underset{i_*}{\overset{i^*}{\rightleftarrows}} \mathrm{Ho}_W(\tilde{\Psi})$$

are adjoint, and that $(*)$ is one of the adjunction arrows. We must define the other adjunction arrow p. 163

$$\beta : i_* i^* \longleftarrow \mathrm{id}_{\mathrm{Ho}_W(\tilde{\Psi})}. \qquad (**)$$

But for this, I must use the

Lemma 2. Let $\mathcal{G}$ be the full subcategory of $\mathcal{M}(\tilde{\Psi})$, consisting of the $\begin{array}{ccc} X_1 & \xleftarrow{q_2} & X_3 \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle q_1} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$ such that $p_1, p_2 \in \mathrm{Fib}_W$ and let $W_{\mathcal{G}}$ be the localizer induced by $W(\tilde{\Psi})$ (termwise W-equivalences). Then the inclusion $\mathcal{G} \longrightarrow \mathcal{M}(\tilde{\Psi})$ induces an equivalence

$$\mathcal{G}W_{\mathcal{G}}^{-1} \xrightarrow{\sim} \mathcal{M}(\tilde{\Psi})W(\tilde{\Psi})^{-1} \overset{\mathrm{def}}{=} \mathrm{Ho}_W(\tilde{\Psi}).$$

I shall for the moment assume this

lemma, just like lemma 1 (p. 249). I am going to consider i^*, i_* as arrows between the categories $\mathcal{F}W_{\mathcal{F}}^{-1}$, $\mathcal{G}W_{\mathcal{G}}^{-1}$. They are induced by the arrows [page 252]

$$\mathcal{F} \underset{i_{0*}}{\overset{i_0^*}{\rightleftarrows}} \mathcal{G}$$

between $\mathcal{F}$ and $\mathcal{G}$ themselves,

$$i_0^* : \quad \mathcal{G} \quad \longrightarrow \quad \mathcal{F} : \qquad \begin{array}{ccc} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2 \end{array} \qquad \longmapsto \qquad \begin{array}{c} X_1 \\ \downarrow \\ S \longleftarrow X_2 \end{array}$$

$$i_{0*} : \quad \mathcal{F} \quad \longrightarrow \quad \mathcal{G} : \qquad \begin{array}{c} X_1 \\ \downarrow \\ S \longleftarrow X_2 \end{array} \qquad \longmapsto \qquad \begin{array}{ccc} X_1 & \longleftarrow & X_1 \times_S X_2 \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2. \end{array}$$

These two functors are adjoint to one another, and compatible with the localizers $W_{\mathcal{F}}$, $W_{\mathcal{G}}$. Thus the functors deduced from i_0^*, i_{0*} by passage to the categories of fractions are also adjoint (cf. lemma p. 248).

Thus, modulo the proof of lemmas 1 and 2, we have proved this: p. 164

[page 253]

Theorem. *Let $(\mathcal{M}, W)$ be a category of models ($W \subseteq Fl\,\mathcal{M}$) satisfying the left factorization axiom (p. 240), thus giving rise to a good theory of W-homotopic fibers. Let*

$$\begin{array}{ccc} & & 1 \\ & \nearrow & \\ \Psi = 0 & & \\ & \searrow & \\ & & 2 \end{array} \qquad \xrightarrow{i} \qquad \begin{array}{ccccc} & & 1 & & \\ & \nearrow & & \searrow & \\ \tilde{\Psi} = 0 & & & & 3 \\ & \searrow & & \nearrow & \\ & & 2 & & \end{array}$$

(open immersions in Cat*), and let us consider the restriction functor*

$$i^* : \mathrm{Ho}_W(\tilde{\Psi}) \longrightarrow \mathrm{Ho}_W(\Psi).$$

This functor admits a right adjoint

$$i_* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi}).$$

This functor i_* is "computed" as follows: its value at $\begin{array}{ccc} & & X_1 \\ & & \downarrow{\scriptstyle p_1} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$ *is canonically isomorphic (in* $\mathrm{Ho}_W(\tilde{\Psi})$*) to* $\begin{array}{ccc} X_1 & \longleftarrow & X_1 \times_S X_2 \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2 \end{array}$*, if we suppose that p_1 or p_2 is a W-fibration. In the general case it is computed by factoring p_1 as*

$$X_1 \xrightarrow{i_1} \bar{X}_1 \xrightarrow{p'_1} S, \qquad i_1 \in W,\ p'_1 \in Fib_W,$$

and by taking $\begin{array}{ccc} \bar{X}_1 & \longleftarrow & \bar{X}_1 \times_S X_2 \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2 \end{array}$*, or by proceeding symmetrically, or by simultaneously factoring p_1 and p_2, and taking* [page 254]

$$\begin{array}{ccc} \bar{X}_1 & \longleftarrow & \bar{X}_1 \times_S \bar{X}_2 \\ \downarrow & & \downarrow \\ S & \longleftarrow & \bar{X}_2. \end{array}$$

The functorial law is obtained with the help of lemma 1 p. 249, which allows us to p. 165 *restrict ourselves to the arguments* $\begin{array}{ccc} & & X_1 \\ & & \downarrow{\scriptstyle p_1} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$ *such that $p_1, p_2 \in Fib_W$, so that we find a functor*

$$\begin{array}{ccc} X_1 & & \\ \downarrow & & \\ S & \longleftarrow & X_2 \end{array} \qquad \longmapsto \qquad \begin{array}{ccc} X_1 & \longleftarrow & X_1 \times_S X_2 \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2, \end{array}$$

compatible with the localizers, and passing to the categories of fractions $\mathrm{Ho}_W(\Psi)$ *and* $\mathrm{Ho}_W(\tilde{\Psi})$ *respectively.* [page 255]

NB By proceeding in a similar, simpler way (with no need to resort to lemma 1), we find the existence of a *left adjoint* $i_!$ to $i^* : \mathrm{Ho}_W(\tilde{\Psi}) \longrightarrow \mathrm{Ho}_W(\Psi)$, subject to the existence of an initial object φ in $\mathcal{M}$. This functor $i_!$ is given by

$$i_! \begin{pmatrix} X_1 \\ \downarrow \\ S \longleftarrow X_2 \end{pmatrix} \quad = \quad \begin{pmatrix} X_1 \longleftarrow \varphi \\ \downarrow \qquad\quad \downarrow \\ S \longleftarrow X_2 \end{pmatrix},$$

the functorial law being evident.

This is not yet the end of the sorites! We must still interpret the composite functor

$$\mathrm{Ho}_W(\Psi) \xrightarrow{i_*} \mathrm{Ho}_W(\tilde{\Psi}) \xrightarrow{\varepsilon_3^*} \mathrm{Ho}_W(e) = \mathrm{Ho}_W,$$

where

$$\varepsilon_3 : e \xrightarrow{\sim} \{3\} \qquad \longrightarrow \qquad \begin{array}{ccccc} & & 1 & & \\ & \swarrow & & \nwarrow & \\ 3 & & & & 0 \\ & \nwarrow & & \swarrow & \\ & & 2 & & \end{array}$$

is the inclusion of vertex 3 in $\tilde{\Psi}$. If $\mathrm{Ho}_W(\Psi)$ is interpreted, in accordance with lemma 1, as $\mathcal{F}W_{\mathcal{F}}^{-1}$, this composite functor is none other than

$$\begin{array}{c} X_1 \\ \downarrow \\ S \longleftarrow X_2 \end{array} \qquad \longmapsto \qquad X_1 \times_S X_2$$

(for $\begin{pmatrix} X_1 \\ {\scriptstyle p_1}\downarrow \\ S \xleftarrow[p_2]{} X_2 \end{pmatrix} \in \mathrm{Ob}\,\mathcal{F}$, i.e. $p_1, p_2 \in \mathrm{Fib}_W$.)

We want to interpret this functor as a W-homotopic $\varprojlim$, i.e. viewed as a "W-homotopic fiber product". To be precise, we have this: p. 166

Corollary 1. *Consider $p_\Psi : \Psi \longrightarrow e$, and the functor* [page 256]

$$p_\Psi^* : \mathrm{Ho}_W(e) = \mathrm{Ho}_W \longrightarrow \mathrm{Ho}_W(\Psi),$$

deduced by passage to the categories of fractions from the functor

$$\begin{array}{ccc} \mathcal{M} & \longrightarrow & \mathcal{M}(\Psi) \\ X & \longrightarrow & \begin{pmatrix} X \\ \downarrow \\ X \longleftarrow X \end{pmatrix}. \end{array}$$

This functor admits the preceding functor $\varepsilon_3^* i_*$ as a right adjoint, i.e.

$$p_{\Psi *} = \varepsilon_3^* i_* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W.$$

Proof. We can regard p_Ψ as the composite

$$\Psi \xrightarrow{i} \tilde{\Psi} \xrightarrow{p_{\tilde{\Psi}}} e,$$

and we therefore have

$$p_{\Psi}^{*} \simeq i^{*}(p_{\tilde{\Psi}})^{*}.$$

Thus the existence and the determination of a right adjoint for p_{Ψ}^{*} will follow from the existence and the determination of right adjoints i_{*}, $(p_{\tilde{\Psi}})_{*}$ of i^{*}, $(p_{\tilde{\Psi}})^{*}$, via

$$(p_{\Psi})_{*} \simeq (p_{\tilde{\Psi}})_{*} i_{*}.$$

Thus corollary 1 follows from the formula [page 257]

$$p_{\tilde{\Psi}\,*} = \varepsilon_{3}^{*},$$

which itself is a special case of the

Lemma. *Let I be in* Cat*, with final object e, consider the inclusion* p. 167

$$\varepsilon : e \longrightarrow I, \qquad \text{and } p : I \longrightarrow e.$$

Let $(\mathcal{M}, W)$ be a category of models, and consider

$$\mathrm{Ho}_{W}(I) \xleftarrow{p^{*}} \mathrm{Ho}_{W}.$$

This functor admits a right adjoint p_{*}, which is isomorphic to ε^{*}.

Proof. Consider the pair of adjoint functors

$$\mathcal{M}(I) \underset{p_{*}=\varprojlim_{I}}{\overset{p^{*}}{\leftleftarrows}} \mathcal{M} \quad ;$$

as I admits the final object e, the functor p_{*} is isomorphic to ε^{*}. These functors p^{*}, ε^{*} are compatible with the localizers $W(I)$, W, so we conclude by the lemma on page 248.

It only remains to prove lemmas 1 and 2. I will do it tomorrow.

Today, I still feel like giving the sorites of W-cartesian squares in a [page 258]

Proposition 19 : *Let* $\begin{array}{ccc} X_1 & \xleftarrow{f_1} & X_3 \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle f_2} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$ *be a commutative square in $\mathcal{M}$. Consider factorizations of p_1, p_2 as*

$$X_1 \xrightarrow{i_1} \bar{X}_1 \xrightarrow{\bar{p}_1} S, \qquad X_2 \xrightarrow{i_2} \bar{X}_2 \xrightarrow{\bar{p}_2} S,$$

$i_1, i_2 \in W$, $\bar{p}_1, \bar{p}_2 \in F_W$, and consider the arrows induced by $X_3 \longrightarrow X_1 \times_S X_2$ (deduced from Q) via $X_1 \times_S X_2 \longrightarrow \begin{array}{c} \bar{X}_1 \times_S X_2 \\ \nearrow \\ \bar{X}_1 \times_S \bar{X}_2 \\ \searrow \\ X_1 \times_S \bar{X}_2 \end{array}$ *, so that we find a commutative diagram*

$$\begin{array}{ccc} & & \bar{X}_1 \times_S X_2 \\ & {\scriptstyle \pi_1}\nearrow & {\scriptstyle j_2}\downarrow \\ X_3 & \underset{\pi_2}{\xrightarrow{\pi_0}} & \bar{X}_1 \times_S \bar{X}_2 \\ & \searrow & {\scriptstyle j_1}\uparrow \\ & & X_1 \times_S \bar{X}_2, \end{array}$$

where $j_1 = i_1 \times_S \mathrm{id}_{\bar{X}_2}$ *and* $j_2 = \mathrm{id}_{\bar{X}_1} \times_S i_2$ *are in* W *(since* i_1 *and* i_2 *are, and* $\bar{X}_1$ *and* $\bar{X}_2$ *are in* $\underline{\mathrm{Fib}}_W S$*). We therefore have*

$$(\pi_1 \in W) \qquad \Longleftrightarrow \qquad (\pi_2 \in W) \qquad\qquad (\Longleftrightarrow \quad \pi_0 \in W),$$

which proves that the condition $\pi_1 \in W$*, which seems to depend on the choice of the factorization through* i_1*, does not depend on it, and likewise for* $\pi_2 \in W$*, and for* $\pi_0 \in W$ *(which seems to depend on both* i_1*, and* i_2*).*

Definition. *When these conditions are satisfied, we say that the square* Q *is* W*-cartesian.*

p. 168 [page 259]

Corollary 1. *If* Q *is* W*-cartesian, then the image of* Q *in* $\mathrm{Ho}_W(\tilde{\Psi})$ *is in the essential image of*

$$i_* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi}),$$

i.e. the canonical arrow in $\mathrm{Ho}_W(\tilde{\Psi})$

$$Q \longrightarrow i_* i^* Q$$

is isomorphism. The converse is true if W *is strongly saturated.*

Corollary 2. *Consider a commutative diagram in* $\mathcal{M}$

$$\begin{array}{ccccc} X'_0 & \longleftarrow & X'_1 & \longleftarrow & X'_2 \\ \downarrow & 1 & \downarrow & 2 & \downarrow \\ X_0 & \longleftarrow & X_1 & \longleftarrow & X_2, \end{array} \tag{D}$$

whence two commutative squares 1, 2, and a composite square

$$\begin{array}{ccc} X'_0 & \longleftarrow & X'_2 \\ \downarrow & 3 & \downarrow \\ X_0 & \longleftarrow & X_2. \end{array}$$

Suppose square 1 is W*-cartesian. Then square 2 is* W*-cartesian if and only if square 3 is.*

Proof. Consider a factorization of $\begin{array}{c} X'_0 \\ \downarrow \\ X_0 \end{array}$ as

$$\begin{array}{ccc} \bar{X}'_0 & \longleftarrow & X'_0 \\ & \searrow & \downarrow \\ & & X_0 \end{array}$$

with $(X'_0 \longrightarrow \bar{X}'_0) \in W$, and $(\bar{X}'_0 \longrightarrow X_0) \in \mathrm{Fib}_W$. We can then replace X'_0 by $\bar{X}'_0$ in (D), obtaining a diagram $(\bar{D})$, and we see at once that each of the squares 1(D), 2(D), 3(D) is W-cartesian if and only if the corresponding square 1$(\bar{D})$, 2$(\bar{D})$ or 3$(\bar{D})$ is. This shows that we can assume $X'_0 \longrightarrow X_0$ to be in Fib_W. Consider then the commutative diagram deduced from (D)

p. 169 [page 260]

$$\begin{array}{ccccccc} X'_0 & \longleftarrow & \tilde{X}'_1 \overset{\mathrm{def}}{=} X'_0 \times_{X_0} X_1 & \overset{\alpha}{\longleftarrow} & X'_1 & \longleftarrow & X'_2 \\ \downarrow & & \downarrow & & & & \downarrow \\ X_0 & \longleftarrow & X_1 & & \longleftarrow & & X_2, \end{array}$$

where $\alpha \in W$ by virtue of the W-cartesian hypothesis on square 1. We see again that square 2 is W-cartesian if and only if the corresponding square with X_1' replaced by $\tilde{X}_1'$ is, as for squares 1, 3, they do not change under this modification. This shows that we can assume square 1 *cartesian.* Consider then the diagram

$$\begin{array}{ccccc} X_0' & \longleftarrow & X_1' & \longleftarrow & \tilde{X}_2' \overset{\text{def}}{=} X_1' \times_{X_1} X_2 \xleftarrow{\beta} X_2' \\ {\scriptstyle \text{Fib}_W}\downarrow & \text{cart.} & \downarrow{\scriptstyle \text{Fib}_W} & \text{cart.} & \downarrow{\scriptstyle \text{Fib}_W} \\ X_0 & \longleftarrow & X_1 & \longleftarrow & X_2. \end{array}$$

Then proposition 19 and the definition show that square 2 is W-cartesian if and only if $\beta \in W$, and likewise for square 3. Which completes the proof.

Corollary 2. *Let* $\begin{array}{ccc} X_1 & \xleftarrow{f_2} & X_3 \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle f_1} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$ *be a W-cartesian square. Then*

$$\begin{aligned} p_1 \in W \quad &\Longrightarrow \quad f_1 \in W \\ p_2 \in W \quad &\Longrightarrow \quad f_2 \in W, \end{aligned}$$

more precisely, if $p_i \in W$, then Q is W-cartesian if and only if $f_i \in W$.

[page 261]

Suppose $p_1 \in W$. Then in the factorization of p_1 as $X_1 \longrightarrow \bar{X}_1 \longrightarrow S$, we can take $\bar{X}_1 = S$. Thus the square is W-cartesian if and only if

$$(X_3 \longrightarrow \bar{X}_1 \times_S X_2 = X_2) \in W,$$

i.e. $f_1 \in W$, q.e.d.

Of course, we cannot in general assert the reverse implications: $f_i \in W$ by no means implies $p_i \in W$.

p. 170

All these results, and the notion of a W-cartesian square, dualize (using the appropriate axiom of right factorization, cf. page 240) into a theory of W-homotopic amalgamated sums, and of W-cocartesian squares. This time we work with the category

$$\Phi \;=\; \begin{pmatrix} & & 1 \\ & \swarrow & \\ 0 & & \\ & \nwarrow & \\ & & 2 \end{pmatrix} \qquad \overset{j}{\hookrightarrow} \qquad \tilde{\Phi} \;=\; \begin{pmatrix} & & 1 & & \\ & \swarrow & & \nwarrow & \\ 0 & & & & 3 \\ & \nwarrow & & \swarrow & \\ & & 2 & & \end{pmatrix},$$

j being a closed immersion. We have a restriction functor

$$j^* : \text{Ho}_W(\tilde{\Phi}) \longrightarrow \text{Ho}_W(\Phi),$$

for which we construct a left adjoint

$$j_! : \text{Ho}_W(\Phi) \longrightarrow \text{Ho}_W(\tilde{\Phi}).$$

[page 262]

And we find a natural notion of a W-cocartesian square, which seems irreproachable from every point of view, and of perfect simplicity.

This leads us to look again at the notion in question in Cat, where it looked very clever, deep and a little obscure, with integrals and all that. This is the moment to realize that "that's all it is". So to prove the

Proposition 20. *Suppose* $\mathcal{M} = \mathrm{Cat}$, *$W$ a basic localizer satisfying W(7 bis). Consider a commutative square in* Cat

$$Q = \begin{pmatrix} S & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & X_3 \end{pmatrix}.$$

For it to be W-cocartesian in the sense of the theory developed here (for general categories of models), it is necessary and sufficient that

$$\varphi_Q : \quad \mathcal{X}_Q = \begin{pmatrix} & & X_1 \\ & \nearrow & \\ \int S & & \longrightarrow \quad X_3 \\ & \searrow & \\ & & X_2 \end{pmatrix}$$

be in W. (NB The interest of this criterion is that it is "intrinsic" to Q, not calling on a factorization of β_1 or of β_2 into $S \xrightarrow{\in \mathrm{Cof}_W} \tilde{X}_i \xrightarrow{\in W} X_i$. Cf. interesting application in section V 3.4., p. 146 ff., th. 4 and corollaries.) p. 171

Proof. Consider a factorization of $S \longrightarrow X_1$ as

[page 263]

$$S \xrightarrow{i_1} \tilde{X}_1 \xrightarrow{p_1} X_1,$$

with i_1 an open Dwyer immersion, $p_1 \in W$. So we have a commutative square

$$\tilde{Q} = \begin{pmatrix} S & \xhookrightarrow{\mathrm{Dw}} & \tilde{X}_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & X_3 \end{pmatrix}$$

and a homomorphism $\tilde{Q} \xrightarrow{\alpha} Q$ which is the identity on the three vertices other than $\tilde{X}_1$, and which is p_1 at $\tilde{X}_1$. We deduce from this a commutative diagram

$$\begin{array}{ccc} \mathcal{X}_Q & \xleftarrow{\mathcal{X}_\alpha = \lambda} & \mathcal{X}_{\tilde{Q}} \\ & {\scriptstyle \varphi_Q} \searrow \quad \swarrow {\scriptstyle \varphi_{\tilde{Q}}} & \\ & X_3, & \end{array}$$

and we have $\lambda \in W$ (for it is a Φ-arrow of fibered categories over Φ, which induces a W-equivalence on each of three fibers.) Thus $\varphi_Q \in W \iff \varphi_{\tilde{Q}} \in W$, on the other hand it is clear that Q is W-cocartesian if and only if $\tilde{Q}$ is. So we may assume that $S \hookrightarrow X_1$ is open (Dwyer) immersion ([134]). Applying this with the roles of X_1 and X_2 interchanged, we see that we may moreover assume that $S \longrightarrow X_2$ is also open Dwyer immersion. Consider then

[page 264]

$$\begin{array}{ccccc} S & \xhookrightarrow{\qquad} & X_1 & & \\ \downarrow & Q' & \downarrow & & \\ X_2 & \xhookrightarrow{\qquad} & X_1 \amalg_S X_2 & \xrightarrow{\pi} & X_3 \\ & & {\scriptstyle \varphi_{Q'}} \uparrow & \nearrow {\scriptstyle \varphi_Q} & \\ & & \mathcal{X}_Q & & \end{array}$$

[134] Here we use W(3) (strong form), i.e. Loc(3 bis) in the notation of XVI. But we can do without it, cf. below.

We have $\varphi_Q = \pi\varphi_{Q'}$, on the other hand we know that $\varphi_{Q'} \in W$ (cor. 3, p. 149). Thus p. 172

$$\pi \in W \qquad \Longleftrightarrow \qquad \varphi_Q \in W,$$

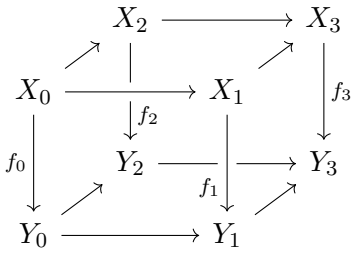
where $\pi \in W$ means that Q is W-cocartesian, q.e.d.

[page 265]

8 Appendix : Return to the axioms W(7), W(7 bis) (pages 75, 76).

([135]). We have seen (prop. 6, p. 78) that W(7 bis) implies W(7). Let us prove a partial converse.

Proposition. *Suppose* W(7 bis) (cocartesian square axiom) *satisfied, and let us place ourselves under the conditions of the statement of* W(7), *and more generally, in the case of a cubical diagram*



where the two squares X and Y (upper and lower face of the cube) are W-cocartesian. Consider the following conditions on this diagram

a) $f_0 \in W$.

b) f_1 *and* $f_2 \in W$.

c) $f_3 \in W$.

Then we have the implications

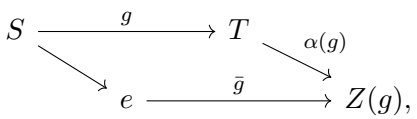
$$a \ \& \ b \Longrightarrow c, \qquad b \ \& \ c \Longrightarrow a.$$

Moreover, if W is stable under direct factors (e.g. W strongly saturated), then we also have

$$a \ \& \ c \Longrightarrow b,$$

thus the conjunction any two among the conditions a) b) c) implies the third.

Proof. Using the functorial factorization of an arbitrary arrow (here the f_i) as pi, p. 173 with i an open Dwyer immersion and $p \in W$ (p. 229), we are reduced to the case where the f_i are open Dwyer immersions. But for any arrow $g : S \longrightarrow T$ which is in Cof_W, let $Z(g)$ be its mapping cone [page 266]

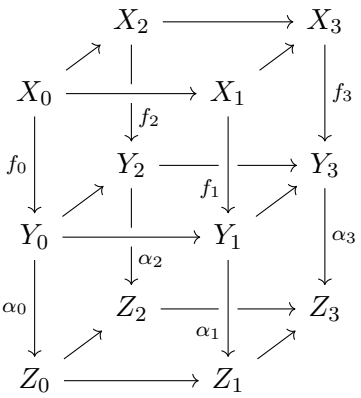


and

$$\alpha(g) : T = \mathrm{but}(g) \longrightarrow Z(g)$$

[135] Provisional reflection, superseded by XVI.

the canonical arrow ([136]). This is functorial in g, so that the given cubical diagram extends to a double cube



with

$$Z_i = Z(f_i),\ \alpha_i = \alpha(f_i) \qquad \text{for } i \in \{0,1,2,3\}.$$

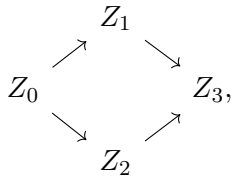
Note that W(7 bis) implies that if $g \in \mathrm{Cof}_W$, we have [page 267]

$$g \in W \qquad \Longleftrightarrow \qquad \bar{g} \in W, \qquad \text{i.e. } Z(g)\ W\text{-aspheric},$$

in particular

$$f_i \in W \qquad \Longleftrightarrow \qquad Z_i\ W\text{-aspheric}.$$

Thus, the statement to be proved about the initial cube reduces to a statement about the square p. 174



namely that the three conditions

a′) Z_0 W-aspheric.

b′) Z_1 and Z_2 W-aspheric.

c′) Z_3 W-aspheric.

giving rise to the implications

$$\text{a}' \ \&\ \text{b}' \Longrightarrow \text{c}', \qquad \text{b}' \ \&\ \text{c}' \Longrightarrow \text{a}',$$

and that if W is stable under direct factors, then

$$\text{a}' \ \&\ \text{c}' \Longrightarrow \text{b}'.$$

This is more or less the corollary on page 76, supplemented on page 77 (corollary), except that there the arrows are assumed to be open immersions, and the square cartesian (hence also cocartesian) in Cat, i.e. $Z_3 = Z_1 \cup Z_2$, $Z_0 = Z_1 \cap Z_2$. But what matters for [page 268]

[136] This is a bit quick, for it is not clear that $X_1' \amalg_{X_0} X_2' \longrightarrow X_1 \amalg_{X_0} X_2$ is W, without already using one of the three ingredients of W(7) (namely $f_0, f_1, f_2 \in W \implies f_3 \in W$). It seems then that our argument works only with an additional hypothesis of this sort …But there is *Brown's lemma,* cf. XVI §3, notably p. 20.

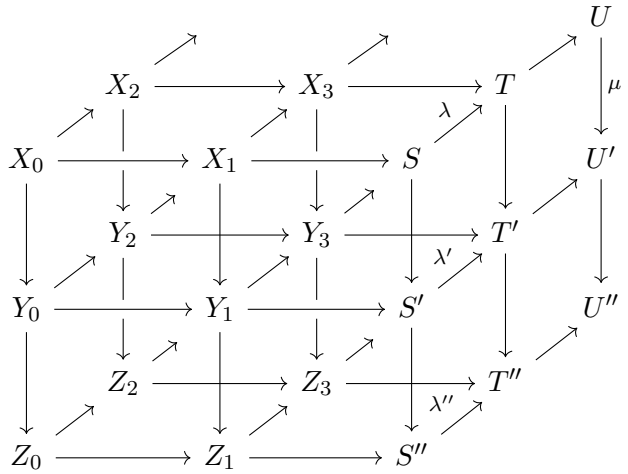
the validity of this statement is (as one might suspect) that Z_* be W-cocartesian. So we must first prove this.

Lemma. *If the cubical diagram p. 265 is such that the faces X, Y are W-cocartesian, and the f_i ($i \in \{0,1,2,3\}$) W-cofibrations, then the square Z of the cofibers of the f_i is itself also W-cocartesian.*

This lemma follows from the theory of the "unfolding of cubical diagrams", relative to a category of models $(\mathcal{M}, W)$ admitting a theory of W-homotopic cofibers, which I really shall have to write! (But the existence for (Cat, W) of such a theory of W-homotopic cofibers, and hence of that of unfoldings of diagrams in $\mathcal{M}$ of type $(\mathbf{\Delta}^1)^I$, I a finite index set, depends only on W(7 bis) - we do not have to call on W(7).)

This "unfolding" is an infinite diagram all of whose infinite sequences, in the three directions of the coordinate axes, are "suspension sequences". I reproduce the useful part of the unfolding. [page 269]

p. 175



The condition that X be W-cocartesian is expressed by $\lambda \in W$, or again by $U \in \mathrm{Asph}_W$ (for U identifies with the homotopy cofiber $Z(\lambda)$ of λ …), likewise Y W-cocartesian is expressed by $U' \in \mathrm{Asph}_W$, and Z W-cocartesian by $U'' \in \mathrm{Asph}_W$. Now $U'' \sim Z(\mu)$, where $\mu : U \longrightarrow U'$ is in W if U, U' are W-aspheric, hence $U'' = Z(\mu)$ is then W-aspheric. This proves the lemma.

This shows that, assuming W(7 bis) (in fact, we have used only the "universal" part

$$\begin{array}{ccc} T_0 & \longrightarrow & T_2 \\ {\scriptstyle i}\searrow & & \searrow{\scriptstyle i'} \\ T_1 & \longrightarrow & T_3 \end{array}$$

of W(7 bis)), namely that in a cocartesian diagram of open immersions, $i \in W \Longrightarrow i' \in W$ (instead of $\Longleftrightarrow$),

the validity of W(7) reduces to the case where the square Y is the trivial square [page 270]

$$\begin{array}{ccc} & e & \\ e & & e \\ & e & \end{array}$$

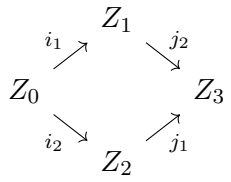
e the final object of Cat.

We can now conclude: We have (assuming the forward part of W(7 bis))

$$\mathrm{a}' \;\&\; \mathrm{b}' \Longrightarrow \mathrm{c}',$$

for if Z_0, Z_1 are $\in \mathrm{Asph}_W$, then $i_1 \in W$, hence $j_1 \in W$, and if in addition $Z_2 \in \mathrm{Asph}_W$,

we therefore have $Z_3 \in \mathrm{Asph}_W$.



If we also have the reverse ("unusual") part of W(7 bis), then

$$\mathrm{b}' \ \& \ \mathrm{c}' \Longrightarrow \mathrm{a}',$$

for if Z_1 and Z_3 are W-aspheric, then $j_2 \in W$, hence $i_2 \in W$, and if in addition Z_2 is W-aspheric, we conclude Z_0 W-aspheric.

Finally, suppose that W is *stable under direct factors,* and W(7 bis) satisfied in forward form. I say that then p. 176

$$\mathrm{c}' \ \& \ \mathrm{a}' \Longrightarrow \mathrm{b}',$$

i.e.

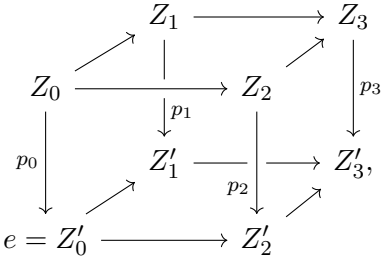
$$Z_0 \text{ and } Z_3 \ \in \mathrm{Asph}_W \qquad \Longrightarrow \qquad Z_1 \text{ and } Z_2 \ \in \mathrm{Asph}_W.$$

Factoring i_1 and i_2 as

$$Z_0 \xrightarrow{i'_1} Z'_1 \xrightarrow{\alpha_1} Z_1,$$
$$Z_0 \xrightarrow{i'_2} Z'_2 \xrightarrow{\alpha_2} Z_2,$$

with i'_1, i'_2 open Dwyer immersions, and $\alpha_1, \alpha_2 \in W$, we are reduced to the case where

i_1, i_2 are *open Dwyer immersions,* and where the square is cocartesian (which implies [page 271] that it is W-cocartesian). Then making the cobase change by $Z_0 \xrightarrow{p_0} Z'_0 = e$, we find a homomorphism of cocartesian squares



where all the faces of the cube are W-cocartesian, and since $p_0 \in W$ (Z_0 being W-aspheric), the other vertical arrows p_1, p_2, p_3 are W, hence the square Z'_* also satisfies Z'_0 and Z'_3 aspheric (and in addition $Z'_0 = e$), and we must prove Z'_1, Z'_2 W-aspheric (whence likewise for Z_1, Z_2). Thus, we are reduced to the particular case where $Z_0 = e$.

But in this case there exists a retraction (trivial!) of Z_1 onto Z_0, hence also of Z_3 onto Z_2. Thus, Z_2 is a direct

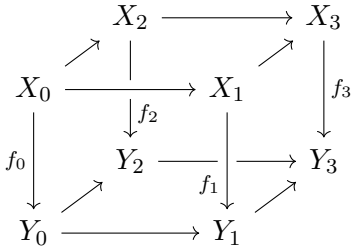
factor of a W-aspheric object, which implies, by the hypothesis on W, that Z_2 is [page 272] W-aspheric. By symmetry, Z_1 is too. We have won!

NB In fact, these arguments are general, and give this:

Theorem. *Let $(\mathcal{M}, W)$ be a model category, admitting a theory of the W-homotopy cofiber* ([137]) *(i.e. we have a factorization of every arrow $f : X \longrightarrow Y$ as pi, with*

[137] But we only know how to verify the existence of such a theory in (Cat, W) if we accept the weak form of W(7), i.e. $f_0, f_1, f_2 \in W \implies f_3 \in W$, or some special case of it. But *Brown's lemma* saves the situation, cf. XVI §3.

$i \in \mathrm{Cof}_W$, $p \in W$.) We therefore have a notion of W-cocartesian square in $\mathcal{M}$ ([138]). Let then $f : X \longrightarrow Y$ be a homomorphism from a W-cocartesian square X to another Y, p. 177
$f = (f_0, f_1, f_2, f_3)$.



Let us consider the conditions on f

a) $f_0 \in W$.

b) f_1 and $f_2 \in W$.

c) $f_3 \in W$.

Then

1°) a & $b \Longrightarrow c$.

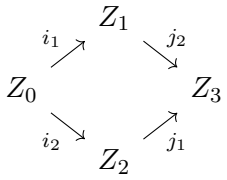
[page 273]

2°) Suppose that $\mathcal{M}$ has a final object e. Then the implication

$$b \ \& \ c \Longrightarrow a$$

for every *homomorphism $f : X \longrightarrow Y$ of W-cocartesian squares is equivalent to the following axiom on W:*

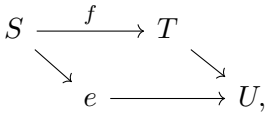
Axiom of conservative W-cocartesian squares ([139]): *If*



is W-cocartesian, then $j_1 \in W \Longrightarrow i_1 \in W$ (hence in fact $i_1 \in W \Longleftrightarrow j_1 \in W$ and $i_2 \in W \Longleftrightarrow j_2 \in W$).

This axiom is equivalent to the conjunction of the following two axioms.

Cons 1. *If we have a cocartesian square* p. 178

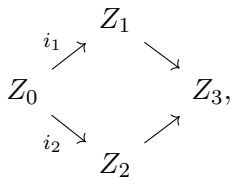


with $f \in \mathrm{Cof}_W$, then $U \in \mathrm{Asph}_W \Longrightarrow f \in W$ (hence $U \in \mathrm{Asph}_W \Longleftrightarrow f \in W$).

[138] to check that the factorization suffices, even if not all objects are fibrant over e. But we can then replace $\mathcal{M}$ by $\mathcal{M}_f$, and develop this formalism of homotopy fibers accordingly …

[139] cf. argument page 78 which shows that the implication b & c $\Longrightarrow$ a entails the conservative axiom.

Cons 2. *If we have a cocartesian square,*



with $i_1, i_2 \in \mathrm{Cof}_W$, then: If Z_1, Z_2, Z_3 are W-aspheric, Z_0 is too.

[page 274]

3°) Suppose that every direct factor of a W-aspheric object is W-aspheric, which is the case in particular if W itself is stable under direct factors, a fortiori if W is strongly saturated. Suppose in addition that $\mathcal{M}$ has final object. Then we have (under the general conditions at the beginning)

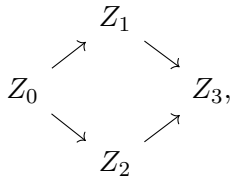
$$c \;\&\; a \Longrightarrow b,$$

i.e. if f_0, f_3 are W, likewise f_1, f_2.

This last part can be unfolded as follows:

Corollary 1. *Under the additional hypothesis 3°) on W, the validity of $c \;\&\; a \Longrightarrow b$ (for every arrow f between W-cocartesian squares) is equivalent to each of the following two other axioms on W (which are therefore equivalent):*

1) If we have a W-cocartesian square



then Z_0 and Z_3 W-aspheric $\Longrightarrow$ Z_1 and Z_2 W-aspheric.

1′) As in 1), with $Z_0 = e$ (in the case where we suppose that $\mathcal{M}$ admits final object e).

[page 275]

Corollary 2. *Suppose that W satisfies the axiom of conservative W-cocartesian squares, that every direct factor of a W-aspheric object is W-aspheric (e.g. W stable under direct factors), finally that $\mathcal{M}$ admits a final object. Then for every homomorphism $f : X \longrightarrow Y$ of W-cocartesian squares, the conjunction of two of the conditions a) b) c) on p. 272 implies the third.*

THE DERIVATORS

p.

ALEXANDRE GROTHENDIECK

Appendix to Chapter XII

Proposition (?) ([1]). *Let X be in* Cat*, with b, c in X such that*

p. 1

a) $b \times c$ is in X.

[page 1]

b) $\forall$ x in X, $b(x)$ ($\overset{\text{def}}{=} \mathrm{Hom}_X(x,b)$) and $c(x)$ are $\neq \emptyset$, i.e. $X/b \longrightarrow X$ and $X/c \longrightarrow X$ are surjective on objects.

Then X is $\mathbf{W}_\infty$-aspheric. ([2])

The fact that b is a greatest element of ObX already ensures 0-connectedness.

For 1-connectedness, let us prove that in the fundamental groupoid $\Pi(X)$, $a \overset{\text{def}}{=} b \times c$ becomes a final element.

$$\begin{array}{ccc} b \times c & \longrightarrow & b \\ \downarrow & & \\ c & & \end{array}$$

For this, let us first prove that $\forall$ x in X, all the elements of $a(x) = b(x) \times c(x)$ are equal in Fl$\Pi(X)$. Indeed, since $b \times c \longrightarrow b$ is isomorphism, hence monomorphism in $\Pi(X)$, this already shows that if $\beta, \beta' \in b(x)$, $\gamma, \gamma' \in c(x)$, whence (β,γ), (β,γ'), $(\beta',\gamma') \in b(x) \times c(x)$, we have $(\beta,\gamma) = (\beta,\gamma')$ in $\Pi(X)$ (since $\mathrm{pr}_1(\beta,\gamma) = \mathrm{pr}_1(\beta,\gamma')$, and likewise (using that pr_2 is isomorphism, hence monomorphism) $(\beta,\gamma') = (\beta',\gamma')$). Thus all the $(\beta,\gamma) \in b(x) \times c(x) = a(x)$ are equal in Fl$\Pi(X)$. To finish proving that a is a final object of $\Pi(X)$, it suffices to see that every arrow *in* $\Pi(X)$ from an $x \in \mathrm{Ob}X$ to a, comes from an arrow $x \longrightarrow a$ in X. But an arrow in $\Pi(X)$ is obtained by a path in X

$$\begin{array}{ccccccccc} x = x_0 & \xrightarrow{u_1} & x_1 & \xleftarrow{u_2} & x_2 & \longrightarrow & \cdots & \xleftarrow{u_{2N}} & x_{2N} = a \\ & \searrow^{\alpha_0} & \downarrow{\scriptstyle \alpha_1} & & \swarrow_{\alpha_2} & & & & \\ & & a. & & & & & & \end{array}$$

Let us take an arrow $\alpha_0 : x \longrightarrow a$ in X, I say that the homotopy class of the path c (i.e. its image in Fl$(\Pi(X))$) is equal to that of α_0. To see this, let us choose for every $i \in [0, 2N]$ an arrow $\alpha_i : x_i \longrightarrow a$. (It exists thanks to hypothesis b)). I say that for each $i \in [0, 2N]$, the triangle

[page 2]

$$\begin{array}{ccc} x_0 & \xrightarrow{c_i} & x_i \\ & \searrow^{\alpha_0} \quad \swarrow_{\alpha_i} & \\ & a & \end{array}$$

p. 2

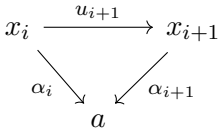
in $\Pi(X)$ is commutative, where c_i denotes the homotopy class of the path from x_0 to x_i defined by the u_j ($j \in [1,i]$) . We do this by induction on i, the assertion is trivial if $i = 0$ (c_0 represents the identity map). The diagram

$$\begin{array}{ccccc} & & c_{i+1} & & \\ & \overset{\frown}{\quad} & & \overset{\frown}{\quad} & \\ x_0 & \xrightarrow{c_i} & x_i & \xrightarrow{u_{i+1}} & x_{i+1} \\ & \searrow^{\alpha_0} \quad \swarrow_{\alpha_i} & & \swarrow_{\alpha_{i+1}} & \\ & & a & & \end{array}$$

[1] Cf. pages 11 ff. for the key counterexamples.

[2] False, cf. below. X is 1-connected, but not necessarily 2-connected.

shows that it suffices to prove that for $i \in [0, 2N-1]$ the triangle



is commutative in $\Pi(X)$. (**NB** u_{i+1} is in the direct or retrograde direction, according as i is even or odd.) But this follows from the fact that the image of $\mathrm{Hom}_X(x_i, a)$ (resp. $\mathrm{Hom}_X(x_{i+1}, a)$) in $\mathrm{Fl}\Pi(X)$ is reduced to a single element.

[page 3]

Thus we find that X *is* 1-*connected.* So there is hope! It remains to prove

$$H^i(X, M) = 0, \qquad i \geq 2, \quad M \text{ constant coefficient.}$$

I have looked at the case when we have this: $b = c$, $\mathrm{Ob}X = \{b, b \times b\}$, $\mathrm{Hom}_X(b \times b, b)$ $=$ { disjoint union of the sets $u \circ \mathrm{pr}_1$ and $u \circ \mathrm{pr}_2$, with $u \in \mathrm{Hom}_X(b, b) = G$ an arbitrary monoid }. Then X° is fibered, via $I \longmapsto b^I$, over the full subcategory S of Ens consisting of $\{0\}$ and $\{0, 1\}$, and thus X is cofibered (in particular proper) over $S^\circ \overset{\text{def}}{=} T$. (NB $S^\circ = T$ has an initial object, but not a final object. Its cohomology with values in an arbitrary presheaf is not always trivial, cf. below.)

We find

$$H^n(X, M) \Longleftarrow H^p(T, \underbrace{R^q f_*(M_X)}_{=\mathcal{H}^q}), \qquad \text{where } f : X \longrightarrow S^\circ$$

. I admit here (provisionally) this:

$$\begin{aligned} &H^p(T, F) = 0 \ , \qquad \text{for } p \geq 2, \text{ any } F \in T^\wedge \ , \\ &H^0(T, F) = \mathrm{Ker}\delta \ , \\ &H^1(T, F) = \mathrm{Coker}\delta \ , \end{aligned}$$

where $\delta : F_0 \longrightarrow F_1$ is defined as the difference $\delta = p_1^* - p_2^*$, where p_1, p_2 are the two projections of $\{0, 1\} \simeq \{0\}^2$ in $T = S^\circ$. Moreover we have

$$(\mathcal{H}^q)_0 = H^q(G, M), \qquad (\mathcal{H}^q)_1 = H^q(G \times G, M),$$

p. 3

and

$$p_1^*, \ p_2^* \ : \ H^q(G, M) \rightrightarrows H^q(G \times G, M) \qquad \text{are the transposes of } \mathrm{pr}_1, \ \mathrm{pr}_2 : G \times G \longrightarrow G.$$

We readily check that

$$H^0(T, \mathcal{H}^q) = 0 \qquad \text{if } q \geq 1,$$

and there remains

[page 4]

$$H^n(X, M) \simeq H^1(T, \mathcal{H}^{n-1}) = \mathrm{Coker}(\delta : H^{n-1}(G) \longrightarrow H^{n-1}(G \times G)).$$

This gives, for $n = 2$,

$$\begin{aligned} H^2(X, M) &\simeq \mathrm{Coker}\Big(\delta : \mathrm{Hom}_{\mathrm{mon}}(G, M) \longrightarrow \underbrace{Hom_{\mathrm{mon}}(G \times G, M)}_{\mathrm{Hom}_{\mathrm{mon}}(G,M) \times \mathrm{Hom}_{\mathrm{mon}}(G,M)} \Big) \\ &\simeq \mathrm{Coker}\Big(u \longmapsto (u, -u) : H \longrightarrow H \times H\Big), \end{aligned}$$

where H is the abelian group $\mathrm{Hom}_{\mathrm{mon}}(G, M)$.

Whence

$$H^2(X, M) \ \simeq \ H = \mathrm{Hom}_{\mathrm{mon}}(G, M).$$

Thus

X is 2-connected if and only if the enveloping *abelian* group of the monoid G is zero.

Thus, if G is an abelian group $\neq 0$, X *is not* 2-connected.

Corollary. $\Psi = \begin{array}{ccc} & \nearrow & b_0 \\ a_0 & & \\ & \searrow & c_0 \end{array}$ *is* not *totally* $\mathbf{W}_\infty$*-aspheric; i.e. if we have a diagram*

$\begin{array}{ccc} & \nearrow & b \\ a & & \\ & \searrow & c \end{array}$ *in a category* $X \in \mathrm{ObCat}$*, such that its amalgamated sum in* $X^\wedge$ *is equal to* e_X*, it does not follow that* X *is* $\mathbf{W}_\infty$*-aspheric.* (But perhaps will be willing to be 1-connected?)

Question. *Are there other totally* $\mathbf{W}_\infty$*-aspheric objects of* Cat*, than the categories having a final object?* I now suspect not – hence that one cannot improve substantially on the already known notion of an open Dwyer immersion. p. 4 [page 5]

In the diagram

$$\begin{array}{ccccc} U & \xhookrightarrow{i} & X & \xleftarrow{j} & T \\ {\scriptstyle p}\downarrow & \text{cocart. and cart.} & {\scriptstyle q}\downarrow & & \vdots\ \text{iso} \\ U' & \xhookrightarrow{i'} & X' & \xleftarrow{j'} & T', \end{array} \qquad (*)$$

let us take

$$\begin{aligned} U = \Psi, \quad & T = \{t\} = e \text{ final category,} \\ & t \text{ final object of } X, \\ & \text{i.e. } H_t = (u \longmapsto \mathrm{Hom}_X(u,t)) = e_U \in \mathrm{Ob}U^\wedge\ ([3]). \end{aligned}$$

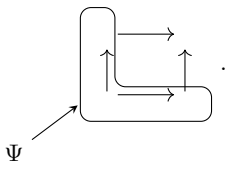
Let us take for U' the category spoken of in the hypothetical (and false) "proposition" p. 1 with $b = c$, more precisely a category with two objects b and $b \times b$, a group (or even a monoid) G of automorphisms of b, and the other arrows those which follow from these, requiring that $b \times b$ really be the cartesian square of b. We find a cofibered category (hence proper) over the category T opposite to that of finite sets of the form $\{0\} \cup \{0,1\}$ (two objects) with fibers B_G and $B_{G\times G}$ respectively (B_π = category with a single object, and endomorphism monoid π). The functor

$$U = \Psi \longrightarrow U'$$

is the one given by the diagram [page 6]

$$\begin{array}{ccc} & \xrightarrow{\mathrm{pr}_1} & b \\ b \times b & & \\ & \xrightarrow{\mathrm{pr}_2} & b \end{array}$$

[3] i.e. X obtained from $U = \Psi$ by adding to it a final object t – thus $X \simeq \mathbf{\Delta}^1 \times \mathbf{\Delta}^1$, p. 5



in U', and hypotheses a), b) of page 1 ensure that the amalgamated sum $\varinjlim$ of this diagram in $U'^{\wedge}$ is $e_{U'}$, i.e. that

$$p_!(e_U) = e_{U'}.$$

It follows that in the amalgamated sum, X' considered in $(*)$, the unique element t' of T' is a final object of X'. We have seen that U' is 1-connected, but not 2-connected if the enveloping abelian group of G (i.e. $G/$(commutator subgroup) in the case where G is a group) is not reduced to 1. The most economical example is obtained by taking $G = \{\pm 1\}$ the group with two elements. This is thus a case where in $(*)$ we have

$$i \in W \quad \text{(and even } i \text{ aspheric)}, \qquad \text{but } i' \notin W,$$

and which has the advantage over the more sophisticated example of page 103 (prop. 10) of involving only finite categories – and even very small ones!

Let us now proceed as in proposition 11, denoting by U'' the category obtained from U' by adding to it a final object, and let

[page 7]

$$p' : U' \longrightarrow U''$$

be the inclusion, so that

$$p'_!(e_{U'}) \quad = \quad (p'p)_!(e_U) = U' \qquad \text{as an object in } U''^{\wedge}$$
$$\text{(subpresheaf of the final presheaf } e_{U''}).$$

Consider

$$\begin{array}{ccccc} U & \xrightarrow{\ i\ } & X & \xleftarrow{\ j\ } & T \\ {\scriptstyle p''=p'p}\downarrow & \text{cocart. and cart.} & {\scriptstyle q''}\downarrow & & \vdots\ \text{iso} \\ U'' & \xrightarrow{\ i''\ } & X'' & \xleftarrow{\ j''\ } & T''. \end{array}$$

This time U, X *and* U'' are $\mathbf{W}_\infty$-aspheric (and even all three contractible), hence i, p'' are in W, yet X'' is not $\mathbf{W}_\infty$-aspheric (hence $i'', q'' \notin \mathbf{W}_\infty$), for in order for it to be so, i.e. for $i'' \in \mathbf{W}_\infty$, it would be necessary (prop. 7, cor. page 85 ([4])) that i'' be $\mathbf{W}_\infty$-aspheric, i.e. that U''/t'' be $\mathbf{W}_\infty$-aspheric. But by construction $U''/t'' \simeq U'$, which is *not* $\mathbf{W}_\infty$-aspheric.

Here is an even simpler example of a functor $f : \Psi \longrightarrow X$ such that $f_!(e_\Psi)$ is not $\mathbf{W}_\infty$-aspheric, i.e. of a diagram

$$\begin{array}{ccccc} & & & & b \\ & & & \overset{\beta_0}{\nearrow} & \\ & & a & & \\ & & & \underset{\gamma_0}{\searrow} & \\ & & & & c \end{array}$$

in a category V, such that the direct limit $L = b \amalg_a c$ *in* $V^{\wedge}$ is a non-aspheric object of $V^{\wedge}$, i.e. V/L is non-aspheric. I know that, for such a counterexample, $a \longrightarrow b$ and $a \longrightarrow c$ must not be monomorphisms. The idea is to take a universal example where neither of these arrows is monomorphism, by taking

p. 6
[page 8]
[page 9]

$$V = \left(\begin{array}{ccccc} & & & & b \\ & & & \overset{\beta}{\nearrow} & \\ 0 & \underset{\mu}{\overset{\lambda}{\rightrightarrows}} & a & & \\ & & & \underset{\gamma}{\searrow} & \\ & & & & c \end{array} \right)$$

[4] not yet proved!

with

$$\beta\lambda = \beta\mu \qquad \text{and} \qquad \gamma\lambda = \gamma\mu$$

the only two nonidentity arrows, besides the four arrows marked β, γ, λ, μ.

We find that

$$b \amalg_a c \xrightarrow{\sim} e_V \qquad \text{(amalgamated sum in } V^\wedge\text{)}.$$

Yet V is not aspheric, nor even 1-connected: it is homotopic to the open subcategory $V_0 = (0 \underset{\mu}{\overset{\lambda}{\rightrightarrows}} a)$, for we have a retraction $V \xrightarrow{r} V_0$ (sending b and c to a), and an arrow $ir \longrightarrow \mathrm{id}_V$ (where $i : V_0 \hookrightarrow V$ is the inclusion). Thus V has the same homotopy type *as* V_0, which is a circle. So it is in our interest to take this V, where $\tilde{V}$ obtained from V by adding a final element e,

$$\tilde{V} = \left(\begin{array}{ccccc} & & & & b & & \\ & & & \nearrow & & \searrow & \\ 0 & \rightrightarrows & a & & & & e \\ & & & \searrow & & \nearrow & \\ & & & & c & & \end{array} \right),$$

as U', for the counterexamples on pages 6 and 7.

[page 10]

2.6. Counterexamples for total asphericity.

I have finally convinced myself that

$$\Psi = \left\{ \begin{array}{ccccc} & & & & b_0 & & \\ & & & \nearrow & & & \\ & & a_0 & & & & \\ & & & \searrow & & & \\ & & & & c_0 & & \end{array} \right\}$$

is *not* totally aspheric (for the localizer $\mathbf{W}_\infty$ – in the present number, we shall assume $W = \mathbf{W}_\infty$), or what amounts to the same thing, that we can have a diagram of type Ψ p. 7

$$\begin{array}{ccccc} & & & & b \\ & & & \overset{\beta}{\nearrow} & \\ a & & & & \\ & & & \underset{\gamma}{\searrow} & \\ & & & & c \end{array} \qquad (*)$$

in a category $Z \in \mathrm{Ob}\,\mathrm{Cat}$, whose direct limit $b \amalg_a c$ in $Z^\wedge$ is the final presheaf e_Z, i.e. such that we have

$$b(z) \amalg_{a(z)} c(z) \xrightarrow{\sim} \text{one-point set}$$

for every object z of Z (where $a(z) = \mathrm{Hom}_Z(z, a)$) without Z being aspheric for all that, nor even 1-connected ([5]). The simplest example I have found, is the "universal" example of a diagram $(*)$ where neither one nor the other of the two arrows $a \longrightarrow b$, $a \longrightarrow c$ is

[5] false – it is necessarily 1-connected ([6]), but not 2-connected.

[6] cf. comment XII, p. 101.

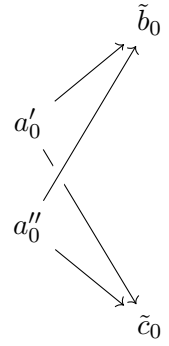
monomorphism (for if it is, then we can prove that Z is indeed aspheric, as we shall see later on). It is the category

$$Z = \left\{ \begin{array}{ccccc} & & & & b_0 \\ & & & \overset{\beta_0}{\nearrow} & \\ 0_0 & \underset{\mu_0}{\overset{\lambda_0}{\rightrightarrows}} & a_0 & & \\ & & & \underset{\gamma_0}{\searrow} & \\ & & & & c_0 \end{array} \right\}$$

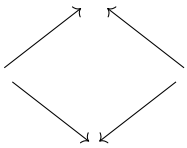
obtained from Ψ by adding to it an object 0_0, connected to a_0 by two distinct arrows λ_0, μ_0, such that [page 11]

$$\beta_0\lambda_0 = \beta_0\mu_0 \qquad \text{and} \qquad \gamma_0\lambda_0 = \gamma_0\mu_0,$$

these two composite arrows being, together with the four arrows β_0, γ_0, λ_0, μ_0 marked on the diagram, the only (six) nonidentity arrows of Z. Ψ immersed in Z as a closed subcategory, with complement the one-point category reduced to 0_0. (NB The closed immersion $\Psi \longrightarrow Z$ is not a Dwyer immersion, i.e. (as Z is the smallest open subcategory of Z containing Ψ) is not coaspheric, for the category $0\backslash\Psi$, cofibered over Ψ with discrete fibers, of cardinality 2 at a_0 and 1 at b_0 and c_0, is the category



i.e. p. 8



,

which has the homotopy type of the circle.) Nor the open immersion $\{0_0\} \longrightarrow Z$, for $\{0_0\}/a_0$ is the discrete category of cardinality 2.

The inclusion $i : Z_0 \longrightarrow Z$ of

$$Z_0 \;=\; (\; 0_0 \underset{\mu_0}{\overset{\lambda_0}{\rightrightarrows}} a_0 \;)$$

into Z is a homotopism ([7]) (for there exists an obvious retraction r of Z onto Z_0 ([7]), and an arrow $ir \longrightarrow \mathrm{id}_Z$), hence Z is homotopic to Z_0, which has the homotopy type of the circle, hence is not 1-connected.

On the other hand, the verification of [page 12]

$$b_0 \amalg_{a_0} c_0 \;=\; e_Z \qquad \text{in } Z^\wedge$$

[7] false, there is no retraction at all! In fact, Z is the *suspension* of the $\rightrightarrows$, which has the homotopy type (topological) of the circle, and it therefore has the homotopy type of the 2-sphere.

is immediate.

We are going to use this arrow

$$\Psi \longrightarrow Z$$

to construct interesting (and heartrending) examples of amalgamated sums

$$\begin{array}{ccc} U & \overset{i}{\hookrightarrow} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ U' & \overset{i'}{\hookrightarrow} & X', \end{array}$$

with i an open immersion, such that we even have

$$i:\ U \longrightarrow \overline{U} \qquad \text{is aspheric, hence in } W,$$

whereas nevertheless we shall have

$$i':\ U' \longrightarrow \overline{U'} \qquad \notin W,$$

and what is better (or worse),

$$p \in W \qquad \text{but} \qquad q \notin W$$

(and possibly, at the same time, $i \in W$ but $i' \notin W$). Moreover (unlike the constructions of propositions 10, 11 (p. 101-107)), the categories considered may all be finite. I am particularly interested in the case where X is a "square" p. 9

$$X = \Phi_n \times \Phi_m \qquad (n, m \geq 2),$$

and where U is its "boundary"

$$U = \dot{X} \overset{\text{def}}{=} (\dot{\Phi}_n \times \Phi_m) \cup (\Phi_n \times \dot{\Phi}_m).$$

[page 13]

$$\begin{array}{ccccccc} 0 & \longrightarrow & a & \longleftarrow & a' & \longrightarrow & \cdots \\ \downarrow & & \downarrow & & \downarrow & & \\ b & \longrightarrow & t_1 & \longleftarrow & \cdots & & \\ \uparrow & & \uparrow & & \uparrow & & \\ b' & \longrightarrow & \uparrow & \longleftarrow & \cdots & & \\ \downarrow & & & & & & \\ \cdots & & & & & & \end{array}$$

Proposition. *Let $n, m \geq 2$, and consider the inclusion $U = \partial X \overset{i}{\hookrightarrow} X$ of the boundary of the "square" $X = \Phi_n \times \Phi_m$ into X. It is a W-aspheric open immersion, to be precise, the U/t ($t \in \dot{U} = \bar{U} \backslash U$) are isomorphic to e (one-point category) or to* $\Phi = \begin{array}{c} \swarrow \\ \nwarrow \end{array}$ *(category with a final element), except for the four elements t_i of the four*

"corners" of $\dot{U}$, where it is the category

$$\tilde{\Psi} = \left(\begin{array}{ccccc} & & & & a' \\ & & & \nearrow & \downarrow \\ & & a & & \\ & \nearrow & & & \\ 0 & & & & \\ & \searrow & & & \\ & & b & \nwarrow & \\ & & & & \downarrow \\ & & & & b' \end{array} \right)$$

having no final element. Thus $U \xrightarrow{i} X$ is not an open Dwyer immersion (but $U \longrightarrow X\backslash\{t_1, t_2, t_3, t_4\}$ is one …). This being so, I say that $i : U \longrightarrow X$ is not a W-cofibration.

p. 10

Proof. Let

$$V \hookrightarrow U$$

be the inclusion into $U = \partial X$ of the corner-category $\begin{array}{ccc} 0 & \longrightarrow & a \\ \downarrow & & \\ b & & \end{array}$ (cf. figure), isomorphic to Ψ. One sees at once that this is a closed Dwyer immersion (the smallest open of U containing Ψ is just the category $\tilde{\Psi} = U/t_1$ from earlier), hence a W-cofibration. If $U \longrightarrow X$ were a W-cofibration, the same would hold for the composite $V \longrightarrow X$. We are going to see that this is not the case.

Let $\overline{V} = \begin{array}{ccc} 0 & \longrightarrow & a \\ \downarrow & & \downarrow \\ b & \longrightarrow & t_1 \end{array}$ be the closure of V in X, and consider the inclusion

[page 14]

$$\overline{V} \longrightarrow X,$$

I say that is a closed Dwyer immersion, i.e. that for every $s \in W\backslash\overline{V}$, where W is the smallest open of X containing $\overline{V}$, i.e.

$$W = \begin{array}{ccccc} 0 & \longrightarrow & a & \longleftarrow & a' \\ \downarrow & & \downarrow & & \downarrow \\ b & \longrightarrow & \downarrow & \longleftarrow & a'' \\ \uparrow & & \uparrow & & \uparrow \\ b' & \longrightarrow & b'' & \longleftarrow & c, \end{array}$$

the category $s\backslash\overline{V}$ has an initial object. (The categories $s\backslash\overline{V}$ that one obtains, from the five possible choices for s, are $\mathbf{\Delta}^1$ (if $s = a'$ or b') or e (for $s = a''$, b'' or c)). Thus $\overline{V} \longrightarrow X$ is a W-cofibration.

This being established, one knows that there exists, for the inclusion (open immersion)

$$V \hookrightarrow \overline{V} \qquad \text{i.e. } \Psi \hookrightarrow \mathbf{\Delta}^1 \times \mathbf{\Delta}^1,$$

a functor $V \longrightarrow V'$, giving rise to a cocartesian square

$$\begin{array}{ccc} V & \xhookrightarrow{i} & \overline{V} \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \overline{p}} \\ V' & \xhookrightarrow{i'} & \overline{V'} \end{array} \qquad (*)$$

with

$$i, p \in W \text{ (for } V,\ V',\ \overline{V}\ W\text{-aspheric), but } i', \overline{p} \notin W \text{ (i.e. } \overline{V'} \textit{ not } W\text{-aspheric).}$$

I recall that one first takes

p. 11

$$V_0' = \left(0' \underset{\mu}{\overset{\lambda}{\rightrightarrows}} 0 \begin{array}{l} \overset{\alpha}{\nearrow} a \\ \underset{\beta}{\searrow} b \end{array} \right) \qquad \text{with} \quad \begin{cases} \alpha\lambda = \alpha\mu \\ \beta\lambda = \beta\mu \end{cases}$$

and the "identical" closed immersion of V into V_0'

$$p_0 : V \longrightarrow V_0'$$

(which is such that

$$p_{0!}(e_V) = e_{V_0'} \quad),$$

[page 15]

then for V' one takes the category obtained by adding a final object to V_0'

$$V' = \left(0' \underset{\mu}{\overset{\lambda}{\rightrightarrows}} 0 \begin{array}{ccc} & \overset{\alpha}{\nearrow} & a & \overset{\rho}{\searrow} & \\ & & & & e \\ & \underset{\beta}{\searrow} & b & \underset{\sigma}{\nearrow} & \end{array} \right) \qquad \text{with} \quad \begin{cases} \alpha\lambda = \alpha\mu \\ \beta\lambda = \beta\mu \\ \rho\alpha = \sigma\beta \end{cases}$$

and for p the composite functor

$$V \xrightarrow{p_0} V_0' \xrightarrow{p_1} V', \qquad V \xrightarrow[p]{} V'$$

where p_1 is the inclusion (open). One then finds the square $(*)$, with $\overline{V'}$ not W-aspheric, or what amounts to the same thing, $i' \notin W$. For since $\overline{V'}\backslash V' \simeq \overline{V}\backslash V$ is the one-point category, $i' \in W$ is equivalent to V'/t W-aspheric ([8]) (cf. cor. 4, p. 149 in XII). Now by construction $V'/t \simeq V_0'$, which is not W-aspheric (in the case, say, where $W \subseteq W_1$, for $\pi_1(V_0') = 0$ and $\pi_2(V_0') \simeq \mathbf{Z}$).

Let us then consider the diagram of cocartesian squares

$$\begin{array}{ccccc} V & \xhookrightarrow{i} & \overline{V} & \hookrightarrow & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \overline{p}} & & \downarrow{\scriptstyle q} \\ V' & \xhookrightarrow{i'} & \overline{V'} & \hookrightarrow & X' \end{array} \quad .$$

[page 16]

Since $\overline{V} \hookrightarrow X$ is closed Dwyer immersion, the square on the right is W-cocartesian, hence $(\overline{p} \in W) \iff (q \in W)$, hence (since $\overline{p} \notin W$) we have

$$p \in W, \qquad \text{but} \qquad q \notin W,$$

p. 12

which indeed shows that $V \longrightarrow X$ is not a W-cofibration. Factoring $V \longrightarrow X$ through U (instead of $\overline{V}$), one finds

$$\begin{array}{ccccc} V & \longrightarrow & U & \longrightarrow & X \\ {\scriptstyle p\in W}\downarrow & & \downarrow{\scriptstyle q_U \in W} & & \downarrow{\scriptstyle q\notin W} \\ V' & \longrightarrow & U' & \longrightarrow & X' \end{array} \quad ,$$

[8] assuming (W 7 bis).

where $q_U \in W$ since $p \in W$ and $V \longrightarrow U$ is Dwyer immersion. This makes explicit that $U \longrightarrow X$ is not a W-cofibration, since

$$q_U \in W, \qquad \text{but} \qquad q \notin W,$$

q.e.d.

Gluing in Cat **of an open and a complementary closed.** [page 17]

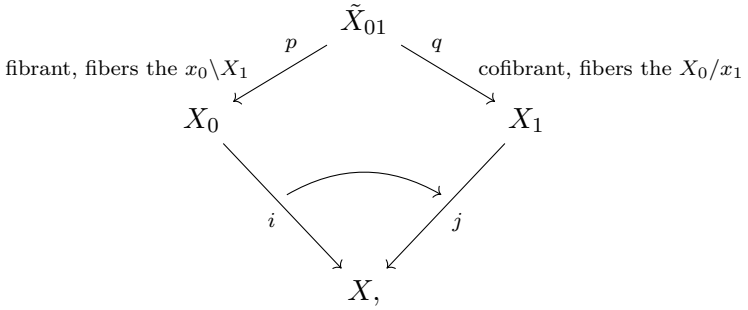
$$X_0 \underbrace{\hookrightarrow}_{\text{open}} X \underbrace{\hookleftarrow}_{\substack{\text{closed}\\ \text{comple-}\\ \text{mentary}}} X_1$$

in Cat,

$$\tilde{X}_{01} \subseteq \mathrm{Fl}(X)$$

be full subcategory of arrows with source in X_0, target in X_1.

Question. *This diagram*

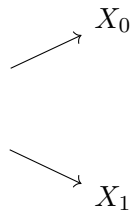


(∗)

is it W-cocartesian? p. 13

OK if i W-aspheric or j W-coaspheric, i.e. $X \longrightarrow \mathbf{\Delta}^1$ smooth or proper – this is a case where one has either $X_0 \xrightarrow{W} X$, or $X_1 \xrightarrow{W} X$. But even assuming i or j in W, and moreover X aspheric (hence X_0 and X aspheric, or X_1 and X aspheric), and $W = \mathbf{W}_\infty$, I have not been able to prove that the diagram is cocartesian, i.e. $q \in W$ resp. $p \in W$ ([9]). (Nor if $X_0 \simeq e$, X, X_1 W-aspheric $\Longrightarrow e\backslash X_1$ W-aspheric, i.e. $X_1 \longrightarrow X$ W-coaspheric ...) But I have at least proved, in all cases, that

$$\mathcal{X} \overset{\text{def}}{=} \int \tilde{X}_{01} \longrightarrow X \qquad (**)$$



is in $\mathbf{W}_1$.

Something *false* (unless i or j in $W = \mathbf{W}_\infty$) if one replaces $\tilde{X}_{01}$ by $\tilde{X}'_{01}$, formed of the triples (x_0, x_1, c), with $x_0 \in \mathrm{Ob}\, X_0$, $x_1 \in \mathrm{Ob}\, X_1$, $c : x_0 \dashrightarrow x_1$ a *path* from x_0 to x_1 in X! We have therefore a "topological" situation entirely peculiar to the category of models Cat, which does not seem to have an analog in categories such as Esp or $\mathbf{\Delta}^\wedge$, where however the notions of open and complementary closed (with a grain of salt in $\mathbf{\Delta}^\wedge$) have a meaning. [page 18]

[9] I assume that W satisfies Loc (6).

The quasi-trivial categorical argument which establishes that $(**)$ is in $\mathbf{W}_1$, i.e. induces an equivalence of categories between $\mathrm{Rev}(X)$ and

$$\mathrm{Rev}(\mathcal{X}) \;\simeq\; \mathrm{Rev}(X_0) \overset{2}{\times}_{\mathrm{Rev}(\tilde{X}_{01})} \mathrm{Rev}(X_1),$$

seems to me to have to extend to higher dimensions, by dint of n-gerbes on X – but the foundations for the theory of n-gerbes are lacking. (And I have not had the courage to spell out the argument even for $n = 2$ (where however GIRAUD provides what is needed) ([10]).) So I conjecture that φ is at the very least in $\mathbf{W}_\infty$, i.e. that the diagram $(*)$ is W-cocartesian if $W \supseteq \mathbf{W}_\infty$ (and satisfies Loc (6 a′)).

[10] But I have nevertheless made a sketch of a verification for $n = 2$, which convinces me that we do indeed have $\varphi \in \mathbf{W}_2$. And I have hardly any doubt that we even have $\varphi \in \mathbf{W}_\infty$.

p. –

THE DERIVATORS

ALEXANDRE GROTHENDIECK

Chapter XIII

Categories of models (1)

p. –
p. 1
[page 1]

1 Quillen couples (Φ, Ψ)

I want first to give a weakened variant of the notion of a Quillen triple (Quillen's "model category"). This will not be an entirely self-dual notion – thus there will be *two* weakened variants, dual to one another.

But even before that, a sorites on the LLP (left lifting property) and RLP (right lifting property) ([1]). Let $\mathcal{M}$ be any category (most often, $\mathcal{M}$ will be large). We consider subsets

$$\Phi,\ \Psi\ \subset\ \mathrm{Fl}(\mathcal{M})$$

and we define maps

$$\boxed{\begin{array}{ccc}\Phi & \longmapsto & \Phi_* \\ \Psi & \longmapsto & \Psi^*\end{array}} \qquad \mathrm{Fl}(\mathcal{M}) \longrightarrow \mathrm{Fl}(\mathcal{M})$$

by

$$\begin{aligned}\Phi_* &= \{\text{set of } p \in \mathrm{Fl}(\mathcal{M}) \text{ that have the RLP with respect to } \Phi\} \\ \Psi^* &= \{\text{set of } i \in \mathrm{Fl}(\mathcal{M}) \text{ that have the LLP with respect to } \Psi\},\end{aligned}$$

and then

$$\boxed{\begin{array}{ccc}\Phi & \longmapsto & \widetilde{\Phi} \\ \Psi & \longmapsto & \overline{\Psi}\end{array}} \qquad \mathrm{Fl}(\mathcal{M}) \longrightarrow \mathrm{Fl}(\mathcal{M})$$

[1] give formal definition, with reference to Quillen.

by

$$\widetilde{\Phi} \;=\; (\Phi_*)^*$$
$$\overline{\Psi} \;=\; (\Psi^*)_* \;.$$

I summarize the most immediate properties of these operations in a [page 2]
Proposition 1.

a) $\Phi \longmapsto \Phi_*$, $\Psi \longmapsto \Psi^*$ *are decreasing maps from* $\mathrm{Fl}(\mathcal{M})$ *to itself, for the order relation* $\subset$, *and* $\Phi \longmapsto \widetilde{\Phi}$, $\Psi \longmapsto \overline{\Psi}$ *increasing maps.*

p. 2

b) *We have*

$$\Phi \;\subset\; \widetilde{\Phi},$$

and for $\Phi = \widetilde{\Phi}$ *to hold, it is necessary and sufficient that* Φ *be of the form* Ψ^*. *We then say that* Φ *is* left closed (^{2}). *The set of subsets of* $\mathrm{Fl}(\mathcal{M})$ *that are left closed is stable under arbitrary intersections* (^{3}), *and* $\widetilde{\Phi}$ *is the smallest closed subset of* $\mathrm{Fl}(\mathcal{M})$ *that contains* Φ *and the largest subset* Φ' *of* $\mathrm{Fl}(\mathcal{M})$ *such that* $\Phi'_* = \Phi_*$ *(or such that* $\Phi'_* \supset \Phi_*$*).*

Symmetric statements for Ψ*: we have*

$$\Psi \;\subset\; \overline{\Psi},$$

we have $\Psi = \overline{\Psi}$ *if and only if* Ψ *of the form* Φ_*, *we then say that* Ψ *is* right closed. *The set of subsets of* $\mathrm{Fl}(\mathcal{M})$ *that are right closed is stable under arbitrary intersections* (^{4}), *and* $\overline{\Psi}$ *is the smallest right closed subset containing* Ψ, *and also the largest subset* Ψ' *of* $\mathrm{Fl}(\mathcal{M})$ *such that* $\Psi'^* = \Psi^*$ *(or such that* $\Psi'^* \supset \Psi^*$*).*

[page 3]

c) *The maps*

$$\Phi \;\longmapsto\; \Phi_*, \qquad\qquad \Psi \;\longmapsto\; \Psi^*$$

establish mutually inverse bijections between the set of left closed subsets Φ *of* $\mathrm{Fl}(\mathcal{M})$, *and the set of right closed subsets* Ψ. *This is an anti-isomorphism of ordered sets. We then have*

$$(\Phi \cap \Phi')_* \;=\; \begin{array}{l} \textit{right closed subset of } \mathrm{Fl}(\mathcal{M}) \textit{ ``generated by } \Phi_*,\ \Phi'_*\textit{''},\\ \text{i.e. } \textit{smallest right closed subset containing } \Phi_*,\\ \Phi'_*, \text{ i.e. } \overline{\Phi_* \cup \Phi'_*} \end{array}$$

, more generally

$$(\bigcap_i \Phi_i)_* \;=\; \overline{\bigcup_i (\Phi_{i*})}$$

for every family $(\Phi_i)_{i\in I}$ *of left closed subsets of* $\mathrm{Fl}(\mathcal{M})$. *Dual formulas*

$$\begin{cases} (\Psi \cap \Psi')^* \;=\; \widetilde{\Psi^* \cup \Psi'^*} \\ (\bigcap_j \Psi_j)^* \;=\; \widetilde{\bigcup_j (\Psi_j^*)} \;. \end{cases}$$

d) *If* $\Phi, \Psi \subset \mathrm{Fl}(\mathcal{M})$, *then we have*

$$\Phi \subset \Psi^* \iff \Psi \subset \Phi_*, \qquad\qquad \textit{we then write } \Phi \longleftrightarrow \Psi,$$

p. 3

this means that Φ *has the LLP for* Ψ, *or that* Ψ *has the RLP for* Φ. [page 4]

Proposition 2. *Let Φ be a left closed subset of* $\mathrm{Fl}(\mathcal{M})$*. Then Φ satisfies the conditions* a) b) c) *below:*

a) Φ *contains the isomorphisms, and is stable under composition, and under cobase change (*i.e. *if we have* $\begin{array}{ccc} A & \xrightarrow{i} & B \\ \downarrow & & \downarrow \\ A' & \xrightarrow{i'} & B' \end{array}$ *a cocartesian square in $\mathcal{M}$, with $i \in \Phi$, we have $i' \in \Phi$).*

b) Φ *stable under direct factors (as a subset of the set of objects of* $\underline{\mathrm{Fl}}(\mathcal{M})$*).*

c) Φ *"stable under ordinal direct limits", in the following sense: Let I be a well-ordered set, $(A_i)_{i\in I}$ a direct system in $\mathcal{M}$, such that the $u_{ji} : A_i \longrightarrow A_j$ are in Φ, and $A_{i_0} \xleftarrow{\sim} \varinjlim_{i<i_0} A_i$ if $i_0 \in I$ is a limit element. Suppose that $A = \varinjlim_I A_i$ exists. Then $\forall\ i_0 \in I$, the canonical arrow $A_{i_0} \longrightarrow A$ is in Φ.*

There is a dual statement for a right closed subset Ψ; it contains the isomorphisms, is stable under composition and under base change, under direct factors, and finally stable under ordinal inverse limits.

We will have a partial converse in accessible categories. It is for the statement of this converse that I have set apart condition *b*) of stability under direct factors, instead of lumping it together with *a*). [page 5]

Definition 1.

a) (For the record.) RLP, LLP, the relation $\Phi \longleftrightarrow \Psi$ between subsets of $\mathrm{Fl}(\mathcal{M})$ (nonsymmetric relation!).

b) (For the record.) Subset of $\mathrm{Fl}(\mathcal{M})$ left closed, right closed.

c) Let $\Phi \longleftrightarrow \Psi$. We say that the pair (Φ, Ψ) has the *Quillen factorization property*, if *every* arrow f of $\mathcal{M}$ factors as

$$f = pi, \qquad i \in \Phi,\ p \in \Psi.$$

We say that (Φ, Ψ) is a *Quillen couple* if it is a pair associated by properties LLP and RLP ($\Phi \longleftrightarrow \Psi$), and they are closed (*i.e.* $\Phi = \Psi^*$, $\Psi = \Phi_*$), and if they have the Quillen factorization property.

I state right now the essential technical result, ensuring the existence of many Quillen couples:

Theorem 1. *Suppose that $\mathcal{M}$ is* accessible, *and stable under (small) direct limits. (There should suffice stability under amalgamated sums, and under filtering* $\varinjlim$*, or even ordinal …). Let Φ_0 be a* small *subset of* $\mathrm{Fl}(\mathcal{M})$*, $\Psi = (\Phi_0)_*$, and Φ a subset of* $\mathrm{Fl}(\mathcal{M})$ *such that*

$$\Phi_0 \ \subset\ \Phi\ \subset \Psi^*\ (= \widetilde{\Phi}_0).$$

p. 4 [page 6]

Then:

[2] Give formal definition.

[3] more precisely, $\bigcap_i (\Psi_i^*) = (\bigcup_i \Psi_i)^*$

[4] more precisely, we have $\bigcap_i (\Phi_{i\,*}) = (\bigcup_i \Phi_i)_*$

a) *For Φ to be left closed,* i.e. $\Phi = \widetilde{\Phi}_0$*, it is necessary and sufficient that it satisfy conditions* a) b) c) *of proposition 2. When this is so, the couple* (Φ, Ψ) *is a Quillen couple.*

b) *Suppose only that Φ satisfies conditions* a), c) *of proposition 2 (excluding condition* b) *on direct factors) – thus we cannot assert that Φ is left closed. Nevertheless,* $(\Phi, \underbrace{\Psi}_{=\Phi_*})$ *satisfies the Quillen factorization property.*

Remark. In the proof, the existence of $\Phi_0 \subset \Phi$, which plays the role of a small set of "generators" (on the left) for Φ, plays an essential role. It would be worth developing the notion of *accessible subsets* of $\mathrm{Fl}(\mathcal{M})$, which would imply the existence of a small set $\Phi_0 \subset \Phi$ such that $\Phi \subset \widetilde{\Phi}_0$, and which would have the property that [page 7]

$$\begin{array}{lcl} \Phi \text{ accessible} & \Longrightarrow & \Phi_* \text{ accessible} \\ \Psi \text{ accessible} & \Longrightarrow & \Psi^* \text{ accessible.} \end{array}$$

Any couple (Φ, Ψ) of associated closed subsets ($\Phi = \Psi^*$, $\Psi = \Phi_*$) and accessible, would thus be a Quillen couple.

2 Weak Quillen categories, and construction thereof (Conjectural theorem-scholie)

Throughout what follows, we give ourselves a *category of models*

$$\begin{cases} (\mathcal{M}, W), \text{ where } W \subset \mathrm{Fl}(\mathcal{M}) \\ \\ W \textit{ weakly saturated } ([5]): \quad \begin{array}{ll} a) & W \text{ contains the isomorphisms.} \\ b) & \text{If two among } f,\ g,\ gf \text{ in } W, \text{ the third also.} \end{array} \end{cases}$$

We want to describe additional data a little weaker than those of a Quillen model category. We suppose given subsets p. 5

$$C,\ F,\ TC,\ TF\ \subset\ \mathrm{Fl}(\mathcal{M}),$$

with the following properties. As a preliminary axiom, I lay down

M 0 Stability of $\mathcal{M}$ under finite $\varprojlim$ and finite $\varinjlim$.

M 1 Lifting-extension property [page 8]

$$C \longleftrightarrow TF, \qquad TC \longleftrightarrow F.$$

M 2 Factorization property for (C, TF), (TC, F).

M 3 Stability of F under composition and under base change, of C under composition and cobase change, F and C contain the isomorphisms.

M 4 Stability of TF under composition and base change, of TC under composition and cobase change, and TF and TC contain the isomorphisms.

[5] This is axiom M 5 of Quillen.

M′ $TC \subset C \cap W$, $TF \subset F \cap W$.

M″ $C \subset \underbrace{\mathrm{Cof}_W}_{W\text{-cofibrations}}$, $F \subset \underbrace{\mathrm{Fib}_W}_{W\text{-fibrations}}$.

In the case of Quillen (Q-*categories of models*), M′ is strengthened to

$$TC = C \cap W, \qquad TF = F \cap W,$$

and there is no axiom M″. Axiom M″ is equivalent to the property of "propriety", corresponding to Q-categories of models that are *proper.*

Proposition 3. *Suppose conditions* M 0 *(for the record),* M 1, M 2, M′, M″ *are satisfied. Then replacing* C, TC *by the saturations for the stability properties of containing the isomorphisms, of being stable under composition and under cobase change, and* F, TF *by those of containing the isomorphisms and of being stable under composition and under base change, we find* C', TC', F', TF' *satisfying all the axioms* M 1 to M 4, M′, M″. p. 6 [page 9]

Proof. Proposition 2 assures us that M 1 continues to be satisfied, and for M 2 this is trivial. M 3 and M 4 will hold by construction. On the other hand, M″ remains valid, since Cof_W has the stabilities required for C', and Fib_W those required for F'. It remains to prove M′, *i.e.*

$$TC' \subset C' \cap W, \qquad TF' \subset F' \cap W,$$

and for this it suffices to see that $C' \cap W$ resp. $F' \cap W$ has the stabilities required for TC' resp. TF'. This follows from the

Lemma. *Let* C *be a subset of* Cof_W. *If* C *contains the isomorphisms, the same is true of* $C \cap W$. *If moreover* C *is stable under composition resp. under cobase change, the same is true* $C \cap W$. *Dual statement for a subset* F *of* Fib_W.

This comes from the fact that

$$\boxed{\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}\ (^6)} \overset{\mathrm{def}}{=} \begin{array}{l}\text{set of arrows } A \longrightarrow B \text{ such that for every} \\ A \longrightarrow A',\ A' \longrightarrow B' = A' \vee_A B \ \text{ is in } W.\end{array}$$

is itself stable under composition and under cobase change, and of course we have, if $C \subset \mathrm{Cof}_W$,

$$C \cap W \ = \ C \cap (\mathrm{Cof}_W \cap W) \ = \ C \cap W^{\mathrm{univ}}.$$

So it remains to prove the

Corollary 1. *Suppose there exist subsets* C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$ *satisfying* M 2, M′ ([7]), M″. *Then* [page 10]

$$\begin{aligned}\mathrm{Cof}_W \cap W &= W^{\mathrm{univ}} \\ \mathrm{Fib}_W \cap W &= W_{\mathrm{univ}}\ (^8).\end{aligned}$$

To see this, let us show e.g. the second (the first will follow from it by duality).

$$\begin{array}{ccc} X & \longleftarrow & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \longleftarrow & Y' \end{array} \qquad\qquad f \in \mathrm{Fib}_W,\ f \in W \Longrightarrow f' \in W.$$

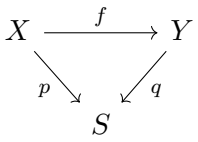
[6] arrows *couniversally* in W, or W*-equivalences that are couniversal.*

[7] We do not need all of M′, only $TC \subset W$, $TF \subset W$. p. 7

[8] Arrows $f : X \longrightarrow Y$ *universally* in W, *i.e.* such that for every $Y' \longrightarrow Y$, $f' : X' = X \times_Y Y' \longrightarrow Y'$ is in W.

We have better:

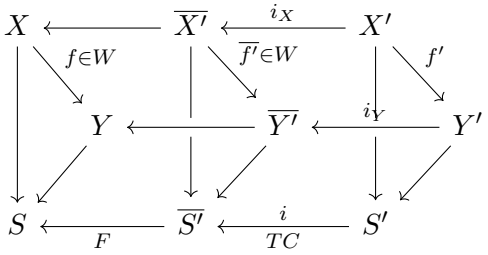
Proposition 4. *Suppose that every arrow of $\mathcal{M}$ factors as pi, with $i \in W$ and $p \in \mathrm{Fib}_W$. Then, if X, Y are in $\underline{\mathrm{Fib}}_W S \subset \mathcal{M}/S$, then every $f : X \longrightarrow Y$ in $\mathcal{M}/S$ which is in W_S is in W_S^{u},* i.e. *it remains in W after any base change $S' \longrightarrow S$.*



$p, q \in \mathrm{Fib}_W$

Dual statement for the arrows of $S\backslash\underline{\mathrm{Cof}}_W \subset S\backslash\mathcal{M}$, such an arrow which is in W is S-couniversally in W, if every arrow of $\mathcal{M}$ factors as qj, with $j \in \mathrm{Cof}_W$ and $q \in W$.

PROOF. [page 11]



We factor $S' \longrightarrow S$ as $S' \xrightarrow{i} \overline{S'} \xrightarrow{p} S$, with $i \in W$, $p \in \mathrm{Fib}_W$, by M 2. Thus $f \in W \Longrightarrow \overline{f'} \in W$. Since $X, Y \in \mathrm{Ob}\,\underline{\mathrm{Fib}}_W S$, we have $\overline{X'}, \overline{Y'} \in \mathrm{Ob}\,\underline{\mathrm{Fib}}_W \overline{S'}$, and since $i : S' \longrightarrow \overline{S'}$ is in W, hence also i_X, i_Y, we have $f' \in W$ by M 5 (weak saturation of W).

Comment on Proposition 3. We can be even more demanding, by also requiring in M 3, M 4 stability under direct factors, and (when only M 1, M 2, M′, M″ are satisfied) by saturating C, TC, F, TF also for this additional stability. Proposition 2 again ensures that M 1 continues to hold (and M 2, M 3, M 4 trivially), we see that this is still so for M′, M″ provided we strengthen the weak saturation axiom M 5 on W by

stability of W under direct factors,

which indeed implies that Cof_W, Fib_W are stable under direct factors, as are W^{univ} and W_{univ}.

It does not seem that axioms M 0 to M 4, plus M′ and M″ suffice to develop the exact sequences of suspension and cosuspension. A tiny little bit more is needed: [page 12]

Definition 2. I call a *left pre-Q-category of models,* the data of $\mathcal{M}$ and of W, C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$, satisfying axioms M 0 to M 5, M′, M″, plus the strengthened form of the first inclusion in M′ p. 8

$$\mathrm{M}'_0\ (\mathrm{g}) \qquad\qquad TC = C \cap W \qquad\qquad (\text{not only } TC \subset C \cap W).$$

Likewise, we define the *right pre-Q-category of models,* by the condition

$$\mathrm{M}'_0\ (\mathrm{d}) \qquad\qquad TF = F \cap W.$$

NB When the two conditions are satisfied at once, then the data of TC, TF already follow from those of W, C, F, and we simply have a "proper" Q-category of models (in the sense of QUILLEN).

Scholie. For me, the introduction of C, TC, F, TF is purely auxiliary – what matters to me are the properties of W. Thus, the existence of C, TC, F, TF satisfying the weak axioms M 0 to M 4, M′, M″, since it gives the factorizations spoken of in proposition 4, entails the remarkable property of W stated in that proposition, and in the corollary [page 13]

which precedes it. Likewise, these factorization properties ([9]) ensure the existence of a good theory of W-homotopic fibers and cofibers without any need for reference in this theory to C, TC, F, TF (whose introduction gives at most some small supplements …). But I do not see how, from the factorization hypotheses alone, to prove the crucial exactness property for W-cartesian, and W-cocartesian squares, necessary to be able to develop the *exact* sequence of cosuspension and of suspension. It is to get this exactness that the auxiliary, and transitory, introduction of the paraphernalia (C, TC, F, TF) is useful, the additional axiom on the left $TC = C \cap W$ ensuring (unless I am mistaken) the exactness of W-cartesian squares, and the axiom on the right $TF = F \cap W$ that of W-cocartesian squares. (Something to be checked with more care than I have done so far.) Once this is obtained, and possibly likewise for the $(\mathcal{M}(I), W(I))$ for every diagram-category I, we can totally forget these auxiliary sets – I feel like forgetting them [page 14] in any case.

My hope is that for a given W, even if we do not manage to find an associated Quillen triple structure (W, C, F) (e.g. because there might not be one), we can find both quintuples (W, C, F, TC, TF) which are left pre-Quillen, and quintuples (*other* quintuples!) which are right pre-Quillen.

Here is the kind of statement in proper form, for the moment conjectural or even entirely heuristic, that I have in mind:

Conjectural theorem-scholie: *We suppose that $\mathcal{M}$ is stable under "arbitrary" $\varinjlim$, and under finite $\varprojlim$, and that W is stable under filtering $\varinjlim$ (in addition to weak saturation).*

p. 9

① *Then* Cof_W *and* W^{univ} *have the stability properties* a) *and* c) *of proposition* 2 (p. 4), *and also the stability property* b) *under direct factors, provided that W itself has it.*

② *Suppose* $\mathcal{M}$ accessible, *so that we have theorem* 1. *Equivalent conditions on W :*

[page 15]

a) *There is a small subset C_0 of* Cof_W, *and TC_0 of* W^{univ}, *such that*

$$TF \overset{\mathrm{def}}{=} C_{0*} \subset W, \qquad (TC_0)_* \overset{\mathrm{def}}{=} F \subset \mathrm{Fib}_W.$$

a′) *There is a quadruple (C, TC, F, TF) of subsets of* $\mathrm{Fl}(\mathcal{M})$, *satisfying conditions* M 1 *to* M 4, M′, M″ *on page* 6 ([10]).

a″) *As in* a′), *assuming* M′ *in strengthened form with* M'_0 (g), i.e. *we have a left pre-Q-category of models structure ($TC = C \cap W$).*

b) *We have*

$$(\mathrm{Cof}_W)_* \subset W, \qquad (W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W$$

(thus $(\mathrm{Cof}_W)_* \subset \mathrm{Fib}_W \cap W$ *since* $(\mathrm{Cof}_W)_* \subset (W^{\mathrm{univ}})_*$ ([11])*).*

b′) *We have*

$$(\mathrm{Cof}_W)_* \subset W_{\mathrm{univ}}, \qquad (W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W.$$

③ *For there to be a quadruple which, with W, defines a right pre-Q-category of models ($TF = F \cap W$)* ([12]), *it is necessary and sufficient that we have* b′) *above, as well as*

$$(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} \quad \textit{is right closed.}$$

[9] in a slightly strengthened form it is true – a *functorial* factorization for f is needed … [page 16]

[10] with C, TC left closed, F, TF right closed.

[11] $W^{\mathrm{univ}} \subset \mathrm{Cof}_W$.

[12] We are still assuming $\mathcal{M}$ accessible in 3° and 4°, with in addition F, TF (and C, TC) right closed (resp. left closed).

④ *The preliminary assumptions on $(\mathcal{M}, W)$, including accessibility, as well as the equivalent conditions in* ②*, are stable under passage from $(\mathcal{M}, W)$ to $(\mathcal{M}(I), W(I))$. But what about the condition in* ③ *? Is it a consequence of the conditions in* ② *(*i.e. *is it equivalent to them) ?*

I am going to sketch the proof of the theorem-scholie, to pin down the technical work that remains to be done.

1°) Is routine verification.

2°) Here are the ± tautological implications between the conditions under consideration :

$$a'' \Longrightarrow a' \overset{?}{\Longrightarrow} a \Longrightarrow b \Longleftarrow b', \qquad (*)$$

but I see that $a' \Longrightarrow a$ is not at all tautological, and perhaps even false after all ! ([13]). We would have to assume that in quadruple (C, F, TC, TF), F and TF are right closed, and more precisely, that $F = (TC)_*$, $TF = C_*$, which will be the case e.g. if we have a "closed" Q-category of models structure ("closed model category"), in the sense of Quillen. The other three implications in $(*)$ really are tautological. p. 10

$a \Longrightarrow a'$. We can in a) assume

$$TC_0 \ \subset \ C_0$$

(replacing C_0 by $C_0 \cup TC_0$ if necessary), then, thanks to 1°) (W^{univ} and Cof_W are left closed), setting

$$\widetilde{TC_0} = TC, \qquad \widetilde{C_0} = C, \qquad \text{[page 17]}$$

we will have $TC \subset W^{\mathrm{univ}}$, $C \subset \mathrm{Cof}_W$. Passing from TC_0 to TC, from C_0 to C, we have lost only the fact that these subsets of $\mathrm{Fl}(\mathcal{M})$ are small. But by theorem 1, setting

$$F \ = \ (TC)_*, \qquad TF \ = \ C_*,$$

the pairs of closed subsets (left – right)

$$(C,\, TF), \qquad (TC,\, F)$$

satisfy the Quillen factorization condition. We will then have properties M 1 to M 4, plus the inclusions $TC \ \subset \ C \cap W$ (since $TC \ \subset \ W^{\mathrm{univ}} \ \subset \ W$, and $TC \ \subset \ C$ by construction), and $C \subset \mathrm{Cof}_W$ of M′ and M″. It remains to check $TF \subset F \cap W$, $F \subset \mathrm{Fib}_W$, but $F \ (= (TC)_*) \ \subset \mathrm{Fib}_W$ by assumption, and $TF \ (= C_*) \ \subset W$ too ; on the other hand $TF \subset F$ follows from $TC \subset C$.

$a' \Longrightarrow a''$. We thus want, in addition to conditions M 1 to M 4 and M′, M″, the condition

$$TC \ = \ C \cap W \ (= C \cap W^{\mathrm{univ}}).$$

The idea is to replace the pair

$$C,\ TC \qquad (C \subset \mathrm{Cof}_W,\ TC \subset C \cap W),$$

giving rise to

$$F = (TC)_*, \qquad TF = C_*, \qquad \text{[page 18]}$$

by

$$C,\ C \cap W \ (\supset TC),$$

which does not change the pair (C, TF), but does change the pair (TC, F), replaced by

$$\underbrace{(TC)'}_{\supset TC} \ = \ \underbrace{C \cap W}_{= C \cap W^{\mathrm{univ}}}, \qquad F' \ = \ \underbrace{(TC')_*}_{\subset F}$$

(we have replaced F by a smaller F') ([14]). The only problem is whether (TC', F') still

[13] That is why I modified it, strengthening a') by . p. 11

[14] NB TC' will still be left closed as an intersection of two left closed subsets.

satisfies the Quillen factorization condition. These are closed subsets (left for TC', right for F') "dual" to one another,

$$TC' = F'^{*}, \qquad F' = TC'_{*} ,$$

but since we do not know whether TC' is of the form $\widetilde{TC'_0}$, with TC'_0 small, we are not sure that the conclusion of theorem 1 applies. But if we assume in a') that C is an *accessible* subset of $\mathrm{Fl}(\mathcal{M})$ (*i.e.*, already being left closed, that $C = \widetilde{C_0}$, C_0 small), then [page 19]
I presume that we will be able to prove that the same is true of $C \cap W$ (W itself being accessible, thanks to the condition of stability under small filtering $\varinjlim$), so theorem 1 really does apply to $TC' = C \cap W$. But if we keep a') as it stands, then we would probably have to prove that a') *implies* a') in the stronger form, with C *accessible.* But since we have $a' \Longrightarrow a$, and we have just established that $a \Longrightarrow a'$ in this stronger form, we win (modulo the conjectural fact assumed about the accessible subsets of $\mathrm{Fl}(\mathcal{M})$).

It thus remains to prove

$$b \implies a, \qquad b \implies b',$$

and it suffices to prove the first implication, since we know that $a) \Longrightarrow a')$, and that a') implies that

$$\mathrm{Fib}_W \cap W \;=\; W_{\mathrm{univ}}$$

(cor. 1, page 10), so $(\mathrm{Cof}_W)_* \subset W$ and $(W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W$ (*i.e.* b)) imply, since $(\mathrm{Cof}_W)_* \subset (W^{\mathrm{univ}})_*$ (because $W^{\mathrm{univ}} \subset \mathrm{Cof}_W$), $(\mathrm{Cof}_W)_* \subset W \cap \mathrm{Fib}_W = W_{\mathrm{univ}}$, which is the part of b') to be established (the other $(W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W$ being tautologically contained in b).

Let us now note that b) is nothing other than the weakened variant of a), when we [page 20]
omit specifying that C_0, TC_0 must be small. So the only problem, for $b \Longrightarrow a$, is to see this :

Conjectural lemma ([15]).

① *If we have a left closed subset* Φ *of* $\mathrm{Fl}(\mathcal{M})$*, such that* $\Phi_* \subset \mathrm{Fib}_W$ *(resp.* $\Phi_* \subset W$*), then there is a subset* of small size $\Phi_0 \subset \Phi$*, such that* $\Phi_{0*} \subset \mathrm{Fib}_W$ *(resp.* $(\Phi_0)_* \subset W$*).*

② *If* C*,* C' *are two left closed subsets of* $\mathrm{Fl}(\mathcal{M})$*, if* C *and* C' *are accessible, the same is true of* $C \cap C'$*.*

I believe this really is true under the assumptions made on $\mathcal{M}$, W – which must p. 12
imply that in a suitable sense, W is an accessible subset of the set $\mathrm{Fl}(\mathcal{M})$ of arrows of the accessible category $\mathcal{M}$. It should follow from this that a subset of $\mathcal{M}$ constructed explicitly in terms of W, such as Fib_W, is itself accessible, and the lemma should come out of that.

Thus, 2° is proved, modulo this accessibility lemma, which I will have to prove ± at the same time as theorem 1 (which I have only stated here).

Let us prove 3°. First suppose there is a "closed" quadruple (C, TC, F, TF), fitting [page 21]
into a "closed Quillen pre-quintuple" which is a right Q-category of models, *i.e.*

$$TF \;=\; F \cap W \;=\; F \cap W_{\mathrm{univ}}.$$

Let us then prove

$$\underbrace{(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}}_{A_W} \qquad \text{is right closed.} \qquad (*)$$

[15] This lemma looks a bit false, but it should be possible to fix that …

Indeed, we have

$$TC \ \subset \ W^{\mathrm{univ}},$$

hence

$$(W^{\mathrm{univ}})_* \ \subset \ (TC)_* \ = \ F,$$

so

$$A = (W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} \ \subset \ F \cap W_{\mathrm{univ}} \overset{\text{(by hyp.)}}{=} \ TF,$$

now TF is right closed and contained in W_{univ}, hence

$$\overline{A} \ \subset \ TF \ \subset \ W_{\mathrm{univ}},$$

and since $(W^{\mathrm{univ}})_*$ is right closed, we also have

$$\overline{A} \ \subset \ (W^{\mathrm{univ}})_*,$$

hence $\overline{A} \subset W_{\mathrm{univ}} \cap (W^{\mathrm{univ}})_* = A$, so A is right closed, q.e.d.

Let us prove the converse, *i.e.* suppose the conditions of 2° are satisfied, *plus* condition $(*)$ above. We must construct the quadruple.

Let us first note, to get our bearings, that the relations b') of 2°), and the transposed relations (taking into account that Cof_W and W^{univ} are left closed, by 1°), namely

$$\begin{cases} (\mathrm{Cof}_W)_* \ \subset \ W_{\mathrm{univ}} \ , \ (W^{\mathrm{univ}})_* \ \subset \ \mathrm{Fib}_W \\ (W_{\mathrm{univ}})^* \ \subset \ \mathrm{Cof}_W \ , \ (\mathrm{Fib}_W)^* \ \subset \ W^{\mathrm{univ}}, \end{cases}$$

[page 22]

giving ([16]) the two square diagrams of inclusions

$$\begin{array}{ccccc} & & W^{\mathrm{univ}} & & \\ & \nearrow & & \searrow & \\ (\mathrm{Fib}_W)^* & & & & \mathrm{Cof}_W \\ & \searrow & & \nearrow & \\ & & (W_{\mathrm{univ}})^* & & \end{array} \qquad ([17]) \qquad \text{(I)} \quad \text{p. 13}$$

and

$$\begin{array}{ccccc} & & (W^{\mathrm{univ}})_* & & \\ & \swarrow & & \leftarrow & \\ \mathrm{Fib}_W & & & & (\mathrm{Cof}_W)_* \ , \\ & \leftarrow^{\alpha} & & \swarrow & \\ & & W_{\mathrm{univ}} & & \end{array} \qquad \text{(II)}$$

of which the first is the transpose (under $\Phi \longmapsto \Phi^*$) of the second, and where the 8 sets of arrows considered, except for Fib_W and W_{univ} (linked by α in square II) are closed (the four sets in I left closed, and the two sets in II not involved in α, right closed).

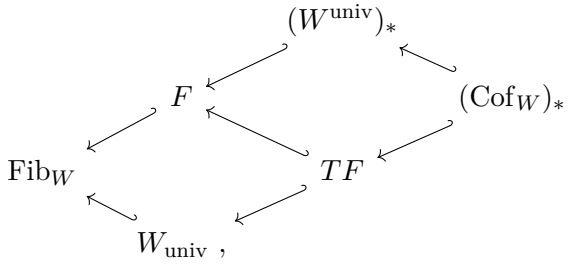
The "closed Quillen pre-quintuples", associated with W, correspond to the choice of left closed subsets TC and C, fitting into diagram (I) as shown in I′ below, and give rise by transposition to F and TF as in II′ below,

[page 23]

$$\begin{array}{ccccc} & & W^{\mathrm{univ}} & & \\ & \nearrow & & \searrow & \\ & TC & & & \mathrm{Cof}_W \\ & \nearrow & & \searrow & \nearrow \\ (\mathrm{Fib}_W)^* & & & & C \\ & \searrow & & \nearrow & \\ & & (W_{\mathrm{univ}})^* & & \end{array} \qquad \text{(I}')$$

[16] taking into account $W^{\mathrm{univ}} \subset \mathrm{Cof}_W$, $W_{\mathrm{univ}} \subset \mathrm{Fib}_W$ and the transposed inclusions.

[17] **NB** We have these squares of inclusions, whenever we have a Quillen system (W, C, F, TC, TF) that is *closed.*



(II′)

the "catch" thus being to choose TC and C large enough that we have

$$\underbrace{(TC)_*}_{=F} \subset \mathrm{Fib}_W, \qquad \underbrace{C_*}_{=TF} \subset W_{\mathrm{univ}} \qquad -$$

and the inclusions stipulated in 2°, b′) ensure that we can indeed find TC and C, if need be taking them as large as possible, $TC = W^{\mathrm{univ}}$, $C = \mathrm{Cof}_W$ – in which case, moreover, we will have

p. 14
[page 24]

$$TC \;=\; C \cap W^{\mathrm{univ}},$$

i.e. we have a left pre-Q-category of models. But now, on the other hand, the point is to ensure that

$$TF \;=\; F \cap W_{\mathrm{univ}},$$

which implies, as we have just seen, (noting $(W^{\mathrm{univ}})_* \subset F$) that

$$\overline{(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}} \;\subset\; \overline{F \cap W_{\mathrm{univ}}} = \overline{TF} = TF \subset W_{\mathrm{univ}}. \qquad (*)$$

Conversely, if $(*)$ is satisfied, let us take

$$F = (W^{\mathrm{univ}})_*$$

as small as possible, *i.e.* $C = W^{\mathrm{univ}}$ as large as possible. We must therefore take

$$TF \;=\; F \cap W_{\mathrm{univ}} \;=\; (W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}},$$

and this set is indeed right closed by hypothesis.

Corollary. *Suppose condition* b′) *of the theorem-scholie is satisfied. The following conditions* a), b) *on W are equivalent (and imply the condition of* 3° *of the theorem-scholie).*

[page 25]

ⓐ $(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} = (\mathrm{Cof}_W)_*$.

ⓑ $\widetilde{W^{\mathrm{univ}} \cup (W_{\mathrm{univ}})^*} = \mathrm{Cof}_W$, i.e. Cof_W *is the left closed subset generated by* W^{univ} *and* $(W_{\mathrm{univ}})^*$,

and these conditions imply

ⓒ *that there exists a structure of a* closed Quillen triple (W, C, F) *including W (thus with $TC = W \cap C$, $TF = W \cap F$) (which will be closed in Quillen's sense if and only if W stable under direct factors – for C and F are already spontaneously stable).*

Indeed, it suffices then to take

p. 15
[page 26]

$$TC \;=\; W^{\mathrm{univ}}, \qquad C \;=\; \mathrm{Cof}_W.$$

But I doubt that the existence of a closed Quillen triple (W, C, F) suffices to imply conditions *a*), *b*) above, or even the weaker condition

$$(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} \;=\; (\mathrm{Cof}_W)_*$$

which means *exactly* that $C = \mathrm{Cof}_W$, $TC = W^{\mathrm{univ}}$, and their transposes $TF = (\mathrm{Cof}_W)_*$, $F = (W^{\mathrm{univ}})_*$ form a closed Quillen category.
Conditions on $(\mathcal{M}, W)$ with

a) $\mathcal{M}$ stable under finite $\varprojlim$ and (small) $\varinjlim$,

b) W stable under filtering $\varinjlim$,

c) $\mathcal{M}$ accessible.

I
- ⓐ existence of a weak Quillen system (W, C, TC, F, TF) (closed if one insists)
- ⇕
- ⓑ existence of a left exact weak Quillen system $(TC = C \cap W)$.
- ⇕
- ⓒ $\underbrace{(\mathrm{Cof}_W)_* \subset W,\ (W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W}_{(*)}$ $(\Longleftrightarrow \quad (\mathrm{Cof}_W)_* \quad \subset \quad W_{\mathrm{univ}},$ $(W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W)$.

⇑

II
- ⓐ existence of a right exact weak Quillen system $(TF = F \cap W)$
- ⇕
- ⓓ We have $(*)$, and moreover $(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}$ right closed.
- ⇕
- ⓑ existence of a closed Quillen system (closed model category), $TC = C \cap W$, $TF = F \cap W$, $F = (TC)_*$, $TF = C_*$, $TC = F^*$, $C = (TF)^*$.
- ⇕
- ⓒ Putting $TC = W^{\mathrm{univ}}$, $F = (TC)_*$, $TF = F \cap W_{\mathrm{univ}}$, $C = (TF)^*$, we find together with W a closed Quillen system.

⇑

III
- $C = \mathrm{Cof}_W$ and $TC = W^{\mathrm{univ}}$ are part (together with W) of a closed Quillen system (thus $F = (W^{\mathrm{univ}})_*$, $TF = (\mathrm{Cof}_W)_*$, $W = TF \circ TC$).
- ⇕
- $(*)$, and moreover $(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} = (\mathrm{Cof}_W)_*$.

⇑

IV
- We have $(*)$ and $\mathrm{Cof}_W = \widetilde{(W_{\mathrm{univ}})^* \cup W^{\mathrm{univ}}}$.
- ⇕
- We have $(*)$ and $(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} = (\mathrm{Cof}_W)_*$.

Case $\mathcal{M} = \mathrm{Cat}$, $W = \mathbf{W}_\infty$. We know that II is satisfied, by Thomason. Thus we have

p. 16

$$(\mathrm{Cof}_W)_* \subset W_{\mathrm{univ}}, \qquad (W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W, \qquad (W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} \text{ right closed.}$$

I have been able to prove the first two inclusions directly (*i.e.* I), but not that $(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}$ is right closed. (I should at least try!) ([18]). We do not have IV, *i.e.* we have

$$(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} \neq (\mathrm{Cof}_W)_*$$

for

$$(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} \not\subset W_{\mathrm{univ}}$$

(contains the $X \longrightarrow e$ with X nonempty and discrete), so we have

$$(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} \underset{\neq}{\supset} \underbrace{(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}}_{\text{is } \textit{not} \text{ the orthogonal of } W^{\mathrm{univ}} \cup (W_{\mathrm{univ}})^*} .$$

[page 27]

3 Quillen category (second version)

We are still given a pair

$$(\mathcal{M}, W), \qquad W \subset \mathrm{Fl}(\mathcal{M}), \qquad W \text{ weakly saturated},$$

$\mathcal{M}$ satisfies M 0 (stability under finite $\varprojlim$ and finite $\varinjlim$).
Definition 1 : A *weak Quillen structure* associated to the category of models $(\mathcal{M}, W)$ is the datum of four subsets (C, TC, F, TF) of $\mathrm{Fl}(\mathcal{M})$, satisfying the following conditions.

M 1 $C \longleftrightarrow TF$, $TC \longleftrightarrow F$ (*i.e.* $TF \subset C_*$, $F \subset (TC)_*$).

M 2 Quillen factorization property for (C, TF) and (TC, F).

M 3 Stability of C under composition and under cobase change, of F under composition and base change – C and F contain the isomorphisms.

M 4 Similar stability for TC and TF.

M′ $TC \subset C \cap W$, $TF \subset F \cap W$.

M″ $C \subset \mathrm{Cof}_W$, $F \subset \mathrm{Fib}_W$.

We say that the Quillen structure is *exact* if it satisfies the additional axiom
M 2 bis $W = TF \circ TC$. p. 17
We say that it is *left exact* if it satisfies the additional axiom
M′$_0$ (g) $TC = C \cap W$ (instead of only $TC \subset C \cap W$),
right exact if it satisfies
M′$_0$ (d) $TF = F \cap W$ (instead of only $TF \subset F \cap W$) ([20]).
Remark. [page 28]

a) A left exact or right exact Quillen structure is exact.

[18] It would suffice to know that $\overline{W_{\mathrm{univ}}} \cap \mathrm{Fib}_W = W_{\mathrm{univ}}$, which would imply that III and IV are equivalent ([19]).

[19] But this is false, e.g. if X nonempty discrete category, $X \longrightarrow e$ is in $\overline{W_{\mathrm{univ}}} \cap \mathrm{Fib}_W$.

[20] **NB** If C, F, W stable under direct factors (thus also $C \cap W$ and $F \cap W$), then left exactness $TC = C \cap W$ *is equivalent* to right exactness $TF = F \cap W$, and implies that C, TC, F, TF are *closed* (on the left for C, TC, on the right for F, TF).

b) If we have a datum (C, TC, F, TF) satisfying M 1, M 2, M′, M'', when we saturate C, TC on the left and F, TF on the right, for the stabilities specified in M 3, M 4, we find a weak Quillen structure. (All the axioms are obvious, except M′, for which I refer to p. 9, 10, in particular prop. 4.) If we had M 2 bis, this will still hold (a fortiori) after saturation, *i.e.* we have an exact Quillen structure.

Definition 1 bis. The weak Quillen structure is called a *Quillen structure* (without qualification) if it is left and right exact, or (as one says) "biexact", then (C, F) already determines (C, TC, F, TF). (That corresponds exactly to the "model categories" of Quillen .) We say that the weak Quillen structure is *closed* if we have

$$C \leftarrow + \rightarrow TF\ , \qquad TC \leftarrow + \rightarrow F\ ,$$

i.e. $TF = C_*$, $C = (TF)^*$ and $F = (TC)_*$, $TC = F^*$ ([21]). A weak Quillen structure that is both biexact (*i.e.* not weak at all!) and closed is called a *perfect* Quillen structure. (That corresponds exactly to the "closed model category" of Quillen.)

Remark. For a weak Quillen structure to be perfect, it suffices that it be biexact and [page 29] that we have

$$C = (TF)^*, \qquad F = (TC)_*$$

(which already implies that $TC = F^*$, $TF = C_*$, *i.e.* that TC is left closed and TF right closed). This is proved in Quillen. It also implies that W is strongly saturated – it's the land of plenty!

Scholie : The pair $TC \longleftrightarrow F$, satisfying the Quillen factorization condition, and satisfying $TC \subset W$, $F \subset \mathrm{Fib}_W$, already allows one to develop a satisfactory theory of W-homotopic fibers in $\mathcal{M}$ – but without an exactness property for W-cartesian squares. Dually, the datum of $C \longleftrightarrow TF$, with factorization, and $TF \subset W$, $C \subset \mathrm{Cofib}_W$, gives a good theory of W-homotopic cofibers, but without an exactness property for W-cocartesian squares. The weak Quillen structure brings together these two data dual to one another, with in addition the additional link between these two data, by $TC \subset C$, $TF \subset F$ (stipulated in M'). This datum already ensures that [p. 18]

$$\underbrace{\mathcal{M}_{\mathrm{cf}}}_{\substack{\text{full subcategory of } \mathcal{M} \text{ consisting of} \\ \text{objects fibrant and} \\ \text{cofibrant at once}}} \qquad W_{\mathrm{cf}}^{-1} \xrightarrow{\sim} \mathcal{M}W^{-1} = \mathrm{Ho}_W$$

is an equivalence, but not that the localization functor [page 30]

$$\mathcal{M}_{\mathrm{cf}} \longrightarrow \mathcal{M}_{\mathrm{cf}} W_{\mathrm{cf}}^{-1}$$

is a simple passage to the quotient on arrows, *i.e.* that if X, Y are in $\mathcal{M}_{\mathrm{cf}}$, then

$$\mathrm{Hom}_{\mathcal{M}}(X, Y) \longrightarrow \underbrace{\mathrm{Hom}_W(X, Y)}_{\overset{\mathrm{def}}{=} \mathrm{Hom}_{\mathrm{Ho}_W}(X,Y)} \qquad (*)$$

is surjective. This is ensured by the exactness axiom M 2 bis,

$$W \ = \ TF \circ TC.$$

With this axiom alone, we have no nice characterization of the equivalence relation in $\mathrm{Hom}_{\mathcal{M}}(X, Y)$ defined by the arrow $(*)$ (for X, Y in $\mathcal{M}_{\mathrm{cf}}$).

[21] This also means that C, TF, TC, F are stable under direct factors. We say that the weak Quillen structure is *left closed* (resp. *right closed*) if C, TC (resp. F, TF) are left closed (resp. right closed).

If we have left exactness,

$$TC = C \cap W \qquad (\Longleftrightarrow\ TC = C \cap W^{\mathrm{univ}}),$$

then this equivalence relation is the one generated by the relation of *right Q-homotopy* ([22], [23]) (implied by left Q-homotopy), and the quotient category thus defined

$$\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{cf}} \ = \ \pi\mathcal{M}_{\mathrm{cf}} \ \simeq \ \mathrm{Ho}_W,$$

maps into $\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}}$,

$$\pi\mathcal{M}_{\mathrm{cf}} \longrightarrow \pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}},$$

and this functor can be interpreted (unless I am mistaken)

as a *right adjoint* (fully faithful) of the localization functor

p. 19
[page 31]

$$\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}} \longrightarrow (\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}})W_{\mathrm{c}}^{-1} \ \simeq \ \mathrm{Ho}_W,$$

and Ho_W is obtained from $\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}}$ by a *calculus of left fractions.* Dually, if we have right exactness, *i.e.* $TF = F \cap W$ ($\Longleftrightarrow TF = F \cap W_{\mathrm{univ}}$), then the relevant equivalence relation on $\mathrm{Hom}_{\mathcal{M}}(X, Y)$, for X, Y in $\mathcal{M}_{\mathrm{cf}}$, is left Q-homotopy (implied by right Q-homotopy), and this time the natural arrow from $\pi_{\mathrm{g}}\mathcal{M}_{\mathrm{cf}} = \pi\mathcal{M}_{\mathrm{cf}} \simeq \mathrm{Ho}_W$ to $\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{f}}$,

$$\underbrace{\pi\mathcal{M}_{\mathrm{cf}}}_{\simeq \mathrm{Ho}_W} \longrightarrow \pi_{\mathrm{d}}\mathcal{M}_{\mathrm{f}}$$

is a *left adjoint* (fully faithful) of the localization functor

$$\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{f}} \longrightarrow \mathcal{M}_{\mathrm{f}}W_{\mathrm{f}}^{-1} \ \simeq \ \mathrm{Ho}_W \qquad ,$$

and the localized category is made explicit by a calculus of right fractions.

In the biexact case (*i.e.* that of genuine Quillen structures), the two Q-homotopy relations in $\mathcal{M}_{\mathrm{cf}}$ (more generally in $\mathrm{Hom}(X, Y)$, if X in $\mathcal{M}_{\mathrm{c}}$ and Y in $\mathcal{M}_{\mathrm{f}}$) coincide, and even become particularly simple. (For if $f, g : X \rightrightarrows Y$, with X cofibrant and Y fibrant, then $f \sim g$ can be tested on *any* "cylinder object" for X, or on *any* "path object" for Y.)

[page 32]

In the case where the weak Quillen structure is left exact, *i.e.* where the relevant equivalence relation in $\mathrm{Fl}\,\mathcal{M}_{\mathrm{cf}}$ is right Q-homotopy, then the exactness axiom is satisfied for W-cartesian squares, but I doubt that the same holds for W-cocartesian squares ([24]). (Unless we make an additional hypothesis, e.g. right exactness of the Quillen structure.) If we have only exactness (but neither left nor right), it seems that we cannot infer from it the exactness either of W-cartesian squares or of W-cocartesian squares, hence not that of cosuspension sequences (loop sequences), nor of suspension sequences. We might therefore think that the exactness condition, all by itself, is not of much use for the homotopy formalism in the categories of fractions Ho_W etc. But we do nevertheless have this: let $\Psi = \ 0 \begin{array}{l} \nearrow 1 \\ \searrow 2 \end{array}$, and suppose that $(\mathcal{M}, W)$ *and* $(\mathcal{M}(\Psi), W(\Psi))$ admit an exact weak Quillen structure. Then in $\mathcal{M}$ the W-cartesian squares are W-exact, so we have

[page 33]

p. 20

[22] It would seem that the relation of right Q-homotopy is an equivalence relation, as soon as Y in $\mathcal{M}_{\mathrm{f}}$ and without assuming the weak Quillen structure to be right or left exact.

[23] **NB** According to Baues, it would seem that on $\mathcal{M}_{\mathrm{cf}}$, this relation of right homotopy is equivalent to the relation of left homotopy, and that the latter does not depend on the cylinder chosen …

[24] It seems nevertheless that this is the case, according to Baues, cf. corresponding marginal annotation page 30.

in $\mathcal{M}$ long exact cosuspension sequences (exact loop sequences) ([25]). (The hypothesis on $(\mathcal{M}(\Psi), W(\Psi))$ serves only to ensure that every arrow of $\mathrm{Ho}_{W(\Psi)} = \mathcal{M}(\Psi)W(\Psi)^{-1}$ is isomorphic to an arrow coming from an arrow of $\mathcal{M}(\Psi)$ itself. But this is an apparently very delicate property, and I know of no other way to obtain it than via the existence of an exact weak Quillen structure.) Dually, if $(\mathcal{M}, W)$ and $(\mathcal{M}(\Phi), W(\Phi))$ admit exact weak Quillen structures, then the W-cocartesian squares in $\mathcal{M}$ are W-exact, and we find (if $\mathcal{M}$ is pointed) long exact suspension sequences.

[page 34] This shows that if, for a given category of models $(\mathcal{M}, W)$, we are interested in "doing homotopical algebra" for all the pairs $(\mathcal{M}(I), W(I))$ at once, with I in Cat, it is enough (if we want to have a good theory of W-homotopic fibers and cofibers in these categories, including the property of W-exactness for W-cartesian squares and for W-cocartesian squares) to have for each $(\mathcal{M}(I), W(I))$ an exact weak Quillen structure, nothing more.

This therefore shows us the interest of developing criteria, for a $(\mathcal{M}, W)$, ensuring that this is indeed so. On this subject, I note that I know how to check the exactness of a Quillen structure *only if* I have left exactness, or right exactness. This means that I must in fact bring out conditions on $(\mathcal{M}, W)$ ensuring (say) that the $(\mathcal{M}(I), W(I))$ admit left exact weak Quillen structures.

[page 35] The weak Quillen structures for the $(\mathcal{M}(I), W(I))$ will be deduced from a weak Quillen structure for $(\mathcal{M}, W)$, in the following way: they will all be *"right closed", i.e.*

$$F(I) \;=\; (TC(I))_*, \qquad TF(I) \;=\; (C(I))_*, \tag{1}$$

in particular (for $I = e$)

$$F \;=\; (TC)_*, \qquad TF \;=\; C_*,$$

which already implies (since $C \subset \mathrm{Cof}_W$, $TC \subset W^{\mathrm{univ}}$)

$$(\mathrm{Cof}_W)_* \;\subset\; W, \qquad (W^{\mathrm{univ}})_* \;\subset\; \mathrm{Fib}_W \tag{2}$$

(whence $(\mathrm{Cof}_W)_* \subset W_{\mathrm{univ}}$). By virtue of (1), to define the weak Quillen structures for the $(\mathcal{M}(I), W(I))$, it suffices to define $C(I)$, $TC(I)$, and even, since we want left exactness, *i.e.*

$$TC(I) \;=\; C(I) \cap W(I) \tag{3}$$

(in particular p. 21

$$TC \;=\; C \cap W \quad), \tag{4}$$

it remains to define $C(I)$. We shall define $C(I)$ "argument by argument", *i.e.* for $f \in \mathrm{Fl}(\mathcal{M}(I))$

$$f \in C(I) \qquad \Longleftrightarrow \qquad \alpha_s^*(f) \in C \;\text{ for every } s \in \mathrm{Ob}\, I, \tag{5}$$

where

$$\alpha_s : e \longrightarrow I \qquad \text{defined by } \alpha_s(e) = s.$$

It follows moreover at once, by (3),

$$f \in TC(I) \qquad \Longleftrightarrow \qquad \alpha_s^*(f) \in TC \;\text{ for every } s \in \mathrm{Ob}\, I. \tag{6}$$

[page 36] We therefore suppose that $(\mathcal{M}, W)$ admits a left exact and right closed weak Quillen structure (C, TC, F, TF), defined in terms of

$$C \;\subset\; \mathrm{Fl}(\mathcal{M}),$$

[25] it seems that I botched it here, and I doubt that the result mentioned is correct.

by

$$TC \;=\; C \cap W, \qquad F \;=\; (TC)_*, \qquad TF = C_*.$$

For this to really give a weak Quillen structure (necessarily left exact and right closed), we simply have to check the following conditions.

ⓘ Factorization for $(C, TF = C_*)$ and $(TC = C \cap W, F = (TC)_*)$ – we shall come back to this point.

ⓘⓘ Stability of C under composition and cobase change, and C contains the isomorphisms (from which the similar stability of $TC = C \cap W$ will follow, taking into account that $C \subset \mathrm{Cof}_W$) (NB The appropriate stabilities for F, TF are trivial by definition $F = (TC)_*$, $TF = C_*$.)

ⓘⓘⓘ $C \subset \mathrm{Cof}_W$, $\underbrace{C_*}_{=TF} \subset W$, $(C \cap W)_* \subset \mathrm{Fib}_W$.

To check, similarly, that $(C(I), TC(I), F(I), TF(I))$ is a weak Quillen structure for $W(I)$, we must check the conditions similar to ⓘⓘⓘ . Now (*ii*) for $C(I)$ follows at once from condition (*ii*) for C. It remains to examine (*iii*) and (*i*).

The inclusions of (*iii*). p. 22

The inclusion [page 37]

$$C \;\subset\; \mathrm{Cof}_W$$

implies that every $f \in C(I)$ is in Cof_W argument by argument. This implies that f is in $\mathrm{Cof}_{W(I)}$, as we see at once from the definition – I make it explicit in the

Lemma 1: *Let $f : X_\bullet \longrightarrow Y_\bullet$ be an arrow of $\mathcal{M}(I)$. For f to be in* $\mathrm{Cof}_{W(I)}$ (resp. *in* $\mathrm{Fib}_{W(I)}$), *it* suffices *that for every s in* $\mathrm{Ob}\, I$, $\alpha_s^*(f)$ *be in* Cof_W (resp. *in* Fib_W). *Likewise (for the record) for a condition for $f \in W(I)$ – there it is necessary and sufficient that $\alpha_s^*(f) \in W \;\forall\, s \in \mathrm{Ob}\, I$.*

We shall apply this same lemma to establish the inclusions

$$C(I)_* \;\subset\; W(I), \qquad (C(I) \cap W(I))_* \;\subset\; \mathrm{Fib}_{W(I)}.$$

So it suffices to see that

$$\begin{aligned} \alpha_s^*(C(I)_*) \;&\subset\; C_* && (\subset\; W) \\ \alpha_s^*((\underbrace{C(I) \cap W(I)}_{(C\cap W)(I)})_*) \;&\subset\; (C \cap W)_* && (\subset\; \mathrm{Fib}_W), \end{aligned}$$

and more generally

$$\alpha_s^*(\Phi(I)_*) \;\subset\; \Phi_* \tag{7}$$

for any (?) subset Φ of $\mathrm{Fl}(\mathcal{M})$, thus defining $\Phi(I) \subset \mathrm{Fl}(\mathcal{M}(I))$ term by term.

Making this explicit, we find the condition [page 38]

$$\alpha_{s!}(\Phi) \subset \Phi(I) \qquad \text{for every } s \in \mathrm{Ob}\, I,$$

for it to work, *i.e.* we want that

$$i \in \Phi \qquad \Longrightarrow \qquad \alpha_{s!}(i) \in \Phi(I), \quad \textit{i.e.}\;\; \alpha_t^*(\alpha_{s!}(i)) \in \Phi \;\; \forall\, t \in \mathrm{Ob}\, I.$$

Now the functor $\alpha_t^* \alpha_{s!}$ is made explicit as

$$A \longmapsto \underbrace{\mathrm{Hom}_I(t,s) \times A}_{\substack{\text{sum of } \mathrm{Hom}_I(t,s) \\ \text{copies of } A \text{ in } \mathcal{M}}}, \qquad \text{p. 23}$$

so we must suppose Φ *stable under direct sums* (of the type considered $\mathrm{Hom}_I(t,s) \times A$). This is indeed the case for $\Phi = C$ and $\Phi = C \cap W$ and for direct sums that are *finite,* thanks to (*ii*). We thus find the

Lemma 2.

a) *Let I in* Cat *be such that direct sums of type* $\mathrm{Hom}_I(t,s)$ *exist in $\mathcal{M}$. (E.g. we suppose that I is* finite, *more generally that the* $\mathrm{Hom}_I(t,s)$ *are finite sets; or that "small" direct sums exist in $\mathcal{M}$.) Let Φ be a subset of* $\mathrm{Fl}(\mathcal{M})$*, stable under this type of direct sums, and let $\Phi(I) \subset \mathrm{Fl}(\mathcal{M}(I))$ be the corresponding subset. Then for every $s \in \mathrm{Ob}\, I$, we have*

[page 39]

$$\alpha_s^*(\Phi(I)_*) \ \subset \ \Phi_*.$$

b) *Suppose that the* $\mathrm{Hom}_I(t,s)$ *are finite (e.g. I finite). Then the hypothesis of* a) *applies for $\Phi = C$ and for $\Phi = C \cap W$, so condition* (iii) *p.* 36 *is satisfied for the structure $(W(I), C(I), TC(I), F(I), TF(I))$ in $\mathcal{M}(I)$. Same conclusion without a finiteness hypothesis on the* $\mathrm{Hom}_I(t,s)$*, when we suppose $\mathcal{M}$ stable under "small" direct sums, and C and W also stable under such direct sums.*

NB Given the scholie above, it is perfectly reasonable to restrict ourselves if need be to categories I that are *finite* – given that the category of these I is stable under the products $I \longmapsto I \times \Psi$, $I \longmapsto I \times \Phi$.

It remains, to be happy, to check the factorization condition ⓘ for (C, TF), (TC, F), but also and above all for the $(C(I), TF(I), TC(I), F(I))$. This means that even starting from a closed Quillen structure (land of Cockaigne) does not get us very far all by itself, [page 40] for lack of knowing that the factorization axiom is preserved, in passing from $(\mathcal{M}, W, \ldots)$ to $(\mathcal{M}(I), W(I), \ldots)$.

It is here that the factorization theorem of no. 1 should prove providential. But to be able to apply it to the pairs $(C(I), C(I)_*)$, $(TC(I), TC(I)_*)$, we need (in addition to the accessibility of $\mathcal{M}$, which is a quite innocuous condition) very strong stability conditions for $C(I)$, $TC(I)$ ([26]). To tell the truth, there remains the stability of C and of $C \cap W$ under ordinal direct limits (which will imply the same stability for the $C(I)$, $(C \cap W)(I) = C(I) \cap W(I)$).

Truth to tell, I see little chance of finding a weak Quillen structure (C, TC, F, TF) associated with W, and satisfying all the stated conditions, including that of stability of C and of $C \cap W$ under ordinal $\varinjlim$, without supposing that W itself is already stable under (small) filtering direct limits.

We can think of using a *functorial* factorization of the arrows in $\mathcal{M}$ as [page 41]

$$\begin{aligned} f = pi \quad \text{with} \quad & i \in TC, \quad p \in F \\ \text{or} \quad & i \in C, \quad p \in TF, \end{aligned}$$

to deduce a similar factorization for the arrows f of $\mathcal{M}(I)$, as p. 24

$$\begin{aligned} f = pi \quad \text{with} \quad & i \in TC(I), \quad p \in F(I) \\ \text{or} \quad & i \in C(I), \quad p \in TF(I), \end{aligned}$$

where $C(I)$, $TC(I)$, $F(I)$, $TF(I)$ are here defined via C, TC, F, TF "argument by argument". But the trouble is that by proceeding in this naive way, we will no longer

[26] Plus the *small* systems of generators for C, TC.

necessarily have $C(I) \longleftrightarrow TF(I)$, $TC(I) \longleftrightarrow F(I)$. We have to shrink $TF(I)$ and $F(I)$ considerably, to have an RLP property with respect to $C(I)$ resp. $TC(I)$.

So I can hardly see any other way out, in this general context, than by assuming the stability of W under filtering $\varinjlim$ (which is after all not very nasty).

[page 42]

Willy-nilly, we come back to the situation considered in paragraph 2. We are going to restate the essentials of what was sensed then:

Theorem 2 (still a little conjectural). *Let $(\mathcal{M}, W)$ be a category of models, with W (as always) weakly saturated, but in addition we assume $\mathcal{M}$ stable under small $\varinjlim$ and under finite $\varprojlim$, and W stable under filtering $\varinjlim$, finally $\mathcal{M}$ accessible. Consider the following conditions:*

a) *There exists a weak Quillen structure for W, left exact, right closed, (C, TC, F, TF).*

a′) *As in* a), *but with in addition C, TC left closed (*i.e. *we have a closed weak Quillen structure), and C, TC of the form $\widetilde{C_0}$, $\widetilde{TC_0}$, where C_0 and TC_0 are small.*

a″) *There exist $C_0 \subset \mathrm{Cof}_W$, $TC_0 \subset W^{\mathrm{univ}}$, small subsets, such that $C_{0*} \subset W$, $(TC_0)_* \subset \mathrm{Fib}_W$.*

b) *We have*

$$(\mathrm{Cof}_W)_* \ \subset \ W_{\mathrm{univ}}, \qquad (W^{\mathrm{univ}})_* \ \subset \ \mathrm{Fib}_W$$

(whence it follows, Cof_W and W^{univ} being left closed, that

$$(W_{\mathrm{univ}})^* \subset \mathrm{Cof}_W, \qquad (\mathrm{Fib}_W)^* \subset W^{\mathrm{univ}} \qquad).$$

[page 43]

These conditions imply that for every I in Cat, *there exists a left exact and closed weak Quillen structure for $W(I)$ in $\mathcal{M}(I)$. Such a structure is defined, in terms of a similar structure C, $TC = C \cap W$, $F = (TC)_*$, $TF = C_*$ for W, by*

$$C(I) \ = \ \Big\{ f \in \mathrm{Fl}(\mathcal{M}(I)) \ \Big| \ \alpha_s^*(f) \in C \ \ \forall\, s \in \mathrm{Ob}I \Big\},$$

then p. 25

$$\begin{aligned} TC(I) \ &= \ C(I) \cap W(I) \ = \ \Big\{ f \in \mathrm{Fl}(\mathcal{M}(I)) \ \Big| \ \alpha_s^*(f) \in TC \ \ \forall\, s \in \mathrm{Ob}I \Big\} \\ F(I) \ &= \ (TC(I))_* \\ TF(I) \ &= \ C(I)_* \,. \end{aligned}$$

Corollary-Scholie. *These conditions imply that for every I in* Cat, $(\mathcal{M}(I), W(I))$ *gives rise to a good theory of $W(I)$-homotopic fibers and cofibers, with $W(I)$-cartesian $W(I)$-exact squares, and $W(I)$-cocartesian $W(I)$-exact squares – hence with long* exact *sequences of cosuspension and of suspension, in the case where $\mathcal{M}$, hence also $\mathcal{M}(I)$, is a category with a zero object.*

[page 44]

I will take care of the proof of the preceding theorem, as well as of the factorization theorem of paragraph 1, when I have my notes on accessible categories and functors and cardinal filtrations. In the meantime, I already feel like making a little more explicit the formalism one arrives at by means of the hypotheses made above. (I am pretty sure at least that condition a'') suffices to imply all the others. And it is indeed in the form a'') that these conditions are going to be verified in practice, I presume, notably in the case where $\mathcal{M} = \mathrm{Cat}$.)

4 The variances u^*, u_*, $u_!$ for $(\mathcal{M}(I), W(I))$ (I variable)

I assume the conditions of the preceding paragraph are satisfied for $(\mathcal{M}, W)$ (cf. theorem p. 42), and choose a closed and left exact weak Quillen structure (C, TC, F, TF) for W, whence for every I in Cat

[page 45]

a closed and left exact weak Quillen structure $(C(I), TC(I), F(I), TF(I))$ for $W(I)$, in $\mathcal{M}(I)$, described on page 43. Now let

$$u \ : \ I' \longrightarrow I$$

be an arrow in Cat, whence an arrow

$$u^* \ : \ (\mathcal{M}(I), W(I)) \ \longrightarrow \ (\mathcal{M}(I'), W(I')).$$

We know, since $\mathcal{M}$ is stable under arbitrary $\varinjlim$ and also (being moreover accessible) under arbitrary $\varprojlim$, that u^* admits left and right adjoint functors,

$$u_!, \ u_* \ : \ \mathcal{M}(I') \ \longrightarrow \ \mathcal{M}(I),$$

and I propose here to summarize the most immediate properties of u^* and of $u_!$, u_* with respect to the weak Quillen structures under consideration. p. 26

Proposition 1: *With the preceding notations, we have the relations:*

$$u^*(C(I)) \subset C(I'), \qquad u^*(TC(I)) \subset TC(I'). \tag{1}$$

$$u_*(F(I')) \subset F(I), \qquad u_*(TF(I')) \subset TF(I). \tag{2}$$

$$\begin{aligned} u^*(F(I)) \subset F(I') &\iff u_!(TC(I')) \subset TC(I) \\ u^*(TF(I)) \subset TF(I') &\iff u_!(C(I')) \subset C(I). \end{aligned} \tag{3}$$

On the other hand, I do not know anything positive to say about the inclusions

$$\begin{aligned} u_*(C(I')) \ &\subset \ C(I), \ \ u_*(TC(I')) \ \subset \ TC(I) \\ u_!(F(I')) \ &\subset \ F(I), \ \ u_!(TF(I')) \ \subset \ TF(I). \end{aligned}$$

Except that for a u that is at all general (u not an isomorphism, say), u_* has no desire

to be compatible with the structures C, TC (a property of homotopical right exactness, morally, since u_* is left exact and in general not at all right exact), nor $u_!$ with the structures F, TF (dual comment to be given, $u_!$ is *right* exact, and does not feel like being "homotopically left exact"). [page 46]

Proof of (1), (2), (3). Relations (1) are obvious by definition. Relations (2), (3) are scarcely less so, in view of the following general sorital property, whose place would be in paragraph 1:

Proposition 2. *Let*

$$C' \underset{u^*}{\overset{u_!}{\rightleftarrows}} C$$

be a pair of adjoint functors $(u_!, u^*)$, and let $\Phi' \subset \mathrm{Fl}(C')$, $\Psi \subset \mathrm{Fl}(C)$. Then

$$(u_!(\Phi') \longleftrightarrow \Psi) \qquad \iff \qquad (\Phi' \longleftrightarrow u^*(\Psi)).$$

Corollary. (This is the same statement, with only the notation changed.) *Suppose that $u^* : C \longrightarrow C'$ admits a* right *adjoint u_* (not necessarily a left adjoint)*

$$C' \underset{u_*}{\overset{u^*}{\rightleftarrows}} C, \qquad \text{p. 27}$$

let $\Phi \subset \mathrm{Fl}(C)$, $\Psi' \subset \mathrm{Fl}(C')$, *then*

$$(u^*(\Phi) \longleftrightarrow \Psi') \qquad \Longleftrightarrow \qquad (\Phi \longleftrightarrow u_*(\Psi')).$$

This is proposition 2 for $C'_0 \underset{u_0^*}{\overset{u_{0!}}{\rightleftarrows}} C_0$, with $C'_0 = C$, $C_0 = C'$, $(u_{0!} : C \longrightarrow C') = u^*$, and $(u_0^* : C' \longrightarrow C) = u_*$.

[page 47]

For the inclusions (2), we use the corollary of proposition 2 for the pairs $\Phi = TC(I)$, $\Psi' = F(I')$, resp. $\Phi = C(I)$, $\Psi' = TF(I')$, and apply (1) which gives $u^*(\Phi) \longleftrightarrow \Psi'$, whence $\Phi \longleftrightarrow u_*(\Psi')$, which is nothing other than (2). For the equivalences (3), we use proposition 2, putting $\Psi = F(I)$, $\Phi' = TC(I')$ resp. $\Psi = TF(I)$, $\Phi' = C(I')$.

Corollary 1 (of proposition 1): *Suppose* $u : I' \longrightarrow I$ *is such that for every* $i \in \mathrm{Ob}I$, *the category* $i\backslash I'$ *is discrete* ([27]) *(e.g.* I' *discrete). Then we have*

$$\begin{cases} u_!(C(I')) \ \subset \ C(I), \ \ u_!(TC(I')) \ \subset \ TC(I) \\ u^*(F(I)) \ \subset \ F(I'), \ \ u^*(TF(I)) \ \subset \ TF(I'). \end{cases}$$

Indeed, the relations on the second line are equivalent (because of (3)) to those on the first. We have to prove that if f in $\mathrm{Fl}\mathcal{M}(I')$ is in $C(I')$ resp. $TC(I')$, $u_!(f)$ is in $C(I)$ resp. $TC(I)$, which also means that $\alpha_s^*(u_!(f))$ is in C resp. TC, for every $s \in \mathrm{Ob}I$. Now the functor

$$\alpha_s^* u_! \ : \ \mathcal{M}(I') \ \longrightarrow \ \mathcal{M}$$

is given explicitly by the well-known formula

$$\alpha_s^*(u_!(X')) \overset{\mathrm{def}}{=} u_!(X')(s) \ \simeq \ \varinjlim_{i' \in s\backslash I'} X'(i').$$

Since

$s\backslash I'$ is discrete, we therefore have the isomorphism functorial in X' in $\mathcal{M}(I')$

[page 48]

$$\alpha_s^*(u_!(X')) \ \simeq \coprod_{(s', x : s \longrightarrow u(s')) \text{ in } \mathrm{Ob}\,(s\backslash I')} X'(s'),$$

so if $f \ : \ X' \longrightarrow Y'$, $\alpha_s^*(u_!(f))$ is made explicit as a direct sum in $\underline{\mathrm{Fl}}(\mathcal{M})$ of arrows of the form

$$f_{s'} \ : \ X'(s') \ \longrightarrow \ Y'(s'),$$

which are in C resp. TC since f is in $C(I')$ resp. $TC(I')$. As C, TC are stable under arbitrary direct sums (C, TC being left closed), we win.

Corollary 2. *Let* I *be in* Cat, *and* $f \in \mathrm{Fl}(\mathcal{M}(I))$. *For* f *to be in* $F(I)$ (resp. *in* $TF(I)$ *),* *it* is necessary *that* $\forall\, s \in \mathrm{Ob}I$, $f_s : X(s) \longrightarrow Y(s)$ *be so.* (But this condition does not look sufficient at all!) *In particular, let* $X \in \mathrm{Ob}\mathcal{M}(I)$, *then* $X \in \mathrm{Ob}\mathcal{M}(I)_f \Longrightarrow X(s) \in \mathrm{Ob}\mathcal{M}_f \ \ \forall\, s \in \mathrm{Ob}I$.

p. 28

We apply corollary 1 to $\alpha_s : e \longrightarrow I$.

Scholie. Proposition 1 (relation (1)) shows that the functor u^* transforms W-cocartesian squares into likewise, long exact sequences of suspension into likewise – this is "the homotopical right exactness" of u^*. We expect the same to hold for cartesian squares and long exact sequences

[27] More generally, it suffices that every connected component Z_α of $i\backslash I'$ have an initial object e_α, so that $\varinjlim_{i\backslash I'} X'(i') \simeq \coprod_{\alpha \in \pi_0(i\backslash I')} X'(e_\alpha)$. This is the case in particular if $I' \longrightarrow I$ is a local isomorphism, e.g. an open immersion.

[page 49]

of cosuspension, but to prove it some work will be needed, which we will do in what follows.

Relations (2) show that the functor u_* transforms W-cartesian squares into likewise, hence long exact sequences of cosuspension into likewise. I do *not* expect the same to hold in general for W-cocartesian squares – *i.e.* that u_* be "homotopically right exact", and not only left exact. But I should at least do some examples (with I', I finite ordered categories).

As for $u_!$, the proposition says nothing very tangible about it. Nevertheless we expect $u_!$ to transform W-cocartesian squares into likewise – that is the least one can expect! Or perhaps I am mistaken, in confusing the functors u_*, $u_!$ considered here, with the similar functors

$$\mathrm{HO}_W(I') \underset{u_*^W}{\overset{u_!^W}{\rightrightarrows}} \mathrm{HO}_W(I),$$

left resp. right adjoints to

$$\mathrm{HO}_W(I') \xleftarrow{u_W^*} \mathrm{HO}_W(I),$$

induced by

$$u^* \ : \ (\mathcal{M}(I), W(I)) \longrightarrow (\mathcal{M}(I'), W(I')).$$

But in general u_*, $u_!$ are not compatible

[page 50]

with the localizers $W(I')$, $W(I)$, so they must not be confused with the functors $u_!^W$, u_*^W. So perhaps I was hasty, in what I said about u_* and $u_!$. I will be obliged to study the situation very closely, when I try to construct the functors u_*^W and $u_!^W$. For the moment, I still have no assurance that the conditions considered for $(\mathcal{M}, W)$ suffice to ensure the existence of u_*^W, $u_!^W$ – not even for I, I' finite ordered sets!

Corollary 1. *Let $X \in \mathcal{M}(I)_{\mathrm{f}}$, then for every $\alpha : s \longrightarrow t$ in I which is* monomorphism, *$X(\alpha) : X(t) \longrightarrow X(s)$ is in F.*

(We apply corollary 1 (p. 47) to $\tilde{\alpha} : \mathbf{\Delta}_1 \longrightarrow I$.) p. 29

Proposition 3. ([28]) *Let X be in $\mathcal{M}(\mathbf{\Delta}_1)$, $X = (X_0 \xleftarrow{\alpha} X_1)$. Then X is in $\mathcal{M}(\mathbf{\Delta}_1)_{\mathrm{f}}$ if and only if X_0, X_1 in $\mathcal{M}_{\mathrm{f}}$ and $\alpha \in F$.*

Corollary 2. *Let $I' \overset{u}{\hookrightarrow} I$ be* open immersion. *Then for $X \in \mathrm{Ob}\,\mathcal{M}(I)_{\mathrm{f}}$, $\varprojlim_I X(i) \longrightarrow \varprojlim_{I'} X(i)$ is in F.*

5 Quillenizable categories of models

I return to the general case of a category of models $(\mathcal{M}, W)$.

Definition 1. We say that $(\mathcal{M}, W)$ is *Quillenizable* if there exists an *exact* weak Quillen structure (C, TC, F, TF) for $\mathcal{M}$.

That is a self-dual condition, but one which I know how to verify in practice only by means of either left or right exactness – thus by a stronger condition which destroys the symmetry. It is to preserve it as far as possible that I introduce the preceding definition, as well as the

Definition 2. Let, in addition to $(\mathcal{M}, W)$, Diag $\subset$ Cat be a full subcategory of Cat. We say that W is *Quillenizable on* Diag, if for every I in Diag, $(\mathcal{M}(I), W(I))$ is Quillenizable. [page 51]

[28] Compare with the lemma p. 63 and its proof (more exactly, the corollary of the lemma).

In practice, I will use this definition in the case where Diag = Cat, or else, that it satisfy at least the stability properties useful for the domain of a derivator: stable under finite $\varprojlim$, under finite sums – possibly under finite $\varinjlim$, etc ([29]).

Example 1: That of paragraphs 3, 4, where we isolated conditions on $(\mathcal{M}, W)$, with $\mathcal{M}$ a "big" category and $\mathcal{M}$ accessible, for W to be Quillenizable on the whole of Cat. This is the case which seems to me at present the most interesting.

Example 2: Let $\mathcal{A}$ be an abelian category, and $\mathcal{M} = K^\bullet(\mathcal{A})$ the category of cochain complexes with values in $\mathcal{A}$, $W = W_{\mathcal{A}}$ the set of quasi-isomorphisms, hence

$$\mathcal{M}W^{-1} \;=\; K^\bullet(\mathcal{A})W_{\mathcal{A}}^{-1} \;\simeq\; D^\bullet(\mathcal{A}), \qquad \text{derived category of } \mathcal{A}.$$

We also define $\mathcal{M}^+$, $\mathcal{M}^-$, $\mathcal{M}^{\mathrm{b}}$, with the induced localizers $W_{\mathcal{A}}^+$, $W_{\mathcal{A}}^-$, $W_{\mathcal{A}}^{\mathrm{b}}$, and the categories of fractions $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, $D^{\mathrm{b}}(\mathcal{A})$. Or again, for I in Cat, [page 52]

$$\mathcal{M}(I) \;=\; K^\bullet(\mathcal{A}(I))$$

p. 30

$$W(I) \;=\; \underbrace{W_{\mathcal{A}(I)}}_{\text{quasi-isomorphisms}} \;\subset\; \mathrm{Fl}(K^\bullet(\mathcal{A}(I))),$$

and likewise for $\mathcal{M}^\alpha(I)$, $W^\alpha(I)$ ($\alpha \in \{+, -, \mathrm{b}\}$), at least *if I is finite.* There are also the subcategories $K^{\geq 0}(\mathcal{A})$ and $K^{\leq 0}(\mathcal{A})$ of cochain resp. chain complexes in the strict sense (with $K^i = 0$ if $i < 0$, resp. if $i > 0$). We will have for arbitrary I

$$\begin{cases} K^{\geq 0}(\mathcal{A})(I) \;\simeq\; K^{\geq 0}(\mathcal{A}(I)) \\ \quad W_{\mathcal{A}}^{\geq 0}(I) \;=\; W_{\mathcal{A}(I)}^{\geq 0}, \end{cases}$$

and likewise for $K^{\leq 0}$, $W^{\leq 0}$.

According to Quillen page 1.2, $(K^+(\mathcal{A}), W_{\mathcal{A}}^+)$ is Quillenizable if $\mathcal{A}$ has enough injectives, $(K^-(\mathcal{A}), W_{\mathcal{A}}^-)$ is so if $\mathcal{A}$ has enough projectives. He says nothing for $(K^\bullet(\mathcal{A}), W_{\mathcal{A}})$ without more – yet one does cohomological algebra with $D^\bullet(\mathcal{A})$ without more, and even without making an assumption of the existence of injectives or projectives. I have somewhat forgotten what Verdier did on this subject – I should all the same look at his thesis (!), or else the exposition of Hartshorne.

[page 53]

At any rate, we have here examples of a quite different type from those considered in 1°. Thus, there is no need to assume that $\mathcal{M}$ is stable under small $\varinjlim$ – $\mathcal{M}$ can even be a small category, it is so if and only if $\mathcal{A}$ itself is. If $\mathcal{A}$ is stable under (small) filtering $\varinjlim$, and is accessible (e.g. has a small generating family, and filtering $\varinjlim$ are exact), then $K^\bullet(\mathcal{A})$ etc. ([30]) have the same properties, and $(K^+(\mathcal{A}), W_{\mathcal{A}}^+)$ is Quillenizable on the whole of Cat ([31]). (This is the case where we know that $\mathcal{A}$ has enough injectives, and cohomological algebra is done mostly by dint of injective resolutions …) In this case we know that the

$$u_* \;:\; D^+(I', \mathcal{A}) \longrightarrow D^+(I, \mathcal{A})$$

exist as right adjoints of the

$$u^* \;:\; D^+(I, \mathcal{A}) \longrightarrow D^+(I', \mathcal{A}),$$

[29] I will assume that Diag contains at least the categories coming from finite ordered sets.

[30] !!

[31] careful, $K^+(\mathcal{A})$ is *not* stable under infinite filtering $\varinjlim$ – so one cannot apply the carpet of paragraphs 2, 3 directly to K^+.

but I am no longer too sure whether we know how to define u_* also on all of $D(I', \mathcal{A})$, and I doubt that $u_!$ exists without an assumption on u (of the sort: finite cohomological dimension). Probably we must always have

$$u_! \ : \ D^-(I', \mathcal{A}) \longrightarrow D^-(I, \mathcal{A}),$$

left adjoint to

$$u^* \ : \ D^-(I, \mathcal{A}) \longrightarrow D^-(I', \mathcal{A}).$$ [page 54]

But I no longer really know whether we should expect to be able to define $u_!$ on all of $D(I', \mathcal{A})$, apart from assumptions like finite cohomological dimension. The sad truth is that I have ± forgotten the theory of derived functors! (I ought to ask Jean Malgoire again to send me canonical texts.) p. 31

Scholie. If $(\mathcal{M}, W)$ is Quillenizable on Diag $\subset$ Cat (and even without its being Quillenizable), it defines a prederivator $\mathbf{D}_{\mathcal{M},W}$ on Diag. We will sometimes set

$$\mathrm{HO}_{\mathcal{M},W}(I) \ \text{ or } \ \mathrm{HO}_W(I) \ = \ \mathbf{D}_{\mathcal{M},W}(I).$$

The Quillenizable hypothesis implies, on each of these homotopy categories, the (relatively weak) "triangulated" structure, coming from the *exact* suspension and cosuspension sequences ([32]). (The Quillenizable character ensures the exactness of these sequences …) The functors $u^* : \mathrm{HO}_W(I) \longrightarrow \mathrm{HO}_W(I')$ are "homotopically exact" in the sense of these "triangulated" (noncommutative) structures, *i.e.* transform exact suspension resp. cosuspension sequences into likewise. [page 55]

Which axioms of derivators are satisfied under the Quillenizable hypothesis alone? **Der** 1 is so without any hypothesis at all on the category of models $(\mathcal{M}, W)$:

$$\mathrm{HO}_W(\coprod_\alpha I_\alpha) \ \simeq \ \prod_\alpha \mathrm{HO}_W(I_\alpha)$$

for a *finite* family of objects I_α in Diag. The localization axiom **Der** 2 is so under the hypothesis that W be *strongly saturated,* and Quillenizable. We use the fact that every arrow of $\mathrm{HO}_W(I)$ comes, up to isomorphism, from an arrow in $\mathcal{M}(I)$. (That was one of the points which remained pending in VI, in the theory of HOT_W, $W \subset \mathrm{Fl(Cat)}$ a basic localizer …)

The axiom **Der** 3, new version , involves a basic localizer $\mathbf{W} \subset \mathrm{Fl(Cat)}$ (not necessarily $\mathbf{W}_\infty$) ([33]). The strongest form would be this: if

$$u \ : \ I' \longrightarrow I$$

is in $\mathbf{W}$, we want the functor u^* to induce an *equivalence*

$$u^*_{\mathrm{lc}} \ : \ \mathrm{HO}^{\mathrm{lc}}_W(I) \xrightarrow{\sim} \mathrm{HO}^{\mathrm{lc}}_W(I').$$

But I doubt that we are equipped to verify so strong an axiom, without having at least a theory of the operations $u_!$ or u_*. One could on the off chance posit this axiom by taking $\mathbf{W} = \mathbf{W}_\alpha \overset{\mathrm{def}}{=}$ the smallest basic localizer satisfying W (1 to 7) (the largest of these localizers being $\mathbf{W}_\infty$). For the moment, the way to verify this axiom (in terms of the datum $(\mathcal{M}, W)$) is for me a ± total mystery. (But I know how to verify it for HOT_W, when $W \subset \mathrm{Fl(Cat)}$ satisfies W (1, 2, 3, 4, 5, …).) [page 56]

I am going to look into it, once I have a reasonable command of the formalism of u_*, $u_!$ relative to the $u : I' \longrightarrow I$ in Diag. The Quillenizable hypothesis alone seems too coarse to ensure the existence of $u_!$ or of u_* in interesting cases. p. 32

[32] In the case where $\mathcal{M}$ is pointed by neutral object. Otherwise, there is the structure associated to cartesian and cocartesian squares. It would be quite interesting to axiomatize such a species of structure.

[33] It is natural here to assume that $\mathbf{W}$ satisfies all the "good axioms" on basic localizers W (1, 2, 3, 4, 5, 6, 7) . [page 57]

6 Quillenizable category of models, and functors $u_!$, u_*

Definition 1:

ⓐ We say that $(\mathcal{M}, W)$ is *uniformly Quillenizable on the left* on Diag, if we can find a weak Quillen structure (C, TC, F, TF) for $\mathcal{M}$ which is *closed,* and which moreover has the following property: for I in Cat, let $C(I)$, $TC(I) \subset \mathrm{Fl}(\mathcal{M}(I))$ be defined by

$$(u : X \longrightarrow Y) \in C(I) \text{ (resp. } TC(I)) \text{ if and only if } \forall\, s \in \mathrm{Ob}\, I,$$
$$u_s : X_s \longrightarrow Y_s \text{ is in } C \text{ (resp. } TC).$$

Let moreover

$$F(I) \;=\; TC(I)_*, \qquad TF(I) \;=\; C(I)_*.$$

The condition in view is that for every I *in* Diag, $(C(I), TC(I), F(I), TF(I))$ is a *closed exact* weak Quillen structure for $W(I)$ ([34]).

ⓑ We say that $(\mathcal{M}, W)$ is *Quillenizable on Diag uniformly and exactly left*, if we have in addition $TC = C \cap W$, *i.e.* if (C, TC, F, TF) is left exact, hence defined by C alone, via

$$TC = C \cap W, \qquad F = (TC)_*, \qquad TF = C_* \qquad ([35])$$

We saw in paragraph 3 that if $\mathcal{M}$ is stable under small $\varinjlim$ and accessible, and W stable under small filtering $\varinjlim$, finally if W admits a weak Quillen structure (C, TC, F, TF) which is closed and left exact – then the conditions of definition 1 are satisfied. p. 33

On the other hand, let us note that proposition 1 page 45 and its corollaries, including proposition 3 (page 50) and its corollaries, are valid provided that the Quillenizations of the $(\mathcal{M}(I), W(I))$ under consideration are defined "uniformly on the left" by a uniform left Quillenization (not necessarily left exact) of $(\mathcal{M}, W)$ on Diag. [page 58]

Uniform left Quillenization will serve us to establish the existence of homotopy direct images u_*, for certain u. Uniform right Quillenization, dually, will have as its main consequence the existence of homotopy $u_!$ for certain $u : I' \longrightarrow I$ in Diag.

I am going to summarize what I have been able to establish so far in this direction, in the

Theorem-scholie: *Let* (C, TC, F, TF) *be a uniform left Quillenization of* W *on* Diag $\subset$ Cat. [page 59]

1) *Let* $u : I' \longrightarrow I$ *be an arrow in* Diag. *We assume* Diag *matched to* $\mathcal{M}$ *in such a way that for* u*, an arrow in* Diag, *the* $\varprojlim$ *in* $\mathcal{M}$ *(of type* I'/i*) required to ensure the existence of*

$$u_* \;:\; \mathcal{M}(I') \longrightarrow \mathcal{M}(I)$$

exist in $\mathcal{M}$*. (This is O.K. if* I*,* I' *finite,* $\mathcal{M}$ *being assumed stable under finite* $\varprojlim$ *in any case.) We know* (prop. 1, page 45) *that* u_* *induces*

$$u_*^{\mathrm{f}} \;:\; \mathcal{M}(I')_{\mathrm{f}} \longrightarrow \mathcal{M}(I)_{\mathrm{f}}\,. \qquad (*)$$

We say that u *is* admissible (on the left) for the given uniform left Quillenization of W on Diag, *if* u_*^{f} *is compatible with the localizers* $W(I')_{\mathrm{f}}$, $W(I)_{\mathrm{f}}$, i.e. *if* $f' : X' \longrightarrow Y'$ *in* $\mathcal{M}(I')$*,* X'*,* Y' *fibrant,* $f' \in W(I')$ *implies* $u_*(f') \in W(I)$*. We say that* Z *in* Diag *is* admissible (for the given uniform left Quillenization of W on Diag …) *if the functor* $Z \longrightarrow e$ *is admissible. This being so:*

[34] (C, TC, F, TF) is then called a *uniform left Quillenization* of $(\mathcal{M}, W)$ on Diag.

[35] We say that $(C, \ldots)$, or simply C, defines a *uniform and left exact Quillenization* of $(\mathcal{M}, W)$ on Diag.

a) *For u to be admissible, it suffices that the I'/s for $s \in \mathrm{Ob}\, I$, be so. NB there is a converse, when I is ordered, hence the $I'/s \longrightarrow I'$ open immersions.*

b) *Every finite ordered set is admissible. More generally (barring error) every ordered set I such that the topological space associated to I° is noetherian (*i.e. *every increasing sequence of open sets of I° is stationary, or again every increasing sequence of closed sets of I is stationary), and if moreover every irreducible closed subset of I° admits a generic point,* i.e. *I° is (not only a noetherian space, but also) a topological space which is* sober. *(For I, this means that every upward filtering open subset of I admits a greatest element.)*

[page 60]

p. 34

c) *In particular, if I', I are finite ordered sets, then every $u : I' \longrightarrow I$ is admissible. The same holds, more generally, if I, I' are ordered sets and I'° is a noetherian sober space.*

2) *Suppose $u : I' \longrightarrow I$ admissible. Consider*

$$u_*^W \ : \ \underbrace{\mathcal{M}(I')_{\mathrm{f}} W(I')_{\mathrm{f}}^{-1}}_{\simeq \mathrm{HO}_W(I')} \ \longrightarrow \ \underbrace{\mathcal{M}(I)_{\mathrm{f}} W(I)_{\mathrm{f}}^{-1}}_{\mathrm{HO}_W(I)},$$

induced by u_*. This functor is right adjoint to $u_W^* : \mathrm{HO}_W(I) \longrightarrow \mathrm{HO}_W(I')$.

[page 61]

Moreover, the canonical homomorphism (for $s \in \mathrm{Ob} I$, $X' \in \mathrm{Ob}\mathrm{HO}_W(I')$)

$$u_*^W(X')(s) \ \simeq \ p_{s*}(X'|I'/s) \ \overset{\mathrm{def}}{=} \ \varprojlim_{I'/s} {}_W(X'|I'/s)$$

(where $p_s : I'/s \longrightarrow e$) is an isomorphism, if we suppose that the I'/s are admissible (we shall then say that u is strictly *admissible).*

Corollary. *On the domain formed by finite ordered sets, the prederivator* HO_W *satisfies (in addition to axioms* **Der** 1, **Der** 2*) axioms* **Der** 4 *and* **Der** 5 *(existence of the direct images u_*, right adjoints of the functors u^*, and standard calculation of the fibers of a direct image).*

Proof of the theorem-scholie.

1°) *a*) is immediate. For *b*), we proceed by induction on the cardinality n of the finite ordered set I, the cases $n \in \{0, 1\}$ being trivial. Suppose then $n \geq 2$, and the theorem proved for the $n' < n$. I distinguish two cases: if I has a greatest element a, then, denoting by $\alpha : e \longrightarrow I$ the functor with value a, by $p : I \longrightarrow e$ the projection, we know that we have

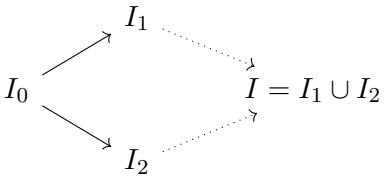
$$p_* \ \simeq \ \alpha^*, \qquad \textit{i.e.} \ \varprojlim_I X_i \simeq X_a,$$

whence the fact that on the whole of $\mathcal{M}(I)$ (not only on $\mathcal{M}(I)_{\mathrm{f}}$), p_* is compatible with the localizers $W(I)$, W. [page 62]

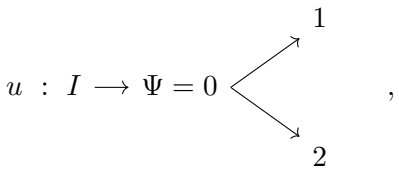
If I has no greatest element, then the set M of its maximal elements has cardinality ≥ 2, we can therefore write it $M = M_1 \amalg M_2$, and we will have

$$I \ = \ I_1 \cup I_2, \qquad \text{where } I_1 = I/M_1, \ I_2 = I/M_2$$

are two open subsets of I, different from I, hence of cardinalities n_1, $n_2 < n$. Consider also $I_0 = I_1 \cap I_2$, it is an open subset of cardinality $< n$. The diagram



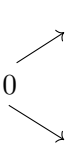
corresponds to an arrow p. 35



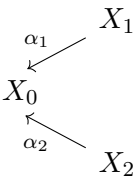
with

$$I_1 = I/1, \qquad I_2 = I/2, \qquad I_0 = I/0.$$

The induction hypothesis and *a*) imply that u is admissible. As a composite of admissible functors is admissible (tautologically, via $(vu)_* \simeq v_* u_*$), we are finally reduced to the [page 63]

Lemma. *The category*  *is admissible.*

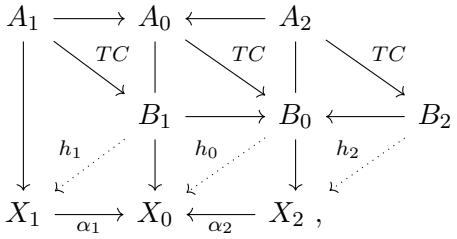
This will follow from the

Corollary ([36]). *Let*  *be in* $\mathcal{M}(\Psi)$. *For* $X_\bullet$ *to be fibrant, it is necessary and sufficient that we have*

a) X_0, X_1, X_2 *are fibrant, and*

b) α_1, $\alpha_2 \in F$.

The necessity of *a*) is a special case of corollary 2, page 48, that of *b*) is a special case of corollary 1 page 50. For sufficiency (which we shall not need for the theorem), we know how to show that in the commutative diagram in solid lines

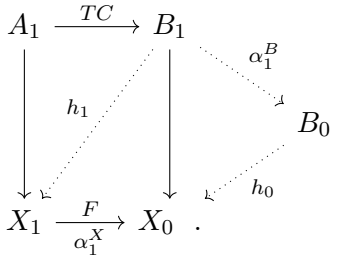


with three arrows marked TC, we can find h_1, h_0, h_2 leaving the diagram commutative (three triangles and two squares to keep commutative). For this, we begin by choosing p. 36

[page 64]

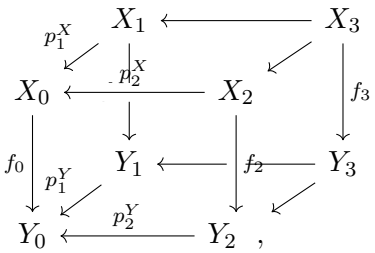
h_0 so as to make the middle vertical triangle commutative – this is possible since X_0 is fibrant, $A_0 \longrightarrow B_0$ in TC.

Next, we see separately for h_1, h_2, that we can find them with each satisfying the two commutativity constraints (1 vertical triangle and one tilted square). For h_1 e.g. existence is seen from the auxiliary square diagram (solid lines)



NB This is the proof of proposition 3, page 50.

The corollary (the "necessary" part) having been seen, the lemma follows from it, for we know that if we have a homomorphism f of cartesian squares X, Y



with

$$p_1^X,\ p_2^X,\ p_1^Y,\ p_2^Y \quad \text{fibrant (even } p_1^X \text{ and } p_1^Y \text{ fibrant is enough),}$$

and

$$f_0,\ f_1,\ f_2 \ \in\ W,$$

then $f_3 \in W$. (This was seen in the theory of W-homotopic fiber products , and the direct verification is immediate.) Of the relevant initial hypothesis that $X_1 \swarrow X_0 \nwarrow X_2$ and $Y_1 \swarrow Y_0 \nwarrow Y_2$ are fibrant, we only need to use a little bit, e.g. that p_1^X and p_1^Y are fibrant, which ensures that the two squares X, Y are W-homotopically cartesian, which is enough then to imply that $f_0,\ f_1,\ f_2 \in W \Longrightarrow f_3 \in W$.

[page 65]

I am going to try to generalize this reasoning, to handle the case of an ordered set I which may be infinite, but such that I° is noetherian and sober, hence every increasing sequence of closed subsets of I, and every decreasing sequence of open subsets, is stationary, and moreover every upward filtering open subset has a greatest element.

p. 37

[36] **NB** There is an obvious variant, characterizing the X such that $X \longrightarrow e$ is in $TF(\Psi)$ as those for which X_0, X_1, X_2 are fibrant and *trivial,* and $\alpha_1,\ \alpha_2 \in TF$.

We want to prove that I is admissible, and by "noetherian induction" we are reduced to proving this: $(*)$ If I is such that for every open subset $I' \subset I$, with $I' \neq I$, I' is admissible, then I too.

$(*)$ granted, let us prove that $I = I_0$ is admissible.

If it were not, by $(*)$ there would exist $I_1 \subset I_0$, $I_1 \neq I_0$, with I_1 not admissible. Likewise, there would be $I_2 \subset I_1$, $I_2 \neq I_1$, not admissible etc. – we would have constructed a strictly decreasing sequence of open subsets I_n of I, contrary to the noetherian character of I°. [page 66]

Let us prove $(*)$, and again distinguish two cases.

a) I° is irreducible, hence has a generic point, *i.e.* a least element, *i.e.* I has a greatest element. This case was already treated earlier – every category with a final object is admissible.

b) I° is reducible, hence a union of two closed subsets I_1°, I_2° distinct from I°, so I is a union of two open subsets I_1, I_2 distinct from I. Then by the hypothesis in $(*)$, I_1, I_2 and $I_0 = I_1 \cap I_2$ are admissible. We then finish as before, being reduced to the

admissibility of $\Psi = 0 \begin{array}{l} \nearrow 1 \\ \searrow 2 \end{array}$!

Let us prove part *c*) of the theorem-scholie, part 1°). This follows from *a*), once we have seen that for $s \in I$, I'/s satisfies the same hypotheses as I'. Now I being ordered, I/s is an open subcategory of I, hence I'/s an open subcategory of I', with $(I'/s)^\circ$ a closed ordered subset of I'°. The latter being noetherian and sober, the same holds for every closed subset, whence the conclusion. [page 67]

Let us prove part 2°) of the theorem-scholie. Consider

$$\begin{array}{ccccc} \mathcal{M}(I')_{\mathrm{c}} & \hookleftarrow & \mathcal{M}(I') & \hookrightarrow & \mathcal{M}(I')_{\mathrm{f}} \\ \uparrow{\scriptstyle u_{\mathrm{c}}^*} & & \uparrow{\scriptstyle u^*} & & \downarrow{\scriptstyle u_*^{\mathrm{f}}} \\ \mathcal{M}(I)_{\mathrm{c}} & \hookleftarrow & \mathcal{M}(I) & \hookrightarrow & \mathcal{M}(I)_{\mathrm{f}} \ , \end{array}$$

where u^* and a fortiori u_{c}^* are compatible with $W(I)$, $W(I')$, and by hypothesis u_* is compatible with $W(I')_{\mathrm{f}}$, $W(I)_{\mathrm{f}}$. Passing to the categories of fractions, we find

p. 38

[page 68]

$$\begin{array}{ccccc} \mathcal{M}(I')_{\mathrm{c}}W(I')_{\mathrm{c}}^{-1} & \xrightarrow{\sim} & \mathrm{HO}_W(I') & \xleftarrow{\sim} & \mathcal{M}(I')_{\mathrm{f}}W(I')_{\mathrm{f}}^{-1} \\ \uparrow{\scriptstyle u_W^*} & & & & \downarrow{\scriptstyle u_*^W} \\ \mathcal{M}(I)_{\mathrm{c}}W(I)_{\mathrm{c}}^{-1} & \xrightarrow{\sim} & \mathrm{HO}_W(I) & \xleftarrow{\sim} & \mathcal{M}(I)_{\mathrm{f}}W(I)_{\mathrm{f}}^{-1} \end{array}$$

and we want to show that u_W^* and u_*^W are adjoint, more precisely find a bifunctorial isomorphism, for F *in* $\mathcal{M}(I)_{\mathrm{c}}$ and G' *in* $\mathcal{M}(I')_{\mathrm{f}}$

$$\mathrm{Hom}_W(u^*F, G') \xrightarrow{\sim} \mathrm{Hom}_W(F, u_*G') \qquad (*)$$

(where Hom_W denotes the Hom in $\mathrm{HO}_W(I')$ resp. $\mathrm{HO}_W(I)$). As F is in $\mathcal{M}(I)_{\mathrm{c}}$, u^*F is in $\mathcal{M}(I')_{\mathrm{c}}$, and as G' is in $\mathcal{M}(I')_{\mathrm{f}}$, u_*G' is in $\mathcal{M}(I)_{\mathrm{f}}$. It follows that the two sides of $(*)$ are quotients of the two sides in the canonical bijection

$$\mathrm{Hom}_{\mathcal{M}(I')}(u^*F, G') \xrightarrow[\sim]{\alpha} \mathrm{Hom}_{\mathcal{M}(I)}(F, u_*G'). \qquad (**)$$

[page 69]

We simply have to show that the latter transforms the relevant equivalence relation R' on the first side, into the one R on the second.

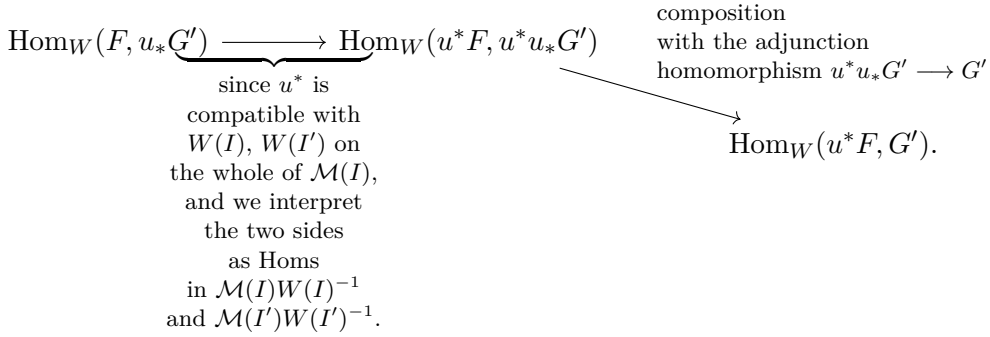
I say that in any case we have

$$R \ \leq \ \alpha(R')$$

without any admissibility hypothesis on u, *i.e.* that the inverse bijection of $(**)$ gives by passage to quotients a homomorphism

$$\mathrm{Hom}_W(u^*F, G') \longleftarrow \mathrm{Hom}_W(F, u_*G'), \qquad (***)$$

necessarily surjective, and moreover functorial in F , variables in $\mathrm{HO}_W(I)$ and $\mathrm{HO}_W(I')$ respectively. Indeed, we define the arrow $(***)$ as the composite



This therefore implies that the functor on $\mathrm{HO}_W(I)$ $F \longmapsto \mathrm{Hom}_{\mathrm{HO}_W(I')}(u^*F, G')$ is, if not representable, at least a term-by-term quotient of the functor represented by u_*G' ([37]). p. 39

But suppose now that u is admissible. Then I am going to define the arrow in the original direction $(*)$ (p. 68) as follows: I choose an arrow [page 70]

$$i' \ : \ u^*F \longrightarrow F' \qquad \text{with } i' \in TC \subset C \cap W, \ F' \in \mathrm{Ob}\,\mathcal{M}(I')_{\mathrm{f}}$$

(so in fact $F' \in \mathrm{Ob}\,\mathcal{M}(I')_{\mathrm{cf}}$, since u^*F is cofibrant), and by adjunction the arrow i' corresponds to

$$i \ : \ F \longrightarrow u_*F'.$$

I consider the diagram (commutative!)

$$\begin{array}{ccc} \mathrm{Hom}(F', G') & \xrightarrow{\ \alpha\ } & \mathrm{Hom}(u_*F', u_*G') \\ \Big\downarrow{\scriptstyle \lambda : f' \longmapsto f' \circ i'} & & \Big\downarrow{\scriptstyle \mu : f \longmapsto f \circ i} \\ \mathrm{Hom}(u^*F, G') & \xrightarrow{\ \text{adj.}\ } & \mathrm{Hom}(F, u_*G') \ . \end{array}$$

We note that we still have F' cofibrant (since u^*F is and $i' \in TC \subset C$), so $\mathrm{Hom}_W(F', G')$ is a quotient of $\mathrm{Hom}(F', G')$, and λ induces an isomorphism

$$\mathrm{Hom}_W(F', G') \xrightarrow{\ \sim\ } \mathrm{Hom}_W(u^*F, G')$$

(since $i' \in TC \subset W$). Since F', $G' \in \mathrm{Ob}\,\mathcal{M}(I')_{\mathrm{f}}$ and u_*^{f} is compatible with $W(I')_{\mathrm{f}}$, $W(I)_{\mathrm{f}}$, it follows that the horizontal arrow α is compatible with the equivalence relations [page 71]

[37] But it is not clear, without an additional hypothesis on the situation, that this representable functor can be chosen in functorial dependence on G', viewed in $\mathrm{HO}_W(I')$.

defined by

$$\mathrm{Hom}(F', G') \longrightarrow \mathrm{Hom}_{W(I')}(F', G') \quad \text{resp.} \quad \mathrm{Hom}(u_* F', u_* G') \longrightarrow \mathrm{Hom}_{W(I)}(u_* F', u_* G') \ .$$

And the same holds for the arrow $\mu : f \longmapsto f \circ i$, trivially. So the adjunction arrow $\mathrm{Hom}(u^* F, G') \longrightarrow \mathrm{Hom}(F, u_* G')$ is compatible with the relations R, R', which finishes proving that $(**)$, page 68, induces a bijection $(*)$, page 68.

This therefore shows

1°) that for a fixed object G' in $\mathrm{HO}_W(I')$ (which we can always suppose, up to isomorphism, to come from $\mathcal{M}(I')_{\mathrm{f}}$), the functor

$$F \ \longmapsto \ \mathrm{Hom}_{\mathrm{HO}_W(I')}(u_W^* F, G')$$

is representable – *i.e.* the functor u_W^* admits a right adjoint, and

2°) that this right adjoint is computed as $u_*(G')$ (when the argument G' is taken fibrant) – since the homomorphism $(***)$, page 69, is *functorial in* F. (A posteriori it follows that it is also functorial in G' …)

The last statement in 2° of the theorem-scholie follows from the computation of u_*^W, taking into account that $I'/s \longrightarrow I'$ is a local isomorphism, so the inverse image functor sends

p. 40
[page 72]

$$\mathcal{M}(I')_{\mathrm{f}} \longrightarrow \mathcal{M}(I'/s)_{\mathrm{f}},$$

so if G' is in $\mathcal{M}(I')_{\mathrm{f}}$,

$$u_*(G')(s) \ \simeq \ p_{s\,*}(G'|I'/s)$$

is a computation of $p_{s\,*}^W(G'|I'/s)$.

This completes the proof of the theorem-scholie on direct images.

Comments.

1) This theorem seems to me far from satisfactory, it is a makeshift. I had hoped, with the hypotheses of paragraph 3, that we would have the existence of u_*^W for *every* arrow $u : I' \longrightarrow I$ in Cat. I now doubt that this is the case, but have not yet succeeded in making a counterexample. The most important test cases for the question of admissibility of a category I seem to me the following:

a) I an infinite discrete category (say countable). The question here comes down to this: let $f_i : X_i \longrightarrow Y_i$ be in W, with X_i, Y_i fibrant, *i.e.* such that $X_i \longrightarrow e$, $Y_i \longrightarrow e$ are in F . Then is $f = \prod f_i : \prod X_i \longrightarrow \prod Y_i$ in W?

[page 73]

b) I the category

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots ,$$

i.e. the ordered category associated with the ordered set $\mathbf{N}$, for the usual order relation $m \leq n$. Thus $\mathcal{M}(I)$ consists of the *inverse* systems in $\mathcal{M}$, indexed by $\mathbf{N}$. One easily proves (proceeding as in the corollary to the lemma p. 63) that $\mathcal{M}(I)_{\mathrm{f}}$ consists of the inverse systems

$$X_0 \longleftarrow X_1 \longleftarrow X_2 \longleftarrow \cdots \longleftarrow X_n \xleftarrow{p_n} X_{n+1} \longleftarrow \cdots$$

such that the X_i are in $\mathcal{M}_{\mathrm{f}}$, and the p_n in F. The question is therefore: if $X_\bullet$, $Y_\bullet$ are fibrant and $f_\bullet : X_\bullet \longrightarrow Y_\bullet$ is such that $f_i \in W$ for $i \in \mathbf{N}$, then is $\varprojlim f_i : X_\infty \longrightarrow Y_\infty$ in W?

p. 41
[page 74]

If the answer is affirmative for **N** (for a W and a given Quillenization), it is also so for every countable discrete category, as one sees at once. But I do not see how to deduce even *a*) from the axioms, say for a closed Quillen triple (closed model category) such that $\mathcal{M}$ is moreover stable under filtering $\varinjlim$, that these are exact, and that there is a small set of generators in $\mathcal{M}$ (so $\mathcal{M}$ is accessible), that W is stable under filtering $\varinjlim$. I should ask THOMASON whether he knows a counterexample.

c) The two preceding examples were ordered categories. As a typical example of a non-ordered category, where I do not see what happens, I point out $I = B_G$, G a group (so a single object, with an automorphism group G). Also the case of $\mathbf{\Delta}[1] = \{\mathbf{\Delta}_0, \mathbf{\Delta}_1\}$ (full subcategory of $\mathbf{\Delta}$) is not clear to me.

2) One would like to be able to treat the existence of $u_!^W$ dually, for want of better. Under the starting hypothesis of paragraph 4 on $(\mathcal{M}, W)$, this therefore leads to a change of tactics. This time, we shall have to assume the existence of a closed Quillenization that is *right exact* for W (which is more demanding, it seems, than such a left exact structure, in the case under consideration), so

$$TF \;=\; F \cap W,$$

and moreover also C, TC such that we have [page 75]

$$C \;=\; \widetilde{C_0}, \qquad TC \;=\; \widetilde{TC_0}$$

with C_0, TC_0 small ([38]). We shall now set

$$\begin{aligned} F(I) \;&=\; \{f \in \mathrm{Fl}\,\mathcal{M}(I) \;\mid\; f_s \in F \quad \forall\, s \in \mathrm{Ob}\, I\} \\ TF(I) \;&=\; \{f \in \mathrm{Fl}\,\mathcal{M}(I) \;\mid\; f_s \in TF \;\; \forall\, s \in \mathrm{Ob}\, I\}, \end{aligned}$$

then

$$\begin{aligned} C(I) \;&=\; TF(I)^* \\ TC(I) \;&=\; F(I)^* \;. \end{aligned}$$

The hope (which I shall have to check later) is that in this way we obtain a right-uniform Quillenization of W, over the whole of Cat. The only problem is the validity of the factorization axioms for $f \in \mathrm{Fl}\,\mathcal{M}(I)$ as $f = pi$, with $i \in TC$, $p \in F$, and also with $i \in C$, $p \in TF$. I have good hope that this is true – this reduces essentially to seeing that $(C(I), TF(I))$ and $(TC(I), F(I))$, say $(\Phi(I), \Psi(I))$, satisfy, in addition to $\Phi(I) = \Psi(I)^*$, $\Psi(I) = \Phi(I)_*$ (*i.e.* $\Psi(I)$ is right closed), and that $\Phi(I)$ is accessible (just like $\Phi = \Phi(e)$). If this works, we shall thus have the existence of $u_!$ for u a functor between finite ordered p. 42 categories, and under slightly more general conditions. (E.g. $u : I' \longrightarrow I$ with I ordered, [page 76] I' ordered and with underlying space noetherian and sober.) Perhaps this time, the hypothesis that W is stable under filtering $\varinjlim$ will be of use after all! Thus, the case of discrete categories, and of left filtering categories such as $\mathbf{N}^\circ$, will no longer pose any problems because of this. So there is hope! Every ordered set which is a filtering $\varinjlim$ of *closed* noetherian and sober ordered subsets, is then W-admissible on the right. But what about categories of type $\Gamma = B_G$ e.g. ? ? ?

[page 77]

[38] One then says that C, TC (as left closed subsets) are "accessible".

7 Cardinal filtrations and accessible subsets of a big category

I refer to the theory of accessible functors and categories in SGA 4, I, §9 (p. 138-178). I give myself a big category (always a $\mathfrak{U}$-category) E with a *cardinal filtration* (increasing)

$$\Big(\mathrm{Filt}^{\pi}(E)\Big)_{\pi\geq\pi_0}, \qquad \pi_0 \text{ a fixed infinite cardinal,}$$

by the strictly full subcategories $\mathrm{Filt}^{\pi}(E)$. We have

a) $\mathrm{Filt}^{\pi}(E)$ essentially small, *i.e.* equivalent to a small category (*i.e.* to a category belonging to $\mathfrak{U}$).

b) E satisfies L_{π_0} (*i.e.* E is stable under $\varinjlim$ of type I, when I is an ordered set large relative to π_0), and $\mathrm{Filt}^{\pi}(E)$ is stable under filtering direct limits indexed by I large relative to π_0, and such that $\mathrm{card}\, I \leq \pi$.

c) For every $\pi \geq \pi_0$, and for every $X \in \mathrm{Ob}\, E$, we have

$$X \;\simeq\; \varinjlim_{i\in I} X_i,$$

where the X_i in $\mathrm{Filt}^{\pi}(E)$, and where I is an ordered set large relative to π. Moreover, if $X \in \mathrm{Ob}\,\mathrm{Filt}^{\pi'}(E)$, with $\pi' > \pi$, we can take I such that $\mathrm{card}\, I \leq \pi'^{\pi}$ (so if $\pi' = 2^{c'}$ with $c' \geq \pi$, whence $(\pi')^{\pi} = (2^{c'})^{\pi} = 2^{c'\pi} = 2^{c'} = \pi'$, we shall have $\mathrm{card}\, I \leq \pi'$). (**NB** If we assume the generalized continuum hypothesis, we shall have $\pi'^{\pi} = \pi'$ for $\pi < \mathrm{cf}(\pi')$, hence $\mathrm{card}\, I \leq \pi'$.) ([39])

Let us consider the representation in *c*), p. 43 [page 78 crossed out] [page 79]

$$X \;=\; \varinjlim_{I} X_i \qquad\qquad \begin{cases} X_i \in \mathrm{Filt}^{\pi}, \\ I \text{ large relative to } \pi. \end{cases}$$

Let

$$\tilde{I} \;=\; \{I' \subset I \mid \mathrm{card}\, I' \leq \pi, \quad I' \text{ large relative to } \pi_0\},$$

endowed with the inclusion order relation. For

$$I' \;\in\; \tilde{I}, \qquad \text{let} \quad X_{I'} = \varinjlim_{I'} X_i \;,$$

this limit exists by L_{π_0} (hypothesis *b*), it is in Filt^{π} by *b*), so we have a direct system

$$(X_{I'})_{I'\in\tilde{I}}, \qquad X_{I'} \in \mathrm{Filt}^{\pi},$$

with

$$\mathrm{card}\,\tilde{I} \;\leq\; (\mathrm{card}\, I)^{\pi},$$

and if $X \in \mathrm{Filt}^{\pi'}$ $(\pi' > \pi)$, taking

$$\mathrm{card}\, I \;\leq\; \pi'^{\pi},$$

we shall have $\mathrm{card}\,\tilde{I} \leq (\pi'^{\pi})^{\pi} = \pi'^{\pi^2} = \pi'^{\pi}$, hence

$$\mathrm{card}\,\tilde{I} \;\leq\; \pi'^{\pi}. \qquad \text{[page 80]}$$

I say that $\tilde{I}$ is itself also *large relative to* π, just like I. Indeed, let

$$(I_\alpha)_{\alpha\in A}, \qquad I_\alpha \in \tilde{I}, \qquad \operatorname{card} A \ \leq\ \pi,$$

let us prove that there exists $I' \in \tilde{I}$ which is an upper bound for the I_α, *i.e.* which contains $I'_0 = \bigcup_{\alpha\in A} I_\alpha$. We shall have $\operatorname{card} I'_0 \leq \pi^2 = \pi$ (since $\operatorname{card} A \leq \pi$ and $\operatorname{card} I_\alpha \leq \pi$ for every α), and it suffices to see this:

Lemma 1: *Let π_0, π be two cardinals, with $\pi \geq \pi_0$, I an ordered set large relative to π, and I'_0 a subset of I of cardinality $\leq \pi$. Then there exists a subset I' of I satisfying the conditions:* ([40])

a) $I' \supset I'_0$.

b) $\operatorname{card} I' \leq \pi$.

p. 44

c) *I' large relative to π_0.*

We are going to try to construct I' by transfinite induction, defining the $I'_\alpha \subset I$ as follows:

$$I'_0 = I'_0\ ,$$
$$I'_\alpha = \textstyle\bigcup_{\beta<\alpha} I'_\beta \quad \text{if } \alpha \text{ is a limit ordinal,}$$
$$I'_{\alpha+1} = \begin{cases} \text{derived from } I'_\alpha \text{ as follows: for every subset } J \text{ of } I'_\alpha \\ \text{such that } \operatorname{card} J \leq \pi_0, \text{ which is not bounded above in} \\ I'_\alpha, \text{ we choose an upper bound } i_J \text{ in } I \text{ (exists, since } I \\ \text{large relative to } \pi, \text{ hence to } \pi_0) \text{ and } I'_{\alpha+1} \text{ is obtained} \\ \text{by adjoining these } i_J \text{ to } I'_\alpha. \end{cases}$$

Let then α_0 be the first ordinal of cardinality $> \pi_0$, and consider [page 81]

$$I' \ =\ I'_{\alpha_0} \ =\ \bigcup_{\substack{\alpha<\alpha_0 \\ \textit{i.e. } \operatorname{card}\alpha\leq\pi_0}} I'_\alpha.$$

Then I' contains I'_0, it is large relative to π_0, but what is its cardinality? Let us prove by transfinite induction that

$$\operatorname{card} I'_\alpha \ \leq \pi \qquad \text{if } \alpha < \alpha_0, \quad \textit{i.e.} \quad \operatorname{card}\alpha \leq \pi_0.$$

This is O.K. for $\alpha = 0$ by the hypothesis on I'_0. If it is true for I'_α, is it true for $I'_{\alpha+1}$? We adjoin to I'_α a set of elements i_J whose cardinality is $\leq \operatorname{card}\mathfrak{P}_{\leq\pi_0}(I'_\alpha)$, hence of cardinality $\leq \operatorname{card}(I'_\alpha)^{\pi_0} \leq \pi^{\pi_0}$, and if we suppose

$$\pi^{\pi_0} \ =\ \pi,$$

we win. Finally suppose that α is a limit ordinal (of cardinality $\leq \pi_0$). So we have

$$I'_\alpha \ =\ \bigcup_{\beta<\alpha} I'_\beta,$$

so

$$\operatorname{card} I'_\alpha \ \leq\ \underbrace{(\operatorname{card}\alpha)\pi}_{\leq\pi\pi=\pi^2=\pi} \qquad \text{since the } \operatorname{card} I'_\beta \leq \pi,$$

p. 45 [page 82]

[39] NB *b*) and *c*) imply that $E = \bigcup_{\pi\geq\pi_0} \operatorname{Filt}^\pi(E)$.

[40] Provided one assumes $\pi > \pi_0$, and moreover $\pi^{\pi_0} = \pi$ (e.g. $\pi = 2^c$ with $c \geq \pi_0$).

we win again.

Finally we find, by the same argument,

$$\operatorname{card} I'_{\alpha_0} \;\leq\; \underbrace{(\operatorname{card}\alpha_0)\pi \leq \pi^2}_{\substack{\text{if } \operatorname{card}\alpha_0 \leq \pi, \\ \textit{i.e.} \text{ if } \pi_0 < \pi}} = \pi,$$

so we win again.

Thus, we have found a second direct system

$$(X_{\tilde{\imath}})_{\tilde{\imath}\in\tilde{I}}$$

of the same type as the initial system, having as direct limit X (that it has the same direct limit is immediate). But what have we gained? This: in $\tilde{I}$ every subset of cardinality $\leq \pi$ and large relative to π_0 is not only *bounded above* in $\tilde{I}$, but even admits a least upper bound there. This follows from

Lemma 2: *Let I be large relative to $\pi > \pi_0$, with $\pi^{\pi_0} = \pi$ (cf. lemma 1). Let* [page 83]
$(I'_\alpha)_{\alpha\in A}$ be a set of subsets of I which are in $\tilde{I}$, i.e.

a) *of cardinality $\leq \pi$, and*

b) *large relative to π_0,*

and suppose that $\operatorname{card} A \leq \pi$ *and that A, ordered by inclusion, is large relative to π_0. Then $I' = \bigcup_{\alpha\in A} I'_\alpha$ is still in $\tilde{I}$,* i.e. *I' is of cardinality $\leq \pi$ (since of cardinality $\leq \pi^2 = \pi$), and moreover, large relative to π_0.* (Immediate verification.)

Thus, we have proved this:

Proposition 1. *Let E be big category, with cardinal filtration* $(\mathrm{Filt}^\pi)_{\pi\geq\pi_0}$. *Let π be a cardinal with*

$$\pi > \pi_0, \qquad \pi^{\pi_0} = \pi \qquad \textit{(e.g. } \pi = 2^c, \textit{ with } c \geq \pi_0\textit{)}$$

Then for every X in E, we can find a direct system $(X_i)_{i\in I}$, with limit X, with

a)
$$\begin{cases} X_i \in \mathrm{Filt}^\pi \;\; \textit{for every } i \in I, \\ I \textit{ large relative to } \pi, \end{cases}$$

satisfying the following additional conditions:

b) *Every subset J of I of cardinality $\leq \pi$, and large relative to π_0, admits a least upper bound i_J in I.*

c) *We have, under these conditions*

$$X_{i_J} \xleftarrow{\sim} \varinjlim_{i\in J} X_i. \qquad \text{p. 46}$$ [page 84]

Moreover, if $X \in \mathrm{Filt}^{\pi'}$ with $\pi' > \pi$, we can take I of cardinality $\leq \pi'^\pi$ (hence of cardinality $\leq \pi'$ if $\pi'^\pi = \pi'$, e.g. if $\pi' = 2^{c'}$, with $c' \geq \pi$).

Definition. We call a direct system $(X_i)_{i\in I}$, satisfying conditions *a*), *b*), *c*) above, a *well-adapted direct system* (to the given cardinal filtration, for the given cardinals π_0, π), and the corresponding representation

$$X \xleftarrow{\sim} \varinjlim X_i$$

a *well-adapted representation* of X in terms of objects of Filt^π.

NB When E is stable under small filtering $\varinjlim$, the situation simplifies considerably. We can then take

$$\tilde{I} \;=\; \{I' \subset I \mid \mathrm{card}\, I' \leq \pi,\ I' \text{ filtering}\}$$

(without burdening ourselves with the condition I' large relative to π_0) and again define the direct system $(X_{I'})_{I' \in \tilde{I}}$ by

$$X_{I'} \;=\; \varinjlim_{i \in I'} X_i\ .$$

We again have [page 85]

$$\mathrm{card}\, \tilde{I} \;\leq\; (\mathrm{card}\, I)^\pi \;\leq\; \pi'^\pi \qquad \text{if} \quad \mathrm{card}\, I \leq \pi'^\pi$$

(in particular, if $X \in \mathrm{Filt}^{\pi'}$, and we take I as indicated in condition c of cardinal filtrations). Moreover, this time, the fact that $\tilde{I}$ is large relative to π is trivial, we even have better, apart from I being filtering:

(∗) *Every* subset $(I'_\alpha)_{\alpha \in A}$ of $\tilde{I}$ which is filtering and of cardinality $\leq \pi$ admits a least upper bound $\tilde{\imath}_A$,

namely the union (which is indeed filtering). Moreover, the direct system $(X_{\tilde{\imath}})_{\tilde{\imath} \in \tilde{I}}$ has the stronger property that

$$X_{\tilde{\imath}_A} \xleftarrow{\sim} \varinjlim_{\tilde{\imath} \in A} X_{\tilde{\imath}}\ .$$

Thus we have

Proposition 1 bis. *Suppose E is stable under small filtering $\varinjlim$, and Filt^π stable under these, where $\pi \geq \pi_0$ is given. Then for every X in E, we can find a direct system $(X_i)_{i \in I}$ with direct limit X, with*

a) *$X_i \in \mathrm{Filt}^\pi$ for every $i \in I$, I an ordered set large relative to π.*

b) *Every filtering subset J of I of cardinality $\leq \pi$ admits a least upper bound i_J in I.*

c) *Under these conditions*

$$X_{i_J} \;=\; \varinjlim_{i \in J} X_i\ .$$

p. 47
[page 86]

Such direct systems will be called *"perfectly adapted"* (to the given cardinal filtration, for the given cardinal $\pi - \pi_0$ no longer figures in it, except as the beginning of the cardinal filtration and nothing more).

Since in practice, we will have to handle mostly E stable under small filtering $\varinjlim$, and cardinal filtrations stable accordingly (strengthening of condition b) of cardinal filtrations), it is this case, and the notion of *perfectly adapted representation* of X as $\varinjlim_I X_i$ ($X_i \in \mathrm{Filt}^\pi$), that I will put forward. The "general" case, where we suppose only E satisfies L_{π_0}, is more convoluted and is handled less prettily ([41]).

Definition 2. Let E be equipped with a strict cardinal filtration $(\underline{\mathrm{Filt}}^\pi)_{\pi \geq \pi_0}$. Let M be a subset of $\mathrm{Ob}\, E$, $\underline{M}$ the full subcategory of E which it defines, and let π be a cardinal $\geq \pi_0$. We say that M is *π-accessible,* or that $\underline{M}$ is *π-accessible*, if the following conditions are satisfied. For every direct system [page 87]

$$(X_i)_{i \in I}$$

[41] We will call a cardinal filtration *strict,* if the $\underline{\mathrm{Filt}}^\pi$ are stable under filtering $\varinjlim$ of cardinality $\leq \pi$ (and these small filtering limits exist in E).

perfectly adapted to the cardinal filtration for π (hence I a filtering ordered set such that every filtering subset J of cardinality $\leq \pi$ has a least upper bound i_J, and $X_{i_J} \xleftarrow{\sim} \varinjlim_{j\in J} X_j$); let I_M be the set of $i \in I$ such that $X_i \in M$. With this understood, we want this: for X ($= \varinjlim X_i$) to be in M, it is necessary and sufficient that I_M be cofinal in I (so that X appears as a filtering $\varinjlim$, with index set I_M large relative to π, of elements of $M \cap \underline{\mathrm{Filt}}^\pi \overset{\mathrm{def}}{=} M^\pi$).

I feel like taking up again the definition of the π-accessibility of M, for a given cardinal filtration, by way of equivalent conditions. For every direct system $(X_i)_{i\in I}$ in E , I keep the notation [page 88]

$$I_M \;=\; \{i \in I \mid X_i \in M\}.$$

I set, for every cardinal $\pi \geq \pi_0$,

$$M^\pi \;=\; M \cap \mathrm{Filt}^\pi.$$

(a$_\pi$) M is the set of objects of the form

$$\varinjlim_{i\in I} X_i \ ,$$

where the X_i are in M^π, and I is large relative to π.

p. 48

(b$_\pi$) Let $(X_i)_{i\in I}$ be a direct system in E perfectly adapted to the cardinal filtration for π, *i.e.* I filtering and every filtering subset J of I of cardinality $\leq \pi$ admits a least upper bound i_J, and $X_{i_J} \xleftarrow{\sim} \varinjlim_{J} X_i$, and finally $X_i \in \mathrm{Filt}^\pi$ for every $i \in I$. Let $X = \varinjlim X_i$. In order to have $X \in M$, it is necessary and sufficient that I_M be cofinal in I.

(c$_\pi$) $\underline{M}^\pi = \underline{M} \cap \underline{\mathrm{Filt}}^\pi$ is a subcategory of $\underline{M}$ generating by strict epimorphisms, $\underline{M}$ is stable under $\varinjlim$ large relative to π, finally for every $X \in M$, $\underline{M}^\pi/X$ is a filtering category.

I presume that these conditions are equivalent. [page 89]

$b_\pi \Longrightarrow a_\pi$. We suppose b) holds, we want to prove a). First that every $X \in M$ is of the stated form $\varinjlim_{I} X_i$, with the $X_i \in M^\pi$ and I large relative to π. By proposition 1, we know that we have a perfect adapted presentation for π, $X = \varinjlim_{i\in I} X_i$. By b) "it is necessary", the subset I_M of I is cofinal in I, so since I is large relative to π, so is I_M, and we will have (setting $J = I_M$)

$$X \;=\; \varinjlim_{i\in J} X_i, \qquad\qquad X_i \in M^\pi, \quad J \text{ large relative to } \pi,$$

O.K. Conversely, let us prove that if X can be put in the form stated in a), it is in M. This will follow from

Lemma 1: *If condition* b) (*or* a)) *is satisfied, M is stable under direct limits large relative to π.*

We must prove that if [page 90]

$$X \;=\; \varinjlim_{i\in I} X_i \ ,$$

with I large relative to π, the X_i in M, then $X \in M$. Condition b) "it is sufficient" ensures this when we suppose in addition the X_i in Filt^π (hence in M^π), so that we will have $I_M = I$, hence I_M cofinal, hence $X \in M$ ([42]). If we do not suppose the X_i in Filt^π, we must work a little more.

Go straight to the part of the proof below, page 96, which proves $b) \Longrightarrow a)$. The rest of the proof will prove the

Lemma 2. *If every element of Filt^π is π-accessible, then b_π) (and even a_π) implies that M is stable under $\varinjlim$ large relative to π.*

Let p. 49 [page 91]

$$\mathcal{C} \;=\; \underline{M}^\pi/\mathcal{X}$$

be the following category. Its objects are the triples

$$(Z, i, \alpha) \quad \text{such that} \quad Z \in M^\pi,\ i \in I,\ \alpha : Z \longrightarrow X_i\ ,$$

the arrows

$$(Z, i, \alpha) \;\longrightarrow\; (Z', i', \alpha')$$

are the pairs (u, λ) with

$$u : Z \longrightarrow Z', \qquad \lambda : i \longrightarrow i', \qquad (\textit{i.e.}\ \text{we have } i \leq i' \text{ and } \lambda \text{ is the unique arrow from } i \text{ to } i'),$$

such that the diagram

$$\begin{array}{ccc} Z & \xrightarrow{\ \alpha\ } & X_i \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \text{transition } \lambda_{i',i}} \\ Z' & \xrightarrow{\ \alpha'\ } & X_{i'} \end{array}$$

commutes. We can therefore define $\mathcal{C}$ as the fiber product in the cartesian square

$$\begin{array}{ccccc} & & \mathcal{C} & \xrightarrow{\text{cofibered}} & I \\ & \overset{\mathcal{Z}}{\swarrow} & \downarrow & & \downarrow{\scriptstyle \mathcal{X}:i\longmapsto X_i} \\ \underline{M}^\pi & \xleftarrow{\ \text{source}\ } & \underline{M}^\pi/E & \xrightarrow[\text{cofibered}]{\text{target}} & E. \end{array}$$

The fiber $\mathcal{C}_i$ of $\mathcal{C}$ at $i \in I$ is M^π/X_i . We have, as in $b) \Longrightarrow$ first half of a), that $\underline{M}^\pi$ is a system of strict generators of $\underline{M}$, and $X_i \in \mathrm{Ob}\,\underline{M}$ by hypothesis, [page 92]

$$X_i \;=\; \varinjlim_{Z \in \mathrm{Ob}\,\underline{M}^\pi/X_i} Z \;=\; \varinjlim_{\mathcal{C}_i} \mathcal{Z}|\mathcal{C}_i,$$

so the given direct system $(X_i)_{i\in I}$ on I with values in E, is deduced from the direct system (not ordered in general) on $\mathcal{C}$, with values in $\underline{M}^\pi$, by "integrating over the fibers" of $\mathcal{C} \longrightarrow I$. Besides, since $\underline{M}^\pi/E \longrightarrow E$ is cofibered, hence proper, the same is true of $\mathcal{C} \longrightarrow I$, so the $\varinjlim$ of $\mathcal{Z}$ over the fibers $\mathcal{C}_i$ $(= \underline{M}^\pi/X_i)$ do form a covariant functor $I \longrightarrow E$, whose direct limit is that of $\mathcal{Z}$ ([43]).

Let, for every $i \in I$, $\tilde{\mathcal{C}}_i$ be the ordered set p. 50 [page 93]

$$\tilde{\mathcal{C}}_i = \{\text{subcategories } \mathcal{C}'_i \text{ of } \mathcal{C}_i = \underline{M}^\pi/X_i \mid \mathrm{card}\,\mathcal{Z}(\mathcal{C}'_i) \leq \pi,\ \ \mathcal{C}'_i \text{ filtering}\}.$$

[42] Not quite – we assume more there, about the direct system. We cannot avoid, even in the case $X_i \in M^\pi\ \forall\, i \in I$, making the argument of the following pages.

[43] **NB** If we admit the validity of a), then it already follows that $\varinjlim_{i\in I} X_i \simeq \varinjlim_{\alpha\in C} Z_\alpha$ is in M. (Stability of M under $\varinjlim$ large relative to π).

I admit for the moment the

Lemma ([44]). *For every $X \in M$, the category $\underline{M^\pi}/X$ is filtering. (Under hypothesis* b), *or that every object of M is of the form stated in* a).*)*

It follows, since $X_i \in M$, that $\mathcal{C}_i$ is filtering, so the union of the $\mathcal{C}'_i \in \tilde{\mathcal{C}}_i$ is $\mathcal{C}_i$; the ordered set $\tilde{\mathcal{C}}_i$ is filtering, and every filtering subset $(\mathcal{C}'_{i_\alpha})_{\alpha\in A}$ of cardinality $\leq \pi$ of $\tilde{\mathcal{C}}_i$ has a supremum $\mathcal{C}'_{i_A}$ there, namely the union of the $\mathcal{C}'_{i_\alpha}$. $\tilde{\mathcal{C}}_i$ is therefore a π-adapted ordered set.

We can regard the ordered sets $\tilde{\mathcal{C}}_i$, for varying $i \in I$, as the fibers of an increasing map of ordered sets

$$p \,:\, \tilde{\mathcal{C}} \;\longrightarrow\; I,$$

by writing

$$\begin{cases} (\mathcal{C}'_i \subset \mathcal{C}_i) \;\leq\; (\mathcal{C}'_j \subset \mathcal{C}_j) \\ \text{if and only if } i \leq j, \\ \text{and the canonical transition functor } \mathcal{C}_i \longrightarrow \mathcal{C}_j \text{ maps } \mathcal{C}'_i \text{ into } \mathcal{C}'_j. \end{cases}$$

We thus see that $\tilde{\mathcal{C}}$ too is filtering and large relative to π.

We define a direct system on $\tilde{\mathcal{C}}$, with values in $\underline{\mathrm{Filt}}^\pi$, by setting [page 94]

$$Z_{\mathcal{C}'_i} \;=\; \varinjlim_{\alpha\,\in\,\mathcal{C}'_i \;(\subset \mathcal{C}_i = \underline{M^\pi}/X_i)} Z_\alpha$$

(where an $\alpha \in \mathcal{C}_i$ consists, I recall, of $Z_\alpha \xrightarrow{u_\alpha} X_i$, with $Z_\alpha \in M^\pi \subset \underline{\mathrm{Filt}}^\pi$). These $Z_{\mathcal{C}'_i}$ are no longer necessarily in M^π, but only in Filt^π. (Then what was the use of defining the $\mathcal{C}_i$ themselves as $\underline{M^\pi}/X_i$, instead of $\underline{\mathrm{Filt}}^\pi/X_i$?)

The direct system

$$(Z_{\mathcal{C}'_i})_{\mathcal{C}'_i \in \tilde{\mathcal{C}}_i}$$

is perfectly well adapted to the cardinal filtration for the cardinal π, and it is immediate that its $\varinjlim$ is isomorphic to $\varinjlim_{\alpha\in \mathrm{Ob}\,\mathcal{C}_i = \underline{M^\pi}/X_i} Z_\alpha = X_i \in M$. Thus by b), the set $\tilde{\mathcal{C}}_M$ of the $\mathcal{C}'_i \in \tilde{\mathcal{C}}_i$ such that $Z_{\mathcal{C}'_i} \in M$ (so $Z_{\mathcal{C}'_i} \in M^\pi$) is cofinal in $\tilde{\mathcal{C}}_i$. It follows therefore that the set $\tilde{\mathcal{C}}_M$ is also cofinal in $\tilde{\mathcal{C}}$.

We also check immediately that

p. 51

[page 95]

$$\varinjlim_{\alpha\in\tilde{\mathcal{C}}} Z_\alpha \;=\; \varinjlim X_i \;=\; X,$$

so, $\tilde{\mathcal{C}}_M$ being cofinal,

$$\varinjlim_{\alpha\in\tilde{\mathcal{C}}_M} Z_\alpha \;=\; X.$$

All in all, we have managed to represent X in the form

$$X \;=\; \varinjlim_{\alpha\in A} X_\alpha,$$

where this time the X_α are *in* M^π (not just in M), and A is large relative to π. But we do not know that every filtering subset B of cardinality $\leq \pi$ of A has a supremum α_B in A, and still less that $X_{\alpha_B} = \varinjlim_{\alpha\in B} X_\alpha$ ([45]). So we cannot yet apply b) directly. If it were only a matter of proving the second half of a), where we already assume the $X_i \in M^\pi$ and I large relative to π, we have gained nothing at all by our labors!

[44] Uses the tacit hypothesis that the elements of Filt^π are π-accessible.

[45] If we assume (a_π), this argument proves at least that M is stable under $\varinjlim$ large relative to π. [page 96]

But let us return to this case

$$X \;=\; \varinjlim X_i \;\Bigg|\; \begin{array}{l} X_i \in M^\pi \\ \\ I \text{ large relative to } \pi, \end{array}$$

and introduce again

$$\tilde{I} \;=\; \{I' \subset I \mid I' \text{ filtering, of cardinality } \leq \pi\},$$

so this time $\tilde{I}$ is indeed adapted to π (filtering, and every filtering subset of cardinality $\leq \pi$ has a supremum), and if we set

$$X_{\tilde{\imath}} \;=\; \varinjlim_{i \in I'_{\tilde{\imath}}} X_i,$$

we see that the direct system

$$(X_{\tilde{\imath}})_{\tilde{\imath} \in \tilde{I}}$$

is perfectly adapted to π and the cardinal filtration. Moreover, I say that the set $\tilde{I}_M$ is *cofinal* in $\tilde{I}$. Indeed, $\tilde{I}_M$ contains the subset $\tilde{I}_0$ consisting of the $I' \in \tilde{I}$ which have a greatest element – and $\tilde{I}_0$ is indeed cofinal! Now, we can apply the "it suffices" in b), which concludes that $X = \varinjlim X_i \in M$, q.e.d. p. 52 [page 97]

We have proved $b \Longrightarrow a$, without using the hypothesis that every object of Filt^π is π-accessible. But we have used this hypothesis to prove that b (or a) implies that M is stable under $\varinjlim$ large relative to π (lemma 2, p. 90).

$a \overset{?}{\Longrightarrow} b$. The "it suffices" in b) is an immediate consequence of a). It remains to prove that if, in the π-adapted direct system $(X_i)_{i \in I}$ with limit X, we have $X \in M$, then I_M is cofinal in I. But by hypothesis a), we will have $X = \varinjlim_J Y_j$, with J large relative to π, the Y_j in M^π. Let us admit that $\mathrm{Filt}^\pi \subset E_\pi$, *i.e.* every element of Filt^π, in particular the X_i, are π-accessible.

I think I will not manage to prove $a_\pi \Longrightarrow b_\pi$, but only this: If π' is a cardinal $> \pi$ (say the successor of π in the ordinal sequence of cardinals), then $a_\pi \Longrightarrow b_{\pi'}$.

This will follow from [page 98]

Lemma 3. *Let $X \simeq \varinjlim_I X_i$, $X \simeq \varinjlim_J Y_j$ be two representations of X in the form of filtering $\varinjlim$, with I large relative to π', J large relative to π, with π, π' two cardinals such that $\pi' > \pi$. We assume $(X_i)_{i \in I}$ π'-adapted,* i.e. *for every filtering subset I' of I of cardinality $\leq \pi'$, I' has a supremum $i_{I'}$, and $X_{i_{I'}} = \varinjlim_{i \in I'} X_i$. Then for every $i \in I$, we can find a subset I' of I, containing i,* large relative to π, *of cardinality $\leq \pi'$, and a subset J' of J, finally an isomorphism $I' \xrightarrow[\sim]{\lambda} J'$ and a strictly increasing map $\mu : J' \longrightarrow I'$ and homomorphisms*

p. 53

Index of notations

LLP, RLP 1
Φ_*, Ψ^* 1
$\widetilde{\Phi}$, $\overline{\Psi}$ 1
$\Phi \longleftrightarrow \Psi$ 3
Cof_W, Fib_W 8
W^{univ} 9

W_{univ} . . . 10
$\underline{\mathrm{Fib}}_W S \subset \mathcal{M}/S$ . . . 10
W_S . . . 10
W_S^{u} . . . 10
$S \backslash \underline{\mathrm{Cof}}_W$ . . . 10
$\Phi \leftarrow\!\!\!| \!\!\rightarrow \Psi$ . . . 28
$\mathcal{M}_{\mathrm{cf}}$, W_{cf} . . . 29
Ho_W . . . 29
Hom_W . . . 30
π_{d}, π . . . 30
$\mathcal{M}_{\mathrm{c}}$ . . . 30
W_{c} . . . 31
π_{g} . . . 31
$\mathcal{M}_{\mathrm{f}}$ . . . 31
$f \sim g$ . . . 32
Ψ, Φ, $\mathcal{M}(\Psi)$, $W(\Psi)$, $\mathcal{M}(\Phi)$, $W(\Phi)$ . . . 33
$\mathcal{M}(I)$, $W(I)$ . . . 34
$F(I)$, $TF(I)$, $C(I)$, $TC(I)$ . . . 35
α_s . . . 35
u^*, u_*, $u_!$ . . . 45
u_W^*, u_*^W, $u_!^W$ . . . 49
$\mathrm{HO}_W(I)$ . . . 49
Diag . . . 50
$\mathrm{HO}_{\mathcal{M},W}(I)$, $\mathbf{D}_{\mathcal{M},W}(I)$ . . . 54
$\mathrm{HO}_W^{\mathrm{lc}}(I)$, u_{lc}^* . . . 56
u_*^{f} . . . 59
u_{c}^* . . . 67
$\mathfrak{U}$ . . . 77
$\mathrm{Filt}^\pi(E)$ . . . 77
L_π . . . 77 p. 54
$\tilde{I}$ . . . 79
$\underline{M}$ . . . 86
I_M . . . 87
M^π . . . 87 p. 55

Terminological index

accessible (left closed subset) . . . 75
admissible (on the left) . . . 59
category of models . . . 7
closed Quillen category . . . 25
left closed, right closed . . . 2
Quillen couple . . . 5
weakly saturated . . . 7
cardinal filtration . . . 77
strict cardinal filtration . . . 86
arrow couniversally in W . . . 9
arrow universally in W . . . 10
large relative to π . . . 77
path object . . . 32

cylinder object 32
π-accessible (subset of a category equipped with a strict cardinal filtration) 86
left, right pre-Q-category of models 12
left, right pre-Quillen 14
closed Quillen pre-quintuple 21
proper 8
Quillen factorization property 5
Q-category of models 8
right Q-category of models 21
"closed" Q-category of models 16
left, right Q-homotopy 30
Quillenizable 50
Quillenizable on Diag 51
Quillenizable on Diag uniformly and left exactly 57
uniformly left Quillenizable on Diag 57
uniform right Quillenization on Diag 58
uniform left Quillenization on Diag 57
uniform and left exact Quillenization on Diag 57
well-adapted representation 84
perfectly adapted representation 86
strictly admissible 61
Quillen structure 28
exact Quillen structure . . . . . 27 p. 56
left exact, right exact Quillen structure 27
weak Quillen structure 27
biexact weak Quillen structure 28
closed, left closed, right closed weak Quillen structure 28
perfect Quillen structure 28
closed Quillen system 26
weak Quillen system 26
left exact, right exact weak Quillen system 26
well-adapted direct system 84
perfectly adapted direct system 86
closed Quillen triple 25
Quillen triple 14
W-cofibration 8
couniversal W-equivalence 9
W-fibration . . . . . 8 p. 57

List of axioms

$\mathcal{M}$ category
$W \subset \mathrm{Fl}(\mathcal{M})$, $\quad C, F, TC, TF \subset \mathrm{Fl}(\mathcal{M})$

M 0 The category $\mathcal{M}$ admits finite inverse limits and finite direct limits.

M 1 The arrows belonging to C have the left lifting property relative to those belonging to TF, and the arrows belonging to F have the right lifting property relative to those belonging to TC.

M 2 Every arrow of $\mathcal{M}$ decomposes into an arrow of C followed by an arrow of TF, and into an arrow of TC followed by an arrow of F.

M 3 The subset F of $\mathrm{Fl}(\mathcal{M})$ is stable under composition and base change, C is stable under composition and cobase change, F and C contain the isomorphisms.

M 4 The subset TF of $\mathrm{Fl}(\mathcal{M})$ is stable under composition and base change, TC is stable under composition and cobase change, TF and TC contain the isomorphisms.

M 5 If in a commutative triangle of $\mathcal{M}$ two of the three arrows are in W, the same holds for the third ; W contains the isomorphisms.

M′ $TC \subset C \cap W, TF \subset F \cap W.$

M″ The arrows belonging to C are W-cofibrations, and the arrows belonging to F are W-fibrations.

M′$_0$(g) $TC = C \cap W.$

M′$_0$(d) $TF = F \cap W.$

M 2 bis $W = TF \circ TC.$

p. 58

Glossary

Section 1

- $\Phi \subset \mathrm{Fl}(\mathcal{M})$

  Φ is *left closed* : there exists $\Psi \subset \mathrm{Fl}(\mathcal{M})$ such that Φ consists of the arrows having the left lifting property relative to the arrows of Ψ.

- $\Psi \subset \mathrm{Fl}(\mathcal{M})$

  Ψ is *right closed* : there exists $\Phi \subset \mathrm{Fl}(\mathcal{M})$ such that Ψ consists of the arrows having the right lifting property relative to the arrows of Φ.

- $\Phi, \Psi \subset \mathrm{Fl}(\mathcal{M})$

  (Φ, Ψ) has the *Quillen factorization property* :

  a) the arrows of Φ have the left lifting property relative to the arrows of Ψ ;

  b) every arrow f of $\mathcal{M}$ factors as

  $$f = pi \quad , \qquad i \in \Phi \ , \quad p \in \Psi \ .$$

- (Φ, Ψ) *Quillen couple* :

  a) (Φ, Ψ) has the Quillen factorization property ;

  b) Φ is left closed and Ψ right closed.

Section 2

- $W \subset \mathrm{Fl}(\mathcal{M})$

  W *weakly saturated* :

  a) W contains the isomorphisms ;

  b) if two among the three arrows of a commutative triangle are in W, the third too.

- $W \subset \mathrm{Fl}(\mathcal{M})$

  A morphism $p : X \longrightarrow Y$ of $\mathcal{M}$ is a W-*fibration* ($p \in \mathrm{Fib}_W$) if for every diagram of cartesian squares

p. 59

$$\begin{array}{ccccc} X & \longleftarrow & X' & \xleftarrow{t} & X'' \\ {\scriptstyle p}\downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & Y' & \xleftarrow[s]{} & Y'' \end{array} \quad ,$$

  if $s \in W$, then $t \in W$.

- $W \subset \mathrm{Fl}(\mathcal{M})$

  A morphism $i : A \longrightarrow B$ of $\mathcal{M}$ is a W-*cofibration* ($i \in \mathrm{Cof}_W$) if for every diagram of cocartesian squares

$$\begin{array}{ccccc} A'' & \xleftarrow{s} & A' & \longleftarrow & A \\ \downarrow & & \downarrow & & {\scriptstyle i}\downarrow \\ B'' & \xleftarrow[t]{} & B' & \longleftarrow & B \end{array} \quad ,$$

  if $s \in W$, then $t \in W$.

- $W \subset \mathrm{Fl}(\mathcal{M})$

  A morphism $p : X \longrightarrow Y$ of $\mathcal{M}$ is *universally in* W, or a W-*universal equivalence* ($p \in W_{\mathrm{univ}}$) if for every cartesian square

$$\begin{array}{ccc} X & \longleftarrow & X' \\ {\scriptstyle p}\downarrow & & {\scriptstyle p'}\downarrow \\ Y & \longleftarrow & Y' \end{array} \quad ,$$

  the arrow p' belongs to W.

- $W \subset \mathrm{Fl}(\mathcal{M})$

  A morphism $i : A \longrightarrow B$ of $\mathcal{M}$ is *couniversally in* W, or a W-*couniversal equivalence* ($i \in W^{\mathrm{univ}}$) if for every cocartesian square

$$\begin{array}{ccc} A' & \longleftarrow & A \\ {\scriptstyle i'}\downarrow & & {\scriptstyle i}\downarrow \\ B' & \longleftarrow & B \end{array} \quad ,$$

  the arrow i' belongs to W.

p. 60

- *Left pre-Q-category of models* (resp. *right*):

  $\mathcal{M}$ and W, C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$ satisfying axioms **M 0** to **M 5**, $\mathbf{M}'$, $\mathbf{M}''$ and $\mathbf{M}'_{\mathbf{0}}(\mathbf{g})$ (resp. $\mathbf{M}'_{\mathbf{0}}(\mathbf{d})$).

- *Q-category of models*, *Quillen triple*:

  model category in the sense of Quillen.

- *"Closed" Q-category of models*, *closed Quillen triple*, *closed Quillen category*, *closed Quillen system*:

  closed model category in the sense of Quillen.

- *Left pre-Quillen* (resp. *right*):

  left pre-Q-category of models (resp. right).

- *Closed Quillen pre-quintuple*:

  W, C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$ satisfying axioms **M 0** to **M 5**, $\mathbf{M}'$, $\mathbf{M}''$, C, TC being left closed and F, TF right closed.

- *Weak Quillen system*:

  W, C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$ satisfying axioms **M 0** to **M 5**, $\mathbf{M}'$, $\mathbf{M}''$.

- *Left exact weak Quillen system* (resp. *right*):

  W, C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$ satisfying axioms **M 0** to **M 5**, $\mathbf{M}'$, $\mathbf{M}''$ and $\mathbf{M}'_{\mathbf{0}}(\mathbf{g})$ (resp. $\mathbf{M}'_{\mathbf{0}}(\mathbf{d})$).

Section 3

- $(\mathcal{M}, W)$ *category of models*:

  $\mathcal{M}$ a category satisfying axiom **M 0**;

  $W \subset \mathrm{Fl}(\mathcal{M})$ weakly saturated.

- *Weak Quillen structure associated with the category of models* $(\mathcal{M}, W)$:

  quadruple (C, F, TC, TF) of subsets of $\mathrm{Fl}(\mathcal{M})$ satisfying axioms **M 1** to **M 5**, $\mathbf{M}'$, $\mathbf{M}''$.

- *Exact weak Quillen structure*:

  weak Quillen structure satisfying in addition axiom **M 2 bis**.

- *Left exact weak Quillen structure* (resp. *right*):

  weak Quillen structure satisfying in addition axiom $\mathbf{M}'_{\mathbf{0}}(\mathbf{g})$ (resp. $\mathbf{M}'_{\mathbf{0}}(\mathbf{d})$).

p. 61

- *Left closed weak Quillen structure* (resp. *right*):

  weak Quillen structure (C, F, TC, TF) such that C, TC (resp. F, TF) are left closed (resp. right closed).

- *Closed weak Quillen structure*:

  weak Quillen structure both left and right closed.

- *Quillen structure*:

  "biexact" weak Quillen structure:

  weak Quillen structure both left and right exact

  = proper Quillen model category.

- *Perfect Quillen structure*:

  "biexact" and closed weak Quillen structure

  = proper closed Quillen model category.

Section 5

- Category of models $(\mathcal{M}, W)$ *Quillenizable*:

  there exists an *exact* weak Quillen structure (C, F, TC, TF) associated with $(\mathcal{M}, W)$.

- $(\mathcal{M}, W)$ *Quillenizable on* Diag $\subset$ Cat:

  for every I in Diag, the category of models $(\mathcal{M}(I), W(I))$ is Quillenizable.

Section 6

- Category of models $(\mathcal{M}, W)$ *uniformly left Quillenizable on* Diag:

  there exists a *closed* weak Quillen structure (C, F, TC, TF) for $(\mathcal{M}, W)$ such that for every I in Diag, if one defines $C(I)$ and $TC(I)$ "argument by argument" (*), and $F(I)$ and $TF(I)$ by the right lifting property with respect to $TC(I)$ and $C(I)$ respectively (†), the quadruple $(C(I), TC(I), F(I), TF(I))$ is a *closed exact* weak Quillen structure for $(\mathcal{M}(I), W(I))$. One then says that (C, F, TC, TF) is a *uniform left Quillenization* of $(\mathcal{M}, W)$ on Diag).

p. 62

- Category of models $(\mathcal{M}, W)$ *Quillenizable on* Diag *uniformly and exactly on the left*:

  $(\mathcal{M}, W)$ admits a uniform left Quillenization on Diag (C, F, TC, TF) such that $TC = C \cap W$ (in other words, such that (C, F, TC, TF) is a *left exact* weak Quillen structure). One then says that (C, F, TC, TF) is a *uniform and left exact Quillenization* of $(\mathcal{M}, W)$ on Diag.

Section 7

- I a filtering ordered set, π a cardinal

  I *large relative to* π:

  every subset of I of cardinality less than or equal to π admits an upper bound in I.

*A morphism of functors $u : X \longrightarrow Y$ from I° to $\mathcal{M}$ being in $C(I)$ (resp. $TC(I)$) if and only if for every object i of I, u_i is in C (resp. TC).

†A morphism of $\mathcal{M}(I)$ being in $F(I)$ (resp. $TF(I)$) if and only if it has the right lifting property with respect to the arrows belonging to $TC(I)$ (resp. $C(I)$).

- E a category

  A *cardinal filtration* of E is an increasing filtration

  $$\Big(\mathrm{Filt}^{\pi}(E)\Big)_{\pi\geq\pi_0} \quad , \qquad \pi_0 \text{ a fixed infinite cardinal,}$$

  of E by strictly full subcategories $\mathrm{Filt}^{\pi}(E)$, indexed by the cardinals $\pi \geq \pi_0$ belonging to a given universe, and satisfying the following conditions:

  a) for every $\pi \geq \pi_0$, $\mathrm{Filt}^{\pi}(E)$ is equivalent to a small category;

  b) the category E admits filtering direct limits indexed by ordered sets I large relative to π_0, and for every $\pi \geq \pi_0$, $\mathrm{Filt}^{\pi}(E)$ is stable under the said limits if $\mathrm{card}\, I \leq \pi$;

  c) For every $\pi \geq \pi_0$, and for every $X \in \mathrm{Ob}\, E$, there exists an isomorphism

  $$X \ \simeq \ \varinjlim_{i\in I} X_i \ ,$$

  where $(X_i)_{i\in I}$ is a filtering direct system in E, indexed by an ordered set I large relative to π, and where the X_i are objects of $\mathrm{Filt}^{\pi}(E)$. Moreover, if $X \in \mathrm{Ob}\,\mathrm{Filt}^{\pi'}(E)$, $\pi' \geq \pi$, one can take I such that $\mathrm{card}\, I \leq \pi'^{\pi}$.

- E a category, $(\mathrm{Filt}^{\pi}(E))_{\pi\geq\pi_0}$ a cardinal filtration of E, $\pi \geq \pi_0$ a cardinal

  A *well-adapted direct system* (to the given cardinal filtration, for the given cardinals π_0, π) is a filtering direct system $(X_i)_{i\in I}$ satisfying the following conditions:

  a) I is large relative to π, and for every $i \in I$, $X_i \in \mathrm{Filt}^{\pi}(E)$;

  b) Every subset J of I of cardinality $\leq \pi$, and large relative to π_0, admits a least upper bound i_J, and we have

  $$X_{i_J} \ \xleftarrow{\sim} \ \varinjlim_{i\in J} X_i \ .$$

  If X is an object of E, a *well-adapted representation* of X, in terms of objects of $\mathrm{Filt}^{\pi}(E)$, is the datum of a well-adapted direct system $(X_i)_{i\in I}$, to the given cardinal filtration, for the cardinals π_0, π, and of an isomorphism

  $$X \ \xleftarrow{\sim} \ \varinjlim_{i\in I} X_i \ .$$

  p. 63

- E a category admitting small filtering direct limits

  A *strict cardinal filtration* of E is a cardinal filtration $(\mathrm{Filt}^{\pi}(E))_{\pi\geq\pi_0}$ such that for every cardinal $\pi \geq \pi_0$, $\mathrm{Filt}^{\pi}(E)$ is stable under filtering direct limits of cardinality $\leq \pi$.

- E a category admitting small filtering direct limits

  $(\mathrm{Filt}^{\pi}(E))_{\pi\geq\pi_0}$ a strict cardinal filtration of E, $\pi \geq \pi_0$ a cardinal

  A *perfectly adapted direct system* (to the given cardinal filtration, for the cardinal π) is a filtering direct system $(X_i)_{i\in I}$ satisfying the following conditions:

  a) I is large relative to π, and for every $i \in I$, $X_i \in \mathrm{Filt}^{\pi}(E)$;

b) Every filtering subset J of I of cardinality $\leq \pi$, admits a least upper bound i_J, and we have

$$X_{i_J} \xleftarrow{\sim} \varinjlim_{i \in J} X_i \ .$$

If X is an object of E, a *perfectly adapted representation* of X, in terms of objects of $\mathrm{Filt}^{\pi}(E)$, is the datum of a perfectly adapted direct system $(X_i)_{i \in I}$, to the given cardinal filtration, for the cardinal π, and of an isomorphism

$$X \xleftarrow{\sim} \varinjlim_{i \in I} X_i \ .$$

p. 64

References

H. J. Baues, "Algebraic Homotopy", Cambridge studies in advanced mathematics, vol. 15 (1989).
R. Hartshorne, "Residues and Duality", Lecture Notes in Mathematics, vol. 20, Springer-Verlag (1966).
D. G. Quillen, "Homotopical Algebra", Lecture Notes in Mathematics, vol. 43, Springer-Verlag (1967).
R. W. Thomason, *Cat as a Closed Model Category*, Cahiers de Topologie et Géométrie Différentielle, Vol. XXI-3, pp. 305-324 (1980).
J.-L. Verdier, "Des catégories dérivées des catégories abéliennes", Astérisque, vol. 239 (1996).
SGA 4, I, "Théorie des Topos et Cohomologie Étale des Schémas", Séminaire de Géométrie Algébrique du Bois-Marie, directed by M. Artin, A. Grothendieck, J.-L. Verdier, Lecture Notes in Mathematics, vol. 269, Springer-Verlag (1972).

p.

THE DERIVATORS

ALEXANDRE GROTHENDIECK

Chapter XIV

h-cartesian and h-cocartesian squares

p.
p. 1

1 Axioms for quadrangulated structures (an aborted attempt)

[page 1]

Let $(\mathcal{M}, W)$ be a category of models admitting a theory of the homotopy fiber. Then we have in $\mathrm{Ho}_W = \mathcal{M}W^{-1}$ a notion of *W-cartesian square* (or W-homotopically cartesian), by which we mean a square diagram

$$\begin{array}{ccc} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{array}$$

in Ho_W coming, up to isomorphism, from a W-cartesian square in $\mathcal{M}$ – thus isomorphic, in $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1, \mathrm{Ho}_W)$, to a *cartesian* square

$$\begin{array}{ccc} Y_1 & \longleftarrow & Y_3 \ (\simeq Y_1 \times_{Y_0} Y_2) \\ {\scriptstyle p_1}\downarrow & & \downarrow \\ Y_0 & \xleftarrow[p_2]{} & Y_2 \end{array} \quad ,$$

with $p_1,\ p_2 \in \mathrm{Fib}_W$. (In fact, it suffices to have p_1 or p_2 in Fib_W to ensure the W-cartesian character of the cartesian square under consideration.) I would like to axiomatize this situation, mainly to make a list of the essential useful properties of this notion, and of the dual notion of W-cocartesian square (or W-homotopically cocartesian), in the case where $(\mathcal{M}, W)$ admits a theory of the homotopy cofiber. The species of structure,

namely that of "quadrangulated categories" (left, right, two-sided) is a variant, in a nonadditive case, of the notion of triangulated category isolated by VERDIER. Like the [page 2]
latter, this notion is in fact too weak – it is *not* "the right one" – but it is of provisional practical interest. The "right" notion (if there is one apart from that of derivator) should work with a finer notion of square and of W-cartesian square, working systematically in $\mathrm{HO}_W(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1) = \mathcal{M}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1)W(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1)^{-1}$ (where the "real" W-cartesian squares live!), and similar categories, such as $\mathrm{HO}_W(\Psi)$, $\mathrm{HO}_W(\Phi)$. But to include them in an axiomatics making no mention of a category of models $(\mathcal{M}, W)$, manifestly amounts to putting one's finger in the gears of the notion of derivator, and that is not my purpose for the moment.

Definition 1. I call a *"category with h-cartesian squares",* or a *"left quadrangulated category",* a category $\mathcal{H}$, equipped with a subset Cart of the set $\mathrm{Hom}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1, \mathcal{H})$ of p. 2
square diagrams

$$\begin{array}{ccc} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{array}$$

in $\mathcal{H}$, squares called *"h-cartesian"* (h as in "homotopically"), this datum being subject [page 3]
to the following axioms, Cart (1) to Cart (…).

Since I am not sure I know all the essential properties in advance, the number of these axioms remains open for the moment.

Cart (1) (Symmetry). If a square

$$\begin{array}{ccc} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{array}$$

is h-cartesian, likewise the symmetric square (obtained by interchanging the roles of X_1 and X_2).

Cart (2) (Trivial squares). Let

$$\begin{array}{ccc} X_1 & \xleftarrow{q_2} & X_3 \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle q_1} \\ X_0 & \xleftarrow[p_2]{} & X_2 \end{array}$$

be a square in $\mathcal{H}$, with p_1 an isomorphism. Then the square is h-cartesian if and only if q_1 is an isomorphism.

Cart (3) (Composite squares). Let there be a commutative diagram

$$\begin{array}{ccccc} X'_0 & \longleftarrow & X'_1 & \longleftarrow & X'_2 \\ \downarrow & \mathrm{I} & \downarrow & \mathrm{II} & \downarrow \\ X_0 & \longleftarrow & X_1 & \longleftarrow & X_2 \end{array} \quad ,$$

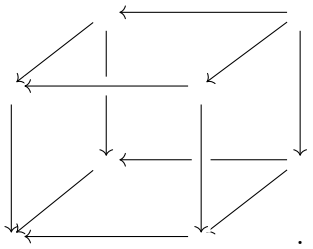
whence the squares I, II and III:

$$\begin{array}{ccc} X'_0 & \longleftarrow & X'_2 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{array}$$

p. 3

(which we call the *composite square* ("horizontal") of the squares I and II). Suppose I h-cartesian. Then II is h-cartesian if and only if III is .

Corollary. *Let there be a commutative cubical diagram, which we can regard as a homomorphism from one square* (*upper face of the cube*) *to another* (*lower face*):



If all the faces of the cube[1] *except the upper face are h-cartesian squares (namely the lower face and the four lateral faces), the same holds for the upper face.*[2] [page 4]

These axioms are in the nature of stability axioms, giving us criteria asserting that certain squares are h-cartesian, if such and such conditions are fulfilled (e.g. that certain other squares are already h-cartesian). I have the impression of having exhausted here the relevant properties of this type (*i.e.* that the others should be deducible from them). At present, I am dealing with properties of existence and uniqueness of h-cartesian squares.

Cart (4) (Fiber product axiom). Every diagram

$$\begin{array}{ccc} X_1 & & \\ \downarrow & & \\ X_0 & \longleftarrow & X_2 \end{array}$$

in $\mathcal{H}$ can be completed to an h-cartesian square

$$\begin{array}{ccc} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{array} \quad .$$

The following axiom (*cf.* its corollary) will ensure among other things that the h-cartesian square in question is *unique up to isomorphism* – but one should note that it is in general not unique "up to unique isomorphism".

Cart (5) (Exactness axiom) Let us consider two h-cartesian squares p. 4

$$X = \begin{pmatrix} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{pmatrix} \quad \text{and} \quad Y = \begin{pmatrix} Y_1 & \longleftarrow & Y_3 \\ \downarrow & & \downarrow \\ Y_0 & \longleftarrow & Y_2 \end{pmatrix}$$

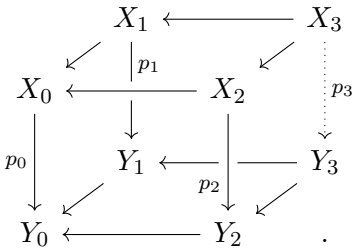
and a homomorphism $p = (p_0, p_1, p_2)$ between the diagrams of type

$$\Phi = \begin{array}{ccc} 1 & & \\ \downarrow & & \\ 0 & \longleftarrow & 2 \end{array}$$

associated with them (*cf.* the solid-line diagram below):

[1] In fact, the lower face, and the two parallel lateral faces (e.g. the front one and the back one) suffice.

[2] **Corollary** (of Cart (3) and Cart (2)): *every square isomorphic to an h-cartesian square is h-cartesian.* [page 5]



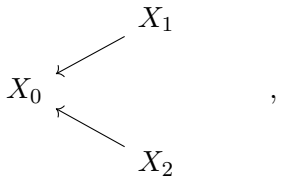
Then:

a) There exists a homomorphism $\bar{p} : X \longrightarrow Y$ inducing p, *i.e.* there exists an arrow p_3 so as to complete the commutative cubical diagram (thus respecting the commutativity constraint for the two squares adjacent to the edge p_3).

b) If moreover p_0, p_1, p_2 are isomorphisms, the same is true of p_3 .

Taking the case where p_0, p_1, p_2 are identities, we find:

Corollary 1. *Under the conditions of* Cart (4)*, if we have* two *h-cartesian squares* X*,* X' *which complete*



there exists a homomorphism from X *to* X' *which induces the identity on the vertices with indices* 0, 1, 2*, and such a homomorphism is always an isomorphism.*

But this isomorphism is not necessarily unique, in other words, an automorphism of an h-cartesian square can induce the identity on the vertices X_0, X_1, X_2, without being the identity.

Corollary 2. *Let us consider a commutative square*

p. 5
[page 6]

$$Y = \begin{pmatrix} Y_1 & \longleftarrow & Y_3 \\ \downarrow & & \downarrow \\ Y_0 & \longleftarrow & Y_2 \end{pmatrix} \quad ,$$

and an object T *of* $\mathcal{H}$*, whence (putting* $\mathrm{Hom}(T, X) = X(T)$*) a commutative square in* (Ens)

$$\begin{array}{ccc} Y_1(T) & \longleftarrow & Y_3(T) \\ \downarrow & & \downarrow \\ Y_0(T) & \longleftarrow & Y_2(T) \end{array} \quad ,$$

and hence

$$Y_3(T) \longrightarrow Y_1(T) \times_{Y_0(T)} Y_2(T) \quad . \qquad (*)$$

If Y *is h-cartesian, this map* $(*)$ *is* surjective *whatever* T *may be.*

We apply Cart (5) to Y and to the constant square X with value T

$$X = \begin{array}{ccc} T & \xleftarrow{\mathrm{id}} & T \\ {\scriptstyle \mathrm{id}}\downarrow & & \downarrow{\scriptstyle \mathrm{id}} \\ T & \xleftarrow{\mathrm{id}} & T \end{array} \quad ,$$

noting that giving $p = (p_0, p_1, p_2)$ from

$$\begin{array}{ccc} T & & Y_1 \\ \downarrow & \text{to} & \downarrow \\ T \longleftarrow T & & Y_0 \longleftarrow Y_2 \end{array}$$

amounts to giving an element of the right-hand side of $(*)$,

$$\begin{array}{ccc} Y_1 & \xleftarrow{p_1} & T \\ \downarrow & \swarrow^{p_0} & \downarrow{\scriptstyle p_2} \\ Y_0 & \longleftarrow & Y_2 \end{array} \quad ,$$

and that the elements of $Y_3(T)$ which map (by $(*)$) to this element are none other than the p_3 whose existence axiom Cart (5) asserts.

N.B. We shall note that the surjective map $(*)$ is in general *not* bijective, *i.e.* the h-cartesian square is in general not cartesian.

N.B. There is a dual notion of *category with h-cocartesian squares,* or *right quadrangulated category.* Most often, the two structures are present simultaneously. We should bring out properties which relate the two dual structures on one and the same category. Thus, unless I am mistaken, in the case $\mathcal{H}$ additive, the two kinds of squares, the h-cartesian and the h-cocartesian ones, should *coincide.* (I shall have to come back to this later.) In the case $\mathcal{H}$ "pointed", there is the adjunction of the functors Σ, Ω linked, one to the right quadrangulated structure, the other to the left quadrangulated structure. And there are, when there are internal or external $\underline{\mathrm{Hom}}$, possible properties of $\underline{\mathrm{Hom}}(X, Y)$ with respect to h-cocartesian quadrangles in X, h-cartesian in Y – but this is hardly a link between the two structures …

p. 6 [page 7]

2 Examples

A first trivial example, is the one where $\mathcal{H}$ is stable under fiber products, and where the h-cartesian squares are defined as the cartesian squares. Axioms Cart (1) to Cart (5) are then well-known, ± tautological properties of the notion of a cartesian square. And in a sense, the general case must reduce to this banal special case.

[page 8]

The only type of other case that I know of, is the one where $\mathcal{H} = \mathcal{M}W^{-1}$, where W is a localizer admitting a theory of the W-homotopy fiber (*cf.* chapter XII). In this case, as we have said, we take as h-cartesian squares in $\mathcal{H}$ those which are isomorphic, in $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1, \mathcal{H})$, to a W-cartesian square in $\mathcal{M}$. We still have to verify axioms Cart (1) to Cart (5). I think that I have little chance of getting there except by means of a

Proposition 1. *Under hypotheses on $(\mathcal{M}, W)$ which will be brought out below, satisfied in any case if $(\mathcal{M}, W)$ is Quillenizable, on the category of finite ordered sets* (*cf.* XIII, p. 51)*, we have this: let*

$$X = \begin{pmatrix} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{pmatrix}$$

be a square in $\mathcal{M}$, and let $\mathrm{ho}_W(X)$ be its image in $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1, \mathcal{H})$. For $\mathrm{ho}_W(X)$ to be h-cartesian, it is necessary (*and it is sufficient, of course*) *that X be W-cartesian .*

p. 7 p. 8 p. 9

THE DERIVATORS

p. —

ALEXANDRE GROTHENDIECK

Chapter XV

Factorization theorems (Models (2))

p. —
p. 1
[page 1]

1 Fundamental construction: the transfinite factorization sequence of an arrow f.

Let $\mathcal{M}$ be a $\mathfrak{U}$-category, stable under "small" (*i.e.* $\mathfrak{U}$-small) filtering $\varinjlim$. Let

$$C_0 \subset \mathrm{Fl}(\mathcal{M}) \tag{1.1}$$

be a *small* set of arrows, we propose to construct for every arrow f of $\mathcal{M}$, a factorization

$$f = p\,i\,, \qquad i \in \widetilde{C_0} = ((C_0)_*)^*\,,\ p \in (C_0)_*\,. \tag{1.2}$$

For this, we are going to construct an ordinal sequence of functors

$$\left\{\begin{array}{rcl} f & \longmapsto & X_\alpha(f) = X_\alpha(C_0, f) \\ X_\alpha : \underline{\mathrm{Fl}}(\mathcal{M}) & \longrightarrow & \mathcal{M} \end{array}\right. \tag{1.3}$$

and functorial homomorphisms

$$i_\alpha(f) : \mathrm{source}\, f \longrightarrow X_\alpha(f)\,, \tag{1.4}$$

$$p_\alpha(f) : X_\alpha(f) \longrightarrow \mathrm{but}\, f\,, \tag{1.5}$$

so that we shall have the diagram (where $X = \mathrm{source}\, f$, $Y = \mathrm{but}\, f$)

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle i_\alpha(f)}\searrow \quad \nearrow{\scriptstyle p_\alpha(f)} & \\ & X_\alpha(f) & \end{array}\,, \tag{1.6}$$

this diagram being commutative

$$f = p_\alpha(f) i_\alpha(f)\,. \tag{1.7}$$

We can also say that we have a section of the canonical functor of "composition", transpose of $\mathbf{\Delta}_1 \longrightarrow \mathbf{\Delta}_2$, given by $0 \longmapsto 0$, $1 \longmapsto 2$:

$$\left\{\begin{array}{l} \underline{\mathrm{Nerf}}_2(\mathcal{M}) \underset{\Phi_\alpha}{\overset{\gamma}{\rightleftarrows}} \underline{\mathrm{Nerf}}_1(\mathcal{M}) = \underline{\mathrm{Fl}}(\mathcal{M})\,, \\ \\ \end{array}\right. \tag{1.8}$$

p. 2
[page 2]

Φ as in "factorization", where

$$\Phi_\alpha(f) \;=\; \{\, X \xrightarrow{i_\alpha(f)} X_\alpha(f) \xrightarrow{p_\alpha(f)} Y \,\} \tag{1.9}$$

with $p_\alpha(f) i_\alpha(f) = f$. The functoriality of 1.4 and 1.5, or that of Φ_α, *i.e.* of 1.9, is expressed by the fact that every commutative square

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle u_X}\downarrow & & \downarrow{\scriptstyle u_Y} \\ X' & \xrightarrow{f'} & Y' \end{array} \tag{1.10}$$

in $\mathcal{M}$, interpreted as an arrow $u = (u_X, u_Y)$ from f to f' in $\underline{\mathrm{Fl}}(\mathcal{M})$, gives rise to the commutative double square

$$\begin{array}{ccccc} X & \xrightarrow{i_\alpha(f)} & X_\alpha(f) & \xrightarrow{p_\alpha(f)} & Y \\ \downarrow{\scriptstyle u_X} & & \downarrow{\scriptstyle X_\alpha(u)} & & \downarrow{\scriptstyle u_Y} \\ X' & \xrightarrow{i_\alpha(f')} & X_\alpha(f') & \xrightarrow{p_\alpha(f')} & Y' \,. \end{array} \tag{1.11}$$

N.B. When we want to make explicit the dependence of these constructions on C_0, we shall write

$$X_\alpha(C_0, f)\,, \qquad i_\alpha(C_0, f)\,, \qquad p_\alpha(C_0, f)\,, \qquad \Phi_\alpha(C_0, f)\,,$$

or

$$X_\alpha(C_0, -) \;\text{ or }\; X_\alpha^{C_0}\,, \qquad i_\alpha(C_0, -) \;\text{ or }\; i_\alpha^{C_0} \qquad \textit{etc.}\,,$$

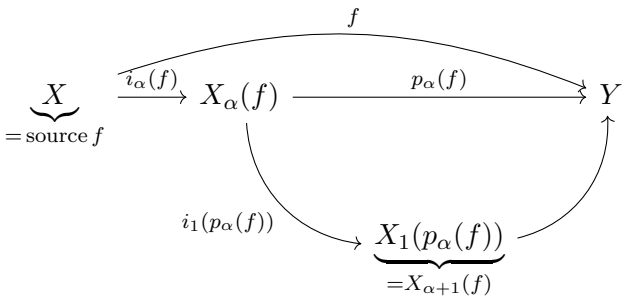
instead of $X_\alpha(f)$, $i_\alpha(f)$, …

Here are the essential properties of the ordinal direct system of the $X_\alpha(f)$ or $\Phi_\alpha(f)$.

1. $X_0(f) = X$, $i_0(f) = \mathrm{id}_X$, $p_0(f) = f$ (*i.e.* the factorization $\Phi_0(f)$ of f is the trivial factorization $X \xrightarrow{\mathrm{id}_X} X \xrightarrow{f} Y$). [page 3]

2.

$$\begin{aligned} X_{\alpha+1}(f) \;&=\; X_1(p_\alpha(f)) \\ i_{\alpha+1}(f) \;&=\; i_1(p_\alpha(f)) \circ i_\alpha(f) \\ p_{\alpha+1}(f) \;&=\; p_1(p_\alpha(f)) \end{aligned}$$



or better

$$\begin{array}{ccccccc}
X & & \xrightarrow{\qquad\qquad f\qquad\qquad} & & & & Y \\
\downarrow{\scriptstyle \mathrm{id}} & & & & & & \downarrow{\scriptstyle \mathrm{id}} \\
X & \xrightarrow{i_\alpha(f)} & X_\alpha(f) & & \xrightarrow{\qquad p_\alpha(f)\qquad} & & Y \\
\downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \mathrm{id}} & & & & \downarrow{\scriptstyle \mathrm{id}} \\
X & \longrightarrow & X_\alpha(f) & \xrightarrow{i_1(p_\alpha(f))} & \underbrace{X_1(p_\alpha(f))}_{=X_{\alpha+1}(f)} & \xrightarrow{p_1(p_\alpha(f))} & Y\,.
\end{array} \tag{1.12}$$

The transition morphism $X_\alpha(f) \longrightarrow X_{\alpha+1}(f)$ is

$$X_\alpha(f) \xrightarrow{\quad i_1(p_\alpha(f))\quad} X_1(p_\alpha(f)) \;=\; X_{\alpha+1}(f)\,.$$

3. If α is a limit ordinal, we have

$$\begin{cases}
X_\alpha(f) &=& \varinjlim\limits_{\alpha'<\alpha} X_{\alpha'}(f). \\
i_\alpha(f) &=& \text{canonical morphism from } X = X_0(f) \text{ into the direct limit.} \\
p_\alpha(f) &=& \text{inverse limit of the } p_{\alpha'}(f) : X_{\alpha'}(f) \longrightarrow Y.
\end{cases}$$

(We can regard the direct limit defining $X_\alpha(f)$ as a limit in $X\backslash\mathcal{M}/Y\ldots$)

The transition morphisms

$$X_{\alpha'}(f) \longrightarrow X_\alpha(f) \qquad (\,\alpha' < \alpha\,)$$

are the canonical morphisms relative to the direct system of the $X_{\alpha'}(f)$.

We can therefore say that the ordinal sequence of functorial factorizations $\Phi_\alpha(f)$ of variable f, is deduced from the functorial factorization $\Phi_1(f)$ of f as [page 4]

$$X \xrightarrow{i_1(f)} X_1(f) \xrightarrow{p_1(f)} Y \qquad (f : X \to Y)$$

by "transfinite iteration on the right", *i.e.* by regarding the operations Φ_α as functors

$$\begin{cases}
p_\alpha :\ \underline{\mathrm{Fl}}(\mathcal{M}) &\longrightarrow& \underline{\mathrm{Fl}}(\mathcal{M}) \\
f &\longmapsto& p_\alpha(f)\,,
\end{cases}$$ [1]

equipped with functorial homomorphisms (over $\mathcal{M}$)

$$\mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})} \longrightarrow p_\alpha\,.$$

Thus, if we consider p. 4

$$\begin{cases}
\underline{\mathrm{Fl}}(\mathcal{M}) & \text{as an object of CAT}/\mathcal{M} \text{ by the functor } f \longmapsto \text{target } f \\
\text{where} & \text{CAT is the category of categories } \textit{elements of } \mathfrak{V}\text{, a universe} \\
& \text{such that } \mathfrak{U} \in \mathfrak{V}\text{, whence Cat} \overset{\text{(def)}}{=} \mathrm{Cat}^{\mathfrak{U}},\ \mathcal{M},\ \underline{\mathrm{Fl}}(\mathcal{M}) \in \mathrm{CAT},
\end{cases}$$

[1] **N.B.** p_α respects the category structures *over* $\mathcal{M}$, via the target functors: $\underline{\mathrm{Fl}}(\mathcal{M}) \longrightarrow \mathcal{M}$, *i.e.* $\underline{\mathrm{Fl}}(\mathcal{M})$ is regarded as an object of CAT/$\mathcal{M}$.

we have an endofunctor p_1 of $\underline{\mathrm{Fl}}(\mathcal{M})$ over $\mathcal{M}$, and a functorial homomorphism

$$i_1 \ : \ \mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})/\mathcal{M}} \longrightarrow p_1 \ , \qquad \textit{i.e.} \qquad \begin{array}{ccccc} X & & \xrightarrow{\quad f \quad} & & Y \\ & {\scriptstyle i_1(f)}\searrow & & \nearrow{\scriptstyle p_1(f)} & \\ & & \underbrace{X_1(f)}_{=\,\mathrm{source}\,p_1(f)} & & \end{array} \ , \tag{1.13}$$

and the description of the ordinal sequence of endofunctors p_α of $\underline{\mathrm{Fl}}\,\mathcal{M}/\mathcal{M}$ follows from this datum alone, taking into account that small filtering limits exist in the fibers of $\underline{\mathrm{Fl}}\,\mathcal{M} \xrightarrow{\text{target}} \mathcal{M}$.

[page 5]

$$\begin{array}{ccccccccccc} X & \xrightarrow{i_1(f)} & X_1(f) & \xrightarrow{i_1(p_1(f))} & X_2(f) & \longrightarrow \cdots \longrightarrow & X_\alpha(f) & \xrightarrow{i_1(p_\alpha(f))} & X_{\alpha+1}(f) & \longrightarrow \cdots \\ & {\scriptstyle f}\searrow & \downarrow{\scriptstyle p_1(f)} & \swarrow{\scriptstyle p_2(f)} & & & & {\scriptstyle p_\alpha(f)} & {\scriptstyle p_{\alpha+1}(f)=p_1(p_\alpha(f))} & \\ & & Y & & & & & & & \end{array} \tag{1.14}$$

So, to finish defining the ordinal sequence of the $X_\alpha(f)$ (equipped with the $i_\alpha(f)$, $p_\alpha(f)$), or of the $p_\alpha(f)$ (equipped with the $i_\alpha(f)$), it remains to describe $X_1(f)$, $i_1(f)$, $p_1(f)$.

The essential property of this triple that we shall need is the following:

(1.15) **Fact**(C_0) (factorization property with respect to C_0).

For every commutative diagram

$$D: \qquad \begin{array}{ccc} A & \longrightarrow & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle f} \\ B & \longrightarrow & Y \end{array} \qquad \text{with } i \in C_0 \ ,$$

whence a solid-arrow diagram

$$\begin{array}{ccccc} A & \longrightarrow & X & & \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle i_D} & \overset{i_1(f)}{\searrow} & \\ B & \longrightarrow & X_D \overset{\text{def}}{=} X \amalg_A B & \xrightarrow{u_D} & X_1(f) \\ & & \downarrow{\scriptstyle p_D} & \underset{p_1(f)}{\searrow} & \\ & & Y \ , & & \end{array}$$

there exists a homomorphism from the factorization $f = p_D i_D$ of f to the factorization $f = p_1(f) i_1(f)$.

p. 5

To formulate this condition, I have used the existence of the amalgamated sums X_D, *i.e.* the condition on C_0:

(1.16) The arrows $i \in C_0$ are cosquarable, *i.e.* for a diagram

[page 6]

$$\begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle i}\downarrow & & \\ B & & \end{array}$$

with $i \in C_0$, the amalgamated sum exists.

But we can also formulate 1.15 as follows, independently of the hypothesis (1.16):

(1.17) For every commutative square

$$\begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle f} \\ B & \xrightarrow{v} & Y \end{array} \quad ,$$

with $i \in C_0$, there exists a solid-arrow commutative square

$$\begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle i_1(f)} \\ B & \xrightarrow{v_1} & X_1(f) \\ & \searrow^{v} & \downarrow{\scriptstyle p_1(f)} \\ & & Y \end{array}$$

such that $v = p_1(f)v_1$.

This has as a consequence the

Corollary 1 (of 1.17). *Let $f \in \mathrm{Fl}(\mathcal{M})$, and α be an ordinal ($\in \mathfrak{U}$). For every commutative square*

$$\begin{array}{ccc} A & \xrightarrow{u_\alpha} & X_\alpha(f) \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle p_\alpha(f)} \\ B & \xrightarrow{v} & Y \end{array}$$

there exists a commutative square

p. 6

$$\begin{array}{ccc} A & \xrightarrow{u_\alpha} & X_\alpha(f) \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle \text{trans.}} \\ B & \xrightarrow{v_1} & X_{\alpha+1}(f) \\ & \searrow^{v} & \downarrow{\scriptstyle p_{\alpha+1}(f)} \\ & & Y \end{array} \qquad p_\alpha(f)$$

such that

$$v \;=\; p_{\alpha+1}(f)\; v_1 \;.$$

Corollary 2. *Let π be a cardinal ($\in \mathfrak{U}$) such that for every arrow $i : A \longrightarrow B$ in C_0, the source A is π-accessible. Let α be a nonzero limit ordinal of cofinality strictly greater than π* (i.e. *such that the ordered set I_α of ordinals $\alpha' < \alpha$ is large relative to π*) – *e.g. α the first ordinal whose cardinality is $> \pi$. Then for every arrow $f \in \mathrm{Fl}(\mathcal{M})$, we have $p_\alpha(f) \in (C_0)_*$.* [page 7]

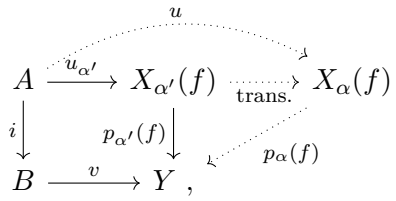
Proof. This means that for every commutative square

$$\begin{array}{ccc} A & \xrightarrow{u} & X_\alpha(f) \\ {\scriptstyle i}\downarrow & \nearrow^{w} & \downarrow{\scriptstyle p_\alpha(f)} \\ B & \xrightarrow{v} & Y \;, \end{array} \qquad (*)$$

with $i \in C_0$, the dotted arrow w (with $wi = u$, $p_\alpha(f)w = v$) exists. Now

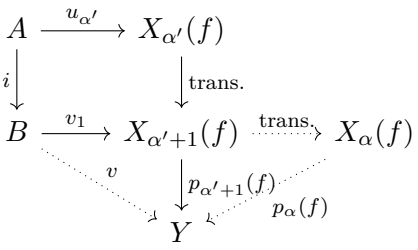
$$X_\alpha(f) = \varinjlim_{\alpha' \in I_\alpha} X_{\alpha'}(f) \;,$$

so A being π-accessible and I_α large relative to π, u comes from an arrow $u_{\alpha'} : A \longrightarrow X_{\alpha'}(f)$, and (if necessary replacing α' by $\alpha'' > \alpha'$ and $u_{\alpha'}$ by the composite $u_{\alpha''} : A \longrightarrow X_{\alpha'}(f) \longrightarrow X_{\alpha''}(f)$) we may suppose $p_{\alpha'}(f)u_{\alpha'} = vi$, *i.e.* that $(*)$ comes from a commutative square



and applying corollary 1, we find

p. 7



with $v = p_{\alpha'+1}(f)v_1$, and we shall take

$$w \;=\; \mathrm{trans}_{\alpha,\alpha'+1} \circ v_1 \ .$$

But we want a factorization of f, an arbitrary arrow in $\mathcal{M}$, as

[page 8]

$$f \;=\; p\ i \ , \qquad \text{with} \;\; p \in (C_0)_* \;\; \textit{and} \;\; i \in \widetilde{C_0} \ ,$$

so we must still state a property, ensuring that the $i_\alpha(f)$ are in $\widetilde{C_0}$. We shall even have something more precise. Let us introduce

(1.18) $C_0' \;=$ smallest subset of $\mathrm{Fl}(\mathcal{M})$ containing C_0, stable under cobase change (whenever it exists).

$C_0'' \;=$ set of arrows which are either in C_0', or amalgamated sums of a possibly empty family of such arrows $A \xrightarrow{i_\alpha} B_\alpha$ with the same source A, $i_\alpha \in C_0'$; the arrow $A \longrightarrow B = \coprod_{\alpha\,(A)} B_\alpha$ is in C_0''.

(Condition 1.16 implies that the elements of C_0' are cosquarable, and this suffices, together with the condition of stability of $\mathcal{M}$ under filtering $\varinjlim$, to ensure that the infinite amalgamated sums envisaged exist in $\mathcal{M}$.)

N.B. C_0'' is *equal* to the smallest set of arrows containing C_0 and stable under cobase change and under small amalgamated sums.

(1.19) $C_0''' \;=$ smallest subset of $\mathrm{Fl}(\mathcal{M})$ containing C_0'', and moreover stable under strict ordinal direct limits, *i.e.* for an ordinal direct system $(A_\alpha)_{\alpha<\alpha_0}$, such that for α a limit ordinal we have $A_\alpha \xleftarrow{\sim} \varinjlim_{\alpha' \in I_\alpha} A_{\alpha'}$, if the transition morphisms $i_{\beta,\alpha} : A_\alpha \longrightarrow A_\beta$ are in C_0''', so too the canonical morphisms

$$A_\alpha \xrightarrow{i_\alpha} \varinjlim_{\beta \in I_{\alpha_0}} A_\beta \ . \quad {}^{2}$$

p. 8

[page 9]

Proposition 2, p. 4 of XIII ensures that we have

$$C_0''' \ \subset\ \widetilde{C_0}\ .$$

This being so, here is the second essential property of the factorization $f = p_1(f)i_1(f)$ (the first being 1.17):

(1.20) For every arrow $f \in \mathrm{Fl}(\mathcal{M})$, we have $i_1(f) \in C_0''$.

It follows at once, by ordinal induction:

(1.21) For every arrow $f \in \mathrm{Fl}(\mathcal{M})$ and every ordinal $\alpha \in \mathfrak{U}$, we have $i_\alpha(f) \in C_0'''$, a fortiori $i_\alpha(f) \in \widetilde{C_0}$.

So if α is large relative to π (*cf.* cor. 2, page 7), the factorization $f = p_\alpha(f)i_\alpha(f)$ satisfies

$$\left\{\begin{array}{l} f = p_\alpha(f)i_\alpha(f), \text{ with } i_\alpha(f) \in C_0''' \subset \widetilde{C_0} = \Phi\ ,\ p_\alpha(f) \in C_{0*} = \Psi, \text{ when } \alpha \text{ is} \\ \text{large relative to } \pi, \text{ the sources of the } i \in C_0 \ \pi\text{-accessible.} \end{array}\right. \tag{1.22}$$

It remains to construct the functor $f \longmapsto X_1(f)$, with $i_1(f)$, $p_1(f)$, satisfying conditions (1.17) and (1.20), under the hypothesis (1.16) that the $i \in C_0$ are cosquarable. We then consider, for $f \in \mathrm{Fl}(\mathcal{M})$ fixed, the set $\Delta(f)$ (or $\Delta(C_0, f)$) of *all* the diagrams D of the form (1.15), with $i \in C_0$, and we put

$$X_1(f)\ =\ \coprod_{D\in\Delta(f)}{}_{(X)}X_D \tag{1.23}$$

(amalgamated sum under X, *i.e.* sum indexed by $\Delta(f)$ in $X\backslash\mathcal{M}$), the morphism $X \xrightarrow{i_1(f)}$ [page 10]
$X_1(f)$ being the costructural morphism, and the morphism $X_1(f) \xrightarrow{p_1(f)} Y$ being the one with the p_D as components. In short, we proceed "as economically as possible" (and as naturally as possible) to ensure the validity of (1.17), in the equivalent form (1.15). On the other hand, by definition of C_0', C_0'', the i_D are in C_0', and $i_1(f)$ is in C_0'', which is (1.20)[3].

To sum up:

Theorem 1. *Let $\mathcal{M}$ be a $\mathfrak{U}$-category stable under filtering $\mathfrak{U}$-direct limits, $C_0 \subset \mathrm{Fl}(\mathcal{M})$ a small set of arrows of $\mathcal{M}$. We assume the arrows $i \in C_0$ to be cosquarable. This ensures the definition of an endofunctor $f \longmapsto p_1(f)$ of $\underline{\mathrm{Fl}}(\mathcal{M})$, and of a functorial homomorphism $i_1 : \mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})} \longrightarrow p_1$,* i.e. *$i_1(f) : f \longrightarrow p_1(f)$, inducing the identity on the target objects of f and of $p_1(f)$,*

$$\begin{array}{ccc} X & \xrightarrow{\ i_1(f)\ } & X_1(f) \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle p_1(f)} & \\ & Y\ , & \end{array}$$

whence by iteration an ordinal direct system of endofunctors $f \longmapsto p_\alpha(f)$ of $\underline{\mathrm{Fl}}(\mathcal{M})$, p. 9
and of homomorphisms $i_\alpha : \mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})} \longrightarrow p_\alpha$, i.e.

$$\begin{array}{ccc} X & \xrightarrow{\ i_\alpha(f)\ } & X_\alpha(f) \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle p_\alpha(f)} & \\ & Y\ , & \end{array}$$

[page 11]

forming an ordinal direct system of factorizations of f functorial in f.

This being so, we have:

a) *For every ordinal α, we have $i_\alpha(f) \in \Phi$.*

b) *Suppose that the sources of the $i \in C_0$ are accessible and then let π be a cardinal in $\mathfrak{U}$ such that they are π-accessible. Then for every nonzero limit ordinal α of cofinality strictly greater than π, we have $p_\alpha(f) \in C_{0*}$, hence*

$$f \;=\; p_\alpha(f)\, i_\alpha(f) \qquad \textit{with} \qquad i_\alpha(f) \in \Phi\,, \qquad p_\alpha(f) \in \Psi\,.$$

We have a slightly more precise result, as follows:

Corollary 1. *Let Φ_0 be a subset of $\mathrm{Fl}(\mathcal{M})$ such that*

$$C_0''' \;\subset\; \Phi_0 \;\subset\; \Phi \qquad (\textit{hence}\;\; C_0 \subset \Phi_0 \subset \Phi)\,, \tag{1.24}$$

where $C_0''' \subset \Phi$ is defined in 1.19, *p.* 8. *For example, if Φ_0 is a subset of $\mathrm{Fl}(\mathcal{M})$ satisfying*

$$C_0 \;\subset\; \Phi_0 \;\subset\; \Phi\,,$$

and stable under cobase change, under small amalgamated sums (cf. (1.18), p. 8), *under strict ordinal $\varinjlim$ (cf.* (1.19), p. 8), *it satisfies* (1.24). (**N.B.** *If $\mathcal{M}$ admits simple amalgamated sums, such Φ_0 exist trivially, it suffices to take $\Phi_0 = \Phi$.*)

This being so, for every arrow f in $\mathcal{M}$, and every ordinal α, we have [page 12]

$$i_\alpha(f) \;\in\; \Phi_0\,.$$

Corollary 2. *Suppose that the sources of the arrows $i \in C_0$ are accessible, and let Φ_0 be as in Corollary* 1 (e.g. $\Phi_0 = C_0'''$). *Then the pair (Φ_0, Ψ) is a Quillen precouple,* i.e.

a) $\Phi_0 \longleftrightarrow \Psi$, i.e. $\Psi \subset \Phi_{0*}$ (*or what amounts to the same thing,* $\Phi_0 \subset \Psi^*$), *and*

b) *we have the Quillen factorization property.*

This therefore implies that Φ $(= \widetilde{C_0} = \widetilde{\Phi_0})$ consists of the arrows which are direct factors of arrows of Φ_0. Thus if Φ_0 is stable under direct factors, we have $\Phi_0 = \Phi$, i.e. *Φ_0 is left closed.*

p. 10

2 Variation of the factorizations $f = p_\alpha i_\alpha$ with the generating subset C_0

I say that the ordinal direct system of functors $p_\alpha = p_\alpha^{C_0} : \underline{\mathrm{Fl}}(\mathcal{M}) \longrightarrow \underline{\mathrm{Fl}}(\mathcal{M})$, equipped with the arrows [page 13]

$$i_\alpha^{C_0} \;:\; \mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})} \longrightarrow p_\alpha^{C_0}\,,$$

depends functorially on C_0, when one regards C_0 as an element of the ordered set of *small* subsets of $\mathrm{Fl}(\mathcal{M})$, consisting of cosquarable arrows (thus of the ordered set

$$\mathfrak{P}_{\mathfrak{U}\text{-small}}(\mathrm{Fl}_{\mathrm{cosquar.}}(\mathcal{M}))\quad).$$

If we have an inclusion

$$C_0 \;\hookrightarrow\; C_1\,, \tag{2.1}$$

[2] $C_0 \subset C_0' \subset C_0'' \subset C_0'''$

[3] One should all the same make explicit the functorial dependence of $X_1(f)$ on f !!!

we deduce from it for every ordinal α a functorial homomorphism

$$u_\alpha \;=\; u_\alpha^{C_1,C_0} \;:\; X_\alpha^{C_0} \;\longrightarrow\; X_\alpha^{C_1}\,, \tag{2.2}$$

making commutative the squares deduced from $f : X \longrightarrow Y$ in $\mathcal{M}$,

$$\begin{array}{ccc} X & \xrightarrow{i_\alpha^{C_0}(f)} & X_\alpha^{C_0}(f) \xrightarrow{p_\alpha^{C_0}(f)} Y \\ {\scriptstyle \mathrm{id}}\downarrow & & \downarrow{\scriptstyle u_\alpha(f)} \qquad\qquad \downarrow{\scriptstyle \mathrm{id}} \\ X & \xrightarrow{i_\alpha^{C_1}(f)} & X_\alpha^{C_1}(f) \xrightarrow{p_\alpha^{C_1}(f)} Y\,, \end{array} \tag{2.3}$$

and compatible with the transition morphisms for variable α,

$$\begin{array}{ccc} X_\alpha^{C_0} & \xrightarrow{\mathrm{trans}_{\beta,\alpha}^{C_0}} & X_\beta^{C_0} \\ {\scriptstyle u_\alpha}\downarrow & & \downarrow{\scriptstyle u_\beta} \\ X_\alpha^{C_1} & \xrightarrow{\mathrm{trans}_{\beta,\alpha}^{C_1}} & X_\beta^{C_1}\,. \end{array} \tag{2.4}$$

To define the u_α, satisfying the preceding conditions, it suffices to define them for $\alpha = 1$ "and to iterate":

[page 14]

$$\begin{array}{ccccc} & & X_1^{C_0}(f) & & \\ & {\scriptstyle i_1^{C_0}(f)}\nearrow & & \searrow{\scriptstyle p_1^{C_0}(f)} & \\ X & & {\scriptstyle u_1(f)}\downarrow & & Y \\ & {\scriptstyle i_1^{C_1}(f)}\searrow & & \nearrow{\scriptstyle p_1^{C_1}(f)} & \\ & & X_1^{C_1}(f) & & . \end{array} \tag{2.5}$$

Now $X_1^{C_0}(f)$ is defined as a direct sum in $X\backslash\mathcal{M}/Y$, relative to an index set $\Delta(C_0, f)$ *contained* in $\Delta(C_1, f)$ – and the family of objects whose direct sum is taken being induced from a (non-small) family whose index set would be $\Delta(\mathrm{Fl}_{\mathrm{cosquar.}}(\mathcal{M}), f)$, namely consisting of *all* the square diagrams

p. 11

$$D: \qquad \begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle f} \\ B & \xrightarrow{v} & Y \end{array} \qquad \text{with } i \text{ cosquarable and nothing more}\,,$$

the definition of the X_D, i_D, p_D associated to D depending only on D, and not on the set C_0 or C_1 of cosquarable arrows which contains i. This said, the arrow $u_1(f)$ in (2.5) is none other than the canonical inclusion of a partial amalgamated sum, indexed by $\Delta(C_0, f)$, into the amalgamated sum corresponding to the larger index set. The functoriality of this arrow $u_1(f)$, for variable f, "is immediate", it follows from the commutativity of the square, associated to a morphism $v = (v_X, v_Y)$ of arrows

[page 15]

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle v_X}\downarrow & & \downarrow{\scriptstyle v_Y} \\ X' & \xrightarrow{f'} & Y'\,, \end{array} \tag{2.6}$$

namely the diagram of maps between index sets

$$\begin{array}{ccc} \Delta(C_0, f) & \xrightarrow{u_1(f)} & \Delta(C_1, f) \\ {\scriptstyle \text{functorial dependence}\atop \text{on } f \text{ of } \Delta(C_0,f)}\downarrow & & \downarrow{\scriptstyle \text{functorial dependence}\atop \text{on } f \text{ of } \Delta(C_1,f)} \\ \Delta(C_0, f') & \xrightarrow{u_1(f')} & \Delta(C_1, f') \end{array} \tag{2.7}$$

(commutativity which amounts to saying that $\Delta(C_0, f)$ may be regarded as a functor in the pair (C_0, f) as an object of the product category $\mathfrak{P}_{\text{sm.}}(\mathrm{Fl}_{\text{cosquar.}}(\mathcal{M})) \times \underline{\mathrm{Fl}}(\mathcal{M})$ …). The commutative square (2.7) is the "indices" part of a commutative square between families of objects $\mathcal{X}(C_0, f)$, $\mathcal{X}(C_0, f')$, $\mathcal{X}(C_1, f)$, $\mathcal{X}(C_1, f')$, indexed by these sets, namely the [page 16] $X_D^{C_0}$ and the $X_D^{C_1}$ relative to the diagrams D involving, either the arrow f, or the arrow f' …

Now the canonical construction of the ordinal direct system p_α, with the $i_\alpha : \mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})} \longrightarrow p_\alpha$, is functorial with respect to the initial datum p_1, i_1 – I do not spell out the detail of the functoriality, although I should …

3 Upper bounds for cardinals

I now suppose that $\mathcal{M}$ is equipped with a *cardinal filtration*

$$(\mathrm{Filt}^\pi)_{\pi \geq \pi_0} \qquad (\pi_0,\ \pi \text{ cardinals in } \mathfrak{U}) \tag{3.1}$$

(SGA 4 I 9.12, p. 149), the Filt^π stable under simple amalgamated sums (whenever they exist), hence also under infinite amalgamated sums indexed by sets of indices of p. 12 cardinality $\leq \pi$. We further suppose that every object of Filt^{π_0} is accessible, and we denote by π_1 the smallest cardinal such that $\pi_1 \geq \pi_0$ and that the objects of Filt^{π_0} are π_1-accessible. Denoting, for every cardinal, by $\mathcal{M}_\pi$ the subset of $\mathrm{Ob}\,\mathcal{M}$ consisting of the π-accessible objects, we will then have for every $\pi \geq \pi_1$ (*loc. cit.* 9.16)

$$\mathcal{M}_\pi \ \subset\ \mathrm{Filt}^\pi \ \subset\ \mathcal{M}_{(\pi^{\pi_0})} \qquad (\text{if } \pi \geq \pi_1)\,, \tag{3.2}$$

thus if $\pi^{\pi_0} = \pi$, e.g. if $\pi = 2^c$ with $c \geq \pi_0$, we will have [page 17]

$$\mathrm{Filt}^\pi \ =\ \mathcal{M}_\pi\,. \tag{3.2 bis}$$

(If we admit the generalized continuum hypothesis, we have $\pi^{\pi_0} = \pi$ for every $\pi > \pi_0$, hence $\mathrm{Filt}^\pi = \mathcal{M}_\pi$ if $\pi \geq \pi_1$ and $\pi > \pi_0$.)

I put, for every cardinal $\pi \geq \pi_0$,

$$\mathrm{Fl}^\pi(\mathcal{M}) \ =\ \{\ f \in \mathrm{Fl}(\mathcal{M}) \mid \text{source and target of } f \text{ are in } \mathrm{Filt}^\pi\ \} \tag{3.3}$$

– which probably gives a cardinal filtration on $\underline{\mathrm{Fl}}(\mathcal{M})$ (but that is not our concern …). My aim now, it is to bring out conditions on $C_0 \subset \mathrm{Fl}_{\text{cosquar.}}(\mathcal{M})$ (small subset of $\mathrm{Fl}(\mathcal{M})$) and on the cardinal π, and the ordinal α, so that we have

$$\begin{aligned} f \in \mathrm{Fl}^\pi(\mathcal{M}) \ &\overset{?}{\Longrightarrow}\ X_\alpha(f) \in \mathrm{Filt}^\pi \\ &\quad (\textit{i.e. } p_\alpha(f) \in \mathrm{Fl}^\pi(\mathcal{M}),\ \text{or again } i_\alpha(f) \in \mathrm{Fl}^\pi(\mathcal{M}))\,. \end{aligned} \tag{3.4}$$

More generally, I propose to give an estimate of the "cardinality" of $X_\alpha(f)$, for fixed C_0, as a function of C_0, of f, of α, more exactly, as a function of the three cardinals $\pi(C_0)$, $\pi(f)$, $\pi(\alpha)$ defined as

$$\begin{cases} \pi(C_0) &= \text{smallest cardinal such that } C_0 \subset \mathrm{Fl}^\pi(\mathcal{M}) \\ \pi(f) &= \pi(\{f\}) \ =\ \text{smallest cardinal such that } f \in \mathrm{Fl}^\pi(\mathcal{M}) \\ \pi(\alpha) &= \mathrm{card}\,\alpha\,. \end{cases} \tag{3.5}$$

Thus I am looking for a cardinal function of cardinal variables φ, such that we have [page 18]

$$X_\alpha(f) \in \mathrm{Filt}^\pi \qquad \text{if} \qquad \pi = \varphi(\underbrace{\pi(C_0)}_{r}, \underbrace{\pi(f)}_{s}, \underbrace{\pi(\alpha)}_{t}). \tag{3.6}$$

The crucial estimate will surely concern the case $\alpha = 1$. Let us note that $X_1^{C_0}(f) = X_1(f)$ is defined as an amalgamated sum (in general infinite) of objects X_D, for a set $\Delta(C_0, f) = \Delta(f)$ of diagrams, namely the commutative diagrams

$$\begin{array}{ccc} A & \longrightarrow & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle f} \\ B & \longrightarrow & Y \end{array}$$

with $i \in C_0$. For a fixed i, and a fixed f, the cardinality of the set of diagrams of this type is bounded above by that of $\mathrm{Hom}(A, X) \times \mathrm{Hom}(B, Y)$. We have $A, B \in \mathrm{Filt}^r$, $X, Y \in \mathrm{Filt}^s$, so we need an upper bound for

p. 13

$$\mathrm{card}\,\mathrm{Hom}_{\mathcal{M}}(A, X), \qquad A \in \mathrm{Filt}^r, \qquad X \in \mathrm{Filt}^s.$$

Here are the essential upper bounds in this direction: [page 19]

Proposition 3.1. *Let π_0 be the initial cardinal of the cardinal filtration $(\mathrm{Filt}^\pi)_{\pi \geq \pi_0}$ under consideration, π_1 the smallest cardinal $\geq \pi_0$ such that the objects of Filt^{π_0} are π_1-accessible, $C_1 = \mathrm{card}\,\underbrace{\widetilde{\underline{\mathrm{Fl}}^{\pi_1}(\mathcal{M})}}_{=\underline{\mathrm{Fl}}(\mathcal{M}^{\pi_1})}$[4]. For every cardinal $\pi \geq \pi_0$, I denote by $\mathcal{M}^\pi$ the strictly full subcategory of $\mathcal{M}$ whose set of objects is Filt^π. Finally, if $\mathcal{C}$ is a category, I denote by $\widetilde{\mathcal{C}}$ the set of isomorphism classes of objects of $\mathcal{C}$.*[5]

1°) *If $r \geq \sup(\pi_1, C_1)$, we have*

$$\mathrm{card}\,(\underbrace{\widetilde{\underline{\mathrm{Fl}}^r(\mathcal{M})}}_{=\underline{\mathrm{Fl}}(\mathcal{M}^r)}) \leq 2^{(r^{\pi_0})} \quad (= 2^r, \text{ if } r = 2^c, \text{ with } c \geq \pi_0). \tag{3.7}$$

2°) *If $r, s \geq \sup(\pi_1, C_1)$, and if $A \in \mathrm{Filt}^r$, $X \in \mathrm{Filt}^s$, we have*

$$\mathrm{card}\,\mathrm{Hom}_{\mathcal{M}}(A, X) \leq s^{(r^{\pi_0})} \qquad \begin{cases} = s^r & \text{if } r = 2^c, \text{ with } c \geq \pi_0, \\ = s & \text{if } s = 2^{c'}, \text{ with } c' \geq r^{\pi_0}. \end{cases} \tag{3.8}$$

$$\mathrm{card}\,\underline{\mathrm{Fl}}(\widetilde{\mathcal{M}^r/X}) \leq s^{(r^{\pi_0})} \tag{3.9}$$

N.B. If we admit the generalized continuum hypothesis, which implies that $\pi^{\pi'} = \pi$ if $\pi > \pi'$ (π an infinite cardinal, $\pi' \geq 1$), then taking r such that $r > \pi_0$, the preceding formulas give (with moreover $r \geq \sup(\pi_1, C_1)$):

$$\mathrm{card}\,\widetilde{\underline{\mathrm{Fl}}^r(\mathcal{M})} \leq 2^r, \tag{3.7 bis}$$

$$\mathrm{card}\,\mathrm{Hom}_{\mathcal{M}}(A, X) \leq s^r \qquad (\text{if } A \in \mathrm{Filt}^r,\ X \in \mathrm{Filt}^s), \tag{3.8 bis}$$

$$\mathrm{card}\,\underline{\mathrm{Fl}}(\widetilde{\mathcal{M}^r/X}) \leq s^r \qquad (\text{if } X \in \mathrm{Filt}^s). \tag{3.9 bis}$$

Without the continuum hypothesis, these formulas are also valid for r of the form 2^c, with $c \geq \pi_0$. It is these last formulas which look the most natural. For (3.8 bis) and (3.9 bis), I point out that the right-hand side s^r is the cardinality of the set $\mathrm{Hom}_{\mathrm{Ens}}(A, X)$, where A, X are two sets such that $\mathrm{card}\,A = r$, $\mathrm{card}\,X = s$. We therefore cannot hope to improve (3.8 bis)! For the

[4] *cf.* below the notation $\widetilde{\mathcal{C}}$. p. 14

[5] Let us point out that we always have $\mathrm{card}\,\widetilde{\underline{\mathrm{Fl}}(\mathcal{C})} \geq \mathrm{card}\,\widetilde{\mathcal{C}}$ since $\widetilde{\mathcal{C}} \hookrightarrow \widetilde{\underline{\mathrm{Fl}}(\mathcal{C})}$ via $\mathrm{cl}(X) \longmapsto \mathrm{cl}(\mathrm{id}_X)$. [page 20]

first formula (3.7 bis), let us note that we have equality when $\mathcal{M}$ is the category of pairs (A, B), A a set and B a subset of A, with the obvious arrows, and we define $\mathcal{M}^\pi$ (for $\pi \geq \pi_0$) as the strictly full subcategory consisting of the (A, B) such that $\operatorname{card} A \leq \pi$.

Proof of 3.1. Let us first prove formula (3.8),

$$\operatorname{card}\operatorname{Hom}(A, X) \;\leq\; s^{(r^{\pi_0})} \qquad \begin{cases} \text{if } A \in \operatorname{Filt}^r,\ X \in \operatorname{Filt}^s, \\ r \geq \pi_1,\ s \geq \sup(\pi_1, C_1). \end{cases}$$

(**N.B.** I shall not use $r \geq C_1$, postulated in 3.1 1°.) I write

$$A \;=\; \varinjlim_{I} A_i\ , \quad \operatorname{card} I \;\leq\; r^{\pi_0}\ , \quad A_i \;\in\; \operatorname{Filt}^{\pi_0}\ ,$$
$$X \;=\; \varinjlim_{J} X_j\ , \quad \operatorname{card} J \;\leq\; s^{\pi_1}\ , \quad X_j \;\in\; \operatorname{Filt}^{\pi_1}\ , \qquad\qquad J \text{ large relative to } \pi_1\ .$$

We shall have

$$\operatorname{Hom}(A, X) \;\simeq\; \varprojlim_{i \in I} \underbrace{\operatorname{Hom}(A_i, \varinjlim_{J} X_j)}_{\overset{(6)}{\simeq} \varinjlim_{j \in J} \underbrace{\operatorname{Hom}(A_i, X_j)}_{\text{of cardinality } \leq C_1}}\ ,$$

thus for every $i \in I$

$$\operatorname{card}\operatorname{Hom}(A_i, X) \;\leq\; \operatorname{card}(J) \times C_1 \;\leq\; s^{\pi_1} \times C_1 \;=\; s^{\pi_1}\ ,$$

since $s \geq C_1$, hence $s^{\pi_1} \geq C_1$, whence

$$\operatorname{card}(\varprojlim_{i \in I} \operatorname{Hom}(A_i, X)) \;\leq\; (s^{\pi_1})^{\operatorname{card} I} \;\leq\; \underbrace{(s^{\pi_1})^{(r^{\pi_0})}}_{s^{\pi_1 r^{\pi_0}}}\ .$$

Since $r \geq \pi_1$, we have $r^{\pi_0} \geq \pi_1$, hence $r^{\pi_0}\pi_1 = r^{\pi_0}$, whence

$$\operatorname{card}\operatorname{Hom}(A, X) \;\leq\; s^{(r^{\pi_0})}\ ,$$

this is indeed formula (3.8). To derive from it an estimate for $\mathcal{M}^r/X$ (with $X \in \operatorname{Filt}^s$), I shall first bound above the cardinality of the isomorphism classes of objects of $\mathcal{M}^r/X$, *i.e.* of the arrows up to isomorphism (in $\mathcal{M}^r$) [page 21]

$$A \;\longrightarrow\; X\ , \qquad X \text{ fixed in } \operatorname{Filt}^s,\ A \text{ variable in } \mathcal{M}^r\ .$$

We shall have

$$A \;=\; \varinjlim_{i \in I} A_i \qquad \text{with } A_i \in \operatorname{Filt}^{\pi_0},\ I \text{ ordered set of cardinality} \leq r^{\pi_0}\ .$$

For fixed I, the cardinality of the set of direct systems indexed by I, in $\mathcal{M}^{\pi_0}/X$, up to isomorphism, can be bounded above as follows: for every $i \in I$, one must choose A_i in p. 15
$\mathcal{M}^{\pi_0}$ (but only its class in $\widetilde{\mathcal{M}^{\pi_0}}$ matters), so this is bounded above by $\operatorname{card}\widetilde{\mathcal{M}^{\pi_0}} \leq C_0$ [(7)], and for all the i, by $\operatorname{card} I \times C_0 \leq r^{\pi_0} \times C_0$, and if $r \geq C_0$, hence $r^{\pi_0} \geq C_0$, we find an upper bound on the possible choices of the system of *objects* A_i in $\mathcal{M}^{\pi_0}$, up to isomorphism, given by r^{π_0}. If in addition we specify the transition morphisms $A_i \longrightarrow A_j$ for $i < j$, this gives for each comparable pair, an element of $I \times I$, a choice bounded above by C_0, hence

[6] since the $A \in \operatorname{Filt}^{\pi_0}$ are π_1-accessible.

[7] Here we have put $C_0 = \operatorname{card}\widetilde{\underline{\mathrm{Fl}}^{\pi_0}}$, hence $C_0 \leq C_1$.

again an upper bound of $\underbrace{\operatorname{card}(I \times I)}_{=(\operatorname{card} I)^2=\operatorname{card} I} \cdot C_0 \leq r^{\pi_0} \cdot C_0 = r^{\pi_0}$. Thus the set of direct systems in question, up to isomorphism, for fixed I, has cardinality $\leq r^{\pi_0} \cdot r^{\pi_0} = r^{\pi_0}$. Finally, the *homomorphisms* of a given direct system into X are bounded above by $\operatorname{card} I \times s^{(r^{\pi_0})}$ (by (3.8), already proved), hence by

$$r^{\pi_0} \cdot s^{(r^{\pi_0})} \;=\; s^{(r^{\pi_0})} \qquad (\text{since } r^{\pi_0} \leq s^{(r^{\pi_0})}) \;.$$

Altogether, for fixed I, the set of direct systems up to isomorphism, indexed by I, with values in $\mathcal{M}^r/X$, is of cardinality bounded above by $\underbrace{r^{\pi_0}}_{\substack{\text{number of}\\ \text{direct systems}\\ \text{in } \mathcal{M}^{\pi_0}\text{, up to}\\ \text{isomorphism}}} \cdot \underbrace{s^{(r^{\pi_0})}}_{\substack{\text{number of}\\ \text{arrows from}\\ \text{such a direct system}\\ \text{into } X}} =$ [page 22]

$s^{(r^{\pi_0})}$.

But we must now let I vary, up to isomorphism, in the category of ordered sets (filtering *etc.*) of cardinality $\leq \lambda = r^{\pi_0}$. We see at once that this set of isomorphism classes (which can be interpreted as a quotient set of a certain set of subsets of a fixed set I_0 of cardinality λ) has cardinality $\leq 2^{\lambda}$. We thus finally find

$$\operatorname{card} \widetilde{\mathcal{M}^r/X} \;\leq\; s^{(r^{\pi_0})} \cdot 2^{(r^{\pi_0})} \;=\; s^{(r^{\pi_0})} \;,$$

on condition that

$$r \;\geq\; \sup(\pi_1, C_0) \;, \qquad s \;\geq\; \sup(\pi_1, C_1) \;.$$

We must prove the similar inequality for card $\underline{\mathrm{Fl}}(\widetilde{\mathcal{M}^r/X})$ (3.9). Now, since for A, B in $\mathcal{M}^r$, we have, if $r \geq \sup(\pi_1, C_1)$,

$$\operatorname{card} \operatorname{Hom}_{\mathcal{M}}(A, B) \;\leq\; r^{(r^{\pi_0})}$$

by virtue of (3.8) where we put $s = r$, we find that

$$\begin{aligned} \operatorname{card} \underline{\mathrm{Fl}}(\widetilde{\mathcal{M}^r/X}) \;&\leq\; \left(\operatorname{card} \widetilde{\mathcal{M}^r/X}\right)^2 \cdot r^{(r^{\pi_0})} \\ &\leq\; s^{(r^{\pi_0})} \cdot r^{(r^{\pi_0})} \\ &=\; s^{(r^{\pi_0})} \qquad \text{if } s \geq r \;. \end{aligned}$$

(We shall look below at the cases $s \leq r$ [8]). This is indeed formula (3.9).

Let us turn to the category $\mathcal{M}^r$ itself. The argument on page 21 shows that the set of isomorphism classes of objects of $\mathcal{M}^r$ indexed by a fixed I, of cardinality $\leq r^{\pi_0}$, as $A = \varinjlim_i A_i$ with $A_i \in \mathrm{Filt}^{\pi_0}$, has cardinality bounded above by r^{π_0} (if $r \geq C_0$). If we let I vary, we find an upper bound of $r^{\pi_0} \cdot 2^{(r^{\pi_0})} = 2^{(r^{\pi_0})}$. We conclude from this as before that the set $\widetilde{\mathrm{Fl}\,\mathcal{M}^r}$ has cardinality bounded above by

p. 16
[page 23]

$$(2^{(r^{\pi_0})})^2 \times r^{(r^{\pi_0})} \qquad (\text{if } r \geq \sup(\pi_1, C_1)) \;,$$

hence by $r^{(r^{\pi_0})}$. I say that we have

$$r^{(r^{\pi_0})} \;=\; 2^{(r^{\pi_0})} \;,$$

indeed, we have $\geq$, and to prove $\leq$, we note that $r < 2^r$, hence $r^{(r^{\pi_0})} \leq (2^r)^{(r^{\pi_0})} = 2^{r \cdot r^{\pi_0}} = 2^{(r^{\pi_0})}$, O.K.! This thus establishes 3.7. This argument in fact proves the

[8] *cf.* cor. 2, page 23.

Lemma 3.2. *Let r be a cardinal, r' a cardinal $\geq r$. Then we have*

$$r^{r'} \;=\; 2^{r'} \qquad (= s^{r'} \text{ for all } 2 \leq s \leq r)\ .$$

Corollary 1. *Let $r \geq \sup(\pi_1, C_1)$, and $A, X \in \mathrm{Filt}^r$. Then*

$$\mathrm{card}\,\mathrm{Hom}_{\mathcal{M}}(A,X) \;\leq\; 2^{(r^{\pi_0})} \qquad (= 2^r \text{ if } r = 2^c,\ \text{with } c \geq \pi_0)\ . \tag{3.10}$$

We apply (3.8), which gives the upper bound $r^{(r^{\pi_0})}$, and lemma 3.2.

We shall note that we find the same upper bound for the cardinality of $\widetilde{\underline{\mathrm{Fl}}(\mathcal{M}^r)}$ as a whole (3.7)! Likewise, (3.9) gives:

Corollary 2. *Let $r \geq \sup(\pi_1, C_1)$, and $X \in \mathrm{Ob}\,\mathcal{M}^r$. Then*

$$\mathrm{card}\,\widetilde{\underline{\mathrm{Fl}}(\mathcal{M}^r/X)} \;\leq\; 2^{(r^{\pi_0})} \qquad (= 2^r \text{ if } r = 2^c,\ \text{with } c \geq \pi_0)\ . \tag{3.11}$$

so this is formula (3.9) for $s \leq r$, left pending.

Let us return to the cardinal bound for the $X_\alpha(f)$, where $f : X \longrightarrow Y$ is in $\mathrm{Fl}^s(\mathcal{M})$, *i.e.* $X, Y \in \mathrm{Filt}^s$. We assume moreover $C_0 \subset \mathrm{Fl}^r(\mathcal{M})$, [page 24]

$$C_0 \;\subset\; \mathrm{Fl}^r(\mathcal{M}) \;=\; \mathrm{Fl}(\mathcal{M}^r)\ , \qquad (f : X \longrightarrow Y) \in \mathrm{Fl}^s(\mathcal{M})\ ,\ \ \textit{i.e. } X, Y \in \mathrm{Filt}^s\ ,$$
$$\text{with } s \geq r\ . \tag{3.10}$$

There is a stupid difficulty, which is that $\mathcal{M}^r$ is equivalent to a small category, but is not itself small. Even if C_0 contained only a single isomorphism class of arrows of $\mathcal{M}^r$, its cardinality could be (for fixed r) arbitrarily large. Thus, in what follows, I must assume the condition of "cardinal non-plethoricity" for C_0:

> $\forall\, i \in C_0$, denoting by $C_0(i)$ the set of $i' \in C_0$ such that $i \simeq i'$ in $\mathrm{Fl}(\mathcal{M})$, I require that $\mathrm{card}\, C_0(i) \leq 2^{(r^{\pi_0})}$ (which is also the value of the upper bound for the cardinality of $\widetilde{\underline{\mathrm{Fl}}^r(\mathcal{M})} = \widetilde{\underline{\mathrm{Fl}}(\mathcal{M}^r)}$). (3.11)

This thus implies that we have $\mathrm{card}\, C_0 \leq (2^{(r^{\pi_0})})^2 = 2^{(r^{\pi_0})}$, p. 17

$$\mathrm{card}\, C_0 \;\leq\; 2^{(r^{\pi_0})}\ , \tag{3.12}$$

which is equivalent to (3.11) (given $C_0 \subset \mathrm{Fl}^r(\mathcal{M})$, postulated in (3.10)).

It follows by (3.8) that we have

$$\mathrm{card}\,\Delta(C_0, f) \;\leq\; \underbrace{\mathrm{card}(C_0)}_{\leq 2^{(r^{\pi_0})}} \cdot s^{(r^{\pi_0})} \cdot s^{(r^{\pi_0})} \;\leq\; s^{(r^{\pi_0})}\ .$$

On the other hand, the X_D whose amalgamated sum under X we take, to obtain $X_1(f)$, are in $\mathcal{M}^s$, if we make the following assumption. [page 25]

> The $\mathcal{M}^\pi = \underline{\mathrm{Filt}}^\pi$ are stable under simple amalgamated sums, and under filtering $\varinjlim$ of cardinality $\leq \pi$ (not necessarily large relative to π_0) , (3.13)

taking into account that $r \leq s$, hence $\mathcal{M}^r \subset \mathcal{M}^s$, hence A, B, X in $X_D = B \amalg_A X$ are in $\mathcal{M}^s$. The *finite* partial amalgamated sums of the X_D are therefore also in $\mathcal{M}^s$, by induction on the cardinality of the index set. The infinite amalgamated sum thus appears as a filtering $\varinjlim$ of objects of $\mathcal{M}^s$, the index set being the set $\mathfrak{P}^*_{\mathrm{f}}\Delta(C_0, f)$ of nonempty finite subsets of $\Delta(C_0, f)$. This set has cardinality bounded above by s', if s'

is an *infinite* cardinal such that $\operatorname{card}\Delta(C_0, f) \leq s'$. We can therefore take $s' = s^{(r^{\pi_0})}$. We have therefore proved, taking into account (3.13) applied to $\mathcal{M}^{s'}$, the

Lemma 3.3. *We assume* (3.10), i.e. $C_0 \subset \mathrm{Fl}^r(\mathcal{M})$, $f \in \mathrm{Fl}^s(\mathcal{M})$, *with* $s \geq r \geq \sup(\pi_1, C_1)$, *and moreover that* C_0 *satisfies the non-plethoricity condition* (3.11) (*which we can always assume, at the cost of reducing* C_0 *without changing* $\widetilde{C_0} = \Phi$ *and* $C_{0*} = \Psi$), *and that the cardinal filtration satisfies the stability condition (harmless!)* (3.13). *Put*

$$s' \;=\; s^{(r^{\pi_0})}\ . \tag{3.14}$$

Then $X_1(C_0, f)$ *is in* $\mathcal{M}^{s'}$. *More generally and more prettily, for fixed* C_0 *and* r (*with* $C_0 \subset \mathrm{Fl}^r(\mathcal{M})$ *and satisfying the non-plethoricity condition* (3.11) *relative to* r), *for every cardinal of the form* [page 26]

$$s' \;=\; s^t\ , \qquad \text{with } t \geq r^{\pi_0},\ s \geq 2 \tag{3.15}$$

(whence $s' \geq 2^{(r^{\pi_0})} > r^{\pi_0} \geq r$), $\underline{\mathrm{Fl}}^{s'}(\mathcal{M})$ *is stable under the functor* $p_1^{C_0} : \underline{\mathrm{Fl}}(\mathcal{M}) \longrightarrow \underline{\mathrm{Fl}}(\mathcal{M})$.

Indeed, if we apply the first part of the lemma to the pair (r, s'), we find that $X_1(f)$ is in $\mathcal{M}^{s''}$, with $s'' = s'^{(r^{\pi_0})}$. But

$$s'' \;=\; s'^{(r^{\pi_0})} \;=\; (s^t)^{(r^{\pi_0})} \;=\; s^{tr^{\pi_0}} \underset{\substack{\text{since } t \geq r^{\pi_0},\\ \text{hence } tr^{\pi_0} = t}}{=} s^t \;=\; s'\ ,$$

so we see that $f \in \mathrm{Fl}^{s'}$ implies $X_1(f) \in \mathcal{M}^{s'}$, or what amounts to the same thing, p. 18
$p_1(f) \in \mathrm{Fl}^{s'}$.

Corollary. *Let* C_0, r, $s' = s^t$ $(t \geq r^{\pi_0},\ s \geq 2)$ *be as above, and let* α *be an ordinal number such that*

$$\operatorname{card}\alpha \;\leq\; s' \;=\; s^t\ .$$

Then $\underline{\mathrm{Fl}}^{s'}(\mathcal{M})$ *is stable under the functor* $p_\alpha = p_\alpha^{C_0}$.

We see this at once by transfinite induction, taking into account lemma 3.3, and the stability assumption (3.13).

To summarize:

Theorem 2. *Let* $\mathcal{M}$ *be a category stable under small filtering* $\varinjlim$, *and endowed with a cardinal filtration* $(\mathrm{Filt}^\pi)_{\pi \geq \pi_0}$, *such that the* Filt^π *are stable under filtering* $\varinjlim$ *of cardinality* $\leq \pi$ *(not necessarily large relative to* π_0*), and under the simple amalgamated sums that exist in* $\mathcal{M}$. *We assume that the elements of* Filt^{π_0} *are accessible, and we denote by* π_1 *the smallest cardinal* $\geq \pi_0$, *such that the objects of* Filt^{π_0} *are* π_1*-accessible. We put* [page 27]

$$c_1 \;=\; \operatorname{card}\mathrm{Fl}(K_1), \qquad K_1 \subset \mathcal{M}^{\pi_1} \text{ a chosen small full skeleton}$$

(where for every $\pi \geq \pi_0$, $\mathcal{M}^\pi$ *denotes the strictly full subcategory of* $\mathcal{M}$ *defined by* Filt^π, *and* $\mathrm{Fl}^\pi(\mathcal{M}) = \mathrm{Fl}(\mathcal{M}^\pi)$*).*

Let C_0 *be a small subset of* $\mathrm{Fl}(\mathcal{M})$, *consisting of cosquarable arrows, and put*

$$\Phi \;=\; \widetilde{C_0} \;=\; \Psi^*\ , \qquad \text{where } \Psi = C_{0*} = \Phi_*\ .$$

Let r *be a cardinal* $\geq \sup(\pi_1, c_1)$, *and such that we have* $C_0 \subset \mathrm{Fl}^r(\mathcal{M})$. *We assume that* C_0 *is irredundant (*i.e. *two isomorphic objects of* C_0 *are equal), or more generally, that* C_0 *satisfies the very harmless non-plethoricity condition* (3.11) *relative to* r. *We consider the transfinite sequence of functors* $p_\alpha = p_\alpha^{C_0}$ *on* $\underline{\mathrm{Fl}}(\mathcal{M})$ (*cf.* th. 1), *giving rise to the functorial factorizations of an* $f \in \underline{\mathrm{Fl}}(\mathcal{M})$, $f : X \longrightarrow Y$, *as*

$$X \;\xrightarrow{i_\alpha(f)}\; X_\alpha(f) \;\xrightarrow{p_\alpha(f)}\; Y\ . \tag{3.16}$$

We then have the following:

1°) *We have (for the record)* $i_\alpha(f) \in \Phi$ *for every* α*, and moreover,* if α is large relative to r^{π_0}, *we also have* $p_\alpha(f) \in \Psi$. *If* r' *is a cardinal* $> r^{\pi_0}$ *(e.g. the successor cardinal of* r^{π_0}*), we can take for* α *the first ordinal whose cardinality is equal to* r'.

2°) *For every cardinal* s *of the form*

$$s \;=\; c^t \qquad \textit{with}\;\; t \geq r^{\pi_0},\; c \geq 2$$

(thus $s = 2^t$ *if we take* $c \leq t$, cf. *lemma 3.2),* $\underline{\mathrm{Fl}}^s(\mathcal{M})$ *is stable under the* p_α *such that* $\operatorname{card}\alpha \leq s$. *If we take e.g. for* α *the first ordinal whose cardinality is* $> r^{\pi_0}$*, then* $\underline{\mathrm{Fl}}^s(\mathcal{M})$ *is stable under* p_α*, which by* 1°*) thus realizes a functorial factorization* (3.16) *with* $i_\alpha(f) \in \Phi$, $p_\alpha(f) \in \Psi$.

PROOF. Everything has been proved, if we recall that the objects of $\mathcal{M}^r$ are r^{π_0}-accessible (*cf.* (3.2)). p. 19 [page 28]

4 Complements on Quillen structures

Definition 4.1. Let $\mathcal{M}$ be a category. By a *Quillen precouple* in $\mathcal{M}$ we mean a pair (Φ, Ψ) of subsets of $\mathrm{Fl}(\mathcal{M})$, satisfying the following two conditions.

a) We have $\Phi \longleftrightarrow \Psi$, *i.e.* $\Psi \subset \Phi_*$, or what amounts to the same, $\Phi \subset \Psi^*$ – or again: for every commutative square in $\mathcal{M}$

$$\begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle i}\downarrow & \overset{w}{\nearrow} & \downarrow{\scriptstyle p} \\ B & \xrightarrow{v} & Y\,, \end{array} \tag{4.1.1}$$

with $i \in \Phi$, $p \in \Psi$, there exists w such that $wi = u$, $pw = v$.

b) Every $f \in \mathrm{Fl}(\mathcal{M})$ factors as

$$f \;=\; p\,i\,, \qquad \text{with } i \in \Phi,\, p \in \Psi\;.$$

We write

$$\Phi \;\leftrightarrows\; \Psi \tag{4.1.2}$$

to indicate that (Φ, Ψ) is a Quillen precouple.

We say that the Quillen precouple is *closed*, if Φ is left closed, Ψ right closed. We shall see below (cor. 2, page 31) that this implies (hence is equivalent to)

$$\Phi \;=\; \Psi^*\,, \qquad \Psi \;=\; \Phi_*\;. \tag{4.1.3}$$

A closed Quillen precouple is also called a Quillen couple. [page 29]

Proposition 4.2. *Let* (Φ, Ψ) *be a Quillen precouple in* $\mathcal{M}$*, and let* $j \in \mathrm{Fl}(\mathcal{M})$[9]. *We then have the equivalent conditions on* j *:*

a) $j \in \widetilde{\Phi}$.

b) $j \in \Psi^*$.

c) j *is a direct factor of an* $i \in \Phi$.

[9] **N.B.** We use only that $\Phi \longleftrightarrow \Psi$, and that j can be put in the form pi, with $i \in \Phi$, $p \in \Psi$.

c′) *j is a direct factor of an $i \in \Phi$ having the same source as j, more precisely, there exists a diagram*

$$\begin{array}{ccc} A & \xrightarrow{\ j\ } & B \\ {\scriptstyle \mathrm{id}_A}\downarrow\uparrow{\scriptstyle \mathrm{id}_A} & & {\scriptstyle q}\downarrow\uparrow{\scriptstyle p} \\ A & \xrightarrow{\ i\ } & B' \,, \end{array} \tag{4.2.1}$$

with $i \in \Phi$, and p. 20

$$qj \;=\; i\,, \qquad pi \;=\; j\,, \qquad pq \;=\; \mathrm{id}_B\,, \tag{4.2.2}$$

so that j is a direct factor of $i \in \Phi$, via (id_A, q), admitting the retraction (id_A, p).

d) *For every commutative solid-arrow diagram*

$$\begin{array}{ccc} A & \xrightarrow{\ j\ } & B \\ & {\scriptstyle i}\searrow \quad \nearrow{\scriptstyle p} & \\ & B' \;\circlearrowleft\; {\scriptstyle q} & \end{array}\,, \tag{4.2.3}$$

i.e. *every factorization of j as pi, if we have $p \in \Psi$, there exists a section q of B' over B in $A\backslash\mathcal{M}$,* i.e. *an arrow $q : B \longrightarrow B'$ such that*

$$pq \;=\; \mathrm{id}_B \qquad \textit{and} \qquad qj \;=\; i\,.$$

Proof of the proposition. It proceeds by the following sequence of implications, all (±) tautological:

$$(c') \implies (c) \implies (a) \implies (b) \implies (d) \implies (c')\,,$$

of which the first four use only the hypothesis $\Phi \longleftrightarrow \Psi$ (from (a) of definition 4.1), and [page 30] only the last, $(d) \Longrightarrow (c')$, calls on the Quillen factorization hypothesis ((b) in 4.1); and even then it suffices to assume it for j only!

$(c') \Longrightarrow (c)$ pure tautology, $(c) \Longrightarrow (a)$ since we know that $\widetilde{\Phi}$ is stable under direct factors (and contains Φ), $(a) \Longrightarrow (b)$ since $\widetilde{\Phi} \subset \Psi^*$ ($\widetilde{\Phi}$ being by definition the smallest left closed subset of $\mathrm{Fl}(\mathcal{M})$ containing Φ, and Ψ^* being left closed), $(b) \Longrightarrow (d)$ since we write the given diagram (4.2.3) (solid arrows) as

$$\begin{array}{ccc} A & \xrightarrow{\ u\ } & X \\ {\scriptstyle j\in\Psi^*}\downarrow & \nearrow{\scriptstyle q} & \downarrow{\scriptstyle p\in\Psi} \\ B & \xrightarrow{\ v\ } & Y\,, \end{array} \tag{$*$}$$

putting $Y = B$, $v = \mathrm{id}_B$ in it, and rebaptizing X as B', and u as i: the hypothesis (b), $j \in \Psi^*$, means that in a solid-arrow diagram $(*)$, there always exists q such that $qj = u$, $pq = v$, and expressing this in the particular case where $v = \mathrm{id}_B$, we find (d).

It remains to prove $(d) \Longrightarrow (c')$, this time using the factorization hypothesis on j, *i.e.* that we can write j in the form pi:

$$\begin{array}{ccc} A & \xrightarrow{\ j\ } & B \\ & {\scriptstyle i}\searrow \quad \nearrow{\scriptstyle p} & \\ & B' \;\circlearrowleft\; {\scriptstyle q} & \end{array}\,,$$

p. 21
[page 31]

with $i \in \Phi$, $p \in \Psi$. By virtue of (d), since $p \in \Psi$, there thus exists $q : B \longrightarrow B'$ satisfying

$$pq \;=\; \mathrm{id}_B\;, \qquad qj \;=\; i\;,$$

which proves (c'), since $i \in \Phi$.

Corollary 1. *Let (Φ, Ψ) be a Quillen precouple. Then we have*

$$\widetilde{\Phi} \;=\; \Psi^* \;=\; \begin{array}{l}\textit{set of objects of } \underline{\mathrm{Fl}}(\mathcal{M}) \textit{ which are direct fac-}\\ \textit{tors of elements of } \Phi\;,\end{array} \tag{4.3}$$

and dually

$$\overline{\Psi} \;=\; \Phi_* \;=\; \begin{array}{l}\textit{set of objects of } \underline{\mathrm{Fl}}(\mathcal{M}) \textit{ which are direct fac-}\\ \textit{tors of elements of } \Psi\;.\end{array} \tag{4.4}$$

Corollary 2. *The following conditions on the Quillen precouple (Φ, Ψ) are equivalent.*

a) *The precouple is* closed, i.e. $\Phi = \widetilde{\Phi}$, $\Psi = \overline{\Psi}$.

b) $\Phi = \Psi^*$, $\Psi = \Phi_*$.

c) *Φ and Ψ are stable under direct factors.*

Now let

$$W \;\subset\; \mathrm{Fl}(\mathcal{M}) \tag{4.5}$$

be a localizer, satisfying the usual conditions

$$\begin{cases} a) & W \text{ contains the isomorphisms.}\\ b) & \text{If } X \xrightarrow{f} Y \xrightarrow{g} Z \text{ in } \mathcal{M}, \text{ if two of } f,\ g,\ gf \text{ are}\\ & \text{in } W, \text{ then the third too.}\end{cases} \tag{4.6}$$

(W mildly saturated). Suppose moreover given four other subsets

$$C\;,\; TC\;, \qquad F\;,\; TF \tag{4.7}$$

of $\mathrm{Fl}(\mathcal{M})$, satisfying the following conditions.

$$\begin{cases} a) & (C, TF) \text{ and } (TC, F) \text{ are Quillen precouples.}\\ b) & TC \subset C \cap W,\ TF \subset F \cap W.\end{cases} \tag{4.8}$$

These properties are spelled out as follows:

[page 32]

M 1 Lifting-extension properties, $C \longleftrightarrow TF$ and $TC \longleftrightarrow F$.

M 2 Factorization property for every arrow f of $\mathcal{M}$ as $f = pi$, with, at our choice, $i \in C$, $p \in TF$, or $i \in TC$, $p \in F$.

M′ $TC \subset C \cap W$, $TF \subset F \cap W$.

This is therefore part of the axioms of a weak Quillen structure for W (XIII, p. 27, def. 1) – the stability axioms M 3, M 4, and the "properness" (propriety) axioms M″ are missing, namely $F \subset \mathrm{Fib}_W$, $C \subset \mathrm{Cof}_W$. p. 22

Proposition 4.3. *Let $\mathcal{M}$ be a category, (C, TF) and (TC, F) two Quillen precouples in $\mathcal{M}$, satisfying*

$$TC \;\subset\; C\;, \qquad TF \;\subset\; F\;, \tag{4.3.1}$$

and let W be a subset of $\mathrm{Fl}(\mathcal{M})$ *of which for the moment we assume only* 4.6, (b)[10]*, and*

$$W \supset TC\ ,\ TF\ , \tag{4.3.2}$$

so we have

$$TC \subset C \cap W\ , \qquad TF \subset F \cap W\ . \tag{4.3.3}$$

Let us consider the following extension-lifting conditions (a), (a′)*, and the factorization conditions* (b), (b′), (c), (d) (*recalling that* $\widetilde{TC} = F^*$, $\overline{TF} = C_*$*, by virtue of proposition* 4.2 (*cor.* 1) *applied to the Quillen precouples* (TC, F), (C, TF)):

a) $C \cap W \subset \widetilde{TC}$ (*hence* $TC \subset C \cap W \subset \widetilde{TC}$).

a′) $F \cap W \subset \overline{TF}$ (*hence* $TF \subset F \cap W \subset \overline{TF}$).

b) $W \subset TF \circ \widetilde{TC}$.

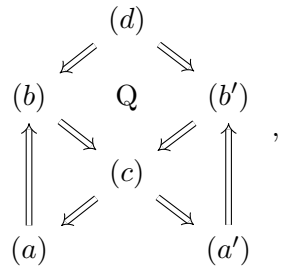
b′) $W \subset \overline{TF} \circ TC$.

c) $W \subset \overline{TF} \circ \widetilde{TC}$.

d) $W \subset TF \circ TC$ (*hence, by* (4.3.2), $W = TF \circ TC$ *if W is stable under composition*).[11]

Then all the conditions except the last are equivalent (*and they are therefore implied by the latter*)*. Thus when TC or TF are stable under direct factors* (*e.g. if TC is left closed, or TF right closed*)*, all the conditions under consideration are equivalent* (*since by corollary* 2 *to proposition* 4.2*, we then have $TC = \widetilde{TC}$ resp. $TF = \overline{TF}$, hence* (b′) *resp.* (b) *is then equivalent to* (d)). [page 33]

Proof. It proceeds by the diagram of implications



, p. 23

where the four implications of the upper square Q are tautological. It thus suffices to establish the implications

$$(c) \implies (a) \implies (b)\ ,$$

the implications $(c) \Longrightarrow (a') \Longrightarrow (b')$ follow from these by duality.

Suppose then (c), *i.e.*

$$W \subset \overline{TF} \circ \widetilde{TC}\ , \tag{$*$}$$

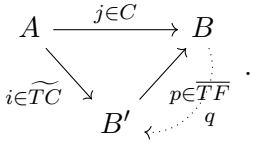
and let us prove (a), *i.e.* that

$$C \cap W \subset \widetilde{TC}\ .$$

But if $j \in C \cap W$, to prove that $j \in \widetilde{TC}$, let us write $j = pi$, $i \in \widetilde{TC}$, $p \in \overline{TF}$, which is possible by the hypothesis $(*)$ and $j \in W$. [page 34]

[10] We use only that if $X \xrightarrow{f} Y \xrightarrow{g} Z$, and if $gf \in W$, then $f \in W \iff g \in W$. We do not need W to contain the isomorphisms, nor to be stable under composition.

[11] **N.B.** Condition (a) also means that $(C \cap W, F)$ is still a Quillen precouple, resp. (b) means that $(C, F \cap W)$ is one – and then this pair of Quillen precouples this time satisfies (d), *i.e.* $W = TF' \circ TC'$.



Let us now use the hypothesis $j \in C$ and $C \longleftrightarrow \overline{TF}$, we thus find (prop. 4.2, $(a) \Longrightarrow (d)$) an arrow $q : B \longrightarrow B'$ such that $pq = \mathrm{id}_B$, $qj = i$, so j appears as a direct factor of i. Since $i \in \widetilde{TC}$ and $\widetilde{TC}$ is stable under direct factors, we conclude $j \in \widetilde{TC}$, *q.e.d.*

$(a) \Longrightarrow (b)$. We thus assume

$$C \cap W \ \subset \ \widetilde{TC} \ , \tag{$*$}$$

and we want to prove (b), *i.e.*

$$W \ \subset \ TF \circ \widetilde{TC} \ .$$

Let then $f \in W$, and let us factor f as pi, with $i \in C$, $p \in TF$. (O.K. because of the Quillen precouple (C, TF)). Since $TF \subset W$, we have $p \in W$, and since $pi \in W$, we conclude $i \in W$ (mild saturation of W – we do have to use it after all!). We thus have $i \in C \cap W$, hence $i \in \widetilde{TC}$ by $(*)$, whence $f \in TF \circ \widetilde{TC}$, *q.e.d.*

Corollary 1. *Suppose TC stable under direct factors. Then all the conditions of proposition* 4.3 *are equivalent, and are equivalent to the relation*

$$C \cap W \ = \ TC \ . \tag{4.3.4}$$

Dually, if TF is stable under direct factors, the conditions of proposition 4.3 *are all equivalent and are equivalent to*

$$F \cap W \ = \ TF \ . \tag{4.3.4$'$}$$

Indeed, by corollary 2 of proposition 4.2, the hypothesis on TC implies that $TC = \widetilde{TC}$, so since $TC \subset C \cap W$, the hypothesis $C \cap W \subset \widetilde{TC}$ $(= TC)$ is equivalent to $C \cap W = TC$. p. 24 [page 35]

Corollary 2. *Suppose C and W stable under direct factors. Then the equivalent conditions of proposition* 4.3 *are equivalent to*

$$C \cap W \ = \ \widetilde{TC} \ . \tag{4.3.5}$$

Dually, if F, W are stable under direct factors, the equivalent conditions of proposition 4.3 *are equivalent to*

$$F \cap W \ = \ \overline{TF} \ . \tag{4.3.5$'$}$$

Indeed, if C and W are stable under direct factors, the same is true of $C \cap W$, and hence if $C \cap W \subset \widetilde{TC}$, since $C \cap W$ contains TC, we conclude $\widetilde{C \cap W} = \widetilde{TC}$. But since $C \cap W \longleftrightarrow F$, $(C \cap W, F)$ is a Quillen precouple, and $C \cap W$ is stable under direct factors, $C \cap W$ is left closed by corollary 2 of proposition 4.2, *i.e.* $C \cap W = \widetilde{C \cap W}$, whence $C \cap W = \widetilde{TC}$, *q.e.d.*

Corollary 3. *Suppose the equivalent conditions* (a) *to* (c) *of* 4.3 *satisfied, and moreover W* stable under direct factors[12]. *Then the preliminary conditions* (4.3.1) *and* (4.3.2) *remain satisfied when one replaces (C, TF) and (TC, F) by their closures*

[12] In fact, it suffices to suppose that we have $W \supset \widetilde{TC}, \overline{TF}$, *i.e.* that every direct factor of an element of W which is either in C, or in F, is still in W. Then $(\widetilde{C}, \widetilde{TC}, \bar{F}, \overline{TF})$ define a closed Quillen structure, and we know that this implies that in fact, W too is stable under direct factors.

$(C', TF') = (\widetilde{C}, \overline{TF})$ *and* $(TC', F') = (\widetilde{TC}, \bar{F})$*, and the equivalent conditions of proposition* 4.3 *are still satisfied for these couples, in the strong forms*

$$\begin{cases} C' \cap W \;=\; TC' & (= \widetilde{TC'}\,) \\ F' \cap W \;=\; TF' & (= \overline{TF'}\,) \\ \quad W \;=\; TF' \circ TC' \;. \end{cases} \tag{4.3.6}$$

The triple (W, C', F') *is a closed Quillen triple* (*corresponding to a "closed model category" of Quillen, at least if one assumes* $\mathcal{M}$ *stable under finite* $\varinjlim$ *and finite* $\varprojlim$*, and that* W *is mildly saturated*).[13] [page 36]

Proof. The relations (4.3.1) imply

$$\widetilde{TC} \;\subset\; \widetilde{C} \qquad \text{and} \qquad \overline{TF} \;\subset\; \bar{F}\;,$$

i.e. p. 25

$$TC' \;\subset\; C'\;, \qquad TF' \;\subset\; F'\;,$$

i.e. (4.3.1) for the new Quillen couples. The relation 4.3.2, $W \supset TC, TF$, implies

$$W \;\supset\; \widetilde{TC}\;,\; \overline{TF}\;, \qquad \textit{i.e.} \qquad W \;\supset\; TC'\;,\; TF'\;,$$

by virtue of corollary 2 to proposition 4.2 and the fact that W is stable under direct factors; so we have (4.3.2) for the new couples. Since by hypothesis we have

$$W \;=\; \overline{TF} \circ \widetilde{TC}\;,$$

this reads

$$W \;=\; TF' \circ TC'\;,$$

so the new couples satisfy the same conditions, which this time, the couples being closed, take the strong form (4.3.6).

Let us prove that the triple (W, C', F') satisfies Quillen's axioms for a closed Quillen triple. We do have, by (4.3.6),

$$\begin{cases} C' \;=\; (TF')^* \;=\; (F' \cap W)^*\;, & F' \;=\; (TC')_* \;=\; (C' \cap W)_*\;, \\ W \;=\; TF' \circ TC' \;=\; C'_* \circ F'^*\;, \end{cases}$$

which corresponds to the definition of Quillen's *closed* model categories: Quillen's axioms M 1, M 2, M 3, M 4 are trivially satisfied (M 3 and M 4 because C and F, TC and TF are left closed resp. right closed). Axiom M 5 is the mild saturation of W. [page 37]

4.4. Here is the idea for constructing more or less "all" the closed Quillen structures (C, TC, F, TF) associated to a given localizer $W \subset \mathrm{Fl}(\mathcal{M})$. We suppose that $\mathcal{M}$ satisfies the conditions of the preceding paragraph, concerning the existence of filtering $\varinjlim$, of a cardinal filtration, accessibility in $\mathcal{M}$ (that of the elements of Filt^{π_0}). We start from a small subset

$$C_0 \;\subset\; \mathrm{Fl}(\mathcal{M})$$

[13] This gives the

Corollary 4. *Let* W *be a mildly saturated localizer, and* (C, TF)*,* (TC, F) *two closed Quillen couples such that* $TC \subset C \cap W$*,* $TF \subset F \cap W$*. Then the following conditions are equivalent.*

a) $TC = C \cap W$.

b) $TF = F \cap W$.

c) $W = TF \circ TC$.

d) (W, C, F) *is a closed Quillen triple.*

such that

$$C_{0*} \subset W \ , \tag{4.4.1}$$

and we set

$$\begin{cases} C = \widetilde{C_0} \ , & TC = C \cap W \ , \\ TF = C_* = C_{0*} \ , & F = (TC)_* \ . \end{cases} \tag{4.4.2}$$

There is only one problem, since we already know that (C, TF) is a Quillen couple by the factorization theorem in §1 (p. 10): we need the factorization for the couple $(TC, F = (TC)_*)$. At that point, the preliminary conditions (4.3.1), (4.3.2) of proposition 4.3 are satisfied:

$$TC \subset C \ , \qquad TF \subset F \ , \qquad W \supset TC \ , \ TF \ ,$$

as well as condition (a), $C \cap W \subset \widetilde{TC}$, since $C \cap W = TC$. So we find, by the equivalence of conditions (a), (a'), (c), and since TC and TF are closed,

$$TF = F \cap W \ , \qquad W = TF \circ TC \ ,$$

on the other hand the hypothesis on W implies that W is stable under direct factors, C is too, being left closed, so $TC = C \cap W$ also, so by proposition 4.2, corollary 2, we have $TC = \widetilde{TC}$, TC is left closed. So we win, by the last assertion of corollary 3, proposition 4.3 (p. 35). p. 26 [page 38]

In fact, independently of the particular conditions on $\mathcal{M}$, W, we have just proved this:

Lemma 4.4.1 (corollary of proposition 4.3.). *Let $\mathcal{M}$ be a category, W a localizer in $\mathcal{M}$ mildly saturated and stable under direct factors. Then the closed Quillen structures (C, F) associated to W are obtained as follows: one takes a subset C_0 of* $\mathrm{Fl}(\mathcal{M})$ *such that*

$$TF \overset{\text{def}}{=} C_{0*} \subset W \ ,$$

one sets

$$C = \widetilde{C_0} = TF^* \ , \qquad TC = C \cap W \ , \qquad F = (TC)_* \ ,$$

and one must manage to get (C, TF) and (TC, F) to satisfy Quillen's factorization condition, i.e. *be Quillen precouples.* (In fact, C, TF and F being closed by construction, and $TC = C \cap W$ being stable under direct factors (C and W being so), they will even be Quillen *couples.*)

The additional hypotheses made earlier on $\mathcal{M}$ served to ensure that if I take C_0 small, then the factorization works for (C, TF) automatically. There remains the problem for TC, F. [page 39]

Let us note that if it works, then $TC = \widetilde{TC} = F^*$, so TC is left closed, and *a fortiori* satisfies the corresponding stability conditions:

$$\begin{cases} a) & \text{Stable under cobase change, under composition, under direct factors (these last two are ensured automatically, since } C, W \text{ are stable).} \\ b) & \text{Stable under strict ordinal } \varinjlim. \end{cases} \tag{4.4.3}$$

The stability hypothesis on W serves (among other things) to guarantee stability (b) for TC (in fact, for this alone it would suffice that W be stable under strict ordinal $\varinjlim$). There remains the stability of $TC = C \cap W$ under cobase change, which is problematic since W is in general *not* stable under cobase change. It is to ensure this stability that I find myself practically compelled here to impose

$$C_0 \subset \mathrm{Cof}_W \ , \tag{4.4.4}$$

I say that with the stability hypothesis made on W, this implies

$$C \overset{\text{def}}{=} \widetilde{C_0} \subset \mathrm{Cof}_W \ . \tag{4.4.5}$$

Indeed, we have:

Proposition 4.4.2. *Let $\mathcal{M}$ be stable under small filtering direct $\varinjlim$, and $W \subset \mathrm{Fl}(\mathcal{M})$ a localizer (mildly saturated)* stable under the said limits. *Then Cof_W satisfies the stability conditions* (4.4.3) ("*left Quillen stability*"), *and is also stable under arbitrary filtering $\varinjlim$.*

p. 27
[page 40]

Indeed, condition (a) is satisfied without any additional condition on W. On the other hand, it is immediate that this condition with respect to filtering $\varinjlim$ implies that Cof_W *is stable under filtering direct limits* in $\underline{\mathrm{Fl}}(\mathcal{M})$.

Corollary 4.4.3. *Under the conditions of proposition* 4.4.2, *and assuming that every arrow factors as a W-cofibration followed by a W-equivalence,* $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$ (*set of co-universal W-equivalences in $\mathcal{M}$) is stable under filtering direct limits, and hence "left stable" in the sense of* (4.4.3).

Corollary 4.4.4. *Under the conditions of proposition* 4.4.2, *suppose moreover that simple amalgamated sums exist in $\mathcal{M}$, and let C_0 be a small subset of* Cof_W *whose elements are accessible, and consider*

$$C \ = \ \widetilde{C_0} \ , \qquad TC \overset{\text{def}}{=} C \cap W \ .$$

Then we have

$$C \ \subset \ \mathrm{Cof}_W \ , \qquad TC \ = \ C \cap W^{\mathrm{univ}} \ \subset \ W^{\mathrm{univ}} \ ,$$

and C and TC are left Q-stable.

Indeed, by theorem 1 of paragraph 1 (p. 10) and its corollary, $\widetilde{C_0}$ is the smallest subset of $\mathrm{Fl}(\mathcal{M})$ containing C_0 and left Q-stable. Since Cof_W is left Q-stable and contains C_0, we therefore have $C = \widetilde{C_0} \subset \mathrm{Cof}_W$. So $TC \overset{\text{def}}{=} C \cap W \subset \mathrm{Cof}_W \cap W \overset{\text{def}}{=} W^{\mathrm{univ}}$, so $TC = C \cap W^{\mathrm{univ}}$. Since W^{univ} is itself also left Q-stable (preceding corollary), the same is true of $C \cap W$, *q.e.d.*

[page 41]

Corollary 4.4.5. *Suppose that every object of $\mathcal{M}$ is accessible. Under the conditions of the preceding corollary, setting*

$$TF \ = \ C_{0*} \ = \ C_* \ , \qquad F \ = \ (TC)_* \ ,$$

the couple (C, TF) is a Quillen couple. For the same to hold for the couple (TC, F), it suffices that there exist a small *subset TC_0 of $C \cap W$, such that $\widetilde{TC_0} = TC$ ($\overset{\text{def}}{=} C \cap W$).*

The first assertion is, as we saw earlier, theorem 1 of paragraph 1. The second follows from this same theorem. We can be more precise in this direction:

Proposition 4.4.6. *Let $\mathcal{M}$ be a category stable under small filtering $\varinjlim$, under amalgamated sums, and all of whose objects are accessible, W a localizer mildly saturated in $\mathcal{M}$, stable under filtering $\varinjlim$, C_0 a* small *subset of* $\mathrm{Fl}(\mathcal{M})$, *such that*

$$C_0 \ \subset \ \mathrm{Cof}_W \ , \qquad \underbrace{C_{0*}}_{(=TF)} \ \subset \ W \ ,$$

[page 42]

let us set

$$C \ = \ \widetilde{C_0} \qquad TC \ = \ C \cap W \ , \qquad TF \ = \ C_* \ , \qquad F \ = \ (TC)_* \ ,$$

and let TC_0 be a small subset of $TC = C \cap W$. Put $F_0 = (TC_0)_*$. We then have

p. 28

$$\begin{cases} C \ \subset \ \mathrm{Cof}_W \ , & TC \ = \ C \cap W^{\mathrm{univ}} \quad (\textit{for the record}) \\ \widetilde{TC_0} \ \subset \ TC \ , & F_0 \supset F \quad (\textit{the latter trivial}) \ , \end{cases} \tag{4.4.6}$$

and the following conditions are equivalent:

a) $\widetilde{TC_0} = TC$ (*we have* $\subset$);

a′) $F_0 = F$ (*we have* $\supset$);

b) $TF = F_0 \cap W$ (*we have* $\subset$)[14];

c) $W = TF \circ \widetilde{TC_0}$ (*we have* $\supset$)[15];

and imply that (W, C, F) *is a closed Quillen triple.*

Proof. The inclusion $\widetilde{TC_0} \subset TC$ follows from the left Q-stability of TC (cor. 4.4.4) and from the fact that $TC_0 \subset TC$ is small (one then applies theorem 1 just as for $C \subset \mathrm{Cof}_W \ldots$). The equivalence of (a), (b), (c) is none other than proposition 4.3 applied to W and the two Quillen couples (C, TF), $(\widetilde{TC_0}, F_0)$ (taking into account theorem 1 of paragraph 1). That we then have a closed Quillen triple was seen in corollary 3 of proposition 4.3 (p. 35). It remains to prove the equivalence of (a) and (a'). We have $(a) \Longrightarrow (a')$, since $(TC_0)_* = (\widetilde{TC_0})_* = F_0$. Conversely, (a') implies $\widetilde{TC_0} = \widetilde{TC}$, hence the factorization theorem for (TC, F) by virtue of the factorization theorem, corollary 1 (p. 11) applied to $TC_0 \subset TC \subset \widetilde{TC_0}$ and to $F_0 = F = (TC_0)_*$.

Thus, the only problem, once C_0 (small) has been found satisfying $C_0 \subset \mathrm{Cof}_W$, $C_{0*} \subset W$, is to find a *small* subset $TC_0 \subset TC$ which "generates $TC = C \cap W$", *i.e.* which satisfies the equivalent conditions (a), (a'), (b), (c) of the preceding proposition. I have spent days trying to prove the existence of TC_0, assuming $\mathcal{M}$ equipped with a cardinal filtration, and W an accessible subset of $\mathcal{M}$, without having succeeded. Of course, all that really matters to us is that (TC, F) satisfy the factorization condition. But I see no way of proving it, except by "generating" TC on the left by a *small* subset TC_0. [page 43]

5 Preliminary technical results on accessible subsets of $\mathrm{Ob}\,\mathcal{M}$, stable under filtering direct limits

Yet I think I can arrive at something usable, by working systematically with C's ($\subset \mathrm{Cof}_W$, left Q-closed, left Q-generated by a *small* subset C_0 of C) which are stable under all filtering direct limits.

I again assume the following conditions on $\mathcal{M}$: p. 29 [page 44]

a) $\mathcal{M}$ stable under small direct limits (and not only filtering ones).

b) $\mathcal{M}$ endowed with a cardinal filtration $(\mathrm{Filt}^\pi)_{\pi \geq \pi_0}$.

c) The objects of $\mathcal{M}$ are accessible (**N.B.** It suffices to assume this for the objects of Filt^{π_0}).

Condition (b) implies that $\mathcal{M}$ contains a small generating family by strict epimorphisms (e.g. a small subcategory $\mathcal{C}$ of $\underline{\mathrm{Filt}}^{\pi_0} = \mathcal{M}^{\pi_0}$ such that the inclusion $\mathcal{C} \longrightarrow \underline{\mathrm{Filt}}^{\pi_0}$ is an equivalence of categories). By SGA 4 I 8.12.7, it follows that $\mathcal{M}$ is stable under small inverse limits. There follows the

Lemma 5.1 (under conditions (a), (b)). *For every object X of $\mathcal{M}$, the set* $\mathrm{Ss}(X)$ *of subobjects of X, endowed with the natural order relation "of inclusion", admits* inf *and* sup *of arbitrary size.*

inf, since these are expressed as inverse limits in $\mathcal{M}$. Now the existence of inf implies that of sup, since $\sup_J = \inf_{J'}$, where J' is the set of upper bounds of J in $\mathrm{Ss}(X)$.

[14] **N.B.** We see easily that $\boxed{TF = F \cap W,}$ but that does not suffice to have $TF = F_0 \cap W$ for a suitable TC_0.

[15] **N.B.** We know that $\boxed{W = TF \circ TC,}$ which is in no way sufficient to imply (c)!

Corollary 5.2. *For every arrow $f : Z \longrightarrow X$ in $\mathcal{M}$, there exists, among the subobjects X' of X through which f factors, a smallest one (which we will denote* Im f*). The morphism $Z \longrightarrow \operatorname{Im} f$ is an epimorphism.* [page 45]

This last assertion comes from this one:

Lemma 5.3. *Let $\mathcal{M}$ be a category in which kernels of double arrows exist. Let $f : Z \longrightarrow Y$ be an arrow of $\mathcal{M}$. Then f is an epimorphism if and only if f does not factor through a subobject that is* strict*, Y_0, of Y different from Y.*

We are going to make an additional (practically innocuous) hypothesis on the Filt^π:

d) Filt^π is stable under direct limits indexed by index categories I such that $\operatorname{card} \mathrm{Fl}(I) \leq \pi$[16]. (It suffices to assume stability under initial object, simple amalgamated sums, and filtering $\varinjlim$ indexed by an ordered set I of cardinality $\leq \pi$.) Moreover, if $f : Z \longrightarrow X$ is an arrow of $\mathcal{M}$ such that Z or X is in Filt^π, then $\operatorname{Im} f \in \mathrm{Filt}^\pi$.

This last condition on Im f splits into two: [page 46]

d_1) Every subobject of an object of Filt^π is in Filt^π.

d_2) If Z is in Filt^π, then every quotient Z' of Z such that $\operatorname{Im} p = Z'$ (where $p : Z \longrightarrow Z'$ is the canonical morphism) is in Filt^π.[17]

Of course, in practice it will always be true that Filt^π is stable under passage to subobjects and quotients of an object (not only the particular quotients at issue in (d_2)). Besides, when every bimorphism (*i.e.* monomorphism *and* epimorphism) of $\mathcal{M}$ is an isomorphism, then (d_2) is indeed equivalent to stability of Filt^π under quotients of objects. p. 30

Proposition 5.3. *Suppose conditions* (a), (b), (d) *are satisfied for $\mathcal{M}$. Let π be a cardinal $\geq \pi_0$, and X an object of $\mathcal{M}$. Let $\mathrm{Ss}_\pi(X)$ be the set of subobjects of X that are in* Filt^π*. This set, ordered by the order relation induced by* $\mathrm{Ss}(X)$*, is large relative to π (and in particular filtering). If, moreover, filtering direct limits of monomorphisms in $\mathcal{M}$ are monomorphisms*[18] *(e.g. if filtering direct limits are left exact), or if the objects of* Filt^π *are π-accessible, we have* $\varinjlim_{\alpha \in \mathrm{Ss}_\pi(X)} Z_\alpha \xrightarrow{\sim} X$[19].

Proof. Let $(Z_\alpha)_{\alpha \in A}$ be a family of subobjects of X such that $Z_\alpha \in \mathrm{Filt}^\pi$, and $\operatorname{card} A \leq \pi$. Let $Z = \coprod_{\alpha \in A} Z_\alpha$, we have $Z \in \mathrm{Filt}^\pi$ by (d_0). The smallest subobject of X that is an upper bound for the Z_α is none other than Im $(Z \longrightarrow X)$, which is in Filt^π by (d_2). Let us prove that $\varinjlim_{\alpha \in \mathrm{Ss}_\pi(X)} Z_\alpha \xrightarrow{\sim} X$ is an isomorphism, when we know that it is a monomorphism. Now [page 47]

$$X = \varinjlim_I X_i \ ,$$

where $(X_i)_{i \in I}$ is a filtering direct system, I an ordered set large relative to π, the X_i in Filt^π. For every $i \in I$, let X'_i be the image of X_i in X. We therefore have $X'_i \in \mathrm{Filt}^\pi$ by (d_2). Using the fact that the $X_i \longrightarrow X'_i$ are epimorphisms, we see at once that $X = \varinjlim_I X'_i$. Consider the map

$$\begin{array}{ccc} I & \longrightarrow & \mathrm{Ss}_\pi(X) \\ i & \longmapsto & X'_i \ . \end{array}$$

[16] This condition will be called (d_0) below.

[17] It would seem that we will not need (d_1), only (d_2). But in fact, (d_1) must be a consequence of (d_2), and (d_2) is a consequence of the first part of (d) (stability under finite $\varinjlim$) if every epimorphism in $\mathcal{M}$ is strict, hence effective (if the Filt^π are stable under finite $\varprojlim$ as well …).

[18] This is an *exactness condition* on $\mathcal{M}$, which I will denote as:

ⓔ Monomorphisms in $\mathcal{M}$ are stable under filtering $\varinjlim$ in $\underline{\mathrm{Fl}}(\mathcal{M})$.

[19] Say also that $\mathrm{Ss}_\pi(X)$ is π-adapted, and the direct system $(Z_\alpha)_\alpha$ as well.

This is an increasing map, let J be its image, a filtering subset, large relative to π, of $\mathrm{Ss}_\pi(X)$, such that

$$\varinjlim_{\alpha\in J} Z_\alpha \xrightarrow{\sim} X \ .$$

We therefore have [page 48]

$$\underbrace{\varinjlim_{\alpha\in J} Z_\alpha}_{\simeq X} \longrightarrow \varinjlim_{\alpha\in \mathrm{Ss}_\pi X} Z_\alpha \xrightarrow{\varphi} X \ ,$$

the composite being the identity. If we know that φ is a monomorphism, it follows that it is an isomorphism (a monomorphism having a section). Now φ will be a monomorphism if every filtering $\varinjlim$ of monomorphisms is a monomorphism, and then we win.

If we assume that the elements of Filt^π are π-accessible, we win as well, for we find: p. 31

Corollary 5.4. *If the objects of* Filt^π *are* π*-accessible (e.g. if* $\pi \geq \pi_1$ *and* $\pi = \pi^{\pi_0}$*, e.g.* $\pi \geq \pi_1$ *and* $\pi = 2^c$*, with* $c \geq \pi_0$*,* cf. (3.2) *p. 16), then for every filtering subset* J *of* $\mathrm{Ss}(X)$ *large relative to* π*, such that* $\varinjlim_{\alpha\in J} Z_\alpha \xrightarrow{\sim} X$*,* J *is cofinal in* $\mathrm{Ss}_\pi(X)$ *(hence* $\varinjlim_{\alpha\in J} Z_\alpha \xrightarrow{\sim} \varinjlim_{\alpha\in \mathrm{Ss}(X)} Z_\alpha$*).*

This is immediate.

[page 49]

Scholie 5.5. When we make on $\mathcal{M}$ the fairly innocuous exactness hypothesis that filtering $\varinjlim$ of monomorphisms are monomorphisms, *or* restrict ourselves to cardinals π such that the objects of Filt^π are π-accessible, we can *canonize* the representation of an object X as $\varinjlim_{i\in I} X_i$, with $X_i \in \mathrm{Filt}^\pi$ and I large relative to π, by taking $I = \mathrm{Ss}_\pi(X)$, and the corresponding direct system. We also have the following pleasant additional property: every filtering subset J of I, of cardinality $\leq \pi$, admits a supremum i_J, and (if we assume that $\mathcal{M}$ satisfies condition (e), page 46, note 18), we will have

$$X_{i_J} = \varinjlim_{i\in J} X_i \ ;$$

in other words, we have a representation of X by a direct system that is (perfectly) π*-adapted* (*cf.* XIII, p. 85, 86).

Now let [page 50]

$$M \subset \mathrm{Ob}\,\mathcal{M}$$

be a subset of $\mathrm{Ob}\,\mathcal{M}$, stable under the isomorphism relation. If M is π-accessible (according to condition b_π of XIII, p. 88), then putting

$$M^\pi = M \cap \mathrm{Filt}^\pi \ ,$$

M^π determines M by the following condition on $X \in \mathrm{Ob}\,\mathcal{M}$.

$$\begin{aligned} &X \in M \iff \text{the subset } \mathrm{Ss}_\pi^M(X) \text{ of } \mathrm{Ss}_\pi(X) \text{ consisting of the } X_\alpha \in \mathrm{Ss}_\pi(X) \\ &\text{such that } X_\alpha \in M \ (\iff X_\alpha \in M^\pi) \text{ is cofinal in } \mathrm{Ss}_\pi(X). \end{aligned} \tag{5.1}$$

Note that when $X \in \mathrm{Filt}^\pi$, this criterion is trivially valid, since then $\mathrm{Ss}_\pi(X)$ admits X as greatest element, and the cofinal subsets are those containing this final object. Thus (5.1) is interesting only for the $X \in \mathrm{Ob}\,\mathcal{M}\backslash\mathrm{Filt}^\pi$. Note that for *every* subset M^π of Filt^π, we can define a subset M of $\mathrm{Ob}\,\mathcal{M}$ by condition (5.1), and we will trivially have

$$M^\pi = M \cap \mathrm{Filt}^\pi \ . \tag{5.2}$$

But probably, although M satisfies (5.1), it is not for all that necessarily π-accessible. p. 32
Indeed, for it to be so, it must be stable under $\varinjlim$ large relative to π of elements of [page 51]

M^π, which implies that every direct factor of an element of M^π is in M, hence also in M^π, according to the stability condition (d) on Filt^π. This is a particular condition on the subset M^π of Filt^π. So, we will have to have a condition on $\mathcal{M}$ stronger than the validity of (5.1), namely:

> For every $X \in \mathrm{Ob}\,\mathcal{M}$ and for every subset J of $\mathrm{Ss}_\pi(X)$, such that
> *a)* J is cofinal in $\mathrm{Ss}_\pi(X)$ and
> *b)* If $J' \subset J$ is a filtering subset of J, of cardinality $\leq \pi$, then its supremum $\alpha_{J'}$ in $\mathrm{Ss}_\pi(X)$ belongs to J,
>
> we have $X \in M$ if and only if $J^M \overset{\text{def}}{=} J \cap \mathrm{Ss}_\pi^M(X)$ is cofinal in J (or equivalently, in $\mathrm{Ss}_\pi(X)$) [20]. (5.3)

If we already admit the "it suffices" of (5.1), then it suffices in (5.3) to assume $X \in M$ and require that J^M then be cofinal in J.

But note the [page 52]

Proposition 5.6.[21] *Suppose π chosen in such a way that the objects of* Filt^π *are π-accessible (e.g. $\pi = 2^c$, with $c \geq \pi_1$,* cf. (3.2), *p. 16). Then, for a subset M of* $\mathrm{Ob}\,\mathcal{M}$, *M is π-accessible if and only if it satisfies criterion* (5.3).

We have seen that this is necessary (tautologically, in view of proposition 5.3 and scholie 5.5, bottom of page 49). Let us prove that this is sufficient, by first establishing the

Lemma 5.7. *Under the conditions of* 5.6 *on π, and condition* (e) *on $\mathcal{M}$* (p. 46), *for every π-adapted representation $X = \varinjlim_{i \in I} X_i$ of an object X of $\mathcal{M}$, the subset I' of those $i \in I$ such that $X_i \longrightarrow X$ is a monomorphism, is cofinal in I. Moreover, I' is stable under suprema in I of filtering subsets J of I' of cardinality $\leq \pi$.*

Let us show how this lemma 5.7 implies the "it suffices" of proposition 5.6. So let $X = \varinjlim_{i \in I} X_i$ be a π-adapted representation of X ($X_i \in \mathrm{Filt}^\pi$, I large relative to π, *etc.*). We must prove that, if 5.3 is valid, then $X \in M$ if and only if the set I_M is cofinal in I.

Assume $X \in M$. Consider the subset $I' \subset I$ described in the lemma. We have a map [page 53]
$I' \longrightarrow \mathrm{Ss}_\pi(X)$, namely $i \longmapsto X'_i = \mathrm{Im}\,(X_i \hookrightarrow X)$, and we have

$$X \simeq \varinjlim_{i \in I} X_i \;\simeq\; \varinjlim_{i \in I'} X_i \;\simeq\; \varinjlim_{i \in I'} X'_i \;\simeq\; \varinjlim_{\alpha \in J} Z_\alpha \ ,$$

where J is the image of I' in $\mathrm{Ss}_\pi(X)$. Since the objects of Filt^π are π-accessible, we conclude that J is cofinal in $\mathrm{Ss}_\pi(X)$. By 5.3, it follows that the set J_M of those $\alpha \in J$ such that $Z_\alpha \in M$ is cofinal in J, hence $\neq \emptyset$. Thus I'_M, which is the inverse image of J_M p. 33
in I', is in any case nonempty. Applying the same result while replacing I' by the subset $I'_{\geq i'_0}$, we see that I'_M is cofinal in I', hence also in I, since I' is cofinal in I; thus $I_M \supset I'_M$ is cofinal in I, *q.e.d.*

Conversely, suppose that I_M is cofinal in I, and prove $X \in M$. The difficulty here comes from the fact that the representation $X = \varinjlim_{i \in I_M} X_i$ is no longer adapted to π – I_M is not necessarily stable in I under suprema (in I) of filtering subsets in I_M, I' of cardinality $\leq \pi$. In short, we must prove that M is stable under $\varinjlim$ large relative to π [page 54]

[20] Instead of J cofinal in $\mathrm{Ss}_\pi(X)$, it even suffices to have $\varinjlim_{\alpha \in J} X_\alpha \xrightarrow{\sim} X$ – the corresponding condition on M thus becomes more demanding. But if the objects of Filt^π are π-accessible, this condition is in fact equivalent to the weaker condition.

[21] False as it stands, true if we assume that M^π is stable under filtering $\varinjlim$.

without more, and I now doubt that this follows from just the hypotheses made here, which only concern direct systems where the transition morphisms are monomorphisms.

Here, I think, is a counterexample. Let us take

$$M \quad = \quad \text{set of objects of } \mathcal{M} \text{ isomorphic to a fixed object } X_0 \in \mathrm{Filt}^\pi, \text{ such that } X_0 \text{ has no endomorphism which is a monomorphism other than the identity.} \tag{5.4}$$

Then condition 5.3 is satisfied. For:

a) If $X \in M$, *i.e.* $X \simeq X_0$, then $X \in \mathrm{Filt}^\pi$, hence $\mathrm{Ss}_\pi(X)$ admits X as final object, and J, being cofinal, contains this element, hence J_M (which also contains it) is cofinal.

And:

b) If J_M is cofinal, hypothesis (5.4) implies that the set of transition morphisms between the Z_α ($\alpha \in J_M$) consists of isomorphisms, hence the direct limit X of the Z_α in question is isomorphic to each of them, hence $X \simeq X_0$, *i.e.* $X \in M$.

Yet, M is π-accessible only if it is stable under $\varinjlim$ large relative to π, which implies that it is stable under *direct factors.* But hypothesis (5.4) in no way rules out that X_0 admits [page 55]
a projector p ($p^2 = p$) that is nontrivial. This will have an image X' in $\mathcal{M}$, which will be a direct factor of X_0 not isomorphic to X_0, *i.e.* not in M.

These difficulties disappear if we assume that M^π is stable under filtering $\varinjlim$ indexed by ordered sets I of cardinality $\leq \pi$. Indeed, in this case, in the preceding proof, the subset I_M is stable under suprema (in I) of subsets J of I_M that are filtering and of cardinality $\leq \pi$. Thus the direct system $(X_i)_{i \in I_M}$ is still π-adapted, and we can apply lemma 5.7 to it, to find a cofinal subset $I'_M \subset I_M$, such that the $X_i \longrightarrow X$ for $i \in I'_M$ are monomorphisms. The image J of I'_M in $\mathrm{Ss}_\pi(X)$ is then a subset of $\mathrm{Ss}_\pi(X)$ such that $\varinjlim_{\alpha \in J} Z_\alpha = X$, hence (by the hypothesis made on π) cofinal in $\mathrm{Ss}_\pi(X)$ (5.4), with $Z_\alpha \in M^\pi$ for every $\alpha \in J$. Thus by 5.3 we have $X \in M$, *q.e.d.*

[page 56]

It remains to prove lemma 5.7. With the hypotheses and notation of the lemma, so let $i_0 \in I$, we must find $i' \in I'$ such that $i' \geq i_0$. Consider the image X'_i of X_i in X, and the inclusion

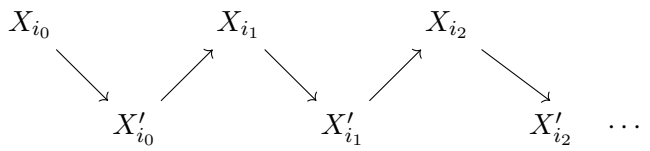
p. 34

$$\underbrace{X'_{i_0}}_{\in \mathrm{Filt}^\pi} \longrightarrow X \simeq \varinjlim X_i \, .$$

Since $X'_{i_0} \in \mathrm{Filt}^\pi$, hence is π-accessible, there is an $i_1 > i_0$ such that $X'_{i_0} \longrightarrow X$ factors through

$$X'_{i_0} \longrightarrow X_{i_1} \, .$$

Applying this same result to X_{i_1}, X'_{i_1} *etc.*, we find an infinite sequence of morphisms



such that the morphisms composed two at a time are, on the upper line, the transitions

$$X_{i_m} \longrightarrow X_{i_n} \qquad (\, i_m < i_n \,) \, ,$$

and on the lower line, the inclusions

$$X'_{i_m} \hookrightarrow X'_{i_n} \qquad (\hookrightarrow X) \, .$$

Let $i_\omega = \sup_n i_n$ in I, hence

$$X_{i_\omega} \;=\; \varinjlim_n X_{i_n} \;\simeq\; \varinjlim_n X'_{i_n} \;\simeq\; X'_{i_\omega}\ ,$$

so $X_{i_\omega} \longrightarrow X$ is a monomorphism.

The last assertion of 5.7 follows at once from the fact that a filtering direct limit of monomorphisms in $\mathcal{M}$ is a monomorphism (condition (e), p. 46). [page 57]

We can make proposition 5.6 obtained in this way more precise:

Proposition 5.7. *Let $\mathcal{M}$ be a category satisfying conditions* (a), (b), (c), (d) (p. 44, 45) and (e) (p. 46). *Let M be a subset of* $\mathrm{Ob}\mathcal{M}$, *stable under isomorphism, π a cardinal such that the objects of* Filt^π *are π-accessible. The following conditions are equivalent:*

a) *M is π-accessible, and stable under filtering $\varinjlim$.*

b) *M is π-accessible, and $M \cap \mathrm{Filt}^\pi$ is stable under filtering $\varinjlim$ of cardinality $\leq \pi$.*

c) *There is a subset M^π of Filt^π, stable under filtering $\varinjlim$ of cardinality $\leq \pi$ and such that for every $X \in \mathrm{Ob}\mathcal{M}$, putting $J = \mathrm{Ss}_\pi(X)$, we have $X \in M$ if and only if the set J_M of the $Z_\alpha \in J$ such that $Z_\alpha \in M^\pi$ is cofinal in J.*

(*Moreover, under condition* (c), *we have* $M^\pi = M \cap \mathrm{Filt}^\pi$.)

p. 35

c′) *There is a subset M_0 of Filt^π such that M is the smallest subset of $\mathcal{M}$ containing M_0 and stable under filtering direct limits.* (*Moreover, we can then take M_0* small.)[22]. [page 58]

Corollary 5.8. *The maps*

$$M \;\longmapsto\; M^\pi = M \cap \mathrm{Filt}^\pi\ ,$$

$$M^\pi \quad\longmapsto\quad \begin{array}{l} \textit{the set of } X \in \mathrm{Ob}\mathcal{M} \textit{ such that the set} \\ \mathrm{Ss}_\pi^{M^\pi}(X) \textit{ of the } Z_\alpha \in \mathrm{Ss}_\pi(X) \textit{ which are in} \\ M^\pi \textit{ is cofinal } (= \textit{smallest subset of } \mathrm{Ob}\mathcal{M} \\ \textit{containing } M^\pi \textit{ and stable under filtering } \varinjlim) \end{array}$$

establish bijections, inverse to one another, between the set of $M \subset \mathrm{Ob}\mathcal{M}$ satisfying the equivalent conditions of proposition 5.7, *and the set of subsets M^π of* Filt^π *stable under filtering $\varinjlim$ of cardinality $\leq \pi$.*

Proof of 5.7. The implication $(b) \Longrightarrow (c)$ is tautological, in view of the definition of the π-accessibility of M, and of proposition 5.3. And $(a) \Longrightarrow (b)$ comes from the stability of Filt^π under filtering $\varinjlim$ of cardinality $\leq \pi$ (condition (d)).

It therefore remains to prove $(c) \Longrightarrow (a)$. (It is here that we must use the special hypothesis on π.) Once this is seen, we shall know the equivalence of (a), (b), (c), and on the other hand, it is immediate that $(a) \Longrightarrow (c')$, by taking $M_0 = M \cap \mathrm{Filt}^\pi$. On the other hand, $(c') \Longrightarrow (c)$ (without hypotheses on π ?), since we can suppose that the subset M_0 of Filt^π is stable under filtering $\varinjlim$ of cardinality $\leq \pi$. But then, knowing the equivalence of (a), (b), (c), whence the corollary at once, it follows that the envelope stable under filtering $\varinjlim$ of M_0 in $\mathcal{M}$ is none other than the accessible subset deduced from M_0 that is in question in (c), and we win. [page 59]

[22] **N.B.** We have

$$\begin{array}{ccc} (a) & \Longleftrightarrow & (b) \\ \Downarrow & & \Downarrow \\ (c') & \overset{?}{\Longrightarrow} & (c) \end{array}\ ,$$

without conditions on π (except $\pi \geq \pi_0$).

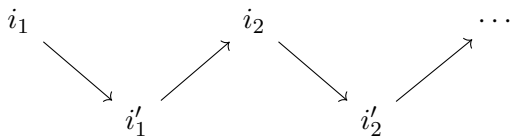
So it all comes down to proving that, if all the objects of Filt^π are π-accessible, then (c) implies (a). Let us first show that (c) implies the membership criterion (5.3) (p. 51), so by proposition 5.6 "revised and corrected" in note 21, it already follows that $(c) \Longrightarrow (b)$, and it will remain to establish $(b) \Longrightarrow (a)$, *i.e.* that (b) implies the stability of M under arbitrary filtering $\varinjlim$ (not only large relative to π).

That (c) implies (5.3) will follow from the following lemma, applied to $I = \mathrm{Ss}_\pi(X)$:

Lemma 5.8. *Let I be an ordered set that is* π-adapted, i.e. *filtering and such that every filtering subset K of I, of cardinality $\leq \pi$, has a supremum i_K in I. I say that a subset J of I is* π-cofinal, *if it is cofinal, and if for every filtering subset K of J of cardinality $\leq \pi$, its supremum i_K in I is in J. (It follows, then, that J too is π-adapted, for the induced order structure.) This said, the intersection of two π-cofinal subsets of I is again π-cofinal.*

p. 36
[page 60]

It is enough to see that it is cofinal, for its stability under suprema in I of filtering subsets of cardinality $\leq \pi$ is tautological (if it is true for some J_α, it is true for $\bigcap_\alpha J_\alpha$). Let $i_0 \in I$, and let us construct an element of $J \cap J'$ that dominates i_0. We construct step by step a double sequence



with the i_α in J, the i'_α in J', and $i_1 \geq i_0$. We thus have

$$\sup i_\alpha \;=\; \sup i'_\alpha \,,$$

and if i is the common value of the two sides, we have $i \in J \cap J'$.

This seen, to prove the validity of (5.3), we note that the subset $J' = \mathrm{Ss}_\pi^{(M^\pi)}(X)$ of $I = \mathrm{Ss}_\pi(X)$, consisting of the $Z_\alpha \subset X$ such that $Z_\alpha \in M^\pi$, is stable under suprema of filtering subsets of cardinality $\leq \pi$, since M^π is stable under $\varinjlim$ of cardinality $\leq \pi$. So if J is a π-cofinal subset of $\mathrm{Ss}_\pi(X)$, $J \cap J'$ is cofinal in J, *i.e.* in I, if and only if J' is cofinal in I (*i.e.* if and only if $X \in M$, by condition (c)). For $J \cap J'$ cofinal $\Longrightarrow$ J' cofinal tautologically, and J' cofinal implies J' π-cofinal, hence $J \cap J'$ π-cofinal by lemma 5.8, hence cofinal.

[page 61]

It remains only to prove that $(b) \Longrightarrow (a)$, *i.e.* the stability of M under arbitrary filtering direct limits, assuming (b). Here again, this should not use the special hypothesis on π, nor even the exactness hypothesis (e) on $\mathcal{M}$, but be a general story of accessibility:

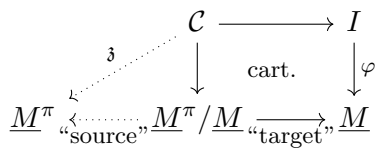
Proposition 5.9. *Let $\mathcal{M}$ be a category satisfying* (a), (b), (c), (d_0) (pages 44, 45), *$\pi \geq \pi_0$ a cardinal, M a π-accessible subset of* $\mathrm{Ob}\mathcal{M}$, i.e. *satisfying condition b_π of* XIII, p. 88. *Then M is stable under filtering $\varinjlim$ if and only if $M^\pi \overset{\text{def}}{=} M \cap \mathrm{Filt}^\pi$ is stable under filtering $\varinjlim$ of cardinality $\leq \pi$.*

I don't feel like spelling out the proof here, having the painful feeling of forever redoing the same proof (here and in XIII). I shall have to come back to this in a more settled situation.

[page 62]

I am going to sketch the proof all the same. Let $X = \varinjlim_I X_i$, I filtering, the X_i in M, we must prove that $X \in M$ (assuming the stability hypothesis on $M^\pi = M \cap \mathrm{Filt}^\pi$).

p. 37



$\mathcal{C}$ is a cofibered category over I, whose fibers are

$$\mathcal{C}_i \simeq M^\pi/X_i ,$$

now since $X_i \in M$, we see, because of the π-accessibility condition, that M^π/X_i is *filtering*. We are going to single this out as

Lemma 5.10. *Let M be a π-accessible subset of $\mathcal{M}$. Then for every $Y \in M$, M^π/Y is a category that is* filtering, *and $\mathrm{Ss}_\pi^{M^\pi}(Y)$ is cofinal in it.*

Let us consider the canonical functor [page 63]

$$I = \mathrm{Ss}_\pi^{M^\pi}(Y) \xrightarrow{\varphi} \underline{M}^\pi/Y = I' ,$$

where the source category I is filtering. By SGA 4 I 8.1.3, this functor is cofinal if and only if it satisfies two conditions F 1, F 2, and then I' is also filtering. Condition F 1 is the usual condition that every $i' \in \mathrm{Ob}I'$ is "dominated" by a $\varphi(i)$, *i.e.* every $Z \longrightarrow Y$, with $Z \in M^\pi$, is "dominated" by a *monomorphism* $Z_\alpha \hookrightarrow Y$, with $Z_\alpha \in M^\pi$, which follows from the fact that it is dominated by $Z_\alpha \hookrightarrow Y$ with $Z_\alpha \in \mathrm{Filt}^\pi$, *i.e.* $Z_\alpha \in \mathrm{Ss}_\pi(Y)$, and that among these Z_α those which are in M^π are cofinal (since $Y \in M$). The other condition[23] is that a double arrow $i' \overset{f'}{\underset{g'}{\rightrightarrows}} \varphi(i)$ can be equalized by an arrow $\varphi(h)$, with [page 64] $h : i \longrightarrow j$. But in fact, for an $i' \in \mathrm{Ob}I'$, *i.e.* a given $Z \longrightarrow Y$ ($Z \in M^\pi$), and an $i \in \mathrm{Ob}I$, *i.e.* a given $Z_\alpha \hookrightarrow Y$ ($Z_\alpha \in M^\pi$), there is at most *one* arrow in $I' = \underline{M}^\pi/Y$ from $Z \longrightarrow Y$ to $Z_\alpha \hookrightarrow Y$. So condition F 2 is trivially satisfied (we take $h = \mathrm{id}_i$).

Corollary 5.11. *If $(X_i)_{i\in I}$ is a filtering direct system, with the X_i in M^π, then the category $\mathcal{C} = \underline{M}^\pi/\underline{M} \times_{\underline{M}} (I, \varphi)$, constructed on page* 62, *is filtering, and the canonical functor (cofibrant) $\mathcal{C} \longrightarrow I$ is cofinal.*

That $\mathcal{C}$ is filtering and $\mathcal{C} \longrightarrow I$ cofinal follows formally from the fact that I is, that the fibers of $\mathcal{C}$ are, and that $\mathcal{C}$ is cofibered over I. (Deserves a lemma, for future reference.)

N.B. We can replace $\mathcal{C}$ by $\mathcal{C}_0$ whose fibers are the

$$\mathrm{Ss}_\pi^{M^\pi}(X_i) \subset \underline{M}^\pi ; \qquad \text{cofinal in } \mathcal{C}_i = \underline{M}^\pi/X_i ,$$

we find a subcategory of $\mathcal{C}$ cofinal in $\mathcal{C}$, and which this time is ordered just like I. [page 65]

We have p. 38

$$X_i = \varinjlim_{Z_\alpha \in \mathcal{C}_i} Z_\alpha ,$$

hence

$$X = \varinjlim_{\alpha \in \mathcal{C}} Z_\alpha ,$$

altogether we have arrived at representing X as a filtering $\varinjlim$ of elements Z_α which are (not only in M, but even) in M^π.

Let us change the notation, taking

$$X = \varinjlim_{i\in I} X_i , \qquad I \text{ filtering ordered set, } X_i \in M^\pi \ \ \forall\, i \in I ,$$

let I' be the set of *filtering* subsets $J \subset I$ such that $\mathrm{card}\, J \leq \pi$, I' ordered by inclusion. Then I' is large relative to π, and we have a direct system

$$(X_J)_{J\in I'} , \qquad X_J = \varinjlim_{i\in J} X_i .$$

[23] Since I is filtering.

By the stability hypothesis on M^π (which has not yet been used), the X_J are in M^π. We therefore have

$$X = \varinjlim_{J \in I'} X_J , \qquad \text{the } X_J \text{ in } M^\pi,\ I' \text{ large relative to } \pi ,$$

so $X \in M$ since M is π-admissible, *q.e.d.*

[page 66]

Definition 5.12.

1) Let $\mathcal{M}$ be a category satisfying conditions (a), (b), (c), (d) (p. 44-46). Let $\pi \geq \pi_0$ be a cardinal. A subset M of $\mathrm{Ob}\mathcal{M}$ is said to be *L-π-accessible,* if it is π-accessible and stable under small filtering $\varinjlim$[24], or, what amounts to the same thing (5.9), if it is π-accessible and if $M^\pi = M \cap \mathrm{Filt}^\pi$ is stable under filtering $\varinjlim$ of cardinality $\leq \pi$. (**N.B.** If the objects of Filt^π are π-accessible, 5.8 tells us that this also means that it suffices for this that there is a subset M_0 of Filt^π such that M is the L-envelope (envelope stable under filtering $\varinjlim$) of M_0 – a condition which is in any case always necessary, by taking $M_0 = M^{\pi}$.)

2) Let M be a subset of $\mathrm{Ob}\mathcal{M}$. We say that it is *L-accessible,* if it satisfies one of the following four equivalent conditions:

   a) There is a cardinal π such that M is L-π-accessible.

   a') M is accessible and L-stable.

   b) There is a small subset M_0 of $\mathrm{Ob}\mathcal{M}$, such that M is the L-envelope of M_0[25].

   b') There is a subset M_0 of $\mathcal{M}$ contained in a subset Filt^π (or, what amounts to the same thing, such that the full subcategory $\underline{M}_0$ of $\mathcal{M}$ is essentially small, *i.e.* equivalent to a small category, such that M is the *L*-envelope of M_0).

The equivalence of (a) and (a'), or of (b) and (b') is more or less tautological. The equivalence of (a) and (b) follows at once from the N.B. included in 1°, taking into account that the cardinals π such that the objects of Filt^π are π-accessible are cofinal in the set of cardinals in $\mathfrak{U}$. (*Cf.* 3.2, page 16).

p. 39

[page 67]

Proposition 5.13. *Every finite union of L-π-accessible subsets of* $\mathrm{Ob}\mathcal{M}$ *is L-π-accessible. The intersection of a family* $(M_\alpha)_{\alpha \in A}$ *of L-π-accessible subsets, with* $\mathrm{card}\, A \leq \pi$*, is L-$\pi$-accessible.*

Proof. The first assertion comes from the fact that a *finite* union of L-stable subsets of M is L-stable, and that we have likewise for the notion of π-accessibility. The two things are moreover proved practically in the same way, using the

Lemma 5.14. *Let I be a filtering ordered set, $(I_\alpha)_{\alpha \in A}$ a* finite *nonempty family of subsets of I, such that $\bigcup I_\alpha = I$. Then there exists $\alpha \in A$ such that I_α is cofinal.*

[page 68]

Proof by contradiction. If for no α, I_α were cofinal, we would conclude from this that for every α there exists an i_α such that $I/i_\alpha \cap I_\alpha = \emptyset$. Let i be an upper bound for the i_α, we will thus have $I/i \cap I_\alpha = \emptyset \ \ \forall\, \alpha \in A$, contrary to the hypothesis that $\bigcup I_\alpha = I$.

The case of an *intersection* of L-π-accessible subsets: it is of course L-stable, so it remains to prove that it is π-accessible, *i.e.* satisfies condition b_π of *loc. cit.*

[page 69]

Let then

$$X = \varinjlim_{i \in I} X_i , \qquad X_i \in \mathrm{Filt}^\pi ,$$

[24] We then say that M is *L-stable.*

[25] This condition does not depend on the chosen cardinal filtration of $\mathcal{M}$.

be a π-adapted representation of X, with I a π-adapted ordered set. Thus we have

$$X \in M_\alpha \iff \text{the set } I_{M_\alpha} \text{ of } i \in I \text{ such that } X_i \in M_\alpha \text{ is cofinal in } I. \qquad (*)$$

Moreover, since $M_\alpha^\pi = M_\alpha \cap \mathrm{Filt}^\pi$ is stable under filtering $\varinjlim$ of cardinality $\leq \pi$, we see at once that the subset I_{M_α} of I is stable under suprema in I of filtering subsets J of I_{M_α} of cardinality $\leq \pi$. Thus I_{M_α} cofinal is equivalent to I_{M_α} *π-cofinal* (*cf.* 5.8, page 59). Let $M = \bigcap_{\alpha\in A} M_\alpha$, I want to prove that

$$X \in M \iff \text{the set } I_M \ (= \bigcap I_{M_\alpha}) \text{ is cofinal in } I. \qquad (**)$$

Now $(*)$ implies that $X \in M$ $(= \bigcap M_\alpha)$ is equivalent to I_{M_α} π-cofinal for every α. So all in all we want to prove

$$I_{M_\alpha} \text{ cofinal in } I \text{ for every } \alpha \iff \bigcap I_{M_\alpha} \text{ is cofinal in } I. \qquad (***)$$

We of course have $\Longleftarrow$, so it remains to prove $\Longrightarrow$. If the index set A is finite, this follows from Lemma 5.8, taking into account that the cofinality of the I_{M_α} implies their π-cofinality. We must therefore simply improve 5.8, by the [page 70]

Lemma 5.15. *Let π be a cardinal, I a π-adapted filtering ordered set, and let $(I_\alpha)_{\alpha\in A}$ be a family of subsets I_α that are π-cofinal* (cf. 5.8), *such that* $\operatorname{card} A \leq \pi$. *Then $\bigcap I_\alpha$ is π-cofinal.* p. 40

It suffices to prove that it is cofinal, for the property of stability under suprema of filtering subsets of cardinality $\leq \pi$ is obvious. Replacing I by I/i_0, and the I_α by their traces on I/i_0 if need be, the question is simply to prove that $\bigcap_\alpha I_\alpha \neq \emptyset$. We are going to construct an element s of $\bigcap I_\alpha$ by transfinite induction. Let us take a well-order on A, and prove by transfinite induction that for every $\alpha \in A$, the set [page 71]

$$J_\alpha \ = \ \bigcap_{\beta<\alpha} I_\beta$$

is cofinal. If we know that J_α is cofinal, the same holds for $J_{\alpha+1} = J_\alpha \cap I_\alpha$, by 5.8. So we are reduced to proving that J_α is cofinal if α is a limit ordinal, thus to proving Lemma 5.15 for a *decreasing* transfinite sequence of π-cofinal subsets J_β $(\beta < \alpha)$ when $\operatorname{card}\alpha \leq \pi$, such that $J_\beta = \bigcap_{\beta'<\beta} J_{\beta'}$ when β is a limit. To prove that the intersection is π-cofinal – and we are reduced to proving that it is nonempty. We are going to construct by transfinite induction an *increasing* sequence of elements

$$i_\beta \ \in \ J_\beta \qquad (\beta \ \leq \ \alpha)\,.$$

If i_β is constructed, since $J_{\beta+1}$ is cofinal, there exists $i_{\beta+1} \geq i_\beta$ in $J_{\beta+1}$, O.K. If β is a limit ordinal, we take

$$i_\beta \ = \ \sup_{\beta'<\beta} i_{\beta'}\ ,$$

i_β is in each J_{β_0} for $\beta_0 < \beta$, since the $i_{\beta'}$ for $\beta_0 \leq \beta' < \beta$ are all in $J_{\beta_0} \supset J_{\beta'}$, hence also their limit, hence

$$i_\beta \ \in \ \bigcap_{\beta_0<\beta} J_{\beta_0} \ = \ J_\beta\,. \qquad [\text{page 72}]$$

So transfinite induction works, and thus gives us an element s_α of J_α !

I am also going to prove in due form the assertion for $\bigcup_{\alpha\in A} M_\alpha$, when A is finite. Setting this time $M = \bigcup_\alpha M_\alpha$, we must prove, using the relations $(*)$ (p. 69), that we have

$$\begin{array}{lll} X \in M & \iff & \text{the set } \ I_M \ \ (= \ \bigcup_\alpha I_{M_\alpha}) \\ (\textit{i.e.}\ \exists\ \alpha \text{ with } X \in M_\alpha) & & \text{is cofinal in } I\ , \end{array} \qquad (**)$$

and using $(*)$ the first term of this equivalence to be established means that there exists α with I_{M_α} cofinal in I. Of course, if one of the I_{M_α} is cofinal in I, a fortiori their union. But conversely, if the union is cofinal, it is filtering and a union of the subsets I_{M_α}, one of which must therefore be cofinal in it by Lemma 5.14, hence also in I, *q.e.d.*

[page 73]

Corollary 5.16 (of Proposition 5.13). *Every finite union of L-accessible subsets of $\mathcal{M}$ is L-accessible. Every intersection $\bigcap M_\alpha$ of a small family of L-accessible subsets of $\mathcal{M}$, is L-accessible.*

p. 41

6 Abortive attempt at a generation criterion for L-accessible subsets of $\mathrm{Fl}(\mathcal{M})$.

Let $\mathcal{M}$ be a category equipped with a cardinal filtration $(\mathrm{Filt}^\pi)_{\pi\geq\pi_0}$, and let S be an object of Cat, let us consider

$$\mathcal{M}(S) \;=\; \underline{\mathrm{Hom}}(S,\mathcal{M})\;,$$

and equip $\mathcal{M}(S)$ with the filtration by the

$$\mathcal{M}^\pi(S) \;=\; \underline{\mathrm{Hom}}(S,\mathcal{M}^\pi = \underline{\mathrm{Filt}}^\pi(\mathcal{M}))\;.$$

I say that this is a cardinal filtration of $\mathcal{M}(S)$. We must check conditions (a), (b), (c) of SGA 4 I 9.12. Conditions (a) (Filt^π essentially small) and (b) (condition L_{π_0}, and stability of Filt^π under $\varinjlim_I$, I filtering large relative to π_0 and of cardinality $\leq\pi$) pass at once from $\mathcal{M}$ to $\mathcal{M}(S)$. Condition (c) is more delicate (representation of an object as a filtering $\varinjlim$ large relative to π of objects of Filt^π). I am not going to redo the work, done in a much more general and more delicate case in SGA 4 I §9, namely in Corollary 9.24 of Theorem 9.22 (p. 177). The only restriction is that there the objects of $\mathcal{M}$ are assumed accessible, and denoting by π_1 the smallest cardinal such that $\pi_1\geq\pi_0$ and that the objects of Filt^{π_0} are π_1-accessible, we restrict ourselves for the cardinal filtration of $\underline{\mathrm{Hom}}(S,\mathcal{M})$ to the cardinals $\boxed{\pi\geq c=2^{c_0},}$ where

[page 74]

$$c_0 \;=\; \sup(\pi_1,\mathrm{card}\,\mathrm{Fl}(S))\;.$$

This is a reasonable and innocuous restriction.

For the moment, I am interested above all in the case where $S=\mathbf{\Delta}_1$, hence in

$$\underline{\mathrm{Hom}}(\mathbf{\Delta}_1,\mathcal{M}) \;=\; \underline{\mathrm{Fl}}(\mathcal{M})\;,$$

in which case we have $c_0=\pi_1$, so it suffices to restrict ourselves to the cardinals $\pi\geq 2^{\pi_1}$ to have a good cardinal filtration on $\underline{\mathrm{Fl}}(\mathcal{M})$.

It is thus the set of the two conditions (b), (c) on $\mathcal{M}$ (p. 44) that is stable under passage to the $\underline{\mathrm{Hom}}(S,\mathcal{M})=\mathcal{M}(S)$. The same holds for condition (a) (stability under small direct limits), and also for condition (d) (p. 45). Likewise the exactness condition (e) (p. 46) on filtering $\varinjlim$ (they must transform monomorphisms into monomorphisms).

[page 75]

To be on the safe side, I will make on $\mathcal{M}$ the assumptions (a) to (e) inclusive (although (e) seems a little artificial to me, we should be able to eliminate it, with a second wind!). These conditions thus have the desirable property of stability under passage to the $\mathcal{M}(S)$, and in particular under passage to $\mathrm{Fl}(\mathcal{M})$. Increasing π_0 if need be, we can therefore assume that the $\mathrm{Fl}^\pi(\mathcal{M})=\mathrm{Hom}(\mathbf{\Delta}_1,\mathcal{M}^\pi)$ $(\pi\geq\pi_0)$ also form a cardinal filtration of $\underline{\mathrm{Fl}}(\mathcal{M})$.

We thus have at our disposal the notions of π-accessible, accessible, L-π-accessible, L-accessible subsets of $\mathrm{Fl}(\mathcal{M}) \;=\; \mathrm{Ob}\underline{\mathrm{Fl}}(\mathcal{M})$. These notions are stable under finite

p. 42

[page 76]

unions and intersections, and the absolute notions (without π) stable under "arbitrary" intersections (*i.e.* for *small* families of subsets). This will be something particularly crucial, the case of an intersection $C \cap W$!

The main result, for which I practiced the scales of the preceding section, is the following:

Theorem 6.1[26]. *Let $\mathcal{M}$ be a category satisfying the stated conditions:*

a) *Stability under small $\varinjlim$ (not necessarily filtering).*

b) *$\mathcal{M}$ equipped with a cardinal filtration* $(\mathrm{Filt}^\pi)_{\pi \geq \pi_0}$.

c) *The objects of $\mathcal{M}$ are accessible.*

d) *The* Filt^π *are stable under passage to subobjects and under $\varinjlim$ indexed by categories I such that* $\mathrm{card}\,\mathrm{Fl}(I) \leq \pi$, *and if $Z \in \mathrm{Filt}^\pi$, every quotient Z' of Z (such that $Z \longrightarrow Z'$ does not factor through a subobject Z'_0 of Z', different from Z') is in* Filt^π.

e) *Filtering direct limits in $\mathcal{M}$ transform monomorphisms into monomorphisms.*

Let C be an L-accessible subset of $\mathrm{Fl}(\mathcal{M}) = \mathrm{Ob}\underline{\mathrm{Fl}}(\mathcal{M})$, i.e. (5.12) *a subset stable under $\varinjlim$ that are filtering, and accessible. Suppose moreover that C is made up of monomorphisms*[27], *and that C contains the* isomorphisms. *Then there exists a small subset $C_0 \subset C$, such that $C \subset \widetilde{C_0}$. More precisely, if π is a cardinal such that C is π-accessible, and if we set $C^\pi = C \cap \mathrm{Fl}^\pi(\mathcal{M})$, then we have* [page 77]

$$C \;\subset\; \widetilde{C^\pi} \;=\; \widetilde{C_0}\;, \tag{6.1.1}$$

where C_0 is any small subset of C^π such that every object of C^π is isomorphic to an object of C_0.

Corollary 6.2. *We have $C_* = C_{0*}$, $\widetilde{C} = \widetilde{C_0}$.*

Corollary 6.3. *Equivalent conditions on the L-accessible subset C of* $\mathrm{Fl}(\mathcal{M})_{\mathrm{mono}}$, *containing the isomorphisms:*

a) $C = \widetilde{C}$.

b) *C is stable under cobase change and under composition.*

b′) *C is stable under cobase change, and under simple amalgamated sums,* i.e. *if $A \longrightarrow B_1$, $A \longrightarrow B_2$ are in C, then $A \longrightarrow B_1 \amalg_A B_2$ as well.*

c) *C^π is stable in* $\mathrm{Fl}^\pi(\mathcal{M}) = \mathrm{Fl}(\mathcal{M}^\pi)$ *under cobase change and under composition.*

c′) *C^π is stable in* $\mathrm{Fl}(\mathcal{M}^\pi)$ *under cobase change and under amalgamated sums* (cf. b′).

When these conditions are satisfied, then (C, C_*) is a Quillen couple, which falls under Theorem 1 *of section* 1 (p. 10).

The last assertion follows from Theorem 6.1 and from the corollary to Theorem 1 of section 1. We have on the other hand tautologically p. 43 [page 78]

$$\begin{array}{ccccc} (a) & \Longrightarrow & (b) & \Longrightarrow & (b') \\ & & \Downarrow & & \Downarrow \\ & & (c) & \Longrightarrow & (c')\;, \end{array}$$

[26] Botched proof, statement perhaps *false* !!

[27] In the end, we get rid of this assumption.

and it remains to prove $(c') \Longrightarrow (a)$, which we will do below.

Let us then prove 6.1. In (6.1.1) the equality $\widetilde{C_0} = \widetilde{C^\pi}$ is tautological. So everything comes down to proving

$$C \subset \widetilde{C^\pi} .$$

One difficulty here is that we do *not* know that $\widetilde{C^\pi}$ is stable under filtering $\varinjlim$ – we know it only for C. Let then

$$A \hookrightarrow B$$

be an arrow in C, let us prove that it is in $\widetilde{C^\pi}$. For this, we are going to construct, by transfinite induction, a *strictly* increasing transfinite sequence of subobjects A_α of B containing $A = A_0$,

$$\underbrace{A}_{=A_0} \overset{i_\alpha}{\hookrightarrow} A_\alpha \overset{i_{\beta,\alpha}}{\hookrightarrow} A_\beta \xrightarrow{j_\beta} B ,$$

such that we have

$$i_\alpha \in \widetilde{C^\pi} , \qquad \text{more generally the } i_{\beta,\alpha} \in \widetilde{C^\pi} , \qquad j_\alpha \in C ,$$

and that we have

$$A_\alpha = \varinjlim_{\beta<\alpha} A_\beta \qquad \text{if } \alpha \text{ limit ordinal} .$$

This already specifies the recursive definition for $\alpha = \alpha_0$ a limit ordinal. Let us note [page 79]
that the A_{α_0} constructed in terms of the A_β $(\beta < \alpha_0)$ does satisfy the conditions that

a) the $i_{\alpha_0,\beta} : A_\beta \longrightarrow A_{\alpha_0}$ are in $\widetilde{C^\pi}$, for $\widetilde{C^\pi}$ is stable under "strict ordinal direct limits" (*cf.* 1.19, p. 8),

b) $j_{\alpha_0} : A_{\alpha_0} \longrightarrow B$ is in C, for it is the filtering direct limit of the system of the $A_\beta \longrightarrow B$ $(\beta < \alpha_0)$ which are in C, and C is stable under these limits.

It remains then to specify how we pass from A_α to $A_{\alpha+1}$. This comes down to proving the

Lemma 6.4. *Let $f : A \hookrightarrow B$ be in C, such that f is not an isomorphism. Then there exists a subobject A_1 of B,* distinct from A, *such that $A \longrightarrow A_1$ is in $\widetilde{C^\pi}$, and $A_1 \longrightarrow B$ in C.*

If this lemma is admitted, this means that the transfinite construction continues as long as we find A_α such that $A_\alpha \neq B$, *i.e.* $A_\alpha \hookrightarrow B$ is not an isomorphism. But since the A_α are all different, if μ is the cardinality of the set of subobjects of B (we know that it is a

small set, because of the hypotheses made on $\mathcal{M}$), the ordinals α for which A_α exists p. 44
satisfy $\operatorname{card}\alpha \leq \mu$. So there must exist an α such that $A_\alpha = B$. But then $i_\alpha : A \longrightarrow A_\alpha$ [page 80]
is equal to f, hence $f \in \widetilde{C^\pi}$, *q.e.d.*

It remains then to prove 6.4. For this, let I be the set of subobjects of B which are in Filt^π, ordered by inclusion. (If B is in Filt^π, A too, so f is already in $C^\pi \subset \widetilde{C^\pi}$, and there is nothing to prove.) It is a filtering set large relative to π (and even π-*adapted*), whose $\varinjlim$ is B (5.3, p. 46). Consider the

$$A_i = B_i \cap A ,$$

they are in Filt^π, as subobjects of objects of Filt^π (condition (d) on the cardinal filtration). Thus, the arrow f appears as a filtering direct limit of the arrows

$$f_i : A_i \longrightarrow B_i . \qquad \text{[page 81]}$$

It is trivial that the system of the A_i is cofinal in $\mathrm{Ss}_\pi(A)$, hence $A = \varinjlim A_i$, whence $f = \varinjlim f_i$.

On the other hand, I say that the direct system of the f_i in $\underline{\mathrm{Fl}}(\mathcal{M})$ is *well adapted to* π. We already know that I is well adapted to π, the sup in I of a filtering subset J of cardinal $\leq \pi$ being the sup in the ordered set $\mathrm{Ss}(B)$ (which sup is in $\mathrm{Ss}_\pi(B)$). It is essential, for our argument, to know that if $i_J = \sup J$, then not only $B_{i_J} = \varinjlim_{i \in J} B_i$, but also $A_{i_J} = \varinjlim_{i \in J} A_i$, so that the direct system of the f_i is π-adapted in $\underline{\mathrm{Fl}}(\mathcal{M})$. It seems ‖
that we therefore need an exactness condition on $\mathcal{M}$ close to (e), namely:

(e′) The filtering $\varinjlim$ functors commute with base change.

(It suffices here in the very particular case of a base change by a monomorphism $A \longrightarrow B$, and of a filtering direct limit of subobjects of B with inclusion morphisms as transitions.) Let us note that ⓔ, (e′) generally arise as a consequence of the condition

(e_0) The filtering $\varinjlim$ in $\mathcal{M}$ are left exact.

With this laid down, we thus see, C being π-accessible and I large relative to π, that [page 82]
the set I_C of $i \in I$ such that $f_i \in C$ is *cofinal* in I. But since $f_i \in \mathrm{Fl}^\pi(\mathcal{M})$, $f_i \in C$ is equivalent to $f_i \in C^\pi$. Since $A \longrightarrow B$ is not an isomorphism, there exists an i_0 such that $B_{i_0} \not\subset A$, then an $i \geq i_0$ such that $f_i \in C^\pi$, and also $B_i \not\subset A$ (since $B_i \supset B_{i_0}$). Thus we have proved this:

(∗) Under the hypotheses of (6.4) and with the hypothesis (e′) on $\mathcal{M}$, there exists a subobject B' of B satisfying the following conditions.

a) $B' \in \mathrm{Filt}^\pi$.

b) $B' \not\subset A$.

c) Setting $A' = B' \cap A$, the inclusion

$$i' \ : \ A' \ \longrightarrow \ B'$$

is in C^π.

Moreover, the set $J \subset \mathrm{Ss}_\pi(B)$ formed by the B' in question is cofinal in $\mathrm{Ss}_\pi(B)$, and even π-cofinal. p. 45

We still have to prove this last point – that J (which we know is cofinal) is even π-cofinal, *i.e.* stable under sup (in $\mathrm{Ss}_\pi(B)$) of filtering subsets $J' \subset J$ such that $\mathrm{card}\, J' \leq \pi$. But this follows at once from the fact that C being stable under
filtering direct limits, $C^\pi = C \cap \mathrm{Filt}^\pi$ is stable under filtering $\varinjlim$ of cardinal $\leq \pi$, by [page 83]
virtue of condition (d) on Filt^π. Consider now the cocartesian square

$$\begin{array}{ccc} A' & \xhookrightarrow{\ i'\ } & B' \\ \downarrow & & \downarrow \\ A & \xrightarrow{\ i\ } & C' = A \amalg_{A'} B' \end{array} \xrightarrow{\ j\ } B \ .$$

Since $i' \in C^\pi$, we have $i \in \widetilde{C^\pi}$ (for $\widetilde{C^\pi}$ stable under cobase change). On the other hand, I need the fact that j is in C, and I do not even know that it is a monomorphism! So we

have to go back to the indexed notation

$$\begin{array}{ccccc} A_i & \xrightarrow{\alpha_i} & B_i & & \\ \downarrow & & \downarrow & & \\ A & \xrightarrow{\beta_i} & C_i & \xrightarrow{\gamma_i} & B\ . \end{array}$$

Since the formation of amalgamated sums commutes with filtering direct limits, we see that $\varinjlim C_i \simeq A \amalg_{\varinjlim A_i} \varinjlim B_i = A \amalg_A B = B$, hence the direct limit of the γ_i is an isomorphism (essentially id_B). On the other hand, the same argument shows that the direct system of the C_i ($i \in J$) in $\mathcal{M}$ is π-adapted, hence also that of the γ_i[28] in $\underline{\mathrm{Fl}}(\mathcal{M})$. Since $\varinjlim \gamma_i$ is an isomorphism, it is in C. Since C is π-accessible, it follows that the set J' formed by the $i \in J$ such that $\gamma_i \in C$ is cofinal as well[29]. So we find i with $\alpha_i \in C^\pi$, $\gamma_i \in C$, which establishes lemma 6.4, and completes the proof of theorem 6.1. [page 84]

7 Construction of closed Quillen triples with $C = \mathrm{Mon} \subset \mathrm{Cof}_W$

Finally, all the beautiful developments of paragraphs 5, 6 do not give me the desired "sweeping result". Perhaps it is false – perhaps the property of generation "~" by a small subset (for accessible, or L-accessible sets) is satisfied much more rarely, than I would have imagined. To do things properly, I would have to work through to the end a few examples, e.g. that of Cof_W, and of W^{univ} in Cat, for a more or less general basic localizer, and especially in the case $W = \mathbf{W}_\infty$. Surely Cof_W and W^{univ} are accessible (I ought to check it nevertheless), and if the hypothetical "theorem" of paragraph 6 is false for them, at least I will know for sure. But there still remains the question: if $C = \widetilde{C}$ is stable under filtering $\varinjlim$, and of the form $\widetilde{C_0}$, C_0 small, and if W is L-accessible as well, then is $C \cap W$ also of the form $\widetilde{TC_0}$, with TC_0 small? I have just tried once again to prove it, assuming moreover $C \subset$ mono, and see no way of getting there. So a counterexample would be needed to dispel these perplexities! And the best would be in Cat, while we're at it. p. 46 [page 85]

There nevertheless remains afloat a tangible positive result, which I will now try to pin down properly and prove:

Theorem 7.1. *Let $\mathcal{M}$ be a category stable under small filtering $\varinjlim$, $M_0 \subset \mathrm{Ob}\,\mathcal{M}$ a subset of $\mathrm{Ob}\,\mathcal{M}$ generating by strict monomorphisms, more precisely we assume this:* [page 86]

> *For every monomorphism $i : A \hookrightarrow B$ in $\mathcal{M}$ which is not an isomorphism, there exists a monomorphism $B' \overset{j}{\hookrightarrow} B$ having the following properties:*
>
> a) *$B' \in M_0$.*
> b) *j does not factor through i.*
> c) *$A' = A \times_B B'$ exists in $\mathcal{M}$, as does $C = A \amalg_{A'} B'$, A' is in M_0, and in the canonical diagram*

$$\begin{array}{ccccc} A' & \xrightarrow{i'} & B' & & \\ \downarrow & & \downarrow & \searrow^{j} & \\ A & \xrightarrow{\alpha} & C & \xrightarrow{\beta} & B \\ & \searrow_{i} & & \nearrow & \end{array} \qquad (7.1.1)$$

[28] *Except* that we do not know that the C_i are in Filt^π – in fact, they will not be in general, since we do not assume that $A, B \in \mathrm{Filt}^\pi$.

[29] *Botched, cf.* preceding annotation.

the arrow β is a monomorphism.

(**N.B.** These conditions are satisfied if we assume, in addition to the condition that M_0 be generating for monomorphisms, that M_0 contains, along with every object B', and for every object A' which is either a quotient, or a subobject of B', an object A'_0 isomorphic to A'; and if moreover $\mathcal{M}$ satisfies the following exactness conditions:

1°) Every arrow f of $\mathcal{M}$ factors as ip, with i a monomorphism, p an epimorphism, and:

2°) If A_1, A_2 are two subobjects of an object B of $\mathcal{M}$, then the intersection $A_0 = A_1 \times_B A_2$ exists, as does the amalgamated sum $C = A_1 \amalg_{A_0} A_2$, and the canonical morphism $C \longrightarrow B$ is a monomorphism.) p. 47 [page 87]

Assume moreover that $\mathcal{M}$ satisfies condition ©, p. 76, i.e. *that every filtering $\varinjlim$ of monomorphisms is a monomorphism.*

1°) *Let* $\mathrm{Mon} \subset \mathrm{Fl}(\mathcal{M})$ *be the set of monomorphisms, and* Mon_0 *the subset of* Mon *formed by the $i \in \mathrm{Mon}$ such that source and target belong to M_0. We then have*

$$\mathrm{Mon} \subset \widetilde{\mathrm{Mon}_0} \quad , \qquad \textit{whence} \quad \begin{cases} \widetilde{\mathrm{Mon}} & = & \widetilde{\mathrm{Mon}_0} \\ (\mathrm{Mon})_* & = & (\mathrm{Mon}_0)_* \,.^{30} \end{cases} \tag{7.1.2}$$

2°) *If M_0 is* small *and if the objects of M_0 are accessible* (e.g. $\mathcal{M}$ satisfies ©, p. 76), *and if* Mon *is stable under cobase change, then there exists a functorial factorization of every arrow f of $\mathcal{M}$ as*

$$f \;=\; p\,i\,,$$

with $i \in \mathrm{Mon}$, $p \in (\mathrm{Mon})_*$. (**N.B.** An arrow $p : X \longrightarrow Y$ of $\mathcal{M}$ is in $(\mathrm{Mon})_*$ if and only if it makes X an *injective* object of $\mathcal{M}/Y$ – thus the factorization assertion ensures that in the induced categories $\mathcal{M}/Y$ "there are enough injectives".) *Moreover, we have*

$$\mathrm{Mon} \;=\; \widetilde{\mathrm{Mon}_0} \quad . \tag{7.1.3}$$

3°) *Let W be a subset of* $\mathrm{Fl}(\mathcal{M})$ *which is* L-accessible (i.e. *accessible, and stable under filtering direct limits*). (*We now assume that $\mathcal{M}$ satisfies the conditions* (a), (b), (c), (d), (e) *on page 76, whence a good theory of L-accessible subsets of* $\mathrm{Ob}\,(\mathcal{M})$ *or of* $\mathrm{Fl}(\mathcal{M})$.) *We assume, more precisely, that W satisfies the conditions:*

[page 88]

$$\begin{cases} \text{a)} & W \textit{ is L-accessible } (\textit{i.e.} \text{ stable under filtering } \varinjlim, \text{ and accessible}). \\ \text{b)} & \textit{Let } \alpha, \beta \textit{ be composable, with } \beta\alpha \textit{ and } \alpha \textit{ in } W, \textit{ then } \beta \in W. \ (\textbf{N.B.} \text{ This condition will be satisfied if } W \text{ is a mildly saturated localizer, and it is to this situation that we are going to apply our statement.}) \\ \text{c)} & \mathrm{Mon} \cap W \textit{ is stable under cobase change.} \end{cases} \tag{7.1.4}$$

and that $\mathcal{M}$ satisfies the additional conditions

$$\begin{cases} \text{e}') & \textit{The filtering } \varinjlim \textit{ commute with base change.} \\ \text{f)} & \textit{The exactness condition } (7.2.1) \textit{ below.} \end{cases} \tag{7.1.5}$$

Let π be a cardinal such that W is π-accessible, and $M_0 \subset \mathrm{Filt}^\pi$ (hence M_0 essentially small). Then we have

$$W \cap \mathrm{Mon} \;\subset\; \widetilde{(W \cap \mathrm{Mon})^\pi} \tag{7.1.6}$$ p. 48

(*where for every subset* C *of* $\mathrm{Fl}(\mathcal{M})$, *we set* $C^\pi = C \cap \mathrm{Fl}^\pi(\mathcal{M})$), *hence*

$$\widetilde{W \cap \mathrm{Mon}} \;=\; (\widetilde{W \cap \mathrm{Mon}})^\pi \;, \tag{7.1.7}$$

hence $\widetilde{W \cap \mathrm{Mon}}$ *is Q-generated by a* small *subset of* $W \cap \mathrm{Mon}$.

Corollary 7.2. *Let* $\mathcal{M}$ *satisfy* (a), (b), (c), (d), (e) *and the additional conditions:*

$$\begin{cases} \text{e}') & \textit{The filtering } \varinjlim \textit{ commute with base change.} \\ \text{f}) & \textit{If } A_1, A_2 \textit{ are two subobjects of an object } B \textit{ of } \mathcal{M}, \textit{ and } A_0 = A_1 \cap A_2, \textit{ then } A_1 \amalg_{A_0} A_2 \hookrightarrow B \textit{ is a monomorphism}\,[31]. \end{cases} \tag{7.2.1}$$

Let W *be a subset of* $\mathrm{Fl}(\mathcal{M})$ *satisfying the following conditions.*

$$\begin{cases} \text{a}) & W \textit{ is a mildly saturated localizer.} \\ \text{b}) & W \textit{ is L-accessible,} \text{ i.e. } \textit{it is stable under filtering } \varinjlim, \textit{ and is accessible.} \\ \text{c}) & \mathrm{Mon} \cap W \textit{ is stable under cobase change,} \text{ i.e. } \textit{if we have a diagram which is} \text{ cocartesian} \\ & \qquad \begin{array}{ccc} A & \xrightarrow{\;i\;} & B \\ {\scriptstyle\alpha}\downarrow & \text{cocart.} & \downarrow{\scriptstyle\beta} \\ A' & \xrightarrow{\;i'\;} & B' \end{array} \\ & \textit{with } i \textit{ a monomorphism belonging to } W, \textit{ then } i' \textit{ is a monomorphism belonging to } W.\,[32] \end{cases} \tag{7.2.3}$$

Under these conditions, we have [page 89]

$$\underbrace{\mathrm{Mon}}_{C} \cap W \;=\; \widetilde{W \cap \mathrm{Mon}} \quad , \qquad \text{i.e. } \mathrm{Mon} \cap W \textit{ is Q-closed on the left} \;, \tag{7.2.4}$$

and p. 49

$$\textit{there exists a small subset } TC_0 \subset \mathrm{Mon}, \textit{ such that } \mathrm{Mon} \cap W = \widetilde{TC_0} \;. \tag{7.2.5}$$

(More precisely, if $\pi \geq \pi_0$ *is a cardinal such that* W *is* π*-accessible, we can take* $TC_0 = (\mathrm{Mon} \cap W)^\pi$ *(or a* small *subset of* $(\mathrm{Mon} \cap W)^\pi$ *having the same isomorphism classes).) Thus* (th. 1 of §1) *the couple* $(TC = \mathrm{Mon} \cap W,\ F = (TC)_*)$ *is a closed Quillen couple, and there is even a* functorial *factorization of every arrow* f *as*

$$f \;=\; p\,i \;; \qquad i \;\in\; TC \;, \qquad p \;\in\; F \;. \tag{7.2.4}$$

[30] **N.B.** Here we have a criterion for injectivity of objects in the categories $\mathcal{M}/Y$ – it suffices to test the injectivity condition by monomorphisms i which are in Mon_0.

[31] Another way of saying it: if $C = \sup_{\mathrm{Ss}(B)}(A_1, A_2)$, the cartesian square

$$\begin{array}{ccc} A_0 & \longrightarrow & A_1 \\ \downarrow & & \downarrow \\ A_2 & \longrightarrow & C \end{array}$$

is also cocartesian.

[32] *Cf.* below for a fourth condition (d) on W, which ensures that W fits into a Quillen triple.

If we set

$$C \;=\; \mathrm{Mon}\;, \qquad TC \;=\; \mathrm{Mon}\cap W\;, \qquad F \;=\; (TC)_*\;, \qquad TF \;=\; C_*\;, \tag{7.2.5}$$

the triple (W, C, F) is a closed Quillen triple, if and only if we have the following additional conditions.

$$\begin{cases} (7.2.6) \quad \text{Ⓖ } \textit{(on } \mathcal{M} \textit{ only)} \quad \mathrm{Mon} \textit{ stable under cobase change.} \\ (7.2.7) \quad \underbrace{TF}_{\overset{\text{def}}{=}(\mathrm{Mon})_*} \subset W \quad \text{i.e. } \textit{if } \mathrm{d}) \textit{ every } f : X \longrightarrow Y \textit{ which makes } X \textit{ an injective object of } \mathcal{M}/Y, \textit{ is in } W \textit{ (hence also in } W_{\mathrm{univ}}\textit{).} \end{cases}$$

Proof of 7.1. We can give a somewhat more delicate statement, by supposing given from the outset, not a subset M_0 of $\mathrm{Ob}\mathcal{M}$, but a subset [page 90]

$$\mathrm{Mon}_0 \;\subset\; \mathrm{Mon}\;, \tag{7.1}$$

satisfying the following condition

> For every monomorphism $i : A \longrightarrow B$ in $\mathcal{M}$ which is not an isomorphism, there exists $i' \in \mathrm{Mon}_0$, $i' : A' \longrightarrow B'$, and an arrow $i' \longrightarrow i$ in $\underline{\mathrm{Fl}}(\mathcal{M})$,
>
> $$\begin{array}{ccc} A' & \xrightarrow{\;i'\;} & B' \\ \downarrow & & \downarrow \\ A & \xrightarrow{\;i\;} & B\;, \end{array} \tag{7.2}$$
>
> such that the corresponding arrow
>
> $$C \;=\; A \amalg_{A'} B' \;\longrightarrow\; B$$
>
> is a monomorphism, and $B' \longrightarrow B$ does not factor through A.

That is what was aimed at by condition (7.1.1) on M_0, which implies (7.2) on Mon_0 defined in terms of M_0 as said in 1°.

In what follows, we suppose given a subset $\mathrm{Mon}_0 \subset \mathrm{Fl}(\mathcal{M})$ satisfying (7.2). In practice, it will often be deduced from a subset M_0 of $\mathrm{Ob}\mathcal{M}$, generating for monomorphisms, as made explicit in 1°, but we do not need the fact that $i' \longrightarrow i$ is a monomorphism (*i.e.* (7.1.1) is stronger than what we need). E.g. if $B' \in M_0$ and $B' \longrightarrow B$ is given, not factoring through A, we shall be able to take $A' = B' \times_B A$ and $i' : A' \longrightarrow B'$ (to ensure $i' \in \mathrm{Mon}_0$, we shall therefore have to suppose M_0 stable under passage to subobjects), and chances are p. 50

that $C \longrightarrow B$ will then indeed be a monomorphism … [page 91]

1°) We must prove

$$\mathrm{Mon} \;\subset\; \widetilde{\mathrm{Mon}_0}$$

assuming (7.2) (and the stability of $\mathcal{M}$ under filtering $\varinjlim$). This will follow directly from the

Lemma 7.3. *Let $\mathcal{M}$ be a category stable under filtering direct limits, and C_0, C subsets of* $\mathrm{Fl}(\mathcal{M})$*. We shall suppose*

ⓐ *C made up of monomorphisms, and*

ⓑ *that the following condition (generalizing* (7.2) *above) is satisfied.*

For every arrow $i : A \hookrightarrow B$ in C such that i is not an isomorphism, there exists $i' : A' \longrightarrow B'$ in $\widetilde{C_0}$, and an arrow $i' \longrightarrow i$ in $\underline{\mathrm{Fl}}(\mathcal{M})$, such that in the diagram

$$\begin{array}{ccccc} A' & \xrightarrow{\quad i' \quad} & B' & & \\ \downarrow & & \downarrow & \searrow & \\ A & \xrightarrow{\alpha} & D \;=\; A \amalg_{A'} B' & \xrightarrow{\beta} & B \;, \\ & \searrow^{i} & & \nearrow & \end{array} \tag{7.3.1}$$

the arrow β is in C, and $B' \longrightarrow B$ does not factor through A (or, what amounts to the same, the subobject of B defined by D and β is not equal to the one defined by A and i).

We suppose moreover

ⓒ *C stable under filtering $\varinjlim$, and*

ⓓ *that for every $B \in \mathrm{Ob}\mathcal{M}$, the set of subobjects of B is small.* (A most innocuous condition! This is a condition *on* $\mathcal{M}$, not involving C).

Then we have

$$C \;\subset\; \widetilde{C_0}\;, \qquad \textit{hence } \widetilde{C} \;\subset\; \widetilde{C_0}\;, \tag{7.3.2}$$

hence, if $C_0 \subset C$, we conclude

[page 92]

$$\widetilde{C} \;=\; \widetilde{C_0}\;. \tag{7.3.3}$$

(This lemma immediately implies the first assertion of 7.1, 1°.) Let us prove this lemma, *i.e.* that every $i \in C$ is in $\widetilde{C_0}$, by constructing by transfinite induction a strictly increasing sequence of subobjects of B, p. 51

$$A_\alpha \overset{j_\alpha}{\hookrightarrow} B\;, \qquad \text{with the } j_\alpha \in C\;,$$

and with:

① $A_0 = A$. We shall prove that $A \xrightarrow{i_\alpha = i_{\alpha,0}} A_\alpha$ in $\widetilde{C_0}$, and there will be a final ordinal α such that $A_\alpha = B$; we shall therefore conclude from this that $i = i_\alpha$ is in $\widetilde{C_0}$.

If i is an isomorphism, we know that $i \in \widetilde{C_0}$, there is nothing to prove (the transfinite construction stops at stage 0). The two steps of the construction, after the initial step 1°, are:

② Let A_α be constructed. If $A_\alpha = B$, the construction is stopped. If $A_\alpha \xrightarrow{j_\alpha} B$ is not an isomorphism, we apply (7.3.1) to $j_\alpha \in C$, to find a $D \overset{j}{\hookrightarrow} B$, thus $j \in C$ and $D \supsetneqq A_\alpha$. Moreover,

$$i_{\alpha+1,\alpha} \;:\; A_\alpha \;\longrightarrow\; A_{\alpha+1} \overset{\mathrm{def}}{=} D \qquad \text{is in } \widetilde{C_0}\;, \tag{7.3.4}$$

since obtained from $i' \in \widetilde{C_0}$ by cobase change.

[page 93]

③° If α is a limit ordinal, we set

$$A_\alpha \;=\; \varinjlim_{\beta<\alpha} A_\beta \;, \tag{7.3.5}$$

and the canonical arrow

$$A_\alpha \;\xrightarrow{j_\alpha}\; B$$

is in C, since the j_β are and C is stable under filtering direct limits.

We have thus completed the description of the transfinite systems. For every α, we have

$$\mathrm{card}\alpha \;\leq\; \mathrm{cardSs}(B) \;,$$

since the map $\beta \longmapsto A_\beta$ from the set $I_\alpha = \{\beta \mid \beta < \alpha\}$ to $\mathrm{Ss}(B)$ is *strictly* increasing. Thus there exists an ordinal α such that $\mathrm{card}\alpha \leq \mathrm{cardSs}(B)$, such that $A_\alpha = B$. But then $i \in \widetilde{C_0}$, as a transfinite direct limit (or composite) of arrows belonging to $\widetilde{C_0}$. (The condition $\forall\ \beta$, $i_{\beta+1,\beta} \in \widetilde{C_0}$ suffices to imply that all the transition morphisms $i_{\beta',\beta}$ are in $\widetilde{C_0}$.)

Let us return to 7.1, 2°. It remains to prove that if Mon is stable under cobase change, then we have

$$\mathrm{Mon} \;=\; \widetilde{\mathrm{Mon}_0} \quad .$$

But since Mon is already stable under composition, and under arbitrary filtering direct limits (hence also under direct factors), it follows by corollary 2 of theorem 1, §1 (p. 11), that $\mathrm{Mon} = \widetilde{\mathrm{Mon}_0}$. The factorization statement is theorem 1 of §1.

Let us prove 7.1, 3°, *i.e.* formula (7.1.6), p. 52 [page 94]

$$W \cap \mathrm{Mon} \;\subset\; (W \cap \widetilde{\mathrm{Mon}})^\pi \;.$$

For this, we again apply lemma 7.3 to

$$C_0 \;=\; (W \cap \mathrm{Mon})^\pi \;, \qquad C \;=\; W \cap \mathrm{Mon} \;.$$

The conditions (a), (b), (c), (d) of the lemma are satisfied. This is clear for all, except for (b), which calls for a non-tautological verification. Let then

$$f \;:\; A \hookrightarrow B \;, \qquad f \in \mathrm{Mon} \cap W,\ f \text{ not an isomorphism} \;.$$

We regard B as a direct limit[33]

$$B \;=\; \varinjlim_{I} B_i \;, \qquad I \;=\; \mathrm{Ss}_\pi(B) \;.$$

For every i, we set

$$A_i \;=\; A \cap B_i \;=\; A \times_B B_i \;,$$

and

$$f_i \;:\; A_i \;\hookrightarrow\; B_i \;,$$

hence

$$f \;=\; \varinjlim_{i \in I} f_i \;.$$

We need to know that the direct system of arrows f_i is π-adapted, which leads us to postulate the following additional exactness condition on $\mathcal{M}$:

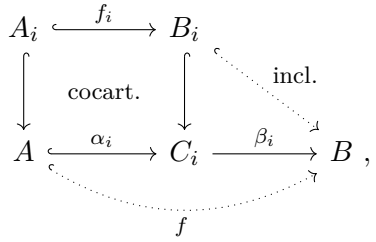
[page 95]

[33] *Of a π-adapted direct system.*

$$e')\quad \text{the filtering } \varinjlim \text{ in } \mathcal{M} \text{ commute with base change ,} \tag{7.4}$$

it even suffices to suppose it in the very special case we have considered here.

Since W is π-accessible, and $(f_i)_{i\in I}$ is π-adapted, since $\varinjlim f_i \in W$, it follows that the set I_W of $i \in I$ such that $f_i \in W$ is cofinal. Since $A \overset{f}{\hookrightarrow} B$ is not an isomorphism, there is therefore an $i \in I_W$ such that $B_i \hookrightarrow B$ does not factor through A. We then have the diagram



with $f_i \in (C \cap W)^\pi = C_0$, hence $\alpha_i \in \widetilde{C_0}$. We need to know that $\alpha_i \in W$, hence an additional condition on W: p. 53

If we have a cocartesian square

$$\begin{array}{ccc} A & \xrightarrow{\ i\ } & B \\ \downarrow & & \downarrow \\ A' & \xrightarrow{\ i'\ } & B' \ , \end{array} \tag{7.5}$$

with $i \in \mathrm{Mon} \cap W$, then $i' \in W$[34].

(I have strengthened (7.1.4) accordingly).

Since by hypothesis $f = \beta_i\alpha_i$ is in W, and $\alpha_i \in W$, it follows by (7.1.4), (b) that $\beta_i \in W$. We shall have to know moreover that β_i is a monomorphism, hence $\beta_i \in \mathrm{Mon} \cap W = C$, which will finish establishing the missing condition (b) of 7.3. This is the exactness condition (7.2.1), which I evidently have to add to the hypotheses in 7.1, 3°. Having done that, lemma 7.3 applies and establishes 7.1, 3°. [page 96]

Proof of 7.2. Let us set

$$C \ = \ \mathrm{Mon}\ , \qquad TC \ = \ C \cap W\ , \qquad TC_0 \ = \ TC^\pi\ ,$$

hence by 7.1, 3°, we have

$$\underbrace{TC_0}_{\substack{\text{essentially}\\ \text{small}}} \ \subset\ TC \ \subset\ \widetilde{TC_0} \ = \ \widetilde{TC}\ ,$$

and by theorem 1, §1 (p. 10-12), cor. 2, to check that $TC = \widetilde{TC_0}$ (*i.e.* TC is Q-closed on the left), it amounts to the same to see that it is Q-stable on the left, *i.e.* stable under composition, under cobase change, under strict ordinal $\varinjlim$, under direct factors. Now the last two are implied by the fact that $TC = C \cap W$ is stable under filtering $\varinjlim$, seeing that W is, and C also by condition ⓔ (p. 76) on $\mathcal{M}$. On the other hand, $C = \mathrm{Mon}$ and W being stable under composition, likewise for $C \cap W$. So there remains cobase [page 97]

[34] **N.B.** This condition is satisfied.

change, and this is ensured by 7.2.3, (c) (which we already needed in 7.1, 3°, to establish $C \cap W \subset \widetilde{TC_0}$). (**N.B.** the hypotheses of 7.2 are practically those of 7.1, 3°, except that we suppose that W is a localizer (understood: mildly saturated), to be precise we have used, up to now, as an additional hypothesis, only the stability of W under composition.)

It remains to establish that $(W, C = \mathrm{Mon}, F = (TC)_*)$ is a closed Quillen triple, if and only if the conditions (7.2.6) and (7.2.7), *i.e.*

$$\begin{cases} C = \mathrm{Mon} \ \text{ stable under cobase change,} \\ TF = C_* = (\mathrm{Mon})_* \subset W, \end{cases}$$

are satisfied. Necessity is obvious, by definition of closed Quillen triples, and for sufficiency, we note that it follows from theorem 7.1, 2°, and from the part of 7.2 already proved, that (C, TF), (TC, F) are closed Quillen couples, and we have moreover p. 54

$$TC \ = \ C \cap W \ , \qquad TF \ \subset \ F \cap W$$

(since $TF \subset W$ by (7.2.7), and $TF \subset F$ since $TC \subset C$). We know (paragraph 4.3, corollary 4, page 36) that it follows that (W, C, F) is a closed Quillen triple. This completes the proof of 7.1 and 7.2. [page 98]

Let us sum up the essentials:

Corollary 7.4.[35] *Let $\mathcal{M}$ be a category satisfying the conditions* (a), (b), (c), (d), (e) (p. 76) *and* (e′), (f) ((7.2.1), p. 88), *finally* (g) (7.2.6), i.e. *stability of the set* Mon *of monomorphisms of $\mathcal{M}$ under cobase change. Let $W \subset \mathrm{Fl}(\mathcal{M})$ be a localizer in $\mathcal{M}$, which is* stable under filtering $\varinjlim$ *and moreover* (*a subtler condition, but in practice innocuous*) accessible. *We put*

$$C \ = \ \mathrm{Mon} \ , \qquad TC \ = \ C \cap W \ , \qquad TF \ = \ C_* \ , \qquad F \ = \ (TC)_* \ .$$

Then:

1°) (C, TF) *is a Quillen couple* (*a statement independent of the datum W*), *and even a Quillen couple with functorial factorization.*

2°) *For (TC, F) to be a Quillen couple, it is necessary and sufficient that TC be stable under cobase change.* (*And then there is functorial factorization.*)[36].

3°) *For (W, C, F) to be a closed Quillen triple, it is necessary and sufficient that the following conditions be satisfied.*

$$\begin{cases} \text{a)} \quad TC \overset{\text{def}}{=} W \cap \mathrm{Mon} \ \textit{stable under cobase change.} \\ \text{b)} \quad TF \overset{\text{def}}{=} (\mathrm{Mon})_* = \textit{maps of injective type} \subset W. \end{cases} \tag{7.4.1}$$

4°) *For (W, C, F) to be a closed and left proper Quillen triple* (i.e. $C \subset \mathrm{Cof}_W$), *it is necessary and sufficient that we have condition* (7.4.1), (b) (i.e. *W contains the morphisms of injective type*), *and moreover*

[page 99]

$C \subset \mathrm{Cof}_W$, *or again: for every cocartesian square*

$$\begin{array}{ccc} A & \xrightarrow{\ i\ } & B \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\beta} \\ A' & \xrightarrow{\ i'\ } & B' \ , \end{array} \tag{7.4.2}$$

with i a monomorphism, then $\alpha \in W \Longrightarrow \beta \in W$.

[35] **N.B.** It is ultimately this corollary that is the main result of the present section.

[36] **N.B.** In cases 1° and 2°, C, $TC = C \cap W$ are Q-generated on the left by *small* subsets.

5°) *For (W, C, F) to be a closed Quillen triple and* proper, *it is necessary and sufficient that the two conditions of 4° be satisfied, and also that we have:* p. 55

$F \subset \mathrm{Fib}_W$, *or again: for every cartesian square*

$$\begin{array}{ccc} X & \xrightarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xrightarrow[p]{} & Y' \end{array} \tag{7.4.3}$$

if $f' \in F$ (i.e. $f' \in (C \cap W)_*$) *and $p \in W$, we have $q \in W$.*

Proof of 7.4. Parts 1°, 2°, 3° are just a summary of what has been established. For 4°, we have to see that the stated conditions imply (7.4.1), (a), now we have $C \subset \mathrm{Cof}_W$, hence $C \cap W \subset \mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$ hence $C \cap W = C \cap W^{\mathrm{univ}}$, which is stable under cobase change, since C and W^{univ} are. As for 5°, it is a tautology, taking 4° into account. [page 100]

8 Application to the categories of models $(A^\wedge, W_A)$

Let A be in Cat, and let us consider in $A^\wedge$ the localizer W_A, inverse image under

$$\begin{cases} i_A : A^\wedge & \longrightarrow & \mathrm{Cat} \\ F & \longmapsto & A/F \end{cases} \tag{8.1}$$

of a given basic localizer W, satisfying the conditions

$$\mathrm{W}(\underbrace{1}_{\text{sat.}}, \underbrace{2}_{\text{loc.}}, \underbrace{3}_{\text{final obj.}}, \underbrace{5}_{\text{conn. } \pi_0}, \underbrace{6}_{\varinjlim}, \underbrace{7\text{ bis}}_{\substack{\text{coc. squares} \\ \text{or "M.V."}}}, \underbrace{8}_{\text{acc.}}), \tag{8.2}$$

where W (8) is the accessibility of W (thus W is L-accessible if it also satisfies W (6)).

To be able however to apply the developments on accessible subsets of a big category in §5, we must verify the conditions (a), (b), (c), (d), (e) of §5, both in Cat, and (especially) in $A^\wedge$.

a) Stability under small filtering $\varinjlim$, O.K.

b) Cardinal filtration. Well known for $A^\wedge = \underline{\mathrm{Hom}}(A^{\mathrm{op}}, \mathrm{Ens})$, starting from that of Ens by the cardinals (*cf.* SGA 4 I, 9.24, p. 177). For Cat, we define a cardinal filtration

$$(\mathrm{Cat}^\pi)_{\pi \geq \pi_0}\ , \qquad \text{where } \pi_0 = \text{cardinal of countable sets}\ , \tag{8.3}$$

by the condition, for A in Cat, p. 56 [page 101]

$$A \in \mathrm{Filt}^\pi(\mathrm{Cat}) \ = \ \mathrm{Cat}^\pi \qquad \Longleftrightarrow \qquad \mathrm{card}\,\mathrm{Fl}(A) \ \leq \ \pi\ . \tag{8.4}$$

We must verify the conditions (a), (b), (c) of *loc. cit.* 9.12. Condition (a) is obvious, (b) is contained in the stronger condition of stability of Cat^π under $\varinjlim_I$, for I a category such that $\mathrm{card}\,\mathrm{Fl}(I) \leq \pi$ (this is condition (d_0) of these notes, p. 45-46). Let us note that Cat^π is also stable under finite $\varprojlim$ (and if π is of the form 2^c, it is even stable under inverse limits $\varprojlim_I$, with $\mathrm{card}\,\mathrm{Fl}(I) \leq c$ ...). Condition (c) is easily verified by the method of §5, noting that for X in Cat, the set

$$\mathrm{Ss}_\pi(X)$$

of subcategories of X which are in Cat^π is upward filtering and large relative to π, and that X is its direct limit. On the other hand, it is also clear that if $X \in \mathrm{Cat}^{\pi'}$ with $\pi' > \pi$, then

$$\operatorname{card} \mathrm{Ss}_\pi(X) \ \leq \ \pi'^{\pi} \ ,$$

which finishes establishing condition (c).

Moreover, condition (d) on p. 45 is immediately verified for the Cat^π, and also for the $A^{\wedge\pi}$ (for $\pi \geq \operatorname{card} \mathrm{Fl}(A)$).

Also that monomorphisms are stable under filtering $\varinjlim$ (condition (e), page 46).

Let us pass to the additional conditions (e'), (f), (g) required in corollary 7.4. Condition (e') (commutation of filtering $\varinjlim$ with base change) is satisfied in $A^\wedge$ and in Cat. Condition (f) says that for two subobjects A_1, A_2 of an object B, if $A_0 = A_1 \cap A_2$, then [page 102]

$$A_1 \amalg_{A_0} A_2 \longrightarrow B$$

is a monomorphism. I doubt that this is true in Cat, and it does not matter, for it is to $A^\wedge$, not to Cat, that we shall want to apply 7.4 – and it is obviously true in $A^\wedge$, and in any topos, being so in (Ens). Finally, stability of monomorphisms under cobase change is surely false in Cat, but true in $A^\wedge$ – so it is O.K.

Since W is stable under filtering $\varinjlim$ (W (6)), and i_A commutes with them, W_A is also stable. Similarly, W_A is L-accessible, by the

Lemma 8.1. *Let $\mathcal{M}$, $\mathcal{M}'$ be two categories satisfying* (a), (b), (c), (d), (e), *and $\varphi : \mathcal{M} \longrightarrow \mathcal{M}'$ a functor commuting with filtering $\varinjlim$. Then for any L-accessible subset M' of $\mathcal{M}'$, $\varphi^{-1}(M')$ is L-accessible.* [page 103]

To be precise, let π be a cardinal such that M' is π-accessible and φ maps $\mathcal{M}^\pi$ into $\mathcal{M}'^\pi$ (for example, for this last condition, it suffices $\pi = 2^c$ for c large enough, cf. loc. cit. 9.14, p. 153). *Then $M = \varphi^{-1}(M')$ is π-accessible.*

Immediate verification.

We therefore have the

Lemma 8.2. *Let A be in* Cat. *Then $(A^\wedge, W_A)$ satisfies the preliminary conditions of corollary* 7.4, *as well as the following conditions.* p. 57

$$\underbrace{\mathrm{Mon}(A^\wedge)}_{\substack{\text{set of}\\ \text{monomorphisms}\\ \text{in } A^\wedge}} \quad \subset \ \mathrm{Cof}_{W_A} \ , \tag{8.2.1}$$

$$\mathrm{Mon} \cap W_A \qquad \text{stable under cobase change} \ , \tag{8.2.2}$$

in other words (*taking into account that* Mon *is stable under cobase change*), *for a cocartesian square*

$$\begin{array}{ccc} A & \xrightarrow{\ i\ } & B \\ & {\scriptstyle\alpha}\searrow & \swarrow{\scriptstyle\beta} \\ & A' & \xrightarrow{\ i'\ } B' \ , \end{array} \tag{8.2.3}$$

with i a monomorphism, if $\alpha \in W_A$ (resp. $i \in W_A$), we have $\beta \in W_A$ (resp. $i' \in W_A$). In fact, we have better:

$$i \ \in \ W_A \iff i' \ \in \ W_A \ , \qquad \alpha \ \in \ W_A \iff \beta \ \in \ W_A \ . \tag{8.2.4}$$

It remains to verify (8.2.4), which was seen in XII … [page 104]

We therefore conclude from 7.4 1°, 2°, 3°:

Corollary 8.3. *Let us put*

$$\begin{cases} C \;=\; \mathrm{Mon}(A^\wedge)\;, & TC \;=\; C \cap W_A\;, \\ F \;=\; (TC)_*\;, & TF \;=\; C_*\;. \end{cases} \tag{8.3.1}$$

Then (C, TF) *and* (TC, F) *are Quillen couples with functorial factorization.* (**N.B.** C *and* TC *being moreover Q-generated on the left by small subsets of* $\mathrm{Fl}(A^\wedge)$.) *For* (W_A, C, F) *to be a closed Quillen triple, it is necessary and sufficient that we have*

$$\underbrace{TF}_{\overset{\text{def}}{=}(\mathrm{Mon})_*} \;\subset\; W_A\;, \tag{8.3.2}$$

i.e. *that every arrow* $f : X \longrightarrow Y$ *of injective type in* $A^\wedge$ (i.e. *which makes* X *an injective object of* $A^\wedge/Y$) *be in* W_A.

When this condition is satisfied, the Quillen triple (W_A, C, F) *is left proper, by* 8.2.1.

It is not clear that it is necessarily proper-plain-and-simple, *i.e.* that we have

$$F \;\subset \mathrm{Fib}_W \qquad (?)\;. \tag{8.3.3}$$

We will come back to this later. For the moment, the essential thing is to examine the condition (8.3.2) on A.

Theorem 8.4. *Let* W *be a basic localizer in* Cat *satisfying* W (1, 2, 3, 6, 7, 8), *notations of* XII, (i.e. *all the axioms considered so far, except perhaps the one* W (4) *on fibrations*), *let us suppose moreover* W *stable under direct factors* (*strengthening* W (1 bis) *of the axiom of mild saturation* W (1)).[37] *Let* A *be an object of* Cat, *and* $W_A = i_A^{-1}(W)$ *the localizer of* $A^\wedge$ *deduced from* W. *With the notations of* 8.3: p. 58 [page 105]

1°) *The following conditions are equivalent.*

a) *We have* (8.3.2),

$$TF \;\subset\; W_A\;,$$

i.e. *every arrow of injective type in* $A^\wedge$ *is in* W_A.

b) *Let*

$$L \;=\; L_A \tag{8.4.1}$$

be the Lawvere element in $A^\wedge$, *hence the presheaf*

$$a \quad\longmapsto\quad \begin{array}{l} \textit{set of subpresheaves of } a \\ (\text{i.e. } \textit{set of opens of } A/a, \\ \text{i.e. } \textit{of sieves of } A/a)\;, \end{array}$$

and let us consider the arrow $L \longrightarrow e_A$ *(final object of* $A^\wedge$*). We want this arrow to be* W_A*-aspheric,* i.e. *to be universally in* W_A *– or what comes to the same thing, that for every* $a \in A$, $L_A \times a$ $(= L_{A/a})$ *be a* W_A*-aspheric object of* $A^\wedge$ *(or, what comes to the same thing, of* $(A/a)^\wedge$*).* [page 106]

c) *There exists an object* I *of* $A^\wedge$ *such that* $I \longrightarrow e_A$ *is* W_A*-aspheric, with two sections* δ_0, δ_1 *of* I *over* e_A, *with* $\mathrm{Ker}(\delta_0, \delta_1) = \emptyset_A$ *(empty presheaf).*

[37] **N.B.** For the validity of 1°, 3° we do not need W (6) ($\varinjlim$) or W (8) (accessibility), nor even W (7) (cocartesian square axiom). The latter is already necessary to ensure part (*a*) of corollary 8.5. But for part 2° of theorem 8.4, we need *all* the axioms of the basic localizer W, except only W (4) .

d) *The localized categories $A/a = B$ are* test categories *for W* (cf. *Pursuing Stacks, §65, especially pages 172-173), which amounts to saying that we have this, for each of the categories $B = A/a$:*

$$\text{The functors}\quad \begin{cases} i_B : & B^\wedge \longrightarrow \mathrm{Cat} \ \ (\ F \longmapsto B/F\ ) \\ j_B : & \mathrm{Cat} \longrightarrow B^\wedge \ \ (\ C \longmapsto (b \longmapsto \mathrm{Hom}(B/b, C))\ ) \end{cases} \tag{8.4.2}$$

(*where j_B is the right adjoint of i_B*) *are both "model preserving"*, i.e.

$$\begin{cases} W_B = i_B^{-1}(W) & \text{(by definition of } W_B\text{, in fact)} \\ W = j_B^{-1}(W_B)\ , \end{cases} \tag{8.4.3}$$

p. 59

and the localized functors

$$\mathrm{Hot}_{B,W} \underset{\overline{j_B}}{\overset{\overline{i_B}}{\rightleftarrows}} \mathrm{Hot}_W$$

are equivalences of categories. Moreover, these equivalences of categories are quasi-inverse to one another, the adjunction morphisms being deduced from the adjunction morphisms for the pair of adjoint functors (i_B, j_B), these adjunction morphisms being moreover in W resp. in W_B.[38]

[page 107]

2°) *If the equivalent conditions of* 1° *are satisfied, then the triple*

$$(W_A\ ,\ C = \mathrm{Mon}_{A^\wedge}\ ,\ F = (C \cap W_A)_*)$$

is a (closed) left proper Quillen triple.

3°) *If the equivalent conditions of* 1° *are satisfied, and if moreover A is W-aspheric, then (and only then) A itself is a* test category *for W, which amounts to saying that the conditions on the category B in* (d) *are satisfied when B is* either *A, or a localized category A/a.*

Corollary 8.5. *The equivalent conditions of* 1° *and of* 3° *are satisfied in each of the following cases (and this for any W, satisfying the stated hypotheses).*

a) *$A = \mathbf{\Delta}$, $\mathbf{\Delta}^\sim$, $\square$, $\square^\sim$ (categories of standard combinatorial simplices, ordered or not, and of standard combinatorial cubes, ordered or not …)*[39].

b) *A is stable under finite products*[40]*, and contains an object I such that I has two sections δ_0, δ_1 over the final object, with δ_0, δ_1 disjoint,* i.e. $\mathrm{Ker}(\delta_0, \delta_1) = \emptyset_{A^\wedge}$ *(empty subpresheaf of the final presheaf).*

[page 108]

[38] **N.B.** Condition (*d*), seemingly very strong, is equivalent to the seemingly more innocuous condition:
d') the functor $j_A : \mathrm{Cat} \longrightarrow A^\wedge$ transforms elements of W into W_A-aspheric arrows;
and it even suffices to require it for $C \longrightarrow e$ in Cat, where C is an object of Cat having a final object.

[39] $\square^\sim$ is essentially the "universal" category A, stable under finite products, and containing a "separating interval" (I, δ_0, δ_1) as in (*b*).

[40] In fact, it suffices that $I \longrightarrow e$ be squarable, *i.e.* that A have a final object, and that the products $I \times a$ ($a \in \mathrm{Ob}\, A$) exist in A.

Case (b) of the corollary falls under condition (c) in 1°. (Moreover, A having a final object, is W-aspheric, so we are under the conditions of 3°.) The cases $\mathbf{\Delta}$, $\mathbf{\Delta}^{\sim}$, $\square$ (and $\square^{\sim}$, which in fact falls under case (b)) are treated in Pursuing Stacks §34 (p. 47-50) .

Proof of theorem 8.4. The essential part, technically, is of course $1^{\circ 41}$. We have the $\pm$ tautological implications

$$(a) \implies (b) \implies (c)\ ,$$

where $(a) \Longrightarrow (b)$ comes from the fact that L is obviously an injective object of $A^{\wedge}$. (The Lawvere object of any topos is an injective object.) It remains to establish

$$(c) \implies (d) \implies (a)\ .$$

That we have $(c) \Longrightarrow (d)$ (and $(c) \Longrightarrow (b)$) is established in Pursuing Stacks, see the summary of the W-test notions in Pursuing Stacks, §65, p. 168-179; the relevant statement here is proposition 3, page 173. p. 60

It remains to establish $(d) \Longrightarrow (a)$ – but now we're on easy street! I note here the following lemma, whose place would rather be at the end of §4 of the generalities on Quillen triples:

Lemma 8.6. *Let $\mathcal{M}$ be a category, (C, TF) a Quillen precouple in $\mathcal{M}$, and $W \subset \mathrm{Fl}(\mathcal{M})$ a subset of $\mathrm{Fl}(\mathcal{M})$ stable under direct factors. Equivalent conditions:* [page 109]

a) $TF \subset W$.

b) *Every arrow f in $\mathcal{M}$ factors as $f = pi$, with $i \in C$, $p \in W$.*

b′) *As in* (b), *with f in TF.*

Proof. The implications $(a) \Longrightarrow (b) \Longrightarrow (b')$ are tautological and do not appeal to the stability hypothesis on W. But for a given $f \in TF$, to see that it is in W, if W is stable under direct factors, it suffices for that particular f to factor as pi, with $i \in C$, $p \in W$. Indeed, we see at once that then f is a direct factor of p (since $i \in C$), and since $p \in W$, we have $f \in W$, *q.e.d.*

Let us return to 8.4, proof of $(d) \Longrightarrow (a)$.

In the end, my plan of proof changes, for although (d) looks very, very strong as a condition, as it stands I do not see, in the end, how to deduce (a) from it. But in fact we can consider that the equivalence of (b), (c), (d) is put here for the record, it is established in *loc. cit.*, so I only have to concern myself with the implication $(c) \Longrightarrow (a)$, which will follow from lemma 8.6, which reduces us to proving that every arrow $f : X \longrightarrow Y$ in $A^{\wedge}$ factors as pi, with i a monomorphism and $p \in W_A$. We use the mapping cone argument , starting from [page 110] [page 111]

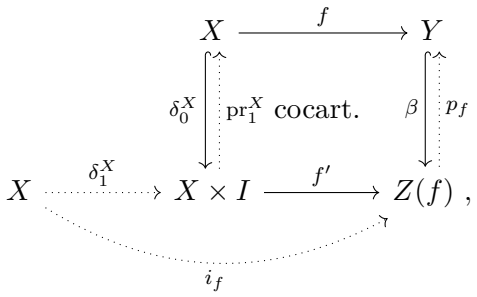
$$\begin{array}{ccc} e_A \amalg e_A & \xrightarrow[\text{mono}]{(\delta_0,\delta_1)} & I \\ & \searrow \quad \swarrow_{(W_A)_{\text{univ}}} & \\ & e_A\ , & \end{array}$$

whence for every X in $A^{\wedge}$, taking the product with X,

$$\begin{array}{ccc} X \amalg X & \xrightarrow[\text{mono}]{(\delta_0^X,\delta_1^X)} & X \times I \\ & \searrow_{(\mathrm{id},\mathrm{id})} \quad \swarrow_{\mathrm{pr}_1^X \in W_A} & \\ & X\ , & \end{array}$$

p. 61

and we construct the mapping cone of f by the cocartesian diagram



p_f deduced from $\mathrm{pr}_1^X : X \times I \to X$ (8.6.1)

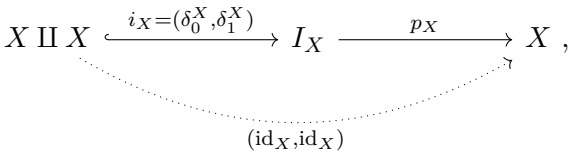
the desired factorization of f is

$$f \;=\; p_f\, i_f \;, \qquad \text{where} \quad i_f = f'\delta_1^X \;.$$

Here, since $\mathrm{pr}_1^X \in W_A$, we conclude that δ_0^X, δ_1^X (which are sections of it) are in W_A, since δ_0^X is a monomorphism, we thus have $\delta_0^X \in TC$, hence (TC being stable under cobase change) $\beta \in W_A$, hence finally p_f (which is a retraction of $Z(f)$ onto Y) is also in W_A. It remains to see that i_f is a monomorphism, which is seen from the more complete diagram (*cf.* XII, p. 230) that I have in front of me: i_f appears as a composite of two monomorphisms $X \longrightarrow Y \amalg X$ and $Y \amalg X \longrightarrow Z(f)$ … Theorem 8.4 is proved. [page 112]

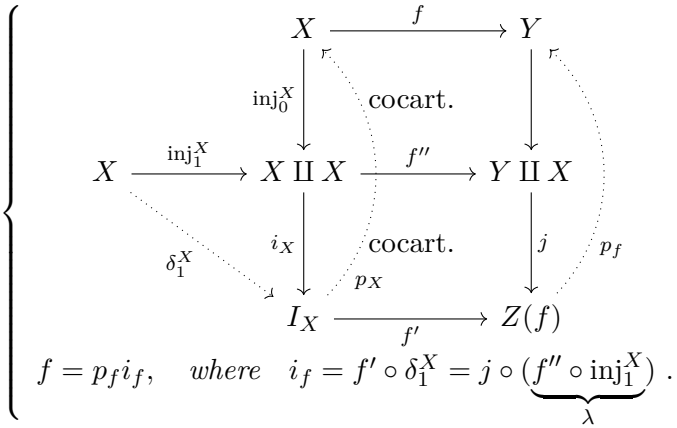
I will all the same isolate the everlasting formal argument (which I have used mostly in dual form) in a

Lemma 8.7. *Let $\mathcal{M}$ be a category, W a mildly saturated localizer in $\mathcal{M}$, C a subset stable under composition and under cobase change. We propose, for X fixed in $\mathcal{M}$, to find a functorial factorization of every arrow $f : X \longrightarrow Y$* (i.e. *of the costructural morphism of a Y in $X\backslash\mathcal{M}$) as $p_f i_f$, with $i_f \in C$, $p_f \in W$. We assume for this that we have a factorization of this type for $X \amalg X \xrightarrow{(\mathrm{id},\mathrm{id})} X$,* i.e. *an object I_X and a factorization* [page 113]



with $i_X \in C$, $p_X \in W$. We then associate to every f the mapping cone $Z(f)$ as in diagram (8.6.1) *(where one must read I_X instead of $X \times I$), and we then have $p_f \in W$, $i_f \in C$ as desired. But one must assume, to know that $i_f \in C$, that $X \in \mathcal{M}_C$ (C-cofibrant objects),* i.e. *that for every Y, $X \longrightarrow Y \amalg X$ is in C. (Moreover, we have tacitly assumed $\mathcal{M}$ stable under finite* $\varinjlim$*.)*

The complete diagram is



(8.7.1)

p. 62

[41] Part 2° follows from it, thanks to corollary 8.3.

[page 114]

We have $i_X \in C$ by hypothesis, hence $j \in C$ (stability of C under cobase change), we have $\lambda : X \longrightarrow Y \amalg X$ in C since X in $\mathcal{M}_C$, hence the composite $i_f = j \circ \lambda$ is in C (stability of C under composition) – we win!

Comments. For the equivalence of (a), (b), (c) in theorem 8.4, 1°, we thus did not have to call on the developments of Pursuing Stacks, which serve only to establish the equivalence of the three conditions in question with the "modelizing" condition (d). To summarize:

1°) The implications

$$(a) \implies ((b) \iff (c) \iff (d))$$

are valid under just the conditions

$$\mathrm{W}(1,\ 2\,\mathrm{bis},\ 3)\ ,$$

and for the equivalence of (a) with the other conditions, it suffices to have

$$\mathrm{W}(1\,\mathrm{bis},\ 2\,\mathrm{bis},\ 3)$$

(where W (1 bis) strengthens W (1) by the direct factor condition, whereas W (2 bis) weakens W (2), requiring only that W-aspheric morphisms be in W). The conclusions of 8.4, 3°, concerning the case where A is W-aspheric, are also valid under the conditions W (1, 2 bis, 3) alone.

[page 115]

2°) Corollary 8.5, referring to conditions (b), (c), (d) of 1° (to the exclusion of (a), which assumes W stable under direct factors), is valid for a W satisfying

$$\mathrm{W}(1,\ 2\,\mathrm{bis},\ 3,\ \underbrace{7\,\mathrm{bis}}_{\substack{\text{square}\\ \text{cocartesian}}}\ )\ .$$

p. 63

3°) For the validity of 8.4, 2°, *i.e.* to have a Quillen triple (W_A, C, F), we need (almost) the whole range of conditions on W,

$$\mathrm{W}(1\,\mathrm{bis},\ 2\,\mathrm{bis},\ 3,\ \underbrace{6}_{\text{limits}},\ \underbrace{7\,\mathrm{bis}}_{\substack{\text{square}\\ \text{cocartesian}}}\ ,\ \underbrace{8}_{\text{accessibility}}\ )^{42}\ ,$$

to the exclusion of W (4), and being content for W (2) with the weakened form W (2 bis) and no more. (Even the connectedness axiom W (5) was not used, but I had initially included it, since there is little chance that we will ever have to work with W's that do not satisfy it! It might moreover be, in the presence of the other axioms, or only of W (1 bis, 2 bis, 3, 7 bis), that if W (5) is *not* satisfied, we have $W = \mathrm{Fl}\,\mathrm{Cat}$ in its entirety, hence $W_A = \mathrm{Fl}\,A^\wedge$!)

[page 116]

9 Revision of the notations, for the axioms of a basic localizer[43]

Loc (1) (Saturation).

[42] **N.B.** We should renumber the axioms W (…).

[43] *Cf.* XVII for a definitive version (?).

a) Isomorphisms are in W.

b) If f, g are composable, and two among f, g, gf are in W, the third as well.

c) If $X \underset{g}{\overset{f}{\rightleftarrows}} Y$, if gf and fg are in W, then f and g as well.

Strengthened variant:

Loc (1 bis) Loc (1) plus

d) W is stable under direct factors.

Loc (2) (Final object axiom). If X in Cat has a final object, then X is W-aspheric, *i.e.* $X \longrightarrow e$ is in W.

Loc (3) (Localization axiom, weak form). Let $f : X \longrightarrow Y$ be such that for every $y \in Y$, X/y is W-aspheric (*i.e.* the morphism $f/y : X/y \longrightarrow Y/y$ deduced from f by base change $Y/y \longrightarrow Y$ is in W). Then $f \in W$.

Strengthened variant of Loc (3):

Loc (3 bis) (Localization axiom, strong form). Let $f : X \longrightarrow Y$ be a morphism in Cat/S, such that $\forall\ s \in S$, $f/s : X/s \longrightarrow Y/s$ (deduced from f by base change $S/s \longrightarrow S$) is in W. Then $f \in W$. (**N.B.** Unless I am mistaken, it suffices to require it if X, Y are split Cat-fibered over S, and f S-cartesian compatible with the splitting – assuming Loc (1, 2)).

p. 64

Loc (4) (Cocartesian square axiom). Let X be in Cat, X_1, X_2 two opens with union X, $X_0 = X_1 \cap X_2$, whence a cocartesian diagram in Cat, [page 117]

$$\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1 \\ {\scriptstyle i_2}\downarrow & & \downarrow{\scriptstyle j_2} \\ X_2 & \xrightarrow{j_1} & X\ . \end{array}$$

We then have $i_1 \in W \iff j_1 \in W$ (and symmetrically $i_2 \in W \iff j_2 \in W$).

Loc (6) [44] (Axiom of W-fibrations). There are variants ⓐ, (a′), ⓑ, (b′), (b″), (c), (c′), *cf.* XII, p. 53-59. The strongest form ⓐ is this:

Let $f : X \longrightarrow Y$ in Cat be a W-fibration in the old sense, *i.e.* such that for every $Y'' \longrightarrow Y'$ in Cat/Y, with Y', Y'' contractible, the arrow $X'' \longrightarrow X'$ obtained by the base change $X \longrightarrow Y$ is in W. Then f is in Fib_W, *i.e.* the base change functor by f, Cat/$Y \longrightarrow$ Cat/X, transforms arrows in W into arrows in W.

Loc (7) (Direct limit axiom). W is stable under filtering direct limits.

Strengthened form of Loc (7):

Loc (7 bis) (L-accessibility). W is stable under filtering $\varinjlim$ *and* accessible, *i.e.* there exists a small subset W_π of Fl(Cat), such that W is the smallest subset of Fl(Cat) stable under filtering $\varinjlim$, containing W_π. [page 118]

Loc (8) [45] (Connectedness axiom). $f \in W \Longrightarrow \pi_0(f)$ bijective.

N.B. All the axioms, except the last (and a smidgen of the next to last) are stability axioms – they all say that W is "large": it contains certain arrows, and if it contains such and such arrows, it contains such and such others. Loc (8), and to a certain extent Loc (7 bis), are "limiting" axioms on the size of W: Loc (8) tells us that $W \subset \mathbf{W}_0$ (= set of $f \in \mathrm{Fl}\,\mathrm{Cat}$ such that $\pi_0(f)$ is bijective), Loc (7 bis) tells us of the existence of a cardinal such that W is equal to the L-envelope of W^π (hence $W \subset \text{L-env}(W^\pi)$).

I have tried to arrange the axioms in order of practical importance. I ended up adding the connectedness axiom at the end, for in the theoretical developments on general localizers, it practically never comes in. But I recall (XII, p. 20, th. 2) that if W satisfies Loc (1, 2, 3, 6), and if Loc (8) is *not* satisfied, then $W = \mathrm{Fl}(\mathrm{Cat})$; thus in the presence of Loc (1, 2, 3, 6), Loc (8) *is equivalent* to $W \neq \mathrm{Fl}(\mathrm{Cat})$ in its entirety.

The axioms that were used for 8.4, 1°, 3° and 8.5, (b) are [page 119]

$$\text{Loc}\,(1\,\text{bis},\ 2,\ 3)\ .$$

Those which are necessary for 8.5, (a) are p. 65

$$\text{Loc}\,(1\,\text{bis},\ 2,\ 3,\ 4)\ .$$

Finally, those necessary for 8.4, 2° are

$$\text{Loc}\,(1\,\text{bis},\ 2,\ 3,\ 4,\ 7\,\text{bis})\ ,$$

and they imply the *strong saturation of* W (*cf.* next paragraph).

I must repair an omission, having somewhat treated with contempt (!) the questions of connected components, up to now.

Proposition 9.1. *Suppose that W satisfies* Loc (1, 2, 3, 4) *and no more. Let $f_i : X_i \longrightarrow Y_i$ ($i \in I$) be a* finite *family of arrows of* Cat. *For $f = \coprod f_i : \coprod X_i \longrightarrow \coprod Y_i$ to be in W, it suffices that $\forall\ i \in I$, we have $f_i \in W$. If W also satisfies* Loc (7) (*direct limit axiom*), *then this conclusion remains valid without a finiteness condition on I.*

Proof. The case of finite I is handled by induction on $n = \mathrm{card}\, I$, we are immediately reduced to the case where $n = 2$[46]. We first see the

Corollary 9.2. *Under the hypothesis* Loc (1, 4), *let $f : X \longrightarrow Y$ be an arrow of* Cat, *and Z an object of* Cat. *Then $f \in W$ if and only if $f \amalg \mathrm{id}_Z : X \amalg Z \longrightarrow Y \amalg Z$ is in W.* [page 120]

Immediate from the cocartesian square, where inj_0^X is an open Dwyer immersion,

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ {\scriptstyle \mathrm{inj}_0^X}\downarrow & & \downarrow{\scriptstyle \mathrm{inj}_0^Y} \\ X \amalg Z & \xrightarrow[f' = f \amalg \mathrm{id}_Z]{} & Y \amalg Z\ , \end{array}$$

which shows that $f \in W \iff f' \in W$.

By iterating this result, we find, if

$$f_1\ :\ X_1 \longrightarrow Y_1\ , \qquad f_2\ :\ X_2 \longrightarrow Y_2$$

are given, that

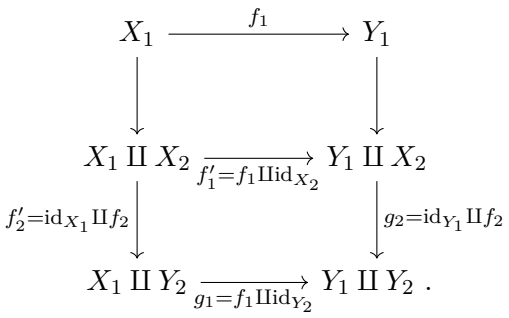
$$f_1\ \in\ W\ ,\ f_2\ \in\ W \quad \Longrightarrow \quad f_1 \amalg f_2\ \in\ W\ ,$$

[44] There will be a more "basic" Loc (5) below.

[45] I have changed the numbering, to take account of a Loc (5) below, which is again a stability axiom .

[46] The infinite case reduces to this under the hypothesis Loc (7), by passage to the limit.

by looking at the diagram of two cocartesian squares:



If f_1 and f_2 are in W, the same is true of f_1', f_2', g_1, g_2, by 9.2, hence also of $f_1 \amalg f_2 = g_2 f_1'$. p. 66

But I see no way of proving the converse, that $f_1 \amalg f_2 \in W \Longrightarrow f_1, f_2 \in W$. Thanks to what has been proved, we are reduced to the case where f_1, f_2 are open Dwyer immersions, but it does not seem that this gets us a hair's breadth further! In short, we need to know that the canonical maps $Y_i \hookrightarrow \coprod Y_i$ are in Fib_W. Since Y_i is a direct summand, we are reduced to the case of a canonical immersion $Y_1 \hookrightarrow Y_1 \amalg Y_2 = Y$: if $f : X \longrightarrow Y$ is in W, is the same true of $X_1 = X \times_Y Y_1 \longrightarrow Y_1$? This is a property which, of course, would be a consequence of the old W (4 a) (XII, p. 53), or just W (4 a′), hence of the current Loc (6 a′). Even the form W (4 c′) = Loc (6 c′), the weakest of all, suffices, as we see by factoring $X \longrightarrow Y$ as $X \xrightarrow{i} X' \xrightarrow{p} Y$, with $i \in W$ (more precisely, $i_1 : X_1 \longrightarrow X_1'$ and $i_2 : X_2 \longrightarrow X_2'$ in W!), and $p \in \mathrm{Parf}_W$. By Loc (6 c′), $f \in W$, hence $p \in W$ implies that the fibers of p are W-aspheric, which implies that p is in W_{univ}, hence $p_1 : X_1' \longrightarrow Y_1$ is in W. [page 121]

So let us note for conscience's sake: [page 122]

Corollary 9.3. *If W satisfies* Loc (1, 2, 3, 6 c′) (*where* Loc (6 c′) *is the weakest of all the variants of the fibration axiom*), *then for any family $(f_i : X_i \longrightarrow Y_i)_{i \in I}$ of arrows in* Cat, *$f = \coprod f_i \in W \Longrightarrow$ the f_i are in W $\forall\ i \in I$*[47]. *And if $i : Y_0 \hookrightarrow Y$ is an arrow which is both an open immersion and a closed immersion* (i.e. *an isomorphism with a direct summand of Y*), *then i is in* Fib_W, i.e. *if $f : X \longrightarrow X'$ is an arrow in* Cat$/Y$, *then $f \in W$ implies $(f_0 : X_0 \longrightarrow X_0') \in W$, where $X_0 = X|Y_0$, $X_0' = X'|Y_0$.*

But it is rather scandalous to have to use such a strong axiom, to conclude from it such an innocuous result! I note this:

Proposition 9.4. *Let $W \subset$ Fl(Cat), satisfying* Loc (1) *and no more. The following conditions are equivalent* (*and implied by* Loc (1, 2, 3, 4, 6 c′)).

a) *If $f_1 : X_1 \longrightarrow Y_1$, $f_2 : X_2 \longrightarrow Y_2$ are two arrows of* Cat, *then*

$$f_1 \amalg f_2 \in W \iff f_1 \in W,\ f_2 \in W\,.$$

b) *The following two conditions on W are satisfied:* [page 123]

1°) *If $f : X \longrightarrow Y$ is in W, then it is so couniversally under $\emptyset$,* i.e. *for every Z in* Cat, *$f \amalg \mathrm{id}_Z : X \amalg Z \longrightarrow Y \amalg Z$ is in W.* (**N.B.** *This is equivalent to: every finite sum of arrows $f_i \in W$ is in W – and likewise even for infinite sums, if* Loc (7), *the limit axiom, is satisfied.*)

[47] In fact, I do not know how to prove the implication in the reverse direction, assuming only Loc (1, 2, 3, 6 c′).

2°) *Let $Y_0 \hookrightarrow Y$ be an inclusion of a direct summand of an object of* Cat. *Then if $X \xrightarrow{f} Y$ is in W, so is $X_0 = X|Y_0 \longrightarrow Y_0$.* (**N.B.** *This is equivalent to the condition that the inclusion of a direct summand is always a W-fibration.*)

Moreover, if $\pi_0(Y)$ is finite, or if W satisfies the limit axiom Loc (7), *it suffices to require 2° for Y_0 a connected component of Y.*

p. 67

c) *For every* finite *family $(f_i)_{i\in I}$ of arrows $f_i : X_i \longrightarrow Y_i$ in* Cat,

$$\coprod f_i \in W \iff f_i \in W \text{ for all } i \in I.$$

Moreover, if W satisfies Loc (7), *then it is unnecessary in* (c) *to restrict ourselves to finite I. (And in any case, the equivalent conditions* (a), (b), (c) *imply for* every *family of arrows $(f_i)_{i\in I}$, $\coprod f_i \in W \Longrightarrow f_i \in W \ \forall\ i \in I$.)*

According to 9.3, these conditions are satisfied assuming Loc (1, 2, 3, 4, 6 c′), but Loc (6 c′) is a steam hammer for a fly! One would rather like to add an axiom to the axioms Loc (1) to Loc (4) and Loc (6) to Loc (8), which together with Loc (8) would be the most innocuous axiom of all.

[page 124]

Loc (5) (Decomposition axiom, weak form). If $(f_i)_{i\in I}$ is a finite family of arrows of Cat, then $f = \coprod f_i$ is in W if and only if the f_i are. (These are the equivalent conditions of 9.4.)[48]

Loc (5 bis) (Decomposition axiom, strong form). Like Loc (5), but without assuming I finite.[49]

Thus, assuming the limit axiom Loc (7), the two preceding axioms are equivalent. Moreover, let us note this:

Corollary 9.5 (of proposition 9.4). *The axiom* Loc (5 bis) *is equivalent to each of the following properties:*

ⓐ *For every Y of* Cat, *decomposed into its connected components Y_i ($i \in I = \pi_0(Y)$), and for every arrow $f : X \longrightarrow Y$ in* Cat, *we have $f \in W$ if and only if the $f_i : X_i = X|Y_i \longrightarrow Y_i$ are in W.*

ⓑ *For Y, Y_i as in* ⓐ, *for every arrow $f : X \longrightarrow X'$ in* Cat$/Y$, *we have $f \in W$ if and only if $\forall\ i \in I$, the morphism $f_i : X_i \longrightarrow X'_i$ over Y_i is in W.*

[page 125]

Corollary 9.6. *If the axiom* Loc (5) *is satisfied, then the connectedness axiom* Loc (8) *is equivalent to this: for every arrow $f : X \longrightarrow Y$ in W, with Y 0-connected, X is 0-connected.*

N.B. The axioms Loc (8) (connectedness) and Loc (5 bis) (of decomposition) seem to be of a similar nature, both being linked to the decomposition of the objects of Cat

p. 68

[48] It may also be convenient to introduce the still weaker condition:

Loc (5 a) If f_1, f_2 are in W, then $f_1 \amalg f_2 \in W$.
This is also condition (b), 1° of proposition 9.4.

Loc (5 b) If $f_1 \amalg f_2$ is in W, then f_1 and f_2 too.
$\iff$ If $Y_0 \subset Y$ is a direct summand, then $Y_0 \hookrightarrow Y$ is in Fib_W.

[49]

Loc (5) *a*) If $f_1, f_2 \in W$, then $f_1 \amalg f_2 \in W$.
b) If $f_1 \amalg f_2 \in W$, then $f_1, f_2 \in W$
($\iff$ *b′*) every direct summand $Y_0 \hookrightarrow Y$ is in Fib_W).

Loc (5 bis) Like Loc (5), but (*a*) replaced by (*a* bis) a family (not necessarily finite).

into connected components. (Nevertheless, I think they are independent.) I presume that we will never really have to work with W which do not satisfy these two axioms! (In addition to Loc (1, 2, 3), which for their part constitute the "vital minimum".) But apparently there is no need for them in the general theoretical developments so far. (But the connectedness axiom Loc (8) could play a role, when one speaks of the reciprocal determination of the structures of contractibility, and of W-asphericity, *cf.* Pursuing Stacks.)

[page 126]

Let us note this:

Proposition 9.7.[50] *Suppose that W satisfies* Loc (1, 2, 3, 5), i.e. *it is a basic localizer satisfying the decomposition condition (weak form). If the axiom* Loc (8) *of connectedness is* not *satisfied, then the subcategory* Hot_W^* *of the* $X \in \mathrm{Cat}$ *which are nonempty is equivalent to the one-point category (hence* Hot_W *is equivalent either to the one-point category, if* $\emptyset \longrightarrow e$ *is an isomorphism in* Hot_W*, or to* $\mathbf{\Delta}_1$ *otherwise*[51]*.)*

This is a variant of th. 2 of XII (p. 20), where we assumed Loc (6 c′), stronger than Loc (5) and even than Loc (5 bis), but with the stronger conclusion that $W = \mathrm{Fl}(\mathrm{Cat})$.

Proof. Let

$$f \ : \ X \longrightarrow Y \ , \qquad f \ \in \ W \ ,$$

be such that $\pi_0(f)$ is not *surjective.* If Y_0 is a connected component which is not in $\mathrm{Im}\pi_0(f)$, then $X_0 = X|Y_0$ is empty, and by Loc (5), $\emptyset \longrightarrow Y_0$ is in W. Thus $\forall\ Z$ in Cat, $\underbrace{\emptyset \times Z}_{=\emptyset} \longrightarrow Y_0 \times Z$ is in W, and for every arrow $g : Z \longrightarrow Z'$, $\mathrm{id}_{Y_0} \times g : Y_0 \times Z \longrightarrow Y_0 \times Z'$ is in W. Taking g such that $\pi_0(g)$ is not injective, e.g. $e \amalg e \longrightarrow e$, we find an arrow $\varphi = \mathrm{id}_{Y_0} \times g$ in W such that $\pi_0(\varphi)$ is not injective.

[page 127]

Let then $\varphi \ : \ I \longrightarrow J$ be in W, with $\pi_0(\varphi)$ not injective. Proceeding as before using Loc (5), we are reduced to the case where in addition J is 0-connected. (**N.B.** In both cases, we could also have invoked 9.6.) Let us choose two elements i_0, i_1 in I in two distinct connected components, let j_0, j_1 be their images in J. We are going to treat (I, i_0, i_1) and (J, j_0, j_1) as "homotopy intervals". The fact that i_0, i_1 belong to distinct components of I implies that two arrows $f_0, f_1 : X \rightrightarrows Y$ are "homotopic with respect to (I, i_0, i_1)", *i.e.* there exists $h \ : \ I \times X \longrightarrow Y$ in Cat such that $f_0 \ = \ h\partial_0^X$, $f_1 = h\partial_1^X$, where

$$\partial_0^X \ , \ \partial_1^X \ : \ X \ \longrightarrow \ I \times X$$

are given by $x \longmapsto (i_\alpha, x)$ $(\alpha \in \{0, 1\})$. To find h, we decompose I as

$$I \ = \ I_0 \amalg I_1 \ , \qquad \text{with} \quad i_0 \in I_0, \ i_1 \in I_1 \ ,$$

whence $I \times X \ \simeq \ (I_0 \times X) \amalg (I_1 \times X)$, and we define h by its restrictions h_0, h_1 to the two pieces $I_0 \times X$, $I_1 \times X$, namely

$$h_0 \ = \ f_0 \circ \mathrm{pr}_2^{I_0} \ , \qquad h_1 \ = \ f_1 \circ \mathrm{pr}_2^{I_1} \ ,$$

where

p. 69

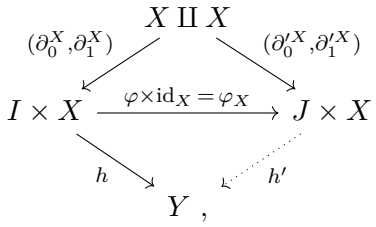
$$\mathrm{pr}_2^{I_\alpha} \ : \ I_\alpha \times X \longrightarrow X \ .$$

[page 128]

[50] *Cf.* stronger version in 9.9 below (rid of the hypothesis Loc (5).)

[51] The first case is also the one where $\exists\ f \in W$ with $\pi_0(f)$ *not surjective,* or again $\emptyset \longrightarrow X$ in W, with X nonempty.

Let us now consider the diagram (solid lines)



where $\partial_0'^X$, $\partial_1'^X$ are defined as ∂_0^X, ∂_1^X, but using $j_0, j_1 \in \mathrm{Ob}J$ instead of $i_0, i_1 \in \mathrm{Ob}I$. If there existed in Cat an arrow h' (dotted) such that $h' \circ \varphi_X = h$, it would follow that h' is a homotopy of f_0 and f_1 along (J, j_0, j_1). Since J is 0-connected, it would follow at once that f_0, f_1 are homotopic in the usual sense in Cat, hence that f_0 and f_1 have the same image in Hot_W. But if one cannot find an arrow h' in Cat, at least one can find it in Hot_W, since $\varphi_X \in W$, hence an isomorphism in Hot_W (and h' in $\mathrm{Hom}_W(J \times X, Y)$ will even be uniquely determined). Thus in Hot_W, we have

$$f_0 \;=\; h'\partial_0'^X \;, \qquad\qquad f_1 \;=\; h'\partial_1'^X \;.$$

To prove that

$$f_0 \;=\; f_1 \qquad \text{in} \quad \mathrm{Hom}_W(X, Y) \;,$$

[page 129]

it suffices to see that

$$\partial_0'^X \;=\; \partial_1'^X \qquad \text{in} \quad \mathrm{Hom}_W(X, J \times X) \;.$$

Now in fact $\partial_0'^X$ and $\partial_1'^X$ are homotopic, for $e \overset{j_0}{\underset{j_1}{\rightrightarrows}} J$ are homotopic (j_0 and j_1 belonging to the same connected component of J), hence also their product with id_X.

One sees therefore that for any two objects X, Y of Cat, all the arrows from X to Y are equal in Hot. The argument given even proves that all the arrows *in* Hot_W from X to Y are equal. This means that the category Hot_W^* (where for any two objects X, Y, $\mathrm{Hom}_{\mathrm{Cat}}(X, Y)$, and a fortiori $\mathrm{Hom}_W(X, Y)$ is $\neq \emptyset$) is equivalent to the one-point category. As for Hot_W itself, whose set of objects is $\{\emptyset\} \amalg \mathrm{Ob}\mathrm{Hot}_W^*$, with $\emptyset$ an initial object, it will therefore be equivalent to the one-point category, or else to $\mathbf{\Delta}_1$, according as $\emptyset \longrightarrow e$ is an isomorphism in Hot_W, or not.

Comment 9.8. If the axioms Loc (1, 2, 3, 5) [(52)] are satisfied, then the connectedness axiom Loc (8) *is equivalent* to each of the following:

a) Hot_W is equivalent neither to e, nor to $\mathbf{\Delta}_1$. [page 130]

b) Hot_W has infinitely many isomorphism classes.

p. 70

c) $f_0 : X_0 \longrightarrow Y_0$ (arrow given in advance in Cat, such that $\pi_0(f_0)$ is not an isomorphism, and $X_0 \neq \emptyset$) is *not* an isomorphism in Hot_W.

d) X_0 and Y_0 are not isomorphic in Hot_W (where X_0, Y_0 are two objects given in advance in Cat, with X_0, Y_0 nonempty and $\mathrm{card}\,\pi_0(X_0) \neq \mathrm{card}\,\pi_0(Y_0)$).

One can therefore say that the connectedness axiom plays the role of a *nontriviality axiom* for the localized category Hot_W of Cat. It will be satisfied automatically for all the basic localizers W that we will be led to construct or to introduce. But we will practically never have to invoke it as an assumption, with a view to certain conclusions.

I finally realize that in proposition 9.7 the assumption Loc (5) is unnecessary: ‖

Proposition 9.9. [(53)] *Let W be a basic localizer in* Cat (i.e. *we have* Loc (1, 2, 3)

[52] In fact, this comment holds even in the absence of the assumption Loc (5), *cf.* 9.9 below.
[53] *Cf.* better statement 9.12 (p. 133 bis).

and nothing more), *plus the condition that the* sum *of* $f_1, f_2 \in W$ *is in* W [(54)]. *If the connectedness axiom is not satisfied, then* Hot_W *is equivalent to the one-point category* (*if* $\emptyset \longrightarrow e$ *is an isomorphism in* Hot_W [(55)]) *or to* $\mathbf{\Delta}_1$ (*in the opposite case*), *and* Hot_W^* ($\overset{\text{def}}{=} \mathrm{Hot}_W \backslash \{\emptyset\}$) *is equivalent to the one-point category.*

The condition on sums ensures that in Hot_W the sum of two objects exists, and that $\mathrm{Cat} \longrightarrow \mathrm{Hot}_W$ commutes with sums of two objects. This said, let [page 131]

$$f \ : \ X \longrightarrow Y \qquad (*)$$

be an arrow such that $f \in W$, but $\pi_0(f)$ not surjective. Let us first suppose $X \neq \emptyset$ (the least obvious case). If Y_0 is a nonempty summand of Y such that f factors through $Y_1 = Y \backslash Y_0$, then f is the sum of

$$f_1 \ : \ X \ = \ X_1 \longrightarrow Y_1 \qquad \text{and} \qquad f_0 \ : \ X_0 \ = \ \emptyset \longrightarrow Y_0$$

in Cat, hence also in Hot_W. Thus we have a sum $f \ = \ f_0 \amalg f_1$ in Hot_W which is an isomorphism, hence for every Z in Hot_W,

$$\mathrm{Hom}_W(Y_0, Z) \times \mathrm{Hom}_W(Y_1, Z) \ \xrightarrow{\sim} \ \underbrace{\mathrm{Hom}_W(X_0, Z)}_{=e} \times \mathrm{Hom}_W(X_1, Z)$$

(**N.B.** $\emptyset$ is an initial object of Hot_W, being so in Cat), *i.e.* the composite

$$Z(Y_0) \times Z(Y_1) \ \xrightarrow{\mathrm{pr}_2} \ Z(Y_1) \ \xrightarrow{Z(f_1)} \ Z(X_1)$$

is bijective. Since $X \ = \ X_1 \neq \emptyset$, hence also $Y_1 \neq \emptyset$, and $Y_0 \neq \emptyset$, the sets $Z(Y_0)$, $Z(Y_1)$, $Z(X_1)$ are $\neq \emptyset$ if $Z \neq \emptyset_{\mathrm{Cat}}$. The composite being bijective, the first arrow pr_2 is injective, which implies that $Z(Y_0)$ is reduced to a point (the two factors $Z(Y_0)$, $Z(Y_1)$ being $\neq \emptyset$). This means that Y_0 is an initial object in Hot_W^*. It follows in particular that $Y_0 \amalg Y_0$ is also an initial object, and the canonical map $Y_0 \amalg Y_0 \longrightarrow Y_0$ is therefore an isomorphism in Hot_W, whereas the induced map on the π_0 is not injective. (It is $\pi_0(Y_0) \amalg \pi_0(Y_0) \longrightarrow \ \pi_0(Y_0)$, with $\pi_0(Y_0)$ nonempty.) This case will be treated below. p. 71 [page 132]

If in $(*)$ X is empty, Y nonempty (since $\pi_0(f)$ not surjective), then Y is an initial object of Hot_W, and one sees again that $Y \amalg Y \longrightarrow Y$ is an isomorphism in Hot_W, although not inducing an isomorphism of the π_0. Proposition 9.9 will therefore follow from the

Lemma 9.10. *Let* W *be a basic localizer and nothing more* (*axioms* Loc (1, 2, 3)). *Suppose there exists* $f : I \longrightarrow J$ *in* Cat *such that* $\mathrm{hot}_W(f)$ *is an isomorphism, but* $\pi_0(f)$ is not injective [(56)]. *Then* Hot_W^* *is equivalent to the one-point category* e, *and* Hot_W *is equivalent to* e (*if* $\emptyset \longrightarrow e$ *is an isomorphism in* Hot_W) *or to* $\mathbf{\Delta}_1$ (*in the opposite case*).

The proof is the one given for 9.7: one proves that if X, Y are two objects of Cat, *nonempty*, then two arrows $f_0, f_1 : X \rightrightarrows Y$ are equal in Hot_W, by a homotopy argument, choosing two points i_0, i_1 of I which are in two distinct connected components, and such that their images j_0, j_1 in J are in the same component. [page 133]

Proposition 9.11. *Let* W *be a basic localizer* (i.e. *we have* Loc (1, 2, 3)). *Then the following conditions are equivalent.*

a) *We have* Loc (5 bis) (*decomposition axiom, strong form*) *and* Loc (8) (*connectedness axiom*).

[54] This condition Loc (5 a) holds in particular if W also satisfies one of the two axioms Loc (4) (*cf.* 9.1), or Loc (6 c′) (*cf.* 9.3).

[55] This also means that there exists $Z \neq \emptyset$ in Cat, such that $\emptyset \longrightarrow Z$ is an isomorphism in Hot_W.

[56] Or else such that I is 0-connected, and J not 0-connected (*i.e.* $\mathrm{card}\,\pi_0(I) \ = \ 1$, $\mathrm{card}\,\pi_0(J) \ \geq \ 2$). This second case is treated by the same method essentially.

b) *We have* Loc (5 bis), *and* Hot_W *is not equivalent to* e *or to* $\mathbf{\Delta}_1$ (cf. *comment* 9.8).

c) *For every* $f : X \longrightarrow Y$, *we have* $f \in W$ *if and only if* $\pi_0(f)$ *is an isomorphism, and for every connected component* X_i *of* X, *if* Y_i *denotes the corresponding connected component of* Y, *the induced arrow* $f_i : X_i \longrightarrow Y_i$ *is in* W.

The reason why I do not isolate this property as one of the axioms Loc (…) is that it is *not* a stability axiom, but an axiom of a mixed nature.

[page 133 bis]

Proposition 9.12. *Let* W *be a basic localizer* (*nothing more*), i.e. *satisfying* Loc (1, 2, 3). *The following conditions are equivalent.*

a) W *satisfies the connectedness axiom* Loc (8), i.e. $f \in W \Longrightarrow \pi_0(f)$ *bijective.*

b) *If* $f, g : X \rightrightarrows Y$ *in* Cat *are such that* $\mathrm{hot}_W(f) = \mathrm{hot}_W(g)$, *then* $\pi_0(f) = \pi_0(g)$.

c) *There exist* X, Y *in* Cat *such that the map* $\mathrm{hot}_W^{X,Y} : \mathrm{Hom}_{\mathrm{Cat}}(X, Y) \longrightarrow \mathrm{Hom}_W(X, Y)$ *is not constant.*

c′) *The subcategory* Hot_W^* *of* Hot_W, *consisting of the nonempty objects, is not equivalent to the one-point category.*

c″) *The category* Hot_W *is equivalent neither to the category* e *nor to the category* $\mathbf{\Delta}_1$.

d) *The set of isomorphism classes of objects of* Hot_W *is infinite.*

PROOF. We have $(a) \Longrightarrow (b)$, since (a) is equivalent to saying that the functor $\pi_0 : \mathrm{Cat} \longrightarrow \mathrm{Ens}$ factors through Hot_W. Moreover the corresponding functor

p. 72

$$\mathrm{Hot}_W \longrightarrow (\mathrm{Ens})$$

is essentially surjective – there is a right inverse up to isomorphism, $I \longmapsto \mathrm{hot}_W(I)$. This proves that (a) implies (d), and we have $(d) \Longrightarrow (c')$, $(d) \Longrightarrow (c'')$ tautologically. A more or less tautological argument shows moreover that if Hot_W^* is equivalent to e, then Hot_W is equivalent to e or to $\mathbf{\Delta}_1$ (for $\emptyset$ is an initial object of Hot_W) – hence $(c'') \Longrightarrow (c')$. Thus we have more or less tautologically

$$\begin{array}{ccccccc} (\mathrm{a}) & \Longrightarrow & (\mathrm{d}) & \Longrightarrow & (\mathrm{c}'') & \Longrightarrow & (\mathrm{c}') \\ \Downarrow & & & & & \nearrow & \\ (\mathrm{b}) & & \Longrightarrow & & (\mathrm{c}) & & \end{array}$$

where the implications $(b) \Longrightarrow (c) \Longrightarrow (c')$ are also trivial . Thus everything comes down to proving that $(c) \Longrightarrow (a)$, or again:

[page 133 ter]

Corollary 9.13. *If the connectedness axiom is not satisfied, then if* X, Y *are two objects of* Cat, *all the arrows* $X \longrightarrow Y$ *in* Cat *are equal in* Hot_W, i.e. *the map* $\mathrm{Hom}_{\mathrm{Cat}}(X, Y) \longrightarrow \mathrm{Hom}_W(X, Y)$ *is constant.*

This therefore implies that all the objects of Cat^*, *i.e.* all the nonempty objects of Cat, are isomorphic in Hot_W, and one finds a canonical isomorphism between them, with transitivity. Let then $\mathcal{H}$ be the quotient category of Cat, having the same objects as Cat, and having exactly one arrow between two nonempty objects of Cat and exactly one arrow from $\emptyset$ to every other object of Cat (and no arrow from a nonempty object of Cat to $\emptyset$). It is therefore a category equivalent to $\mathbf{\Delta}_1$. We have seen that $\mathrm{Cat} \longrightarrow \mathrm{Hot}_W$ factors as

$$\mathrm{Cat} \xrightarrow{\text{can.}} \mathcal{H} \longrightarrow \mathrm{Hot}_W \ .$$

If Cat $\longrightarrow \mathcal{H}$ transforms every arrow belonging to W into an isomorphism, then it is immediate that $\mathcal{H}$ is a solution to the universal problem defining Hot_W, hence that $\mathcal{H} \longrightarrow \mathrm{Hot}_W$ is an isomorphism. This also means that W does not contain an arrow $\emptyset \longrightarrow X$ with X nonempty (in which case W contains all arrows). Thus in this case $\mathrm{Hot}_W \simeq \mathbf{\Delta}_1$. In the opposite case, let $\mathcal{H}'$ be the category obtained from $\mathcal{H}$ by inverting the arrows $\emptyset \longrightarrow X$, *i.e.* the 1-connected groupoid on the set of objects ObCat. It is no longer a quotient category of Cat, but we still have a factorization

$$\mathrm{Cat} \xrightarrow{\mathrm{epi}} \mathcal{H} \xrightarrow{\mathrm{loc.}} \mathcal{H}' \longrightarrow \mathrm{Hot}_W \ ,$$

and this time it is $\mathcal{H}'$ that is a solution of the universal problem (Cat $\longrightarrow \mathcal{H}'$ transforming arrows of W into isomorphisms). Thus in this case, $\mathcal{H}' \longrightarrow \mathrm{Hot}_W$ is an isomorphism, and Hot_W is equivalent to e. In both cases, Hot_W^* is equivalent to e. p. 73 [page 133 quater]

Thus, proposition 9.12 follows from corollary 9.13, which it is now a matter of proving.

Let $f : I \longrightarrow J$ in Cat be such that $\pi_0(f)$ is not injective. We have seen in the proof of 9.7 how one deduces from this (by choosing two elements i_0, i_1 in two distinct connected components of I, such that $j_0 = f(i_0)$ and $j_1 = f(i_1)$ are in the same connected component of J) that for X, Y in Cat, two arrows $f_0, f_1 : X \rightrightarrows Y$ are always equal in Hot_W (*cf.* lemma 9.10).

Let now

$$f \ : \ I \longrightarrow J$$

in Cat be such that $\pi_0(f)$ is not surjective, hence $J = J_0 \amalg J_1$, with f factoring through J_0, and $J_1 \neq \emptyset$. Let us then consider the discrete category with two objects,

$$\varepsilon \ = \ \{0, 1\} \ ,$$

and two arrows

$$g_0, \ g_1 \ : \ J \ \rightrightarrows \ \varepsilon \ ,$$

g_0 being constant with value 0, and g_1 being equal to 0 on J_0, to 1 on J_1. We have

$$g_0 f \ = \ g_1 f \ ,$$

hence, since $f \in W$,

$$g_0 \ = \ g_1 \qquad \text{in} \quad \mathrm{Hot}_W \ .$$

One concludes from this, by taking a point j_1 in J_1, and the corresponding inclusion $\alpha_{j_1} : e \longrightarrow J$, that

$$g_0 \alpha_{j_1} \ = \ g_1 \alpha_{j_1} \ ,$$

i.e. the two sections of ε over e are equal in Hot_W. Since the functor Cat $\longrightarrow \mathrm{Hot}_W$ commutes with finite products , it follows that for X in Cat, the two arrows [page 133 quinquies]

$$X \ \overset{\partial_0^X}{\underset{\partial_1^X}{\rightrightarrows}} \ X \times \varepsilon \ = \ X \amalg X$$

are equal in Hot_W. If then

$$f_0, \ f_1 \ : \ X \ \rightrightarrows \ Y$$

are two arrows in Cat, they are of the form

$$f_0 \ = \ h\partial_0^X \ , \qquad f_1 \ = \ h\partial_1^X \ ,$$

with

$$h \ = \ (f_0, f_1) \ : \ X \amalg X \ \longrightarrow \ Y \ ,$$

and since $\partial_0^X = \partial_1^X$ in Hot_W, one concludes $f_0 = f_1$ in Hot_W, *q.e.d.* p. 74 [page 134]

10 Application to the derivator HOT_W

Theorem 10.1. *Let W be a localizer in* Cat *satisfying the conditions* Loc (1, 2, 3, 4, 7 bis) (*axioms of strong-form saturation* (*direct factors*), *of the final object, of weak-form localization, of the cocartesian square, of L-accessibility* [(57)]). *Let* HOT_W *be the prederivator on* (Cat) *associated to* (Cat, W) *in the usual way,*

$$I \longmapsto \mathrm{Cat}(I)\,W(I)^{-1} \;=\; \underline{\mathrm{Hom}}(I^\circ, \mathrm{Cat})\,W(I)^{-1}\,. \tag{10.1.1}$$

One then has the following.

1°) *For every arrow $u : I \longrightarrow J$ in* Cat, *the functor $u_!$ (left adjoint to u^*) exists* (**N.B.** For this one does not need Loc (4) or Loc (7 bis), it suffices that W be a basic localizer), *and it is computed in the standard way* (*axioms* Der 4 bis *and* Der 5 bis). *The functor u_*, right adjoint to u^*, exists* [(58)] [*if u is of the form $v \times \mathrm{id}_S$, where $v : I_0 \longrightarrow J_0$ is an arrow between finite ordered categories, and S an arbitrary object of* Cat], *and this functor u_* is computed in the standard way* (*axioms* Der 4 *and* Der 5). *Thus one has a theory of W-cartesian squares and of W-cocartesian squares in the categories* $\mathrm{HOT}_W(S)$, *and of long sequences of suspension and of cosuspension in the categories* $\mathrm{HOT}^\bullet_W(S)$. (*More* [page 135] *precisely, these are theories relative to the induced derivators* $\mathrm{HOT}_{W,S}$, $\mathrm{HOT}^\bullet_{W,S}$ *on S, $I \longmapsto \mathrm{HOT}_W(S \times I)$ resp. $I \longmapsto \mathrm{HOT}^\bullet_W(S \times I)$ …). Moreover, the cocartesian squares and the cartesian squares relative to the induced derivators* $\mathrm{HOT}_{W,S}$ *are* exact, i.e. *the long suspension and cosuspension sequences are exact* (*axioms of right and left exactness* Der 6 bis *and* Der 6).

2°) *The prederivator* HOT_W *also satisfies the axiom* Der 1 *of finite decomposition* [(59)] (*for $I = \coprod_\alpha I_\alpha$*), *and the axiom* Der 2 *of localization. Moreover, W is strongly saturated,* i.e. *$f \in W \iff \mathrm{hot}_W(f)$ isomorphism.*

3°) *Concerning the axiom* Der 3 *and its variants* (*cf.* I, pages 111-112, 125), *one has this:*

a) [(60)]. *Let $f : S' \longrightarrow S$ be an arrow in* Cat. *For it to be a HOT_W-equivalence,* i.e. *for the functor $f^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$ to induce an* equivalence *between the full subcategories formed by the objects* constant *on S, S', it is necessary and sufficient that f be in W. The corresponding statement in the induced derivators* $\mathrm{HOT}_{W,I}$ *is also valid, for $I \neq \emptyset$. Thus a fortiori,* HOT_W *and* $\mathrm{HOT}_{W,I}$ *satisfy the final object axiom,* [page 136] i.e. *if S in* Cat *has a final object, then $S \longrightarrow e$ is a HOT_W-equivalence, and even a* p. 75 *$\mathrm{HOT}_{W,I}$-equivalence for every I in* Cat.

b) *Suppose that W satisfies* Loc (1 ter, 2, 3, 6 b) (*thus the axiom* Loc (6 b) *of W-fibrations, variant* b). *Then for every $f : S' \longrightarrow S$ in W, $\mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S')$ is an* equivalence of categories (*the converse being clear by* (a)). *Likewise for*

$$\mathrm{HOT}^{\mathrm{lc}}_{W,I}(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_{W,I}(S') \qquad (\text{i.e. for } \mathrm{HOT}^{\mathrm{lc}/I}_W(S \times I) \longrightarrow \mathrm{HOT}^{\mathrm{lc}/I}_W(S' \times I)\ ).$$

In other words, for the derivator HOT_W *and for all the* $\mathrm{HOT}_{W,I}$ (**D** *being one of the these derivators*) [(61)], *every* **D***-equivalence is a* **D***-equivalence which is* 2*-complete,* i.e. *for $f : S' \longrightarrow S$, for $f^* : \mathbf{D}(S) \longrightarrow \mathbf{D}(S')$ to induce an equivalence between the subcategories $\mathbf{D}^{\mathrm{lc}}(S) \longrightarrow \mathbf{D}^{\mathrm{lc}}(S')$, it already suffices that it induce a fully faithful functor on the full subcategories formed by the constant objects, and this is equivalent (if $I \neq \emptyset$) to the condition that $f \in W$.*

[page 137]

[57] **N.B.** Loc (1 bis), *i.e.* $W = W^\natural$, is already implied by the condition that W be stable under filtering $\varinjlim$.

[58] *Always ! Cf.* XIX, p. 29.

[59] Without any axiom on W !

[60] Here one needs only Loc (1 ter, 2, 3) (Loc (1 ter) is strong saturation, *i.e.* $f \in W \iff \mathrm{hot}_W(f)$

Comment. In this summary statement of what I know how to prove at present for HOT_W, I have somewhat mixed the various sources – what is put "for the record" from VI (and which does not depend on Loc (4, 7 or 7 bis)), *i.e.* what depends essentially on the construction of $u_!$ given in VI (thus what concerns $u_!$ in 1°, *minus* right exactness, and all that concerns 3°, (a), (b)).

The proof of 10.1 will follow from the developments of the following section , taking into account the main result of section 8 (th. 8.4), and of section 7 (cor. 7.4), which are the "hard" ingredients of Theorem 10.1.

isomorphism).

[61] To be checked carefully in the case $\mathbf{D} = \mathrm{HOT}_{W,I}$.

THE DERIVATORS

p. [unnumbered]

ALEXANDRE GROTHENDIECK

Chapter XVI
The basic localizers

p. [unnumbered]
p. 1

1 Axioms Loc(1) to Loc(3) (and variants): basic localizers

[page 1]

I feel like taking up again the exposition of the axioms useful for a basic localizer

$$W \subset \mathrm{Fl}(\mathrm{Cat}),$$

guiding myself this time more by the geometric meaning of the axioms, than by the ± technical aliases one encounters (or does not encounter) in checking them. I am therefore revising in this spirit the presentation of XV §9.

Loc (1) (*saturation*).

a) Iso $\subset W$.

b) If two among f, g, gf are in W, the third too.

c) If gf and fg are in W, f and g too.

The conditions (*a*) and (*b*) alone, I am going to call them (for a general $W \subset \mathrm{Fl}\,\mathcal{M}$) *weak saturation* of the localizer. The conditions (*a*), (*b*), (*c*), *mild saturation* ("mild saturation"). Other variants:

Loc (1 bis) The conditions (*a*), (*b*), (*c*) above, plus

d) W stable under direct factors.

This is of course implied by:

Loc (1 ter) W is *strongly saturated, i.e.* if $f \in \mathrm{Cat}$ is such that $\mathrm{hot}_W(f)$ is an isomorphism, then $f \in W$.

Loc (2) (*Final object*). If X in Cat has a final object, then X is W-aspheric, *i.e.* $X \longrightarrow e$ is in W.

Sometimes, this axiom is a consequence of the following weakened variant.

Loc (2_0) (Segment axiom). $\mathbf{\Delta}^1$ is W-aspheric.

[page 2]

Loc (3) (*localization,* weak form). Let $f : X \longrightarrow S$ be an arrow in Cat which is W*-aspheric, i.e.* such that $\forall\ s \in \mathrm{Ob}\,S$, the localized category X/s over S at s, $X/s = X \times_S S/s$, is W-aspheric. Then $f \in W$.[1]

p. 2

Recall 1. Loc (1, 2, 3) imply that we have

$$W = W^\circ \qquad \textit{i.e. } f \in W \iff f^\circ \in W,$$

where we denote by $X \longmapsto X^\circ$, $f \longmapsto f^\circ$ the functor of passage to the opposite category, functor

$$X \longmapsto X^\circ : \mathrm{Cat} \xrightarrow{\sim} \mathrm{Cat}$$

which is an *automorphism* of Cat (thus commutes with all $\varinjlim$ and $\varprojlim$). It follows that the set of axioms Loc (1, 2, 3) is equivalent to Loc (1, 2^{opp}, 3^{opp}), where Loc (2^{opp}) is the "dual" axiom of Loc (2), namely that every object of Cat having an initial object is W-aspheric, and likewise Loc (3^{opp}) the axiom that every $f : X \longrightarrow S$ which is W-*coaspheric, i.e.* such that the colocalized categories $s\backslash X$ of X over S are W-aspheric, is in W.

Definition 1.1. By a *basic localizer* we mean a subset W of Fl(Cat) satisfying the axioms Loc (1, 2, 3) (*i.e.* mild saturation, final object axiom, localization axiom).

Recall 1.2. If W is a basic localizer, then every *homotopism* $f : X \longrightarrow Y$ of Cat is in W (it is essentially for this reason that we imposed (c) in mild saturation Loc (1)). If $f, g : X \rightrightarrows Y$ are homotopic (*i.e.* belong to the same connected component of $\underline{\mathrm{Hom}}(X, Y)$), then $\mathrm{hot}_W(f) = \mathrm{hot}_W(g)$ and $f \in W \iff g \in W$. [page 3]

Here is the strong form of Loc (3).

Loc (3 bis) (*Localization,* strong form). Let S be in Cat and $f : X \longrightarrow Y$ a morphism in Cat$/S$ such that "f is a W-equivalence locally over S", *i.e.* for every $s \in \mathrm{Ob}\, S$, the induced morphism

$$f/s : X/s \longrightarrow Y/s$$

for the localized categories of X, Y over S at s, is in W. Then $f \in W$.

The axiom Loc (3) is nothing but the special case of Loc (3 bis) corresponding to the case where $Y = S$.

Proposition 1.3. *Let $W \subset$ Fl(Cat). Then we have the following equivalences*

$$\text{Loc (1, 2, 3)} \iff \text{Loc (1, } 2_0\text{, } 3')$$
$$\text{Loc (1, 2, 3 bis)} \iff \text{Loc (1, } 2_0\text{, } 3' \text{ bis)},$$

where the axioms Loc ($3'$) *and* Loc ($3'$ bis) *are formulated as follows:*

Loc ($3'$) ⓐ (Axiom of Cat-fibrations, weak direct form). Let S be in Cat, X a split fibered category over S, such that the fibers X_s ($s \in \mathrm{Ob}\, S$) are W-aspheric. Then $X \longrightarrow S$ is in W. [page 4]

ⓑ Let X be in Cat$/S$, then $i_X : X \longrightarrow \Phi_S(X)$ is in W[2].

Loc ($3'$ bis) ⓐ (Axiom of Cat-fibrations, strong direct form). Let S be in Cat, X, Y two split fibered categories over S, $f : X \longrightarrow Y$ a cartesian S-functor compatible with the splittings, such that $\forall\ s \in \mathrm{Ob}\, S$, the induced functor on the fibers p. 3

$$f_s : X_s \longrightarrow Y_s$$

is in W. Then $f \in W$.

ⓑ As in Loc ($3'$) (b).

[1] In a final version, one should interchange the notations for Loc (3) and for Loc (3 bis), since the latter appears to me to be the more important of the two - it is satisfied for all the localizers that have been used (to my knowledge), and is valid for every localizer of the form $W_{\mathbf{D}}$, where $\mathbf{D}$ is a derivator (or just a prederivator satisfying Der (4,5)).

[2] (b) concerns the *comparison* of an arbitrary X over S, with the split fibered category $\Phi_S(X)$ over S.

N.B. In fact, assuming Loc (1, 2), the statements Loc (3′), Loc (3′ bis) imply the similar statements without splittings, and without even assuming f cartesian in Loc (3′ bis), *cf.* proof below.

PROOF. *a*) Loc (1, 2, 3) $\Longrightarrow$ Loc (1, 2_0, 3′). It suffices to prove Loc (3′). Let us prove Loc (3′ *a*). For this, under the conditions of Loc (3′ *a*), let us prove that $X \longrightarrow S$ is W-coaspheric, *i.e.* that the categories

$$s\backslash X = X \times_S s\backslash S$$

are W-aspheric. It will indeed follow from this that $f \in W$ by virtue of Loc (3) applied to f° and of $W = W^\circ$. But let us consider the canonical inclusion [page 5]

$$i_s : X_s \hookrightarrow s\backslash X,$$

it follows from the fact that $p : X \longrightarrow S$ is Cat-fibrant that this functor is W-aspheric (it even has a right adjoint, *i.e.* the categories X_s/u, where $u \in \mathrm{Ob}\,(s\backslash X)$, have a final object). Thus by Loc (3) we have $i_s \in W$, and since X_s is W-aspheric by hypothesis, likewise $s\backslash X$, q.e.d.

Let us prove Loc (3′ *b*): We note that $i_X : X \longrightarrow \Phi_S(X)$ is W-*coaspheric* (it even has a left adjoint), hence i_X° is W-aspheric, hence in W, hence $i_X \in W$ since $W = W^\circ$.

a bis) Loc (1, 2, 3 bis) $\Longrightarrow$ Loc (1, 2_0, 3′ bis). This time we have to prove only Loc (3′ bis). For (*b*), this is already done. Remains to prove (*a*). If f is as in the statement Loc (3′ bis *a*), let us prove that for every s in S, the induced morphism

$$s\backslash f : s\backslash X \longrightarrow s\backslash Y$$

is in W. Since $W = W^\circ$, we will therefore have $(s\backslash f)^\circ \in W$, and applying Loc (3 bis) to f°, we find $f^\circ \in W$, hence $f \in W$, and we will be done. But $s\backslash f$ fits into the commutative square [page 6]

$$\begin{array}{ccc} X_s & \xrightarrow{f_s} & Y_s \\ {\scriptstyle i_s^X}\downarrow & & \downarrow{\scriptstyle i_s^Y} \\ s\backslash X & \xrightarrow{s\backslash f} & s\backslash Y, \end{array}$$

where, as we saw in (*a*), i_s^X and i_s^Y are in W. By hypothesis we have $f_s \in W$, hence p. 4
$s\backslash f \in W$, q.e.d.

b) Loc (1, 2_0, 3′) $\Longrightarrow$ Loc (1, 2, 3). Let us first prove Loc (2). Applying Loc (3′ *a*) to the projection

$$\mathbf{\Delta}^1 \times X \longrightarrow X \qquad (\text{where } X \in \mathrm{Ob}\,\mathrm{Cat}),$$

which is split fibered with fibers isomorphic to $\mathbf{\Delta}^1$, hence W-aspheric by Loc (2_0), we see that this projection is in W. This allows us to apply the standard homotopy arguments (*cf.* XII, p. 1 …) to conclude that a homotopism is in W, hence a contractible category is W-aspheric, a fortiori a category with a final object (or with an initial object) is W-aspheric, *i.e.* we have Loc (2).

Let us prove Loc (3). Since Loc (1, 2, 3) $\Longleftrightarrow$ Loc (1, 2°, 3°), it suffices to prove Loc (3°), since we have proved Loc (2°). Let

$$f : X \longrightarrow S \qquad W\text{-}co\text{aspheric},$$

let us prove $f \in W$. Let us introduce the *fibered* category over S whose fibers are the $s\backslash X$, namely $\Phi_S(X)$, and consider the canonical inclusion

$$i_X : X \longrightarrow \Phi_S(X). \qquad [\text{page 7}]$$

By Loc (3′ *b*) it is in W. On the other hand, the fibers $s\backslash X$ of $\Phi_S(X)$ are W-aspheric by hypothesis, hence by Loc (3′ *a*) $\Phi_S(X) \longrightarrow S$ is in W. Since $i_X \in W$, it follows that $(f : X \longrightarrow S) \in W$, q.e.d.

b bis) Loc (1, 2_0, 3′ bis) $\Longrightarrow$ Loc (1, 2, 3 bis). We have already proved Loc (1, 2, 3) (since Loc (3′ bis) contains Loc (3′) as a special case). Let us therefore prove Loc (3 bis) assuming all that already. It will again suffice to prove it in dual form, since

$$\text{Loc (1, 2, 3 bis)} \iff \text{Loc}(1, 2^\circ, 3 \text{ bis}^\circ),$$

and Loc (2°) is known from Loc (1, 2, 3) $\iff$ Loc (1, 2°, 3°). It thus remains to prove that if [page 8]

$$f : X \longrightarrow Y \qquad \text{in Cat}/S$$

is such that

$$s\backslash f : s\backslash X \longrightarrow s\backslash Y$$

is in W for every $s \in \text{Ob}\, S$, then $f \in W$. Let us then consider the commutative square

$$\begin{array}{ccc} X & \xrightarrow{i_X} & \Phi_S(X) \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \Phi_S(f)} \\ Y & \xrightarrow{i_Y} & \Phi_S(Y), \end{array}$$

where $i_X, i_Y \in W$ by Loc (3′ bis *b*). The hypothesis on f means that $\Phi_S(f)$ induces W-equivalences on the fibers, hence by Loc (3′ bis) $\Phi_S(f) \in W$, hence $f \in W$ by Loc (1), q.e.d.

p. 5

2 Axioms Loc (4) and Loc (5) (of direct sums, of connectedness)[3]

Loc (4) (Axiom of *direct sums*). For a family $(f_i)_{i\in I}$ of arrows in Cat, if $f = \coprod f_i$, we have $f \in W$ if and only if $f_i \in W$ for every $i \in I$.

This axiom splits into two, into the "if" and the "only if", which I am going to formulate in the most economical form:

Proposition 2.1. *The axiom* Loc (4) *of sums is equivalent to the conjunction of the following two axioms:*

Loc (4 a) Let $(f_i)_{i\in I}$ be a family of arrows in Cat, if the $f_i \in W$, then $f = \coprod f_i \in W$. [page 9]

Loc (4 b) Let S be in Cat, and $S_0 \hookrightarrow S$ a direct summand (subcategory both open and closed). Let X be in Cat/S. If $X \longrightarrow S$ is in W, then so is $X_0 = X|S_0 \longrightarrow S_0$. (In other words, the inclusion $S_0 \hookrightarrow S$ is a "W-fibration", *i.e.* is in Fib_W.)

Corollary 2.2. *Consider the variant of* Loc (4 a) *obtained by restricting to finite families. (It is equivalent to* Loc (4 a) *when W is stable under* $\varinjlim$ *that are filtering,* cf. *axiom* Loc (8) *below). This variant is equivalent to*

Loc (4 a_0) Every object X of Cat is "W-cofibrant", *i.e.* $\emptyset \longrightarrow X$ is in Cof_W, *i.e.* for every $f : Y \longrightarrow Y'$ which is in W, the corresponding morphism $\text{id}_X \amalg f : X \amalg Y \longrightarrow X \amalg Y'$ is in W.

[3] We are dealing here with axioms that one can regard as being of a more or less "trivial" nature. In all cases known to me, they are indeed trivially satisfied.

Reminder 2.3. In any case, if W is a basic localizer (which we shall assume from now on, unless otherwise stated), the *finite* analogue of Loc (4 a) for products is true, *i.e.* if we have a *finite* family of arrows $(f_i)_{i\in I}$ in Cat, if the f_i are in W, then $f = \prod f_i \in W$. The analogue for an infinite family is satisfied for practically no basic localizer.

[page 10]

Proposition 2.4. *Let W be a basic localizer. Then finite products exist in* Hot_W*, and the functor* $\mathrm{hot}_W : \mathrm{Cat} \longrightarrow \mathrm{Hot}_W \overset{\mathrm{def}}{=} (\mathrm{Cat})W^{-1}$ *commutes with them. If W satisfies* Loc (4 a) (resp. Loc (4 a_0)), *then arbitrary sums* (resp. *finite*) *exist in* Hot_W*, and the functor* hot_W *commutes with them. Moreover, in this case sums are distributive with respect to products (since this is so in* Cat).

The proof follows simply from the general result:

Lemma 2.5. *Let $(\mathcal{M}, W)$ be a category of models (W any subset of* $\mathrm{Fl}\,\mathcal{M}$*). Suppose that products* (resp. *finite products,* resp. *binary products*) *exist in $\mathcal{M}$, and that moreover for every family $(f_i)_{i\in I}$ of arrows of $\mathcal{M}$* (resp. *every finite family,* resp. *binary family*) *$f_i \in W$ for every $i \in I$ implies $f = \prod f_i \in W$. Then products of the type envisaged exist in* $\mathrm{Ho}_W = \mathcal{M}W^{-1}$*, and the functor* $\mathrm{ho}_W : \mathcal{M} \longrightarrow \mathcal{M}W^{-1}$ *commutes with them. Finally, if $\mathcal{M}$ admits a final object e, then e is also a final object of* Ho_W.

p. 6

[page 11]

We deduce from this the dual statements concerning direct sums and initial objects.

The proof is a special case of the following very useful result, applied to the pair of adjoint functors

$$\mathcal{M} \underset{\substack{\text{products}\\ \text{in } \mathcal{M}\\ \text{indexed by } I}}{\overset{\text{diag}}{\rightleftarrows}} \mathcal{M}^I \qquad :$$

Lemma 2.6[(4)]. *Let $(\mathcal{M}, W)$, $(\mathcal{M}', W')$ be two categories of models, and*

$$\mathcal{M} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \mathcal{M}'$$

a pair of adjoint functors compatible with the localizers, i.e. *such that*

$$\varphi(W) \subset W', \qquad \psi(W') \subset W.$$

Then the localized functors

$$\mathcal{M}W^{-1} \underset{\bar\psi}{\overset{\bar\varphi}{\rightleftarrows}} \mathcal{M}'W'^{-1}$$

are also adjoint, the adjunction arrows for $\bar\varphi\bar\psi$ and $\bar\psi\bar\varphi$ being deduced from those for $\varphi\psi$ and $\psi\varphi$ by composition with the localization functors $\mathcal{M} \longrightarrow \mathcal{M}W^{-1}$, $\mathcal{M}' \longrightarrow \mathcal{M}'W'^{-1}$.

[page 12]

Loc (5) (Axiom of *connectedness* or of *non-triviality*).

Let $f \in W$, then $\pi_0(f)$ is bijective.

This property of W is in constant use in practice, when we work with W instead of restricting ourselves to constructing W's and playing around with the axioms and their stability under passage to $\underline{\mathrm{Hom}}(I, M)$'s (where here $\mathcal{M} = \mathrm{Cat}$). On the other hand, we can also regard it as an axiom of *non-triviality* for Hot_W. I recall in this connection:

Proposition 2.7. *Let W be a basic localizer. Equivalent conditions:*

a) *W satisfies the connectedness axiom* Loc (5).

p. 7

b) *The functor* $\pi_0 : \mathrm{Cat} \longrightarrow \mathrm{Ens}$ *factors through a functor*

$$\mathrm{Cat} \dashrightarrow \mathrm{Hot}_W \xrightarrow{\pi_0} \mathrm{Ens}.$$
$$\pi_0$$

c) *The category* $\mathrm{Hot}_W^* = \mathrm{Hot}_W \backslash \{\emptyset\}$ *is not equivalent to the one-point category.*

c′) *The category* Hot_W *is equivalent neither to* e *nor to* $\mathbf{\Delta}^1$.[5]

For the proof, and for other equivalent forms of the connectedness axiom, I refer to XV 9.12 (p. 133 bis). For the name "connectedness axiom", I can offer the following explanation 2.8 and 2.9. [page 13]

Corollary 2.8. *If the connectedness axiom* Loc (5) *is satisfied, then the* W*-aspheric objects are* 0*-connected* (*tautology*), *and the converse is true if* W *is strongly saturated.* (*Or also, if* W *satisfies the axiom of fibrations* Loc (7) *below, since by* XII, th. 2, p. 20, *in this case,* Loc (5) *is equivalent to* $W \neq \mathrm{Fl}(\mathcal{M})$ *…*).

For if the connectedness axiom is not satisfied, then for every nonempty X in Cat, $X \longrightarrow e$ is an isomorphism, hence (W being strongly saturated) $X \longrightarrow e$ is in W, *i.e.* X is W-aspheric, hence there are W-aspheric objects that are not 0-connected. Let us also note:

Corollary 2.9. *Suppose that* W *satisfies the axiom* Loc (4) *of direct sums* (*or only* Loc (4 b)). *For* W *to satisfy the connectedness axiom, it is necessary and sufficient that for every arrow* $f : X \longrightarrow Y$ *with* Y 0*-connected, if* f *is in* W*, then* X *is* 0*-connected.* (*This is necessary of course without assuming* Loc (4 b) *or anything at all.*)

2.7. Let us now consider the composite functor [page 14]

$$\mathrm{Ens} \xrightarrow{I \longmapsto I_{\mathrm{Cat}}} \mathrm{Cat} \xrightarrow{\mathrm{hot}_W} \mathrm{Hot}_W, \qquad I \longmapsto I_{\mathrm{Hot}_W} \text{ or } I_W, \tag{2.7.1}$$

where the first functor is the one that assigns to the set I the discrete category I_{Cat} having I as its set of objects. (Most often, we shall identify I_{Cat} with I in the notation, just as for an ordered set J, one identifies J and the "ordered category" defined by J. Here I can be regarded as equipped with the discrete order.) We have, moreover

$$I_{\mathrm{Cat}} \simeq I \times e \quad (\overset{\mathrm{def}}{=} \text{sum of } I \text{ copies of } e), \tag{2.7.2}$$

so when Loc (4 a) is satisfied, so hot_W commutes with direct sums, we shall have a canonical isomorphism

$$I_W \simeq I \times e_W \quad \begin{array}{l}(\text{where } e_W \text{ is the final object of } \mathrm{Hot}_W,\\ \textit{i.e. } e \text{ regarded as an object of } \mathrm{Hot}_W).\end{array} \tag{2.7.3}$$

Proposition 2.8. *Suppose the connectedness axiom* Loc (5) *is satisfied, whence a functor* p. 8

$$\pi_0 : \mathrm{Hot}_W \longrightarrow \mathrm{Ens}. \tag{2.8.1}$$

This functor is left adjoint *to the functor* $I \longmapsto I_W$ (2.7.1), i.e. *we have a bijection functorial in* (X, I) *in* $\mathrm{Hot}_W \times \mathrm{Ens}$:

[4] *cf.* XII, lemma on page 248. [page 15]

[5] **NB** If W is strongly saturated (condition Loc (1 ter)), then the conditions opposite (*i.e.* Loc (5)) are equivalent to this:

d) W *is equal neither to* $Fl(\mathrm{Cat})$ *in its entirety, nor to* $Fl(\mathrm{Cat}^*) \cup \{\mathrm{id}_\emptyset\}$ (i.e. $Fl(\mathrm{Cat}) \backslash W$ *is neither empty, nor equal to the set of arrows* $\emptyset \longrightarrow X$ *with* $X \neq \emptyset$).

If W satisfies Loc (7 D_{W_ω}), then W satisfies Loc (5) if and only if $W \neq \mathrm{Fl}(\mathrm{Cat})$ (*cf.* XII, th. 2, p. 20).

$$\mathrm{Hom}_{\mathrm{Ens}}(\pi_0(X), I) \simeq \mathrm{Hom}_W(X, I_W). \tag{2.8.2}$$

The functor π_0 is also right adjoint *to the* same *functor $I \longmapsto I_W$,* i.e. *we have a functorial bijection*

$$\mathrm{Hom}_{\mathrm{Ens}}(I, \pi_0(X)) \simeq \mathrm{Hom}_W(I_W, X). \tag{2.8.3}$$

Taking $I = e$ (one-point set), we find the:

Corollary 2.9. *We have the isomorphism functorial in X in* Hot_W

$$\boxed{\mathrm{Hom}_W(e_W, X) \simeq \pi_0(X).} \tag{2.8.4}$$

Comment 2.10. The adjunction formula (2.8.2) is, moreover, familiar, whenever we have a category $\mathcal{H}$ (such as Cat, or Hot_W *etc.*) equipped with a functor π_0, $\pi_0(X)$ corresponding to the intuition of a decomposition of an object into "connected components". (I am not going to spell out here the general conditions under which one defines such a functor.) On the other hand, the adjunction in the reverse direction of these same two functors is a phenomenon quite special to categories of the type Hot_W. If we take into account the description of I_W as $I \times e_W$, the unusual adjunction formula (2.8.3) is equivalent, of course, to formula (2.8.4) - it is deduced from it by "raising to the power I". We can reverse the steam, and in a general category $\mathcal{H}$, *define* a functor $\pi_0 : \mathcal{H} \longrightarrow \mathrm{Ens}$ by formula (2.8.4), [page 16]

$$\pi_0(X) \overset{\mathrm{def}}{=} \mathrm{Hom}(e_{\mathcal{H}}, X) \qquad (= \Gamma_{\mathcal{H}}(X), \text{ with a familiar notation}),$$

where $e_{\mathcal{H}}$ is a final object of $\mathcal{H}$. We shall therefore have, putting $I_{\mathcal{H}} = I \times e_{\mathcal{H}}$, the formula

$$\mathrm{Hom}(I_{\mathcal{H}}, X) \xrightarrow{\sim} \mathrm{Hom}(I, \pi_0(X))$$

(*i.e.* formula (2.8.3)), saying that we have a pair $(I_{\mathcal{H}}, \pi_0)$ of adjoint functors. Suppose on the other hand that for every set I, the canonical adjunction map

$$I \longrightarrow \pi_0(I_{\mathcal{H}}) = \mathrm{Hom}_{\mathcal{H}}(e_{\mathcal{H}}, e_{\mathcal{H}} \times I)$$

is *bijective*[6], *i.e.* that $I \longmapsto I_{\mathcal{H}}$ is *fully faithful.* We then define, for every I, a canonical map

p. 9

$$\begin{array}{ccc} \mathrm{Hom}(X, I_{\mathcal{H}}) & \longrightarrow & \mathrm{Hom}(\pi_0(X), I) \\ & {\scriptstyle \pi_0^{X, I_{\mathcal{H}}}} \searrow \quad \swarrow {\scriptstyle \sim} & \\ & \mathrm{Hom}(\pi_0(X), \pi_0(I_{\mathcal{H}})). & \end{array}$$

The "bi-adjunction" of the functors π_0 and $I \longmapsto I_{\mathcal{H}}$ amounts to saying that this arrow is bijective, or again, that the arrow $\pi_0^{X, I_{\mathcal{H}}}$: [17]

$$\mathrm{Hom}(X, I_{\mathcal{H}}) \longrightarrow \mathrm{Hom}(\Gamma_{\mathcal{H}}(X), \Gamma_{\mathcal{H}}(I_{\mathcal{H}}) \simeq I)$$

is bijective.

Corollary 2.11. *The functor $I \longmapsto I_W$ from* Ens *to* Hot_W *is fully faithful,* i.e. *the adjunction homomorphism*

$$\pi_0(I_W) \longrightarrow I$$

is bijective. (*Or again, π_0 is a localization functor.*)

[6] that tends to be ultrafalse if $\mathcal{H}$ is additive. On the other hand, true if $\mathcal{H}$ is a category with sums distributive with respect to products (or even universal), whose final object is *connected* …

Indeed, this bijectivity is obvious when one works in Cat, and follows from this at once in Hot_W.

Proof of 2.8. The adjunction formula (2.8.2) follows from the corresponding formula for Cat, obvious, and from lemma 2.6 on adjoint functors between categories of models. The adjunction in the other direction reduces to formula (2.8.4). We have canonical arrows

$$\mathrm{Hom}_W(e_W, X) \underset{\beta}{\overset{\alpha}{\rightleftarrows}} \pi_0(X),$$

where α is $\pi_0^{e_W,X}$, given that $\pi_0(e_W) = \pi_0(e_{\mathrm{Cat}}) = e_{\mathrm{Ens}}$, and $\mathrm{Hom}(e_{\mathrm{Ens}}, \pi_0(X)) \simeq \pi_0(X)$. The arrow β is the composite arrow

$$\mathrm{Ob}\, X \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Cat}}(e_{\mathrm{Cat}}, X) \xrightarrow{\mathrm{hot}_W} \mathrm{Hom}_W(e_W, X)$$

noting that this arrow passes to the quotient as $\pi_0(X) \longrightarrow \mathrm{Hom}_W(e_W, X)$, since two [18]
homotopic maps from e_{Cat} to X are equal in Hot_W. I say that

$$\beta\alpha = \mathrm{id}\ , \qquad\qquad \alpha\beta = \mathrm{id}\ .$$

The relation $\alpha\beta = \mathrm{id}$ amounts to saying that the arrow $i_x : e_{\mathrm{Cat}} \longrightarrow X$ defined by $x \in \mathrm{Ob}\, X$, gives $\pi_0(i_x) : e_{\mathrm{Ens}} \longrightarrow \pi_0(X)$ whose value is the class of x in $\pi_0(X)$, this is tautological. To prove that $\beta\alpha = \mathrm{id}$, it then suffices to prove that β is surjective, *i.e.* the

Corollary 2.12. *For every X in* Cat, *the map*

$$\underbrace{\mathrm{Hom}_{\mathrm{Cat}}(e_{\mathrm{Cat}}, X)}_{\simeq \mathrm{Ob}\, X} \longrightarrow \mathrm{Hom}_W(e_W, X)$$

is surjective (*assuming the connectedness axiom* Loc (5)). p. 10

Proof of 2.12. If $X = \emptyset$, the assertion amounts to saying that $\mathrm{Hom}_W(e_W, \emptyset) = \emptyset$, but more generally, we have the

Corollary 2.13. (Assuming Loc (5).) *Let X be nonempty in* Cat, *then* $\mathrm{Hom}_W(X, \emptyset) = \emptyset$. (*Thus $\emptyset$ is an initial object that is* strict *in* Hot_W.)

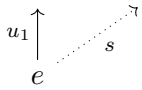
Indeed, if we had $X \longrightarrow \emptyset$ in Hot_W, we would deduce $\pi_0(X) \longrightarrow \emptyset$, or $\pi_0(X) = \emptyset$, [page 13']
absurd!

There remains the case where X is nonempty. This follows from the general fact :

Lemma 2.14. *Let $\mathcal{M}$ be a category having an initial object that is* strict (*if* $\mathrm{Hom}_{\mathcal{M}}(X, \emptyset_{\mathcal{M}}) \neq \emptyset$, *we have $X = \emptyset_{\mathcal{M}}$*), *a final object e, and such that for every $X \neq \emptyset_{\mathcal{M}}$, $X \longrightarrow e$ has a section,* i.e. $\mathrm{Hom}(e, X) \neq \emptyset$. *Let $W \subset \mathrm{Fl}\, \mathcal{M}$*

1°) *not containing any arrows $\emptyset_{\mathcal{M}} \longrightarrow X$ with $X \neq \emptyset_{\mathcal{M}}$, and*

2°) *such that for every diagram* $X_1 \xleftarrow{u_2} X_2$ *with* $u_2 \in W$ $\exists$ s in $\mathcal{M}$ *such that*



$u_2 s = u_1$

in $\mathcal{M}W^{-1}$.

Then

a) *$\emptyset_{\mathcal{M}}$ is an initial object that is* strict *in $\mathcal{M}W^{-1}$.*

b) *For every $X \neq \emptyset_{\mathcal{M}}$, $\mathrm{Hom}_{\mathcal{M}}(e, X) \longrightarrow \mathrm{Hom}_W(e, X)$ is surjective.*

PROOF. That $\emptyset_{\mathcal{M}}$ is an initial object of $\mathcal{M}W^{-1}$ has been seen (last assertion of lemma 2.5, dual form), let us prove that it is strict, *i.e.* that if $\mathrm{Hom}_W(X, \emptyset_{\mathcal{M}}) \neq \emptyset$, we have $X = \emptyset_{\mathcal{M}}$. Indeed, if we had $u : X \longrightarrow \emptyset_{\mathcal{M}}$ in $\mathcal{M}W^{-1}$, u would be represented by

$$X = X_0 \xleftarrow{W} X_1 \longrightarrow X_2 \xleftarrow{W} \cdots X_{2n-2} \longrightarrow X_{2n-1} \xleftarrow{W} X_{2n} = \emptyset_{\mathcal{M}}.$$

We proceed by induction on the half-length of this chain to see that $X = \emptyset_{\mathcal{M}}$. Indeed, the hypothesis on W implies that $X_{2n-1} = \emptyset_{\mathcal{M}}$, then the one that $\emptyset_{\mathcal{M}}$ is strict implies $X_{2n-2} = \emptyset_{\mathcal{M}}$, whence the conclusion.

Let us prove the surjectivity of $\mathrm{Hom}_{\mathcal{M}}(e, X) \longrightarrow \mathrm{Hom}_W(e, X)$ for $X \neq \emptyset_{\mathcal{M}}$, by showing that every reduced path of length 2n+1 is "homotopic" to a reduced path of length $2n - 1$: [page 14']

$$e = X_0 \longrightarrow X_1 \xleftarrow{W} X_2 \longrightarrow X_3 \longleftarrow \cdots \longleftarrow X_{2n} \longrightarrow X_{2n+1} = X.$$

Indeed, we can replace the section

$$e = X_0 \xrightarrow[u_1]{} X_1 \xleftarrow[u_2]{W} X_2 \xrightarrow[u_3]{} X_3 \xleftarrow{W} X_4 \qquad (s : e \to X_2)$$

by

$$e \xrightarrow[u_3 s]{} X_3 \xleftarrow{W} \cdots, \qquad \text{p. 11}$$

where s is a section of X_2 over e such that

$$u_2 s = u_1 \qquad in \qquad \mathcal{M}W^{-1}\ .$$

This therefore proves by induction on n that the path is homotopic to a path of length 1, $e \longrightarrow X$, q.e.d.

We must therefore, in the present case, check 1°) (this follows from Loc (5)), and 2°)

$$\begin{array}{ccc} X_1 & \xleftarrow{u_2 \in W} & X_2 \\ {\scriptstyle u_1}\uparrow & \nearrow_{s} & \\ e & & \end{array} \quad .$$

But since $u_2 \in W$, we have $\pi_0(u_2)$ an isomorphism, hence surjective, hence $\exists\ s$ such that $u_2 s$ is homotopic to u_1 in Cat, which implies $u_2 s = u_1$ in Hot_W, q.e.d.

Final comment. The connectedness axiom Loc (5) differs from all the other axioms Loc (1) to Loc (8) (but with the exception of Loc (8 bis) on accessibility) in that it is not a stability axiom. The other axioms Loc (1) to Loc (8) (and also Loc (8 bis) on accessibility and even the later axioms of exactness [7]) are satisfied if $W = \mathrm{Fl}\,\mathcal{M}$ - the connectedness axiom is therefore the only one that stands in the way of the *triviality* of the homotopy theory defined by W. This is to say that it cannot be a consequence of any of the others. On the other hand, the axiom Loc (4) of direct sums is not independent of the others. Let us point out in this connection (reminder) : [page 15']

Loc (4 a_0)	is a consequence of the cocartesian square axiom Loc (6 a) (direct form).
Loc (4 a)	is a consequence of the preceding Loc (6 a) + the axiom of filtering direct limits Loc (8).
Loc (4 b)	is a consequence of the fibration axiom Loc (7). Thus
Loc (4)	is a consequence of Loc (6 a, 7, 8).

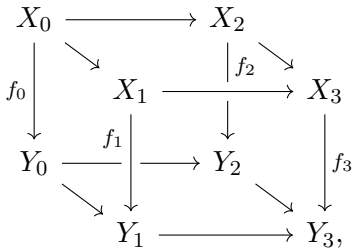
p. 12

[7] there will in fact be no need for them, thanks to the providential theory of K. S. BROWN! [page 16']

3 Axiom Loc (6) of the cocartesian square (or of "Mayer-Vietoris", or of W-cofibrations) [8]

I see several variants, and am not yet totally clear in my mind about the mutual implications. I begin with the more complicated variant, but which we will need in any case :

Loc 6 (Axiom of W-cofibrations, or 5-lemma, or axiom of homomorphisms of cocartesian squares). Let us consider in Cat a commutative cubical diagram



a homomorphism $f = (f_0, f_1, f_2, f_3)$ from a square diagram X to another Y. We suppose that X comes from an object X_3 of Cat from two opens X_1, X_2 of X_3 *covering* X_3, and from their intersection $X_0 = X_1 \cap X_2$, and likewise for Y. (Let us call these diagrams in Cat MV diagrams ("*open*") - "*Mayer-Vietoris*".) We then have the following [9] :

a) $f_0, f_1, f_2 \in W \Longrightarrow f_3 \in W$ [10].

b) $f_1, f_2, f_3 \in W \Longrightarrow f_0 \in W$.

c) $f_0, f_3 \in W \Longrightarrow f_1, f_2 \in W$ [11].

[12].

[13].

N.B. I distinguish these conditions as Loc (6 a), Loc (6 b), Loc (6 c). I presume that we will never have to make use of Loc (6 b), without also assuming Loc (6 a), but the reverse is surely not the case. Thus, $W_0 = \{f \in \mathrm{Fl}\,\mathrm{Cat} \mid \pi_0(f) \text{ bijective}\}$ satisfies a), but not b). I presume (without having checked) that it is the same for the W_n, $n \in \mathbf{N}$, of "*n-homotopy equivalence*".

p. 13
[page 17']

It could be that Loc (6 a) and Loc (6 a, b) are equivalent to the following weaker, and simpler, form. (If so, it is in the simplest form, of course, that they should be stated.)

[8] It is the name "*axiom of W-cofibrations*" that strikes me at present as the best, – it is this name that seems to me best to capture the conceptual and not technical meaning of the axiom.

[9] It is good to introduce properties of the diagram

1°) $f_0 \in W$.

2°) f_1 and f_2 in W.

3°) f_3 in W.

and to formulate parts a) b) c) of Loc (6) as follows

a) $1° + 2° \Longrightarrow 3°$.

b) $2° + 3° \Longrightarrow 1°$.

c) $1° + 3° \Longrightarrow 2°$.

[10] N.B. it seems that in Loc (6 a), the statement suffices on assuming f_0, f_2 to be *isomorphisms, cf.* p. 22.

[11] One should not include c in Loc 6 *cf.* p. 26.

[12] N.B. We will see below (3.3) that Loc 6 c already implies Loc (6 a, b), *i.e.* all of Loc (6). And if W satisfies Loc (1 bis), *i.e.* stable under direct factors, then Loc (6 a, b) $\Longrightarrow$ Loc (6 c).

[13] I am going to denote by Loc (6) the axiom Loc (6 a, b), without including c in it.

Loc 6′ (Axiom of cocartesian squares, or of Mayer-Vietoris). Let

$$\begin{array}{ccc} X_0 & \xrightarrow{i_2} & X_2 \\ {\scriptstyle i_1}\searrow & & \searrow{\scriptstyle j_1} \\ & X_1 & \xrightarrow{j_2} X_3 \end{array}$$

be an MV diagram in Cat. Then

a) If $i_1 \in W$, $j_1 \in W$.

b) If $j_1 \in W$, $i_1 \in W$.

Proposition 3.1. [page 18']

$$\begin{array}{lcl} \text{Loc (6 a)} & \Longrightarrow & \text{Loc (6′ a)}, \\ \text{Loc (6 b)} & \Longrightarrow & \text{Loc (6′ b)}, \\ \text{Loc (6 c)} & \Longrightarrow & \text{Loc (6′ a, b)} \ ^{(14)}. \end{array}$$

This is seen on the cubical diagram

(Q)

$$\begin{array}{ccccc} X_0 & \xrightarrow{i_2} & X_2 & & \\ {\scriptstyle i_1}\searrow & & \uparrow{\scriptstyle j_2} \ \searrow{\scriptstyle j_1} & & \\ {\scriptstyle f_0=\mathrm{id}}\uparrow & X_1 & \xrightarrow{j_2} & X_3 & \\ & \uparrow{\scriptstyle f_1=i_1} & & \downarrow{\scriptstyle f_2=\mathrm{id}} & \\ X_0 & \xrightarrow{\mathrm{id}} & X_2 & \xrightarrow{\mathrm{id}} & \uparrow{\scriptstyle f_3=j_1} \\ & \searrow{\scriptstyle i_2} & & \searrow & \\ & & X_0 & \xrightarrow{i_2} & X_2, \end{array}$$

where the f_0, f_1, f_2, f_3 of the statement of Loc (6) become id_{X_0}, i_1, id_{X_2}, j_1. Loc (6 a) therefore gives $i_1 \in W \Longrightarrow j_1 \in W$, *i.e.* Loc (6′ a), and Loc (6 c) $j_1 \in W \Longrightarrow i_1 \in W$, *i.e.* Loc (6′ b). For Loc (6 b) $\Longrightarrow$ Loc (6′ b), we use (*cf.* XII, p. 78) the cube

p. 14

(Q′)

$$\begin{array}{ccccc} X_0 & \xrightarrow{i_2} & X_2 & & \\ \downarrow{\scriptstyle i_1} & \searrow{\scriptstyle i_1} & \downarrow{\scriptstyle j_1} & \searrow{\scriptstyle j_1} & \\ & X_1 & \xrightarrow{j_2} & X_3 & \\ X_1 & \xrightarrow{\mathrm{id}} & \downarrow & \downarrow & \\ & \searrow{\scriptstyle \mathrm{id}} & X_1 & \xrightarrow{j_2} X_3 & \downarrow{\scriptstyle \mathrm{id}} \\ & & \searrow{\scriptstyle j_2} & \searrow{\scriptstyle \mathrm{id}} & \\ & & X_1 & \xrightarrow{j_2} & X_3, \end{array}$$

where $f_0 = i_1$, $f_1 = \mathrm{id}_{X_1}$, $f_2 = j_1$, $f_3 = \mathrm{id}_{X_3}$, hence

$$j_1 \in W \Longrightarrow i_1 \in W, \quad \textit{i.e.}\ \text{Loc (6 b)} \Longrightarrow \text{Loc (6′ b)},$$

and if we have Loc (6 c), we see that $i_1 \in W \Longrightarrow j_1 \in W$, *i.e.* Loc (6′ a).

Comment 3.2. Loc (6 a) appears to me as the "vital minimum" for establishing the existence of a W-homotopic theory of cofibers in Cat. For this, one must indeed have at hand a serviceable criterion ensuring that an arrow $i : A \longrightarrow B$ in Cat is in Cof_W.

We find, when we have Loc (6 a) (in dual form, *i.e.* for the *closed ones,* it's all the same), that the *open Dwyer immersions* are in Cof_W, which suffices to have *a good* [page 19]

[14] N.B. we even have Loc (6 c) $\Longrightarrow \underbrace{\text{Loc 6 (a, b)}}_{\text{Loc (6)}}$.

theory of W-homotopic cofibers, in which MV squares are W-cocartesian (something we absolutely insist on).

As for Loc (6 b), if I have Loc (6′ b) + Loc (6 a) instead, I should manage to prove Loc (6). In XII Appendix (pages 265 ff.) I prove, it seems, that (with the present notation) Loc (6′) implies Loc (6), and even Loc (6 c) if W is stable under direct factors (condition Loc (1 bis)). But the proof is incomplete, since I tacitly assume that the Dwyer immersions are in Cof_W, something I had only established using axiom W (7), in fact using Loc (6 a) (which is part of W (7)).

But here the idea comes to me of the lemma of K. S. Brown, cited by Baues, which [page 20] I am going to give again here:

Lemma 3.2.1. (K. S. Brown) *Let $(\mathcal{M}, W)$ be a localizer (W mildly saturated), equipped with a set $C \subset \mathrm{Fl}\,\mathcal{M}$ satisfying the conditions:*

1) *C stable under composition and under cobase change* (N.B. *one has to assume the existence of the amalgamated sums* $\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1 \\ {\scriptstyle i_2}\searrow & & \searrow \\ & X_2 & \longrightarrow X_3 \end{array}$ *only if $i_1 \in C$*), *contains the isomorphisms.*

2) *$\mathcal{M}$ contains an initial object, $\emptyset_{\mathcal{M}}$, and for every object X of $\mathcal{M}$, $\emptyset_{\mathcal{M}} \longrightarrow X$ is in C.* (N.B. (1) *and* (2) *imply that $\mathcal{M}$ is stable under finite sums.*)

3) *Every arrow f of $\mathcal{M}$ factors as pi, with $i \in C$, $p \in W$.*

4) *$C \cap W$ is stable under cobase change,* i.e. *if*

$$\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1 \\ {\scriptstyle i_2}\searrow & & \searrow \\ & X_2 & \xrightarrow[j_1]{} X_3 \end{array} \qquad (*)$$

is cocartesian and $i_1 \in C \cap W$, then $j_1 \in C \cap W$. (N.B. *$j_1 \in C$ is already contained in* (1), *it remains to require $j_1 \in W$.*) p. 15

Under these conditions, we have $C \subset \mathrm{Cof}_W$.

One can also state the lemma as follows

Corollary 3.2.2. *Suppose that C satisfies* (1), (2), (3). *Then the following conditions on C are equivalent.* [page 21]

a) *Condition* (4) *above,* i.e. *stability of $C \cap W$ under cobase change.*

a′) $C \cap W \subset W^{\mathrm{univ}}$.

b) $C \subset \mathrm{Cof}_W$.

b′) *For every cocartesian square*

$$\begin{array}{ccc} X_0 & \xrightarrow{i} & X_1 \\ {\scriptstyle p}\searrow & \text{coc.} & \searrow{\scriptstyle q} \\ & X_0' \xrightarrow{j} & X_1' \end{array}$$

with $i \in C$, $p \in W$, we have $q \in W$.

PROOF. We have

$$\begin{aligned} \mathrm{b} &\iff \mathrm{b}' \\ \mathrm{a} &\iff \mathrm{a}' \end{aligned}$$

since C is stable under cobase change. We have $\mathrm{b} \Longrightarrow \mathrm{a}'$ since $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$ thanks to hypothesis 3°). We have $\mathrm{a} \Longrightarrow \mathrm{b}'$ by Brown's lemma.

We are going to apply this to the case $\mathcal{M} = \mathrm{Cat}$, $C =$ open Dwyer immersions. Conditions 1), 2) do not depend on W, they are satisfied, and 3) as well as soon as W is a basic localizer. As for 4), it is a consequence of the old W (7 bis), *i.e.* Loc (6′) (we apparently need Loc (6′ a *and* b).) We conclude $C \subset \mathrm{Cof}_W$ - it's wonderful! (But I don't know how to prove the lemma …) [page 22]

This legitimizes the proof given in XII *loc. cit.*, and we get the

Proposition 3.3. *The condition* Loc (6′) (*cocartesian square axiom, strong form* (a) *and* (b)) is equivalent *to* Loc (6 a, b) (*axiom of homomorphisms of cocartesian squares*) *and is* implied *by* Loc (6 c). *If W satisfies* Loc (1 bis), i.e. *is stable under direct factors, then* Loc (6′) is equivalent *to* Loc (6 c) *and to* Loc (6 a b c).

Comment 3.4. Thus, we get a theory of W-homotopic cofibers *either* under hypothesis Loc (6′) (equivalent to the more complicated Loc (6)), or under the weaker (and more complicated) hypothesis Loc (6 a).

In the first case, it is a theory of a very special nature, since the W-cocartesian squares are "reversible" (*i.e.* we have not only $i_1 \in W \Longrightarrow j_1 \in W$, but also the reverse implication).

There remains then the question whether Loc (6′ a) suffices for a theory of W-homotopic cofibers, or what surely amounts to the same thing, whether it already implies Loc (6 a) (hence is equivalent to it). This hangs entirely on the *question*:

does the axiom Loc (6′ a) (cocartesian square axiom, direct form) imply that the set of open Dwyer immersions which are in W (W-trivial Dwyer immersions) is stable under cobase change? In fact, we see that for this it would suffice to verify the axiom of the cube (or of homomorphisms of cocartesian squares), direct form Loc (6 a), in the case where f_0 and f_2 are identities, hence to prove that then $f_1 \in W \Longrightarrow f_3 \in W$ (which would directly give $\mathrm{Dw} \subset \mathrm{Cof}_W$). But I see no way of proving it … p. 16 [page 23] ? ‖

Suppose now that condition Loc (6 a) is satisfied, so that we have a "good theory of W-homotopic cofibers", hence also a theory of W-cocartesian squares (*cf.* XII, p. 258-262). There are two questions that arise, concerning the validity, with these weaker hypotheses, of the developments of XII:

1°) Given a commutative square

$$\begin{array}{ccc} X_0 & \longrightarrow & X_2 \\ & \searrow \quad \searrow & \\ & X_1 \longrightarrow & X_3 \end{array}$$

in Cat, is it true that this square is W-cocartesian if and only if the arrow

$$\text{canonical} \int X_0 \to \begin{matrix} X_1 \\ \nearrow \\ \\ \searrow \\ X_2 \end{matrix} \longrightarrow X_3 \text{ is in } W \text{ ?} \quad (15)$$

[page 24]

2°) Is it still true that if A is in Cat, and $u : F \hookrightarrow G$ a homomorphism in $A^\wedge$, then $u \in \mathrm{Cof}_{W_A}$, and if $u \in W$, $u \in W_A^{\mathrm{univ}}$, hence do we have

$$\mathrm{Mono}_{A^\wedge} \subset \mathrm{Cof}_{W_A}, \qquad \mathrm{Mono}_{A^\wedge} \cap W_A \subset W_A^{\mathrm{univ}} \ ? \ {}^{(16)}$$

But 1°) is proved in XII, prop. 20, p. 262 - I have just checked that the proof is valid under the present hypotheses [(17)], provided that I check that corollary 3, p. 149 in XII is valid, *i.e.*

1° bis) If $\begin{array}{ccc} X_0 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & X \end{array}$ is an MV square in Cat, then $\int X_0 \nearrow X_1 \searrow X_2 \longrightarrow X$ is in W (hence, assuming 1°), the square is W-cocartesian).

To check this, I am led to check that theorem 4 in loc. cit. (p. 146) is still valid. p. 17

I have just done so, and have annotated theorem 4 and its corollaries accordingly (some results (cor. 4 and cor. 5) using W (7 bis) in its full strength, *i.e.* Loc (6′), *i.e.* Loc (6 a *and* b). But corollary 3 of theorem 4 is O.K.

I have also just checked the validity of 2°), by checking the proof of the corollary of theorem 5 (XII, p. 168) and annotating it accordingly.

3.5. Let us sum up: we have [page 25]

$$\text{Loc (6 c)} \implies \text{Loc (6)} \iff \text{Loc (6}') \tag{3.5.1}$$

and

$$\text{If } W = W^{\natural}, \text{ we have Loc (6 c)} \iff \text{Loc (6)} \iff \text{Loc (6}'). \tag{3.5.2}$$

On the other hand

$$\text{Loc (6 a)} \implies \text{Loc (6}' \text{ a)}, \tag{3.5.3}$$

and I do not know whether the reverse implication

$$\text{Loc (6}' \text{ a)} \overset{?}{\implies} \text{Loc (6 a)}$$

is valid [(18)]. To establish Loc (6 a), it suffices in any case to do so when f_0 and f_2 are isomorphisms.

The axiom Loc (6 a) seems to be the "vital minimum" for W to give rise to a "good theory of W-homotopic cofibers", hence also of W-cocartesian squares. More precisely, I feel like proving this:

Proposition 3.6. *The axiom* Loc (6 a) *is equivalent to the conjunction of the following two properties.*

p. 18

[page 26]

Loc (6 A) Cat has "enough W-cofibrations", *i.e.* every arrow f of Cat factors as $f = pi$, with $i \in \mathrm{Cof}_W$ and $p \in W$. (This also means that Cat equipped with (W, Cof_W) is a category with cofibrations of K. S. Brown - Anderson, and all the results of Brown-Quillen theory are applicable. In particular, we have a good theory of W-cocartesian squares.)

Loc (6 B) If X in Cat is the union of two opens U, V, with intersection S, then the square

$$\begin{array}{ccc} S & \longrightarrow & U \\ \downarrow & & \downarrow \\ V & \longrightarrow & X \end{array}$$

is W-cocartesian [19].

On the other hand, condition Loc (6 A) can be replaced by the condition (apparently weaker)

Loc (6 A′) Every Dwyer morphism is in Cof_W.

Corollary 3.7. *The condition* Loc (6), i.e. Loc (6 a, b) (*or again the equivalent condition* Loc (6′)) *is equivalent to the conjunction of conditions* Loc (6 A) (*or* Loc (6 A′)) *and* Loc (6 B) *above*, *and of the condition*

Loc (6 C) For every W-cocartesian square

$$\begin{array}{ccc} X_0 & \xrightarrow{i} & X_1 \\ \downarrow & & \downarrow \\ X_2 & \xrightarrow{j} & X \end{array}$$

in Cat, $j \in W \Longrightarrow i \in W$.

On the other hand, I suspect that Loc (6 c) is hardly ever satisfied unless $W = W^{\natural}$ (the most common case of course), in which case Loc (6 c) *is equivalent* to Loc (6). There is probably no reason to include it in Loc (6), any more than $W = W^{\natural}$ in Loc (1). [page 27]

p. 19

4 Axiom Loc (7) of W-fibrations

4.1. Recall the notion of a *weak W-fibration* (XII p. 51, 52): the strongest version of the 8 equivalent conditions for $f : X \longrightarrow S$ to be a weak fibration is this.

---

[15] Better, in all cases, when we have a diagram $X_0 \nearrow X_1 \searrow X_2$, its integral $\int$ is a W-homotopic amalgamated sum of said diagram (assuming Loc (6 a) and no more, and even only Loc (6′ a)).

[16] N.B. Loc (6′ a) suffices in place of (Loc (6 a)).

[17] But it uses in addition Loc (3 bis). But with another simpler proof that I have just added (p 264), we can do without it (and we get a more complete view of the situation).

[18] This is the case in the presence of Loc (8, 9).

[19] In fact, Loc (6 B) is a consequence of Loc (6 A), which therefore is *equivalent* to Loc (6 a). (*Cf.* sketch of proof in 4.4.15, pages 66-68.)

(Ff) $\forall\ S'' \longrightarrow S'$ in Cat/S, with S', S'' *contractible,* the corresponding morphism $X'' = X \times_S S'' \longrightarrow X' = X \times_S S'$ is in W.

When W is strongly saturated, then this implies the condition, apparently stronger:

(Ff′) $\forall\ S'' \longrightarrow S'$ in Cat/S, such that $S'' \longrightarrow S'$ is a homotopism, $X'' \longrightarrow X'$ is in W.

I denote by Fibf_W the set of weak fibrations. I summarize in a diagram of inclusions the main classes of arrows in Cat that we have encountered, akin to classes of "fibrations" (for W, in a suitable sense):

[page 28]

$$\begin{array}{ccccccccc} & & & & \mathrm{Fib}_W & & & & \\ & & & & \downarrow & & & & \\ \mathrm{Parf}_\omega & \longrightarrow & \mathrm{Parf}_W & \longrightarrow & \mathrm{Fibf}_W & \longleftarrow & \underbrace{\mathrm{Catfib}^W}_{\mathcal{F}_W} & \longleftarrow & \underbrace{\mathrm{Catfib}^\omega}_{\mathcal{F}_\omega}\ , \end{array}$$

where Fib_W denotes the W-fibrations ($f : S' \longrightarrow S$ is in Fib_W if and only if the base change functor $X \longmapsto X' = X \times_S S'$ from Cat/S to Cat/S' is compatible with the localizers W_S and $W_{S'}$), Catfib the Cat-fibrations, Catfib^W (resp. Catfib^ω) the Cat-fibrations such that the base change functors are in W (resp. are in W_ω, *cf.* def. below) [20], Parf_W the W-perfect morphisms, *i.e.* W-smooth and W-proper, finally Parf_ω the W_ω-perfect morphisms, where W_ω is the smallest localizer satisfying Loc (1, 2, 3). Among the kinds of "fibrations" considered, it is only this last one, together with Catfib^ω, that is independent of W.

As such, it seems to me to play a role similar, dual, to that of Dwyer morphisms, and I presume that one can make them play a similar role. In any case, we know that every arrow f of Cat factors as pi, with $i \in W_\omega$ and $p \in \mathrm{Parf}_\omega$, hence $i \in W$ for every W satisfying Loc (1, 2, 3). We also know that Catfib^ω satisfies the strong factorization statement, f factors as $f = pi$, $i \in W_\omega$, $p \in \mathrm{Catfib}_0^\omega$. [21]

[page 29]

Let us also note that Parf_ω and Parf_W, as well as Fib_W, are stable not only under base change (this is the case for the 5 classes considered, which moreover contain all isomorphisms), but also under composition. (Moreover, Fib_W is stable under direct factors, and I presume that the same holds for Liss_W and for Prop_W, hence for Parf_W, in particular for Parf_{W_ω}, but I have not taken the trouble to check it.) On the other hand, a priori Fibf_W is not in general stable under composition, and Catfib^W, Catfib^ω have no reason either to be stable under composition [22].

p. 20

4.2. Finally, it seems more "illuminating" to introduce, instead of the well-determined basic localizer W_ω, a basic localizer

$$W_0 \subset \mathrm{Fl}(\mathrm{Cat}),$$

which will play the role of a sort of "reference localizer" for formulating certain properties of W. We will always assume that

$$W \supset W_0.$$

[page 30]

Depending on the case, W_0 may be W_ω, or some other basic localizer defined as the "smallest" satisfying given stability axioms. Or it may be $\mathbf{W}_\infty$ (the usual weak equivalences), when one is interested in localizers that contain $\mathbf{W}_\infty$ (the only ones, apparently,

[20] N.B. $\mathrm{Catfib}_0^W = \mathrm{Catfib} \cap \mathrm{Fibf}_W$, $\mathrm{Catfib}_0^\omega = \mathrm{Catfib} \cap \mathrm{Fibf}_{W_\omega}$

[21] Thus the factorization property holds for all these classes, except perhaps Fib_W.

[22] this should be made explicit by examples.

with which one has worked up to now in homotopy). It may also be W, and the statements obtained by specializing the results obtained to the case where $W = W_0$ are by no means tautological. The diagram on page 28 is then replaced by the following diagram of inclusions, more illuminating for our purpose:

$$\begin{array}{ccccc} & & \mathrm{Fib}_W & & \\ & & \downarrow & & \\ \mathrm{Parf}_W & \longrightarrow & \mathrm{Fibf}_W & \longleftarrow & \mathcal{F}_W \\ \uparrow & & \uparrow & & \uparrow \\ \mathrm{Parf}_{W_0} & \longrightarrow & \mathrm{Fibf}_{W_0} & \longleftarrow & \mathcal{F}_{W_0} \\ & & \uparrow & & \\ & & \mathrm{Fib}_{W_0}, & & \end{array} \tag{4.2.1}$$

where for every basic localizer W, I put for short

$$\mathcal{F}_W = \mathrm{Catfib}^W,$$

which thus also defines $\mathcal{F}_{W_0}$. The class Fib_W should be regarded as the focus of our attention; most of the properties of W that we are going to review here, will be expressed by an inclusion of the form [page 31]

$$\mathcal{F} \subset \mathrm{Fib}_W,$$

where $\mathcal{F}$ is one of the seven other classes of diagram (4.2.1); with the exception at most of the case $\mathcal{F} = \mathrm{Fib}_{W_0}$, *i.e.* of the inclusion $\mathrm{Fib}_{W_0} \subset \mathrm{Fib}_W$. As a property of W, this inclusion seems very weak and for that reason not of much use, except of course in the case where $\mathrm{Fib}_{W_0} \longrightarrow \mathrm{Fibf}_{W_0}$ is an equality, so that the inclusion considered is equivalent to $\mathrm{Fibf}_{W_0} \subset \mathrm{Fib}_W$ (and implies that Parf_{W_0} and $\mathcal{F}_{W_0}$ are also contained in Fib_W). This on the other hand, as we shall see, is a fairly strong and very useful property of W, implying in particular the fact that $(\mathrm{Cat}, W, \mathrm{Fib}_W)$ is a category with fibrations.

For purposes of later reference, I am going to formulate now the various main aspects of *the W-fibration axiom* Loc (7). The formulation which I had arrived at (*cf.* I, VI, XII) used the notion which I now call "*weak* W-fibrations" (the former W-fibration, now dethroned!), and in its strongest form was written as the inclusion [page 32]

p. 21

$$\mathrm{Fibf}_W \subset \mathrm{Fib}_W.$$

This notion of weak W-fibration, just like that of W-perfect morphism, however useful they may be, nevertheless seems to me a little technical, and I prefer to give a formulation that calls only on more familiar notions, more elementary in appearance. This is why I am going to replace Fibf_W by the subclass $\mathcal{F}_W = \mathrm{Catfib}^W$, which (we shall see) in fact comes to the same thing.

This class $\mathcal{F}_W$ is extremely natural, and even imposes itself in the yoga of derivators. Indeed, in the construction of $\mathrm{HOT}_W(S)$ for S in Cat, one passes to a category of fractions of the category [page 33]

$$\mathrm{Cat}(S) = \underline{\mathrm{Hom}}(S^\circ, \mathrm{Cat}),$$

by a localizer $W(S)$ (which we had generally denoted by W_S^{g} or W_S^{u} …). This category $\mathrm{Cat}(S)$ is on the other hand isomorphic to the category

$$\mathrm{Fibsc}\, S$$

of fibered and split categories over S, with as arrows the S-functors which respect the splitting (and in particular cartesian) (*cf.* VI, and especially the diagram p. 175). We have seen (*loc. cit.*) that the localized category $\mathrm{HOT}_W(S)$ is equivalent to the localized category of the very closely related category

$$\mathrm{Fib}\, S \qquad \text{(which I would now prefer to denote by } \mathrm{Catfib}\, S \text{ or } \mathrm{Catfib}_S\text{)},$$

formed by the *fibered* categories over S, with as arrows only the cartesian functors. (One should note however that the canonical functor $\mathrm{Fibsc}\, S \longrightarrow \mathrm{Fib}\, S$ is faithful but not fully, nevertheless it induces an equivalence on the localized categories, defining in the obvious way, by W-equivalence fiber by fiber, *i.e.* as W_S^{u}, the relevant localizer on $\mathrm{Fib}\, S$.) Thus for this reason of principle, and moreover throughout the development of the theory of HOT_W, the category $\mathrm{Fib}\, S$ and thereby, the fibrant functors, have played an essential role. The class of arrows which we denote here by $\mathcal{F}_W$ or Catfib^W (one could also denote it by Fib^W, were it not for the risk of confusion with Fib_W) is then interpreted very naturally starting from there. An arrow $f : X \longrightarrow S$ is in $\mathcal{F}_W$ if and only if it is a) fibrant, and thereby defines in a particularly direct way an object of $\mathrm{HOT}_W(S)$, since up to S-equivalence, X/S comes from an object of $\mathrm{Fibsc}\, S$ or again of $\mathrm{Cat}(S)$, and b) the object in $\mathrm{HOT}_W(S)$ defined by f is in $\mathrm{HOT}_W^{\mathrm{lc}}(S)$, *i.e.* in the essential image of the strictly full subcategory $\mathrm{Fibsc}_0^W S$ of $\mathrm{Fibsc}\, S$ (under the canonical functor $\mathrm{Fibsc} \longrightarrow \mathrm{HOT}_W(S)$), p. 22 or what comes to the same thing (at least if W is strongly saturated) that the functor

$$\begin{array}{ccc} S^\circ & \longrightarrow & \mathrm{Hot}_W \\ s & \longmapsto & \mathrm{hot}_W(X_s) \end{array}$$

be a "locally constant" functor. Thus, we shall regard the $f \in \mathcal{F}_W$ as the most economical and most natural arrows in Cat, which define locally constant relative W-homotopy types.

Let us also note that all the classes of arrows (4.2.1) are stable under base change (and they contain the isomorphisms). The categories Fib_W, Parf_W and their W_0 variants are also stable under composition, on the other hand, this is not the case for Fibf_W and $\mathcal{F}_W$ and their W_0 variants, except under special axioms on W or on W_0. It follows at once that if $\mathcal{F}$ is one of these classes, the condition

$$\mathcal{F} \subset \mathrm{Fib}_W$$

is equivalent to this one:

For every cartesian diagram

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow[p]{} & S', \end{array} \qquad (4.2.2)$$

with $f \in \mathcal{F}$ and $p \in W$, we have $q \in W$.

(N.B. We also have $f' \in \mathcal{F}$, by the stability recalled above.)

4.3. Here then (*cf.* below) is the relevant axiom, Loc (7) with its three aspects a, b, c [23]. We have seen (and shall see again) that we have the implications

$$\mathrm{a} \qquad \Longrightarrow \qquad \mathrm{b} \qquad \Longrightarrow \qquad \mathrm{c} \qquad (\Longrightarrow \qquad \mathrm{F}).$$

[23] I finally added a fourth F to it (for "fibration").

When we refer to Loc (7) without further specification, it will be understood that this is the strongest form Loc (7 a), which thus implies the other two. No less is needed to be completely at ease. But I recall that the weakened variant Loc (7b) is sufficient to develop the theory of the functors $f_!$ for the categories $\mathrm{HOT}_W(S)$ [24] (*cf.* VI).

[page 37]

This is moreover a surprising thing, since via the general yoga of derivators associated with categories of models, one would rather expect that to have left adjoints $f_!$ of the inverse image functors f^*, one must make hypotheses on W of type *W-cofibration* (essentially, Loc (6 a)). It seems unlikely to me that Loc (7 b), or even Loc (7), implies

p. 23

Loc (6 a), or even its weaker variant Loc (6′ a) (axiom of the cocartesian square, or of Mayer-Vietoris, weak form). If on the other hand these axioms are not *satisfied,* whereas Loc (7) is, there is a chance that this will give an example where there is a formalism of the $f_!$, without the left exactness axiom being satisfied.

As for the form Loc (7 c), weaker still, it is nevertheless sufficient (as we shall see below) to ensure that Cat has enough W-fibrations (a result which is by no means obvious,

[page 38]

and could only become apparent to me after I had learned of the substantial lemma of K. S. Brown, in his beautiful work where he introduces categories with fibrations and with cofibrations).

We will also give below some very simple cases, where the condition Loc (7 b) already implies Loc (7) (W strongly saturated, *i.e.* Loc (1 ter), and satisfying Loc (3 bis)), or even where Loc (7 c) implies Loc (7) (the case where W satisfies the direct limit axiom Loc (8)). I point out in passing that I know of no basic localizer satisfying the axiom Loc (8) (which already implies $W = W^{\natural}$, *i.e.* Loc (1 bis)), without W being strongly saturated (*i.e.* even satisfying Loc (1 ter)).

Loc (7)

a) We have $\mathcal{F}_W \subset \mathrm{Fib}_W$, *i.e.* for every cartesian diagram (4.2.2) in Cat, with $f \in \mathcal{F}_W$ (*i.e.* f Cat-fibrant and weakly W-fibrant) and $p \in W$, we have $q \in W$.

b) For every commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & S & \end{array}$$

with $p, q \in \mathcal{F}_W$ and $f \in W$, we have $f \in W_S^{\mathrm{u}}$ (or, what comes to the same thing since X, Y are W-smooth over S, $f_s \in W$ for every $s \in \mathrm{Ob}\, S$) [25].

[page 39]

c) We have

$$\mathcal{F}_W \cap W \subset W^{\mathrm{u}},$$

in other words, for every arrow $f : X \longrightarrow S$ in Cat such that $f \in \mathcal{F}_W$ (one could call these arrows Cat-W-fibrations, it being understood that they are also Cat-fibrations, but not necessarily W-fibrations), if $f \in W$, then $f \in W_S^{\mathrm{u}}$ (or, what comes to the same thing, the fibers X_s of f ($s \in \mathrm{Ob}\, S$) are W-aspheric).

p. 24

[24] Here my memory is playing a trick on me - for this one has no need at all of the already very strong axiom Loc (7b), but only of Loc (3 bis), which does seem to be satisfied for all "reasonable" basic localizers.

[25] **N.B.** The proof of $B(\mathcal{F}_W) \Longrightarrow B(\mathrm{Fibf}_W)$ in fact shows that it suffices to postulate b) when f is cartesian, and even when X, Y are split over S and f compatible with the splitting.

F) [26] Cat has sufficiently many W-fibrations, *i.e.* Cat equipped with (W, Fib_W) is a category with fibrations (proper, and even a Brown one, *i.e.* $X \longrightarrow e$ is in Fib_W for every X in Cat).

It is clear that c is the special case of b obtained by taking $Y = S$, $q = \mathrm{id}_S$, so b $\Longrightarrow$ c. The implication a $\Longrightarrow$ b is less evident, it is established first in XII, and we will return to it below (4.4.2 below). The rest of the present paragraph is devoted to studying various variants of the preceding conditions a) b) c) and the implications between them.

4.4. Let, more generally, [page 40]

$$\mathcal{F} \subset \mathrm{Fl}(\mathrm{Cat})$$

be a set of arrows in Cat, thinking more particularly of the case where $\mathcal{F}$ is one of the seven sets in diagram (4.2.1) (p. 30) other than Fib_W. We can formulate three properties for $\mathcal{F}$ similar to those formulated in Loc (7 a b c) in the case of $\mathcal{F}_W$, let us denote these properties

$$\mathrm{A}(\mathcal{F})\,,\ \mathrm{B}(\mathcal{F})\,,\ \mathrm{C}(\mathcal{F})\,,\ ^{27}$$

thus

$$\begin{cases} \mathrm{A}(\mathcal{F}) & : \quad \mathcal{F} \subset \mathrm{Fib}_W. \\ \mathrm{B}(\mathcal{F}) & : \quad \mathrm{Fl}(\mathcal{F}_S) \cap W_S \subset W_S^{\mathrm{u}} \\ & \quad \text{(where } \mathcal{F}_S \text{ denotes the full subcategory of } \mathrm{Cat}/S \\ & \quad \text{consisting of the } f : X \longrightarrow S \text{ with } f \in \mathcal{F}). \\ \mathrm{C}(\mathcal{F}) & : \quad \mathcal{F} \cap W \subset W^{\mathrm{u}}. \end{cases} \tag{4.4.1}$$

It is clear that we have $\mathrm{B}(\mathcal{F}) \Longrightarrow \mathrm{C}(\mathcal{F})$, and I am going to give a condition on $\mathcal{F}$ which implies that we also have $\mathrm{A}(\mathcal{F}) \Longrightarrow \mathrm{B}(\mathcal{F})$.

We will apply it in the following cases: [page 41]

$$\mathcal{F} \ = \ \mathcal{F}_W,\ \mathrm{Fibf}_W,\ \mathrm{Parf}_W,\ \mathcal{F}_{W_0},\ \mathrm{Fibf}_{W_0}. \tag{4.4.2}$$

I point out moreover that we will see below (cor. 4.4.10) that

$$\mathrm{A}(\mathcal{F}_W) \iff \mathrm{A}(\mathrm{Fibf}_W), \qquad \mathrm{A}(\mathcal{F}_{W_0}) \iff \mathrm{A}(\mathrm{Fibf}_{W_0}), \tag{4.4.3}$$

and likewise for the variants B, C, so that for the implication we want to establish, there is no need to distinguish the cases of $\mathcal{F}_W$ and Fibf_W, of $\mathcal{F}_{W_0}$ and Fibf_{W_0}. Moreover it will suffice to look at the sets $\mathcal{F}_{W_0}$, Fibf_{W_0}, Parf_{W_0}, since the case of the sets relative to W is a special case of these (taking $W_0 = W$). [28]

Lemma 4.4.1. [29] *Suppose that $\mathcal{F}$ satisfies the following condition: every arrow f of* Cat *factors as $f = pi$, with $i \in W$, $p \in \mathcal{F}$. Then the condition* $\mathrm{A}(\mathcal{F})$ *on $\mathcal{F}$ implies that in* Cat *there are "enough W-fibrations", and consequently* (*by a formal argument well known*), *that we have* $\mathrm{B}(\mathrm{Fib}_W)$, *and a fortiori* (*since by hypothesis $\mathcal{F} \subset \mathrm{Fib}_W$*) $\mathrm{B}(\mathcal{F})$. p. 25

N.B. Indeed it is clear that if we have $\mathcal{F} \subset \mathcal{F}'$, then [page 42]

$$\mathrm{A}(\mathcal{F}') \Longrightarrow \mathrm{A}(\mathcal{F}) \qquad \text{and likewise for B, C.} \tag{$*$}$$

[26] added after the fact.

[27] **N.B.** These conditions make sense if $\mathcal{F} \subset \mathrm{Fl}(\mathcal{M})$, if $(\mathcal{M}, W)$ is an arbitrary category of models.

[28] This will thus make a priori $3 \times 6 = 18$ different conditions, but because of (4.4.3) there are (up to equivalence) only $4 \times 3 = 12$, and in fact two among these four triples of conditions reduce to a single one (for $\mathcal{F} = \mathrm{Parf}_W$ and $\mathcal{F} = \mathrm{Parf}_{W_0}$), so that in all only *eight* different conditions remain, which we summarize in the summary diagram (4.4.7.1) on page 48. There are even only *four,* A_{W_0}, B_{W_0}, C_{W_0}, D_{W_0}, as functions of W_0, the other four A, B, C, D are obtained by specializing to the case where $W_0 = W$.

[29] Holds in an arbitrary category of models.

Now the preliminary condition on $\mathcal{F}$ is indeed satisfied in the case of a set of one of the forms $\mathcal{F} = \mathcal{F}_W$, Fibf_W, Parf_W. It suffices to see this in the case of the subsets $\mathcal{F}_W$, Parf_W of Fibf_W, *i.e.* that every arrow f of Cat factors as

$$f = pi,$$

with $i \in W$, and p being, as desired, in Parf_W, or in $\mathcal{F}_W$. Now in the case of Parf_W, this is what is established in VII (Categories of paths (2)) thanks to the formalism of the ‖
$\underline{\mathrm{Ch}}(S) \longrightarrow S \times S$. To find a factorization with $f = qj$, $j \in W$, $q \in \mathcal{F}_W$, it suffices to see that p factors in this way as $p = q_0 j_0$, we will have

$$f = \underbrace{q_0}_{\in \mathcal{F}_W}\ \underbrace{(j_0 i)}_{\in W},$$

so we can assume $f \in \mathrm{Parf}_W$. But if X is W-perfect over S, hence a weak W-fibered [page 43]
object over S, then in the factorization

$$\begin{array}{ccc} X & \xrightarrow{\ i\ } & \Phi_S(X) \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle p} & \\ & S & \end{array}$$

with $i \in W$ and $p \in \mathrm{Catfib}$ (*cf.* VI), we know that p is also a weak W-fibration, *i.e.* $p \in \mathcal{F}_W$, whence the conclusion. Thus we find the

Corollary 4.4.2. *Suppose that $\mathcal{F}$ is one of the six sets* (4.4.2) (i.e. *one of the sets in diagram* (4.2.1), *excluding Fib_W and Fib_{W_0}). Then we have*

$$\mathrm{A}(\mathcal{F}) \implies \mathrm{B}(\mathcal{F}) \implies \mathrm{C}(\mathcal{F})\,.$$

Lemma 4.4.3. *Let $(\mathcal{M}, W)$ be a category of models, and let $\mathcal{F} \subset \mathrm{Fl}(\mathcal{M})$ satisfy the following conditions*

1) *$\mathcal{F}$ stable under base change, composition, contains the isomorphisms.*

2) *$\mathcal{M}$ has a final object, and for every $X \in \mathrm{Ob}\,\mathcal{M}$, $X \longrightarrow e$ is in $\mathcal{F}$.*

3) *Every arrow f of $\mathcal{M}$ factors as $f = pi$, with $i \in W$, $p \in \mathcal{F}$ (this is the condition of lemma* 4.4.1).

Then the conditions $\mathrm{A}(\mathcal{F})$, $\mathrm{B}(\mathcal{F})$, $\mathrm{C}(\mathcal{F})$ *on $\mathcal{F}$ are equivalent.* p. 26

Indeed, by 4.4.1, it remains to prove the implication

$$\mathrm{C}(\mathcal{F}) \implies \mathrm{A}(\mathcal{F}),$$

i.e. this: if $\mathcal{F}$ also satisfies the condition [page 44]

4) $\mathcal{F} \cap W \subset W^{\mathrm{u}}$ (or what amounts to the same thing, since $\mathcal{F}$ is stable under base change, that $\mathcal{F} \cap W$ is stable under base change),

then we conclude from this

$$\mathcal{F} \subset \mathrm{Fib}_W.$$

Now this is none other than the lemma of K. S. Brown, in the form dual to the one given above (3.2.1, page 20).

Among the six sets of arrows considered in (4.4.2), the hypotheses 1) 2) 3) are satisfied for Parf_W and Parf_{W_0}, to the exclusion of the other four. (The others satisfy all the conditions, with the sole exception of stability under composition [30]). We thus find:

Corollary 4.4.4. *The conditions* A(Parf_W), B(Parf_W), C(Parf_W) *are equivalent* (we denote this condition on W by D). *Likewise the conditions* A(Parf_{W_0}), B(Parf_{W_0}), C(Parf_{W_0}) *are equivalent.*

(We will denote the condition obtained on the pair $W \supset W_0$ of basic localizers by D_{W_0}.) [page 45]

Thus we have (since $\mathrm{Parf}_{W_0} \subset \mathrm{Parf}_W$)

$$\mathrm{D} = \mathrm{D}_W \implies \mathrm{D}_{W_0}\,, \tag{4.4.4}$$

and if $W_0' \subset W_0$, we will have

$$\mathrm{D}_{W_0} \implies \mathrm{D}_{W_0'}\,. \tag{4.4.5}$$

Corollary 4.4.5. *Let us denote by* A, B, C *respectively the conditions* Loc (7 a), Loc (7 b), Loc (7 c). *Then we have the implications*

$$\mathrm{A} \implies \mathrm{B} \implies \mathrm{C}\,. \tag{4.4.5.1}$$

Moreover, if we denote (*for a set $\mathcal{F}$ of arrows of* Cat) *by* $\mathrm{Comp}(\mathcal{F})$ *the condition on $\mathcal{F}$ that $\mathcal{F}$ be stable under composition, we have the equivalences*

$$\mathrm{A} \iff (\mathrm{C} + \mathrm{Comp}(\mathcal{F}_W)) \iff (\mathrm{C} + \mathrm{Comp}(\mathrm{Fibf}_W)). \tag{4.4.5.2}$$

Proof. The first assertion is contained in corollary 4.4.2, applied to the case $\mathcal{F} = \mathcal{F}_W$. On the other hand, if $\mathcal{F}_W$ satisfies $\mathrm{Comp}(\mathcal{F}_W)$, it satisfies conditions 1) 2) 3) of Brown's lemma 4.4.3, so combined with C, it implies A. It remains, for the first equivalence, to prove that p. 27

$$A \implies \mathrm{Comp}(\mathcal{F}_W).$$

But by definition we have [page 46]

$$\mathcal{F}_W = \mathrm{Catfib} \cap \mathrm{Fibf}_W \tag{$*$}$$

and we know that Catfib is stable under composition. So $\mathcal{F}_W$ is too, provided that the composite gf of two arrows in $\mathcal{F}_W$ is in Fibf_W. Now by hypothesis A, which means $\mathcal{F}_W \subset \mathrm{Fib}_W$, we have $f, g \in \mathrm{Fib}_W$, hence $gf \in \mathrm{Fib}_W$, since Fib_W is stable under composition, a fortiori $gf \in \mathrm{Fibf}_W$.

On the other hand, the preceding formula ($*$) shows that $\mathrm{Comp}(\mathrm{Fibf}_W) \implies \mathrm{Comp}(\mathcal{F}_W)$, so we get for free the implication $\Longleftarrow$ in the second equivalence (4.4.5.2). The implication in the opposite direction then amounts to

$$\mathrm{A} \implies \mathrm{Comp}(\mathrm{Fibf}_W),$$

which will be evident thanks to the result (given below) [page 47]

$$\mathrm{A} = \mathrm{A}(\mathcal{F}_W) \iff \mathrm{A}(\mathrm{Fibf}_W) \qquad \text{and likewise for B, C,}$$

which thus means that

$$\mathcal{F}_W \subset \mathrm{Fib}_W \qquad \text{implies} \qquad \mathrm{Fibf}_W \subset \mathrm{Fib}_W \qquad (\text{so } \mathrm{Fibf}_W = \mathrm{Fib}_W),$$

[30] **N.B.** For Fib_{W_0}, it is the conditions 1) 2) that hold, and on the other hand, condition 3) that is a problem. Let us denote it by $\mathrm{F}(W_0)$. If it is satisfied (*i.e.* if Cat has enough W_0-fibrations), then we also have equivalence of the conditions A(Fib_{W_0}), B(Fib_{W_0}), C(Fib_{W_0}).

now Fib_W being stable under composition, the same is true of Fibf_W, q.e.d.

Corollary 4.4.6. *Suppose that $\mathcal{F}_{W_0}$ is stable under composition, which is the case in particular (by* 4.4.5.2 *applied to W_0 instead of W) if W_0 satisfies* Loc (7). *Then the conditions* $\mathrm{A}(\mathcal{F}_{W_0})$, $\mathrm{B}(\mathcal{F}_{W_0})$, $\mathrm{C}(\mathcal{F}_{W_0})$ *(which we also denote simply by* A_{W_0}, B_{W_0}, C_{W_0}*) are equivalent. (**N.B.** If we do not make the preceding hypothesis on W_0, we will only have* $\mathrm{A}_{W_0} \Longrightarrow \mathrm{B}_{W_0} \Longrightarrow \mathrm{C}_{W_0}$, *by* 4.4.2.)

4.4.7. We have now gone through all the conditions of "fibration" type on W, or on the pair (W, W_0), which have seemed useful to me. I am going to lay them out here in a (more or less) summary diagram:

[page 48]
p. 28

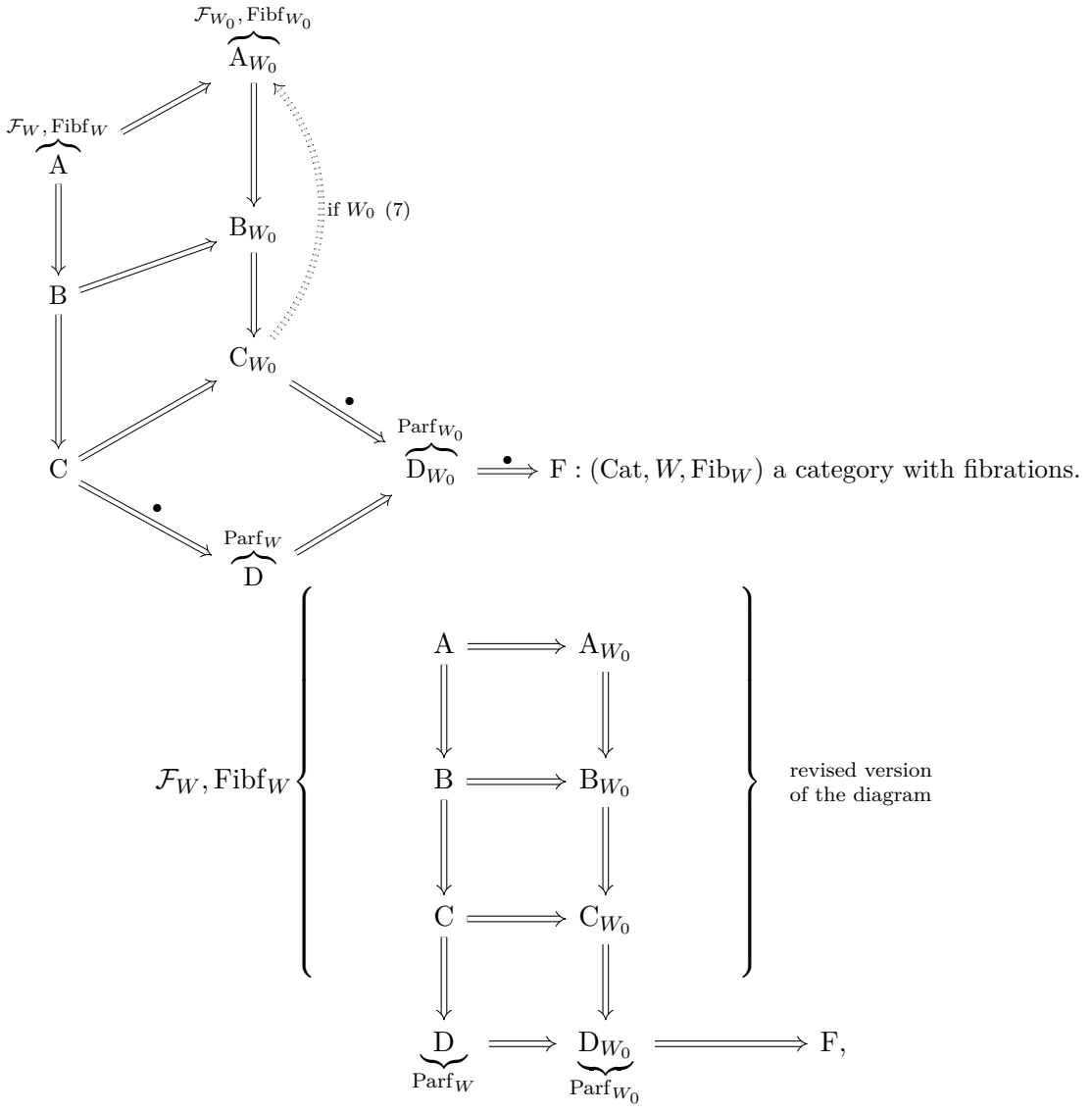


diagram which I am going to comment on. All the solid-line implications in the diagram have already been obtained or are tautological, except for the three $\overset{\bullet}{\Longrightarrow}$ to which I will return. The dotted implication $\mathrm{C}_{W_0} \dashrightarrow \mathrm{A}_{W_0}$ is a conditional implication, subject to the condition marked on the diagram as W_0 (7), which means: W_0 satisfies Loc (7). This implication, *i.e.* the equivalence of A_{W_0}, B_{W_0}, C_{W_0} under the condition W_0 (7), was seen in 4.4.6. I have marked, above the two groups (A, B, C) and (A_{W_0}, B_{W_0}, C_{W_0}) as well as above D and D′ (which each correspond to such a group of conditions $\mathrm{A}(\mathcal{F})$, $\mathrm{B}(\mathcal{F})$, $\mathrm{C}(\mathcal{F})$ all equivalent, with $\mathcal{F} = \mathrm{Parf}_W$ or $\mathcal{F} = \mathrm{Parf}_{W_0}$) the sets of arrows [page 49]

which give rise to them respectively: it will be noted that there are two for the two triples named first, A, B, C being definable via $\mathcal{F}_W$ or Fibf_W indifferently, and likewise A_{W_0}, B_{W_0}, C_{W_0} via $\mathcal{F}_{W_0}$ or Fibf_{W_0}, as already stated (4.4.3) and as we will see below (cor. 4.4.10, pages 54-55). On the other hand, the reduced triples D and D_{W_0} are each defined by just one $\mathcal{F}$, namely by Parf_W and

Parf_{W_0} respectively. (In all, this brings in six distinct sets $\mathcal{F}$, namely the six sets of diagram (4.2.1) other than Fib_W and Fib_{W_0}.) p. 29

The non-tautological implication $\mathrm{C} \overset{\bullet}{\Longrightarrow} \mathrm{D}$ is a special case of the implication $\mathrm{C}_{W_0} \overset{\bullet}{\Longrightarrow} \mathrm{D}_{W_0}$, which is contained in what has just been announced, namely $\mathrm{C}_{W_0}(\mathcal{F}_{W_0}) \Longrightarrow \mathrm{C}_{W_0}(\mathrm{Fibf}_{W_0})$, taking into account that $\mathrm{Parf}_{W_0} \subset \mathrm{Fibf}_{W_0}$ (*cf.* (∗), p. 42). Taking $W_0 = W$, we thus find the implication $\mathrm{A} \Longrightarrow \mathrm{D}$, which is however contained in the other implications of the diagram, as a composite in the chain of implications [page 50]

$$\mathrm{A} \Longrightarrow \mathrm{B} \Longrightarrow \mathrm{C} \Longrightarrow \mathrm{D} \Longrightarrow \mathrm{D}_{W_0} \overset{\bullet}{\Longrightarrow} \mathrm{F}.$$

It remains, in these general comments, to say a word about this last implication

$$\mathrm{D}_{W_0} \overset{\bullet}{\Longrightarrow} \mathrm{F},$$

i.e. about the fact that the weakest of all the conditions A to D_{W_0} considered here already implies that Cat has enough W-fibrations, *i.e.* is a category with fibrations for (W, Fib_W). This is clear when we take D_{W_0} in the form $\mathrm{A}(\mathrm{Parf}_{W_0})$, *i.e.*

$$\mathrm{Parf}_{W_0} \subset \mathrm{Fib}_W.$$

As every arrow f of Cat factors as $f = pi$, with

$$i \in W_0 \subset W \quad \text{and} \quad p \in \mathrm{Parf}_{W_0} \subset \mathrm{Fib}_W\,,$$

we find the factorization with $i \in W$, $p \in \mathrm{Fib}_W$.

4.4.8. To finish establishing the implications of the diagram p. 48, it only remains to establish the equivalences [page 51]

$$\begin{cases} \mathrm{A}(\mathcal{F}_{W_0}) & \Longleftrightarrow & \mathrm{A}(\mathrm{Fibf}_{W_0}) \\ \mathrm{B}(\mathcal{F}_{W_0}) & \Longleftrightarrow & \mathrm{B}(\mathrm{Fibf}_{W_0}) \\ \mathrm{C}(\mathcal{F}_{W_0}) & \Longleftrightarrow & \mathrm{C}(\mathrm{Fibf}_{W_0}), \end{cases} \tag{4.4.8.1}$$

which will give the similar equivalences for W, on taking $W_0 = W$.

The first equivalence (4.4.8.1) will be a consequence of the following (± tautological) lemma:

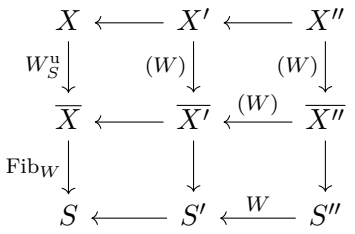
Lemma 4.4.9. *Let $(\mathcal{M}, W)$ be a category of models, $\mathcal{F}, \mathcal{G} \subset \mathrm{Fl}(\mathcal{M})$, such that every $f \in \mathcal{F}$, $f : X \longrightarrow S$, can be written $f = pi$, with $i \in W_S^u$, $p \in \mathcal{G}$. Then* $\mathrm{A}(\mathcal{G}) \Longrightarrow \mathrm{A}(\mathcal{F})$, i.e.

$$\mathcal{G} \subset \mathrm{Fib}_W \implies \mathcal{F} \subset \mathrm{Fib}_W, \tag{4.4.9.1}$$

and likewise $\mathrm{C}(\mathcal{G}) \Longrightarrow \mathrm{C}(\mathcal{F})$, i.e.

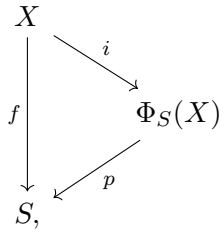
$$\mathcal{G} \cap W \subset W^u \implies \mathcal{F} \cap W \subset W^u. \tag{4.4.9.2}$$

Proof. A) Indeed, if $\mathcal{G} \subset \mathrm{Fib}_W$, the factorization $f = pi$ of f is such that $p \in \mathrm{Fib}_W$. Now a composite pi, with $p \in \mathrm{Fib}_W$ and $i \in W_S^u$, is in Fib_W, as one verifies at once on the diagram of cartesian squares p. 30



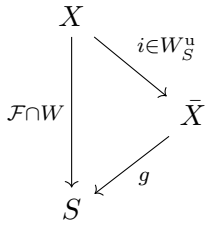
on which one reads that $X'' \longrightarrow X'$ is in W, q.e.d.

This lemma applies in particular to the case where $\mathcal{F} = \mathrm{Fibf}_{W_0}$, $\mathcal{G} = \mathcal{F}_{W_0}$, for f factors as [page 52]



and we know that $i \in W_{0S}^{\mathrm{u}} \subset W_S^{\mathrm{u}}$, and $p \in \mathcal{F}_{W_0} \subset \mathcal{F}_W$, which are the desired conditions.

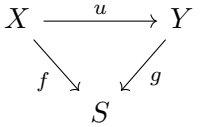
C) Similar proof: Let $f \in \mathcal{F} \cap W$, factoring as $f = gi$, with $i \in W_S^{\mathrm{u}}$, $g \in \mathcal{G}$, we will then have $g \in W$, hence $g \in \mathcal{G} \cap W \subset W^{\mathrm{u}}$ (by hypothesis C($\mathcal{G}$)), a fortiori we have $g \in W_S^{\mathrm{u}}$, hence $f \in W_S^{\mathrm{u}}$ as a composite of two arrows of W_S^{u}, and as the target of f is S itself, this means $f \in W^{\mathrm{u}}$, q.e.d.



I do not see a way of also proving

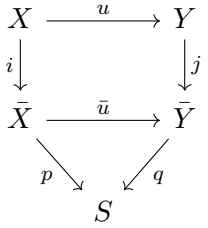
$$\mathrm{B}(\mathcal{G}) \qquad \Longrightarrow \qquad \mathrm{B}(\mathcal{F}),$$

with only the hypotheses of the preceding lemma on the pair $\mathcal{F}, \mathcal{G}$. It would be necessary that for every S in $\mathcal{M}$, and every arrow $u : X \longrightarrow Y$ in $\mathcal{M}/S$, we could insert the diagram obtained in $\mathcal{M}$ [page 53]



into a commutative diagram below

p. 31



(4.4.9.3)

with

$$f = pi, \qquad g = qj, \qquad i, j \in W_S^{\mathrm{u}}, \qquad p, q \in \mathcal{G}.$$

This will in any case be so if for every S, and a variable object X of $\mathcal{M}/S$, we can find a factorization *functorial in* X of its structural morphism f_X as $p_X i_X$, with $i_X \in W_S^{\mathrm{u}}$ and $p_X \in \mathcal{G}$, *i.e.* if i, p are obtained from a functor

$$\varphi : \mathcal{F}_S \longrightarrow \mathcal{M}/S \text{ [31]},$$

and a morphism of functors

$$i : \underbrace{\mathrm{inc}_{\mathcal{F}_S}}_{\substack{\text{inclusion}\\ \text{functor}}} \longrightarrow \varphi.$$

Assuming the existence of a diagram having the stated properties, for every arrow u in $\mathcal{F}_S$, or only every arrow u which is also in W_S, we can now prove that $u \in W_S \Longrightarrow u \in W_S^{\mathrm{u}}$, provided that the similar property is true for $\mathcal{G}$ in place of $\mathcal{F}$.

Indeed, the hypothesis on $u \in W$ implies $\bar{u} \in W$, and the hypothesis on $\mathcal{G}$ then implies $\bar{u} \in W_S^{\mathrm{u}}$. As W_S^{u} is a localizer in $\mathcal{M}/S$, we do conclude $u \in W_S^{\mathrm{u}}$, q.e.d. [page 54]

Thus we have obtained

Corollary 4.4.10. *Suppose, under the conditions of lemma* 4.4.9, *that every triangular commutative diagram* $\begin{array}{ccc} X & \xrightarrow{u} & Y \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle g} & \\ & S & \end{array}$ *can be inserted into a diagram of type* (4.4.9.3) *as above* (*with the explicit relations on the arrows occurring in it*). *Then*

$$\mathrm{B}(\mathcal{G}) \implies \mathrm{B}(\mathcal{F}).$$

Thus if moreover $\mathcal{G} \subset \mathcal{F}$, *so that* $\mathrm{A}(\mathcal{F}) \Longrightarrow \mathrm{A}(\mathcal{G})$ *tautologically, and likewise for* B *and for* C, *we find*

$$\begin{aligned} \mathrm{A}(\mathcal{F}) &\iff \mathrm{A}(\mathcal{G}) \\ \mathrm{B}(\mathcal{F}) &\iff \mathrm{B}(\mathcal{G}) \\ \mathrm{C}(\mathcal{F}) &\iff \mathrm{C}(\mathcal{G}). \end{aligned}$$

We can apply this corollary in the case that concerns us here, with

$$\mathcal{G} \overset{\text{def}}{=} \mathcal{F}_{W_0} \subset \mathcal{F} \overset{\text{def}}{=} \mathrm{Fibf}_{W_0},$$

seeing that in this case we do have a *functorial* factorization of an $f \in \mathrm{Fibf}_{W_0}$, $f : X \longrightarrow S$, as $f = pi$, $i \in W_{0S}^{\mathrm{u}} \subset W_S^{\mathrm{u}}$, $p \in \mathcal{F}_{W_0} = \mathcal{G}$, namely the one recalled on page 43: [page 55] p. 32

$$\begin{array}{ccc} X & \xrightarrow{i_X} & \bar{X} = \Phi_S(X) \\ & {\scriptstyle f=f_X}\searrow \quad \swarrow{\scriptstyle p_X} & \\ & S. & \end{array}$$

This thus proves the announced equivalences (4.4.8.1) and finishes establishing the validity of the implications of the diagram on page 48.

These implications, with the sole exception of the dotted implication $\mathrm{C}_{W_0} \dashrightarrow \mathrm{A}_{W_0}$ (which assumes that we have W_0 (7)), make no appeal to any particular hypothesis on the basic localizers W and W_0 (apart from the inclusion $W \supset W_0$). But we can, using lemma 4.4.10, give remarkable conditional implications, under the sole condition that W, or W_0, satisfy the innocuous *axiom of limits* Loc (8), and even only in the weak form of stability of localizers under $\varinjlim$ of *sequences*. Let us first give them in technical form, before unfolding the consequences in a theorem-scholie: [page 56]

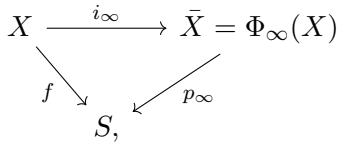
[31] $\mathcal{F}_S$ is the full subcategory of $\mathcal{M}/S$ consisting of the X such that $f_X \in \mathcal{F}$.

Corollary 4.4.11. *Suppose that W_0 satisfies axiom* Loc (8), *or only* Loc (8 bis) (*stability under* $\varinjlim$ *of sequences*). *Then* $\mathrm{D}_{W_0} \Longrightarrow \mathrm{A}_{W_0}$ (*hence the conditions* A_{W_0}, B_{W_0}, C_{W_0}, D_{W_0} *are equivalent*). *In particular* (*taking* $W_0 = W$), *if W satisfies* Loc (8 bis), *then we have* $\mathrm{D}_W \Longrightarrow \mathrm{A}_W$ (*hence the conditions* A_W, B_W, C_W, D_W *are equivalent*).

Indeed, the implication to be proved can also be written

$$\mathrm{A}(\mathrm{Parf}_{W_0}) \iff \mathrm{A}(\mathrm{Fibf}_{W_0}).$$

The non-tautological implication $\Longrightarrow$ is covered by lemma 4.4.9. We must check that assuming Loc (8 bis) for W_0, every $f \in \mathrm{Fibf}_{W_0}$, $f : X \longrightarrow S$, factors as $f = pi$, with $i \in W_{0S}^{\mathrm{u}}$, $p \in \mathrm{Parf}_{W_0}$. Now we know[32] (VI, p. 193 ff.) that axiom Loc (8 bis) implies [page 57] that $f \in \mathrm{Fibf}_{W_0}$ factors as



(category of paths infinite on one side, relativized over S, *i.e.* of pairs (x, c), with $x \in \mathrm{Ob}X$ and $c : f(x) \longrightarrow s$ an infinite path in S, with source $f(x)$.)

with $p_\infty \in \mathrm{Parf}_{W_0}$ (as we want), and $i_\infty \in W_{0S}^{\mathrm{u}}$, and as $W_{0S}^{\mathrm{u}} \subset W_S^{\mathrm{u}}$, we do have $i_\infty \in W_S^{\mathrm{u}}$.

Corollary 4.4.12. *Suppose that W* and *W_0 satisfy* Loc (8 bis). *Then we have* $\mathrm{D}_{W_0} \Longrightarrow \mathrm{A}$ (*hence the eight conditions* A, B, C, D, A_{W_0}, B_{W_0}, C_{W_0}, D_{W_0} *are equivalent*).

This means indeed that p. 33

$$\mathrm{A}(\mathrm{Parf}_{W_0}) \Longrightarrow \mathrm{A}(\mathrm{Fibf}_W),$$

and this is again covered by lemma 4.4.9. Indeed, the above factorization of an arrow f does not depend on the choice of a localizer, and p_∞ is W-perfect for *every* localizer satisfying Loc (8 bis), hence in particular for W_0. And the arrow i_∞ is in W_S^{u}, since the arrow f under consideration is in Fibf_W.

I am going to recapitulate the results obtained in this section in a theorem, which [page 58] will constitute the main result of the present paragraph on the axioms of W-fibrations.

Theorem 4.4.13. *Let W be a basic localizer in* Cat. *For every localizer W_0 contained in W, we have made explicit a set of four conditions on the pair W_0, W, which for W fixed, depend on W_0 alone, denoted*

$$\mathrm{A}_{W_0},\ \mathrm{B}_{W_0},\ \mathrm{C}_{W_0},\ \mathrm{D}_{W_0}, \tag{4.4.13.1}$$

and defined respectively as

$$\mathrm{A}(\mathcal{F}_{W_0}),\ \mathrm{B}(\mathcal{F}_{W_0}),\ \mathrm{C}(\mathcal{F}_{W_0}), \qquad \textit{and}$$

$$\mathrm{A}(\mathrm{Parf}_{W_0})\Big(\iff \mathrm{B}(\mathrm{Parf}_{W_0}) \iff \mathrm{C}(\mathrm{Parf}_{W_0})\Big),$$

with the notations A(·), B(·), C(·) *introduced in* (4.4.1). *Taking $W_0 = W$, we find the conditions on W*

$$\mathrm{A},\ \mathrm{B},\ \mathrm{C},\ \mathrm{D} \tag{4.4.13.2}$$

with $\mathrm{A} = \mathrm{A}_W$, $\mathrm{B} = \mathrm{B}_W$, $\mathrm{C} = \mathrm{C}_W$, $\mathrm{D} = \mathrm{D}_W$, *of which the first three are none other than* Loc (7 a), Loc (7 b), Loc (7 c) (cf. 4.3, *page* 38).

① *For these conditions we have the following implications.*

[page 59]

a) *We have the implications*

$$\mathrm{A}_{W_0} \implies \mathrm{B}_{W_0} \implies \mathrm{C}_{W_0} \implies \mathrm{D}_{W_0}, \tag{4.4.13.3}$$

(*whence in particular*

$$\mathrm{A}_{W} \implies \mathrm{B}_{W} \implies \mathrm{C}_{W} \implies \mathrm{D}_{W}\ ). \tag{4.4.13.4}$$

b) *If* $W'_0 \subset W_0$ *is a basic localizer contained in* W_0, *we have* (*tautologically*)

$$\mathrm{A}_{W_0} \implies \mathrm{A}_{W'_0}, \qquad \mathrm{B}_{W_0} \implies \mathrm{B}_{W'_0}, \qquad \mathrm{C}_{W_0} \implies \mathrm{C}_{W'_0}.$$

We thus find, by (a) *and* (b) (*taking there for the pair* (W, W_0) *as* (W_0, W'_0)),

the rectangle of implications p. 34

$$\mathcal{F}_W, \mathrm{Fibf}_W \left\{ \begin{array}{ccc} \mathrm{A} & \Longrightarrow & \mathrm{A}_{W_0} \\ \Downarrow & & \Downarrow \\ \mathrm{B} & \Longrightarrow & \mathrm{B}_{W_0} \\ \Downarrow & & \Downarrow \\ \mathrm{C} & \Longrightarrow & \mathrm{C}_{W_0} \\ \Downarrow & & \Downarrow \\ \underbrace{\mathrm{D}}_{\mathrm{Parf}_W} & \Longrightarrow & \underbrace{\mathrm{D}_{W_0}}_{\mathrm{Parf}_{W_0}} \end{array} \right\} \mathcal{F}_{W_0}, \mathrm{Fibf}_{W_0}, \tag{4.4.13.5}$$

where we have recalled, next to the braces, what the set, or the sets $\mathcal{F} \subset \mathrm{Fl}(\mathrm{Cat})$ *are which define the conditions considered (as one of the conditions* $\mathrm{A}(\mathcal{F})$, $\mathrm{B}(\mathcal{F})$, $\mathrm{C}(\mathcal{F})$ *or* $\mathrm{D}(\mathcal{F}) = \mathrm{A}(\mathcal{F})$ *in the case where the three conditions are equivalent). The weakest of the eight conditions considered is* D_{W_0}, *the strongest* A. (*And for* W_0 *variable, we find the weakest condition of all by taking* D_{W_ω}, *where* W_ω *is the smallest of the basic localizers in* Cat.)

c) *We have moreover, for the weakest condition* D_{W_0}, *the implication*

$$\mathrm{D}_{W_0} \implies \mathrm{F},$$

where $\mathrm{F} = \mathrm{Loc}$ (7 F) *is the condition that* Cat *has enough* W*-fibrations,* i.e. *that it is a category of fibrations for* (W, Fib_W). [page 60]

② *If* W *and* W_0 *satisfy* Loc (8 bis) *(axiom of countable limits), then* $\mathrm{D}_{W_0} \implies \mathrm{A}$, i.e. *the 8 conditions considered are equivalent. Thus if* W *satisfies* Loc (8 bis), *considering the smallest basic localizer* W_L *satisfying* Loc (8 bis) *(so that* $W \supset W_L$*), we see that the eight conditions*

$$\mathrm{A},\ \mathrm{B},\ \mathrm{C},\ \mathrm{D},\ \mathrm{A}_{W_L},\ \mathrm{B}_{W_L},\ \mathrm{C}_{W_L},\ \mathrm{D}_{W_L}$$

[32] and this is an entirely unexpected find, a little like the lemma of K. S. Brown …

are equivalent [33]. *In particular, under the same (quite innocuous) hypothesis on* W, *the conditions* Loc (7) = Loc (7 a), Loc (7 b) *and* Loc (7 c) *are equivalent, and they are also equivalent to the condition* $\mathrm{D} = \mathrm{A}(\mathrm{Parf}_W)$, i.e. *to*

$$\mathrm{Parf}_W \subset \mathrm{Fib}_W \qquad (\textit{special case of Loc (7 a)}),$$

and to its two seemingly weaker variants $\mathrm{B}(\mathrm{Parf}_W)$, $\mathrm{C}(\mathrm{Parf}_W)$ *(special case of* Loc (7 b) and of Loc (7 c) *respectively).*

③ *Suppose that* W_0 *satisfies* Loc (8 bis) *(without any particular hypothesis on* W*). Then the four conditions* A_{W_0}, B_{W_0}, C_{W_0}, D_{W_0} *are equivalent, and the diagram of implications of* 1°) *(p.* 59*) therefore reduces to the linear diagram* p. 35 [page 61]

$$\mathrm{A} \implies \mathrm{B} \implies \mathrm{C} \implies \mathrm{D} \implies \mathrm{D}_{W_0} \ (\iff \mathrm{A}_{W_0} \ldots),$$

whose first term is Loc (7 a) = Loc (7), *and the last is equivalent to the inclusion*

$$\mathcal{F}_{W_0} \subset \mathrm{Fib}_W,$$

special case of A = Loc (7 a), *or again to the inclusion (apparently stronger)*

$$\mathrm{Fibf}_{W_0} \subset \mathrm{Fib}_W,$$

(special case of the equivalent form

$$\mathrm{Fibf}_W \subset \mathrm{Fib}_W,$$

of the axiom Loc (7 a)).

④ *Finally, we have the following more technical conditional implications.*

a) *If* W_0 *satisfies Loc (7), then the conditions* A_{W_0}, B_{W_0}, C_{W_0} *are equivalent, so diagram (4.4.13.5) reduces to the diagram*

$$\begin{array}{ccc} \mathrm{A} & & \\ \Downarrow & & \\ \mathrm{B} & & \\ \Downarrow & & \\ \mathrm{C} & \Longrightarrow & \mathrm{C}_{W_0} \\ \Downarrow & & \Downarrow \\ \mathrm{D} & \Longrightarrow & \mathrm{D}_{W_0}\ . \end{array}$$

b) *If* W *is strongly saturated* (*axiom* Loc (1 ter), *which is satisfied in the great majority of cases, and is even imperative for constructing a prederivator satisfying the localization axiom* Der 2) *and satisfies* Loc (3 bis) (*strong localization axiom,* cf. §1), *then we have* $\mathrm{B} \Longrightarrow \mathrm{A}$, *hence* [page 62]

$$\mathrm{A} \iff \mathrm{B},$$

i.e. Loc (7 a) = Loc (7) *is equivalent to* Loc (7 b).

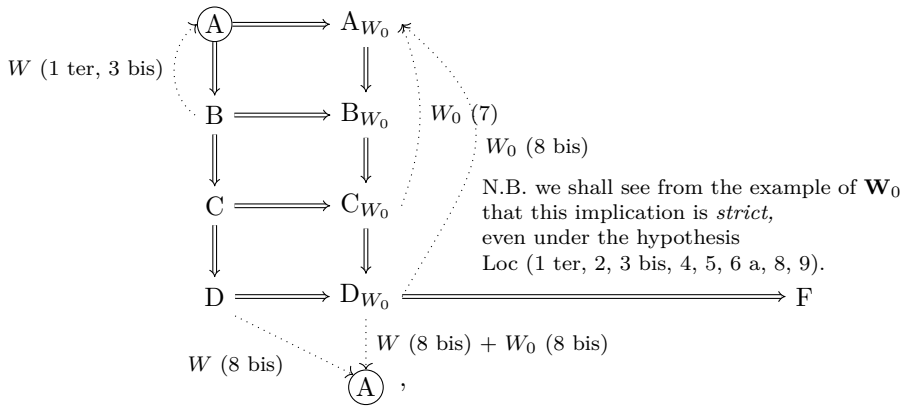
p. 36

PROOF. Everything stated has been seen in the preceding pages, with the sole exception of 4°, b. But this implication B $\Longrightarrow$ A for W strongly saturated is proved in XII (prop. 1, p. 54). I recall that the proof makes essential use of the formalism of the prederivator HOT_W (constructed and studied mainly in VI), and the theory of the functors $f_!$, and in particular the result (valid if W satisfies Loc (3 bis, 7 b)) that for an arrow $f : S' \longrightarrow S$ in W, the corresponding arrow

$$f^*_{\mathrm{lc}} : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S')$$

is an equivalence of categories. So the implication B $\Longrightarrow$ A, too, is a result which is not at all obvious.

4.4.14. I summarize the implications obtained so far, including the conditional implications, in the following summary diagram: [page 63]



where I have circled the strongest and most important condition of all, A = Loc (7), which appears at two opposite ends of the diagram.

I would like to make one more comment on the ingredients of the results proved here, and in particular of the equivalence of the eight conditions A to D_{W_0}, when W and W_0 satisfy Loc (8 bis): [page 64]

1) The theory of *categories of paths* $\underline{\mathrm{Ch}}(X)$ (VII), relying on that of *smooth, proper, perfect morphisms.*

2) The lemma of K. S. BROWN (3.2.1, page 20).

3) For the implication B $\Longrightarrow$ A, *i.e.* Loc (7 b) $\Longrightarrow$ Loc (7) (under the hypothesis W (1 ter, 3 bis)), I use the theory of the prederivator Hot_W (VI), and more particularly that of the functors $f_!$ (defined under the sole hypothesis W (3 bis), strengthening the automatic condition W (3) (forming part of the definition of a basic localizer)).

It is clear that I would not have managed to see clearly, without the fundamental work of K. S. BROWN. Yet there still remain open questions. Thus, if we do not assume that W satisfies Loc (8 bis), but if need be Loc (3 ter) (and I have not yet had occasion to work with a localizer which does not satisfy it …), is it nevertheless true that we still have the implication p. 37

$$\mathrm{D} \qquad \Longrightarrow \qquad \mathrm{A},$$

so that the conditions A, B, C, D are equivalent (and in particular Loc (7 c) already implies Loc (7) in all its force)? It is clear in all this that I have not applied myself [page 65]

[33] N.B. The weakest form, $\mathrm{D}_{W_{\mathrm{L}}}$, of these conditions, is the relation $\boxed{\mathrm{Parf}_{W_{\mathrm{L}}} \cap W \subset W^{\mathrm{u}}.}$

enough to developing examples and counterexamples. Not to mention that it seems to me that all the basic localizers on Cat which we encounter "in practice" already satisfy Loc (8) and a fortiori Loc (8 bis), so for them we do indeed have the expected implication. The same remark applies to the hypothetical implication

$$\mathrm{D}_{W_0} \overset{?}{\Longrightarrow} \mathrm{A},$$

under hypothesis Loc (8 bis) on W_0 and on W. An obvious candidate for a counterexample already to the first question would be W_ω = smallest basic localizer on Cat, of which it is almost obvious that it does *not* satisfy Loc (8) nor even Loc (8 bis). But there is no reason why it should satisfy F either, so that (if it does not satisfy it, hence not D either) the implication D $\Longrightarrow$ A is indeed valid. This brings us to the question, a more substantial one, whether we might not have the implication [page 66]

$$\mathrm{F} \overset{?}{\Longrightarrow} \mathrm{A}\ {}^{(34)},$$

at least under suitable fairly innocuous conditions on W, of the sort Loc (1 ter, 3 bis, 8 bis).

This is probably the most substantial and the most interesting question which is raised by the diagram on page 63, and which no longer calls for the introduction of an auxiliary basic localizer W_0. In view of Brown's lemma, it does not seem to me absolutely ruled out that such a miracle should indeed take place!

4.4.15 (Digression on the axiom Loc (6) of cofibrations.) This draws my attention to the in some sense dual question: If Cat has enough W-*co*fibrations, *i.e.* if equipped with (W, Cof_W) it is a category with cofibrations, does this already imply the axiom Loc (6 a) (which had been isolated, essentially, to ensure that this is indeed so). Proposition 3.6. (p. 25) tells us that it amounts to the same thing to prove that, if W satisfies Cof, then the open M. V. squares are W-cocartesian. But it seems to me now that we can prove this, under the sole additional condition Loc (3 bis) [35], which ensures [page 67] the existence of the functors $f_!$, and in particular the construction of Holim of a diagram p. 38

$$\begin{array}{ccc} & X_1 & \\ \nearrow & & \\ X_0 & & \\ \searrow & & \\ & X_2 & \end{array} \quad \text{as} \int \begin{array}{ccc} & X_1 & \\ \nearrow & & \\ X_0 & & \\ \searrow & & \\ & X_2 & \end{array} \quad . \text{ So to say that an open } \mathcal{M} \text{ diagram } \begin{array}{ccccc} & & X_1 & & \\ & \nearrow & & \searrow & \\ X_0 & & & & X \\ & \searrow & & \nearrow & \\ & & X_2 & & \end{array}$$

is W-cocartesian, must amount to saying that the canonical arrow

$$\int \begin{array}{ccc} & X_1 & \\ \nearrow & & \\ X_0 & & \\ \searrow & & \\ & X_2 & \end{array} \longrightarrow \quad X$$

is in W; at least if W is assumed strongly saturated. Now this has indeed been seen in XII 3.4 (theorem 4 and its *corollary* 3, p. 149). In loc. cit. I state the conclusions under the hypothesis which I called W (7 bis) (the present Loc (6)), but I have just checked that for its part *in the forward direction,* hence also for its corollaries 1 to 3, we do not need this hypothesis (as I indicate in marginal annotations).

4.16. The result I have just proved (without carefully checking the proof) seems to [page 68] me as delicate, as it is satisfying. This makes the hesitant "conjecture" of 4.4.14, namely that

$$\mathrm{F} \overset{?}{\Longrightarrow} \mathrm{A} \qquad \text{hence} \qquad \mathrm{F} \Longleftrightarrow \mathrm{A}$$

[34] This is false, *cf.* below, the implication Loc (7 D_{W_ω}) $\Longrightarrow$ Loc (7 F) is *strict.*

[35] and perhaps, in addition, Loc (1 ter), *i.e.* W strongly saturated.

assuming that W satisfies Loc (1 ter, 3 bis, 8 bis), decidedly more inviting. For one begins to get used to the most astounding miracles …

Yet I have just checked that this conjecture is false, using the example of

$$\mathbf{W}_0 = \{f \in \mathrm{Fl}(\mathrm{Cat}) \mid \pi_0(f) \text{ iso}\},$$

which satisfies F, but not A, nor even D_{W_ω}. Thus the implication

$$\mathrm{D}_{W_\omega} \implies \mathrm{F}$$

and a fortiori the implications

$$\begin{cases} \text{Loc (7 a)} \implies \mathrm{F},\ \text{Loc (7 b)} \implies \mathrm{F},\ \text{Loc (7 c)} \implies \mathrm{F} \\ \mathrm{D}_{W_0} \implies \mathrm{F} \end{cases}$$

are *strict,* even when restricting ourselves to basic localizers W satisfying (like $\mathbf{W}_0$)

$$\text{Loc (1 ter, 2, 3 bis, 4, 5, 6 a, 8, 9)}\ ^{(36)}.$$

That $\mathbf{W}_0$ satisfies this set of axioms is checked by AQT . (**N.B.** That it satisfies Loc (1 ter, 2, 3 bis) is already a consequence of the fact that $\mathbf{W}_0$ is of the form $W_{\mathbf{D}}$, where $\mathbf{D}$ ($= \mathbf{D}_{\mathbf{W}_0}$ in this case) is a prederivator satisfying Der (3, 4, 5), which is indeed the case for $\mathbf{D}_{\mathbf{W}_0}$, which is a genuine *derivator.*) We must check that every f in Cat factors as $f = pi$, with $i \in \mathbf{W}_0$, p a $\mathbf{W}_0$-fibration. Now if $f : X \longrightarrow Y$, $I = \pi_0(X)$, we take $\bar{X} = \coprod_{\alpha \in I} Y_{f(\alpha)}$, whence a factorization $f = pi$, where $\pi_0(i)$ is an isomorphism, *i.e.* $i \in \mathbf{W}_0$, and where $p : \bar{X} \longrightarrow Y$ is, over the connected component Y_j of Y, given by the product $\bar{X}|Y_j \simeq Y_j \times \underbrace{\pi_0(f)^{-1}(j)}_{\text{discrete category}}$. The morphism p is indeed a $\mathbf{W}_0$-fibration, for thanks to Loc (4), it suffices to check this on each Y_j, and there it is deduced from an arrow $I_j \longrightarrow e$ (a $\mathbf{W}_0$-fibration) by base change $Y_j \longrightarrow e$. p. 39 [page 69]

On the other hand, that $\mathbf{W}_0$ does *not* even satisfy D_{W_ω} has already been seen (I, p. 87), by realizing the map of the n-covering $f : S^1 \longrightarrow S^1$ in Cat by a W_ω-perfect morphism (in fact, by a locally trivial Cat-fibration), a very easy thing[37].

I think that by now the whole set of different variants of the axiom of W-fibrations is more or less understood, except that it is still not clear, without the axiom Loc (8 bis), whether the implications in the diagram [page 70]

$$\begin{array}{ccc} \mathrm{A} & \Longrightarrow & \mathrm{A}_{W_0} \\ \Downarrow & & \Downarrow \\ \mathrm{B} & \Longrightarrow & \mathrm{B}_{W_0} \\ \Downarrow & & \Downarrow \\ \mathrm{C} & \Longrightarrow & \mathrm{C}_{W_0} \\ \Downarrow & & \Downarrow \\ \mathrm{D} & \Longrightarrow & \mathrm{D}_{W_0} \end{array}$$

are strict or not. But it seems to me that all the interesting basic localizers encountered so far satisfy Loc (8), and even Loc (1 ter, 2, 3 bis, 4, 5, 6 a, 8, 9) (perhaps even Loc

[36] **N.B.** $\mathbf{W}_0$ does *not* satisfy Loc (6 b), nor Loc (7 D_{W_ω}) (*cf.* below), on the other hand Loc (7 F).

[37] Another way of seeing it: we *know* (XII) that W (5, 7 D_W) $\implies W \subset \mathbf{W}_\infty$, now we have $\mathbf{W}_\infty \subset \mathbf{W}_0$, and the inclusion is *strict!* ?

(7 F) - I would have to test on the $\mathbf{W}_n$), so the unresolved perplexity seems to me of no great consequence.

[page 71]

p. 40

5 Limits axiom Loc (8), and the accessibility axiom Loc (9)

It is the axiom:

Loc (8) *W is stable under filtering $\varinjlim$.*

A useful weakened variant is the following:

Loc (8 bis) *W is stable under countable $\varinjlim$.*

Explicitly, if we have two filtering direct systems in Cat

$$\underline{X} = (X_i)_{i\in I}$$
$$\underline{Y} = (Y_i)_{i\in I},$$

and an arrow

$$\underline{f} = (f_i)_{i\in I} : \underline{X} \longrightarrow \underline{Y},$$

given by $f_i : X_i \longrightarrow Y_i$, putting $X = \varinjlim X_i$, $Y = \varinjlim Y_i$, $f = \varinjlim f_i : X \longrightarrow Y$, Loc (8) tells us that if the $f_i \in W$, then $f \in W$. Here I is an arbitrary (small) filtering category, [page 72] and after replacing it by a cofinal category if necessary, we may suppose I an ordered set. The axiom Loc (8 bis) corresponds to the case where $I = \mathbf{N}$.

Note that thanks to the construction of the category of paths, we have a factorization, functorial in f, of every arrow $f : X \longrightarrow Y$ of Cat as $f = pi$:

$$X \xrightarrow{i=i_f} \bar{X}(f) = \bar{X} \xrightarrow{p=p_f} Y,$$

with $i \in W_\omega$, $p \in \mathrm{Parf}_{W_\omega}$, a fortiori $i \in W$, $p \in \mathrm{Parf}_W$. This functor $f \longmapsto (\bar{X}(f), p_f)$ commutes with direct limits. Thus, if we apply this to each of the $f_\alpha : X_\alpha \longrightarrow Y_\alpha$, factoring it as

$$X_\alpha \xrightarrow{i_\alpha} \bar{X}_\alpha \xrightarrow{p_\alpha} Y_\alpha,$$

we find a direct system of diagrams of type $\mathbf{\Delta}^2$ (factoring the f_i), and on passing to the $\varinjlim$ we find

$$X \xrightarrow{i} \bar{X} \xrightarrow{p} Y,$$

which is none other than the canonical factorization of $f = \varinjlim f_i$ as pi, with $i \in W_\omega$, [page 73] $p \in \mathrm{Parf}_{W_\omega}$ (N.B. Whereas neither W_ω, nor Parf_{W_ω} are stable under filtering $\varinjlim$, for sure!). We therefore have $i \in W$, and the condition $f \in W$, *i.e.* $pi \in W$, is therefore equivalent to $p \in W$. This reduces us, for checking the axiom Loc (8) (resp. Loc (8 bis)), to the case where the f_i are in Parf_{W_ω}.

Suppose now that W satisfies D_{W_ω} (the weakest of all the variants considered here of the axiom of W-fibrations Loc (7), with the sole exception of F). Then we have

$$p \in W \qquad \Longleftrightarrow \qquad \text{the fibers of } p \text{ are } W\text{-aspheric},$$

the implication $\Longrightarrow$ resulting from the hypothesis D_{W_ω} $\mathrm{Parf}_{W_\omega} \cap W \subset W^{\mathrm{u}}$, and $\Longleftarrow$ p. 41 resulting from the sorites of W-smooth morphisms *etc.*, via Loc (3). Likewise, the hypothesis that the $f_i \in W$ is equivalent to the one that the fibers of the p_i are W-aspheric. On the other hand, it is immediate that every fiber of p is isomorphic to the $\varinjlim_I$ of the [page 74] corresponding fibers of the p_α. In this way, we find:

Proposition 5.1. *Suppose that W satisfies the axiom of W-fibrations* Loc (7 D_{W_ω}) (*which is therefore the case if it satisfies one of the stronger axioms* Loc (7 a *or* b *or* c)).

For W to satisfy Loc (8) (resp. *to satisfy* Loc (8 bis)), *it is necessary and sufficient that the set of objects in* Cat *which are W-aspheric be stable under filtering* $\varinjlim$ (resp. *under* $\varinjlim$ *of countable sequences*).

5.2. Note that the category Cat is a paratopos, a fortiori it is *accessible* (*cf.* XVIII, §8), and the same therefore holds for $\underline{\mathrm{Fl}}(\mathrm{Cat})$. Thus we have in $\mathrm{Fl}(\mathrm{Cat}) = \mathrm{Ob}\,\underline{\mathrm{Fl}}(\mathrm{Cat})$ the notion of an *accessible subset.* We shall therefore say that a subset W of $\mathrm{Fl}\,\mathrm{Cat}$ (more generally of a category $\mathrm{Fl}(\mathcal{M})$, where $\mathcal{M}$ is an accessible category) is *accessible,* if it is so with respect to the ambient category $\underline{\mathrm{Fl}}\,\mathrm{Cat}$. This being so, the axiom Loc (9) is the following. [page 75]

Loc (9) W is an *accessible* subset of $\mathrm{Fl}(\mathrm{Cat}) = \mathrm{Ob}\,\underline{\mathrm{Fl}}(\mathrm{Cat})$.

5.3. Let us recall that this somewhat technical notion simplifies considerably, if we suppose that W satisfies the limits axiom Loc (8). Then W is accessible if and only if there exists a *small* subset W_0 of $\mathrm{Fl}(\mathrm{Cat})$, such that W is the "L_1-envelope" of W_0, *i.e.* the smallest subset of $\mathrm{Fl}(\mathrm{Cat})$ containing W_0 and L_1-stable (*i.e.* stable under filtering $\varinjlim$). If this is so, it is clear that we can take W_0 of the form

$$W_\pi = W \cap \mathrm{Fl}_\pi(\mathrm{Cat}),$$

where π is a suitable infinite cardinal, and where $\mathrm{Fl}_\pi(\mathrm{Cat}) \overset{\mathrm{def}}{=} \mathrm{Fl}(\underline{\mathrm{Cat}}_\pi)$, with $\underline{\mathrm{Cat}}_\pi$ denoting the strictly full subcategory of Cat consisting of the objects X such that $\mathrm{card}\,X \leq \pi$ (where by definition, $\mathrm{card}\,X = \mathrm{card}\,\mathrm{Fl}\,X$).

Note that the filtration we have just described on Cat (indexed by the small infinite cardinals) is none other than its canonical filtration, *i.e.* $\underline{\mathrm{Cat}}_\pi$ is the full subcategory of Cat consisting of the π-accessible objects of Cat, and likewise in $\underline{\mathrm{Fl}}(\mathrm{Cat})$. We thus find, thanks to the developments in XVIII, §13, the following characterization, which I recall here for reference (restricting myself for simplicity to the case where W satisfies Loc (8)): [page 76]

Proposition 5.3.1.[38] *Let W be a subset of* $\mathrm{Fl}(\mathrm{Cat})$ *stable under filtering* $\varinjlim$, *and let, for every infinite cardinal π, $W_\pi = W \cap \mathrm{Fl}_\pi(\mathrm{Cat})$, and $\underline{W}$, $\underline{W}_\pi$ be the corresponding full subcategories of* $\underline{\mathrm{Fl}}(\mathrm{Cat})$. *We say that W is* π-accessible *if W is the L-envelope of W_π (thus W is accessible if and only if there exists an infinite cardinal π such that W is π-accessible). This being so, the following conditions are equivalent:*

p. 42

a) *W is π-accessible.*

b) *For every direct system $(f_i)_{i\in I}$, L_1-π-adapted in* $\underline{\mathrm{Fl}}\,\mathrm{Cat}$ (cf. *below*), *with direct limit f, we have $f \in W$ if and only if the full subcategory I_W of the $i \in \mathrm{Ob}I$ such that $f_i \in W$ is cofinal in I.* [page 77]

b′) (*When we have chosen, for every $f \in \mathrm{Fl}(\mathrm{Cat})$, an L_1-π-adapted representation of f as a* $\varinjlim$): *If $f \in W$, then the full subcategory I_W defined in* b) *is cofinal in I.*

Let us recall here the notion of an L_1-π-adapted direct system. The conditions are:

1°) I is large relative to π, and stable under filtering $\varinjlim$ of cardinal $\leq \pi$.

2°) The functor $i \longmapsto x_i : I \longrightarrow \underline{\mathrm{Fl}}(\mathrm{Cat})$ commutes with the $\varinjlim$ in question.

3°) The x_i are in $\underline{\mathrm{Fl}}(\mathrm{Cat})_\pi \overset{\mathrm{def}}{=} \underline{\mathrm{Fl}}_\pi(\mathrm{Cat})$.

Note that in the case of a "parabelian paratopos" such as Cat or $\underline{\mathrm{Fl}}\,\mathrm{Cat}$, there is a canonical way, π being given, to represent an object f as the $\varinjlim$ of an L_1-π-adapted direct system. Here,

$$f : X \longrightarrow Y$$ [page 78]

being given, we consider the set I of pairs $(X_i \subset X, Y_i \subset Y)$ of subcategories of X and Y, such that we have $X_i, Y_i \in \mathrm{Ob}\,\mathrm{Cat}_\pi$ (*i.e.* $\mathrm{card}\,(X_i), \mathrm{card}\,(Y_i) \leq \pi$) and that $f(X_i) \subset Y_i$. It is immediate that I is filtering, for the obvious order relation

$$(X_i, Y_i) \subset (X_j, Y_j) \qquad \Longleftrightarrow \qquad X_i \subset X_j \ \text{ and } \ Y_i \subset Y_j,$$

and even large relative to π, that I is stable under filtering $\varinjlim$ of cardinal $\leq \pi$, and that the functor

$$i \longmapsto (f_i : X_i \longrightarrow Y_i) : I \longrightarrow \underline{\mathrm{Fl}}\,\mathrm{Cat}$$

commutes with the said $\varinjlim$, finally the f_i are in $\underline{\mathrm{Fl}}(\mathrm{Cat})_\pi$ by construction. Thus we have an L_1-π-adapted direct system, and it is clear that its direct limit is canonically isomorphic to f. This having been seen:

Corollary 5.3.2. *The condition that W be π-accessible therefore means that for every $f : X \longrightarrow Y \in W$, and for every subcategory X_i of X and Y_i of Y, of cardinal $\leq \pi$ (with $f(X_i) \subset Y_i$, but in fact this is unnecessary, after enlarging Y_i if necessary), there exist subcategories $X_j \supset X_i$ and $Y_j \supset Y_i$ of cardinal $\leq \pi$, such that f induces $f_j : X_j \longrightarrow Y_j$* [page 79] *and that we have $f_j \in W$.*

The notion of accessibility of W seems to me as well understood as that of limits, I no longer see any mystery in it. On the other hand, it seems to me that all the basic localizers used to date satisfy it (just as they do the axiom Loc (8)). Here, I shall confine myself to establishing:

Theorem 5.4.[39] *The basic localizer $\mathbf{W}_\infty$ ("usual" weak equivalences) satisfies the axiom* Loc (9) *(in fact, it satisfies* all *the axioms seen so far, namely* Loc (1 ter, 2, 3 bis, 4, 5, 6, 7, 8, 9)). *More precisely, $\mathbf{W}_\infty$ is $\aleph_0$-accessible (where $\aleph_0$, or π_0, is the first infinite cardinal).* p. 43

I do not return here to the verification of the other axioms, and confine myself to proving Loc (9), accessibility, using what may be useful to me. I confine myself to [page 80] indicating the proof (rather technical) in its broad outlines. For the moment, W is again an arbitrary basic localizer, except that it satisfies Loc (8).

Lemma 5.5.[40]

a) *For the criterion* 5.3.1 b) *of π-accessibility of W to be satisfied, it is necessary and sufficient that it be so when the f_i are W_ω-perfect.*

b) *Suppose that $Parf_{W_\omega} \cap W \subset W^u$,* i.e. *that W satisfies* Loc (7 D_{W_ω}) (*the weakest of the variants of the axiom of W-fibrations, except only for* Loc (7 F)). *Suppose moreover (to be on the safe side) that W satisfies . W is π-accessible if and only if the set* $\mathrm{Asph}_W \subset \mathrm{Ob}\,\mathrm{Cat}$ *of W-aspheric objects of* Cat *is π-accessible. Or again, that the subset* $\mathrm{Asph}_W \subset \mathrm{Ob}\,\mathrm{Cat}_{0\text{-connected}}$ *be π-accessible, where* $\mathrm{Cat}_{0\text{-connected}}$ *denotes the full subcategory of* Cat *consisting of* 0*-connected objects.*

Essentially, the proof is the same as that of 5.1, a little more technical. The essential tool is still the functorial factorization $f = pi$ of an arrow of Cat, with $i \in W_\omega$, $p \in \mathrm{Parf}_{W_\omega}$, and the property of commutation with filtering $\varinjlim$ of the said factorization. [page 81]

Let us now return to the case of $W = \mathbf{W}_\infty$. We must prove that $\mathrm{Asph}_{\mathbf{W}_\infty} \subset \mathrm{Ob}\,\mathrm{Cat}_{0\text{-connected}}$ is π_0-accessible. Thus to prove the criterion corresponding to 5.3.1 b),

[38] *cf. loc. cit.* 13.36.

[39] Well deserves the name of *theorem,* especially if we include in it the whole list of Loc axioms mentioned opposite ! This theorem is not at all obvious, starting from the cohomological definition of $\mathbf{W}_\infty$ - it is a result ("well known") which seems deep to me.

[40] We suppose that W satisfies Loc (8).

placing ourselves now in $\mathrm{Cat}_{0\text{-connected}}$ instead of $\mathrm{Fl}\,\mathrm{Cat}$. We therefore have

$$X = \varinjlim_{i \in I} X_i,$$

an L_1-π-admissible representation of X $\mathbf{W}_\infty$-aspheric, to prove that for every i in I, $\exists j$ in I and $i \longrightarrow j$, with X_j $\mathbf{W}_\infty$-aspheric.

I admit here the AQT lemma:

Lemma 5.6. *Let* $\mathbf{W}_0 = \{f \in \mathrm{Fl}\,\mathrm{Cat} \mid \pi_0(f)$ *isomorphisme*$\}$. *Then* $\mathbf{W}_0$ *is* π_0*-accessible (and* L_1*-stable). A fortiori, the set of* 0*-connected objects of* Cat *is* π_0*-accessible.*

This reduces us, after replacing I by a cofinal category if necessary, to supposing that the X_i are 0-connected. (But I forgot that this reduction has already been made! So one must view 5.6 as an ingredient

of the proof of 5.5 b).)

p. 44

[page 82]

Let x be an object (point) of X, represented by a point x_{i_0} of X_{i_0}; after restricting to the cofinal category $i_0\backslash I$, let x_i be its image in X_i. Let us consider (via my standard construction of categories of paths) the $\Omega^1(X, x)$, $\Omega^1(X_i, x_i)$. We find

$$\Omega^1(X, x) \simeq \varinjlim_{I} \Omega^1(X_i, x_i),$$

and one checks that this is again an L_1-π_0-adapted direct system. Iterating, we find for $n \in \mathbf{N}^*$

$$\Omega^n(X, x) \simeq \varinjlim_{I} \Omega^n(X_i, x_i),$$

L_1-π_0-adapted direct limit. Using 5.6, we conclude from this that for every n, the subset (let us suppose I an ordered set, what the devil, that's enough!)

$$I_n = \{i \in I \mid \Omega^n(X_i, x_i) \text{ is 0-connected}\}$$

is cofinal in I, moreover it is stable under filtering $\varinjlim$ of cardinality $\leq \pi_0$. Thus

$$I_\infty \overset{\text{def}}{=} \bigcap I_n$$

is also cofinal in I (*cf.* XVIII, 13.10.2, p. 302).

[page 83]

Now $i \in I_\infty$ means that all the $\Omega^n(X_i, x_i)$ are 0-connected, which also means that X_i is $\mathbf{W}_\infty$-aspheric (since it is already 0-connected) (*cf.* XII, cor. p. 190). This completes the proof of theorem 5.4.

6 The condition Loc (5, 7 D_{W_ω}) implies $W \subset \mathbf{W}_\infty$

I am going to give here again, for the record, the proof of theorem 1 of XII:

Theorem 6.1. *Let* W *be a basic localizer satisfying* D_{W_ω} *(where* W_ω *is the smallest of all basic localizers). Then we have* $W = \mathrm{Fl}\,\mathrm{Cat}$, or $W \subset \mathbf{W}_\infty$ *(according as* W *does not satisfy the connectedness axiom* Loc (5)*, or does satisfy it.*)

I will give the proof assuming that W satisfies Loc (5). For the proof of the fact that in the contrary case, we have $W = \mathrm{Fl}(\mathrm{Cat})$, I refer to XII, th. 2, p. 20. (See also the very close result, more general and weaker, recalled in prop. 2.7, page 12.)

Lemma 6.2. *Let* W*,* W' *be two basic localizers,* W *satisfying* Loc (7 D_{W_ω}). *For* $W \subset W'$ *to hold, it is necessary and sufficient that* $\mathrm{Asph}_W \subset \mathrm{Asph}_{W'}$.

[page 84]

It suffices of course to prove "it suffices", so let us suppose that every W-aspheric object is W'-aspheric, and let us prove that $W \subset W'$. So let $f : X \longrightarrow Y$ be in W, let us prove $f \in W'$. For this, I use the canonical factorization $f = pi$,

$$X \xrightarrow{i \in W_\omega} \bar{X} \xrightarrow{p \in \mathrm{Parf}_{W_\omega}} Y.$$

We therefore also have $i \in W$, and since $f = pi \in W$, we have $p \in W$, hence $p \in$ p. 45
$\mathrm{Parf}_{W_\omega} \cap W$. Now D_{W_ω} means that this set is contained in W^{u}, so we have $p \in W^{\mathrm{u}}$, and in particular the fibers of p are W-aspheric. They are therefore W'-aspheric, and since p is W_ω-perfect and a fortiori W'-perfect, it follows that $p \in W'$. We also have $i \in W_\omega \subset W'$, hence $i \in W'$, hence also $f = pi \in W'$, q.e.d.

Let us then return to the proof of 6.1. By the lemma, it suffices to prove that if X in Cat is W-aspheric, it is $\mathbf{W}_\infty$-aspheric. But by the hypothesis Loc (5) we already know that X is 0-connected, and choosing a point x in X, its being $\mathbf{W}_\infty$-aspheric is expressed by the condition that the $\Omega^n(X, x)$ (for $n \geq 1$) are 0-connected (*cf.* XII, cor. p. 190 [page 85]
already cited). For this, because of Loc (5) it suffices to prove that they are W-aspheric, and by induction it suffices to prove that $\Omega^1(X, x)$ is. Now by definition of $\Omega^1(X, x)$, this is the fiber at x of an arrow in Cat

$$E = \underline{\mathrm{Ch}}(X, x, -) \xrightarrow{p \in \mathrm{Parf}_{W_\omega}} X,$$

where the source E is W_ω-aspheric, a fortiori W-aspheric. Since X is as well, we therefore have $p \in W$, hence $p \in \mathrm{Parf}_{W_\omega} \cap W \subset W^{\mathrm{u}}$, hence p has W-aspheric fibers, and in particular $\Omega^1(X, x)$ is W-aspheric, q.e.d.

Corollary 6.3. (Characterization of $\mathbf{W}_\infty$.) $\mathbf{W}_\infty$ *is the largest basic localizer satisfying* Loc (5, 7 D_{W_ω}) - *hence also the largest basic localizer satisfying* Loc (5, 7).

I think that we will find below an "internal" characterization of $\mathbf{W}_\infty$, via a "descent axiom" (very strong) Loc (10), as the unique basic localizer satisfying Loc (5, 7 D_{W_ω}, [page 86]
10) (or again Loc (5, 7, 10)).

6.4. *Comment on the axioms for W-fibrations of the type* Loc (7 …), *and in particular on* Loc (7 D_{W_ω}). *Conjectural relation with the* "simple homotopy types" *of* WHITEHEAD.

a) It seems to me that *all* the basic localizers with which we have tacitly worked up to now (the $\mathbf{W}_n$ ($n \in \mathbf{N}^*$), homological and cohomological equivalence, rational equivalence …) correspond to localizers $W \supset \mathbf{W}_\infty$. By theorem 6.1, it therefore follows that *none* of them, apart from $\mathbf{W}_\infty$ itself, satisfies the very weakest version D_{W_ω} of axiom Loc (7 …), with the sole exception of Loc (7 F). Thus, the localizers W which satisfy it are with the exception of $\mathbf{W}_\infty$ itself, of a quite new type - localizers which are strictly contained in $\mathbf{W}_\infty$.

b) This raises, of course, the question of the existence of basic localizers contained in $\mathbf{W}_\infty$ and distinct from $\mathbf{W}_\infty$. I have not even proved that $W_\omega \neq \mathbf{W}_\infty$, although I [page 87]
have not the slightest doubt that this is indeed the case. A fortiori, I do not know whether the smallest basic localizer satisfying Loc (7 D_{W_ω}) is different from $\mathbf{W}_\infty$. I would be quite particularly interested in the smallest localizer $\tilde{\mathbf{W}}_\infty \subset \mathbf{W}_\infty$ satisfying all the stability axioms Loc (1 ter, 2, 3 bis, 4, 6, 7, 8) (**N.B.** it satisfies Loc (5) necessarily, p. 46
since $\tilde{\mathbf{W}}_\infty \subset \mathbf{W}_\infty \subset \mathbf{W}_0$, moreover Loc (7) is equivalent, in the presence of Loc (8), to the very weak form Loc (7 D_{W_ω}): $\mathrm{Parf}_{W_\omega} \cap W \subset W^{\mathrm{u}}$. Finally, one shows that Loc (4) is a consequence of the other axioms, *cf.* below.) I now conjecture that $\tilde{\mathbf{W}}_\infty$ (which gives rise to a homotopy formalism just as complete as the everlasting "weak equivalence") is indeed different from $\mathbf{W}_\infty$. And I suspect that the category $\mathrm{Hot}_{\tilde{\mathbf{W}}_\infty}$ deserves to be [page 88]
called the category of *"simple homotopy types"* (in the style of WHITEHEAD)[41]. A first

[41] Botched (?), *cf.* comment on page 92.

decisive test is to check that the principal cohomological invariant of WHITEHEAD of a simple homotopy type, namely an element in a $K(\mathbf{Z}[\pi_1])$ (if I remember correctly[42]) - or something close to it, can already be defined in terms of the objects in $\mathrm{Hot}_{\bullet\,\tilde{\mathbf{W}}_\infty}$ (the pointed Hot). And secondly, if one associates to a finite "simplicial mock-up" (describing combinatorially a finite triangulation) the ordered set of its simplices, viewed as an object of Cat, and the corresponding object in $\mathrm{Hot}_{\tilde{\mathbf{W}}_\infty}$, the latter does not change[43], up to canonical isomorphism, when one performs on this "polyhedron" under consideration an elementary operation in the style of WHITEHEAD. This moreover suggests trying to [page 89] define such elementary operations in the whole of Cat, or at least in the subcategory of (small) ordered sets, in the hope of arriving (perhaps) at a more direct construction of the localizer relevant to simple homotopy type.

c) I end with a comment on the meaning of axiom Loc (7 D_{W_ω}), as opposed to the weaker axiom Loc (7 F) which only ensures the existence of a "structure with fibrations" (W, Fib_W) on Cat. I also point out in passing that Loc (7 F), just like Loc (5) and like Loc (9), and unlike all the other axioms reviewed up to now, is *not* a stability axiom: it asserts, not that certain arrows (under suitable hypotheses) are in W, but the existence of a factorization. Thus it is not at all clear that the intersection of two basic localizers [page 90] satisfying Loc (7 F) satisfies it.

Loc (7 D_{W_ω}) is exactly what is needed to ensure that the canonical factorization of an arrow $f : X \longrightarrow Y$ of Cat as $f = pi$, with $i \in W_\omega$, $p \in \mathrm{Parf}_{W_\omega}$, is also a factorization of the type stipulated by the factorization axiom Loc (7 F): we have (of course) $i \in W$, but also $p \in \mathrm{Fib}_W$ (since D_{W_ω} is equivalent to its strong form $\mathrm{Parf}_{W_\omega} \subset \mathrm{Fib}_W$). This also means that the canonical factorization of f can be used to "compute" its homotopy fibers, not only at the points of Y, but also with respect to a base change morphism $Y' \longrightarrow Y$.

In the absence of this axiom, assuming only Loc (7 F), we know that there exists a [page 91] theory of W-homotopy fibers, but we have a priori no practical means of "computing" the p. 47 said fibers, for lack of a manageable criterion sufficient to ensure $p \in \mathrm{Fib}_W$. I therefore have the impression that axiom Loc (7 F) is going to be of meager use. (It is true that the example of $\mathbf{W}_0$ ought to make me cautious! I would have to look at the case of the $\mathbf{W}_n$, whether they satisfy Loc (7 F) …). To my mind, axiom Loc (7 D_{W_ω}) is the exact "bare minimum" for being at ease in handling W-homotopy fibers. It is remarkable that this implies that one must have $W \subset \mathbf{W}_\infty$, and that this at the same time excludes *all* the variants considered to date of the notion of homotopy type!

6.5. Finally, the relation glimpsed with the "simple homotopy types" of WHITEHEAD [page 92] vanishes[44]: indeed, the latter are strictly finer than ordinary homotopy types (*strict*), and Whitehead's invariant is *not* an invariant of ordinary homotopy type. It is true that when one considers the simplicial skeleton of a triangulated polyhedron as an object of Cat, ordinary (continuous) homotopy between two such polyhedra is interpreted, in Cat, not (it seems to me) by the existence of a homotopism between the corresponding objects of Cat, but by their being isomorphic in $\mathrm{Hot}_{\mathbf{W}_\infty} \overset{\mathrm{def}}{=} \mathrm{Cat}(\mathbf{W}_\infty)^{-1}$. But the method of simplicial approximations of BROUWER shows that if the polyhedra X and Y are homotopic, then there exist simplicial subdivisions X' and Y' of X and of Y which are "combinatorially homotopic" (in a suitable sense). So I presume that the simplicial mock-ups of X' and Y' are then homotopic in Cat. It would follow that X and Y, I mean [page 93] their mock-ups, are indeed homotopic in Cat, if it is true that a simplicial subdivision

[42] no, it is a finer invariant …

[43] This is obvious even for W_ω, since the homotopy type *in the strict sense does not change!* - No, there is confusion here between topological homotopy and combinatorial homotopy, *cf.* p. 92 …

[44] But perhaps not after all - the question remains open.

(barycentric subdivision would suffice, iterating …) X' of X has a mock-up in Cat which is homotopic to it. The question raised here seems interesting to me, clearly one must find the answer to it. It already arises for any ordered set I, which we will suppose finite to simplify, taking the set (ordered by inclusion) $I' = \mathrm{sd}(I)$ of totally ordered subsets of I. We know that the topological realizations of I and of $\mathrm{sd}(I)$ are canonically isomorphic, but are I and $\mathrm{sd}(I)$ homotopic in Cat? We do have a canonical increasing map

$$f : \mathrm{sd}(I) \longrightarrow I,$$

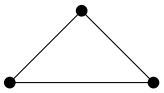
which associates to every totally ordered subset of I its greatest element. But is there an increasing map in the opposite direction, which is a homotopism inverse to the preceding one up to homotopy? I have just checked in the case

[page 94]

missing

[page 95]

of the boundary of the 2-simplex



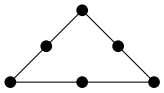
or



,

barycentric subdivision

p. 48



or



(hexagon), that for every increasing map $g : I \longrightarrow \mathrm{sd}(I)$, its topological realization is homotopic to a constant map, so g cannot be a homotopism in Cat (for this would imply that its topological realization is a homotopism).

This therefore shows that the notion of homotopism in Cat is *strictly finer* than that for the topological realizations, which once again restores substance to the hope that Whitehead's invariant could be defined as a homotopy invariant on Cat, or at least on a suitable full subcategory, such as that of ordered sets (possibly restricted to be finite), or of simplicial mock-ups. And even, that the set of Wh-homotopy types (simple homotopy types) can be canonically embedded in a suitable Hot_W, where W would be a basic [page 96] localizer strictly finer than $\mathbf{W}_\infty$, e.g. (who knows?) $\tilde{\mathbf{W}}_\infty$. The first piece of work one would like to do is to construct such a W *ad hoc*, so that Whitehead's invariant can be written as a functor on $\mathrm{Hot}_{\bullet W}$. To begin with, one would already have to make sure that this invariant can be viewed as an invariant of homotopy type, *i.e.* that it is interpreted as a functor on $\mathrm{Hot}_\bullet\mathrm{Cat} \simeq \mathrm{Cat}(\underbrace{\text{homotop}}_{\substack{\text{set of}\\ \text{homotopisms}}})^{-1}$. This work will perhaps also suggest the definition of "higher Whitehead invariants" (involving the π_i, not only the π_1), which could perhaps serve (with the one of dimension 1) to describe the hypothetical localizer W ("of Whitehead").

[page 97]

7 Comments on the axioms

7.1. The main reason for being of a basic localizer W is to define a theory of homotopy, and more precisely a prederivator HOT_W, of which one hopes that it will even be a derivator, or that it will at least satisfy some of the axioms of derivators. The axioms Loc (1) to Loc (9) are for the most part oriented toward this objective, except that Loc (1, 2, 3) constitutes an absolute "bare minimum", to be able to begin working with W. The main questions which arise seem to me to be the following:

1) When is Cat equipped with (W, Cof_W), resp. with (W, Fib_W), a category with cofibrations resp. with fibrations, so that the results of K. S. BROWN are applicable?

2) Under what conditions, on W and on f, do $f_!$ and f_* exist? And when do we have, in addition, exactness of cocartesian resp. cartesian squares?

[page 98]

3) More generally, in terms of the axioms imposed on W, which are the axioms Der (…) that are satisfied for the prederivator HOT_W? (Which could also depend on the choice of the domain Diag $\subset$ Cat. The prederivator is defined a priori on the whole of Cat). In particular,

p. 49

4) when is HOT_W really and truly a derivator on all of Cat, *i.e.* when does it satisfy all the axioms?

A question which seems to me less important (in particular than 1°) is the following:

1 bis) When does W fit into a closed Quillen triple (W, C, F) (preferably proper).

I shall go over these questions one by one.

7.2. We have seen (but without my having made a careful check of the converse) that (W, Cof_W) defines a structure with cofibrations if the axiom *Loc (6 a)* is satisfied, and that this condition is also necessary if Loc (3 bis) is satisfied[45].

Then the open Dwyer immersions (or, at one's choice, the closed Dwyer immersions) are in Cof_W, and the standard mapping cone construction defines, for every arrow f in Cat, a *functorial* factorization

[page 99]

$$f = pi \quad \begin{array}{l} i \text{ open Dwyer immersion (hence in } \mathrm{Cof}_W), \\ p \in W. \end{array}$$

This, plus the construction of the functor $f_!$, and in particular of W-amalgamated sums $\int X_0 \begin{array}{c} \nearrow X_1 \\ \searrow X_2 \end{array}$, given in VI (theory of HOT_W) and XII, gives us perfect mastery (it seems to me) over the "left homotopy structure" defined by W, under the sole axiom Loc (6 a) (Cube axiom, direct form). I recall that the functors $f_!$ are constructed in VI under the sole hypothesis *Loc (3 bis)* (f any arrow of Cat), but unless I am mistaken, the construction and the W-homotopic interpretation of $\int X_0 \begin{array}{c} \nearrow X_1 \\ \searrow X_2 \end{array}$ given in XII do not even depend on this hypothesis[46] - but I cannot guarantee it. It seems to me that XII (and in particular the long §5 on amalgamated sums) is written under the hypothesis that W (1, 2, 3), *i.e.* Loc (1, 2, 3bis), are satisfied.

[page 100]

7.3. It is a rather extraordinary thing that the functors $f_!$ for HOT_W exist, under the sole hypothesis *Loc (3 bis)* (localization axiom, strong form) - and they are "computed" in

[45] Check all the same whether Loc (3 bis) is not already used in the forward direction! **O.K.**

[46] *cf.* XII, page 264

a magically simple way! I am pretty sure that under this hypothesis alone, *the exactness axiom for cocartesian squares* is *not* satisfied(47). I see no way of checking it, except via BROWN's theory, under the hypothesis *Loc (6 a)* (*cf.* 7.2).

But under the sole condition Loc (3 bis) the standard calculation of the fibers of an $f_!(\xi)$ is valid, in other words the prederivator HOT_W satisfies Der (4*, 5*) - and moreover Der (6*) (exactness "on the left") if W satisfies Loc (6 a). p. 50 [page 101]

7.4. Let us come to the question whether Cat equipped with (W, Fib_W) is a category with fibrations. This is the axiom *Loc (7 F),* but I know scarcely any way of checking it (apart from the very, very special case $W = \mathbf{W}_0$) other than by means of at least the axiom *Loc (7 D_{W_ω}),* or one or another of the stronger variants of Loc (7). Loc (7 D_{W_ω}) has been discussed at length above, *cf.* in particular 6.4 (pages 89-91). In my view, this axiom ensures a mastery over the right W-homotopic structure (W-homotopic fibers, exact cosuspension sequences) comparable to that of Loc (6 a) for the left structure. In [page 102] particular, we have the *functorial* factorization of every arrow f of Cat as

$$f = pi, \qquad i \in W_\omega \subset W,\ p \in \mathrm{Parf}_{W_\omega} \subset \mathrm{Fib}_W.$$

But *the axiom Loc (7 D_{W_ω}) seems to me to be of a considerably stronger nature than Loc (6 a),* which (unless I am mistaken) is satisfied for all the localizers that have been considered so far. (But I admit not having looked at the case of the $\mathbf{W}_n$, $n \in \mathbf{N}$, except $\mathbf{W}_0$.) On the other hand, the axiom Loc (7 D_{W_ω}) implies $W \subset \mathbf{W}_\infty$ (theorem 6.1), a truly draconian relation (*cf.* comment 6.4 a).

Besides, the axiom Loc (7 D_{W_ω}) alone, as far as I know, does not imply the existence of f_* for *every* arrow in Cat, but only for arrows of the form $f \times \mathrm{id}_X : S \times X \longrightarrow T \times X$, [page 103] where $f : S \longrightarrow T$ is a morphism between finite ordered sets. Or more generally, between ordered sets S, T such that S° is artinian, *i.e.* S noetherian, and $\mathrm{Esp}(S^\circ)$ is noetherian (this last hypothesis being unnecessary under the axiom Loc (8) on limits, *cf.* XVII, th. 3.1, p. 50). So this is not the result dreamed of.

7.5. The only case I know where f_* exists for *every* arrow in Cat, is the case where W satisfies

$$\textit{Loc (6 a, 8, 9)}\ {}^{(49)}\ {}^{(50)}, \qquad (*)$$

that is to say the axiom of weak W-cofibrations, already mentioned at length and finally those of limits and accessibility, *i.e.* W is L_1-accessible in $\underline{\mathrm{Fl}}(\mathrm{Cat})$. The relevant key result is given in XV 8.4 (p. 105-107), which must be combined with XV 7.4 (p. 98-99) to make good use of it, and to deduce theorem 10.1 (XV, p. 134-135). Better than the existence of the f_*, the surprisingly weak set of axioms I have just spelled out suffices [page 104] to ensure that HOT_W is a *derivator* on all of Cat, in the full sense of the term, *i.e.* that it satisfies Der (1, 2, 3, 4, 5, 6, 4*, 5*, 6*). This is obtained by replacing (thanks to XV 8.4) Cat by $A^\wedge$, where A is a suitable test category, W by $W_A \subset \mathrm{Fl}\, A^\wedge$, and proving that W_A fits into a closed Quillen triple (W, C, F), with $C = \mathrm{Mono}(A^\wedge)$, which determines F as $(C \cap W_A)_*$. We then apply XV 7.4 to this situation (results summarized in XIX, th. 2.3 (p. 19), 2.6 (p. 25), 2.9 (p. 28). We recover the existence of the $f_!$ (with a method of calculation quite different from the one in Cat!), and in addition we find the existence of f_* (plus the exactness in cor. 5 of W-cartesian squares) - which for me, for a good while, was *the* big problem for the theory of HOT_W!). p. 51 [page 105]

I find it staggering that one arrives at such a strong result with such ridiculously weak axioms. More particularly, that one has all the desirable properties of the right homotopy

47 An example would be needed!

49 Check whether the condition Loc (3 bis) is not also needed in addition to Loc (6 a, 8, 9). Check made, it is O.K. (without Loc (3 bis)).

50 **N.B.** On further reflection, it appears that even Loc (6′ a, 8, 9) suffices.

structure, without having to make any hypothesis of the W-fibration type on W! These axioms do *not* imply the axiom Loc (7) (equivalent to Loc (7 D_{W_ω}) in the presence of Loc (8)), as we have seen from the example of $\mathbf{W}_0$. Do they at least imply Loc (7 F), *i.e.* the existence of enough W-fibrations? I would not venture any conjecture on this subject! But one must bear in mind that in the absence of Loc (7) (and even assuming Loc (7 F)), [page 106] while knowing that the right homotopy structure associated with W is perfect in every respect, one has a priori no mastery over it, in particular for computing W-homotopic fibers and constructing W-cartesian squares. Even on reducing all these constructions to $A^\wedge$ (taking, if one insists, $A = \mathbf{\Delta}$, *i.e.* reducing to the semi-simplicial context), though one knows that there is a *functorial* factorization $f = pi$ with $i \in \mathrm{TC} \subset W_A$, $p \in F \subset \mathrm{Fib}_{W_A}$, I see no concrete way of making it explicit. Unless the consideration of infinite paths, which (in the absence of Loc (7)) turns out to be inoperative in Cat saves the day in $\mathbf{\Delta}^\wedge$? ?

A question which ought to be easy to clear up, but on which I do not want to linger [page 107] just now.

Let us note in passing that Quillen's strong saturation result, for closed Quillen triples, shows that the set of axioms $(*)$ implies the axiom Loc (1 ter), *i.e.* that W is *strongly saturated.*

Let us also note that unlike what happens for the $f_!$, I know how to prove the existence of f_*, and in particular of cosuspension sequences, only under conditions (either Loc (7 D_{W_ω}), or Loc (6a, 8, 9)) which already ensure that the W-cartesian squares are exact, and consequently the cosuspension sequences exact.

I think that in the preceding pages, I have said all that I know how to say that is relevant, concerning questions 1, 2, 4 raised in 7.1. These answers seem satisfactory to [page 108] me - more satisfactory than I would have dared hope during the first two months of my cogitations! There remain questions 3 and 1 bis, which I shall now examine.

7.6. On the different derivator axioms: under what conditions on W is one of these p. 52 axioms satisfied for HOT_W[51]? As for axioms Der (4, 5, 6, 4*, 5*, 6*), I think I have said all that I know how to say about them, and have perfectly manageable criteria for validity, in the preceding pages. Let us summarize nevertheless:

structure with cofibrations	:	Loc (6) (+ Loc (3 bis) ?)
structure with fibrations	:	Loc (7 D_{W_ω})
Der (4*, 5*)	:	Loc (3 bis)
Der (4*, 5*, 6*)	:	Loc (3 bis, 6a) or Loc (6′ a, 8, 9)
Der (4, 5, 6) (on Cat) (and all the rest !)	:	Loc (6′ a, 8, 9)
Der (4, 5, 6) (on Diag = finite ordered sets)	:	Loc (7 D_{W_ω}) (or only Loc (7 F)).

There remain the axioms Der (1) (decomposition), Der (2) (localization), Der (3) (unit object), which I shall now go over. One checks quasi-tautologically that the axiom [page 109] Der 1 is satisfied with no condition on W (not even Loc (2, 3)!) - in fact, it is satisfied for every prederivator $\mathbf{D}_{(\mathcal{M},W)}$ defined by a category of models.

The localization axiom Der (2), on the other hand, is delicate in nature. It is valid, more generally, for a category of models $(\mathcal{M}, W)$ (defining a prederivator $\mathbf{D} = \mathbf{D}_{(\mathcal{M},W)}$) and an object S of Cat, when $(\mathcal{M}(S), W(S))$ is associated with a structure with fibrations or with cofibrations, which by K. S. Brown implies that every arrow of $\mathrm{HO}_W(S) = \mathcal{M}(S)W(S)^{-1}$ is "realized" by an arrow of $\mathcal{M}(S)$ - and then the result is obvious. The

[51] For axioms Der (1) to Der (5*), *cf.* I, p. 67-68. The axioms Der (6, 6*) are the exactness axioms, whose necessity became apparent to me only later.

condition I have just stated is satisfied when $(\mathcal{M}, W)$ itself is associated with a structure with cofibrations or fibrations, and if moreover we assume that we are in one of the following two cases: [page 110]

a) S is a finite ordered set (which leads us to restrict ourselves, for want of anything better, to Diag = subcategory of Cat formed by the finite ordered sets).

b) The factorization stipulated in the factorization axiom $f = pi$ (with $i \in \mathrm{Cof}$, $p \in W$ (cofibration case) or $i \in W$, $p \in \mathrm{Fib}$ (fibration case)) can be chosen functorial in f.

Now, in the case of Cat, in the cases where one knows how to prove that W is associated with a structure with cofibrations (axiom *Loc (6 a)*) or with fibrations (axiom *Loc (7 D_{W_ω})*), one does indeed find a functorial factorization. Thus each of these axioms implies Der (2).

There remains the contractibility axiom: if S in Cat is contractible, is it HOT_W-aspheric, *i.e.* is the functor [page 111]

$$\mathrm{HOT}_W(e) = \mathrm{Hot}_W \longrightarrow \mathrm{HOT}_W(S)$$

fully faithful? When one has the theory of $f_!$ "Cat version" (thus under the sole axiom *Loc (3 bis)*), one finds the following (noted in VI, p. 202)[52]: p. 53

> Let $f : S \longrightarrow T$ be an arrow in Cat. For it to be a HOT_W-equivalence, *i.e.* for $f^* : \mathrm{HOT}_W(T) \longrightarrow \mathrm{HOT}_W(S)$ to induce a *fully faithful* functor on the subcategory of constant objects of $\mathrm{HOT}_W(T)$, it is necessary and sufficient that $\mathrm{hot}_W(f) : \mathrm{hot}_W(S) \longrightarrow \mathrm{hot}_W(T)$ be an isomorphism. Thus it *suffices* that $f \in W$, and this condition is also necessary if W is strongly saturated (axiom Loc (1 ter)).

To ease my conscience, I spell out the proof here. The existence of the functors $f_!$, in particular of the homology objects $H_\bullet^{\mathrm{HOT}_W}(S, \xi)$ (or simply $H_\bullet^W(S, \xi)$) for $\xi \in \mathrm{HOT}_W(e) = \mathrm{Hot}_W$, makes it possible to interpret homologically the condition of HOT_W-equivalence, as meaning that for every $\xi = \mathrm{hot}_W(X)$ in Hot_W, the canonical arrow in Hot_W [page 112]

$$H_\bullet^W(S, \xi) \longrightarrow H_\bullet^W(T, \xi) \qquad (*)$$

is an isomorphism. Now the calculation of the $f_!$ (in the case of $p : S \longrightarrow e$ and of $T \longrightarrow e$) shows us that the two sides of $(*)$ are none other (up to isomorphism) than $\mathrm{hot}_W(X \times S)$ and $\mathrm{hot}_W(X \times T)$, the arrow being induced by

$$\mathrm{id}_X \times f : X \times S \longrightarrow X \times T. \qquad (*)$$

Thus the condition under consideration means that this arrow is an isomorphism in Hot_W for every X in Cat. Taking $X = e$, one sees that this implies that $\mathrm{hot}_W(f)$ is an isomorphism, and this suffices, since the functor $\mathrm{hot}_W : \mathrm{Cat} \longrightarrow \mathrm{Hot}_W$ commutes with finite products (Loc (1, 2, 3) suffices) and the arrow $(*)$ in Hot_W identifies with [page 113]

$$\mathrm{id}_{\mathrm{hot}_W(X)} \times \mathrm{hot}_W(f)\,.$$

This therefore shows that if S is a W-aspheric object, *i.e.* such that $S \longrightarrow e$ is in W (and a fortiori, if S is contractible), then $S \longrightarrow e$ is HOT_W-aspheric.

[52] under Loc (3 bis).

To summarize, we get the table

$$\begin{cases} \text{Der (1)} & : & \text{with no hypothesis on } W \\ \text{Der (2)} & : & \text{Loc (6 a) } \textit{or} \text{ Loc (7 D}_{W_\omega}\text{)} \\ \text{Der (3)} & : & \text{Loc (3 bis) } \textit{or} \text{ Loc (6' a, 8, 9)} \end{cases} \quad (53).$$

7.7. It only remains to justify that Loc (6′ a, 8, 9) implies Der (3). But then, as we have seen, HOT_W can be defined by a very special Quillen triple. But even for a merely special Quillen triple $(\mathcal{M}, W)$, the prederivator $\mathbf{D}$ which it defines satisfies (in addition to Der (4*, 5*, 6*)) the conditions Der (1) (trivially), Der (2)

(*cf.* p. 110, case b)) and *in addition* Der (3). (Thus in the case of a *very* special Quillen triple, the prederivator obtained is indeed a derivator.) I will spell out the proof here. p. 54 [page 114]

I will denote by $(\mathcal{M}, \Sigma)$ the given category of models ($\Sigma \subset \mathrm{Fl}\,\mathcal{M}$), by $\mathbf{D} = \mathbf{D}_{\mathcal{M},\Sigma}$ the corresponding prederivator, and by

$$W = W_{\mathbf{D}}$$

the associated localizer in Cat, consisting of the "**D**-equivalences". We see at once that W is strongly saturated (true for any $W_{\mathbf{D}}$, $\mathbf{D}$ a prederivator), *i.e.* satisfies Loc (1 ter), a fortiori Loc (1), on the other hand it satisfies Loc (3 bis*), as a consequence of the fact that $\mathbf{D}$ satisfies Der (4*, 5*). A fortiori it satisfies Loc (3*), and to see that every X contractible in Cat is W-aspheric, *i.e.* $X \longrightarrow e$ in W, we know (*cf.* §1) that it suffices to [page 115] prove that this is so if X has an initial object (axiom Loc (2*) dual to axiom Loc (2)). At that point, we will know that W is a basic localizer, and moreover it satisfies Loc (3* bis), hence also Loc (3 bis) (strong form of the localization axiom).

Thus *we must prove* that if S has an initial object, then S is W-aspheric, *i.e.* $S \longrightarrow e$ is in W, *i.e.* $p^* : \mathbf{D}(e) \longrightarrow \mathbf{D}(S)$ is fully faithful. It amounts to the same thing to say that the adjunction arrow

$$p_! p^* \xi \longrightarrow \xi$$

is an isomorphism, for every ξ in $\mathbf{D}(e)$. Now the computation of $p_!(\eta)$ (for $\eta \in \mathrm{Ob}\,\mathbf{D}(S)$, *i.e.* $\eta \in \mathrm{Ob}\,\mathcal{M}(S)$) is done as follows: we choose an arrow

$$\eta' \xrightarrow{\Sigma_S} \eta$$

with η' "cofibrant" (for the relevant cofibration structure on $\mathcal{M}(S)$), and we set [54] [page 116]

$$p_!(\mathrm{ho}_{\Sigma,S}(\eta)) = \mathrm{ho}_\Sigma(p_!(\eta')).$$

When $\eta = p^*(\xi)$, the relevant adjunction arrow in $\mathrm{Ho}_\Sigma = \mathbf{D}(e)$ is obtained by composition

$$p_!(\eta') \longrightarrow p_!(p^*(\xi)) \xrightarrow{\text{adj.}} \xi, \qquad (*)$$

where here $p_!$ is still taken in the ordinary sense ($\mathcal{M}(S) \longrightarrow \mathcal{M}(e) = \mathcal{M}$). Now since S has an initial object 0, we have

$$\begin{aligned} p_!(\eta') &\simeq \eta'(0) \\ p_!(p^*(\xi)) &\simeq p^*(\xi)(0) \simeq \xi, \end{aligned}$$

[53] **N.B.** We shall see below that Loc (6 a, 8, 9) already implies Loc (3 bis).

[54] operation $p_!$ at the level of $\mathcal{M}(S)$.

and the adjunction arrow in $(*)$ is the identity, the first arrow is the arrow $\eta'(0) \longrightarrow \eta(0)$ induced by $\eta' \longrightarrow \eta$, it is in Σ, hence $(*)$ is in Σ, hence it is an isomorphism in $\mathrm{Ho}_\Sigma = \mathbf{D}(e)$, q.e.d.

The same argument works, moreover, if $\mathcal{M}$ is only a category with cofibrations or with fibrations, but then restricting oneself to the case where S is a finite ordered set.

I will set out separately what we have just obtained: p. 55

[page 117]

Proposition 7.7.1. *Let $(\mathcal{M}, W)$ be a category of models, associated to a closed Quillen triple which is* special *(W, C, F). Then the prederivator $\mathbf{D} = \mathbf{D}_{\mathcal{M},W}$ on* Cat *satisfies the axioms* Der (1, 2, 3, 4*, 5*, 6*)*. If moreover the Quillen triple is* very *special (thus in particular $C = \mathrm{Mono}(\mathcal{M})$), then it satisfies in addition* Der (4, 5, 6), i.e. *it is a derivator in the full sense of the term. On the other hand, the localizer $W = W_{\mathbf{D}}$ in* Cat *associated to* **D** *(in the case of a special Quillen triple and no more) is a* basic localizer, *satisfying in addition* Loc (1 ter, 3 bis).

N.B. What is said here for the case of a very special Quillen triple (which does not call upon the theory of basic localizers in Cat) shows once again that if W a basic localizer in Cat satisfies Loc (6′ a, 8, 9), then HOT_W is a genuine derivator. We saw moreover in 7.6 (page 111) that we then have $W = W_{\mathrm{HOT}_W}$ [55], hence it follows: [page 118]

Corollary 7.7.2: *We have the implication*

$$\text{Loc (6′ a, 8, 9)} \implies \text{Loc(1 ter, 3 bis)}\ [56].$$

Let us also note, for the record:

Corollary 7.7.3. *Let $(\mathcal{M}, W)$ be a category of models, associated to a category with cofibrations. Then on the domain* Diag *consisting of finite ordered sets, the prederivator $\mathbf{D} = \mathbf{D}_{\mathcal{M},W}$ satisfies the axioms* Der (1, 2, 3, 4*, 5*, 6*), *and dually. Thus if W is associated both to a cofibration structure and to a fibration structure,* **D** *is a derivator on* Diag.

then the base change arrow [page 119]

$$p^* f_! \longrightarrow f'_! q^*$$

between functors $\mathrm{HOT}_W(X) \longrightarrow \mathrm{HOT}_W(S')$ is an isomorphism.

This is practically trivial, considering, in the case f W-smooth, that if Z over X is W-smooth over X, it is so over S; and in the case p W-proper, using the last statement ("omission") in ⑥, p. 199.

7.8. I finish by putting in the spotlight what, at present, seems to me the main substance of all these "reminders" and as it were the quintessence of the theory of basic localizers developed up to now: p. 56

$$\begin{cases} \text{Loc (3 bis)} & \implies & \text{Der (4*, 5*)} \\ \text{Loc (3 bis, 6 a)} & \implies & \text{Der (1, 2, 3, 4*, 5*, 6*)} \\ \text{Loc (6′ a, 8, 9)} & \implies & \text{all the axioms of derivators (+ Loc (1 ter)).} \end{cases}$$

I consider the axioms Loc (3 bis), Loc (8), Loc (9) as "harmless" axioms - all the localizers one defines by ordinary cohomological or homotopical means, satisfy them. There remains the axiom Loc (6 a) or Loc (6′ a) of W-cofibrations (weak form), which visibly plays a crucial role in the theory of HOT_W. On the other hand, the axioms Loc (4) (decomposition), Loc (5) (connectedness) and Loc (7) and its variants (and of course also Loc (6 b)) do not enter in at all! This is particularly impressive, as regards the axiom [page 120]

[55] But this was under the hypothesis Loc (3 bis)! Beware of the vicious circle.

[56] Loc (3 bis) not proved, probably false.

of W-fibrations Loc (7) and its variants (all equivalent to Loc (7) in the presence of Loc (8), with the sole exception of Loc (7 F)). As we saw, Loc (7) is not at all implied by Loc (6 a, 8, 9) ($\mathbf{W}_0$ does not satisfy it), at most it could be that Loc (7 F) is. The same is true of the connectedness axiom Loc (5), as one sees from the example of $W = \mathrm{Fl}(\mathrm{Cat})$, which satisfies all the axioms seen so far, with the sole exception of Loc (5). As for the [page 121] question of the independence of the axiom Loc (4) from Loc (6 a, 8, 9), we are going to examine this kind of question in the following paragraph.

The set of axioms Loc (6′ a, 8, 9) now seems to me so important, that I feel like giving it a name, and calling a *special* (basic) *localizer* in Cat [57], a basic localizer which satisfies them. These are thus also the ones for which we know that HOT_W is a genuine *derivator.* If one cannot dismiss as "pathological" a structure as rich and delicate as a derivator, it follows at once that one cannot dismiss special localizers in Cat either (even if one affects to look down on the general basic localizers!). Indeed, W is recovered uniquely with the [page 122] help of $\mathbf{D} = \mathrm{HOT}_W$, or of any derivator equivalent to it, as the set of $\mathbf{D}$-equivalences in Cat [58]: it is the basic localizer (corresponding to a notion of "universal" equivalence, *i.e.* in the category Cat of all the elementary "spaces") canonically associated to the derivator. It is therefore by no means a sufficiently fine fixed reference localizer, such as $\mathbf{W}_\infty$, that is fit to play a role!

This brings us again to a question of the kind already raised: if $W \subset \mathbf{W}_\infty$ is the smallest localizer which satisfies Loc (6′ a, 8) (or Loc (3 bis, 6′ a, 8)), can we say a) that W is even *special*, *i.e.* satisfies Loc (9) (accessibility) as well?, and moreover b) do we [page 123] have $W \neq \mathbf{W}_\infty$? I am convinced that this is the case (there is less merit for that W in being different from $\mathbf{W}_\infty$, than for the $\tilde{\mathbf{W}}_\infty$ considered in 6.4 (p. 87)). p. 57

7.9. Omission: question 1 bis (p. 98), on the existence of a closed Quillen structure on Cat, corresponding to W. The only result I know in this direction is that of Thomason, explicitly constructing such a Quillen structure (moreover proper), in the case where $W = \mathbf{W}_\infty$. I have not really looked at the proof of Thomason, and do not know whether it can be extended to more general basic localizers, for example to those satisfying Loc (6′ a, 8, 9). If this were so, it would moreover prove that these localizers have enough [page 124] cofibrations and enough fibrations, hence that they satisfy Loc (6 a) and Loc (7 F). (As for Loc (6 a), it seems that one can prove it directly very simply, *cf.* the following paragraph. But here again, one would have to look at some examples (homological or cohomological equivalences, $\mathbf{W}_n$, *i.e.* the homotopy n-equivalences), and see to begin with whether Loc (7 F) is satisfied.)

It does not seem to me that the theory of Thomason has been up to now [...] to be very useful, technically speaking. Psychologically, I think it has been important, for giving confidence that Cat really is an excellent category of models for Hot. I was really delighted and stimulated to learn of this theorem. But the notion of a Dwyer map (which I learned in 1983 in the article of Thomason) has been, technically, much more important for me: it was exactly what was missing from my panoply of "homotopical" notions and intuitions in Cat …

[57] I hesitate whether I should not add the axiom Loc (3 bis), which ensures that $W = W_{\mathbf{D}}$, $\mathbf{D} = \mathrm{HOT}_W$.
[58] But this is proved only assuming Loc (3 bis).

p. –

THE DERIVATORS

ALEXANDRE GROTHENDIECK

Chapter XVII

Categories with fibrations and cofibrations (Models (3))

p. –
p. 1
[page 1]

1 Preliminaries[1]

Proposition 1. *Let $\mathcal{M}$ be a category stable under arbitrary* $\varinjlim$, $C \subset \mathrm{Fl}(\mathcal{M})$ *containing the isomorphisms, stable under cobase change and under composition* of transfinite length. *Let, for every small ordered set I, $C_I \subset \mathrm{Fl}(\mathcal{M}(I^\circ))$ be the set of arrows $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ such that $\forall\ i_0 \in I$, in the diagram*

$$\begin{array}{ccc} X'_{i_0} = \varinjlim_{i<i_0} X_i & \longrightarrow & X_{i_0} \\ {\scriptstyle \alpha_{i_0} = \varinjlim_{i<i_0} u_i}\downarrow & \text{cocart.} & {\scriptstyle \beta_{i_0}}\downarrow \\ Y'_{i_0} = \varinjlim_{i<i_0} Y_i & \longrightarrow & \tilde{Y}_{i_0} \\ & & {\scriptstyle \tilde{u}_{i_0}}\downarrow \\ & & Y_{i_0}\ , \end{array} \qquad u_{i_0} : \tilde{Y}_{i_0} \longrightarrow Y_{i_0}$$

(defined as an amalgamated sum) (∗)

the arrow $\tilde{u}_{i_0}$ is in C[2]*. I say that if I is* well ordered, *then $u_\bullet \in C_I$ implies*

a) *that $\varinjlim u_i : \varinjlim X_i \longrightarrow \varinjlim Y_i$ is in C, and*

b) $\forall\ i \in I$, $(u_i : X_i \longrightarrow Y_i) \in C$[3].

[1] **N.B.** These preliminaries will be completely taken up again and superseded in section 2. Not to be read!

[2] Condition stable under passage from I to an *open* ordered subset I' of I.

[3] Say also that we have

b′) *If $i' \leq i$, then in the diagram similar to* (∗),

.

Corollary. *Let $W \subset \mathrm{Fl}(\mathcal{M})$ be a mildly saturated localizer, such that $TC \overset{\text{def}}{=} C \cap W$ is stable under cobase change, and under composition* of transfinite length. *Let $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ be in C_I. Then $u_\bullet \in W_{I^\circ}$ (i.e. $\forall\, i \in I$, we have $u_i \in W$) if and only if $u_\bullet \in (TC)_I$, i.e. if and only if for every $i \in I$, the arrow $\tilde{u}_i$ in $(*)$ is in TC, or what amounts to the same thing, in W (since by hypothesis it is already in C). If this condition is satisfied, then $\varinjlim u_i : \varinjlim X_i \longrightarrow \varinjlim Y_i$ is in W.* [page 2 does not exist] p. 2

[page 3]

Proof of the corollary. Suppose $u_\bullet \in (TC)_I$, then by the proposition, $\varinjlim_{i \in I} u_i$ is in TC, or what amounts to the same thing, in W. On the other hand, applying this same result to the direct systems induced on $I_{<i_0}$, we find that in $(*)$, the arrow α_{i_0} is in TC, hence also β_{i_0} by cobase change, hence also $u_{i_0} = \tilde{u}_{i_0}\beta_{i_0}$. So it remains to prove that conversely, if we assume the $u_{i_0} \in W$, then the $\tilde{u}_{i_0}$ are themselves also in W (i.e. in $TC = C \cap W$), or what amounts to the same thing, that the β_{i_0} are in W, or again in TC. But since TC is stable under cobase change, it suffices for this to prove that $\alpha_{i_0} \in TC$, or again $\alpha_{i_0} \in W$. So we are reduced to proving that

c) $u \in W_{I^\circ} \cap C_I \Longrightarrow (\varinjlim_{i<i_0} u_i) \in W$ for every $i_0 \in I$,

so, changing notation, to proving . We will prove a) b) c) together by transfinite induction on the ordinal number α such that $I \simeq I_\alpha = \{\text{set of ordinals } i < \alpha\}$.

[page 4]

1°) Case $\alpha = 0$, i.e. $I = \emptyset$: it is trivial.

2°) If it's ok for I_α, also for $I_{\alpha+1} = \{\beta \mid \beta \leq \alpha\}$. Here, $I_{\alpha+1}$ having a greatest element α, a) is contained in b) (for $i = \alpha$, greatest element of $I_{\alpha+1}$.) To prove b), it remains to prove that $u_\alpha \in C$. We look at the diagram $(*)$ for $i_0 = \alpha$. By the induction hypothesis (form a), applied to I_α, we have $\alpha_{i_0} \in C$, hence by cobase change $\beta_{i_0} \in C$, hence by composition $u_{i_0} = \tilde{u}_{i_0}\beta_{i_0} \in C$. To prove c), it remains to prove that $\varinjlim_{i<\alpha} u_i \in W$. But by the induction hypothesis applied to $I_\alpha = \{i \mid i < \alpha\}$, we already know that $(u_i)_{i<\alpha} \in (TC)_{I_\alpha}$, hence by the proved part of the corollary or of the proposition, $\varinjlim_{i<\alpha} u_i \in TC$.

3°) If α is a limit ordinal, and if a) b) c) hold for I_β with $\beta < \alpha$, they also hold for I_α. This time, b) is contained in the induction hypothesis, we must prove a), i.e. that

[page 5]

$$u_\alpha \overset{\text{def}}{=} \varinjlim_{i<\alpha} u_i \;:\; \underbrace{X_\alpha}_{\overset{\text{def}}{=} \varinjlim_{i<\alpha} X_i} \longrightarrow \underbrace{Y_\alpha}_{\overset{\text{def}}{=} \varinjlim_{i<\alpha} Y_i}$$

is in C. As for c), it is already contained in the induction hypothesis. So it finally remains to prove

$$u_\alpha \;\in\; C$$

when we assume that the proposition is proved for the ordinals $\beta < \alpha$.

$$\begin{array}{ccc} X'_{i'} & \longrightarrow & X'_i \\ \downarrow & \text{cocart.} & \downarrow \\ Y'_{i'} & \longrightarrow & \tilde{Y}'_i \\ & & \downarrow{\scriptstyle u_{i',i}} \\ & & Y'_i\,, \end{array}$$

we have $u_{i',i} \in C$ (we recover $(*)$ by taking $i' = i_0$, $i = i_0 + 1$). Case b) is obtained by taking $i' = 0$, i arbitrary, whence $(X'_i \longrightarrow Y'_i) \in C$. Applying it to $i+1$ instead of i, we find $(u_i : X_i \longrightarrow Y_i) \in C$.

We are going to construct a third ordinal system on I_α, by introducing for every $0 \leq i < \alpha$, i.e. $i \in I_\alpha$, Z_i by the cocartesian square p. 3

$$\begin{array}{ccc} X_i & \longrightarrow & X_\alpha \\ \downarrow & \text{cocart.} & \downarrow \\ Y_i & \longrightarrow & Z_i \ . \end{array}$$

The transition arrows between the Z_i are obvious, and we will have $\varinjlim_i Z_i \xrightarrow{\sim} Y_\alpha$. Thus, $u_\alpha : X_\alpha (= Z_{-1}) \longrightarrow Y_\alpha$ is decomposed into a transfinite composite of morphisms [page 6]

$$Z_i \xrightarrow{v_{i+1}} Z_{i+1} \qquad (-1 \leq i < \alpha) \tag{$*$}$$

and

$$\varinjlim_{i<i_0} Z_i \xrightarrow{v_{i_0}} Z_{i_0} \qquad i_0 \text{ limit ordinal } < \alpha. \tag{$**$}$$

(**N.B.** The case where i_0 is *not* a limit ordinal is none other than $(*)$ for $i = i_0 - 1$.) Everything comes down to proving that these arrows are in C (hence their transfinite composite too).

Case $(*)$.

$$\begin{array}{ccccc} X_i & \longrightarrow & X_{i+1} & \longrightarrow & X_\alpha \\ {\scriptstyle u_i}\downarrow & \text{cocart.} & {\scriptstyle v_{i+1}}\downarrow & \text{cocart.} & \downarrow \\ Y_i & \longrightarrow & \tilde{Y}_{i+1} & \longrightarrow & Z_i \\ & & {\scriptstyle \tilde{u}_{i+1}}\downarrow & \text{cocart.} & \downarrow{\scriptstyle v_{i+1}} \\ & & Y_{i+1} & \longrightarrow & Z_{i+1} \ . \end{array}$$

[4]

Since $\tilde{u}_{i+1} \in C$, $v_{i+1} \in C$ by cobase change, OK. We must however treat separately the case $i = -1$, i.e. the arrow $X_\alpha \longrightarrow Z_0$, which is included in

$$\begin{array}{ccc} X_0 & \longrightarrow & X_\alpha = Z_{-1} \\ {\scriptstyle u_0}\downarrow & \text{cocart.} & \downarrow{\scriptstyle v_0} \\ Y_0 & \longrightarrow & Z_0 \ , \end{array}$$

where $u_0 \in C$, hence $v_0 \in C$, OK.

Case $(**)$. If $i_0 < \alpha$ is a limit ordinal, let us inspect $(**)$. We have by passage to the limit

$$Z'_{i_0} \overset{\text{def}}{=} \varinjlim_{i<i_0} Z_i \ \simeq \ Y'_{i_0} \amalg_{X'_{i_0}} X_\alpha$$

(cf. diagram $(*)$ for the definition of X'_{i_0}, Y'_{i_0}), fitting into the following diagram grafted onto $(*)$

$$\begin{array}{ccccc} X'_{i_0} & \longrightarrow & X_{i_0} & \longrightarrow & X_\alpha \\ {\scriptstyle \alpha_{i_0}}\downarrow & \text{cocart.} & {\scriptstyle \beta_{i_0}}\downarrow & \text{cocart.} & \downarrow \\ Y'_{i_0} & \longrightarrow & \tilde{Y}_{i_0} & \longrightarrow & Z'_{i_0} \\ & & {\scriptstyle \tilde{u}_{i_0}}\downarrow & \text{cocart.} & \downarrow{\scriptstyle v_{i_0}} \\ & & Y_{i_0} & \longrightarrow & Z_{i_0} \ . \end{array}$$

p. 4 [page 7]

[4] To give oneself confidence – one could treat the general case right away.

We have $\tilde{u}_{i_0} \in C$ by hypothesis, hence $v_{i_0} \in C$, which completes the proof.

Proposition 2. *$(\mathcal{M}, C)$ and where applicable W, as in proposition 1 and its corollary, I a locally finite ordered set, i.e. such that $\forall\ i_0 \in I$, I/i_0 is finite. Let*

$$u_\bullet\ : X_\bullet \longrightarrow Y_\bullet$$

an arrow in[5] *$\mathcal{M}(I^\circ)_{\mathrm{c}}$ = full subcategory of direct systems on I, with values in $\mathcal{M}$, such that $\underbrace{\emptyset_{\mathcal{M}(I^\circ)}}_{\substack{\text{initial}\\ \text{object}\\ \text{of } \mathcal{M}(I^\circ)}} \longrightarrow Z_\bullet$ is in C_I (defined in prop. 1), i.e. such that $\forall\ i_0 \in I$, the arrow*

$$Z'_{i_0}\ =\ \varinjlim_{i<i_0} Z_i\ \longrightarrow\ Z_{i_0}$$

is in C, and suppose $u_\bullet \in C_I$. Then

a) $\varinjlim_{i\in I} u_\bullet : X_I \overset{\text{def}}{=} \varinjlim_{i\in I} X_i \longrightarrow Y_I \overset{\text{def}}{=} \varinjlim_{i\in I} Y_i$ *is in C,* [page 8]
and

b) *$\forall\ i_0 \in I$, we have $\alpha_{i_0}, \beta_{i_0} \in C$ in diagram $(*)$ p. 1.*

Corollary. *Moreover (assuming W given as in prop. 1, cor. 1), the following conditions on a $u_\bullet$ between C-cofibrant $X_\bullet$, $Y_\bullet$ are equivalent.*

1°) *$u_\bullet \in (TC)_I$, i.e. the $\tilde{u}_{i_0} \in TC$ ($\forall\ i_0 \in I$).*

2°) *$u_\bullet \in C_I \cap W_{I^\circ}$, i.e. $\forall\ i_0 \in I$, $\tilde{u}_{i_0} \in C$, $u_{i_0} \in W$.*

3°) *For every $i_0 \in I$, we have*

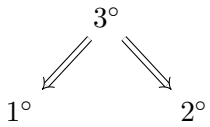
$$\alpha_{i_0},\ \beta_{i_0},\ \tilde{u}_{i_0},\ u_{i_0}\ \in\ TC$$

(i.e. $\alpha_{i_0}, \tilde{u}_{i_0} \in TC$, which implies $\beta_{i_0} \in TC$, then $u_{i_0} \in TC$ by cobase change and by composition).

And these conditions imply

4°) *that $u_I : X_I \longrightarrow Y_I$ is in TC.*

Let us see how or to what extent the corollary follows from the proposition. We have of course p. 5



tautologically. Besides, the proposition applied to TC instead of C shows that we have $1° \Longrightarrow 3°$ & 4°, for by applying it to the sets $I_{<i_0}$ we find that $\alpha_{i_0} \in TC\ \forall\ i_0 \in I$. So proposition 2 implies

$$(1° \Longleftrightarrow 3°) \Longrightarrow 2°\ (\&\ 4°).$$

So to prove $2° \Longrightarrow 3°$, it would remain to prove that 2° suffices to imply 4° (and this for any locally finite ordered set I) – for applying this to the $I_{<i_0}$, we find that 2° implies [page 9]

[5] in fact, the hypothesis that $X_\bullet$, $Y_\bullet$ in $\mathcal{M}(I^\circ)_{\mathrm{c}}$ is unnecessary, both in proposition 2 and in its two corollaries.

$\alpha_{i_0} \in TC = C \cap W$ (not only $\alpha_{i_0} \in C$, according to proposition 2), hence $\beta_{i_0} \in TC \subset W$, hence $\tilde{u}_{i_0} \in W$ by saturation, hence $\tilde{u}_{i_0} \in TC$, whence 3°.

Note on the other hand that b) in proposition 2 is a consequence of a), applied to the restriction of $u_\bullet$ to $I_{<i_0}$. All in all, there are therefore only the two things to prove

$$\begin{cases} \text{a)} \quad u_\bullet \in C_I \implies u_I \in C \\ \text{b)} \quad u_\bullet \in C_I \cap W \implies u_I \in C \cap W \ (\overset{\text{def}}{=} TC) \\ \qquad\qquad\qquad\quad (\text{and } u_\bullet \in (TC)_I) \end{cases} \tag{$*$}$$

We shall prove this by reduction to the special case of proposition 1, already proved.

Besides, I also feel like proving the

Corollary 2. *$\mathcal{M}$, C (and possibly W). The hypotheses on $X_\bullet, Y_\bullet \in \mathcal{M}(I^\circ)_c$ and on $u_\bullet \in C_I$, finally on I locally finite, are those of proposition 2. For every subset J of I, let*

$$u_J \overset{\text{def}}{=} \varinjlim_{i \in J} u_i \ : \ X_J \overset{\text{def}}{=} \varinjlim_{i \in J} X_i \longrightarrow Y_J \overset{\text{def}}{=} \varinjlim_{i \in J} Y_i \ .$$

Let $I' \subset I''$ be two subsets open *in I, and consider the diagram*

[page 10]

$$\begin{array}{ccc} X_{I'} & \xrightarrow{\tau_X} & X_{I''} \\ {\scriptstyle \alpha = u_{I'}}\downarrow & \text{cocart.} & \downarrow{\scriptstyle \beta} \\ Y_{I'} & \xrightarrow{\tilde{\tau}_Y} & \tilde{Y}_{I''} \\ & {\scriptstyle \tau_Y}\searrow & \downarrow{\scriptstyle \tilde{u}} \\ & & Y_{I''} \ . \end{array} \qquad u_{I''} \qquad \text{(defined as an amalgamated sum)} \tag{$*$}$$

Then we have

p. 6

$$\begin{cases} u_{I'}, u_{I''} \in C \text{ (hence also } \beta \in C) \text{ (i.e. for every } J \text{ open in } I, \\ u_J : X_J \longrightarrow Y_J \text{ is in } C)^6, \text{ and } \tau_X, \tau_Y \in C \text{ (hence also } \tilde{\tau}_Y \in C) \\ \text{(i.e. for } I' \subset I'' \text{ an inclusion of open subsets, and } X_\bullet \text{ } C\text{-cofibrant in } \mathcal{M}(I^\circ), \\ \text{we have } X_{I'} \longrightarrow X_{I''} \text{ in } C)^7, \text{ and finally, even} \\ \qquad\qquad \boxed{\tilde{u} \in C.} \end{cases}$$

N.B. The case of proposition 2 in the form ($*$) a), page 9, (i.e. also the fact that the $u_{I'}$, $u_{I''}$, … are in C) is obtained by taking $I'' = I$, $I' = \emptyset$, the diagram becomes

$$\begin{array}{ccc} \emptyset & \longrightarrow & X_I \\ {\scriptstyle \text{id}}\downarrow & \text{cocart.} & \downarrow{\scriptstyle \text{id}} \\ \emptyset & \longrightarrow & X_I \\ & & \downarrow{\scriptstyle \tilde{u} = u_I} \\ & & Y_I \ , \end{array}$$

[6] This is of course contained in prop. 2, applied to $u_\bullet|_J$.

[7] It is for this assertion $\tau_X \in C$ (whence $\tilde{\tau}_Y, \tau_Y \in C$, provided $\tilde{u} \in C$) that we need the hypothesis $X_\bullet \in \mathcal{M}(I^\circ)_c$, $\tilde{u} \in C$ true without this hypothesis.

and the conclusion is therefore $u_I \in C$[8]. The *definition* of C_I, i.e. the condition $u \in C_I$, is recovered by requiring $\tilde{u} \in C$ in $(*)$ above, in the case $I'' = I/i_0$, $I' = I_{<i_0}$. It is remarkable that this definition of C_I gives a conclusion as strong as $\tilde{u} \in C$ for *every* pair [page 11] $I' \subset I''$ of open subsets of I.

It is in the end in this form that we are going to prove proposition 2 – it is *this* now that seems to me *the* right statement in this direction. With the supplement, of course, that if moreover $u_\bullet \in W_{I^\circ}$, i.e. the u_i are in W, then $u_\bullet$ is even in $(TC)_I$, or again simply [p. 7] that the $u_{I'}$ *are in* TC (whence it will therefore follow that $\tilde{u}$ in $(*)$, p. 10, is also in TC).

We are going to proceed by transfinite construction. We may of course suppose that $I'' = I$. We are nevertheless going to prove first the special case where $I' = \emptyset$, i.e. prove at least that $u_I \in C$. We construct by transfinite induction a strictly increasing sequence of open subsets

$$I_\alpha \ \subset \ I,$$

ending at

$$I_{\alpha_0} \ = \ I,$$

by setting

$$\begin{cases} I_0 \ = \ \emptyset\,, & \\ I_\alpha \ = \ \bigcup_{\beta<\alpha} I_\beta & \text{if } \alpha \text{ a limit ordinal}\,, \\ I_{\alpha+1} \ = \ I_\alpha \cup (I/s_\alpha) & \text{if } I_\alpha \neq I, \text{ and choosing } s_\alpha \in I \backslash I_\alpha\,. \end{cases}$$

(We end up having $I_{\alpha_0} = I$, of course.) We set [page 12]

$$\begin{aligned} X_\alpha \ = \ X_{I_\alpha} \ &= \ \varinjlim_{i \in I_\alpha} X_i \\ Y_\alpha \ = \ Y_{I_\alpha} \ &= \ \varinjlim_{i \in I_\alpha} Y_i \\ u_\alpha \ = \ u_{I_\alpha} \ &: \ X_\alpha \longrightarrow Y_\alpha\ , \end{aligned}$$

homomorphism of transfinite direct systems. We shall have

$$X_{\alpha_0} \ = \ X_I, \qquad Y_{\alpha_0} \ = \ Y_I, \qquad u_{\alpha_0} \ = \ u_I,$$

and we must therefore prove that $u_{\alpha_0} \in C$. For this, we apply proposition 1, so it suffices to prove that $(u_\alpha) \in C_{I_{\alpha_0+1}}$ (where $I_{\alpha_0+1} = \{\text{ordinals } i \mid i < \alpha_0 + 1, \text{ i.e. } i \leq \alpha_0\}$). So we must prove that the arrows $\tilde{u}_{i_0}$ in $(*)$ p. 1 (where this time the i, i_0 are ordinals, $i_0 \leq \alpha_0$) are in C. I am going to distinguish the two cases where $\alpha = i_0$ is a limit ordinal, which is trivial for then $X_{i_0} \xleftarrow{\sim} \varinjlim_{i<i_0} X_i$, $Y_{i_0} \xleftarrow{\sim} \varinjlim_{i<i_0} Y_i$, and $\tilde{u}_{i_0} = \mathrm{id}$, and the case α not a

[8] The case $(Y_{I'} \xrightarrow{\tau_Y} Y_{I''}) \in C$ is obtained by taking $X_\bullet = \emptyset_\bullet$, we shall have

$$\begin{array}{ccc} \emptyset & \xrightarrow[\sim]{\tau_X} & \emptyset \\ \downarrow & \text{cocart.} & \downarrow \\ Y_{I'} & \xrightarrow{\mathrm{id}} & \tilde{Y}_{I''} = Y_{I'} \\ & \searrow^{\tau_Y} & \downarrow{\scriptstyle \tilde{u}=\tau_Y} \\ & & Y_{I''}\ . \end{array}$$

limit, i.e.

$$\begin{array}{ccc} X_{I_{\alpha-1}} & \longrightarrow & X_{I_\alpha} \\ {\scriptstyle u_{\alpha-1}}\downarrow & & {\scriptstyle \beta_\alpha}\downarrow \\ Y_{I_{\alpha-1}} & \longrightarrow & \tilde{Y}_{I_\alpha} \\ & & {\scriptstyle \tilde{u}_\alpha}\downarrow \\ & & Y_{I_\alpha}\ , \end{array} \qquad u_\alpha$$

to prove that $\tilde{u}_\alpha \in C$. When we suppose in addition $u_\bullet \in W_{I^\circ}$, i.e. the $u_i \in W$, we must in addition prove that $\tilde{u}_\alpha \in TC = C \cap W$. Proposition 2 and its corollary 1 will follow from this. To prove $\tilde{u}_\alpha \in C$ resp. $\tilde{u}_\alpha \in TC$, we may suppose $I_\alpha = I$ (changing notation if need be). We therefore set

p. 8 [page 13]

$$\begin{aligned} I_\alpha &= I, \qquad I_{\alpha-1} = I' \\ I/s_{\alpha-1} &= I_0, \quad I_0 \cap I' = I'_0\ , \end{aligned}$$

so that we have a commutative diagram of open inclusions

$$\begin{array}{ccc} I_0 \cap I' = I'_0 & \hookrightarrow & I_0 \\ \downarrow & & \downarrow \\ I' & \hookrightarrow & I = I_0 \cup I'. \end{array}$$

We shall use only the fact that I_0, hence I'_0, is finite, i.e. that in the pair $(I' \subset I)$, I can be written in the form $I = I' \cup I_0$, with I_0 an open subset that is *finite,* to conclude from this that in the diagram

$$\begin{array}{ccc} X_{I'} & \xrightarrow{\tau^X_{I',I}} & X_I \\ {\scriptstyle u_{I'}}\downarrow & \text{cocart.} & {\scriptstyle v_{I',I}}\downarrow \\ Y_{I'} & \longrightarrow & \tilde{Y}_I \\ & {\scriptstyle \tau^Y_{I',I}}\searrow & {\scriptstyle \tilde{u}=u_{I',I}}\downarrow \\ & & Y_I \end{array} \qquad u_I$$

we have $\tilde{u} \in C$ resp. $\tilde{u} \in TC = C \cap W$.

We need general notation, on the one hand for an inclusion $J \subset K$ of open subsets of I, giving rise to the diagram

[page 14]

$$\begin{array}{ccc} X_J & \xrightarrow{\tau^X_{J,K}} & X_K \\ {\scriptstyle u_J}\downarrow & \text{cocart.} & {\scriptstyle \tilde{u}_{J,K}}\downarrow \\ Y_J & \longrightarrow & Y_{J,K} \\ & {\scriptstyle \tau^Y_{J,K}}\searrow & {\scriptstyle u_{J,K}}\downarrow \\ & & Y_K\ , \end{array} \qquad u_K \qquad \text{(formerly } \tilde{Y}_K\text{)} \tag{1}$$

and for inclusions of three nested open subsets $J \subset K \subset L$ of I,

p. 9

$$\begin{array}{ccccc}
X_J & \longrightarrow & X_K & \longrightarrow & X_L \\
{\scriptstyle u_J}\downarrow & \text{cocart} & {\scriptstyle \tilde{u}_{J,K}}\downarrow & \text{cocart} & {\scriptstyle \tilde{u}_{J,L}}\downarrow \\
Y_J & \longrightarrow & Y_{J,K} & \longrightarrow & Y_{J,L} \\
 & & {\scriptstyle u_{J,K}}\downarrow & \text{cocart} & {\scriptstyle u^L_{J,K}}\downarrow \\
 & & Y_K & \longrightarrow & Y_{K,L} \\
 & & & & {\scriptstyle u_{K,L}}\downarrow \\
 & & & & Y_L\ ,
\end{array}
\qquad
\begin{array}{l}
\\
\\
\\
\\
\tilde{u}_{K,L} \\
\\
u_{J,L}
\end{array}
\tag{2}$$

giving rise to a new arrow $u^L_{J,K}$, deduced from $u_{J,K}$ by the cobase change $Y_{J,K} \longrightarrow Y_{J,L}$. [page 15]
We shall use this diagram in the two cases where $(J \subset K \subset L)$ is one of the two triples

$$\begin{cases} I'_0 \hookrightarrow I_0 \hookrightarrow I \\ I'_0 \hookrightarrow I' \hookrightarrow I, \end{cases}$$

arising from the cocartesian square (in Cat) (MV square)

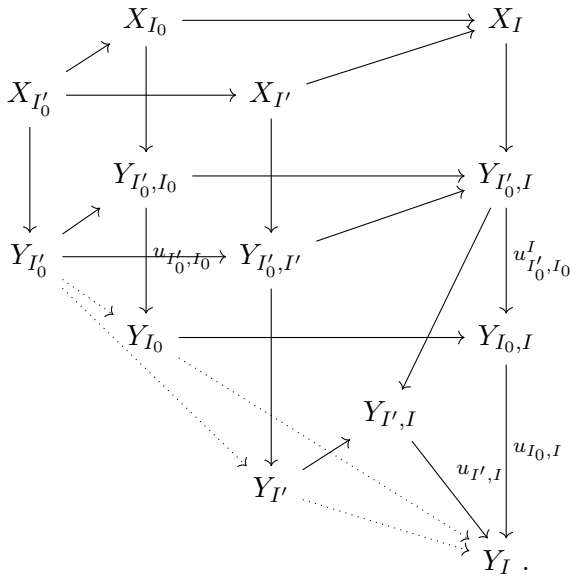
$$\begin{array}{ccc}
I'_0 & \hookrightarrow & I_0 \\
\downarrow & & \downarrow \\
I' & \hookrightarrow & I\ .
\end{array}
\tag{3}$$

We must insert the two corresponding diagrams of type $(*)$ into one and the same diagram. We shall note in this connection that the squares in $\mathcal{M}$ deduced from (3) by applying to it $J \longmapsto X_J$ and $J \longmapsto Y_J$

$$\left.\begin{array}{ccc}
(4_X) & \begin{array}{ccccc}
X_{I'_0} & & \longrightarrow & & X_{I_0} \\
 & \searrow & & & \\
 & & X_{I'} & \xrightarrow{u_{I_0}} & X_I \\
{\scriptstyle u_{I'_0}}\downarrow & & \downarrow & & \downarrow \\
Y_{I'_0} & & \xrightarrow{u_{I'}} & & Y_{I_0} \\
 & \searrow & & & \\
 & & Y_{I'} & \longrightarrow & Y_I
\end{array} & \begin{array}{c} \\ \\ \\ u_I \\ \\ \\ \end{array} \\
(4_Y) & & & &
\end{array}\right\} \ (4)$$

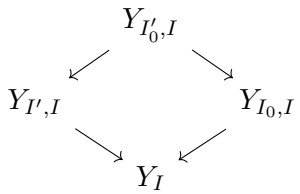
are cocartesian in $\mathcal{M}$. (Transitivity of the $\varinjlim$ …).

We have a homomorphism $(u_{I'_0}, u_{I_0}, u_{I'}, u_I)$ deduced from u from the square (4_X) [page 16]
into the square (4_Y), and this homomorphism too fits into a diagram, including the two avatars of (2) already mentioned. Here is the relevant diagram: p. 10

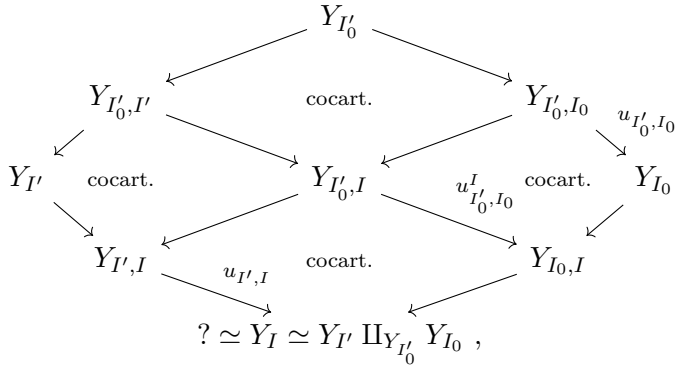


(5 [(9)])

The upper cube is the cube in $\mathcal{M}$ with 6 cocartesian faces deduced from the three arrows from $X_{I'_0}$ to X_{I_0}, $X_{I'}$, $Y_{I'_0}$. We shall need to know that the *base* of the cube is a cocartesian square. The front part of the diagram is the avatar of (2) corresponding to the inclusions $I'_0 \subset I' \subset I$, the back part the avatar for the inclusion $I'_0 \subset I_0 \subset I$. So the two other vertical squares, below the upper cube, are also cocartesian, as is the slanting dotted square, which is none other than the base (4_Y) of diagram (4). From these features of diagram (5) it follows easily (although I put in a good day of hit-or-miss scribbling, before finally realizing it …) that the diamond-shaped square in solid lines [page 17]



is also cocartesian. We see it trivially on the unfolded diagram p. 11



(6)

which is finally the relevant diagram (diagram (5) was of no use, it is a wrong end of things that I took hold of). We see on diagram (6) that to prove $u_{I',I} \in C$ resp. TC, it suffices to prove $u_{I'_0,I_0} \in C$ resp. TC. But this is an assertion concerning the inclusion of [page 18]

[9] unfathomable diagram, to be replaced by diagram (6) below.

ordered sets $I'_0 \subset I_0$ that are *finite.* We are thus reduced to proving the following lemma, where I_0 becomes I:

Lemma. *Let I be a finite ordered set, $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ an arrow in $\mathcal{M}(I^\circ)$, where $\mathcal{M}$ is a category equipped with a set $C \subset \mathrm{Fl}(\mathcal{M})$ of arrows such that C is stable under cobase change and under composition, and contains the isomorphisms. We suppose $u_\bullet \in C_I$, i.e. that for every $i \in I$, the arrow $u_{I_{<i},I/i}$ is in C. Then for every open subset I' of I, the arrow $u_{I',I}$ is in C* [(10)].

Corollary. *If moreover $W \subset \mathrm{Fl}(\mathcal{M})$ is a mildly saturated localizer, such that $TC \overset{\text{def}}{=} W \cap C$ is stable under cobase change (and is so in any case under composition, and contains the isomorphisms), then we have*

$$C_I \cap W_{I^\circ} \;=\; (TC)_I,$$

thus if $u_\bullet \in C_I$ (cf. above) and the $u_i \in W$, then $u_\bullet \in (TC)_I$, hence applying what is said for C to TC, we have $u_{I',I} \in TC$ for every open subset I' of I.

[page 19]

We shall prove the lemma and its corollary at the same time, proceeding by induction on $\operatorname{card} I = n$. If $n = 0$, i.e. $I = \emptyset$, the statements are trivial. By transitivity, using diagrams of type (2), we are reduced to the case where $I = I' \cup \{i_0\}$, where $i_0 \in I \backslash I'$. We again set $I_0 = I/i_0$, $I'_0 = I_0 \cap I'$, and the reasoning we have just carried out shows us that it suffices to prove that $u_{I'_0,I_0} \in C$ resp. $\in TC$. If $I_0 \neq I$, this follows from the induction hypothesis. Otherwise, i_0 is a final object of I, and $I' = I_{<i_0}$, and then $u_{I',I} \in C$ is included in the definition of C_I. Remains then to prove that $u_{I',I} \in TC$ if the $u_i \in W$. We have the diagram

p. 12

$$\begin{array}{ccc} X_{I'} & \longrightarrow & X_I \\ {\scriptstyle u_{I'}}\downarrow & \text{cocart} & \downarrow{\scriptstyle \tilde{u}_{I',I}} \\ Y_{I'} & \longrightarrow & Y_{I',I} \\ & & {\scriptstyle u_{I',I}}\downarrow \\ & & Y_I \end{array} \qquad u_I$$

and the induction hypothesis implies $u_{I'} \in TC$, hence $\tilde{u}_{I',I} \in TC$ by cobase change. On the other hand, the arrow $u_I : X_I \longrightarrow Y_I$ identifies with $X_{i_0} \longrightarrow Y_{i_0}$ (i_0 being a final element of I), so it is in W. Thus by saturation, $u_{I',I} \in W$, hence $u_{I',I} \in TC$, q.e.d.

[page 20]

Thus, proposition 2 is proved, as well as its corollary 1 – and we see that if in proposition 2 and its corollary, we suppose I *finite,* then it is unnecessary in the hypotheses for C, resp. $TC = C \cap W$, to suppose stability under composition that is *transfinite,* ordinary composition suffices. But it remains to prove corollary 2, which I have just proved in the case I *finite,* without having to appeal to a "transfinite" hypothesis. We must now prove that it is still OK in the locally finite case, supposing C stable under transfinite composition. (There is no longer any need to return to the case of W, for we now already know that $C_I \cap W_{I^\circ} = (TC)_I$, by corollary 1 already proved.)

The proof is essentially the same as for the special case already treated $I' = \emptyset$, $I'' = I$, and we could just as well have written up the proof in this setting, with the help of a strictly increasing transfinite sequence of open subsets in I, I_α, with $\boxed{I_0 = I',}$ $I_{\alpha_0} = I''$ $(= I)$, and

[page 21]

$$\begin{aligned} I_\alpha &= I_{\alpha-1} \cup I/s_{\alpha-1} && \text{if } \alpha \text{ nonlimit, } I_{\alpha-1} \neq I, \text{ choosing } s_{\alpha-1} \in I \backslash I_{\alpha-1}, \\ I_\alpha &= \textstyle\bigcup_{\beta<\alpha} I_\beta && \text{if } \alpha \text{ limit,} \end{aligned}$$

[10] **N.B.** Of course, $u_{J,K} \in C$ if $J \subset K$ are two open subsets of I, as we see by applying the lemma to K instead of I.

and again setting

$$X_\alpha \;=\; X_{I_\alpha}, \qquad Y_\alpha \;=\; Y_{I_\alpha}, \qquad u_\alpha = u_{I_\alpha} \;:\; X_\alpha \longrightarrow Y_\alpha.$$

Let α_0 be such that $I_{\alpha_0} = I$ (terminal index of the sequence), we must prove that in

$$\begin{array}{ccc} X_0 & \longrightarrow & X_{\alpha_0} \\ \downarrow & & \downarrow \\ Y_0 & \longrightarrow & Y_{0,\alpha_0} \\ & & \downarrow{\scriptstyle u_{0,\alpha_0}} \\ & & Y_{\alpha_0}\ , \end{array}$$

we have

p. 13

$$u_{0,\alpha_0} \;\in\; C.$$

[page 22]

Taking into account the marginal supplement to proposition 1 (which is proved in the same way …), we are reduced to proving that the homomorphism of the ordinal systems under consideration is in $C_{I_{\alpha_0+1}}$ (where I_{α_0+1} is the set of ordinals $\leq \alpha_0$). Now, that this is so is proved exactly as earlier, except that we must first make a verification of (∗) p. 1 ($\tilde{u}_\bullet \in C$) for $i_0 = 0$. Then diagram (∗) p. 1 is

$$\begin{array}{ccc} \emptyset & \longrightarrow & X_0 = X_{I'} \\ \downarrow{\scriptstyle \text{iso}} & & {\scriptstyle \text{iso}}\downarrow{\scriptstyle \text{id}} \\ \emptyset & \longrightarrow & X_0 = X_{I'} \\ & & \downarrow{\scriptstyle u_0 = u_{I'}} \\ & & Y_0 = Y_{I'}\ , \end{array}$$

and we must therefore verify $u_0 = u_{I'} \in C$. Now this is contained in proposition 2 itself, already proved, which we must therefore use. The rest is proved as before. Thus, corollary 2 of proposition 2 is also proved.

Comments: There is an uneasiness, for proposition 1 and proposition 2 (as well as their corollaries) have essentially the same conclusion, with hypotheses on I which are different: well-ordered in one case, locally finite in another. The natural idea is to give a common statement, which covers both at once, with I artinian, i.e. every decreasing sequence is stationary. What is needed is a hypothesis on I on the one hand, on $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ in $\mathcal{M}(I^\circ)$ on the other hand, which ensures this:

[page 23]

(∗) $\forall\ i_0 \in I$, and every open subset J' of $J = I/i_0$, the arrow $u_{J',J}$ in the diagram

$$\begin{array}{ccc} X_{J'} & \longrightarrow & X_J \\ {\scriptstyle u_{J'}}\downarrow & {\scriptstyle \text{cocart}}\tilde{\Psi}_{J',J}\downarrow & \searrow \\ Y_{J'} & \longrightarrow & Y_{J',J} \quad u_J \\ & & {\scriptstyle u_{J',J}}\downarrow \\ & & Y_J \end{array}$$

is in C.

p. 14

Question: is it correct that it suffices, for the validity of (∗), to suppose

[page 24]

a) $u_\bullet \in C_I$, i.e. $u_{J',J} \in C$ when $J = I/i_0$, $J' = J\backslash\{i_0\}$, and

b) *I locally finite?*

Lemma. *Let $P \subset I$ be the set of $i_0 \in I$ such that for every open subset K of $J = I/i_0$, the arrow $u_{K,J}$ is in C. This said, let $i_0 \in I$ be such that every $i < i_0$ is in P. Then $i_0 \in P$.* (We assume $u_\bullet \in C_I$, but I arbitrary.)

Indeed, let $K \subset J = I/i_0$ be an open subset. If $K = J$, it is trivial that $u_{K,J} \in C$ (it is an isomorphism). If $K \neq J$, we have

$$K \ \subset \ J' \overset{\text{def}}{=} J\backslash\{i_0\} \ \subset \ J,$$

and to prove that $u_{K,J} \in C$, it suffices to prove that $u_{K,J'}, u_{J',J} \in C$. Now $u_{J',J} \in C$ by the hypothesis $u_\bullet \in C_I$. And $u_{K,J'} \in C$ by what has been proved for the case of J', taking into account that by the hypothesis on i_0, every $i \in J'$ (i.e. every $i < i_0$) is in P.

Corollary. *If I is artinian, $P = I$.*

For any subset P of I having the property just stated ($I_{<i_0} \subset P \Longrightarrow i_0 \in P$) is equal to I. If indeed there were $x_0 \notin P$, there would exist $x_1 < x_0$ with $x_1 \notin P$, and by induction we would construct a strictly decreasing sequence in I, absurd.

[page 25]

2 Description of $C_I \subset \mathrm{Fl}(\mathcal{M}(I^\circ))$, for I an ordered set

Let $\mathcal{M}$ be a category, stable under (small) $\varinjlim$, $C \subset \mathrm{Fl}(\mathcal{M})$ ("cofibrations"). We assume

$$\begin{cases} \text{a)} \quad C & \text{contains the isomorphisms and is stable} \\ & \text{under cobase change.} \\ \text{b)} \quad C & \text{stable under transfinite composition.} \end{cases} \tag{2.1}$$

Let I be a (small) ordered set, and consider

$$\mathcal{M}(I^\circ) \ = \ \underline{\mathrm{Hom}}(I, \mathcal{M}),$$

we are going to define a subset

$$C_I \ \subset \ \mathrm{Fl}(\mathcal{M}(I^\circ))$$

as follows, for $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ an arrow in $\mathcal{M}(I^\circ)$:

$$u_\bullet \in C_I \quad \Longleftrightarrow \quad \begin{array}{l} \forall\ i_0 \in I \text{ and for every open subset } J \subset I/i_0 = K, \\ \text{the arrow } u_{J,K} \text{ is in } C. \end{array} \tag{2.2}$$

Here is the definition of $u_{J,K}$. For every open subset J of I, we set

$$\begin{cases} X_J \ = \ \varinjlim_{i \in J} X_i, & Y_J \ = \ \varinjlim_{i \in J} Y_i, \\ u_J : X_J \ \longrightarrow \ Y_J, & u_J \ = \ \varinjlim_{i \in J} u_i, \end{cases} \tag{2.3}$$

and we consider, for $K \subset J$, the diagram

p. 15

[page 26]

$$\begin{array}{ccc} X_K & \xrightarrow{\tau^X_{K,J}} & X_J \\ {\scriptstyle u_K}\downarrow & \text{cocart.} & \downarrow{\scriptstyle \tilde{u}_{K,J}} \\ Y_K & \xrightarrow[\tau^Y_{K,J}]{} & Y_{K,J} \overset{u}{=} Y_K \amalg_{X_K} X_J \\ & & {\scriptstyle u_{K,J}}\downarrow \\ & & Y_J \ . \end{array} \tag{2.4}$$

Theorem 2.1. *Let $\mathcal{M}$ be a $\mathfrak{U}$-category stable under small $\varinjlim$, $C \subset \mathrm{Fl}(\mathcal{M})$, satisfying the stability conditions* (2.1) *(under cobase change and under transfinite composition), I a small ordered set, $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ an arrow in $\mathcal{M}(I^\circ)$.*

1°) *We assume $u_\bullet \in C_I$ (cf.* (2.2)*). Let $K \subset J$ be two open subsets of I. Then $u_{K,J}$ (defined in* (2.4)*) is in C.*

2°) *Suppose I is artinian. In order to have $u_\bullet \in C_I$, i.e. for $u_{K,J}$ to be in C for every pair $(K \subset J)$ of open subsets such that J has a greatest element, i.e. is of the form I/i_0 (i.e. also, by 1°), for* every *pair of open subsets $K \subset J$), (it is necessary and) it suffices that this be so in the particular case where $J = I/i_0$, $J' = I'/i_0 \overset{\text{def}}{=} I/i_0 - \{i_0\} = \{i \in I \mid i < i_0\}$.*

[page 27]

3°) *Suppose again I is artinian*[11]*. Let $W \subset \mathrm{Fl}(\mathcal{M})$ be a mildly saturated localizer, such that $TC \overset{\text{def}}{=} C \cap W$ also satisfies the stability conditions* (2.1) *(under cobase change, under transfinite composition). The following conditions on an arrow $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ in $\mathcal{M}(I^\circ)$ are equivalent.*

a) *$u_\bullet \in (TC)_I$, i.e. for every $i_0 \in I$ and K an open subset of $J = I/i_0$, the arrow $u_{K,J}$ is in TC.*

b) *$u_\bullet \in C_I \cap W_{I^\circ}$, i.e. for every $i_0 \in I$, we have $u_{i_0} \in W$, and for every open subset K of $I/i_0 = J$, we have $u_{K,J} \in C$.*

Let us give the main corollaries right away.

Corollary 2.2. *Suppose $(u_\bullet : X_\bullet \longrightarrow Y_\bullet) \in C_I$. Then*

a) *$\forall\ i \in I$, $u_i : X_i \longrightarrow Y_i$ is in C.*

b) *The arrow*

$$u_I \ : \ X_I \longrightarrow Y_I \tag{2.2.1}$$

is in C. More generally, for every open subset J of I, the arrow

$$u_J \ : \ X_J \longrightarrow Y_J \tag{2.2.2}$$

is in C.

For a), we apply 1°) (or simply the *definition* of C_I) in the case of the inclusion of open subsets $\emptyset \hookrightarrow I/i$, since we have

p. 16
[page 28]

$$u_{\emptyset,I/i} \ \simeq \ u_i \ .$$

For b), we apply 1°) to the inclusion $\emptyset \longrightarrow I$, since

$$u_{\emptyset,I} \ \simeq \ u_I \ ,$$

and more generally

$$u_{\emptyset,J} \ \simeq \ u_J \ .$$

Corollary 2.3. *Suppose $u_\bullet \in C_I \cap W_{I^\circ}$. Then for every inclusion of open subsets $K \subset J$ of I, we have $u_{K,J} \in TC$, and in particular (taking $K = \emptyset$), we have $u_J \in TC$. Finally, for every $i \in I$, we have $u_i \in TC$.*

Corollary 2.4. *Let $X_\bullet$ in $\mathcal{M}(I^\circ)_{\mathrm{c}}$ be a C-cofibrant direct system, i.e. such that $\emptyset_I \longrightarrow X_\bullet$ is in C_I, i.e. such that for every $i_0 \in I$ and every open subset K of $J = I/i_0$,*

[11] I do not know whether this hypothesis is necessary, cf. comment p. 39.

the transition arrow $\tau_{K,J} : X_K = \varinjlim_{i\in K} X_i \longrightarrow X_J \simeq X_{i_0}$ *is in* C. *Then for* every *pair of open subsets* $K \subset J$, *the arrow*

$$\tau_{K,J} \ : \ \varinjlim_{i\in K} X_i \longrightarrow \varinjlim_{i\in J} X_i$$

is in C[12]. *In particular (taking* $K = I/k$, $J = I/j$ *for* $k \leq j$*), if* $k \leq j$ *are in* I, *the transition morphism*

$$\tau_{k,j} \ : \ X_k \longrightarrow X_j$$

is in C. *Finally, if* $W \subset \mathrm{Fl}(\mathcal{M})$ *is as in* (2.1. 3°), *and if* $\emptyset_I \longrightarrow X$ *is in* W_{I°, *i.e. the* [page 29]
X_i *are* W*-coaspheric, then* $X_\bullet$ *is in* $\mathcal{M}(I^\circ)_{TC}$ *(where* $TC = C \cap W$*), and for every open subset* J *of* I, $X_J = \varinjlim_{i\in J} X_i$ *is in* $\mathcal{M}_{TC}$, *hence* X_J *is* W*-coaspheric, and for every pair of open subsets* $K \subset J$, *the transition arrow* $\tau_{K,J}$ *is in* TC.

Let us point out finally the following complement to theorem 2.1, which will follow from its proof :

Corollary 2.5. *Suppose that the topological space associated with* I *(whose open subsets are the open subsets of* I, *and the* I/i *form a basis of open subsets) is* noetherian, *i.e. every increasing sequence of open subsets is stationary. Then all the conclusions in theorem* 2.1 *and its corollaries remain valid, assuming only the stability of* C *or of* $TC = C \cap W$ *under simple composition (instead of transfinite composition), in addition to the other two stabilities.*

PROOF of theorem 2.1 and of corollary 2.5. p. 17

2.1. 1°) We may assume $J = I$, $K \subset I$. We construct a strictly increasing transfinite sequence of open subsets I_α, with [page 30]

$$\begin{cases} I_0 &= K, \\ I_{\alpha+1} &= I_\alpha \cup I/s_\alpha \quad \text{where } s_\alpha \in I\backslash I_\alpha \text{ (if } I_\alpha \neq I\text{, otherwise, the transfinite construction stops)}, \\ I_\alpha &= \bigcup_{\beta<\alpha} I_\beta \quad \text{if } \alpha \text{ a limit ordinal.} \end{cases}$$

There exists an α_0 such that we have $I_{\alpha_0} = I$. We then see as in paragraph 1 that [page 31]
for every α, we have

$$u_{I_\alpha, I_{\alpha+1}} \ \in \ C, \qquad (*)$$

for it is deduced from $u_{(I/s_\alpha)\cap I_\alpha, I/s_\alpha}$ by cobase change. Cf. diagram (6), p. 17, which ought to be extracted as a lemma, stripped of the technical peculiarities of the situation, i.e. a lemma on homomorphisms of cocartesian squares :

Lemma 2.6. *Let there be a homomorphism of cocartesian squares* $X_\bullet$, $Y_\bullet$, *in a category* $\mathcal{M}$ *with amalgamated sums (**N.B.** it suffices that the amalgamated sums occurring above exist) :*

$$\begin{array}{ccccc} X_0 & \longrightarrow & X_1 & & \\ \downarrow{\scriptstyle f_0} & \searrow & \downarrow & \searrow & \\ & X'_0 & \longrightarrow & X'_1 & \\ & \downarrow{\scriptstyle f'_0} & & \downarrow{\scriptstyle f_1} & \\ Y_0 & \longrightarrow & Y_1 & & \downarrow{\scriptstyle f'_1} \\ & \searrow & & \searrow & \\ & & Y'_0 & \longrightarrow & Y'_1 \ . \end{array} \qquad (2.6.1)$$

[12] and conversely, characterizes the C-cofibrant objects.

Let Z_0, Z_1 be the two lateral squares and consider, for every commutative square $Z = \begin{array}{ccc} X & \longrightarrow & X' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y' \end{array}$ *, the arrow u_Z of the diagram*

$$\begin{array}{ccccc} X & \longrightarrow & X' & & \\ \downarrow & \text{cocart.} & \downarrow & & \\ Y & \longrightarrow & \tilde{Z} & \xrightarrow[u_Z]{} & Y' \ . \end{array}$$

Then the arrow u_{Z_1} is deduced from u_{Z_0} by cobase change, more precisely, we have a p. 18 [page 32]
cocartesian square, where $T = \begin{array}{ccc} X_0 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ Y_0 & \longrightarrow & Y_1 \end{array}$ *and* $T' = \begin{array}{ccc} X'_0 & \longrightarrow & X'_1 \\ \downarrow & & \downarrow \\ Y'_0 & \longrightarrow & Y'_1 \end{array}$ *denote the back and front faces of the cube* (2.6.1) :

$$\begin{array}{ccc} \tilde{Z}_0 & \xrightarrow{u_{Z_0}} & Y'_0 \\ \downarrow & \text{cocart.} & \downarrow \\ \tilde{Z}_1 & \xrightarrow{u_{Z_1}} & Y'_1 \ , \end{array} \qquad (*)$$

which we see on the base of the diagram

$$\begin{array}{ccccc} X_0 & \longrightarrow & X_1 & & \\ {\scriptstyle f_0}\downarrow & \searrow & \downarrow & \searrow & \\ Y_0 & \longrightarrow & X'_0 & \longrightarrow & X'_1 \\ & \searrow & \downarrow{\scriptstyle f_1} & & \downarrow \\ & & Y_1 & \longrightarrow & \tilde{Z}'_1 \\ & & \tilde{Z}'_0 & \xrightarrow{f'_0} & \\ & & {\scriptstyle u_{Z_0}}\searrow & \text{cocart.} & \swarrow{\scriptstyle u_{Z_1}} \\ & & & Y'_0 \longrightarrow Y'_1 \ , & \end{array} \qquad (**)$$

in which the cube is a totally cocartesian cube, deduced from the cocartesian square X by "exponentiating" it by the arrow $f_0 : X_0 \longrightarrow Y_0$.

The base of the cube, and the square composed of this one and of $(*)$ (base of the [page 33]
polyhedral diagram $(**)$) being cocartesian, the same holds for $(*)$.

Lemma 2.7. *Let $(I_\alpha)_{\alpha \leq \alpha_0}$ be an increasing ordinal sequence of open subsets of I. Then $u_{I_0,I_{\alpha_0}} : Y_{I_0,I_{\alpha_0}} \longrightarrow Y_{I_{\alpha_0}}$ is a transfinite composite, indexed by the set of ordinals $0 \leq \alpha < \alpha_0$, of a sequence of arrows v_α, where each v_α is deduced from $u_{I_\alpha,I_{\alpha+1}}$ by cobase change.*

To see this, we introduce the transfinite sequence of objects $Z_\alpha = Y_{I_\alpha,I_{\alpha_0}}$, defined by the diagram

p. 19

$$\begin{array}{ccc} X_{I_\alpha} & \longrightarrow & X_{I_{\alpha_0}} \\ \downarrow & & \downarrow \\ Y_{I_\alpha} & \longrightarrow & Y_{I_\alpha,I_{\alpha_0}} = Z_\alpha \\ & & \downarrow \\ & & Y_{I_{\alpha_0}} \ , \end{array}$$

we have $Z_0 = Y_{I_0,I_{\alpha_0}}$, $Z_{\alpha_0} = Y_{I_{\alpha_0},I_{\alpha_0}} \simeq Y_{I_{\alpha_0}}$, $Z_\alpha = \varinjlim_{\beta<\alpha} Z_\beta$ if α a limit ordinal. So it [34]
remains to determine the $v_{\alpha,\alpha+1} : Z_\alpha \longrightarrow Z_{\alpha+1}$ (for $\alpha < \alpha_0$), which we see from the diagram

$$\begin{array}{ccccc}
X_{I_\alpha} & \longrightarrow & X_{I_{\alpha+1}} & \longrightarrow & X_{I_{\alpha_0}} \\
\downarrow & & \downarrow & & \downarrow \\
Y_{I_\alpha} & \longrightarrow & Y_{I_\alpha,I_{\alpha+1}} & \longrightarrow & Y_{I_\alpha,I_{\alpha_0}} = Z_\alpha \\
 & & \downarrow{\scriptstyle u_{I_\alpha,I_{\alpha+1}}} & & \downarrow{\scriptstyle v_{\alpha,\alpha+1}} \\
 & & Y_{I_{\alpha+1}} & \longrightarrow & Y_{I_{\alpha+1},I_{\alpha_0}} = Z_{\alpha+1} \ .
\end{array}$$

Let us resume the proof of 2.1. We are going to prove 1° by the two preceding lemmas. Let us note that when the space underlying I is noetherian, then the transfinite construction is necessarily finite, so we do not need stability of C under infinite transfinite composites.

Let us prove 2°. We therefore assume that for every $i_0 \in I$, we have $u_{I_{<i_0},I/i_0} \in C$, [35]
we want to deduce from this that $u_\bullet \in C_I$, i.e. $u_{J,I/i_0} \in C$ for *every* open set.

Lemma 2.8. (Under the artinian hypothesis on I.) *Let $P \subset I$ be the set of $i_0 \in I$ such that $u_\bullet|I/i_0$ is in C_{I/i_0}, i.e. that for every $i \leq i_0$, and every open set K of I/i, we have $u_{K,I/i} \in C$. Let $i_0 \in I$ be such that $I_{<i_0} \subset P$. Then $i_0 \in P$.*

Proof. It remains to prove that if $K \subset I/i_0 = J$ is an open set, then $u_{K,J} \in C$. We may assume $K \neq I/i_0$, i.e. $K \subset I_{<i_0} \overset{\text{def}}{=} J'$. But then by the initial hypothesis on $u_\bullet$, we have $u_{J',J} \in C$, and on the other hand, by the hypothesis on i_0, $u_\bullet|J'$ is in $C_{J'}$. So by theorem 2.1. 1°, we have $u_{K,J'} \in C$. So by transitivity $u_{K,J} \in C$, q.e.d.

Corollary 2.9. *If I is artinian, then $P = I$, i.e. $u_\bullet \in C_I$.*

This therefore proves 2.1. 2°. Let us note that if $\mathrm{Esp}(I)$ is noetherian, the same is true of every open set (such as J' above), so for the conclusion of the lemma, no need to [36]
assume C stable under infinite composition.

Let us prove 2.1. 3°. We have a) $\Longrightarrow$ b), for by 2.1. 1°, $u_\bullet \in (TC)_I$ implies that for p. 20
$i \in I$, $u_i : X_i \longrightarrow Y_i$ is in TC (cf. cor. 2.2 a). So it remains to prove b) $\Longrightarrow$ a), i.e. if $u_\bullet \in C_I \cap W_{I^\circ}$, then $u_\bullet \in (TC)_I$, i.e. the $u_{K,I/i_0}$ are in TC. We already know that $u_{K,I/i_0} \in C$ by hypothesis $u_\bullet \in C_I$, so we must prove that it is in W. We are reduced to proving that $u_K \in W$, i.e. $u_K \in TC$, for by the diagram (where $J = I/i_0$)

$$\begin{array}{ccc}
X_K & \longrightarrow & X_J \\
{\scriptstyle u_K}\downarrow & \text{cocart.} & \downarrow{\scriptstyle \tilde{u}_{K,J}} \\
Y_K & \longrightarrow & Y_{K,J} \\
 & & {\scriptstyle u_{K,J}}\downarrow \\
 & & Y_J \ ,
\end{array} \qquad u_J$$

$u_K \in TC$ implies $\tilde{u}_{K,J} \in TC$, and since $u_J \simeq u_{i_0} \in W$ by hypothesis, it indeed follows that $u_{K,J} \in W$ by saturation. But we are going to go about it differently:

Lemma 2.9. *I an arbitrary ordered set, C, W as in th.* 2.1, 1° *and* 3°, *$u_\bullet : X_\bullet \longrightarrow Y_\bullet$* [37]
in $C_I \cap W_{I^\circ}$. Let P be the set of $i_0 \in I$ such that $u_\bullet|I/i_0$ is in $(TC)_{I/i_0}$. If $i_0 \in I$ is such that $I_{<i_0} \subset P$, then $i_0 \in P$.

Indeed, it remains to prove that for every open set K of $J = I/i_0$, $K \neq J$, we have $u_{K,J} \in TC$. But for this, it suffices to see that $u_K, u_J \in W$ (as we had just seen). Now $u_J \simeq u_{i_0} \in W$ by hypothesis. On the other hand $u_\bullet|K \in (TC_K)$ since $K \subset I_{<i_0} \subset P$, so by theorem 2.1, 1°, we conclude that $u_K \in TC$.

If now we assume I artinian, it follows from 2.9 that $P = I$, so $u_\bullet \in (TC)_I$, q.e.d.

This completes the proof of theorem 2.1, and also of corollary 2.5, seeing that for the proof of lemma 2.9 as well, the infinite hypotheses on C, $C \cap W$ are unnecessary if $\mathrm{Top}(I)$ is noetherian, hence its open sets are also noetherian.

Remarks.

1) In 2.1. 2°, the hypothesis I artinian is not superfluous, as we see by taking $I = \mathbf{N}^{\mathrm{opp}} \simeq \mathbf{Z}^{\leq 0}$, and for an arrow [38]

$$\emptyset_{\mathcal{M}(I)} \longrightarrow Y_\bullet \ .$$

To say that this arrow is in C_I means that the transition morphisms $Y_i \xrightarrow{u_i} Y_{i+1}$ are in C, *and* that the Y_i are C-cofibrant, i.e. $\emptyset_{\mathcal{M}} \longrightarrow Y_i$ is in C, for every i. But the condition considered in 2.1. 2° is only that the u_i are in C. So by taking for $Y_\bullet$ a constant direct system with value Y, with Y not C-cofibrant, i.e. $\emptyset_{\mathcal{M}} \longrightarrow Y$ not in C, we have a counterexample.

With the same I, we see that $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ is in C_I if and only if

a) the $u_i \in C$, and

b) the $u_{i,i+1} : Y_{i,i+1} \longrightarrow Y_{i+1}$ are in C.

If moreover the u_i are in W, hence in TC, we see as above that $u_{i,i+1} \in TC$ for every i, hence $u_\bullet \in (TC)_I$. So here we still have p. 21

$$(TC)_I \ = \ C_I \cap W_{I^\circ} \ ,$$

although I is not artinian. I do not know whether the artinian hypothesis in 2.1. 3° is superfluous or not.

2) Let $u_\bullet \in \mathrm{Fl}(\mathcal{M}(I^\circ))$[13], so $u_\bullet$ is an object of $\underline{\mathrm{Fl}}(\mathcal{M}(I^\circ)) \simeq \mathcal{M}(I^\circ \times \mathbf{\Delta}^1) = \mathcal{M}((I \times (\mathbf{\Delta}^1)^{\mathrm{opp}})^{\mathrm{opp}})$. We check that for this object to be C-cofibrant, it is necessary and sufficient that

1°) $X_\bullet$ be C-cofibrant, and

2°) $u_\bullet \in C_I$ (which implies that $Y_\bullet$ is also cofibrant).

[39]

3 Categories of models with cofibrations

Let $\mathcal{M}$ be a category equipped with two sets of arrows

$$W, \ C \ \subset \ \mathrm{Fl}(\mathcal{M}) \ .$$

We assume that $\mathcal{M}$ contains an initial object, and:

Cof$_0$ W is mildly saturated, i.e. contains the isomorphisms and if f, g are composable, and if two among f, g, gf are in W, the third too.

Cof$_1$ C contains the isomorphisms, is stable under composition and under cobase change.

Cof$_2$ $TC \overset{\mathrm{def}}{=} C \cap W$ is stable under cobase change.

[13] $I^\circ = I^{\mathrm{opp}}$.

Cof$_3$ Every arrow f in $\mathcal{M}$ is written

$$f \;=\; pi$$

with $i \in C$, $p \in W$.

These are the axioms of K. S. Brown (dual form), who moreover assumes that the objects of $\mathcal{M}$ are C-cofibrant, i.e. $\emptyset_{\mathcal{M}} \longrightarrow X$ is in C. I will not make this assumption. The subcategory $\mathcal{M}_c$ of $\mathcal{M}$ formed of the C-cofibrant objects satisfies the axioms of [40]
Brown, moreover one proves that

$$\mathrm{Ho}_W(\mathcal{M}_c) \xrightarrow{\sim} \mathrm{Ho}_W(\mathcal{M})$$

is an equivalence. Thus the results of Brown concerning $\mathcal{M}_c$, equipped with the sets of arrows induced by W, C, are applicable to $\mathrm{Ho}_W(\mathcal{M})$ and to its avatars (such as $\mathrm{Ho}_W(S\backslash\mathcal{M})$, $\mathrm{Ho}_W(\mathcal{M}/T)$ and $\mathrm{Ho}_W(S\backslash\underset{u_0}{\mathcal{M}}/T)$, where $u_0 : S \longrightarrow T$ is given). It turns out that for the most part, the standard homotopy results (in particular exact sequence of cofibrations relating to the homotopy operations "on the right" (or should one say on the left? yet these are questions of right exactness, not left exactness)) are valid. I will have to return to this in greater detail later on. In particular, there is a good theory of W-cocartesian squares (and *in* $\mathcal{M}_c$, there are enough W-cofibrations[14], …), and moreover, p. 22
these are *exact.* The category $\mathrm{Ho}_W(\mathcal{M}) \simeq \mathrm{Ho}_W(\mathcal{M}_c)$ "is computed" by a calculus of [41]
left fractions in $\pi\mathcal{M}_c$, where $\pi\mathcal{M}_c$ is the category obtained from $\mathcal{M}_c$ by replacing the $\mathrm{Hom}_{\mathcal{M}_c}(X,Y) = \mathrm{Hom}_{\mathcal{M}}(X,Y)$ in it by the quotient by a suitable homotopy relation:

$f, g : X \rightrightarrows Y$ are QB-homotopic (in the sense of *Quillen-Brown*) if there exists an $X' \xrightarrow{\alpha} X$ in W (and we may assume $X' \in \mathrm{Ob}\,\mathcal{M}_c$ as well) such that $f\alpha \overset{1}{\sim} g\alpha$ in the sense of Quillen, i.e. there exists a cylinder object

$$\begin{array}{ccc} I_{X'} & \xrightarrow{\in W} & X' \\ {\scriptstyle \in C}\uparrow & \nearrow & \\ X' \vee X' & & \end{array}$$

for X', such that $(f\alpha, g\alpha) \;:\; X' \vee X' \longrightarrow Y$ factors through an $h' : I_{X'} \longrightarrow Y$.

It is from this criterion that the (left) exactness of W-cocartesian squares follows.

I want to see how a cofibration structure on $\mathcal{M}$ extends to the categories $\mathcal{M}(I^\circ) = \underline{\mathrm{Hom}}(I, \mathcal{M})$. We have in $\mathcal{M}(I^\circ)$ the set W_{I°, and C_I (it is probably better to denote it [42]
also by C_{I°) defined in the preceding paragraph. It is a matter of checking the axioms. The axiom Cof_0 is immediate, and well known for I an arbitrary category.

$\mathrm{Cof}_1 : C_{I^\circ}$ contains the isomorphisms, that is clear, because C contains them. Stable under composition $X_\bullet \xrightarrow{u} Y_\bullet \xrightarrow{v} Z_\bullet$, let us look at $K \subset J$, put $w = v \circ u$,

$$\begin{array}{ccccccc} X_K & \longrightarrow & X_J & & & & \\ \downarrow & & \downarrow & & & & \\ Y_K & \longrightarrow & Y_{K,J} & \xrightarrow{u_{K,J}} & Y_J & & \\ \downarrow & & \downarrow & & \downarrow & & \\ Z_K & \longrightarrow & \bullet & \xrightarrow{u'_{K,J}} & Z_{K,J} & \xrightarrow{v_{K,J}} & Z_J\ , \\ & & \underbrace{\qquad\qquad\qquad\qquad}_{w_{K,J}} & & & & \end{array}$$

[14] More precisely, by K. S. Brown, *in* $\mathcal{M}_c$, the objects of C are W_c-cofibrations.

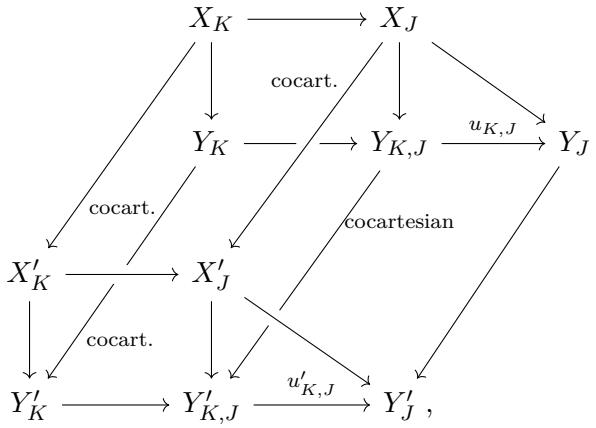
$w_{K,J}$ is the composite of $v_{K,J}$, which is in C, and $u'_{K,J}$, which is too since obtained from $u_{K,J}$ by cobase change.

(**N.B.** We have already made the argument with a single $u_\bullet$, and *three* open sets $J \subset K \subset L$. It would be appropriate to introduce the notion of a C-square $\begin{array}{ccc} X_0 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & X_3 \end{array}$, as a square such that $X_1 \amalg_{X_0} X_2 \longrightarrow X_3$ is in C (**N.B.** notion symmetric for the symmetry functors of squares), and say that the composite of two C-squares is a C-square.) [43] p. 23

Cobase change

$$\begin{array}{ccc} X_\bullet & \xrightarrow{u} & Y_\bullet \\ \downarrow & & \downarrow \\ X'_\bullet & \xrightarrow{u'} & Y'_\bullet \end{array} .$$

Let us consider $K \subset J$, we have



where the three lateral squares marked "cocart." are cocartesian, as is the square in an inclined plane. It follows, first that the fourth lateral square of the cube is itself also cocartesian, then that the square having two sides $u_{K,J}$, $u'_{K,J}$ is cocartesian. (**N.B.** compare the similar diagram p. 32 – there would be a common lemma to extract, distinguishing two cases …) [44]

Cof_2. To prove that $C_{I^\circ} \cap W_{I^\circ}$ is stable under cobase change, I see no way of doing it, if I do not already know that we have

$$C_{I^\circ} \cap W_{I^\circ} \;=\; (TC)_{I^\circ} \;, \qquad (*)$$

and then we apply the preceding result (where W did not enter) to TC instead of C. But to have $(*)$, I must assume that we are in one of the following two cases (so as to be able to apply theorem 2.1 and its corollary 2.5).

a) I finite, or more generally I artinian *and* $\mathrm{Top}(I)$ is noetherian.

b) I artinian, and moreover C and TC are stable under transfinite composition.

Cof_3. The factorization axiom seems to me a priori the most delicate[15]. But in the end this comes out without too much trouble. Let there be given

$$f \;:\; X_\bullet \longrightarrow Y_\bullet$$

p. 24 [45]

in $\mathcal{M}(I^\circ)$, let us consider the quadruples

$$(J,\ Z,\ i,\ p)\ ,$$

where $J \subset I$ is an open set, Z an object of $\mathcal{M}(J^\circ)$, and i, p are arrows

$$\begin{cases} X|J \xrightarrow{\ i\ } Z \xrightarrow{\ p\ } Y|J \\ \text{with } i \in C_{J^\circ},\ p \in W_{J^\circ}. \end{cases}$$

We order the quadruples by extension. If we have a totally ordered set, of $(J_\alpha, Z_\alpha, i_\alpha, p_\alpha)$, putting $J = \bigcup J_\alpha$, we find unique Z, i, p on J which extend the given Z_α, i_α, p_α, moreover we will have $i \in C_{J^\circ}$, $p \in W_{J^\circ}$, since membership in C_{J° resp. W_{J° is a local question on J. So by Zorn, we are reduced to proving that if (J, Z, i, p) is maximal, we have $J = I$ – whence the desired factorization $f = pi$. So to prove that if $J \neq I$, the quadruple is not maximal. Let us consider the nonempty closed set $I \backslash J$. I is assumed artinian, there is a *minimal element* i_0 of $I \backslash J$. Let us put [46]

$$J' \ = \ J \cup \{i_0\} \ = \ J \cup I/i_0,$$

and let us show that we can extend the triple Z, i, p to Z', i', p' on J'. Putting $K' = I/i_0$, $K = J \cap K' = I_{<i_0}$, we thus have

$$\begin{array}{ccc} K = J \cap K' & \hookrightarrow & K' = I/i_0 \\ \downarrow & & \downarrow \\ J & \hookrightarrow & J' = J \cup K'. \end{array}$$

We then see that the extensions of (Z, i, p) from J to J' correspond to those of $(Z|K, i|K, p|K)$ to K'. For this, we note that for an arrow φ between objects of $\mathcal{M}(J'^\circ)$, this arrow is in $C_{J'^\circ}$ resp. $W_{J'^\circ}$, if and only if its restrictions to J and to K' are.

We are thus reduced to the standard case of an ordered set with final object e_0, an ordered set which we will denote by I, and the ordered set I' of objects $i < e_0$, i.e. $I' = I \backslash \{e_0\}$. We have $f : X_\bullet \longrightarrow Y_\bullet$, an arrow in $\mathcal{M}(I^\circ)$, a factorization of $f|I'$ as $p' \circ i'$ with $i' \in C_{I'^\circ}$, $p' \in W_{I'^\circ}$, and it is a matter of extending it to a similar factorization of [47] f. We note that giving an object $X_\bullet$ of $\mathcal{M}(I^\circ)$ is equivalent to giving an object $X'_\bullet$ of $\mathcal{M}(I'^\circ)$ (the restriction of X to I') and an arrow

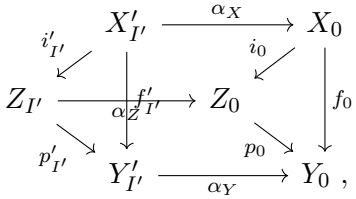
$$X'_{I'} \xrightarrow{\ \alpha_X\ } X_{e_0} = X_0$$

in $\mathcal{M}$, where $X'_{I'} = \varinjlim_{i' \in I'} X'_{i'}$. Thus giving the arrow $f : X_\bullet \longrightarrow Y_\bullet$ amounts to giving $f' = f|I' : X'_\bullet \longrightarrow Y'_\bullet$, and an arrow $f_0 : X_0 \longrightarrow Y_0$ making the square commutative p. 25

$$\begin{array}{ccc} X'_{I'} & \xrightarrow{\ \alpha_X\ } & X_0 \\ {\scriptstyle f'_{I'}}\downarrow & & \downarrow{\scriptstyle f_0} \\ Y'_{I'} & \xrightarrow{\ \alpha_Y\ } & Y_0\ . \end{array}$$

[15] The argument that follows applies to other factorization situations, for example for a given Quillen precouple (Φ, Ψ) in $\mathcal{M}$.

Likewise, finding a factorization of f as pi, which extends a given factorization $f' = p'i'$ on I', amounts to giving a commutative diagram



(∗)

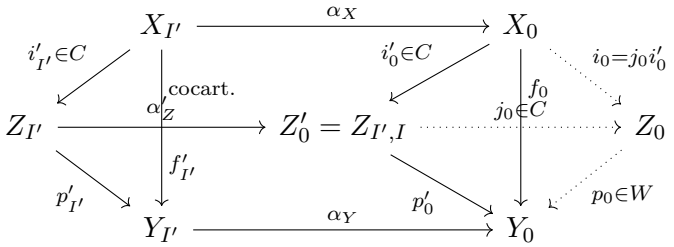
where the object Z_0, and the three arrows α_Z, i_0, p_0 are to be determined. If we have $i' \in C_{I'^\circ}$, $p' \in W_{I'^\circ}$, then we conclude by theorem 2.1. 1° (cor. 2.2) that [48 blank] [49]

$$i'_{I'} \ \in \ C \ ,$$

and if we want $i \in C_{I^\circ}$, $p \in W_{I^\circ}$, it only remains to require

$$p_0 \in W, \qquad (i'_{I',I} : \underbrace{Z_{I'} \amalg_{X_{I'}} X_0}_{Z'_0} \longrightarrow Z_0) \in C,$$

by theorem 2.1. 2°. We will then construct the diagram (∗), from the similar diagram below



(with Z_0 of (∗) replaced by $Z'_0 = Z_{I',I}$) by factoring the canonical arrow $p'_0 : Z'_0 \longrightarrow Y_0$ as

$$p'_0 \ = \ p_0 j_0 \ , \qquad \text{with } p_0 \in W, \ j_0 \in C \ ,$$

and putting

$$i_0 \ = \ j_0 i'_0 \ ,$$

which completes the construction.

We have thus proved the

Theorem 3.1. *Let $(\mathcal{M}, W, C)$ be a category with cofibrations (cf. page 39), $\mathcal{M}$ stable under finite $\varinjlim$. For any ordered set I, let us consider the corresponding tuple $(\mathcal{M}(I^\circ), W_{I^\circ}, C_{I^\circ})$, where C_{I° is defined in paragraph 2 (cf. on this point theorem 2.1). Suppose that I is an* artinian *ordered set (e.g. I locally finite, i.e. the I/i finite). Suppose moreover that we are in one of the following two cases:* p. 26 [50]

a) *The space* $\mathrm{Esp}(I)$ *is noetherian, i.e. every increasing sequence of open sets of I is stationary (e.g. I finite).*

b) *C and $TC = C \cap W$ are stable, not only under ordinary composition, but also under transfinite composition. In this case, we also assume $\mathcal{M}$ stable under small filtering $\varinjlim$ (hence under arbitrary small $\varinjlim$).*

Under these conditions, $(\mathcal{M}(I^\circ), W_{I^\circ}, C_{I^\circ})$ is also a category with cofibrations.

Proposition 3.2. *Let $u : I \longrightarrow I'$ be an increasing map between ordered sets, whence a pair of adjoint functors*

$$\mathcal{M}(I^\circ) \underset{u^*}{\overset{u_!}{\rightleftarrows}} \mathcal{M}(I'^\circ) \quad . \tag{3.2.1}$$

[51]

a) *The functor u^* is compatible with the localizers $W_{I'^\circ}$, W_{I°. If I is discrete, or u étale, then u^* maps $C_{I'^\circ}$ into C_{I°. The same is still true when u is locally cofiltering, i.e. if for every $i' \in \mathrm{Ob}\, I'$, the category $i'\backslash I$ is locally cofiltering, i.e. its connected components are cofiltering (condition* $\mathrm{PS_g}$ 1 *on $i'\backslash I$)* [16].

b) *In all cases, $u_!$ transforms C_{I° into $C_{I'^\circ}$, TC_{I° into $TC_{I'^\circ}$, finally $\mathcal{M}(I^\circ)_c$ into $\mathcal{M}(I'^\circ)_c$.*

Proof. Let us prove a), in the most awkward case, where we assume only that u is locally weakly cofiltering. We apply II 7.4 (p. 92), which tells us that $\forall i \in \mathrm{Ob}\, I$, considering the morphism induced by u [52]

$$I/i \longrightarrow I'/i' \qquad (\text{where } i' = u(i)),$$

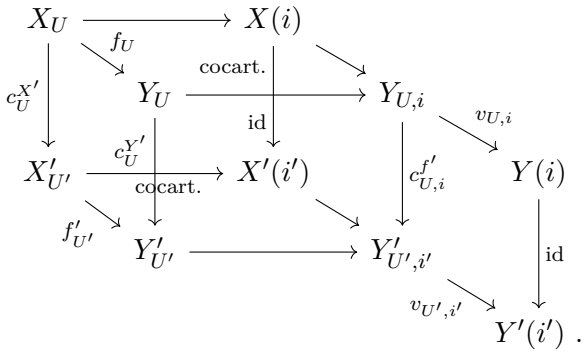
this is *(globally)* weakly cofiltering, and that for every open set U of I/i, denoting by U' the open set generated by its image, the induced morphism $v : U \longrightarrow U'$ is still weakly cofiltering, and a fortiori *cofinal*. Let then $f' : X' \longrightarrow Y'$ be an arrow in $C_{I'^\circ}$, let us prove that

$$\underbrace{u^*(f')}_{f} \;:\; \underbrace{u^*(X')}_{X} \longrightarrow \underbrace{u^*(Y')}_{Y}$$

is in C_{I°, i.e. that for every $i \in I$, and U an open set of I/i, in the canonical diagram p. 27

$$\begin{array}{ccc} X_U & \longrightarrow & X(i) \\ \downarrow & \text{cocart.} & \downarrow \\ Y_U & \longrightarrow & Y_{U,i} \\ & & \downarrow{\scriptstyle v_{U,i}} \\ & & Y(i)\ , \end{array} \qquad (*)$$

the arrow $v_{U,i}$ is in C. The diagram $(*)$ maps into the similar diagram for $f' : X' \longrightarrow Y'$, [53] $i' = u(i)$, and the open set $U' \subset I'/i'$ generated by $u(U)$:



$(**)$

By cofinality, the two vertical arrows $c_U^{X'}$ and $c_U^{Y'}$ are isomorphisms, in addition to $X(i) \xrightarrow{id} X'(i')$, so the fourth arrow of the cube, $c_{U,i}^{f'}$, is an isomorphism. Since $v_{U',i'}$ is in C, $v_{U,i}$ too, we win.

Let us prove b). So let

$$f \;:\; X \longrightarrow Y$$

[16] no need to speak of "weakly" locally cofiltering, since these are ordered sets where there is no need to distinguish between the strong and weak notions.

be an arrow in C_I, and let

$$f' \;=\; u_!(f) \;:\; u_!(X) = X' \longrightarrow u_!(Y) = Y',$$

let us prove that it is in $C_{I'^\circ}$, hence that for two open sets $U' \subset V'$ of I' (where we can [54] assume that $V' = I'/i'$), in the base-diagram of $(**)$, $v_{U',V'}$ is in C. Let $U = u^{-1}(U')$, $V = u^{-1}(V')$, and let us consider the diagram similar to $(**)$, defined by the two inclusions of open sets $U' \subset V'$, $U \subset V$. This time, since the formation of $u_!$ commutes with restrictions to open sets, by transitivity of $\varinjlim$, the three vertical arrows $c^X_{U'}$, $c^Y_{U'}$, $c^X_{V'}$ are isomorphisms, hence so is $c^f_{U',V'}$. In the same way, the arrow $Y_V \longrightarrow Y'_{V'}$ (in place of $Y(i) \longrightarrow Y'(i')$) is an isomorphism. Since $v_{U,V} \in C$, the same is true of $v_{U',V'}$, which proves that $f' \in C_{I'^\circ}$. We proceed in the same way for TC_I. Finally, the assertion that $u_!$ maps $\mathcal{M}(I^\circ)_c$ into $\mathcal{M}(I'^\circ)_c$ follows from the definitions and from $u_!(C_{I^\circ}) \subset C_{I'^\circ}$.

Corollary 3.3 (of prop. 3.2 and th. 3.1). *Let $u : I \longrightarrow I'$ be as in* 3.2. *Suppose moreover I artinian, and that we are for $\mathcal{M}$, W, C, I under one of the conditions* a), [p. 28] b) *of theorem* 3.1. *Then $u_!$ induces an arrow* [55]

$$u_{!\,\mathrm{cof}} \;:\; \mathcal{M}(I^\circ)_c \longrightarrow \mathcal{M}(I'^\circ)_c \;\subset\; \mathcal{M}(I'^\circ)$$

which transforms arrows of $W(I^\circ)$ into arrows of $W(I'^\circ)$ (i.e. compatible with the localizers). It thus induces a functor on the localized categories, whence a

$$u_!^W \;:\; \mathrm{Ho}_W(I^\circ) \longrightarrow \mathrm{Ho}_W(I'^\circ) \qquad {}^{(17)}. \tag{3.3.1}$$

Proof. By theorem 3.1, we know that $\mathcal{M}(I^\circ)$, $W(I^\circ)$, C_{I° is a structure of a category of models with cofibrations. We also know that $TC_{I^\circ} = C_{I^\circ} \cap W(I^\circ)$, by th. 2.1. 3° (p. 27). Thus by 3.2, $u_!$ maps $C_{I^\circ} \cap W(I^\circ)$ into $TC_{I'^\circ} = C_{I'^\circ} \cap W(I'^\circ)$, and a fortiori, into arrows in $W(I'^\circ)$. Moreover, by the theory of categories with cofibrations, we find that every arrow f between cofibrant objects is a composite $f = pi$, where $i \in C_{I^\circ}$, and where p is in $W(I^\circ)$, and moreover a *left inverse* of an arrow in $C_{I^\circ} \cap W(I^\circ)$, i.e. of an arrow of $(TC)_{I^\circ}$. When $f \in W(I^\circ)$, then $i \in W(I^\circ)$, hence $i \in W(I^\circ) \cap C_{I^\circ} = (TC)_{I^\circ}$. [56] Altogether, every arrow $f \in W(I^\circ)_c$ is a composite pi in $\mathcal{M}(I^\circ)_c$, where $i \in (TC)_{I^\circ}$, and p is a left inverse of a $j \in (TC)_{I^\circ}$. Since $u_!$ transforms $(TC)_{I^\circ}$ into $(TC)_{I'^\circ}$, it transforms f into $f' = p'i'$, with $i' \in (TC)_{I'^\circ} \subset W_{I'^\circ}$, and p' a left inverse of a $j' = u_!(j)$ which is also in $W_{I'^\circ}$. But then $p' \in W(I'^\circ)$, hence also $f' = p'i' \in W_{I'^\circ}$, which proves that $u_!(W(I^\circ)_c) \subset W(I'^\circ)$. The last assertion comes from the following lemma, applied to $(\mathcal{M}(I^\circ), W(I^\circ), C_{I^\circ})$ in place of $(\mathcal{M}, W, C)$:

Lemma 3.4. *Let $(\mathcal{M}, W, C)$ be a category with cofibrations. Then the inclusion $\mathcal{M}_c \subset \mathcal{M}$ induces an equivalence*

$$\mathcal{M}_c W_c^{-1} \xrightarrow{\sim} \mathcal{M}W^{-1} = \mathrm{Ho}_W \;.$$

We will give the proof (by AQT) later.

This completes the proof of 3.3.

3.5. The question arises whether the functor $u_!^W$ of (3.3.1) is indeed a left adjoint to [57] the functor in the opposite direction u_W^*, induced by u^*. Let us note that this is the case if u^* transforms $\mathcal{M}(I'^\circ)_c$ into $\mathcal{M}(I^\circ)_c$, and when we assume that I' is also artinian and also satisfies one of the hypotheses a) b) of theorem 3.1. For then $(\mathcal{M}(I^\circ), W(I^\circ), C_{I^\circ})$

[17] **N.B.** This arrow is defined, even without assuming I' ordered, since $u_{!\,\mathrm{cof}}$ still transforms arrows of $W(I^\circ)$ into arrows of $W(I'^\circ)$. Cf. comment p. 86.

is a category with cofibrations by theorem 3.1, we can apply the preceding lemma to it, so u_W^* and $u_!^W$ are realized from a pair of functors

$$\mathcal{M}(I^\circ)_{\mathrm{c}} \underset{u_{\mathrm{c}}^*}{\overset{u_!^{\mathrm{c}}}{\rightleftarrows}} \mathcal{M}(I'^\circ)_{\mathrm{c}} \quad ,$$

which are visibly adjoint to one another (since induced by a pair of adjoint functors $u_!$, u^*), and moreover compatible with the localizers. So we can apply the adjoint functor lemma for the localized categories. To sum up: p. 29

Corollary 3.5. *Suppose that I, I' are artinian, and that we are in one of the following two cases:*

a) *The spaces* $\mathrm{Esp}(I)$, $\mathrm{Esp}(I')$ *are noetherian (e.g. I, I' finite).* [page 58]

b) *C and $TC = C \cap W$ are stable under transfinite composition.*

Suppose moreover that u^* maps $\mathcal{M}(I'^\circ)_{\mathrm{c}}$ into $\mathcal{M}(I^\circ)_{\mathrm{c}}$, which is the case in particular if u is locally cofiltering, i.e. the localized ordered sets $i'\backslash I$ are locally cofiltering. Then the functor $u_!^W$ constructed in (3.3.1) is left adjoint to u_W^*.

Remark 3.6. When we have a pair of adjoint functors

$$\mathcal{C} \underset{u^*}{\overset{u_!}{\rightleftarrows}} \mathcal{C}' \quad ,$$

with the localizers W, W' of $\mathcal{C}$, $\mathcal{C}'$, u^* compatible with the localizers, and if moreover we have a full subcategory $\mathcal{C}_0$ of $\mathcal{C}$ such that

a) $u_!|\mathcal{C}_0$ transforms arrows of W into arrows of W', and

b) the inclusion $\mathcal{C}_0 \hookrightarrow \mathcal{C}$ induces an *equivalence* $\mathcal{C}_0 W_0^{-1} \xrightarrow{\sim} \mathcal{C}W^{-1}$ (where $W_0 = W \cap \mathrm{Fl}(\mathcal{C}_0)$),

we find two canonical functors

$$\mathcal{C}W^{-1} \underset{u_W^*}{\overset{u_!^W}{\rightleftarrows}} \mathcal{C}'W'^{-1} \quad ,$$

where $u_!^W$ is defined only up to canonical isomorphism.

Contrary to what Anderson claims, it is not true in general that these two functors are adjoint, without additional hypotheses. Here is a counterexample: we take $\mathcal{C} = \mathcal{C}'$, $u_! = u^* = \mathrm{id}_{\mathcal{C}}$, $W' = \mathrm{Is}(\mathcal{C})$, $W = \mathrm{Fl}(\mathcal{C})$, so u_W^* is the canonical functor [page 59]

$$\underbrace{\Pi\mathcal{C}}_{\text{fundamental groupoid}} \longleftarrow \mathcal{C} \ .$$

Suppose that $\mathcal{C}$ is 1-connected, i.e. $\Pi\mathcal{C}$ is equivalent to e. Let $e \in \mathrm{Ob}\,\mathcal{C}$ be such that $\mathrm{End}_{\mathcal{C}}(e)$ is the trivial group, and take $\mathcal{C}_0 = \{e\}$. Then the hypothesis on $\mathcal{C}_0$ is satisfied. But if $u_!^W$ were left adjoint to u_W^*, e would be an initial object of $\mathcal{C}$. For example if $\mathcal{C}$ has a final object e (which implies that $\mathcal{C}$ is 1-connected and e rigid), this final object would have to be initial if Anderson were right with "it is clear that …".

Nevertheless: [page 60]

Theorem 3.7. *Let $(\mathcal{M}, W, C)$ be a category with cofibrations stable under finite $\varinjlim$, and $u : I \longrightarrow I'$ an increasing map between artinian ordered sets. We suppose that we are under one of the following hypotheses.* p. 30

a) $\mathrm{Esp}(I)$, $\mathrm{Esp}(I')$ *are noetherian, e.g.* I, I' *finite.*

b) C *and* $TC \overset{\text{def}}{=} C \cap W$ *are stable under transfinite composition, and* $\mathcal{M}$ *stable under small filtering* $\varinjlim$.

Then the functor $u_!^W : \mathrm{Ho}_W(I^\circ) \longrightarrow \mathrm{Ho}_W(I'^\circ)$ *constructed in* (3.3.1) *is left adjoint to the functor* u_W^* *in the opposite direction.*

Proof. We use the well-known diagram

$$\begin{array}{ccc} & & I \\ & & \downarrow{\scriptstyle \partial_1} \\ I & \underset{\mathrm{pr}_1}{\overset{\partial_0 \text{ (open)}}{\rightleftarrows}} & I \times \mathbf{\Delta}^1 \quad \overset{w}{\curvearrowright} \\ {\scriptstyle u}\downarrow & \text{cocart.} & \downarrow \\ I' & \underset{v}{\overset{\text{open imm.}}{\rightleftarrows}} & C(u) \end{array}$$

to factor u as

$$u \;=\; vw \;,$$

where w is a *closed immersion,* and v is obtained from $\mathrm{pr}_1 : I \times \mathbf{\Delta}^1 \longrightarrow I$ by push-out. It is a retraction of the mapping cone C(u) onto the open I', whose closed complement is identified with I via w.

The gluing functor between I', I [page 61]

$$H \;:\; I'^\circ \times I \;\longrightarrow\; \mathrm{Ens}$$

is given by

$$\underbrace{H(i',i)}_{=\mathrm{Hom}_{C(u)}(i',i)} \;=\; \mathrm{Hom}_{I'}(i',u(i)) \;. \qquad (*)$$

The retraction v from $C(u)$ to I' maps i to $u(i)$, and an arrow $i' \longrightarrow i$ corresponding to $i' \longrightarrow u(i)$ in I' via $(*)$, is transformed by v into that same arrow $i' \longrightarrow u(i)$. Let us calculate, for a given i'_0, the category $i'_0\backslash\Big(C(u),\ v : C(u) \longrightarrow I'\Big)$, of pairs of an object ξ of $C(u)$ and an arrow $i'_0 \longrightarrow v(\xi)$ in I'. We see that $i'_0\backslash C(u)$ admits the open $i'_0\backslash I'$, with the closed complement $i'_0\backslash I$,

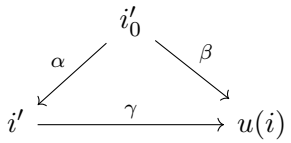
$$\begin{array}{ccccc} & & & & \text{closed} \\ & & & & \text{complement} \\ & & & & \text{of } I' \\ I' & \underset{\text{open}}{\hookrightarrow} & C(u) & \hookleftarrow & I \\ & & \downarrow & & \\ & & I' \;, & & \end{array}$$

these two categories being glued by the bifunctor

$$\begin{array}{rcl} (i'_0\backslash I')^\circ \times (i'_0\backslash I) & \longrightarrow & \mathrm{Ens} \\ \Big((\alpha : i'_0 \longrightarrow i'), (\beta : i'_0 \longrightarrow u(i), i)\Big) & \longmapsto & \mathrm{Hom}_{i'_0\backslash I'}(i', u(i)) \end{array}$$

induced by $(*)$; i.e. the arrows from $(\alpha : i'_0 \longrightarrow i') \in \mathrm{Ob}\,(i'_0\backslash I')$ to $(\beta : i'_0 \longrightarrow u(i)) \in i'_0\backslash I$ p. 31

are the arrows $i' \xrightarrow{\gamma} u(i)$ such that the triangle



commutes.

If $i' = i'_0$, $\alpha = \mathrm{id}_{i'_0}$, then this commutativity amounts to $\gamma = \beta$, which means that there is one and only one arrow from α to (i, β). This therefore shows that $(i'_0, \mathrm{id}_{i'_0})$ is an initial object of $i'_0\backslash C(u)$. We have therefore proved the lemma: [page 62]

Lemma 3.8. *Let $u : I \longrightarrow I'$ be an arrow in* (Cat)*, let us factor it as*

$$I \xrightarrow{w} C(u) \xrightarrow{v} I' \ ,$$

where $C(u)$ is the mapping cone defined on page 60*, w a closed immersion with complement I', v the canonical retraction of $C(u)$ onto I'. This being so, for every $i'_0 \in \mathrm{Ob}\, I'$, the category $i'_0\backslash(C(u), v)$ admits an initial object – hence $v : C(u) \longrightarrow I'$ is (as thoroughly as can be) cofiltering.*

We can now prove 3.7. First of all, the hypotheses made on I, I' carry over to $C(u)$ (artinian hypothesis, and possibly noetherian hypothesis on $\mathrm{Esp}(C(u))$). So we can apply the preceding results, in particular 3.5, to the arrows v, w. The preceding lemma indeed shows that v satisfies the conditions of 3.5, so the functor $v_!^W$ constructed in 3.3.1 is left adjoint to v_W^*. Consider on the other hand the functor $w_!^W$ constructed in 3.3.1, I say that the same conclusion applies. This comes from the fact that w is a closed immersion. Indeed, in this case it is immediate that [page 63]

$$w_! \ : \ \mathcal{M}(I^\circ) \ \longrightarrow \ \mathcal{M}(I'^\circ), \qquad \text{where } J = I' = C(u),$$

is compatible with the localizers, since we have

$$\begin{cases} w_!(X)(j) \ \simeq \ X(j) & \text{if } j \in I, \\ w_!(X)(j) \ \simeq \ \emptyset_{\mathcal{M}} & \text{if } j \in J\backslash I, \end{cases}$$

by the calculation of the fibers

$$w_!(X)(j) \ \simeq \ \varinjlim_{i \in I/j} X(i) \ .$$

So the definition of $w_!^W$ given in 3.3.1 amounts in fact to passing directly to the localized categories for $w_!$, without having to restrict ourselves to $\mathcal{M}(I^\circ)_{\mathrm{c}}$, and the adjoint functor lemma tells us that this functor is left adjoint to w_W^*.

Since we obviously have [p. 32] [page 64]

$$u_!^W \ \simeq \ v_!^W \circ w_!^W \ , \qquad u_W^* \ \simeq \ w_W^* \circ v_W^* \ ,$$

we conclude from this that $u_!^W$ is indeed left adjoint to u_W^*, q.e.d.

Corollary 3.9. *Let $(\mathcal{M}, W, C)$ be a category with cofibrations. Then the prederivator $\mathbf{D}_W$, restricted to the category* Diag_0 *consisting of the categories I that are* ordered*, noetherian and such that* $\mathrm{Esp}(I^\circ)$ *is noetherian, satisfies the axioms* Der 1 (decomposition), Der 2 (localization), Der 3 (final object), Der 4 bis (existence of $u_!$), Der 5 bis (mode of calculation of the $u_!$), Der 6 (exactness of W-cocartesian squares). *If C, $TC = C \cap W$ are stable under transfinite composition, we have the same result on the category* Diag *consisting of the noetherian ordered categories I, nothing more.* [page 65]

3.10. Thus, to have a derivator in the full sense of the term, either on Diag_0, or on Diag, all that is missing is Der 4 (existence of the u_*), Der 5 (calculation of the u_*) and Der 6 (exactness of W-cartesian squares). We are going to obtain them in principle with the aid of a *category with fibrations* structure $(\mathcal{M}, F, W)$ associated to $(\mathcal{M}, W)$, i.e. the kind of structure dual to the one introduced on page 39. These two structures do not have to be related to each other in any way other than by the fact that they correspond to the same W. They are considerably easier to construct than Quillen structures (W, C, F), where C, F must be related in an extremely strict way, with the properties $C \longleftrightarrow \underbrace{TF}_{=F\cap W}$, $\underbrace{TC}_{=C\cap W} \longleftrightarrow F$ (of which there is no mention in the axiomatics of K. S. Brown), and factorization for (C, TF) and (TC, F).

But returning to the construction of the $u_!^W$, there remains the problem of constructing them for an arrow $u : I \longrightarrow I'$ *whatsoever* in Cat. [page 66]

To tell the truth, the construction of the functor

$$u_!^W \ : \ \mathrm{Ho}_W(I^\circ) \ \longrightarrow \ \mathrm{Ho}_W(I'^\circ) \tag{3.3.1}$$

is valid without supposing that I' is an ordered category – it can be any category. This comes from the fact that the functor induced by $u_!$,

$$u_{!\,\mathrm{cof}} \ : \ \mathcal{M}(I^\circ)_\mathrm{c} \ \longrightarrow \ \mathcal{M}(I'^\circ)$$

is still compatible with the localizers, as we see by the calculation of the fibers

$$u_!(X)(i') \ \simeq \ \varinjlim_{i\in I/i'} X(i) \ .$$

Thus, we are reduced to the cases where $I' = e$, at the cost of replacing I by I/i', and noting that ($I/i' \longrightarrow I$ being étale) the inverse image of X on I/i' is cofibrant (cf. 3.2.a).

But there arises again the perplexity whether the functor (3.3.1) thus constructed is indeed left adjoint to u_W^*. We can think of using once again the same factorization as $u = vw$, by using the mapping cone. The difficulty, however, is that now $J = C(u)$ is no longer an ordered category, so we do not have a ready-made definition of $C_{J^\circ} \subset \mathrm{Fl}(\mathcal{M}(J^\circ))$, giving rise to a subcategory $\mathcal{M}(J^\circ)_\mathrm{c}$ etc. In fact, it would seem that we are practically paralyzed, so long as we do not manage to construct on $\mathcal{M}(J^\circ)$ also a category with cofibrations structure, behaving well with respect to the functors $v_!$. The "proof" by Anderson of the existence of $u_!$ for *every* arrow in Cat, is a total flop (even for $I \longrightarrow e$, I a finite ordered set – but there I was able to set things right), for he reduces to the case where I is ordered and locally finite, and he assumes without further ado that *in this case* $u_!^W$ is left adjoint to u_W^*, and *moreover,* that if u is $\mathbf{W}_0$-connected, hence $u^* : \mathcal{M}(I'^\circ) \longrightarrow \mathcal{M}(I^\circ)$ fully faithful, that the same is true of u_W^* obtained by passing to the localized categories. Now even if we grant that $u_!^W$ is left adjoint to u_W^*, for lack of having an adjunction morphism close at hand "down below" (in $\mathrm{Ho}_W(I'^\circ)$), obtained from the adjunction morphism in $\mathcal{M}(I^\circ)$, it is not at all clear that u_W^* is fully faithful, i.e. $u_!^W$ a localization functor. Counterexample 3.6 clearly shows that, even if we have adjunction (i.e. if e is an initial object of $\mathcal{C} = \mathcal{C}'$), the functor $u^* = \mathrm{id}_\mathcal{C}$ may well be fully faithful (and even an equivalence of categories), the localized functor has no reason to be so: this would prove that every object of Cat having an initial object is equivalent to e! [page 67] p. 33 [page 68]

3.11. Let us say that we have a *strict category with cofibrations,* if in the factorization axiom Cof_3 (cf. p. 39), the factorization can be taken functorially in f. It amounts to $\pm$ the same thing to say that the factorization axiom remains valid, when we think of a category $\mathcal{M}(I)$ for I in Cat, defining $W(I)$ and $C(I)$ there (not to be confused with C_I) fiber by fiber. This amounts to the same thing at least if $\mathcal{M}$ is equivalent to a [page 69]

small category. Otherwise, we must assume functoriality for every *small* full subcategory of $\underline{\mathrm{Fl}}(\mathcal{M})$. It is better to define the strict category with cofibrations by this stability property.

We then see that the prederivator $\mathbf{D} = \mathbf{D}_{\mathcal{M}}$ on Cat is a derivator *relative to* $\mathcal{D} = \mathrm{Diag}_0$ defined above (noetherian ordered categories I, such that $\mathrm{Esp}(I^\circ)$ is noetherian), or even relative to $\mathcal{D} = \mathrm{Diag}$ (noetherian ordered categories I, nothing more) when C and TC are stable under transfinite composition. By this I mean that $u_!$ exists when u is an arrow of $\mathcal{D}$, and that this remains valid when we pass to an induced derivator $\mathbf{D}_Z$ (Z arbitrary in Cat) ; in other words, that $v_!$ exists when v is of the form $\mathrm{id}_Z \times u$, where $u : I \longrightarrow I'$ is an arrow of $\mathcal{D}$, and that the calculation is done *over* I' according to the rule Der 5 bis. This therefore means that the base change arrow for the $f_!$ is an isomorphism in the case that follows: [page 70]

$$\begin{array}{ccc} I \times Z & \longleftarrow & (i'\backslash I) \times Z \\ \downarrow{\scriptstyle u\times \mathrm{id}_Z} & & \downarrow{\scriptstyle \mathrm{pr}_2} \\ I' \times Z & \longleftarrow & \{i'\} \times Z \end{array} \quad ,$$

where i' is an arbitrary object of I'.

The *strict* hypothesis for a category with cofibrations seems fairly innocuous to me, in most cases known to me, it is satisfied. But even with this hypothesis, I see no way to prove the existence of $u_!$ for *every* arrow of Cat, and this is of course a very awkward situation. One would very much like to isolate conditions that ensure the validity of Der 4 bis and of Der 5 bis (existence of $u_!$ for *every* arrow of Cat, and way of computing the fibers …). [page 71]

p. 34

4 Special categories with cofibrations

4.1. I now suppose that

ⓐ $\mathcal{M}$ is stable under filtering $\varinjlim$, and

ⓑ that its objects are accessible,

so that we are under the conditions for applying the factorization theorem (XV, §1). We suppose:

ⓒ given $W, C \subset \mathrm{Fl}(\mathcal{M})$, satisfying the conditions Cof_1, Cof_2 of page 39. For Cof_3, cf. below.

ⓓ C, $TC \overset{\mathrm{def}}{=} C \cap W$ *stable under transfinite composition* (this is the case if the structure comes from a closed Quillen structure (W, C, F)).

$$TF \overset{\mathrm{def}}{=} C_* \ \subset \ \mathrm{Fl}(\mathcal{M}) \ , \tag{4.1}$$

it follows from the factorization theorem that every arrow f of $\mathcal{M}$ factors as

$$f \ = \ pi \ , \qquad p \in TF, \ i \in C \ ,$$

provided we make the hypothesis

ⓔ

$$\exists\, C_0 \subset C, \ \textit{small} \text{ subset of } C, \text{ such that } C \subset \tilde{C}_0, \text{ i.e. } TF = (C_0)_*. \tag{4.3}$$

Under these conditions, if we suppose

ⓕ

$$TF \subset W \ , \tag{4.4}$$

then the factorization axiom Cof_3 is satisfied. Conversely, if Cof_3 is satisfied, and if W is stable under direct factors, then we have 4.4. Thus if W is stable under direct factors, Cof_3 is equivalent to (4.4). [page 72]

4.2. It seems to me that these conditions, in practice, will hardly ever be fulfilled, unless W (which is given in the very first place, satisfying Cof_0 [(18)]), is stable under $\varinjlim$ that are filtering. At that point, to construct the cofibration structure, we start from C_0 a small subset of $\mathrm{Fl}(\mathcal{M})$, taken so that we have p. 35

$$(C_0)_* \ \subset \ W \ , \tag{4.5}$$

we take for C at choice the saturation C_0' of C_0 under cobase change and transfinite composition, or $C = \tilde{C}_0$ = set of direct factors of C_0'. The conditions Cof_1 (stabilities of C), Cof_3 (factorization) are satisfied automatically, and it remains to take care of Cof_2, which comes down to the stability of $C \cap W$ under cobase change (stability under composition and that it contain the isomorphisms, being immediate). This is a delicate condition, and the only standard way I see to ensure it, is by supposing

$$C_0 \ \subset \ \mathrm{Cof}_W \ ; \tag{4.6}$$

the stability of W under filtering $\varinjlim$ (hence also under direct factors) then ensures that we also have [page 73]

$$C \ \subset \ \mathrm{Cof}_W \ , \tag{4.7}$$

and consequently

$$\begin{cases} C \cap W \subset \mathrm{Cof}_W \cap W = W^{\mathrm{univ}}, \\ \text{hence} \ \ TC \overset{\mathrm{def}}{=} C \cap W = C \cap W^{\mathrm{univ}} \ , \end{cases} \tag{4.8}$$

which ensures the stability of $C \cap W$ under cobase change. The stability of TC under transfinite composition follows from that of C, and from the stability of W under transfinite composition.

4.3. I return to the general conditions ⓐ to ⓕ of 4.1, without necessarily supposing W stable under filtering $\varinjlim$. I now want to extend the cofibration structure to the $\mathcal{M}(I)$, I arbitrary in Cat. For this we do not (I think) have to use the developments of sections 2, 3, but we will make the connection with these later on.

Let us denote by I_0 the discrete category having the same objects as I, whence a canonical functor

$$v = v_I \ : \ I_0 \longrightarrow I \ . \tag{4.9}$$

The idea is to define the structure of $\mathcal{M}(I)$ via the evident structure of $\mathcal{M}(I_0) \simeq \mathcal{M}^{I_0}$, by playing on the pair of adjoint functors

$$\mathcal{M}(I_0) \underset{\psi = v^*}{\overset{\varphi = v_!}{\rightleftarrows}} \mathcal{M}(I) \quad .$$

We are going to present things in a slightly more general way.

Definition 4.4. I call a *special* cofibration structure a structure $(\mathcal{M}, W, C)$ satisfying conditions a) to f) of 4.1, which I reformulate: [page 74]

ⓐ $\mathcal{M}$ stable under small direct limits, and the objects of $\mathcal{M}$ are accessible.

[18] i.e. weak saturation. p. 36

ⓑ W is weakly saturated (i.e. Cof_0 of p. 39), C stable under cobase change and under composition, $TC \overset{\text{def}}{=} C \cap W$ stable under cobase change (and under composition automatically).

ⓑ′ C and TC are stable even under composition that is *transfinite.*

ⓒ $\exists$ small subset $C_0 \subset C$ such that $C \subset \tilde{C}_0$ ($\Longrightarrow \tilde{C} = \tilde{C}_0$).

ⓓ $TF \overset{\text{def}}{=} C_* \subset W$ (which implies, together with ⓐ to ⓒ, the factorization axiom Cof_3 of page 39).

4.5. We are going to give a general recipe for extending such a structure, when we have a pair of adjoint functors

$$\mathcal{M} \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}' \quad ,$$

$\mathcal{M}$ being equipped with a special cofibration structure (W, C), and $\mathcal{M}'$ satisfying the preliminary condition ⓐ. We will then define the subsets W', C' of $\mathrm{Fl}(\mathcal{M}')$ by [page 75]

$$\begin{cases} W' &= (v^*)^{-1}(W) \\ C' &= \text{closure stable under cobase change} \\ & \qquad \text{and transfinite composition of } v_!(C), \end{cases} \tag{4.5.1}$$

thus

$$v_!(C) \ \subset \ C' \ \subset \ (v_!(C))^{\sim} = \ \tilde{C}' \ . \tag{4.5.2}$$

I say that (W', C'), and at choice $(W', \tilde{C}')$, is a special cofibration structure on $\mathcal{M}'$. We are going to check the various conditions of 4.4. Condition a) is already assured.

b) [(19)]. The weak saturation of W' follows from that of W, via (4.5.1). The stability of C' under cobase change and under composition (even transfinite) is assured by definition of C', and likewise for $\tilde{C}'$ (being Q-closed on the left). Stability of $TC' \overset{\text{def}}{=} C' \cap W'$ under cobase change? It suffices to see that if we have a cocartesian square

$$\begin{array}{ccc} X'_0 & \xrightarrow{i'} & X'_1 \\ \downarrow & \text{cocart.} & \downarrow \\ X'_2 & \xrightarrow{j'} & X'_3 \end{array}$$

in $\mathcal{M}'$, with i' in $TC' = C' \cap W'$, then $j' \in W'$ (for we already know $j' \in C'$), i.e. $v^*(j') \in W$. We are going to suppose that v^* *commutes with amalgamated sums,* and we are thus reduced to proving that $v^*(i') \in TC$. Thus we must prove the ||

Lemma 4.5.2. *Under the conditions* (4.5.3) *and* (4.5.4) *below, we have* p. 37 [page 76]

$$v^*(C') \ \subset \ C \ , \qquad v^*(TC') \ \subset \ TC \ ,$$

and if $C = \tilde{C}$, we have $v^*(\tilde{C}') \subset C$, $v^*(T\tilde{C}') \subset TC$.

For $v^*(C') \subset C$, we must first prove

$$v^*(v_!(C)) \ \subset \ C \ , \tag{4.5.3}$$

and I see no way – we have to take it as a hypothesis. Moreover, that v^* commutes with arbitrary $\varinjlim$ (cobase change already admitted – now we need filtering direct limits)

$$v^* \text{ commutes with small } \varinjlim \text{ of any kind} \tag{4.5.4}$$

[19] **N.B.** It is better to start by proving c).

to conclude from (4.5.3) that $v^*(C' = \text{closure of } v_!(C) \text{ under cobase change and transfinite composition})$ is also in C. Passing to $v^*(TC') \subset TC$, by what precedes, $v^*(C') \subset C$, it suffices to prove $v^*(TC') \subset W$, but $v^*(TC') \subset v^*(W') \subset W$ by definition of W'.

Let us suppose $C = \tilde{C}$, i.e. C stable under direct factors, and let us prove $v^*(\tilde{C}') \subset C$, $v^*(T\tilde{C}') \subset TC$. By c) below, $\tilde{C}'$ is formed of the direct factors of elements of C', hence [page 77] $v^*(C') \subset C$ implies $v^*(\tilde{C}') \subset C$ since C stable under direct factors.

Let us finally prove $v^*(T\tilde{C}') \subset TC \overset{\text{def}}{=} C \cap W$. As $T\tilde{C}' = \tilde{C}' \cap W'$, this follows from $v^*(\tilde{C}') \subset C$, and from $v^*(W') \subset W$.

Let us prove b′). The stability of C' and of $\tilde{C}'$ under transfinite composition is assured, it remains to prove the stability of $TC' = C' \cap W'$ and of $T\tilde{C}' = \tilde{C}' \cap W'$. It suffices to see that the transfinite composite f in question is in W', hence that $v^*(f) \in W$, but this follows from the fact that v^* commutes with transfinite composition by (4.5.4), and that v^* takes TC' and $T\tilde{C}'$ into TC.

Let us prove c) for C' (hence also for $\tilde{C}'$). Let C_0 be a small subset of C such that $C \subset \tilde{C}_0$, i.e. C is contained in the closure of C_0 under the three operations of cobase change, of transfinite composition, of direct factors. We then see that $v_!(C)$ is in the closure of $v_!(C_0)$ under the operations, hence also C', and we can then take $C'_0 = v_!(C_0)$.

Let us finally prove d), it is the same assertion for C' and for $\tilde{C}'$, since

$$C'_* \;=\; \tilde{C}'_* \;(\overset{\text{def}}{=}\; TF'\;)\;,$$

we must therefore prove [page 78]

$$TF' \overset{\text{def}}{=} C'_* \;\subset\; W'\;.$$

This will follow from the [p. 38]

Lemma 4.5.3. *We have* $TF' = (v^*)^{-1}(TF)$.

This is formal from the definition of TF' as

$$TF' \;=\; (v_!(C))_*\;,$$

of TF as

$$TF \;=\; C_*\;,$$

and from the adjunction of $v_!$, v^*. As $TF \subset W$, we conclude

$$TF' \;=\; (v^*)^{-1}(TF) \;\subset\; (v^*)^{-1}(W) = W'\;,$$

q.e.d.

Thus we have proved this:

Proposition 4.6. *Let us consider a pair of adjoint functors*

$$\mathcal{M} \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}'\quad,$$

where $\mathcal{M}$ *is equipped with a cofibration structure that is* special (W, C). *We further suppose the following conditions satisfied.*

$$\begin{aligned}
&1^\circ) \quad \mathcal{M}' \text{ stable under small } \varinjlim \text{ , its elements are accessible.}\\
&2^\circ) \quad v^* \text{ commutes with small } \varinjlim\ . \qquad (4.6.1)\\
&3^\circ) \quad v^*(v_!(C)) \subset C.
\end{aligned}$$

Let us consider the subsets W', C' *of* $\mathrm{Fl}(\mathcal{M}')$ *defined by* [page 79]

$$\begin{cases} W' &= (v^*)^{-1}(W), \\ C' &= \textit{closure of } v_!(C) \textit{ under cobase change and under composition} \\ & \quad \textit{that is transfinite.} \end{cases} \tag{4.6.2}$$

Then $(\mathcal{M}', W', C')$ is a special cofibration structure. If moreover $C = \tilde{C}$, i.e. C is stable under direct factors, then $(\mathcal{M}', W', \tilde{C}')$ is also a special cofibration structure [20].

4.7. I would like, with a view to statements of type 3.7 (establishing the existence of functors of type $u_!^W$), to have a hold on $TC' = C' \cap W'$, and on $T\tilde{C}' = \tilde{C}' \cap W'$, to be precise, I would like to have the relation

$$v_!(TC) \ \subset TC' \subset v_!(TC)\tilde{}\,, \tag{4.7.1}$$

or even the slightly stronger relation

$$\underbrace{v_!(TC) \ \subset TC'} \overset{\text{tautol.}}{\subset} \underbrace{T\tilde{C}' \ \subset (v_!(TC))\tilde{}}\,, \tag{4.7.2}$$

which decomposes into two

p. 39

$$\begin{cases} v_!(TC) & \subset & TC' \\ T\tilde{C}' & \subset & (v_!(TC))\tilde{}. \end{cases} \tag{4.7.3}$$

The less delicate of the two is the first, $v_!(TC) \subset TC'$, which amounts, by virtue of the definitions, implying [page 80]

$$v_!(TC) \subset v_!(C) \subset C'\,,$$

to

$$v_!(TC) \ \subset \ W'\,,$$

i.e. by definition of W', to

$$v^* v_!(TC) \ \subset \ TC\,, \tag{4.7.4}$$

which is a hypothesis of type (4.6.1) 3°), to be added to the hypotheses (4.6.1).

The hypothetical relation

$$\underbrace{T\tilde{C}'}_{=\tilde{C}'\cap W'} \ \subset \ (v_!(TC))\tilde{}, \quad \text{or only} \quad TC' \subset (v_!(TC))\tilde{}, \tag{$*$}$$

appears to me to be of a distinctly more delicate nature. Since $T\tilde{C}' \supset TC' \supset v_!(TC)$, it is equivalent to the *equality* of $(T\tilde{C}')_*$ resp. $(TC')_*$ and $(v_!(TC))_* = (v^*)^{-1}(TC)_*$. Let us put

$$\begin{cases} (TC)_* & \overset{\text{def}}{=} & F \\ (TC')_* & \overset{\text{def}}{=} & F' \ \supset \ (T\tilde{C}')_* \ = \ F'_\sim\,. \end{cases} \tag{4.7.5}$$

We therefore have, by virtue of the inclusion

$$v_!(TC) \ \subset \ TC' \ (\subset \ T\tilde{C}')\,,$$

the relation

$$F'_\sim \ \subset \ F' \ \subset \ (v^*)^{-1}(F)\,, \tag{4.7.6}$$

and the question is therefore whether we have equality $F' = (v^*)^{-1}(F)$, or even $F'_\sim = (v^*)^{-1}(F)$ (whence $F'_\sim = F'$).

[20] **N.B.** This is a general fact for special categories with cofibrations, and should be made explicit below (cf. 5.1). The hypothesis on C is unnecessary here (or rather, it suffices instead that W be stable under direct factors *and* under transfinite composition, which is not a more cheering hypothesis …). [page 81]

In fact, we show fairly easily that we have $F'_{\sim} = F'$, i.e. $T\tilde{C}' \overset{\text{def}}{=} \tilde{C}' \cap W' \subset (C' \cap W')\tilde{}$ – it is a general story about special categories with cofibrations [21]. But I hardly see how to prove (under suitable hypotheses) that $F' = (v^*)^{-1}(F)$. I shall have to come back to this if need be [22]. For the moment, I shall confine myself to reaping the fruit of 4.6 in a particular case:

Proposition 4.8. *Let $(\mathcal{M}, W, C)$ be a special category with cofibrations* (definition 4.4, page 74)*. Let I be an object of* Cat*, I_0 the associated discrete category, let us consider $v : I_0 \longrightarrow I$ and*

p. 40

$$\mathcal{M}(I_0) \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}(I) \quad ,$$

finally let us endow $\mathcal{M}(I_0) = \mathcal{M}^{I_0}$ with the structure of a category with cofibrations, product *of those of the factors, i.e. let us define $W(I_0)$, $C(I_0)$ "fiber by fiber". (This coincides with the general definition given in paragraph 2, for I_0 endowed with the discrete order structure, and makes $\mathcal{M}(I_0)$ a special category with cofibrations.) Then the couple $(v_!, v^*)$ satisfies the conditions of proposition* 4.6*, as well as the condition supplementary to* (4.6.1)*,*

[page 82]

$$4^\circ) \qquad v^*(v_!(TC(I_0))) \subset TC(I_0)\ . \tag{4.8.1}$$

Thus we find a structure of a special category with cofibrations on $\mathcal{M}(I)$, by putting

$$W(I) \ = \ (v^*)^{-1}(W(I_0)) \tag{4.8.2}$$

(this is in fact equivalent to the usual definition of $W(I)$) and

$$C(I) \quad = \quad \begin{array}{l}\textit{closure of } v_!(C(I_0)) \textit{ under the operations of cobase}\\ \textit{change and transfinite composition.}\end{array} \tag{4.8.3}$$

We have moreover under these conditions:

$$\underbrace{TF(I)}_{\overset{\text{def}}{=}(C(I))_*} \ = \ (v^*)^{-1}\Big(\ \underbrace{TF(I_0)}_{\substack{\overset{\text{def}}{=}(C_{I_0})_*,\\ \textit{is also defined}\\ \textit{fiber by fiber}}} \ \Big)\ , \tag{4.8.8}$$

$$\left\{\begin{array}{rcl} v_!\big(\overbrace{(TC)(I_0)}^{C(I_0)\cap W(I_0)}\big) & \subset & TC(I) \\ F(I) \overset{\text{def}}{=} (TC(I))_* & \subset & (v^*)^{-1}(F(I_0)). \end{array}\right. \quad {}^{(23)} \tag{4.8.9}$$

Proof. We already know that $\mathcal{M}(I)$ is stable under small $\varinjlim$, and that every element is accessible, as follows at once from the appropriate properties of $\mathcal{M}$. It is also clear that v^* commutes with small $\varinjlim$. It remains to prove (4.6.1. 3°), and 4° in (4.8.1), i.e. that $v^*v_! : \mathcal{M}(I_0) \longrightarrow \mathcal{M}(I_0)$ maps C into C, TC into TC. Now we have

$$v_!(F_0)(i) \ = \ \varinjlim_{(\alpha,j)\in i\backslash I_0} F_0(j) \qquad {}^{(24)}$$

[page 83]

and $i\backslash I_0$ is the discrete category over I_0, whose fiber at $j \in \mathrm{Ob}\, I_0 \ = \ \mathrm{Ob}\, I$ is $\mathrm{Hom}_I(i,j)$:

$$i\backslash I_0 \ = \ \coprod_{j\in \mathrm{Ob}\, I} (\text{discrete category associated to } \mathrm{Hom}_I(i,j))\ ,$$

thus

[21] cf. prop. 5.1 below.

[22] see however prop. 5.8 below.

[23] These inclusions should give $TC(I) = (v_!(TC(I_0)))\tilde{}$ and $F(I) = (v^*)^{-1}(F(I_0))$, but I do not know how to prove it without making stronger hypotheses, of the type *special Quillen category.*

p. 41

[24] where $j \in \mathrm{Ob}\, I_0 = \mathrm{Ob}\, I$, and $\alpha : i \longrightarrow v(j) = j$.

$$v_!(F_0)(i) \;=\; \coprod_{j\in \mathrm{Ob}\, I=\mathrm{Ob}\, I_0} \underbrace{F_0(j)^{\mathrm{Hom}_I(i,j)}}_{\substack{\text{direct sum}\\ \text{of } \mathrm{Hom}_I(i,j)\\ \text{copies of } F_0(j)}},$$

so we find

$$v^*v_!F_0(\underbrace{\quad i \quad}_{\in \mathrm{Ob}\, I_0=\mathrm{Ob}\, I}) \;=\; v_!(F_0)(i) \;=\; \text{  } \quad .$$

Since C and TC are stable under arbitrary direct sums (being so under finite direct sums, because of the finite stability conditions on C and TC, and also under infinite sums because of the axiom of transfinite stability …[25]), it does indeed follow that $v^*v_!$ maps $C(I_0)$ and $TC(I_0)$ into themselves.

Thus, the first assertion of 4.8 follows from 4.6. Moreover, (4.8.8) and (4.8.9) were seen in 4.7, under the general conditions of 4.6 and assuming in addition the 4° of (4.7.4).

[page 84]

5 Relation of special categories with cofibrations to special Quillen categories

Let $(\mathcal{M}, W, C)$ be a special category with cofibrations. I introduce

$$TF \;=\; C_*, \qquad TC \;=\; C\cap W, \qquad F \;=\; (TC)_*\,, \tag{5.1}$$

so we have

$$\begin{array}{ccc} TC & \hookrightarrow & C \\ \updownarrow & & \updownarrow \\ F & \hookleftarrow & TF \end{array}\quad, \tag{5.2}$$

(C, TF) being a right closed Quillen precouple. We then know (XV 4.2 cor. 1, page 31) that $\tilde{C}$ consists of the direct factors of elements of C. Let us put

$$T\tilde{C} \;=\; \tilde{C}\cap W\,, \tag{5.3}$$

whence

$$TC \;\subset\; T\tilde{C}\,. \tag{5.4}$$

Proposition 5.1 *Let $(\mathcal{M}, W, C)$ be a special category with cofibrations. We have $T\tilde{C} \subset \tilde{(TC)}$ (and even $T\tilde{C} \subset (TC)^\natural$ [26]), or what amounts to the same thing by virtue of (5.4), TC and $T\tilde{C}$ have the same left Q-closure:*

$$\tilde{(T\tilde{C})} = \tilde{(TC)} \qquad \textit{(and even } T\tilde{C} \subset (TC)^\natural\textit{)}, \tag{5.5}$$

whence p. 42

$$(\tilde{C})_* = C_*\,(= TF), \qquad (T\tilde{C})_* = (TC)_*\,(= F) \qquad {}^{(27)}. \tag{5.6}$$

Moreover, if W stable under direct factors (or if merely every direct factor of an arrow of TC is in W), and if moreover W is stable under transfinite composition, then $(\mathcal{M}, W, \tilde{C})$ is a special category with cofibrations. Moreover, $T\tilde{C}$ is left Q-saturated, hence (by 5.5)

$$T\tilde{C} \;=\; (TC)^\natural\,. \tag{5.7}$$

[page 85]

[25] deserves a general lemma …

[26] **N.B.** For a subset A of $\mathrm{Fl}(\mathcal{M})$, $A^\natural$ denotes the set of direct factors of elements of A.

[27] These relations do not depend on the hypothesis W stable under direct factors.

PROOF. Let us prove

$$T\tilde{C} \ \subset \ (TC)\tilde{},$$

and more precisely, that every element of $T\tilde{C}$ is a direct factor of an element of TC:

$$T\tilde{C} \subset (TC)^{\natural} .$$

Let then $f \in T\tilde{C} \overset{\text{def}}{=} \tilde{C} \cap W$, let us consider their factorization of f as pi, with $i \in C$, $p \in TF$. Since $TF \subset W$, we have $p \in W$, hence $i \in W$, hence $i \in C \cap W \overset{\text{def}}{=} TC$. Since $f \in \tilde{C}$, $p \in TF$, there exists $q : Y \longrightarrow X'$ such that $pq = \mathrm{id}_Y$, $qf = i$,

$$\begin{array}{ccc} X & \xrightarrow{f \in \tilde{C} \cap W} & Y \\ & {\scriptstyle i \in C} \searrow \quad \overset{q}{\swarrow} \quad \nearrow {\scriptstyle p \in TF} & \\ & X' & \end{array}$$

so f appears as a direct factor of $i \in TC$, q.e.d.

Let us prove that $(\mathcal{M}, W, \tilde{C})$ is a special category with cofibrations (def. 4.4, page 74). Condition a) on $\mathcal{M}$ settled, condition b), the part on W, too. The stability of $\tilde{C}$ under cobase change and under composition OK, also (for b′) the stability under transfinite composition. Stability of $T\tilde{C} = \tilde{C} \cap W$ under cobase change: it remains to prove that if we have a cocartesian square

$$\begin{array}{ccc} X & \xrightarrow{\ i\ } & Y \\ \downarrow & \text{cocart.} & \downarrow \\ X' & \xrightarrow{\ i'\ } & Y' \end{array} ,$$

if $i \in \tilde{C} \cap W$, then $i' \in W$ (for we already know that $i' \in \tilde{C}$). But the preceding proof shows that it is a direct factor *in* $X \backslash \mathcal{M}$ of an arrow in TC, so i' is a direct factor of an arrow of $TC \subset W$. So if W is stable under direct factors, $i' \in W$.

This finishes proving b). For b′), it remains to see that $T\tilde{C}$ is stable under transfinite composition. I see no other way of seeing it, than by assuming that W itself is stable under transfinite composition (which seems scarcely less demanding than the stability of W under $\varinjlim$). This finishes proving b, b′). The condition c) on C implies the same condition on $\tilde{C}$. Finally, since $\tilde{C}_* = C_*$, the condition d) is the same for C and $\tilde{C}$. This finishes proving that $(\mathcal{M}, W, \tilde{C})$ is a special category with cofibrations. (**N.B.** we needed two additional hypotheses on W, stability under direct factors and under transfinite composition, to establish that $T\tilde{C}$ is stable under cobase change and under transfinite composition). The last assertion of 5.1 and the relation (5.7) which results from it, amounts to saying that $T\tilde{C} = \tilde{C} \cap W$ is stable under direct factors, which is obvious by the same property of $\tilde{C}$, and the hypothesis made on W of stability under direct factors (or only that the direct factors of elements of $T\tilde{C}$ are in W, hence in $T\tilde{C}$).

p. 43
[page 86]

Proposition 5.2. *Let $(\mathcal{M}, W, C)$ be a special category with cofibrations. Then with the notations of* (5.1)*, we have*

[page 87]

$$W \ = \ TF \circ TC, \tag{5.8}$$

$$TF \ = \ F \cap W. \tag{5.9}$$

PROOF. Let $f \in W$, let us write $f = pi$, with $i \in C$, $p \in TF$, hence $p \in W$, hence $i \in W$, hence $i \in C \cap W = TC$, whence (5.8).

Let us prove (5.9). We have $TF \subset F \cap W$, let us prove conversely that an $f \in F \cap W$ is in TF. Let us write $f = pi$, with $p \in TF$, $i \in TC$.

$$\begin{array}{ccc} X & \xrightarrow{f \in F \cap W} & Y \\ & {\scriptstyle i \in TC} \searrow \quad \nearrow {\scriptstyle p \in TF} & \\ & X' & \end{array}$$

Since $f \in F$, $i \in TC$, and $TC \longleftrightarrow F$ by definition of F, $\exists\, r : X' \longrightarrow X$ such that $ri = \mathrm{id}_X$, $fr = p$. So f is a direct factor of $p \in TF$, and since $TF = C_*$ is stable under direct factors, we conclude that $f \in TF$.

Corollary 5.3. *Let $(\mathcal{M}, W, C)$ be a special category with cofibrations. For (W, C, F), $(F \overset{\text{def}}{=} (TC)_*)$ to be a Quillen triple in $\mathcal{M}$, it is necessary and sufficient that the couple (TC, F) satisfy the factorization axiom (i.e. that it be a Quillen precouple). For this Quillen structure to be closed, it is necessary and sufficient that C be stable under direct factors.*

PROOF. The necessity of the factorization axiom for (TC, F), in order to have a Quillen triple, is tautological. For sufficiency, already having the two Quillen precouples (C, TF), (TC, F) with [page 88]

p. 44

$$\begin{array}{ccc} TC & \hookrightarrow & C \\ \updownarrow & & \updownarrow \\ F & \hookrightarrow & TF\ , \end{array} \tag{5.2}$$

and $TC = C \cap W$, the fact that (W, C, F) is a Quillen triple is corollary 4 p. 36 of XV: The conditions $TC = C \cap W$, $TF = F \cap W$, $W = TF \circ TC$ are equivalent, and equivalent to the fact that (W, C, F) is a Quillen triple.

By Quillen's definition of triples that are *closed,* by the properties

$$\underbrace{W = TF \circ TC}_{\substack{\text{true for all}\\ \text{the Quillen}\\ \text{triples}}}, \qquad C = (TF)^*, \qquad \underbrace{F = (TC)_*}_{\substack{\text{true in}\\ \text{the present case}}}\ ,$$

it remains to express that $C = (TF)^*$, which is equivalent to C stable under direct factors (cf. loc. cit. cor. 1, p. 31), q.e.d.

Remark 5.4. The statements 5.2 and 5.3, and the relation $T\tilde{C} \subset (TC)^\natural$ in proposition 5.1, do not call upon the whole hypothesis that $(\mathcal{M}, W, C)$ be a special category with cofibrations, but only that we have a category with cofibrations, with $C \longleftrightarrow \underbrace{TF}_{\overset{\text{def}}{=} C_*} \subset W$, i.e. a category with W, C satisfying Cof_0, Cof_1, Cof_2 of page 39, plus $TF \overset{\text{def}}{=} C_* \subset W$, plus the factorization axiom for (C, TF) (which implies Cof_3). [page 89]

Corollary 5.5. *Let $(\mathcal{M}, W, C)$ be a special category with cofibrations, such that there exists a* small *subset $TC_0 \subset TC = C \cap W$, with*

$$TC \ \subset\ \tilde{TC_0}\ , \tag{5.5.1}$$

or what amounts to the same thing, $(TC)\tilde{} = (TC_0)\tilde{}$, i.e.

$$\underbrace{F}_{\overset{\text{def}}{=}(TC)_*} \ =\ (TC_0)_*\ . \tag{5.5.2}$$

Then (W, C, F) is a Quillen triple. This triple is closed if and only if C is stable under direct factors.

Indeed, the factorization theorem in XV §1 implies that under the hypothesis made on $TC_0 \subset TC$, the pair (TC, F) satisfies the factorization condition. Indeed, by conditions b, b′ of definition 4.4, TC has the desired stability properties (contains the isomorphisms, is stable under cobase change and under transfinite composition).

Definition 5.6. The Quillen triples deduced as in 5.5 from a special category with cofibrations are called *special.* Thus a Quillen triple (W, C, F) in $\mathcal{M}$ is special if it satisfies conditions a) to d) in definition (4.4), concerning (W, C) exclusively, plus the following two conditions. [page 90]

p. 45

ⓒ′ ∃ small subset TC_0 of TC, such that $TC \subset (TC_0)^{\tilde{}}$.

ⓔ $F = (TC)_*$.

Corollary 5.7. *Let (W, C, F) be a Quillen triple that is* closed *in $\mathcal{M}$. For this triple to be special, it is necessary and sufficient that it satisfy the following conditions.*

ⓐ *$\mathcal{M}$ stable under small $\varinjlim$, and its elements are accessible.*

(c, c′) *There exists a small subset C_0 of C, TC_0 of TC, such that we have*

$$C = C_0^{\tilde{}}\,, \qquad TC = (TC_0)^{\tilde{}}\,. \tag{5.7.1}$$

Indeed, all the other conditions a) to e), including c′) of 5.6, are automatically satisfied.

Proposition 5.8. *Let us consider a pair of adjoint functors* [page 91]

$$\mathcal{M} \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}'\,, \tag{5.8.1}$$

where $\mathcal{M}$ is equipped with a special closed Quillen triple $(W, C, F = (TC)_*)$, with $TC = C \cap W$ (cf. definition 5.6). *We suppose the following conditions satisfied* (compare prop. 4.6, relations (4.6.1), and (4.7.4)):

$$\begin{cases} \text{a)} & \mathcal{M}' \text{ stable under small } \varinjlim \text{ , its elements are accessible.} \\ \text{b)} & v^* \text{ commutes with small } \varinjlim\,. \\ \text{c)} & v^*(v_!(C)) \subset C,\ v^*(v_!(TC)) \subset TC. \end{cases} \tag{5.8.2}$$

Let us define (W', C', F') in $\mathcal{M}'$ (compare prop. 4.6, notably 4.6.2) *by*

$$\begin{cases} W' &= (v^*)^{-1}(W) \\ C' &= v_!(C)^{\tilde{}} \\ F' &= (TC')_*\,, \qquad \text{where } TC' \overset{\text{def}}{=} C' \cap W'\,. \end{cases} \tag{5.8.3}$$

Then (W', C', F') is a closed special Quillen triple in $\mathcal{M}'$, and moreover we have

$$F' = (v^*)^{-1}(F)\,, \qquad TF' = (v^*)^{-1}(TF)\,, \tag{5.8.4}$$

$$TC' = (v_!(TC))^{\tilde{}}\,. \tag{5.8.5}$$

Proof. By 4.6, we already know that $(\mathcal{M}', W', C')$ is a special category with cofibrations. (**N.B.** What is denoted C' here, was denoted $\tilde{C}'$ in proposition 4.6. We must use the last assertion of 4.6, where we assume C stable under direct factors, OK since (W, C, F) assumed closed.) To prove that $(W', C', F' \overset{\text{def}}{=} (TC')_*)$ is a Quillen triple that is special, it remains, by 5.3 and definition 5.6, to prove that there exists a small subset TC'_0 of TC', such that [page 92] p. 46

$$(\ TC'_0 \ \subset\ )\ TC' \ \subset\ (TC'_0)^{\tilde{}}\,. \tag{5.8.6}$$

(This Quillen triple will moreover be closed, since C' is Q-closed on the left by definition.) So let TC_0 be a small subset of TC such that

$$TC \ \subset\ (TC_0)^{\tilde{}}\,, \tag{5.8.7}$$

which exists by the hypothesis on (W, C, F). We shall take

$$TC'_0 \overset{\text{def}}{=} v_!(TC_0)\,. \tag{5.8.8}$$

We saw in 4.7 that we have

$$v_!(TC) \ \subset \ TC' \tag{5.8.9}$$

(follows from the hypothesis $v^*(v_!(TC)) \subset TC$ in (5.8.2. c)), a fortiori

$$TC'_0 \ = \ v_!(TC_0) \ \subset \ TC' \ ,$$

it remains to prove [page 93]

$$TC' \ \overset{?}{\subset} \ (TC'_0)\tilde{} \ , \tag{5.8.10}$$

which is equivalent to $(TC')\tilde{} \subset (TC'_0)\tilde{}$, or again to

$$(TC'_0)_* \ \overset{?}{\subset} \ (TC')_* \ \overset{\text{def}}{=} \ F' \ , \tag{5.8.10 bis}$$

whereas we have a priori the reverse inclusion (deduced from $TC'_0 \subset TC'$)

$$F' \ \subset \ (TC'_0)_* \ . \tag{5.8.11}$$

Now by adjunction, we have

$$\begin{aligned} (\ \underbrace{TC'_0}_{v_!(TC_0)}\ )_* \ &= \ (v_!(TC_0))_* \\ &= \ (v^*)^{-1}(\ \underbrace{(TC_0)_*}_{=(TC)_* \overset{\text{def}}{=} F}\ ) \\ &= \ (v^*)^{-1}(F) \ , \end{aligned}$$

so (5.8.11) is written

$$F' \ \subset \ (v^*)^{-1}(F) \ , \qquad \text{i.e.} \ \ v^*(F') \ \subset \ F \tag{5.8.12}$$

(compare (4.7.6)), and we must show that this inclusion is an equality (on which I had gotten stuck p. 80, 81 …), which is none other than the first of the two relations (5.8.4). But it is in the initial form (5.8.10) (transposed from (5.8.4)) that we are going to prove the thing, finally. Note that we have p. 47 [page 94]

$$\underbrace{(v_!(TC_0))\tilde{}}_{TC'_0} \ = \ (v_!(TC))\tilde{} \ ,$$

since TC is the closure of TC_0 under the operations of cobase change, of transfinite composition, and of direct factors, and $v_!$ commutes with these operations. So we must simply show

$$TC' \ \subset \ (v_!(TC))\tilde{} \tag{5.8.12 bis}$$

(where there is no longer any question of TC_0). But the factorization theorem applies to the pair

$$\Big((v_!(TC))\tilde{}, \ (v_!(TC))_* = (v^*)^{-1}(F)\Big) \ ,$$

since $(v_!(TC))\tilde{} = (TC'_0)\tilde{}$, TC'_0 small. So if we have $f' \in TC' = C' \cap W'$, let us factor f' as $p'i'$, with

$$i' \ \in \ (v_!(TC))\tilde{} \ , \qquad p' \ \in \ (v^*)^{-1}(F) \ . \tag{5.8.13}$$

$$\begin{array}{ccc} X' & \xrightarrow[q']{f' \in C' \cap W'} & Y' \\ & {\scriptstyle i'} \searrow \quad \nearrow {\scriptstyle p'} & \\ & X'_1 & \end{array} \tag{$*$}$$ [page 95]

Note that we have

$$(v_!(TC))\tilde{} \ \subset \ TC' \ (\overset{\text{def}}{=} \ C' \cap W' \ ) \tag{5.8.14}$$

for $v_!(TC) \subset TC' \subset C'$, hence $(v_!(TC))\tilde{} \subset \tilde{C}' = C'$, and it remains to prove that $v_!(TC)\tilde{} \subset W'$, i.e. $v^*((v_!(TC))\tilde{}) \subset W$. Now I say that we even have

$$v^*((v_!(TC))\tilde{}) \ \subset \ TC \ ,$$

i.e.

$$v_!(TC)\tilde{} \ \subset \ (v^*)^{-1}(TC) \ . \tag{5.8.15}$$

This follows from the fact that TC is Q-closed, hence Q-saturated on the left, which implies, since v^* commutes with small $\varinjlim$, that $(v^*)^{-1}(TC)$ is so as well. On the other hand, it contains $v_!(TC)$, hence also the left Q-saturation of $v_!(TC)$. But the latter is *equal* to the Q-closure $(v_!(TC))\tilde{}$, by the factorization theorem, seeing that $TC'_0 \subset v_!(TC) \subset (TC'_0)\tilde{}$, with TC'_0 small.

Thus, (5.8.13) and (5.8.14) imply $i' \in TC' = C' \cap W'$, hence $i' \in W'$, whence $p' \in W'$, i.e. $v^*(p') \in W$, and since by hypothesis (5.8.13) we have $v^*(p') \in F$, we have $v^*(p') \in TF$, i.e.

p. 48

[page 96]

$$p' \ \in \ (v^*)^{-1}(TF) \ = \ TF' \ (\overset{\text{def}}{=} \ C'_* \ ) \ ,$$

the equality $(v^*)^{-1}(TF) = TF'$, i.e.

$$(v^*)^{-1}(C_*) \ = \ (v_!(C))_* \ ,$$

resulting from the adjunction of $v_!$, v^*. Altogether, we therefore have

$$p' \ \in \ TF' \ (\ = \ C'_* \ ) \ , \tag{5.8.16}$$

and since $f' \in C'$, diagram $(*)$, page 94, shows us that there exists

$$q' \ : \ Y' \longrightarrow X'_1 \ ,$$

with $p'q' = \mathrm{id}$, $q'f' = i'$. So f' is a direct factor of $i' \in v_!(TC)\tilde{}$, so we have $f' \in v_!(TC)\tilde{}$, whence the relation

$$TC' \ \subset \ v_!(TC)\tilde{} \ ,$$

which finishes proving that (W', C', F') is a special and closed Quillen triple.

The two relations (5.8.4) were proved along the way (the second being essentially trivial, and the first representing the main difficulty in the proof). Remains to establish (5.8.5). But we already know (5.8.9) that

$$v_!(TC) \ \subset \ TC' \ ,$$

whence

$$v_!(TC)\tilde{} \ \subset \ TC',$$

since TC' is Q-closed on the left (or because that was proved in (5.8.14)), we have already proved the inclusion in the reverse direction (5.8.12 bis), an assertion equivalent to (5.8.4) (and which represented the main difficulty).

[page 97]

Remark 5.9. If we do not assume that the structure on $\mathcal{M}$ is closed, then the proof given of 5.8 shows that we nevertheless find on $\mathcal{M}'$ a special Quillen structure, by defining C' this time as in 4.6. But I doubt that we will ever have to work with special Quillen triples which are not also closed. In any case, if W is stable under small filtering $\varinjlim$[28] (and I doubt that one finds special Quillen triples outside this case), then if (W, C, F) is a special Quillen triple, the same is true of $(W, \tilde{C} = C^{\natural}, F)$, and I presume that we have every interest in replacing the initial triple by this one.

p. 49

[28] which implies stability under direct factors, and also implies the same property for $W' \overset{\text{def}}{=} (v^*)^{-1}(W)$.

[page 98]

6 Application to the construction of derivators (associated with special Quillen triples)

Theorem 6.1.[29] *Let (W, C, F) be a special closed Quillen triple on the category $\mathcal{M}$* (cf. def. 5.6, p. 90, and cor. 5.7). *For every I in* Cat*, let us consider the category*

$$\mathcal{M}(I) \;=\; \underline{\mathrm{Hom}}(I^\circ, \mathcal{M})\;, \tag{6.1.1}$$

and let us consider

$$W(I),\; F(I) \;\subset\; \mathrm{Fl}(\mathcal{M}(I))\;,$$

defined by

$$\begin{array}{l} u : X \longrightarrow Y \text{ in } \mathcal{M}(I) \text{ is in } W(I) \text{ (resp. } F(I)) \text{ if and only} \\ \text{if } \forall\, i \in \mathrm{Ob}I,\; u(i) : X(i) \longrightarrow Y(i) \text{ is in } W \text{ (resp. } F). \end{array} \tag{6.1.2}$$

a) *There exists a closed Quillen structure (necessarily unique) in $\mathcal{M}(I)$, having $W(I)$ and $F(I)$ as the sets of quasi-isomorphisms and of fibrations respectively. We thus also have*

$$\begin{cases} TF(I) \;=\; F(I) \cap W(I) \;=\; \{u \in \mathrm{Fl}(\mathcal{M}(I)) \mid u(i) \in TF \;\; \forall\, i \in \mathrm{Ob}I\}, \\ C(I) \;=\; TF(I)^*. \end{cases} \tag{6.1.3}$$

This Quillen structure is special.

b) *Let I_0 be the discrete category having the same set of objects as I, and* [page 99]

$$v_I \;\; \textit{or} \;\; v \;:\; I_0 \longrightarrow I \tag{6.1.4}$$

the canonical functor. Let us denote by

$$C(I) \;=\; (TF(I))^*, \qquad TC(I) \;=\; C(I) \cap W(I) \;=\; F(I)^* \tag{6.1.5}$$

the sets of cofibrations resp. of trivial cofibrations in $\mathcal{M}(I)$, and let us use the same notations $C(I_0)$, $TC(I_0)$ for I_0. We then have

$$\begin{array}{l} u : X_0 \longrightarrow Y_0 \text{ in } \mathcal{M}(I_0) \text{ is in } C_0 \text{ (resp. } TC_0) \text{ if and only if} \\ \text{for every } i \text{ in } \mathrm{Ob}I,\; u(i) : X_0(i) \longrightarrow Y_0(i) \text{ is in } C \text{ (resp. } TC). \end{array} \tag{6.1.6}$$

Finally we have

$$\begin{cases} C(I) \;=\; (v_!(C(I_0)))\check{} \\ TC(I) \;=\; (v_!(TC(I_0)))\check{}. \end{cases} \tag{6.1.7}$$

c) *Let*

$$u \;:\; I \longrightarrow I'$$

be an arrow in Cat*, let us consider the couple of adjoint functors*

$$\mathcal{M}(I) \underset{u^*}{\overset{u_!}{\rightleftarrows}} \mathcal{M}(I') \quad . \tag{6.1.8}$$

This couple satisfies Quillen's conditions for couples of adjoint functors between Quillen categories (Hom. Alg. Chp. I, §4, th. 3, p. 4.5), *so that they define by passage to the localized categories a couple of adjoint functors*

p. 50 [page 100]

$$\mathrm{Ho}_W(I) \underset{u^*_W}{\overset{u^W_!}{\rightleftarrows}} \mathrm{Ho}_W(I') \quad . \tag{6.1.9}$$

[29] This is the main result in this part XVII of the notes.

We have, more precisely, the relations

$$\begin{cases} u_!(C(I)) \ \subset \ C(I') \\ u_!(TC(I)) \ \subset \ TC(I') \ , \end{cases} \tag{6.1.10}$$

whence

$$(u_!(\mathcal{M}(I)_{\mathrm{c}})) \ \subset \ \mathcal{M}(I')_{\mathrm{c}} \ , \tag{6.1.11}$$

and moreover

$$\begin{array}{l} \textit{for every arrow } f \textit{ in } \mathcal{M}(I)_{\mathrm{c}}\textit{, such that } f \in W(I)\textit{, we have} \\ u_!(f) \in W(I')\textit{, i.e. } (u_!)_{\mathrm{c}} : \mathcal{M}(I)_{\mathrm{c}} \longrightarrow \mathcal{M}(I')_{\mathrm{c}} \textit{ is compatible} \\ \textit{with the localizers.} \end{array} \tag{6.1.12}$$

On the other hand we have

$$\begin{cases} u^*(F(I')) \ \subset \ F(I) \\ u^*(TF(I')) \ \subset \ TF(I) \\ u^*(W(I')) \ \subset \ W(I) \ , \end{cases} \tag{6.1.13}$$

hence

$$\begin{cases} u^*(\mathcal{M}(I')_{\mathrm{f}}) \ \subset \ \mathcal{M}(I)_{\mathrm{f}} \\ u_{\mathrm{f}}^* : \mathcal{M}(I')_{\mathrm{f}} \longrightarrow \mathcal{M}(I)_{\mathrm{f}} \textit{ is compatible with the localizers.} \end{cases} \tag{6.1.14}$$

[page 101]

d) *Suppose that I is an ordered category, and let us consider the set $C_I \subset \mathrm{Fl}(\mathcal{M}(I))$ defined in §2*[30] (cf. page 25, cf. also th. 2.1, p. 26, and its corollaries). *We then have*

$$\begin{cases} C_I \ \supset \ C(I) \\ TC_I \ \supset \ TC(I) \ , \end{cases} \tag{6.1.15}$$

and we have equality when I is an ordered set that is Noetherian.

Proof. Assertion a) is a special case of 5.8, applied to the couple of adjoint functors

$$\mathcal{M}(I_0) \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}(I) \quad .$$

The conditions (5.8.2) are satisfied, this was seen in (4.8). Proposition 5.8 then tells us of the existence of a closed Quillen triple structure on $\mathcal{M}(I)$, whose set of quasi-isomorphisms $W(I)$ and set of fibrations $F(I)$, are given by the first formula (5.8.3) and by the first formula (5.8.4), which is equivalent to the definitions (6.1.2). (**N.B.** The fact that $\mathcal{M}(I_0)$, with the obvious structure $W(I_0)$, $C(I_0)$, $F(I_0)$ (product), is indeed a special closed Quillen category, is ± tautological.) p. 51 [page 102]

Let us prove b). The relation (6.1.6) is tautological. The relations (6.1.7) are contained in 5.8 (cf. second relation in (5.8.3) for $C' = v_!(C)\tilde{}$, i.e. $C(I) = (v_!(C(I_0)))\tilde{}$, and (5.8.5) for $TC(I) = (v_!(TC(I_0)))\tilde{}$).

Let us prove c). The first assertion is, by Quillen, a special case of the relations (6.1.10) to (6.1.14), it is these that have to be established. The relations (6.1.13) and consequently (6.1.14) are tautological by virtue of the definitions, so it is a matter of establishing *(6.1.10)* and (6.1.12).

[30] In paragraph 2, this was denoted C_{I° instead of C_I.

For the formulas (6.1.10), I use the commutative diagram

$$\begin{array}{ccc} I & \xleftarrow{\ v\ } & I_0 \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle u_0} \\ I' & \xleftarrow{\ v'\ } & I'_0\ , \end{array}$$

and I note that the functors $u_!^{\mathcal{M}}$, $u_{0!}^{\mathcal{M}}$, commuting with small direct limits, behave well [page 103] with respect to the operation

$$\Phi \longmapsto \Phi^{\check{}},$$

when Φ is a subset of $\mathrm{Fl}(\mathcal{M}(I))$ resp. of $\mathrm{Fl}(\mathcal{M}(I_0))$ which is Q-accessible, i.e. such that there exists a *small* subset $\Phi_0 \subset \Phi$ with $\Phi \subset \tilde{\Phi}_0$, more precisely, we have

$$u_!(\Phi^{\check{}}) \ \subset\ (u_!(\Phi))^{\check{}}.$$

This noted, we have by b)

$$C(I) \ =\ (v_!(C(I_0)))^{\check{}},$$

whence

$$u_!(C(I)) \ \subset\ (\underbrace{u_!(v_!(C(I_0)))}_{(uv)_!(C(I_0))})^{\check{}},$$

and the right-hand side can also be written

$$(v'_!u_{0!}(C(I_0)))^{\check{}} \overset{?}{\subset}\ (v'_!(C(I'_0)))^{\sim}\ =\ C(I')\ .$$

Thus everything comes down to showing that

$$u_{0!}(C(I_0)) \ \subset\ C(I'_0)\ , \tag{6.1.16}$$

which follows at once from the computation of $u_{0!}$ by means of direct sums, and from the fact that C is stable under arbitrary direct sums. This thus proves $u_!(C(I)) \subset C(I')$, and we prove in the same way $u_!(TC(I)) \subset TC(I')$, which reduces in the same way to the special case

$$u_{0!}(TC(I_0)) \ \subset\ TC(I'_0)\ . \tag{6.1.17}$$

It remains to prove (6.1.12), and we proceed as for 3.3 (p. 54), using the fact that an p. 52 arrow f in $\mathcal{M}(I)_{\mathrm{c}}$ which is in $W(I)$, can be written in $\mathcal{M}(I)_{\mathrm{c}}$ in the form $f = pi$, [page 104] where $i \in TC(I)$, and where p is a left inverse (retraction) of an arrow $j \in TC(I)$. We will therefore have $u_!(f) = u_!(p)u_!(i)$, and $u_!(i) \in TC(I')$ by (6.1.10), and $u_!(p)$ is a left inverse of $u_!(j) \in TC(I')$. We then have $u_!(i), u_!(j), u_!(p) \in W(I')$, whence $u_!(f) \in W(I')$, q.e.d.

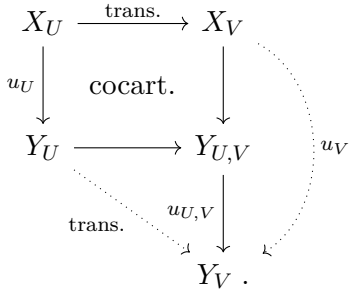
Let us finally prove d). In view of the definition of C_I, to establish $C(I) \subset C_I$, it suffices to see this:

Corollary 6.2.[31] *Let I be arbitrary in* Cat*, let $U \subset V$ be two closed sets in I (i.e. open sets in I°), and let $f : X_\bullet \longrightarrow Y_\bullet$ be an arrow in $C(I)$ (resp. in $TC(I)$). Let us also put* [page 105]

$$X_U \ =\ \varinjlim_{i\in U} X(i), \qquad X_V \ =\ \varinjlim_{i\in V} X(i)\ ,$$

[31] **N.B.** This result generalizes, under the general conditions in 6.2.5.

and likewise for Y, and let us consider the diagram



(6.2.1)

Then the arrow $u_{U,V}$ is in C (resp. TC).

Proof of 6.2. Let us admit that the restriction of an arrow of $C(I)$ resp. $TC(I)$ to a closed set V of I is in $C(V)$ resp. $TC(V)$ (cf. … below). This allows us to assume $V = I$. Giving U, which is an open set of I°, is equivalent to giving an arrow $v : I^\circ \longrightarrow \mathbf{\Delta}^1$ (taking the fiber at 0), or again to giving an arrow $u : I \longrightarrow (\mathbf{\Delta}^1)^\circ \simeq \mathbf{\Delta}^1$ (taking the fiber at 1). Now

$$u \ : \ I \longrightarrow \mathbf{\Delta}^1 \ , \qquad U \ = \ I_1 \stackrel{\text{def}}{=} u^{-1}(\{1\}) \ . \tag{6.2.2}$$

We then have [page 106]

$$X_U \ = \ u_!(X)(1), \quad X_I \ = \ u_!(X)(0) \ ,$$

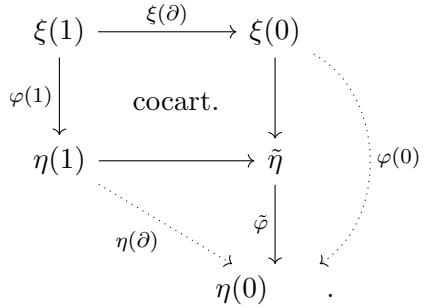
and the transition morphism $X_U \longrightarrow X_I$ is deduced from the arrow $0 \stackrel{\partial}{\longrightarrow} 1$ in $\mathbf{\Delta}^1$, and likewise for Y. Let

$$\xi \ = \ u_!(X), \qquad \eta \ = \ u_!(Y) \ ,$$

so the diagram (6.2.1) is identified with the diagram deduced from p. 53

$$\varphi \ : \ \xi \longrightarrow \eta \ , \qquad \varphi \ = \ u_!(f) \ ,$$

as



(6.2.3)

Now by 6.1.10 applied to $u : I \longrightarrow \mathbf{\Delta}^1$, the relation $f \in C(I)$ resp. $f \in TC(I)$ implies $\varphi \in C(\mathbf{\Delta}^1)$ resp. $\varphi \in TC(\mathbf{\Delta}^1)$. Thus to establish 6.1, we are reduced to the special case where $I = \mathbf{\Delta}^1$, $U = \{1\}$, $V = I$, hence to proving the

Lemma 6.2.4. *Let $\varphi : \xi \longrightarrow \eta$ be an arrow in $\mathcal{M}(\mathbf{\Delta}^1)$. If $\varphi \in C(\mathbf{\Delta}^1)$ (resp. $\varphi \in TC(\mathbf{\Delta}^1)$), then the arrow $\tilde{\varphi}$ in the diagram* (6.2.3) *is in C (resp. TC).* [page 107]

More generally, and more precisely:

Lemma 6.2.5. *Let $\mathcal{M}$ be a category stable under amalgamated sums, and Φ a subset Q-closed on the left* (we will take $\Phi = C$, and $\Phi = TC$, in the case of lemma 6.2.4). *Let $\Psi = \Phi_*$, and let us consider, for every I in* Cat, *$\Phi(I)$, $\Psi\langle I\rangle$ in $\mathcal{M}(I)$, defined by:*

$$\begin{aligned} &u : X \longrightarrow Y \text{ in } \mathcal{M}(I) \text{ is in } \Psi\langle I\rangle \text{ if and only if } \forall\, i \in \mathrm{Ob}I, \\ &u(i) : X(i) \longrightarrow Y(i) \text{ is in } \Psi, \end{aligned} \tag{6.2.5.1}$$

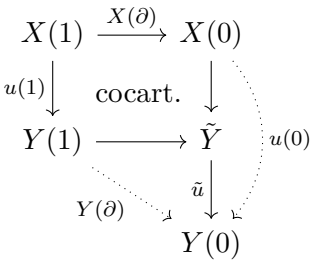
$$\Phi(I) \;=\; \Psi\langle I\rangle^*. \tag{6.2.5.2}$$

Let us take the case $I = \mathbf{\Delta}^1 = \{0 \xrightarrow{\partial} 1\}$, and let $u : X \longrightarrow Y$ be an arrow in $\mathcal{M}(\mathbf{\Delta}^1)$, corresponding to a commutative square

$$\begin{array}{ccc} X(1) & \xrightarrow{X(\partial)} & X(0) \\ {\scriptstyle u(1)}\downarrow & & \downarrow{\scriptstyle u(0)} \\ Y(1) & \xrightarrow{Y(\partial)} & Y(0)\ . \end{array} \tag{6.2.5.3}$$

Let us consider the corresponding diagram

p. 54
[page 108]



. (6.2.5.4)

For us to have $u \in \Phi(\mathbf{\Delta}^1)$, it is necessary and sufficient that we have

$$u(0),\ u(1),\ \tilde{u}\ \in\ \Phi\ . \tag{6.2.5.5}$$

I have already checked this (in dual form), but can no longer find the place (perhaps I did not take notes). I don't feel like looking for the proof here.

To finish proving 6.1. d), i.e. the equality $C(I) = C_I$ for I noetherian, I feel like proving something still more general, as follows:

Corollary 6.3. *Let $(\mathcal{M}, \Phi, \Psi)$ and $I \longmapsto \Phi(I) \subset \mathrm{Fl}(\mathcal{M}(I))$ be as in* 6.2.5. *Let I be an ordered set, and let us also consider Φ_I defined (in the dual situation) on page* 25 (2.2) *(but with dual notation …). I say that we have*

$$\Phi_I\ \supset\ \Phi(I)\ , \tag{6.3.1}$$

and we have equality if I is noetherian, i.e. I° is artinian.

[page 109]

Proof. The proof of $\Phi(I) \subset \Phi_I$ is the one we have just given. It remains to prove that if I is noetherian, we have $\Phi_I \subset \Phi(I)$. Given $i : A \longrightarrow B$ in Φ_I, we must prove that $i \in \Phi(I)$, i.e. that for a commutative diagram in $\mathcal{M}(I)$,

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & X \\ {\scriptstyle i}\downarrow & \overset{h}{\nearrow} & \downarrow{\scriptstyle p} \\ B & \xrightarrow[\beta]{} & Y \end{array} \tag{6.3.2}$$

with $p \in \Psi\langle I\rangle$, i.e. $p(i) : X(i) \longrightarrow Y(i)$ in Ψ for every i, $\exists\ h : B \longrightarrow X$ leaving the diagram commutative. We are going to return to the dual notation, working in $\mathcal{M}(I^\circ)$, with I artinian. We consider the set $\mathcal{H}$ of pairs (U, h_U) of an open set U of I and an $h_U : B|U \longrightarrow X|U$, making the restriction to U of (6.3.2) commutative. We order $\mathcal{H}$ by extension, we note that every totally ordered subset (U_α, h_{U_α}) in $\mathcal{H}$ has a least upper bound, namely $U = \bigcup_\alpha U_\alpha$, h_U defined by the h_{U_α} on the U_α.

(**N.B.** The set $\mathcal{H}$ is not empty, for $(\emptyset, h_\emptyset)$ is in $\mathcal{H}$ – but this is contained in the preceding statement that every totally ordered subset of $\mathcal{H}$ has a least upper bound in it

p. 55
[page 110]

…). By Zorn, there exists a maximal element (U, h_U). To prove that $U = I$, and we will be done. Or again that if (U, h_U) such that $U \neq I$, it is *not* maximal. Let $F = I\backslash U \neq \emptyset$ be the complementary closed set. By the artinian hypothesis on I, F has a minimal element i_0. Let

$$U' \;=\; U \cup \overbrace{I/i_0}^{V} \;=\; U \cup \{i_0\}\ ,$$

we are going to extend h_U to $h_{U'}$ on U', we will be done. For this, let $V_0 = V \cap U = V\backslash\{i_0\}$, we are reduced to extending the restriction h_{V_0} of h_U to V_0, from V_0 to V. All told, we may suppose $I = I/i_0$ equipped with a final element i_0, $U = I\backslash\{i_0\}$, and show that a $(U, h_U) \in \mathcal{H}$ extends to I. Let A', B', X', Y' be the restrictions of A, B, X, Y to U (which we may also denote by I'). The datum of A is equivalent to that of A', that of $A_0 = A(i_0) \in \mathrm{Ob}\,\mathcal{M}$, and of an arrow from

$$A'_{I'} \overset{\text{def}}{=} \varinjlim_{i' \in I'} A'(i') \xrightarrow{\lambda_A} A_0 \ (\simeq A_I)\ ,$$

and likewise for B, X, Y. More precisely, the category $\mathcal{M}(I^\circ)$ can be interpreted as that [page 111]
of triples $X = (X', X_0, \lambda_X)$, where $X' \in \mathcal{M}(I'^\circ)$, $X_0 \in \mathcal{M}$, and $\lambda_X : X'_{I'} \longrightarrow X_0$. In these terms, the datum of the diagram (6.3.2) is equivalent to that of a similar square diagram

$$\begin{array}{ccc} A' & \xrightarrow{\alpha'} & X' \\ {\scriptstyle i'}\downarrow & \overset{h'}{\nearrow} & \downarrow{\scriptstyle p'} \\ B' & \xrightarrow[\beta']{} & Y' \end{array} \tag{6.3.3}$$

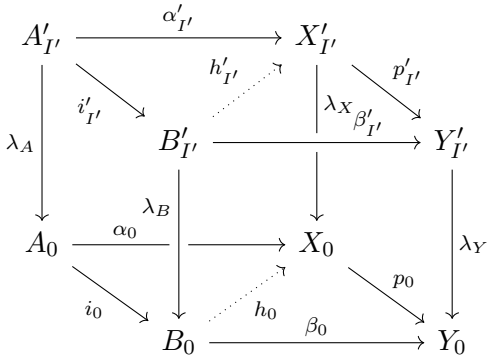
in $\mathcal{M}(I'^\circ)$, whence a diagram in $\mathcal{M}$

$$\begin{array}{ccc} A'_{I'} & \xrightarrow{\alpha'_{I'}} & X'_{I'} \\ {\scriptstyle i'_{I'}}\downarrow & \overset{h'_{I'}}{\nearrow} & \downarrow{\scriptstyle p'_{I'}} \\ B'_{I'} & \xrightarrow[\beta'_{I'}]{} & Y'_{I'}\ , \end{array} \tag{6.3.4}$$

of another square diagram *in* $\mathcal{M}$

$$\begin{array}{ccc} A_0 & \xrightarrow{\alpha_0} & X_0 \\ {\scriptstyle i_0}\downarrow & & \downarrow{\scriptstyle p_0} \\ B_0 & \xrightarrow[\beta_0]{} & Y_0\ , \end{array} \tag{6.3.5}$$

and finally of a homomorphism $\lambda = (\lambda_A, \lambda_B, \lambda_X, \lambda_Y)$ from the square (6.3.4) to the square (6.3.5): p. 56
[page 112]



32 (6.3.6)

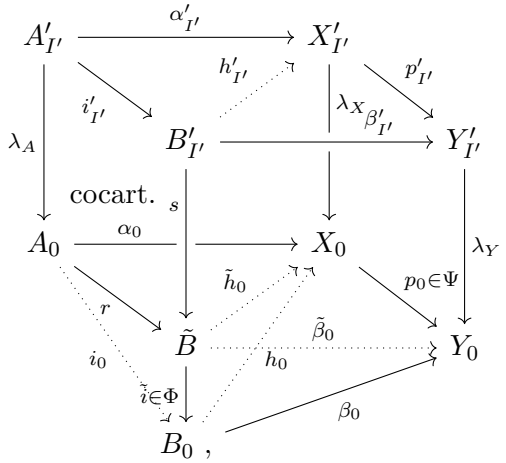
On the other hand, the question of extending $(I', h') \in \mathcal{H}$ to (I, h), amounts to that of finding $h_0 : B_0 \longrightarrow X_0$ in $\mathcal{M}$, so that together with $h'_{I'}$ this again makes a commutative diagram of (6.3.6), which amounts to saying that on the one hand the two triangles formed in the base square are commutative:

$$p_0 h_0 \;=\; \beta_0, \qquad h_0 i_0 \;=\; \alpha_0 \;, \tag{6.3.7}$$

and on the other hand the diagonal square defined by $h'_{I'}$, h_0 is commutative (which expresses that we have an arrow h from B to X):

$$\lambda_X h'_{I'} \;=\; h_0 \lambda_B \;. \tag{6.3.8}$$

But the diagram (6.3.6), where the densely dotted arrow $h'_{I'}$ is already in hand, and h_0 in spaced dots remains to be found, gives rise to the diagram, in which is brought out the arrow $\tilde{i} \in \Phi$ which expresses the hypothesis $i \in \Phi_I$, in view of $i' \in \Phi_{I'}$: [page 113]



(6.3.9)

where the densely dotted arrow $\tilde{h}_0$ from $\tilde{B} \overset{\text{def}}{=} A_0 \amalg_{A'_{I'}} B'_{I'}$ to X_0, is defined by the commutativity conditions p. 57

$$\tilde{h}_0 r \;=\; \alpha_0, \qquad \tilde{h}_0 s \;=\; \lambda_X h'_{I'} \;, \tag{6.3.10}$$

which is legitimate, seeing that

$$\begin{aligned} \alpha_0 \lambda_A \;&=\; (\lambda_X h'_{I'}) i'_{I'} \\ &=\; \lambda_X (h'_{I'} i'_{I'}) \\ &=\; \lambda_X \alpha'_{I'}. \end{aligned}$$

As for the arrow $\tilde{\beta}_0 : \tilde{B} \longrightarrow Y_0$, it is given by the commutativity condition [page 114]

$$\tilde{\beta}_0 r \;=\; p_0 \alpha_0, \qquad \tilde{\beta}_0 s \;=\; \lambda_Y \beta'_{I'} \;, \tag{6.3.11}$$

which is again legitimate, since

$$(p_0 \alpha_0) \lambda_A \;=\; (\lambda_Y \beta'_{I'}) i'_{I'}$$

[32] **N.B.** The condition $p \in \Psi\langle I \rangle$ is expressed by $p' \in \Psi\langle I' \rangle$ and $\boxed{p_0 \in \Psi.}$

by virtue of the commutativity of the cube (6.3.6). To establish the full commutativity of the diagram (6.3.9) (solid lines, and densely dotted lines for $h'_{I'}$, $\tilde{h}_0$ and $\tilde{\beta}_0$ – excluding h_0 not yet constructed), we must still verify on the one hand the relations

$$\begin{cases} \tilde{h}_0 r \;=\; \alpha_0, \quad p_0\tilde{h}_0 \;=\; \tilde{\beta}_0, \\ \lambda_X h'_{I'} \;=\; \tilde{h}_0 s \end{cases} \qquad {}^{33}, \tag{6.3.12}$$

establishing the commutativity of the upper cube with the diagonal square (diagram of type (6.3.6), the dotted arrow h_0 included in it), on the other hand the relations

$$\begin{cases} \tilde{i} r \;=\; i_0 \qquad (\text{and } \tilde{i} s = \lambda_B \text{ for the record}) \\ \beta_0 \tilde{i} \;=\; \tilde{\beta}_0 \ , \end{cases} \tag{6.3.13}$$

where the first line is already known by the definition of $\tilde{B}$ and of $\tilde{i}$. The second reduces [page 115]
to the two relations

$$\begin{aligned} (\beta_0\tilde{i})r \;&=\; \beta_0(\tilde{i}r) \\ &=\; \beta_0 i_0 \\ &=\; p_0\alpha_0 \\ &=\; \tilde{\beta}_0 r \ , \end{aligned}$$

OK by commutativity of the base square of (6.3.5),

$$\begin{aligned} (\beta_0\tilde{i})s \;&=\; \beta_0(\tilde{i}s) \\ &=\; \beta_0\lambda_B \\ &=\; \lambda_Y\beta'_{I'} \\ &=\; \tilde{\beta}_0 s \ , \end{aligned}$$

OK by commutativity of the front square in (6.3.6). p. 58

So it remains to verify the second relation (6.3.12),

$$p_0\tilde{h}_0 \;=\; \tilde{\beta}_0 \ ,$$

which reduces to

$$\begin{aligned} (p_0\tilde{h}_0)r \;&=\; p_0(\tilde{h}_0 r) \\ &=\; p_0\alpha_0 \\ &=\; \tilde{\beta}_0 r \ , \end{aligned}$$

OK by (6.3.11),

$$\begin{aligned} (p_0\tilde{h}_0)s \;&=\; p_0(\tilde{h}_0 s) \\ &=\; p_0(\lambda_X h'_{I'}) \\ &=\; (p_0\lambda_X)h'_{I'} \\ &=\; (\lambda_Y p'_{I'})h'_{I'} \\ &=\; \lambda_Y(p'_{I'}h'_{I'}) \\ &=\; \lambda_Y\beta'_{I'} \\ &=\; \tilde{\beta}_0 s \ , \end{aligned}$$

OK.

[33] **N.B.** the first and the third of these relations are already known, cf. (6.3.10).

This being settled, I say that the constraints (6.3.7), (6.3.8) on h_0 amount to the commutativity of the two triangles in the square

$$\begin{array}{ccc} \tilde{B} & \xrightarrow{\tilde{h}_0} & X_0 \\ {\scriptstyle \tilde{i}\in\Phi}\downarrow & \overset{h_0}{\nearrow} & \downarrow{\scriptstyle p_0\in\Psi} \\ B_0 & \xrightarrow[\beta_0]{} & Y_0 \ , \end{array} \tag{6.3.14}$$

i.e. to the two relations [page 116]

$$h_0\tilde{i} \ = \ \tilde{h}_0 \ , \qquad p_0h_0 \ = \ \beta_0 \ . \tag{6.3.15}$$

The second of these relations is none other than the first relation (6.3.7), and it remains to see that the first relation (6.3.15) is equivalent to the two remaining relations in (6.3.7), (6.3.8) taken together:

$$h_0i_0 \ = \ \alpha_0 \ , \qquad \lambda_X h'_{I'} \ = \ h_0 \underbrace{\lambda_B}_{\tilde{i}s} \tag{6.3.16}$$

Now the relation in question $h_0\tilde{i} = \tilde{h}_0$ is equivalent to the set of the two relations

p. 59

$$\begin{aligned} (h_0\tilde{i})r &= h_0(\tilde{i}r) \\ &= h_0i_0 \\ &\overset{?}{=} \alpha_0 \\ &= \tilde{h}_0r, \end{aligned}$$

equivalent to $h_0i_0 = \alpha_0$,

$$\begin{aligned} (h_0\tilde{i})s &= h_0(\tilde{i}s) \\ &= h_0\lambda_B \\ &\overset{?}{=} \lambda_X h'_{I'} \\ &= \tilde{h}_0s, \end{aligned}$$

equivalent to $h_0\lambda_B = \lambda_X h'_{I'}$, that's OK.

This being so, in the square (6.3.14), the hypotheses $i \in \Phi_I$, $p \in \Psi\langle I\rangle$ imply $\tilde{i}_0 \in \Phi$, $p_0 \in \Psi$. Thus the desired arrow h_0 exists, q.e.d.

This completes the proof of corollary 6.3, and thereby also, that of the last part, d), of 6.1. We are happy !

Proposition 6.4.[34] *Let $\mathcal{M}$ be a category, and $\Phi \longleftrightarrow \Psi$ a Q-closed couple in Q-duality ($\Phi = \Psi^*$, $\Psi = \Phi_*$). Let $u : I \longrightarrow I'$ be an arrow in* Cat, *whence a sequence of adjoint functors* [page 117]

$$\mathcal{M}(I) \ \begin{array}{c} \xrightarrow{u_!} \\ \xleftarrow[u^*]{} \\ \xrightarrow[u_*]{} \end{array} \ \mathcal{M}(I') \quad .$$

We define $\Phi(I)$, $\Psi\langle I\rangle$, $\Phi(I')$, $\Psi\langle I'\rangle$ as in 6.2.5. *We then have:*

ⓐ

$$\begin{cases} u_!(\Phi(I)) \ \subset \ \Phi(I') \ , \\ u^*(\Psi\langle I'\rangle) \ \subset \ \Psi\langle I\rangle \ . \end{cases} \tag{6.4.1}$$

ⓑ *We have the equivalence*

$$u^*(\Phi(I')) \subset \Phi(I) \qquad \Longleftrightarrow \qquad u_*(\Psi\langle I\rangle) \subset \Psi\langle I'\rangle \ . \tag{6.4.2}$$

ⓒ *The equivalent relations* (6.4.2) *hold in each of the following three cases:*

[34] Subject to the existence of the relevant $\varinjlim$ and $\varprojlim$ in $\mathcal{M}$. We may assume for simplicity $\mathcal{M}$ stable under $\varinjlim$ and $\varprojlim$.

1°) *u is coétale.*

2°) *I is discrete.*

3°) *For every $i' \in \mathrm{Ob}\, I'$, the connected components of I/i' admit a final object.*

Corollary 6.5. *Let (W, C, F) be a special closed Quillen triple in $\mathcal{M}$, and let us define, for every I in* Cat*, $C(I)$, $TC(I)$ and $F(I)$, $TF(I)$ as in theorem* 6.1. a)*. We then have*

p. 60

[page 118]

the equivalences

$$\Big(u^*(C(I')) \ \subset \ C(I)\Big) \quad \Longleftrightarrow \quad \Big(u_*(TF) \ \subset \ TF'\Big)\,, \tag{6.5.1}$$

$$\Big(u^*(TC(I')) \ \subset \ TC(I)\Big) \quad \Longleftrightarrow \quad \Big(u_*(F) \ \subset \ F'\Big)\,. \tag{6.5.2}$$

The equivalent conditions (6.5.1) *as well as* (6.5.2) *are satisfied when we are in one of the cases* 1°) 2°) 3°) *of* 6.4. c).

Proof of 6.4. The assertions ⓐ , ⓑ are ± tautological, by adjunction, compare XIII, prop. 2, and cor., page 46. To prove c), it suffices to treat case 3°), for $1° \Longrightarrow 3°$, $2° \Longrightarrow 3°$. But the computation of

$$u_*(X)(i') \ \simeq \ \varprojlim_{i \in I/i'} X(i)$$

shows that if the connected components $(I/i')_\alpha$ of I/i' each have a final object e_α, then

$$u_*(X)(i') \ \simeq \ \prod_\alpha X(e_\alpha)\,.$$

Now Ψ, being right closed, is stable under arbitrary products, so $X \in \Psi\langle I\rangle$ implies that the $u_*(X)(i')$ are in Ψ, so $u_*(X)$ is in $\Psi\langle I'\rangle$, q.e.d.

THE DERIVATORS

p.

ALEXANDRE GROTHENDIECK

Chapter XVIII

Accessible categories and sets

p.

p. 1

4 Paratopos and the operation $\mathcal{M} \boxtimes \mathcal{N}$

[page 50]

4.1 Pseudo-topos. Problem of the product.

I call a *pseudo-$\mathfrak{U}$-topos category*, or simply a *pseudo-topos*, a $\mathfrak{U}$-category $\mathcal{M}$ stable under $\mathfrak{U}$-direct limits, and which admits a *small* subcategory $\mathcal{M}_0$ that is *strictly generating* (i.e. generating by strict epimorphisms). Then $\mathcal{M}$ also admits $\mathfrak{U}$-inverse limits (SGA 4 I 8.12.7). (But the example of (Gr), cf. loc. cit. 8.12.9, shows that $\mathcal{M}^\circ$ is not necessarily, for all that, a pseudo-$\mathfrak{U}$-topos, for lack of $\mathcal{M}$ admitting a small subcategory that is strictly *co*generating. But theorem 2.6 gives conditions on $\mathcal{M}$ (mainly of the exactness type) for [page 51] $\mathcal{M}$ to admit enough injectives – which will then imply the existence of a small strictly cogenerating subcategory, hence that $\mathcal{M}^\circ$ is also a pseudo-topos ([1])).

I visualize pseudo-topos categories as being the category of "sheaves" on an object $\mathcal{X}$, visualized as a sort of "space" which I distinguish from the category $\mathcal{M} = \mathrm{Fais}(\mathcal{X})$ of sheaves on $\mathcal{X}$. It is $\mathcal{X}$ that will be for this the "pseudo-topos". Pseudo-topoi form a 2-category, the variance in $\mathcal{X}$ being opposite to that in $\mathrm{Fais}(\mathcal{X}) = \mathcal{M}$. To be precise, if $\mathcal{X}$ corresponds to $\mathcal{M}$, $\mathcal{Y}$ to $\mathcal{N}$, a morphism f from $\mathcal{X}$ to $\mathcal{Y}$ will be given by a functor

$$f^* \ : \ \mathcal{N} \longrightarrow \mathcal{M} \tag{4.1.1}$$

in the opposite direction, commuting with arbitrary direct $\varinjlim$ ([2]). (**N.B.** if $\mathcal{X}$, $\mathcal{Y}$ are even topoi, i.e. $\mathcal{M}$, $\mathcal{N}$ topos categories, and if we want f to be a morphism of topoi, we will require in addition that f^* be in addition *left exact*. We will say that a *morphism of pseudo-topoi is exact* if in addition f^* is left exact.)

By virtue of the dual of loc. cit. 8.12.7, this is equivalent to the fact that f^* admits [page 52] a right adjoint f_*,

$$f_* \ : \ \mathcal{M} \longrightarrow \mathcal{N} \ , \tag{4.1.2}$$

which therefore commutes with small $\varprojlim$. The datum of f_* satisfying this property therefore determines f^* up to unique isomorphism. But in general, it is not enough to give oneself such an f_*, to be able to assert that it corresponds to an f^*, i.e. that it admits a left adjoint ; for this, we would have to know that the functors $\mathcal{M} \longrightarrow \mathrm{Ens}$ which commute with small $\varprojlim$ are representable, which will be the case if $\mathcal{M}$ admits a small subcategory that is *co*generating, i.e. if $\mathcal{M}^\circ$ is also a pseudo-topos category.

It is probably wise to proceed as in the case of topoi, and to call a *morphism of* p. 2 *pseudo-topoi* $f : \mathcal{X} \longrightarrow \mathcal{Y}$ a quadruple

$$(f^*, f_*, \alpha, \beta) \tag{4.1.3}$$

of two adjoint functors (f^*, f_*), and of the adjunction morphisms. When f^* is an [page 53]

[1] This is the case in particular if $\mathcal{M}$ is a topos.

[2] Say here that the objects of $\mathcal{M}$ are visualized as generalized "opens" of $\mathcal{X}$. We have $\mathcal{M} \xrightarrow{\sim} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ, \mathrm{Ens})$, i.e. $\mathcal{M}$ is indeed interpreted as the "sheaves" on this category $\mathcal{M}$ of opens …

equivalence, which is equivalent to f_* an equivalence, or α and β isomorphisms, we will say that we have an *equivalence of pseudo-topoi*, and for all practical purposes, we will then be able to regard $\mathcal{X}$ and $\mathcal{Y}$ as essentially identical objects.

For two pseudo-topoi $\mathcal{X}$, $\mathcal{Y}$, I feel like introducing (as in the case of topoi) a product pseudo-topos

$$\mathcal{X} \times \mathcal{Y} \,, \tag{4.1.4}$$

which would have the properties of a 2-product in the 2-category of pseudo-topoi ([3]). I should specify that by definition

$$\begin{aligned} \underline{\mathrm{Hom}}_{\mathrm{pstop}}(\mathcal{X},\mathcal{Y}) &\simeq \underline{\mathrm{Hom}}_{!}(\mathcal{F}(\mathcal{Y}),\mathcal{F}(\mathcal{X}))^{\circ} \\ &\simeq \underline{\mathrm{Hom}}^{!'}(\mathcal{F}(\mathcal{X}),\mathcal{F}(\mathcal{Y})), \end{aligned} \tag{4.1.5}$$

where the subscript ! (resp. the superscript !) indicates the full subcategory in the category $\underline{\mathrm{Hom}}$ under consideration consisting of the functors which commute with small $\varinjlim$ (resp. with small $\varprojlim$), and the accent in the $\underline{\mathrm{Hom}}^{!'}$ in the last member indicates that we require in addition that the functor in question have a left adjoint. [page 54]

So let

$$\mathcal{M} \;=\; \mathcal{F}(\mathcal{X}) \,, \qquad \mathcal{N} \;=\; \mathcal{F}(\mathcal{Y}) \,, \tag{4.1.6}$$

and let $\mathcal{Z}$ be a third pseudo-topos, described by

$$\mathcal{P} \;=\; \mathcal{F}(\mathcal{Z}) \,. \tag{4.1.7}$$

We want to make explicit, up to equivalence, the product category $\underline{\mathrm{Hom}}_{\mathrm{pstop}}(\mathcal{Z},\mathcal{X}) \times \underline{\mathrm{Hom}}_{\mathrm{pstop}}(\mathcal{Z},\mathcal{Y})$, i.e.

$$\underline{\mathrm{Hom}}_{!}(\mathcal{M},\mathcal{P})^{\circ} \;\times\; \underline{\mathrm{Hom}}_{!}(\mathcal{N},\mathcal{P})^{\circ}. \tag{4.1.8}$$

I am going to introduce for this a category p. 3 [page 55]

$$\mathcal{M} \boxtimes \mathcal{N} \tag{4.1.9}$$

by the formula

$$\mathcal{M} \boxtimes \mathcal{N} \;\simeq\; \underline{\mathrm{Hom}}^{!!}(\mathcal{M}^{\circ} \times \mathcal{N}^{\circ}, \mathrm{Ens}) \,, \tag{4.1.10}$$

where the sign $\underline{\mathrm{Hom}}^{!!}$ denotes the full subcategory of the category $\underline{\mathrm{Hom}}$, consisting of the bi-contrafunctors

$$F \;:\; \mathcal{M}^{\circ} \times \mathcal{N}^{\circ} \longrightarrow \mathrm{Ens} \,, \tag{4.1.10.1}$$

such that for every x in $\mathcal{M}$, the functor

$$y \longmapsto F(x,y) \;:\; \mathcal{N}^{\circ} \longrightarrow \mathrm{Ens} \tag{4.1.10.2}$$

is in $\underline{\mathrm{Hom}}^{!}(\mathcal{N}^{\circ},\mathrm{Ens})$, i.e. transforms small $\varinjlim$ into $\varprojlim$, and that, symmetrically, for every y in $\mathcal{N}$, the functor

$$x \longmapsto F(x,y) \;:\; \mathcal{M}^{\circ} \longrightarrow \mathrm{Ens} \tag{4.1.10.3}$$

is in $\underline{\mathrm{Hom}}^{!}(\mathcal{M}^{\circ},\mathrm{Ens})$. But this also means that these functors are *representable*; so we also find

$$\mathcal{M} \boxtimes \mathcal{N} \xleftarrow{\sim} \underline{\mathrm{Hom}}^{!}(\mathcal{N}^{\circ},\mathcal{M}) \tag{4.1.11}$$

and

$$\mathcal{M} \boxtimes \mathcal{N} \xleftarrow{\sim} \underline{\mathrm{Hom}}^{!}(\mathcal{M}^{\circ},\mathcal{N}) \tag{4.1.12}$$

[page 56]

using (in each of these formulas) on both sides the equivalences of categories

$$\begin{cases} \mathcal{M} & \xrightarrow{\sim} & \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ, \mathrm{Ens}) \\ \mathcal{N} & \xrightarrow{\sim} & \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ, \mathrm{Ens})\ . \end{cases} \tag{4.1.13}$$

We can also say that the category $\mathcal{M} \boxtimes \mathcal{N}$ is equivalent to that of *pairs* of functors

$$\begin{cases} \varphi & : & \mathcal{M}^\circ & \longrightarrow & \mathcal{N} \\ \psi & : & \mathcal{N}^\circ & \longrightarrow & \mathcal{M}\ , \end{cases} \tag{4.1.14}$$

equipped with two adjunction morphisms

$$\begin{cases} \psi\varphi^\circ & \xleftarrow{\alpha} & \mathrm{id}_{\mathcal{M}} \\ \varphi\psi^\circ & \xleftarrow{\beta} & \mathrm{id}_{\mathcal{N}} \end{cases} \tag{4.1.15}$$

satisfying the two usual compatibilities expressing the adjunction of φ° and of ψ (or, equivalently, of ψ° and of φ). The functor

$$(\varphi, \psi) \longmapsto \varphi \qquad (\text{resp. } (\varphi, \psi) \longmapsto \psi)$$

gives an equivalence of the category of these pairs with $\underline{\mathrm{Hom}}^!(\mathcal{M}^\circ, \mathcal{N})$ (resp. with $\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ, \mathcal{M})$), so that we have a commutative square of canonical equivalences of functors:

p. 4
[page 57]

$$\begin{array}{ccc} & \mathcal{M} \mathbin{\widetilde{\boxtimes}} \mathcal{N} & \\ \swarrow^{\sim} & & \searrow^{\sim} \\ \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ, \mathcal{N}) & & \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ, \mathcal{M}) \\ \searrow_{\sim} & & \swarrow_{\sim} \\ & \underbrace{\mathcal{M} \boxtimes \mathcal{N}}_{\overset{\mathrm{def}}{=} \underline{\mathrm{Hom}}^{!!}(\mathcal{M}^\circ \times \mathcal{N}^\circ, \mathrm{Ens})} & \end{array} \tag{4.1.16}$$

I say that $\mathcal{M} \boxtimes \mathcal{N}$ *is a pseudo-topos category*. It is clear, addition, that it admits small $\varprojlim$, which are computed argument by argument in $\underline{\mathrm{Hom}}(\mathcal{M}^\circ \times \mathcal{N}^\circ, \mathrm{Ens})$, or at one's choice in $\underline{\mathrm{Hom}}^!(\mathcal{M}^\circ, \mathcal{N})$, or $\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ, \mathcal{M})$. How to construct small $\varinjlim$? There is no question of invoking an argument using cogenerators. Besides, one must also find a small strictly generating subcategory. Evidently, there is something nontrivial to prove!

4.2 Muddles over $\mathcal{M} \boxtimes \mathcal{N}$

Let us return to the general situation of a pseudo-topos $\mathcal{X}$, described by the pseudo-topos category $\mathcal{M}$. If $\mathcal{N}$ is any category, we define the category

[page 58]

$$\mathrm{Fais}(\mathcal{X}, \mathcal{N}) \qquad \text{or} \qquad \mathcal{F}(\mathcal{X}, \mathcal{N})$$

of sheaves on $\mathcal{X}$, with values in $\mathcal{N}$, by

$$\mathcal{F}(\mathcal{X}, \mathcal{N}) \overset{\mathrm{def}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ, \mathcal{N})\ . \tag{4.2.1}$$

[3] It turns out in the end that what we construct is *not* a 2-product in the 2-category under consideration, so it is better to denote it by a sign such as $\mathcal{X} \boxtimes \mathcal{Y}$. But if $\mathcal{X}$, $\mathcal{Y}$ are two topoi, $\mathcal{X} \boxtimes \mathcal{Y}$ is a 2-product in the 2-category of topoi.

We thus have a fully faithful functor

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{def}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) \hookrightarrow \mathrm{Hom}^!(\mathcal{N}^\circ,\overline{\mathcal{M}}) , \tag{4.2.2}$$

where $\overline{\mathcal{M}}$ denotes the subcategory of $\mathcal{M}^\wedge$ (**N.B.** $\mathcal{M}^\wedge$ is not in general a $\mathcal{U}$-category never mind that) formed by the *representable* functors $\mathcal{M}^\circ \longrightarrow \mathrm{Ens}$ (which is a $\mathfrak{U}$-category, equivalent to $\mathcal{M}$ by the canonical inclusion $\mathcal{M} \hookrightarrow \overline{\mathcal{M}}$). Thus we have up to unique isomorphism a factorization of (4.2.2) as

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{fully faithful}}{\hookrightarrow} \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathcal{M}) , \tag{4.2.3}$$

the essential image consisting of the functors

$$\psi \ : \ \mathcal{N}^\circ \longrightarrow \mathcal{M}$$

such that for every x in $\mathcal{M}$,

$$y \longmapsto \mathrm{Hom}_{\mathcal{M}}(x,\psi(y)) \ : \ \mathcal{N}^\circ \longrightarrow \mathrm{Ens}$$

is representable, i.e. of functors which admit a *left adjoint*. When $\mathcal{N}$ itself is a pseudo-topos category, this condition is automatic (it follows from the hypothesis that ψ commutes with small $\varinjlim$). Thus in this case, (4.2.3) is an equivalence of categories. p. 5 [page 59]

Now let S (S as in "site") be a *small* subcategory of $\mathcal{M}$, generating by strict epimorphisms. We know that this means that the composite functor

$$\mathcal{M} \overset{\sim}{\longrightarrow} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathrm{Ens}) \xrightarrow{\text{restriction to } S} S^\wedge \ (\overset{\text{def}}{=} \underline{\mathrm{Hom}}(S^\circ,\mathrm{Ens}) \ ) \tag{4.2.4}$$

is fully faithful, and this implies more generally, that for any category $\mathcal{N}$, the restriction functor

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{def}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) \overset{\text{fully faithful}}{\hookrightarrow} \underline{\mathrm{Hom}}(S^\circ,\mathcal{N}) \tag{4.2.5}$$

is fully faithful. Indeed, the second member, via the fully faithful embedding

$$\mathcal{N} \overset{\text{fully faithful}}{\hookrightarrow} \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathrm{Ens}) \hookrightarrow \mathcal{N}^\wedge ,$$

embeds into

$$\underline{\mathrm{Hom}}(S^\circ,\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathrm{Ens})) \simeq \mathrm{Hom}^!(\mathcal{N}^\circ,S^\wedge) ,$$

and the full faithfulness of (4.2.5) is therefore equivalent to that of the composite functor

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{def}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) \longrightarrow \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,S^\wedge) \simeq \underline{\mathrm{Hom}}(S^\circ,\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathrm{Ens})) . \tag{4.2.6}$$

Now we have commutativity in the square

[page 60]

$$\begin{array}{ccc} \mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{def}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) & \xrightarrow{(4.2.5)} & \underline{\mathrm{Hom}}(S^\circ,\mathcal{N}) \\ \Big\downarrow{\scriptstyle (4.2.3)}\ \text{fully faithful} & & \Big\downarrow\ \text{fully faithful} \\ \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathcal{M}) & \xrightarrow[\text{fully faithful}]{(4.2.4)} & \underbrace{\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,S^\wedge)}_{\simeq \underline{\mathrm{Hom}}(S^\circ,\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathrm{Ens}))} \end{array} \tag{4.2.7}$$

(where the vertical arrows are equivalences, if $\mathcal{N}$ itself is a pseudo-topos category). Let us note on the other hand that we have an equivalence

$$\underline{\mathrm{Hom}}(S^\circ,\mathcal{N}) \overset{\sim}{\longleftarrow} \underline{\mathrm{Hom}}^!(S^{\wedge\circ},\mathcal{N}) \overset{\text{def}}{=} \mathcal{F}(\mathrm{Top}(S),\mathcal{N}) , \tag{4.2.8}$$

where $\mathrm{Top}(S)$ is the pseudo-topos (in fact, a topos) defined by the pseudo-topos category $S^\wedge$. For this, let us note that we have

$$\underline{\mathrm{Hom}}^!(S^{\wedge\circ},\mathcal{N}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_!(S^\wedge,\mathcal{N}^\circ)^\circ ,$$

and on the other hand the restriction functor to $S \hookrightarrow S^\wedge$ gives

$$\underline{\mathrm{Hom}}_!(S^\wedge,\mathcal{N}^\circ) \xrightarrow{\sim} \underline{\mathrm{Hom}}(S,\mathcal{N}^\circ) ,$$

whence

p. 6

$$\underline{\mathrm{Hom}}^!(S^{\wedge\circ},\mathcal{N}) \xrightarrow{\sim} (\underline{\mathrm{Hom}}(S,\mathcal{N}^\circ))^\circ \xrightarrow{\sim} \underline{\mathrm{Hom}}(S^\circ,\mathcal{N}) , \qquad \texttt{[page 61]}$$

which is none other than (4.2.8). Thus:

Proposition 4.2.1. *Let $\mathcal{M} = \mathcal{F}(\mathcal{X})$ be a pseudo-topos category, equipped with a strictly generating full subcategory S, $\mathcal{N}$ any $\mathfrak{U}$-category, then the following "restriction to S" functor is fully faithful.*

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\mathrm{def}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) \longrightarrow \underline{\mathrm{Hom}}(S^\circ,\mathcal{N}) \xleftarrow{\sim} \underbrace{\underline{\mathrm{Hom}}^!(S^{\wedge\circ},\mathcal{N})}_{\overset{\mathrm{def}}{=}\mathcal{F}(\mathrm{Top}(S),\mathcal{N})} , \qquad (4.2.1.1)$$

and it fits into the commutative square (4.2.7) of fully faithful functors.

Corollary 4.2.2. *Suppose that $\mathcal{N} = \mathcal{F}(\mathcal{Y})$ is also a pseudo-topos category, equipped with a strictly generating full subcategory T. Then*

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \simeq \mathcal{F}(\mathcal{Y},\mathcal{M}) \simeq \underbrace{\underline{\mathrm{Hom}}^{!!}(\mathcal{M}^\circ\times\mathcal{N}^\circ,\mathrm{Ens})}_{=\mathcal{M}\boxtimes\mathcal{N}} ,$$

and the restriction functor

$$\begin{aligned} \mathcal{M}\boxtimes\mathcal{N} \overset{\mathrm{def}}{=} \underline{\mathrm{Hom}}^{!!}(\mathcal{M}^\circ\times\mathcal{N}^\circ,\mathrm{Ens}) &\longrightarrow \underline{\mathrm{Hom}}(S^\circ\times T^\circ,\mathrm{Ens}) \\ &\simeq (S\times T)^\wedge \\ &\overset{\mathrm{def}}{=} \mathcal{F}(\mathrm{Top}(S\times T)) \end{aligned} \qquad (4.2.2.1)$$

is fully faithful.

The first assertion was seen in 4.1. For the full faithfulness of (4.2.2.1), we note that this functor is interpreted, up to equivalence of categories, as the composite of the fully faithful functor (4.2.1.1), and of the functor analogous for $\mathcal{N}$ (hence it too fully faithful)

[page 62]

$$\mathcal{F}(\mathcal{Y},S^\wedge) \overset{\mathrm{def}}{=} \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,S^\wedge) \xrightarrow{\text{restr. to } T} \underbrace{\underline{\mathrm{Hom}}(T^\circ,S^\wedge)}_{\underline{\mathrm{Hom}}(S^\circ\times T^\circ,\mathrm{Ens})} , \qquad (4.2.2.2)$$

taking into account the symmetry equivalence

$$\mathcal{F}(\mathcal{Y},S^\wedge) \overset{\mathrm{def}}{=} \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,S^\wedge) \simeq \underline{\mathrm{Hom}}^!(S^{\wedge\circ},\mathcal{N}) . \qquad (4.2.2.3)$$

All in all, when $\mathcal{M}$, $\mathcal{N}$ are both pseudo-topos categories, equipped with strictly generating subcategories S resp. T, we have a canonical fully faithful embedding

$$\mathcal{M}\boxtimes\mathcal{N} \xhookrightarrow{\rho} (S\times T)^\wedge , \qquad (4.2.9)$$

which we subsequently interpret as $(f_S^{\mathcal{M}}\times f_T^{\mathcal{N}})_*$, where

p. 7

$$\begin{cases} f_S^{\mathcal{M}} : \mathcal{X} \longrightarrow \mathrm{Top}S \\ f_T^{\mathcal{N}} : \mathcal{Y} \longrightarrow \mathrm{Top}T \end{cases} \qquad (4.2.10)$$

are the morphisms of pseudo-topoi, given by the fully faithful embeddings (direct image functors)

$$\begin{cases} (f_S^{\mathcal{M}})_* \ : \ \mathcal{M} \ \longrightarrow \ S^\wedge \\ (f_T^{\mathcal{N}})_* \ : \ \mathcal{N} \ \longrightarrow \ T^\wedge \ , \end{cases} \tag{4.2.11}$$

which do indeed admit left adjoints [page 63]

$$\begin{cases} (f_S^{\mathcal{M}})^* \ : \ S^\wedge \ \longrightarrow \ \mathcal{M} \\ (f_T^{\mathcal{N}})^* \ : \ T^\wedge \ \longrightarrow \ \mathcal{N} \ , \end{cases} \tag{4.2.12}$$

namely the canonical extensions (by commutation with small $\varinjlim$) of the inclusion functors $S \longhookrightarrow \mathcal{M}$, $T \longhookrightarrow \mathcal{N}$.

The essential question, now, is evidently to construct a left adjoint for the functor (4.2.9). It amounts to the same thing, for (s,t) in $S \times T$, to define an object

$$s \boxtimes t \ \in \ \mathrm{Ob}\, \mathcal{M} \boxtimes \mathcal{N} \ , \tag{4.2.13}$$

such that we have, for F varying in $\mathcal{M} \boxtimes \mathcal{N}$, an isomorphism functorial in F,

$$\underbrace{\rho(F)(s,t)}_{\overset{\text{def}}{=} F(s,t)} \ \simeq \ \mathrm{Hom}(s \boxtimes t, F) \ .$$

Thus we must prove this:

Lemma 4.2.3 ([4]). *Let $\mathcal{M}$, $\mathcal{N}$ be two pseudo-topos categories, and $s \in \mathrm{Ob}\,\mathcal{M}$, $t \in \mathrm{Ob}\,\mathcal{N}$, and let us consider on $\mathcal{M} \boxtimes \mathcal{N} \overset{\text{def}}{=} \underline{\mathrm{Hom}}^{!!}(\mathcal{M}^\circ \times \mathcal{N}^\circ, \mathrm{Ens})$ the functor $F \longmapsto F(s,t)$. This functor is* representable *(by an object which I denote by $s \boxtimes t$).*

The natural idea would be to define $s \boxtimes t$ by

$$(s \boxtimes t)(x,y) \ = \ s(x) \times t(y) \ = \ \mathrm{Hom}_{\mathcal{M}}(x,s) \times \mathrm{Hom}_{\mathcal{N}}(y,t) \ ,$$

which does define a functor [page 64]

$$\mathcal{M}^\circ \times \mathcal{N}^\circ \longrightarrow \mathrm{Ens} \ .$$

Unfortunately, for one of the arguments fixed, say for y fixed, this functor is *not* compatible with inverse limits, e.g. it does not transform the final object of $\mathcal{M}^\circ$ corresponding to the initial object of $\mathcal{M}$ into a final object of Ens, but into $t(y) = \mathrm{Hom}_{\mathcal{N}}(y,t)$. Nevertheless, I think that $s \boxtimes t$ must indeed be definable as the product of two objects of $\mathcal{M} \boxtimes \mathcal{N}$, depending functorially on s resp. on t ([5]). We are thus led right now, whether we like it or not, to try to define canonical functors, commuting with small $\varinjlim$, p. 8

$$\mathcal{M} \longrightarrow \mathcal{M} \boxtimes \mathcal{N} \longleftarrow \mathcal{N} \tag{4.2.13}$$

(which will be the inverse image functors for the canonical projections $\mathcal{X} \times \mathcal{Y} \to \mathcal{X}$, $\mathcal{X} \times \mathcal{Y} \to \mathcal{Y}$)

([6]).

[4] Not proved; perhaps false!

[5] also false, cf. below.

[6] The case where $\mathcal{M}$, $\mathcal{N}$ are topos-categories, shows us that these functors should be described as $x \longmapsto x \boxtimes e_{\mathcal{N}}$, $y \longmapsto e_{\mathcal{M}} \boxtimes y$. But it is not clear that we have outside the case of topoi, and even granting the existence of the $x \boxtimes y$, the formula (valid for a topos) $x \boxtimes y \simeq (x \boxtimes e_{\mathcal{N}}) \times (e_{\mathcal{M}} \boxtimes y)$.

4.3 The operation $\mathcal{M} \boxtimes \mathcal{N}$ and the accessibility axiom

I spent the day yesterday, trying to establish the existence of the $x \boxtimes y$ in $\mathcal{M} \boxtimes \mathcal{N}$, without succeeding – at least not without making an additional accessibility hypothesis, about which it is becoming obvious now that it has to be made in any case. I am going all the same to state the positive of my perplexities. [page 65]

Proposition-perplexity 4.3.1. *Let $\mathcal{M}$, $\mathcal{N}$ be two pseudo-topos categories, whence $\mathcal{M} \boxtimes \mathcal{N}$ as in 4.1. Let $S \subset \mathcal{M}$, $T \subset \mathcal{N}$ be two small strictly generating subcategories, and let us consider the corresponding fully faithful inclusion:*

$$\underbrace{\mathcal{M} \boxtimes \mathcal{N}}_{\overset{\text{def}}{=} \underline{\mathrm{Hom}}^{!!}(\mathcal{M}^\circ \times \mathcal{N}^\circ, \mathrm{Ens})} \overset{\varphi\,?}{\hookleftarrow} \xrightarrow[\alpha]{} S^\wedge \boxtimes T^\wedge \;\simeq\; (S \times T)^\wedge . \tag{4.3.1.1}$$

Let us consider the following conditions:

a) $\mathcal{M} \boxtimes \mathcal{N}$ is a pseudo-topos.

b) For $x \in \mathrm{Ob}\,\mathcal{M}$, $y \in \mathrm{Ob}\,\mathcal{N}$, $x \boxtimes y \in \mathrm{Ob}\,\mathcal{M} \boxtimes \mathcal{N}$ exists, i.e. the functor (commuting with small $\varprojlim$)

$$F \;\longmapsto\; F(x, y)\ , \qquad \mathcal{M} \boxtimes \mathcal{N} \longrightarrow \mathrm{Ens} \tag{4.3.1.2}$$

is representable.

b′) As in b), with $x \in \mathrm{Ob}\,S$, $y \in \mathrm{Ob}\,T$.

c) The functor α (which commutes with small $\varprojlim$) admits a left adjoint

$$\varphi \;:\; (S \times T)^\wedge \longrightarrow \mathcal{M} \boxtimes \mathcal{N}\ .$$

We then have the implications

$$b \implies b' \iff c \implies a\ . \tag{4.3.1.3}$$

Moreover, if c is satisfied, we have this: for $s \in S$, $t \in T$, let us consider p. 9

$$(s, t) \;\in\; (S \times T) \;\subset\; (S \times T)^\wedge \;\simeq\; S^\wedge \boxtimes T^\wedge\ ,$$

which is also written out as

$$\begin{cases} (s,t) \;=\; \mathrm{pr}_1^*(s) \times \mathrm{pr}_2^*(t) \;\simeq\; s \overset{S^\wedge, T^\wedge}{\boxtimes} t \in \mathrm{Ob}\,(S \times T)^\wedge\ , \\ \textit{i.e. } (s,t)(s',t') \;=\; s(s') \times t(t') \;=\; \mathrm{Hom}_S(s', s) \times \mathrm{Hom}_T(t', t) \\ \qquad\qquad\qquad\qquad\quad =\; \mathrm{Hom}_{S \times T}((s', t'), (s, t))\ , \end{cases} \tag{4.3.1.4}$$

$$\begin{array}{ccc} S \times T & \xrightarrow{\mathrm{pr}_2} & T \\ \Big\downarrow{\scriptstyle \mathrm{pr}_1} & & \\ S\ , & & \end{array}$$

we have [page 66]

$$s \overset{\mathcal{M}, \mathcal{N}}{\boxtimes} t \;\simeq\; \varphi(s, t)\ , \tag{4.3.1.4}$$

and $\varphi(S \times T) = \{s \boxtimes t \mid s \in S,\ t \in T\}$ *is a strictly generating subset of* $\mathcal{M} \boxtimes \mathcal{N}$ *(*[7]*).*
N.B. I know of no example where the conditions under consideration are not satisfied, but nevertheless now doubt that even the weakest always is, i.e. that $\mathcal{M} \boxtimes \mathcal{N}$ is a pseudo-topos as soon as $\mathcal{M}$ and $\mathcal{N}$ are.
Proof of 4.3.1. The implication b $\Longrightarrow$ b′ is tautological. The implication b′ $\Longleftrightarrow$ c follows from the
Lemma 4.3.2. *Let* $\mathcal{E}$, X *be* $\mathfrak{U}$*-categories with* X *small,* $\mathcal{E}$ *stable under small* $\varinjlim$,

$$\alpha \ :\ \mathcal{E} \longrightarrow X^{\wedge}$$

a functor. For α *to admit a left adjoint* φ*, it is necessary and sufficient that for every* $x \in \mathrm{Ob}\, X$*, the functor*

$$\begin{cases} \xi & \longmapsto & \mathrm{Hom}_{X^{\wedge}}(x, \alpha(\xi)) \ = \ \alpha(\xi)(x) \\ \mathcal{E} & \longrightarrow & \mathrm{Ens} \end{cases}$$

be representable.
This is obviously necessary. Sufficiency follows from the fact that $X \subset X^{\wedge}$ is strictly generating in $X^{\wedge}$, i.e. that every object F of $X^{\wedge}$ is a direct limit $F = \varinjlim_{I} F_i$, the F_i in X (we take $I = X/F$), whence p. 10 [page 67]

$$\begin{aligned} \mathrm{Hom}(F, \alpha(\xi)) \ &\simeq \ \varprojlim_{I} \ \underbrace{\mathrm{Hom}(F_i, \alpha(\xi))}_{\mathrm{Hom}(\rho(F_i), \xi)} \\ &\simeq \ \mathrm{Hom}(\varinjlim_{I} \rho(F_i), \xi) \ , \end{aligned}$$

so the functor $\xi \longmapsto \mathrm{Hom}(F, \alpha(\xi))$ is represented by $\rho(F) \overset{\mathrm{def}}{=} \varinjlim_{I} \rho(F_i)$.
Let us prove c $\Longrightarrow$ a. For the existence of small $\varinjlim$ in $\mathcal{M} \boxtimes \mathcal{N}$, and the existence of a strictly generating subset, we invoke the general lemma:
Lemma 4.3.3. *Let* $\alpha : \mathcal{E} \hookrightarrow \mathcal{F}$ *be a fully faithful functor admitting a left adjoint* ρ*.*

ⓐ *If the* $\varinjlim$ *of a given type* I *exist in* $\mathcal{F}$*, they exist in* $\mathcal{E}$*, and are computed as*

$$\varinjlim_{I}{}_{\mathcal{E}}\, \xi_i \ = \ \rho(\varinjlim_{I}{}_{\mathcal{F}}\, \alpha(\xi_i)) \ . \qquad (4.3.3.1)$$

ⓑ *If* U *is a small strictly generating subset of* $\mathcal{F}$*,* $\rho(U)$ *is a small strictly generating subset in* $\mathcal{E}$*.*

Indeed, if ξ denotes the right-hand side , and η a variable object of $\mathcal{E}$, we will have

$$\begin{aligned} \mathrm{Hom}_{\mathcal{E}}(\xi, \eta) \ &\overset{\mathrm{adj.}}{\simeq} \ \mathrm{Hom}_{\mathcal{F}}(\varinjlim_{I} \alpha(\xi_i), \alpha(\eta)) \\ &\simeq \ \varprojlim_{I} \ \underbrace{\mathrm{Hom}_{\mathcal{F}}(\alpha(\xi_i), \alpha(\eta))}_{\simeq \mathrm{Hom}_{\mathcal{E}}(\xi_i, \eta) \quad \text{since } \alpha \text{ fully faithful}} \ . \\ &\simeq \ \varprojlim_{I} \mathrm{Hom}_{\mathcal{E}}(\xi_i, \eta) \ , \end{aligned}$$

[page 68]

[7] **N.B.** Of course we have $(s,t) = s \overset{S^{\wedge}, T^{\wedge}}{\boxtimes} t$.

which proves that ξ is a direct limit of the ξ_i in $\mathcal{E}$ ([8]).
This proves that (assuming c)) $\mathcal{M} \boxtimes \mathcal{N}$ is stable under small $\varinjlim$, so it remains to prove that $\mathcal{M} \boxtimes \mathcal{N}$ admits a small strictly generating family. This will follow from b) in the lemma, since $S \times T$ strictly generates $(S \times T)^\wedge$. This proves at the same time the last assertion of the proposition, subject to proving formula (4.3.1.4). We must therefore prove an isomorphism functorial in $F \in \mathrm{Ob}\, \mathcal{M} \boxtimes \mathcal{N}$,

$$\mathrm{Hom}(\rho(s,t), F) \;\simeq\; F(s,t)\ ,$$

but by adjunction we have

$$\mathrm{Hom}(\rho(s,t), F) \;\simeq\; \mathrm{Hom}((s,t), \alpha(F)) \;=\; \alpha(F)(s,t) \;=\; F(s,t)\ ,$$

OK.

p. 11
[page 69]

Proposition 4.3.4. *Let $\mathcal{M}$ be a $\mathfrak{U}$-category. The following conditions are equivalent.*

a) $\mathcal{M}$ is a $\mathfrak{U}$-pseudo-topos category, i.e. it is stable under $\mathfrak{U}$-direct limits, and admits a $\mathfrak{U}$-small strictly generating subcategory S.

b) There exist a $\mathfrak{U}$-small category S, and a functor $i : \mathcal{M} \hookrightarrow S^\wedge$ that is fully faithful, *having a left adjoint.*

Proof. We have a) $\Longrightarrow$ b), for it suffices to take for S in b) the one specified in a), the hypothesis on S implies that the canonical functor

$$\mathcal{M} \;\longrightarrow\; S^\wedge\ , \qquad \text{composite of } \mathcal{M} \hookrightarrow\; \mathcal{M}^\wedge \xrightarrow{\text{restriction}} S^\wedge$$

is fully faithful. That it admits a left adjoint follows from lemma 4.3.2. The implication b) $\Longrightarrow$ a) follows from lemma 4.3.3.

Corollary 4.3.5. *Let $\mathcal{M}$ be a $\mathfrak{U}$-category. For $\mathcal{M}$ to be a $\mathfrak{U}$-pseudo-topos that is* accessible *(i.e. all of whose objects are accessible), it is necessary and sufficient that there exist S in* Cat *and a functor fully faithful and* accessible $\mathcal{M} \overset{\alpha}{\hookrightarrow} S^\wedge$, *admitting a left adjoint ρ.*

Proof. It suffices to note that if S, α are as in 4.3.4 b), then α is accessible if and only if the $\rho(S)$ are made up of accessible objects in $\mathcal{M}$; as this subset is strictly generating, we know that this implies that *every* object of $\mathcal{M}$ is accessible. (Cf. SGA 4 I 9.9.)

[page 70]

I am going to try to give a more complete statement:

Theorem 4.3.6. *Let $\mathcal{M}$ be a $\mathfrak{U}$-category. Equivalent conditions:*

a) $\mathcal{M}$ stable under small $\varinjlim$, and it admits a small full subcategory S, strictly generating, and consisting of accessible objects of $\mathcal{M}$.

a′) $\mathcal{M}$ is a weakly accessible $\mathfrak{U}$-pseudo-topos category, i.e. satisfies a), with in addition every *element of $\mathcal{M}$ (not only those of S) accessible in $\mathcal{M}$.*

b) There exist a small category S, and a functor $\alpha : \mathcal{M} \hookrightarrow S^\wedge$ that is fully faithful and accessible, *such that α admits a left adjoint.*

[page 71]

c) $\mathcal{M}$ is stable under small $\varinjlim$, is accessible, and admits a cardinal filtration $(\mathrm{Filt}^\pi \mathcal{M})_{\pi \geqslant \pi_0}$.

[8] To prove b), we note that the full faithfulness of α implies that ρ is essentially surjective; as ρ commutes with small $\varinjlim$ and every object of $\mathcal{F}$ is of the form $\varinjlim_I x_i$, the x_i in U, we will have $\rho(x) = \varinjlim_I \rho(x_i)$, the $\rho(x_i)$ in $\rho(U)$, we win.

d) There exist a cardinal π_1, a small category $\mathcal{M}_{\pi_1}$, stable under direct limits of cardinality $\leqslant \pi_1$, such that we have an equivalence of categories

$$\mathcal{M} \xleftarrow{\sim} \mathrm{Ind}_{\pi_1} \mathcal{M}_{\pi_1} , \tag{4.3.6.1}$$

where the right-hand side denotes the full subcategory of $\mathrm{Ind}(\mathcal{M}_{\pi_1})$ *consisting of the ind-objects $(X_i)_{i\in I}$ of $\mathcal{M}_{\pi_1}$ such that I is large relative to π_1.*

Proof of 4.3.6 ([9]). We have already seen $\mathrm{a}' \Longrightarrow \mathrm{a} \Longrightarrow \mathrm{b} \Longrightarrow \mathrm{a}'$, i.e. the equivalence of a, a′, b, let PTA be this condition. We are going to prove the implications

$$\mathrm{PTA} \implies \mathrm{c} \implies \mathrm{d} \implies \mathrm{PTA} ,$$

which will establish the theorem. p. 12

Let us prove PTA $\Longrightarrow$ c. Let S be a small strictly generating full subcategory. Thus, for every x in $\mathcal{M}$ we have [page 72]

$$x = \varinjlim_{s\in S/x} s \tag{4.3.6.2}$$

(cf. SGA 4 I 7.2 (i)). Let $\pi_0 = \operatorname{card} \mathrm{Fl}(S)$, and for every cardinal $\pi \geqslant \pi_0$, let

$$\begin{aligned} \mathrm{Filt}^{\pi}\mathcal{M} \;=\; & \text{full subcategory of } \mathcal{M} \text{ consisting of objects } x \text{ such that} \\ & \operatorname{card}(\mathrm{Fl}(S/x)) \leqslant \pi . \end{aligned} \tag{4.3.6.3}$$

I want to prove that we thus get a cardinal filtration on $\mathcal{M}$, i.e. that conditions a) b) c) of SGA 4 I 9.12 are satisfied. This is proved in loc. cit. 9.13 under slightly different hypotheses, the condition of accessibility of $\mathcal{M}$ being replaced by the one (much less pleasant!) that the strict epimorphic families of arrows in $\mathcal{M}$ are strict universal epimorphic families. This condition is used only in the proof of condition b), namely the stability of $\mathrm{Filt}^{\pi}\mathcal{M}$ under $\varinjlim_{I}$ of cardinality $\leqslant \pi$ (without, moreover, having to assume that I is filtering, nor a fortiori large relative to π_0). It must moreover be said that the definition of $\mathrm{Filt}^{\pi}\mathcal{M}$ given in loc. cit. is not quite the same as (4.3.6.3), but it is [page 73]

$$\begin{aligned} \mathrm{Filt}'^{\pi}\mathcal{M} \;=\; & \text{full subcategory of } \mathcal{M} \text{ consisting of objects } x \text{ such that there} \\ & \text{exists a strict epimorphic family } x_i \xrightarrow{\alpha_i} x \text{ of car-} \\ & \text{dinality } \leqslant \pi, \text{ with } x_i \in S \;\forall\; i \in I. \end{aligned} \tag{4.3.6.3′}$$

Thus we have

$$\mathrm{Filt}^{\pi}\mathcal{M} \subset \mathrm{Filt}'^{\pi}\mathcal{M} ,$$

but in loc. cit. it is proved that for x in $\mathrm{Filt}'^{\pi}\mathcal{M}$, we have

$$\operatorname{card}\mathrm{Fl}(S/x) \leqslant \pi^{\pi_0} ,$$

whence

$$\mathrm{Filt}^{\pi}\mathcal{M} \subset \mathrm{Filt}'^{\pi}\mathcal{M} \subset \mathrm{Filt}^{\pi^{\pi_0}}\mathcal{M} ,$$

so if $\pi = \pi^{\pi_0}$, e.g. $\pi = 2^c$ with $c \geqslant \pi_0$, then we get

$$\mathrm{Filt}^{\pi}\mathcal{M} = \mathrm{Filt}'^{\pi}\mathcal{M} .$$

So the two filtrations look ± equivalent. But the proof given of the stability of Filt'^{π} really seems to make essential use of the universality condition on strict epimorphic families (hence effective epimorphic, with the hypotheses made implying the existence of small $\varprojlim$). So I must, willy-nilly, do the work over again! p. 13 [page 74]

Condition a): Filt^π equivalent to a small category. This is immediate, since by (4.3.6.2), Filt^π consists of $\varinjlim_I$ in $\mathcal{M}$ of objects of S (a small category), with I such that $\mathrm{card}\,\mathrm{Fl}\,I \leqslant \pi$.

Condition b): Filt^π stable under $\varinjlim_I$, when $\mathrm{card}\,\mathrm{Fl}\,I \leqslant \pi$ ([10]). So let $(x_i)_{i\in I}$, where

$$\begin{array}{ccc} I & \longrightarrow & \mathcal{M} \\ i & \longmapsto & x_i \end{array}$$

a direct system in $\mathcal{M}$, with x_i in Filt^π. Let us consider the fiber product

$$\begin{array}{ccccc} I & \xleftarrow{\text{cofibered}} & S/I = J & & \\ {\scriptstyle\varphi}\downarrow & & \downarrow & \overset{\psi}{\nearrow} & \\ \mathcal{M} & \xleftarrow[\text{Cat-cofibered}]{\text{target}} & S/\mathcal{M} & \xrightarrow{\text{source}} & S \hookrightarrow \mathcal{M}\ , \end{array} \tag{4.3.6.4}$$

where J is Cat-cofibered over I, its fiber at $i \in \mathrm{Ob}\,I$ being isomorphic to S/x_i, so $\mathrm{card}\,\mathrm{Fl}\,J_i \leqslant \pi$. Since by hypothesis $\mathrm{card}\,\mathrm{Fl}\,I \leqslant \pi$, we conclude that $\mathrm{card}\,\mathrm{Fl}\,J \leqslant \pi^3 = \pi$. (**N.B.** The set of arrows $v : j_0 \longrightarrow j_1$ of J over an arrow $u : i_0 \longrightarrow i_1$ of I is "contained" in the product set $\mathrm{Ob}\,J_{i_0} \times \mathrm{Fl}\,J_{i_1}$, by the map

$$v \longmapsto (\mathrm{source}\,v, v' : u_*(j_0) \longrightarrow j_1)\ ,$$

hence of cardinality bounded above by π^2.) We then have

$$x \overset{\text{def}}{=} \varinjlim_I x_i \ = \ \varinjlim_J \underbrace{\xi_j}_{=\psi(j)} \qquad ([11])\ , \tag{[page 75]}$$

where this time the limit is a limit of objects of S, taken in $\mathcal{M}$. It follows that $x \in \mathrm{Filt}'^\pi$, this is not quite $x \in \mathrm{Filt}^\pi$. (That implies $x \in \mathrm{Filt}^{\pi^{\pi_0}}$, so $x \in \mathrm{Filt}^\pi$ if $\pi = \pi^{\pi_0}$, e.g. $\pi = 2^c$ with $c \geqslant \pi_0$.) But this is the moment to use the fact that the objects of S are accessible, and we will take π_0 such that the objects of S are π_0-accessible, and this time we will take I filtering and large relative to π_0 all the same (to restrict ourselves to what is required in condition b) of loc. cit.). We must prove that $\mathrm{card}\,(\mathrm{Fl}(S/x)) \leqslant \pi$ (which means $x \in \mathrm{Filt}^\pi$). But since the objects of S are π_0-accessible and I large relative to π_0, we see at once that

$$S/x \ \simeq \ \varinjlim_i S/x_i\ ,$$

whence

$$\mathrm{Fl}\,S/x \ \simeq \ \varinjlim_i \mathrm{Fl}\,S/x_i\ ,$$

p. 14

hence

$$\mathrm{cardFl}(S/x) \ \leqslant \ \underbrace{(\mathrm{card}I)}_{\leqslant\pi}\ \underbrace{(\sup_i \mathrm{cardFl}(S/x_i))}_{\leqslant\pi} \ \leqslant \pi^2 = \pi\ ,$$

q.e.d.

[9] For a complete proof (the one given here is botched from a certain point on), cf. §4.6 (p. 139-148).

[10] when I filtering and large relative to π_0 – I prove it only in this case, alas!

[11] We use the distributivity property of $\varinjlim$, relative to a Cat-cofibrant functor [page 76]

$$\begin{array}{ccc} J & \longrightarrow & I \\ & \searrow & \\ & & \mathcal{M}\ , \end{array}$$

made explicit above.

Let us prove c) ([12]), i.e. that for every $\pi' > \pi \geqslant \pi_0$ and every $x \in \mathrm{Ob}\mathrm{Filt}^{\pi'}$, one can write

$$x \;=\; \varinjlim_{I} x_i \qquad I \text{ large relative to } \pi, \text{ the } x_i \text{ in } \mathrm{Filt}^{\pi}, \ \mathrm{card} I \leqslant \pi'^{\pi}. \tag{4.3.6.5}$$

The proof goes as in loc. cit. (p. 151-152), slightly changing the definition of I, as denoting the set of full subcategories i of S/x which are filtering, of cardinality $\leqslant \pi$ *and moreover large relative to* π_0 ([13]). (We may suppose S/x filtering by enlarging S so that S is stable in $\mathcal{M}$ under small finite $\varinjlim$, hence the categories S/x too, which implies that they are filtering. It is)

We have thus proved PTA $\Longrightarrow$ c).

Let us prove c) $\Longrightarrow$ d). Let

$$\pi_1 \;=\; 2^c \qquad \text{with} \qquad c \;\geqslant\; \pi_0 \,.$$

We then know that

$$\mathrm{Filt}^{\pi_1}\mathcal{M} \;=\; \mathcal{M}_{\pi_1} \overset{\text{def}}{=} \text{ full subcategory of } \mathcal{M} \text{ of the } \pi_1\text{-accessible objects of } \mathcal{M}$$

(loc. cit. 9.16), which implies in particular that $\mathcal{M}_{\pi_1}$ is stable under $\varinjlim$ of cardinality $\leqslant \pi_1$, by virtue of loc. cit. 9.9 (page 145). Moreover, by loc. cit. 9.18 b) (p. 156), we have [page 77]

$$\mathrm{Ind}_{\pi_1}(\mathcal{M}_{\pi_1}) \xrightarrow{\sim} \mathcal{M} \,,$$

which proves d).

Let us prove d) $\Longrightarrow$ PTA. By loc. cit. 9.18 a) (p. 156), we see that the objects of $\mathcal{M}_{\pi_1}$ are π_1-accessible in $\mathrm{Ind}_{\pi_1}(\mathcal{M}_{\pi_1})$, on the other hand it is clear that $\mathcal{M}_{\pi_1}$ is a strictly generating full subcategory of $\mathrm{Ind}_{\pi_1}(\mathcal{M}_{\pi_1})$, so it remains to see that $\mathrm{Ind}_{\pi_1}(\mathcal{M}_{\pi_1})$ is stable under small $\varinjlim$, i.e. the

Lemma 4.3.7. *Let $\mathcal{M}_\pi$ be a $\mathfrak{U}$-category stable under direct limits of cardinality $\leqslant \pi$, where π is a given infinite cardinal. Then $\mathrm{Ind}_\pi\mathcal{M}_\pi$ is stable under small direct limits. Moreover, $\mathcal{M} = \mathrm{Ind}_\pi\mathcal{M}_\pi$, equipped with $\mathcal{M}_\pi \longrightarrow \mathcal{M}$, is a solution of a 2-universal problem, namely that for every category $\mathcal{N}$ stable under small direct limits, the restriction functor induces an equivalence* p. 15

$$\underline{\mathrm{Hom}}_{!}(\mathcal{M},\mathcal{N}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{!\,\pi}(\mathcal{M}_\pi,\mathcal{N}) \,, \tag{4.3.7.1}$$

where the sign ! (resp. $!\,\pi$) denotes the full subcategory of the category $\underline{\mathrm{Hom}}$*, formed by the functors which commute with small* $\varinjlim$ *(resp. with direct limits of cardinality $\leqslant \pi$).*

Let us first prove that the $\varinjlim$ of cardinality $\leqslant \pi$ *in* $\mathcal{M}_\pi$ are also $\varinjlim$ *in* $\mathcal{M} = \mathrm{Ind}_\pi\mathcal{M}_\pi$ ([14]). We thus have [page 78]

$$x \xleftarrow{\sim} \varinjlim_{I} x_i \qquad \textit{in } \mathcal{M}_\pi \,,$$

i.e.

$$\mathrm{Hom}(x,y) \xrightarrow{\sim} \varprojlim_{I} \mathrm{Hom}(x_i,y) \qquad \text{if } y \in \mathrm{Ob}\mathcal{M}_\pi \,, \tag{$*$}$$

let us prove that this remains true for y in $\mathcal{M}$, i.e. y of the form

$$y \;=\; \varinjlim_{J} y_j \,, \qquad J \text{ large relative to } \pi, \text{ the } y_j \text{ in } \mathcal{M}_\pi.$$

[12] *Botched proof*, only works if $\pi^{\pi_0} = \pi$!!

[13] It is not certain that I is large relative to π, unless one assumes $\pi^{\pi_0} = \pi$!

[14] But *beware*, $\mathcal{M}_\pi \longrightarrow \mathrm{Ind}(\mathcal{M}_\pi)$ does *not* commute with these $\varinjlim$!

We have indeed

$$\begin{aligned}\mathrm{Hom}(x,y) &\simeq \varinjlim_J \mathrm{Hom}(x,y_j)\\ &\overset{(*)}{\simeq} \varinjlim_J \varprojlim_I \mathrm{Hom}(x_i,y_j)\\ &\overset{\text{SGA 4 I 9.8}}{\simeq} \varprojlim_I \varinjlim_J \mathrm{Hom}(x_i,y_j)\\ &\simeq \varprojlim_I \mathrm{Hom}(x_i,y)\ ,\end{aligned}$$

altogether

$$\mathrm{Hom}(x,y) \simeq \varprojlim_I \mathrm{Hom}(x_i,y),$$

q.e.d.

To prove that the small $\varinjlim$ are representable in $\mathcal{M} = \mathrm{Ind}_\pi(\mathcal{M}_\pi)$, as they are in $\mathrm{Ind}(\mathcal{M}_\pi)$ (SGA 4 I 8.9.5 b, we use the hypothesis that the *finite* $\varinjlim$ are representable in $\mathcal{M}_\pi$), it remains to see that $\mathrm{Ind}_\pi(\mathcal{M}_\pi)$ is stable in $\mathcal{M}_\pi$ under these limits.

[page 79]

4.4 Study of $\mathrm{Ind}_\pi(C)$ (for C a small category stable under $\varinjlim$ of cardinality $\leqslant \pi$)

4.4.1. In the end, the proof of Lemma 4.3.7 is not so obvious as I anticipated, and I must devote some technical preliminaries to it, which I am going to collect here. Throughout what follows, C represents a $\mathfrak{U}$-category equivalent to a small category.

p. 16

Proposition 4.4.2. *Let $F \in C^\wedge$ be an ind-representable functor, i.e. such that C/F is filtering. Let, on the other hand, $i \longmapsto x_i$ be a functor $I \longrightarrow C$, where I is a small filtering category, and let*

$$\alpha\ :\ x\ =\ \varinjlim_I {}_{C^\wedge} x_i \longrightarrow F \qquad ([15]) \tag{4.4.2.1}$$

*an inverse system of arrows $x_i \longrightarrow F$ (**N.B.** the direct limit x is taken in $C^\wedge$, and is computed argument by argument), which is equivalent to a factorization $\varphi_\alpha : I \longrightarrow C/F$ of $I \longrightarrow C$ as*

$$I \xrightarrow{\varphi_\alpha} C/F \underset{\psi}{\xleftarrow{can.}} C\ .$$

That said, for α (4.4.2.1) to be an isomorphism in $\mathrm{Ind}'(C) \subset C^\wedge$ (full subcategory of ind-representable functors on C), it is necessary and sufficient that φ_α be cofinal*, or what amounts to the same thing, I being filtering, that it satisfy the two usual conditions:*

F 1) *$\forall\, \xi \in \mathrm{Ob}C/F$, $\exists\, i \in \mathrm{Ob}I$ and an arrow $\xi \longrightarrow \varphi_\alpha(i)$.*

F 2) *$\forall\, \xi \in \mathrm{Ob}C/F$, and $i \in \mathrm{Ob}I$, and every double arrow*

[page 80]

$$\xi \rightrightarrows \varphi_\alpha(i)\ ,$$

there exists an arrow $i \xrightarrow{u} j$ in I such that $\varphi_\alpha(u)$ equalizes the double arrow.

[15] Giving α is also equivalent to giving a homomorphism of ind-objects $(x_i)_{i\in I} \longrightarrow F$, F being considered as a (generalized) ind-object indexed by C/F.

PROOF. If φ_α is cofinal, we conclude from this by definition that $\varinjlim_I \psi\varphi_\alpha \longrightarrow \varinjlim_{C/F} \psi$ is an isomorphism, now this arrow is none other than α. Conversely, suppose that this arrow is an isomorphism, hence $C/F \xleftarrow{\sim} C/\varinjlim x_i$, and it remains to see that the canonical functor

$$I \longrightarrow C/\varinjlim_{C^\wedge} x_i$$

is cofinal, i.e. satisfies F 1) and F 2). The verification is ± tautological, using the fact that for every y in C, we have

$$\mathrm{Hom}(y, \varinjlim_{C^\wedge} x_i) \xleftarrow{\sim} \varinjlim_{\mathrm{Ens}} \mathrm{Hom}(y, x_i),$$

and the "calculation" of filtering direct limits in Ens.

Corollary 4.4.3. *Let π be an infinite ordinal. For F to be in $\mathrm{Ind}'_\pi(C)$ (essential image of $\mathrm{Ind}_\pi(C)$ in $C^\wedge$), it is necessary and sufficient that there exist an ordered category I large relative to π, and a cofinal functor $I \longrightarrow C/F$.*

We must therefore give conditions, on a category J $(= C/F)$, for there to exist an ordered category I large relative to π, and a cofinal functor $I \longrightarrow J$.

Such a category will be called *large relative to π*. (It is immediate that in the case [page 81] where J is itself ordered, this terminology is consistent with that already adopted for ordered sets, i.e. that if there exists an increasing cofinal map $\varphi : I \longrightarrow J$, with I large relative to π, then J is itself large relative to π.) Such a category is automatically filtering p. 17 (SGA 4 I 8.1.3 b)). Let us also note that if we have a cofinal functor $J \longrightarrow J'$, if J is large relative to π, J' is too (by transitivity of cofinality).

Proposition 4.4.4. *Let J be a $\mathfrak{U}$-category, π an infinite cardinal. Equivalent conditions:*

a) J is large relative to π, i.e. there exists an ordered set large relative to π and a cofinal functor $\varphi : I \longrightarrow J$.

b) J satisfies the following two conditions:

PF$_\pi$ 1 *Ord J is large relative to π, i.e. for every family $(y_\alpha)_{\alpha\in A}$ of objects of J, with $\mathrm{card}A \leqslant \pi$, there exists an upper bound y in J (i.e. a y and arrows $y_\alpha \xrightarrow{v_\alpha} y$).*

PF$_\pi$ 2 *For y_0, y_1 in J and every set $(u_\alpha)_{\alpha\in A}$ of arrows $y_0 \xrightarrow{u_\alpha} y_1$ with $\mathrm{card}A \leqslant \pi$, there exists an arrow $y_1 \xrightarrow{u'} y_2$ which equalizes them, i.e. such that $u'u_\alpha = u'u_\beta$ $\forall\, \alpha, \beta \in A$. (It suffices to require this when $\mathrm{card}A \geqslant 2$.)*

c) For every subset A of $\mathrm{Fl}J$, with $\mathrm{card}A \leqslant \pi$, exists a subcategory J' of J, with $\mathrm{card}\mathrm{Fl}J' \leqslant \pi$, such that $A \subset \mathrm{Fl}J'$ and J' has a final object.

d) Let I be the set of subcategories J' of J such that $\mathrm{card}J' \leqslant \pi$ and having a final object

and let us order I by inclusion. Then I is large relative to π, moreover the [page 82] *canonical functor $I \longrightarrow J$ ($J' \longmapsto e_{J'}$ = final object of J') is cofinal* ([16],[17]).

([19])

PROOF.

[16] Introduce $I(J)$ before the statement of the proposition. **N.B.** The cofinality condition $I(J) \longrightarrow J$ is not automatic, as we see by taking $J = B_G$, $G = \{\pm 1\}$.

[17] **N.B.** The cofinality hypothesis is superfluous if J is rigid, i.e. every endomorphism of an object of J is the identity.

[19] Other conditions:

a $\Longrightarrow$ b immediate verification, using the criterion F 1, F 2 for cofinality for I filtering.

b $\Longrightarrow$ c by AQT .

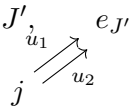
c $\Longrightarrow$ d by AQT .

p. 18

d $\Longrightarrow$ a Let us consider the ordered set large relative to π constructed in the statement of d), and the *functor*

$$I \longrightarrow J$$

given by $J' \longmapsto e_{J'}$ = final object of J'. (Verification that it is a functor immediate.) This functor is surjective, hence satisfies the cofinality condition F 1. Condition F 2 also follows from d) by looking at the three objects J', $j \xrightarrow{u_1} e_{J'}$, $j \xrightarrow{u_2} e_{J'}$ of I,



and by taking a J'' which is an upper bound for all three, then $e_{J'} \longrightarrow e_{J''}$ equalizes u_1, u_2 ([20]).

Corollary 4.4.5. *Let $F \in \mathrm{Ob}C^\wedge$. For F to belong to $\mathrm{Ind}'_\pi(C)$, it is necessary and sufficient that C/F satisfy the set of four equivalent conditions of 4.4.4.*

Corollary 4.4.6. *Let J be a category stable under $\varinjlim$ of cardinality $\leqslant \pi$. Then J is large relative to π.*

Indeed, conditions PF_π 1, PF_π 2 are visibly satisfied.

[page 83]

4.4.6. I give myself a functor

$$J \xrightarrow{\varphi} \mathrm{Ind}'_\pi C \ , \tag{4.4.6.1}$$

J a small category, and we look for a $\varinjlim$ in $\mathrm{Ind}'_\pi C$, postulating if need be the existence of $\varinjlim$ of cardinality $\leqslant \pi$ in C, and the hypothesis $\mathrm{card}\mathrm{Fl}J \leqslant \pi$. The most natural procedure for reducing the question of the existence of $\varinjlim$ in $\mathrm{Ind}'_\pi C$, to the similar question in C, is made explicit in SGA 4 I 8.8.3. We consider the fully faithful inclusion

$$C \xhookrightarrow{\alpha} \mathrm{Ind}'_\pi C \ , \tag{4.4.6.2}$$

which commutes with $\varinjlim$ of cardinality $\leqslant \pi$ (those which are representable in C, cf. p. 78 and reference to SGA 4 I 9.8), whence

$$\underline{\mathrm{Hom}}(J, C) \xhookrightarrow{\alpha} \underline{\mathrm{Hom}}(J, \mathrm{Ind}'_\pi C) \tag{4.4.6.3}$$

also fully faithful, and commuting with $\varinjlim$ of cardinality $\leqslant \pi$ (representable in the first member). This arrow fits into the commutative square relative to the datum of a $j \in \mathrm{Ob}J$, and of the inclusion functors

$$e \xrightarrow{\varepsilon_j} J \ , \qquad \varepsilon_j(e) \ = \ j \ ; \tag{4.4.6.4}$$

p. 19

[page 84]

$$\begin{array}{ccc} \underline{\mathrm{Hom}}(J,C) & \xrightarrow{\alpha^J} & \underline{\mathrm{Hom}}(J,\mathrm{Ind}'_\pi C) \\ \downarrow{\scriptstyle \varepsilon_j^*} & & \downarrow{\scriptstyle \varepsilon_j^{*\prime}} \\ C & \xrightarrow{\alpha} & \mathrm{Ind}'_\pi(C) \end{array} \qquad \begin{array}{c} \ni \varphi \\ \downarrow \\ \ni \varphi(j)\ , \end{array} \qquad (^{21}) \qquad (4.4.6.2)$$

which induces a functor

$$\underbrace{\underline{\mathrm{Hom}}(J,C)/\varphi}_{\stackrel{\mathrm{def}}{=} I} \xrightarrow{\varepsilon_j^\varphi} C/\varphi(j)\ . \qquad (4.4.6.3)$$

If C is stable under $\varinjlim$ relative to a set K of the type of conditions $\leqslant \pi$ (e.g. finite $\varinjlim$), the same holds for $\underline{\mathrm{Hom}}(J,C)$, and also for $\underline{\mathrm{Hom}}(J,C)/\varphi$, and for $C/\varphi(j)$, by virtue of the fact that α^J and α commute with the $\varinjlim$ of the type envisaged, moreover (4.4.6.3) commutes with the limits in question. Let us therefore suppose C stable at least under finite $\varinjlim$, so the source and target categories in (4.4.6.3) are filtering. (The fact that the functor commutes with finite $\varinjlim$ does not seem useful in the present context.) This will serve us to have the convenient cofinality criterion SGA 4 I 8.1.3 b) in terms of F 1, F 2.

Let us recall that the cofinality of the functors (4.4.6.3), when it holds (?), implies [page 85] the fact that φ is a direct limit of the composite functor

$$\begin{array}{ccccc} I & \xrightarrow{\psi_J} & \underline{\mathrm{Hom}}(J,C) & \xrightarrow{\alpha^J} & \underline{\mathrm{Hom}}(J,\mathrm{Ind}'_\pi(C)) \\ \| & & \vdots{\scriptstyle \varepsilon_j^*} & & \vdots{\scriptstyle \varepsilon_j^{*\prime}} \\ \underline{\mathrm{Hom}}(J,C)/\varphi & & & & \\ {\scriptstyle \varepsilon_j^\varphi}\downarrow & & & & \\ C/\varphi(j) & \xrightarrow{\psi_j} & C & \xrightarrow{\alpha} & \mathrm{Ind}'_\pi(C)\ , \end{array} \qquad (4.4.6.4)$$

along the filtering category I, more precisely that φ is the $\varinjlim$ "argument by argument in j", which means that for every $j \in \mathrm{Ob}\, J$, $\varphi(j) = \varepsilon_j^{*\prime}(\varphi)$ is a direct limit of $\varepsilon_j^{*\prime}(\alpha^J\psi) = (\alpha\psi_j)\varepsilon_j^\varphi$. (Now we do indeed know that $\varphi(j)$ is the limit of $\alpha\psi$, since $\varphi(j) \in \mathrm{Ob}\, C^\wedge$ is ind-representable, hence also that of $(\alpha\psi_j)\varepsilon_j^\varphi$ if ε_j^φ is cofinal.)

Thus, admitting the cofinality of (4.4.6.3) and interpreting ψ_J in (4.4.6.4) [page 86]

$$\psi_J\ :\ I \longrightarrow \underline{\mathrm{Hom}}(J,C)$$

as a functor

$$I \times J \xrightarrow{\psi} C\ ,$$

we find the "indicial representation" of φ

$$\varphi(j)\ =\ \underbrace{\varinjlim_{i\in I} \psi(i,j)}_{\substack{\text{direct limit} \\ \text{taken in } \mathrm{Ind}'_\pi(C) \\ \text{of objects of } C \hookrightarrow \mathrm{Ind}'_\pi(C)}}\ ,$$

p. 20

e) For every set A with $\mathrm{card} A \leqslant \pi$, $\mathrm{diag}_J : J \longrightarrow J^A$ is cofinal.

e′) (If A_0 is a given set, of cardinality $= \pi$.) $J \longrightarrow J^{A_0}$ is cofinal.

See also 4.9.6.9.1 and 4.9.6.9.2, pages 237, 238.

[20] A bit brief, the case where $e_{J'} = j$ escapes this argument, and requires that one

[21] The functors ε_j^*, $\varepsilon_j^{*\prime}$ are the evaluation functors at j.

whence

$$\underbrace{\varinjlim_{j\in J} \operatorname{Ind}'_\pi C\varphi(j)}_{\text{subject to existence}} = \varinjlim_{j\in J}\varinjlim_{i\in I}\psi(i,j) = \varinjlim_{i\in I}\varinjlim_{j\in J}\psi(i,j)\ ,$$

all direct limits being taken in $\operatorname{Ind}'_\pi(C)$. But if the $\varinjlim$ of type J are representable in C, and $\operatorname{card} J \leqslant \pi$, we know that the $\varinjlim_{j\in J}\psi(i,j)$ *in* C is also their $\varinjlim$ in $\operatorname{Ind}'_\pi C$, and we therefore find that $\varinjlim_{j\in J}\varphi(j)$ exists in $\operatorname{Ind}'_\pi(C)$ if and only if $\varinjlim_{i\in I}\Psi(i)$ exists in $\operatorname{Ind}'_\pi(C)$, where for $i \in I$, we set

$$\Psi(i) \ = \ \varinjlim_{j\in J}\psi(i,j) \qquad \textit{in } C.$$

Let us summarize what we have obtained: [page 87]

Lemma 4.4.7. *Let C be a small category, π an infinite cardinal, J a category such that* $\operatorname{card}\operatorname{Fl} J \leqslant \pi$, *$\varphi : J \longrightarrow \operatorname{Ind}'_\pi(C)$ a functor. Suppose the $\varinjlim$ of type J to exist in C, and that the functors ε_j^φ ($j \in \operatorname{Ob} J$) of (4.4.6.3) are* cofinal *(which is made explicit in a simple way especially if source and target are filtering, something established if C is stable under finite $\varinjlim$). Then φ has a "canonical indicial representation"*

$$\psi \ : \ I \times J \longrightarrow C \tag{4.4.7.1}$$

with

$$I \ = \ \underline{\operatorname{Hom}}(J, C)/\varphi\ , \tag{4.4.7.2}$$

i.e. we have the isomorphism functorial in $j \in \operatorname{Ob} J$

$$\varphi(j) \ \simeq \ \varinjlim_{I}\psi(i,j) \qquad ([22])\ . \tag{4.4.7.3}$$

Let us set for every $i \in \operatorname{Ob} I$

$$\Psi(i) \ = \ \varinjlim_{J}\psi(i,j) \qquad ([23])\ . \tag{4.4.7.4}$$

Then $\varinjlim_{j\in J}\varphi(j)$ (in $\operatorname{Ind}'_\pi C$) exists if and only if the $\varinjlim_{i\in I}\Psi(i)$ exists (in $\operatorname{Ind}'_\pi C$ as well), [page 88]
and the two are equal.

Corollary 4.4.8. *Suppose that C is stable under $\varinjlim$ of cardinality $\leqslant \pi$ (not only under those of type J). Then $\underline{\operatorname{Hom}}(J, C)$ and $I = \underline{\operatorname{Hom}}(J, C)/\varphi$ have the same property, I is large relative to π, and $\varinjlim_{j\in J}\varphi(j) \ = \ \varinjlim_{i\in I}\Psi(i)$ exists in $\operatorname{Ind}'_\pi(C)$, subject to cofinality of the functors (4.4.6.3).*

Indeed it suffices to verify this p. 21

Lemma 4.4.8.1. *Let I be a category large relative to π, and $\Psi : I \longrightarrow C$ a functor. Then $\varinjlim_i \Psi(i)$ exists in $\operatorname{Ind}'_\pi(C)$, and it is the direct limit in $C^\wedge$, or again "it is" the ind-object large relative to π represented by Ψ.* (In other words ([24]), $\operatorname{Ind}'_\pi(C) \subset C^\wedge$ is a strictly full subcategory *stable* under $\varinjlim$ large relative to π.)

[page 89]

[22] direct limit in Ind'_π of objects of C.

[23] direct limit taken in C, but valid also in $\operatorname{Ind}'_\pi C$.

[24] This is an abuse, this stability is a distinctly stronger statement! I prove it on page 99.

This lemma is a tautology: if F denotes the direct limit in $C^\wedge$, it is clear by definition (taking a cofinal functor $I_0 \longrightarrow I$, with I_0 *ordered* and large relative to π, so that the $\varinjlim$ can be taken along I_0) that F is in $\mathrm{Ind}'_\pi(C)$, and then it is a fortiori a $\varinjlim$ in $\mathrm{Ind}'_\pi C$.

To verify the existence of the $\varinjlim$ of cardinality $\leqslant \pi$ in $\mathrm{Ind}'_\pi C$, in the case where C itself is stable under the said limits, it therefore suffices, thanks to Corollary 4.4.8, to prove the cofinality of the functors (4.4.6.3). This will follow from the

Lemma 4.4.9. *Let C be a small category, π an infinite cardinal, J a small category such that for all $j, j' \in \mathrm{Ob}\, J$, we have* $\mathrm{card}\,\mathrm{Hom}(j, j') \leqslant \pi$. *Under these conditions, for every $j \in \mathrm{Ob}\, J$, the functor (4.4.6.3)* admits a left adjoint, *hence a fortiori it is cofinal (since the categories colocalized over $C/\varphi(j)$ have an initial object, a fortiori they are* 0*-connected).*

[page 90]

Proof of 4.4.9. Let us return to the diagram (4.4.6.2), from which we deduce (4.4.6.3), we are going to rewrite it in a simplified way:

$$\begin{array}{ccc} B & \xrightarrow{\beta} & B' \\ {\scriptstyle\lambda}\,\hat{\uparrow}\,\downarrow{\scriptstyle\varepsilon} & & {\scriptstyle\lambda'}\,\hat{\uparrow}\,\downarrow{\scriptstyle\varepsilon'} \\ C & \xrightarrow{\gamma} & C' \end{array} \qquad \begin{array}{c} \ni y'_0 \\ \top \\ \downarrow \\ \ni z'_0\ . \end{array} \tag{4.4.9.1}$$

I say that ε, ε' have *left adjoints* λ resp. λ', and that the canonical "base change" homomorphism

$$\lambda'\gamma \longrightarrow \beta\lambda \tag{4.4.9.2}$$

is an *isomorphism.* Indeed, the existence of the adjoints of ε, ε', namely of the functors $\varepsilon_{j!}$ and $\varepsilon'_{j!}$ relative to the inclusion

$$\varepsilon_j \ : \ e \longrightarrow J$$

and to the functor categories $\underline{\mathrm{Hom}}(J, C)$ resp. $\underline{\mathrm{Hom}}(J, \mathrm{Ind}'_\pi C)$, resulting from the existence of sums of type $\mathrm{Hom}(j, j')$ ($j' \in \mathrm{Ob}\, J$) in C (already established), *and in* $\mathrm{Ind}'_\pi C$ *;* and the fact that the functor $C \longrightarrow \mathrm{Ind}'_\pi C$ commutes with the said sums (something also already established, since $\mathrm{card}\,(\mathrm{Hom}(j, j')) \leqslant \pi$) then implies that (4.4.9.2) is an isomorphism. Subject to the pending verification (the existence of sums of cardinality $\leqslant \pi$ in $\mathrm{Ind}'_\pi(C)$), Lemma 4.4.9 will follow from the more general result:

[page 91]

p. 22

Lemma 4.4.10. *Consider a commutative square (4.4.9.1) (solid lines) of categories, where ε admits a left adjoint λ, ε' a left adjoint λ', and the base change arrow (4.4.9.2) is an isomorphism. Then for every y'_0 in B', setting $\varepsilon'(y'_0) = z'_0$, the functor induced by $(\varepsilon, \varepsilon')$,*

$$\overline{B} \ = \ B/y'_0 \xrightarrow{\bar{\varepsilon}} \overline{C} = C/z'_0\ ,$$

admits a left adjoint $\overline{\lambda}$, given by

$$\overline{\lambda}(\gamma(z) \xrightarrow{v} z'_0) \quad = \quad \Big(\lambda'\gamma(z) \simeq \beta(\lambda(z)) \xrightarrow{\lambda'(v)} \lambda'(z'_0) \Big\| \ \lambda'\varepsilon'(y'_0) \Big\downarrow_{\text{adj.}} \ y'_0\Big)\ .$$

($\overline{\lambda}(v) = \text{composite adj}\circ\lambda'(v)$)

The proof is carried out by a tedious AQT (I confess not to have pushed to the last detail …).

[page 92]

To finish the proof of 4.4.9, there remains to prove one last (?)

Lemma 4.4.11. *Let C be a small category, stable under sums of cardinality $\leqslant \pi$. Then $\mathrm{Ind}'_\pi C$ is stable under sums of cardinality $\leqslant \pi$ (and, for the record, the functor $\alpha : C \longrightarrow \mathrm{Ind}'_\pi C$ commutes with them).*

Applying Corollary 4.4.8 to Lemma 4.4.7, we are reduced to proving the cofinality of (4.4.6.3) when J is discrete, plus the fact that $\underline{\mathrm{Hom}}(J, C)/\varphi$ is large relative to π, in this case. But giving φ amounts to giving a family $(\varphi_j)_{j\in J}$ of objects of $\mathrm{Ind}'_\pi(C)$, and the functor (4.4.6.3) is made explicit as the projection

$$\prod_{j\in J}(C/\varphi_j) \longrightarrow C/\varphi_j$$

from the total product onto one of the factors. So we are reduced to this:

[page 93]

Lemma 4.4.12. *Let $(C_j)_{j\in J}$ be a family of small categories large relative to π, indexed by a small set J. Then $\prod C_j$ is large relative to π, and the projections (for $j_0 \in J$) $\prod C_j \longrightarrow C_{j_0}$ are cofinal.*

Let us first prove cofinality, which is the most important for us. This is obvious if $J = \{j_0\}$, otherwise we have $J = J' \cup \{j_0\}$, and $\prod_{j\in J} C_j \simeq C' \times C_{j_0}$, where $C' = \prod_{j\in J'} C_j$, so we must see that $C' \times C_{j_0} \longrightarrow C_{j_0}$ is cofinal, i.e. that every category $\xi\backslash(C' \times C_{j_0})$ (for $\xi \in \mathrm{Ob}\, C_{j_0}$) is 0-connected, now it is isomorphic to $C' \times (\xi\backslash C_{j_0})$, and $\xi\backslash C_{j_0}$ being 0-connected, it remains to see that C' is 0-connected. We should be careful in this regard that an *infinite* product of 0-connected categories is not necessarily 0-connected, so cofinality is not completely tautological! But if we prove that the category $\prod_{j\in J} C_j$ is large relative to π, hence filtering, this implies that it is 0-connected, and we can apply this same result to $C' = \prod_{j\in J'} C_j$, and we win.

p. 23

[page 94]

So we are reduced to proving that $\prod C_j$ is large relative to π. This can be seen either from the criterion $\mathrm{PF}_\pi\, 1$, $\mathrm{PF}_\pi\, 2$ of 4.4.4 b, or by reducing to the case where the C_j are ordered, but that looks longer. Let us note that we prove in the same way that a (small) product of filtering categories is again a filtering category.

Remarks.

1. This argument in no way shows (even if C is stable under $\varinjlim$ of cardinality $\leqslant \pi$) that for every $\varphi \in \underline{\mathrm{Hom}}(J, \mathrm{Ind}(C))$, the functor similar to (4.4.6.3)

$$\underline{\mathrm{Hom}}(J, C)/\varphi \longrightarrow C/\varphi(j)$$

is cofinal, and I doubt that it always is. The difficulty comes from the fact that $C \longrightarrow \mathrm{Ind}\, C$ does not commute with sums of cardinality $\leqslant \pi$, only with finite sums, so Lemma 4.4.10 cannot be invoked.

[page 95]

2. We have just proved, via 4.4.8 and 4.4.9 (whose proof is completed with 4.4.12), that if C is a small category stable under $\varinjlim$ of cardinality $\leqslant \pi$, then $\mathrm{Ind}_\pi C$ is also stable under these limits. Moreover, the direct systems of this type J in $\mathrm{Ind}_\pi C$ admit an indicial representation

$$\psi \ : \ I \times J \longrightarrow C \ ,$$

with I large relative to π. And we can always suppose, at the cost of taking $I' \longrightarrow I$ cofinal with I' an ordered set large relative to π, that I is moreover an ordered set.

That is the means then, for the proof of the

Theorem 4.4.13. *Let C be a category equivalent to a small category, and stable under $\varinjlim$ of cardinality $\leqslant \pi$, where π is a given infinite cardinal. Then:*

[page 96]

a) (For the record.) The canonical functor

$$C \longrightarrow \mathrm{Ind}_\pi(C) \xrightarrow{\sim} \mathrm{Ind}'_\pi(C)$$

commutes with $\varinjlim$ *of cardinality* $\leqslant \pi$.

b) (For the record.) The direct systems of cardinality $\leqslant \pi$ *in* $\mathrm{Ind}_\pi(C)$ *(i.e. the functors* $\varphi : J \longrightarrow \mathrm{Ind}_\pi(C)$*, with* $\operatorname{card}\mathrm{Fl}(J) \leqslant \pi$*) admit an indicial representation*

$$\psi \ : \ I \times J \longrightarrow C \ ,$$

with I *an ordered set large relative to* π*, and admit a direct limit in* $\mathrm{Ind}_\pi C$ *which are computed by*

$$\underbrace{\varinjlim_{j\in J} \varphi(j)}_{\text{in } \mathrm{Ind}_\pi(C)} \ \simeq \ \text{“} \varinjlim_{i\in I} \text{”}\Psi(i)$$

with p. 24

$$\Psi(i) \ = \ \varinjlim_{j\in J}{}_C \, \psi(i,j) \ .$$

c) $\mathrm{Ind}_\pi C$ *is stable under small direct limits* ([21]).

It remains to establish this last point. Since we already have finite $\varinjlim$ by b, it remains to prove the existence of small filtering $\varinjlim$, or, if preferred, of infinite sums, which will be nicer in fact. But we can just as well treat the general case of a $\varinjlim$ of arbitrary type J (J small). For this, let [page 97]

$$I \ = \ I(J) \ = \ \begin{array}{l}\text{set of subcategories } J' \text{ of } J \text{ such that} \\ \operatorname{card}\mathrm{Fl}(J') \leqslant \pi,\end{array}$$

let us order I by inclusion. It is obvious that $I \neq \emptyset$ (for $\emptyset \in I$), and even that I *is large relative to* π, for if $(J'_\alpha)_{\alpha\in A}$ with $\operatorname{card}(A) \leqslant \pi$ is a family of elements of I, then the subcategory of J generated by the J'_α is still in I (immediate verification), and is an upper bound for the J'_α. In fact, the consideration of $I(J)$ associated with J will allow us to prove this, which generalizes 4.4.13 c), taking b) into account:

Corollary 4.4.14 ([22]). *Let* $\mathcal{M}$ *be a category,* π *an infinite cardinal such that*

a) $\mathcal{M}$ *is stable under* $\varinjlim$ *of cardinality* $\leqslant \pi$*, and* [page 98]

b) $\mathcal{M}$ *is stable under* $\varinjlim$ *over ordered sets* I *large relative to* π.

Then $\mathcal{M}$ *is stable under small* $\varinjlim$.

([23]).

Indeed, if $\varphi : J \longrightarrow \mathcal{M}$ is given, we construct from it

$$\psi \ : \ I \ = \ I(J) \ \longrightarrow \ \mathcal{M}$$

by

$$\psi(J') \ = \ \varinjlim_{J'} \varphi(J') \ ,$$

[21] also add, in d), corollary 4.4.16, page 101.

[22] Compare also 4.11.3, page 270.

[23] **Corollary 4.4.14.1.** $\mathcal{M}$ being as in 4.4.14, and $f : \mathcal{M} \longrightarrow \mathcal{E}$ being a functor which commutes with $\varinjlim$ of cardinality $\leqslant \pi$, and with small $\varinjlim$ large relative to π, then f commutes with small $\varinjlim$.

which exists by the hypothesis on $\mathcal{M}$. It is then immediate that $\varinjlim_J \varphi$ exists if and only if $\varinjlim_I \psi$ exists, now this existence is assured by the fact that I is large relative to π, and the hypothesis on $\mathcal{M}$.

To finish proving theorem 4.4.13, we are thus reduced to the particular case of direct limits $\varinjlim_J \varphi(j)$, when $\varphi : J \longrightarrow \mathrm{Ind}'_\pi C$ is given, with J an ordered set large relative to π. But this will follow from the more precise result: p. 25 [page 99]

Corollary 4.4.15 ([24]). *Let C be a small category. Then the strictly full subcategory $\mathrm{Ind}'_\pi C$ of $C^\wedge$ is stable in $C^\wedge$ under $\varinjlim$ large relative to π, in other words, these $\varinjlim$ exist in $\mathrm{Ind}'_\pi(C)$, and the inclusion functor commutes with them.*

Proof. Let I be an ordered set, large relative to π, and $i \longmapsto F_i$ a functor $I \longrightarrow \mathrm{Ind}'_\pi(C)$, $F = \varinjlim_I F_i$ in $C^\wedge$, let us prove that $F \in \mathrm{Ob}\,\mathrm{Ind}'_\pi(C)$, i.e. (4.4.3) that C/F is large relative to π. I use criterion b) of 4.4.4, we have to check $\mathrm{PF}_\pi 1$, $\mathrm{PF}_\pi 2$. Let

$$x_\alpha \xrightarrow{u_\alpha} F = \varinjlim F_i$$

be objects of C/F indexed by A with $\mathrm{card}\, A \leqslant \pi$. Each u_α comes from $x_\alpha \longrightarrow F_{i_\alpha}$, and taking a common upper bound for the i_α (I large relative to π), we find an $i \in I$ and $v_\alpha : x_\alpha \longrightarrow F_i$. Since F_i is in $\mathrm{Ind}'_\pi(C)$, i.e. C/F_i large relative to π, considering the v_α as objects of C/F_i, they are bounded above by an object $x \longrightarrow F_i$ of C/F_i. We thus have

$$\begin{array}{ccccc} x_\alpha & \longrightarrow & F_{i_\alpha} & & \\ \downarrow & & \downarrow & \searrow & \\ x & \longrightarrow & F_i & \longrightarrow & F\,, \end{array}$$

which shows that the $x_\alpha \longrightarrow F$ are bounded above by $x \longrightarrow F$, which proves $\mathrm{PF}_\pi 1$. Let us prove $\mathrm{PF}_\pi 2$, i.e. let $x \xrightarrow{u} F$, $y \xrightarrow{v} F$ be two objects of C/F, and $(f_\alpha)_{\alpha \in A}$ a family of cardinality $\leqslant \pi$ of arrows from u to v. We know that u, v come from $u_i : x \longrightarrow F_i$, $v_i : y \longrightarrow F_i$ for i large enough. [page 100]

$$\begin{array}{ccc} x & \xrightarrow{f_\alpha} & y \\ & {\scriptstyle u}\searrow \quad \swarrow{\scriptstyle v} & \\ & F & \end{array} \qquad \begin{array}{ccc} x & \xrightarrow{f_\alpha} & y \\ & {\scriptstyle u_i}\searrow \quad \swarrow{\scriptstyle v_i} & \\ & F_i & \end{array}$$

For every α, we can choose $i_\alpha \geqslant i$ such that $v_i f_\alpha$ and u_i become equal on composing with $F_i \longrightarrow F_{i_\alpha}$, and since $\mathrm{card}\, A \leqslant \pi$ and I large relative to π, there exists an upper bound i' for the i_α. All told, we can assume (replacing i by i') that $v_i f_\alpha = u_i$ for every α. Then u_i, v_i being considered as objects of C/F_i, the f_α are arrows of C/F_i from u_i to v_i. Since C/F_i is large relative to π (by the hypothesis $F_i \in \mathrm{Ob}\,\mathrm{Ind}'_\pi(C)$), we find that there exists an equalizer $(y, v_i) \xrightarrow{g} (z, w_i)$.

$$\begin{array}{ccccc} x & \overset{(f_\alpha)}{\rightrightarrows} & y & \xrightarrow{g} & z \\ & {\scriptstyle u_i}\searrow & \downarrow{\scriptstyle v_i} & \swarrow{\scriptstyle w_i} & \\ & & F_i & & \end{array}$$

p. 26

[24] compare with 4.7.3.1, which ought to be made explicit here.

Then the image of g in C/F (via $F_i \longrightarrow F$) equalizes the arrows f_α, as arrows of C/F, which proves $\mathrm{PF}_\pi\,2$ and completes the proof of 4.4.15, and thereby also that of theorem 4.4.13, and of lemma 4.3. This also completes the proof of the first assertion of lemma 4.3.7 (p. 77), which completes the proof of theorem 4.3.6 (p. 70-71). It remains [page 101] to establish the universal property stated in 4.3.7, which I am going to recall:

Corollary 4.4.16 ([25]). *Let C be a category equivalent to a small category, stable under $\varinjlim$ of cardinality $\leqslant \pi$, where π is a given infinite cardinal, and let us consider the canonical inclusion functor*

$$C \xrightarrow{\alpha} \mathrm{Ind}_\pi(C)\,. \tag{4.4.16.1}$$

This functor is 2-universal for functors $f : C \longrightarrow \mathcal{E}$, where $\mathcal{E}$ is a $\mathcal{U}$-category stable under small $\varinjlim$, and f a functor commuting with these direct limits. In other words, we have

1°) α has the preceding properties, i.e. $\mathrm{Ind}_\pi(C)$ is stable under $\varinjlim$, and the functor α commutes with $\varinjlim$ of cardinality $\leqslant \pi$.

2°) For every $\mathfrak{U}$-category $\mathcal{E}$ stable under small direct limits, the functor

$$\begin{array}{rcl} g & \longmapsto & g \circ \alpha \\ \underline{\mathrm{Hom}}_!(\mathrm{Ind}_\pi C, \mathcal{E}) & \longrightarrow & \underline{\mathrm{Hom}}_{!\,\pi}(C, \mathcal{E}) \end{array} \tag{4.4.16.2}$$

is an equivalence of categories, where the subscript $!$ (resp. $!\,\pi$) indicates the strictly full subcategory of the category $\underline{\mathrm{Hom}}$ under consideration, formed by the functors which commute with small $\varinjlim$ (resp. with those of cardinality $\leqslant \pi$). [page 102]

We can thus also say that $\mathrm{Ind}_\pi(C)$ plays the role of an "envelope" stable under small $\varinjlim$ of the category C, when the $\varinjlim$ of cardinality $\leqslant \pi$, which already exist in C, are respected by the embedding of C in its envelope. It is a by no means tautological result that it suffices for this to add to C some $\varinjlim$ *large relative to* π of objects of C, and that more precisely, the envelope category is realized by $\mathrm{Ind}_\pi C$ (or, if preferred and less plethorically, by $\mathrm{Ind}'_\pi C \subset C^\wedge$).

Proof. 1°) is contained in 4.4.13 a) and c). Let us therefore prove 2°). It is obvious that α is a fully faithful functor whose essential image strictly generates $\mathcal{M} = \mathrm{Ind}_\pi C$, p. 27 whence it follows that the functor (4.4.16.2) is fully faithful, being induced by the fully faithful functor

$$\alpha^*_{\mathcal{E}} \;:\; \underline{\mathrm{Hom}}_!(\mathcal{M}, \mathcal{E}) \;\longrightarrow\; \underline{\mathrm{Hom}}(C, \mathcal{E})$$

on the full subcategories under consideration. It thus remains to show that (4.4.16.2) is [page 103] essentially surjective. So let $f : C \longrightarrow \mathcal{E}$ be in $\underline{\mathrm{Hom}}_{!\,\pi}(C, \mathcal{E})$, let us consider its canonical extension

$$\begin{cases} Lf \;:\; \mathrm{Ind}(C) \;\longrightarrow\; \mathcal{E} \\ Lf((X_i)_{i\in I}) \;=\; \varinjlim\limits_{i\in I} f(X_i) \end{cases} \tag{4.4.16.3}$$

(cf. SGA 4 I 8.7.3), and let

$$\psi(f) \;=\; Lf|\,\mathrm{Ind}_\pi(C)\,. \tag{4.4.16.4}$$

Let us recall that Lf commutes with small filtering $\varinjlim$, but since the inclusion functor

$$\mathrm{Ind}_\pi C \;\longrightarrow\; \mathrm{Ind}(C) \tag{4.4.16.5}$$

does not commute with these $\varinjlim$, it is not clear a priori that $\psi(f)$ commutes with the said limits, nor a fortiori that it commutes with small $\varinjlim$ without more. But we

[25] Could be incorporated as part d) in theorem 4.4.13, for it is an important complement, and not so obvious after all.

have seen that (4.4.16.5) commutes with $\varinjlim$ *large relative to* π, which is contained in corollary 4.4.15 (since the *filtering* direct limits in $\mathrm{Ind}(C)$ are the limits *in* $C^\wedge$): to say that $\mathrm{Ind}'_\pi(C) \longrightarrow \mathrm{Ind}'(C)$ commutes with a certain type of filtering $\varinjlim$, amounts to saying that this is so for the composite with $\mathrm{Ind}'_\pi(C) \hookrightarrow C^\wedge$, which is established by 4.4.15. From this it follows that $\psi(f)$ *commutes with* $\varinjlim$ *large relative to* π. To establish that $g = \psi(f)$ commutes with small $\varinjlim$, it thus remains to see that it commutes with [page 104] $\varinjlim$ of cardinality $\leqslant \pi$, by virtue of 4.4.14.1 (margin page 98). To establish this, we use theorem 4.4.13 b, representing a functor $\varphi : J \longrightarrow \mathrm{Ind}_\pi(C)$ (where $\mathrm{card}\,\mathrm{Fl}J \leqslant \pi$) in indexed form by

$$\Phi \ : \ I \times J \ \longrightarrow \ C \ , \tag{4.4.16.6}$$

I being an ordered set large relative to π. We thus have

$$\varphi(j) \ = \ \varinjlim_{i \in I} {}_{\mathrm{Ind}_\pi C}\, \Phi(i,j) \ ,$$

and since $g = \psi(f)$ commutes with $\varinjlim$ of type I by what we have seen, we will have

$$g(\varphi(j)) \ = \ \varinjlim_{i \in I} {}_{(\mathcal{E})}\ g(\Phi(i,j)) \ ,$$

whence

$$\varinjlim_{j \in J} {}_{(\mathcal{E})}\, g(\varphi(j)) \ = \ \varinjlim_{j \in J} \varinjlim_{i \in I} \underbrace{g(\Phi(i,j))}_{= f(\Phi(i,j))} \qquad (\text{limits in } \mathcal{E}) \ . \tag{$*$}$$

p. 28

On the other hand, we have

$$\begin{aligned} \varinjlim_{j \in J} {}_{\mathrm{Ind}_\pi}\, \varphi(j) \ &\simeq \ \varinjlim_{j \in J} \varinjlim_{i \in I} \Phi(i,j) \\ &\simeq \ \varinjlim_{i \in I} \ \underbrace{\varinjlim_{j \in J} \Phi(i,j)}_{\substack{\text{can be computed} \\ \text{in } C\text{, say} \\ \Psi(i) \in \mathrm{Ob}\, C}} \ , \end{aligned}$$

thus

$$\varinjlim_{j \in J} {}_{\mathrm{Ind}_\pi}\, \varphi(j) \simeq \varinjlim_{i \in I} \ \Psi(i) \ , \tag{$**$}$$

whence by applying g, which commutes with $\varinjlim$ of type I, and now recalling that [page 105] $g|C \simeq f$,

$$g(\varinjlim_{J} {}_{\mathrm{Ind}_\pi C}\, \varphi(j)) \ \simeq \ \varinjlim_{i \in I} \ f(\Psi(i)) \ . \tag{$***$}$$

We have on the other hand

$$\Psi(i) \ \overset{\mathrm{def}}{=} \ \varinjlim_{j \in J} \ \Phi(i,j) \ , \qquad\qquad \varinjlim \ \text{ in } C \ ,$$

and since f commutes with $\varinjlim$ of type J (for $\mathrm{card}\,\mathrm{Fl}J \leqslant \pi$), we conclude from this

$$f(\Psi(i)) \ \simeq \ \varinjlim_{j \in J} \ f(\Phi(i,j)) \ ,$$

thus by $(***)$

$$g(\varinjlim_{j \in J} \ \varphi(j)) \ \simeq \ \varinjlim_{i \in I} \varinjlim_{j \in J} \ f(\Phi(i,j)) \ . \tag{$****$}$$

By composing with $(*)$ and using again the commutativity of $\varinjlim_I$ with $\varinjlim_J$, we thus find

$$g(\varinjlim \varphi(j)) \xleftarrow{\sim} \varinjlim g(\varphi(j)) ,$$

which proves that g commutes with direct limits of type J, and finishes proving that g is $\underline{\mathrm{Hom}}_!(\mathrm{Ind}_\pi(C), \mathcal{E})$. Since $g|C \simeq f$, we conclude that the functor (4.4.16.2) is essentially surjective, which finishes proving 4.4.16. Moreover, we have constructed an explicit quasi-inverse functor, ψ, fitting into the commutative diagram [page 106]

$$\begin{array}{ccccc}
\underline{\mathrm{Hom}}_{!\,\pi}(C,\mathcal{E}) & \xrightarrow[(\sim)]{\psi} & \underline{\mathrm{Hom}}_!(\mathrm{Ind}_\pi(C),\mathcal{E}) & \longrightarrow & \underline{\mathrm{Hom}}(\mathrm{Ind}_\pi(C),\mathcal{E}) \\
\big\downarrow \text{fully faithful incl.} & & \big\downarrow \text{fully faithful} & \nearrow_{R\ \text{(restriction)}} & \\
\underset{\text{(cf. SGA 4 I 8.7.3)}}{\underline{\mathrm{Hom}}(C,\mathcal{E})} & \xrightarrow[\sim]{L} & \underline{\mathrm{Hom}}_{!\,\mathrm{filt}}(\mathrm{Ind}(C),\mathcal{E}) & &
\end{array} \quad . \tag{4.4.16.6}$$

p. 29

4.5 Canonical cardinal filtration of a $\mathfrak{U}$-paratopos category

Definition 4.5.1. By a $\mathfrak{U}$-*paratopos category* we mean a category $\mathcal{M}$ which is a $\mathfrak{U}$-pseudo-topos and accessible, i.e. which satisfies the equivalent conditions of theorem 4.3.6 (p. 70). We say that it is a π-$\mathfrak{U}$-*paratopos category*, if it is π-accessible, i.e. if it admits a small generating subcategory whose objects are π-accessible, or what amounts to the same thing, if $\mathcal{M}$ is equivalent to a category of the form $\mathrm{Ind}'_\pi(\mathcal{M}_\pi)$, where $\mathcal{M}_\pi$ is a small category (or: a category equivalent to a small category), and stable under $\varinjlim$ of cardinality $\leqslant \pi$. (And we can take for $\mathcal{M}_\pi$ the full subcategory of $\mathcal{M}$ consisting of the objects which are π-accessible).

In what follows, we are going, more generally, to consider a category C equivalent to a small category, stable under $\varinjlim$ of cardinality $\leqslant \pi_0$ ([30]), and the category [page 107]

$$\mathcal{M} \overset{\mathrm{def}}{=} \mathrm{Ind}'_{\pi_0} C \ \subset\ C^\wedge , \tag{4.5.1}$$

where π_0 is a given infinite cardinal. I propose to introduce on $\mathcal{M}$ a canonical cardinal filtration:

Proposition 4.5.2 ([31]). *Let $F \in \mathrm{Ob}\,\mathcal{M} \subset \mathrm{Ob}\,C^\wedge$, and let π be a cardinal $\geqslant \pi_1 = \sup(\pi_0, \mathrm{card}\,\mathrm{Fl}\,C_0)$, where C_0 is a small full subcategory of C such that $C_0 \overset{\sim}{\hookrightarrow} C$. The following conditions are equivalent:*

a) $\mathrm{card}\,\mathrm{Fl}(C_0/F) \leqslant \pi$.

a′) $\mathrm{card}\,\mathrm{Ob}\,(C_0/F) \leqslant \pi$.

a″) For every x in C_0, $\mathrm{card}\,F(x) \leqslant \pi$.

b) There exists a small filtering category I (not necessarily ordered), large relative to *π_0, with* $\mathrm{card}\,I \leqslant \pi$, *and a direct system $(x_i)_{i\in I}$ in C, such that we have*

$$F \ \simeq\ \varinjlim_{i\in I} x_i , \tag{4.5.2.1}$$

where the $\varinjlim$ is taken in $\mathcal{M}$, or what amounts to the same thing (by virtue of 4.4.15, I being large relative to π_0) in $C^\wedge$, i.e. F is isomorphic to the ind-representable functor defined by the ind-object $\underline{x} = (x_i)_{i\in I}$ of C.

[30] There is no need to assume this right away.

[31] Is proved (and is true) only if $\pi^{\pi_0} = \pi$, see correction 4.5.6, page 116.

c) F is π-accessible in $\mathcal{M}$.

c′) F is π-accessible in $C^\wedge$.

PROOF. We have the ± tautological implications [page 108]

$$\begin{array}{ccccc} & & \mathrm{a}' & \Longrightarrow & \mathrm{a}'' \\ \mathrm{a} & \Rightarrow & & & \\ & & \mathrm{b} & \Longrightarrow & \mathrm{c}' \Longrightarrow \mathrm{c}\ , \end{array} \qquad (*)$$

where $\mathrm{a}' \Longrightarrow \mathrm{a}''$ comes from the canonical inclusion $F(x) = \mathrm{Hom}_{\mathcal{M}}(x, F) \hookrightarrow \mathrm{Ob}\, C/F$, p. 30
$\mathrm{a} \Longrightarrow \mathrm{b}$ is obtained by taking $I = C_0/F$ and noting that I is stable under direct limits of cardinality $\leqslant \pi_0$ (C being so), hence is large relative to π_0 ([32]). We have $\mathrm{b} \Longrightarrow \mathrm{c}'$ by virtue of the fact that the strictly full subcategory $C^\wedge_\pi$ of $C^\wedge$ defined by

$$\left\{ \begin{array}{lcl} C^\wedge_\pi & = & \text{strictly full subcategory of } C^\wedge \text{ consisting of the} \\ & & F \in C^\wedge \text{ which are } \pi\text{-accessible in } C^\wedge. \end{array} \right. \qquad (4.5.2.2)$$

is stable under $\varinjlim$ of cardinality $\leqslant \pi$ (SGA 4 I 9.9). Finally $\mathrm{c}' \Longrightarrow \mathrm{c}$ comes from the fact that the inclusion functor $\mathcal{M} \hookrightarrow C^\wedge$ is fully faithful and π_0-accessible, i.e. commutes with direct limits large relative to π_0, and a fortiori with those large relative to π. By the implications of the diagram above, it therefore suffices to prove that we have

$$\mathrm{c} \overset{([33])}{\Longrightarrow} a \qquad \text{and} \qquad \mathrm{a}'' \Longrightarrow \mathrm{a}\ .$$

Let us prove $\mathrm{a}'' \Longrightarrow \mathrm{a}$. Let us consider the map [page 109]

$$\mathrm{Fl}\, C/F \longrightarrow \mathrm{Fl}\, C \qquad (*)$$

induced by the étale functor $C' = C/F \longrightarrow C$, then the set of arrows above an arrow $\alpha : x \longrightarrow y$ of C is in one-to-one correspondence with $\mathrm{Ob}\, C'_y = F(y)$, since such an arrow α' is known when one knows its target, which is an arbitrary object of $F(y)$ ($C' \longrightarrow C$ being étale, i.e. fibrant with discrete fibers). Thus the fiber of $(*)$ at α is of cardinality $= \mathrm{card}\, F(y) \leqslant \pi$, so we have

$$\mathrm{card}\, \mathrm{Fl}\, C/F \leqslant \pi \underbrace{\mathrm{card}\, \mathrm{Fl}\, C}_{\leqslant \pi_1} \leqslant \pi \underbrace{\pi_1}_{\leqslant \pi} \leqslant \pi^2 = \pi\ ,$$

which proves a.

Let us finally prove $\mathrm{c} \Longrightarrow \mathrm{a}$. For this, we are going to use the first inclusion $\mathcal{M}_\pi \subset \mathrm{Filt}^\pi\, \mathcal{M}$ of SGA 4 I 9.16, which applies as soon as we prove the

Corollary 4.5.3 ([34]). *Under the conditions of 4.5.2, let, for every cardinal $\pi \geqslant \pi_1$,* $\mathrm{Filt}^\pi(\mathcal{M})$ *be the strictly full subcategory of $\mathcal{M}$ consisting of the $F \in \mathrm{Ob}\, \mathcal{M}$ which satisfy condition a) of 4.5.2. Then the family of* $(\mathrm{Filt}^\pi)_{\pi \geqslant \pi_1}$ *is a cardinal filtration of $\mathcal{M}$.* [page 110]

Condition a) of loc. cit. 9.12 (the $\underline{\mathrm{Filt}}^\pi$ are equivalent to small categories) is obvious. Let us prove b), i.e. the stability of $\underline{\mathrm{Filt}}^\pi$ under filtering $\varinjlim F = \varinjlim F_i$ *large relative to* π_0, of cardinality $\leqslant \pi$. Since $\mathcal{M} \hookrightarrow C^\wedge$ commutes with these limits, for x in C

$$F(x) = \varinjlim_i F_i(x)\ ,$$

p. 31

whence $\operatorname{card} F(x) \leqslant \operatorname{card} \operatorname{Ob} I \cdot \pi \leqslant \pi \cdot \pi = \pi$, which (by the equivalence a $\iff$ a'') implies $F \in \operatorname{Ob} \operatorname{Filt}^\pi$. Let us prove condition c) of loc. cit. For this, we will need the fact, more precise than b), which we have just established:

Corollary 4.5.4 ([35]). *For every $\pi \geqslant \pi_1$, Filt^π is stable under $\varinjlim$ (in $\mathcal{M}$) of cardinality $\leqslant \pi$ (not necessarily filtering, a fortiori not necessarily large relative to π_0).*

It suffices, by virtue of what is already proved of lemma 4.4.14 ([36]), to show that Filt^π is stable under $\varinjlim$ *of cardinality* $\leqslant \pi_0$. If $\varphi : J \longrightarrow \operatorname{Filt}^\pi$ is a functor with $\operatorname{card} J \leqslant \pi_0$, we put it in indexed form

$$\psi \ : \ I \times J \ \longrightarrow \ C \ ,$$

thanks to 4.4.13 b, with I large relative to π_0.

I recall that we could take [page 111]

$$I \ = \ \underline{\operatorname{Hom}}(J, C)/\varphi \ ,$$

where $\underline{\operatorname{Hom}}(J, C) \hookrightarrow \underline{\operatorname{Hom}}(J, \mathcal{M} = \operatorname{Ind}'_{\pi_0} C)$, at the cost of not requiring I to be ordered. We would like to prove that $\operatorname{card} \operatorname{Fl} I \leqslant \pi$. The trouble is that we only arrive at the upper bound

$$\operatorname{card} \operatorname{Fl} I \ \leqslant \ \pi^{\pi_0} \ .$$

(Even if J is discrete, we do no better, since then

$$I \ \simeq \ \prod_{j \in J} C/\varphi(j) \ , \qquad \text{whence} \quad \operatorname{Fl} I = \prod_{j \in J} \operatorname{Fl} C/\varphi(j) \ ,$$

and if $\operatorname{card} \operatorname{Fl} C/\varphi(j) = \pi$ for every j, we find $\operatorname{card} \operatorname{Fl} I = \pi^{\pi_0}$ – there is no way to do better !)

So I cannot manage to prove 4.5.4 directly, which would have come in handy for proving condition c) of loc. cit. But to tell the truth, we have done here the work of the proof of theorem 4.3.6, part b $\implies$ c – we have already proved there that the filtration we have just constructed (by condition a) of 4.5.2) is a cardinal filtration. Condition c) of SGA 4 I 9.12 is proved on page 76. But I am going to redo the proof more seriously. [page 112]

So let $F \in \operatorname{Ob} \mathcal{M}$, consider

$$J \ = \ C/F \ ,$$

it is a category stable under $\varinjlim$ of cardinality $\leqslant \pi_0$, hence large relative to π_0, and we have

$$F \ \simeq \ \varinjlim_{j \in J} x_j \qquad (\varinjlim \text{ in } \mathcal{M})$$

denoting by $j \longmapsto x_j$ the canonical functor $C/F \longrightarrow C$. Then let I be the set of subcategories J' of J which are *of cardinality* $\leqslant \pi$ *and large relative to* π_0, I ordered by inclusion, and for every $J' \in I$, let us put p. 32

$$X_{J'} \ = \ \varinjlim_{j \in J'} x_j \ .$$

As J' is large relative to π_0, the direct limit in $\mathcal{M}$ is also a limit in $C^\wedge$, whence it follows at once, since $\operatorname{card} \operatorname{Ob} J' \leqslant \operatorname{card} \operatorname{Fl} J' \leqslant \pi$, that $X_{J'}$ is in Filt^π. Then

$$J' \ \longmapsto \ X_{J'} \ \in \ \operatorname{Ob} \operatorname{Filt}^\pi$$

[32] We do not need this stability, in any case C_0/F is large relative to π_0, by virtue of 4.4.3, p. 80.

[33] is proved (and is true) only if $\pi^{\pi_0} = \pi$.

[34] Not proved, condition c) of cardinal filtrations perhaps not satisfied. Probably *false* in general.

[35] Proved only if $\pi^{\pi_0} = \pi$! Probably *false* in general.

[36] We need a variant of lemma 4.4.14, for the $\varinjlim$ of cardinality $\leqslant \pi'$, which is proved in the same way.

is a functor

$$I \longrightarrow \mathrm{Filt}^{\pi} ,$$

whose direct limit is that of the x_j, hence isomorphic to F. On the other hand, it is immediate that I is large relative to π, i.e. that if $(J'_\alpha)_{\alpha\in A}$ is a family of elements of I, [page 113] it is bounded above by a $J' \in I$. This follows from the

Lemma 4.5.5 ([37]). *Let π, π_0, with $\pi > \pi_0$, be infinite cardinals, and let J be a category large relative to π_0. Then for every subset Φ of $\mathrm{Fl}\, J$, such that $\mathrm{card}\,\Phi \leqslant \pi$, there exists a subcategory J' of J* large relative to π_0, *such that $\mathrm{card}\, J' \leqslant \pi$ ([38]), and $\mathrm{Fl}\, J' \supset \Phi$.*

This lemma being admitted, we have therefore represented an arbitrary object F of $\mathcal{M}$ as a $\varinjlim_{i\in I} X_i$, with I an ordered set large relative to π, the X_i in Filt^{π}. It remains to prove that if $\pi' > \pi$ and $F \in \mathrm{Filt}^{\pi'}$, then we can take I such that $\mathrm{card}\, I \leqslant \pi'^{\pi}$. But $F \in \mathrm{Filt}^{\pi'}$ means that $J = C/F$ satisfies $\mathrm{card}\,\mathrm{Fl}\, J \leqslant \pi'$. Then the set of subsets of cardinality $\leqslant \pi$ of $\mathrm{Fl}\, J$ has a cardinality bounded above by π'^{π}, a fortiori $\mathrm{card}\, I \leqslant \pi'^{\pi}$, which finishes proving that $(\mathrm{Filt}^{\pi})_{\pi\geqslant\pi_1}$ is a cardinal filtration on $\mathcal{M}$.

To prove the lemma, we begin by saturating Φ under finite composition, we find a [page 114] subcategory J'_0 of J such that $\mathrm{card}\,\mathrm{Fl}\, J'_0 \leqslant \pi$ … But I notice that to prove the lemma, I again need $\pi^{\pi_0} = \pi$ – an envelope of J' large relative to π_0 is going to have a cardinality bounded above (at best) by π^{π_0}.

Let us take e.g.

$$J = \mathfrak{P}(S) \qquad \text{(ordered set)}$$

with $\mathrm{card}\, S = \pi > \pi_0$, so J is large relative to π_0. Let us consider the map

$$S \longrightarrow J = \mathfrak{P}(S) , \qquad s \longmapsto \{s\} ,$$

and let us take for $\Phi \subset \mathrm{Fl}\, J$ the set of $\mathrm{id}_{\{s\}}$. Thus we are looking for a subset J' of J, containing S, and large relative to π_0 for the order induced by J. We see at once that the smallest subset J' that does the job is the set $\mathfrak{P}_{\pi_0}(S)$ of subsets of S of cardinality $\leqslant \pi_0$. But we find

$$\mathrm{card}\, J' \cdot \pi_0! = \pi^{\pi_0} ,$$

where p. 33

$$\begin{aligned} \pi_0! &= \mathrm{card}\,\mathrm{Aut}\,(E_0) \qquad (E_0 \text{ a set of cardinality } \pi_0) \\ &= \pi_0^{\pi_0} \\ &= 2^{\pi_0} , \end{aligned}$$ [page 115]

so

$$\mathrm{card}\, J' \cdot 2^{\pi_0} = \pi^{\pi_0} . \qquad (*)$$

On the other hand, it is immediate that $\mathrm{card}\, J' \geqslant \mathrm{card}\,\mathfrak{P}(E_0) = 2^{\pi_0}$ (where E_0 is a subset of S of cardinality π_0), so the left-hand side of $(*)$ is equal to $\mathrm{card}\, J'$, so we find

$$\mathrm{card}\, J' = \pi^{\pi_0} ,$$

sorry !

Finally, I have convinced myself that proposition 4.5.2, and the preceding corollaries and lemmas by which I tried to establish it, is true only for those cardinals $\pi > \pi_0$ such that

$$\pi^{\pi_0} = \pi ,$$

[37] Is proved *only* if $\pi^{\pi_0} = \pi$! Probably false in general.

[38] This is OK if we require only $\mathrm{card}\, J' \leqslant \pi^{\pi_0}$.

which, assuming the generalized continuum hypothesis, is equivalent to

$$\pi^{\pi_0} \ < \ 2^{\pi} \ (\ = \ \pi^{\pi} \ )$$

(or again to $\pi^{\pi_0} \neq 2^{\pi} \ (= \pi^{\pi})$). I am going to give here a corrected statement. [page 116]

Proposition 4.5.6.

(1°) *Under the general hypotheses of 4.5.2 on C, π_0, π_1, π, the conditions a, a′, a″, b, c′ of this proposition are equivalent (provided $\pi \geqslant \pi_1$) ([39]), and imply condition c (F is π-accessible). When $\pi^{\pi_0} = \pi$ (in particular when $\pi = 2^c$, with $c \geqslant \pi_0$), then all the conditions a to c′ of 4.5.2 are equivalent.*

(2°) *More generally and more precisely, for $\pi \geqslant \pi_1$ given, let us consider the conditions on π:*

1) $\pi^{\pi_0} = \pi$.

2) $\mathrm{Filt}^{\pi}(\mathcal{M})$ (defined in 4.5.3) is stable under $\varinjlim$ of cardinality $\leqslant \pi$.

3) $\forall$ F in $\mathcal{M}$, we have $F \simeq \varinjlim_{I} X_i$, with I an ordered set large relative to π, the X_i in $\mathrm{Filt}^{\pi}(\mathcal{M})$. Moreover, if $\pi' > \pi$ and $F \in \mathrm{Filt}^{\pi'}(\mathcal{M})$, we can take I of cardinality $\leqslant \pi'^{\pi}$.

4) As in 3), without the bound on cardinalities, and assuming only I a category large relative to π (not necessarily an ordered category).

5) Condition c) of 4.5.2 implies the other conditions (they are equivalent, by 1°), in other words

$$\mathcal{M}_{\pi} \ \subset \ \mathrm{Filt}^{\pi} \qquad \textit{(so } \mathcal{M}_{\pi} = \mathrm{Filt}^{\pi} \textit{ , since we know that } \mathrm{Filt}^{\pi} \subset \mathcal{M}_{\pi}) \ ,$$

where $\mathcal{M}_{\pi}$ denotes the strictly full subcategory of $\mathcal{M}$ formed of the π-accessible objects. p. 34 [page 117]

This being so, we have the implications

$$1) \implies 2) \implies 3) \implies 4) \implies 5) \ ,$$

and in the particular case where $\mathcal{M} = (\mathrm{Ens})$, C = full subcategory formed of the sets of cardinal $\leqslant \pi_0$, we also have 5) $\implies$ 1), i.e. all these conditions are equivalent.

Proof.

1) $\implies$ 2). Cf. proof of 4.5.4.

2) $\implies$ 3). The category $J = \mathrm{Filt}^{\pi}/F$ is stable under $\varinjlim$ of cardinal $\leqslant \pi$, Filt^{π} being so, hence J is large relative to π, moreover $F = \varinjlim_{j \in J} X_j$, where $j \longmapsto X_j$ is the source functor $\mathrm{Filt}^{\pi}/F \longrightarrow \mathrm{Filt}^{\pi}$. Whence the desired representation of F, replacing J if need be by a cofinal ordered set I, e.g. by taking for I the set of subcategories J of I such that $\mathrm{card}\,\mathrm{Fl}\,J \leqslant \pi$. The bound $\mathrm{card}\,I \leqslant \pi'^{\pi}$, for $F \in \mathrm{Ob}\,\mathrm{Filt}^{\pi'}$, is obtained by AQT .

The implication 3) $\implies$ 4) is tautological.

4) $\implies$ 5) ([40]), for if $F \in \mathrm{Ob}\,\mathcal{M}_{\pi}$, we find by 4) that F is a direct factor of an object of Filt^{π}, and Filt^{π} being stable under direct factors (argument by argument), whence

[39] **N.B.** For this equivalence, the hypothesis that C be stable under $\varinjlim$ of cardinality $\leqslant \pi$ is unnecessary, and also for c′) to imply c), and to be equivalent to it if $\pi = \pi^{\pi_0} \geqslant \pi_1$, cf. 4.7.1, p. 161. [page 118]

[40] compare lemma 4.5.14, page 133.

$F \in \mathrm{Ob}\,\mathrm{Filt}^{\pi}$.

Finally, if $\mathcal{M} = \mathrm{Ens}$, $C = \mathrm{Ens}(\pi_0)$. We note that

$$\mathrm{Filt}^{\pi} \;=\; \{\; X \in \mathrm{Ob}\,\mathrm{Ens} \mid \mathrm{card}\,(X)^{\pi_0} \;\leqslant\; \pi \;\}\;.$$

On the other hand, we easily find

$$\begin{aligned}(\mathrm{Ens})_{\pi} \;&\overset{\mathrm{def}}{=}\; \text{full subcategory of (Ens) formed}\\ &\qquad\text{of the } \pi\text{-accessible objects}\\ &=\; \{X \in \mathrm{Ob}\,\mathrm{Ens} \mid \mathrm{card}\,X \leqslant \pi\}\;.\end{aligned}$$

Thus we have $(\mathrm{Ens})_{\pi} \subset \mathrm{Filt}^{\pi}$ if and only if $\pi^{\pi_0} \leqslant \pi$, i.e. $\pi^{\pi_0} = \pi$, q.e.d. ([41]).

Corollary 4.5.7. *For every cardinal $\pi \geqslant \pi_1$, we have*

$$\mathrm{Filt}^{\pi} \;\subset\; \mathcal{M}_{\pi} \;\subset\; \mathrm{Filt}^{\pi^{\pi_0}}\;. \tag{4.5.7.1}$$

The inclusion $\mathrm{Filt}^{\pi} \subset \mathcal{M}_{\pi}$ is the implication a $\Longrightarrow$ c of part 1°) of 4.5.6. To prove $\mathcal{M}_{\pi} \subset \mathrm{Filt}^{\pi^{\pi_0}}$, we put $\pi' = \pi^{\pi_0}$ and note that $\pi'^{\pi_0} = \pi^{\pi_0 \cdot \pi_0} = \pi^{\pi_0} = \pi'$. Thus we can apply 1° to π' instead of π, and we find that $\mathcal{M}_{\pi'} = \mathrm{Filt}^{\pi'}$. Since $\mathcal{M}_{\pi} \subset \mathcal{M}_{\pi'}$, we conclude $\mathcal{M}_{\pi} \subset \mathrm{Filt}^{\pi'}$, q.e.d. p. 35

I forgot to prove 1°) in 4.5.6. In view of the diagram of implications (∗), p. 108, and the implication a″ $\Longrightarrow$ a already proved (p. 109), it remains only to prove [page 119]

$$\mathrm{c}' \;\Longrightarrow\; \mathrm{a}''\;,$$

i.e. that F π-accessible *in* $C^{\wedge}$ implies $\mathrm{card}\,F(x) \leqslant \pi$ $\forall$ $x \in C$. This will follow from the somewhat more general result:

Lemma 4.5.8.

1°) In the category $\mathcal{E} = (\mathrm{Ens})$, the π-accessible objects (π an infinite cardinal) are the sets X such that $\mathrm{card}\,X \leqslant \pi$.

2°) ([42]). *Let C be a category equivalent to a small category, such that $\forall$ $x, y \in \mathrm{Ob}\,C$, we have* $\mathrm{card}\,\mathrm{Hom}(x, y) \leqslant \pi$. *Let $\mathcal{E}$ be a $\mathfrak{U}$-category stable under $\varinjlim$ large relative to π, and weakly π-accessible (i.e. admitting a generating subcategory formed of π-accessible objects), and stable under products of cardinal $\leqslant \pi$. Then for an object F of $\underline{\mathrm{Hom}}(C, \mathcal{E})$ to be π-accessible, it is* necessary *that $\forall$ $x \in \mathrm{Ob}\,C$, $F(x)$ be a π-accessible object of $\mathcal{E}$, and this condition is also sufficient if* $\mathrm{card}\,\mathrm{red}\,\mathrm{Fl}(C) \leqslant \pi$ ([43]).

Proof. 1°) $\mathrm{card}\,X \leqslant \pi \Longrightarrow X$ π-accessible in $\mathcal{E} = (\mathrm{Ens})$, for $X = \coprod_{x \in X}\{x\}$, and we apply SGA 4 I 9.9. Conversely, suppose X π-accessible, and write $X \simeq \varinjlim_{i \in I} X_i$, where I is the set of subsets of X which are of cardinal $\leqslant \pi$. Since I is large relative to π, the identity map $X \xrightarrow{\mathrm{id}} X$ factors through a set X_i, hence $X_i = X$, hence $\mathrm{card}\,X = \mathrm{card}\,X_i \leqslant \pi$, q.e.d.

2°) Consider, for $x \in \mathrm{Ob}\,C$, the functor [page 120]

$$e \xrightarrow{\varepsilon_x} C\;,$$

[41] Cf. proof of part 1°) of 4.5.6, below (bottom of page and following page).

[42] Cf. generalization 4.14.18, p. 391 b) and c).

[43] even *without* any condition of stability of $\mathcal{E}$ under products of cardinal $\leqslant \pi$.

whence a functor

$$\varepsilon_x^* \ : \ \underline{\mathrm{Hom}}(C, \mathcal{E}) \longrightarrow \mathcal{E} \ , \qquad F \longmapsto F(x) \ ,$$

admitting a right adjoint

$$\varepsilon_{x\,*} \ : \ \mathcal{E} \ \longrightarrow \ \underline{\mathrm{Hom}}(C, \mathcal{E}) \ ,$$

given by

$$\varepsilon_{x\,*}(E)(y) \ = \ E^{\mathrm{Hom}(y,x)}$$

(subject to the existence of products of cardinal $\leqslant \pi$). This functor $\varepsilon_{x\,*}$ is π-accessible by virtue of loc. cit. 9.8.

This being so, let F be a π-accessible object of $\underline{\mathrm{Hom}}(C, \mathcal{E})$, we see that $\varepsilon_x^*(F)$ is π-accessible, ε_x^* having a π-accessible right adjoint:

$$E \ \longmapsto \ \mathrm{Hom}_{\mathcal{E}}(\underbrace{F(x)}_{\varepsilon_x^*(F)}, E) \ \simeq \ \mathrm{Hom}_{\underline{\mathrm{Hom}}(C,\mathcal{E})}(F, \varepsilon_{x\,*}(E))$$

is the composite of the π-accessible functor $\varepsilon_{x\,*}$, and the π-accessible covariant functor represented by F in $\underline{\mathrm{Hom}}(C, \mathcal{E})$. p. 36

Conversely, if the $F(x)$ are π-accessible, let us prove that F is so, provided $\mathrm{card}\,\mathrm{Fl}(C) \leqslant \pi$. Done by AQT , I am not going to spell it out here.

Proposition 4.5.9. *Let C be a category equivalent to a small category, and let $\mathcal{M} = \mathrm{Ind}'_{\pi_0}(C)$, π_0 a given infinite cardinal. Assume C stable under $\varinjlim$ of cardinal $\leqslant \pi_0$. For every cardinal $\pi \geqslant \pi_0$, let $\mathcal{M}_\pi \supset C$ be the full subcategory of $\mathcal{M}$ formed of the π-accessible objects of $\mathcal{M}$. Let, on the other hand, $\pi_1 = \sup(\pi_0, \mathrm{card}\,\mathrm{Fl}\,C)$. Then $(\mathcal{M}_\pi)_{\pi \geqslant \pi_1}$ is a cardinal filtration of $\mathcal{M}$ ([44]), and moreover, for $\pi \geqslant \pi_0$, $\mathcal{M}_\pi$ is stable in $\mathcal{M}$ under $\varinjlim$ of cardinal $\leqslant \pi$.* [page 121]

Proof. The stability of $\mathcal{M}_\pi$ is a reminder, cf. SGA 4 I 9.9. Since we have

$$\mathcal{M}_\pi \ \subset \ \mathrm{Filt}^{\pi^{\pi_0}}$$

for $\pi \geqslant \pi_1$ (cf. 4.5.7), we therefore see that the $\mathcal{M}_\pi$ are equivalent to small categories (condition a) for cardinal filtrations). Condition b) for cardinal filtrations is contained in the stability property of the $\mathcal{M}_\pi$ which we have just recalled. Finally let us prove condition c). So let $\pi \geqslant \pi_0$, and let x be in $\mathcal{M}$, consider the category

$$I \ = \ \mathcal{M}_\pi / x \ ,$$

it is stable under $\varinjlim$ of cardinal $\leqslant \pi$, by the similar stability of $\mathcal{M}_\pi$, hence I is large relative to π and equivalent to a small category, on the other hand, we have [page 122]

$$x \ = \ \varinjlim_{i \in I} x_i \ ,$$

the x_i in $\mathcal{M}_\pi$ ($\varinjlim$ in $\mathcal{M}$ of the functors $I = \mathcal{M}_\pi/x \longrightarrow \mathcal{M}_\pi$). Replacing I if need be by a cofinal ordered category, this proves the qualitative part of c. Suppose finally $x \in \mathrm{Ob}\,\mathcal{M}_{\pi'}$, where $\pi' > \pi$, let us show that we then have

$$\mathrm{card}\,I_0, \ \mathrm{card}\,I' \ \leqslant \ \pi'^{\pi} \qquad ([45], [46]) \ , \tag{4.5.9.1}$$

which will finish proving that we have a cardinal filtration. We need a bound on the cardinals.

[44] **N.B.** All the conditions for cardinal filtrations are satisfied with the exception, when $\pi \geqslant \pi_0$, of that of the bound on cardinals $\mathrm{card}\,I \leqslant \pi'^{\pi}$ condition c).

[45] where I_0, I'_0 are skeletal subcategories of I, I' such that $I_0 \hookrightarrow I$, $I'_0 \hookrightarrow I'$ are equivalences .

[46] We only prove it if $\pi \geqslant \pi_1$.

Lemma 4.5.10 ([47]). *Let π be a cardinal $\geqslant \pi_1$. Then we have*

$$\operatorname{card} \operatorname{Fl}((\operatorname{Filt}^{\pi})_0) \leqslant 2^{\pi} . \tag{4.5.10.1}$$

Indeed we have, denoting by $C^{\wedge}(\pi)$ the strictly full subcategory of $C^{\wedge}$ formed of the $F : C^{\circ} \longrightarrow \operatorname{Ens}$ with $\operatorname{card} F(x) \leqslant \pi$ for every x in C (i.e. F factors through $\operatorname{Ens}(\pi)$), and by $C^{\wedge}(\pi)_0$ an exhaustive skeletal subcategory, p. 37
[page 123]

$$\operatorname{card} C^{\wedge}(\pi)_0 \leqslant 2^{\pi} , \tag{4.5.10.2}$$

whence (4.5.10.1), since for $\pi \geqslant \pi_1$, $\operatorname{Filt}^{\pi}$ is a full subcategory of $C^{\wedge}(\pi)$. To establish (4.5.10.2), we note that the set of isomorphism classes of $\operatorname{Ens}(\pi)$ is identified with the set of cardinals $\leqslant \pi$, which set is of cardinal $\leqslant \pi$. Replacing $\operatorname{Ens}(\pi)$ by $\operatorname{Ens}(\pi)_0$, C by C_0, we must bound $\operatorname{card} \operatorname{Fl}(\underline{\operatorname{Hom}}(C_0, \operatorname{Ens}(\pi)_0))$. Now $\operatorname{Ob} \underline{\operatorname{Hom}}(C_0, \operatorname{Ens}(\pi)_0)$ is contained in the set of maps from $\operatorname{Fl} C_0$ to $\operatorname{Fl}(\operatorname{Ens}(\pi)_0)$, the first is bounded by π_1, the second by $\pi^{\pi} \times \pi \times \pi = \pi^{2\pi} = \pi^{\pi} = 2^{\pi}$ (**N.B.** $\pi < 2^{\pi}$, whence $\pi^{\pi} \leqslant (2^{\pi})^{\pi} = 2^{\pi^2} = 2^{\pi}$), hence $\operatorname{Ob} \underline{\operatorname{Hom}}$ bounded by $(2^{\pi})^{\pi_1} = 2^{\pi\pi_1} = 2^{\pi}$. On the other hand, if F, G are in $\underline{\operatorname{Hom}}(C_0, \operatorname{Ens}(\pi))$, $\operatorname{Hom}(F, G)$ is bounded by $(\pi^{\pi})^{\pi_1} = \pi^{\pi} = 2^{\pi}$, hence $\operatorname{Fl} \underline{\operatorname{Hom}}$ is bounded by $2^{\pi} \times 2^{\pi} \times 2^{\pi}$, q.e.d.

Corollary 4.5.11. *Let $F \in \operatorname{Ob} \operatorname{Filt}^{\pi}$, $F' \in \operatorname{Ob} \operatorname{Filt}^{\pi'}$, with $\pi' \geqslant \pi \geqslant \pi_1$. Then*

$$\operatorname{card} \operatorname{Hom}(F, F') \leqslant \pi'^{\pi} . \tag{4.5.11.1}$$

It is essentially the same proof, slightly simpler. (We should have started there.) [page 124]

Corollary 4.5.12. *If $\pi' \geqslant \pi \geqslant \pi_1$, we have for $x \in \operatorname{Ob} \mathcal{M}_{\pi'}$*

$$\operatorname{card} (\mathcal{M}_{\pi}/x)_0 \leqslant \pi'^{(\pi^{\pi_0})} . \tag{4.5.12}$$

Indeed,

$$\operatorname{card} \operatorname{Ob} (\mathcal{M}_{\pi}/x)_0 \overset{\text{by 4.5.11}}{\leqslant} \underbrace{\underbrace{\operatorname{card} \operatorname{Ob} (\mathcal{M}_{\pi})_0}_{\leqslant 2^{(\pi^{\pi_0})} \text{ by 4.5.10}} \times (\pi'^{\pi_0})^{\pi^{\pi_0}}}_{= \pi'^{(\pi^{\pi_0})}} ,$$

i.e.

$$\operatorname{card} \operatorname{Ob} (\mathcal{M}_{\pi}/x)_0 \leqslant \pi'^{(\pi^{\pi_0})} ,$$

whence

$$\operatorname{card} \operatorname{Fl}(\mathcal{M}_{\pi}/x)_0 \leqslant \pi'^{(\pi^{\pi_0})} \times \pi'^{(\pi^{\pi_0})} \times (\pi'^{\pi_0})^{(\pi^{\pi_0})} \leqslant \pi'^{(\pi^{\pi_0})} .$$

This therefore gives (returning to 4.5.9.1)

$$\operatorname{card} I_0 \leqslant \pi'^{(\pi^{\pi_0})} .$$

On the other hand, I' can be defined in terms of I_0 as the set of subcategories of I_0 which are of cardinal $\leqslant \pi$, hence their number is bounded by $(\operatorname{card} I_0)^{\pi} \leqslant (\pi'^{\pi^{\pi_0}})^{\pi} = \pi'^{\pi\pi^{\pi_0}} = \pi'^{(\pi^{\pi_0})}$.

But this is not quite the required bound, since

$$\pi'^{\pi} \leqslant \pi'^{(\pi^{\pi_0})} \qquad ([48]),$$

and since it is possible, a priori, that the inequality is strict! Apparently, the "bounds" p. 38
[page 125]

[47] Give a common statement with 4.5.11, and first prove 4.5.11.

[48] **N.B.** We have equality if we have $\pi' = 2^{c'}$ with $c' \geqslant \pi^{\pi_0}$, or $\pi = 2^{c}$ with $c \geqslant \pi_0$.

$$\mathcal{M}_\pi \ \subset\ \mathrm{Filt}^{\pi^{\pi_0}}\ , \qquad \mathcal{M}_{\pi'} \ \subset\ \mathrm{Filt}^{\pi'^{\pi_0}}$$

are too crude. So it is a matter of getting a better grasp of the categories $\mathcal{M}_\pi$, for $\pi \geqslant \pi_0$ ([49]). I am going to venture a

Proposition 4.5.13. *The hypotheses and notations are those of 4.5.9. Let π be a cardinal $\geqslant \pi_0$, and let $x \in \mathrm{Ob}\,\mathcal{M}$. Consider the conditions:*

a) There exists a category I, with $\mathrm{card}\,\mathrm{Fl}\,I \leqslant \pi$, *and a functor $i \longmapsto x_i : I \longrightarrow C$, such that x is isomorphic to $\varinjlim x_i$.*

a′) As in a), with I filtering.

a″) As in a), with I ordered and filtering.

b) There exists a cofinal *functor $I \longrightarrow C/x$, with* $\mathrm{card}\,\mathrm{Fl}\,I \leqslant \pi$.

b′) The filtering category C/x is "essentially of cardinal $\leqslant \pi$", i.e. there exists an ordered category filtering I, $\mathrm{card}\,\mathrm{Fl}\,I \leqslant \pi$, *and a cofinal functor $I \longrightarrow C/x$.*

b″) As in b′), with I moreover ordered.

c) There exists a strict epimorphic family of arrows $(u_\alpha)_{\alpha\in A}$, $x_\alpha \longrightarrow x$, with $x_\alpha \in \mathrm{Ob}\,C$ for every $\alpha \in A$, indexed by a set A such that $\mathrm{card}\,A \leqslant \pi$.

d) We have $x \in \mathrm{Ob}\,\mathcal{M}_\pi$, i.e. x is π-accessible in $\mathcal{M}$.

This being so: [page 126]

(1°) *The first six conditions a, a′, a″, b, b′, b″ are equivalent, say $P(\pi)$, and imply the last two,*

$$\begin{array}{ccc} & P(\pi) & \\ \swarrow\!\!\swarrow & & \searrow\!\!\searrow \\ c & & d\ . \end{array}$$

If $\pi \geqslant \pi_1$, then $P(\pi)$ is equivalent to c.

(2°) *Let* $\mathrm{Filt}'^{\pi} \subset \mathcal{M}$ *be the strictly full subcategory of $\mathcal{M}$ formed of the objects which satisfy condition $P(\pi)$ above, and* Filt^{π} *the strictly full subcategory defined in 4.5.3 (p. 109). We then have*

$$\mathrm{Filt}^{\pi} \ \subset\ \mathrm{Filt}'^{\pi} \ \subset\ \mathcal{M}_\pi\ , \tag{4.5.13.1}$$

and for there to be equality of the three terms, it suffices that we have $\pi^{\pi_0} = \pi$. (Cf. prop. 4.5.6, page 116). Moreover, Filt'^{π} *is stable in $\mathcal{M}$ under* sums *of cardinal $\leqslant \pi$.*

(3°) *The following conditions on C, π_0, π are equivalent.*

p. 39

1) Filt'^{π} *is stable in $\mathcal{M}$ under $\varinjlim$ of cardinality $\leqslant \pi$.*

[49] But we can use 4.5.10, 4.5.11 to conclude that for $x \in \mathrm{Ob}\,\mathcal{M}_{\pi'} \subset \mathrm{Ob}\,\mathrm{Filt}^{\pi'^{\pi_0}}$, we have $\mathrm{card}\,\mathrm{Fl}(C_0/x) \leqslant (\pi'^{\pi_0})^{\pi_0} = \pi'^{\pi_0}$, we will therefore have $x = \varinjlim_J x_j$ ($x_j \in \mathrm{Ob}\,C$, $\varinjlim$ filtering, $\mathrm{card}\,\mathrm{Fl}\,J \leqslant \pi'^{\pi_0}$), and introducing the set I of subcategories $J' \subset J$ which are of cardinal $\leqslant \pi$, and putting $x_{J'} = \varinjlim_{j\in J'} x_j$, we find $x = \varinjlim_{J'\in I} x_{J'}$, where this time I is an *ordered set* large relative to π, the $x_{J'} \in \mathcal{M}_\pi$, and $\mathrm{card}\,I \leqslant (\mathrm{card}\,J)^{\pi} \leqslant (\pi'^{\pi_0})^{\pi} = \pi'^{\pi\pi_0} = \pi'^{\pi}$, this is the right bound!

1′) Filt'^{π} *is stable under cokernels of double arrows.*

2) Every object x *of* $\mathcal{M}$ *is of the form* $\varinjlim_I x_i$*, where* x_i *in* Filt'^{π}*,* I *large relative to* π*.*

3) Condition d) of 1°) implies $P(\pi)$ *(hence is equivalent to it), i.e. we have*

$$\mathcal{M}_\pi \;=\; \mathrm{Filt}'^{\pi}\,.$$

These conditions are satisfied in the following three cases: [page 127]

A) $\pi^{\pi_0} = \pi$ *(as a reminder, this is contained in 2°).*

B) The strict epimorphic families in $\mathcal{M}$ *are strict universal epimorphic families.*

C) Filtering direct limits in $\mathcal{M}$ *commute with base change (which is the case in particular if the filtering* $\varinjlim$ *are left exact)* ([50]).

Remark. The filtration of $\mathcal{M}$ by the Filt'^{π} is none other than the one considered in SGA 4 I 9.13, where one makes in addition assumption B) on $\mathcal{M}$, which afterwards was no longer so much to my taste. At any rate, according to (4.5.13.1), this filtration comes closer to the "good" filtration by the $\mathcal{M}_\pi$, than the one I had still been inclined not so long ago to prefer to it, the one by the Filt^{π}. It has moreover over the latter the advantage, in the case B) considered, of being a cardinal filtration (which is not necessarily the case for Filt^{π}, at least if there exist infinite cardinals $\pi \geqslant \pi_0$ with $\pi^{\pi_0} \neq \pi$). I now suspect that in general (and if there exist infinite cardinals $\pi > \pi_0$ such that $\pi^{\pi_0} \neq \pi$), outside conditions A) B) C), the filtration by the (Filt'^{π}) is *not* a cardinal filtration. In any case, [page 128] 4.5.6 and the present proposition, 3°, show that if either $(\mathrm{Filt}^{\pi})_{\pi\geqslant\pi_2}$, or $(\mathrm{Filt}'^{\pi})_{\pi\geqslant\pi_2}$ is a cardinal filtration (π_2 being a given cardinal $\geqslant \pi_1$), then it coincides with the filtration by the $\mathcal{M}_\pi$. So apparently, in all the cases where I know how to construct a cardinal filtration, on a paratopos at least (i.e. on a category $\mathcal{M}$ stable under small $\varinjlim$, and all of whose objects are accessible – the existence of a cardinal filtration implying moreover that of a small strictly generating subcategory …), this is none other than the one by the $\mathcal{M}_\pi$ – something I had not noticed when writing SGA 4 I 9.

Proof of 4.5.13.

Let us prove 1°. We have the tautological implications

$$\begin{array}{ccccc} \mathrm{b}'' & \Longrightarrow & \mathrm{b}' & \Longrightarrow & \mathrm{b} \\ \Downarrow & & \Downarrow & & \Downarrow \\ \mathrm{a}'' & \Longrightarrow & \mathrm{a}' & \Longrightarrow & \mathrm{a}\,, \end{array} \qquad ([51])\,,$$

and to prove the equivalence of these six conditions, it suffices to prove

$$\mathrm{a} \;\Longrightarrow\; \mathrm{b}''\,.$$

If we have [p. 40] [page 129]

$$x \;\simeq\; \varinjlim_I x_i\,, \qquad \mathrm{card}\,\mathrm{Fl}\,I \;\leqslant\; \pi\,,$$

[50] I suppose that this condition amounts to the same condition on C, for the filtering $\varinjlim$ of cardinality $\leqslant \pi$.

[51] taking into account that C is strictly generating in $\mathcal{M}$, hence $x \simeq \varinjlim_{i\in C/x} x_i$.

let I' be the set of finite *subdiagrams* $J \subset I$ of I, ordered by inclusion, so I' is a filtering ordered set, and from the fact that C is stable under finite $\varinjlim$, we find a functor

$$J \longmapsto x_J \;=\; \varinjlim_{i \in J} x_i \;:\; I' \longrightarrow C \;,$$

and we will have

$$x \;\simeq\; \varinjlim_{J \in I'} x_J \;.$$

On the other hand, we have

$$\begin{aligned} \operatorname{card} I' \;&=\; \operatorname{card} \mathfrak{P}_f(\; \underbrace{\operatorname{Fl}(I)}_{\text{of card} \leqslant \pi} \;) \\ &\leqslant\; \pi + \pi^2 + \;\cdots\; + \pi^n + \;\ldots \\ &\leqslant\; \pi \aleph_0 \;=\; \pi \;, \end{aligned}$$

q.e.d. To establish 1°, it remains to establish

$$\begin{array}{ccc} & P(\pi) & \\ \swarrow\!\!\swarrow & & \searrow\!\!\searrow \\ \mathrm{c} & & \mathrm{d} \;, \end{array}$$

and the reverse implication $\mathrm{c} \Longrightarrow P(\pi)$ when $\pi \geqslant \pi_1$. But $P(\pi) \Longrightarrow \mathrm{c}$ is a tautology, and $P(\pi) \Longrightarrow \mathrm{d}$ since $\mathcal{M}_\pi$ is stable under $\varinjlim$ of cardinality $\leqslant \pi$. It remains to prove that $\mathrm{c} \Longrightarrow P(\pi)$ if $\pi \geqslant \pi_1$.

Suppose then that we have a strict epimorphic family [page 130]

$$x_\alpha \longrightarrow x \qquad (\alpha \in A,\ \operatorname{card} A \leqslant \pi) \;,$$

this therefore means that x is the direct limit of a direct system indexed by the category B having as set of objects $A \amalg (A \times A)$, the nonidentity arrows being represented by the canonical arrows

$$\begin{array}{ccc} & x_\alpha \times_x x_\beta & \\ {\scriptstyle p_{\alpha,\beta}} \swarrow & & \searrow {\scriptstyle q_{\alpha,\beta}} \\ x_\alpha & & x_\beta \;. \end{array}$$

(**N.B.** B is not an ordered category, for for the object (α, α), represented by $x_\alpha \times_x x_\alpha$, there are *two* arrows into the object α, represented by x_α. But we obviously have $\operatorname{card} \operatorname{Fl}(B) \leqslant \pi$.) If we knew that C is stable under binary products and under passage to subobjects (here the $x_\alpha \times_x x_\beta \;\subset\; x_\alpha \times x_\beta$), we would have proved condition a). Failing that, let us again use the fact that C is generating by epimorphisms (even strict ones) and choose for every pair (α, β) an epimorphic set of arrows:

$$\underbrace{x_{\alpha,\beta,\gamma}}_{\in \operatorname{Ob} C_0} \xrightarrow{u_{\alpha,\beta,\gamma}} x_\alpha \times_x x_\beta \;,$$

where C_0 is a reduced exhaustive subcategory of C. The arrow $u_{\alpha,\beta,\gamma}$ is known when we know the composites p. 41 [page 131]

$$\begin{array}{ccc} & & x_\alpha \\ & {\scriptstyle v_{\alpha,\beta,\gamma}} \nearrow & \\ x_{\alpha,\beta,\gamma} & & \\ & {\scriptstyle w_{\alpha,\beta,\gamma}} \searrow & \\ & & x_\beta \;, \end{array}$$

so for $x_{\alpha,\beta,\gamma} \in \mathrm{Ob}\, C_0$ fixed, the possible choices for $u_{\alpha,\beta,\gamma}$ have cardinality bounded by $\pi_1\pi_1 = \pi_1$, moreover we have $\mathrm{card}\,\mathrm{Ob}\, C_0 \leqslant \pi_1$, so the total number of arrows $v_{\alpha,\beta,\gamma}$, $w_{\alpha,\beta,\gamma}$ (for α, β fixed) is bounded by $\pi_1 \cdot \pi_1 = \pi_1$. So the total number of arrows $u_{\alpha,\beta,\gamma}$, for α, β, γ variable, is bounded by $\pi \cdot \pi_1 = \pi$ (if $\pi \geqslant \pi_1$). So x is represented by a category I having as objects the disjoint union of A and the set of triple indices α, β, γ, and for each such index, exactly two nonidentity arrows $v_{\alpha,\beta,\gamma}$, $w_{\alpha,\beta,\gamma}$ with this object as source, and going respectively into x_α, x_β. We thus have $\mathrm{card}\,\mathrm{Fl}\, I \leqslant \pi$, and we have a functor $I \longrightarrow C$, $\alpha \longmapsto x_\alpha$, $(\alpha,\beta,\gamma) \longmapsto x_{\alpha,\beta,\gamma}$, such that $x = \varinjlim_{i\in I} x_i$. This completes the proof of 1°).

Let us prove 2°. The first inclusion (4.5.13.1) is tautological, the second merely expresses $P(\pi) \Longrightarrow \mathrm{d}$. The fact that $\pi^{\pi_0} = \pi$ implies equality of the extreme terms is a reminder of 4.5.6. It remains to prove that Filt'^{π} is stable in $\mathcal{M}$ under sums of cardinality $\leqslant \pi$. But if $(x_\alpha)_{\alpha\in A}$ is a family of objects of Filt'^{π}, with $\mathrm{card}\, A \leqslant \pi$, and if $x = \coprod_{\alpha\in A} x_\alpha$, taking for every x_α a functor $\varphi_\alpha : I_\alpha \longrightarrow C$, with $\mathrm{card}\,\mathrm{Fl}\, I_\alpha \leqslant \pi$, such that x_α is the $\varinjlim$ in $\mathcal{M}$ of this functor, we find that x is the $\varinjlim$ of the functor $\varphi : I \longrightarrow C$ defined by the φ_α, where $I = \coprod_\alpha I_\alpha$ (sum in Cat). It is obvious that $\mathrm{card}\,\mathrm{Fl}(I) \leqslant \pi \cdot \pi = \pi$, which completes the proof of 2°. [page 132]

Let us prove 3°. We have the tautological implications $1 \Longrightarrow 1'$, and $1' \Longrightarrow 1$, since Filt'^{π} is stable under sums of cardinality $\leqslant \pi$. Thus $1 \Longleftrightarrow 1'$, and implies that for every x in $\mathcal{M}$, $I = \mathrm{Filt}'^{\pi}/x$ is stable under $\varinjlim$ of cardinality $\leqslant \pi$, hence is large relative to π. Since $x \simeq \varinjlim_{i\in I} x_i$, this proves that $1 \Longrightarrow 2$. The implication $2 \Longrightarrow 3$ is standard, and follows from $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_\pi$, I am going to spell out in a [page 133]

Lemma 4.5.14. *Let $\mathcal{M}$ be a category stable under small $\varinjlim$, π an infinite cardinal, $\mathcal{M}_0$ a strictly full subcategory of $\mathcal{M}$, stable under direct factors. Consider the conditions*

a) $\mathcal{M}_0$ is a strictly generating subcategory, and stable in $\mathcal{M}$ under $\varinjlim$ of cardinality $\leqslant \pi$.

b) For every object of $\mathcal{M}$, we have an isomorphism

$$x \simeq \varinjlim_{i\in I} x_i\ ,$$

the x_i in $\mathcal{M}_0$, I large relative to π.

p. 42

b′) As in b), with I a category that is ordered.

c) $\mathcal{M}_\pi \subset \mathcal{M}_0$ (where $\mathcal{M}_\pi$ is the strictly full subcategory of $\mathcal{M}$ consisting of the π-accessible objects).

We thus have the implications

$$a \Longrightarrow b \Longleftrightarrow b' \Longrightarrow c\ ,$$

and if we assume $\mathcal{M}_0 \subset \mathcal{M}_\pi$, and $\mathcal{M}_\pi$ a strictly generating subcategory of $\mathcal{M}$, then the four conditions a, b, b′, c are equivalent, and equivalent to $\mathcal{M}_0 = \mathcal{M}_\pi$ ([52]).

Finally (returning to the proof of 4.5.13, 3°), the implication 3) $\Longrightarrow$ 1) comes from the fact that $\mathcal{M}_\pi$ is stable under $\varinjlim$ of cardinality $\leqslant \pi$. [page 134]

It remains to prove that these conditions, i.e. the stability of Filt'^{π} under cokernels of double arrows, is satisfied in cases A) B) C), case A) being moreover already known. Case B) follows immediately from the transitivity of strict universal epimorphic families,

[52] cf. proof of proposition 4.5.6, 4 $\Longrightarrow$ 5, page 117.

and this is moreover also contained explicitly (stability of Filt'^{π} under $\varinjlim$ of cardinality $\leqslant \pi$) in SGA 4 I 9.13 (if not in the statement, at least it is pointed out in the proof). When writing SGA 4 I 9, I therefore knew that we have $\mathcal{M}_\pi \subset \mathrm{Filt}'^{\pi}$ for every π, but the inclusion, more obvious however in the case at hand, $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_\pi$ (when we assume the objects of C π_1-accessible) had apparently escaped me ?

It remains to deal with case C). (I have the impression that it is a little more general than case B), i.e. that condition B) on $\mathcal{M}$ implies C), but not conversely. Thus, condition C) is verified as soon as $\mathcal{M}$ corresponds to a species of algebraic structure defined in terms [page 135] of $\varprojlim$ *which are finite*, i.e. when $\mathcal{M}$ is of the form $\mathrm{Ind}(C)$, C stable under $\varinjlim$ *which are finite*, whereas I doubt that condition B) is always satisfied for these categories. For example in the category Cat (which is a paratopos of this species), the filtering $\varinjlim$ are left exact, but I do not believe that the strict epimorphic families are universal. Let us take e.g. two group homomorphisms $G_0 \xrightarrow{u_0} G$, $G_1 \xrightarrow{u_1} G$ such that $G_0 * G_1 \longrightarrow G$ is surjective, then this two-element family is strictly epimorphic in Gr and in Cat ([53]), but is not necessarily universally strictly epimorphic, even if e.g. $G_0 = G_1 = H$, $G = H \times H$, and u_0, u_1 are injections $H \rightrightarrows H \times H$, given by $u_0(h) = (h, 1)$, and $u_1(h) = (1, h)$. If I take the diagonal map $H \longrightarrow H \times H$, the inverse images of u_0, u_1 are the same object of $\mathcal{M}/H$, namely $e \longrightarrow H$, and this family is strictly epimorphic only if H is the trivial group.)

Case C) will follow from the more precise result:

Corollary 4.5.15. *Under the general conditions of 4.5.13, suppose that in $\mathcal{M}$ the* $\varinjlim$ that are filtering *commute with base change (e.g. the filtering* $\varinjlim$ *are exact). Then* [page 136] *for every $\pi \geqslant \pi_1$, Filt'^{π} consists of the $x \in \mathrm{Ob}\,\mathcal{M}$ satisfying the following condition.*

p. 43

c′) There exists a family of arrows $(u_\alpha)_{\alpha\in A}$, $u_\alpha : x_\alpha \longrightarrow x$, in $\mathcal{M}$, which is universally strictly epimorphic *(i.e. covering for the canonical topology of $\mathcal{M}$), with $x_\alpha \in \mathrm{Ob}\,C$ $\forall\ \alpha \in A$, and* $\mathrm{card}\,A \leqslant \pi$.

Proof : we must prove that this condition is equivalent to condition c) of 4.5.13 1°). Since we trivially have c′ $\Longrightarrow$ c, it remains to show that c $\Longrightarrow$ c′. But we saw in 4.5.13 1°) that c implies a″, i.e. that we have an isomorphism

$$x \;=\; \varinjlim_{i\in I} x_i \ ,$$

the x_i in C, and I a *filtering ordered* set, with $\mathrm{card}\,I \leqslant \pi$. It remains to see that if x in $\mathcal{M}$ is a *filtering* direct limit of objects x_i, then the set of arrows $x_i \longrightarrow x$ is strictly epimorphic *universally.* Now this is immediate, for for $x' \longrightarrow x$, we have $x' = \varinjlim_i x'_i$ (by assumption C), where $x'_i = x' \times_x x_i$, so the family $x'_i \longrightarrow x'$ is strictly epimorphic, q.e.d.

Let us now prove that in case C, the condition of 3°, 1 is satisfied, i.e. that Filt'^{π} is stable under $\varinjlim$ of cardinality $\leqslant \pi$. This follows at once from the [page 137]

Corollary 4.5.16. (of 4.5.15). *Under the conditions of 4.5.15 (i.e. condition C) of 4.5.13, 3°), let x be an object of $\mathcal{M}$, and suppose that there exists a family* strictly epimorphic universally *$(x_\alpha \longrightarrow x)_{\alpha\in A}$ of arrows of $\mathcal{M}$, with $x_\alpha \in \mathrm{Ob}\,\mathrm{Filt}'^{\pi}$ for every $\alpha \in A$, and* $\mathrm{card}\,A \leqslant \pi$. *Then $x \in \mathrm{Ob}\,\mathrm{Filt}'^{\pi}$.*

Indeed, the assumption on x_α is equivalent, by virtue of 4.5.15, to the existence of a strict universal epimorphic family of arrows $(x_{\alpha\beta} \longrightarrow x_\alpha)_{\beta\in B_\alpha}$, with $\mathrm{card}\,B_\alpha \leqslant \pi$ and the $x_{\alpha\beta}$ in C. Let $B = \coprod_{\alpha\in A} B_\alpha$, we then have $\mathrm{card}\,B \leqslant \pi^2 = \pi$, and on the other hand,

[53]!, this is *false* in general, and also false in the particular case below for nontrivial H.

the family of composite arrows $x_{\alpha\beta} \longrightarrow x_\alpha \longrightarrow x$ indexed by B is universally strictly epimorphic (by transitivity of this type of family), hence $x \in \mathrm{Ob}\,\mathrm{Filt}'^{\pi}$, q.e.d.

With all that, we have still not given a characterization of $\mathcal{M}_\pi$ itself, which is better than Filt^π and Filt'^{π} from the categorical point of view, for only the $\mathcal{M}_\pi$ define a cardinal filtration, and moreover they have the desirable stability properties in this context, for [page 138] $\varinjlim$ of cardinality $\leqslant \pi$. The only nontautological criterion I manage to extract in the general case (for every $\pi \geqslant \pi_0$, and without any additional exactness assumption on $\mathcal{M}$) is the following, much less "explicit" or "computational" than the very concrete criteria for Filt^π and for Filt'^{π}:

Proposition 4.5.17. *Let π be a cardinal $\geqslant \pi_0$. Then $\mathcal{M}_\pi$ is equal to the smallest strictly full subcategory of $\mathcal{M}$, containing C, and stable under $\varinjlim$ of cardinality $\leqslant \pi$.* (It is important to note that it *is not enough* to say: stable under $\varinjlim_I$, with I *large relative to* π_0, and $\mathrm{card}\, I \leqslant \pi$. We then find Filt^π, which risks being a strictly smaller category than $\mathcal{M}_\pi$. It is also the category of the $\varinjlim_I x_i$, with $x_i \in C\ \forall\ i \in \mathrm{Ob}\, I$, and $\mathrm{card}\,\mathrm{Fl}\, I \leqslant \pi$ – since that category is Filt'^{π}.)

Proof. Let $\mathcal{M}'$ be the category of the statement, we thus have $\mathcal{M}' \subset \mathcal{M}_\pi$, since $\mathcal{M}_\pi$ contains C and is stable under the $\varinjlim$ of cardinality $\leqslant \pi$. Conversely, lemma 4.5.14, a $\Longrightarrow$ c.

p. 44

[page 139]

4.6 Characterization of paratopos categories (proof of 4.3.6)

It is time to return to the proof of Theorem 4.3.6 (p. 70-71), the one I had given being incomplete (and erroneous – I asserted that $(\mathrm{Filt}^\pi)_{\pi\geqslant\pi_0}$ is a cardinal filtration, which is false if one assumes only $\pi^{\pi_0} = \pi$ for every $\pi \geqslant \pi_1$).

We had seen the equivalence of conditions a, a′, b, namely PTA. I am going to finish the proof by the logical scheme

$$\mathrm{PTA} \implies \mathrm{d}) \implies \mathrm{c}) \implies \mathrm{PTA}\ . \qquad (*)$$

Let us prove PTA $\Longrightarrow$ d), i.e. that if $\mathcal{M}$ is an accessible pseudo-topos, then there exists an infinite cardinal π_0, and a small category C_0 stable under $\varinjlim$ of cardinality $\leqslant \pi_0$, and an equivalence of categories

$$\mathcal{M} \xrightarrow{\sim} \mathrm{Ind}'_{\pi_0}(C_0)\ . \qquad (4.6.1)$$

More precisely, we have this:

Proposition 4.6.1. *Let $\mathcal{M}$ be a paratopos, i.e. a $\mathfrak{U}$-category $\mathcal{M}$ stable under small $\varinjlim$, admitting a full generating subcategory S equivalent to a small category, consisting of accessible objects of $\mathcal{M}$. Let π_0 be a cardinal such that the objects of S are π_0-accessible (i.e. $S \subset \mathcal{M}_{\pi_0}$), and let C be the smallest strictly full subcategory of $\mathcal{M}$, containing S and stable under $\varinjlim$ of cardinality $\leqslant \pi_0$. Then*

1°) C is equivalent to a small category.

2°) $C = \mathcal{M}_{\pi_0}$, strictly full subcategory of $\mathcal{M}$ consisting of the π_0-accessible objects. [page 140]

3°) The canonical functor

$$\mathrm{Ind}'_{\pi_0}(C) \underset{\sim}{\to} \mathcal{M}$$

(defined up to unique isomorphism by the condition that its composite with the inclusion $C \hookrightarrow \mathrm{Ind}'_{\pi_0}(C)$ be isomorphic to the inclusion $C \longrightarrow \mathcal{M}$, and that φ

commute with small $\varinjlim$ *, or again with filtering* $\varinjlim$ *large relative to* π_0 *– cf. 4.4.16, page 101) is an equivalence of categories* ([54]).

PROOF. 1°) We are going to construct C by transfinite induction, setting

$$\begin{aligned} C_0 &= \text{strictly full subcategory of } \mathcal{M} \text{ generated by } S. \\ C_{\alpha+1} &= \text{strictly full subcategory of } \mathcal{M} \text{ generated by the } \varinjlim_I \text{ in} \\ &\quad \mathcal{M} \text{ of objects of } C_\alpha, \text{ with } I \text{ a category such that } \operatorname{card} \operatorname{Fl} I \leqslant \pi_0. \\ C_\alpha &= \textstyle\bigcup_{\alpha'<\alpha} C_{\alpha'} \quad \text{if } \alpha \text{ is a limit ordinal.} \end{aligned}$$

This uniquely defines the C_α (for every small ordinal α). By the definition of C, we see by transfinite induction that the C_α are contained in C. Let α_0 be the first ordinal such that $\operatorname{card} \alpha_0 > \pi_0$, so the set I_{α_0} of ordinals $\alpha < \alpha_0$ is large relative to π_0. I say that we have $C_{\alpha_0} = C$. For this, it suffices to see that C_{α_0} is stable under $\varinjlim$ of cardinality $\leqslant \pi_0$. But if I is a category such that $\operatorname{card} \operatorname{Fl} I \leqslant \pi_0$, and $\varphi : I \longrightarrow C_{\alpha_0} = \bigcup_{\alpha<\alpha_0} C_\alpha$ a functor, then the definition of α_0 implies that φ factors as $\varphi_\alpha : I \longrightarrow C_\alpha$, with $\alpha < \alpha_0$ ([55]). Thus $\varinjlim \varphi \in \operatorname{Ob} C_{\alpha+1} \subset \operatorname{Ob} C_{\alpha_0}$, which shows the required stability. p. 45 [page 141]

To see that C is equivalent to a small category, it suffices to see this for the C_α, proceeding by transfinite induction. This follows from the following lemma, immediate:

Lemma 4.6.1.1.

1°) The direct limit C of a filtering direct system $(C_i)_{i\in I}$ of categories C_i equivalent to small categories is equivalent to a small category. If $\operatorname{card} \operatorname{red}(C_i) \leqslant \pi$ *for every i, and* $\operatorname{card} I \leqslant \pi$*, then* $\operatorname{card} \operatorname{red}(C) \leqslant \pi$ *(π being an infinite cardinal). Here, for any category $\mathcal{A}$, I put* $\operatorname{card} \mathcal{A} = \operatorname{card} \operatorname{Fl} \mathcal{A}$*,* $\operatorname{card} \operatorname{red}(\mathcal{A}) = \operatorname{card}(\mathcal{A}_0)$*, where $\mathcal{A}_0$ is a reduced category equivalent to $\mathcal{A}$.* [page 142]

2°) Let $C_0 \subset \mathcal{M}$ be a full subcategory of a category $\mathcal{M}$, π_0 an infinite cardinal, and let C_0' be the strictly full subcategory of $\mathcal{M}$ generated by the direct limits in $\mathcal{M}$ (representable in $\mathcal{M}$) of functors $I \longrightarrow C_0$, with I a category such that $\operatorname{card} I \leqslant \pi_0$*. If C_0 is equivalent to a small category, the same holds for C_0'. More precisely, if* $\operatorname{card} \operatorname{red}(C_0) \leqslant \pi_1$*, π_1 an infinite cardinal, then* $\operatorname{card} \operatorname{red}(C_0') \leqslant \pi_1^{\pi_0}$ ([56]).

Corollary 4.6.1.2. *$\mathcal{M}$ and C_0 being as in 2°) above, let C be the smallest strictly full subcategory of $\mathcal{M}$ containing C_0, and stable under* $\varinjlim$ *of cardinality $\leqslant \pi_0$ (those which are representable in $\mathcal{M}$). If C_0 is equivalent to a small category, the same holds for C. More precisely, if* $\operatorname{card} \operatorname{red}(C_0) \leqslant \pi_1$*, where π_1 is an infinite cardinal, then we have* $\operatorname{card} \operatorname{red}(C) \leqslant \pi_1^{\pi_0}$ ([57]). [page 143]

3°) is proved by the preceding transfinite proof, we prove by transfinite induction that

$$\operatorname{card} \operatorname{red} C_\alpha \ \leqslant\ \pi_1^{\pi_0}$$

for every α. This is OK for $\alpha = 0$ by hypothesis (since $\pi_1 \leqslant \pi_1^{\pi_0}$). If it is true for C_α, then 2°) ensures that

$$\operatorname{card} \operatorname{red} C_{\alpha+1} \ \leqslant\ (\pi_1^{\pi_0})^{\pi_0} \ =\ \pi_1^{\pi_0 \cdot \pi_0} \ =\ \pi_1^{\pi_0}\ ,$$

[54] thus we also have $\operatorname{Ind}'_{\pi_0}(C_0) \xrightarrow{\sim} \mathcal{M}$, where C_0 is a *small* category equivalent to C.

[55] A bit short. One must invoke the fact that I_{α_0} is large relative to π_0.

[56] This bound is not proved and probably false if one does not make somewhat stricter assumptions.

[57] bound not proved, probably false, cf. annotation above.

finally, if α is a limit ordinal $< \alpha_0$, we will have $C_\alpha = \bigcup_{\alpha'<\alpha} C_{\alpha'}$, hence

$$\operatorname{card}\operatorname{red} C_\alpha \leqslant \pi_0 \cdot \pi_1^{\pi_0} = \pi_1^{\pi_0}$$

(since $\pi_1^{\pi_0} \geqslant \pi_0$), by virtue of 1° above, and taking into account the fact that $\operatorname{card} I_\alpha \leqslant \pi_0$. Let π_0' be the successor cardinal of π_0, then the same bound gives us

$$\operatorname{card}\operatorname{red} C_{\alpha_0} \leqslant \pi_0' \pi_1^{\pi_0} .$$

But we have $\pi_1^{\pi_0} \geqslant 2^{\pi_0} > \pi_0$, hence $\pi_1^{\pi_0} \geqslant \pi_0'$, hence $\pi_1^{\pi_0} \geqslant \pi_0'$, hence $\pi_0'\pi_1^{\pi_0} = \pi_1^{\pi_0}$, whence p. 46

$$\operatorname{card}\operatorname{red} C_{\alpha_0} \leqslant \pi_1^{\pi_0} ,$$

which is the announced bound.

Let us prove 4.6.1 2°, i.e. $C = \mathcal{M}_{\pi_0}$. We have $C \subset \mathcal{M}_{\pi_0}$, since by hypothesis $S \subset \mathcal{M}_{\pi_0}$, and we know that $\mathcal{M}_{\pi_0}$ is stable under $\varinjlim$ of cardinality $\leqslant \pi_0$. On the other hand, we have $\mathcal{M}_{\pi_0} \subset C$, by virtue of 4.5.14 a $\Longrightarrow$ c applied to C, which is strictly generating in $\mathcal{M}$, S being so, and $C \supset S$, and stable under $\varinjlim$ of cardinality $\leqslant \pi_0$ by construction. [page 144]

Let us finally prove 3°. This is equivalent to the following:

Lemma 4.6.1.3. *Let $\mathcal{M}$ be a $\mathfrak{U}$-category stable under* small $\varinjlim$ *, π_0 a cardinal, $\mathcal{M}_0$ a subcategory of $\mathcal{M}$* strictly generating, *stable under $\varinjlim$ of cardinality $\leqslant \pi_0$, and consisting of π_0-accessible objects. (By virtue of 4.6.1 2° already proved, applied to $S = \mathcal{M}_0$, if $\mathcal{M}_0$ is strictly full, we therefore have $\mathcal{M}_0 = \mathcal{M}_{\pi_0}$.) Then the canonical functor*

$$\operatorname{Ind}'_{\pi_0}(\mathcal{M}_0) \longrightarrow \mathcal{M}$$

is an equivalence of categories.

The fact that this functor is fully faithful follows at once from the hypothesis that the objects of $\mathcal{M}_0$ are π_0-accessible:

$$\begin{aligned}\operatorname{Hom}_{\mathcal{M}}(\varinjlim_{i\in I} \underbrace{X_i}_{\in\mathcal{M}_0}, \varinjlim_{j\in J} \underbrace{Y_j}_{\in\mathcal{M}_0}) &\simeq \varprojlim_{i\in I} \underbrace{\operatorname{Hom}_{\mathcal{M}}(X_i, \varinjlim_{j\in J} Y_j)}_{\varinjlim_{j\in J} \operatorname{Hom}(X_i,Y_j)} \\ &\simeq \underline{\operatorname{Hom}}_{\operatorname{Ind}_{\pi_0}(\mathcal{M}_0)}((X_i),(Y_j)) .\end{aligned}$$

It remains to see that this functor is essentially surjective. Since $\mathcal{M}_0$ is strictly generating, it follows that every $x \in \mathcal{M}$ is written [page 145]

$$x = \varinjlim_{i\in I} x_i ,$$

where $I = \mathcal{M}_0/x$, and the $x_i \in \mathcal{M}_0$. The hypothesis that $\mathcal{M}_0$ is stable under $\varinjlim$ of cardinality $\leqslant \pi_0$ implies the same property for $I = \mathcal{M}_0/x$, so I is large relative to π_0. Thus $(x_i)_{i\in I}$ defines an object of $\operatorname{Ind}_{\pi_0}\mathcal{M}_0$, whose image in $\mathcal{M}$ is isomorphic to x, q.e.d. (Compare SGA 4 I 9.18.)

Corollary 4.6.2. *Under the conditions of 4.6.1, let us take for π_0* the smallest *cardinal such that the objects of S are π_0-accessible. For every $\pi \geqslant \pi_0$, let $\mathcal{M}_\pi$ be the strictly full subcategory of $\mathcal{M}$ consisting of the π-accessible objects, which is also (by 4.6.1 2°) the closure of C in $\mathcal{M}$ stable under $\varinjlim$ of cardinality $\leqslant \pi$.*

1°) Let $\pi_1 = \operatorname{card}\operatorname{red} S$. We then have

$$\operatorname{card}\operatorname{red} \mathcal{M}_\pi \leqslant \pi_1^{\pi} \qquad (= 2^{\pi} \text{ if } \pi \geqslant \pi_1 \geqslant 2). \tag{4.6.2.1}$$

p. 47
[page 146]

2°) $(\mathcal{M}_\pi)_{\pi\geqslant\pi_1}$ is a cardinal filtration of $\mathcal{M}$, where $\pi_1 = \sup(\pi_0, \operatorname{card} S)$.

Proof. The bound (4.6.2.1) is contained in 4.6.1.2, where π_0 has become π. If $\pi \geqslant \pi_1$, we have

$$\pi_1^\pi \;=\; 2^\pi \qquad (*)$$

for we have $\pi_1 \geqslant 2$, whence $\pi_1^\pi \geqslant 2^\pi$, on the other hand, we have $\pi_1 \leqslant 2^{\pi_1}$, whence $\pi_1^\pi \leqslant (2^{\pi_1})^\pi = 2^{\pi_1\pi} = 2^\pi$, since $\pi_1\pi = \pi$ for $\pi \geqslant \pi_1$; whence $(*)$.

Assertion 2° is included for the record; by virtue of $\mathcal{M} \simeq \operatorname{Ind}'_{\pi_0}\mathcal{M}_{\pi_0}$, we may assume that we have $\mathcal{M} = \operatorname{Ind}'_{\pi_0}(\mathcal{M}_{\pi_0})$, and we can then apply 4.5.9. One would moreover like to say that $(\mathcal{M}_\pi)_{\pi\geqslant\pi_0}$ is a cardinal filtration, without restricting oneself to $\pi \geqslant \pi_1$. This is true except for the bound on cardinals in condition c) of the definition of cardinal filtrations (SGA 4 I 9.12):

$$\underbrace{x}_{\in \operatorname{Ob}\mathcal{M}_{\pi'}} \;=\; \varinjlim_{i\in I} \underbrace{x_i}_{\in \operatorname{Ob}\mathcal{M}_\pi}\;, \qquad\qquad I \text{ large relative to } \pi \text{ and } \operatorname{card} I \leqslant \pi'^\pi$$

for $\pi' \geqslant \pi$. One must assume, for this bound, that $\pi \geqslant \pi_1$.

[page 147]

We can now finish the proof of 4.3.6, by proving the last two implications $(*)$ (p. 139). We have d $\Longrightarrow$ c, i.e. if $\mathcal{M} \simeq \operatorname{Ind}'_{\pi_0}(C)$, C stable under direct limits of cardinality $\leqslant \pi_0$, and C equivalent to a small category, then $\mathcal{M}$ is stable under small $\varinjlim$ (4.4.13, p. 95-96), its elements are accessible, and it admits a cardinal filtration (cf. 4.5.9). But one must check all the same that the elements of C are π_0-accessible. This is contained in SGA 4 I 9.18, which asserts that the full faithfulness of the functor $\operatorname{Ind}_{\pi_0}(C) \longrightarrow \mathcal{M}$ *is equivalent* to the fact that the elements of C are π_0-accessible in $\mathcal{M}$. The proof in loc. cit. was not clear, the "it suffices" was seen bottom of page 144. The "it is necessary" does not seem clear to me, with the generality given in loc. cit. – and now I see that it is even false, as one sees by taking for C the unit category e, so $\operatorname{Ind}'_{\pi_0}C \simeq e$, and the assertion affirming that if e is fully embedded in $\mathcal{M}$ (thus by a *rigid* object of $\mathcal{M}$, i.e. an object a such that $\operatorname{End} a = \{\mathrm{id}_a\}$), then a is $\aleph_0$-accessible. This is false in the category of rings, where $\mathbf{R}$ is a rigid object not $\aleph_0$-accessible!

[page 148]

In the case we are concerned with, $\mathcal{M} = \operatorname{Ind}'_{\pi_0}C$, one must clearly use the explicit calculation of the $\varinjlim$ large relative to π_0 in $\operatorname{Ind}'_{\pi_0}C \subset C^\wedge$, they are indeed calculated "argument by argument", cf. 4.4.15.

This completes the proof of d $\Longrightarrow$ c in $(*)$, page 139. Finally the implication c $\Longrightarrow$ PTA is tautological, the existence of a cardinal filtration implying that of a small strictly generating subcategory. This completes the proof of 4.3.6.

I am going to give again a slightly more precise formulation of 4.3.6., specifying the cardinals, and adding to it an additional condition, inspired by SGA 4 I 9.11 (revised and corrected, cf. an annotation in the copy of SGA 4 I ([58])).

p. 48

Theorem 4.6.3. *Let $\mathcal{M}$ be a $\mathfrak{U}$-category.*

(1°) *The following conditions are equivalent for an infinite cardinal π_0:*

- *a_{π_0}) $\mathcal{M}$ is a $\mathfrak{U}$-π_0-paratopos category, i.e. $\mathcal{M}$ is stable under small $\varinjlim$, and contains a small full generating subcategory S, consisting of π_0-accessible objects of $\mathcal{M}$.*

[page 149]

- *b_{π_0}) There exist a small category S, and a fully faithful and π_0-accessible functor $\mathcal{M} \xrightarrow{i} S^\wedge$, admitting a left adjoint.*

[58] "cf. simpler result above, if C is *strictly* generating"

Corollary. *Let F admit subcategory generating by strict epimorphisms, consisting of accessible objects, then* every *object of F is accessible (use 7.2. (i) b)).*

c_{π_0}) *There exists a small category C, stable under $\varinjlim$ of cardinality $\leqslant \pi_0$, such that $\mathcal{M} \simeq \mathrm{Ind}'_{\pi_0}(C)$.*

c'_{π_0}) *As c), with C only equivalent to a small category.*

2°) These conditions imply that $\mathcal{M}$ is stable under small $\varinjlim$ and small $\varprojlim$, and the condition:

e_{π_0}) *For every category I large relative to π_0, the functor*

$$L = \varinjlim_I : \underline{\mathrm{Hom}}(I, \mathcal{M}) \longrightarrow \mathcal{M}$$

is left exact (hence exact).

This condition implies in particular the following:

$\alpha(\pi_0)$ *On the category of double arrows of $\mathcal{M}$, the functor* Ker *(with values in $\mathcal{M}$) is π_0-accessible.*

$\beta(\pi_0)$ *For every arrow $y \longrightarrow x$ in $\mathcal{M}$, the base change functor $\mathcal{M}/x \longrightarrow \mathcal{M}/y$ is π_0-accessible.*

([59]).

[page 150]

3°) The following conditions are equivalent.

a) $\mathcal{M}$ is a $\mathfrak{U}$-paratopos category, i.e. is stable under small $\varinjlim$, and contains a small full subcategory S generating by strict epimorphisms, and consisting of accessible objects.

a′) As a), but requiring that $\mathcal{M}$ be weakly accessible, i.e. that every *object of $\mathcal{M}$ be accessible.*

b) There exist a small category S and a fully faithful and accessible functor $\mathcal{M} \hookrightarrow S^\wedge$ admitting a left adjoint.

ⓒ *There exist an infinite cardinal π_0, a small category S admitting direct limits of categories I with* $\mathrm{card}\,\mathrm{Fl}\,I \leqslant \pi_0$, *and an equivalence $\mathcal{M} \simeq \mathrm{Ind}'_{\pi_0}(S)$.*

c′) As c, assuming only S equivalent to a small category.

d) $\mathcal{M}$ is stable under small $\varinjlim$, is weakly accessible, i.e. all its objects are accessible, and admits a cardinal filtration.

p. 49

ⓔ ([60]). *$\mathcal{M}$ is stable under small $\varinjlim$, contains a small strictly generating full subcategory, and there exists an infinite cardinal π_0 such that condition e_{π_0} of 2°) is satisfied.*

ⓔ′ *As e, but requiring, instead of e_{π_0}, the weaker condition conjunction of $\alpha(\pi_0)$ and $\beta(\pi_0)$ in 2°.*

Proof. Part 1° has already been proved. The first assertion in 2°), namely the stability of $\mathcal{M}$ under small $\varprojlim$, is well known (follows from stability under small $\varinjlim$ [page 151]

[59] **Question.** Condition e_{π_0} (or even, the pair $\alpha(\pi_0) + \beta(\pi_0)$ plus the existence of a small strictly generating subcategory (and the stability of $\mathcal{M}$ under small $\varinjlim$), does it imply that $\mathcal{M}$ is a π_0-paratopos ? I presume not.)

[60] This is the prettiest intrinsic characterization of paratopoi in the "economy" mode, the other, the structural statement, is ⓒ .

, and from the existence of a small generating family, cf. SGA 4 I 8.12.7, dual version of the marginal annotation). The validity of condition e_{π_0} follows from the fact that it holds in $C^\wedge$ (where C is the category mentioned in 1°, c_{π_0}), and from the fact that the $\varprojlim$ in $\mathcal{M}$, and the $\varinjlim$ large relative to π_0 in $\mathcal{M}$, are computed in $C^\wedge$. We moreover find in the same way :

Corollary 4.6.4. *If $\mathcal{M}$ is a $\mathfrak{U}$-π_0-paratopos category, then for every cardinal $\pi \geqslant \pi_0$, and every category J such that* $\operatorname{card} J \leqslant \pi$*, the functor*

$$\underline{\mathrm{Hom}}(J, \mathcal{M}) \xrightarrow{\varprojlim_J} \mathcal{M}$$

is π-accessible, or what amounts to the same thing, for every small category I large relative to π, the functor

$$\underline{\mathrm{Hom}}(I, \mathcal{M}) \xrightarrow{\varinjlim_I} \mathcal{M}$$

commutes with the $\varprojlim$ of cardinality $\leqslant \pi$.

On the other hand, the fact that e_{π_0} implies $\alpha(\pi_0)$ and $\beta(\pi_0)$ is immediate. [page 152]

Part 3°), as regards the equivalence of conditions a) to d), is contained in 1°, and the equivalence with d) has also been seen (the only non-tautological thing being that condition c), namely $\mathcal{M} \simeq \mathrm{Ind}_{\pi_0}(C)$ with C small and stable under $\varinjlim$ of cardinality $\leqslant \pi_0$, implies the existence of a cardinal filtration, which was seen in 4.5.9, and reaffirmed in 4.6.2 2° above). On the other hand, that these conditions a) to d) imply ⓔ , and that ⓔ implies e′, was seen in 2°. So it remains to see that the conditions of e′ imply a), and for this, it remains to establish that the elements of $\mathcal{M}$ are accessible. But this is what is established in SGA 4 I 9.11, except that the statement of condition c), second case, is botched, and should be replaced by condition $\beta(\pi_0)$ (condition $\alpha(\pi_0)$ appearing in condition a) of loc. cit.).

Remark 4.6.5. The developments of sections 4.4, 4.5, 4.6 considerably sharpen and strengthen, in many respects, the theory of cardinal filtrations of SGA 4 I 9. This is so, at least, in the case of $\mathfrak{U}$-categories $\mathcal{M}$ stable under small $\varinjlim$. In loc. cit. one works with slightly more general $\mathfrak{U}$-categories, those satisfying the *condition L* : there exists an infinite cardinal π_0 such that $\mathcal{M}$ is stable under small $\varinjlim$ *large relative to* π_0. I presume however that one should be able to improve in a similar way the theory of *accessible* cardinal filtrations, taking inspiration from corollary 9.18 of loc. cit., establishing an equivalence of categories [page 153] p. 50

$$\mathcal{M} \simeq \mathrm{Ind}_{\pi_0}(C) ,$$

for a cardinal π_0 and a suitable small category C. Here, one can no longer assume C stable under this or that type of $\varinjlim$. Nevertheless, 4.4.15 tells us that without any further hypothesis on C, $\mathcal{M} \overset{\text{def}}{=} \mathrm{Ind}_{\pi_0}(C)$ does satisfy L_{π_0}, i.e. is stable (and even stable in $C^\wedge$)

under small $\varinjlim$ large relative to π_0. [page 154]

This seen, I feel like calling *accessible* a category $\mathcal{M}$ which admits a cardinal filtration, and which is moreover weakly accessible every object of which is accessible. And to venture a

Conjecture 4.6.6 ([61]). *Let $\mathcal{M}$ be a $\mathfrak{U}$-category. Equivalent conditions :*

a) $\mathcal{M}$ satisfies condition L, and admits a small strictly generating category consisting of accessible objects.

[61] All this will be made considerably more precise in the theory which will be developed in sections 4.7, 4.16.

⇑ *b) There exists a small category C, and an infinite cardinal π_0, such that $\mathcal{M}$ is equivalent to the category* $\mathrm{Ind}'_{\pi_0}(C)$.

⇑ *c) $\mathcal{M}$ is "accessible", i.e. is weakly accessible (every object accessible) and admits a cardinal filtration.*

We have b $\Longrightarrow$ a $\Longleftarrow$ c tautologically, and c $\Longrightarrow$ b was seen in loc. cit. 9.18, so we have the sequence

$$\mathrm{c} \implies \mathrm{b} \implies \mathrm{a}\,,$$

so it would remain to establish a $\Longrightarrow$ c.

[page 155]

The work accomplished (and the sweat shed) in 4.5 will moreover save us, when the time comes, from quite natural errors in the description of a cardinal filtration, starting from a strictly generating subcategory S : it is neither Filt^{π}, nor Filt'^{π}, but something subtler – perhaps one must take the category $\mathcal{M}_\pi$ again, or else saturate S under the direct limits of cardinality $\leqslant \pi$ which are willing to be representable in $\mathcal{M}$: we then find a $\mathrm{Filt}''^{\pi} \subset \mathcal{M}_\pi$, and if these Filt''^{π} do indeed give a cardinal filtration, we must necessarily have $\mathcal{M}_\pi \subset \mathrm{Filt}''^{\pi}$, hence finally $\mathrm{Filt}''^{\pi} = \mathcal{M}_\pi$. So I expect that in any event, the cardinal filtration (if it exists) will be given by the $\mathcal{M}_\pi$. Condition a) of loc. cit. 9.12 ($\mathcal{M}_\pi$ equivalent to a small category), and the bound on cardinals in loc. cit. c), should not cause any difficulties, and condition b) is OK. The difficulty

[page 156]

comes from the fact that $\mathcal{M}_\pi$ not even being stable under sums and cokernels of double arrows (when these are not represented in $\mathcal{M}$), the category $\mathcal{M}_\pi/x$ (for x in $\mathcal{M}$) has no reason to be filtering, as far as I can see, nor a fortiori to be large relative to π. p. 51 So it is not too clear how then to represent x as $\varinjlim_I x_i$, the x_i in $\mathcal{M}_\pi$ and I large relative to π.

It must indeed be said that in SGA 4 I 9, I give no *example* of a cardinal filtration on a category where arbitrary small $\varinjlim$ would not be representable (cf. loc. cit. 9.13). Perhaps after all it is against nature to want them beyond ? ? Perhaps one must wait until forced to it, to try to develop a theory outside the framework of paratopoi.

[page 157]

4.6.7. We are now in a position to reformulate the litanies of somewhat technical conditions a) to f) of theorem 2.6 (p. 25 in XVIII) concerning the "special on the left" character of the set $\mathrm{Mon}(\mathcal{M}) \subset \mathrm{Fl}(\mathcal{M})$:

Proposition 4.6.7 ([62]). *The preliminary conditions on the $\mathfrak{U}$-category $\mathcal{M}$, ensuring the validity of theorem XVIII 2.6, in a very slightly modified form (namely by modifying in 2.6 d) the condition of stability of* Filt^{π} *under quotients, into an exactness property d) below), are equivalent to the set of the following six conditions a) to f).*

$$\begin{cases} \text{a)} & \mathcal{M} \text{ is stable under small } \varinjlim\,. \\ \text{b)} & \mathcal{M} \text{ admits a small strictly generating subcategory.} \\ \text{c)} & \text{Filtering direct limits in } \mathcal{M} \text{ are exact.} \end{cases}$$

These conditions imply that $\mathcal{M}$ is a $\mathfrak{U}$-paratopos category *(cf. 4.6.3, 3°), and in particular that the objects of $\mathcal{M}$ are* accessible, *and that $\mathcal{M}$ admits a cardinal filtration $(\mathcal{M}_\pi)_{\pi \geqslant \pi_0}$ such that for every $\pi \geqslant \pi_0$, $\mathcal{M}_\pi$ is stable under $\varinjlim$ of cardinality $\leqslant \pi$ (in fact, we will have $\mathcal{M}_\pi$ = full subcategory of the π-accessible objects of $\mathcal{M}$). Also, they imply the stability of $\mathcal{M}$ under small $\varprojlim$. On the other hand, the following three conditions are stability conditions for monomorphisms.*

[62] Perhaps one could call these categories "*parabelian paratopos*" categories ? ?

d) Let $f : X \longrightarrow Y$ be an arrow in $\mathcal{M}$ and $R = X \times_Y X \rightrightarrows X$, then $X/R \longrightarrow Y$ is a monomorphism – in other words, f factors as a composite ip, with i monomorphic and p an effective epimorphism.

[page 158]

e) If A_1, A_2 are two subobjects of an object B of $\mathcal{M}$, and $A_0 = A_1 \cap A_2 \overset{\text{def}}{=} A_1 \times_B A_2$, then $A_1 \amalg_{A_0} A_2 \longrightarrow B$ is a monomorphism. (Note the close kinship of conditions d) and e).)

f) The set $\mathrm{Mon}(\mathcal{M})$ of monomorphisms in $\mathcal{M}$ is stable under cobase change.

In fact, it is not clear that these conditions, and in particular d), suffice to entail the stability of the $\mathcal{M}_\pi$ under quotients (only under effective quotients), as postulated in 2.6 d). But in the statement and the proof given in XV cor. 7.4 (p. 98-99), we were content with a somewhat weaker form of this stability property, restricting ourselves to the quotients Z' of $Z \in \mathrm{Ob}\,\mathcal{M}_\pi$ *such* that there is no subobject Z'_0 of Z', distinct from Z', and such that $Z \longrightarrow Z'$ factors through Z'_0 ; we therefore required that $Z' \in \mathrm{Ob}\,\mathcal{M}_\pi$. But if condition d) is satisfied, it follows at once that a quotient Z' of Z having this additional property is necessarily effective, from which it does follow that $Z' \in \mathrm{Ob}\,\mathcal{M}_\pi$.

p. 52

Corollary 4.6.8. *Conditions d), e), f) of 4.6.7 are satisfied if $\mathcal{M}$ is an abelian category. So the set of conditions a) to f) is satisfied in each of the following two cases :*

[page 159]

1) $\mathcal{M}$ is an abelian category where filtering direct limits are representable and exact, and admitting a small strictly generating category ([63]).

2) $\mathcal{M}$ is a topos.

Remark 4.6.9. I am not sure I know other examples where the conditions of 4.6.7 are verified, than those noted in 4.6.8 above ([64]). One will note that case 1°) mentioned there is that of Tohoku, where I establish the existence of "enough injective objects" in such an abelian category. In Tohoku, I confine myself to assuming the existence of a small subcategory, *generating for monomorphisms*, a condition apparently weaker than the one stated in 4.6.8 1°). But in an abelian category, the two notions (and their various variants in SGA 4 I 7.1) are equivalent by virtue of SGA 4 I 7.3 (i).

[page 160]

On the other hand, it seems that conditions e) f) are practically never verified for the categories defined by the species of algebraic structure "definable by finite products" and nothing more, such as groups, rings (of all kinds), etc. But the category of *commutative* groups, being abelian, is an exception. (One can moreover interpret it as $\mathcal{M} \boxtimes \mathcal{M}$, where $\mathcal{M} = (\mathrm{Gr})$, and one can then wonder whether there are some generalizations to paratopos categories of this form, $\mathcal{M}$ being a paratopos category …)

[page 161]

4.7 $\mathrm{Ind}_\pi C$ (C arbitrary) and generalized cardinal filtrations.

I feel the need to launch into developments on this subject, to arrive at a better understanding of the notion of the *accessible part* of a paratopos, and of the stability theorem for paratopoi under certain key operations on them.

In what follows, C denotes a category equivalent to a small category, C_0 a skeletal exhaustive subcategory, π_0 an infinite cardinal. We set

$$\mathcal{M} \;=\; \mathrm{Ind}_{\pi_0} C \;\subset\; C^\wedge \qquad ([65])\,. \tag{4.7.1}$$

[63] This is the case studied in "Tohoku".

[64] or which follow trivially from them (products etc.)

We propose to introduce on $\mathcal{M}$ various $\pm$ "cardinal" filtrations, by corresponding properties of an object F of $\mathcal{M} \subset C^\wedge$, which I am now going to introduce, taking inspiration from 4.5.2 (p. 107) and 4.5.6 (p. 116).

Proposition 4.7.1. *Set $\pi_1 = \sup(\pi_0, \operatorname{card} C_0)$. Let $\pi \geqslant \pi_0$ be a cardinal, and let us consider the following conditions (depending on π) on an object F of $\mathcal{M}$:*

a) $\operatorname{card} C_0/F \leqslant \pi$ *(i.e.* $\operatorname{card} \mathrm{Fl}\, C_0/F \leqslant \pi$*).*

a′) $\operatorname{card} \mathrm{Ob}\, C_0/F \leqslant \pi$.

a″) $\operatorname{card} F(x) \leqslant \pi$ *for every $x \in \mathrm{Ob}\, C$ (***N.B.** *it suffices to require this for x in C_0).*

p. 53
[page 162]

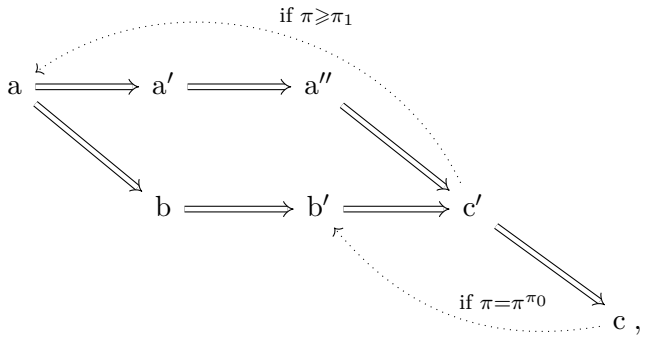
*b) We have $F = \varinjlim_{i \in I} x_i$, with the $x_i \in \mathrm{Ob}\, C$, I a category (not necessarily ordered) large relative to π_0, and such that $\operatorname{card} I \leqslant \pi$. (***N.B.** *by virtue of 4.4.2 (p.79), for given I, this is equivalent to the existence of a cofinal functor $I \longrightarrow C/F$, or again $I \longrightarrow C_0/F$.)* ([66]).

b′) We have $F \in \mathrm{Ob}\, \mathrm{Filt}'^{\pi}$, where Filt'^{π} is the smallest strictly full subcategory of $\mathcal{M}$ containing C, and stable under $\varinjlim$ large relative to π_0 and of cardinality $\leqslant \pi$.

c) We have $F \in \mathrm{Ob}\, \mathcal{M}_\pi$, i.e. F is a π-accessible object of $\mathcal{M}$.

c′) We have $F \in \mathrm{Ob}\, C^\wedge_\pi$, i.e. F is a π-accessible object of $C^\wedge$.

This being set, we have the following diagram of implications.



where the two dotted implications, $c' \Longrightarrow a$ and $c \Longrightarrow b$, are conditional subject to the condition $\pi \geqslant \pi_1$, resp. $\pi = \pi^{\pi_0}$.

[page 163]

Corollary 4.7.2.

1°) If $\pi \geqslant \pi_1$, the six conditions a, a′, a″, b, b′, c′ are equivalent, and the condition in question (which can be written $F \in \mathrm{Ob}\, \mathrm{Filt}^\pi$ or $F \in \mathrm{Ob}\, \mathrm{Filt}'^{\pi}$ indifferently) implies c :

$$\mathrm{Filt}^\pi \;=\; \mathrm{Filt}'^{\pi} \;\subset\; \mathcal{M}_\pi \qquad \text{if } \pi \geqslant \pi_1 \tag{4.7.2.1}$$

(whereas in general, for $\pi \geqslant \pi_0$, we have

$$\mathrm{Filt}^\pi \;\subset\; \mathrm{Filt}'^{\pi}\ (\subset C^\wedge_\pi \cap \mathcal{M}\,) \;\subset\; \mathcal{M}_\pi \qquad \text{if } \pi \geqslant \pi_0). \tag{4.7.2.2}$$

[65] **N.B.** I now denote by $\mathrm{Ind}C$, $\mathrm{Ind}_\pi C$ what was previously denoted $\mathrm{Ind}'C$, $\mathrm{Ind}'_\pi C$.

[66] **N.B.** We shall denote by $\mathrm{Filt}^\pi \subset \mathcal{M}$ the strictly full subcategory of $\mathcal{M}$ consisting of the F satisfying this condition.

2°) When $\pi = \pi^{\pi_0}$, e.g. when π is of the form 2^c, with $c \geqslant \pi_0$, then conditions b, b′, c, c′ are equivalent, i.e. we have

$$\mathrm{Filt}^{\pi} \;=\; \mathrm{Filt}'^{\pi} \;=\; (C^{\wedge})_{\pi} \cap \mathcal{M} \;=\; \mathcal{M}_{\pi}\,. \tag{4.7.2.3}$$

3°) When $\pi \geqslant \pi_1$ and $\pi = \pi^{\pi_0}$, then the seven conditions a), a′), a″), b) , b′), c) , c′) are equivalent, so in particular

$$\mathrm{Filt}^{\pi} \;=\; \mathrm{Filt}'^{\pi} \;=\; C_{\pi}^{\wedge} \cap \mathcal{M} \;=\; \mathcal{M}_{\pi} \qquad (\pi \geqslant \pi_1 \text{ and } \pi = \pi^{\pi_0}). \tag{4.7.2.4}$$

Proof of 4.7.1. The implications $\mathrm{a} \Longrightarrow \mathrm{a}' \Longrightarrow \mathrm{a}''$ are tautological, as is $\mathrm{b} \Longrightarrow \mathrm{b}'$. The implication $\mathrm{b}' \Longrightarrow \mathrm{c}'$ comes from the stability of $C_{\pi}^{\wedge}$ under $\varinjlim$ of cardinality $\leqslant \pi$, and from the fact that the $\varinjlim$ of the type under consideration (in particular, large relative to π_0) *in* $\mathcal{M}$ are the same as in $C^{\wedge}$, i.e. are computed "argument by argument" (cf. 4.4.15, p. 99). This, together with the fact that the inclusion $\mathcal{M} \hookrightarrow C^{\wedge}$ is fully faithful, also implies $\mathrm{c}' \Longrightarrow \mathrm{c}$. p. 54

The implication $\mathrm{a}'' \Longrightarrow \mathrm{c}'$ is seen as in 4.5.6 (p. 119) by applying lemma 4.5.8 1° and 2°, to the case of $C^{\wedge} = \underline{\mathrm{Hom}}(C^{\circ}, \mathcal{E})$, where $\mathcal{E} = (\mathrm{Ens})$. The reverse conditional implication $\mathrm{c}' \Longrightarrow \mathrm{a}''$, for $\pi \geqslant \pi_1$, is seen as in loc. cit., by applying this lemma 4.5.8. So it still remains to establish the three implications [page 164]

$$\begin{cases} \mathrm{a} & \Longrightarrow & \mathrm{b} & \\ \mathrm{a}'' & \Longrightarrow & \mathrm{a} & \text{if } \pi \geqslant \pi_1 \\ \mathrm{c} & \Longrightarrow & \mathrm{b} & \text{if } \pi = \pi^{\pi_0}. \end{cases} \tag{$*$}$$

But $\mathrm{a} \Longrightarrow \mathrm{b}$ is tautological, on taking $I = C_0/F$, since we know that C_0/F is large relative to π_0 and that $F = \varinjlim_{i \in I} x_i$, practically by definition of $\mathcal{M} = \underbrace{\mathrm{Ind}_{\pi_0}(C)}_{\simeq \mathrm{Ind}_{\pi_0} C_0} \subset C^{\wedge}$. So it remains to establish the two conditional implications of $(*)$. The implication $\mathrm{a}'' \Longrightarrow \mathrm{a}$ for $\pi \geqslant \pi_1$ is established as in 4.5.2 (cf. page 109), it is essentially sorital. It remains to establish

$$\mathrm{c} \Longrightarrow \mathrm{b} \qquad \text{if } \pi \;=\; \pi^{\pi_0}\,. \tag{$*$}$$

We will need the auxiliary result:

Lemma 4.7.3. *Let $\pi \geqslant \pi_0$ be such that $\pi^{\pi_0} = \pi$ (thus in fact $\pi > \pi_0$). For every object F of $\mathcal{M}$, we have* [page 165]

$$F \;=\; \varinjlim_{i \in I} x_i \qquad (^{67})\,. \tag{4.7.3.1}$$

If in addition $\pi' > \pi$ and $F \in \mathrm{Ob}\,\mathrm{Filt}^{\pi'}$, then we can take I such that

$$\mathrm{card}\, I \;\leqslant\; \pi'^{\pi}\,. \tag{4.7.3.2}$$

Indeed, we have

$$F \;=\; \varinjlim_{j \in J} x_j \qquad \text{with } J = C_0/F \qquad (^{68})\,,$$

with J large relative to π_0. Then let

$$I \;=\; \text{set of subcategories } J' \text{ of } J \textit{ large relative to } \pi_0, \text{ and such that } \mathrm{card}\, J' \leqslant \pi. \tag{4.7.3.3}$$

[67] the x_i in Filt^{π}, I an ordered set large relative to π.

[68] where J can also be replaced by a category cofinal in C_0/F large relative to π_0. p. 55

I say that I (ordered by inclusion) *is large relative to* π ([69]). This follows at once from lemma 4.5.5, p. 113, (stated at first, erroneously, assuming only $\pi > \pi_0$ and not $\pi^{\pi_0} = \pi$). Let us then put, for every $J' \in I$,

$$x_{J'} = \varinjlim_{j \in J'} x_j ,$$

we thus have

$$x_{J'} \in \mathrm{Ob}\,\mathrm{Filt}^{\pi}$$

by definition of I and of Filt^{π}. On the other hand,

$$J' \longmapsto x_{J'} : I \longrightarrow \mathrm{Filt}^{\pi}$$

is a functor, hence [page 166]

$$\varinjlim_{J' \in I} x_{J'} \simeq \varinjlim_{j \in J} x_j \simeq F ,$$

whence the first assertion of 4.7.3. There remains the bound on cardinals for $F \in \mathrm{Ob}\,\mathrm{Filt}^{\pi'}$. In this case, replacing C_0/F if need be by a cofinal category large relative to π_0, $\mathrm{card}\, J \leqslant \pi'$. We then see at once that $\mathrm{card}\, I \leqslant \pi'^{\pi}$, which is the announced bound.

We can now prove the implication $(*)$, page 164, i.e.

$$\mathcal{M}_{\pi} \subset \mathrm{Filt}^{\pi} \qquad (\text{for } \pi = \pi^{\pi_0}).$$

So let F be in $\mathcal{M}_{\pi}$, i.e. π-accessible, let us write F in the form (4.7.3.1), with I large relative to π. By definition of π-accessibility, applied to $F \xrightarrow{\mathrm{id}} F \simeq \varinjlim F_i$, we see that F is a direct factor of an F_i, hence of an object of Filt^{π}. Not knowing that Filt^{π} is stable under direct factors, we cannot conclude from this that F is in Filt^{π} (so I cannot prove c $\Longrightarrow$ b), but only that F is in Filt'^{π}, which, for its part, is stable under $\varinjlim$ large relative to π_0, of given cardinality $\pi > \pi_0$, hence also stable under direct factors. This completes the proof of 4.7.1, and at the same time we get: [page 167]

Corollary 4.7.4. *If $\pi \geqslant \pi_0$ is such that $\pi^{\pi_0} = \pi$, then we have*

$$\mathrm{Filt}'^{\pi} = (\mathrm{Filt}^{\pi})^{\natural} , \tag{4.7.4.1}$$

i.e. Filt'^{π} *consists of the objects F of $\mathcal{M}$ which are direct factors of an object of* Filt^{π}.

Indeed, since $\mathrm{Filt}^{\pi} \subset \mathrm{Filt}'^{\pi}$, and since $(\mathrm{Filt}^{\pi})^{\natural} \subset \mathrm{Filt}'^{\pi}$, and since Filt'^{π} is stable under direct factors, we have $(\mathrm{Filt}^{\pi})^{\natural} \subset \mathrm{Filt}'^{\pi}$ for every $\pi \geqslant \pi_0$. But conversely, since $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_{\pi}$, the preceding argument shows that for $\pi = \pi^{\pi_0}$, we have $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_{\pi} \subset (\mathrm{Filt}^{\pi})^{\natural}$, which proves 4.7.4.

Remark 4.7.4.1 . Of course, by 4.7.2, 1°, as soon as $\pi \geqslant \pi_1$, we even have $\mathrm{Filt}^{\pi} = \mathrm{Filt}'^{\pi}$ (without having to assume $\pi = \pi^{\pi_0}$), hence *if* $\pi > \pi_0$ (which will be the case if $\pi_1 > \pi_0$, i.e. $\pi_1 \neq \pi_0$). Filt'^{π} being stable under direct factors, Filt^{π} is so too, (4.7.4.1) is satisfied trivially. But if $\pi = \pi_1 = \pi_0$, we have $\mathrm{Filt}^{\pi} = \mathrm{Filt}^{\pi_0} = \mathrm{Filt}'^{\pi_0} =$ subcategory of functors *representable* in C (equivalent to C), and the latter is equal to $C^{\natural}$ in $\mathcal{M}$ only if C is Karoubian, cf. … below.

4.7.5. For the proof of 4.7.1 to be complete, I would still have to give a proof of 4.5.5, p. 113, which is the key technical lemma for establishing the p. 56 [page 168]

Lemma 4.7.5.1 (used in the proof of 4.7.3). *Let π_0, π be two infinite cardinals, with $\pi \geqslant \pi_0$ and $\pi^{\pi_0} = \pi$ (hence $\pi > \pi_0$). Let J be a category large relative to π_0. Then the set I, ordered by inclusion, of subcategories J' of J which are large relative to π_0, and such that* $\mathrm{card}\, J' \leqslant \pi$, *is large relative to π. (Moreover, if* $\mathrm{card}\, J \leqslant \pi'$, *we have* $\mathrm{card}\, I \leqslant \pi'^{\pi}$.*)*

[69] cf. proof 4.7.5, page 168-171.

So we must show the

Corollary 4.7.5.2 ([70]) (this is lemma 4.5.5 not yet proved). *Under the conditions of 4.7.5, let $K \subset \mathrm{Fl}J$ be such that* $\operatorname{card} K \leqslant \pi$. *Then there exists $J' \in I$ such that $K \subset J'$.*

We know, π being infinite, that the category K' generated by K is also of cardinality $\leqslant \pi$, so we can suppose that K "is" a subcategory of J. We are going to construct a $J' \in I$ containing K, by constructing by transfinite recursion an increasing sequence K_α of subcategories of J, with $\operatorname{card} K_\alpha \leqslant \pi$, indexed by the ordinals α such that $\boxed{\operatorname{card}\alpha \leqslant \pi}$:

[page 169]

1°) $K_0 = K$.

2°) $K_\alpha = \bigcup_{\beta<\alpha} K_\beta$ $(= \varinjlim K_\beta)$ if α is a limit ordinal (hence $\operatorname{card} K_\alpha \leqslant \pi$, since the $\operatorname{card} K_\beta \leqslant \pi$).

3°) $K_{\alpha+1}$ obtained from K_α in the following way. If K_α is large relative to π_0, we take $K_{\alpha+1} = K_\alpha$ (hence, in view of 2°, we will have $K_{\alpha'} = K_\alpha$ for $\alpha' \geqslant \alpha$ – the transfinite construction stops …). Otherwise, $K_{\alpha+1}$ is obtained as follows:

   a) for every subset A of cardinality $\leqslant \pi_0$ of $\operatorname{Ob} K_\alpha$, we choose an upper bound x_A in J (we feel like taking it in K_α, if there is an upper bound in K_α);

   b) for every pair x, y of objects of K_α, and every subset of cardinality $\leqslant \pi_0$ of $\operatorname{Hom}_{K_\alpha}(x,y)$, we choose an arrow $u_B : y \longrightarrow z_B$ in J, which equalizes the $v_\beta \in B$ (and we take this arrow in K_α, when possible);

   c) we take for $K_{\alpha+1}$ the smallest subcategory containing the id_{x_A} and the u_B.

We must check that

$$\operatorname{card} K_{\alpha+1} \leqslant \pi .$$

The number of possible choices for $A \in \mathfrak{P}_{\leqslant\pi_0}(\operatorname{Ob} K_\alpha)$ is $\leqslant \pi^{\pi_0} = \pi$, the number of possible choices for $B \in \mathfrak{P}_{\pi_0}(\mathrm{Fl}\, K_\alpha)$ is also $\leqslant \pi^{\pi_0} = \pi$, so the number of arrows of the form id_{x_A} or u_B is $\leqslant \pi + \pi = \pi$. But then the subcategory $K_{\alpha+1}$ of J generated by this set of arrows is of cardinality $\leqslant \pi$. Then let α_0 be the least ordinal such that

[page 170]

p. 57

$$\operatorname{card}\alpha_0 > \pi_0 ,$$

we thus have

$$\operatorname{card}\alpha_0 = \text{ successor cardinal of } \pi_0 \leqslant \pi ,$$

since $\pi^{\pi_0} = \pi$, whence $\pi > \pi_0$ (for $\pi^{\pi_0} = \pi_0^{\pi_0} \geqslant 2^{\pi_0} > \pi_0$). Thus, α_0 is among the admissible ordinals α (i.e. such that $\operatorname{card}\alpha \leqslant \pi$). I say that K_{α_0} *is large relative to* π_0 (which will prove 4.7.5.2, on taking $J' = K_{\alpha_0}$). This follows immediately from the fact that

$$K_{\alpha_0} = \bigcup_{\alpha<\alpha_0} K_\alpha \qquad (\textit{increasing} \text{ union}),$$

and that the set of $\alpha < \alpha_0$ is *large relative to* π_0 ([71]). So every set of objects or arrows of K_{α_0}, of cardinality $\leqslant \pi_0$, is contained in some K_α with $\alpha < \alpha_0$.

This implies, if it is a set of objects, that there is a common upper bound in $K_{\alpha+1}$, a fortiori in K_{α_0}. If it is a subset of a $\operatorname{Hom}_{K_\alpha}(x,y)$, that there is an equalizer $y \longrightarrow z$ in $K_{\alpha+1}$, a fortiori in K_{α_0}. Thus the criterion PF_{π_0} 1,2 of 4.4.4 (page 89) is satisfied by K_{α_0}, which completes the proof of 4.7.5.2.

[page 171]

[70] Compare with the proof of 4.6.1.2.

[71] spell this out in a

Lemma. *Let π_0 be a cardinal, $\alpha_0 = \alpha_{\pi_0'}$ (π_0' the successor of π_0) the least ordinal such that* $\operatorname{card}\alpha_0 > \pi_0$, *then $I_{\alpha_0} = \{\alpha \mid \alpha < \alpha_0\}$ is large relative to π_0.*

Corollary 4.7.6 (of 4.7.1). *For every cardinal $\pi \geqslant \pi_0$, we have*

$$\underbrace{\mathrm{Filt}'^{\pi}}_{\text{if } \pi \geqslant \pi_1 \atop = \mathrm{Filt}^{\pi}} \subset \mathcal{M}_\pi \subset \underbrace{\mathrm{Filt}'^{\pi^{\pi_0}}}_{\text{if } \pi^{\pi_0} \geqslant \pi_1 \atop = \mathrm{Filt}^{\pi^{\pi_0}}} \qquad (= (\mathrm{Filt}^{\pi^{\pi_0}})^{\natural} \text{ by cor. 4.7.4}), \tag{4.7.6.1}$$

therefore if $\pi \geqslant \pi_1$, we also have

$$\mathrm{Filt}^{\pi} \subset \mathcal{M}_\pi \subset \mathrm{Filt}^{\pi^{\pi_0}} \qquad (\text{if } \pi^{\pi_0} \geqslant \pi_1). \tag{4.7.6.2}$$

Indeed, $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_\pi$ is none other than the implication b′ $\Longrightarrow$ c of 4.7.1, on the other hand, we have $\pi \leqslant \pi^{\pi_0}$, whence

$$\mathcal{M}_\pi \subset \mathcal{M}_{\pi^{\pi_0}} = \mathrm{Filt}'^{\pi^{\pi_0}},$$

where the last equality comes from $\mathcal{M}_{\pi'} = \mathrm{Filt}'^{\pi'}$ if $\pi'^{\pi_0} = \pi'$, i.e. from (4.7.2.3) (page 163), putting $\pi' = \pi^{\pi_0}$, taking into account that $\pi'^{\pi_0} = (\pi^{\pi_0})^{\pi_0} = \pi^{\pi_0 \cdot \pi_0} = \pi^{\pi_0} = \pi'$.

Proposition 4.7.7. *Consider C, C_0, $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, $\pi_1 = \sup(\pi_0, \operatorname{card} \mathrm{Fl}\, C_0)$ as before. For every cardinal $\pi \geqslant \pi_0$, let $\mathcal{F}^\pi$ be the strictly full subcategory Filt'^{π} (resp. $(C^\wedge)_\pi \cap \mathcal{M}$, resp. $\mathcal{M}_\pi$) in $\mathcal{M}$. Then the increasing filtration $(\mathcal{F}^\pi)_{\pi \geqslant \pi_0}$ of $\mathcal{M}$ by the subcategories $\mathcal{F}^\pi$ satisfies the following conditions.* [page 172]

a) For every $\pi \geqslant \pi_0$, $\mathcal{F}^\pi$ is equivalent to a small category.

b) $\mathcal{M}$ satisfies L_{π_0} (i.e. is stable under filtering $\varinjlim$ large relative to π_0), and for every $\pi \geqslant \pi_0$, $\mathcal{F}^\pi$ is stable under filtering limits large relative to π_0 and of cardinality $\leqslant \pi$. p. 58

c) If $\pi \geqslant \pi_0$ is such that $\boxed{\pi^{\pi_0} = \pi,}$ *then for every F in $\mathcal{M}$, we have an isomorphism*

$$F = \varinjlim_{i \in I} x_i ,$$

the x_i in $\mathcal{F}^\pi$, and I an ordered category large relative to π. If in addition $\pi' > \pi$ and $F \in \mathrm{Ob}\, \mathcal{F}^{\pi'}$, we can take I such that

$$\operatorname{card} I \leqslant \pi'^{\pi} \qquad \underline{\text{if}}\ \pi'^{\pi} \geqslant \pi_1. \tag{4.7.7.1}$$

d) If $\pi \geqslant \pi_0$, we have $\mathcal{F}^\pi \subset \mathcal{M}_\pi$, i.e. the objects of $\mathcal{F}^\pi$ are π-accessible.

N.B. These are exactly, apart from d), the conditions a), b), c) for cardinal filtrations of SGA 4 I 9.7.12, except only that in c), we restrict ourselves here to the ordinals π such that $\pi^{\pi_0} = \pi$ ([72]). The additional condition d) is an "accessibility condition", cf. below. [page 173]

Proof of 4.7.7. a) Since we have

$$\mathrm{Filt}'^{\pi} \subset (C^\wedge)_\pi \cap \mathcal{M} \subset \mathcal{M}_\pi ,$$

we are reduced to proving that $\mathcal{M}_\pi$ is equivalent to a small category. We are reduced to the case $\pi \geqslant \pi_1$, $\pi = \pi^{\pi_0}$, in which case we have $\mathcal{M}_\pi = \mathrm{Filt}^{\pi}$. In fact, we find:

Corollary 4.7.7.1 ([73]). *If $\pi \geqslant \pi_1$, we have*

$$\begin{cases} \operatorname{card} \operatorname{red} \mathrm{Filt}^{\pi} \leqslant 2^{\pi} \\ \operatorname{card} \operatorname{red} \mathrm{Filt}'^{\pi} \leqslant \operatorname{card} \operatorname{red} \mathcal{M}_\pi \leqslant 2^{(\pi^{\pi_0})} \qquad (= 2^{\pi} \text{ if } \pi^{\pi_0} = \pi). \end{cases}$$

The second bound follows from the first and from

$$\mathcal{F}^\pi \subset \mathcal{M}_\pi \subset \mathrm{Filt}^{\pi^{\pi_0}} \qquad (\text{for } \pi^{\pi_0} \geqslant \pi_1) \qquad ([74]). \tag{4.7.7.2}$$

[page 174]

For the first, we use condition a″) of 4.7.1 (which characterizes Filt^{π} for $\pi \geqslant \pi_1$). I pass over the details of the verification.

Let us prove 4.7.7 b). The condition L_{π_0} is known, there remains the stability of $\mathcal{F}^{\pi}$. In the case of Filt'^{π}, it is established by definition. In the cases $C_{\pi}^{\wedge} \cap \mathcal{M}$ and $\mathcal{M}_{\pi}$, it follows from the stability of the notion of π-accessibility under $\varinjlim$ of cardinality $\leqslant \pi$, either in $C^{\wedge}$ or in $\mathcal{M}$ (taking into account that $C^{\wedge}$ and $\mathcal{M}$ have a generating family C formed of π_0-accessible objects, hence π-accessible).

Let us prove c): this is essentially lemma 4.7.3, taking into account that we have

$$\mathrm{Filt}^{\pi} \ \subset \ \mathcal{F}^{\pi}\ ,$$

except that one must take care with the cardinal bound, $\operatorname{card} I \leqslant \pi'^{\pi}$. But if $F \in \operatorname{Ob} \mathcal{F}^{\pi'}$, p. 59 since by (4.7.6.1)

$$\mathcal{F}^{\pi'} \ \subset \ \underbrace{\mathcal{M}_{\pi'}}_{\subset \mathcal{M}_{\pi'^{\pi}} \subset \mathrm{Filt}'^{\pi'^{\pi}}} \ \subset \ \mathrm{Filt}'^{\pi'^{\pi_0}},$$

hence a fortiori

$$\mathcal{F}^{\pi'} \ \subset \ \mathrm{Filt}'^{\pi'^{\pi}} \ \underbrace{=}_{\text{if } \pi'^{\pi} \geqslant \pi_1} \ \mathrm{Filt}^{\pi'^{\pi}},$$

whence, by 4.7.3, with π' replaced by $\pi'^{\pi} = \pi''$, the possibility of taking I with

$$\operatorname{card} I \ \leqslant \ \pi''^{\pi} \ = \ (\pi'^{\pi})^{\pi} \ = \ \pi'^{\pi \cdot \pi} \ = \ \pi'^{\pi},$$

which completes the proof.

Remark 4.7.7.2. I was obliged, in the statement of condition c), to bring in a second cardinal π_1, and this for lack of knowing whether, for π'' such that $\pi'' = \pi''^{\pi_0}$, $\mathrm{Filt}^{\pi''}$ is stable under direct factors. (To do things properly, one would have to either prove that it is stable, hence that we have in 4.7.1 c $\Longrightarrow$ b, i.e. $\mathcal{M}_{\pi} \subset \mathrm{Filt}^{\pi}$ (and not only c $\Longrightarrow$ b′, i.e. $\mathcal{M}_{\pi} \subset \mathrm{Filt}'^{\pi}$) if $\pi = \pi^{\pi_0}$, or else make a counterexample!) This somewhat strange ingredient, we shall alas be obliged to drag it along in the definition of generalized cardinal filtrations, if we do not want to forbid ourselves to regard in the present case $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$ as such a filtration, and to allow ourselves only to regard $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_1}$ as such, which would seem ridiculous to me. In practice (even if theoretically the introduction of π_1 in the description of a generalized cardinal filtration were indeed necessary) we shall hardly have cases where it is necessary to make mention of π_1 in condition c. Indeed, $\pi \geqslant \pi_0$ and $\pi^{\pi_0} = \pi$ imply $\pi \geqslant 2^{\pi_0}$, and then $\pi'^{\pi} \geqslant 2^{(2^{\pi_0})}$. If therefore $\pi_1 \leqslant 2^{(2^{\pi_0})}$, i.e. $\operatorname{card} C_0 \leqslant 2^{(2^{\pi_0})}$, then the condition $\pi'^{\pi} \geqslant \pi_1$ in c) is automatically satisfied, and there is therefore no need to mention it. Now it is very rare to have to deal with small categories C_0, whose cardinality is $>$ the power of the continuum c, and a fortiori, such that $\operatorname{card} C_0 > 2^{c}$. If on the other hand $\operatorname{card} C_0 \leqslant 2^{c}$, since $2^{c} \leqslant 2^{(2^{\pi_0})}$ (since $\pi_0 \geqslant \aleph_0$), we shall indeed have $\pi_1 \leqslant 2^{(2^{\pi_0})}$, and we can rest easy. [page 175] [page 176]

Definition 4.7.8. Let $\mathcal{M}$ be a $\mathfrak{U}$-category. We call a $\mathfrak{U}$-*generalized cardinal filtration* (or simply generalized cardinal filtration, when $\mathfrak{U}$ is understood) the datum of an infinite cardinal π_0, and of a filtration $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$ of $\mathcal{M}$ by strictly full subcategories $\mathcal{F}^{\pi}$, indexed by the $\mathfrak{U}$-cardinals $\geqslant \pi_0$, and satisfying conditions a) b) c) stated in 4.7.7, where in c) we put

$$\pi_1 \ = \ \sup(\pi_0, \operatorname{card} \operatorname{red} \mathcal{F}^{\pi_0})\,. \tag{4.7.8.1}$$

[72] and *in addition*, in the bound (4.7.7.1), to the cardinals $\pi' > \pi$ such that we also have $\pi'^{\pi} \geqslant \pi_1$ (which is automatic if we have $\pi \geqslant \pi_1$, which is the case, given that $\pi = \pi^{\pi_0} \geqslant 2^{\pi_0}$, if we have $2^{\pi_0} \geqslant \pi_1$.)

[73] Compare with 4.7.19.

[74] Cf. (4.7.6.2).

(Thus if $\operatorname{card}\operatorname{red}\mathcal{F}^{\pi_0} \leqslant 2^{(2^{\pi_0})}$, there is no need to take account of the condition $\pi'^{\pi} \geqslant \pi_1$, cf. 4.7.7.1.) We say that the generalized cardinal filtration (fcg) is *accessible*, if it satisfies in addition condition d). We say that it is *canonical* if we even have

$$\mathcal{F}^{\pi} = \mathcal{M}_{\pi} \tag{4.7.8.2}$$

for every $\pi \geqslant \pi_0$.

Proposition 4.7.9. *Let $(\mathcal{F}^{\pi})_{\pi\geqslant\pi_0}$ be an fcg on the $\mathfrak{U}$-category $\mathcal{M}$, and $\pi > \pi_0$ a cardinal such that $\pi^{\pi_0} = \pi$. Then we have*

p. 60

[page 177]

$$\begin{cases} \mathcal{M}_{\pi} \subset \mathcal{F}^{\pi} \quad \textit{(if } \pi = \pi^{\pi_0}\textit{)}, \\ \textit{hence } \mathcal{M}_{\pi} = \mathcal{F}^{\pi} \textit{ if moreover the fcg is } \text{accessible}. \end{cases} \tag{4.7.9.1}$$

Proof . Applying condition c) for fcg to an $F \in \operatorname{Ob}\mathcal{M}_{\pi}$, we find that F is a direct factor of an $x_i \in \operatorname{Ob}\mathcal{F}^{\pi}$. Since $\mathcal{F}^{\pi}$ is stable under $\varinjlim$ large relative to π_0 and of cardinality $\leqslant \pi$, and since *in addition* $\pi > \pi_0$, we conclude that:

Lemma 4.7.9.1 . *If $\pi > \pi_0$ ([76]), $\mathcal{F}^{\pi}$ is stable under direct factors.*

Whence it follows that $F \in \mathcal{F}^{\pi}$, whence (4.7.9.1).

Corollary 4.7.10. *Suppose the fcg accessible. For every cardinal $\pi \geqslant \pi_0$, we have*

$$\mathcal{F}^{\pi} \subset \mathcal{M}_{\pi} \subset \mathcal{F}^{\pi^{\pi_0}} \,(= \mathcal{M}_{\pi^{\pi_0}}). \tag{4.7.10.1}$$

Indeed, $\mathcal{F}^{\pi} \subset \mathcal{M}_{\pi}$ by the definition of accessible fcg, and $\mathcal{M}_{\pi} \subseteq \mathcal{M}_{\pi^{\pi_0}}$ (since $\pi \leqslant \pi^{\pi_0}$), now $\mathcal{M}_{\pi^{\pi_0}} = \mathcal{F}^{(\pi^{\pi_0})}$ by virtue of (4.7.9.1) applied to $\pi' = \pi^{\pi_0}$ instead of π (since $\pi'^{\pi_0} = \pi'$), whence (4.7.10.1). We shall note that this inclusion generalizes (4.7.9.1), it gives $\mathcal{F}^{\pi} = \mathcal{M}_{\pi}$ if $\pi = \pi^{\pi_0}$.

4.7.11. It seems to me that the only useful generalized cardinal filtrations are those which are *accessible.* In what follows, I am going to confine myself to this case. Since in that case $\mathcal{F}^{\pi} \subset \mathcal{M}_{\pi}$, we conclude from this that the canonical functor

[page 178]

$$\operatorname{Ind}_{\pi}(\mathcal{F}^{\pi}) \hookrightarrow \mathcal{M} \tag{4.7.11.1}$$

extending the inclusion $\mathcal{F}^{\pi} \subset \mathcal{M}$, so as to commute with $\varinjlim$ large relative to π, is *fully faithful.* Moreover, when $\pi^{\pi_0} = \pi$, condition c) for fcg tells us among other things that this functor is essentially surjective, hence that it is an *equivalence of categories*:

$$\operatorname{Ind}_{\pi}(\mathcal{F}^{\pi}) \xrightarrow{\sim} \mathcal{M} \qquad (\text{if } \pi = \pi^{\pi_0}). \tag{4.7.11.2}$$

Proposition 4.7.12. *Let $(\mathcal{F}^{\pi})_{\pi\geqslant\pi_0}$ be an accessible fcg (fcga) on the $\mathfrak{U}$-category $\mathcal{M}$, and let π be a cardinal $\geqslant \pi_0$. The following conditions are equivalent.*

a) The fully faithful functor (4.7.11.1) is essentially surjective, i.e. is an equivalence of categories.

b) For every $F \in \operatorname{Ob}\mathcal{M}$, $\mathcal{F}^{\pi}/F$ is a filtering category and large relative to π.

When these conditions are satisfied, and if $\pi > \pi_0$, we have

$$\mathcal{F}^{\pi} = \mathcal{M}_{\pi}. \tag{4.7.12.1}$$

p. 61

Proof. The equivalence of a) and b) follows at once from 4.4.2 and from the definition of categories large relative to π (p. 81). That these conditions imply $\mathcal{F}^{\pi} = \mathcal{M}_{\pi}$, i.e. $\mathcal{M}_{\pi} \subset \mathcal{F}^{\pi}$ in the case where $\pi > \pi_0$, is seen as for (4.7.9.1) by using Lemma 4.7.9.1.

Definition 4.7.13.

[page 179]

[76] But $\mathcal{F}^{\pi_0}$ is not necessarily stable under direct factors, as we see by taking $\mathcal{M} = \operatorname{Ind}_{\pi_0}(C)$, C a small non-Karoubian category, and the cardinal filtration for the $\operatorname{Filt}'^{\pi}$ – we note that $\operatorname{Filt}'^{\pi_0} = \operatorname{Filt}^{\pi_0} \simeq e$, since a category large relative to π_0 and of cardinality $\leqslant \pi_0$ is cofinal to e.

1°) A cardinal $\pi \geqslant \pi_0$ is called a *useful cardinal* for the fcga $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$, if it satisfies the equivalent conditions a) and b) of 4.7.12, and if in addition in the case where $\pi = \pi_0$, we have $\mathcal{F}^\pi = \mathcal{M}_\pi$. (Thus if π is a useful cardinal, we always have $\mathcal{F}^\pi = \mathcal{M}_\pi$.) In the contrary case, we say that π is a *superfluous cardinal* (for the given fcga).

2°) A cardinal π is called *regular* for the given fcg, if it is useful, and if in addition the cardinal bound stated in (4.7.7.1) is valid.

4.7.14 Thus, all cardinals $\pi \geqslant \pi_0$ such that $\pi^{\pi_0} = \pi$ are useful, and even regular for the given cardinal filtration. Let us note that if all cardinals $\pi \geqslant \pi_0$ are useful, then we have $\mathcal{F}^\pi = \mathcal{M}_\pi$ for every $\pi \geqslant \pi_0$, hence the fcga is canonical. Let us also note that in all cases, if $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$ is an fcga, then $(\mathcal{M}_\pi)_{\pi\geqslant\pi_0}$ is one too.
Indeed, condition a) for fcg (cf. 4.7.7) follows from the same condition for $(\mathcal{F}^\pi)$, and from [page 180]
(4.7.10.1), $\mathcal{M}_\pi \subset \mathcal{F}^{\pi^{\pi_0}}$. Condition b) is a general property of π-accessibility, taking into account that $(\mathcal{F}^{\pi_0})_0$ is a small generating subcategory consisting of π_0-accessible objects of $\mathcal{M}$. Condition d) is satisfied tautologically, and it remains to check condition c). But we have seen that for $\pi = \pi^{\pi_0}$, we have $\mathcal{M}_\pi = \mathcal{F}^\pi$ (4.7.9.1), hence condition c) for $(\mathcal{M}_\pi)$ follows from the same condition for $(\mathcal{F}^\pi)$. Thus, we find:

Proposition 4.7.15. *If $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$ is an fcga on the $\mathfrak{U}$-category $\mathcal{M}$, so is $(\mathcal{M}_\pi)_{\pi\geqslant\pi_0}$. Moreover, every useful (resp. regular) cardinal for the first filtration, is so for the second (called the* associated canonical *fcg of the given cardinal filtration).*

The last assertion follows from the fact that if π is useful for $(\mathcal{F}^\pi)$, then we have $\mathcal{F}^\pi = \mathcal{M}_\pi$.

4.7.16. It seems to me that one can say that the canonical fcg associated with a given fcga is always "better" than the latter (if the latter is not canonical, i.e. if the two [page 181]
are not equal), because of the following two facts:

a) precisely because it is *canonical*, i.e. depends on no other choice than that of the cardinal π_0 alone, and

b) because there are a priori more useful cardinals and regular cardinals for the canonical fcg.

So all in all, it seems to me that the cardinal filtrations the only useful ones are those which are canonical.
Let us note that a cardinal filtration (in the sense of SGA 4 I 9, which I retain here) p. 62
which is in addition *accessible* (i.e. $\mathcal{F}^\pi \subset \mathcal{M}_\pi$ for every $\pi \geqslant \pi_0$), is nothing other than a canonical fcga, such that *all* cardinals are regular (a fortiori useful) for the said filtration. I presume (without however having constructed an example) that the canonical cardinal filtrations of categories of the form $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, C a small *Karoubian* category (so that π_0 is a useful cardinal) are not necessarily fc (not generalized, i.e. in the SGA sense), i.e. I presume that it happens that there are cardinals $\pi \geqslant \pi_0$ not regular, and even superfluous.
I must finally study what happens when we change the initial cardinal π_0 of an fcga. [page 182]

Proposition 4.7.17. *Let $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$ be an fcga on the $\mathfrak{U}$-category $\mathcal{M}$, and let π_0' be a cardinal $\geqslant \pi_0$. Then $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0'}$ is an fcga on $\mathcal{M}$, which is canonical if the one we start with is canonical. For a cardinal $\pi \geqslant \pi_0'$, it is useful for $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0'}$ if and only if it is so for $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$. For it to be regular for $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0'}$, it* suffices *that it be so for $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$, and this is also necessary when we assume in addition that $2^\pi \geqslant \pi_1' \overset{\text{def}}{=} \sup(\pi_0', \operatorname{card}\operatorname{red}\mathcal{F}^{\pi_0'})$, a fortiori when $\pi \geqslant \pi_1'$.*

Proof. Conditions a) b) are trivially satisfied by the second filtration, being satisfied by the first, as is d) if it is by the first. The same is true of condition c) if $\pi = \pi^{\pi_0'}$, which

indeed implies $\pi^{\pi_0} = \pi$, since in general

$$\pi \leqslant \pi^{\pi_0} \leqslant \pi^{\pi_0'}.$$

It remains to examine the cardinal bound

$$\operatorname{card} I \leqslant \pi'^{\pi} \qquad \text{if } F \in \operatorname{Ob} \mathcal{F}^{\pi'} \text{ and } \pi'^{\pi} \geqslant \pi_1',$$

where this time $\pi_1' = \sup(\pi_0', \operatorname{card} \operatorname{red} \mathcal{F}^{\pi_0'})$. In any event, we have $\pi_1' \geqslant \pi_1$, hence $\pi'^{\pi} \geqslant \pi_1'$ implies $\pi'^{\pi} \geqslant \pi_1$, hence we can apply condition c) for fcg for $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$, to conclude that it is still verified. So in short, *all* the conditions for fcg (resp. fcga) are trivially verified for the *restricted* fcg of the first. Likewise, the fact that a cardinal $\pi' \geqslant \pi_0'$ is useful is trivially independent of which of the two filtrations we consider, it depends only on the datum of the cardinal π' and of the subcategory $\mathcal{F}^{\pi'}$ of $\mathcal{M}$. There remains the cardinal bound, in the case where we are interested in the *regularity* of π' for the fcga considered. The bound required is the same, but the precondition for the validity of this bound, namely [page 183]

$$\pi'^{\pi} \geqslant \pi_1' \qquad \text{resp.} \qquad \pi'^{\pi} \geqslant \pi_1,$$

is not the same. Since $\pi_1' \geqslant \pi_1$, we see that the precondition is less restrictive in the case of the first filtration, than of the second, whence the "it suffices" in the last assertion. But when $2^{\pi} \geqslant \pi_1'$, even the more restrictive one is automatically verified, since $\pi'^{\pi} \geqslant 2^{\pi} \geqslant \pi_1'$, and then the regularity of π for the second filtration implies the regularity for the first, q.e.d. [page 184]

Corollary 4.7.18. *Suppose π_0 regular for the fcga ([77]). If we restrict the fcga $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$ to the cardinals $\pi \geqslant \pi_0'$, where $\pi_0' \geqslant \pi_1$, then we find an fcga where the cardinal bound (for $\pi = \pi^{\pi_0}$) is valid without the precondition $\pi'^{\pi} \geqslant \pi_1'$; more precisely, we then have* p. 63

$$\pi_1' \leqslant 2^{(\pi_0'^{\pi_0})} \qquad \textit{(a fortiori } \pi_1' \leqslant 2^{(2^{\pi_0'})}\textit{)}, \tag{4.7.18.1}$$

hence the precondition $\pi'^{\pi} \geqslant \pi_1'$ is automatically verified if $\pi \geqslant \pi_0$, $\pi = \pi^{\pi_0'}$, $\pi' > \pi$ (cf. remark 4.7.7.2).

This is therefore equivalent to the following cardinal bound (applied to the case of $\pi = \pi_0'$).

Lemma 4.7.19. *Suppose π_0 regular for the fcga $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$. Then for every cardinal $\pi \geqslant \pi_1$ ($\overset{\text{def}}{=} \sup(\pi_0, \operatorname{card} \operatorname{red} \mathcal{F}^{\pi_0})$), we have*

$$\operatorname{card} \operatorname{red} \mathcal{F}^{\pi} \leqslant 2^{(\pi^{\pi_0})}, \tag{4.7.19.1}$$

and if $\pi' \geqslant \pi_1$ and $F \in \operatorname{Ob} \mathcal{F}^{\pi'}$, we have

$$\operatorname{card} \operatorname{red} \mathcal{F}^{\pi}/F \leqslant \pi'^{(\pi^{\pi_0})} \qquad ([78]). \tag{4.7.19.2}$$

Proof of the lemma. Since π_0 is regular, putting $C = \mathcal{F}^{\pi_0} = \mathcal{M}_{\pi_0}$, we find $\mathcal{M} \simeq \operatorname{Ind}_{\pi_0}(C)$, hence we can assume $\mathcal{M} = \operatorname{Ind}_{\pi_0}(C)$, and apply the results proved for categories of this form. Since $\mathcal{F}^{\pi} \subset \mathcal{M}_{\pi}$, $\mathcal{F}^{\pi'} \subset \mathcal{M}_{\pi'}$, the bounds (4.7.19.1) and (4.7.19.2) are contained in [page 185]

$$\left\{ \begin{array}{rcll} \operatorname{card} \operatorname{red} \mathcal{M}_{\pi} & \leqslant & 2^{(\pi^{\pi_0})} & \\ \operatorname{card} \operatorname{red} \mathcal{M}_{\pi}/F & \leqslant & \pi'^{(\pi^{\pi_0})} & \text{if } F \in \operatorname{Ob} \mathcal{M}_{\pi'}, \text{ with} \\ & & & \pi' \geqslant \pi \geqslant \pi_1. \end{array} \right\} \text{ for } \pi \geqslant \pi_1 \qquad ([79]). \qquad (*)$$

[77] It suffices for this that π_0 be useful, i.e. $\mathcal{M} \xleftarrow{\sim} \operatorname{Ind}_{\pi_0}(\mathcal{F}^{\pi_0} = \mathcal{M}_{\pi_0})$, this implies regular, cf. 4.8.5.

[78] **N.B.** I do not know whether we can replace in these bounds π^{π_0} by π_0.

The first of these bounds is nothing other than the second bound in 4.7.7.1. Let us note that we do indeed have

$$2^{(\pi^{\pi_0})} \leqslant 2^{(2^{\pi})},$$

which indeed follows from

$$\pi^{\pi_0} \leqslant 2^{\pi},$$

which we prove by noting that $\pi < 2^{\pi}$, whence $\pi^{\pi_0} \leqslant (2^{\pi})^{\pi_0} = 2^{\pi\cdot\pi_0} = 2^{\pi}$ since $\pi \geqslant \pi_0$. It remains to establish the second bound

$$\operatorname{card}\operatorname{red}\mathcal{M}_{\pi}/F \leqslant \pi'^{(\pi^{\pi_0})} \qquad \text{if } F \in \operatorname{Ob}\mathcal{M}_{\pi'}, \text{ with } \pi' \geqslant \pi \geqslant \pi_1\,. \tag{$*$}$$

(We do not need it for 4.7.18, but it will probably be useful elsewhere.) We use $\mathcal{M}_{\pi'} \subset \operatorname{Filt}^{\pi'^{\pi_0}}$ if $\pi' \geqslant \pi_1$ (4.7.6.2), hence if $F \in \operatorname{Ob}\mathcal{M}_{\pi'}$, we have

p. 64
[page 186]

$$\operatorname{card} F(x) \leqslant \pi'^{\pi_0} \qquad \forall\, x \in \operatorname{Ob} C,$$

thus if $G \in \operatorname{Ob}\mathcal{M}_{\pi} \subset \operatorname{Ob}\operatorname{Filt}^{\pi^{\pi_0}}$, whence

$$\operatorname{card} G(x) \leqslant \pi^{\pi_0} \qquad \forall\, x \in \operatorname{Ob} C,$$

we find

$$\operatorname{card}\operatorname{Hom}(G,F) \leqslant \underbrace{\underbrace{\operatorname{card}(\operatorname{Ob} C)}_{\leqslant\pi_1}\cdot(\pi'^{\pi_0})^{(\pi^{\pi_0})}}_{\pi'^{\pi_0}\cdot\pi^{\pi_0}=\pi'^{(\pi^{\pi_0})}},$$

whence

$$\operatorname{card}\operatorname{Ob}\mathcal{M}_{\pi}/F \leqslant \underbrace{\underbrace{\operatorname{card}(\operatorname{Ob}\mathcal{M}_{\pi})}_{\leqslant 2^{(\pi^{\pi_0})}}\cdot\pi'^{(\pi^{\pi_0})}}_{\gamma=\pi'^{\pi^{\pi_0}}},$$

and finally

$$\operatorname{card}\operatorname{Fl}\mathcal{M}_{\pi}/F \leqslant \underbrace{\underbrace{\gamma\cdot\gamma}_{\gamma=\pi'^{(\pi^{\pi_0})}}\cdot\underbrace{\operatorname{card}(\operatorname{Fl}\mathcal{M}_{\pi})}_{\leqslant 2^{(\pi^{\pi_0})}}}_{\pi'^{(\pi^{\pi_0})}},$$

which is the announced upper bound (the same as the one proved in 4.5.12, the proof being essentially the same).

4.7.20. A canonical fcg is called *maximal* (or *strictly canonical*) if it is not deduced from a strictly more extensive canonical fcg $(\mathcal{F}^{\pi})_{\pi\geqslant\pi_{00}}$ (thus with $\pi_{00} < \pi_0$). So if $\mathcal{M}$ admits an fcga, then it even admits a strictly canonical fcg, and the latter is moreover unique. We will come back to this in the next section.

[page 187]

4.8 $\mathfrak{U}$-accessible categories

Proposition 4.8.1. *Let $\mathcal{M}$ be a $\mathfrak{U}$-category, π_0 an infinite cardinal. Consider the following conditions:*

a) $\mathcal{M}$ satisfies L_{π_0} (i.e. is stable under small $\varinjlim$ large relative to π_0), and contains a small full subcategory S strictly generating, *consisting of π_0-accessible objects.*

[79] Compare with bounds $\operatorname{card}\operatorname{red}\operatorname{Filt}^{\pi} \leqslant 2^{\pi}$, $\operatorname{card}\operatorname{red}\operatorname{Filt}^{\pi}/F \leqslant \pi'^{\pi}$, if $F \in \operatorname{Ob}\operatorname{Filt}^{\pi'}$, with $\pi' \geqslant \pi \geqslant \pi_1$, bounds more pleasant than those of (4.7.19). We should all the same make sure on an example whether the nicer bounds might not be valid in the general case.

b) There exists a small category C, and an equivalence of categories

$$\mathcal{M} \simeq \mathrm{Ind}_{\pi_0}(C).$$

c) $\mathcal{M}$ admits an accessible generalized cardinal filtration $(\mathcal{F}^\pi)_{\pi \geqslant \pi_0}$, having π_0 as initial cardinal, and such that π_0 is a useful cardinal for the said filtration.

p. 65

c′) As in c), with π_0 a regular cardinal (and not only useful) for the accessible generalized cardinal filtration in question, and this filtration being moreover assumed canonical. [page 188]

We then have the diagram of implications

$$c' \Longleftrightarrow c \Longleftrightarrow b \Longrightarrow a.$$

4.8.1.1. I must here make the terminology precise, as to what I mean by a (full) subcategory C *strictly generating* in $\mathcal{M}$. For this, I refer to SGA 4 I 7.2, (i), to the equivalent conditions b) and c) of loc. cit.:

ⓑ $\forall\, F \in \mathrm{Ob}\,\mathcal{M}$, the canonical arrow

$$\varinjlim_{i \in C/F} x_i \longrightarrow F$$

is an isomorphism.

ⓒ The canonical functor

$$\begin{array}{ccc} \mathcal{M} & \longrightarrow & C^\wedge \\ F & \longmapsto & \big(\underbrace{x}_{\in \mathrm{Ob}\, C} \longmapsto \underbrace{\mathrm{Hom}_{\mathcal{M}}(x, F)}_{\in \mathrm{Ob}(\mathrm{Ens})}\big) \end{array}$$

is fully faithful.

These conditions imply:

a) C is generating in $\mathcal{M}$ by strict epimorphisms, i.e. for every F in $\mathcal{M}$, the family of arrows $x \longrightarrow F$ with source in C, with target F, is a strict epimorphic family.

And when $\mathcal{M}$ is stable under fiber products, this condition a) is equivalent to the preceding ones. In loc. cit. I assert

this equivalence without a restrictive hypothesis on $\mathcal{M}$ – but in a marginal annotation I assert that this assertion is *false*, outside the hypothesis of the existence of fiber products in $\mathcal{M}$. Yet I have just checked the proof given in loc. cit. of the implication a $\Longrightarrow$ b, and it is correct. But perhaps the annotation results from a confusion between condition a), and the following condition, weaker a priori, and which resembles it like a brother: [page 189]

a′) There exists a family of arrows $u_i : x_i \longrightarrow F$, with source in C, and with target F, which is a strict epimorphic family (cf. SGA 4 I 10.3).

One feels like saying that when one *enlarges* a strict epimorphic family of arrows, by adding other arrows with the same target, it must remain a strict epimorphic family ("a fortiori") – which gives the impression that a′) must imply a), hence be equivalent to it. But if one tries to prove this hypothetical property of stability for strict epimorphic families, one sees that one only gets there if one assumes that the initial family is moreover p. 66 [page 190]

a *universal epimorphic family* (if fiber products exist in $\mathcal{M}$), or (subject to the preceding proviso) if for every $x' \longrightarrow F$, the sieve of $\mathcal{M}/F'$ inverse image of the sieve of $\mathcal{M}/F$ generated by the $x_i \longrightarrow F$ is an "epimorphic sieve". This does indeed suggest that there must be cases where condition a') is satisfied, but not condition a) (equivalent to b and to c).

But we are then confronted with the very unpleasant situation: if C is a strictly generating subcategory of $\mathcal{M}$, it does not thereby follow that a full subcategory C' of $\mathcal{M}$ such that $C' \supset C$ is also strictly generating! Yet I have always used previously, without thinking twice about it, that if C is strictly generating, the same is true of $C' \supset C$. So I shall have to be very attentive now, and check whether this ± tacit error renders certain [page 191] proofs inoperative, or even, makes certain statements false. Also, I would need to give an example where C is strictly generating, but $C' \supset C$ is not.

Finally, on reflection, these feared troubles disappear. I am going to state a result in due form:

Proposition 4.8.2. *Let C, $\mathcal{M}$ be two $\mathfrak{U}$-categories, $i : C \longrightarrow \mathcal{M}$ a functor, whence a functor*

$$\begin{cases} \varphi : \mathcal{M} \longrightarrow C^\wedge \\ \varphi(X) = (x \longmapsto \mathrm{Hom}_{\mathcal{M}}(i(x), X)) \end{cases} \tag{4.8.2.1}$$

(composite functor $\mathcal{M} \xrightarrow{can.} \mathcal{M}^\wedge \underbrace{\xrightarrow{i^*}}_{\substack{i.e.\ composition \\ with\ i}} C^\wedge$). For every X in $\mathcal{M}$, we consider the category $C/X \simeq C/\varphi(X)$, and the canonical injection (in $\mathcal{M}^\vee$)

$$\varinjlim_{x \in C/X} i(x) \longrightarrow X. \tag{4.8.2.2}$$

1°) Consider the following conditions.

- ⓐ *For every $X \in \mathrm{Ob}\,\mathcal{M}$, the family of arrows $i(x) \longrightarrow X$, indexed by $\mathrm{Ob}\,C/X$, is a strict epimorphic family.*
- ⓑ *For every $X \in \mathrm{Ob}\,\mathcal{M}$, the canonical arrow 4.8.2.2. is an isomorphism, i.e. makes X a $\varinjlim$ of the canonical functors $C/X \longrightarrow \mathcal{M}$.* [page 192]
- ⓒ *The functor φ (4.8.2.1) is fully faithful.*

We then have the implications

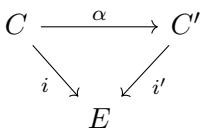
$$\mathrm{b} \Longleftrightarrow \mathrm{c} \Longrightarrow \mathrm{a},$$

and if we assume i fully faithful, then the three conditions a, b, c *are equivalent.* (**N.B.** We say that the functor $i : C \longrightarrow \mathcal{M}$ is *strictly generating*, if the equivalent conditions b, c are satisfied. If C is a subcategory of $\mathcal{M}$ (not necessarily full), we p. 67 say that C is *strictly generating* if the inclusion functor is strictly generating, i.e. satisfies the equivalent conditions b), c). For this, it is therefore necessary that condition a) be satisfied, i.e. that for every X in $\mathcal{M}$, the family of arrows $x \longrightarrow X$ with source in C, with target X, be a strict epimorphic family, and this condition is also sufficient when C is a full subcategory of $\mathcal{M}$.)

2°) . Suppose that i is strictly generating ([81]). Then for every object X of $\mathcal{M}$, the family of arrows $i(x) \longrightarrow X$, indexed by $\mathrm{Ob}\,C/X$, is not only a strict epimorphic family (condition a) in 1°), but moreover, every family of arrows with target X, including the preceding family, is also a strict epimorphic family.

[page 193]

3°) Consider a commutative diagram of functors of $\mathfrak{U}$-categories:



If the functor i is strictly generating ([82]), and if moreover i' is fully faithful, then i' is strictly generating.

Corollary 4.8.3. *Let $\mathcal{M}$ be a $\mathfrak{U}$-category, C a strictly generating subcategory, C' a* full *subcategory of $\mathcal{M}$ which contains C, then C' is strictly generating.*

This is indeed a special case of 4.8.2 3°.

[page 194]

Proof of 4.8.2. The proof of 1°) is that of SGA 4 I 7.2 (i). (Where one assumes that C is a full subcategory of $\mathcal{M}$, but the proof given in fact proves the statement given here.) Let us prove 2°). Let us therefore add to the family of

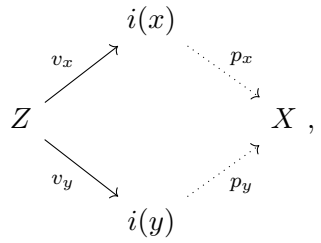
$$i(x) \longrightarrow X \qquad (x \in \mathrm{Ob}\, C/X)$$

a family

$$X_\alpha \xrightarrow{p_\alpha} X \qquad (\alpha \in A)$$

of new arrows, and prove that the total family thus obtained is still a strict epimorphic family. We must therefore prove that if Y is an object of $\mathcal{M}$, and if we have a family of arrows $u_x : i(x) \longrightarrow Y$, $u_\alpha : X_\alpha \longrightarrow Y$, satisfying the usual compatibility condition for pairs of arrows:

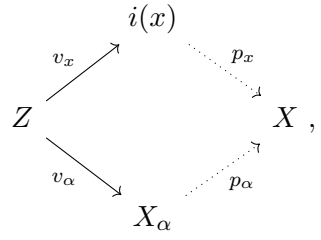
(v_x, v_y)



or

p. 68

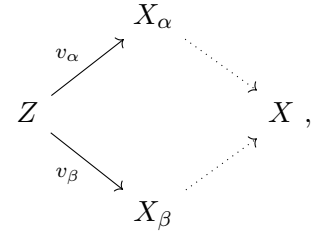
(v_x, v_α)



or

[page 195]

(v_α, v_β)



[81] It is even enough that it satisfy the weaker condition a) of 1°).

[82] In fact, it is enough that it satisfy the weaker condition a) of 1°).

namely the relations

$$\begin{cases} u_x v_x = u_y v_y \\ u_x v_x = u_\alpha v_\alpha \\ u_\alpha v_\alpha = u_\beta v_\beta, \end{cases} \qquad (*)$$

then there exists an arrow

$$u : X \longrightarrow Y$$

(necessarily unique, by property a) of strict epimorphicity of the initial family indexed by C/X) such that we have

$$\begin{cases} up_x = u_x & \text{for every } x \in \operatorname{Ob} C/X \\ up_\alpha = u_\alpha & \text{for every } \alpha \in A. \end{cases} \qquad (*)$$

Now by property a), there exists a unique $u : X \longrightarrow Y$, in such a way that the first of the relations $(*)$ is satisfied, for every $x \in \operatorname{Ob} C/X$. So it is only a matter of proving that for this u, we have for every α the relation

[page 196]

$$up_\alpha = u_\alpha.$$

Now, applying property a) to X_α instead of X, we see that the family of arrows

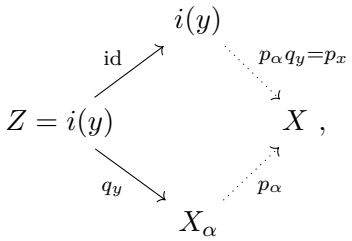
$$i(y) \xrightarrow{q_y} X_\alpha$$

(indexed by C/X_α) is a strict epimorphic family, a fortiori epimorphic. Thus to prove $up_\alpha = u_\alpha$, it is enough to prove that we have

$$(up_\alpha)q_y = u_\alpha q_y \qquad (**)$$

for every y in C/X_α. But this is a consequence of compatibility $(*)$ (second formula), applied to the following case,

p. 69



we therefore have, setting $x = (y, p_\alpha q_y) \in \operatorname{Ob} C/X$,

$$\underbrace{u_x \circ \mathrm{id}_{i(y)}}_{u_x} \;=\; u_\alpha q_y \ ,$$

so to prove $(**)$, it remains to verify

$$u_x \;=\; \underbrace{(up_\alpha)q_y}_{u(p_\alpha q_y) = up_x} \quad ,$$

which is indeed verified by definition of u_x. This finishes proving 2°. [page 197]

Let us prove 3°. Since i' is fully faithful, by 1°, to prove that i' is strictly generating, it suffices to establish that for every object X of $\mathcal{M}$, the family of arrows $i'(x') \longrightarrow X$, indexed by $\operatorname{Ob} C'/X$, is a strict epimorphic family. Now this family contains the family

of $\underbrace{i'(\alpha(x))}_{=i(x)} \longrightarrow X$, indexed by $\mathrm{Ob}\, C/X$. Thus it follows from the hypothesis on i, and from 2°, that this family is a strict epimorphic family. We've won! Whew …

Proof of 4.8.1. We have

$$\mathrm{c}' \implies \mathrm{c} \implies \mathrm{b} \implies \mathrm{a}\,,$$

the first implication tautological, the second resulting from the fact that π_0 a useful cardinal for the fcga under consideration, implies that

$$\mathcal{M} \xleftarrow{\sim} \mathrm{Ind}_{\pi_0}(\mathcal{F}^{\pi_0})\ ,$$

$\mathcal{F}^{\pi_0}$ equivalent to a small category, whence b).

Finally b) $\implies$ a) follows from the fact that $\mathrm{Ind}_{\pi_0}(C)$ is stable under $\varinjlim$ large relative to π_0 (4.4.15), and that C is a strictly generating subcategory of $\mathrm{Ind}_{\pi_0}(C)$, made up of π_0-accessible objects. We need to spell out the [page 198]

Lemma 4.8.4. *Let C be a $\mathfrak{U}$-category, π_0 an infinite cardinal. Then C is strictly generating in $\mathcal{M} = \mathrm{Ind}\, C$ and in $\mathcal{M} = \mathrm{Ind}_{\pi_0} C$.*

Indeed, the canonical functor $\mathcal{M} \longrightarrow C^\wedge$ is fully faithful, and we apply criterion c) of 4.8.2. [page 199]

To finish proving 4.8.1, it therefore remains to establish the implication

$$\mathrm{a} \implies \mathrm{c}'\,.$$

Let us suppose then that $\mathcal{M}$ satisfies L_{π_0}, and admits a small strictly generating subcategory S, made up of π_0-accessible objects, hence a canonical functor p. 70

$$\mathrm{Ind}_{\pi_0} S \longrightarrow \mathcal{M}\ ,$$

extending the inclusion $S \hookrightarrow \mathcal{M}$, so as to commute with the $\varinjlim$ large relative to π_0, and because $S \subset \mathcal{M}_{\pi_0}$, this functor is *fully faithful.* But how to prove that it is essentially surjective? This will not be true, moreover, without a further hypothesis on S. The natural idea is to saturate S under the $\varinjlim$ of cardinality $\leqslant \pi_0$ representable in $\mathcal{M}$, and in particular under direct factors, but even then (and even admitting that $\mathcal{M}_{\pi_0}$ itself is equivalent to a small category, and taking $C = \mathcal{M}_{\pi_0}$), it is nevertheless not clear that the fully faithful functor $\mathrm{Ind}_{\pi_0} C \longrightarrow \mathcal{M}$ is essentially surjective, which would also amount to saying that for every F in $\mathcal{M}$, C/F is large relative to π_0. It is in fact not even clear that C/F is filtering. [page 200]

So in the end, I doubt that we have a) $\implies$ c′) or just a) $\implies$ b), the situation is visibly not well understood.

But let us prove all the same

$$\mathrm{b} \implies \mathrm{c}'\,.$$

For this, we may assume $\mathcal{M} = \mathrm{Ind}_{\pi_0} C$, C small. We then know that the filtration of $\mathcal{M}$ by the $(\mathcal{M}_\pi)_{\pi \geqslant \pi_0}$ is an fcga (prop. 4.7.7 and definition 4.7.8), and even an fcg that is *canonical.* Moreover, I say that the cardinal π_0 is *useful* and even regular for this filtration (which will finish proving c′). We have a slightly more precise result:

Corollary 4.8.5. *Let C be a category equivalent to a small category, π_0 an infinite cardinal, $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C) \subset C^\wedge$. Then:* [page 201]

1°) π_0 is a cardinal useful *and even regular for the canonical fcg $(\mathcal{M}_\pi)_{\pi \geqslant \pi_0}$ of $\mathcal{M}$.*

2°) $\mathcal{M}_{\pi_0} = C^\natural$ (full subcategory of $\mathcal{M} = \mathrm{Ind}_{\pi_0} C$ made up of the direct factors in $\mathcal{M}$ of objects of C), and $\mathcal{M}_{\pi_0}$ is a Karoubi envelope of C.

([54], [55]).

PROOF. p. 71

1°) We already know that

$$\mathrm{Ind}_{\pi_0}\mathcal{M}_{\pi_0} \longrightarrow \mathcal{M} \qquad (*)$$

is fully faithful, and to see that it is essentially surjective, we note that we have

$$\mathcal{M} \simeq \mathrm{Ind}_{\pi_0} C \hookrightarrow \mathrm{Ind}_{\pi_0}\mathcal{M}_{\pi_0} \longrightarrow \mathcal{M}\ ,$$

the composite being isomorphic to the identity functor, is essentially surjective, hence so is $(*)$. This therefore proves that π_0 is a useful cardinal for the filtration under consideration, let us prove that this cardinal is even regular, hence that for

$$F \in \mathrm{Ob}\,\mathcal{M}_\pi\ , \qquad \text{with } \pi > \pi_0,\ \pi^{\pi_0} \geqslant \pi_1$$

we have

$$F = \varinjlim_I x_i\ , \qquad x_i \in \mathrm{Ob}\,\mathcal{M}_{\pi_0},\ I \text{ large relative to } \pi_0,\ \mathrm{card}\, I \leqslant \pi^{\pi_0}.$$

For this, we use (4.7.6.2), page 171, [page 202]

$$\mathcal{M}_\pi \subset \mathrm{Filt}^{\pi^{\pi_0}} \qquad (\text{if } \pi \geqslant \pi_0,\ \pi^{\pi_0} \geqslant \pi_1),$$

and the definition of $\mathrm{Filt}^{\pi'}$ (for $\pi' \geqslant \pi_0$) shows us that this is the conclusion sought.

Let us then prove 2°). Since $\mathcal{M}_{\pi_0}$ is stable under the finite direct $\varinjlim$ which are representable in $\mathcal{M}$ (SGA 4 I 9.9), it is stable under direct factors, and since it contains C, we therefore have $C^\natural \subset \mathcal{M}_{\pi_0}$. The reverse inclusion $\mathcal{M}_{\pi_0} \subset C^\natural$ is seen in the standard way, by writing an object F of $\mathcal{M}_{\pi_0}$ as $\varinjlim_I x_i$, the x_i in C, I large relative to π_0, then F is a direct factor of one of the x_i.

The fact that $C^\natural$ is a *Karoubi envelope* of C in $\mathcal{M} \subset C^\wedge$ comes from the fact that $\mathcal{M}$ is stable under direct factors, and that C is a full subcategory of $\mathcal{M}$, cf. SGA 4 I (?…).

4.8.6. Open questions. We have thus proved 4.8.1 revised and corrected. There remains the question:

1) Give an example where 4.8.1 a) holds, but not b) etc. ([56]).

2) Bring out serviceable additional conditions which, joined to a), imply b), hence c, c′.

Let us also note for the record the following questions. [page 203]

[54] **Corollary 4.8.5.1.** *If C is Karoubian, then $\mathcal{M}_{\pi_0} \subset C^\wedge$ is the category made up of the functors* representable *in $C \overset{\sim}{\hookrightarrow} \mathcal{M}_{\pi_0}$.*

[55] **4.8.5.2.** Spell this out too: Let $u : C \longrightarrow C'$ be a functor, with C, C' equivalent to small categories. Equivalent conditions:

a) u induces $\mathrm{Kar}\, C \xrightarrow{\sim} \mathrm{Kar}\, C'$.

a′) u fully faithful, and every object of C' is isomorphic to a direct factor of an object of the essential image.

b) $\mathrm{Ind}(u) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C')$ is an equivalence.

c) $\mathrm{Ind}_{\pi_0}(u) : \mathrm{Ind}_{\pi_0}(C) \longrightarrow \mathrm{Ind}_{\pi_0}(C')$ is an equivalence (π_0 a given infinite cardinal).

d) $u_! : C^\wedge \longrightarrow C'^\wedge$ is an equivalence.

d′) $u_* : C'^\wedge \longrightarrow C^\wedge$ is an equivalence .

d″) $u^* : C'^\wedge \longrightarrow C^\wedge$ is an equivalence.

[56] It suffices to find a nonregular cardinal, for the canonical fcg of an $\mathcal{M} = \mathrm{Ind}_{\pi_0} C$.

3) Let $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, and $\pi > \pi_0$. Then is Filt^{π} stable under direct factors? Is this true at least if $\pi = \pi^{\pi_0}$, or what amounts to the same thing (cf. 4.7.3), do we have in this case $\mathrm{Filt}^{\pi} = \mathrm{Filt}'^{\pi}$? If so, it would be unnecessary in the description 4.7.7 of cardinal filtrations, in the bound on cardinals (4.7.7.1), $\mathrm{card}\, I \leqslant \pi'^{\pi}$, to put in the restrictive hypothesis $\pi'^{\pi} \geqslant \pi_1$.

To have an affirmative answer to this question, for a given π, it would suffice (and probably be necessary) to have an affirmative answer to this one: Let $\mathcal{D}$, $\mathcal{D}_0$ be two categories large relative to π_0, $\mathcal{D}_0$ a direct factor of $\mathcal{D}$, $\mathcal{D}$ admitting a cofinal category I large relative to π_0, of cardinality $\leqslant \pi$. Then is the same true of $\mathcal{D}_0$? If p is a retraction of $\mathcal{D}$ onto $\mathcal{D}_0$, the difficulty is that I do not know whether p is a cofinal functor. In fact, I see that this is not always the case, even if $\mathcal{D}$ is finite. Let us take e.g. for $\mathcal{D}_0$ a category p. 72 having final object e, and an arrow $x \longrightarrow e$ which is not an isomorphism, and admitting [page 204] a section (e.g. $\mathcal{D}_0$ the full subcategory of $\mathbf{\Delta}$ formed by $\mathbf{\Delta}_0$, $\mathbf{\Delta}_1$: $\mathbf{\Delta}_0 \underset{\kappa}{\overset{\partial_0}{\underset{\partial_1}{\rightrightarrows}}} \mathbf{\Delta}_1$). Let $\mathcal{D}$ be obtained from $\mathcal{D}_0$ by adding a final object $\tilde{e}$. Then $\mathcal{D}_0$, $\mathcal{D}$ are filtering, and we get a retraction of $\mathcal{D}$ onto $\mathcal{D}_0$, sending $\tilde{e}$ to x. This functor is not cofinal, for a cofinal functor must send $\tilde{e}$ to e.

More generally, if $\mathcal{D}_0$ is a category such that there exists an arrow

$$\alpha : e_{\mathcal{D}_0^{\wedge}} \longrightarrow x_0 \qquad (*)$$

in $\mathcal{D}_0^{\wedge}$, with $x_0 \in \mathrm{Ob}\, \mathcal{D}_0$, then we see that $\mathcal{D}_0$ is large relative to π_0 whatever the infinite cardinal π_0, and that on the other hand, again denoting by $\mathcal{D}$ the category obtained from $\mathcal{D}_0$ by adding a final object to it, there is a retraction of $\mathcal{D}$ onto $\mathcal{D}_0$ which sends $e_{\mathcal{D}}$ to x_0. This retraction is cofinal only if x_0 is a final object of $\mathcal{D}_0$, i.e. if and only if $(*)$ is an isomorphism. Let us suppose this is not so. But there is a cofinal functor from the subcategory I_0 of $\mathcal{D}_0$, such that $\mathrm{Ob}\, I_0 = \{x_0\}$, $\mathrm{Fl}\, I_0 = \mathrm{End}_{I_0}(x_0) = \{p, \mathrm{id}_{x_0}\}$, where p is the composite $x_0 \xrightarrow{\text{can.}} e_{\mathcal{D}_0^{\wedge}} \xrightarrow{\alpha} x_0$. So no counterexample this way!

[page 205]

4) Let us consider $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, C small, and let us consider the canonical filtration $(\mathcal{M}_{\pi})_{\pi \geqslant \pi_0}$. Give examples where there are non-useful cardinals, or better, where for every cardinal π, exists a cardinal $\pi' > \pi$ which is non-useful. (This will give the example asked for in 1°.) On the other hand, give examples of useful cardinals which are not regular – if possible, as before, such cardinals as large as we want.

Definition 4.8.7. Let $\mathcal{M}$ be a $\mathfrak{U}$-category, π_0 an infinite cardinal. We say that $\mathcal{M}$ is π_0-*accessible* if $\mathcal{M}$ is equivalent to a category $\mathrm{Ind}_{\pi_0}(C)$, C a small category, i.e. if it satisfies the equivalent conditions b, c, c′ of 4.8.1 (implying condition a), probably strictly weaker). We say that $\mathcal{M}$ is accessible if there exists a cardinal such that $\mathcal{M}$ is π_0-accessible (**N.B.** if π_0 is such a cardinal, then every cardinal $\pi > \pi_0$ such that $\pi^{\pi_0} = \pi$, in particular the cardinals π'^{π_0}, with $\pi' \geqslant 2$, have the same property), or what amounts [page 206] to the same thing by 4.8.1, if $\mathcal{M}$ admits an fcga, or again a canonical fcg.

Comment 4.8.8. This definition is in contradiction with the terminology introduced in SGA 4 I 9.4 (p. 114); we shall henceforth designate by the names "weakly π_0-accessible category", "weakly accessible category" the notions introduced in 9.4 – i.e. the categories admitting a full generating subcategory (not even strictly generating) whose objects are π_0-accessible (resp. accessible).

Let us note that if $\mathcal{M}$ is π_0-accessible (resp. accessible), the condition of 4.8.1 tells us that $\mathcal{M}$ is weakly π_0-accessible (resp. weakly accessible), and even that there exists a small full *strictly* generating subcategory, which is made up of π_0-accessible objects (resp. p. 73

accessible). I very much doubt that this last necessary condition is also sufficient for $\mathcal{M}$ to be π_0-accessible (resp. accessible). But by 4.6.3.3 (p. 150), this is so if we assume $\mathcal{M}$ stable under small $\varinjlim$, or what amounts to the same thing, assuming L_{π_0}, stable under direct limits of cardinality $\leqslant \pi_0$.

[page 207]

Proposition 4.8.9. *Let $\mathcal{M}$ be an accessible $\mathfrak{U}$-category (4.8.7), π_0 a cardinal such that $\mathcal{M}$ satisfies L_{π_0}. Equivalent conditions:*

a) $\mathcal{M}$ is a paratopos category (cf. 4.6.3, 3°, p. 150).

b) $\mathcal{M}$ is stable under small direct limits.

b′) $\mathcal{M}$ is stable under direct limits of cardinality $\leqslant \pi_0$.

b″) (If $\mathcal{M}$ is equivalent to a categories $\mathrm{Ind}_{\pi_0} C$, with C a small category that is Karoubian.*) The category C is stable (in itself) under direct limits of cardinality $\leqslant \pi_0$.*

c) $\mathcal{M}$ is stable under small inverse limits.

c′) (Under the hypothesis $\mathcal{M} \simeq \mathrm{Ind}_{\pi_0}(C)$, and when we give ourselves a cardinal π such that $\pi = \pi^{\pi_0} \geqslant \pi_1 = \sup(\pi_0, \mathrm{card}\,\mathrm{red}\,C)$.) $\mathcal{M}_\pi$ is stable under finite $\varprojlim$, and $\mathcal{M}$ is stable under small infinite products (or again, under filtering $\varprojlim$).

c″) There exists an infinite cardinal π, and a small category $\mathcal{D}$ stable under finite $\varprojlim$, such that $\mathcal{M} \simeq \mathrm{Ind}_\pi(\mathcal{D})$, moreover $\mathcal{M}$ stable under infinite products (or again under filtering $\varprojlim$).

([57], [58]).

Proof. The following implications are tautological, or already ± known:

$$\mathrm{a} \Longrightarrow \mathrm{b} \xRightarrow{\text{cf. 4.4.14}} \mathrm{b}' \xRightarrow{\bullet} \mathrm{b}''$$
$$\mathrm{a} \Longrightarrow \mathrm{c} \left(\xRightarrow{\bullet} \mathrm{c}' \Longrightarrow \mathrm{c}'' \right),$$

where only the dot-marked implications $\mathrm{b}' \Longrightarrow \mathrm{b}''$ and $\mathrm{c} \Longrightarrow \mathrm{c}'$ call for a proof. For $\mathrm{b}' \Longrightarrow \mathrm{b}''$, we note that the hypothesis C Karoubian implies that $C \simeq \mathcal{M}_{\pi_0}$ (4.8.5, 2°), now we know that $\mathcal{M}_{\pi_0}$ is stable in $\mathcal{M}$ under direct limits of cardinality $\leqslant \pi_0$ (and these are representable in $\mathcal{M}$ by b'), hence $\mathcal{M}_{\pi_0}$ admits $\varinjlim$ of the type under consideration, hence C too.

[page 208]

$\mathrm{c} \Longrightarrow \mathrm{c}'$. Let us note that if $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, then the $\varprojlim$ representable in $\mathcal{M}$ are computed in $C^\wedge$. It follows that if $\pi \geqslant \pi_1$, then Filt^π is stable under finite $\varprojlim$ (Filt^π then being given by condition a″ of 4.7.1), on the other hand, if in addition $\pi^{\pi_0} = \pi$, π is a useful cardinal, hence $\mathcal{M} \simeq \mathrm{Ind}_\pi(\mathcal{D})$, where $\mathcal{D} = \mathrm{Filt}^\pi = \mathcal{M}_\pi$. It now suffices to see that $\mathrm{Ind}_\pi(\mathcal{D})$ is then stable under finite $\varprojlim$, to conclude that the same holds for $\mathcal{M}$, hence that $\mathcal{M}$ is stable under arbitrary inverse limits, if it is under small products or under small filtering $\varprojlim$, whence c′. It therefore remains to prove the following lemma.

Lemma 4.8.10. *Let C be a small category, π_0 an infinite cardinal, $\pi_1 = \sup(\pi_0, \mathrm{card}\,red\,C)$. Suppose $\mathrm{Ind}_{\pi_0} C$ is stable under finite $\varprojlim$ (resp. and of a certain type). Then for every $\pi \geqslant \pi_1$, Filt^π is stable under the said $\varprojlim$. (Thus also $\mathcal{M}_\pi = \mathrm{Filt}^\pi$, if $\pi = \pi^{\pi_0}$.) More generally, let J be a small category such that $\mathrm{Ind}_{\pi_0} C$*

p. 74

[page 209]

[57] **N.B.** The equivalence of c′, c″ with the other conditions is not established (here, but only in 4.11 below.)

[58] **N.B.** The criteria c′, c″ seem to me more technical and less useful .

is stable under $\varprojlim$ *of type* J*, and let* $c = \operatorname{card} \mathrm{red}\, J$*, suppose that* $\pi^{\max(1,c)} = \pi$ *(e.g.* c *finite). Then* Filt^π *is stable in* $\mathcal{M}$ *under* $\varprojlim$ *of type* J*. (Thus the same holds for* $\mathcal{M}_\pi = \mathrm{Filt}^\pi$*, if in addition* $\pi^{\pi_0} = \pi$*.)*

We use criterion a″ of 4.7.1, namely $\operatorname{card} F(x) \leqslant \pi$ for every $x \in C$, for membership in Filt^π. We may suppose J reduced, hence $\operatorname{card} \mathrm{Ob}\, J \leqslant c$. Since the inclusion functor $\mathrm{Ind}_{\pi_0}(C) \longrightarrow C^\wedge$ commutes with arbitrary $\varprojlim$, the computation of $\varprojlim_{j \in J} x_j$ in $C^\wedge$ fiber by fiber shows that in $C^\wedge$, we have

$$X \overset{\text{def}}{=} \varprojlim_{j \in J} x_j \hookrightarrow \underbrace{\prod_{j \in \mathrm{Ob}\, J} x_j}_{\text{product in } C^\wedge} ,$$

whence

$$\operatorname{card} X(x) \leqslant \prod_{j \in \mathrm{Ob}\, J} \underbrace{\operatorname{card} x_j(x)}_{\leqslant \pi} \leqslant \pi^{\max(1,c)} ,$$

whence (with no other hypothesis on π than $\pi \geqslant \pi_1$)

$$\varprojlim_{j \in J} x_j \in \mathrm{Ob}\, \mathrm{Filt}^{\pi^{\max(1,c)}} , \tag{4.8.10.1}$$

whence the desired conclusion if $\pi^{\max(1,c)} = \pi$, q.e.d.

[page 210]

We are going to prove the additional implications

$$\mathrm{b}'' \implies \mathrm{b} , \qquad \mathrm{c} \implies \mathrm{b} , \qquad \mathrm{c}'' \overset{?}{\implies} \mathrm{c} ,$$

which completes the proof.

The implication b″ ⟹ a is contained in 4.6.3.

For the implication c ⟹ b, we note that the stability of $\mathcal{M}$ under small $\varinjlim$ is equivalent to the assertion that every small inverse limit of representable functors $\mathcal{M} \xrightarrow{\xi_i}$ Ens is representable. Now the objects of $\mathcal{M}$ being accessible ($\mathcal{M}$ being accessible), the representable functors ξ_i are accessible, hence so is their $\varprojlim$ (SGA 4 I 9.7). On the other hand, the functor ξ is left exact, as a $\varprojlim$ of representable hence left exact functors. Thus the fact that ξ is representable follows from the

Lemma 4.8.11. *Let* $\mathcal{M}$ *be an accessible category, stable under finite* $\varprojlim$*,* $\mathcal{M} \longrightarrow$ Ens *a functor. For* ξ *to be prorepresentable, it is necessary and sufficient that it be left exact and accessible. Thus (SGA 4 I 8.12.8.1) if* $\mathcal{M}$ *is stable under small* $\varprojlim$ *,* ξ *is representable if and only if it commutes with the said* $\varprojlim$ *, and is accessible.*

The "it is necessary" is included as a reminder, we have just seen it as a consequence of the fact that every object of $\mathcal{M}$ is accessible. The "it is sufficient" is essentially SGA 4 I 8.13.2.b), except that in loc. cit. one supposes $\mathcal{M}$ equipped with an fc that is *strict.* But the proof given in loc. cit. also shows for an fcga, choosing in loc. cit. (p. 137) π such that ξ is π-accessible, and moreover π is useful for the generalized cardinal filtration under consideration.

p. 75

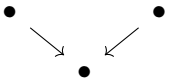
[page 211]

It remains to prove c″ $\overset{?}{\implies}$ c, subject to this really being true. This would be a consequence of the following lemma (where $\mathcal{D}$, π are replaced by C, π_0).

Lemma 4.8.12 (?) ([88])**.** *Let* C *be a small category,* π_0 *an infinite cardinal. If* C *is stable under finite* $\varprojlim$*, then* $\mathrm{Ind}_{\pi_0} C$ *as well.*

The analogous assertion for $\mathrm{Ind}(C)$, instead of $\mathrm{Ind}_\pi C$, is established in SGA 4 I 8.9.1 c). But it is not clear without more that $\mathrm{Ind}_\pi C$ is stable in $\mathrm{Ind}(C)$ under finite $\varprojlim$,

[page 212]

in the case under consideration. More precisely, if $F = \varprojlim_{j\in J} F_j$, the F_j in $\mathrm{Ind}_\pi C$, F in $\mathrm{Ind}(C) \subset C^\wedge$, we know that C/F is filtering and we have to see that it is filtering large relative to π_0. We are reduced to the three cases where $J = \emptyset$, $J = e \amalg e$, and $J =$  . The first two cases are essentially trivial (for e, and a product of two categories large relative to π_0, are large relative to π_0), the third case on the other hand does not look trivial a priori. This is visibly related to the question of indicial representations of functors $J \longrightarrow \mathrm{Ind}_{\pi_0}(C)$, for $\mathrm{card}\,\mathrm{red}\,J \leqslant \pi_0$ – i.e. the validity of 4.4.13 b) outside the hypothesis that C is stable under $\varinjlim$ of cardinality $\leqslant \pi_0$. We will have to come back to this very soon, for this question now appears to me to be a key question, for the stability properties of the notion of accessible category (without any hypothesis of existence of $\varinjlim$ or $\varprojlim$).

Proposition 4.8.13 ([89]). *Let C be a category equivalent to a small category, and let $\pi_0 \geqslant \mathrm{card}\,\mathrm{red}\,C$ be an infinite cardinal. Then $\mathcal{M} = \mathrm{Ind}_{\pi_0} C = \mathcal{M}_{\pi_0}$ (i.e. all the objects of $\mathrm{Ind}_{\pi_0} C$ are π_0-accessible), and $\mathcal{M}$ is a Karoubi envelope of C.*

Let us point out at once the following corollaries. [page 213]

Corollary 4.8.14 ([90]). *Let $\mathcal{M}$ be a $\mathfrak{U}$-category. Equivalent conditions:*

a) $\mathcal{M}$ is accessible, and is equivalent to a small category.

a′) $\mathcal{M}$ is accessible, and there exists an infinite cardinal π_0 such that all the objects of $\mathcal{M}$ are π_0-accessible (i.e. $\mathcal{M} = \mathcal{M}_{\pi_0}$).

b) $\mathcal{M}$ is equivalent to a small category, and is Karoubian, i.e. stable under direct factors ([91]).

The equivalence of a) and a′) is clear, since $\mathcal{M}$ admits a canonical generalized cardinal filtration $(\mathcal{M}_\pi)_{\pi\geqslant\pi_0}$, consisting of categories equivalent to small categories (condition a) for fcg), and such that every such subcategory is contained in one of the $\mathcal{M}_\pi$ (follows from condition b) for fcg). Since $\mathcal{M}_{\pi_0}$ is equivalent to a small category, and is Karoubian (being stable under filtering $\varinjlim$ large relative to π_0), it is also clear that a′) $\Longrightarrow$ b). It remains to prove b) $\Longrightarrow$ a′), now this follows at once from 4.8.13 applied to $\mathcal{M}$, which moreover shows that for π_0 one can take any infinite cardinal such that $\pi_0 \geqslant \mathrm{card}\,\mathrm{red}\,\mathcal{M}$. p. 76

4.8.15. Proof of 4.8.13. We are going to prove: [page 214]

Lemma 4.8.15.1. *Under the conditions of 4.8.13, let $F \in \mathrm{Ob}\,\mathcal{M}$, $x \in \mathrm{Ob}\,C$, then*

$$\mathrm{card}\,F(x) \;\leqslant\; \pi_0 \;. \qquad (*)$$

This therefore means, by 4.7.1, a″ $\Longrightarrow$ b for $\pi \geqslant \pi_1$, taking into account that here $\pi_1 = \pi_0$, $\pi = \pi_0$, that we have $F \in \mathrm{Filt}^{\pi_0}$, whence a fortiori $F \in \mathcal{M}_{\pi_0}$, which implies $\mathcal{M} = \mathcal{M}_{\pi_0}$, i.e. proposition 4.8.13, taking 4.8.5 into account. So it all comes down to proving $(*)$.

Let us write

$$F \;=\; \varinjlim_{i\in I} x_i \;, \qquad \text{the } x_i \in \mathrm{Ob}\,C,\ I \text{ ordered set large relative to } \pi_0,$$

whence

$$F(x) \;=\; \varinjlim_{i\in I} \mathrm{Hom}_C(x, x_i) \;.$$

[88] Lemma not proved here – cf. proof 4.11.7 (unfinished).

[89] *Not proved.* But OK if $\pi_0 > \mathrm{card}\,\mathrm{red}\,C$, cf. 4.15.10, p. 462.

[90] *Not proved.* Yes after all, cf. 4.15.10, and 4.15.11, pages 462-463.

[91] **N.B.** This implies that $\mathcal{M}$ and $\mathcal{M}^\circ$ are accessible, and I conjecture that the converse is true as well.

We may suppose

$$\operatorname{card} C \ \leqslant \ \pi_0 \ ,$$

replacing C by an equivalent reduced category if need be.

For every $\xi \in \operatorname{Ob} C$, let

$$I_\xi \ = \ \{i \in I \mid x_i = \xi\} \ ,$$

then

$$I \ = \ \bigcup_{\xi \in \operatorname{Ob} C} I_\xi \ ,$$

a union of a family of subsets of I of cardinality $\operatorname{card} \operatorname{Ob} C \leqslant \pi_0$. Since I is large relative to π_0, we see that this implies that one of the I_ξ is cofinal in I. Replacing I by I_ξ if need be, we may suppose $I_\xi = I$, i.e. that all the x_i are equal to the same element ξ. Replacing C by the full subcategory $C_0 = \{\xi\}$ if need be, we may suppose C reduced to a single element. (For if we prove that F, as an object of $\operatorname{Ind}_{\pi_0}(C_0)$, is a direct factor of an object of C_0 (i.e. of ξ), the same will hold for F as an object of $\operatorname{Ind} C$). Thus we may suppose [page 215]

$$C \ = \ \{\xi\} \ ,$$

thus C given by the monoid

$$M \ = \ \operatorname{End}_C(\xi) \ , \qquad \operatorname{card} M \ \leqslant \pi_0 \ ,$$

and giving the direct system $(x_i)_{i \in I}$ amounts to giving a system of elements

$$\alpha_{ji} \ \in \ M \qquad (i, j \in I, \ i \leqslant j)$$

satisfying

$$\alpha_{ii} \ = \ 1 \ , \qquad \alpha_{kj}\alpha_{ji} \ = \ \alpha_{ki} \ \text{ if } i \leqslant j \leqslant k.$$

p. 77

We have

$$F(\xi) \ = \ \varinjlim_{i \in I} M_i \qquad \left\{ \begin{array}{l} \text{where } M_i = M \ \forall\, i \in I, \text{ and } M_i \longrightarrow M_j \\ \text{is given by} \\ \qquad\qquad u \longmapsto \alpha_{ji} u \ . \end{array} \right.$$

I have been able to reduce, by similar arguments ([92]), to the case where the monoid $M = \operatorname{End}(\xi)$ is a monoid of monomorphisms, i.e. [page 216]

$$uv \ = \ uv' \ \implies \ v \ = \ v' \qquad \text{if } u, v, v' \in M \ ,$$

which implies that the transition arrows in the direct system of the $M_i = M$ are monomorphisms. It follows that one can have $\operatorname{card} F(\xi) \leqslant \pi_0$ (I being large relative to π_0) only if α_{ji_0} is an isomorphism for $j \geqslant i_0$, and i_0 large enough, i.e. if the induced direct system for $j \geqslant i_0$ is constant, a fortiori that the direct system under consideration is essentially constant. But I have just spent nearly four hours trying to prove this point, without succeeding! This seems more delicate to prove than I had expected, but I still believe that it really is true.

[page 217]

[92] For these reductions, cf. 4.15.8 (p. 453-463). The ultimate reduction is found in 4.15.8.9 (p. 461), and its corollary 4.15.10.

4.9 The indicial representation theorem for $\mathrm{Ind}_\pi(C)$

4.9.1 . Let C be a category equivalent to a small category, π_0 an infinite cardinal,

$$\mathcal{M} \;=\; \mathrm{Ind}_{\pi_0}(C) \;\subset\; C^\wedge \;, \tag{4.9.1}$$

J a category equivalent to a small category. Consider the fully faithful inclusion $C \hookrightarrow \mathcal{M}$, and the corresponding inclusion of $\mathfrak{U}$-categories

$$\underline{\mathrm{Hom}}(J,C) \;\hookrightarrow\; \underline{\mathrm{Hom}}(J,\mathcal{M}) \qquad \text{(fully faithful)}. \tag{4.9.2}$$

We know, since $C \subset \mathcal{M}_{\pi_0}$, that the objects of $\underline{\mathrm{Hom}}(J,C)$ give π_0-*accessible* objects of $\underline{\mathrm{Hom}}(J,\mathcal{M})$, at least if we suppose

$$\mathrm{card}\,\mathrm{red}\,J \;\leqslant\; \pi_0 \;, \tag{4.9.3}$$

which we shall do henceforth. (Cf. 4.5.8, 2°, p. 119.) As the right-hand side of (4.9.2) satisfies L_{π_0} ($\mathcal{M}$ satisfies it), it follows that this functor (4.9.2) extends to

$$\mathrm{Ind}_{\pi_0}(\underline{\mathrm{Hom}}(J,C)) \;\hookrightarrow\; \underline{\mathrm{Hom}}(J,\underbrace{\mathcal{M}}_{=\mathrm{Ind}_{\pi_0}(C)}) \;, \tag{4.9.3}$$

and this functor is *fully faithful* thanks to the accessibility result we have just stated. I p. 78
am interested in establishing cases where this functor is essentially surjective, i.e. is an *equivalence* of categories. Let us note that for

$$\varphi \;:\; J \longrightarrow \mathcal{M}$$

to belong to the essential image, it is necessary and sufficient that there exist a functor

$$\Phi \;:\; I \;\longrightarrow\; \underline{\mathrm{Hom}}(J,C) \;,\; I \text{ large relative to } \pi_0 \text{ (and we may suppose } I \text{ ordered)} \;, \tag{4.9.4}$$

such that we have [page 218]

$$\varphi \;\xleftarrow{\sim}\; \varinjlim_{i\in I} \Phi(i) \;, \tag{4.9.5}$$

or what amounts to the same thing, such that we have

$$\varphi(j) \;\xleftarrow{\sim}\; \varinjlim_{i\in I} \Phi(i,j) \;, \qquad \text{for } j \in J \;, \tag{4.9.6}$$

where we have put

$$\Phi(i,j) \;=\; \Phi(i)(j) \;,$$

thus interpreting Φ as a functor

$$\Phi \;:\; I \times J \;\longrightarrow\; C \;; \tag{4.9.7}$$

the functorial law of $\varphi(j)$ given by (4.9.6) being evident. In other words, the φ in the essential image of (4.9.3) are those which admit an *indicial representation* by a functor Φ in the form (4.9.4), or in the form (4.9.7), with *in addition* I large relative to π_0. I recall in this connection the

Proposition 4.9.2. *Under the preceding conditions on C, π_0, J (in particular* $\mathrm{card}\,\mathrm{red}\,J \leqslant \pi_0$*), if $\varphi : J \longrightarrow \mathcal{M} \overset{\mathrm{def}}{=} \mathrm{Ind}_{\pi_0}(C)$, for φ to belong to the essential image of the fully faithful functor (4.9.3), it is necessary and sufficient that it satisfy the following two conditions:* [page 219]

a) $\underline{\mathrm{Hom}}(J,C)/\varphi$ *is large relative to* π_0.

b) For every $j \in J$, *the functor*

$$\underline{\mathrm{Hom}}(J,C)/\varphi \longrightarrow C/\varphi(j) \qquad \begin{cases} \textit{functor between filtering categories large relative to } \pi_0, \textit{ by virtue of a),} \end{cases} \tag{4.9.2.1}$$

induced by the functor

$$\begin{cases} \underline{\mathrm{Hom}}(J,C) & \longrightarrow & C \\ F & \longmapsto & F(j)\ , \end{cases} \tag{4.9.2.2}$$

is cofinal.

The main result in the present section is the following: p. 79

Theorem 4.9.3. *Under the general conditions of 4.9.1 for* C, π, J *(in particular* $\operatorname{card}\operatorname{red} J \leqslant \pi$*), the fully faithful functor (4.9.3) is an equivalence of categories in each of the following two cases:*

a) C *is stable under countable filtering* $\varinjlim$.

b) There exists a cardinal $\mu < \pi$, *and a cardinal* $\mu' > \mu$ (**N.B.** *We may suppose that* μ' *is the successor of* μ, *hence* $\mu' \leqslant \pi$*), such that* C *is stable under direct limits large relative to* μ, *and of cardinality* μ'.

([94]).

Remark 4.9.4.

a) This theorem, case a), contains theorem 4.4.13, part b) (p. 95-96), where one supposes that C is stable under direct limits of cardinality $\leqslant \pi_0$, the hypothesis a) of 4.9.3 being considerably weaker.

b) I presume that in the absence of a restrictive hypothesis of type a) or b) above, the fully faithful functor (4.9.3) is not always essentially surjective, even in the particular case where J is a finite category, e.g. $J = B_G$, $G = \{\pm 1\}$. For a counterexample in the very closely related case where we take $\mathcal{M} = \mathrm{Ind}(C)$ (instead of $\mathrm{Ind}_{\pi_0}(C)$), and where we consider the functor (fully faithful if J is a category equivalent to a *finite* category) [page 220]

$$\mathrm{Ind}(\underline{\mathrm{Hom}}(J,C)) \longrightarrow \underline{\mathrm{Hom}}(J, \mathrm{Ind}(C)),$$

I refer to exercise SGA 4 I 8.8.8.

The main application of theorem 4.9.3 consists in the

Corollary 4.9.5. *Let* $\mathcal{M}$ *be an* accessible *category (4.8.7),* J *a category equivalent to a small category. Then the category* $\mathcal{M}(J^\circ) = \underline{\mathrm{Hom}}(J, \mathcal{M})$ *is accessible. More precisely, let* π_0 *be a* useful *cardinal for* $\mathcal{M}$ *(i.e. such that* $\mathrm{Ind}_{\pi_0}(\mathcal{M}_{\pi_0}) \xrightarrow{\sim} \mathcal{M}$, *i.e. a cardinal such that* $\mathcal{M}$ *is* π_0*-accessible), let* π *be an infinite cardinal such that* $\pi^{\pi_0} = \pi \geqslant \operatorname{card}\operatorname{red} J$ ([95]), *i.e.* $\pi = c^{\pi_0}$, *where* c *is any cardinal* $c \geqslant \sup(\pi_0, \operatorname{card}\operatorname{red} J)$, *then* $\underline{\mathrm{Hom}}(J, \mathcal{M})$ *is* π*-accessible, more precisely we have an equivalence of categories*

$$\underline{\mathrm{Hom}}(J,\mathcal{M}) \xleftarrow{\sim} \mathrm{Ind}_\pi(\underline{\mathrm{Hom}}(J, \mathcal{M}_\pi))\ . \tag{4.9.5.1}$$

[94] Common case: there exists a limit ordinal α, such that 1) $\operatorname{card}\alpha \leqslant \pi$. 2) The direct limits type $I_\alpha = \{\beta \mid \beta < \alpha\}$ are representable in C. [page 221]

[95] We thus have $\pi = \pi^{\pi_0} \geqslant 2^{\pi_0} > \pi_0$.

We know indeed, with the assumptions made on π_0, π, that π is useful for $\mathcal{M}$, i.e. that

$$\mathcal{M} \xleftarrow{\sim} \mathrm{Ind}_\pi(\mathcal{M}_\pi) \ ,$$

on the other hand, $\mathcal{M}_\pi$ is stable under $\varinjlim$ large relative to π_0 and of cardinality $\leqslant \pi$. We may therefore apply 4.9.3 b), taking $C = \mathcal{M}_\pi$, π and π_0 as here (hence $\pi_0 < 2^{\pi_0} \leqslant \pi$), and $\pi_0' = \pi$.

4.9.6 ([96]). The rest of the present section 4.9 is devoted to the proof of 4.9.3, i.e. to verifying, for $J \longrightarrow \mathcal{M}$, conditions a), b) of 4.9.2. In all that follows, we therefore fix a p. 80

$$\varphi \ : \ J \longrightarrow \mathcal{M} \ = \ \mathrm{Ind}_\pi(C) \ . \tag{4.9.6.1}$$

We may suppose J reduced, hence small and $\mathrm{card}\, J \leqslant \pi_0$. We shall consider the functor deduced from the functor (4.9.2.1)

$$\underline{\mathrm{Hom}}(J, C)/\varphi \longrightarrow \prod_{j \in \mathrm{Ob}\, J} C/\varphi(j) \ , \tag{4.9.6.2}$$

from which the functors in question are deduced by composing with the projection functors

$$\prod C/\varphi(j) \longrightarrow C/\varphi(j_0) \ . \tag{4.9.6.3}$$

Since the factor categories $C/\varphi(j)$ are filtering (and even large relative to π), their product is also filtering, and even large relative to π (immediate), and likewise for the partial products. It follows at once that the projections (4.9.6.3) are cofinal, by the [page 222]

Lemma 4.9.6.1. *Let I, I' be two categories with I' 0-connected (e.g. filtering), then the projection functor $pr_1 : I \times I' \longrightarrow I$ is cofinal.*

Indeed, for $i \in \mathrm{Ob}\, I$, we have

$$i\backslash(I \times I') \ \simeq \ (i\backslash I) \times I' \ ,$$

now $i\backslash I$ and I' being 0-connected, the same is true of their product.

Lemma 4.9.6.2 ([97]). *Let $(I_\alpha)_{\alpha \in A}$ be a family of* filtering *categories (resp. large relative to π). Then $I_A \overset{\mathrm{def}}{=} \prod_{\alpha \in A} I_\alpha$ is filtering (resp. large relative to π), and for every subset $B \subset A$, the canonical projection $I_A \longrightarrow I_B$ is cofinal.*

([98]). The stability of the property "filtering" or "large relative to π" by AQT . We have, if $B' = A\backslash B$, $I_A \simeq I_B \times I_{B'} \longrightarrow I_B$, and we apply lemma 4.9.6.1 to conclude that this functor is cofinal, $I_{B'}$ being filtering, hence 0-connected.

Since the functor (4.9.6.3) is cofinal, to establish that (4.9.2.1) is so, it suffices to prove that the functor (4.9.6.2) is so, which we shall do. Let us note moreover that the said functor is *faithful*, for it fits into the commutative square [page 223]

$$\begin{array}{ccc} \underline{\mathrm{Hom}}(J, C)/\varphi & \longrightarrow & \prod C/\varphi(j) \\ \downarrow & & \downarrow \\ \underline{\mathrm{Hom}}(J, C) & \longrightarrow & C^{\mathrm{Ob}\, J} \ , \end{array}$$

where the vertical arrows and the second horizontal arrow are faithful. Ultimately, to p. 81

[96] Preliminary results (for the proof of 4.9.3).

[97] Beware, it is not true that an infinite product of 0-connected categories is 0-connected!

[98] Cf. the elegant proof below, 4.9.6.6.

establish theorem 4.9.3, we shall study the *faithful* functor (4.9.6.2)

$$\begin{array}{l} f \ : \ \underline{\mathrm{Hom}}(J,C)/\varphi \ \longrightarrow \ \prod_{j\in J_0} C/\varphi(j) \\ (\text{where we have put } J_0 = \mathrm{Ob}\, J,\ J_1 = \mathrm{Fl}\, J)\ , \end{array} \qquad (*)$$

and we shall prove:

ⓐ This faithful functor $(*)$ is *cofinal.*

ⓑ The source category $\underline{\mathrm{Hom}}(J,C)/\varphi$ is large relative to π_0.

For this we shall bring out an auxiliary lemma on the cofinality of *faithful* functors:

Lemma 4.9.6.3 ([99]). *Let $f : I \longrightarrow I'$ be a functor, and let us consider the conditions:* [page 224]

a) I filtering, f cofinal, i.e. (in the presence of I filtering) f satisfies the familiar conditions (F 1), (F 2).

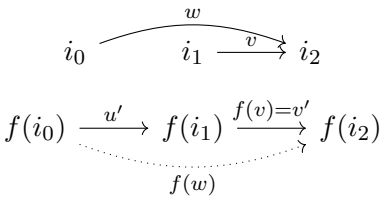
b) I' is filtering, and the $i'\backslash I$ (for $i' \in \mathrm{Ob}\, I'$) are filtering.

b′) I' 0-connected, satisfies PF (2) ([100]) (equalization of double arrows), and the categories $i'\backslash I$ are filtering for every $i' \in \mathrm{Ob}\, I'$.

b″) I' is filtering, the $i'\backslash I$ pseudo-filtering, i.e. satisfy condition PF $1'$.

c) I' filtering, and f satisfies F 1, as well as the following condition.

(F 3) *For every pair (i_0, i_1) of objects of I and every arrow $f(i_0) \xrightarrow{u'} f(i_1)$, there exist an object i_2 of I, and an arrow $v' : f(i_1) \longrightarrow f(i_2)$, such that v' and $v'u'$ can be lifted to arrows $i_1 \xrightarrow{v} i_2$ and $i_0 \xrightarrow{w} i_2$ (i.e. there exist i_2 and $v : i_1 \longrightarrow i_2$, $w : i_0 \longrightarrow i_2$ such that we have $f(v)u' = f(w)$).*



1) We have the implications

$$a \iff b \iff b' \implies b'' \iff c\ ,$$

and if f is faithful *(e.g. I ordered), the five conditions a), b), b′), b″), c) are equivalent.*

2) Let π be an infinite cardinal, and let us consider the conditions: p. 82 [page 225]

a_π) I large relative to π, f cofinal, i.e. F 1, F 2.

b_π) I' large relative to π, the $i'\backslash I$ ($i' \in \mathrm{Ob}\, I'$) also.

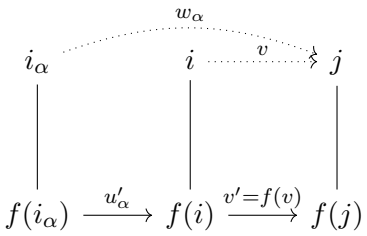
b_π'') I' large relative to π, the $i'\backslash I$ ($i' \in \mathrm{Ob}\, I'$) are π-pseudo-filtering, i.e. satisfy PF ${1_\pi}'$.

c_π) I' large relative to π, and f satisfies F 1 and

[99] This lemma is part of the general sorites on cofinality and filtering categories.

[100] e.g. PF 2 is satisfied if I' is *ordered*, without having to suppose in advance I' filtering, only 0-connected.

(F 3_π) ([101]). *For every family $(i_\alpha)_{\alpha\in A}$ of objects of I, with* card $A \leqslant \pi$, *and $i \in \mathrm{Ob}\, I$, and a family of arrows $u'_\alpha : f(i_\alpha) \longrightarrow f(i)$, exists $j \in \mathrm{Ob}\, I$ and an arrow $v' : f(i) \longrightarrow f(j)$, such that v'* and *the $v'u'_\alpha$ come from arrows of I, i.e. there exist arrows $v : i \longrightarrow j$, $w_\alpha : i_\alpha \longrightarrow j$ such that $f(v)u'_\alpha = f(w_\alpha)\ \forall\ \alpha \in A$.*



This being laid down, we have

$$a_\pi \iff b_\pi \implies c_\pi \implies b_\pi'' \ ,$$

and if f is faithful, or I ordered, the four conditions a_π, b_π, b_π'', c_π are equivalent.

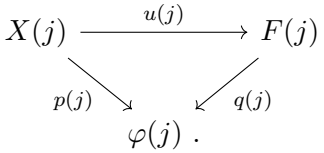
([102]).

4.9.6.4. N.B. In the case of interest to us, where f is the functor $(*)$ (p. 223), we know that I' is large relative to π. We would like to establish that I is so and that f is cofinal, i.e. condition a_π. We shall prove it by taking c_π (which implies a_π since f is faithful). It therefore remains to prove, for the functor f in $(*)$, p. 223, the following two conditions: [page 226]

F 1) For every system of arrows in $\mathcal{M}$

$$X(j) \xrightarrow{p(j)} \varphi(j)\ , \qquad j \in J_0 = \mathrm{Ob}\, J,\ \text{the } X(j) \text{ in } \mathrm{Ob}\, C\ ,$$

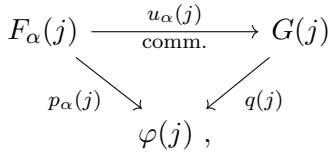
there exists a functor $F : J \longrightarrow C$ and $F \xrightarrow{q} \varphi$, finally a system of arrows $X(j) \xrightarrow{u(j)} F(j)$ in the $C/\varphi(j)$, i.e. making the diagrams commutative



F 3_π) For every system $(F_\alpha)_{\alpha\in A}$ (card $A \leqslant \pi$) of elements in $\underline{\mathrm{Hom}}(J, C)$, and arrows

$$F_\alpha \xrightarrow{p_\alpha} \varphi \qquad \text{in } \underline{\mathrm{Hom}}(J, \mathcal{M}),$$

and for G in $\underline{\mathrm{Hom}}(J, C)$ and $G \xrightarrow{q} \varphi$ in $\underline{\mathrm{Hom}}(J, \mathcal{M})$, given a system of arrows $u_\alpha(j)$,

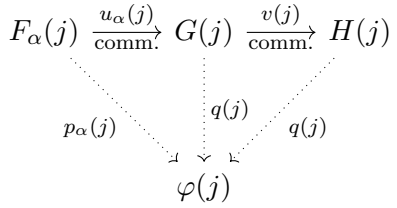


$u_\alpha(j)$ arrow in $C/\varphi(j)$, p. 83

there exists H in $\underline{\mathrm{Hom}}(J, C)$ and $H \xrightarrow{q} \varphi$ in $\underline{\mathrm{Hom}}(J, \mathcal{M})$ and an arrow $G \xrightarrow{v} H$ in $\underline{\mathrm{Hom}}(J, C)/\varphi$, such that for every $\alpha \in A$, the system of composite arrows

[101] **N.B.** The condition F 3_π is tautologically satisfied if f is *fully faithful.*

[102] Proof of the proposition given in 4.9.6.7, page 228. [page 227]



(for $j \in J_0 = \operatorname{Ob} J$) comes from a *natural transformation* $w_\alpha : F_\alpha \longrightarrow H$.

To prove 4.9.3, we are thus reduced to proving the two preceding propositions F 1), F 3_π), under one of the two conditions a) or b) of 4.9.3.

Corollary 4.9.6.5 (of 4.9.6.3). *Let $(f_\alpha : I_\alpha \longrightarrow I'_\alpha)$ be a family of cofinal functors, the source categories being filtering. Then the product functor $f = \prod f_\alpha : I = \prod_\alpha I_\alpha \longrightarrow I' = \prod_\alpha I'_\alpha$ is cofinal.*

Indeed, we must verify that the categories $i'\backslash I$, for $i' \in \operatorname{Ob} I'$, are 0-connected. Now if $i' = (i'_\alpha)_{\alpha\in A}$, we have

$$i'\backslash I \;\simeq\; \prod_\alpha (i'_\alpha \backslash I_\alpha)\ ,$$

and by 4.9.6.3, 1°) a $\Longrightarrow$ b, the $i'_\alpha\backslash I_\alpha$ are filtering, hence their product is filtering (cf. 4.9.6.2), hence 0-connected, q.e.d. [page 228]

Corollary 4.9.6.6. *Let π be an infinite cardinal. Every product of categories large relative to π is large relative to π.*

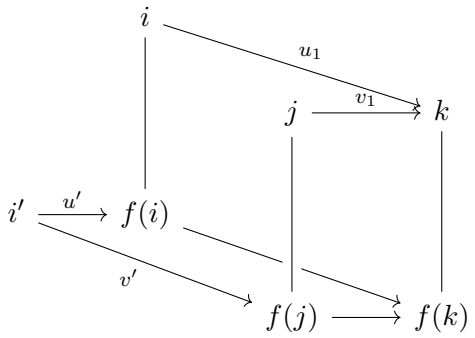
Let $I = \prod_{\alpha\in A} I_\alpha$, let B be a set of cardinality $\leqslant \pi$, we must prove that $I \xrightarrow{\operatorname{diag}_I} I^B$ is cofinal. Now this functor is identified with the product of the similar functors $I_\alpha \longrightarrow I_\alpha^B$, which are cofinal, the source categories being moreover filtering. Thus we may conclude by corollary 4.9.6.5.

4.9.6.7. Proof of 4.9.6.3. (Part 1°.)

a $\Longrightarrow$ b. We already know that a) implies that I' is filtering (SGA 4 I 8.1.3 b). Remains to prove that for every $i' \in \operatorname{Ob} I'$, $i'\backslash I$ is filtering ([103]). Two conditions PF 1′ and PF 2 to verify (in addition to the fact that $i'\backslash I \neq \emptyset$, which is known by cofinality of f, implying that $i'\backslash I$ is 0-connected).

1) Let (i, u'), (j, v') be two elements of $i'\backslash I$, we must see that they are bounded above by a third element. Now I being filtering, i, j are bounded above by k. We would therefore be done if we knew that $f(u_1)u' = f(v_1)v'$, say w', for then (k, w') would do.

p. 84



[page 229]

[103] Short-circuit: we already know that $i'\backslash I$ is 0-connected, on the other hand it is *locally* filtering, since coétale over I which is filtering, hence it is filtering.

But we apply F 2 for f to the double arrow in question $i' \rightrightarrows f(k)$, we find $w : k \longrightarrow \bar{k}$,

$$\begin{array}{ccc} k & \xrightarrow{w} & \bar{k} \\ \Big| & & \Big| \\ i' \rightrightarrows f(k) & \longrightarrow & f(\bar{k}) \end{array}$$

such that

$$\underbrace{f(w)(f(u_1)u')}_{=f(wu_1)u'} = \underbrace{f(w)(f(v_1)v')}_{f(wv_1)v'} ,$$

and replacing k by $\bar{k}$, and u_1, v_1 by wu_1, wv_1, we find what we want.

2) PF (2) for $i'\backslash I$, i.e. we give ourselves a diagram opposite ,

$$\begin{array}{ccccc} & & i & \underset{u_2}{\overset{u_1}{\rightrightarrows}} & j & \overset{w}{\cdots\!\!\rightarrow} & j' \\ & & \Big| & & \Big| & & \\ i' & \xrightarrow{u'} & f(i) & \underset{f(u_2)}{\overset{f(u_1)}{\rightrightarrows}} & f(j) & \overset{f(w)}{\cdots\!\!\rightarrow} & f(j') \end{array} ,$$

with $f(u_1)u' = f(u_2)u'$, say v' (double arrow in $i'\backslash I$ from (i, u') to (j, v')), we want to equalize it on the right. Now I being filtering, there exists $w : j \longrightarrow j'$ such that $wu_1 = wu_2$, now this arrow defines an arrow $(j, v') \longrightarrow (j', f(w)v')$, which equalizes the given double arrow.

b $\Longrightarrow$ b′ is clear.

b) $\Longrightarrow$ a). Since the $i'\backslash I$ are filtering, hence 0-connected, f is cofinal, it remains to prove that I is filtering, hence to check PF 0 (I 0-connected), PF 1, PF 2 for I. I is 0-connected, since I' is, and f cofinal, i.e. $\mathbf{W}_0$-coaspheric, whence PF 0.

[page 230]

Let us prove PF 1, i.e. that if $\begin{array}{ccc} & & i_1 \\ & \nearrow & \\ i_0 & & \\ & \searrow & \\ & & i_2 \end{array}$ is given, we can complete it to a commutative square.

$$\begin{array}{ccccc} & & i_1 & & \\ & \overset{u_1}{\nearrow} & & \overset{v_1}{\searrow} & \\ i_0 & & & & i_3 \\ & \underset{u_2}{\searrow} & & \underset{v_2}{\nearrow} & \\ & & i_2 & & \end{array}$$

p. 85

$$\begin{array}{ccccc} & & f(i_1) & & \\ & \overset{f(u_1)}{\nearrow} & & \overset{f(v_1)}{\searrow} & \\ f(i_0) & & & & f(i_3) \\ & \underset{f(u_2)}{\searrow} & & \underset{f(v_2)}{\nearrow} & \\ & & f(i_2) & & \end{array}$$

But let us consider the objects $\xi_1 = (i_1, f(u_1))$, $\xi_2 = (i_2, f(u_2))$ of $f(i_0)\backslash I$, and using PF 1′ for this category, we find $\xi_3 = (i_3, w')$ in the said category, and arrows from ξ_1, ξ_2 to ξ_3, i.e. $v_1 : i_1 \longrightarrow i_3$, such that $\underbrace{f(v_1)f(u_1)}_{f(v_1u_1)} = w'$, $\underbrace{f(v_2)f(u_2)}_{f(v_2u_2)} = w'$, hence

$$f(v_1u_1) \;=\; f(v_2u_2)\ .$$

Thus, we have a double arrow $i_0 \overset{v_1u_1}{\underset{v_2u_2}{\rightrightarrows}} i_3$, equalized by f. If we prove PF 2 for I, we find $i_3 \xrightarrow{\alpha} \bar{i}_3$ such that $\alpha(v_1u_1) = \alpha(v_2u_2)$, i.e. $(\alpha v_1)u_1 = (\alpha v_2)u_2$, and replacing i_3 by $\bar{i}_3$, v_1, v_2 by αv_1, αv_2 if necessary, we have won.

It remains to prove PF 2 for I, i.e. to equalize a double arrow

$$i_0 \overset{u_1}{\underset{u_2}{\rightrightarrows}} i_1\ .$$

Now I' satisfying PF 2, we can equalize the double arrow $f(u_1)$, $f(u_2)$ by an arrow $f(i_1) \xrightarrow{v'} i'$ in I', and with $i'\backslash I$ nonempty, we can suppose that i' is of the form $f(i_2)$. [page 231]

$$\begin{array}{ccccc}
i_0 & \overset{u_1}{\underset{u_2}{\rightrightarrows}} & i_1 & \longrightarrow & i_2 \\
\big| & & & & \\
f(i_0) & \overset{f(u_1)}{\underset{f(u_2)}{\rightrightarrows}} & f(i_1) & \xrightarrow{v'} & f(i_2)
\end{array}$$

Let us admit for a moment condition F 3 of c), which we are going to apply to (i_1, i_2, v'). There thus exist $i_2 \xrightarrow{v} \bar{i}_2$, $w : i_1 \longrightarrow \bar{i}_2$ such that we have $f(v)v' = f(w)$. It follows that $f(w)$ equalizes $f(u_1)$ and $f(u_2)$, i.e. that we have $f(wu_1) = f(wu_2)$, i.e. the initial double arrow is replaced by $(\bar{u}_1 = wu_1, \bar{u}_2 = wu_2)$, such that $f(\bar{u}_1) = f(\bar{u}_2)$. In short, we are reduced to the case of a double arrow (u_1, u_2) such that $f(u_1) = f(u_2)$, say v'. But then let $\xi_1 = (i_1, v')$, $\xi_0 = (i_0, \mathrm{id}_{f(i_0)})$, we thus have a double arrow $(u_1, u_2) : \xi_0 \rightrightarrows \xi_1$ in $f(i_0)\backslash I$. This category is filtering, so this double arrow is equalizable on the right, which implies at once that the double arrow (v_1, v_2) *in* I is also equalizable on the right. [page 232]

So it remains to prove that b′) implies condition F 3. But let us consider the two objects $\xi_0 = (i_0, \mathrm{id}_{f(i_0)})$, $\xi_1 = (i_1, u')$ of $f(i_0)\backslash I$, and use the filtering condition PF 1 for this category, so we find an object $\xi_2 = (i_2, v')$, and two arrows $\begin{array}{ccc} \xi_0 & \overset{w}{\searrow} & \\ & & \xi_2 \\ \xi_1 & \underset{v}{\nearrow} & \end{array}$, i.e. $i_0 \xrightarrow{w} i_2$ and $i_1 \xrightarrow{v} i_2$ such that we have $f(w)\mathrm{id}_{f(i_0)} = f(v)u' = v'$, whence $f(w) = f(v)u'$, which is condition F 3. p. 86

$$\begin{array}{ccccc}
 & & w & & \\
 & \overset{\cdots\cdots}{\frown} & & \overset{\cdots\cdots}{\frown} & \\
i_0 & & i_1 & \overset{v}{\cdots\cdots\rightarrow} & i_2 \\
\big| & & \big| & & \big| \\
f(i_0) & \xrightarrow{u'} & f(i_1) & & f(i_2) \\
 & \underset{\cdots\cdots}{\smile} & & \underset{\cdots\cdots}{\smile} & \\
 & & v' & &
\end{array}$$

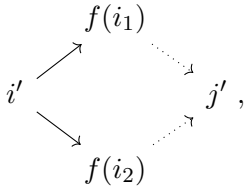
Thus we have established

$$\text{a} \iff \text{b} \iff \text{b}' ,$$

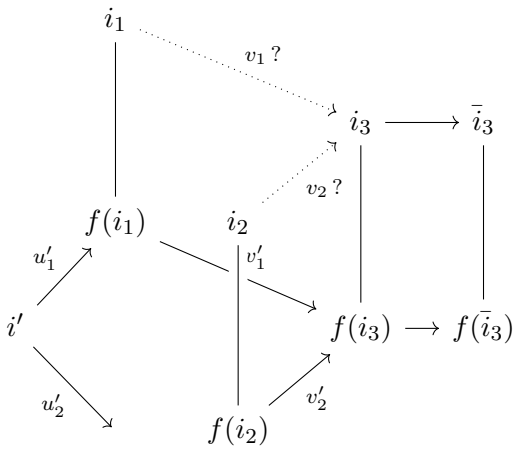
let us now prove

a $\Longrightarrow$ c. We already know that a) implies I' filtering, and it remains to establish F 3. Now I being filtering, there exist arrows $i_0 \xrightarrow{w} i_2 \xleftarrow{v} i_1$. We do not necessarily have [page 233] $f(w) = f(v)u'$, we have here a double arrow $f(i_0) \rightrightarrows f(i_2)$ in I'. But by condition F 2 for f, there exists $i_2 \xrightarrow{\alpha} \bar{i}_2$ in I such that $f(\alpha)$ equalizes the said double arrow, hence $f(\underbrace{\alpha w}_{\bar{w}}) = f(\underbrace{\alpha v}_{\bar{v}})u'$, and replacing i_2 by $\bar{i}_2$, and w, v by $\bar{w} = \alpha w$, $\bar{v} = \alpha v$ if necessary, we win.

It remains to prove (for part 1°) that if f is *faithful*, then c) $\Longrightarrow$ b). It thus remains to prove that c) implies that for i' in I', $i'\backslash I$ is filtering. We already know by F 1 that $i'\backslash I$ is nonempty, it remains to prove conditions PF 1′, PF 2 for $i'\backslash I$. With I' filtering, we can complete



and by F 1 we can suppose j' of the form $f(i_3)$. p. 87

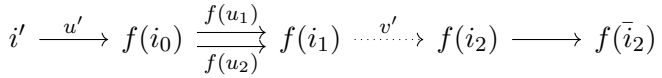


Applying F 3 to i_1, i_3, v'_1, we can suppose (replacing i_3 by $\bar{i}_3$ if necessary) that v'_1 comes from a $v_1 : i_1 \longrightarrow i_3$. Applying F 3 again to i_2, i_3, v'_2, we can suppose that v'_2 also comes from a $v_2 : i_2 \longrightarrow i_3$. Then $\xi_3 = (i_3, v'_1u'_1 = v'_2u'_2)$ is a common upper bound of $\xi_1 = (i_1, u'_1)$, $\xi_2 = (i_2, u'_2)$. [page 234]

Let us finally prove PF 2 for $i'\backslash I$, so we have a diagram opposite with $f(u_1)u' =$

$f(u_2)u'$ (double arrow in $i'\backslash I$), we want to equalize in $i'\backslash I$.





But I' being filtering, we can equalize $(f(u_1), f(u_2))$ by $f(i_1) \longrightarrow j'$, and because of F 1 we can suppose $j' = f(i_2)$. Applying F 3 once more to (i_1, i_2, v'), we find $i_1 \longrightarrow \bar{i}_2$ such that $f(w)$ equalizes $f(u_1)$, $f(u_2)$:

$$f(wu_1) \;=\; f(wu_2)\ .$$

But f being *faithful*, this implies $wu_1 = wu_2$, and we have thus equalized the double arrow of $i'\backslash I$ by $w : (i_1, -) \longrightarrow (\bar{i}_2, -)$.

4.9.6.8. N.B.

1°) If we do not suppose f to be faithful, we see that c) nevertheless implies that the $i'\backslash I$ satisfy PF 1′ (they are therefore pseudo-filtering and 0-connected). So f is cofinal.

[page 235]

We see, by the same argument, that I satisfies PF 1′, i.e. is pseudo-filtering. Thus, if we introduce the condition:

(b″) I' is filtering, the $i'\backslash I$ ($i' \in \mathrm{Ob}\, I'$) are pseudo-filtering (i.e. satisfy PF 1′) (hence f cofinal).

We find the implications

$$(\mathrm{a} \iff \mathrm{b} \iff \mathrm{b}') \implies (\mathrm{b}'' \iff \mathrm{c})\ ,$$

and the equivalent conditions c), b″) imply that I is also pseudo-filtering, and that for every double arrow $\xi_0 \underset{u_2}{\overset{u_1}{\rightrightarrows}} \xi_1$ in an $i'\backslash I$ or in I, there exists an arrow $\xi_1 \xrightarrow{v} \xi_2$ in the same category, such that vu_1 and vu_2 : $\xi_0 \rightrightarrows \xi_2$ have *the same image in* I'. So if f is faithful, then these conditions c, b″ imply that I and the $i'\backslash I$ also satisfy the double arrow condition PF 2, hence are filtering, so in this case the five conditions a) b) b′) c) b″) are equivalent. But also when I is ordered, we know that the double arrow condition in I is automatically satisfied.

p. 88

[page 236]

2°) Let us also note that in condition b′), we cannot drop the hypothesis that I' satisfies PF 2 (in addition to I' 0-connected and the $i'\backslash I$ filtering). To see this, let us take $I' = I$, $f = \mathrm{id}$, so hypothesis a ($\iff$ b $\iff$ b′) means that I is filtering, f being of course cofinal. But weakened b′ means that I is 0-connected and the $i\backslash I$ are filtering, and we know that this does *not* imply that I is filtering, even if we suppose I pseudo-filtering. A simple example is given by I defined by

$$\mathrm{Ob}\, I \;=\; \mathbf{N}\ , \qquad \mathrm{Hom}_I(i,j) \;=\; \begin{cases} \emptyset & \text{if } \; i \;>\; j \\ \{\mathrm{id}_i\} & \text{if } \; i \;=\; j \\ \{\pm 1\} & \text{if } \; i \;<\; j\ , \end{cases}$$

the composition of arrows being carried out by multiplication of the signs $\varepsilon_{k,j}\varepsilon'_{j,i} = (\varepsilon\varepsilon')_{k,i}$ (putting $\mathrm{id}_i = (+1)_{i,i}$). The categories $i\backslash I$ are isomorphic to one another and are *ordered* and pseudo-filtering, hence filtering, on the other hand, I is pseudo-filtering, yet it is not filtering, for it does not satisfy PF 2.

3°) On the other hand, if f is not faithful, then c) (equivalent to b'') does not imply a), as we see by taking $I' = e$; then b″) is equivalent to I pseudo-filtering, which does not imply that I is filtering.

4.9.6.9. Proof of 4.9.6.3, part 2°.

[page 237]

This is essentially "the same" proof as for part 1°).

$a_\pi) \Longrightarrow b_\pi)$. We can short-circuit the direct proof, by looking, for a set A of cardinality $\leqslant \pi$, at the diagram

$$\begin{array}{ccc} I & \xrightarrow{\delta_A} & I^A \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f^A} \\ I' & \xrightarrow{\delta'_A} & I'^A \end{array}$$

. Since f is cofinal, I filtering, f^A is also cofinal (4.9.6.5), on the other hand, I large relative to π means that δ_A is cofinal ($\forall$ A with $\operatorname{card} A \leqslant \pi$). So $f^A\delta_A = \delta'_A f$ is cofinal, hence $\delta'_A f$ and f are, hence δ'_A also, which means that I' is large relative to π.

It remains to prove that for i' in I', $i'\backslash I$ is large relative to π. This follows from the fact that $i'\backslash I \xrightarrow{p} I$ is coétale, that $i'\backslash I$ is 0-connected, and from the

Lemma 4.9.6.9.1. *Let I be a filtering category (resp. large relative to π). Then for every J over I coétale over I and 0-connected, J is filtering (resp. large relative to π).*

The filtering case is already known (cf. II …). For the "large relative to π" case, we p. 89 are going to use the

Corollary 4.9.6.9.2. *Let J be a category. For J to be large relative to π, it is necessary and sufficient that it be 0-connected, and satisfy PF_π 2 (cf. page 81), as well as the following variant of PF_π 1′.* [page 238]

PF_π 1) For every object x_0 of J, $x_0\backslash J$ satisfies PF_π 1′, i.e. for every family of arrows $x_0 \xrightarrow{u_\alpha} y_\alpha$ ($\alpha \in A$, $\operatorname{card} A \leqslant \pi$), there exists an arrow $x_0 \xrightarrow{v} z$ and arrows $v_\alpha : y_\alpha \longrightarrow z$, with $v_\alpha u_\alpha = v$ $\forall\, \alpha \in A$.

Proof. If J is large relative to π, then PF_π 1) follows at once from the conjunction of PF_π 1′ and PF_π 2, hence the "only if". Conversely, suppose that J satisfies PF_π 1, PF_π 2, and is 0-connected, we must prove that it satisfies PF_π 1′. But we already know that the conditions in question, which imply PF 1, PF 2, 0-connected, imply that J is filtering. Thus if we have a family $(x_\alpha)_{\alpha\in A}$, $\operatorname{card} A \leqslant \pi$, choosing an $x_0 \in \operatorname{Ob} J$, we can for every α find y_α dominating x_0 and x_α, hence arrows $\begin{array}{ccc} & x_\alpha & \\ \nearrow & & \searrow \\ x_0 & & y_\alpha \end{array}$, we can apply PF_π 1 to them, hence arrows $y_\alpha \longrightarrow z$, and by composing, arrows $x_\alpha \longrightarrow z$, q.e.d.

Let us now prove the lemma. It suffices (by the preceding corollary) to prove that J satisfies PF_π 1 and PF_π 2. Now PF_π 1 is immediate, since $\forall\, x_0$ in J, $x_0\backslash J \xrightarrow{\sim} y_0\backslash I$, where y_0 is the image of x_0 in I. As for PF_π 2, if we have $x_0 \overset{u_\alpha}{\rightrightarrows} x_1$ in J, hence $g(x_0) \overset{g(u_\alpha)}{\rightrightarrows} g(x_1)$, we equalize in I by $g(x_1) \xrightarrow{w} y$, but since g is coétale, $\exists\, x_1 \xrightarrow{v} x_2$ in J such that $g(v) = w$, so we have $g(vu_\alpha) = g(vu_\beta)$ for $\alpha, \beta \in A$, and since g is faithful, we conclude from this that $vu_\alpha = vu_\beta$, so v equalizes $(u_\alpha)_{\alpha\in A}$, q.e.d. [page 239]

Let us prove $b_\pi) \Longrightarrow a_\pi)$: If I' and the $i'\backslash I$ are large relative to π, then I is too (and f is cofinal, the $i'\backslash I$ being 0-connected, since filtering). Since we already know that I is 0-connected (I' being so and f cofinal), it remains, by 4.9.6.9.2, to verify PF_π 1 and PF_π 2.

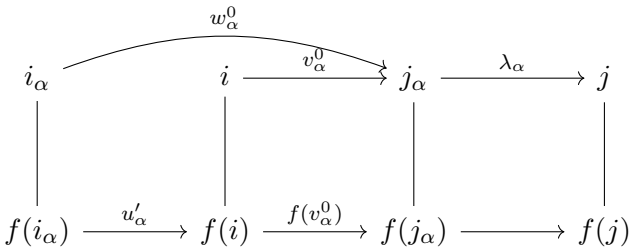
[page 240]

The proof given in 1° for the similar filtering case (implication $b' \Longrightarrow$ a, pages 231-232), applies mutatis mutandis. (Property F 3 has already been established in part 1°), as a consequence of b, and can therefore be used freely.)

Thus we have proved $a_\pi \Longleftrightarrow b_\pi$. Let us prove $a_\pi \Longrightarrow c_\pi$. We already know that a_π implies that I' is large relative to π, and of course condition F 1, it remains to establish the validity of condition F 3_π. By F 3, $\forall\, \alpha \in A$ we can find j_α in I and $w^0_\alpha : i_\alpha \longrightarrow j_\alpha$, $v^0_\alpha : i \longrightarrow j_\alpha$ such that

$$f(v^0_\alpha)u'_\alpha \;=\; f(w^0_\alpha)\ .$$

p. 90



Since I satisfies PF 1_π (cf. 4.9.6.9.2), there exist j in I and $\lambda_\alpha : j_\alpha \longrightarrow j$, such that $\lambda_\alpha v^0_\alpha = \lambda$ is independent of α, then we have for every $\alpha \in A$

$$f(\lambda)u'_\alpha \;\Big(=\; f(\lambda_\alpha)\underbrace{f(v^0_\alpha)u'_\alpha}_{f(w^0_\alpha)}\Big) \;=\; f(\underbrace{\lambda_\alpha w^0_\alpha}_{w_\alpha})\ ,$$

so $\lambda : i \longrightarrow j$ and the $w_\alpha = \lambda_\alpha w^0_\alpha : i_\alpha \longrightarrow j$ satisfy the desired conditions.

$c_\pi \Longrightarrow b''_\pi$. We must prove that if c_π is satisfied, then the $i'\backslash I$ satisfy PF $1'_\pi$.

[page 241]

Let then $\xi_\alpha = (i_\alpha, u'_\alpha)$ ($\alpha \in A$, $\operatorname{card} A \leqslant \pi$) be a family of elements of $i'\backslash I$, since I' satisfies PF 1_π (cf. 4.9.6.9.2), there exist j' in I' and arrows $u''_\alpha : f(i_\alpha) \longrightarrow j'$, such that $u''_\alpha u'_\alpha = u'$ independent of α. Since f satisfies F 1, we can assume that j' is of the form $f(j)$. Applying F 3_π to the i_α, j, u''_α, we find $v : j \longrightarrow k$ and $w_\alpha : i_\alpha \longrightarrow k$, such that

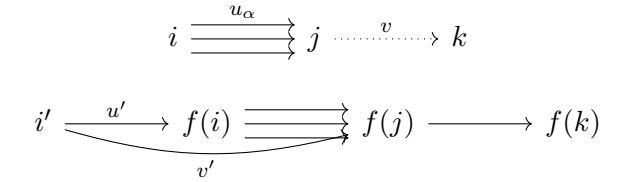
$$f(v)u''_\alpha \;=\; f(w_\alpha) \qquad \forall\, \alpha \in A\ ,$$

hence

$$f(w_\alpha)u'_\alpha \;=\; f(v)\,\underbrace{u''_\alpha u'_\alpha}_{u'} \;=\; f(v)u' \qquad \text{independent of } \alpha,$$

say w', so the w_α define an arrow from the ξ_α to $\xi = (k, w')$, q.e.d.

$b''_\pi \Longrightarrow b_\pi$ if f is faithful. We are reduced to proving that if b''_π is satisfied, f faithful, then the categories $i'\backslash I$, which we already know satisfy PF $1'_\pi$, also satisfy PF 2_π. Let then $\xi = (i, u')$, $\eta = (j, v')$ be two objects of $i'\backslash I$, and $(u_\alpha)_{\alpha\in A}$ a family of arrows $\xi \longrightarrow \eta$ (with $\operatorname{card} A \leqslant \pi$), hence arrows in I such that $f(u_\alpha)u' = v'\ \forall\, \alpha \in A$.



[page 242]

p. 91

Since I' is large relative to π, we can equalize by $f(j) \longrightarrow k'$, and since $k'\backslash I \neq \emptyset$, we can assume k' of the form $f(k)$. Now by the implication $b'' \Longrightarrow a \Longrightarrow c$ of part 1°) (cf. page 235) and since $b''_\pi \Longrightarrow b''$, we already know that f satisfies F 3. Applying it to j, k and $f(j) \longrightarrow f(k)$, we can assume that $f(j) \longrightarrow f(k)$ comes from $v : j \longrightarrow k$. We shall therefore have $f(v)f(u_\alpha) = f(v)f(u_\beta)\ \forall\, \alpha, \beta \in A$, so $f(vu_\alpha) = f(vu_\beta)$. Since f is faithful, this implies that $vu_\alpha = vu_\beta$, i.e. v equalizes the u_α. Besides, v can be regarded as an arrow *in* $i'\backslash I$, from (j, v') to $(k, f(v)v')$. Q.e.d.

4.9.6.10. When I is *ordered*, then the four conditions a_π, b_π, c_π, b''_π are equivalent, since the $i'\backslash I$ are ordered, and it is clear that then $b''_\pi \Longleftrightarrow b_\pi$.

[page 243]

4.10 The indicial representation theorem: proof

We are first going to reformulate theorem 4.9.3 in a more general form, in terms of *cofibered* categories. Let

$$\mathcal{D} \longrightarrow J \tag{4.10.1}$$

be a cofibrant functor, where J is equivalent to a small category, and likewise for the fiber categories $\mathcal{D}_j$ ($j \in \mathrm{Ob}\, J$). We give ourselves an infinite cardinal π. Let

$$\mathcal{M}_j \;=\; \mathrm{Ind}_\pi(\mathcal{D}_j) \qquad (j \in \mathrm{Ob}\, J)\ , \tag{4.10.2}$$

then the $\mathcal{M}_j$ are the fibers of a cofibered category $\mathcal{N}$ over J, which I also denote by $\mathrm{Ind}_\pi(\mathcal{D}/J)$, so we put

$$\mathcal{N} \;=\; \mathrm{Ind}_\pi(\mathcal{D}/J)\ . \tag{4.10.3}$$

We thus have a fully faithful inclusion of cofibered categories over J

$$\mathcal{D} \hookrightarrow \mathcal{N}\ ,$$

hence an inclusion

$$\underline{\Gamma}(\mathcal{D}/J) \longrightarrow \underline{\Gamma}(\mathcal{N}/J)\ . \tag{4.10.4}$$

In the case where $\mathcal{D}$ is of the form $C \times J$, then $\mathcal{N}$ is of the form $\mathcal{M} \times J$, with $\mathcal{M} = \mathrm{Ind}_\pi(C)$, and the inclusion (4.10.4) is none other (up to isomorphism) than [page 244]

$$\underline{\mathrm{Hom}}(J, C) \longrightarrow \underline{\mathrm{Hom}}(J, \mathcal{M})\ .$$

Returning to the general case, we also see that:

1°) The second member of (4.10.4) is stable under $\varinjlim$ large relative to π, which are computed fiber by fiber.

2°) The first member of (4.10.4) consists of π-accessible objects of the second.

Hence a *fully* faithful functor

$$\mathrm{Ind}_\pi(\underline{\Gamma}(\mathcal{D}/J)) \hookrightarrow \underline{\Gamma}(\mathcal{N}/J)\ . \tag{4.10.4}$$

.

Theorem 4.10.1. *Suppose that we have* p. 92

$$\mathrm{card}\,\mathrm{red}\, J \;\leqslant\; \pi\ , \tag{4.10.1.1}$$

and that there exists a limit ordinal α, such that:

1) $\mathrm{card}\,\alpha \leqslant \pi$. [page 245]

2) The direct limits of type $I_\alpha = \{\beta \mid \beta < \alpha\}$ are representable in the categories $\mathcal{D}_j$ ($j \in \mathrm{Ob}\, J$), and for every arrow $\alpha : j \longrightarrow j'$ in J , $\alpha_* : \mathcal{D}_j \longrightarrow \mathcal{D}_{j'}$ commutes with the said limits.

Then the canonical fully faithful functor (4.10.4) is an equivalence of categories.

This will be contained in the slightly more precise result:

Corollary 4.10.2. *The hypotheses being those of 4.10.1, with in addition*

$$\mathrm{card}\, J \ \leqslant\ \pi \tag{4.10.2.1}$$

(which is equivalent to (4.10.1.1) if J is reduced, a case to which we can of course reduce in 4.10.1). Let $\varphi \in \mathrm{Ob}\, \underline{\Gamma}(\mathcal{N}/J)$, and let us consider the canonical faithful *functor:*

$$\underline{\Gamma}(\mathcal{D}/J)/\varphi \longrightarrow \prod_{j\in J} \mathcal{D}_j/\varphi(j)\ . \tag{4.10.2.2}$$

Then the source category $\underline{\Gamma}(\mathcal{D}/J)/\varphi$ is large relative to π, and the functor in question is cofinal.

[page 246]

From this, the theorem follows in an essentially sorital way, just as in the preceding special case 4.9.3 (cf. page 223) ([105]).

To establish the corollary, since the target category of (4.10.2.2) is large relative to π, as a product of such categories (cf. 4.9.6.6, page 228), and the functor faithful, it suffices by virtue of 4.9.6.3, 2°, to verify conditions F 1 and F 3_π (page 225) for the functor in question.

We are going to interpret the functor (4.10.2.2) in terms of a cofibered category

$$\mathcal{D}_\varphi \longrightarrow J\ , \tag{4.10.5}$$

whose fibers are the $\mathcal{D}_j/\varphi(j)$. We note that if $\alpha : j \longrightarrow j'$ is an arrow of J, then we have

$$\begin{cases} \alpha_* : \mathcal{D}_j \ \longrightarrow\ \mathcal{D}_{j'} \\ \text{and}\ \ \alpha_*(\varphi(j)) \ \longrightarrow\ \varphi(j')\ , \end{cases}$$

hence a composite functor

p. 93

[page 247]

$$\mathcal{D}_j/\varphi(j) \ \longrightarrow\ \mathcal{D}_{j'}/\alpha_*(\varphi(j)) \ \longrightarrow\ \mathcal{D}_{j'}/\varphi(j')\ ,$$

with transitivity up to canonical isomorphism etc. We note that we have a canonical isomorphism

$$\underline{\Gamma}(\mathcal{D}_\varphi/J) \ \simeq\ \underline{\Gamma}(\mathcal{D}/J)/\varphi\ ,$$

so that the faithful canonical functor (4.10.2.2) is identified with the functor

$$\underline{\Gamma}(\mathcal{D}_\varphi/J) \longrightarrow \prod_{j\in \mathrm{Ob}\, J} (\mathcal{D}_\varphi)_j\ . \tag{4.10.6}$$

Corollary 4.10.2 will then be a consequence of the following more general result:

Corollary 4.10.3. *Let $\mathcal{E} \longrightarrow J$ be a cofibrant functor, and π an infinite cardinal. We assume:*

a) $\mathrm{card}\, J \leqslant \pi$.

b) The fiber categories $\mathcal{E}_j$ are equivalent to small categories, and large relative to π.

c) There exists a limit ordinal α such that:

[page 248]

1°) $\operatorname{card}\alpha \leqslant \pi$.

2°) The direct limits of type I_α are representable in the categories $\mathcal{E}_j$, and the cobase change functors $\lambda_* : \mathcal{E}_j \longrightarrow \mathcal{E}_{j'}$ commute with them. (This is the case e.g. if the countable *filtering direct limits are representable in the $\mathcal{E}_j$, and if the α_* commute with them; or if there exists a cardinal $\pi_0 < \pi$ and a cardinal $\pi_0' > \pi_0$, such that the direct limits large relative to π_0 and of cardinality $\leqslant \pi_0'$ are representable in the $\mathcal{E}_j$, and the α_* commute with them.)*

Under these conditions, the faithful canonical functor

$$\underline{\Gamma}(\mathcal{E}/J) \longrightarrow \prod_{j\in \operatorname{Ob} J} \mathcal{E}_j \tag{4.10.3.1}$$

is cofinal, and the source category is large relative to π.

It is ± immediate that the hypotheses of 4.10.1 imply that those of 4.10.3 are satisfied for $\mathcal{E} = \mathcal{D}_\varphi \longrightarrow J$ (4.10.5). For this, we use the [page 249]

Lemma 4.10.4. *Let C be a category, and π an infinite cardinal. Then the inclusion functor $C \longrightarrow \mathcal{M} \overset{\text{def}}{=} \operatorname{Ind}_\pi C$ commutes with $\varinjlim$ of cardinality $\leqslant \pi$.*

Indeed, let I be such that $\operatorname{card} I \leqslant \pi$, and $(x_i)_{i\in I}$ a direct system in C, having a limit x there. On the other hand, let $X = \varinjlim_{I'} \xi_{i'}$ be an object of $\operatorname{Ind}_\pi C$, with I' large relative to π. We have

thus p. 94

$$\begin{aligned}
\operatorname{Hom}_{\mathcal{M}}(x, X) &\simeq \varinjlim_{I'} \operatorname{Hom}(x, \xi_{i'}) \\
&\simeq \varinjlim_{I'} \varprojlim_{I} \operatorname{Hom}(x_i, \xi_{i'}) \\
&\overset{\text{SGA 4 I 9.8}}{\simeq} \varprojlim_{I} \underbrace{\varinjlim_{I'} \operatorname{Hom}(x_i, \xi_{i'})}_{\simeq \operatorname{Hom}(x_i, X)} \\
&\simeq \varprojlim_{I} \operatorname{Hom}(x_i, X)\ ,
\end{aligned}$$

which proves that x is also a $\varinjlim$ of the x_i in the whole of $\mathcal{M}$.

Thus, we are reduced to proving 4.10.3. Applying 4.9.6.3, 2°, we are brought back to [250] proving, for the functor under consideration, conditions F 1 and F 3_π (page 225).

In the arguments that follow, we are going to use conditions a), b) of 4.10.3, but not, for the moment, condition c.

Lemma 4.10.5. *Under the conditions of 4.10.3 a) b) (excluding c), for (4.10.3.1) to be cofinal and $\Gamma(\mathcal{E}/J)$ large relative to π, it is necessary and sufficient that the functor under consideration satisfy condition F 1, recalled below, and its variant F′ 1 stated below:*

F 1) *For every system $(X(j))_{j\in \operatorname{Ob} J}$ of elements $X(j) \in \operatorname{Ob}\mathcal{E}_j$, exists a* section *$G$ of $\mathcal{E}$ over J, and a system of arrows*

$$X(j) \longrightarrow G(j) \qquad (j \in ObJ)\ . \tag{4.10.5.1}$$

F′1) *Under the conditions of the preceding* F 1*, suppose in addition a section F of $\mathcal{E}$ over J and a system of arrows* [251]

[105] A bit short ! The argument ought to be spelled out with care – there are two or three pages' worth of it !

$$F(j) \longrightarrow X(j) \qquad (j \in \operatorname{Ob} J)\ ,$$

then G and the system of arrows (4.10.5.1) can be chosen in such a way that the composite arrows

$$F(j) \longrightarrow X(j) \longrightarrow G(j)$$

define a homomorphism of sections

$$F \longrightarrow G$$

(in the category $\underline{\Gamma}(\mathcal{E}/J)$).

([106]).

Proof. We have to see that F 1 & F′ 1 is equivalent to F 1 & F 3_π. The implication $\Longleftarrow$ is ± immediate, it is $\Longrightarrow$ that we shall need, so

$$\text{F 1 \& F}'\ 1 \quad \Longrightarrow \quad \text{F } 3_\pi\ .$$

For this, let us restate F 3_π in the present context. We are given a family $(F_\alpha)_{\alpha\in A}$, $\operatorname{card} A \leqslant \pi$, of objects of $\underline{\Gamma}(\mathcal{E}/J)$, an object F of the same category, and for every j in p. 95
$\operatorname{Ob} J$, a system of arrows

$$F_\alpha(j) \xrightarrow{u_\alpha(j)} F(j)\ ,$$

but without supposing that for α fixed, j variable

this defines a homomorphism of sections $F_\alpha \longrightarrow F$. We seek a section G of $\mathcal{E}$ over J, and a homomorphism of sections $F \longrightarrow G$, such that for every α, the system of [252]
composites

$$F_\alpha(j) \longrightarrow F(j) \longrightarrow G(j)$$

defines a homomorphism of sections $F_\alpha \longrightarrow G$. So let $\lambda : j \longrightarrow j'$ be an arrow of J, let us consider the squares (*noncommutative*)

$$\begin{array}{ccc} F_\alpha(j) & \xrightarrow{u_\alpha(j)} & F(j) \\ {\scriptstyle F_\alpha(\lambda)}\downarrow & ! & \downarrow{\scriptstyle F(\lambda)} \\ F_\alpha(j') & \xrightarrow{u_\alpha(j')} & F(j') \end{array} \qquad (*)$$

(indexed by $\alpha \in A$). For fixed j' in J, the number of these squares is bounded above by $\operatorname{card} \operatorname{Ob}(J/j') \times \operatorname{card} A \leqslant \pi \cdot \pi = \pi$. It follows, since $\mathcal{E}_{j'}$ is large relative to π, that there exists an object $X(j')$ of $\mathcal{E}_{j'}$, and an arrow $v(j') : F(j') \longrightarrow X(j')$ in such a way that for every $\alpha \in A$, and every $\lambda \in \operatorname{Ob} J/j'$, the square deduced from $(*)$

by composing with $F(j') \longrightarrow X(j')$ is commutative. We make this choice for every object j' of J. Let us then apply F′ 1 to F and to the system of arrows $F(j) \xrightarrow{v(j)} X(j)$, [253]
we find G and a system of arrows

$$w(j)\ :\ X(j) \longrightarrow G(j)$$

in such a way that the composite arrows

$$w(j)v(j) : \qquad \begin{array}{ccc} F(j) & \longrightarrow & G(j) \\ & \searrow \quad \nearrow & \\ & X(j) & \end{array}$$

[106] **N.B.** Conditions F 1 and F′ 1 are equivalent to saying that the functor under consideration satisfies conditions F1 and F3, seeing also (by virtue of 4.9.6.3, 1°) that $\underline{\Gamma}(\mathcal{E}/J)$ is *filtering* without more, and the faithful functor (4.10.3.1) cofinal. This already implies that $\underline{\Gamma}(\mathcal{E}/J)$ is large relative to π.

define a homomorphism of sections. I say that this homomorphism has the desired property for F 3_π, i.e. that for every $\alpha \in A$, the system of composites

$$F_\alpha(j) \longrightarrow F(j) \longrightarrow G(j)$$

(for $j \in \operatorname{Ob} J$ variable) defines a homomorphism of sections $F_\alpha \longrightarrow G$. Indeed, for this it suffices to consider, for an arrow λ of J, the diagram p. 96
[254]

$$\begin{array}{ccccccc} F_\alpha(j) & \xrightarrow{u_\alpha(j)} & F(j) & \xrightarrow{v(j)} & X(j) & \xrightarrow{w(j)} & G(j) \\ {\scriptstyle F_\alpha(\lambda)}\downarrow & ! & {\scriptstyle F(\lambda)}\downarrow & & \text{comm.} & & \downarrow{\scriptstyle G(\lambda)} \\ F_\alpha(j') & \xrightarrow{u_\alpha(j')} & F(j') & \xrightarrow{v(j')} & X(j') & \xrightarrow{w(j')} & G(j') \ . \end{array}$$

Lemma 4.10.5 being proved, everything thus comes down to proving condition F 1, and its complement F′ 1, which will be proved in the same breath.

4.10.6. So we are given the system of objects

$$X(j) \ \in \ \operatorname{Ob} \mathcal{E}_j \qquad (j \in \operatorname{Ob} J) \ , \tag{4.10.6.1}$$

and as the case may be, an object F of $\underline{\Gamma}(\mathcal{E}/J)$, and a system of arrows

$$F(j) \xrightarrow{u_j} X(j) \ , \tag{4.10.6.2}$$

and we propose to construct G in $\underline{\Gamma}(\mathcal{E}/J)$, and a system of arrows

$$X(j) \xrightarrow{w(j)} G(j) \ , \tag{4.10.6.3}$$

in such a way that (if we have the additional datum of the $u(j)$) the composite arrows

$$F(j) \xrightarrow{u(j)} X(j) \xrightarrow{w(j)} G(j)$$

are functorial in j. [255]

We are going to make a transfinite construction on the ordinals α such that

$$\operatorname{card} \alpha \ \leqslant \ \pi$$

of a direct system

$$(X_\alpha(j))_\alpha$$

of objects of $\prod_{j \in \operatorname{Ob} J} \mathcal{E}_j$. We are first going to ignore the possible datum (4.10.6.2). We shall put

$$X_0(j) \ = \ X(j) \ , \tag{4.10.6.4}$$

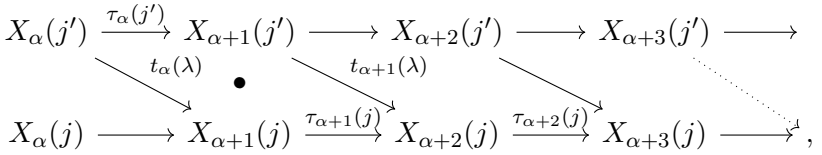
and we are going to give the prescription for passing from X_α to $X_{\alpha+1}$. The construction will also include, for every arrow $\lambda : j' \longrightarrow j$ of J and every α, the datum of

$$X_\alpha(j') \xrightarrow{t_\alpha(\lambda)} X_{\alpha+1}(j) \ , \qquad \text{arrow (in } \mathcal{E}) \text{ over } \lambda, \tag{4.10.6.5}$$

and these arrows, and the transition arrows $X_\alpha(j) \xrightarrow{\tau_\alpha(j)} X_{\alpha+1}(j)$, must satisfy the following three conditions: p. 97
[256]

(i) If λ is an identity arrow $\mathrm{id}_j : j \longrightarrow j$, then $\boxed{t_\alpha(\lambda) = \tau_\alpha(j).}$

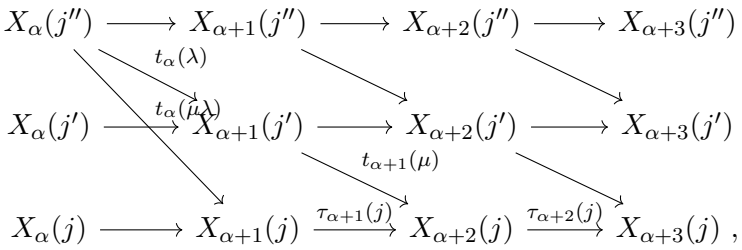
(ii) In the diagram



$\tau_{\alpha+2}(j)$ "makes the square $\bullet$ commutative", i.e. equalizes $t_{\alpha+1}(\lambda)\tau_\alpha(j')$ and $\tau_{\alpha+1}(j)t_\alpha(\lambda)$:

$$\boxed{\tau_{\alpha+2}(j)t_{\alpha+1}(\lambda)\tau_\alpha(j') \;=\; \tau_{\alpha+2}(j)\tau_{\alpha+1}(j)t_\alpha(\lambda)\;.}$$

(iii) If we have two consecutive arrows $\lambda : j'' \longrightarrow j'$, $\mu : j' \longrightarrow j$, we thus have the diagram



$$\boxed{\tau_{\alpha+2}(j)t_{\alpha+1}(\mu)t_\alpha(\lambda) \;=\; \tau_{\alpha+2}(j)\tau_{\alpha+1}(j)t_\alpha(\mu\lambda)}$$

Thus, to define the $X_{\alpha+1}(j)$, we have to take account of all these constraints. So: [257]

1°) For fixed j, $X_{\alpha+1}(j)$ must be the target of the arrows

$$X_\alpha(j') \xrightarrow{t_\alpha(\lambda)} X_{\alpha+1}(j)\;,$$

for the $\lambda : j' \longrightarrow j$ in $\operatorname{Ob} J/j$ – including for $\lambda = \mathrm{id}_j$, in which case $t_\alpha(\lambda) : X_\alpha(j) \longrightarrow X_{\alpha+1}(j)$ will play the role of the transition morphism $\tau_\alpha(j)$.

2°) If α is of the form $\alpha' + 2$, i.e. if it is neither a limit ordinal, nor the successor of a limit ordinal, then we require in addition, for the transition morphisms $\tau_\alpha(j)$, that they equalize the pairs of arrows provided for in (ii) and (iii) above.

Since the set of these data in 1°), or of these constraints in 2°), has cardinality $\leqslant \pi$, and the category $\mathcal{E}_j$ is large relative to π, we see that we can indeed find $X_{\alpha+1}(j)$ and $t_\alpha(\lambda) : X_\alpha(j') \longrightarrow X_{\alpha+1}(j)$, so as to satisfy (if $\alpha = \alpha'+2$) the constraints recalled in 2°. This completes the description of the construction of $X_{\alpha+1}(\bullet)$, in terms of $X_\beta(\bullet)$ ($\beta \leqslant \alpha$) and the $t_\beta(\lambda)$ ($\beta < \alpha$). [258]

We must finally define $X_\alpha(\bullet)$ for α a limit ordinal. This time, we ask it only to receive the $X_\beta(\bullet)$ for $\beta < \alpha$. Since the set I_α of these β has cardinality $\leqslant \pi$, and the $\mathcal{E}_j$ are large relative to π, we can indeed find the $X_\alpha(j)$.

We have thus proved the essential part of

Lemma 4.10.7. *Under the conditions of 4.10.3 a) b), and for a given system $X(\bullet) \in \operatorname{Ob} \prod_{j\in \operatorname{Ob} J} \mathcal{E}_j$, we can find a direct system $(X_\alpha(j))_\alpha$ in $\prod \mathcal{E}_j$, indexed by the ordinals α such that $\operatorname{card}\alpha \leqslant \pi$, and for every arrow $\lambda : j' \longrightarrow j$ in J arrows $t_\alpha(\lambda) : X_\alpha(j') \longrightarrow X_{\alpha+1}(j)$, in such a way that we have the following conditions.* p. 98 [259]

1°) $X_0(\bullet) = X(\bullet)$.

2°) The conditions (i) (ii) (iii) (p. 256).

3°) If we are given F in $\mathrm{Ob}\,\underline{\Gamma}(\mathcal{E}/J)$, and arrows $u(j) : F(j) \longrightarrow X(j) = X_0(j)$, we may suppose that for every arrow $\lambda : j' \longrightarrow j$ in J, we have

$$\begin{array}{ccccc} F(j') & \xrightarrow{u(j')} & X_0(j') & & \\ {\scriptstyle F(\lambda)}\downarrow & & & \searrow^{t_0(\lambda)} & \\ F(j) & \xrightarrow{u(j)} & X_0(j) & \xrightarrow{\tau_0(j)} & X_1(j) \end{array} \qquad \text{commutative.} \qquad (*)$$

It only remains to prove this last complement 3°. This is a constraint only on the [260]
construction of $X_1(\bullet)$ and of the $t_0(\lambda)$. So suppose that we have constructed $X_1(\bullet)$ without taking this constraint into account. For fixed j, we must therefore "make commutative" all these squares, i.e. all these double arrows $(t_0(\lambda)u(j'), \tau_0(j)u(j)F(\lambda))$ with target $X_1(j)$. Since the set of these double arrows has cardinality $\mathrm{card}\,\mathrm{Ob}\,(J/j) \leqslant \pi$, we can find $X_1(j) \longrightarrow X_1'(j)$ in $\mathcal{E}_j$, equalizing all these double arrows, so at the cost of replacing $X_1(j)$ by $X_1'(j)$, we win.

It is finally time to appeal to hypothesis c) of 4.10.3. So let α be a limit ordinal, as [261]
stipulated in this condition. Since $\mathrm{card}\,\alpha \leqslant \pi$, a fortiori $\mathrm{card}\,\beta \leqslant \pi$ if $\beta < \alpha$, we thus have a direct system

$$(X_\beta(\bullet))_{\beta \in I_\alpha}$$

in $\prod_{j\in J} \mathcal{E}_j$. Since the $\varinjlim$ of type I_α exist in $\mathcal{E}_j$, we can therefore consider

$$G(j) \;=\; \varinjlim_{\beta \in I_\alpha} X_\beta(j) \;\in\; \mathrm{Ob}\,\mathcal{E}_j \;.$$

If $\lambda : j' \longrightarrow j$ is an arrow of J, we deduce from it a λ-arrow

$$G(\lambda) \;:\; G(j') = \varinjlim X_\beta(j') \longrightarrow G(j) \simeq \varinjlim X_{\beta+1}(j)$$

by $G(\lambda) = \varinjlim t_\beta(\lambda)$. We must take note that the $t_\beta(\lambda)$ do not define a homomorphism of [262]
direct systems, but by condition (ii), p. 256, they define a homomorphism of ind-objects, so

a homomorphism on the $\varinjlim$ ([106]). Condition (i) ensures that $\lambda \longmapsto G(\lambda)$ transforms p. 99
identities into identities, and condition (iii) that we have $G(\mu\lambda) = G(\mu)G(\lambda)$. Thus G appears as a section of $\mathcal{E}$ over J. On the other hand, we have arrows

$$X(j) \overset{\mathrm{def}}{=} X_0(j) \longrightarrow G(j) = \varinjlim_{\beta \in I_\alpha} X_\beta(j) \;,$$

which proves F (1). On the other hand, condition 3°) of the lemma proves the validity of F′ (1). This completes the proof of 4.10.3, and thereby of 4.10.1 and of 4.9.3.

Corollary 4.10.8 (of theorem 4.10.1). *Let $\mathcal{N} \longrightarrow J$ be a cofibrant functor, such that* [page 263]
the fibers are accessible categories (4.8), the cobase change functors $\lambda_* : \mathcal{N}_{j'} \longrightarrow \mathcal{N}_j$ accessible, finally J equivalent to a small category. Then the category $\mathcal{M} = \underline{\Gamma}(\mathcal{N}/J)$ is accessible. More precisely, let π_0 be an infinite cardinal such that the categories $\mathcal{N}_j$ are π_0-accessible, and that the functors λ_* are π_0-accessible, and let π be an infinite cardinal such that we have:

ⓐ $\pi^{\pi_0} = \pi$.

[106] Careful, it is here that we tacitly use the fact that the functor $\lambda_* : \mathcal{E}_{j'} \longrightarrow \mathcal{E}_j$ commutes with $\varinjlim$ of type I_α, which implies that the inclusions $\mathcal{E}_j \hookrightarrow \mathcal{E}$ commute with this type of $\varinjlim$.

ⓑ $\forall\ \lambda : j' \longrightarrow j$ *in* J, $\lambda_* : \mathcal{N}_{j'} \longrightarrow \mathcal{N}_j$ *maps* $(\mathcal{N}_{j'})_{\pi_0}$ *into* $(\mathcal{N}_j)_{\pi_0}$.

ⓒ $\operatorname{card}\operatorname{red} J \leqslant \pi$.

Then $\mathcal{M}$ *is* π*-accessible, and more precisely, we have*

$$\mathcal{M} \ \simeq\ \operatorname{Ind}_\pi(\mathcal{M}_\pi)\ ,$$

where $\mathcal{M}_\pi$ *is the full subcategory of* $\mathcal{M}$ *consisting of the sections* F *of* $\mathcal{N}$ *over* J *such that we have* $F(j) \in (\mathcal{N}_j)_\pi\ \forall\ j \in \operatorname{Ob} J$, $(\mathcal{N}_j)_\pi$ *denoting the full subcategory of* $\mathcal{N}_j$ *consisting of the* π*-accessible objects of* $\mathcal{N}_j$. [page 264]

Indeed, the hypothesis at the beginning implies the existence of π_0 and of π satisfying the stated conditions. Conditions a) and b) on π, plus the fact that the functors λ_* are π_0-accessible, imply that the strictly full subcategory $\mathcal{N}_\pi$ of $\mathcal{N}$ whose fibers are the $(\mathcal{N}_j)_\pi$, is a cofibered subcategory, and that moreover $\mathcal{N}$ is J-equivalent to $\operatorname{Ind}_\pi(\mathcal{N}_\pi/J)$. On the other hand, the fibers of $\mathcal{N}_\pi/J$ are stable under $\varinjlim$ large relative to π_0 and of cardinality $\leqslant \pi$, so the "ordinal" condition of 4.10.1 is satisfied (taking $\mathcal{D} = \mathcal{N}_\pi$ there). Since moreover $\operatorname{card} J \leqslant \pi$, we conclude by 4.10.1 that $\mathcal{M} \simeq \operatorname{Ind}_\pi(\mathcal{M}_\pi)$.

Corollary 4.10.9. *Let* $\mathcal{N} \longrightarrow J$ *be a "cofibration in paratopoi" (*[107]*), i.e. a cofibrant functor such that the fibers are paratopos categories, and the cobase change functors are inverse image functors (resp. direct image functors) for morphisms of paratopoi, i.e. they commute with (small)* $\varinjlim$ *(resp. with small* $\varprojlim$ *). Assume furthermore* J *equivalent to a small category. Then* $\underline{\Gamma}(\mathcal{N}/J)$ *is a paratopos category.* [page 265]

Indeed, it is an accessible category by virtue of 4.10.8, on the other hand the hypothesis on the λ_* implies at once that $\underline{\Gamma}(\mathcal{N}/J)$ is stable under small $\varprojlim$ (resp. under small $\varprojlim$). Thus $\underline{\Gamma}(\mathcal{N}/J)$ is a paratopos category by virtue of 4.8.9 b $\Longrightarrow$ a (resp. c $\Longrightarrow$ a). p. 100

Corollary 4.10.10. *Let* $\mathcal{N} \longrightarrow J$ *be a category that is* fibered *(not cofibered!), such that the fibers are paratopos categories and the functors of base change are direct image functors for morphisms of paratopoi (i.e. accessible functors, commuting with small* $\varprojlim$ *). Assume furthermore* J *equivalent to a small category. Then* $\underline{\Gamma}(\mathcal{N}/J)$ *is a paratopos category.* [page 266]

Indeed, the hypothesis implies that for every arrow $\lambda : j' \longrightarrow j$ in J, the functor $\lambda^* : \mathcal{N}_j \longrightarrow \mathcal{N}_{j'}$ admits a left adjoint $\lambda_! : \mathcal{N}_{j'} \longrightarrow \mathcal{N}_j$, whence it follows that $\mathcal{N} \longrightarrow J$ is also cofibrant, the cobase change functors commuting with small $\varinjlim$. Thus we can apply 4.10.9, which gives the desired result.

Question 4.10.11. *Let* $\mathcal{N} \longrightarrow J$ *be a functor that is* fibrant, *such that the* $\mathcal{N}_j$ *are accessible categories, and the base change functors* λ^* *accessible,* J *being equivalent to a small category. I conjecture that under these conditions,* $\underline{\Gamma}(\mathcal{N}/J) \overset{\text{def}}{=} \underline{\operatorname{Hom}}_J(J, \mathcal{N})$ *is accessible, and that the same is true of* $\underline{\operatorname{Hom}}\,\underline{\operatorname{cart}}_J(J, \mathcal{N})$. [page 267]

This is what is proved in SGA 4 I 9.24 under slightly stronger hypotheses (with the help of the old notion too strong for my taste, of cardinal filtrations). I would be surprised if the (somewhat longish) proof of loc. cit. did not apply to the general case. But we would like to have a statement that covers this one and 4.10.8, in the case of a functor $\mathcal{N} \longrightarrow J$ that would be assumed neither fibrant nor cofibrant, but …(???).

4.11 Application to the existence of limits in a category of the form $\operatorname{Ind}_{\pi_0}(C)$.

Proposition 4.11.1. *Let* C *be a category equivalent to a small category,* π *an infinite* [page 268]

[107] ambiguous term, cf. the double explanation.

cardinal, put $\mathcal{M} = \mathrm{Ind}_\pi(C)$. *Let* Φ *be a set of categories of reduced cardinality* $\leqslant \pi$. *We propose to examine whether* $\mathcal{M}$ *is "stable under* $\varinjlim$ *of type* Φ*" (i.e. stable under* $\varinjlim$ *of type* I*, whatever* $I \in \Phi$ *may be). It suffices for this that* C *is stable under* $\varinjlim$ *of type* Φ*, and that moreover there exists a limit ordinal* α *with* $\operatorname{card}\alpha \leqslant \pi$ *and such that* C *is stable under* $\varinjlim$ *of type* I_α *(e.g. such that* $I_\alpha \in \Phi$*). Conversely, if* $\mathcal{M}$ *is stable under* $\varinjlim$ *of type* Φ*, and if we assume moreover* C *Karoubian (i.e. stable under direct factors), then* C *is stable under* $\varinjlim$ *of type* Φ. *In either case, the inclusion functor* $C \hookrightarrow \mathcal{M}$ *commutes with the* $\varinjlim$ *under consideration.*

([108]).

Proof. a) Sufficient condition. We apply 4.9.3 to conclude that for every category p. 101
J of reduced cardinality $\leqslant \pi$ (in particular for $J \in \Phi$), every functor $\varphi : J \longrightarrow \mathcal{M} =$ [page 269]
$\mathrm{Ind}_\pi(C)$ admits an indicial representation $\Phi : J \times I \longrightarrow C$, with I large relative to π. This, plus the stability of C under $\varinjlim$ of type J, suffices to establish the existence of the $\varinjlim$ of φ.

The converse comes from the fact that if C is Karoubian, then its essential image in $\mathcal{M}$ is none other than $\mathcal{M}_\pi$, which is stable in $\mathcal{M}$ under any $\varinjlim$ of cardinality $\leqslant \pi$ representable in $\mathcal{M}$. If $\mathcal{M}$ is stable under $\varinjlim$ of type Φ, the same is true of $\mathcal{M}_\pi$, hence of C, q.e.d.

Corollary 4.11.2. *For* $\mathcal{M}$ *to be stable under filtering* $\varinjlim$ *(resp. under small* $\varinjlim$*), it suffices that* C *be so under filtering* $\varinjlim$ *(resp. under* $\varinjlim$*)* of cardinality $\leqslant \pi$, *and this is also necessary if* C *is Karoubian.*

Suppose C stable under filtering $\varinjlim$ (resp. under $\varinjlim$ of cardinality $\leqslant \pi$), then by [page 270]
4.11.1 $\mathcal{M}$ is stable under this same type of direct limits. On the other hand, it is stable under $\varinjlim$ large relative to π. It thus suffices to prove the

Lemma 4.11.3. *Let* $\mathcal{M}$ *be a category,* π *a cardinal. Suppose that* $\mathcal{M}$ *is stable under filtering* $\varinjlim$ *of cardinality* $\leqslant \pi$ *(resp. under* $\varinjlim$ *of cardinality* $\leqslant \pi$*, resp. under sums of cardinality* $\leqslant \pi$*),* and *under* $\varinjlim$ *large relative to* π. *Then* $\mathcal{M}$ *is stable under small filtering* $\varinjlim$ *(resp. under small* $\varinjlim$ *, resp. under small sums).*

This lemma is proved like 4.4.14, page 97.

There remains the converse in 4.11.2, which follows from the similar assertion in 4.11.1.

Corollary 4.11.4. *When is* $\mathcal{M} = \mathrm{Ind}_\pi C$ *stable under sums? If there exists an ordinal* α *such that* $\operatorname{card}\alpha \leqslant \pi$*, and such that* C *is stable under* $\varinjlim$ *of type* I_α*, then it suffices that* C *be stable under sums of cardinality* $\leqslant \pi$. *This condition is also necessary if* C *is Karoubian.*

([109]).

Is proved like 4.11.2, using 4.11.3.

Proposition 4.11.5 ([110]). *Let* Φ *be a set of categories* I *equivalent to finite categories, and "rigid", i.e. for every* $i \in \mathrm{Ob}\, I$, $\underline{\mathrm{Hom}}_I(i,i) = \{\mathrm{id}_i\}$. *Let* C *be a category equivalent to a small category,* π *an infinite cardinal,* $\mathcal{M} = \mathrm{Ind}_\pi C$. *For* $\mathcal{M}$ *to be stable under* $\varinjlim$ [page 271]
(resp. under $\varprojlim$ *) of type* Φ*, it* suffices *that* C *be so. In the non-resp. case, if* C *is Karoubian, this condition is also necessary.*

This last assertion is contained in the similar assertion in 4.11.1. As for the first assertion, this follows in the usual way from the

Lemma 4.11.6 ([111]). *Let* J *be a category equivalent to a finite category,* J *rigid,* C *a category equivalent to a small category,* π *an infinite cardinal. Then every functor*

[108] Make explicit here the corollaries of types 4.13.20, 4.13.21 (p. 341).
[109] For other variants of 4.11.1, cf. 4.13.19 (p. 340) and corollaries.
[110] ?, proof not finished.
[111] ?, proof not spelled out is perhaps false!

$J \longrightarrow \mathcal{M} \overset{\text{def}}{=} \mathrm{Ind}_\pi C$ *admits an indicial representation* $J \times I \longrightarrow C$*, with* I *large relative to* π*, in other words, the canonical fully faithful functor* p. 102 [page 272]

$$\mathrm{Ind}_\pi(\underline{\mathrm{Hom}}(J, C)) \longrightarrow \underline{\mathrm{Hom}}(J, \mathcal{M})$$

deduced from the inclusion $C \hookrightarrow \mathcal{M}$*, hence* $\underline{\mathrm{Hom}}(J, C) \hookrightarrow \underline{\mathrm{Hom}}(J, \mathcal{M})$ *(cf. 4.9), is an equivalence of categories.* (**N.B.** I do not know whether this is so without a rigidity hypothesis on J.)

Proof. This is proved like the similar result for $\mathrm{Ind}(C)$ instead of $\mathrm{Ind}_\pi C$, cf. SGA 4 I 8.8.5.

Corollary 4.11.7 ([112]). *For* $\mathcal{M} = \mathrm{Ind}_\pi C$ *to be stable under* $\varinjlim$ *(resp. under* $\varprojlim$ *) that are* finite, *it* suffices *that* C *be so. And this is also necessary in the non-resp. case, when* C *is assumed Karoubian.* [page 273]

4.12 Accessible functors between accessible categories

Proposition 4.12.1 ([113]). *Let* $\mathcal{M}$*,* $\mathcal{N}$ *be two accessible categories (4.8), and* $f : \mathcal{M} \longrightarrow \mathcal{N}$ *an accessible functor. Then there exist infinite cardinals* π_0*,* π_1 *such that* $\pi_1 \geqslant \pi_0$ *and such that for every cardinal* $\pi \geqslant \pi_1$*, we have*

$$f(\mathcal{M}_\pi) \subset \mathcal{N}_{\pi^{\pi_0}} , \tag{4.12.1.1}$$

and in particular

$$f(\mathcal{M}_\pi) \subset \mathcal{N}_\pi \qquad \text{if } \pi = \pi^{\pi_0}. \tag{4.12.1.2}$$

More precisely, we can take for π_0 *a useful cardinal for* $\mathcal{M}$ *and such that* f *is* π_0*-accessible (e.g. the smallest cardinal of this type). Then let* π_1 *be a cardinal such that we have* $f(\mathcal{M}_{\pi_0}) \subset \mathcal{N}_{\pi_1}$*,* $\pi_1 \geqslant \mathrm{card}\,\mathrm{red}\,\mathcal{M}_{\pi_0}$*, and finally such that* $\mathcal{N}_{\pi_1}$ *is strictly generating subcategory of* $\mathcal{N}$*. The pair* (π_0, π_1) *satisfies the stated conditions.* [page 274]

Proof. Indeed we will have

$$\mathcal{M} \xleftarrow{\sim} \mathrm{Ind}_{\pi_0}(\mathcal{M}_{\pi_0})$$

and f, commuting with $\varinjlim$ large relative to π_0, will be known (up to unique isomorphism) by its restriction to $\mathcal{M}_{\pi_0}$, which factors through $\mathcal{N}_{\pi_1}$,

$$f|\mathcal{M}_{\pi_0} : \mathcal{M}_{\pi_0} \longrightarrow \mathcal{N}_{\pi_1} .$$

To prove (4.12.1.1), putting $\pi' = \pi^{\pi_0} \geqslant \pi$, and noting that $\pi'^{\pi_0} = \pi'$,

$$f(\mathcal{M}_\pi) \subset f(\mathcal{M}_{\pi'}) \overset{?}{\subset} \mathcal{N}_{\pi'} ,$$

we see that it suffices to prove (4.12.1.2) (for π'). So we can assume $\pi = \pi^{\pi_0}$ and prove (4.12.1.2). But since $\pi_1 \geqslant \mathrm{card}\,\mathrm{red}\,\mathcal{M}_{\pi_0}$, we know that if π is such that $\pi \geqslant \pi_1$, $\pi^{\pi_0} = \pi$, then $\mathcal{M}_\pi$ consists of the $\varinjlim_I x_i$, the $x_i \in \mathcal{M}_{\pi_0}$, and I large relative to π_0, and $\mathrm{card}\, I \leqslant \pi$. p. 103

We then have (f being π_0-accessible) [page 275]

$$f(x) = \varinjlim_I f(x_i) ,$$

[112] subject to the validity of 4.11.5.

[113] Would go back to 4.8.

the $f(x_i)$ are in $\mathcal{N}_{\pi_1}$, hence in $\mathcal{N}_\pi$, now $\mathcal{N}_\pi$ stable under $\varinjlim$ of cardinality $\leqslant \pi$ (since $\mathcal{N}$ is generated by $\mathcal{N}_{\pi_1}$, consisting of objects that are π_1-accessible and a fortiori π-accessible, cf. SGA 4 I 9.9). Hence $f(x) \in \mathcal{N}_\pi$, q.e.d.

Corollary 4.12.2. *Assume that π_0 is also useful for $\mathcal{N}$, and that $\pi_1 \geqslant \operatorname{card} \operatorname{red} \mathcal{N}_{\pi_0}$. Let π be a cardinal $\geqslant \pi_1$, and such that $\pi^{\pi_0} = \pi$. Then f is identified, up to equivalence of categories and isomorphism of functors, with the functor*

$$\operatorname{Ind}(f_\pi) \ : \ \operatorname{Ind}_\pi(\mathcal{M}_\pi) \ \longrightarrow \ \operatorname{Ind}_\pi(\mathcal{N}_\pi) \ ,$$

where $f_\pi : \mathcal{M}_\pi \longrightarrow \mathcal{N}_\pi$ is the functor induced by f.

Proposition 4.12.3 ([114]). *Let C be a category equivalent to a small category, $\mathcal{N}$ a $\mathfrak{U}$-category, π_0 an infinite cardinal. We assume that $\mathcal{N}$ satisfies L_{π_0} (stability under $\varinjlim$ large relative to π_0). Let $\mathcal{M} = \operatorname{Ind}_{\pi_0} C$, $\underline{\operatorname{Hom}}_{!\,\pi_0}(\mathcal{M},\mathcal{N})$ the strictly full subcategory of $\underline{\operatorname{Hom}}(\mathcal{M},\mathcal{N})$ consisting of the $f : \mathcal{M} \longrightarrow \mathcal{N}$ that are π_0-accessible. Then the restriction functor* [page 276]

$$\underline{\operatorname{Hom}}_{!\,\pi_0}(\mathcal{M},\mathcal{N}) \ \longrightarrow \ \underline{\operatorname{Hom}}(C,\mathcal{N}) \tag{4.12.3.1}$$

is an equivalence of categories. We find a quasi-inverse functor as the composite

$$\underline{\operatorname{Hom}}(C,\mathcal{N}) \ \longrightarrow \ \underline{\operatorname{Hom}}(\underbrace{\operatorname{Ind}_{\pi_0}(C)}_{\mathcal{M}},\operatorname{Ind}_{\pi_0}(\mathcal{N})) \ \longrightarrow \ \underline{\operatorname{Hom}}(\mathcal{M},\mathcal{N}) \ , \tag{4.12.3.2}$$

where the second arrow is the composition functor $F \longmapsto L \circ F$ with the direct limit functor $L : \operatorname{Ind}_{\pi_0} \mathcal{N} \longrightarrow \mathcal{N}$. The composite functor (4.12.3.2) has image contained in $\underline{\operatorname{Hom}}_{!\,\pi_0}(\mathcal{M},\mathcal{N})$ and induces an equivalence between this last category and $\underline{\operatorname{Hom}}(C,\mathcal{N})$.

In other words, we can say that the canonical inclusion functor [page 277]

$$C \ \longrightarrow \ \operatorname{Ind}_{\pi_0} C \ = \ \mathcal{M}$$

is 2-universal *for functors from C into categories $\mathcal{N}$ satisfying L_{π_0}, the latter being regarded as* 0*-objects of a* 2*-category where the* 1*-objects are the functors $\mathcal{N} \longrightarrow \mathcal{N}'$ that are π_0-accessible, i.e. commute with $\varinjlim$ large relative to π_0.*

This statement is the analogue, for arbitrary small C, of 4.4.16 (page 101), valid for C stable under $\varinjlim$ of cardinality $\leqslant \pi_0$. There are of course other variants, in relation to commuting with other types of $\varinjlim$ (filtering $\varinjlim$, sums, finite direct limits etc.) considered in the preceding section 4.11.

The proof of 4.12.3 is sorital.

Corollary 4.12.4. *Let $\mathcal{M}$ and $\mathcal{N}$ be two accessible categories, and let π_0 be an infinite cardinal such that $\mathcal{M}$ is π_0-accessible, i.e. such that π_0 is a useful cardinal for $\mathcal{M}$. Let $\pi_1 = \sup(\pi_0, \operatorname{card} \operatorname{red} \mathcal{M}_{\pi_0})$. Then for every cardinal $\pi \geqslant \pi_1$ useful for $\mathcal{M}$ (and in particular, for every $\pi \geqslant \pi_1$ such that $\pi^{\pi_0} = \pi$), the category of functors $\mathcal{M} \longrightarrow \mathcal{N}$ that are π-accessible is itself accessible.* p. 104 [page 278]

Indeed, we will have $\mathcal{M} \simeq \operatorname{Ind}_\pi \mathcal{M}_\pi$, whence by 4.12.3

$$\underline{\operatorname{Hom}}_{!\,\pi}(\mathcal{M},\mathcal{N}) \xrightarrow{\sim} \underline{\operatorname{Hom}}(\mathcal{M}_\pi,\mathcal{N}),$$

now $\underline{\operatorname{Hom}}(\mathcal{M}_\pi,\mathcal{N})$ is accessible by 4.9.5 (p. 220). We can moreover make this point more precise. Suppose π_0 chosen so that π_0 is useful not only for $\mathcal{M}$, but for $\mathcal{N}$ as well. Recall that we have $\operatorname{card} \operatorname{Fl} \mathcal{M}_\pi \leqslant 2^\pi$ (cf. 4.7.7, page 173).

Let us therefore put [page 279]

$$\pi' = 2^\pi,$$

[114] goes back to 4.7.

we therefore have $\pi'^{\pi_0} = 2^{\pi\pi_0} = 2^\pi = \pi'$, and $\operatorname{card}\operatorname{red}\mathcal{M}_\pi \leqslant \pi'$. Thus 4.9.5 gives us that

$$\underline{\mathrm{Hom}}(\mathcal{M}_\pi, \mathcal{N}) \simeq \mathrm{Ind}_{\pi'}(\underline{\mathrm{Hom}}(\mathcal{M}_\pi, \mathcal{N}_{\pi'})),$$

whence:

Corollary 4.12.5. *Under the conditions of 4.12.4, if π_0 is useful for $\mathcal{M}$* and *for $\mathcal{N}$, then for $\pi \geqslant \pi_1$, $\underline{\mathrm{Hom}}_{!\,\pi}(\mathcal{M}, \mathcal{N})$ is π'-accessible, where $\pi' = 2^\pi$, more precisely, we have an equivalence of categories*

$$\underline{\mathrm{Hom}}_{!\,\pi}(\mathcal{M}, \mathcal{N}) \simeq \mathrm{Ind}_{\pi'}(\underline{\mathrm{Hom}}(\mathcal{M}_\pi, \mathcal{N}_{\pi'})).$$

Remark 4.12.6. If $\mathcal{M}$, $\mathcal{N}$ are accessible, the category $\underline{\mathrm{Hom}}_{\mathrm{acc}}(\mathcal{M}, \mathcal{N})$ of accessible functors is not always accessible. If $\mathcal{N}$ satisfies L_{π_0}, this category satisfies L_{π_0}, and the $\varinjlim$ large relative to π_0 are computed argument by argument.

If we could find a subcategory (equivalent to a small category) $\mathcal{H}_\pi$ of $\mathcal{H} = \underline{\mathrm{Hom}}_{\mathrm{acc}}(\mathcal{M}, \mathcal{N})$, for a cardinal $\pi \geqslant \pi_0$ such that every object F of $\mathcal{H}$ could be written $\varinjlim_I F_i$, the F_i in $\mathcal{H}_\pi$, I large relative to π, then if the objects of $\mathcal{H}_\pi$ are π'-accessible functors from $\mathcal{M}$ to $\mathcal{N}$, the same would hold for every F in $\mathcal{H}$. Thus there would exist an infinite cardinal π' such that every accessible functor from $\mathcal{M}$ to $\mathcal{N}$ is π'-accessible. But unless I am mistaken, if e.g. $\mathcal{M} = \mathcal{N} = \mathrm{Ens}$, there is no such cardinal. I do not want to dwell here on this question, and make a counterexample. [page 280] [page 281]

4.13 Accessible sets of objects of an accessible category

Definition 4.13.1. Let $\mathcal{M}$ be an accessible category, S a subset of $\mathrm{Ob}\,\mathcal{M}$, $\underline{S}$ the full subcategory of $\mathcal{M}$ defined by S. We say that S is an *accessible subset* of $\mathrm{Ob}\,\mathcal{M}$, or again that $\underline{S}$ is a *subcategory "accessible in $\mathcal{M}$"*, if p. 105

a) $\underline{S}$ is strictly full, i.e. for x, y two isomorphic objects of $\mathcal{M}$, $x \in S$ implies $y \in S$,

b) $\underline{S}$ is accessible,

c) the inclusion functor $\underline{S} \longrightarrow \mathcal{M}$ is accessible.

Let π_0 be a useful cardinal for $\mathcal{M}$, i.e. such that $\mathcal{M} \simeq \mathrm{Ind}_{\pi_0}(\mathcal{M}_{\pi_0})$, let $\pi_1 = \sup(\pi_0, \operatorname{card}\operatorname{red}\mathcal{M}_{\pi_0})$. On the other hand let π'_0 be a cardinal such that

$\underline{S}$ (assumed accessible in $\mathcal{M}$) is π'_0-*accessible*, i.e. π'_0-useful for $\underline{S}$, and moreover such that $f : \underline{S} \hookrightarrow \mathcal{M}$ is π'_0-accessible. Let $\pi'_1 \geqslant \pi'_0$ be a cardinal such that $\underline{S}_{\pi'_0} \subset \mathcal{M}_{\pi'_1}$, $\pi'_1 \geqslant \operatorname{card}\operatorname{red}\underline{S}_{\pi'_0}$. It follows from 4.12.1 that for every cardinal [page 282]

$$\pi \geqslant \pi'_1, \qquad \text{such that } \pi^{\pi'_0} = \pi, \tag{$*$}$$

we must have

$$\underline{S}_\pi \subset \mathcal{M}_\pi, \qquad \text{hence } \underline{S}_\pi \subset \mathcal{M}_\pi \cap \underline{S},$$

moreover, since $\underline{S} \hookrightarrow \mathcal{M}$ is π'_0-accessible and a fortiori π-accessible, and fully faithful, it follows at once that $\mathcal{M}_\pi \cap S \subset S_\pi$, whence

$$S_\pi = \mathcal{M}_\pi \cap S.$$

Moreover, condition $(*)$ implies that π is useful for $\underline{S}$, hence [page 283]

$$\underline{S} \xleftarrow{\sim} \mathrm{Ind}_\pi(\underline{S}_\pi).$$

(Moreover, *if* $\pi'_0 \geqslant \pi_0$, the relation $\pi^{\pi'_0} = \pi$ implies a fortiori $\pi^{\pi_0} = \pi$, and if $\pi'_1 \geqslant \pi_1$, we see that $(*)$ implies that π is also useful for $\mathcal{M}$, i.e.

$$\mathcal{M} \xleftarrow{\sim} \mathrm{Ind}_\pi(\mathcal{M}_\pi).$$

On the other hand, the inclusion functor $\underline{S} \hookrightarrow \mathcal{M}$ is obtained essentially from $\underline{S}_\pi \hookrightarrow \mathcal{M}_\pi$ by passing to the categories Ind_π. ([116]).)

This proves among other things the "only if" in the following proposition:

Proposition 4.13.2. *Let $\mathcal{M}$ be an accessible category, π_0 an infinite cardinal useful for $\mathcal{M}$, S a subset of* $\mathrm{Ob}\,\mathcal{M}$, *$\underline{S}$ the corresponding full subcategory of $\mathcal{M}$.*

[page 284]

a) For S to be accessible, it is necessary and sufficient that there exist an infinite cardinal π and a subset $S(\pi)$ of $\mathcal{M}_\pi$, such that S consists of the objects of $\mathcal{M}$ that are isomorphic to an object of the form $\varinjlim_{i\in I} x_i$, with the x_i in $S(\pi)$ and I large relative to π. In other words, it is necessary and sufficient that there exist an infinite cardinal π and a full subcategory $\underline{S}(\pi)$ of $\mathcal{M}_\pi$, such that $\underline{S}$ is the essential image of the canonical fully faithful functor

$$\mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M} \qquad (\xleftarrow{\sim} \mathrm{Ind}_\pi(\mathcal{M}_\pi)\ ([117])), \tag{4.13.2.1}$$

extending the canonical inclusion $\underline{S}(\pi) \hookrightarrow \mathcal{M}$ (cf. 4.12.3).

p. 106

b) If S is accessible, we can find cardinals π'_0, π'_1 such that $\pi'_1 \geqslant \pi'_0 \geqslant \pi_0$ (where π_0 is a given cardinal useful for $\mathcal{M}$) and such that every cardinal $\pi \geqslant \pi'_1$ with $\pi^{\pi'_0} = \pi$ [page 285] *satisfies the conditions of a), taking $S(\pi) = S \cap \mathrm{Ob}\,\mathcal{M}_\pi$, i.e. $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$. Under these conditions, π is useful for $\underline{S}$* and *for $\mathcal{M}$, $S(\pi)$ is a* strictly *full subcategory of $\mathcal{M}_\pi$, stable in $\mathcal{M}_\pi$ under direct factors (or what amounts to the same thing, $\mathcal{M}_\pi$ being Karoubian, a Karoubian subcategory of $\mathcal{M}_\pi$).*

c) We can take π'_0 to be the smallest cardinal $\geqslant \pi_0$ useful for $\underline{S}$ and such that $\underline{S} \hookrightarrow \mathcal{M}$ is π'_0-accessible, and $\pi'_1 = \sup(\pi'_0, \mathrm{card}\,\mathrm{red}\,\underline{S}_{\pi'_0})$.

Part b) as well as c) has already been proved, which implies the "only if" in part a). It remains to prove the "if" in a). But the hypothesis implies that

$$\underline{S} \xleftarrow{\sim} \mathrm{Ind}_\pi(S_\pi),$$

which implies that $\underline{S}$ is accessible and even π-accessible, and it also implies that $\underline{S}$ is a *strictly* full subcategory of $\mathcal{M}$. It remains to see

[page 286]

the inclusion functor is accessible, in fact it is π-accessible, since (4.13.2.1) is (cf. 4.12.3).

Definition 4.13.3. Let $\mathcal{M}$, $S \subset \mathrm{Ob}\,\mathcal{M}$, $\underline{S} \subset \mathcal{M}$ be as in 4.12.1, π an infinite cardinal. We say that S is a *π-accessible* subset of $\mathrm{Ob}\,\mathcal{M}$, or that $\underline{S}$ is a π-accessible subcategory in $\mathcal{M}$, if S, π satisfy the conditions of 4.13.2 a), i.e. $\underline{S}$ is the essential image of $\mathrm{Ind}_\pi(S(\pi))$, where $S(\pi)$ is a full subcategory of $\mathcal{M}_\pi$.

Note that, at the cost of enlarging $S(\pi)$ to $S'(\pi) = \natural$-closure of $S(\pi)$ in $\mathcal{M}_\pi$, we can suppose that $S(\pi)$ is a *strictly full and Karoubian* subcategory

[page 287]

[116] "optional".

[117] if π useful for $\mathcal{M}$.

of $\mathcal{M}_\pi$. We then know that, $S(\pi)$ being Karoubian, $S(\pi) \longrightarrow \underbrace{(\mathrm{Ind}_\pi(S(\pi))}_{\pi}$ is an

equivalence of categories. Since $\mathrm{Ind}_\pi(S(\pi)) \simeq \underline{S}$ and since $S(\pi)$ is strictly full in $\mathcal{M}_\pi$, hence also in $\mathcal{M}$ and a fortiori in $\underline{S}$, that we have

$$\underline{S}(\pi) = \underline{S}_\pi, \qquad \text{subcategory of } \underline{S} \text{ of the } \pi\text{-accessible objects of } S. \tag{4.13.1}$$

I say that we will also have

$$\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi. \tag{4.13.2}$$

As c is clear, it remains to see that $\underline{S} \cap \mathcal{M}_\pi \subset \underline{S}(\pi)$, now it is immediate that $\underline{S} \cap \mathcal{M}_\pi \subset S_\pi$, and we conclude by (4.13.1). We thus find:

Proposition 4.13.4. *Let $\mathcal{M}$ be an accessible category, π an infinite cardinal. Let Σ_π be the set of*

strictly full subcategories $\underline{S}(\pi)$ of $\mathcal{M}_\pi$ that are Karoubian, or what amounts to the same thing, stable in $\mathcal{M}_\pi$ under direct factors. Let Σ'_π be the set of subcategories of $\mathcal{M}$ that are π-accessible in $\mathcal{M}$. Consider the maps [page 288] p. 107

$$\Sigma_\pi \underset{\beta_\pi}{\overset{\alpha_\pi}{\rightleftarrows}} \Sigma'_\pi \tag{4.13.4.1}$$

defined as follows: If $\underline{S}(\pi) \in \Sigma_\pi$, $\alpha_\pi(\underline{S}(\pi))$ is the essential image of the fully faithful functor $\mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M}$. If $\underline{S} \in \Sigma'_\pi$, $\beta_\pi(\underline{S}) = \underline{S} \cap \mathcal{M}_\pi$. The maps α_π, β_π are bijections, inverse to one another.

We have not had to suppose that π is useful for $\mathcal{M}$, nor that π is bounded below by a cardinal useful for $\mathcal{M}$, i.e. that we have $\pi \geqslant \pi_0$, where π_0 is the smallest cardinal useful for $\mathcal{M}$.

4.13.5. Let $\underline{S}$ be a *strictly full* subcategory of $\mathcal{M}$ (accessible category), π'_0 an infinite cardinal. To say that $\underline{S}$ is π'_0-accessible in $\mathcal{M}$ means that the following conditions are satisfied. [page 289]

1) π'_0 is useful for $\underline{S}$, i.e. $\underline{S} \overset{\sim}{\leftarrow} \mathrm{Ind}_{\pi'_0}(\underline{S}_{\pi'_0})$.

2) $\underline{S} \hookrightarrow \mathcal{M}$ is π'_0-accessible.

3) $\underline{S}_{\pi'_0} \hookrightarrow \mathcal{M}_{\pi'_0}$ (which implies that $S_{\pi'_0}$ is equivalent to a small category).

Suppose these conditions satisfied, and let π be a cardinal $> \pi'_0$. Is $\underline{S}$ π-accessible in $\mathcal{M}$, i.e. are the similar conditions satisfied for π? This is obvious for 2°), but may be false for 1°, 2° . But let us call on 4.13.2 c). Put

$$\pi'_1 = \sup(\pi'_0, \operatorname{card}\operatorname{red} S_{\pi'_0}, \underbrace{\pi_0}_{\substack{\text{smallest infinite}\\ \text{cardinal useful}\\ \text{for } \mathcal{M}}}).$$

Then for $\pi \geqslant \pi'_1$ such that $\pi^{\pi'_0} = \pi$, $\underline{S}$ is π-accessible in $\mathcal{M}$ (4.13.2 b and c). Suppose e.g. [page 290]

$$\pi'_0 \geqslant \pi_0$$

(in practice this will always be satisfied) and moreover

$$\pi'_0 \geqslant \operatorname{card}\operatorname{red} \mathcal{M}_{\pi_0},$$

i.e. suppose

$$\pi'_0 \geqslant \sup(\pi_0, \operatorname{card}\operatorname{red} \mathcal{M}_{\pi_0}) = \pi_1.$$

Then

$$\pi_1' = \sup(\pi_0', \operatorname{card}\operatorname{red} S_{\pi_0'}) \leqslant \sup(\pi_0', \operatorname{card}\operatorname{red} \mathcal{M}_{\pi_0'}) \leqslant 2^{(\pi_0'^{\pi_0})}$$

(4.7.7.1, p. 177). So we find this:

Corollary 4.13.6. *Let π_0' be a cardinal $\geqslant \pi_1 = \sup(\pi_0, \operatorname{card}\operatorname{red}\mathcal{M}_{\pi_0})$ ([118]), and let $\underline{S}$ be a π_0'-accessible subcategory in $\mathcal{M}$. Then for every cardinal π such that $\pi \geqslant 2^{(\pi_0'^{\pi_0})}$, and such that we have $\pi^{\pi_0'} = \pi$ (i.e. π of the form $c^{\pi_0'}$, with c a cardinal $\geqslant 2^{(\pi_0'^{\pi_0})}$), $\underline{S}$ is π-accessible in $\mathcal{M}$. (Moreover, these cardinals are regular for $\underline{S}$ and for $\mathcal{M}$, hence $\underline{S} \simeq \operatorname{Ind}_\pi(\underline{S}_\pi)$, $\mathcal{M} \simeq \operatorname{Ind}_\pi(\mathcal{M}_\pi)$).*

([119], [120]).

Proposition 4.13.7 ([121]). *Let $\mathcal{M}$ be an accessible category, $(S_\alpha)_{\alpha\in A}$ a family of subsets of $\operatorname{Ob}\mathcal{M}$, indexed by a small set A, $(\underline{S}_\alpha)_{\alpha\in A}$ the corresponding family of accessible subcategories in $\mathcal{M}$. Let $S = \bigcap_{\alpha\in A} S_\alpha$, $\underline{S} = \bigcap_{\alpha\in A}\underline{S}_\alpha$, so that $\underline{S}$ is the full subcategory of $\mathcal{M}$ defined by $S \subset \operatorname{Ob}\mathcal{M}$. This being so, if the S_α are accessible, so is S – or again, if the $\underline{S}_\alpha$ are accessible in $\mathcal{M}$, so is $\underline{S}$.* p. 108 [page 291]

This follows from the more precise result:

Corollary 4.13.8 ([122]). *With the notation of 4.13.7 and 4.13.6, let, for every $\alpha \in A$, π_α' be a cardinal $\geqslant \pi_1$ such that S_α is π_α'-accessible. Let*

$$\pi_0' = \sup_{\alpha\in A} \pi_\alpha'$$
$$\pi_0'' = \sup(\operatorname{card} A, \sup_\alpha 2^{(\pi_\alpha'^{\pi_0})})$$

and let $\pi \geqslant \pi_0''$ be a cardinal such that $\pi^{\pi_0'} = \pi$, i.e. of the form $c^{\pi_0'}$, with $c \geqslant \pi_0'$. Then S is π-accessible (i.e. $\underline{S}$ is π-accessible in $\mathcal{M}$). In particular, S is $2^{\pi_0''}$-accessible. ([123]).

Proof. Let $\alpha \in A$. Since $\pi_\alpha' \leqslant \pi_0'$ and $\pi^{\pi_0'} = \pi$, we have a fortiori [page 292]

$$\pi^{\pi_\alpha'} = \pi, \qquad \text{moreover } \pi \geqslant \pi_0'' \geqslant 2^{(\pi_\alpha'^{\pi_0})},$$

therefore by 4.13.6, $\underline{S}_\alpha$ is π-accessible in $\mathcal{M}$. On the other hand, since $\underline{S}_\alpha$ is π_α'-accessible, it is stable in $\mathcal{M}$ under $\varinjlim$ large relative to π_α', a fortiori under $\varinjlim$ large relative to π_0' ($\geqslant \pi_\alpha'$). Moreover, $\pi^{\pi_0'} = \pi$ implies $\pi \geqslant 2^{\pi_0'} > \pi_0'$. Thus 4.13.8 follows from the

Corollary 4.13.9. *Let $\pi > \pi_0$ be a cardinal such that $\underline{S}_\alpha$ is π-accessible in $\mathcal{M}$ for every $\alpha \in A$, and such that* $\operatorname{card} A \leqslant \pi$. *Suppose moreover that there exists a cardinal π_0' such that $\pi_0 \leqslant \pi_0' < \pi$, such that the $\underline{S}_\alpha$ are stable in $\mathcal{M}$ under $\varinjlim$ large relative to π_0'. Then $\underline{S} = \bigcap_{\alpha\in A}\underline{S}_\alpha$ is π-accessible in $\mathcal{M}$.*

([124]).

Put, for every $\alpha \in A$,

$$\underline{S}_\alpha(\pi) \overset{\text{def}}{=} \underline{S}_\alpha \cap \mathcal{M}_\pi$$
$$\underline{S}(\pi) \overset{\text{def}}{=} \underline{S} \cap \mathcal{M}_\pi = \bigcap \underline{S}_\alpha(\pi).$$

Each $\underline{S}_\alpha(\pi)$ is stable in $\mathcal{M}$ under $\varinjlim$ large relative to π_0' and of cardinal $\leqslant \pi$ (since this is so for $\underline{S}_\alpha$ and $\mathcal{M}_\pi$) (so the same is true of their intersection

$\underline{S}(\pi)$. A fortiori, $\underline{S}(\pi)$ is a strictly full and Karoubian subcategory of $\mathcal{M}_\pi$.) [page 293] p. 109

[118] **N.B.** π_0 is the smallest cardinal useful for $\mathcal{M}$.

[119] Continue here with 4.13.23 etc.

[120] Put here lemma 4.16.7.11 noted in the margin p. 292.

[121] Move it after 4.13.23.

[122] Cf. more reasonable and more correct reformulation page 302, 302 bis.

[123] This looks a bit outlandish! We must moreover suppose that π is not a limit cardinal!

[124] This would be the moment to formulate here Lemma 4.16.7.11 (p. 488), which should even be moved up with the first generalities on π-accessible subsets to as early as before 4.13.7 (immediately after 4.13.6).

On the other hand, each $\underline{S}_\alpha$ being stable in $\mathcal{M}$ under $\varinjlim$ large relative to π_0', the same holds for their intersection $\underline{S}$. A fortiori, the latter is stable under $\varinjlim$ large relative to π, so the canonical fully faithful functor $\mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M}$ factors through S as

$$\mathrm{Ind}_\pi\, \underline{S}(\pi) \hookrightarrow \underline{S}\ ,$$

we must prove that this functor is essentially surjective. So let $x \in S = \mathrm{Ob}\,\underline{S}$, it all comes down to proving that

$$\varinjlim_{i \in \underline{S}(\pi)/x} x_i \xrightarrow{\sim} x\ , \qquad (*)$$

and that $\underline{S}(\pi)/x$ is large relative to π.

Note that, since $\pi_0' \geqslant \pi_0$, and $\pi^{\pi_0'} = \pi$, we will have $\pi^{\pi_0} = \pi$, so π is a useful cardinal for $\mathcal{M}$, so $I = \mathcal{M}_\pi/x$ is large relative to π, and we have

$$\varinjlim_{i \in I = \mathcal{M}_\pi/x} x_i \xrightarrow{\sim} x\ . \qquad (**)$$

Thus, if we admit that $\underline{S}(\pi)/x$ is large relative to π, the fact that $(*)$ is an isomorphism is equivalent to saying that the canonical functor [page 294]

$$J \overset{\text{def}}{=} \underline{S}(\pi)/x \hookrightarrow \mathcal{M}_\pi/x \ = \ I$$

is *cofinal* (cf. 4.4.2, p. 79). But we see at once that this functor is an inclusion functor of a strictly full subcategory (stable under $\varinjlim$ large relative to π_0', of cardinality $\leqslant \pi$).

Note now that the arrow $(**)$ similar to $(*)$, with $\underline{S}(\pi)$ replaced by $\underline{S}_\alpha(\pi)$, is indeed an isomorphism, since $x \in \mathrm{Ob}\,\underline{S} \subset \mathrm{Ob}\,\underline{S}_\alpha$, and $\underline{S}_\alpha$ is π-accessible. This is interpreted as the fact that the inclusion functor

$$J_\alpha \overset{\text{def}}{=} \underline{S}_\alpha(\pi)/x \hookrightarrow \mathcal{M}_\pi/x \ = \ I$$

is cofinal. This is again an inclusion of a strictly full subcategory J_α of I, stable under $\varinjlim$ large relative to π_0' and of cardinality $\leqslant \pi$, and we have

$$J \ = \ \bigcap J_\alpha\ .$$

Thus the desired condition will follow from the [page 295]

Lemma 4.13.10 ([125],[126]). *Let π_0' and $\pi > \pi_0'$ be infinite cardinals, I a category equivalent to a small category, $(J_\alpha)_{\alpha \in A}$ a family of subcategories J_α of I. We assume:*

ⓐ *I is large relative to π ([127]).*

ⓑ *I is stable under $\varinjlim$ of cardinality $\leqslant \pi$, and large relative to π_0' ([128]).*

p. 110

ⓒ *$\forall\, r \in A$, J_r is stable in I under the direct limits envisaged in b) (in particular, J_r is a* strictly full *subcategory of I).*

ⓓ *$\forall\, r \in A$, J_r is a* cofinal *subcategory of I (*large relative to π*).*

ⓔ $\mathrm{card}\, A \leqslant \pi$.

Let $J = \bigcap_{r \in A} J_r$. Then J is still cofinal in I, and large relative to π (and moreover stable in I under $\varinjlim$ of cardinality $\leqslant \pi$ and large relative to π_0').

[page 296]

In other words, if I is large relative to π, the intersection J of a family $(J_\alpha)_{\alpha\in A}$ of strictly full subcategories cofinal and large relative to π, is still cofinal and large relative to π, *provided that* $\operatorname{card} A \leqslant \pi$, and that we postulate a stability property of I and of the J_α in I, for direct limits of cardinality $\leqslant \pi$ and large relative to a suitable given cardinal $\pi'_0 < \pi$.

4.13.10.1. Note moreover that if $f : J \longrightarrow I$ is a cofinal *fully faithful* functor, with I large relative to π, then J is automatically large relative to π. This follows e.g. from 4.9.6.3 2° (p. 225) by the criterion c_π (since for f fully faithful, condition F 3_π is tautologically satisfied). We can also see it thus: Let B be a set with $\operatorname{card} B \leqslant \pi$, we must prove that $J \longrightarrow J^B$ is cofinal, now we have

$$\begin{array}{ccc} J & \xrightarrow{\text{cof. ?}} & J^B \\ {\scriptstyle\text{cof.}}\downarrow & & \downarrow{\scriptstyle\text{cof.}} \\ I & \xrightarrow{\text{cof.}} & I^B\ , \end{array}$$

where three of the four arrows of the square are cofinal, $J^B \longrightarrow I^B$ being so as a product of cofinal functors between *filtering* categories (cf. 4.9.6.5, page 227) – indeed we already know that J is filtering.

Thus the composite [page 297]

$$J \longrightarrow J^B \xrightarrow{\text{cof.}} I^B$$

is cofinal. But since the second functor $J^B \longrightarrow I^B$ is fully faithful, it follows at once that $J \longrightarrow J^B$ is cofinal (for $\xi\backslash J$ for $\xi \in \operatorname{Ob} J^B$ is the same, up to isomorphism, whether we consider ξ as an object of J^B, or of I^B).

Thus in the statement of condition d) of the lemma, it is unnecessary to specify that J_α is large relative to π, and likewise in the conclusion about J.

Since $J \hookrightarrow I$ is fully faithful, to see that this functor is cofinal, it suffices to see that it satisfies F 1, i.e. that for every $i \in \operatorname{Ob} I$, there exists $j \in J$ with $i \longrightarrow j$. We are going to construct j as the limit of an ordinal direct system of objects i_α of I, indexed by the ordinals α such that $\operatorname{card}\alpha < \pi$. Denote by α_π the smallest ordinal such that p. 111
$\operatorname{card}\alpha_\pi = \pi$. The ordered set I_{α_π} of ordinals $\alpha < \alpha_\pi$ being large relative to π'_0 ([129]) and [page 298]
of cardinality π, we will therefore be able to take the limit $\varinjlim_{\alpha\in I_{\alpha_\pi}} i_\alpha$ in I.

Construction of $(i_\alpha)_{\alpha\in I_{\alpha_\pi}}$. Let

$$\alpha_\pi \ = \ \text{smallest ordinal } \alpha \text{ such that } \operatorname{card}\alpha = \pi, \tag{1}$$

so this is the ordinal characterized by

$$\begin{cases} \operatorname{card}\alpha_\pi \ = \ \pi \\ \alpha < \alpha_\pi \implies \operatorname{card}\alpha \ < \ \pi\ , \end{cases} \tag{2}$$

or again by the fact that the ordered set I_{α_π} of ordinals $\alpha < \alpha_\pi$ is *perfectly well ordered*. Thus, I_{α_π} *is large relative to* π' *for every cardinal* $\pi' < \pi$ ([130]). In conditions b) c) of

[125] We must assume that π is not a limit cardinal, to be able to conclude that I is stable under $\varinjlim$ of type I_{α_π}, and the J_r stable in I under these limits.

[126] Cf. more general form of the lemma in 4.13.10.2, p. 302.

[127] It suffices that I be large relative to every $\pi' < \pi$.

[128] Here stability under $\varinjlim$ of type I_{α_π} suffices, cf. page 302 .

[129] false if π is a limit cardinal.

[130] False if π is a limit cardinal.

4.13.10, it will suffice to assume I stable under $\varinjlim$ of type I_{α_π}, and the J_r ($r \in A$) stable in I under this type of limits.

Lemma 4.13.10.1. *Let π be an infinite cardinal, A a nonempty set such that* $\operatorname{card} A \leqslant \pi$. *Then there exists a map* [page 299]

$$\varphi \,:\, I_{\alpha_\pi} \longrightarrow A$$

such that for every $r \in A$, $\varphi^{-1}(\{r\})$ is a cofinal subset of I_{α_π}, or what is more precise, $\operatorname{card} \varphi^{-1}(\{r\}) = \pi$.

Indeed, it suffices to consider a set A' of cardinality π (e.g. $A' = I_{\alpha_\pi}$), the map

$$\psi \;=\; \mathrm{pr}_1 \,:\, A \times A' \longrightarrow A\,,$$

and to give oneself a bijection

$$A \times A' \xrightarrow{\sim} I_{\alpha_\pi}$$

which exists since

$$\operatorname{card}(A \times A') \;=\; \operatorname{card} A \cdot \operatorname{card} A' \;=\; \operatorname{card} A' \;=\; \pi \;=\; \operatorname{card} I_{\alpha_\pi}\,,$$

since $0 \neq \operatorname{card} A \leqslant \operatorname{card} A' = \pi$ and π infinite. We assume φ chosen as in 4.13.10.1, and we will construct the direct system $(i_\alpha)_{\alpha \in I_{\alpha_\pi}}$ in such a way that we have

$$i_\alpha \;\in\; J_{\varphi(\alpha)}\,.$$

Setting

$$j \;=\; \varinjlim_{\alpha \in I_{\alpha_\pi}} i_\alpha\,,$$

for every r, since

$$I_r \overset{\text{def}}{=} \varphi^{-1}(r) \;\subset\; I_{\alpha_\pi}$$ p. 112 [page 300]

is cofinal in I_{α_π}, we will therefore have

$$j \;=\; \varinjlim_{\alpha \in I_r} i_\alpha\,,$$

and by virtue of $(*)$, the i_α for $\alpha \in I_r$ are in J_r, so by passage to the limit, using hypothesis c), we find that $j \in J_r$, so

$$j \;\in\; \bigcap_{r \in A} J_r \;=\; J\,,$$

and since we will choose

$$i_0 \;=\; i\,, \qquad (**)$$

we will indeed have found $i \longrightarrow j$ with $j \in J$. So it suffices to construct the direct system (i_α) satisfying $(*)$ and $(**)$.

If the direct system has been constructed up to the stage $i_\alpha \in \operatorname{Ob} J_{\varphi(\alpha)}$, we continue with

$$i_\alpha \longrightarrow i_{\alpha+1} \;\in\; \operatorname{Ob} J_{\varphi(\alpha+1)}\,,$$

which is possible since $J_{\varphi(\alpha+1)}$ is cofinal. So it remains to construct i_α for α a limit ordinal. Now the set of i_β for $\beta < \alpha$ is of cardinality $< \pi$, and since by a) I is large relative to π (it suffices that it be so relative to every cardinal $< \pi$), there exists an $i \in \operatorname{Ob} I$ which dominates all the i_β. Let us choose [page 301]

$$i_\beta \xrightarrow{v_\beta} i\,.$$

We will not have a priori commutativity in

$$i_{\beta'} \xrightarrow{\tau_{\beta,\beta'}} i_\beta \xrightarrow{v_\beta} i\ , \qquad i_\beta \xrightarrow[v_{\beta'}]{} i$$

but since the set of double arrows $i_{\beta'} \overset{v_\beta\tau_{\beta,\beta'}}{\underset{v_{\beta'}}{\rightrightarrows}} i$ is of cardinality $< \pi$, I being large relative to π, there exists $i \longrightarrow i'$ which equalizes them all. Since $J_{\varphi(\alpha)}$ is cofinal in I, there exists $i'' \in J_{\varphi(\alpha)}$, and $i' \longrightarrow i''$. We take $i_\alpha = i''$, with the composite homomorphisms

$$i_\beta \xrightarrow{v_\beta} i \longrightarrow i' \longrightarrow i'' \qquad i_\beta \xrightarrow[\tau_{\alpha,\beta}]{} i''$$

as transition morphisms. This completes the proof of 4.13.10 and thereby of 4.13.9, 4.13.8. p. 113 [page 302]

Remark 4.13.10.2. Here is what we really had to use, to establish the conclusion of 4.13.10, in terms of I, π, $(J_r)_{r\in A}$ (and without introducing any π'_0).

(a′) I is large relative to every cardinal $\pi' < \pi$.

(b′) I is stable under $\varinjlim$ of type I_{α_π}.

(c′) $\forall\, r \in A$, J_r is stable in I under $\varinjlim$ of type I_{α_π}.

(d′) $\forall\, r \in A$, J_r is cofinal in I.

(e′) $\operatorname{card} A \leqslant \pi$.

(**N.B.** d′) and e′) are none other than d) and e) of 4.13.10, whereas a′), b′), c′) are weakened versions of a), b), c) respectively.)

4.13.8 (reformulation). *Let, for $r \in A$, $\pi''_r \geqslant \pi'_r \geqslant \pi_0$ (= given useful cardinal for $\mathcal{M}$) be as specified in 4.13.2 b) (and c), let $\pi' = \sup_{r\in A} \pi'_r$, $\pi'' = \sup_{r\in A} \pi''_r$, so that for every cardinal π with $\pi^{\pi'} = \pi \geqslant \pi''$, $\underline{S}_r$ is π-accessible in $\mathcal{M}$ (and π useful for $\underline{S}_r$* and *for $\mathcal{M}$). Let c be a* non-limit *cardinal such that $c \geqslant \operatorname{card} A$ and $c > \pi'$ (e.g. the successor of $\sup(\operatorname{card} A, \pi')$). Let us take π such that*

$$\pi^{\pi'} \ = \ \pi \ \geqslant \ \sup(\pi'', c)\ .$$

Then $\underline{S} = \bigcap_{r\in A} \underline{S}_r$ is π-accessible.

Indeed, if α_c is the first ordinal of cardinality c, since c is not a limit, I_{α_c} is large relative to every cardinal $c' < c$, in particular relative to π'. Now for every $x \in \operatorname{Ob}\mathcal{M}$, $I = \mathcal{M}_\pi/x$ and the $J_r = \underline{S}_r(\pi)/x$ are stable under $\varinjlim$ large relative to π' and of cardinality $\leqslant \pi$, and the J_r stable in I for these limits, so this is in particular the case for $\varinjlim$ of type I_{α_c}. We can therefore apply result 4.13.10.2 to conclude that if the J_r are cofinal in I (which is large relative to π, π being useful for $\mathcal{M}$), then $J = \underline{S}(\pi)/x$ is so as well. This then implies, as we have seen, that $\underline{S}$ is π-accessible in $\mathcal{M}$. [page 302 bis]

4.13.10.3. We can reformulate the preceding reformulation, using the terminology introduced below (4.13.23):

Let c be a cardinal, α_c the smallest ordinal α such that $\operatorname{card}\alpha = c$, π a cardinal $\geqslant c$. We assume

(∗) *π is useful for $\mathcal{M}$, and $\mathcal{M}$ is stable under $\varinjlim$ of type I_{α_c}. (This second condition is the case e.g. if c is* non-limit *and $c > \pi_0$, π_0 being a cardinal such that $\mathcal{M}$ satisfies 2_{π_0}, e.g. a* useful *cardinal for $\mathcal{M}$ ([131]).*

Then the set of subcategories $\underline{S}$ of $\mathcal{M}$ which are α_c-π-accessible in $\mathcal{M}$ (i.e. which are π-accessible and stable in $\mathcal{M}$ under $\varinjlim$ of type I_{α_c}) is stable under intersection of cardinality $\leqslant c$. More generally, if Φ is a set of categories such that

1°) $I \in \Phi \Longrightarrow \operatorname{card} I \leqslant \pi$ and

2°) $I_{\alpha_c} \in \Phi$,

then the set of subcategories of $\mathcal{M}$ which are Φ-π-accessible in $\mathcal{M}$, is stable under intersection of cardinality $\leqslant c$.

Definition 4.13.11 ([132]). Let $\mathcal{M}$ be an accessible category, π_0 a useful cardinal for $\mathcal{M}$, $\pi > \pi_0$ a cardinal. A category I is said to be π_0-π-*adapted* if

p. 114
[page 303]

a) I is equivalent to a small category,

b) I is large relative to π, and

c) I is stable under $\varinjlim$ of cardinality $\leqslant \pi$, and large relative to π_0 ([133]).

A direct system $(x_i)_{i\in I}$ of type I in $\mathcal{M}$ is said to be π_0-π-*adapted*, if I is π_0-π-adapted, i.e. satisfies conditions a) b) c), and if moreover we have

d) the functor $i \longmapsto x_i : I \longrightarrow \mathcal{M}$ commutes with the $\varinjlim$ (large relative to π_0 and of cardinality $\leqslant \pi$) considered in c), and

e) the x_i are in $\mathcal{M}_\pi$.

Proposition 4.13.12. *Let $\mathcal{M}$ be a π_0-accessible category, and $\pi > \pi_0$ a useful cardinal for $\mathcal{M}$. Then every $x \in \operatorname{Ob}\mathcal{M}$ admits a representation as $x \xleftarrow{\sim} \varinjlim_{j\in J} x_j$, where the direct system $(x_j)_{j\in J}$ in $\mathcal{M}$ is π_0-π-adapted. More precisely, one can take $J = \mathcal{M}_\pi/x$.*

[page 304]

Indeed, the verification of properties a) to d) for the canonical direct system, with $\varinjlim$ isomorphic to x, is immediate.

The preceding proposition does not say whether one can take an ordered set for J. In this case, condition c) means that for every subset J' of J large relative to π_0 and of cardinality $\leqslant \pi$, J admits a least upper bound $i_{J'}$ in J, and condition d) that we then have $x_{i_{J'}} \xleftarrow{\sim} \varinjlim_{i\in J'} x_i$.

The natural idea, to construct a π_0-π-adapted direct system $(x_j)_{j\in J}$, with J ordered, starting from the one we have canonically, with $J = \mathcal{M}_\pi/x$, is to take for I the set, ordered by inclusion, of subcategories J' of J_0 (small full subcategory of J equivalent to J) which are of cardinality $\leqslant \pi$ and large relative to π_0. (Compare (4.7.3.3), page 165). This ordered set is indeed (small and) *large relative to* π, if we assume that $\pi^{\pi_0} = \pi$ (cf. proof 4.7.5, p. 168-171).

Lemma 4.13.13. *Let π_0, $\pi > \pi_0$ be infinite cardinals such that $\pi^{\pi_0} = \pi$, and J a small category, large relative to π. Let I be the set of subcategories J' of J which are large relative to π_0, and of cardinality $\leqslant \pi$. Then I is π_0-π-adapted.*

[page 305]

[131] For then I_{α_c} is large relative to π_0.

[132] Recast with 4.13.23 etc.

[133] **N.B.** If $\pi = \pi_0$, condition c) is equivalent to saying that I is stable under direct factors.

It remains to see that for every ordered set A large relative to π_0 and of cardinality $\leqslant \pi$, if $r \longmapsto J_r$ is an increasing family of elements of I indexed by A, then

$$J \overset{\text{def}}{=} \bigcup_{r\in A} J_r \;=\; \varinjlim_{r\in A} J_r$$

is in I, i.e. is a subcategory of J of cardinality $\leqslant \pi$ (which is clear) and *large relative to* π_0. This will follow from the

Lemma 4.13.13.1. *Let $(J_r)_{r\in A}$ be a direct system of categories large relative to π_0, indexed by an ordered set A large relative to π_0. Then $J = \varinjlim_{r\in A} J_r$ is large relative to π_0.* p. 115

We must see that if B is a set of cardinality $\leqslant \pi_0$, then the diagonal functor [page 306]

$$\delta \;:\; J \longrightarrow J^B \qquad (*)$$

is cofinal. Now since A is large relative to π_0, $\operatorname{card} B \leqslant \pi_0$, products indexed by B commute with the $\varinjlim$ of type A (SGA 4 I 9.8 applied to Cat and to π_0). Thus the arrow $(*)$ is interpreted as the direct limit of the similar arrows

$$\delta_r \;:\; J_r \longrightarrow J_r^B \;, \qquad (**)$$

which are cofinal, hence δ is cofinal as a direct limit of cofinal functors.

Lemma 4.13.13.2. *Under the hypotheses of 4.13.13, let $\mathcal{M}$ be a category satisfying L_{π_0}, and $(x_j)_{j\in J}$ (where $J \longrightarrow \mathcal{M}$) a direct system in $\mathcal{M}$, indexed by J. For every $J' \in I$, put*

$$x_{J'} \;=\; \varinjlim_{j\in J'} x_j$$

(which exists in $\mathcal{M}$, since $\mathcal{M}$ satisfies L_{π_0}). Then the $(x_{J'})_{J'\in I}$ form a direct system in $\mathcal{M}$, and $I \longrightarrow \mathcal{M}$ commutes with direct limits large relative to π_0 and of cardinality $\leqslant \pi$. If $\mathcal{M}$ is π_0-accessible, and the x_j ($j \in \operatorname{Ob} J$) are in $\mathcal{M}_\pi$, then $(x_{J'})_{J'\in I}$ is a π_0-π-adapted direct system in $\mathcal{M}$. [page 307]

All this is essentially trivial, and gives:

Corollary 4.13.14 (of 4.13.12). *Under the conditions of 4.13.12, if $\pi^{\pi_0} = \pi$, one can for every $x \in \operatorname{Ob}\mathcal{M}$ find a representation $x \simeq \varinjlim_{i\in I} x_i$, where $(x_i)_{i\in I}$ is a π_0-π-adapted direct system with I* an ordered set. *Moreover, if $\pi' > \pi$ and if $x \in \mathcal{M}_{\pi'}$, one can take I such that we have*

$$\operatorname{card} I \;\leqslant\; \pi'^{\pi} \;.$$

This last addendum is immediate, via the usual estimates. [page 308]

Theorem 4.13.15 ([134]). *Let $\mathcal{M}$ be a π_0-accessible category, and $\pi > \pi'_0 \geqslant \pi_0$ such that $\pi^{\pi_0} = \pi$ ([135]). Let $\underline{S}$ be a strictly full subcategory of $\mathcal{M}$, and let $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$. The following conditions are equivalent.*

ⓐ *$\underline{S}$ is π-accessible in $\mathcal{M}$ (4.13.3) and $\underline{S}$ is stable in $\mathcal{M}$ under direct limits large relative to π'_0.*

p. 116

a′) *$\underline{S}$ is π-accessible in $\mathcal{M}$, and $\underline{S}(\pi)$ is stable in $\mathcal{M}_\pi$ under direct limits large relative to π'_0 and of cardinality $\leqslant \pi$ ([136]).*

[134] Cf. more general and more symmetric version of theorem 4.13.15 in 4.13.22.

[135] One can restrict oneself to assuming π useful for $\mathcal{M}$ (instead of $\pi^{\pi_0} = \pi$), then one finds that a, a′, b, c are equivalent and imply c′, but are perhaps not equivalent to c′.

[136] give a name to this last condition, say $\mathrm{L}(\pi'_0, \pi)$ (for $\underline{S}(\pi)$ in $\mathcal{M}_\pi$).

b) *For every object x of $\mathcal{M}$, we have this: the strictly full subcategory $\underline{S}(\pi)/x$ of $J = \mathcal{M}_\pi/x$ is cofinal in $\mathcal{M}_\pi/x$ if and only if $x \in \mathrm{Ob}\,\underline{S}$. Moreover, $\underline{S}(\pi)$ is stable in $\mathcal{M}_\pi$ under $\varinjlim$ large relative to π_0' and of cardinality $\leqslant \pi$.*

ⓒ *For every object x of $\mathcal{M}$ and for every representation $x \simeq \varinjlim_{i\in I} x_i$, where $(x_i)_{i\in I}$ is a π_0'-π-adapted direct system, (4.13.11), we have this: the strictly full subcategory $I_{\underline{S}}$ of I formed by the $i \in \mathrm{Ob}\,I$ such that $x_i \in \mathrm{Ob}\,\underline{S}$, is cofinal in I if and only if $x \in \mathrm{Ob}\,\underline{S}$. Moreover, $\underline{S}(\pi)$ is stable in $\mathcal{M}_\pi$ under $\varinjlim$ large relative to π_0' and of cardinality $\leqslant \pi$.*

[page 309]

c′) *As in d) , with I an ordered set.*

Corollary 4.13.16. *The hypotheses on $\mathcal{M}$, π_0, $\underline{S}$ being those of 4.13.15, for there to exist cardinals $\pi > \pi_0' \geqslant \pi_0$ satisfying the conditions $\pi^{\pi_0'} = \pi$ (a fortiori $\pi^{\pi_0} = \pi$) and the equivalent conditions a) to d′) of 4.13.15, it (is necessary and it) suffices that $\underline{S}$ be accessible in $\mathcal{M}$. More precisely, if $\underline{S}$ is accessible in $\mathcal{M}$, let $\pi_0' \geqslant \pi_0$ be a cardinal such that $\underline{S}$ is π_0'-accessible in $\mathcal{M}$, and let π be a cardinal $> \pi_0'$ such that $\pi^{\pi_0'} = \pi$. Then $(\underline{S}, \pi_0', \pi)$ satisfies the equivalent conditions of 4.13.15.*

Indeed, the "it is necessary" is obvious from condition a) of 4.13.15. For the "it suffices" we note that $\underline{S}$ π_0'-accessible in $\mathcal{M}$ implies that $\underline{S}$ is stable in $\mathcal{M}$ under $\varinjlim$ large relative to π_0'. Thus to verify a), it suffices to verify that $\underline{S}$ is π-accessible in $\mathcal{M}$. Now since $\underline{S}$ is π_0'-accessible and $\pi > \pi_0'$ satisfies $\pi^{\pi_0'} = \pi$, we conclude from this that $\underline{S}$ is [page 310] π-accessible, i.e. $\underline{S} \xleftarrow{\sim} \mathrm{Ind}_\pi(\underline{S}_\pi)$. Thus by 4.13.3, it remains to see that $\underline{S}_\pi \subset \mathcal{M}_\pi$, or what amounts to the same thing, that $\underline{S}_\pi \subset \underline{S}(\pi) \overset{\text{def}}{=} \underline{S} \cap \mathcal{M}_\pi$. (In fact, we shall even have equality, for it is clear that $\underline{S}(\pi) \subset \underline{S}_\pi \ldots$). Now $\underline{S}(\pi)$ is a subcategory of $\underline{S}$ stable under $\varinjlim$ large relative to π_0' and of cardinality $\leqslant \pi$ (since $\underline{S}$ and $\mathcal{M}_\pi$ are stable under these limits *in* $\mathcal{M}$, and these limits in $\underline{S}$ and in $\mathcal{M}$ are likewise), and $\underline{S}(\pi)$ contains $\underline{S}_{\pi_0'}$. Thus $\underline{S}(\pi)$ contains $\underline{S}_\pi$, by 4.7.1 (page 161), implication c) $\Longrightarrow$ b′), applied to $\underline{S}$, π_0', π instead of $\mathcal{M}$, π_0, π. This therefore proves corollary 4.13.16.

Proof of 4.13.15. Here are the ± tautological implications,

$$\mathrm{a} \Longrightarrow \mathrm{a}' \overset{\bullet}{\Longrightarrow} \mathrm{c} \Rightarrow \mathrm{c}', \quad \mathrm{c} \Rightarrow \mathrm{b}\,, \tag{$*$}$$

which I am going to spell out a little. We have a $\Longrightarrow$ a′ since if a is satisfied, $\underline{S}$ *and* [page 311] $\mathcal{M}_\pi$ are stable *in* $\mathcal{M}$ under $\varinjlim$ large relative to π_0' and of cardinality $\leqslant \pi$, hence also their intersection $\underline{S}(\pi)$, which is therefore also stable in $\mathcal{M}_\pi$ under the said limits. The p. 117 implication c $\Longrightarrow$ c′ is a tautology, and c $\Longrightarrow$ b as well, in view of 4.13.12, which tells us that the canonical direct system $(x_i)_{i\in I}$, with $I = \mathcal{M}_\pi/x$, with limit x is π_0-π-adapted. (**N.B.** Since $\pi^{\pi_0} = \pi$ and $\pi_0' \geqslant \pi_0$, we also have $\pi^{\pi_0} = \pi$, hence π is useful for $\mathcal{M}$, so that 4.13.12 applies.)

It remains to prove the "± tautology" a′ $\Longrightarrow$ c. So let us take a π_0'-π-adapted representation

$$x \simeq \varinjlim_{i\in I} x_i\,,$$

let us prove that

$$x \in \mathrm{Ob}\,\underline{S} \qquad \Longleftrightarrow \qquad I_{\underline{S}} \text{ cofinal in } I.$$

The implication $\Longleftarrow$ comes from the fact that, $\underline{S}$ being π-accessible in $\mathcal{M}$, is stable in $\mathcal{M}$ under $\varinjlim$ large relative to π ([137]). It remains to prove that if $x \in \mathrm{Ob}\,\underline{S}$, then $I_{\underline{S}}$ is cofinal in I. So let us consider the diagram of categories [page 312]

$$\begin{array}{ccc} I & \xrightarrow{\varphi} & J = \mathcal{M}_\pi/x \\ \uparrow & & \uparrow\alpha \\ I' = I_{\underline{S}} & \xrightarrow{\varphi'} & J' = \underline{S}(\pi)/x \ , \end{array}$$

this is a cartesian square ($I' = I_{\underline{S}}$ is the inverse image under $\varphi : I \longrightarrow J$ of the strictly full subcategory J' of J). We have the following properties

1) I is large relative to π, and φ cofinal ($\Longrightarrow J$ large relative to π).
2) I stable under $\varinjlim$ large relative to π_0' and of cardinality $\leqslant \pi$, and φ commutes with these $\varinjlim$.
3) J' is stable under $\varinjlim$ of the preceding type and α commutes with them ([138]).
4) J' cofinal in J.

We have just concluded that $I' = \varphi^{-1}(J')$ is cofinal in I. As it is a full subcategory (and even strictly full), it suffices to prove property F 1, i.e. that for every $i \in \mathrm{Ob}\,I$, exists $i' \in \mathrm{Ob}\,I'$ and an arrow $i \longrightarrow i'$. [page 313]

We are going to construct i' again by transfinite construction, on the ordinal interval I_{α_π}, by constructing a direct system

$$(i_\alpha)_{\alpha \in I_{\alpha_\pi}} \ , \qquad i_\alpha \in \mathrm{Ob}\,I \ ,$$

of objects of I, and a direct system $(j_\alpha)_{\alpha \in I_{\alpha_\pi}}$, $j_\alpha \in \mathrm{Ob}\,J'$, transition arrows denoted $\tau_{\beta,\alpha} : i_\alpha \longrightarrow i_\beta$, $\tau'_{\beta,\alpha} : j_\alpha \longrightarrow j_\beta$, finally for every α in I_{α_π} arrows

$$\varphi(i_\alpha) \xrightarrow{u_\alpha} j_\alpha \xrightarrow{v_\alpha} \varphi(i_{\alpha+1}) \ (\xrightarrow{u_{\alpha+1}} j_{\alpha+1} \ ) \qquad (*)$$

(with $\tau'_{\alpha+1,\alpha} : j_\alpha \longrightarrow j_{\alpha+1}$ and $\varphi(\tau_{\alpha+1,\alpha}) : \varphi(i_\alpha) \longrightarrow \varphi(i_{\alpha+1})$)

in J, such that

$$\begin{cases} v_\alpha u_\alpha &= \varphi(\tau_{\alpha+1,\alpha}) \\ u_{\alpha+1} v_\alpha &= \tau'_{\alpha+1,\alpha} \end{cases} \qquad ([139]) \ . \qquad (**)$$

p. 118

Let us show how the construction of such a system of objects and arrows with [page 314]

$$i_0 \ = \ i \qquad (***)$$

will imply the existence of $i' \in \mathrm{Ob}\,I'$ and of $i \longrightarrow i'$. Indeed, since I_{α_π} is large relative to π_0' (it is large relative to every cardinal $< \pi$), there exists

$$i' \overset{\text{def}}{=} \varinjlim_{\alpha \in I_{\alpha_\pi}} i_\alpha$$

[137] We use here, for $\Longleftarrow$, of the hypothesis $(x_i)_{i \in I}$ π_0-π-adapted, only the sole fact that I is large relative to π.

[138] This is where the stability hypothesis made in a$'$) comes in, for by a$'$), $\underline{S}$ is π-accessible in $\mathcal{M}$.

[139] cf. also $(***)$, next page.

in I, and we have a transition arrow

$$i \;=\; i_0 \longrightarrow i' \;,$$

so it remains to prove that $i' \in \operatorname{Ob} I'$, i.e. $\varphi(i') \in \operatorname{Ob} J'$. Now φ commuting with the $\varinjlim$ of type I_{α_π}, we shall have

$$\varphi(i') \;=\; \varinjlim \varphi(i_\alpha) \;.$$

I still need compatibilities

$$\begin{array}{ccc} \varphi(i_\alpha) & \xrightarrow{u_\alpha} & j_\alpha \\ {\scriptstyle \varphi(\tau_{\beta,\alpha})}\downarrow & & \downarrow{\scriptstyle \tau'_{\beta,\alpha}} \\ \varphi(i_\beta) & \xrightarrow{u_\beta} & j_\beta \end{array} \quad \text{and} \quad \begin{array}{ccc} j_\alpha & \xrightarrow{v_\alpha} & \varphi(i_{\alpha+1}) \\ {\scriptstyle \tau'_{\beta,\alpha}}\downarrow & & \downarrow{\scriptstyle \varphi(\tau_{\beta+1,\alpha+1})} \\ j_\beta & \xrightarrow{v_\beta} & \varphi(i_{\beta+1}) \end{array} \qquad (***)$$

forgotten earlier.
The compatibilities $(***)$ imply that we have homomorphisms [page 315]

$$\varphi(i') \underset{v}{\overset{u}{\rightleftarrows}} j \qquad \text{where } j \overset{\text{def}}{=} \varinjlim_{\alpha \in I_{\alpha_\pi}} j_\alpha \quad (\text{limit } \textit{in } J', \text{ hence in } J) \;,$$

and the compatibilities $(**)$ imply that

$$vu \;=\; \mathrm{id}_{\varphi(i')} \;, \qquad uv = \mathrm{id}_j \;,$$

hence that u, v are isomorphisms inverse to each other. Since J' is a *strictly* full subcategory of J, it does indeed follow that $\varphi(i') \in J'$, i.e. $i' \in I'$, q.e.d.

So everything comes down to constructing the

$$\underbrace{i_\alpha}_{\in \operatorname{Ob} I} \;,\; \tau_{\beta,\alpha} \;, \qquad \underbrace{j_\alpha}_{\in \operatorname{Ob} J'} \;,\; \tau'_{\beta,\alpha} \;, \qquad u_\alpha \;,\; v_\alpha \;,$$

satisfying the constraints $(**)$, $(***)$, $(****)$. We are going to make the construction p. 119
in three steps.

① Start:

$$i_0 \;=\; i \;.$$

Since J' is cofinal in J, exists $j_0 \in \operatorname{Ob} J'$ and

$$\varphi(i_0) \xrightarrow{u_0} j_0 \;.$$

② Suppose that the i_β, j_β have been constructed for $\beta < \alpha$, as well as the $\tau_{\beta',\beta} : i_\beta \longrightarrow i_{\beta'}$, $\tau'_{\beta',\beta} : j_\beta \longrightarrow j_{\beta'}$ for $\beta, \beta' < \alpha$, finally the $u_\beta : \varphi(i_\beta) \longrightarrow j_\beta$, $v_\beta : j_\beta \longrightarrow \varphi(i_{\beta+1})$ for $\beta < \alpha$ resp. for $\beta + 1 < \alpha$, so as to satisfy the constraints $(**)$, $(***)$, i.e. [page 316]

commutativities in

$$\left\{\begin{array}{ll}
\textcircled{1}\quad \varphi(i_\beta) \xrightarrow{u_\beta} j_\beta \xrightarrow{v_\beta} \varphi(i_{\beta+1}),\ \varphi(i_\beta) \xrightarrow{\varphi(\tau_{\beta+1,\beta})} \varphi(i_{\beta+1}) & \\
\textcircled{2}\quad j_\beta \xrightarrow{v_\beta} \varphi(i_{\beta+1}) \xrightarrow{u_{\beta+1}} j_{\beta+1},\ j_\beta \xrightarrow{\tau'_{\beta+1,\beta}} j_{\beta+1} & \text{if } \beta+1<\alpha \\
\textcircled{3}\quad \begin{array}{ccc} \varphi(i_\beta) & \xrightarrow{u_\beta} & j_\beta \\ {\scriptstyle \varphi(\tau_{\beta',\beta})}\downarrow & & \downarrow{\scriptstyle \tau'_{\beta',\beta}} \\ \varphi(i_{\beta'}) & \xrightarrow{u_{\beta'}} & j_{\beta'} \end{array} & \text{if } \beta<\beta'<\alpha \\
\textcircled{4}\quad \begin{array}{ccc} j_\beta & \xrightarrow{v_\beta} & \varphi(i_{\beta+1}) \\ {\scriptstyle \tau'_{\beta',\beta}}\downarrow & & \downarrow{\scriptstyle \varphi(\tau_{\beta'+1,\beta+1})} \\ j_{\beta'} & \xrightarrow{v_{\beta'}} & \varphi(i_{\beta'+1}) \end{array} & \text{if } \beta<\beta' \text{ and } \beta'+1<\alpha\ ,
\end{array}\right.$$

we must construct

[page 317]

1) $i_\alpha \in \operatorname{Ob} I$, and a homomorphism $(i_\beta)_{\beta\in I_\alpha} \longrightarrow i_\alpha$, i.e. $\tau_{\alpha,\beta} : i_\beta \longrightarrow i_\alpha$ $(\beta \in I_\alpha)$, compatible with the translations $\tau_{\beta',\beta}$ for $\beta<\beta'<\alpha$. p. 120

2) $j_\alpha \in \operatorname{Ob} J'$, and a homomorphism $(j_\beta)_{\beta\in I_\alpha} \longrightarrow j_\alpha$, i.e. $\tau'_{\alpha,\beta} : j_\beta \longrightarrow j_\alpha$ $(\beta \in I_\alpha)$, compatible with the translations $\tau'_{\beta',\beta}$ for $\beta<\beta'<\alpha$.

3) An arrow
$$u_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha\ .$$

4) If α is not a limit ordinal, i.e. $\alpha = \alpha_0+1$, an arrow
$$v_{\alpha_0} : j_{\alpha_0} \longrightarrow \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha)\ ;$$

and this so as to satisfy the constraints ①②③④ when we assume $\beta+1 \leqslant \alpha$ (instead of $\beta+1<\alpha$) in ① and ②, and $\beta' \leqslant \alpha$ (instead of $\beta'<\alpha$) in ③, and $\beta'+1 \leqslant \alpha$ (instead of $\beta'+1<\alpha$) in ④. So for the constraints 1°, 2°, 4°, we need take them into account *only* if α is not a limit ordinal, so $\alpha = \alpha_0+1$, and setting there

$$\beta' \ =\ \alpha_0\ .$$

As for the constraint 3°, we must take it into account for $\beta' = \alpha$. So in all, we must require

[page 318]

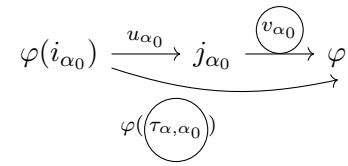
$$(3_\alpha)(\beta)\qquad \begin{array}{ccc} \varphi(i_\beta) & \xrightarrow{u_\beta} & j_\beta \\ {\scriptstyle \varphi(\tau_{\alpha,\beta})}\downarrow & & \downarrow{\scriptstyle \tau'_{\alpha,\beta}} \\ \varphi(i_\alpha) & \xrightarrow{u_\alpha} & j_\alpha \end{array} \qquad \text{commutative for } \beta<\alpha\ ([140]),$$

(by restriction on α), and if $\alpha = \alpha_0+1$ is a non-limit ordinal, we must require in addition

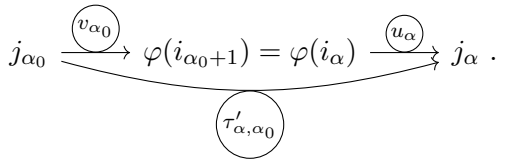
$$(4_\alpha)(\beta)\qquad \begin{array}{ccc} j_\beta & \xrightarrow{v_\beta} & \varphi(i_{\beta+1}) \\ {\scriptstyle \tau'_{\alpha_0,\beta}}\downarrow & & \downarrow{\scriptstyle \varphi(\tau_{\alpha,\beta+1})} \\ j_{\alpha_0} & \xrightarrow{v_{\alpha_0}} & \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha) \end{array} \qquad \beta<\alpha_0$$

[140] **N.B.** It suffices to require it for $\beta = \alpha_0$, in the case $\alpha = \alpha_0+1$, cf. below, p. 323.

(1_α)



(2_α)



p. 121

In these compatibilities, I have circled the arrows $\tau_{\alpha,\beta}$, $\tau'_{\alpha,\beta}$ $(\beta < \alpha)$, $\Big($ $\tau_{\alpha,\beta+1}$ $(\beta+1 < \alpha)$ $\Big)$, v_{α_0}, $\Big($ τ_{α,α_0} $\Big)$, u_α, $\Big($ τ'_{α,α_0} $\Big)$, which were introduced in the data 1) 2) 3) 4), page 317.

[page 319]

We are going to distinguish then the two cases.

2 A) $\alpha = \alpha_0 + 1$.

To construct i_α and the $\tau_{\alpha,\beta} : i_\beta \longrightarrow i_\alpha$, it suffices to construct i_α and

$$\tau_{\alpha,\alpha_0} \ : \ i_{\alpha_0} \ \longrightarrow \ i_\alpha = i_{\alpha_0+1} \ .$$

Similarly for j_α and the $\tau'_{\alpha,\beta}$, it suffices to construct

$$\tau'_{\alpha,\alpha_0} \ : \ j_{\alpha_0} \ \longrightarrow \ j_\alpha \ .$$

We must moreover give ourselves

$$j_{\alpha_0} \ \xrightarrow{v_{\alpha_0}} \ \varphi(\ \underbrace{i_\alpha}_{=i_{\alpha_0+1}}\ ) \ \xrightarrow{u_\alpha} j_\alpha \ ,$$

so in all two objects i_α, j_α (in I and in J' respectively) and four arrows τ_{α,α_0}, τ'_{α,α_0}, v_{α_0}, u_α. We need four compatibilities (page 318), where the last two $3_\alpha(\beta)$, $4_\alpha(\beta)$ depend on a parameter $\beta < \alpha$, i.e. $\beta \leqslant \alpha_0$, or on $\beta < \alpha_0$.

We are first going to deal with the first two compatibilities 1_α, 2_α, in terms of the datum of

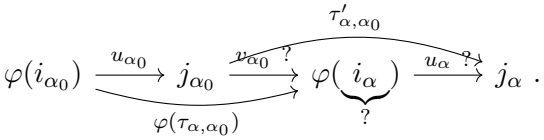
[page 320 struck out]

[page 321]

$$i_{\alpha_0} \ \in \ \mathrm{Ob}\, I \ , \qquad j_{\alpha_0} \ \in \ \mathrm{Ob}\, J' \ , \qquad u_{\alpha_0} \ : \ \varphi(i_{\alpha_0}) \ \longrightarrow \ j_{\alpha_0} \ .$$

The two desired compatibilities for the two objects i_α, j_α and the four arrows τ_{α,α_0}, τ'_{α,α_0}, v_{α_0}, u_α are indicated in the diagram

$(*)$



We note that since φ is cofinal, $\exists\ i_\alpha = i \in \mathrm{Ob}\, I$ and $j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i)$. But we want the composite

$$w \ : \ \varphi(i_{\alpha_0}) \ \xrightarrow{u_{\alpha_0}} \ j_{\alpha_0} \ \xrightarrow{v_{\alpha_0}} \ \varphi(i)$$

to be of the form $\varphi(\tau)$, where $\tau = \tau_{\alpha,\alpha_0} : i_{\alpha_0} \longrightarrow i_\alpha$ is an arrow in I. Since φ is a cofinal functor of filtering categories, we know that we can find $i \xrightarrow{\rho} i'$ in I, such that

p. 122

[page 322]

the composite arrow $\varphi(i_{\alpha_0}) \xrightarrow{w} \varphi(i) \xrightarrow{\varphi(\rho)} \varphi(i')$ is of the form $\varphi(\tau)$. So at the cost of replacing i by i', and v_{α_0} by the composite $v'_{\alpha_0} : j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i) \xrightarrow{\varphi(\rho)} \varphi(i')$, we can suppose that $v_{\alpha_0} u_{\alpha_0}$ is of the form $\varphi(\tau)$. This thus gives us in principle the object $i = i_\alpha$, and the arrows $v_{\alpha_0} : j_{\alpha_0} \longrightarrow \varphi(i)$ and $\tau_{\alpha,\alpha_0} : i_{\alpha_0} \longrightarrow i_\alpha$, so as to satisfy the first compatibility in $(*)$. It remains to construct j_α, τ'_{α,α_0}, u_α so as to satisfy the second compatibility $(*)$. Now J' being cofinal in J, there exists j_α in J' and an arrow $u_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha$. On the other hand, since $J' \longrightarrow J$ is fully faithful, the composite arrow $j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i_\alpha) \xrightarrow{u_\alpha} j_\alpha$ in J, between objects of J', comes from a unique arrow $\tau'_{\alpha,\alpha_0} : j_{\alpha_0} \longrightarrow j_\alpha$ in J'.

Thus, we have constructed the two objects and the four desired arrows, so as to satisfy the constraints 1_α, 2_α of page 318. Are the constraints $3_\alpha(\beta)$, $4_\alpha(\beta)$ then satisfied automatically? [page 323]

$$\begin{array}{ccc} \varphi(i_\beta) & \xrightarrow{u_\beta} & j_\beta \\ \downarrow & & \downarrow \\ \varphi(i_{\alpha_0}) & \xrightarrow{u_{\alpha_0}} & j_{\alpha_0} \\ {\scriptstyle \varphi(\tau_{\alpha,\alpha_0})}\downarrow & & \downarrow{\scriptstyle \tau'_{\alpha,\alpha_0}} \\ \varphi(i_\alpha) & \xrightarrow{u_\alpha} & j_\alpha \end{array} \qquad \beta < \alpha, \text{ i.e. } \beta \leqslant \alpha_0 \ .$$

The upper square is commutative by the induction hypothesis, for the total (composite) square to be commutative, it suffices that the lower square be so. We thus have a double arrow $(\tau'_{\alpha,\alpha_0} u_{\alpha_0}, u_\alpha \varphi(\tau_{\alpha,\alpha_0})) : \varphi(i_{\alpha_0}) \rightrightarrows j_\alpha$, it may not be trivial, but we know, J' being filtering, that we can equalize it by an arrow $j_\alpha \longrightarrow \bar{j}_\alpha$ in J'. So at the cost of replacing j_α by $\bar{j}_\alpha$, u_α by the composite $\bar{u}_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha \longrightarrow \bar{j}_\alpha$ and τ'_{α,α_0} by the composite $\bar{\tau}'_{\alpha,\alpha_0} : j_{\alpha_0} \longrightarrow j_\alpha \longrightarrow \bar{j}_\alpha$ (without touching i_α, v_{α_0}, τ_{α,α_0}), we can obtain commutativity, whence $3_\alpha(\beta)$. [page 324]

There remains the constraint $4_\alpha(\beta)$. For this, let us consider the composite diagram

$$\begin{array}{ccc} j_\beta & \xrightarrow{v_\beta} & \varphi(i_{\beta+1}) \\ {\scriptstyle \tau'_{\alpha_0,\beta}}\downarrow & & \downarrow{\scriptstyle \varphi(\tau_{\alpha,\beta+1})} \\ j_{\alpha_0} & \xrightarrow{v_{\alpha_0}} & \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha) \ . \end{array}$$

If α_0 is not a limit ordinal, but of the form $\alpha_1 + 1$, we can again reduce to the case where $\beta = \alpha_1$, proceeding as above. But we can proceed in all cases, without such an artifice (!), by considering the double arrows

$$(*) \qquad j_\beta \underset{v_{\alpha_0}\tau'_{\alpha_0,\beta}}{\overset{\varphi(\tau_{\alpha,\beta+1})v_\beta}{\rightrightarrows}} \varphi(i_\alpha)$$

deduced from the preceding square. Since the set of these double arrows is of cardinality $\operatorname{card} \alpha < \pi$, and J is large relative to π (it would suffice that it be so for the cardinals $\pi' < \pi$), we can find an arrow $\varphi(i_\alpha) \xrightarrow{u} j$ in J which equalizes all these double arrows. Since φ is cofinal, we can suppose that j is of the form $\varphi(i)$. Since φ is a cofinal functor of filtering categories, there then exists $i \xrightarrow{\rho} i'$ in I, such that the composite $\varphi(i_\alpha) \xrightarrow{u} \varphi(i) \xrightarrow{\varphi(\rho)} \varphi(i')$ is of the form $\varphi(\sigma)$, for $\sigma : i_\alpha \longrightarrow i'$. In all, we have found $\sigma : i_\alpha \longrightarrow i'$, such that $\varphi(\sigma)$ $(= \varphi(\rho)u) : \varphi(i_\alpha) \longrightarrow \varphi(i')$ equalizes all the double arrows considered in $(*)$, p. 324. So it is necessary to replace i_α by $\bar{i}_\alpha = i'$, v_{α_0} by the composite $j_{\alpha_0} \longrightarrow \varphi(i_\alpha) \xrightarrow{\varphi(\sigma)} \varphi(\bar{i}_\alpha)$, τ_{α,α_0} by the composite $\bar{\tau}_{\alpha,\alpha_0} : i_{\alpha_0} \xrightarrow{\tau_{\alpha,\alpha_0}} i_\alpha \xrightarrow{\sigma} \bar{i}_\alpha$. [page 325] p. 123 [page 326]

To sum up, here is the recipe for constructing, first $\boxed{i_\alpha,\ v_{\alpha_0},\ \tau_{\alpha,\alpha_0},}$ and *then* $\boxed{j_\alpha,\ u_\alpha,\ \tau'_{\alpha,\alpha_0}}$.

We proceed first as on page 321, 322 to construct the first triple, so as to satisfy the first compatibility $(*)$ (p. 321), i.e. 1_α. We then "correct" this choice of i_α, v_{α_0}, τ_{α,α_0}, by replacing i_α by $\bar{i}_\alpha$ for $\sigma : i_\alpha \longrightarrow \bar{i}_\alpha$, v_{α_0} by $\bar{v}_{\alpha_0} = \varphi(\sigma)v_{\alpha_0}$, and τ_{α,α_0} by $\bar{\tau}_{\alpha,\alpha_0} = \sigma\tau_{\alpha,\alpha_0}$, so as to obtain the compatibilities $4_\alpha(\beta)$ for $\beta < \alpha_0$, as we have just indicated. We will no longer touch these choices of i_α, v_{α_0}, τ_{α,α_0}, in the rest of the construction. We construct $(j_\alpha, u_\alpha, \tau'_{\alpha,\alpha_0})$ so as to satisfy the second compatibility $(*)$ (p. 321), as indicated on page [page 327]
322. Then, we correct this choice, by replacing j_α by $\bar{j}_\alpha$, with $\rho : j_\alpha \longrightarrow \bar{j}_\alpha$, and u_α by $\bar{u}_\alpha = \rho u_\alpha$, τ'_{α,α_0} by $\bar{\tau}'_{\alpha,\alpha_0} = \rho\tau'_{\alpha,\alpha_0}$, so as to satisfy the constraints $3_\alpha(\beta)$ of page 318.

This completes the construction in the case where $\alpha = \alpha_0 + 1$.

Case 2 B: α is a limit ordinal.

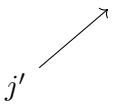
So we must construct the data 1) 2) 3) of page 317 (to the exclusion of datum 4), so as to satisfy the sole constraint $3_\alpha(\beta)$.

We begin to construct i_α and the $\tau_{\alpha,\beta}$ in I. This is possible, for $I_\alpha = \{\beta \mid \beta < \alpha\}$ is of cardinality $\pi_\alpha < \pi$ – it therefore suffices to suppose I large relative to the cardinals $\pi' < \pi$.

We then likewise construct j_α in J' and the $\tau'_{\alpha,\beta}$ $(\beta < \alpha)$, using the fact that J' is [page 328]
large relative to $\operatorname{card}\alpha = \operatorname{card} I_\alpha$. We must moreover construct an arrow

$$u_\alpha \ : \ \varphi(i_\alpha) \ \longrightarrow \ j_\alpha \ .$$

This is possible, for J' being cofinal in J, there exists j' in J' and $\varphi(i_\alpha) \longrightarrow j'$, and J' being filtering, there exists $\bar{j}_\alpha$ and arrows $j_\alpha \longrightarrow \bar{j}_\alpha$, so at the cost of



replacing j_α by $\bar{j}_\alpha$, the $\tau'_{\alpha,\beta}$ by $\bar{\tau}'_{\alpha,\beta}$, we do indeed find an arrow $u_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha$. We must then deal with the

compatibilities $3_\alpha(\beta)$, hence double arrows p. 124

$$\varphi(i_\beta) \underset{u_\alpha\varphi(\tau_{\alpha,\beta})}{\overset{\tau'_{\alpha,\beta}u_\beta}{\rightrightarrows}} j_\alpha \ .$$

As the set of these double arrows has cardinality $\operatorname{card}\alpha < \pi$, and J' is large relative to the said cardinal, we thus find an arrow [page 329]

$$\rho : j_\alpha \longrightarrow j$$

in J which equalizes them all. As J' is cofinal, we can suppose j in J', say $\bar{j}_\alpha$. Thus, at the cost of replacing j_α by $\bar{j}_\alpha$, and the $\tau'_{\alpha,\beta}$ by $\bar{\tau}'_{\alpha,\beta} = \rho\tau'_{\alpha,\beta}$, finally $u_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha$ by $\bar{u}_\alpha = \rho u_\alpha : \varphi(i_\alpha) \longrightarrow \bar{j}_\alpha$, we satisfy all the desired constraints.

I summarize the result ("± tautological"!) proved on pages 313-329.

Key lemma 4.13.17. *Let us consider a cartesian square in* Cat

$$\begin{array}{ccc} I & \xrightarrow{\ \varphi\ } & J \\ \uparrow & & \uparrow{\scriptstyle \psi} \\ I' & \longrightarrow & J' \ . \end{array}$$

We make the following assumptions: [page 330]

1) φ and ψ are cofinal, ψ is the inclusion of a strictly full subcategory J' of J, I is filtering (hence J and J' as well).

We seek conditions for the inclusion of the strictly full subcategory $I' \longrightarrow I$ to be cofinal (in which case we see at once that I' is filtering, and $I' \longrightarrow J'$ is also cofinal). For this, we suppose that there exists a limit ordinal α_0 *having the following additional properties:*

2) I and J' are stable under $\varinjlim$ of type I_{α_0} $(= \{\beta \mid \beta$ ordinal $< \alpha_0\})$, and the functors φ, ψ commute with these limits.

3) For every $\alpha \in I_{\alpha_0}$, I is large relative to $\operatorname{card}\alpha$*. (Thus, φ being cofinal, J is also large relative to* $\operatorname{card}\alpha$*, and likewise, ψ being fully faithful, and cofinal, J' is large relative to* $\operatorname{card}\alpha$*.)*

Under these conditions, I' is cofinal in I (hence also large relative to $\operatorname{card}\alpha$*, for every $\alpha \in I_{\alpha_0}$).*

The proof is the one given above. We did not have to use that the ordinal $\alpha_0 = \alpha_\pi$ was of this particular kind, only that it was a limit ordinal.

This lemma implies, as we saw on page 312, the implication $\mathrm{a}' \Longrightarrow \mathrm{c}$ in the diagram of implications $(*)$ on page 310, which is thus now proved. p. 125 [page 331]

To finish proving theorem 4.13.15, it remains to prove the implications

$$\begin{array}{ccc} \mathrm{c}' & & \\ & \searrow & \\ & & \mathrm{a}' \Longrightarrow \mathrm{a}\ . \\ & \nearrow & \\ \mathrm{b} & & \end{array}$$

Let us therefore prove (b or c′) $\Longrightarrow$ a′. It remains to prove that $\underline{S}$ is π-accessible in $\mathcal{M}$, i.e. that the fully faithful functor

$$\mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M} \qquad (*)$$

has essential image $\underline{S}$. Thus we must prove that if $x \in \mathrm{Ob}\,\mathcal{M}$, then

$$x \in \mathrm{Ob}\,\underline{S} \quad \overset{(?)}{\Longleftrightarrow} \quad \begin{array}{l} x \text{ is in the essential image of } \mathrm{Ind}_\pi(\underline{S}(\pi)), \text{ i.e. } x \\ \text{is isomorphic to } \varinjlim_{i\in I} x_i, \text{ the } x_i \text{ in } \mathrm{Ob}\,\underline{S}(\pi), I \text{ large} \\ \text{relative to } \pi. \end{array} \qquad (*)$$

(Let us first suppose that $\underline{S}$ satisfies b) ([141]). Thus we know that [page 332]

$$x \in \mathrm{Ob}\,\underline{S} \qquad \Longleftrightarrow \qquad \underline{S}(\pi)/x \text{ is cofinal in } \mathcal{M}_\pi/x\ . \qquad (**)$$

We want to deduce the equivalence $(*)$ from the equivalence $(**)$.

Let us prove $\Longrightarrow$ in $(*)$, i.e. let us suppose $x \in \mathrm{Ob}\,\underline{S}$, and prove that x is in the essential image of $\mathrm{Ind}_\pi\,\underline{S}(\pi)$. By virtue of $(**)$, $\underline{S}(\pi)/x$ is a subcategory (strictly full) cofinal in $\mathcal{M}_\pi/x$, thus since $\mathcal{M}_\pi/x$ is large relative to π, $\underline{S}(\pi)/x$ is as well. As we know that

$$x \;=\; \varinjlim_{i\in\mathcal{M}_\pi/x} x_i$$

[141] Clumsy, we can treat the two cases b) and c) at the same time, cf. bottom of page 333.

(π being a useful cardinal for $\mathcal{M}$), it follows by cofinality that we also have

$$x \;=\; \varinjlim_{i\in \underline{S}(\pi)/x} x_i \;,$$

and since $\underline{S}(\pi)/x$ is large relative to π, this implies that x belongs to the essential image of $\mathrm{Ind}_\pi(\underline{S}(\pi))$.

Let us prove $\Longleftarrow$ in $(*)$, i.e. let us suppose that x is in the essential image of $\mathrm{Ind}_\pi(\underline{S}(\pi))$, p. 126
[page 333]

$$x \;=\; \varinjlim_{i\in I} x_i \;, \qquad \text{the } x_i \text{ in } \underline{S}(\pi),\ I \text{ large relative to } \pi.$$

Then the composite functor $i \longmapsto x_i$,

$$I \;\longrightarrow\; \underline{S}(\pi)/x \;\hookrightarrow\; \mathcal{M}_\pi/x$$

is cofinal (cf. 4.4.2, page 79), hence $\underline{S}(\pi)/x \longrightarrow \mathcal{M}_\pi/x$ satisfies condition F 1 (since the composite satisfies it), and since $\underline{S}(\pi)/x \longrightarrow \mathcal{M}_\pi/x$ is fully faithful, this already implies that this functor is cofinal. Thus by $(**)$ we conclude from this $x \in \mathrm{Ob}\,\underline{S}$, q.e.d.

Let us now suppose that S satisfies c$'$ (instead of b), and let us again prove the equivalence $(*)$. For this, let us write

$$\square \qquad x \;=\; \varinjlim_{i\in I} x_i \qquad \begin{cases} \text{the } x_i \in \mathrm{Ob}\,\mathcal{M}_\pi,\ I \textit{ ordered set},\ (x_i)_{i\in I} \text{ is} \\ \pi_0\text{-}\pi\text{-adapted } ([142]). \end{cases}$$

which is possible since π is useful for $\mathcal{M}$ $\Big)$. Let us define $I_{\underline{S}}$ as in the statement of c), hence as the inverse image of $\underline{S}(\pi)/x \hookrightarrow \mathcal{M}_\pi/x$ under $I \longrightarrow \mathcal{M}_\pi/x$. By hypothesis d$'$) , we have

$$x \in \mathrm{Ob}\,\underline{S} \qquad \Longleftrightarrow \qquad I_{\underline{S}} \text{ is cofinal in } I \;. \tag{$**'$}$$

Using $(**')$, let us again prove $(*)$ (page 331). In fact, we shall use only that we have the isomorphism $\square$, with I large relative to π, and that we have the equivalence $(**')$, without having to use that I is ordered. Thus the proof also holds in case c). [page 334]

Let us prove $\Longrightarrow$ in $(*)$, i.e. let us suppose $x \in \mathrm{Ob}\,\underline{S}$, let us prove that x is in the essential image of $\mathrm{Ind}_\pi(\underline{S}(\pi))$. By virtue of $(**')$, $I_{\underline{S}}$ is cofinal in I, and since $I_{\underline{S}} \longrightarrow I$ is a fully faithful inclusion, it follows that $I_{\underline{S}}$ is also large relative to π, on the other hand we shall have by cofinality

$$x \;=\; \varinjlim_{i\in I_{\underline{S}}} x_i \;,$$

where this time the x_i are in $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$, whence it follows that x is in the essential image of $\mathrm{Ind}_\pi(\underline{S}(\pi))$.

Let us prove $\Longleftarrow$ in $(*)$, i.e. let us suppose that we have

$$x \;=\; \varinjlim_{j\in J} x_j \;, \qquad \text{the } x_j \text{ in } \underline{S}(\pi),\ J \text{ large relative to } \pi \;.$$

Invoking 4.4.2 once again, we find that $J \longrightarrow \mathcal{M}_\pi/x$ is cofinal, and since this functor factors through the fully faithful inclusion functor $\underline{S}(\pi)/x \hookrightarrow \mathcal{M}_\pi/x$, it follows as before that the latter is cofinal. We shall admit for a moment that this implies that the inverse image $I_{\underline{S}}$ of this strictly full subcategory, under the likewise cofinal functor $I \longrightarrow \mathcal{M}_\pi/x$, p. 127
[page 335]

[142] **N.B.** That exists, thanks to 4.13.14.

is cofinal in I. This being so, the equivalence $(**')$ (p. 333) implies $x \in \mathrm{Ob}\,\underline{S}$, which is what we wanted.

Thus it remains to prove that if

$$J' \;=\; \underline{S}(\pi)/x \hookrightarrow \mathcal{M}_\pi/x \;=\; J$$

is cofinal, then $I' = I_{\underline{S}} \longrightarrow I$ is as well. It seems that we really do have to invoke again the stability conditions $\mathrm{L}(\pi_0, \pi)$ made in d′) which had not been necessary for case c) and the (a bit) nasty key lemma 4.13.17, where we shall take (as in the proof written before [page 336] the statement of this lemma) $\alpha = \alpha_\pi \overset{\text{def}}{=}$ smallest ordinal α such that $\mathrm{card}\,\alpha = \pi$. Thus I_α has cardinality π, and is large relative to every cardinal $\pi' < \pi$, and in particular relative to π'_0. Assumption 2°) of 4.13.17 is satisfied for I, by definition of a π_0-π-adapted direct system, and for J', thanks to assumption $\mathrm{L}(\pi_0, \pi)$ on $\underline{S}(\pi)$. Likewise, assumption 3°) (I large relative to every cardinal $< \pi$) is satisfied by definition of π_0-π-adapted systems – I is even large relative to π. Assumption 1°) has already been seen.

We have thus proved (by a circular route)

$$\mathrm{a} \Longrightarrow \Big(\mathrm{a}' \Longleftrightarrow \mathrm{c} \Big\langle \begin{matrix} \mathrm{c}' \\ \\ \mathrm{b} \end{matrix} \Big) ,$$

having to invoke the nasty key lemma for the *two* implications

$$\mathrm{a}' \implies \mathrm{c} \qquad \text{and} \qquad \mathrm{c}' \implies \mathrm{a}'$$

(hence for two implications *in opposite directions*, a strange thing …).

It only remains still to prove the implication [page 337]

$$\mathrm{a}' \implies \mathrm{a}\,,$$

which is essentially sorital.

This completes the proof of the accessibility criterion 4.13.15 for a full subcategory $\underline{S}$ of $\mathcal{M}$.

We are going to give a somewhat decanted variant of 4.13.15, which (with slightly weaker assumptions) captures the essential part of its substance. Statement 4.13.15 is an almost immediate consequence of it:

Corollary 4.13.18 (of 4.13.15). *Let $\mathcal{M}$ be an accessible category, π a useful cardinal for $\mathcal{M}$, α_0 a* limit ordinal *such that we have* $\mathrm{card}\,\alpha_0 \leqslant \pi$. *Let $I_{\alpha_0} = \{\alpha \mid \alpha < \alpha_0\}$, and suppose that $\mathcal{M}$ is stable under $\varinjlim$ of type I_{α_0}.* p. 128 [page 338]

Let then $\underline{S}$ be a strictly full subcategory of $\mathcal{M}$, $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$. Let us denote by $L(\alpha_0)$ the condition: $\underline{S}(\pi)$ is stable in $\mathcal{M}_\pi$ (or what amounts to the same thing, in $\mathcal{M}$) under $\varinjlim$ of type I_{α_0}. Then the following conditions are equivalent.

a) $\underline{S}$ is π-accessible in $\mathcal{M}$, and is stable in $\mathcal{M}$ under $\varinjlim$ of type I_{α_0}.

a′) $\underline{S}$ is π-accessible and $\underline{S}(\pi)$ satisfies $L(\alpha_0)$.

b) $\underline{S}(\pi)$ satisfies $L(\alpha_0)$, and for every $x \in \mathrm{Ob}\,\mathcal{M}$, we have this: $x \in \mathrm{Ob}\,\underline{S}$ if and only if the strictly full subcategory $\underline{S}(\pi)/x$ of $\mathcal{M}(\pi)/x$ is cofinal in $\mathcal{M}(\pi)/x$.

c) $\underline{S}(\pi)$ *satisfies* $L(\alpha_0)$*, moreover for every* $x \in \mathrm{Ob}\,\mathcal{M}$*, and every representation*

$$x \;\simeq\; \varinjlim_{i\in I} x_i$$

which is "α_0*-*π*-adapted", i.e. such that* [page 339]

1°) *I equivalent to a small category, and large relative to* π*,*

2°) *the* $\varinjlim$ *of type* I_{α_0} *exist in* I*,*

3°) *the functor* $I \longrightarrow \mathcal{M}$*,* $i \longmapsto x_i$*, commutes with the* $\varinjlim$ *of type* I_{α_0}*, and finally*

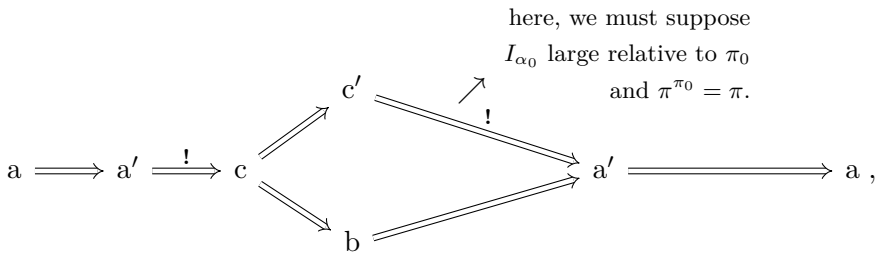
4°) *the* $x_i \in \mathrm{Ob}\,\mathcal{M}_\pi$

(compare 4.13.11), we have this (where we denote by $I_{\underline{S}}$ *the strictly full subcategory of* I *formed by the* i *such that* $x_i \in \mathrm{Ob}\,\underline{S}$*): we have* $x \in \mathrm{Ob}\,\underline{S}$ *if and only if* $I_{\underline{S}}$ *is cofinal in* I*.*

*c′) (*If I_{α_0} *is large relative to* π_0*, and* $\pi^{\pi_0} = \pi$*, to be able to invoke 4.13.14, cf. p. 335.) As in c), but assuming* I *ordered.*

([143]).

Proof. It is carried out following the same logical diagram as that of 4.13.15,



where we have marked with the sign ! the two implications which are "nasty", in the sense that they call on the key lemma 4.13.17 (pages 329-330), which is proved by a somewhat twisted transfinite construction. The proof of the seven implications of the diagram is essentially the one given on the preceding pages 311-337,

with the exception of $\mathrm{a}' \Longrightarrow \mathrm{a}$, which is a little simpler in the present case. We are going to bring it out in the form of a somewhat more general lemma. (**N.B.** The proof of $\mathrm{a}' \Longrightarrow \mathrm{a}$ in 4.13.15 had been omitted as "sorital"!, cf. page 337 …) p. 129 [page 340]

Lemma 4.13.19. *Let* $\mathcal{M}$ *be an accessible category, and* $\underline{S}$ *an accessible subcategory in* $\mathcal{M}$*,* π *a useful cardinal for* $\mathcal{M}$ *and such that* $\underline{S}$ *is* π*-accessible, i.e.* $\underline{S}$ *is the essential image of* $\mathrm{Ind}_\pi(\underline{S}(\pi))$ *under the fully faithful canonical functor* $\mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M}$*, where* $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$*. We assume that there exists a limit ordinal* α*, such that* $\mathrm{card}\,\alpha \leqslant \pi$ *and such that* $\mathcal{M}_\pi$ *and* $\underline{S}(\pi)$ $(= \underline{S}_\pi)$ *are stable under* $\varinjlim$ *of type* I_α*, and that the inclusion functor* $\underline{S}(\pi) \longrightarrow \mathcal{M}_\pi$ *commutes with them. Let* Φ *be a set of categories such that* $\forall\, I \in \Phi$*, we have* $\mathrm{card}\,\mathrm{red}\, I \leqslant \pi$*, and suppose that* $\mathcal{M}_\pi$ *is stable under* $\varinjlim$ *of type* Φ*, and that* $\underline{S}(\pi)$ *is stable in* $\mathcal{M}_\pi$ *under the said limits (or what amounts to the same thing, that it is stable in itself under these* $\varinjlim$*, and that the inclusion functor commutes with them). Then* $\mathcal{M}$ *and* $\underline{S}$ *are stable under* $\varinjlim$ *of type* Φ*, and the inclusion functor* $\underline{S} \longrightarrow \mathcal{M}$ *commutes with them, i.e.* $\mathcal{M}$ *is stable under these limits, and* $\underline{S}$ *stable in* $\mathcal{M}$ *under these same limits.*

Proof. We apply 4.11.1 (p. 267-268) separately to $\mathcal{M} \simeq \mathrm{Ind}_\pi\,\mathcal{M}_\pi$ and to $\underline{S} \simeq \mathrm{Ind}_\pi\,\underline{S}(\pi)$, to conclude that $\mathcal{M}$ and $\underline{S}$ are separately stable under $\varinjlim$ of type Φ. It [page 341]

[143] **N.B.** If we do not suppose I_{α_0} large relative to π_0 and $\pi^{\pi_0} = \pi$, then all the conditions from a) to c) (a, a′, b, c) are equivalent and *imply* c, but are perhaps not equivalent to it.

remains to see that the functor $\underline{S} \longrightarrow \mathcal{M}$ commutes with them. This is seen as in the proof of 4.11.1, more generally we get this:

Lemma 4.13.20 (cor. to 4.11.1). *Let $f_0 : C \longrightarrow C'$ be a functor between categories equivalent to small categories, π an infinite cardinal, $f : \mathcal{M} \overset{\text{def}}{=} \mathrm{Ind}_\pi C \longrightarrow \mathcal{M}' \overset{\text{def}}{=} \mathrm{Ind}_\pi C'$ the canonical extension $\mathrm{Ind}_\pi(f_0)$ of f_0. Let Φ and α be as in 4.11.1, and suppose that $I_\alpha \in \Phi$ and that f_0 commutes with $\varinjlim$ of type Φ. Then f commutes with these same direct limits.* ([144]).

Corollary 4.13.21. *Under the conditions of 4.13.20, suppose that Φ contains the categories of cardinality $\leqslant \pi$ (resp. the filtering categories of cardinality $\leqslant \pi$, resp. the filtering categories large relative to π_0 of cardinality $\leqslant \pi$, resp. the discrete categories of cardinality $\leqslant \pi$). Then f commutes with small $\varinjlim$ (resp. with small filtering $\varinjlim$, resp. with small filtering $\varinjlim$ large relative to π_0, resp. with small sums). (Converse clear and "left to the reader".)*

Seen as in 4.4.14 and 4.4.14.1 (p. 97, 98).

Here at last, it seems to me, is the definitive form (?) of 4.13.15, 4.13.18. [page 342]

Theorem 4.13.22 ([145]). *Let $\mathcal{M}$ be an accessible category, π an infinite cardinal, useful for $\mathcal{M}$, and Φ a set of categories satisfying the two conditions:*

1°) $J \in \Phi$ implies $\mathrm{card}\,\mathrm{red}\, J \leqslant \pi$.

2°) There exists a limit ordinal α_0 *with $I_{\alpha_0} \in \Phi$.*

We assume $\mathcal{M}$ stable under $\varinjlim$ of type Φ. Let $\underline{S}$ be a strictly full *subcategory of $\mathcal{M}$, and $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$. Let us denote by $L(\Phi)$ the condition on $\underline{S}(\pi)$: $\underline{S}(\pi)$ is stable in $\mathcal{M}$* p. 130 *(or what amounts to the same thing, in $\mathcal{M}_\pi$) under $\varinjlim$ of type Φ. This being said, the following conditions on $\underline{S}$ are equivalent:*

ⓐ *$\underline{S}$ is π-accessible in $\mathcal{M}$, and stable in $\mathcal{M}$ under $\varinjlim$ of type Φ.*

a′) *$\underline{S}$ is π-accessible in $\mathcal{M}$, and $\underline{S}(\pi)$ satisfies $L(\Phi)$.*

[page 343]

b) *$\underline{S}(\pi)$ satisfies $L(\Phi)$, and for every $x \in \mathrm{Ob}\,\mathcal{M}$, we have this: $x \in \mathrm{Ob}\,\underline{S}$ if and only if the strictly full subcategory $\underline{S}(\pi)/x$ of $\mathcal{M}(\pi)/x$ is cofinal in $\mathcal{M}(\pi)/x$.*

ⓒ *$\underline{S}(\pi)$ satisfies $L(\Phi)$, and for every $x \in \mathrm{Ob}\,\mathcal{M}$ and every representation $x \simeq \varinjlim_{i \in I} x_i$ which is Φ-π-adapted (cf. definition below), we have this: $x \in \mathrm{Ob}\,\underline{S}$ if and only if $I_{\underline{S}}$ is cofinal in I, where $I_{\underline{S}}$ is the strictly full subcategory of I formed by the $i \in \mathrm{Ob}\, I$ such that $x_i \in \mathrm{Ob}\,\underline{S}$ (or again, $x_i \in \mathrm{Ob}\,\underline{S}(\pi)$).* **N.B.** *We say that a direct system $(x_i)_{i \in I}$ in $\mathcal{M}$ is Φ-π-adapted if*

1) I equivalent to a small category, and I large relative to π,

2) the $\varinjlim$ of type Φ exist in I,

3) the functor $i \longmapsto x_i$ from I to $\mathcal{M}$ commutes with $\varinjlim$ of type Φ, and finally

4) the $x_i \in \mathrm{Ob}\,\mathcal{M}_\pi$.

c′) *(If $\pi^{\pi_0} = \pi$, where π_0 is a cardinal $\leqslant \pi$ useful for $\mathcal{M}$, and if $J \in \Phi$ implies that J is large relative to π_0.) As in c), with I* ordered.

[144] It is better to take functors $f_0 : C \longrightarrow \mathcal{N}$, and $\mathcal{N}$ that f_0 commutes with $\varinjlim$ of type Φ, so the same is true of its extension $\mathrm{Ind}_\pi(C, \mathcal{N}) \longrightarrow \mathcal{N}$. (**N.B.** We assume that $\mathcal{N}$ satisfies $\overrightarrow{\mathrm{L}_\pi}$.)

[145] cf. variant 4.13.36, page 358.

([146],[147]).

Definition 4.13.23 ([148]). Let $\mathcal{M}$ be an accessible category, π a cardinal, Φ a set of categories, satisfying the two conditions:

a) $J \in \Phi \Longrightarrow \operatorname{card} \operatorname{red} J \leqslant \pi$.

b) There exists a limit ordinal α_0, such that $I_{\alpha_0} \in \Phi$ ([149]).

Let S be a subset of $\operatorname{Ob}\mathcal{M}$, $\underline{S}$ the full subcategory it defines. We say that S is a Φ-π-*accessible subset of* $\operatorname{Ob}\mathcal{M}$, or that $\underline{S}$ is a Φ-π-*accessible subcategory in* $\mathcal{M}$, if the following conditions are satisfied. [page 344]

1°) π is useful for $\mathcal{M}$, $\mathcal{M}$ is stable under $\varinjlim$ of type Φ ([150]).

2°) $\underline{S}$ is a π-accessible subcategory in $\mathcal{M}$.

3°) $\underline{S}$ is stable in $\mathcal{M}$ under $\varinjlim$ of type Φ.

([151]).

N.B. When the preliminary condition 1°) *on* $\mathcal{M}$ is satisfied, the validity of conditions 2°) 3°) is therefore equivalent to the condition that $\underline{S}$ be *strictly* full in $\mathcal{M}$, and that in addition $\underline{S}$ satisfy condition a) of theorem 4.13.22. On the other hand, this condition is equivalent to each of conditions a′, b, c of this same theorem, and also to the last condition c′, provided that we have $\pi^{\pi_0} = \pi$ (for a cardinal $\pi_0 < \pi$ useful for $\mathcal{M}$) and that $J \in \Phi$ implies that J is large relative to π_0. p. 131

Conceptually, it is the definition given in 4.13.23, i.e. condition ⓐ of 4.13.22, which seems the clearest, or its immediate variant a′. But from the practical point of view, of working with accessible subsets, it is by property ⓒ which seems the most convenient. The main weight of theorem 4.13.22 is indeed the implication a $\Longrightarrow$ c (via a $\Longrightarrow$ a′ $\overset{!}{\Longrightarrow}$ c). [page 345] So one must above all see this theorem as giving a remarkable criterion for membership in $\operatorname{Ob}\underline{S}$ of an object x of $\mathcal{M}$ which is presented by a Φ-π-adapted direct system:

$$x \in \operatorname{Ob}\underline{S} \qquad \Longleftrightarrow \qquad I_{\underline{S}} \text{ cofinal} , \tag{$*$}$$

and in this equivalence, it is the implication $\Longrightarrow$ which is the non-sorital part, and remarkable.

Proposition 4.13.24. *Let $\mathcal{M}$, π, Φ be as in 4.13.23 ([152]), we assume that condition 1° of 4.13.23 is satisfied (π useful for $\mathcal{M}$, $\mathcal{M}$ stable under $\varinjlim$ of type Φ). Let ${\Sigma'}_\pi^\Phi$ be the set of subcategories $\underline{S}$ of $\mathcal{M}$ which are Φ-π-accessible in $\mathcal{M}$, and Σ_π^Φ the set of subcategories $\underline{S}(\pi)$ of $\mathcal{M}_\pi$ which are strictly full, and stable in $\mathcal{M}_\pi$ under $\varinjlim$ of type Φ. (These subcategories are necessarily Karoubian, since stable under $\varinjlim$ of type I_{α_0}, hence also under direct factors.) Then the bijections (4.13.4.1) (page 288) induce mutually inverse bijections*

$$\Sigma_\pi^\Phi \rightleftarrows {\Sigma'}_\pi^\Phi \quad .$$

[page 346]

[146] **N.B.** When the restrictive conditions opposite are not satisfied, then a, a′, b, c are still equivalent, and imply c′, but are perhaps not equivalent to c′).

[147] **N.B.** One should restate the auxiliary conditions (If …), cf. p. 359-360.

[148] **N.B.** It would be worthwhile to move this definition up (*without* b), even before 4.13.15.

[149] If it is to give a definition, one can dispense with condition b).

[150] Preliminary condition on $\mathcal{M}$.

[151] **N.B.** When $\Phi = \{I_{\alpha_0}\}$, we also say that $\underline{S}$ is α_0-π-accessible in $\mathcal{M}$. If Φ consists of the categories of cardinality $\leqslant \pi$ and large relative to π_0, we say that $\underline{S}$ is π_0-π-accessible in $\mathcal{M}$.

[152] **N.B.** The condition $I_{\alpha_0} \in \Phi$ is not needed.

Proposition 4.13.25. *Let $\mathcal{M}$, π, Φ be as in 4.13.23, with $I_{\alpha_\pi} \in \Phi$ (α_π being the smallest ordinal of cardinality π). Then the set Σ'^{Φ}_{π} of subcategories of $\mathcal{M}$ Φ-π-accessible in $\mathcal{M}$ is stable under intersection of families of cardinality $\leqslant \pi$* ([153]).

(**N.B.** The bijection $\Sigma^{\Phi}_{\pi} \xrightarrow{\sim} \Sigma'^{\Phi}_{\pi}$ of 4.13.24 respects the natural order relations of inclusion, hence transforms intersections into intersections.)

Seen as in 4.13.7 (which is contained in the present statement), using lemma 4.13.10.

4.13.26. Let $\mathcal{M}$, π, Φ be as in 4.13.23, and let $\underline{S}$ be a *strictly full* subcategory of $\mathcal{M}$, $f : \underline{S} \longrightarrow \mathcal{M}$ the inclusion functor. The condition that $\underline{S}$ be Φ-π-accessible in $\mathcal{M}$ breaks up into:

1) π is useful for $\mathcal{M}$ *and* for $\underline{S}$.

2) $\mathcal{M}$ and $\underline{S}$ are stable under $\varinjlim$ of type Φ.

3) f commutes with the said limits.

4) f is π-accessible.

5) $f(\underline{S}_\pi) \subset \mathcal{M}_\pi$.

Let then $\pi' > \pi$ be another cardinal, when can we be sure that if $\underline{S}$ is Φ-π-accessible in $\mathcal{M}$, it is also Φ-π'-accessible? Conditions 2) 3) 4) are visibly stable under passage from π to π'. Conditions 1) and 5) remain. We know that if $\pi'^{\pi} = \pi'$, then condition 1) remains valid for π'. More generally, if π_0 is a cardinal $\leqslant \pi$ such that π_0 is useful for $\mathcal{M}$ and for $\underline{S}$, then if $\pi' > \pi_0$ is such that $\pi'^{\pi_0} = \pi'$, then π' is also useful for $\mathcal{M}$ and for $\underline{S}$. If we admit the GCH (generalized continuum hypothesis), this is so for every $\pi' > \pi_0$, hence for every $\pi' > \pi$.

p. 132
[page 347]

Condition 5) remains, for which an answer is given by 4.12.1 (p. 273). If $\pi_0 \leqslant \pi$ is chosen so that f is π_0-accessible (e.g. $\pi_0 = \pi$), and if

$$\pi'_1 = \operatorname{card}\operatorname{red}\underline{S}_{\pi_0} \qquad (\leqslant \operatorname{card}\operatorname{red}\underline{S}(\pi))$$
$$\pi''_1 = \sup(\pi'_1, \pi) ,$$

we see that if $\pi' \geqslant \pi''_1$ (i.e. $\pi' \geqslant \pi'_1, \pi$) is such that $\pi'^{\pi_0} = \pi'$, then we do indeed have

$$f(\underline{S}_{\pi'}) \ \subset \ \mathcal{M}_{\pi'} .$$

If then in addition $\pi' > \pi \geqslant \pi_0$ (as assumed at the start), and if we assume the GCH, then the condition $\pi'^{\pi_0} = \pi'$ is automatic, and only the condition $\pi' \geqslant \pi''_1$ remains, i.e. in the end $\pi' \geqslant \operatorname{card}\operatorname{red}\underline{S}_{\pi_0}$. To sum up:

[page 348]

Proposition 4.13.27. *Under the conditions of 4.13.23 (without condition b) on Φ), let $\underline{S}$ be a subcategory of $\mathcal{M}$ which is Φ-π-accessible. Let $\pi_0 \leqslant \pi$ be a cardinal which is useful for $\underline{S}$ and for $\mathcal{M}$, and such that $f : \underline{S} \longrightarrow \mathcal{M}$ is π_0-accessible, i.e. such that $\underline{S}$ is stable in $\mathcal{M}$ under $\varinjlim$ large relative to π_0 (one can e.g. take $\pi_0 = \pi$). Then for every cardinal $\pi' > \pi$ such that*

a) $\pi'^{\pi_0} = \pi'$ and

b) $\pi' \geqslant \operatorname{card}\operatorname{red}\underline{S}_{\pi_0}$,

$\underline{S}$ is Φ-π'-accessible in $\mathcal{M}$. Thus if $\pi' > \pi$ is such that $\pi'^{\pi} = \pi'$, $\pi' > \operatorname{card}\operatorname{red}\underline{S}_{\pi}$, then $\underline{S}$ is Φ-π'-accessible. (It suffices e.g. to take $\pi' = 2^c$, with $c \geqslant \sup(\pi, \operatorname{card}\operatorname{red}\underline{S}(\pi))$.) If we admit the GCH, it therefore suffices to have $\pi' > \sup(\pi, \operatorname{card}\operatorname{red}\underline{S}(\pi))$.

[153] If π is not a limit cardinal. Otherwise, one must say: families of cardinality $< \pi$. Cf. 4.13.10.3, where I give a slightly more general corrected statement.

[page 349]

Proposition 4.13.28 ([154]). *Let $\mathcal{M}$, π, Φ be as in 4.13.23. Suppose in addition that $J \in \Phi$ implies that J is* filtering. *Then the set $\tilde{\Sigma}_\pi^\Phi$ of Φ-π-accessible subsets of $\mathcal{M}$ is stable under finite unions. More generally, let λ be a cardinal $< \pi$ such that $\forall\ J \in \Phi$, J is large relative to λ. Then the set $\tilde{\Sigma}_\pi^\Phi$ is stable under unions $(S_r)_{r\in A}$, indexed by sets A of cardinality $\leqslant \lambda$.*

Corollary 4.13.29. *The set of accessible subsets of $\operatorname{Ob}\mathcal{M}$ is stable under "small" unions – i.e. if $(S_r)_{r\in A}$ is a small family of accessible subsets of $\operatorname{Ob}\mathcal{M}$, then $\bigcup_{r\in A} S_r$ is also accessible.*

The proof of 4.13.28 rests on the following lemma (I do not spell out the proof):

Lemma 4.13.30. *Let I be a category large relative to π, and $(J_r)_{r\in A}$ a family of full subcategories such that $\operatorname{card} A \leqslant \pi$, such that $\bigcup_{r\in A} \operatorname{Ob} J_r$ is cofinal in $\operatorname{Ord}(I)$, i.e. for every $i \in \operatorname{Ob} I\ \exists\ j \in \bigcup_{r\in A} \operatorname{Ob} J_r$, and an arrow $i \longrightarrow j$. Then there exists $r \in A$ such that J_r is cofinal in I.* p. 133 [page 350]

Theorem 4.13.31. *Let $\mathcal{M}$, $\mathcal{N}$ be accessible categories, $f : \mathcal{M} \longrightarrow \mathcal{N}$ an accessible functor, $T \subset \operatorname{Ob}\mathcal{N}$ a subset of $\mathcal{N}$, $\underline{T} \subset \mathcal{N}$ the corresponding strictly full subcategory of $\mathcal{N}$, $S = f^{-1}(T)$, $\underline{S} = f^{-1}(\underline{T})$ the associated full subcategory of $\mathcal{M}$. If T is accessible, then S is accessible; in other words, if $\underline{T}$ is accessible in $\mathcal{N}$, $\underline{S}$ is accessible in $\mathcal{M}$.*

This follows from the more precise statement:

Corollary 4.13.32. *Let π_0 and $\pi > \pi_0$ be cardinals having the following properties:* [page 351]

a) $\underline{T}$ is π_0-π-accessible in $\mathcal{N}$ (4.13.23).

b) π_0 is useful for $\mathcal{M}$ (it is for $\mathcal{N}$, because of a)).

c) $f(\mathcal{M}_\pi) \subset \mathcal{N}_\pi$.

d) $\mathcal{M}$ satisfies L_{π_0}, and f is π_0-accessible.

(The existence of such a pair (π_0, π) follows at once from 4.13.16 and from 4.12.1.) Then $\underline{S}$ is π_0-π-accessible in $\mathcal{M}$.

Since f commutes with $\varinjlim$ large relative to π_0 by virtue of d), and $\underline{T}$ is stable under these limits by a) (cf. 4.13.23, 3°), the same is therefore true of $\underline{S}$, i.e. 3° of definition 4.13.23 is satisfied. On the other hand, condition 1° (π useful for $\mathcal{M}$, $\mathcal{M}$ stable under $\varinjlim$ large relative to π_0 and of cardinality $\leqslant \pi$) is verified by b) and d). Thus it remains to verify condition 2°, i.e. that $\underline{S}$ is π-accessible. Let us put

$$\begin{aligned} \underline{T}(\pi) &= \underline{T} \cap \mathcal{N}_\pi \\ \underline{S}(\pi) &= \underline{S} \cap \mathcal{M}_\pi \quad \underbrace{\subset}_{\text{thanks to c)}} \quad f^{-1}(\underline{T}(\pi))\ . \end{aligned}$$

Since $\underline{S}$ is stable in $\mathcal{M}$ under $\varinjlim$ large relative to π_0 (which exist in $\mathcal{M}$ by d), and a fortiori under $\varinjlim$ large relative to π, we have a fully faithful canonical functor

$$\operatorname{Ind}_\pi \underline{S}(\pi) \longrightarrow \underline{S}\ .$$

Besides, $\underline{T}$ being strictly full in $\mathcal{N}$, $\underline{S}$ is strictly full in $\mathcal{M}$, and it remains to verify that the preceding functor is essentially surjective. Let then $x \in \operatorname{Ob}\underline{S}$. We will have, thanks to b), [page 352]

$$x = \varinjlim_I x_i\ , \qquad x_i \in \mathcal{M}_\pi,\ I = \mathcal{M}_\pi/x\ .$$

Besides, the preceding direct system is π_0-π-adapted, i.e. (p. 339):

[154] To be moved further down, after 4.13.31, which is the first important application of 4.13.22, whereas 4.13.28 is proved without having to use 4.13.22 or its variants.

1) I equivalent to a small category, I large relative to π (follows from the fact that π is useful for $\mathcal{M}$, cf. ⓑ).

p. 134

2) The direct limits large relative to π_0 and of cardinality $\leqslant \pi$ exist in I (follows from L_{π_0} on $\mathcal{M}$ (cf. ⓓ)).

3) The functor $\xi : I \longrightarrow \mathcal{M}$ commutes with them.

4) The x_i are in $\mathcal{M}_\pi$.

Since f commutes with the $\varinjlim$ large relative to π_0 (cf. ⓓ), a fortiori with those large relative to π, we therefore have

$$f(x) \simeq \varinjlim_I f(x_i) ,$$

I say that the direct system $(f(x_i))_{i\in I}$ in $\mathcal{N}$ is π_0-π-adapted. Conditions 1), 2) are *ne varietur* (concern only I), the functor $I \longrightarrow \mathcal{N}$ is the composite $I \xrightarrow{\xi} \mathcal{M} \xrightarrow{f} \mathcal{N}$, hence commutes with the $\varinjlim$ large relative to π_0 and of cardinality $\leqslant \pi$ ([155]), since this is so for ξ and f, finally the $f(x_i)$ are in $\mathcal{N}_\pi$ since by ⓒ $f(\mathcal{M}_\pi) \subset \mathcal{N}_\pi$.

Moreover $f(x) \in \mathrm{Ob}\,\underline{T}$ (since $x \in \mathrm{Ob}\,\underline{S}$, where $\underline{S} \overset{\text{def}}{=} f^{-1}(\underline{T})$). Thus we can apply 4.13.22 a $\Longrightarrow$ c to the situation $(\mathcal{N}, \pi_0, \pi, \underline{T})$ and to the direct system $(f(x_i))_{i\in I}$, to conclude that the subcategory I' of I consisting of the i such that $f(x_i) \in \mathrm{Ob}\,\underline{T}$, i.e. such that $x_i \in \underline{S}$, hence $x_i \in \underline{S}(\pi)$, is *cofinal* in I. Thus we have [page 353]

$$x \simeq \varinjlim_{i\in I'} x_i ,$$

with this time the x_i in $\underline{S}(\pi)$, and I' large relative to π (since cofinal and full in I, which is large relative to π), which proves that x is in the essential image of $\mathrm{Ind}_\pi(\underline{S}(\pi))$, q.e.d.

Question 4.13.33. We have just seen that if $f : \mathcal{M} \longrightarrow \mathcal{N}$ is an accessible functor between accessible categories, then the inverse image of an accessible subset of $\mathrm{Ob}\,\mathcal{N}$ is an accessible subset of $\mathrm{Ob}\,\mathcal{M}$. I conjecture that the similar statement, not for direct images, but for *essential images*, is true, i.e. that for every subcategory $\underline{S}$ of $\mathcal{M}$ accessible in $\mathcal{M}$, its essential image in $\mathcal{N}$ is accessible in $\mathcal{N}$ ([156]). In this question, we can assume that $\underline{S} = \mathcal{M}$, and the question comes down to this: does every accessible functor $f : \mathcal{M} \longrightarrow \mathcal{N}$ between accessible categories factor into an (accessible) *essentially surjective* functor, and an accessible fully faithful functor $\mathcal{M} \longrightarrow \underline{T} \longrightarrow \mathcal{N}$ (where $\underline{T}$ is itself accessible) ([157]) ? If so, this would seem to me perhaps the deepest result (perhaps the only truly deep one) in category theory. [page 354]

Here is a question that seems to me very close. Let $\mathcal{M}$ be an accessible category, A a small full subcategory. Suppose that $\mathcal{M}$ satisfies L_{π_0} (e.g. π_0 useful for $\mathcal{M}$), and let $\underline{S}$ be the smallest strictly full subcategory of $\mathcal{M}$ containing A, and stable under limits large relative to π_0. I conjecture that $\underline{S}$ is accessible in $\mathcal{M}$.

Likewise, if Φ is a small set of small categories, such that $\mathcal{M}$ is stable under $\varinjlim$ of type Φ, we can take the envelope $\underline{S}$ of A for the $\varinjlim$ large relative to π_0 *and* the $\varinjlim$ of type Φ, and I again conjecture that $\underline{S}$ is accessible. In this direction, we have at least the following very partial result. [page 355]

p. 135

Proposition 4.13.34. *Let $\mathcal{M}$ be an accessible category stable under small filtering $\varinjlim$, and let $\underline{S}$ be a strictly full subcategory of $\mathcal{M}$. Equivalent conditions:*

[155] this commutation should have a name, condition $\mathrm{L}(\pi_0, \pi)$ on a functor.

[156] False, even if we take the $\natural$-closure of the essential image ("$\natural$-essential image").

[157] We must at least restrict ourselves to essentially $\natural$-surjective – but even that is too optimistic.

a) $\underline{S}$ *is accessible in* $\mathcal{M}$*, and stable in* $\mathcal{M}$ *under small filtering* $\varinjlim$.

b) There exists a small full subcategory A *of* $\mathcal{M}$*, such that* $\underline{S}$ *is the smallest strictly full subcategory of* $\mathcal{M}$ *containing* A *and stable under small filtering* $\varinjlim$.

We have a) $\Longrightarrow$ b), for if $\underline{S}$ is π-accessible in $\mathcal{M}$, it is immediate (without any other [page 356] stability condition on $\underline{S}$) that $\underline{S}$ is the smallest strictly full subcategory of $\mathcal{M}$ containing $\underline{S}(\pi) \overset{\text{def}}{=} \underline{S} \cap \mathcal{M}_\pi$ and stable under $\varinjlim$ large relative to π. If then we already know that $\underline{S}$ is even stable under *all* small filtering $\varinjlim$, it follows that $\underline{S}$ is the smallest subcategory of $\mathcal{M}$ having this stronger stability property, and containing A.

Let us prove b $\Longrightarrow$ a. Let π be a useful cardinal for $\mathcal{M}$, and such that $A \subset \mathcal{M}_\pi$. Let B be the smallest strictly full subcategory of $\mathcal{M}$ containing A and stable under filtering $\varinjlim$ of cardinality $\leqslant \pi$. Thus we have $B \hookrightarrow \mathcal{M}_\pi$ and $B \subset \underline{S}$, and $\underline{S}$ is also the L-envelope of B. Besides B is stable under direct factors. Let us then consider the essential image $\underline{S}'$ [page 357] of $\mathrm{Ind}_\pi(B)$ in $\mathcal{M}$. We have $\underline{S}' \subset \underline{S}$, and everything comes down to proving that $\underline{S}' = \underline{S}$, from which it will indeed follow that $\underline{S}$ is π-accessible in $\mathcal{M}$, a fortiori accessible in $\mathcal{M}$, whence a). To prove that $\underline{S}' = \underline{S}$, by definition of $\underline{S}$ as the L-envelope of A, it suffices to see that $\underline{S}'$ is stable under small filtering $\varinjlim$. Now this follows from the fact that $B \subset \mathcal{M}_\pi$ is stable in $\mathcal{M}_\pi$ under filtering $\varinjlim$ of cardinality $\leqslant \pi$, and from 4.13.19, which implies at least, since $B = \underline{S}' \cap \mathcal{M}_\pi$ (by virtue of 4.13.4) that $\underline{S}'$ is stable under filtering $\varinjlim$ *of cardinality* $\leqslant \pi$. That it is even stable under arbitrary small filtering $\varinjlim$ follows from this, and from the fact that it is so under $\varinjlim$ large relative to π. (Cf. 4.13.21 …)

Definition 4.13.35. Let $\mathcal{M}$ be an accessible category, stable under small filtering [page 358] $\varinjlim$, S a subset of $\mathrm{Ob}\,\mathcal{M}$, $\underline{S}$ the corresponding full subcategory in $\mathcal{M}$. We say that S is *L-accessible*, or that $\underline{S}$ is L-accessible in $\mathcal{M}$, if $\underline{S}$ is accessible in $\mathcal{M}$, and in addition stable under small filtering $\varinjlim$ ([158]).

Proposition 4.13.36. *Let* $\mathcal{M}$*,* S*,* $\underline{S}$ *be as in 4.13.35, and let* π *be a useful cardinal for* $\mathcal{M}$*. Suppose* $\underline{S}$ *stable in* $\mathcal{M}$ *under small filtering* $\varinjlim$*, or only* $\underline{S}(\pi) \overset{\text{def}}{=} \underline{S} \cap \mathcal{M}_\pi$ *stable under small filtering* $\varinjlim$ *of cardinality* $\leqslant \pi$*. Equivalent conditions:*

a) $\underline{S}$ *is* π*-accessible. (It will automatically be stable under small filtering* $\varinjlim$*, hence L-accessible.)*

b) For every $x \in \mathrm{Ob}\,\mathcal{M}$*, we have* $x \in \mathrm{Ob}\,\underline{S}$ *if and only if* $\underline{S}(\pi)/x$ *is cofinal in* $\mathcal{M}_\pi/x$*.* p. 136

c) For every $x \in \mathrm{Ob}\,\mathcal{M}$*, and every representation* $x \simeq \varinjlim_{i \in I} x_i$ *which is L-*π*-adapted (cf. page 343, c), we have this:* $x \in \mathrm{Ob}\,S$ *if and only if* $I_{\underline{S}}$ *is cofinal in* I*.* [page 359]

c′) As in c), with I *ordered.*

c″) (If we have chosen, for every $x \in \mathrm{Ob}\,\mathcal{M}$*, one L-*$\pi$*-adapted representation* $x \simeq \varinjlim_{I} x_i$*):* $x \in \mathrm{Ob}\,\underline{S} \Longleftrightarrow$ *the set* $I_{\underline{S}}$ *is cofinal in* I*.*

Proof. This is essentially the statement of 4.13.22 in the special case where Φ = set of *filtering* categories of cardinality $\leqslant \pi$, but where, to simplify, we have assumed beforehand that $\underline{S}(\pi)$ satisfies L(Φ). The equivalence of a) b) c) is contained in 4.13.22, as is c $\Longrightarrow$ c′. But for c′ $\Longrightarrow$ a, the auxiliary hypothesis $\pi^{\pi_0} = \pi$ and that the $J \in \Phi$ will be large

[158] **N.B.** The set of L-accessible subsets of $\mathrm{Ob}\,\mathcal{M}$ is stable under small intersections, and under *finite* unions. The inverse image of an L-accessible subset under a functor ([159]) $f : \mathcal{M} \longrightarrow \mathcal{N}$ of accessible categories is L-accessible.

[159] an L_0*-functor*: which commutes with small filtering $\varinjlim$.

relative to π_0, is not satisfied here, and we need an independent proof, modeled on that of pages 333-336. For this it suffices, at the place □, page 333, to invoke the

Lemma 4.13.37. *Let $\mathcal{M}$ be a category that is accessible and stable under filtering* [page 360] *$\varinjlim$, π a useful cardinal for $\mathcal{M}$. Then for every $x \in \operatorname{Ob}\mathcal{M}$, we have a representation*

$$x \;=\; \varinjlim_{i\in I} x_i$$

L-π-adapted (i.e. with

1) I equivalent to a small category, and I large relative to π,

2) I stable under filtering $\varinjlim$ of cardinality $\leqslant \pi$,

3) $I \longrightarrow \mathcal{M}$ commutes with these $\varinjlim$,

4) $x_i \in \operatorname{Ob}\mathcal{M}_\pi \;\forall\, i \in I$),

with in addition I ordered.

Indeed, we start from

$$x \;=\; \varinjlim_{j\in J} x_j \;, \qquad J \;\simeq\; \underbrace{(\mathcal{M}_\pi/x)_0}_{\substack{\text{reduced subcategory}\\ \text{equivalent to } \mathcal{M}_\pi/x}} \;.$$

We define I as the set of *filtering* subcategories J' of J, of cardinality $\leqslant \pi$, I ordered by inclusion, and we set, for $J' \in I$,

$$x_{J'} \;=\; \varinjlim_{j\in J'} x_j \;.$$

It is immediate that we thus obtain a direct system in $\mathcal{M}$, indexed by I.

On the other hand, it is clear that I is small, I say that I is stable under filtering [page 361] $\varinjlim$ of cardinality $\leqslant \pi$, and that the functor $J' \longmapsto x_{J'} : I \longrightarrow \mathcal{M}$ commutes with them, finally that the $x_{J'}$ are in $\mathcal{M}_\pi$. It remains to see that I is large relative to π, i.e. that if we have $(J'_r)_{r\in A}$, with $\operatorname{card} A \leqslant \pi$, the J'_r in I, then $\exists\, J' \in I$ which dominates the J'_r. Let K be the subcategory of J generated by the J'_r, we easily see that $\operatorname{card} K \leqslant \pi$. Everything comes down to proving:

Lemma 4.13.38 ([160]). *Let J be a category large relative to π, K a subcategory of J* p. 137 *of cardinality $\leqslant \pi$. Then there exists a subcategory K' that is* filtering *in J, containing K, of cardinality $\leqslant \pi$. If we assume that we are not, in the case where J contains a maximal object, and that K contains all the maximal objects of J, we can even take K'* [page 362] *such that $\operatorname{Ob} K' = \operatorname{Ob} K \cup \{e\}$, where e is a* final object *of K', and K is a full subcategory of K'. (Thus if $K' \neq K$, K' is isomorphic to the right cone constructed on K, "obtained from K by adding a final object to it".)*

Proof. If K has a final object, we take $K' = K$. Otherwise, since J is large relative to $\operatorname{card}\operatorname{Ob} K$, there exists a $j \in \operatorname{Ob} J$ which dominates all the objects of K. Let us choose, for every $k \in \operatorname{Ob} K$, an arrow

$$\alpha_k \;:\; k \longrightarrow j \;.$$

For every arrow

$$u \;:\; k' \longrightarrow k$$

[160] **N.B.** For the proof of 4.13.37, it suffices to prove 4.13.38 in the case where J is ordered, in which case the contortions at the end are unnecessary.

in K, let us consider the double arrow

$$(\alpha_k \circ u, \alpha_{k'}) \ : \ k' \rightrightarrows j \ .$$

The set of these double arrows, with target j, is of cardinality $\leqslant \operatorname{card} \operatorname{Fl} K \leqslant \pi$, so J being large relative to π, there exists an arrow $\rho : j \longrightarrow e$ in J which equalizes all these double arrows. Let $K' \subset J$, having as objects $\operatorname{Ob} K \cup \{e\}$, and having as arrows, in addition to those of K, the arrows $\rho\alpha_k : k \longrightarrow e$ (for $k \in \operatorname{Ob} K$), and id_e. *If* $e \notin \operatorname{Ob} K$, we see that this is indeed a subcategory, admitting e as final object and K as full subcategory. If we had $e \in \operatorname{Ob} K$, and if there exists an object e' of J which dominates e and which is not in K, we immediately reduce to the preceding case, replacing e by e'. [page 363]

(([161]). Thus there remains the case where there exists an object e in K which is a greatest element in K, and such that every object of J dominating e is in K. This implies, J being filtering, that e is a greatest object of J. (For if $x \in J$, $\exists\, x'$ a common upper bound for x and e, hence x' is in K, hence dominated by e, which therefore dominates x.) We see as before that there exists an arrow $p : e \longrightarrow j$ in J, which equalizes all the double arrows $k \rightrightarrows e$ with target e *in* K. Composing with $j \longrightarrow e$ if necessary, we can assume $j = e$. Thus we have an endomorphism p of e in J, such that $pu = pv$ for every double arrow $k \overset{u}{\underset{v}{\rightrightarrows}} e$ in K, with target e.

For $k \in \operatorname{Ob} K$, let [page 364]

$$f_k \ : \ k \longrightarrow e$$

be the common value of the pu, for $u \in \operatorname{Hom}_K(k, e)$. We see at once that if we have $k' \xrightarrow{v} k$ in K, then

$$f_{k'} \ = \ f_k \circ v \ .$$

In particular, taking $k' = e$, we find

$$f_k \circ v \ = \ f_e \,(= \ p \,) \qquad \text{if } v : e \longrightarrow k \ . \qquad \text{p. 138}$$

Let then K' be the subcategory of J having the same objects as K, and as arrows, in addition to those of K, the composite arrows $k \xrightarrow{u} e \xrightarrow{p^r} e \xrightarrow{v} k'$, with u, v in K, and $r \geqslant 1$, i.e. the composite arrows $k \xrightarrow{f_k} e \xrightarrow{p^{r'}} e \xrightarrow{v} k'$, with $k, k' \in \operatorname{Ob} K$, $v \in \operatorname{Fl} K$, $r' \geqslant 0$. We must see that it is stable under composition (to indeed have a subcategory). This is clear for vu if u or v are in $\operatorname{Fl} K$. If u nor v is, we must take a composite

$$k \xrightarrow{u_1} e \underset{r \geqslant 1}{\xrightarrow{p^r}} e \xrightarrow{u_2} k' \xrightarrow{v_1} e \underset{s \geqslant 1}{\xrightarrow{p^s}} e \xrightarrow{v_2} k'' \ ,$$

with u_1, u_2, v_1, v_2 in $\operatorname{Fl} K$, $r, s \geqslant 1$. Now we have

$$p(v_1 u_2) \ = \ p \ , \qquad \text{whence} \qquad p^s(v_1 u_2) \ = \ p^{s-1}(p v_1 u_2) \ = \ p^{s-1} p \ = \ p^s \ ,$$

so the composite is

$$k \xrightarrow{u_1} e \xrightarrow{p^{r+s}} e \xrightarrow{v_2} k'' \ ,$$

which is therefore indeed in $\operatorname{Fl} K'$.

Thus, K' is a subcategory of J, but is it filtering? Condition PS $1'$ is obvious, since e is a greatest element of J, there remains condition PS 2 of the double arrow. We easily see that this condition is satisfied in K', if and only if it is satisfied for every double arrow [page 365]

$$e \overset{p^r}{\underset{p^s}{\rightrightarrows}} e \qquad \text{with } r, s \geqslant 0 \ .$$

[161] The construction which follows is contained in the more general and more precise statement 4.13.39.

(Indeed, we immediately reduce to the double arrows with target e, and note that every arrow of K' with target e can be written as a composite $k \xrightarrow{u} e \xrightarrow{p^r} e \xrightarrow{v} e$, with $r \geqslant 0$ and u, v arrows in K. Composing with $e \xrightarrow{p} e$, we then find, since $pv = p$, $k \xrightarrow{u} e \xrightarrow{p^{r+1}} e$, that is $k \xrightarrow{f_k} e \xrightarrow{p^r} e$.) But we see at once that the possibility of equalizing (p^r, p^s) in K' is equivalent to the existence of $t \geqslant 0$ such that $p^{r+t} = p^{s+t}$. Thus K' is filtering if and only if $r \longmapsto p^r$ is a stationary sequence (take $r = 0$, $s = 1$, whence $p^r = p^{r+1}$ for r large). This is not necessarily the case. [page 366]

Let us therefore set $K = K_0$, $K' = K_1$, and construct by induction an increasing sequence of subcategories K_i ($i \in \mathbf{N}$), with $\mathrm{Ob}\, K_i = \mathrm{Ob}\, K$, and (for $i \geqslant 1$) K_i being obtained from K_{i-1} by adding to it an arrow $p_i : e \longrightarrow e$, as above. Thus every double arrow in a K_i can be equalized in K_{i+1}. Let us take $K_\infty = \varinjlim K_i$, K_∞ is a *filtering* subcategory of J, containing K and of cardinality $\leqslant \pi$, q.e.d.)

We can make this construction more precise as follows: p. 139

Corollary 4.13.39. *Under the conditions of 4.13.38, suppose that K contains an object e maximal in J. Let Σ be the set of endomorphisms p of e in J, such that for every double arrow $k \overset{u}{\underset{v}{\rightrightarrows}} e$ in K, we have $pu = pv$. Then Σ is stable under composition, and there exists a nonempty subset Σ_∞ of Σ, such that for $p, p' \in \Sigma_0$, exists $q \in \Sigma_0$ such that* [page 367] *$qp = qp'$ (i.e. such that the subcategory J_0 of J whose set of objects is $\{e\}$, and set of arrows $\{\mathrm{id}_e\} \cup \Sigma_\infty$, is filtering). If Σ_0 is chosen thus, let $\mathcal{F}$ be the union of $\mathrm{Fl}\, K$ and of the set of composite arrows*

$$k \xrightarrow{u} e \xrightarrow{p} e \xrightarrow{v} k' \ , \qquad (*)$$

with $u, v \in \mathrm{Fl}\, K$, $p \in \Sigma_\infty$. Then $\mathcal{F}$ is the set of arrows of a subcategory K' of J, having the same objects as K, and admitting e as greatest element. Moreover, K' contains K and is filtering.

Proof. That Σ is stable under composition is clear. To construct Σ_∞ having the stated properties, we note that, J being large relative to π, Σ is nonempty ([162]). We construct by induction a sequence of elements

$$p_0 \ , \ p_1 \ , \ \cdots \ , \ p_n \ , \ \cdots$$

of Σ, taking for p_0 any element of Σ, and for $i \geqslant 1$, taking $p_i \in \Sigma$ such that for every double arrow (p, p') of the submonoid M_{i-1} of $\mathrm{End}_J(e)$ generated by $(p_j)_{0 \leqslant j \leqslant i-1}$, we have $p_i p = p_i p'$. Such a p_i exists, for taking the set of double arrows of the preceding form (p, p'), with $p, p' \in M_{i-1}$, and of those of the form $(u, v) : k \rightrightarrows e$, with $u, v \in \mathrm{Fl}\, K$, the set of all these double arrows has cardinality $\leqslant \pi$ (M_{i-1} being countable, hence of cardinality [page 368] $\leqslant \pi$), hence there exists an arrow $e \longrightarrow j$ in J which equalizes them all (J being large relative to π). Composing if necessary with $j \longrightarrow e$ (e being a maximal object of J), we find the desired $p_i : e \longrightarrow e$.

Let then Σ_0 be the multiplicative subset of Σ generated by the p_i ($i \in \mathbf{N}$), direct limit of the similar sets Σ_i generated by the p_j for $j \in [0, i]$. If p, p' are in Σ_∞, they are in one and the same Σ_i, hence we will have $p_{i+1} p = p_{i+1} p'$, which proves that Σ_∞ has the required filtering property.

Let us now prove that $\mathcal{F}$ is the set of arrows of a subcategory K' of J, having $\mathrm{Ob}\, K$ as set of objects. It remains to establish stability under composition vu. But if u and v are in $\mathrm{Fl}\, K$, or if one of the two is in $\mathrm{Fl}\, K$, the other not (hence of the form $(*)$), this stability is evident. Remains to see the case where neither nor v is in $\mathrm{Fl}\, K$, hence both

[162] cf. argument below for construction of p_i.

are of the form $(*)$:

$$k \xrightarrow{u} e \xrightarrow{p} e \xrightarrow{v} k' \xrightarrow{u'} e \xrightarrow{p'} e \xrightarrow{v'} k'' ,$$

with $u, v, u', v' \in \mathrm{Fl}\, K$, $p, p' \in \Sigma_\infty$. Since $(u'v, \mathrm{id}_e)$ is a double arrow of K with target e, we have

p. 140
[page 369]

$$p'u'v \;=\; p'\mathrm{id}_e \;=\; p' ,$$

hence the composite in question is also written

$$k \xrightarrow{u} e \xrightarrow{p'p} e \xrightarrow{v'} k'' ,$$

where $p'p \in \Sigma_\infty$ since Σ_∞ is stable under composition, hence it is in $\mathcal{F}$, q.e.d.

It is clear that K' contains K, and having the same objects as K which admits e as greatest element, that e is a greatest element of K'. Remains to prove that K' is filtering. Since K' has a greatest element, it remains to see that every double arrow $k \rightrightarrows l$ in K' can be equalized in K'. Composing if necessary with $l \longrightarrow e$, we can suppose $l = e$. Now we see that every arrow $k \longrightarrow e$ in K' is of the form

$$k \xrightarrow{u} e \xrightarrow{p} e \xrightarrow{v} e ,$$

with $u, v \in \mathrm{Fl}\, K$ and $p \in \Sigma_\infty \cup \{\mathrm{id}_e\}$. Composing if necessary with $p_0 \in \Sigma_\infty$ fixed, this arrow becomes (since $p_0 v = p_0 \mathrm{id}_e = p_0$)

$$k \xrightarrow{u} e \xrightarrow{p_0 p} e , \qquad \text{i.e. } p_0 p u .$$

For an arrow $v'p'u' : k \longrightarrow e$, we find $p_0 p' u'$. Let then $q \in \Sigma_0$ be such that $q(p_0 p) = q(p_0 p')$, say r (q exists since $p_0 p, p_0 p' \in \Sigma_\infty$). Thus $(q(p_0 p u), q(p_0 p' u')) = (ru, ru')$, and we have $ru = ru'$ since $r \in \Sigma_\infty$. All told, qp_0 equalizes the two arrows vpu and $v'p'u'$, q.e.d.

4.14 Accessibility criteria for big categories

[page 370]

Theorem 4.14.1. *Let $\mathcal{C}$ be a category equivalent to a small category, Φ, Ψ two small sets of categories, equivalent to small categories. For every category I, we denote by $I_\bullet$ (resp. ${}_\bullet I$) the right cone (resp. left) on I, obtained from I by adding a final (resp. initial) object $e = e_I$. Let Λ_1 (resp. Λ_2) be a set of pairs (I, φ) (resp. (I, ψ)), where $I \in \Phi$ (resp. $I \in \Psi$), and where*

$$\varphi \;:\; I_\bullet \longrightarrow \mathcal{C} \qquad (\text{resp. } \psi \;:\; {}_\bullet I \longrightarrow \mathcal{C}\;) \tag{4.14.1.1}$$

is a functor. For every category $\mathcal{N}$ stable ([163]), under $\varinjlim$ of type Φ and under $\varprojlim$ of type Ψ we denote by

$$\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C}, \mathcal{N}) \;\subset\; \underline{\mathrm{Hom}}(\mathcal{C}, \mathcal{N}) \tag{4.14.1.2}$$

the strictly full subcategory of $\underline{\mathrm{Hom}}(\mathcal{C}, \mathcal{N})$ formed of the functors $f : \mathcal{C} \longrightarrow \mathcal{N}$ having the following property:

a) $\forall\ (I, \varphi) \in \Lambda_1$, the composite functor

$$I_\bullet \xrightarrow{\varphi} \mathcal{C} \xrightarrow{f} \mathcal{N}$$

is I-left exact, i.e. makes $(f\varphi)(e_I)$ a $\varinjlim_I$ of $f\varphi|I$.

[163] This stability property is not necessary for the definition of $\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C}, \mathcal{N})$, and it is almost certain that the theorem is valid without making this assumption on $\mathcal{N}$. Cf. 4.14.16, where we see that it is in any case unnecessary for Ψ.

p. 141
[page 371]

b) For every $(I,\psi)\in\Lambda_2$, the composite functor

$$\bullet I \xrightarrow{\psi} \mathcal{C} \xrightarrow{f} \mathcal{N}$$

is I-right exact, i.e. makes $(f\psi)(e_I)$ an inverse limit of $(f\psi)|I$.

With these notations: if $\mathcal{N}$ is an accessible category, the same is true of $\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N})$.

Comment. We will use this theorem chiefly when we have $\Lambda_2=\emptyset$ or $\Lambda_1=\emptyset$, in which case we write $\underline{\mathrm{Hom}}_{\Lambda_1}(\mathcal{C},\mathcal{N})$, resp. $\underline{\mathrm{Hom}}^{\Lambda_2}(\mathcal{C},\mathcal{N})$, for the category $\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N})$. We will apply the theorem chiefly when $\mathcal{C}$ itself is stable under $\varinjlim$ of type Φ and under $\varprojlim$ of type Ψ, and when Λ_1 (resp. Λ_2) is already formed of functors φ (resp. ψ) which are I-left exact (resp. right exact). Thus $\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N})$ is then formed of the $f:\mathcal{C}\longrightarrow\mathcal{N}$ which commute with certain $\varinjlim$ of type Φ, and with certain $\varprojlim$ of type Ψ. In particular, we find:

[page 372]

Corollary 4.14.2. *Let $\mathcal{C}$ be a category equivalent to a small category, $\mathcal{N}$ a $\mathfrak{U}$-category, Φ and Ψ two small sets of categories I, equivalent to small categories. We suppose $\mathcal{C}$ and $\mathcal{N}$ stable under $\varinjlim$ of type Φ, under $\varprojlim$ of type Ψ. Let $\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{C},\mathcal{N})$ be the strictly full subcategory of $\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$ formed of the $f:\mathcal{C}\longrightarrow\mathcal{N}$ which commute with $\varinjlim$ of type Φ and with $\varprojlim$ of type Ψ. Then, if $\mathcal{N}$ is accessible, the same is true of $\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{C},\mathcal{N})$.*

Here are some remarkable special cases of this corollary:

Corollary 4.14.3. *Let $\mathcal{M}$, $\mathcal{N}$ be two accessible categories, and π_0 an infinite cardinal, such that $\mathcal{M}$ satisfies L_{π_0}. Let $\underline{\mathrm{Hom}}_{\pi_0}(\mathcal{M},\mathcal{N})$ be the strictly full subcategory of $\underline{\mathrm{Hom}}(\mathcal{M},\mathcal{N})$ formed of the π_0-accessible functors. Then $\underline{\mathrm{Hom}}_{\pi_0}(\mathcal{M},\mathcal{N})$ is accessible.*

(The case where π_0 is useful for $\mathcal{M}$ has already been treated.)

[page 373]

Proof. In the general case, let $\pi>\pi_0$ be a useful cardinal for $\mathcal{M}$, so that we have

$$\mathcal{M}\xleftarrow{\sim}\mathrm{Ind}_\pi(\mathcal{C}) \qquad \text{with } \mathcal{C}=\mathcal{M}_\pi\ .$$

We can suppose $\mathcal{M}=\mathrm{Ind}_\pi(\mathcal{C})$. The fact that $\mathcal{M}$ satisfies L_{π_0} implies that $\mathcal{C}$ is stable under $\varinjlim$ large relative to π_0, and of cardinality $\leqslant\pi$. Let Φ be the set of categories of this nature. Φ is not small, but there exists a small $\Phi_0\subset\Phi$, such that every $I\in\Phi$ is isomorphic to an $I_0\in\Phi_0$, which will suffice for the validity of 4.14.1 and 4.14.2 (where here we will put $\Psi=\emptyset$). We can consider $\underline{\mathrm{Hom}}_{!\,\pi_0}(\mathcal{M},\mathcal{N})$ as a strictly full subcategory of $\underline{\mathrm{Hom}}_{!\,\pi}(\mathcal{M},\mathcal{N})\xrightarrow{\sim}\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$ (cf. 4.12.3, p. 275). For a functor $f:\mathcal{M}\longrightarrow\mathcal{N}$ which is π-accessible, to say that it is even π_0-accessible amounts to saying that its restriction to $\mathcal{C}$, i.e. its image in $\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$, commutes with $\varinjlim$ of type Φ. (Cf. 4.13.20, page 341 ([164]), where we take $\alpha=\alpha_\pi$, noting that J_{α_π} large relative to π_0 since $\pi>\pi_0$).

Thus we find

[page 374]

$$\underline{\mathrm{Hom}}_{!\,\pi_0}(\mathcal{M},\mathcal{N})\xrightarrow{\sim}\underline{\mathrm{Hom}}_{\Phi}(\mathcal{C},\mathcal{N})\ ,$$

and corollary 4.14.2 gives the desired conclusion.

p. 142

Corollary 4.14.4. *Let $\mathcal{M}$, $\mathcal{N}$ be two accessible categories, Φ and Ψ two sets of categories as in 4.14.2, we suppose that $\mathcal{M}$ and $\mathcal{N}$ satisfy L_{π_0}, and stable under $\varinjlim$ of type Φ and under $\varprojlim$ of type Ψ. Let $\underline{\mathrm{Hom}}^{\Psi}_{\Phi,\pi_0}(\mathcal{M},\mathcal{N})$ be the strictly full subcategory of $\underline{\mathrm{Hom}}(\mathcal{M},\mathcal{N})$ formed of the π_0-accessible functors which commute with $\varinjlim$ of type Φ, and with $\varprojlim$ of type Ψ. Then $\underline{\mathrm{Hom}}^{\Psi}_{\Phi,\pi_0}(\mathcal{M},\mathcal{N})$ is accessible.*

Proof. This time, we choose $\pi>\pi_0$ such that we have $\mathrm{card\,red}\, I\leqslant\pi$ for every I in Φ or in Ψ. The category under consideration is then interpreted (putting again $\mathcal{C}=\mathcal{M}_\pi$) as

[page 375]

[164] The reference is not quite adequate. We should gather in a separate paragraph all the statements of this type, some of which use the indicial representation theorem.

$$\underline{\mathrm{Hom}}^{\Psi}_{\Phi,\pi_0}(\mathcal{M},\mathcal{N}) \xrightarrow{\sim} \underline{\mathrm{Hom}}^{\Psi}_{\Phi'}(\mathcal{C},\mathcal{N}) \ ,$$

where Φ' is formed of the categories which are either in Φ, or which are large relative to π_0 and of cardinality $\leqslant \pi$. It therefore suffices to apply 4.14.2 once more.

Corollary 4.14.5. *Let us consider the following three sets:*

a) Φ_1 *is formed of all categories equivalent to small categories.*

b) Φ_2 *as a), but restricting to filtering categories.*

c) Φ_3 *formed of the discrete categories, and of those which are large relative to* π_0 *(π_0 a given infinite cardinal).*

Let Φ be one of the sets of categories Φ_1, Φ_2, Φ_3, $\Phi_2 \cup \Phi_3$ or the union of one of these and a small set Φ_0 of essentially small categories ([165]). Let Ψ be a small set of categories. Let $\mathcal{M}$, $\mathcal{N}$ be accessible categories, stable under $\varinjlim$ of type Φ, and under $\varprojlim$ of type Ψ. Then $\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{M},\mathcal{N})$ is accessible.

PROOF. Let $\pi'_0 \geqslant \pi_0$ be a useful cardinal for $\mathcal{M}$, π a cardinal having the following properties. [page 376]

a) $\pi \geqslant \sup(\pi'_0, \operatorname{card}\operatorname{red}\mathcal{M}_{\pi'_0})$, $\pi > \pi_0$.

b) $\pi^{\pi'_0} = \pi$.

c) $I \in \Phi_0 \cup \Psi \Longrightarrow \operatorname{card}\operatorname{red} I \leqslant \pi$.

Let $\mathcal{C} = \mathcal{M}_\pi$, whence $\mathcal{M} \simeq \mathrm{Ind}_\pi(\mathcal{C})$. In all the cases under consideration, Φ contains the set of essentially small categories which are large relative to π_0, a fortiori those which are large relative to π, hence the category $\underline{\mathrm{Hom}}^{\Psi}_{\Phi}$ is interpreted as a strictly full subcategory of

$$\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N}) \ .$$

If on the other hand Φ_π denotes the set of $I \in \Phi$ such that $\operatorname{card} I \leqslant \pi$, we find

$$\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{M},\mathcal{N}) \ \simeq \ \underline{\mathrm{Hom}}^{\Psi}_{\Phi_\pi}(\mathcal{C},\mathcal{N}) \ ,$$

and we apply 4.14.2.

In particular, we find: p. 143

Corollary 4.14.6. *Let $\mathcal{M}$, $\mathcal{N}$ be two accessible categories, suppose $\mathcal{M}$ stable under small $\varinjlim$ (i.e. $\mathcal{M}$ is a paratopos category). Then the category $\underline{\mathrm{Hom}}_!(\mathcal{M},\mathcal{N})$ of functors $f : \mathcal{M} \longrightarrow \mathcal{N}$ which commute with small $\varinjlim$ is accessible. If $\mathcal{N}$ is itself also a paratopos, $\underline{\mathrm{Hom}}_!(\mathcal{M},\mathcal{N})$ is a paratopos.* [page 377]

PROOF. The first assertion is the special case of 4.14.5 corresponding to the case where $\Phi = \Phi_1$, $\Psi = \emptyset$. The second assertion follows from it, since $\underline{\mathrm{Hom}}_!(\mathcal{M},\mathcal{N})$ is visibly stable under small $\varinjlim$, hence it is a paratopos being accessible (4.8.9, page 207).

Corollary 4.14.7. *Let Φ, Ψ, $\mathcal{M}$ be as in 4.14.5. Then for every accessible category $\mathcal{N}$, the category* [page 378]

$$\underline{\mathrm{Hom}}^{\Phi}_{\Psi}(\mathcal{M}^\circ,\mathcal{N}) \ \simeq \ (\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{M},\mathcal{N}^\circ))^\circ$$

is accessible.

[165] We could give here a statement which includes 4.14.4 by adding a

d) Φ_4 formed of the categories large relative to π_0,

and take Φ of the form of Φ_1, Φ_2, Φ_4, Φ_3, $\Phi_3 \cup \Phi_2$, $\Phi_3 \cup \Phi_4$, or the union of these and a Φ_0.

PROOF. We proceed as for 4.14.5, to make explicit

$$\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{M},\mathcal{N}^\circ) \xrightarrow{\sim} \underline{\mathrm{Hom}}^{\Psi}_{\Phi_\pi}(\mathcal{C},\mathcal{N}^\circ)\ ,$$

whence

$$\underline{\mathrm{Hom}}^{\Phi}_{\Psi}(\mathcal{M}^\circ,\mathcal{N}) \xrightarrow{\sim} \underline{\mathrm{Hom}}^{\Phi_\pi}_{\Psi}(\mathcal{C}^\circ,\mathcal{N})\ ,$$

and we apply 4.14.2 to $\mathcal{C}^\circ$, Ψ, Φ_π.

Corollary 4.14.8. *Let $\mathcal{M}$ and $\mathcal{N}$ be two paratopos categories. Then*

$$\mathcal{M}\boxtimes\mathcal{N} \ \simeq\ \underline{\mathrm{Hom}}^{!}(\mathcal{M}^\circ,\mathcal{N}) \qquad \textit{(cf. 4.2 and 4.3)}$$

is a paratopos.

Indeed, by virtue of 4.14.7 (with $\Phi=\Phi_1$, $\Psi=\emptyset$), it is an accessible category.

On the other hand, it is immediate that it is stable under small $\varprojlim$, $\mathcal{N}$ being so. [page 379] Thus we conclude again by 4.8.9 (page 207) that it is a paratopos category.

Corollary 4.14.9. *Let $\mathcal{M}$, $\mathcal{N}$ be two paratopos categories, and $\underline{\mathrm{Hom}}^{!}(\mathcal{M},\mathcal{N})$ the category of functors $\mathcal{M}\longrightarrow\mathcal{N}$ which commute with small $\varprojlim$. We then have an equivalence of categories*

$$\begin{aligned}\underline{\mathrm{Hom}}^{!}(\mathcal{M},\mathcal{N}) &\xrightarrow{\sim} \underline{\mathrm{Hom}}_{!}(\mathcal{N},\mathcal{M})^\circ\\ f &\longmapsto \textit{left adjoint of } f\ ,\end{aligned}$$

! ! *and consequently the category* **opposite** *to $\underline{\mathrm{Hom}}^{!}(\mathcal{M},\mathcal{N})$ is accessible.*

This follows from the

Lemma 4.14.10 (for the record). *Let $\mathcal{M}$ be pseudo-topos category, $f:\mathcal{M}\longrightarrow\mathcal{N}$ a functor. For f to admit a left adjoint (resp. a right adjoint), it is necessary and sufficient that f commute with small $\varprojlim$ (resp. with small $\varinjlim$).*

This is equivalent to the special case:

p. 144 [page 380]

Corollary 4.14.11. *Let $\mathcal{M}$ be a paratopos, and f a covariant (resp. contravariant) functor from $\mathcal{M}$ to* Ens*. For f to be representable, it is necessary and sufficient that f commute with $\varprojlim$.*

This has been seen – the case of contrafunctors being the most obvious (cf. 4.1 – it suffices that $\mathcal{M}$ be a paratopos, and even that $\mathcal{M}$ admit a small generating full subcategory, cf. SGA 4 I 8.12.7 (for the dual statement). In the case of a covariant functor, the difficulty comes from the fact that $\mathcal{M}$ does not necessarily admit a small cogenerating subcategory. One must use the existence of a (generalized) cardinal filtration, cf. loc. cit. 8.13.2.)

After all these fine corollaries, we are ripe for tackling the proof of 4.14.1!

PROOF OF 4.14.1. We know that $\mathcal{M}=\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$ is accessible (4.9.5, p. 220), and [page 381] $\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N})$ is a strictly full subcategory of this category. It suffices to prove that it is an accessible subcategory in $\mathcal{M}$ (4.13.1, p. 281). For every $\xi=(I,\varphi)\in\Lambda_1$, we have a functor

$$L_\xi\ :\ \mathcal{M}\ =\ \underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})\longrightarrow \underline{\mathrm{Fl}}(\mathcal{N})\ ,$$

given by

$$L_\xi(f)\ =\ \text{canonical arrow}\ \varinjlim_{i\in I} f\varphi(i)\longrightarrow f\varphi(e_I)\ .$$

I say that the functor L_ξ is *accessible.* Indeed, let π_0 be a cardinal such that $\mathcal{N}$ satisfies L_{π_0}, the same therefore holds for $\mathcal{M}=\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$ and for $\underline{\mathrm{Fl}}(\mathcal{N})\overset{\mathrm{def}}{=}\underline{\mathrm{Hom}}(\mathbf{\Delta}_1,\mathcal{N})$, the limits in question are computed argument by argument. I say that L_ξ is π_0-accessible, i.e. commutes with $\varinjlim$ large relative to π_0. It suffices to see that this is so for the functors composed with the functors

$$\underline{\mathrm{Fl}}(\mathcal{N})\ \underset{\rho}{\overset{\sigma}{\rightrightarrows}}\ \mathcal{N}$$

"source" and "target", i.e. the functors

$$f \longmapsto \varinjlim_{i\in I} f\varphi(i) \qquad \text{and} \qquad f \longmapsto f\varphi(e_I)\ .$$

Now the first functor, $f \longmapsto \varinjlim_{i\in I} f\varphi(i)$, is the composite of [page 382]

$$f \longmapsto f\circ\varphi_I\ :\ \underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N}) \longrightarrow \underline{\mathrm{Hom}}(I,\mathcal{N})$$

(which commutes with the $\varinjlim$ that are representable in $\mathcal{N}$, and in particular with the $\varinjlim$ large relative to π_0) and of the functor

$$\varinjlim_{I}\ :\ \underline{\mathrm{Hom}}(I,\mathcal{N}) \longrightarrow \mathcal{N}\ ,$$

which also commutes with direct limits representable in $\mathcal{N}$, and in particular with those large relative to π_0. As for the functor $f \longmapsto (f\circ\varphi)(e_I) = f(\varphi(e_I))$, it is clear that it too commutes with the $\varinjlim$ representable in $\mathcal{N}$, so it is π_0-accessible.

This being so, let us note that p. 145

$$\underline{\mathrm{Hom}}_{\Lambda_1}(\mathcal{C},\mathcal{N}) \ =\ \bigcap_{\xi\in\Lambda_1} \underline{\mathrm{Hom}}_{\{\xi\}}(\mathcal{C},\mathcal{N})\ ,$$

and that $\underline{\mathrm{Hom}}_\xi(\mathcal{C},\mathcal{N})$ is the inverse image, under the preceding accessible functor L_ξ, of the strictly full subcategory $\underline{\mathrm{Is}}(\mathcal{N})$ of $\underline{\mathrm{Fl}}(\mathcal{N})$ consisting of the *isomorphisms* in $\mathcal{N}$. If we prove that the latter is accessible in $\underline{\mathrm{Fl}}(\mathcal{N})$, it follows that $\underline{\mathrm{Hom}}_\xi(\mathcal{C},\mathcal{N})$ is accessible in [page 383] $\mathcal{M}$ (4.13.31, page 350). Thus, Λ_1 being small, it follows that $\underline{\mathrm{Hom}}_{\Lambda_1}(\mathcal{C},\mathcal{N})$ is accessible in $\mathcal{M}$ (4.13.7, page 291).

We prove in the same way that $\underline{\mathrm{Hom}}^{\Lambda_2}(\mathcal{C},\mathcal{N})$ is accessible in $\mathcal{M} = \underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$. Here, we must consider the functors

$$L'_\xi\ :\ \mathcal{M} = \underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N}) \longrightarrow \underline{\mathrm{Fl}}(\mathcal{N})$$

defined by

$$L'_\xi\ =\ \text{canonical arrow } f\varphi(e'_I) \longrightarrow \varprojlim_{i\in I} f\varphi(i)\ .$$

To prove that this functor is accessible, we proceed as before, taking however π_0 large enough that we have

$$\begin{cases} I\in\Psi \implies \operatorname{card}\operatorname{red} I \leqslant \pi_0. \\ \mathcal{N}_{\pi_0} \text{ is generating in } \mathcal{N}. \end{cases}$$

I say that then L'_ξ is π_0-accessible: indeed, proceeding as before for L_ξ, we are reduced to seeing that

$$\varprojlim_{I} : \underline{\mathrm{Hom}}(I,\mathcal{N}) \longrightarrow \mathcal{N}$$

is π_0-accessible. Now this is contained in SGA 4 I 9.8.

Thus, we have proved that the two strictly full subcategories [page 384]

$$\underline{\mathrm{Hom}}_{\Lambda_1}(\mathcal{C},\mathcal{N})\ , \qquad \underline{\mathrm{Hom}}^{\Lambda_2}(\mathcal{C},\mathcal{N})$$

of $\mathcal{M} \overset{\text{def}}{=} \underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$ are accessible in $\mathcal{M}$. The same therefore holds for their intersection (4.13.7)

$$\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N}) \ =\ \underline{\mathrm{Hom}}_{\Lambda_1}(\mathcal{C},\mathcal{N}) \cap \underline{\mathrm{Hom}}^{\Lambda_2}(\mathcal{C},\mathcal{N})\ ,$$

q.e.d.

It remains to prove the

Lemma 4.14.12. *Let $\mathcal{N}$ be an accessible category. Then the category $\underline{\mathrm{Fl}}(\mathcal{N}) \overset{\mathrm{def}}{=} \underline{\mathrm{Hom}}(\mathbf{\Delta}_1, \mathcal{N})$ is accessible (4.9.5), and the strictly full subcategory $\underline{\mathrm{Is}}(\mathcal{N})$ consisting of the invertible arrows in $\mathcal{N}$ is accessible in $\underline{\mathrm{Fl}}(\mathcal{N})$. (The same holds for the strictly full subcategory $\underline{\mathrm{Mon}}(\mathcal{N})$ consisting of the monomorphisms in $\mathcal{N}$.)*

Proof. Let $\mathcal{C}$ be a small strictly generating subcategory of $\mathcal{N}$, and consider the small family of functors

$$h_x \;=\; (y \longmapsto \mathrm{Hom}(x,y)) \;:\; \mathcal{N} \longrightarrow \mathrm{Ens}\ ,$$

whence functors

p. 146

[page 385]

$$\underline{\mathrm{Fl}}(h_x) \;:\; \underline{\mathrm{Fl}}(\mathcal{N}) \longrightarrow \underline{\mathrm{Fl}}(\mathrm{Ens})\ .$$

For an arrow $u : y \longrightarrow z$ in $\mathcal{N}$ to be an isomorphism (resp. a monomorphism), it is necessary and sufficient that for every $x \in \mathrm{Ob}\,\mathcal{C}$, $h_x(u)$ be so. Thus, $\underline{\mathrm{Is}}(\mathcal{N})$ (resp. $\underline{\mathrm{Mon}}(\mathcal{N})$) appears as the intersection of the inverse images, under the functors $\underline{\mathrm{Fl}}(h_x)$ above, of the subcategory $\underline{\mathrm{Is}}(\mathrm{Ens})$ (resp. $\underline{\mathrm{Mon}}(\mathrm{Ens})$) of $\underline{\mathrm{Fl}}(\mathrm{Ens})$. This reduces us to the case where $\mathcal{N} = (\mathrm{Ens})$. But if $\mathcal{C}$ denotes the strictly full subcategory of Ens consisting of the *finite* sets, these form a strictly generating family consisting of objects of *finite presentation*, and since Ens is stable under small $\varinjlim$, hence under filtering $\varinjlim$, it follows that we have

$$\mathrm{Ind}(\mathcal{C}) \longrightarrow (\mathrm{Ens})\ , \qquad (*)$$

whence at once (SGA 4 I 8.8.5, p. 104)

[page 386]

$$\mathrm{Ind}(\underline{\mathrm{Fl}}(\mathcal{C})) \xrightarrow{\sim} \underline{\mathrm{Fl}}(\mathrm{Ens})\ . \qquad (*)$$

So the desired accessibility result follows from the following two lemmas:

Lemma 4.14.13. *The preceding equivalence $(*)$ induces equivalences*

$$\begin{array}{rcl} \mathrm{Ind}(\underline{\mathrm{Is}}(\mathcal{C})) & \xrightarrow{\sim} & \underline{\mathrm{Is}}(\mathrm{Ens}) \\ \mathrm{Ind}(\underline{\mathrm{Mon}}(\mathcal{C})) & \xrightarrow{\sim} & \underline{\mathrm{Mon}}(\mathrm{Ens}) \end{array}$$

(where $\mathcal{C} = (\mathrm{Ensf})$ is the full subcategory of (Ens) consisting of the finite *sets.)*

Lemma 4.14.14 ([166]). *Let $\mathcal{C}$ be a category equivalent to a small category. Then:*

a) $\mathcal{M} \overset{\mathrm{def}}{=} \mathrm{Ind}(\mathcal{C})$ is an accessible category.

b) For every full subcategory $\mathcal{D}$ in $\mathcal{C}$, the essential image of $\mathrm{Ind}\,\mathcal{D}$ in $\mathcal{M} = \mathrm{Ind}(\mathcal{C})$ is an L-accessible subcategory in $\mathcal{M}$.

Proof of the two lemmas immediate and "left to the reader". The b) in 4.14.14 comes from the fact that the functor $\mathrm{Ind}(\mathcal{D}) \longrightarrow \mathrm{Ind}(\mathcal{C})$ coming from a functor $\mathcal{D} \longrightarrow \mathcal{C}$ (here an inclusion functor) is an L-functor (i.e. commutes with small filtering $\varinjlim$), and a fortiori is accessible (and even $\aleph_0$-accessible).

[page 387]

4.14.15. One cannot help wondering whether for $\mathcal{N}$ accessible, the strictly full subcategory $\underline{\mathrm{Ép}}(\mathcal{N})$ in $\underline{\mathrm{Fl}}(\mathcal{N})$, consisting of the epimorphisms in $\mathcal{N}$, is accessible in $\underline{\mathrm{Fl}}(\mathcal{N})$. It is immediate that if $\mathcal{N}$ satisfies L_{π_0}, then $\underline{\mathrm{Ép}}(\mathcal{N})$ is stable in $\underline{\mathrm{Fl}}(\mathcal{N})$ under the $\varinjlim$ of type L_{π_0}, a fortiori under direct factors. It would therefore have a hard time not being accessible in $\underline{\mathrm{Fl}}(\mathcal{N})$. Let us note all the same:

Corollary 4.14.15.1 (of 4.14.12). *Let $\mathcal{N}$ be an accessible category stable under amalgamated sums $y \amalg_x z$ ([167]). Then $\underline{\mathrm{Ép}}(\mathcal{N})$ is an accessible subcategory in $\underline{\mathrm{Fl}}(\mathcal{N})$.*

p. 147

[166] **N.B.** Up to equivalence, one finds the accessible categories $\mathcal{M}$ stable under small filtering $\varinjlim$, and such that $\mathcal{M}_{\mathrm{pf}}$ (subcategory of objects of *finite* presentation) is strictly generating.

[page 388]

[167] I am almost sure that this hypothesis is superfluous. Compare page 390.

Let us indeed consider the functor

$$\begin{array}{ccc} \gamma \ : \ \underline{\mathrm{Fl}}(\mathcal{N}) & \longrightarrow & \underline{\mathrm{Fl}}(\mathcal{N}) \\ (x \longrightarrow y) & \longmapsto & (y \amalg_x y \xrightarrow{\text{codiag}} y) \ . \end{array}$$

This functor commutes with the types of $\varinjlim$ that are representable in $\mathcal{N}$, in particular, it is accessible. On the other hand, $\underline{\mathrm{Ép}}(\mathcal{N})$ is the inverse image under γ of the strictly full subcategory $\underline{\mathrm{Is}}(\mathcal{N})$ of $\underline{\mathrm{Fl}}(\mathcal{N})$. So the corollary follows from the fact that by 4.14.12 the latter is accessible in $\underline{\mathrm{Fl}}(\mathcal{N})$.

4.14.16. I presume that 4.14.1 remains valid without assuming $\mathcal{N}$ stable under the $\varinjlim$ of type Φ and under the $\varprojlim$ of type Ψ. If we have an essentially small category I, the question is whether the strictly full subcategory $\underline{\mathrm{Hom}}_{\square}(I_\bullet, \mathcal{N})$ of $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{N})$, consisting of the functors satisfying condition a) of 4.14.1, is accessible in $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{N})$, and likewise for $\underline{\mathrm{Hom}}^{\square}({}_\bullet I, \mathcal{N})$. For this second case, it looks OK, for if $\mathcal{C}$ is a small strictly generating subcategory of $\mathcal{N}$, and if for every $x \in \mathrm{Ob}$ we again consider the functor [page 389]

$$h_x \ : \ \mathcal{N} \longrightarrow \mathrm{Ens} \ ,$$

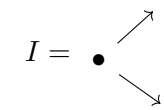
whence

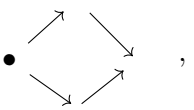
$$\bar{h}_x \ : \ \underline{\mathrm{Hom}}({}_\bullet I, \mathcal{N}) \longrightarrow \underline{\mathrm{Hom}}({}_\bullet I, \mathrm{Ens}) \ ,$$

we see that $\underline{\mathrm{Hom}}^{\square}({}_\bullet I, \mathcal{N})$ is the intersection of the inverse images of the subcategories $\underline{\mathrm{Hom}}^{\square}({}_\bullet I, \mathrm{Ens})$ of $\underline{\mathrm{Hom}}({}_\bullet I, \mathrm{Ens})$. So the latter being accessible in $\underline{\mathrm{Hom}}({}_\bullet I, \mathrm{Ens})$, by 4.14.1 ((Ens) being stable under small $\varprojlim$), it does indeed follow that $\underline{\mathrm{Hom}}^{\square}({}_\bullet I, \mathcal{N})$ is so in $\underline{\mathrm{Hom}}({}_\bullet I, \mathcal{N})$, q.e.d.

The case of $\underline{\mathrm{Hom}}_{\square}(I_\bullet, \mathcal{N})$ in $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{N})$ seems more delicate, in the case where [page 390]

$$I = \quad \bullet \quad , \qquad \text{thus} \qquad I_\bullet = \quad \bullet \quad ,$$





if the envisaged statement is true, we will conclude from it that $\underline{\mathrm{Ép}}(\mathcal{N})$ is accessible in $\underline{\mathrm{Fl}}(\mathcal{N})$, for every accessible category, for it is the inverse image of $\underline{\mathrm{Hom}}_{\square}(I_\bullet, \mathcal{N})$ under the canonical (accessible) functor

$$\begin{array}{ccc} \underline{\mathrm{Fl}}(\mathcal{N}) & \longrightarrow & \underline{\mathrm{Hom}}(I_\bullet, \mathcal{N}) \\ (x \xrightarrow{u} y) & \longmapsto & \left(\begin{array}{ccccc} & & y & & \\ & \overset{u}{\nearrow} & & \overset{\mathrm{id}_y}{\searrow} & \\ x & & & & y \\ & \underset{u}{\searrow} & & \underset{\mathrm{id}_y}{\nearrow} & \\ & & y & & \end{array} \right) \ . \end{array}$$

This is to say that the envisaged generalization of 4.14.1 is not idle, but is probably useful.

Theorem 4.14.17 ([168]). *Let $p : \mathcal{E} \longrightarrow \mathcal{C}$ be a cofibrant or fibrant functor, with $\mathcal{C}$ essentially small (i.e. equivalent to a small category), and such that the fiber categories are accessible, and the cobase change (cofibrant case) or base change (fibrant case) functors associated with the arrows u of $\mathcal{C}$ are accessible. Then the category $\underline{\Gamma}(\mathcal{E}/\mathcal{C})$ of sections of $\mathcal{E}$ over $\mathcal{C}$ is accessible, and for every $s \in \mathrm{Ob}\,\mathcal{C}$, the "evaluation at s" functor $\varepsilon_s : \underline{\Gamma}(\mathcal{E}/\mathcal{C}) \longrightarrow \mathcal{E}_s$ is accessible.* p. 148 [page 391]

Corollary 4.14.18. *Let π_0 be an infinite cardinal such that the fiber categories satisfy the condition L_{π_0}, and that the cobase change (resp. base change) functors between them likewise satisfy L_{π_0}. (Such a π_0 exists.)*

[168] Cf. generalization in theorem page 422, th. 4.14.26.

ⓐ *Then* $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$ *satisfies the condition* $\underline{\mathrm{L}}_{\pi_0}$*, and the functors* ε_s *too, i.e. the* $\varinjlim$ *large relative to* π_0 *in* $\underline{\Gamma}(\mathcal{E}/\mathcal{C})$ *are computed argument by argument.*

ⓑ *Let* $\pi \geqslant \sup(\pi_0, \operatorname{card}\operatorname{red}\mathcal{C})$ *be a cardinal. For an object* f *of* $\mathcal{M}$ *to belong to* $\mathcal{M}_\pi$*, it* suffices *that* $\forall\ s \in \operatorname{Ob}\mathcal{C}$*,* $f(s) \overset{\text{def}}{=} \varepsilon_s(x)$ *be in* $\mathcal{E}_{s\,\pi}$*. (Compare 4.5.8, page 119.)*

ⓒ *Suppose* π_0 *chosen so as* [page 392]
to satisfy the following additional conditions: π_0 *is useful for* $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$*, and for each of the fiber categories* $\mathcal{E}_s$ *(*$s \in \operatorname{Ob}\mathcal{C}$*) (such a* π_0 *exists). Let*

$$\pi_1 \ = \ \sup(\pi_0, \operatorname{card}\operatorname{red}\mathcal{C}, \operatorname{card}\operatorname{red}\mathcal{M}_{\pi_0})\ .$$

Then for every cardinal $\pi \geqslant \pi_1$ *and such that* $\pi^{\pi_0} = \pi$*, we have*

$$\mathcal{M}_\pi \ = \ \left\{ \begin{array}{l} \text{strictly full subcategory of } \mathcal{M} \text{ consisting of the } f \in \\ \operatorname{Ob}\mathcal{M} \text{ such that } f(s) \in \operatorname{Ob}\mathcal{E}_{s\,\pi}\ \forall\ s \in S\ . \end{array} \right.$$

If moreover π *is chosen so that we have*

$$\pi \ \geqslant \ \sup_{s \in \operatorname{Ob}\mathcal{C}} \operatorname{card}\operatorname{red}\mathcal{E}_{s\,\pi_0}\ ,$$

then the strictly full subcategory $\mathcal{E}(\pi)$ *of* $\mathcal{E}$ *such that* $\operatorname{Ob}\mathcal{E}(\pi) = \bigcup_{s \in \operatorname{Ob}\mathcal{C}} \operatorname{Ob}\mathcal{E}_{s\,\pi}$ *is a* $\mathcal{C}$*-cofibered (resp.* $\mathcal{C}$*-fibered) subcategory of* $\mathcal{E}$*, and we have*

$$\underline{\Gamma}(\mathcal{E}/\mathcal{C})_\pi \xleftarrow{\sim} \underline{\Gamma}(\mathcal{E}(\pi)/\mathcal{C}),$$

thus (π *being useful for* $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$*) we have an equivalence of categories*

$$\underline{\Gamma}(\mathcal{E}/\mathcal{C}) \xleftarrow{\sim} \operatorname{Ind}_\pi(\underline{\Gamma}(\mathcal{E}(\pi)/\mathcal{C}))\ .$$

All these assertions are easily proved in the order in which they are given, using 4.14.17 and the general results already proved. Of course, we can begin by replacing $\mathcal{C}$ by an equivalent reduced subcategory $\mathcal{C}_0$ and $\mathcal{E}$ by $\mathcal{E}_0 = \mathcal{E}|\mathcal{C}$, so as to reduce (in b) and c)) to the case where $\operatorname{card}\mathcal{C} \leqslant \pi$. p. 149 [page 393]

Corollary 4.14.19. *Under the conditions of 4.14.17, the subcategory* $\underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}/\mathcal{C})$ *of* $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$ *(case* p *cofibrant), resp. the subcategory* $\underline{\Gamma}_{\mathrm{cart}}(\mathcal{E}/\mathcal{C})$ *(case* p *fibrant) is accessible in* $\mathcal{M}$*, hence accessible and the inclusion functor accessible.*

(**N.B.** These two categories are strictly full in $\underline{\Gamma}(\mathcal{E}/\mathcal{C})$, so the assertion that they are accessible in it *is equivalent* to the last assertion.)

Proof of 4.14.19. We may assume $\mathcal{C}$ reduced, hence small. Let us consider the case where p is cofibrant. For every arrow $\lambda : s \longrightarrow t$ in $\mathcal{C}$, let us consider the functor

$$\varphi_\lambda \ : \ \mathcal{M} \longrightarrow \underline{\mathrm{Fl}}(\mathcal{E}_t)$$

defined by

$$\varphi_\lambda(f) \ = \ (\lambda_*(f(s)) \xrightarrow{f(\lambda)^\natural} f(t))\ . \qquad \text{[page 394]}$$

It is an accessible functor, since its composite with the source and target functors are the functors $\lambda_*\varepsilon_s$ and ε_t, hence accessible, ε_s, ε_t and λ_* being accessible. Let us consider the strictly full subcategory

$$\underline{\mathrm{Is}}(\mathcal{E}_t) \ \subset \ \underline{\mathrm{Fl}}(\mathcal{E}_t)\ ,$$

we know that it is accessible in $\underline{\mathrm{Fl}}(\mathcal{E}_t)$ (4.14.12). Thus its inverse image under φ_λ is accessible in $\mathcal{M}$, and consequently so is

$$\underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}/\mathcal{C}) \ \overset{\text{def}}{=} \ \bigcap_{\lambda \in \mathrm{Fl}\mathcal{C}} \varphi_\lambda^{-1}(\underline{\mathrm{Is}}(\mathcal{E}_t))$$

as the intersection of a *small* family of subcategories accessible in $\mathcal{M}$, q.e.d. The case where p is fibrant is treated in a similar way, by introducing the functors

$$\psi_\lambda \ : \ \mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C}) \longrightarrow \underline{\mathrm{Fl}}(\mathcal{E}_s)$$

given by

$$\psi_\lambda(f) \ = \ (f(s) \xrightarrow{f(\lambda)^\flat} \lambda^* f(t)) \ .$$

Corollary 4.14.20. *Let π_0 be an infinite cardinal chosen as in 4.14.18 a).* [page 395]

ⓐ *Then $\mathcal{M}_{\mathrm{c}} = \underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}/\mathcal{C})$ (resp. $\mathcal{M}_{\mathrm{c}} = \underline{\Gamma}_{\mathrm{cart}}$) is L_{π_0}-stable in $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$, or what amounts to the same thing (as it is strictly full), it satisfies L_{π_0}, and the inclusion functor satisfies L_{π_0}. The $\varinjlim$ large relative to π_0 in this category are computed argument by argument.*

ⓑ *For every cardinal $\pi \geqslant \pi_0$, every object of $\mathcal{M}_{\mathrm{c}}$ which is π-accessible in $\mathcal{M}$ is so (tautologically) in $\mathcal{M}_{\mathrm{c}}$. If $\pi \geqslant \sup(\pi_0, \operatorname{card}\operatorname{red}\mathcal{C})$, then (by 4.14.18 b) every object f of $\mathcal{M}_{\mathrm{c}}$ such that $f(s) \in \operatorname{Ob}\mathcal{E}_{s\,\pi}\ \forall\ s \in S$ is in $\mathcal{M}_\pi$ and in $\mathcal{M}_{\mathrm{c}\,\pi}$.*

ⓒ *Suppose π_0 chosen as in 4.14.18 c, and in addition that $\mathcal{M}_{\mathrm{c}}$ is π_0-accessible in $\mathcal{M}$.* p. 150
(Such a π_0 exists.) We thus have, for $\pi \geqslant \pi_1 = \sup(\pi_0, \operatorname{card}\operatorname{red}\mathcal{C}, \operatorname{card}\operatorname{red}\mathcal{M}_{\pi_0})$ such that $\pi^{\pi_0} = \pi$,

$$(\mathcal{M}_{\mathrm{c}})_\pi \ = \ \mathcal{M}_\pi \cap \mathcal{M}_{\mathrm{c}} \ = \quad \begin{array}{l} \textit{Strictly full subcategory of } \mathcal{M}_{\mathrm{c}} \textit{ consisting of the } f \\ \textit{such that } f(s) \in \operatorname{Ob}\mathcal{E}_{s\,\pi}\ \forall\ s \in \operatorname{Ob}\mathcal{C}. \end{array}$$

If in addition π is chosen so that we have $\pi \geqslant \sup_{s\in\operatorname{Ob}\mathcal{C}} \operatorname{card}\operatorname{red}\mathcal{E}_{s\,\pi_0}$, then

[page 396]

$$\left\{ \begin{array}{lrcl} & \mathcal{M}_{\mathrm{c}\,\pi} & \xleftarrow{\sim} & \underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}(\pi)/\mathcal{C}) \\ \textit{resp.} & \mathcal{M}_{\mathrm{c}\,\pi} & \xleftarrow{\sim} & \underline{\Gamma}_{\mathrm{cart}}(\mathcal{E}(\pi)/\mathcal{C}) \ , \end{array} \right.$$

hence (π being useful for $\mathcal{M}_{\mathrm{c}}$) an equivalence of categories

$$\begin{array}{rrcl} & \underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}/\mathcal{C}) & \xleftarrow{\sim} & \mathrm{Ind}_\pi(\underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}(\pi)/\mathcal{C})) \\ \textit{resp.} & \underline{\Gamma}_{\mathrm{cart}}(\mathcal{E}/\mathcal{C}) & \xleftarrow{\sim} & \mathrm{Ind}_\pi(\underline{\Gamma}_{\mathrm{cart}}(\mathcal{E}(\pi)/\mathcal{C})). \end{array}$$

Proof. Everything is clear in the order in which it is given using the two preceding corollaries.

Corollary 4.14.21. *Let us consider a diagram of functors*

$$\begin{array}{ccccc} \mathcal{E}_1 & & & & \mathcal{E}_2 \\ & \searrow^{p_1} & & \swarrow_{p_2} & \\ & & \mathcal{E}_0 \ , & & \end{array} \qquad (*)$$

the categories and the functors being accessible.

a) The 2-fiber product

$$\mathcal{M}_0 \ = \ \mathcal{E}_1 \overset{2}{\times}_{\mathcal{E}_0} \mathcal{E}_2$$

is accessible, and the canonical functors $\mathcal{M} \longrightarrow \mathcal{E}_1$, $\mathcal{M} \longrightarrow \mathcal{E}_2$ are accessible. [page 397]

b) ([169]). The category $\mathcal{M}$ of triples (x_1, x_2, u) with $x_1 \in \operatorname{Ob}\mathcal{E}_1$, $x_2 \in \operatorname{Ob}\mathcal{E}_2$, and

$$u \ : \ p_1(x_1) \ \longrightarrow \ p_2(x_2) \ ,$$

is accessible, and the canonical functors $\mathcal{M} \longrightarrow \mathcal{E}_1$, $\mathcal{M} \longrightarrow \mathcal{E}_2$ are accessible.

[169] This is contained in 4.14.22.1 above (p. 406), applied to $\Pi = \mathcal{E}_1 \times \mathcal{E}_2$, $\Pi' = \mathcal{E}_0$, and $F, G : \Pi \rightrightarrows \Pi'$ given by $F = p_1 \circ \mathrm{pr}_1$ and $G = p_2 \circ \mathrm{pr}_2$. On the other hand, a) and c) follow easily from b).

c) The strictly full subcategory $\mathcal{M}_0$ of $\mathcal{M}$ is accessible in $\mathcal{M}$.

PROOF. Let $\mathcal{C}$ be the category

$$\begin{array}{ccc} 1 & & 2 \\ & \searrow \quad \swarrow & \\ & 0\ , & \end{array}$$

and $\mathcal{E}$ the cofibered category over $\mathcal{C}$ defined by the diagram $(*)$. We are under the conditions of 4.14.17. The category $\Gamma = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$ is interpreted as the category of quintuples p. 151
$(x_0, x_1, x_2, u_1, u_2)$, with $x_i \in \mathrm{Ob}\,\mathcal{E}_i$ for $i \in \{0,1,2\}$, and $u_1, u_2 \in \mathrm{Fl}\,\mathcal{E}_0$, [page 398]

$$\begin{array}{ccccc} p_1(x_1) & & & & p_2(x_2) \\ & \searrow^{u_1} & & \swarrow_{u_2} & \\ & & x_0\ . & & \end{array} \qquad (**)$$

By 4.14.18, this category $\underline{\Gamma}$ is accessible. The category $\mathcal{M}$ is equivalent to the strictly full subcategory Γ_1 of Γ, consisting of the quintuples for which we have $u_2 \in \underline{\mathrm{Is}}\mathcal{E}_0$. We have indeed functors quasi-inverse to each other

$$\mathcal{M} \underset{\beta}{\overset{\alpha}{\rightleftarrows}} \Gamma_1$$

given by

$$\begin{aligned} \alpha(x_1, x_2, u) &= (p_2(x_2), x_1, x_2, u, \mathrm{id}_{p_2(x_2)}) \\ \beta(x_0, x_1, x_2, u_1, u_2) &= (x_1, x_2, u_2^{-1}u_1)\ , \end{aligned}$$

we then have

$$\begin{aligned} \beta\alpha &= \mathrm{id}_{\mathcal{M}} \\ \alpha\beta &\xrightarrow{\sim} \mathrm{id}_{\Gamma_1}\ . \end{aligned}$$

On the other hand, Γ_1 is obviously accessible in Γ, as the inverse image of $\underline{\mathrm{Is}}(\mathcal{E}_0)$ (which is accessible in $\underline{\mathrm{Fl}}(\mathcal{E}_0)$ under the accessible functor [page 399]

$$\underline{\Gamma} \longrightarrow \underline{\mathrm{Fl}}(\mathcal{E}_0)$$

given by $(x_0, x_1, x_2, u_1, u_2) \longmapsto u_2$. This proves that $\mathcal{M}$, which is equivalent to Γ_1, is also accessible, and the accessibility of the canonical functor $\mathcal{M}_0 \longrightarrow \mathcal{E}_i$ $(i \in \{1,2\})$ follows from that of $\Gamma \longrightarrow \mathcal{E}_i$, implying that of the composite $\Gamma_1 \hookrightarrow \Gamma \longrightarrow \mathcal{E}_i$. This therefore proves b). To prove a), it suffices to see that $\mathcal{M}_0$ is accessible in $\mathcal{M}$, now it is again the inverse image of $\underline{\mathrm{Is}}(\mathcal{E}_0) \subset \underline{\mathrm{Fl}}(\mathcal{E}_0)$ under the canonical accessible functor

$$\mathcal{M} \longrightarrow \underline{\mathrm{Fl}}(\mathcal{E}_0)\ ,$$

so $\mathcal{M}_0$ is indeed accessible in $\mathcal{M}$, as the inverse image of a subcategory accessible in $\underline{\mathrm{Fl}}(\mathcal{E}_0)$. We can moreover also interpret $\mathcal{M}_0$, up to equivalence of categories, as [page 400]

$$\mathcal{M}_0 \simeq \underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}/\mathcal{C})\ ,$$

the canonical functors $\mathcal{M}_0 \longrightarrow \mathcal{E}_i$ $(i \in \{1,2\})$ being interpreted as evaluation functors, which gives a) directly, as a special case of 4.14.19.

4.14.22. PROOF OF 4.14.17. The case where p is cofibrant has already been proved p. 152
(th. 4.10.8, p. 263). We are going to see that the case where p is fibrant reduces to the preceding theorem (in a very special case), using to the full the general results on accessible parts of an accessible category. Let us note that assertions a) and b) of [page 401]

corollary 4.14.18 are proved in AQT fashion without having to appeal to 4.14.17. Part a) already implies that the evaluation functors $\varepsilon_s : \mathcal{M} \longrightarrow \mathcal{E}_s$ are accessible. It therefore remains to prove that $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$ is accessible, in the case where p is *fibrant*. We shall moreover assume that $\mathcal{C}$ is small (replacing $\mathcal{C}$ by an equivalent small full subcategory if need be).

Let

$$\begin{aligned} \Pi &= \prod_{s \in \mathrm{Ob}\,\mathcal{C}} \mathcal{E}_s \\ \Pi' &= \prod_{\lambda \in \mathrm{Fl}\,\mathcal{C}} \mathcal{E}_{s(\lambda)} \ , \end{aligned}$$

where $s(\lambda)$ denotes the *source* s of the arrow $\lambda : s \longrightarrow t$ in $\mathcal{C}$. Π, Π' are accessible categories, as (small) products of accessible categories. (Trivial special case of 4.10.8, for a discrete index category.) We are going to define two functors

$$\Pi \underset{G}{\overset{F}{\rightrightarrows}} \Pi'$$

by defining for every $\lambda : s_0 \longrightarrow t_0$ in $\mathrm{Fl}\,\mathcal{C}$ two functors [page 402]

$$\left\{ \begin{array}{lcll} \Pi & \underset{G_\lambda}{\overset{F_\lambda}{\rightrightarrows}} & \mathcal{E}_{s(\lambda)=s_0} & \\ F_\lambda & = & \mathrm{pr}_{s_0} & \text{projection of } \Pi = \textstyle\prod_s \mathcal{E}_s \text{ onto the factor } \mathcal{E}_{s_0} \\ G_\lambda & = & \lambda^* \mathrm{pr}_{t_0} & \text{composite } \Pi \xrightarrow{\mathrm{pr}_{t_0}} \mathcal{E}_{t_0} \xrightarrow{\lambda^*} \mathcal{E}_{s_0} \end{array} \right. \qquad ({}^{170}, {}^{171}) \ .$$

Let $f \in \mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$, then for every $\lambda \in \mathrm{Fl}\,\mathcal{C}$, f defines an arrow

$$\alpha_\lambda(f) \ : \quad \underbrace{f(s)}_{=F_\lambda(\varepsilon(f))} \quad \xrightarrow{f(\lambda)^\flat} \quad \underbrace{\lambda^*(f(t))}_{=G_\lambda(\varepsilon(f))} \ ,$$

functorial in f. We have a functor

$$\varepsilon \ = \ (\varepsilon_s)_{s \in \mathrm{Ob}\,\mathcal{C}} \ : \ \mathcal{M} \ \longrightarrow \ \Pi \ ,$$

and for every $\lambda \in \mathrm{Fl}\,\mathcal{C}$ a functorial homomorphism

$$\alpha_\lambda \ : \ F_\lambda \circ \varepsilon \ \longrightarrow \ G_\lambda \circ \varepsilon \ ,$$

whence a functorial homomorphism

$$\alpha \ = \ (\alpha_\lambda)_{\lambda \in \mathrm{Fl}\,\mathcal{C}} \ : \ F \circ \varepsilon \ \longrightarrow \ G \circ \varepsilon \ .$$

Let then $\mathcal{M}'$ be the following category: p. 153 [page 403]

$$\begin{aligned} \mathcal{M}' \quad = \quad & \text{category of pairs } (X, u), \text{ with } X \text{ in } \mathrm{Ob}\,\Pi, \\ & \text{and } u : F(X) \longrightarrow G(X) \text{ an arrow in } \Pi'. \end{aligned}$$

The functorial homomorphism $\alpha : F \circ \varepsilon \longrightarrow G \circ \varepsilon$ defines a functor

$$\mathcal{M} \ \longrightarrow \ \mathcal{M}' \ ,$$

[170] It is immediate that the F_λ, G_λ, hence F and G, are *accessible* functors.

[171] **N.B.** We assume chosen for every $\lambda \in \mathrm{Fl}\,\mathcal{C}$ an inverse image functor $\lambda^* : \mathcal{E}_t \longrightarrow \mathcal{E}_s$. We assume that if λ is an identity arrow, id_s, choose $\lambda^* = \mathrm{id}_{\mathcal{E}_s}$.

and it is immediate that this functor is fully faithful. In fact, the category $\mathcal{M}'$ is isomorphic to the category consisting of the pairs (f_0, f_1) of two maps

$$\begin{array}{lccccc} f_0: & \mathrm{Ob}\,\mathcal{C} & \longrightarrow & \mathrm{Ob}\,\mathcal{E} & s & \longmapsto & f_0(s) \\ f_1: & \mathrm{Fl}\,\mathcal{C} & \longrightarrow & \mathrm{Fl}\,\mathcal{E} & \lambda & \longmapsto & f_1(\lambda) \end{array}$$

in such a way that f_1 and f_0 are compatible with the source and target maps, i.e. that they define a homomorphism of *diagrams* $f : \mathrm{diag}\,\mathcal{C} \longrightarrow \mathrm{diag}\,\mathcal{E}$, and that this homomorphism is a right inverse of $\mathrm{diag}(p) : \mathrm{diag}\,\mathcal{E} \longrightarrow \mathrm{diag}\,\mathcal{C}$.

The preceding functor $\mathcal{M} \hookrightarrow \mathcal{M}'$ is interpreted as the functor which associates to [page 404] every section f of $\mathcal{E}$ over $\mathcal{C}$, the corresponding morphism of diagrams. It is therefore the inclusion of a *strictly full* subcategory of $\mathcal{M}'$. Thus to prove that $\mathcal{M}$ is accessible, it suffices to prove:

a) $\mathcal{M}'$ is accessible.

b) $\mathcal{M}$ is accessible in $\mathcal{M}'$.

Let us first prove b) (admitting a). The subcategory $\mathcal{M}$ of $\mathcal{M}'$ is defined by two types of conditions, on an object $f = (f_0, f_1)$ of $\mathcal{M}'$:

1°) $\forall\ s \in \mathrm{Ob}\,\mathcal{C}$, $f_1(\mathrm{id}_s) = \mathrm{id}_{f_0(s)}$.

2°) If $s \xrightarrow{\lambda} t \xrightarrow{\mu} u$ are two composable arrows in $\mathcal{C}$, then

$$f_1(\mu\lambda) \;=\; f_1(\mu) f_1(\lambda)\ .$$

These conditions are made explicit the one and the equality of a double arrow in $\mathcal{E}_s$, namely [page 405]

$$f_0(s) \underset{\mathrm{id}_{f_0(s)}}{\overset{f_1(\mathrm{id}_s)}{\rightrightarrows}} f_0(s)$$

in case 1°), and

$$f_0(s) \underset{(f_1(\mu)f_1(\lambda))^\flat}{\overset{f_1(\mu\lambda)^\flat}{\rightrightarrows}} (\mu\lambda)^*(f_0(u))$$

in case 2°). In either case, we have a functor

$$\mathcal{M}' \xrightarrow{\gamma_s \text{ resp. } \gamma_{\lambda,\mu}} \underline{\mathrm{Hom}}(I, \mathcal{E}_s)$$

in the category of double arrows of $\mathcal{E}_s$ (I denoting the double-arrow category of type p. 154 $0 \rightrightarrows 1$), a functor which is (trivially) accessible. Let then $I \xrightarrow{i} \mathbf{\Delta}_1$ be the canonical functor, and

$$\underbrace{\underline{\mathrm{Hom}}(\mathbf{\Delta}_1, \mathcal{E}_s)}_{\simeq \underline{\mathrm{Fl}}(\mathcal{E}_s)} \hookrightarrow \underline{\mathrm{Hom}}(I, \mathcal{E}_s)$$

the functor $u \longmapsto u \circ i$, which is a strictly fully faithful functor, having as its image the subcategory of double arrows (u, v) such that $u = v$. As the inclusion functor is accessible and $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1, \mathcal{E}_s)$ and $\underline{\mathrm{Hom}}(I, \mathcal{E}_s)$ accessible (a special case of 4.14.1, or also of 4.10.8!), we see that the strictly full subcategory of $\underline{\mathrm{Hom}}(I, \mathcal{E}_s)$ of double arrows equalized in $\mathcal{E}_s$, [page 406] is accessible in $\underline{\mathrm{Hom}}(I, \mathcal{E}_s)$. So, its inverse image in $\mathcal{M}'$ is accessible. So the intersection of these inverse images for $s \in \mathrm{Ob}\,\mathcal{C}$ and $(\lambda, \mu) \in \mathrm{Fl}_2(\mathcal{C})$ varying, is so as well ($\mathcal{C}$ being small).

So everything comes down to proving that $\mathcal{M}'$ is accessible, which is contained in the

Lemma 4.14.22.1. *Let* Π, Π' *be two accessible categories,* $F, G : \Pi \rightrightarrows \Pi'$ *two accessible functors, and* $\mathcal{M}'$ *the category of pairs* (X, u)*, with* $X \in \mathrm{Ob}\,\Pi$ *and* $u : F(X) \longrightarrow G(X)$ *an arrow of* Π'*. Then* $\mathcal{M}'$ *is accessible, and the canonical functor* $\mathcal{M}' \longrightarrow \Pi$ *is accessible.*

Proof of the Lemma. The datum of Π, Π', F, G defines a cofibered category $\mathcal{F}$ over the category $I = (0 \rightrightarrows 1)$ above. The category $\underline{\Gamma}(\mathcal{F}/I)$ is interpreted, up to isomorphism, as the category of triples (x, u, v) with $x \in \mathrm{Ob}\,\Pi$, $x' \in \mathrm{Ob}\,\Pi'$, and u and v arrows [page 407]

$$\begin{array}{ccc} F(x) & & G(x) \\ & {\scriptstyle u}\searrow \quad \swarrow{\scriptstyle v} & \\ & x' & \end{array}$$

in Π'. The category $\mathcal{M}'$ is equivalent to the strictly full subcategory of $\underline{\Gamma}(\mathcal{F}/I)$, consisting of triples (x, u, v) such that v is an isomorphism, i.e. to the subcategory of $\underline{\Gamma}(\mathcal{F}/I)$, the inverse image of $\underline{\mathrm{Is}}(\Pi')$ under the functor $(x, u, v) \longmapsto v$,

$$\underline{\Gamma}(\mathcal{F}/I) \longrightarrow \underline{\mathrm{Fl}}(\Pi') .$$

Now by 4.10.8, $\underline{\Gamma}(\mathcal{F}/I)$ is accessible, and the preceding functor is of course accessible. Finally, by 4.14.12, $\underline{\mathrm{Is}}(\Pi')$ is accessible in the accessible category $\underline{\mathrm{Fl}}(\Pi')$. So its inverse image $\mathcal{M}'$ in $\underline{\Gamma}(\mathcal{F}/I)$ is accessible in $\underline{\Gamma}(\mathcal{F}/I)$, hence accessible. So the same holds for the equivalent category, q.e.d. [page 408]

Thus, thanks to the general results on accessible parts of accessible categories, theorem 4.14.17, in the case where p is fibrant, is reduced to the cofibrant case, in the very special case where the index category is $I = (0 \rightrightarrows 1)$. One could moreover make a similar reduction in the case where p is cofibrant, by a proof practically identical. In other words, theorem 4.10.8 reduces to the case where the index category is $I = (0 \rightrightarrows 1)$. But I suspect that the proof of the theorem in this special case, is not essentially simpler than in the general case. p. 155

4.14.23. One may ask about finding a statement ensuring the accessibility of a category $\underline{\Gamma}(\mathcal{E}/\mathcal{C})$, which covers the two cases considered in 4.14.17. Here is what I can propose. I am going to make the following assumption on the functors $p : \mathcal{E} \longrightarrow \mathcal{C}$ (in addition to that of $\mathcal{C}$ being essentially small): [page 409]

H) For every $\lambda \in \mathrm{Fl}\mathcal{C}$, the category $\mathcal{E}_\lambda = \mathcal{E} \times_{\mathcal{C}} (\mathbf{\Delta}_1, i_\lambda)$, where $i_\lambda : \mathbf{\Delta}_1 \longrightarrow \mathcal{C}$ is the functor defined by λ, is accessible, and the canonical functor $\mathcal{E}_\lambda \xrightarrow{p_\lambda} \mathbf{\Delta}_1$ is accessible.

Note that if $\lambda : s \longrightarrow t$ in $\mathcal{C}$ is such that $\mathcal{E}_\lambda$ and p_λ are accessible, then $\mathcal{E}_s$ and $\mathcal{E}_t$ are accessible, since isomorphic to the inverse image categories under $p_\lambda : \mathcal{E}_\lambda \longrightarrow \mathbf{\Delta}_1$ of the subcategories $\{0\}$ resp. $\{1\}$ of Δ_1, which are accessible in the accessible category $\mathbf{\Delta}_1$, so $\mathcal{E}_{\lambda_0}$ and $\mathcal{E}_{\lambda_1}$ are accessible in $\mathcal{E}_\lambda$, hence are accessible, hence also $\mathcal{E}_s$ and $\mathcal{E}_t$. Thus, assumption H) implies that the fiber categories are accessible. Note moreover: [page 410]

Lemma 4.14.23.1. *Suppose that* $p : \mathcal{E} \longrightarrow \mathcal{C}$ *is cofibrant or fibrant, and let* $(\lambda : s \longrightarrow t) \in \mathrm{Fl}(\mathcal{C})$*. For* $\mathcal{E}_\lambda$ *to be accessible and* $\mathcal{E}_\lambda \xrightarrow{p_\lambda} \mathbf{\Delta}_1$ *accessible, it is necessary and sufficient that* $\mathcal{E}_s$*,* $\mathcal{E}_t$ *be accessible and* $\lambda_* : \mathcal{E}_s \longrightarrow \mathcal{E}_t$ *resp.* $\lambda^* : \mathcal{E}_t \longrightarrow \mathcal{E}_s$ *be an accessible functor.*

Proof. We may suppose that $\mathcal{C} = \mathbf{\Delta}_1$, and that $\mathcal{E}$ is the cofibered (resp. fibered) category over $\mathbf{\Delta}_1$ defined by a functor $\varphi : \mathcal{E}_0 \longrightarrow \mathcal{E}_1$ (resp. $\psi : \mathcal{E}_1 \longrightarrow \mathcal{E}_0$), and to show that $\mathcal{E}$ and $p : \mathcal{E} \longrightarrow \mathbf{\Delta}_1$ are accessible if and only if $\mathcal{E}_0$ and $\mathcal{E}_1$ and φ (resp. ψ) are.

Let us prove the "only if". We have already seen that if $\mathcal{E}$, p are accessible, $\mathcal{E}_0$ and $\mathcal{E}_1$ are accessible in $\mathcal{E}$, hence accessible, and the inclusion functors [page 411]

$$\mathcal{E}_0 \hookrightarrow \mathcal{E} \hookleftarrow \mathcal{E}_1$$

accessible. To say that p is cofibrant (resp. fibrant) means that the inclusion functor $\mathcal{E}_1 \hookrightarrow \mathcal{E}$ admits a left adjoint $\bar{\varphi}$ (resp. that the functor $\mathcal{E}_0 \hookrightarrow \mathcal{E}$ admits a right adjoint $\tilde{\psi}$). Then φ (resp. ψ) is the composite $\mathcal{E}_0 \longrightarrow \mathcal{E} \xrightarrow{\bar{\varphi}} \mathcal{E}_1$ (resp. the composite $\mathcal{E}_1 \hookrightarrow \mathcal{E} \xrightarrow{\tilde{\psi}} \mathcal{E}_0$). So to prove that φ (resp. ψ) is accessible, it suffices to prove that $\bar{\varphi}$ (resp. $\tilde{\psi}$) is. This is a special case of the

Lemma 4.14.23.2. *Let $\mathcal{E}$, $\mathcal{E}'$ be two accessible categories ([172]), and $\mathcal{E} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{E}'$ a pair of adjoint functors between $\mathcal{E}$ and $\mathcal{E}'$. Then F and G are accessible.*

[page 412]

Indeed, F is accessible, since it commutes with arbitrary small $\varinjlim$, a fortiori it satisfies L_{π_0} (where π_0 is such that $\mathcal{E}$, $\mathcal{E}'$ satisfy L_{π_0}). Let us prove that G is accessible. If S is a small generating set of elements in $\mathcal{E}$, so that the small family of functors

p. 156

$$h_\xi \;=\; (x \longmapsto \mathrm{Hom}_{\mathcal{E}}(\xi, x)) \;:\; \mathcal{E} \;\longrightarrow\; \mathrm{Ens}$$

is conservative, to prove that G is accessible, it suffices to prove that the $h_\xi \circ G$ ($\xi \in S \subset \mathrm{Ob}\,\mathcal{E}$) are. Now

$$h_\xi \circ G(x') \;=\; \mathrm{Hom}(\xi, G(x')) \;\simeq\; \mathrm{Hom}_{\mathcal{E}'}(F(\xi), x')\ ,$$

and since $F(\xi)$ is an accessible object of $\mathcal{E}'$, the functor

$$x' \;\longmapsto\; (h_\xi \circ G)(x') \;\simeq\; \mathrm{Hom}_{\mathcal{E}'}(F(\xi), x')$$

is indeed accessible, q.e.d.

This proves the "only if" in 4.14.23.1. Let us prove the "if", i.e. suppose $\mathcal{E}_s$, $\mathcal{E}_t$ accessible, and φ (resp. ψ) accessible.

[page 413]

In both cases, $\mathcal{E}_0$ is an open of $\mathcal{E}$, $\mathcal{E}_1$ is the complementary closed, and $\mathcal{E}$ is the category "glued" with the help of the bifunctor

$$H \;:\; \mathcal{E}_0^\circ \times \mathcal{E}_1 \;\longrightarrow\; \mathrm{Ens}$$

given by

$$\begin{cases} H(x_0, x_1) \;=\; \mathrm{Hom}(\varphi(x_0), x_1) \\ \qquad \text{resp.} \\ H(x_0, x_1) \;=\; \mathrm{Hom}(x_0, \psi(x_1))\ , \end{cases}$$

according as we are in the case p cofibrant or p fibrant. We are going to use the

Proposition 4.14.24 ([173]). *Let $\mathcal{E}$ be a category, $\mathcal{E}_0$ an open subcategory, $\mathcal{E}_1$ the complementary closed subcategory. Equivalent conditions:*

a) $\mathcal{E}$ is accessible, and $\mathcal{E} \longrightarrow \mathbf{\Delta}_1$ (with fibers $\mathcal{E}_0$, $\mathcal{E}_1$) is accessible.

b) $\mathcal{E}$ is accessible, and $\mathcal{E}_0$, $\mathcal{E}_1$ are accessible in $\mathcal{E}$.

c) $\mathcal{E}_0$, $\mathcal{E}_1$ are accessible, and for every $x_0 \in \mathrm{Ob}\,\mathcal{E}_0$, the functor

$$x_1 \;\longmapsto\; \underline{\mathrm{Hom}}_{\mathcal{E}}(x_0, x_1) \;:\; \mathcal{E}_1 \;\longrightarrow\; \mathrm{Ens}$$

is accessible, and there exists π such that for every $x_1 \in \mathrm{Ob}\,\mathcal{E}_1$, the functor

[page 414]

$$x_0 \;\longmapsto\; \underline{\mathrm{Hom}}_{\mathcal{E}}(x_0, x_1)^\circ \;:\; \mathcal{E}_0 \;\longrightarrow\; (\mathrm{Ens})^\circ$$

is π-accessible.

p. 157

Let us admit this lemma , let us finish the proof of 4.14.23.1. We suppose $\mathcal{E}_0$, $\mathcal{E}_1$ and φ (resp. ψ) accessible, we want to prove condition a) of 4.14.24. For this, it suffices to prove condition c). Now we have, in the case p cofibrant,

$$\mathrm{Hom}_{\mathcal{E}}(x_0, x_1) \;\simeq\; \mathrm{Hom}_{\mathcal{E}_1}(\varphi(x_0), x_1) \;,$$

this is indeed accessible in x_1, since $\mathcal{E}_1$ is accessible, and π-accessible in x_0 if φ is, as a composite of φ with $\mathcal{E}_1 \longrightarrow \mathrm{Ens}^\circ$, $\mathrm{Hom}(-, x_1)$, which is. In the case p fibrant, we have

$$\mathrm{Hom}_{\mathcal{E}}(x_0, x_1) \;\simeq\; \mathrm{Hom}_{\mathcal{E}_0}(x_0, \psi(x_1)) \;,$$

this is accessible in x_1, as the composite of the accessible functor $\psi : \mathcal{E}_1 \longrightarrow \mathcal{E}_0$, and the likewise accessible functor $\mathrm{Hom}(x_0, -) : \mathcal{E}_0 \longrightarrow \mathrm{Ens}$, which is π-accessible in x_0 tautologically.

To finish the proof of 4.14.23.1, it therefore remains to prove proposition 4.14.24. We have already noted that we have a $\Longrightarrow$ b, and b $\Longrightarrow$ c clear, since the functor considered $\mathcal{E}_1 \longrightarrow \mathrm{Ens}$ is the composite $\mathcal{E}_1 \overset{\text{incl.}}{\hookrightarrow} \mathcal{E} \xrightarrow{\mathrm{Hom}_{\mathcal{E}}(x_0, -)} \mathrm{Ens}$, is accessible as a composite of two accessible functors. So it remains to prove c $\Longrightarrow$ a. Let then π_0 be a useful cardinal [page 415] for $\mathcal{E}_0$, and such that the functors

$$x_0 \;\longmapsto\; H(x_0, x_1) \;:\; \mathcal{E}_0 \longrightarrow \mathrm{Ens}^\circ$$

are π_0-accessible, let π be a cardinal $\geqslant \sup(\pi_0, \mathrm{card}\,\mathrm{red}\,\mathcal{E}_{0\,\pi_0})$ such that $\pi^{\pi_0} = \pi$ (hence useful for $\mathcal{E}_0$), useful for $\mathcal{E}_1$, and such that the functors

$$x_1 \;\longmapsto\; H(x_0, x_1) \;:\; \mathcal{E}_1 \longrightarrow \mathrm{Ens}$$

are π-accessible, for $x_0 \in \mathrm{Ob}\,\mathcal{E}_{0\,\pi_0}$. Let $\mathcal{E}(\pi)$ be the strictly full subcategory of $\mathcal{E}$, such that

$$\mathrm{Ob}\,\mathcal{E}(\pi) \;=\; \mathrm{Ob}\,\mathcal{E}_{0\,\pi} \cup \mathrm{Ob}\,\mathcal{E}_{1\,\pi} \;.$$

I say that

1°) $\mathcal{E}$ satisfies L_π, and $\mathcal{E}_0$ and $\mathcal{E}_1$ are stable in $\mathcal{E}$ under L_π, the functors $\mathcal{E}_0 \longrightarrow \mathcal{E}$, $\mathcal{E}_1 \longrightarrow \mathcal{E}$, $\mathcal{E} \longrightarrow \mathbf{\Delta}_1$ satisfy L_π ([174]).

2°) $\mathcal{E}(\pi) = \mathcal{E}_\pi$ (whence a fully faithful functor

$$\mathrm{Ind}_\pi(\mathcal{E}(\pi)) \;\longrightarrow\; \mathcal{E} \;).$$

3°) The said functor is essentially surjective, hence an equivalence of categories. [page 416]

Note that 2°) and 3°) imply that $\mathcal{E}$ is accessible, and 1°) that $\mathcal{E} \longrightarrow \mathbf{\Delta}_1$ is accessible, whence condition a) of the lemma we are to establish. So everything comes down to proving 1°, 2°, 3° above.

1°) Since $\mathcal{E}_1$ is closed in $\mathcal{E}$, it is clear that $\mathcal{E}_1$ is stable under all $\varinjlim$ representable in $\mathcal{E}$, and more precisely, that for a functor $I \longrightarrow \mathcal{E}_1$, its direct limit in $\mathcal{E}_1$ exists if and p. 158

[172] **N.B.** It suffices that $\mathcal{E}$ admit a small generating subcategory, and that the objects of $\mathcal{E}'$ be accessible.

[173] cf. more general statements and nice proof, 4.15.1 and 4.15.4 (p. 425, 432).

[174] it suffices that $\mathcal{E}_0$ and $\mathcal{E}_1$ satisfy L_π, and that

$$x_0 \;\longmapsto\; H(x_0, x_1) \;:\; \mathcal{E}_0 \longrightarrow \mathrm{Ens}^\circ$$

satisfy L_π $\forall\, x \in \mathrm{Ob}\,\mathcal{E}$.

only if it exists in $\mathcal{E}$, and the two are isomorphic. Suppose I large relative to π, and let $\varphi : I \longrightarrow \mathcal{E}$, $i \longmapsto x_i$. Let us distinguish two cases.

a) φ takes its values in $\mathcal{E}_0$. Let x_0 be its direct limit *in* $\mathcal{E}_0$. I say that this is also a direct limit in $\mathcal{E}$. We must prove that for $x_1 \in \mathrm{Ob}\,\mathcal{E}_1$, the arrow

$$\mathrm{Hom}_{\mathcal{E}}(x_0, x_1) \longrightarrow \varprojlim_{i\in I} \mathrm{Hom}_{\mathcal{E}}(x_i, x_1)$$

is isomorphic. Now this is the case by the assumption on π, [page 417]

$$\xi \longmapsto H(\xi, x_1) \ : \ \mathcal{E}_0 \longrightarrow (\mathrm{Ens})^\circ$$

satisfies L_π for every x_1 in $\mathrm{Ob}\,\mathcal{E}_1$.

b) φ does not factor through $\mathcal{E}_0$, i.e. $\exists\ i_0 \in \mathrm{Ob}\, I$ with $\varphi(i_0) \in \mathrm{Ob}\,\mathcal{E}_1$. Then we have $\varphi(i) \in \mathrm{Ob}\,\mathcal{E}_1$ for every $i \in \mathrm{Ob}\, I[i_0]$, where $I[i_0]$ denotes the strictly full subcategory of I consisting of the i such that $i \geqslant i_0$. This category is cofinal in I, hence large relative to π like I, and for $\varinjlim_{I} x_i$ to exist, it is necessary and sufficient that $\varinjlim_{I[i_0]} x_i$ exist, which is the case by the cases a) already treated.

This therefore proves at the same time that $\mathcal{E}$ satisfies L_π, and that $\mathcal{E}_0$ and $\mathcal{E}_1$ are L_π-stable. The proof shows at the same time that the functor $p : \mathcal{E} \longrightarrow \mathbf{\Delta}_1$ satisfies L_π.

2°) From this it follows at once that $\mathcal{E}_\pi \subset \mathcal{E}(\pi)$. Let us prove that we also have

$$\mathcal{E}(\pi) \ \subset \ \mathcal{E}_\pi \ ,$$

and let us again distinguish two cases, for $x \in \mathrm{Ob}\,\mathcal{E}(\pi)$, according as x is in $\mathcal{E}_{0\,\pi}$ or $\mathcal{E}_{1\,\pi}$. [page 418]
If it is in $\mathcal{E}_{1\,\pi}$, to prove that

$$\mathrm{Hom}_{\mathcal{E}}(x, \underbrace{\varinjlim_{i} y_i}_{y}) \xleftarrow{\sim} \varprojlim_{i} \mathrm{Hom}_{\mathcal{E}}(x, y_i) \qquad (*)$$

(for a direct system $(y_i)_{i\in I}$ in $\mathcal{E}$ large relative to π), if the y_i are in $\mathcal{E}_0$, hence $y \in \mathcal{E}_0$, then both sides are empty, and this is OK. Otherwise, we may suppose (replacing I by an $I[i_0]$ if need be) that the y_i are in $\mathcal{E}_1$, and then this is OK too, since y is also the $\varinjlim$ in $\mathcal{E}_1$, by 1°).

If x is in $\mathcal{E}_{0\,\pi}$, then $(*)$ is clear if the y_i are in $\mathcal{E}_0$. Otherwise, we may again suppose that the y_i are in $\mathcal{E}_1$, and $(*)$ reads

$$H(x, \varinjlim y_i) \xleftarrow{\sim} \varprojlim H(x, y_i) \ ,$$

for $x \in \mathrm{Ob}\,\mathcal{E}_{0\,\pi}$, the y_i in $\mathcal{E}_1$, I large relative to π. In other words, we must prove that the functor

$$x_1 \longmapsto H(x_0, x_1) \ : \ \mathcal{E}_1 \longrightarrow \mathrm{Ens}$$

is π-accessible, when x_0 is in $\mathcal{E}_{0\,\pi}$. Now by the hypothesis on π, it is π-accessible if p. 159
x_0 is in $\mathcal{E}_{0\,\pi_0}$. But if x_0 is in $\mathcal{E}_{0\,\pi}$, we will have [page 419]

$$x_0 \ = \ \varinjlim_{i\in I} x_{0\,i} \qquad \left\{ \begin{array}{l} I \ \text{large relative to} \ \pi_0, \\ \text{the} \ x_{0\,i} \ \text{in} \ \mathcal{E}_{0\,\pi_0}, \\ \mathrm{card}\, I \leqslant \pi \end{array} \right.$$

(for π_0 is useful for $\mathcal{E}_0$ by hypothesis, and π satisfies $\pi^{\pi_0} = \pi \geqslant \mathrm{card}\ \mathrm{red}\mathcal{E}_{0\,\pi_0}$). As $x_0 \longmapsto H(x_0, x_1) : \mathcal{E}_0 \longrightarrow \mathrm{Ens}^\circ$ satisfies L_{π_0}, for every x_1 in $\mathcal{E}_1$, we will therefore have

$$H(x_0) \ \simeq \ \varprojlim_{i\in I} H(x_{0\,i})$$

(where for x_0 in $\mathcal{E}_0$, $H(x_0)$ denotes the functor $x_1 \longmapsto H(x_0, x_1) : \mathcal{E}_1 \longrightarrow (\mathrm{Ens})$). As the functors $H(x_{0\,i})$ are π-accessible, and $\operatorname{card} I \leqslant \pi$, it follows that their $\varprojlim_I H(x_0)$ is π-accessible (SGA 4 I 9.6 (ii)). This therefore completes the proof of 2°).

3°). To prove that $\mathrm{Ind}_\pi(\mathcal{E}(\pi)) \longrightarrow \mathcal{E}$ is essentially surjective, it suffices to see that every $x \in \mathrm{Ob}\,\mathcal{E}_0$ and every $x \in \mathrm{Ob}\,\mathcal{E}_1$ is in the essential image. Now this follows from the [page 420] hypothesis that π is useful for $\mathcal{E}_0$ and $\mathcal{E}_1$, and from 1°) which implies that $\mathcal{E}_0 \longrightarrow \mathcal{E}$ and $\mathcal{E}_1 \longrightarrow \mathcal{E}$ satisfy L_π.

This completes the proof of proposition 4.14.24, and likewise the following result, more precise than the implication c) $\Longrightarrow$ a).

Corollary 4.14.25. *Let $\mathcal{E}$ be a category, $\mathcal{E}_0$ an open subcategory, $\mathcal{E}_1$ the complementary closed subcategory,*

$$H \;=\; ((x_0, x_1) \longmapsto \mathrm{Hom}_{\mathcal{E}}(x_0, x_1)) \;:\; \mathcal{E}_0^\circ \times \mathcal{E}_1 \;\longrightarrow\; \mathrm{Ens}$$

the gluing functor.

1°) Let π be an infinite cardinal. In order that $\mathcal{E}$ and $\mathcal{E}_0$ satisfy L_π, and the inclusion functor $\mathcal{E}_0 \hookrightarrow \mathcal{E}$ likewise, it is necessary and sufficient that $\mathcal{E}_0$ and $\mathcal{E}_1$ satisfy L_π, and that for every x_1 in $\mathcal{E}_1$, the functor $x_0 \longmapsto H(x_0, x_1) : \mathcal{E}_0 \longrightarrow \mathrm{Ens}^\circ$ satisfy L_π. Then $\mathcal{E}_1$ and $\mathcal{E}_1 \hookrightarrow \mathcal{E}$ also satisfy L_π, and we have [page 421]

$$\mathcal{E}_\pi \;\subset\; \mathcal{E}(\pi) \overset{\text{def}}{=} \begin{array}{l} \textit{strictly full subcategory of } \mathcal{E} \textit{ such that} \\ \mathrm{Ob}\,\mathcal{E}(\pi) = \mathrm{Ob}\,\mathcal{E}_{0\,\pi} \cup \mathrm{Ob}\,\mathcal{E}_{1\,\pi}. \end{array}$$

2°) Suppose that condition c) of 4.14.24 is satisfied, and let π_0 be an infinite cardinal such that π_0 is useful for $\mathcal{E}_0$, and that the functors $x_0 \longmapsto H(x_0, x_1) : \mathcal{E}_0 \longrightarrow \mathrm{Ens}^\circ$ defined by the $x_1 \in \mathrm{Ob}\,\mathcal{E}_1$ are π_0-accessible. Let π be a cardinal such that $\pi^{\pi_0} = \pi \geqslant \sup(\pi_0, \operatorname{card} \mathrm{red}\mathcal{E}_{0\,\pi_0})$, and such that for every $x_0 \in \mathrm{Ob}\,\mathcal{E}_{0\,\pi_0}$, the functor $x_1 \longmapsto H(x_0, x_1) : \mathcal{E}_1 \longrightarrow \mathrm{Ens}$ is π-accessible. Then we have

$$\mathcal{E}_\pi \;=\; \mathcal{E}(\pi)\ .$$

p. 160

3°) If π_0 and π are as in 2°), and if moreover π is useful for $\mathcal{E}_1$, then π is useful for $\mathcal{E}$, hence (by virtue of 2°)

$$\mathcal{E} \;\overset{\sim}{\longleftarrow}\; \mathrm{Ind}_\pi(\mathcal{E}(\pi))\ .$$

We can now state the sought-for generalization of 4.14.17:

Theorem 4.14.26 (generalization of 4.14.17). *Let $p : \mathcal{E} \longrightarrow \mathcal{C}$ be a functor, with $\mathcal{C}$ small. We assume that p satisfies hypothesis H) of page 409, which is equivalent (using 4.14.24) to the following:* [page 422]

H′) $\forall\ s \in \mathrm{Ob}\,\mathcal{C}$, the fiber category $\mathcal{E}_s$ is accessible, and for every $\lambda : s \longrightarrow t$ in $\mathrm{Fl}\,\mathcal{C}$, the functor

$$\left\{ \begin{array}{rcl} H_\lambda : \mathcal{E}_s^\circ \times \mathcal{E}_t & \longrightarrow & \mathrm{Ens} \\ (x_0, x_1) & \longmapsto & \mathrm{Hom}_\lambda(x_0, x_1) \end{array} \right.$$

satisfies the following two conditions:

a) $\forall\ x_0 \in \mathrm{Ob}\,\mathcal{E}_s^\circ$, $x_1 \longmapsto H_\lambda(x_0, x_1) : \mathcal{E}_t \longrightarrow \mathrm{Ens}$ is accessible.

b) There exists an infinite cardinal π_λ, such that for every $x_1 \in \mathrm{Ob}\,\mathcal{E}_t$, the functor $x_0 \longmapsto H_\lambda(x_0, x_1) : \mathcal{E}_s \longrightarrow (\mathrm{Ens})^\circ$ is π_λ-accessible.

[page 423]

This being so, the category $\underline{\Gamma}(\mathcal{E}/\mathcal{C})$ of sections of $\mathcal{E}$ over $\mathcal{C}$ is accessible, and the functors $\varepsilon_s : \underline{\Gamma}(\mathcal{E}/\mathcal{C}) \longrightarrow \mathcal{E}_s$ are accessible. We can be more precise in a way similar to 4.14.18 (statement in due form left to the reader).

Proof of 4.14.26. It is carried out exactly like that of 4.14.17 (p. 400 ff.), but is conceptually simpler. We assume $\mathcal{C}$ small, and introduce the categories

$$\Pi \;=\; \prod_{s \,\in\, \mathrm{Ob}\,\mathcal{C}} \mathcal{E}_s \;, \qquad \Pi' \;=\; \prod_{\lambda \,\in\, \mathrm{Fl}\,\mathcal{C}} \mathcal{E}_\lambda \;,$$

where

$$\mathcal{E}_\lambda \;=\; \mathcal{E} \times_{\mathcal{C}} (\mathbf{\Delta}_1, i_\lambda) \;,$$

and the two canonical functors

$$\Pi \underset{G}{\overset{F}{\rightrightarrows}} \Pi'$$

with components

$$\begin{aligned} F_\lambda \;&=\; i_\lambda \circ \mathrm{pr}_{s(\lambda)} \\ G_\lambda \;&=\; j_\lambda \circ \mathrm{pr}_{b(\lambda)} \;, \end{aligned}$$

where for $\lambda : s \longrightarrow t$, we set

$$s(\lambda) \;=\; s \;, \qquad b(\lambda) \;=\; t \;,$$

and i_λ, j_λ are the canonical inclusions

$$\begin{array}{ccccc} \mathcal{E}_s & \xrightarrow{\;i_\lambda\;} & \mathcal{E}_\lambda & \xleftarrow{\;j_\lambda\;} & \mathcal{E}_t \;. \\ \big\downarrow \wr & & \big\downarrow p_\lambda & & \big\downarrow \wr \\ (\mathcal{E}_\lambda)_0 & & & & (\mathcal{E}_\lambda)_1 \\ & & \mathbf{\Delta}_1 & & \end{array}$$

We denote by $\mathcal{M}'$ the category

p. 161
[page 424]

$$\mathcal{M}' \;=\; \begin{array}{l} \text{category of pairs } (X, u), \text{ with} \\ X \in \mathrm{Ob}\,\Pi,\; u : F(X) \longrightarrow G(X), \end{array}$$

which is constructible by 4.14.22.1 is accessible (for Π and Π' and the functors F and G are, by hypothesis H) made on p). We can interpret it as the category of sections of $\mathrm{Diag}(\mathcal{E})$ over $\mathrm{Diag}(\mathcal{C})$. We have a strictly fully faithful functor

$$\mathcal{M} \;\overset{\mathrm{def}}{=}\; \underline{\Gamma}(\mathcal{E}/\mathcal{C}) \;\hookrightarrow\; \mathcal{M}' \;,$$

which lets us interpret $\mathcal{M}$ as a strictly full subcategory of $\mathcal{M}'$. We then see as in the case of 4.14.17 that this subcategory is accessible in $\mathcal{M}'$ – this is essentially sorital. It follows that $\mathcal{M}$ is accessible and that the functors $\mathcal{M} \longrightarrow \mathcal{E}_s$ are also, as composites of

$$\mathcal{M} \overset{\mathrm{acc.}}{\hookrightarrow} \mathcal{M}' \xrightarrow{\mathrm{acc.}} \Pi \xrightarrow{\mathrm{pr}_s \quad \mathrm{acc.}} \mathcal{E}_s \;,$$

q.e.d.

[page 425]

4.15 Accessible category over another

Proposition 4.15.1. *Let $f : \mathcal{E} \longrightarrow \mathcal{C}$ be a functor with $\mathcal{C}$ small. We assume in addition that condition H (p. 409) is satisfied, which we make explicit here in the equivalent form H':*

H 1) For every $s \in \mathrm{Ob}\,\mathcal{C}$, $\mathcal{E}_s$ is accessible.

H 2) For every $\lambda \in \mathrm{Fl}\,\mathcal{C}$, $\lambda : s \longrightarrow t$, there exists an infinite cardinal π_λ such that for every $x_1 \in \mathrm{Ob}\,\mathcal{E}_t$, the functor

$$x_0 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1) \ : \ \mathcal{E}_s \longrightarrow (\mathrm{Ens})^\circ$$

is π_λ-accessible ($\mathcal{E}_s$ satisfying in addition L_{π_λ}).

H 3) For every $\lambda \in \mathrm{Fl}\,\mathcal{C}$, $\lambda : s \longrightarrow t$, and for every $x_0 \in \mathrm{Ob}\,\mathcal{E}_s$, the functor

$$x_1 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1) \ : \ \mathcal{E}_t \longrightarrow (\mathrm{Ens})$$

is accessible.

Then the inclusion functors $i_s : \mathcal{E}_s \longrightarrow \mathcal{E}$ are accessible. Moreover, if $\mathcal{C}$ "has no projectors" (i.e. if every endomorphism p of an object s of $\mathcal{C}$ such that $p^2 = p$, is the identity), then $\mathcal{E}$ is accessible, and $f : \mathcal{E} \longrightarrow \mathcal{C}$ is accessible.

This follows from the more precise result:

Corollary 4.15.2 ([175]). *The hypotheses on $f : \mathcal{E} \longrightarrow \mathcal{C}$ are those of 4.15.1 ($\mathcal{C}$ small, and f satisfies $H' = H$ 1) 2) 3)).*

p. 162
[page 426]

a) Let $s \in \mathrm{Ob}\,\mathcal{C}$ and let π_1 be an infinite cardinal such that for every $\lambda : s \longrightarrow t$ (arrow of $\mathcal{C}$ with source s), we have $\pi_1 \geqslant \pi_\lambda$ (where π_λ is defined in the statement of H 2). Then the functor $i_s : \mathcal{E}_s \longrightarrow \mathcal{E}$ is π_1-accessible.

b) We assume $\mathcal{C}$ has no projectors. Let π_1 be a cardinal $\geqslant \sup(\pi_\lambda, \mathrm{card}\,\mathcal{C})$. Then $\mathcal{E}$ satisfies L_{π_1}, and $f : \mathcal{E} \longrightarrow \mathcal{C}$ is π_1-accessible.

c) Suppose again that $\mathcal{C}$ has no projectors (e.g. a "rigid" category, e.g. a preordered category). Let π_0, π_1 be cardinals such that:

① *$\forall\ s \in \mathrm{Ob}\,\mathcal{C}$, π_0 is useful for $\mathcal{E}_s$ and $\pi_0 \geqslant \sup_{\lambda \in \mathrm{Fl}\,\mathcal{C}} \pi_\lambda$ (such a π_0 exists thanks to H 1 and to the fact that $\mathcal{C}$, hence $\mathrm{Ob}\,\mathcal{C}$ small).*

② *$\pi_1 \geqslant \sup(\pi_0, \mathrm{card}\,\mathcal{C}, \sup_{\lambda \in \mathrm{Fl}\,\mathcal{C}} \pi_\lambda, \sup_{s \in \mathrm{Ob}\,\mathcal{C}} \mathrm{card}\ \mathrm{red}\mathcal{E}_{s\,\pi_0})$.*

③ *$\forall \lambda \in \mathrm{Fl}\,\mathcal{C}$, and $x_0 \in \mathrm{Ob}\,\mathcal{E}_{s,\pi_0}$, the functor $H_\lambda(x_0)$ of H 3, $x_1 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1) : \mathcal{E}_t \longrightarrow \mathrm{Ens}$ is π_1-accessible.*

Let π be a cardinal such that $\pi^{\pi_0} = \pi \geqslant \pi_1$. Then by b), $\mathcal{E}$ satisfies L_{π_1}, hence L_π, and $f : \mathcal{E} \longrightarrow \mathcal{C}$ is π_1-accessible. Moreover,

$$i_s(\mathcal{E}_{s\,\pi}) \ \subset \ \mathcal{E}_\pi \ ,$$

or again

$$\mathcal{E}(\pi) \ \subset \ \mathcal{E}_\pi \ , \qquad (*)$$

where $\mathcal{E}(\pi)$ is the full subcategory of $\mathcal{E}$ such that

$$\mathrm{Ob}\,\mathcal{E}(\pi) \ = \ \bigcup_{s \in \mathrm{Ob}\,\mathcal{C}} \mathrm{Ob}\,(\mathcal{E}_{s\,\pi}) \ .$$

[175] For a) and b) we use only H 1 and H 2, to the exclusion of H 3.

As $\mathcal{E}$ satisfies L_{π_1} by b), hence a fortiori L_π, we therefore have by $(*)$ a functor fully faithful

$$\mathrm{Ind}_\pi(\mathcal{E}(\pi)) \xrightarrow{\sim} \mathcal{E} ,$$

and this functor is an equivalence of categories (hence $\mathcal{E}$ is accessible).

d) In fact, under the hypotheses of c) on $\mathcal{C}$, π_0, π_1, π, we have $\mathcal{E}(\pi) = \mathcal{E}_\pi$ (not only $\mathcal{E}(\pi) \subset \mathcal{E}_\pi$).

Proof of 4.15.1 and 4.15.2. It is clear that 4.15.1 is contained in 4.15.2 a) and c). We will therefore prove 4.15.2. [page 427]

a) Let $(x_i)_{i\in I}$ be a direct system in $\mathcal{E}_s$, with I large relative to π_1, and x its direct limit in $\mathcal{E}_s$ (which exists, for $\mathcal{E}_s$ satisfies L_{π_λ}, hence L_{π_1}). We must prove that it is a limit in $\mathcal{E}$, hence that for every $y \in \mathrm{Ob}\,\mathcal{E}$, the canonical arrow

$$\mathrm{Hom}_{\mathcal{E}}(x, y) \longrightarrow \varprojlim_I \mathrm{Hom}_{\mathcal{E}}(x_i, y) \qquad (*)$$

is an isomorphism. Now $\mathrm{Hom}_{\mathcal{E}}(x, y)$ and the $\mathrm{Hom}_{\mathcal{E}}(x_i, y)$ are objects of $\mathrm{Ens}/\mathrm{Hom}_{\mathcal{C}}(s, t)$, where $s = f(x)$, $t = f(y)$, and the transition arrows are arrows in $\mathrm{Ens}/\mathrm{Hom}_{\mathcal{C}}(s, t)$. This shows that the bijectivity of $(*)$ comes down to that of p. 163

$$\mathrm{Hom}_\lambda(x, y) \longrightarrow \varprojlim_I \mathrm{Hom}_\lambda(x_i, y) \qquad (*)$$

for every $\lambda \in \mathrm{Hom}_{\mathcal{C}}(f(x), f(y))$. For λ fixed, and y fixed in $\mathrm{Ob}\,\mathcal{E}_t$, the bijectivity in $(*)$, for every direct system large relative to π_1 in $\mathcal{E}_s$, with limit x, means that the functor

$$x \longmapsto \mathrm{Hom}_\lambda(x, y) : \mathcal{E}_s \longrightarrow (\mathrm{Ens})^\circ$$

is π_1-accessible. Now this is indeed the case, by the hypothesis on π_1, whatever the arrow $\lambda : s \longrightarrow t$ with source s, and whatever $y \in \mathrm{Ob}\,\mathcal{E}_t$.

Let $(x_i)_{i\in I}$ be a direct system large relative to π_1, and $(s_i)_{i\in I}$ the image of this system in $\mathcal{C}$. We will prove below the [page 428]

Lemma 4.15.3 ([176],[177]). *Let $\mathcal{C}$ be a category, π a cardinal, such that* $\mathrm{card}\,\mathcal{C} \leqslant \pi$, *and that $\mathcal{C}$ has no projectors. Then for every direct system $(s_i)_{i\in I}$ large relative to π in $\mathcal{C}$, there exists a cofinal subset J in I such that the induced system $(s_i)_{i\in J}$ is* constant, *i.e. such that the s_i are equal to an $s \in \mathcal{C}$, and that the transition morphisms are all equal to* id_s.

But watch out, we rely in proving 4.15.3 on theorem 4.8.13 (page 212-216), which I still have not proved. In the case where $\mathcal{C}$ is rigid (i.e. $\mathrm{End}_{\mathcal{C}}(s) = \{\mathrm{id}_s\}\ \forall\ s \in \mathrm{Ob}\,\mathcal{C}$), the proof is however immediate, cf. below.

To prove that $x = \varinjlim_I x_i$ exists in $\mathcal{E}$, and commutes with f, we can replace I by a cofinal subcategory. This brings us back, thanks to the lemma, to the case where $(s_i)_{i\in I}$ is constant, hence to the case where the direct system comes from a direct system in a fiber category $\mathcal{E}_s$. But by a), we know that its limit x in $\mathcal{E}_s$ is a limit in $\mathcal{E}$, whence the existence of the latter, and commutation with f, since $\varinjlim s_i = s$ ($\varinjlim$ of a constant filtering direct system). [page 429]

c) We must prove that if $x \in \mathrm{Ob}\,\mathcal{E}_s$ is π-accessible in $\mathcal{E}_s$, it is so in $\mathcal{E}$, i.e. that for every direct system $(y_i)_{i\in I}$ large relative to π, the canonical arrow

$$\varinjlim_i \mathrm{Hom}(x, y_i) \longrightarrow \mathrm{Hom}(x, y) \qquad (**)$$

[176] Cf. proof page 452.

[177] subject to the validity of 4.8.13 (But this is now established, in a sufficient form, in 4.15.10).

is bijective. Now as by 2°) $\pi \geqslant \pi_1 \geqslant \sup(\operatorname{card}\mathcal{C}, \sup_\lambda \pi_\lambda)$, it follows again from the lemma that, possibly replacing I by a cofinal subcategory, we can assume that the direct system $(y_i)_{i\in I}$ comes from a system in a fiber category $\mathcal{E}_t$, and that y is its $\varinjlim$ in $\mathcal{E}_t$. But then the sets $\operatorname{Hom}(x,y)$ and $\operatorname{Hom}(x,y_i)$, and the transition morphisms between them, are again in $\operatorname{Ens}/\operatorname{Hom}(s,t)$, and we are reduced

to prove that for every $\lambda : s \longrightarrow t$, the arrow p. 164 [page 430]

$$\varinjlim_i \operatorname{Hom}_\lambda(x,y_i) \longrightarrow \operatorname{Hom}_\lambda(x,y)$$

is bijective. For fixed x and λ, and varying direct systems in $\mathcal{E}_t$, large relative to π, this means that the functor

$$H_\lambda(x) = (y \longmapsto \operatorname{Hom}_\lambda(x,y)) : \mathcal{E}_t \longrightarrow \operatorname{Ens}$$

is π-accessible, and this for any

$$x \in \operatorname{Ob}\mathcal{E}_{s\,\pi} .$$

Now this is the case if $x \in \operatorname{Ob}\mathcal{E}_{s\,\pi_0}$, by virtue of hypothesis 3°) (then $H_\lambda(x_0)$ is π_1-accessible, and a fortiori π-accessible). We have to pass from there to the case where $x \in \operatorname{Ob}\mathcal{E}_{s\,\pi}$. Now since π_0 is useful for $\mathcal{E}_s$ by 1°, we have

$$x = \varinjlim_{j\in J} x_j , \qquad x_j \in \operatorname{Ob}\mathcal{E}_{s\,\pi_0},\ J \text{ large relative to } \pi_0 ,$$

and since by 2°, $\pi^{\pi_0} = \pi \geqslant \pi_1 \geqslant \sup(\pi_0, \operatorname{card}\operatorname{red}\mathcal{E}_{s\,\pi_0})$, we conclude from this that we can take J so that

$$\operatorname{card} J \leqslant \pi .$$

On the other hand, by virtue of $\pi \geqslant \pi_1 \geqslant \pi_\lambda$, the functor

$$\begin{cases} x_0 \longmapsto H_\lambda(x_0) \\ \mathcal{E}_s \longrightarrow (\underline{\operatorname{Hom}}(\mathcal{E}_t, \operatorname{Ens}))^\circ \end{cases}$$

is π_0-accessible, so we have

$$H_\lambda(x) \simeq \varprojlim_{j\in J} H_\lambda(x_j) .$$

Since $x_j \in \operatorname{Ob}\mathcal{E}_{s\,\pi_0}$, $H_\lambda(x_j)$ is a π_1-accessible functor, and since $\operatorname{card} J \leqslant \pi$, the $\varprojlim_{j\in J} H_\lambda(x_j)$ is also π-accessible (cf. SGA 4 I 9.8). [page 431]

Thus we have proved $\mathcal{E}_{s\,\pi} \subset \mathcal{E}_\pi$ for every $s \in \operatorname{Ob}\mathcal{C}$, i.e. the inclusion

$$\mathcal{E}(\pi) \subset \mathcal{E}_\pi ,$$

whence the fully faithful functor

$$\operatorname{Ind}_\pi(\mathcal{E}(\pi)) \longrightarrow \mathcal{E} , \qquad (*)$$

since $\mathcal{E}$ satisfies L_π. It remains to prove that it is essentially surjective. But if $x \in \operatorname{Ob}\mathcal{E}$, $s = f(x)$, we have $x \in \operatorname{Ob}\mathcal{E}_s$, and since π_0 is useful for $\mathcal{E}_s$ by 1°), and $\pi^{\pi_0} = \pi > \pi_0$, π is also useful for $\mathcal{E}_s$, so we have *in* $\mathcal{E}_s$

$$x = \varinjlim_I x_i , \qquad \text{the } x_i \in \mathcal{E}_{s\,\pi} \subset \mathcal{E}(\pi),\ I \text{ large relative to } \pi ,$$

and by a) (since $\pi \geqslant \pi_1 \geqslant \sup_{\lambda\in \mathrm{Fl}\mathcal{C}} \pi_\lambda$) x is also a limit of the x_i in $\mathcal{E}$, so it is in the p. 165 [page 432]

essential image of $(*)$.

d) It remains to prove that we have $\mathcal{E}_\pi \subset \mathcal{E}(\pi)$. But we know (by 4.8. …), since $\mathcal{E} \simeq \mathrm{Ind}_\pi(\mathcal{E}(\pi))$, that $\mathcal{E}_\pi = \mathcal{E}(\pi)^\natural$, i.e. that for $x \in \mathrm{Ob}\,\mathcal{E}_\pi$, $\exists\, y \in \mathcal{E}(\pi)$ and a projector p in y, such that x is isomorphic to the image of p in $\mathcal{E}$. Now $f(p)$ is a projector in $f(y) = s$, so by the hypothesis on $\mathcal{C}$, $f(p) = \mathrm{id}_s$, i.e. p is a projector in $\mathcal{E}_s$. Since therefore x is in $\mathcal{E}_s$, and

even in $\mathcal{E}_{s\,\pi}$, since $\mathcal{E}_{s\,\pi}$ is stable under direct factors in $\mathcal{E}_s$. This proves that $x \in \mathrm{Ob}\,\mathcal{E}(\pi)$, q.e.d.

We can present 4.15.1 in a more congenial form.

Corollary 4.15.4. *Let $f : \mathcal{E} \longrightarrow \mathcal{C}$ be a functor of $\mathfrak{U}$-categories. The following conditions are equivalent:*

a) For every functor $\mathcal{C}' \longrightarrow \mathcal{C}$, with $\mathcal{C}'$ small and without projectors, $\mathcal{E}' \overset{\text{def}}{=} \mathcal{E} \times_{\mathcal{C}} \mathcal{C}'$ is accessible, and $f' : \mathcal{E}' \longrightarrow \mathcal{C}'$ is accessible.

b) As in a), but with $\mathcal{C}' = \mathbf{\Delta}_1$, in other words, this is condition H p. 409: for every $\lambda \in \mathrm{Fl}\,\mathcal{C}$, $\mathcal{E}_\lambda = \mathcal{E} \times_{\mathcal{C}} (\mathbf{\Delta}_1, i_\lambda)$ is accessible, and $\mathcal{E}_\lambda \longrightarrow \mathbf{\Delta}_1$ is accessible.

c) The condition $H' = H\ 1 + H\ 2 + H\ 3$, made explicit in the statement of 4.15.1.

Proof. It is done in a circle,

$$\mathrm{a} \implies \mathrm{b} \implies \mathrm{c} \implies \mathrm{a}\ ,$$

the implication a $\Longrightarrow$ b being tautological, b $\Longrightarrow$ c being almost so (cf. 4.14.24, a $\Longrightarrow$ b $\Longrightarrow$ c). In fact, conditions b and c are conditions on the categories $\mathcal{E}_\lambda \overset{f_\lambda}{\longrightarrow} \mathbf{\Delta}_1$ for $\lambda \in \mathrm{Fl}\,\mathcal{C}$, hence stable under base change. To prove

c $\Longrightarrow$ a, we can therefore assume that $\mathcal{C}' = \mathcal{C}$, $\mathcal{C}' \longrightarrow \mathcal{C}$ the identity functor, and prove [page 433] that if c) is satisfied and if $\mathcal{C}$ is small and without projectors, then $\mathcal{E}$ is accessible and f accessible. Now this is what 4.15.1 asserts.

Remark 4.15.5. If $f : \mathcal{E} \longrightarrow \mathcal{C}$ is a fully faithful functor, it is immediate that condition H is satisfied (for $\mathcal{E}_\lambda \longrightarrow \mathbf{\Delta}_1$ will be fully faithful, so $\mathcal{E}_\lambda$ is equivalent to e or to $\mathbf{\Delta}_1$, hence accessible, and every functor from $\mathcal{E}_\lambda$ to any category is accessible …). But if we assume $\mathcal{C}$ small *without* assuming $\mathcal{C}$ without projectors, it does not follow that $\mathcal{E}$ is accessible, or what amounts to the same thing (being essentially small), that $\mathcal{E}$ is Karoubian. (The most trivial example is the one where $\mathcal{E} = \mathcal{C}$, $f = \mathrm{id}_{\mathcal{C}}$, $\mathcal{E}$ is Karoubian only if $\mathcal{C}$ is!) But even assuming $\mathcal{C}$ a small Karoubian category, it does not necessarily follow that $\mathcal{E}$ is. Indeed, *every* small category $\mathcal{E}$ (Karoubian or not) embeds as a full subcategory in its Karoubian envelope $\mathcal{C}$ – so it suffices to take $\mathcal{E}$ non-Karoubian. The example

that is most economical is [page 434]

$$\mathrm{Ob}\,\mathcal{E} = \{0\}\ , \qquad \mathrm{Fl}(\mathcal{E}) = \{\mathrm{id}_0, p\}\ , \qquad \text{with } p^2 = p \neq \mathrm{id}_0\ ,$$

the Karoubi envelope of $\mathcal{E}$ is given by [p. 166]

$$P = \left\{ 1 \underset{\sigma}{\overset{\rho}{\leftrightarrows}} 0 \circlearrowright p \qquad \begin{array}{l} \rho\sigma = \mathrm{id}_1 \\ \sigma\rho = p \neq \mathrm{id}_0 \end{array} \quad \Longrightarrow\ p^2 = p \right\}.$$

This raises the question of finding a statement similar to 4.15.1, giving simple serviceable conditions, stronger than H′ = H (1+2+3), on $f : \mathcal{E} \longrightarrow \mathcal{C}$, for $\mathcal{E}$ to be accessible and f accessible, assuming $\mathcal{C}$ *Karoubian.* Or again, serviceable necessary and sufficient conditions (stronger than 4.15.4) on $f : \mathcal{E} \longrightarrow \mathcal{C}$, so that every *small Karoubian* category

$\mathcal{C}'$, and every functor $\mathcal{C}' \longrightarrow \mathcal{C}$, the category $\mathcal{E}' = \mathcal{E} \times_{\mathcal{C}} \mathcal{C}'$ be accessible, and the functor $\mathcal{E}' \longrightarrow \mathcal{C}'$ accessible. A natural idea is that it would suffice for this conclusion to be true when $\mathcal{C}'$ is one of the two categories $\mathbf{\Delta}_1$, P (where P is the category with two objects, with three nonidentity arrows, defined above). But I doubt that this is so ([178]).

We will all the same try a somewhat very special statement in this direction:

[page 435]

Proposition 4.15.6 ([180]). *Let $f : \mathcal{E} \longrightarrow \mathcal{C}$ be a functor of $\mathfrak{U}$-categories. We assume:*

a) f is fibrant or cofibrant ([181]).

b) The fiber categories $\mathcal{E}_s$ ($s \in \mathrm{Ob}\,\mathcal{C}$) are accessible, and for every arrow $\lambda : s \longrightarrow t$ in $\mathcal{C}$, the base change functor $\lambda^* : \mathcal{E}_t \longrightarrow \mathcal{E}_s$ (resp. cobase change $\lambda_* : \mathcal{E}_s \longrightarrow \mathcal{E}_t$) is accessible.

c) $\mathcal{C}$ is essentially small and Karoubian.

Then $\mathcal{E}$ is accessible, and f is accessible.

Proof. We have already seen that conditions a) and b) imply H′, i.e. H (1+2+3). Besides, if $\mathcal{C}'$ is a category and $\mathcal{C}' \xrightarrow{\sim} \mathcal{C}$ an equivalence, so then $\mathcal{E}' \overset{\text{def}}{=} \mathcal{E} \times_{\mathcal{C}} \mathcal{C}' \longrightarrow \mathcal{E}$ is an equivalence, so the desired conclusion for $\mathcal{E}$, f is equivalent to the same conclusion for $\mathcal{E}'$, $f' : \mathcal{E}' \longrightarrow \mathcal{C}'$. So we can assume that $\mathcal{C}$ is small (and even reduced, if one insists). By virtue of 4.15.2 a), we thus see that the inclusion functors $\mathcal{E}_s \longrightarrow \mathcal{E}$ are accessible (and even π_1-accessible) as soon as $\pi_1 \geqslant \sup_{\lambda \in \mathrm{Fl}\,\mathcal{C}} \pi_\lambda$. We also see

by AQT that we can find cardinals π_0, π_1 with the following properties:

[page 436]
p. 167

(1°) π_0 is useful for $\mathcal{E}_s$, $\forall\ s \in \mathrm{Ob}\,\mathcal{C}$.

2°) $\pi_1 \geqslant \sup(\pi_0, \mathrm{card}\,\mathcal{C}, \sup_{\lambda \in \mathrm{Fl}\,\mathcal{C}} \pi_\lambda, \sup_{s \in \mathrm{Ob}\,\mathcal{C}} \mathrm{card}\,\mathrm{red}\,\mathcal{E}_{s\,\pi_0})$.

(3°) $\forall\ (\lambda : s \longrightarrow t) \in \mathrm{Fl}\,\mathcal{C}$ and $x_0 \in \mathrm{Ob}\,\mathcal{E}_{s\,\pi_0}$, the functor

$$H_\lambda(x_0)\ :\ (x_1 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1))\ :\ \mathcal{E}_t\ \longrightarrow\ (\mathrm{Ens})$$

is π_1-accessible. (**N.B.** If f is cofibrant, this means that $\forall\ \lambda \in \mathrm{Fl}\,\mathcal{C}$, $\lambda : s \longrightarrow t$, $\boxed{\lambda_*(\mathcal{E}_{s\,\pi_0})\ \subset\ \mathcal{E}_{t\,\pi_1}}$. If f is fibrant, it suffices to require that $\forall\ \lambda \in \mathrm{Fl}\,\mathcal{C}$, $\lambda^* : \mathcal{E}_t \longrightarrow \mathcal{E}_s$ be π_1-accessible.) But in this case, we also want $\boxed{\lambda^*(\mathcal{E}_{t\,\pi_0})\ \subset\ \mathcal{E}_{s\,\pi_1}}$.

(4°) The functors λ^* (case f fibrant) resp. λ_* (case f cofibrant) are π_1-accessible.

Let π then be a cardinal such that

$$\pi\ =\ \pi^{\pi_0}\ \geqslant\ \pi_1\ , \qquad (*)$$

[178] That should work on the other hand if we take the categories $0 \longrightarrow 1$,

$$2 \longrightarrow \underbrace{1 \underset{\sigma}{\overset{\rho}{\leftrightarrows}} 0 \circlearrowright p}_{P} \quad \text{and} \quad 2 \longleftarrow \underbrace{1 \underset{\sigma}{\overset{\rho}{\leftrightarrows}} 0 \circlearrowright p}_{P} \qquad (^{179})\ ,$$

3 objects and 5 nonidentity arrows

thanks to 4.15.7, page 440.

[179] these categories represent respectively the functors Cat $\longrightarrow$ Ens assigning to a category $\mathcal{C}$ the diagrams in $\mathcal{C}$ consisting of (x, y, p, u), with p a projector on x , and $u : y \longrightarrow x^\natural$ (i.e. $u : y \longrightarrow x$ with $u = pu$), resp. $u : x^\natural \longrightarrow y$ (i.e. $u : x \longrightarrow y$ with $u = up$).

[180] *False?*

[181] The fibrant case is not proved, perhaps *false* – and likewise for the cofibrant case!

and let $\mathcal{E}(\pi)$ be the (strictly) full subcategory of $\mathcal{E}$ defined by

$$\mathrm{Ob}\,\mathcal{E}(\pi) \;=\; \bigcup_{s\,\in\,\mathrm{Ob}\,\mathcal{C}} \mathrm{Ob}\,\mathcal{E}_{s\,\pi}\ .$$

Thanks to $(*)$, and to $1°$, $4°$ and to

$$\begin{cases} \pi_1 \;\geqslant\; \sup_s(\mathrm{card}\,\mathrm{red}\,\mathcal{E}_{s\,\pi_0}) & \text{(contained in } 2°\text{)} \\ \lambda^*(\mathcal{E}_{t\,\pi_0}) \;\subset\; \mathcal{E}_{s\,\pi_1} \qquad \text{resp.} \qquad \lambda_*(\mathcal{E}_{s\,\pi_0}) \;\subset\; \mathcal{E}_{t\,\pi_1} & \text{(contained in } 3°\text{)}\ , \end{cases}$$

we see that $\mathcal{E}(\pi)$ is a

fibered subcategory, resp. a cofibered subcategory, over $\mathcal{C}$ (the functor $\mathcal{E}(\pi) \longrightarrow \mathcal{E}$ thus being cartesian resp. cocartesian). Up to $\mathcal{C}$-equivalence, $\mathcal{E}$ is then reconstructed from $\mathcal{E}(\pi)$, by passing to the $\mathrm{Ind}_\pi(\mathcal{E}_{s\,\pi})$, which form a fibered resp. cofibered system over $\mathcal{C}$ via the functors $\mathrm{Ind}_\pi(\lambda_\pi^*)$ resp. $\mathrm{Ind}_\pi(\lambda_*^\pi)$ … [page 437]

Let us prove that $\mathcal{E}$ satisfies L_π and that $f : \mathcal{E} \longrightarrow \mathcal{C}$ is π-accessible ([182]). Since $\mathcal{C}$ is Karoubian, and $\pi \geqslant \pi_1 \geqslant \mathrm{card}\,\mathcal{C}$, $\mathcal{C}$ satisfies L_π and the ind-objects of $\mathcal{C}$ large relative to π are essentially constant (4.8.13). Let I then be large relative to π, and $I \longrightarrow \mathcal{E}$, $i \longmapsto x_i$, be a functor, we have to construct a $\varinjlim$ in $\mathcal{E}$, using the already known $\varinjlim s$ of the $s_i = f(x_i)$. We will take an x in $\mathcal{E}_s$. Let us first take the case where f is cofibrant. We see that the $\beta_{i\,*}(x_i)$ form a direct system in $\mathcal{E}_s$, since for $i \leqslant j$, we have

$$\beta_{i\,*}(x_i) \;=\; \beta_{j\,*}\underbrace{\beta_{ji\,*}(x_i)}_{\longrightarrow\, x_j} \;\longrightarrow\; \beta_{j\,*}(x_j)\ .$$

$$\begin{array}{ccc} x_i & \xrightarrow{\quad\alpha_i\quad} & x\,? \\ s_i & \xrightarrow{\quad\beta_i\quad} & s \\ & {\scriptstyle\beta_{ji}}\searrow \quad \nearrow{\scriptstyle\beta_j} & \\ & s_j & \end{array}$$

We set

$$x \;=\; \varinjlim_I \beta_{i\,*}(x_i)\ , \qquad \varinjlim \ \textit{in}\ \mathcal{E}_s\ .$$ p. 168 [page 438]

We know that this is a $\varinjlim$ in $\mathcal{E}$, hence that for every y in $\mathcal{E}$, we have

$$\mathrm{Hom}(x,y) \xrightarrow{\sim} \varprojlim_I \mathrm{Hom}(\beta_{i\,*}(x_i), y)\ ,$$

or what amounts to the same thing, for every arrow $\lambda : s \longrightarrow t$, and y in $\mathrm{Ob}\,\mathcal{E}_t$, we have

$$\underbrace{\mathrm{Hom}_\lambda(x,y)}_{=\,\mathrm{Hom}_{\mathcal{E}_t}(\lambda_* x, y)} \;\simeq\; \varprojlim \mathrm{Hom}_{\mathcal{E}_t}(\underbrace{\lambda_*\beta_{i\,*}(x_i)}_{(\lambda\beta_i)_*(x_i)}, y)\ .$$

This also follows from the fact that λ_* is π-accessible (being π_1-accessible) … We easily check that this is a $\varinjlim$ of the x_i in $\mathcal{E}$. And of course, this limit commutes with f by construction.

But in the case where f is fibrant, I do not see how to go about constructing a limit in $\mathcal{E}_s$. It is true that we can assume that all the s_i are equal to s, but the transitions β_{ji} are not the identity, nor even isomorphisms. We would have to take $\mathcal{C} = P$ to try to understand – I am not going to insist ([183]).

[182] Here we use only $\pi \geqslant \mathrm{card}\,\mathrm{red}\,\mathcal{C}$.

[183] I have checked that if f is fibrant, then $\mathcal{E}$ is *Karoubian*. So there is hope … [page 439]

In the case where f is fibrant, let us admit (as an additional hypothesis if need be) that $\mathcal{E}$ satisfies L and that f is accessible, and take π large enough for $\mathcal{E}$, f to satisfy L_π. We can then finish the proof of 4.15.6. We have to prove

$$\mathcal{E}(\pi) \ \subset \ \mathcal{E}_\pi \ ,$$

hence that if $x \in \mathrm{Ob}\,\mathcal{E}_{s\,\pi}$, and $(y_i)_{i\in I}$ is a direct system in $\mathcal{E}$ large relative to π, with limit y, we have

$$\varinjlim \mathrm{Hom}_{\mathcal{E}}(x, y_i) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(x, y) \ .$$

So we have to prove that every arrow $x \xrightarrow{u} y$ factors, in an essentially unique way u_i, through a y_i. But the arrow $s \xrightarrow{\lambda = f(u)} t$ in $\mathcal{C}$ factors, in an essentially unique way, through some $\lambda_i : s \longrightarrow t_i$. So we are reduced to finding a *λ_i-morphism* $x \longrightarrow y_i$ for large i, which factors. In the fibrant case, this means that we are looking for $x \longrightarrow \lambda_i^*(y_i)$ which factors $x \longrightarrow \lambda^*(y)$, now the $\lambda_i^*(y_i)$ form a direct system in $\mathcal{E}_s$, and its limit *in* $\mathcal{E}_s$ is $\lambda^*(y)$ (*to be checked*). Since x is in $\mathcal{E}_{s\,\pi}$, the given hom $x \longrightarrow \lambda^*(y)$ factors in
an essentially unique way through a $\lambda_i^*(y_i)$. [page 440]

This time, however, it is the case where f is cofibrant that poses a problem. In fact, I cannot work it out.

Thus, we do find a functor

$$\mathrm{Ind}_\pi(\mathcal{E}(\pi)) \ \longrightarrow \ \mathcal{E} \ ,$$

which is essentially surjective, but for lack of knowing that $\mathcal{E}(\pi) \subset \mathcal{E}_\pi$ in the cofibrant case, it is not clear that this functor is an equivalence of categories. p. 169

4.15.7. I must postpone the proof of lemma 4.15.3 (p. 428), assuming 4.8.13. (Not yet proved – I am going to come back below to the proof of 4.8.13.) We are going to use the

Lemma 4.15.7.1. *Let $\mathcal{C}$ be a category, $\underline{x} = (x_i)_{i\in I}$ an ind-object of $\mathcal{C}$. For $\underline{x}$ to be essentially constant, i.e. isomorphic, in* $\mathrm{Ind}(\mathcal{C})$*, to the constant ind-object defined by an object*

of $\mathcal{C}$, it is necessary and sufficient that there exist a cofinal functor $\varphi : J \longrightarrow I$, with J filtering, in such a way that the reindexed system $(x_j)_{j\in J}$, with $x_j = x_{\varphi(j)}$, have the following property: for every $j \in J$, one can find a projector p_j in x_j, in such a way that: [page 441]

1°) For every arrow $\lambda : j \longrightarrow j'$ in J, we have commutativity in

$$\begin{array}{ccc} x_j & \xrightarrow{x_\lambda} & x_{j'} \\ \downarrow{\scriptstyle p_j} & & \downarrow{\scriptstyle p_{j'}} \\ x_j & \xrightarrow{x_\lambda} & x_{j'} \end{array}$$

(where the x_λ are the transition arrows), i.e. $p_{j'}x_\lambda = x_\lambda p_j$.

2°) For every $j \in \mathrm{Ob}\,J$, there exists an arrow $\lambda : j \longrightarrow j'$ in J such that $p_{j'}x_\lambda = x_\lambda p_j = x_\lambda$.

3°) For every $j \in \mathrm{Ob}\,J$, the projector p_j in x_j admits an image $x_j^\natural$.

4°) For every arrow $\lambda : j \longrightarrow j'$ in J, the arrow $x_\lambda^\natural : x_j^\natural \longrightarrow x_{j'}^\natural$ induced by x_λ (thanks to 1°) is an isomorphism.

[page 442]

Moreover, we can suppose J of the form $i_0\backslash I$, the cofinal functor $J \longrightarrow I$ being the canonical functor $i_0\backslash I \longrightarrow I$ (i_0 being a suitable fixed object of I)

Let us note at once two corollaries.

Corollary 4.15.7.2. *Suppose that $\mathcal{C}$ is without projectors. Then $\underline{x} = (x_i)_{i\in I}$ is an essentially constant ind-object if and only if there exists a cofinal functor, $J \longrightarrow I$, with J filtering ([184]), in such a way that the reindexed direct system $(x_j)_{j\in J}$ is isomorphic* (in the category $\underline{\mathrm{Hom}}(J, \mathcal{C})$) *to a* constant *direct system (i.e. defined by a constant functor $J \longrightarrow \mathcal{C}$: all the x_j equal to one and the same x, and the transition morphisms identities).*

Sufficiency is clear, for necessity we apply 4.15.7.1, which tells us that we can reindex so as to satisfy conditions 1°) to 4°) (for suitable projectors p_j in the x_j). But the hypothesis on $\mathcal{C}$ implies that $p_j = \mathrm{id}_{x_j}\ \forall\, j \in \mathrm{Ob}\, J$, and conditions 1°, 2°, 3° are satisfied tautologically for such a choice, condition 4° means that the transition morphisms are isomorphisms. But it is immediate that for a given direct system $(x_j)_{j\in J}$, with J 1-*connected* (e.g. J filtering), the condition that the x_λ are isomorphisms is necessary and sufficient for this system to be isomorphic to a constant direct system, whence the conclusion.

p. 170
[page 443]

Corollary 4.15.7.3. *Suppose that I is large relative to π_0, where π_0 is an infinite cardinal such that $\pi_0 \geqslant \mathrm{card}\, \mathcal{C}$. Then $(x_i)_{i\in I}$ is essentially constant if and only if we can find a cofinal functor $J \longrightarrow I$ with J large relative to π_0 (and ordered if we like) ([185]), in such a way that the reindexed system $(x_j)_{j\in J}$ satisfies the following conditions:*

1°) The map $j \longmapsto x_j$ from $\mathrm{Ob}\, J$ to $\mathrm{Ob}\, \mathcal{C}$ is constant, i.e. there exists an $x_0 \in \mathrm{Ob}\, \mathcal{C}$ such that $x_j = x_0\ \forall\, j \in \mathrm{Ob}\, J$.

2°) There exists a projector p in x_0 such that for $j < j'$ in J, we have $\tau_{j',j} = p$.

3°) The projector p admits an image in $\mathcal{C}$.

Proof. These conditions obviously imply those of 4.15.7.1, taking $p_j = p\ \forall\, j \in \mathrm{Ob}\, J$, at least if J has no greatest element (which causes a difficulty for 2°). But if J has a final object, the same is true of I, and it is clear then that every direct system indexed by I is essentially constant. So, in these cases, conditions 1°) 2°) 3°) above imply that $\underline{x}$ is essentially constant. Conversely, suppose that $\underline{x}$ is essentially constant, and let us prove [page 444]
that we can reindex as indicated. The case where I has a final object is trivial, we shall take $J = e$, $p = \mathrm{id}$. So suppose that I has no final object. Since I large relative to π_0 and $\pi_0 \geqslant \mathrm{card}\, \mathrm{Ob}\, \mathcal{C}$, we see that there exists an $x_0 \in \mathrm{Ob}\, \mathcal{C}$ such that the full subcategory $I' = I_{x_0}$ of I, consisting of the i such that $x_i = x_0$, is cofinal in I. Since $(x_i)_{i\in I}$ is essentially constant, the same is true of $(x_{i'})_{i'\in I'}$, and we can apply criterion 4.15.7.1 to it, with $J \longrightarrow I'$ cofinal. The proof will show moreover that if I' is large relative to π_0 (an infinite cardinal with no other conditions), we can suppose J large relative to π_0 ([186]). Moreover, it is clear that we can suppose J ordered. Since $\pi_0 \geqslant \mathrm{card}\, \mathrm{Hom}(x_0, x_0)$, we see again that there exists a projector p in x_0 such that the subset $J_1 = J'$ of J, consisting of the j such that $p_j = p$, is cofinal. Being cofinal, it too is large relative to π_0. So we can suppose that all the p_j are equal to one and the same p. Conditions 1°-4° become: ‖

1°) $p\tau_{j'j} = \tau_{j'j}p$ for $j \leqslant j'$.

2°) For every j, exists $j' \geqslant j$ with

$$p\tau_{j'j} \;=\; \tau_{j'j}p \;=\; \tau_{j'j}\ .$$

[184] and we can suppose $J = i_0\backslash I \xrightarrow{\text{can.}} I$

[185] The proof (slightly rearranged) shows that we can take for J a subcategory (not full) of a category $i_1\backslash I$, for a suitable object i_1 of I. Thus we have $\mathrm{card}\, J \leqslant \pi$ if $\mathrm{card}\, I \leqslant \pi$, π being a given infinite cardinal.

[186] take this into account in the proof of 4.15.7.1.

3°) The image $x_0^\natural$ of the projector p in x_0 exists.

4°) The endomorphism $\tau_{j'j}^\natural$ of $x_0^\natural$ is an isomorphism, for $j \leqslant j'$.

Now we have the [page 445]

Lemma 4.15.7.4. *Let $\mathcal{C}$ be a category, x_0 an object of $\mathcal{C}$, p a projector in x_0, τ an endomorphism of x_0 satisfying the relations*

$$\tau p \;=\; p\tau \;=\; \tau\;, \tag{$*$}$$

and such that the automorphism of $x_0^\natural \overset{\text{def}}{=} \operatorname{Im} p$ (defined e.g. as an object of $\mathcal{C}^\wedge$, $x_0^\natural = \operatorname{Ker}(\mathrm{id}_{x_0}, p)$) be the identity. Then we have $\tau = p$. p. 171

Proof of 4.15.7.4. We are reduced at once, by the standard procedure, to the case where $\mathcal{C} = \mathrm{Ens}$, where the verification is immediate: the endomorphisms τ of the set E $(= x)$ which satisfies $(*)$ correspond one-to-one to the endomorphisms u of $E^\natural = pE$, via

$$\begin{cases} u \;\longmapsto\; \sigma u \rho \\ \tau \;\longmapsto\; \tau | E^\natural \end{cases} \qquad (\text{where} \quad E^\natural \underset{\rho}{\overset{\sigma}{\rightleftarrows}} E \quad \text{are the canonical arrows})\;,$$

so $\tau | E^\natural = \mathrm{id} \Longrightarrow \tau = \sigma\rho = p$.

Let us choose a $j_0 \in J$, and consider the homomorphisms

$$\tau_{j,j_0}^\natural \;:\; x_0^\natural \longrightarrow x_0^\natural$$

induced by the τ_{j,j_0} with $j \geqslant j_0$. Here again, since the set of possible isomorphisms is of cardinality $\leqslant \pi_0$, and $j_0\backslash J$ is large relative to π_0, we see that there exists $J' \subset j_0\backslash J$, [page 446] cofinal in $j_0\backslash J$, hence in J, such that for $j \in J'$, the $\tau_{j,j_0}^\natural$ are equal to one and the same $\tau : x_0^\natural \xrightarrow{\sim} x_0^\natural$. So if $j', j'' \in J'$, with $j' \leqslant j''$, we have

$$\underbrace{\tau_{j''j_0}^\natural}_{\tau} \;=\; \tau_{j''j'}^\natural \underbrace{\tau_{j'j_0}^\natural}_{\tau}\;,$$

whence

$$\tau_{j''j'}^\natural \;=\; \mathrm{id}_{x_0^\natural}\;,$$

since the arrow τ is an isomorphism. So the direct system reindexed by J', condition 4° is replaced by the stricter condition $\boxed{4°\text{ bis})\ \tau_{j',j}^\natural = \mathrm{id}_{x_0^\natural}.}$ But condition 2° then becomes, thanks to 4.15.7.4,

$$\boxed{2°\text{ bis) For every } j, \text{ exists } j' \geqslant j \text{ with } \tau_{j',j} = p}\;.$$

So, we can suppose 1°, 2° bis, 3°, 4° bis (we again denote by J, instead of J', the indexing ordered set large relative to π_0).

Let K be the ordered set defined by

[page 447]

$$\begin{cases} \operatorname{Ob} K \;=\; \operatorname{Ob} J \\ \text{if } j, j' \in \operatorname{Ob} J, \text{ we have } j \overset{K}{\leqslant} j' \text{ if and only if } j \overset{\mathrm{J}}{=} j', \text{ or } j \overset{J}{<} j' \text{ and } \tau_{j',j} = p\,. \end{cases}$$

It is immediate that this is an *order relation* on J (since $p^2 = p$), finer than the initial order relation. The direct system induced on K satisfies conditions 1°, 2°, 3° of 4.15.7.3. p. 172 On the other hand, $K \longrightarrow J$ is cofinal, hence so is the composite $K \longrightarrow J \longrightarrow I$. It [page 448]

remains to prove that K is large relative to π_0, hence that every subset $K' \subset K$ such that $\operatorname{card} K' \leqslant \pi_0$ has an upper bound in K. It has an upper bound *in* J, say j. By hypothesis 2 bis, there exists $j' \geqslant j$ such that $\tau_{j'j} = p$, and we can suppose $j' > j$. (Indeed, if $\tau_{j'j} = p$, we see that the initial form 2°) of condition 2° bis that we have $\tau_{j''j} = p$ for every $j'' \geqslant j'$, so if j is not a final object of J, we can suppose $j' > j$. But J has no final object, for otherwise I would have one, a case we had eliminated. So there exists j' such that $j' \overset{K}{>} j$.) Now we have this (which we apply to the case $i \in K'$):

$$i \leqslant j \text{ and } j \overset{K}{<} j' \qquad \Longrightarrow \qquad i \overset{K}{<} j' .$$

$$\underbrace{x_i}_{=x_0} \xrightarrow{\tau_{j,i}} x_j \xrightarrow{\tau_{j',j}=p} x_{j'} ,$$

$$\tau_{j'i} \;=\; \tau_{j'j}\tau_{ji} = \underbrace{p\tau_{ji}}_{\overset{1°}{=}\tau_{ji}p} \overset{?}{=} p ,$$

we have indeed

$$\left.\begin{aligned} \tau_{j',i} &= p\tau_{j',i} \\ \tau_{j',i} &= \tau_{j',i}p \end{aligned}\right\} \quad \overset{\text{by } 4.15.7.4}{\Longrightarrow} \quad \tau_{j',i} = p .$$

This completes the proof of corollary 4.15.7.3.

Corollary 4.15.7.5 (of 4.15.7.3). *Let $(x_i)_{i\in I}$ be an essentially constant direct system large relative to $\pi_0 \geqslant \operatorname{card}\mathcal{C}$, in $\mathcal{C}$* without projectors. *Then there exists $J \longrightarrow I$ cofinal, with J ordered large relative to π_0, and such that the direct system induced on J is* constant.

Let us prove lemma 4.15.7.1. We must make explicit the existence of two arrows in $\operatorname{Ind}(\mathcal{C})$,

$$z \underset{\beta}{\overset{\alpha}{\rightleftarrows}} \text{"}\varinjlim_{i\in I}\text{"} x_i \;=\; \underline{x} ,$$

with $\beta\alpha = \mathrm{id}_z$, $\alpha\beta = \mathrm{id}_{\underline{x}}$. The datum

of α is equivalent to that of an element of $\varinjlim_{i\in I} \operatorname{Hom}(z, x_i)$, and is therefore given by [page 449]

an

$$\alpha_{i_0} \;:\; z \longrightarrow x_{i_0} ,$$

the datum of β is equivalent to that of an element of $\varprojlim_{i\in I} \operatorname{Hom}(x_i, z)$, i.e. of arrows

$$\beta_i \;:\; x_i \longrightarrow z ,$$

with the usual compatibility condition. At the cost of replacing I by I', with $I' \longrightarrow I$ cofinal, we can suppose I ordered (and large relative to π_0, if the initial I was), and the compatibility condition on the β_i reads

$$\beta_j\tau_{ji} = \beta_i \qquad \text{if } i \leqslant j \text{ in } I,$$

where $\tau_{ji} : x_i \longrightarrow x_j$ denotes the transition arrow. We must make explicit, in terms of these data, the conditions

$$\beta\alpha = \mathrm{id}_z, \quad \alpha\beta \;=\; \mathrm{id}_{\underline{x}} .$$

p. 173

$$\begin{array}{ccccc} z & \xrightarrow{\alpha_{i_0}} & x_{i_0} & \xrightarrow{\tau_{i,i_0}} & x_i \\ & \searrow & \downarrow{\scriptstyle \beta_{i_0}} & \swarrow & \\ & & z & & \end{array} \qquad \beta_i$$

The relation $\beta\alpha = \mathrm{id}_z$ is equivalent to

$$\beta_{i_0}\alpha_{i_0} \;=\; \mathrm{id}_z \;, \qquad (*_{i_0})$$

it implies moreover, on setting for $i \geqslant i_0$,

$$\alpha_i \;\overset{\text{def}}{=}\; \tau_{i,i_0}\alpha_{i_0} \;:\; z \longrightarrow x_i \;,$$

that we have for $i \geqslant i_0$

$$\beta_i\alpha_i \;=\; \mathrm{id}_z \qquad \forall\, i \geqslant i_0 \;. \qquad (*_i)$$

Let us then set [page 450]

$$p_i \;=\; \alpha_i\beta_i \;:\; x_i \longrightarrow x_i \;,$$

it is clear that we have

$$\begin{cases} p_i^2 \;=\; p_i \qquad (p_i \text{ a projector}) \\ \alpha_i \;:\; z \longrightarrow x_i \qquad \text{induces an isomorphism} \\ \qquad z \xrightarrow{\sim} x_i^\natural \;\overset{\text{def}}{=}\; \mathrm{Im}\, p_i \;=\; \mathrm{Ker}(\mathrm{id}_{x_i}, p_i) \;. \end{cases}$$

The desired cofinal ordered set J is $i_0\backslash I$, it is filtering, and is large relative to π_0 if I is. To tell the truth, the reduction to the case I ordered is unnecessary, we can take $J = i_0\backslash I$ directly, it is immediate that J is filtering (I being so), and $i_0\backslash I \longrightarrow I$ cofinal, finally that if I is large relative to an infinite cardinal π_0, the same is true of $i_0\backslash I$.

Let us express that $\alpha\beta = \mathrm{id}_{\underline{x}}$; now we see that $\alpha\beta$ is defined by the endomorphism $(p_j)_{j\in J}$ of the direct system $(x_j)_{j\in J}$ – the fact that this really is an endomorphism of this system, i.e. that we have relation 1°) of 4.15.7.1,

$$p_{j'}x_\lambda \;=\; x_\lambda p_j \qquad \text{if } \lambda : j \longrightarrow j' \text{ in } J,$$

is immediate, it reads

$$\alpha_{j'}\underbrace{\beta_{j'}x_\lambda}_{\beta_j} \;=\; \underbrace{x_\lambda\alpha_j}_{\alpha_{j'}}\beta_j \;, \qquad \text{now both sides are equal to } \alpha_{j'}\beta_j.$$

Saying that it is the identity, is expressed by condition 2°) on $(x_j, p_j)_{j\in J}$. Condition 3° is clear by construction, p_j admitting an image isomorphic to z via α_j. On the other hand, one sees away that for an arrow $\lambda : j \longrightarrow j'$ of J, one has commutativity in p. 174 [page 451]

$$\begin{array}{ccc} x_j & \xrightarrow{x_\lambda} & x_{j'} \\ \uparrow & & \uparrow \\ x_j^\natural & \xrightarrow{x_\lambda} & x_{j'}^\natural \\ \wr\uparrow{\scriptstyle \overline{\alpha_j}} & & \wr\uparrow{\scriptstyle \overline{\alpha_{j'}}} \\ z & \xrightarrow[\sim]{\mathrm{id}_z} & z \;, \end{array}$$

where $\overline{\alpha_j}$, $\overline{\alpha_{j'}}$ are the isomorphisms defined by α_j, $\alpha_{j'}$ respectively. This proves that $x_\lambda^\natural$ is an isomorphism.

This proves that if $(x_i)_{i\in I}$ is essentially constant, there exists $J \longrightarrow I$ as indicated. The converse is also true, for one can suppose $J = I$, since $(x_i)_{i\in I}$ and $(x_j)_{j\in J}$ define isomorphic ind-objects. Then the direct system of the $x_j^\natural$ has as transition morphisms isomorphisms, and as J is filtering hence 1-connected, one easily deduces from this that it is isomorphic to a constant direct system, with value z. Choosing such an isomorphism [page 452]

$$z \xrightarrow[\sim]{\overline{\alpha_i}} x_i^\natural \ ,$$

one finds $z \xrightarrow{\alpha} \underline{x}$, and the system of the p_i defines some

$$x_i \underset{\beta_i \overset{\text{def}}{=} \overline{\alpha_i}^{-1}\overline{p_i}}{\overset{\overline{p_i}}{\longrightarrow} x_i^\natural \overset{\overline{\alpha_i}^{-1}}{\longrightarrow}} z \ ,$$

and one easily sees that one will have

$$\beta\alpha \ = \ id_z \ , \qquad \alpha\beta \ = \ \mathrm{id}_{\underline{x}} \ ,$$

q.e.d.

I CAN NOW PROVE 4.15.3. It is indeed essentially the conjunction of 4.15.7.5 (p. 448), and of 4.8.13, which tells us essentially that if $\mathcal{C}$ is *Karoubian* (a condition satisfied if $\mathcal{C}$ is without projectors), then every direct system in $\mathcal{C}$, large relative to π_0, where $\pi_0 \geqslant \operatorname{card} \operatorname{red}\mathcal{C}$, is essentially constant,

q.e.d.

4.15.8. I want to return here to the unfinished proof of 4.8.13 (p. 212-216). I take up the proof where I had left it (p. 215). One must prove that

p. 175
[page 453]

$$F(\xi) \ = \ \varinjlim_{i \in I} M_i$$

has cardinality $\leqslant \pi_0$. I announced that one can reduce to the case where M is a monoid of monomorphisms, i.e. that $uv = uv' \Longrightarrow v = v'$ (for $u, v, v' \in M$). I am going to make this argument explicit.

For every $i \in I$, let

$$N_i \ = \ \left\{ u \in M \ \middle| \ \begin{array}{l} \text{the set } I_{i,u} \text{ of } j \in I \text{ such that} \\ j \geqslant i, \ \alpha_{ji} = u \text{ is cofinal} \\ \text{in } I, \text{ i.e. for every } j_0 \in I, \\ \text{there exists } j \in I_{i,u}, \text{ with } j \geqslant j_0 \end{array} \right\} . \tag{4.15.8.1}$$

Since I, hence $i\backslash I$ is large relative to π_0, and $\operatorname{card} M \leqslant \pi_0$, one sees at once that

$$N_i \ \neq \ \emptyset \ . \tag{4.15.8.2}$$

(If on the other hand $i' \geqslant i$, one finds

$$N_i \ = \ N_{i'}\alpha_{i'i} \ . \tag{4.15.8.3}$$

The inclusion $N_{i'}\alpha_{i'i} \subset N_i$ is ± tautological, let us prove $N_i \subset N_{i'}\alpha_{i'i}$. Let $u \in N_i$, the set $J = I_{i,u} \cap (i'\backslash I)$ of $j \geqslant i'$ such that $\alpha_{ji} = u$ is cofinal in I, hence large relative to π_0.

$$i \underset{u=\alpha_{ji}}{\overset{}{\longrightarrow} i' \overset{\alpha_{ji'}}{\longrightarrow}} j$$

Thus there exists a $v \in M$ such that the set J' of $j \in J$ such that $\alpha_{ji'} = v$ is cofinal in J, hence in I. Since this set J' is contained in $I_{i',v}$, this implies that $I_{i',v}$ is cofinal in I, i.e. $v \in N_{i'}$. On the other hand one has, for $j \in J'$,

[page 454]

$$\underbrace{\alpha_{ji'}}_{v}\alpha_{i'i} \ = \ \underbrace{\alpha_{ji}}_{u} \ ,$$

whence $u = v\alpha_{i'i}$ with $v \in N_{i'}$, whence $u \in N_{i'}\alpha_{i'i}$, q.e.d. $\big)$

For $i \in I$, let

$$\alpha_i \ : \ \underbrace{\mathrm{Hom}(\xi, \underbrace{x_i}_{=\xi})}_{=M} \ \longrightarrow \ \underbrace{\mathrm{Hom}(\xi, F) \ = \ F(\xi)}_{\simeq \varinjlim_i \mathrm{Hom}(\xi, x_i)}$$

be the canonical morphism. One finds this: p. 176

(4.15.8.4) If $u \in N_i$, then the equivalence relation R_i on M defined by α_i ($v \overset{R_i}{\sim} v' \iff \alpha_i(v) = \alpha_i(v')$) is equal to that defined by the map $v \longmapsto uv$ from M to M, i.e.

$$\alpha_i(v) \ = \ \alpha_i(v') \qquad \Longleftrightarrow \qquad uv \ = \ uv' \ .$$

This is immediate, since $I_{i,u}$ being cofinal in I, one has $\alpha_i(v) = \alpha_i(v')$ if and only if $\exists\, j \in I_{i,u}$ with $\alpha_{ji}v = \alpha_{ji}v'$, now $\alpha_{ji} = u$.

For $u \in M$, let us denote by R_u the equivalence relation on M defined by $v \longmapsto uv$. The set of equivalence relations on obtained in this way has cardinality $\leqslant \mathrm{card}\, M \leqslant \pi_0$, [page 455] hence the set of equivalence relations R_i ($i \in I$) also has cardinality $\leqslant \pi_0$. This shows, I being large relative to π_0, that there exists an equivalence relation of the form R_u ($u \in M$), and a cofinal set I' in I, such that $\forall\ i \in I'$ one has $R_i = R_u$. This shows that, at the cost of replacing I by the cofinal ordered set I', that one can suppose that one has

$$R_i \ = \ R_u \qquad \forall\, i \in I \ . \tag{4.15.8.5}$$

Let then

$$E \ = \ M/R_u \ = \ M/R_i \qquad (\forall\ i \in I) \ ,$$

one therefore sees that the $\alpha_{ji} \in M$ (for $i \leqslant j$) are such that the map

$$\tau_{\alpha_{ji}} \ : \ v \ \longmapsto \ \alpha_{ji}v \ : \ M \ \longrightarrow \ M$$

is compatible with the equivalence relation R_u, hence passes to the quotient as a map

$$\overline{\alpha_{ji}} \ : \ E \ \longrightarrow \ E \ ,$$

which is moreover *injective.*

Let M' be the submonoid of M formed of the $u \in M$ which have the preceding stability property, as well as the injectivity property we have just stated, and $\overline{M'}$ the image monoid of M' in the monoid

$$\mathrm{Mon}(E) \ \subset \ \mathrm{End}_{\mathrm{Ens}}(E)$$

of injective maps from E to itself. One therefore sees that the given direct system, defined by the α_{ji}, takes its values in the submonoid M' of M (i.e. of the corresponding [page 456] subcategory of the category B_M), and that the direct limit $F(\xi)$ which interests us (to prove $\mathrm{card}\, F(\xi) \leqslant \pi_0$) is isomorphic to the direct limit of the direct system of a system of sets all equal to E, the transition morphisms being the $\overline{\alpha_{ji}} \in \overline{M'} \subset \mathrm{Mon}(E)$. Moreover, one will have $\mathrm{card}\, \overline{M'} \leqslant \mathrm{card}\, M \leqslant \pi_0$. The desired result $\mathrm{card}\, F(\xi) \leqslant \pi_0$ will then follow from the

Key lemma 4.15.8.1. *Let M be a monoid of monomorphisms (i.e. such that $uv = uv' \Longrightarrow v = v'$, for $u, v, v' \in M$), and let $(\alpha_{ji})_{i \leqslant j}$ be a direct system in B_M, indexed by an ordered set I large relative to π_0, where π_0 is a cardinal which is infinite and such that* $\mathrm{card}\, M \leqslant \pi_0$. *Then there exists an $i_0 \in I$ such that for every $i \geqslant i_0$, α_{i,i_0} is an isomorphism.* p. 177 [page 457]

If one applies this lemma to the case of $\overline{M'}$ and of the $\overline{\alpha_{ji}}$, one therefore finds that there exists an $i_0 \in I$ such that the $\overline{\alpha_{ji_0}}$ are isomorphisms. This implies that $F(\xi) \simeq \varinjlim_{i} \underbrace{E_i}_{E}$ is isomorphic to E_{i_0}, hence has cardinality equal to $\operatorname{card} E_{i_0} = \operatorname{card} E \leqslant \pi_0$, which will complete the proof.

Moreover, statement 4.8.13 implies 4.15.8.1, for in the direct system of the $(M_i = M, \alpha_{ji})$, with $\varinjlim$ equal to $F(\xi)$, if there did *not* exist an i_0 as stipulated in 4.15.8.1, then one would easily conclude from this that one would have $\operatorname{card} F(\xi) > \pi_0$, contrary to 4.8.13. We are going to bring this out in a lemma:

[page 458]

Lemma 4.15.8.2. *Let $(E_i)_{i\in I}$ be a direct system of sets, indexed by an ordered set I large relative to π_0. We suppose that there exists $i_0 \in I$ such that*

a) $i \geqslant i_0$ implies $\operatorname{card} E_i = \pi_0$, *and*

b) $j \geqslant i \geqslant i_0$ implies that $u_{ji} : E_i \longrightarrow E_j$ is a monomorphism. Suppose moreover

c) $\forall\ i \geqslant i_0\ \exists\ j > i$ such that u_{ji} is not *an isomorphism (i.e. is not surjective). Let $E_\infty = \varinjlim_{i\in I} E_i$. Then* $\operatorname{card} E_\infty > \pi_0$.

This lemma therefore implies the *equivalence* of 4.8.13 and the "key lemma" 4.15.8.1. To prove 4.15.8.2, we are going to prove something a little more precise:

Corollary 4.15.8.3. *Under the conditions of 4.15.8.2, let α_0 be the first ordinal such that* $\operatorname{card} \alpha_0 > \pi_0$, *and let I_{α_0} be the set of ordinals $\beta < \alpha$ (an ordered set which is large relative to π_0). There then exists a strictly increasing map*

$$I_{\alpha_0} \longrightarrow I\ , \qquad \alpha \longmapsto i_\alpha\ , \qquad \textit{mapping } 0 \textit{ to } i_0\ ,$$

such that for every $\alpha \in I_{\alpha_0}$, one has

[page 459]

$$\underbrace{u_{i_{\alpha+1},i_\alpha}}_{\substack{\textit{it is a monomorphism,}\\ \textit{by b, since } i_{\alpha+1}>i_\alpha\geqslant i_0}} \qquad \textit{is not an isomorphism } (^{187}).$$

One then has

$$E'_\infty \overset{\text{def}}{=} \underbrace{\varinjlim_{\alpha\in I_{\alpha_0}} E_{i_\alpha}} \hookrightarrow \varinjlim_{i\in I} E_i \overset{\text{def}}{=} E_\infty$$

and

$$\operatorname{card} E'_\infty \geqslant \underbrace{\operatorname{card} I_{\alpha_0}}_{=\ \textit{successor of } \pi_0} > \pi_0\ ,$$

whence $\operatorname{card} E_\infty \geqslant \operatorname{card} E'_\infty > \pi_0$.

Proof. The construction of the i_α by transfinite induction is immediate, using c) and the fact that I is large relative to π_0, and that for every $\alpha \in I_{\alpha_0}$, the set I_α of $\beta < \alpha$ has cardinality $\leqslant \pi_0$. The fact that $E'_\infty \longrightarrow E_\infty$ is injective follows from b). The fact that

p. 178

$$\operatorname{card} E'_\infty \geqslant \operatorname{card} I_{\alpha_0}$$

follows from the fact that E'_∞ is an increasing union of subsets identified with E_{i_α}, and for every α one can find a

$$\xi_\alpha \in E_{i_\alpha} \setminus \bigcup_{\beta<\alpha_0} E_{i_\beta}\ .$$

[page 460]

The map

$$\begin{aligned} \alpha &\longmapsto \xi_\alpha \\ I_{\alpha_0} &\longrightarrow E'_\infty \end{aligned}$$

[187] *better*: $\forall\ \alpha \in I_{\alpha_0}\ \varinjlim_{\beta\in I_\alpha} E_{i_\beta} \longrightarrow E_{i_\alpha}$ is not surjective.

is injective, hence $\operatorname{card} E'_\infty \geqslant \operatorname{card} I_{\alpha_0}$.

Corollary 4.15.8.4. *For given M and π_0, for the key lemma 4.15.8.1 to be valid, it is necessary and sufficient that it be so in the case where $I = I_{\alpha_0}$, α_0 being the first ordinal such that $\operatorname{card}\alpha_0 > \pi_0$. It amounts to the same thing that one cannot find a direct system $(\alpha_{ji})_{j\geqslant i}$ indexed by I_{α_0}, with values in M, such that for $i < j$, one has α_{ji} noninvertible in M.*

[page 461]

I am going to summarize what has been obtained in a summary statement:

Theorem 4.15.9. *Let $\mathcal{C}$ be a small category, π a cardinal such that $\pi \geqslant \operatorname{card}\mathcal{C}$. The following conditions are equivalent (and I conjecture that they are always satisfied):*

a) $\mathcal{M} \overset{\text{def}}{=} \operatorname{Ind}_\pi\mathcal{C}$ *is reduced to* $\mathcal{M}_\pi$.

a′) *(If $\mathcal{C}$ is Karoubian.) Every direct system $\underline{x} = (x_i)_{i\in I}$ in $\mathcal{C}$, large relative to π, is essentially constant.*

a″) *For every direct system $(x_i)_{i\in I}$ in $\mathcal{C}$, large relative to π, there exists a cofinal functor $J \longrightarrow I$, with J an ordered set large relative to π, and such that the reindexed system $(x_j)_{j\in J}$ satisfies the following conditions:*

1) All the x_j are equal to one and the same element x of $\mathcal{C}$.

2) There exists a projector p in x, such that one has $\alpha_{j'j} = p$ for $j < j'$ $(j, j' \in J)$, where $\alpha_{j'j} : x_j \longrightarrow x_{j'}$ is the transition arrow.

b) *For every $x \in \operatorname{Ob}\mathcal{C}$, and for every quotient monoid M' of $M = \operatorname{End}_{\mathcal{C}}(x)$, such that M' is a monoid of monomorphisms (i.e. $uv = uv' \Longrightarrow v = v'$ if $u, v, v' \in M'$), and for every direct system large relative to π with values in $B_{M'}$ (defined by a system of elements $(\alpha_{ji})_{i\leqslant j,\ i,j\in I}$ of an ordered set I large relative to π, with $\alpha_{kj}\alpha_{ji} = \alpha_{ki}$ for $i \leqslant j \leqslant k$), there exists $i_0 \in I$ such that α_{ii_0} is invertible for $i \geqslant i_0$, i.e. α_{ji} is invertible for $j \geqslant i \geqslant i_0$.*

[page 462]

b′) *M' being a monoid of monomorphisms as in b) (hence $\operatorname{card} M' \leqslant \operatorname{card}\mathcal{C} \leqslant \pi$), let moreover α_0 be the smallest ordinal such that $\operatorname{card}\alpha_0 > \pi$, and let $I = I_{\alpha_0}$ be the ordered set of ordinals $\alpha < \alpha_0$ (which set is large relative to π). There exists* no *direct system in $B_{M'}$, such that for $i < j$ in I, α_{ji} is a noninvertible element in M' (hence $u \longmapsto \alpha_{ji}u : M' \longrightarrow M'$ is injective but not surjective).*

Let us point out at once the

p. 179

Corollary 4.15.10. *The preceding equivalent conditions are satisfied, if $\pi > \operatorname{card}\mathcal{C}$.*

Indeed, let α_1 be the smallest ordinal such that $\operatorname{card}\alpha_1 > \operatorname{card}\mathcal{C}$. Thus $\pi_1 \overset{\text{def}}{=} \operatorname{card}\alpha_1$ is the successor of $\pi_0 \overset{\text{def}}{=} \operatorname{card}\mathcal{C}$, and we have $\pi \geqslant \pi_1$, hence (if α_0 is as specified in b′ above) $\operatorname{card}\alpha_0 > \pi_1$, hence $\alpha_1 < \alpha_0$. Let (with the notation of b′), for $\alpha \in I$, i.e. $\alpha < \alpha_0$,

$$M_\alpha \;=\; \varinjlim_{i<\alpha} \underbrace{M_i}_{=M} \;,$$

we thus see, since the α_{ji} are injective and not surjective, that

$$\operatorname{card} M_\alpha \;\geqslant\; \operatorname{card}\alpha \;,$$

and that for $\alpha < \beta$, we have

[463]

$$M_\alpha \;\hookrightarrow\; M_\beta \qquad \text{injective} \;.$$

Let us apply this to the case where $\alpha = \alpha_1$, thus we find

$$\operatorname{card} M_{\alpha_1} \;\geqslant\; \operatorname{card}\alpha_1 \;=\; \pi_1 \;,$$

on the other hand, taking $\beta = \alpha_1 + 1$, whence $M_\beta \simeq M$, we find an injection

$$M_{\alpha_1} \;\hookrightarrow\; M \;,$$

whence

$$\pi_1 \leqslant \operatorname{card} M_{\alpha_1} \leqslant \operatorname{card} M \leqslant \pi_0 ,$$

a contradiction, which establishes b′.

Corollary 4.15.11. *Let $\mathcal{C}$ be an essentially small category, π a cardinal such that $\pi > \operatorname{card}\operatorname{red}\mathcal{C}$. Let $\mathcal{M} \overset{\text{def}}{=} \operatorname{Ind}_\pi \mathcal{C}$, then $\mathcal{M} = \mathcal{M}_\pi$, and it is a Karoubi envelope of $\mathcal{C}$.*

We may assume $\mathcal{C}$ reduced, hence $\operatorname{card}\mathcal{C} < \pi$, and we apply the preceding corollary.

Thus, the disputed result 4.8.13 (p. 212) is finally proved in a slightly weaker form, with $\pi > \operatorname{card}\operatorname{red}\mathcal{C}$ instead of $\pi \geqslant \operatorname{card}\operatorname{red}\mathcal{C}$. By the same token, the applications of 4.8.13 we have made so far (in particular lemma 4.15.3).

[464]

4.16 Examples and questions

4.16.0. I had thought that every "reasonable" functor (or more precisely, every "logically accessible" functor) between accessible categories ought to be accessible. This is unfortunately not true, since a composite of two Hom functors

$$\mathrm{Ens} \longrightarrow \mathrm{Ens} , \qquad E \longmapsto \underbrace{\mathrm{Hom}(\mathrm{Hom}(E, I), J)}_{J^{(I^E)}}$$

is not accessible, if I, J are both of cardinality $\geqslant 2$. Taking e.g. $I = J = \{0, 1\}$, we find the functor p. 180

$$E \longmapsto \mathfrak{P}(\mathfrak{P}(E)) ,$$

which is not accessible.

4.16.1. Second disillusionment: I had thought that for every accessible category $\mathcal{M}$, every strictly full subcategory $\mathcal{N}$ stable under direct factors and "reasonable" (e.g. "logically accessible") ought to be accessible in $\mathcal{M}$. This is not so, even in the particular case of the inverse image of the accessible subcategory in Ens [465]

$$\underbrace{\mathrm{Ens}^*}_{=\{E\in\mathrm{Ens}\,|\,E\neq\emptyset\}} \subset \mathrm{Ens}$$

under an accessible *representable* contrafunctor

$$\begin{array}{ccc} \mathcal{M} & \longrightarrow & (\mathrm{Ens})^\circ \\ x & \longmapsto & \mathrm{Hom}_{\mathcal{M}}(x, z) , \end{array}$$

i.e. the open (hence strictly full) subcategory $\mathcal{M}_z$ of $\mathcal{M}$ consisting of the objects x such that $\mathrm{Hom}_{\mathcal{M}}(x, z) \neq \emptyset$, is not always accessible in $\mathcal{M}$. Here, I have not arrived at as elementary an example (cf. appendix) – I took $\mathcal{M} = (k\text{-comm. alg.})$, k an infinite field and z the initial object k of $\mathcal{M}$ – I found that $\mathcal{M}_z$ is not L_π-stable in $\mathcal{M}$, whatever the cardinal π – but this example used that $\mathfrak{U} = \mathfrak{U}_1$, the smallest universe containing $\mathbf{N}$. I do not know whether for every other universe, one can find a field k of cardinality $\pi \in \mathfrak{U}$ large enough, so that every π-ultrafilter on a set $E \in \mathfrak{U}$ is trivial. Perhaps such an example, where not the construction of the pair $(\mathcal{M}, \mathcal{N})$, but the proof of the fact that $\mathcal{N}$ is not accessible in $\mathcal{M}$, appeals to particular hypotheses on the ambient universe, should not be considered as (altogether!) "logically accessible". To be really convincing, one would need an example of a more elementary nature. [466]

There remains the possibility that there is a metamathematical theorem ensuring that every logically accessible pair $(\mathcal{M}, \mathcal{N})$, with $\mathcal{M}$ accessible, and $\mathcal{N}$ a strictly full subcategory L_π-stable in $\mathcal{M}$ for π large enough, is accessible in $\mathcal{M}$ ([188]). This would

[188] This is false, even for a closed subcategory, cf. 4.16.7.10 below.

apply in particular to the case of a *closed* subcategory: every closed subcategory (that is logically accessible) would be accessible in $\mathcal{M}$. The simplest test example would be that of the closed complement $\mathcal{M}'_z$ of the open subcategory $\mathcal{M}_z$ above ([189]), i.e. the category of the $x \in \mathrm{Ob}\,\mathcal{M}$ such that $\mathrm{Hom}_{\mathcal{M}}(x,z) = \emptyset$, or again the inverse image of the accessible [467]
closed subcategory

$$e \;=\; \{\emptyset\} \;\subset\; (\mathrm{Ens})$$

reduced to $\{\emptyset\}$, under the representable functor $x \longmapsto \mathrm{Hom}(x,z) : \mathcal{M} \longrightarrow \mathrm{Ens}^\circ$. It could thus be that such a category is always accessible in $\mathcal{M}$. But I have been unable to prove p. 181
it, even in the particular case above, where $\mathcal{M} = (k\text{-comm. alg.})$, k being an infinite commutative field. The difficulty is (in all cases) the following: to find, for fixed z, a cardinal $\pi > \pi_0$ such that for every $x \in \mathrm{Ob}\,\mathcal{M}$, and every system (or *one* system) π_0-π-adapted $(x_i)_{i\in I}$ with direct limit x (π_0 a given cardinal, useful for $\mathcal{M}$), *if* $\mathrm{Hom}_{\mathcal{M}}(x,z)$ ($\simeq \varprojlim \mathrm{Hom}_{\mathcal{M}}(x_i,z)$) is empty, then $\exists\, i_0 \in I$ such that $\mathrm{Hom}_{\mathcal{M}}(x_{i_0},z)$ is already empty. In the example $\mathcal{M} = (k\text{-comm. alg.})$, one must therefore prove that one can find a cardinal π such that for every k-algebra A with $\mathrm{Hom}_{\mathcal{M}}(A,k) = \emptyset$, there exists a sub-$k$-algebra A_i of cardinality $\leqslant \pi$, such that $\mathrm{Hom}_k(A_i,k) = \emptyset$ ([190]). I tend to believe that this must be true (and even without assuming that k is a field, and also taking the condition [468]
$\mathrm{Hom}_{\mathcal{M}}(A,Z) = \emptyset$, where Z is an arbitrary object of $\mathcal{M}$). This sort of question seems surprisingly delicate …

4.16.2. It is possible, more generally, that the following question has an affirmative answer: Let

$$f \;:\; \mathcal{M} \longrightarrow \mathcal{N}^\circ$$

be an accessible functor, with $\mathcal{M}, \mathcal{N}$ accessible categories, and let $\underline{T}$ be a subcategory of $\mathcal{N}$ accessible in $\mathcal{N}$, and such that there exists a cardinal π having the following properties: $\mathcal{N}^\circ$ satisfies L_π (i.e. $\mathcal{N}$ stable under $\varprojlim$ large relative to π), and $\underline{T}$ is stable in $\mathcal{N}$ under these same limits. Then is $\underline{S} \overset{\mathrm{def}}{=} f^{-1}(\underline{T})$ accessible in $\mathcal{M}$ ([191])? This generalizes the case $\mathcal{N} = (\mathrm{Ens})$, $\underline{T} = \{\emptyset\}$, f a functor represented by $z \in \mathrm{Ob}\,\mathcal{M}$, considered just now. With the same type of example, but taking $\underline{T} = (\mathrm{Ens})^*$ instead of $\underline{T} = \{\emptyset\}$, we see that the [469]
additional stability condition on $\underline{T}$ is not superfluous. (**N.B.** It serves to ensure that $\underline{S}$ is L_π-stable in $\mathcal{M}$ for large π.)

4.16.3. Related question: I had hoped that for an accessible functor between accessible categories

$$f \;:\; \mathcal{N} \longrightarrow \mathcal{M}\ ,$$

the $\natural$-essential image, consisting of the $y \in \mathrm{Ob}\,\mathcal{M}$ which are isomorphic to a direct factor of an object of the form $f(x)$, with $x \in \mathrm{Ob}\,\mathcal{N}$, would be accessible in $\mathcal{M}$. The example above (not at all elementary!) of the subcategory $\mathcal{M}_z$ of $\mathcal{M}$ (consisting of the $x \in \mathrm{Ob}\,\mathcal{M}$ such that $\mathrm{Hom}_{\mathcal{M}}(x,z) \neq \emptyset$) shows that this is not so. Indeed, $\mathcal{M}_z$ is the essential image, and also the $\natural$-essential image, of the functor

$$\mathcal{M}/z \longrightarrow \mathcal{M}\ ,$$

now $\mathcal{M}/z$ is accessible, and the preceding functor accessible, if $\mathcal{M}$ is accessible (verification by AQT , cf. below, more general statement).

We get the impression that the reason why we have such counterexamples, is that [470]
we have $\natural$-essential images which are not L_π-stable in $\mathcal{M}$ for any cardinal $\pi \in \mathfrak{U}$. This leads to modifying the conjecture as follows: let π be a cardinal such that $\mathcal{M}$ satisfies L_π, and let $\underline{S}(\pi)$ be the smallest strictly full subcategory of $\mathcal{M}$, L_π-stable in $\mathcal{M}$, and p. 182

[189] **N.B.** it is accessible in $\mathcal{M}$ if and only if $\mathcal{M}_z$ is!

[190] This is false, cf. marginal annotation p. 466.

[191] False, with $\mathcal{N} = \mathrm{Ens}$ and a representable functor $\mathcal{M} \longrightarrow \mathrm{Ens}^\circ$, taking $\underline{T} = \{\emptyset\}$, cf. marginal

containing the essential image of f. (One could call $\underline{S}(\pi)$ the L_π-essential image.) Then $\underline{S}(\pi)$ is accessible in $\mathcal{M}$. Moreover, the (decreasing) family of the $\underline{S}(\pi)$, for varying π, is stationary, i.e. there exists a cardinal π such that $\pi' \geqslant \pi$ implies $\underline{S}(\pi') = \underline{S}(\pi)$. It is then clear that $\underline{S}(\pi)$ is the smallest subcategory of $\mathcal{M}$ that is accessible in $\mathcal{M}$ and contains the essential image of f. The same question arises for a functor $g^\circ : \mathcal{N}^\circ \longrightarrow \mathcal{M}$, where $g : \mathcal{N} \longrightarrow \mathcal{M}^\circ$ is an accessible functor, $\mathcal{N}$ and $\mathcal{M}$ accessible. I conjecture that the answer is again affirmative.

[471] Let $\mathcal{M}$ be a category satisfying condition L, and admitting a small strictly generating full subcategory $\mathcal{C}$, consisting of accessible objects. We thus have a fully faithful and accessible functor

$$\mathcal{M} \hookrightarrow \mathcal{C}^\wedge ,$$

$\mathcal{M}$ L_π-stable in $\mathcal{C}^\wedge$ for suitable π. If $\overline{\mathcal{M}}$ is the essential image, so that $\mathcal{M} \simeq \overline{\mathcal{M}}$, it is thus a strictly full subcategory L_π-stable in $\mathcal{C}^\wedge$. So we expect it to be accessible in $\mathcal{C}^\wedge$ (if the situation described is "logically accessible"), or what amounts to the same, $\mathcal{M}$ to be accessible. (Compare 4.8.1, p. 187.) But it should not be difficult ([192]), by making a number $\pi_{\mathfrak{U}}$ of choices (where $\pi_{\mathfrak{U}} = \operatorname{card} \mathfrak{U}$), to make examples all the same ("logically inaccessible") of a strictly full subcategory $\mathcal{M}$ of $\mathcal{C}^\wedge$ that is L_π-stable, but not accessible in $\mathcal{C}^\wedge$. This $\mathcal{M}$ thus admits a small strictly generating subcategory consisting of accessible objects ([193]) and satisfies L_π, without however being accessible.

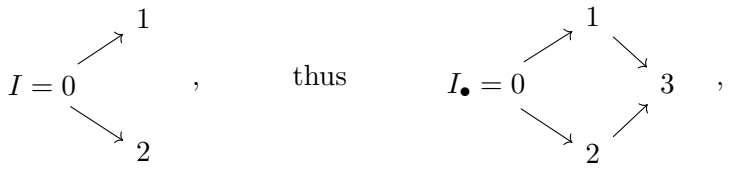
[472] **4.16.4** ([194]). Let $\mathcal{M}$ be a category such that $\mathcal{M}$ and $\mathcal{M}^\circ$ are accessible. I conjecture that then $\mathcal{M}$ is essentially small. More generally, suppose $\mathcal{M}$ accessible, and that $\mathcal{M}$ admits a small subcategory which is *strictly cogenerating*. Then is $\mathcal{M}$ essentially small? More generally, if $\mathcal{M}$ admits a small strictly generating subcategory, and also a small strictly cogenerating subcategory, i.e. if $\mathcal{M}$ and $\mathcal{M}^\circ$ are both equivalent to full subcategories of a $\mathcal{C}^\wedge$, with $\mathcal{C}$ small, then is $\mathcal{M}$ essentially small?

4.16.5. Let I be a small category, $I_\bullet$ the right cone on I (obtained by adding a final object e), $\mathcal{M}$ an accessible category,

$$\underline{\mathrm{Hom}}_\square(I_\bullet, \mathcal{M}) \hookrightarrow \underline{\mathrm{Hom}}(I_\bullet, \mathcal{M})$$

the strictly full subcategory of the (accessible) category $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{M})$, consisting of the functors f which make $f(e)$ the direct limit in $\mathcal{M}$ of the $f(i)$ ($i \in \mathrm{Ob}\, I$). This subcategory is L_π-stable for large π, so there is reason to conjecture that it is accessible in $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{M})$. [473]

(I recall that we saw in 4.14 that this is OK if the $\varinjlim$ of type I are representable in $\mathcal{M}$. And that an affirmative answer, in the case where p. 183



implies that the subcategory $\underline{\mathrm{Ép}}(\mathcal{M})$ in $\underline{\mathrm{Fl}}(\mathcal{M})$ is accessible in $\underline{\mathrm{Fl}}(\mathcal{M})$.)

N.B. If $\mathcal{M}$ admits a small *co*generating subcategory, then the question falls under conjecture 4.16.2 concerning an accessible functor $\mathcal{M} \longrightarrow \mathcal{N}^\circ$, here $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{M}) \longrightarrow \underline{\mathrm{Hom}}(I^\circ \times \mathcal{C}, \mathrm{Ens})^\circ \ldots$

---

annotations p. 466, 467.

[192] OK by taking a *closed* $\underline{S}$ not accessible in $\mathcal{M}$ of an accessible category $\mathcal{M}$, which we embed in a $\mathcal{C}^\wedge$. But these $\underline{S}$ do *not* have a small strictly generating family!

[193] This is not clear, for this the inclusion functor would have to be a left adjoint!

[194] move down after 4.16.5.

4.16.6. There remains the question of the relative accessibility of a category $\mathcal{E}$ over a small category $\mathcal{C}$. I now think, thanks to 4.15.9, and its corollary 4.15.10, that the following statement must "fall out" by AQT :

Proposition 4.16.6.1. *Let $f : \mathcal{E} \longrightarrow \mathcal{C}$ be a functor, with $\mathcal{C}$ a small Karoubian category. We assume that f satisfies condition $H' = H$ (1+2+3) of 4.15.2 (page 415), where H 2, H 3 suffice, for every arrow λ* other than an identity *of $\mathcal{C}$. We assume moreover that for a category $\mathcal{C}'$ of one of the two types we are going to describe, and a functor $\mathcal{C}' \longrightarrow \mathcal{C}$, $\mathcal{E}' = \mathcal{E} \times_{\mathcal{C}} \mathcal{C}'$ is accessible, and $\mathcal{E}' \longrightarrow \mathcal{C}'$ accessible.*

[474]

(*We take for $\mathcal{C}'$ the category*

$$\mathcal{Q} = \left\{ \begin{array}{l} \xi_0 = \xi^{\natural} \underset{\rho}{\overset{\sigma}{\rightleftarrows}} \xi \circlearrowright p \\ \qquad\qquad \downarrow u = u_0\rho \\ \qquad\quad \eta \\ \quad \nwarrow u_0 \end{array} \right. \qquad \begin{array}{l} \rho\sigma = \mathrm{id}_{\xi_0} \\ \sigma\rho = p\ , \quad \textit{whence} \left\{ \begin{array}{lcl} p^2 & = & p \\ p\sigma & = & \sigma \\ \rho p & = & \rho \end{array} \right. \\ u = u_0\rho\ , \quad \textit{whence} \left\{ \begin{array}{lcl} up & = & u \\ u\sigma & = & u_0 \end{array} \right. \end{array}$$

(three objects ξ, ξ_0, η and five nonidentity arrows σ, ρ, p, u_0, u, linked by the relations above), or the category (isomorphic to Q°)

$$\mathcal{Q}' = \left\{ \begin{array}{l} \xi_0 \underset{\rho}{\overset{\sigma}{\rightleftarrows}} \xi \circlearrowright p \\ \quad \nwarrow u_0 \quad \uparrow u \\ \qquad \eta \end{array} \right. \qquad \begin{array}{l} \left. \begin{array}{lcl} \rho\sigma & = & \mathrm{id}_{\xi_0} \\ \sigma\rho & = & p \end{array} \right\} \Longrightarrow \left\{ \begin{array}{lcl} p^2 & = & p \\ p\sigma & = & \sigma \\ \rho p & = & \rho \end{array} \right. \\ u = \sigma u_0 \quad \Longrightarrow \left\{ \begin{array}{lcl} pu & = & u \\ \rho u & = & u_0 \end{array} \right. \end{array}$$

We restrict ourselves to functors $\mathcal{Q} \longrightarrow \mathcal{C}$, $\mathcal{Q}' \longrightarrow \mathcal{C}$ such that p is transformed into an arrow other than an identity *of $\mathcal{C}$ ([195]) (thus the condition relative to $\mathcal{Q}$, $\mathcal{Q}'$ becomes empty, if $\mathcal{C}$ has no projectors).*)

Under these conditions, $\mathcal{E}$ is accessible, and $f : \mathcal{E} \longrightarrow \mathcal{C}$ is accessible.

We then find the

p. 184

[page 475]

Corollary 4.16.6.2. *Let $f : \mathcal{E} \longrightarrow \mathcal{C}$ be a functor. Equivalent conditions:*

a) For every category $\mathcal{C}'$ that is small *and* Karoubian*, and every functor $\mathcal{C}' \longrightarrow \mathcal{C}$, $\mathcal{E}' = \mathcal{E} \times_{\mathcal{C}} \mathcal{C}'$ is accessible, and the functor $\mathcal{E}' \longrightarrow \mathcal{C}'$ is accessible.*

b) As in a), but $\mathcal{C}'$ being one of the three categories Δ_1, $\mathcal{Q}$, $\mathcal{Q}'$ (where $\mathcal{Q}$, $\mathcal{Q}'$ are defined in 4.16.6.1 above).

b') ([196]). (If $\mathcal{C}$ is without projectors.) As in b), but restricting to the case $\mathcal{C}' = \mathbf{\Delta}_1$ (to the exclusion of $\mathcal{Q}$, $\mathcal{Q}'$).

Corollary 4.16.6.3. *Suppose $f : \mathcal{E} \longrightarrow \mathcal{C}$ is fibrant or cofibrant. Then the preceding equivalent conditions imply that:*

1°) The $\mathcal{E}_s$ are accessible ($s \in \mathrm{Ob}\,\mathcal{C}$).

2°) For every arrow $\lambda : s \longrightarrow t$ in $\mathcal{C}$, the functor $\lambda^* : \mathcal{E}_t \longrightarrow \mathcal{E}_s$ (resp. $\mathcal{E}_s \longrightarrow \mathcal{E}_t$) is accessible.

[page 476]

On the other hand, the collection of these conditions 1°, 2°) is equivalent to the validity of b) above, in the case $\mathcal{C}' = \mathbf{\Delta}_1$*, or again to the validity of a) above, when* $\mathcal{C}'$ *is* without projectors.

Question 4.16.6.4. *Do the preceding conditions 1° and 2° imply those of 4.16.6.2? In other words, if the base category* $\mathcal{C}$ *is equal to* $\mathcal{Q}$ *or to* $\mathcal{Q}'$*, then do these conditions imply that* $\mathcal{E}$ *is accessible, and* $f : \mathcal{E} \longrightarrow \mathcal{C}$ *accessible?*

It seems stupid not to know the answer!

4.16.6.5. I feel like calling a category *accessible over* $\mathcal{C}$, a category $\mathcal{E}$ over $\mathcal{C}$ satisfying the equivalent conditions of 4.16.6.2 – at least in the case where $\mathcal{C}$ is assumed small. For this terminology to be really satisfactory, we would have to know how to prove that these conditions imply this: for every accessible category $\mathcal{M}$ and every accessible functor $\mathcal{M} \longrightarrow \mathcal{C}$, the category $\mathcal{E} \times_{\mathcal{C}} \mathcal{M}$ is accessible, and the functor $\mathcal{E} \times_{\mathcal{C}} \mathcal{M} \longrightarrow \mathcal{M}$ is accessible.

[page 476bis]

The idea is that one would call (under a smallness hypothesis) *accessible category* $\mathcal{E}$ over another $\mathcal{C}$, a category having the property we have just stated – but now assuming in addition that $\mathcal{C}$ satisfies L_π, I imagine, for every accessible functor $\mathcal{M} \longrightarrow \mathcal{C}$, with $\mathcal{M}$ accessible, $\mathcal{E} \times_{\mathcal{C}} \mathcal{M}$ is accessible and the canonical functor $\mathcal{E} \times_{\mathcal{C}} \mathcal{M} \longrightarrow \mathcal{E}$ is accessible.

[page 477]

4.16.7. Return to the closed subcategories of an accessible category (4.16.1).

Proposition 4.16.7.1. *Let* $\mathcal{M}$ *be an accessible category,* $(\underline{S}_i)_{i\in I}$ *a family of closed subcategories accessible in* $\mathcal{M}$*. We do not assume that* $\operatorname{card} I$ *is small. Let* $\underline{S}$ *be the strictly full subcategory of* $\mathcal{M}$ *such that* $\operatorname{Ob}\underline{S} = \bigcup_{i\in I} \operatorname{Ob}\underline{S}_i$*. Then for* $\underline{S}$ *to be accessible* p. 185 *in* $\mathcal{M}$*, it is necessary and sufficient that there exist a* small *subset* $J \subset I$*, such that* $\operatorname{Ob}\underline{S} = \bigcup_{i\in J} \operatorname{Ob}\underline{S}_i$*. (In particular, if* I *is ordered, and* $i \longmapsto \underline{S}_i$ *increasing and large relative to every small cardinal, it is necessary and sufficient that there exist an* $i \in I$ *such that* $\underline{S} = \underline{S}_i$*, i.e. such that* $j \geqslant i$ *implies* $\underline{S}_j = \underline{S}_i$*.)*

This is obviously sufficient, by virtue of 4.13… Let us prove conversely that if $\underline{S}$ is accessible in $\mathcal{M}$, then the preceding condition is satisfied. Let π be a cardinal $\geqslant \pi_0$ such that $\underline{S}$ is π-accessible in $\mathcal{M}$, hence

$$\underline{S} \xleftarrow{\sim} \operatorname{Ind}_\pi \underline{S}(\pi) , \qquad \text{where } \underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi.$$

[page 478]

Let $\underline{S}(\pi)_0$ be a skeletal subcategory of $\underline{S}(\pi)$, such that $\underline{S}(\pi)_0 \hookrightarrow \underline{S}(\pi)$ is an equivalence, so that $\underline{S}(\pi)_0$ is small. For every $x \in \operatorname{Ob}\underline{S}(\pi)_0$, let $i_x \in I$ be such that $x \in \operatorname{Ob}\underline{S}_{i_x}$. Let J be the set of all the i_x, which is a small subset of I, and let $\underline{S}_J$ be the strictly full subcategory of $\mathcal{M}$ with set of objects $\bigcup_{j\in J} \operatorname{Ob}\underline{S}_j$. Thus $\underline{S}(\pi)_0$ is contained in $\underline{S}_J$, hence also $\underline{S}(\pi)$, since $\underline{S}_J$ is strictly full:

$$\underline{S}(\pi) \hookrightarrow \underline{S}_J .$$

Moreover $\underline{S}_J$ is accessible in $\mathcal{M}$ (loc. cit.), and is closed in $\mathcal{M}$ as a union of closed subcategories, *a fortiori it satisfies* L_π. Thus, since $\underline{S}(\pi) \subset \underline{S}_J$, the essentially surjective functor $\operatorname{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \underline{S}$ factors through $\underline{S}_J$. Hence $\underline{S}_J \hookrightarrow \underline{S}$ is essentially surjective, and since $\underline{S}_J$ is strictly full, we conclude that $\underline{S}_J = \underline{S}$, q.e.d.

[page 479]

Proposition 4.16.7.2. *Let* $\mathcal{M}$ *be an accessible category,* x *an object of* $\mathcal{M}$*, and* ${}_x\mathcal{M}$ *the closed subcategory of* $\mathcal{M}$ *generated by* x*, consisting of the* $y \in \operatorname{Ob}\mathcal{M}$ *such that* $\operatorname{Hom}(x, y) \neq \emptyset$*, i.e. the essential image of* $x\backslash\mathcal{M} \longrightarrow \mathcal{M}$*. Then* ${}_x\mathcal{M}$ *is a subcategory of* $\mathcal{M}$ *accessible in* $\mathcal{M}$*.*

[195] I am not sure if this is reasonable – to be checked with care!

[196] ?, to be checked carefully!

Indeed, it is the inverse image of the subcategory Ens* accessible in Ens under the representable (hence accessible) functor

$$h_x \ : \ y \longmapsto \mathrm{Hom}(x, y) \ : \ \mathcal{M} \ \longrightarrow \ \mathrm{Ens} \ ,$$

hence is accessible in $\mathcal{M}$ by virtue of 4.13…

Corollary 4.16.7.3. *Let $\mathcal{M}$ be an accessible category, $\underline{S}$ a closed subcategory. For $\underline{S}$ to be accessible, it is necessary and sufficient that there exist a* small *subset J of $\underline{S}$, such that* $\mathrm{Ob}\,\underline{S} = \bigcup_{x\in J} \mathrm{Ob}\,({}_x\mathcal{M})$.

Indeed, we have

$$\mathrm{Ob}\,\underline{S} \ = \ \bigcup_{x\in \mathrm{Ob}\,\underline{S}} \mathrm{Ob}\,({}_x\mathcal{M}) \ ,$$

i.e. $\underline{S}$ appears as the union of the family (not necessarily small) of closed subcategories [page 480] ${}_x\mathcal{M}$ of $\mathcal{M}$ (indexed by $x \in \mathrm{Ob}\,\underline{S}$), accessible in $\mathcal{M}$ by virtue of 4.16.7.2. Thus our assertion is a particular case of 4.16.7.1.

One can also, for every small subset J of $\mathrm{Ob}\,\underline{S}$, consider the closed subcategory $\underline{S}_J \subset \underline{S}$ p. 186 of $\mathcal{M}$ generated by J. The set I of the subsets J in question, ordered by inclusion, is large relative to every cardinal $\pi \in \mathfrak{U}$, and $\underline{S}$ is an increasing union of the $\underline{S}_J$. On the other hand, the proposition tells us that $\underline{S}$ is accessible in $\mathcal{M}$ if and only if there exists a $J \in I$, such that $\underline{S}_J = \underline{S}$, i.e. such that $\underline{S}_J = \underline{S}_{J'}$ for $J \subset J'$.

Proposition 4.16.7.4. *Let $\mathcal{M}$ be an accessible category. The following conditions are equivalent:*

a) There exists a subcategory $\underline{S}$ of $\mathcal{M}$ closed in $\mathcal{M}$, and not accessible in $\mathcal{M}$.

b) Let I be the set of ordinals α such that $\mathrm{card}\,\alpha \in \mathfrak{U}$*, i.e. of the ordinals α which are* [page 481] *in $\mathfrak{U}$. Thus $I = I_{\alpha_0}$, where α_0 is the smallest ordinal which does not belong to $\mathfrak{U}$, and I is large relative to every small cardinal π, i.e. relative to every cardinal in $\mathfrak{U}$ ([197]). The condition under consideration is the following:*

There exists a transfinite sequence

$$\underline{x} \ = \ (x_\alpha)_{\alpha\in I}$$

of elements of $\mathrm{Ob}\,\mathcal{M}$*, such that we have*

$$\mathrm{Hom}(x_\beta, x_\alpha) \ = \ \emptyset \qquad\qquad \text{if } \beta < \alpha.$$

To be precise, for such a transfinite sequence, let $\underline{S}$ or ${}_{\underline{x}}\mathcal{M}$ be the closed subcategory of $\mathcal{M}$ generated by the family of the x_α, i.e. such that $\mathrm{Ob}\,\underline{S} = \bigcup_{\alpha\in I} \mathrm{Ob}\ {}_{x_\alpha}\mathcal{M} = \bigcup_{\alpha\in I} x_\alpha \setminus \mathrm{Ob}\,\mathcal{M}$*. Then ${}_{\underline{x}}\mathcal{M}$ is a closed subcategory not accessible in $\mathcal{M}$, and every closed subcategory not accessible in $\mathcal{M}$ is obtained in this way.*

Proof. Let us consider a sequence $\underline{x} = (x_\alpha)_{\alpha\in I}$ as above. If J is a small subset of I, there exists an $\alpha \in I$ which is an upper bound for the $\beta \in J$, hence $\mathrm{Hom}(x_\beta, x_\alpha) = \emptyset$, for $\beta \in J$, which means that x_α does not belong to any of the $\underline{S}_\beta = {}_{x_\beta}\mathcal{M}$, for $\beta \in J$. [page 482] Thus $\underline{S}$ is not the union of a small subfamily of the family of the ${}_{x_\alpha}\mathcal{M}$, hence by virtue of 4.16.7.1, $\underline{S}$ is not accessible in $\mathcal{M}$.

Conversely, let $\underline{S}$ be such a subcategory in $\mathcal{M}$. Let us consider a well-order on $\mathrm{Ob}\,\underline{S}$ (objects s_α), and construct the sequence of the x_α ($\geqslant s_\alpha$) by transfinite induction ([198]).

[197] introduce $I = I_{\mathfrak{U}}$ before the statement of the proposition.

[198] **N.B.** We have $\mathrm{card}\,\underline{S} \notin \mathfrak{U}$, otherwise, $\underline{S}$ would be small, hence accessible in $\mathcal{M}$, contrary to the hypothesis.

1) We choose x_0 = first element of $\mathrm{Ob}\,\underline{S}$ (exists, since $\mathrm{Ob}\,\underline{S} \neq \emptyset$, i.e. $\underline{S} \neq \emptyset$, since otherwise, $\underline{S}$ would be accessible in $\mathcal{M}$).

2) If the x_β ($\beta < \alpha$) have been constructed, we take for x_α the first element of $\underline{S}$ bounding the x_β and which does not belong to the closed subcategory of $\mathcal{M}$ generated by the x_β ($\beta < \alpha$). (This closed subcategory is $\subset \underline{S}$, and $\neq \underline{S}$, since $\underline{S}$ is not accessible in $\mathcal{M}$.) Thus we do have

$$\mathrm{Hom}(x_\beta, x_\alpha) \;=\; \emptyset \qquad\qquad \text{if } \beta < \alpha.$$

Thus, the transfinite induction continues over all of I. We see by transfinite induction p. 187
that we have, for every $\alpha \in I$, [page 483]

$$s_\alpha \;\leqslant\; x_\alpha$$

(since $x_\alpha > x_\beta \geqslant s_\beta$ for $\beta < \alpha$, hence $x_\alpha > s_\beta$ for every $\beta < \alpha$, which is equivalent to $x_\alpha \geqslant s_\alpha$). Moreover, we also find, by transfinite induction, that

$$\beta \;<\; \alpha \qquad \Longrightarrow \qquad \begin{array}{l} s_\beta \text{ belongs to the closed subcat-} \\ \text{egory } \underline{S}_\alpha \text{ of } \mathcal{M} \text{ generated by the} \\ x_{\beta'} \text{ with } \beta' < \alpha. \end{array} \quad , \qquad (*)$$

since by construction x_α is the *first* of the elements of $\underline{S}$ which is *not* in the said subcategory $\underline{S}_\alpha$ of $\mathcal{M}$. This shows that every s_β belongs to the closed subcategory ${}_{\underline{x}}\mathcal{M}$ of $\mathcal{M}$ generated by the family of *all* the x_α, so we find that $\underline{S} \subset {}_{\underline{x}}\mathcal{M} \subset \underline{S}$, hence $\underline{S} = {}_{\underline{x}}\mathcal{M}$, q.e.d.

Proposition 4.16.7.5. *For a given universe $\mathfrak{U}$ (containing $\mathbf{N}$ as an element), the following conditions are equivalent:*

a) There exists a $\mathfrak{U}$-accessible category $\mathcal{M}$ having a closed subcategory not accessible in $\mathcal{M}$.

[page 484]

b) There exists a category $\mathcal{C} \in \mathfrak{U}$, and a transfinite sequence of elements of

$$\mathcal{C}^\wedge \;\overset{\text{def}}{=}\; \underline{\mathrm{Hom}}(\mathcal{C}^\circ, \mathrm{Ens}_\mathfrak{U}) \;,$$

$\underline{x} = (x_\alpha)_{\alpha \in I}$, indexed by the set of all ordinals $\alpha \in \mathfrak{U}$, and such that

$$\mathrm{Hom}(x_\alpha, x_\beta) \;=\; \emptyset \qquad\qquad \textit{if } \alpha < \beta.$$

Proof. We have b) $\Longrightarrow$ a) by taking $\mathcal{M} = \mathcal{C}^\wedge$, by virtue of the preceding proposition. We see likewise a) $\Longrightarrow$ b), by embedding $\mathcal{M}$ in a category $\mathcal{C}^\wedge$ ($\mathcal{C}$ being a small strictly generating subcategory of $\mathcal{M}$).

I have not been able ([199]) to construct a sequence $(x_\alpha)_{\alpha \in I}$ as required in this proposition, nor to prove that such a sequence does not exist, even in a particular case such as $\mathcal{M} = (k\text{-comm. alg.})$, k a commutative ring. I have not even proved that one cannot find a transfinite sequence of *fields* k_α, having the stated properties.

A fortiori, I have not been able to clear up the question whether an $\underline{\mathrm{L}}$-stable subcat- [page 485]
egory of an accessible category $\mathcal{M}$ is always accessible in $\mathcal{M}$ ([200]). It could be that there is no such case. It could also be that there are some, even with a closed subcategory (which is stable under every type of $\varinjlim$ representable in $\mathcal{M}$, except for the $\varinjlim$ over an empty index category ...).

For *open* subcategories, I recall that I constructed (in the appendix) an example of such a subcategory, not accessible in $\mathcal{M}$. Concerning open subcategories, I point out here the statement (almost tautological): p. 188

[199] But yes, in the end, cf. 4.16.7.10 below.

[200] This is false, cf. below. [page 486]

Proposition 4.16.7.6 ([201]). *Let $\mathcal{M}$ be an accessible category, $\underline{S}$ an* open *subcategory of $\mathcal{M}$. For $\underline{S}$ to be accessible in $\mathcal{M}$, it is necessary and sufficient that it be $\underline{L}$-stable in $\mathcal{M}$, i.e. that there exist a cardinal π_0 such that $\mathcal{M}$ satisfies L_{π_0}, and that $\underline{S}$ be stable in $\mathcal{M}$ under $\varinjlim$ of type L_{π_0}. More precisely, if this condition is satisfied, then for every cardinal π such that $\pi \geqslant \pi_0$ and that π is useful for $\mathcal{M}$, $\underline{S}$ is π-accessible in $\mathcal{M}$. Or again: if π is a useful cardinal for $\mathcal{M}$, then $\underline{S}$ is π-accessible in $\mathcal{M}$ if and only if it is L_π-stable in $\mathcal{M}$.*

This last assertion follows at once from the following corollary, itself tautological.

Corollary 4.16.7.7. *Let π be a cardinal as indicated above, let $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$, and let $x \in \mathrm{Ob}\,\mathcal{M}$ and*

$$x \simeq \varinjlim_{i \in I} x_i \qquad \begin{cases} x_i \in \mathrm{Ob}\,\mathcal{M}_\pi \\ I \text{ large relative to } \pi \end{cases}$$

be a representation of x as the essential image of an object of $\mathrm{Ind}_\pi \mathcal{M}_\pi$. Then $x \in \mathrm{Ob}\,\underline{S}$ $\iff x_i \in \mathrm{Ob}\,\underline{S}(\pi)\ \forall\, i \in I$.

I have finally cleared up the question of the accessibility of open subcategories and [page 487]
closed subcategories of an accessible category:

Proposition 4.16.7.8. *Let $\mathcal{M}$ be an accessible category, $\underline{S}$ a closed subcategory of $\mathcal{M}$, $\underline{T}$ the open subcategory complementary to $\underline{S}$, π an infinite cardinal useful for $\mathcal{M}$. The following conditions are equivalent:*

a) $\underline{S}$ is π-accessible in $\mathcal{M}$.

a′) $\underline{S}$ is the closed subcategory generated by $\underline{S}(\pi) \overset{\text{def}}{=} \underline{S} \cap \mathcal{M}_\pi$, i.e. for $x \in \mathrm{Ob}\,\mathcal{M}$, we have $x \in \mathrm{Ob}\,\underline{S}$ if and only if $\exists\, \xi \in \mathrm{Ob}\,\underline{S}(\pi)$ such that $\mathrm{Hom}_{\mathcal{M}}(\xi, x) \neq \emptyset$.

b) $\underline{T}$ is π-accessible in $\mathcal{M}$.

b′) $\underline{T}$ is $\underline{L}_\pi$-stable in $\mathcal{M}$.

Let us note at once the

Corollary 4.16.7.9. *For $\mathcal{M}$, $\underline{S}$, $\underline{T}$ as above, equivalent conditions:*

a) $\underline{S}$ is accessible in $\mathcal{M}$.

a′) There exists a small *subset A of $\mathrm{Ob}\,\mathcal{M}$, such that $\underline{S}$ is the closed subcategory ${}_A\mathcal{M}$ generated by A, i.e. $\underline{S} = \bigcup_{\xi \in A} {}_\xi\mathcal{M}$, where ${}_\xi\mathcal{M}$ is the closed subcategory of $\mathcal{M}$ consisting of the $x \in \mathrm{Ob}\,\mathcal{M}$ such that $\mathrm{Hom}(\xi, x) \neq \emptyset$.*

b) $\underline{T}$ is accessible in $\mathcal{M}$. [page 488]

p. 189

b′) $\underline{T}$ is $\underline{L}$-stable in $\mathcal{M}$.

Corollary 4.16.7.10. *Let $\mathcal{M}$ be an accessible category. The following conditions are equivalent* ([202]).

a) There exists a closed subcategory of $\mathcal{M}$, not accessible in $\mathcal{M}$.

b) There exists an open subcategory in $\mathcal{M}$, not accessible in $\mathcal{M}$.

[201] This proposition and its corollary are contained in 4.16.7.8 below, there is no need to make a separate statement of them.

[202] Example where these conditions are satisfied: $\mathcal{M} = (k\text{-comm. alg.})$, k an infinite commutative field, $\mathfrak{U}$ being the smallest universe such that $\mathbf{N} \in \mathfrak{U}$.

c) There exists a transfinite sequence

$$\underline{x} \;=\; (x_\alpha)_{\alpha\in I}$$

of objects of $\mathcal{M}$, indexed by the set I of ordinals of the reference universe $\mathfrak{U}$, such that we have

$$\mathrm{Hom}(x_\alpha, x_\beta) \;=\; \emptyset \qquad \textit{for } \alpha < \beta.$$

Indeed, the equivalence of a) and b) is clear from the preceding corollary, and that of a) and c) is none other than 4.16.7.4.

Proof of 4.16.7.8. We shall have to use the

Lemma 4.16.7.11. *Let $\mathcal{M}$ be an accessible category, π a useful cardinal for $\mathcal{M}$, $\underline{S}$ a full subcategory of $\mathcal{M}$, $\underline{S}(\pi) \overset{\text{def}}{=} \underline{S} \cap \mathcal{M}_\pi$, whence a fully faithful functor*

$$\varphi \;:\; \mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M}\ .$$

[page 489]

a) Let $x \in \mathrm{Ob}\,\mathcal{M}$. For x to belong to the essential image of the functor φ, it is necessary and sufficient that the full subcategory $\underline{S}(\pi)/x$ of the category $\mathcal{M}_\pi/x$ (which is filtering and even large relative to π) be cofinal (hence also large relative to π).

b) For $\underline{S}$ to be π-accessible in $\mathcal{M}$, it is necessary and sufficient that for every $x \in \mathrm{Ob}\,\mathcal{M}$, we have

$$x \;\in\; \mathrm{Ob}\,\underline{S} \qquad \Longleftrightarrow \qquad \underline{S}(\pi)/x \textit{ is cofinal in } \mathcal{M}_\pi/x$$

(or again, that $\underline{S}$ satisfy L_π, and that for every x in $\mathcal{M}$, we have

$$x \;\in \mathrm{Ob}\,\underline{S} \qquad \Longrightarrow \qquad \underline{S}(\pi)/x \textit{ cofinal in } \mathcal{M}_\pi/x.)$$

Proof. Part b) is a tautological consequence of a), which I shall prove. To say that x is in the essential image of φ, means that

$$x \;\simeq\; \varinjlim_{i\in I} x_i\ , \qquad (*)$$

with I large relative to π and the x_i in $\underline{S}(\pi)$. We know that $(*)$ implies (I being filtering, and $\mathcal{M}$ being equivalent to $\mathrm{Ind}_\pi(\mathcal{M}_\pi)$ since π useful for $\mathcal{M}$) that $I \longrightarrow \mathcal{M}_\pi/x$ is cofinal (4.4.2, page 79). Now this functor factors through the fully faithful inclusion functor $\underline{S}(\pi)/x \longrightarrow \mathcal{M}_\pi/x$, which is cofinal if and only if it satisfies condition F 1. It is immediate that this condition is satisfied when it is satisfied for the composite functor in question, which is indeed the case by hypothesis. p. 190

Conversely, suppose that $\underline{S}(\pi)/x$ is cofinal in $\mathcal{M}_\pi/x$. Since we have [page 490]

$$x \xleftarrow{\sim} \varinjlim_{\mathcal{M}_\pi/x} x_i\ ,$$

π being useful for $\mathcal{M}$, we will also have by cofinality

$$x \xleftarrow{\sim} \varinjlim_{\underline{S}(\pi)/x} x_i\ ,$$

now $\mathcal{M}_\pi/x$ being large relative to π, the same holds for the cofinal full subcategory $\underline{S}(\pi)/x$. Thus x is in the essential image of $\mathrm{Ind}_\pi\,\underline{S}(\pi)$, q.e.d.

[page 491]

Lemma 4.16.7.12. *Let I be a filtering category, J a closed subcategory, K the complementary open subcategory. We then have the equivalences*

$$J \neq \emptyset \iff J \text{ cofinal} \iff K \text{ not cofinal} \iff K \neq I\ .$$

Proof immediate.
I return to the proof of 4.16.7.8. We apply Lemma 4.16.7.11 to $\mathcal{M}$, π, and to the two subcategories $\underline{S}$, $\underline{T}$ of $\mathcal{M}$, noting that for $x \in \mathrm{Ob}\,\mathcal{M}$, $\underline{S}(\pi)/x$ is a closed subcategory of $\mathcal{M}_\pi/x$, whose open complement is $\underline{T}(\pi)/x$, so that we can apply Lemma 4.16.7.12. We find:

$$\begin{aligned}
\underline{S}\ \pi\text{-accessible} &\overset{4.16.7.11}{\iff} \forall\ x\ \in\ \mathrm{Ob}\,\mathcal{M},\ \text{we have } (x\ \in\ \mathrm{Ob}\,\underline{S} \iff \underline{S}(\pi)/x \text{ cofinal in } \mathcal{M}_\pi/x) \\
&\iff \forall\, x \in \mathrm{Ob}\,\mathcal{M},\ \text{we have } \Big(\underbrace{x \notin \mathrm{Ob}\,\underline{S}}_{\text{(i.e. } x \in \mathrm{Ob}\,\underline{T}\text{, by 4.16.7.12)}} \\
&\qquad \iff \underbrace{\underline{S}(\pi)/x \text{ not cofinal in } \mathcal{M}_\pi/x}_{\text{(i.e. } \underline{T}(\pi)/x \text{ cofinal in } \mathcal{M}_\pi/x\text{, by 4.16.7.12)}} \Big) \\
&\overset{4.16.7.11}{\iff} \underline{T} \text{ is } \pi\text{-accessible.}
\end{aligned}$$

We thus have the equivalence of a) and b). On the other hand, $\underline{S}(\pi)/x$ cofinal in $\mathcal{M}_\pi/x$ means simply $\underline{S}(\pi)/x \neq \emptyset$ (4.16.7.12), so $\underline{S}$ π-accessible means that $\underline{S}$ consists of the $x \in \mathrm{Ob}\,\mathcal{M}$ such that $\underline{S}(\pi)/x \neq \emptyset$, which also means that $\underline{S}$ is the closed subcategory in $\mathcal{M}$ generated by $\underline{S}(\pi)$, whence a $\iff$ a$'$. On the other hand, the equivalence b $\iff$ b$'$ is none other than 4.16.7.6 (last formulation).

[page 492]

4.17 Return to paratopoi, and to the operations $\mathcal{M} \boxtimes \mathcal{N}$, $\underline{\mathrm{Hom}}_!(\mathcal{M}, \mathcal{N})$

p. 191
[page 1]

Appendix

Ultraproducts of fields, and accessible categories

Let $(k_\xi)_{\xi \in E}$ be a family of fields, indexed by a set E. (Let Φ be a nontrivial filter in E, i.e. a subset $\Phi \subset \mathfrak{P}(E)$ such that:

a) $\emptyset \notin \Phi$, i.e. $A \in \Phi \Longrightarrow A \neq \emptyset$.

b) $A, B \in \Phi \Longrightarrow A \cap B \in \Phi$.

c) If $A \in \Phi$, $A \subset B$, then $B \in \Phi$.

d) $\Phi \neq \emptyset$ ($\iff E \in \Phi$).

(This amounts to saying that Φ is a closed subset of the ordered set $\mathfrak{P}(E)$, *left filtering*, and does not contain the least element of $\mathfrak{P}(E)$, i.e. not equal to all of $\mathfrak{P}(E)$.)) For every subset J of E, put

$$k_J\ =\ \prod_{\xi \in J} k_\xi\ ,$$

we thus have an inverse system

$$(k_J)_{J\in\mathfrak{P}(E)}\ , \qquad \text{i.e. } J\subset J' \text{ gives } k_{J'}\longrightarrow k_J\ ,$$

and the system

$$(k_J)_{J\in\Phi}$$

is a direct system of rings, indexed by the filtering ordered set Φ^o. Put

$$k_\Phi \ = \ \varinjlim_{J\in\Phi} k_J\ .$$

If Ψ is a filter finer than Φ, i.e. such that $\Psi\supset\Phi$, we find a canonical homomorphism [page 2]

$$k_\Phi\longrightarrow k_\Psi\ ,$$

so if $\mathcal{F}(E)=\mathcal{F}$ is the set of filters on E, ordered by inclusion, we find a direct system

$$\Phi\longmapsto k_\Phi\ :\ \mathcal{F}\longrightarrow \mathrm{Ann}\ .$$

We have not yet used that the k_ξ are fields, in fact it would suffice that they be objects of a category $\mathcal{A}$ stable under small products and under small filtering $\varinjlim$. Take now $\mathcal{A}=\mathrm{Ann}$ (category of unital rings, not necessarily commutative), and suppose that the k_ξ are fields.

Proposition 1. *Suppose that the k_ξ ($\xi\in E$) are* fields ([203]). *For k_Φ to be a field, it is necessary and sufficient that Φ be an ultrafilter, i.e. that for every decomposition $E=A\amalg B$, we have $A\in\Phi$ or $B\in\Phi$.* (NB If $A\in\Phi$, we have $B\notin\Phi$, since otherwise, we would have $A\cap B\in\Phi$, hence $\emptyset\in\Phi$, absurd.) p. 192

Necessity: Let, for every subset A of E,

$$f_A\in k_E \qquad \text{be defined by} \qquad f_A(\xi)\ =\ \begin{cases} 1_{k_\xi} & \text{if}\quad \xi\in A \\ 0_{k_\xi} & \text{if}\quad \xi\in E\backslash A\ . \end{cases}$$

Let [page 3]

$$f_{A,\Phi}\ \in k_\Phi$$

be the image of f_A under the canonical homomorphism $k_E\longrightarrow k_\Phi$. If $E=A\amalg B$, we will have

$$f_A\ +\ f_B\ =\ 1\ , \qquad f_A\, f_B\ =\ 0\ ,$$

whence

$$f_{A,\Phi}\ +\ f_{B,\Phi}\ =\ 1\ , \qquad f_{A,\Phi}\, f_{B,\Phi}\ =\ 0\ .$$

If k_Φ is a field, this implies that we have $f_{A,\Phi}=0$ or $f_{B,\Phi}=0$. But if e.g. $f_{A,\Phi}=0$, we see that this means that there exists a $J\in\Phi$ such that $f_A|J=0$, which also means that $J\subset B$. Likewise, if $f_{B,\Phi}=0$, we will have $J\subset A$. Thus Φ is an ultrafilter.

Conversely, suppose that Φ is an ultrafilter. Let us prove that k_Φ is a field, i.e. that for $f\in k_\Phi$, $f\neq 0\Longrightarrow f$ invertible. Note that f comes from an element of a k_J ($J\in\Phi$), hence is the canonical image of a $g\in k_E$. To say that $f=0$ means that $\exists\ J\in\Phi$ such that $g|J=0$. The opposite $f\neq 0$ therefore means that for every $J\in\Phi$, we have $g|J\neq 0$. Let then A be the set of $\xi\in E$ such that $g(\xi)\neq 0$, $B=E\backslash A$ the set of $\xi\in E$ such that $g(\xi)=0$. We thus have $B\notin\Phi$ since $f\neq 0$, and since Φ is an ultrafilter, we therefore [page 4]

[203] NB This condition is necessary if one wants k_Φ to be a field for every ultrafilter.

have $A \in \Phi$. But let $g' \in k_E$ be defined by $g'(\xi) = g(\xi)^{-1}$ if $\xi \in A$, $g'(\xi) = 1$ if $\xi \in B$, and let f' be the image of g' in k_Φ. We then see that we have

$$f\ f' \ = \ f'\ \ f \ = \ 1\ ,$$

which indeed shows that k_Φ is a field.
Suppose now that all the k_ξ ($\xi \in E$) are equal to one and the same field k. Then the k_J form an inverse system of k-algebras, the k_Φ a direct system of k-algebras. If k is commutative, the same holds for the k_J, k_Φ, and in particular for Φ an ultrafilter, k_Φ is an *extension* of the field k.
Let us return to the general case, but the k_ξ commutative. Let $\mathcal{F}_0 \subset \mathcal{F}$ be the set of ultrafilters on E. Note that for every $\Phi \in \mathcal{F}$, $k_E \longrightarrow k_\Phi$ is surjective. Thus if $\Phi \in \mathcal{F}_0$, k_Φ identifies with a field quotient of the ring k_E.
For every ring $\mathcal{A}$, let $\mathrm{Max}(\mathcal{A})$ be the maximal spectrum, i.e. the set of maximal ideals of $\mathcal{A}$. We thus find a canonical map

p. 193
[page 5]

$$\left\{\begin{array}{lcl} \mathcal{F}_0 & \longrightarrow & \mathrm{Max}(k_E) \\ \Phi & \longmapsto & \mathrm{Ker}(k_E \longrightarrow k_\Phi) \ \stackrel{\mathrm{def}}{=}\ \mathfrak{m}_\Phi\ . \end{array}\right. \qquad (*)$$

Proposition 2. *The preceding map is* bijective.
Let us first prove that it is injective, this follows at once from the
Corollary 1 ([204]). *Let, for $\Phi \in \mathcal{F}_0$,*

$$\chi_\Phi\ :\ k_E \longrightarrow k_\Phi$$

be the canonical map. Then for $A \in \mathfrak{P}(E)$, we have

$$A\ \notin\ \Phi \qquad \Longleftrightarrow \qquad \chi_\Phi(f_A)\ =\ 0\ , \quad \textit{i.e. } f_A \in \mathfrak{m}_\Phi$$

(i.e. $A \in \Phi \iff \chi_\Phi(f_A) \neq 0$, i.e. $f_A \notin \mathfrak{m}_\Phi$) ([205]).
Indeed, $\chi_\Phi(f_A) = 0$ means that there exists $B \in \Phi$ such that $f_A|B = 0$, i.e. such that $B \subset E\backslash A$; but this is equivalent to $E\backslash A \in \Phi$, hence to $A \notin \Phi$ since Φ is an ultrafilter, q.e.d.
Remains to prove that the map $(*)$ is surjective. So let $\mathfrak{m}$ be a maximal ideal of k_E, and let

[page 6]

$$\Phi\ \stackrel{\mathrm{def}}{=}\ \{A \in \mathfrak{P}(E)\ |\ \chi(f_A)\ =\ 1\}\ ,$$

where $\chi : k_E \longrightarrow k_E/\mathfrak{m} = \kappa(\mathfrak{m})$ is the canonical map. I say that Φ is an ultrafilter. Indeed:

a) $\emptyset \notin \Phi$ since $f_\emptyset = 0$, whence $\chi(f_\emptyset) = 0 \neq 1$ in $\kappa(\mathfrak{m}) = k_E/\mathfrak{m}$.

b) If $A, B \in \Phi$, we have $A \cap B \in \Phi$, since $f_{A\cap B} = f_A \cdot f_B$, whence $\chi(f_{A\cap B}) = \chi(f_A)\chi(f_B) = 1$.

c) If $A \subset B$ and $A \in \Phi$, then $B \in \Phi$. Indeed, we have $f_A = f_A f_B$, whence $\chi(f_A) = \chi(f_A)\chi(f_B)$, and since $\chi(f_A) = 1$, we find $\chi(f_B) = 1$.

---

[204] **NB** For arbitrary $\Phi \in \mathcal{F}$, $\mathfrak{m}_\Phi \stackrel{\mathrm{def}}{=} \mathrm{Ker}(\chi_\Phi : k_E \longrightarrow k_\Phi)$ determines Φ, since

$$A\ \in\ \Phi \qquad \Longleftrightarrow \qquad \chi_\Phi(f_A)\ =\ 1$$

(but one must assume the $K_\xi \neq 0$, i.e. $0_\xi \neq 1_\xi$ ...).

[205] **NB** We thus have

$$(\chi_\Phi(f_A)\ \neq\ 0) \qquad \Longleftrightarrow \qquad (\chi_\Phi(f_A)\ =\ 1)\ .$$

d) We have $E \in \Phi$, i.e. $\chi(f_E) = 1$, now $f_E = 1$, so $\chi(f_E) = 1$.

Thus, Φ is a filter – we have used only the fact that $k_E/\mathfrak{m}$ is a ring $\neq 0$ (to conclude that $\emptyset \notin \Phi$). Let us prove that if $k_E/\mathfrak{m}$ is an integral domain, Φ is an *ultrafilter*, i.e.:

e) If $A \notin \Phi$, then $B \overset{\text{def}}{=} E\backslash A \in \Phi$.

Indeed, we have $f_A + f_B = 1$, $f_A f_B = 0$, whence
ⓐ $\chi(f_A) + \chi(f_B) = 1$, and p. 194
ⓑ $\chi(f_A)\chi(f_B) = 0$.
$\chi(f_A) \neq 1$, we conclude from ⓐ $\chi(f_B) \neq 0$, whence by ⓑ $\chi(f_A) = 0$, whence from ⓐ $\chi(f_B) = 1$, whence $B \in \Phi$.

Let us finally prove that $\mathfrak{m} = \mathfrak{m}_\Phi$. Since $\mathfrak{m}_\Phi$ is a maximal ideal, it suffices to prove [page 7]

$$\mathfrak{m}_\Phi \subset \mathfrak{m} .$$

Now if $A \in \mathfrak{P}(E)$ is such that $A \notin \Phi$, or $f_A \in \mathfrak{m}_\Phi$ (cor. 1), we also have $f_A \in \mathfrak{m}$. Thus it suffices to prove the

Lemma. *Let $\Phi \in \mathcal{F}_0$. Then $\mathfrak{m}_\Phi$ is the ideal generated by the f_A for $A \in \mathfrak{P}(E)\backslash\Phi$. More precisely, the f_A are in $\mathfrak{m}_\Phi$, and every $f \in \mathfrak{m}_\Phi$ is of the form ff_A, with A as above.*

Indeed, $f \in \mathfrak{m}_\Phi$ means that there exists $B \in \Phi$ such that $f|B = 0$. But this also means that $f = ff_A$, where $A = E\backslash B$, whence $A \notin \Phi$.

This proves proposition 2, and at the same time:

Corollary 2. *Every quotient ring of $k_E = \prod_{\xi\in E} k_\xi$ which is an integral domain is a field.*

Let us now note that if $\Phi \in \mathcal{F}$ is any filter on E, then k_Φ is a quotient ring of k_E, hence

$$\operatorname{Max}(k_\Phi) \hookrightarrow \underbrace{\operatorname{Max}(k_E)}_{\simeq \mathcal{F}_0} . \qquad (*)$$

This being so, we find:

Corollary 3 ([206]). *The preceding inclusion identifies* $\operatorname{Max}(k_\Phi)$ *with the set of ultrafilters Ψ on E which are* finer *than Φ.*

Indeed, if Ψ is such an ultrafilter, we have the commutative diagram [page 8]

$$k_E \xrightarrow{\chi_\Phi} k_\Phi \xrightarrow[\chi_\Psi]{} k_\Psi \quad , \qquad (**)$$

which shows that $\mathfrak{m}_\Psi$ is in the image of $(*)$. Conversely, suppose that $\Psi \in \mathcal{F}_0$ comes from $\operatorname{Max}(k_\Phi)$, i.e. that we have $\mathfrak{m}_\Phi \subset \mathfrak{m}_\Psi$ (where $\mathfrak{m}_\Phi$ is the kernel of χ_Φ), or again that we have a factorization $(**)$ of χ_Ψ. Let us prove that Ψ is finer than Φ, i.e. that $\Psi \supset \Phi$. So let $A \in \Phi$, let us prove that we have $A \in \Psi$. But $A \in \Phi$ implies fact that $\chi_\Phi(f_A) = 1$, hence its image in k_Ψ is equal to 1, hence we have $A \in \Psi$ by corollary 1.

Let us now suppose that all the k_ξ are equal one and the same ring $\boxed{k \neq 0}$. Thus p. 195
k_J, k_Φ are "rings under k" $k \longrightarrow k_J \longrightarrow k_\Phi$, moreover $k \longrightarrow k_J$ (if $J \neq \emptyset$) and $k \longrightarrow k_\Phi$ [page 9]
are injective. We ask the question when $k \longrightarrow k_\Phi$ is *bijective*, i.e. surjective. Since $k_\Phi \simeq k_E/\mathfrak{m}_\Phi$, this therefore means that $k_E = k + \mathfrak{m}_\Phi$, i.e. that every $f \in k_E$ can be written

$$f = c + g \qquad \begin{cases} c \in k , & \text{i.e. } c(\xi) = \underbrace{c_0}_{\in k} \ \forall \xi \in E , \\ g \in \mathfrak{m}_\Phi & \text{i.e. } \exists A \in \Phi \text{ with } g|A = 0 . \end{cases}$$

[206] Works for any system of rings $(k_\xi)_{\xi\in E}$ *which are nonzero*, and $\Phi, \Psi \in \mathcal{F}$, Ψ finer than Φ, i.e. $\Phi \subset \Psi$ if and only if $\mathfrak{m}_\Phi \subset \mathfrak{m}_\Psi$ (no need for Ψ to be an ultrafilter).

But for $f \in k_E$, $c \in k$ and $A \in \Phi$ fixed, the existence of $g \in k_E$ such that $g|A = 0$ and $f = c + g$ is equivalent to $f|A = c_A$ (constant function on A, with value c), i.e. $A \subset f^{-1}(\{c\})$. For f, c fixed, the existence of $A \in \Phi$ satisfying the relation is equivalent to $f^{-1}(\{c\}) \in \Phi$. And for f fixed, the existence of c is equivalent to that of a fiber E_c of $f : E \longrightarrow k$, which is in Φ. Thus:

Proposition 3. *Let $\Phi \in \mathcal{F}$. For $k \longrightarrow k_\Phi$ to be surjective, it is necessary and sufficient that for every family of subsets* essentially disjoint *$(E_\xi)_{\xi \in k}$ of E indexed by k, with union E, there exist a $\xi \in k$ such that $E_\xi \in \Phi$. A fortiori, it* is necessary *that Φ be an ultrafilter (since* $\operatorname{card} k \geqslant 2$ *by hypothesis).*

Indeed, giving such a family $(E_\xi)_{\xi \in k}$ is equivalent to giving a map $E \longrightarrow k$, i.e. an element of k_E. [page 10]

Scholie. The property we have just made explicit for a filter Φ on E, depends on k only through $\pi = \operatorname{card} k$. We can also state it by saying that for every *partition* $(E_i)_{i \in I}$ of E into a set I of subsets such that $\operatorname{card} I \leqslant \pi$, there is one (and only one) of the E_i which is in Φ. We then say that Φ is a *π-ultrafilter.*

If $\pi \geqslant 2$ is finite, this property is equivalent to saying that Φ is an ultrafilter. Besides, whatever π, the ultrafilters Φ *which are trivial* (i.e. such that there exists $\xi \in E$ with $\{\xi\} \in \Phi$, or again such that $\Phi = \{A \in \mathfrak{P}(E) \mid \xi \in A\}$) are π-ultrafilters. If π is infinite, it is not clear that there exist nontrivial π-ultrafilters. We know that such exist on E *only* if $\operatorname{card} E \notin \mathfrak{U}_1$, where $\mathfrak{U}_1$ is the smallest universe such that $\mathbf{N} \in \mathfrak{U}_1$, i.e. such that $\mathfrak{U}$ contains as an element an infinite set. [page 11]

If $\mathfrak{U} = \mathfrak{U}_1$, and if k is an infinite field, e.g. $k = \mathbf{Q}$, then for every set $E \in \mathfrak{U}$ and for every *nontrivial* ultrafilter Φ on E, k_Φ is a *nontrivial* extension of k. But if $\mathfrak{U} = \mathfrak{U}_\alpha$ is an arbitrary universe, and π the first cardinal in $\mathfrak{U}$ which does not belong to a strictly smaller universe $\mathfrak{U}' \subset \mathfrak{U}$, I presume that for every $E \in \mathfrak{U}$, every π-ultrafilter on E is trivial.

p. 196

Application to an example of a nonaccessible subcategory.

Let $\mathcal{M}$ be a $\mathfrak{U}$-accessible category, and $a \in \operatorname{Ob} \mathcal{M}$, whence a functor

$$\begin{aligned} f^o \ : \ \mathcal{M}^o \ &\longrightarrow \ \mathrm{Ens} \\ x \ &\longmapsto \ \mathrm{Hom}_{\mathcal{M}}(x, a) \ \overset{\text{def}}{=} \ a(x) \ , \end{aligned}$$

a functor commuting with arbitrary $\varprojlim$, hence such that $f : \mathcal{M} \longrightarrow (\mathrm{Ens})^o$ commutes with arbitrary $\varinjlim$, and in particular f is accessible. Let us consider the strictly full subcategory [page 12]

$$\underbrace{\mathrm{Ens}^*}_{=\mathrm{Ens}\setminus\{\emptyset\}} \ \subset \ \mathrm{Ens}$$

of nonempty sets, we know that it is accessible, but it is not stable under $\varprojlim$ of type L_π, however large we take π.

Let us consider

$$\underbrace{\underline{S} \ = \ \underline{S}_a}_{\overset{\text{def}}{=} f^{-1}(\mathrm{Ens}^*)} \ \subset \ \mathcal{M} \ ,$$

strictly full subcategory of $\mathcal{M}$ consisting of the $x \in \operatorname{Ob} \mathcal{M}$ such that

$$a(x) \ \neq \ \emptyset \ , \qquad \text{i.e. } x \leqslant a \text{ for the canonical preorder of } \mathcal{M} \ .$$

It is therefore an *open* subcategory of $\mathcal{M}$. Its closed complement $\underline{S}' = \underline{S}'_a$ is the closed subcategory which is the inverse image, under f, of the strictly full subcategory of Ens

reduced to the single object $\{\emptyset\}$. This subcategory of Ens is the standard final category. It is stable in Ens under arbitrary $\varinjlim$, and under arbitrary filtering $\varprojlim$. Thus, $\underline{S}'$ is stable in $\mathcal{M}$ under arbitrary filtering $\varinjlim$. Thus, $\underline{S}'$ is stable in $\mathcal{M}$ under arbitrary filtering $\varinjlim$, in particular, it is L_π-stable for π sufficiently large. Let us make this explicit: [page 13]

$$\underbrace{\underline{S}' = \underline{S}'_a}\ \subset\ \mathcal{M}$$

$$\Big\| \text{ def} \qquad \text{closed subcategory of } \mathcal{M}$$

$$f^{-1}(\{\emptyset\}) = \text{consisting of the } x \in \operatorname{Ob}\mathcal{M} \text{ such that } a(x) = \emptyset.$$

Since $\underline{S}'$, being closed in $\mathcal{M}$, is stable under nonempty $\varinjlim$ and in particular L_π-stable for π large, there is more reason to expect $\underline{S}'$ to be accessible in $\mathcal{M}$ than for its complement $\underline{S}$, for which it is not clear that it has this stability property, but only stability under nonempty $\varprojlim$, and in particular under direct factors.

In fact, we shall see that it can happen that $\underline{S}$ and $\underline{S}'$ are both nonaccessible in $\mathcal{M}$:

Proposition 4. *Let $\mathfrak{U}$ be a universe, containing an infinite cardinal π_0 such that for every $E \in \mathfrak{U}$, every π_0-ultrafilter on E is trivial. Let then $k \in \mathfrak{U}$ be a field of cardinality $\pi'_0 \geqslant \pi_0$, and let $\mathcal{M}$ be the category of commutative k-algebras. Thus $\mathcal{M}$ is* p. 197
a $\mathfrak{U}$-paratopos, admitting k as initial object. Let $\underline{S}'$ be the strictly full subcategory, stable [page 14]
under small nonempty $\varinjlim$, consisting of the objects A of $\mathcal{M}$ such that $\operatorname{Hom}_{\mathcal{M}}(A,k) = \emptyset$, and $\underline{S}$ the complementary strictly full subcategory, consisting of the objects A such that $\operatorname{Hom}_{\mathcal{M}}(A,k) \neq \emptyset$. Then $\underline{S}'$ is a subcategory closed *in $\mathcal{M}$, and $\underline{S}$ the complementary open subcategory, hence $\underline{S}$ (just as $\underline{S}'$) is stable under direct factors. This being so:*

a) There is no cardinal π such that $\underline{S}$ is L_π-stable in $\mathcal{M}$, a fortiori $\underline{S}$ is not accessible in $\mathcal{M}$.

b) ([207]). Although $\underline{S}'$ is L_0-stable in $\mathcal{M}$ (i.e. stable under filtering $\varinjlim$, and even, as we have said, under every nonempty $\varinjlim$), $\underline{S}'$ is nevertheless not accessible in $\mathcal{M}$.

Proof. Assertion a) will follow from the

Corollary 1. *Under the conditions of proposition 4, let $E \in \mathfrak{U}$ be an infinite set, Φ_0 the filter of complements of finite subsets of E. Then $\operatorname{Hom}_{\mathcal{M}}(k_{\Phi_0}, k) = \emptyset$, a fortiori, for every filter Φ on E finer than Φ_0, we have $\operatorname{Hom}_{\mathcal{M}}(k_\Phi, k) = \emptyset$.*

Here we have put, of course, for every filter Φ on E, [page 15]

$$k_\Phi = \varinjlim_{J \in \Phi} k_J\ .$$

We know that for every commutative k-algebra $\mathcal{A}$, $\operatorname{Hom}(\mathcal{A}, k)$ is in 1-1 correspondence with the subset of $\operatorname{Max}(\mathcal{A})$ consisting of the maximal ideals $\mathfrak{m}$ of $\mathcal{A}$ such that $k \longrightarrow \mathcal{A}/\mathfrak{m}$ is an isomorphism. If $\mathcal{A}$ is of the form k_Φ, Φ a filter on E, this set is in one-to-one correspondence with the set of π'_0-ultrafilters Ψ on E finer than Φ, where $\pi'_0 = \operatorname{card}(k)$ (proposition 3, and cor. 3 of prop. 2, page 7). Since $\pi'_0 \geqslant \pi_0$, such an ultrafilter is trivial by hypothesis. Thus we find

$$\operatorname{Hom}_{\mathcal{M}}(k_\Phi, k) \simeq \text{ set of } \xi \in E \text{ such that } \xi \in \textstyle\bigcap_{A \in \Phi} A\ . \qquad (*)$$

But saying that Φ is finer than Φ_0 also means that for every $\xi \in E$, we have $E\backslash\{\xi\} \in \Phi$, i.e. $\exists\, A \in \Phi$ with $A \subset E\backslash\{\xi\}$, i.e. $\xi \notin \bigcap_{A\in\Phi} A$. Hence we then have $\operatorname{Hom}_{\mathcal{M}}(k_\Phi, k) = \emptyset$.

[207] ?, not proved.

The conclusion of corollary 1 is expressed by the relation

$$k_\Phi \notin \operatorname{Ob} \underline{S} , \qquad \text{i.e. } k_\Phi \in \operatorname{Ob} \underline{S}' ,$$

on the other hand, we have by definition

$$k_\Phi = \varinjlim_{J \in \Phi} k_J ,$$

and it is clear (since these $J \in \Phi$ are $\neq \emptyset$), that the k_J in question are in $\underline{S}$. Thus, [page 16]
for a filter Φ as in corollary 1, we find a direct system of objects of $\underline{S}$, indexed by Φ^o, whose direct limit is not in $\underline{S}$. Thus a) of proposition 4 follows by noting that for every cardinal $\pi \in \mathfrak{U}$, we can find a set $E \in \mathfrak{U}$, and a filter Φ on E finer than Φ_0, such that the ordered set Φ^o is large relative to π. To find such a Φ, it suffices to take E of cardinality $\pi' > \pi$, and for Φ the set of complements of subsets of E of cardinality $\leqslant \pi$, so that Φ^o p. 198
is isomorphic to the ordered set of subsets of E of cardinality $\leqslant \pi$, which is indeed large relative to π. This proves a) of proposition 4.

Let us now prove b). Let us note the

Lemma. *Let $\mathcal{M}$ be an accessible category stable under $\varinjlim$ of type I_α, α a given limit ordinal, $\underline{S}'$ a closed subcategory, π a useful cardinal for $\mathcal{M}$, such that* $\operatorname{card} \alpha \leqslant \pi$. *For* [page 17]
every $x \in \operatorname{Ob} \underline{S}'$, let us choose an ω-π-adapted representation (ω being the first infinite ordinal)

$$x \simeq \varinjlim_{i \in I_x} x_i$$

(thus I_x large relative to π, stable under $\varinjlim$ of type I_α, $i \longmapsto x_i : I \longrightarrow \mathcal{M}$ commutes with the said limits, finally the $x_i \in \operatorname{Ob} \mathcal{M}_\pi$). For $\underline{S}'$ to be π-accessible in $\mathcal{M}$, it is necessary and sufficient that for every $x \in \operatorname{Ob} \underline{S}'$, there exist an $i \in I_x$ such that $x_i \in \operatorname{Ob} \underline{S}'$.

Indeed, note that $\underline{S}'$, being closed, is stable in $\mathcal{M}$ under every nonempty $\varinjlim$ representable in $\mathcal{M}$, and in particular under $\varinjlim$ of type I_α. So we can apply 4.13.22 (p. 342) to $\mathcal{M}$, $\underline{S}'$, π and to $\Phi = \{I_\omega\}$. Then a) $\Longrightarrow$ c) implies that if $\underline{S}'$ is π-accessible in $\mathcal{M}$, then for every $x \in \operatorname{Ob} \underline{S}'$, the set $I_{x,S'}$ of the $i \in I_x$ such that $x_i \in \operatorname{Ob} \underline{S}'$ is cofinal in I_x. This implies that it is nonempty. (And conversely too, $\underline{S}'$ being a closed subcategory in $\mathcal{M}$, since if $x_i \in \underline{S}'$, then for $j \in \operatorname{Ob} I_x$ with $i \leqslant j$, we have $x_j \in \underline{S}'$, i.e. $I_{x,\underline{S}'}$ contains the essential image of $i \backslash I_x$, which is cofinal in I …) This proves the "only if". Let us prove the "if". So let us prove that $\underline{S}'$ is equal to the essential image of $\operatorname{Ind}_\pi(\underline{S}'(\pi))$. It contains it, since $\underline{S}'$ is stable under $\varinjlim$ of type L_π.

So it remains to prove that every $x \in \operatorname{Ob} \underline{S}'$ is in the essential image. Now the [page 18]
hypothesis assures us that $x \simeq \varinjlim_{i \in I_x} x_i$ is isomorphic to $\varinjlim_{i \in I_{x,\underline{S}'}} x_i$ (for $I_{x,\underline{S}'}$ is cofinal in I_x), where this time the x_i are in $\underline{S}'$. As I_x large relative to π, likewise the cofinal full subcategory $I_{x,\underline{S}'}$, so we win.

Corollary. *Under the preceding conditions, if $\underline{S}'$ is π-accessible, the same is true of the complementary open subcategory $\underline{S}$,* provided *that $\underline{S}$ is stable in $\mathcal{M}$ under $\varinjlim$ of type L_π.*

We must prove that $\underline{S}$ is equal to the essential image of $\operatorname{Ind}_\pi \underline{S}(\pi)$. The "provided" hypothesis ensures that this essential image is $\subset \underline{S}$. Conversely, every $x \in \operatorname{Ob} \underline{S}$ is in this essential image. We will have

$$x \simeq \varinjlim_{i \in I} x_i , \qquad x_i \in \operatorname{Ob} \mathcal{M}_\pi,\ I = \mathcal{M}_\pi / x, \text{ large relative to } \pi ,$$

and since $x \in \operatorname{Ob} \underline{S}$ and $\underline{S}$ open, the x_i are in $\underline{S}$, hence in $\underline{S}(\pi)$, OK. [page 19]

In the case which concerns us, where $\mathcal{M} = (k\text{-alg.})$, as $\underline{S}$ is precisely *not* stable in $\mathcal{M}$ under L_π (whatever π) (that is what made us conclude that $\underline{S}$ is not accessible in $\mathcal{M}$), we therefore cannot infer from this inaccessibility of $\underline{S}$, that of $\underline{S}'$. Let us note the

p. 199

Corollary. *Let π be an infinite cardinal. For the closed subcategory $\underline{S}'$ of $\mathcal{M}$ (consisting of commutative k-algebras A such that $\mathrm{Hom}_{\mathcal{M}}(A, k) = \emptyset$) to be π-accessible, it is necessary and sufficient that for every k-algebra A (of cardinality $> \pi$), such that $\mathrm{Hom}_{\mathcal{M}}(A, k) = \emptyset$, there exist a subalgebra A_i* of cardinality $\leqslant \pi$, *such that* $\mathrm{Hom}_{\mathcal{M}}(A_i, k) = \emptyset$.
(We apply the lemma, taking for α the first infinite ordinal.)

Question. *Does this hold, simply taking $\pi \geqslant \mathrm{card}\, k$?* The question of the existence of a cardinal π, having the property indicated above, seems delicate to me, and I do not know the answer. If one wants to construct a counterexample for every π, the preceding developments suggest taking for A an infinite product of nontrivial extensions of k, or a ring k_Φ as in the proof of proposition 4 a).

THE DERIVATORS

p. (unnumbered)

ALEXANDRE GROTHENDIECK

Chapter XIX

Models (4)

p. (unnumbered)
p. 1
[page 1]

1 Reminders on Quillen couples (special Quillen couples)

I am going to take up again the notations and terminology for Quillen's LLP and RLP relations. I write (for a category $\mathcal{M}$, and $\Phi, \Psi \subset \mathrm{Fl}(\mathcal{M})$)

$$\Phi \;\text{---}\!\!\rightarrowtail\; \Psi \qquad \Longleftrightarrow \qquad (\Phi \subset \Psi^*) \qquad \Longleftrightarrow \qquad (\Psi \subset \Phi_*)\ , \tag{1.1}$$

i.e. Φ has the LLP with respect to Ψ, or again Ψ the RLP with respect to Φ. I also say that (Φ, Ψ) is a *couple* which is Q-*orthogonal.* This notion is not symmetric, but I sometimes allow myself (exceptionally) to write

$$\Psi \;\leftarrowtail\!\!\text{---}\; \Phi \qquad \text{as a synonym for} \qquad \Phi \;\text{---}\!\!\rightarrowtail\; \Psi\ . \tag{1.2}$$

Φ is Q-*orthogonal on the left* to Ψ, Ψ is Q-*orthogonal on the right* to Φ, and Q-orthogonality on the left (resp. on the right) is denoted by the half-arrowhead pointing downward (resp. upward). I write

$$\begin{cases} \Phi \;\leftarrow\!\!\text{---}\; \Psi & \Longleftrightarrow \quad (\Phi = \Psi^*) \\ \Phi \;\text{---}\!\!\rightarrow\; \Psi & \Longleftrightarrow \quad (\Psi = \Phi_*) \\ \Phi \;\leftarrow\!\!\text{---}\!\!\rightarrow\; \Psi & \Longleftrightarrow \quad (\Phi = \Psi^* \ \text{ and } \ \Psi = \Phi_*)\ . \end{cases} \tag{1.3}$$

These three cases of strengthened Q-orthogonality thus respectively imply $\Phi = \tilde{\Phi}$ resp. $\Psi = \bar{\Psi}$ resp. $\Phi = \tilde{\Phi}$ and $\Psi = \bar{\Psi}$. We say that Φ is Q-*left closed* if $\Phi = \tilde{\Phi}$, that Ψ is Q-*right closed* if $\Psi = \bar{\Psi}$, that the Q-orthogonal couple (Φ, Ψ) is Q-*left closed* (resp. *right*) if Φ (resp. Ψ) is left Q-closed (resp. right). We say that it is Q-*closed* if the two conditions

[page 2]

are satisfied simultaneously. When no confusion is to be feared, I shall sometimes say simply that Φ is Q-closed, or that Ψ is Q-closed, without specifying whether it is on the left or on the right. But for a Q-orthogonal couple (Φ, Ψ) (where I prefer to write it $\Phi \;\text{---}\!\!\rightarrowtail\; \Psi$), Q-closed always means left and right Q-closed.

When a Q-orthogonal couple $\Phi \;\text{---}\!\!\rightarrowtail\; \Psi$ has Quillen's factorization property, I indicate it by the vertical bar crossing the horizontal, thus

$$\Phi \;\rightarrowtail\!\!\!\!\mid\; \Psi\ , \tag{1.4}$$

and we define in the same way the meaning of the graphisms

$$\begin{cases} \Phi \;\rightarrowtail\!\!\!\!\mid\; \Psi \\ \Phi \;\leftarrow\!\!\!\!\mid\; \Psi \\ \Phi \;\rightarrowtail\!\!\!\!\mid\!\!\rightarrow\; \Psi \\ \Phi \;\leftarrow\!\!\!\!\mid\!\!\rightarrow\; \Psi\ . \end{cases} \tag{1.5}$$

p. 2
[page 3]

A Q-*orthogonal* Q-*factorizing* couple $\Phi \rightarrowtail\!\!\!\!\!\!\leftarrow \Psi$ is called a *Quillen precouple.* If it is moreover Q-closed, i.e. if we have $\Phi \longleftrightarrow \Psi$, we say that it is a *Quillen couple.*

Φ is said to be Q-*left saturated* if it contains the isomorphisms and is stable under cobase change, under (small) transfinite composition, and under direct factors. Dually Ψ is said to be Q-*right saturated* if it

contains the isomorphisms, is stable under base change, under transfinite composition, and under direct factors. If Φ is left Q-saturated, it is stable under (small) direct sums, i.e. if the $f_i : A_i \longrightarrow B_i$ ($i \in I$, I small) are in Φ, $f = \coprod f_i : \coprod A_i \longrightarrow \coprod B_i$ also, and likewise for (small) finite or infinite amalgamated sums (when we work in an $A\backslash\mathcal{M}$ instead of $\mathcal{M}$). We have the dual statement for Ψ, when Ψ is right Q-saturated.

In fact, there are weaker notions of Q-saturation than the one I have just given: Q-*simple saturation* (on the left: Φ contains the isomorphisms, stable under binary composition, under cobase change), Q-*transfinite saturation* (we require in addition stability under transfinite composition), finally Q-*total saturation,* or Q-saturation for short (when we require in addition stability under direct factors). We have the implications

$$\begin{cases} \text{left Q-closed} \\ \Downarrow \\ \text{Q-saturated (total)} \\ \Downarrow \\ \text{transfinitely Q-saturated} \\ \Downarrow \\ \text{simply Q-saturated.} \end{cases} \tag{1.6}$$

For $\Phi \subset \mathrm{Fl}(\mathcal{M})$, we have its (left) Q-closure $\tilde{\Phi}$ (which is the smallest left Q-closed subset of $\mathrm{Fl}(\mathcal{M})$ containing Φ) and likewise its (total) left Q-saturation, its transfinite Q-saturation, its simple Q-saturation

with the tautological inclusions [page 4]

$$\Phi \subset \underbrace{\Phi'}_{\substack{\text{Q-saturation} \\ \text{simple on the left}}} \subset \underbrace{\Phi''}_{\substack{\text{Q-saturation} \\ \text{transfinite on the left}}} \subset \underbrace{\Phi'''}_{\substack{\text{Q-saturation} \\ \text{total on the left}}} \subset \underbrace{\tilde{\Phi}}_{\substack{\text{Q-closure} \\ \text{on the left}}} , \tag{1.7}$$

and dually for Ψ. Between Φ and Φ' there is also a Φ'_0,

$$\Phi \subset \Phi'_0 \subset \Phi' , \tag{1.8}$$

which is the Q-*saturation under cobase change* (also containing, by definition, all isomorphisms). To construct all these saturations, we proceed as follows:

1) Φ'_0 = set of arrows $i : A \longrightarrow B$ fitting into a cocartesian square p. 3

$$\begin{array}{ccc} A_0 & \xrightarrow{i_0} & B_0 \\ \downarrow & & \downarrow \\ A & \xrightarrow{i} & B , \end{array}$$

with $i_0 \in \Phi$ or i_0 an identity (in which case we find for i the isomorphisms).

2) Φ' = set of multiple composite arrows

$$i_n\, i_{n-1} \cdots i_1 ,$$

with $i_\alpha \in \Phi'_0$.

3) Φ'' = set of transfinite composites of arrows $\in \Phi'_0$.

We feel like putting

$$\Phi''' = \Phi''^{\natural} \overset{\text{def}}{=} \text{set of direct factors of elements of } \Phi'',$$

and it is a fact that $\Phi''^{\natural} \subset \Phi'''$ trivially, but it is not clear in general that we have equality, for lack of being able to show that $\Phi''^{\natural}$ is stable, not only under direct factors, but also under cobase change and under transfinite composition. However, let us recall: [page 5]

Theorem 1.1. *Suppose $\mathcal{M}$ stable under small $\varinjlim$, and that its elements are accessible. Suppose moreover that Φ is small, more generally, that there exists a small subset $\Phi_0 \subset \Phi$ such that $\Phi \subset \tilde{\Phi_0}$, hence*

$$\Psi \overset{\text{def}}{=} \Phi_* = (\Phi_0)_* \quad {}^{1}. \tag{1.1.1}$$

Under these conditions, if Φ''_0 denotes the transfinite left Q*-saturation of Φ_0, then we have $\Phi''_0 \rightarrowtail\!\!\!\!\!\rightarrow \Psi$, i.e. every arrow f of $\mathcal{M}$ factors as $f = pi$, with $i \in \Phi''_0$, $p \in \Psi$ [2]. A fortiori, if Φ'' denotes the transfinite left* Q*-saturation of Φ, we have*

$$\Phi'' \rightarrowtail\!\!\!\!\!\rightarrow \Psi .$$

Corollary 1.2. *Under the conditions of the theorem, we have*

$$\Phi''' \text{ (total left Q-saturation of } \Phi) = \underbrace{\Phi''^{\natural}}_{\substack{\text{set of} \\ \text{direct factors of elements} \\ \text{of } \Phi''}} = \tilde{\Phi} . \tag{1.2.1}$$

More generally (in appearance only!), if Φ_1 is a subset of Fl($\mathcal{M}$) *which is transfinitely left* Q*-saturated, and such that*

$$\Phi_0 \subset \Phi_1 \subset \tilde{\Phi} = \tilde{\Phi_0} ,$$

then p. 4 [page 6]

a) *one has $\Phi_1 \rightarrowtail\!\!\!\!\!\rightarrow \Psi$ and*

b) $\Phi_1^{\sim} = \Phi^{\sim} = \Phi_1^{\natural}$.

This corollary follows from the theorem by virtue of XV prop. 4.2, of which I recall here the essentials:

Proposition 1.3. *Let $\Phi \rightarrowtail\!\!\!\!\!\rightarrow \Psi$. Then $\Phi^{\sim} = \Phi^{\natural} = \Psi^*$, and dually one has $\Psi^- = \Psi^{\natural} = \Phi_*$.*

Let us also note, in this line of thought:

Corollary 1.4. *Let $\Phi \rightarrowtail \Psi$, and let W be an arbitrary subset of* Fl($\mathcal{M}$)*. In order to have $\Phi \subset W^{\natural}$, it suffices that every arrow of $\mathcal{M}$ factor as $f = pi$, with $i \in W$, $p \in \Psi$. (And this condition is obviously also necessary if $\Phi \rightarrowtail\!\!\!\!\!\rightarrow \Psi$, and W stable under direct factors.)*

One says that a subset Φ_0 of Φ is Q-*generating on the left* for Φ if it satisfies the condition

$$\Phi \subset \Phi_0^{\sim} , \tag{1.9}$$

and one says that Φ is a *special* subset if Φ admits a *small* left Q-generating subset. If $\Phi_0 \subset \Phi$ is small, and if $\mathcal{M}$ satisfies the conditions of theorem 1.1, then Φ_0 is a Q-generating subset (on the left) if and only if its total Q-saturation (on the left) Φ_0''' [page 7]

contains Φ (a condition apparently stronger than $\Phi \subset \Phi_0^{\sim}$, since in general one has $\Phi_0''' \subset \Phi_0^{\sim}$). This therefore also means, via 1.2, that every element of Φ is a direct factor of a transfinite composite of arrows obtained from

$$\Phi_0 \cup (\text{Identities of } \mathcal{M})$$

by cobase change.

I recall also the ± tautological sorites of adjoint functors:

Proposition 1.5. *Let*

$$\mathcal{M}' \underset{v^*}{\overset{v_!}{\leftrightarrows}} \mathcal{M} \tag{1.5.1}$$

be a pair of adjoint functors, let

$$\Phi \ \subset \ \mathrm{Fl}(\mathcal{M}) \ , \qquad \Psi' \ \subset \ \mathrm{Fl}(\mathcal{M}') \ ,$$

then one has

$$(v_!(\Phi) \longmapsto \Psi') \qquad \Longleftrightarrow \qquad (\Phi \longmapsto v^*(\Psi')) \ . \tag{1.5.2}$$

Corollary 1.6. *For every subset* Φ *of* $\mathrm{Fl}(\mathcal{M})$,

$$(v_!(\Phi))_* \ = \ (v^*)^{-1}(\Phi_*) \ . \tag{1.6.1}$$

Indeed, applying 1.5 to Φ and to $\{f'\}$, where $f' \in \mathrm{Fl}(\mathcal{M}')$, one finds

$$f' \in (v_!(\Phi))_* \qquad \Longleftrightarrow \qquad v^*(f') \in \Phi_* \ ,$$

which is equivalent to (1.6.1). Dually: p. 5

Corollary 1.7. *For every subset* Ψ' *of* $\mathrm{Fl}(\mathcal{M}')$*, one has*

$$(v^*(\Psi'))^* \ = \ (v_!)^{-1}(\Psi'^*). \tag{1.7.1}$$

Indeed, if $f \in \mathrm{Fl}(\mathcal{M})$, one has by (1.5.2)

$$f \in (v^*(\Psi'))^* \qquad \Longleftrightarrow \qquad v_!(f) \in \Psi'^* \ .$$

(3).

A Q-orthogonal couple $\Phi \longmapsto \Psi$ is said to be *special* if it satisfies the following conditions.

$$\begin{cases} \text{a)} & \mathcal{M} \text{ is stable under small } \varinjlim \text{, the objects of } \mathcal{M} \text{ are accessible.} \\ & \text{(One then says that } \mathcal{M} \text{ is "special".)} \\ \text{b)} & \text{The couple is closed: } \Phi \longleftrightarrow \Psi. \\ \text{c)} & \Phi \text{ is } \textit{special.} \end{cases} \tag{1.10}$$

[page 8]

This therefore implies, by 1.1, that $\Phi \longleftrightarrow \Psi$ is a Quillen couple. If $\mathcal{M}$ satisfies a), the special Q-orthogonal couples in $\mathcal{M}$, or again the special Quillen couples, thus correspond one-to-one to the subsets Φ of $\mathrm{Fl}(\mathcal{M})$ which are *special* and left closed (or what amounts to the same thing, left Q-saturated), by associating to such a Φ the couple $\Phi \longleftrightarrow \Phi_*$.

This being so:

Corollary 1.8. *Suppose that* $\mathcal{M}$*,* $\mathcal{M}'$ *satisfy condition* a) *of* (1.10)*, and let* (Φ, Ψ) *be a special Quillen couple in* $\mathcal{M}$*. Then* $\Big(\Phi' = v_!(\Phi)^{\sim}, \Psi' = (v^*)^{-1}(\Psi)\Big)$ *is a special Quillen couple in* $\mathcal{M}'$*.*

Indeed, by corollary 1.6 one has

$$\Phi'_* \ = \ (v_!(\Phi))_* \ = \ (v^*)^{-1}(\Psi) \ = \ \Psi' \ ,$$

hence $\Phi' \longleftrightarrow \Psi'$. It remains to prove that Φ' is special. Let Φ_0 be a small subset of Φ ‖ such that $\Phi \subset \Phi_0^{\sim}$, put $\Phi'_0 \ = \ v_!(\Phi_0) \subset \Phi'$, it suffices to see that one has

[1] **N.B.** We thus have $\Phi \longmapsto \Psi$, a fortiori $\Phi_0 \longmapsto \Psi$.

[2] There is even *functorial* factorization in f, a property which deserves a notation, e.g. $\Phi_0'' \rightarrowtail\!\!\!\!\!\rightarrow \Psi$.

[3] give here 1.9 page 9. [page 9]

$$v_!(\Phi) \ \subset \ v_!(\Phi_0)^\sim \ , \qquad (*)$$

from which it will follow that one has $\Phi' \overset{\text{def}}{=} (v_!(\Phi))^\sim \subset v_!(\Phi_0)^\sim$. But one has

$$v_!(\Phi_0)^\sim \ = \ (v_!(\Phi_0)_*)^* \ ,$$

and p. 6

$$v_!(\Phi_0)_* \ = \ (v^*)^{-1}(\underbrace{\Phi_{0\,*}}_{\Psi}) \ = \ (v^*)^{-1}(\Psi) \ \overset{\text{def}}{=} \ \Psi' \ ,$$

hence

$$v_!(\Phi_0)^\sim \ = \ \Psi'^* \ = \ \Phi' \ = \ v_!(\Phi)^\sim \ ,$$

whence $(*)$. This argument does not appeal to the "special" hypothesis on $\mathcal{M}$, and can be made explicit in the

Corollary 1.9. [(4)] *Under the general conditions of* 1.5 *($v_! : \mathcal{M} \longrightarrow \mathcal{M}'$ being a functor admitting a right adjoint), let Φ_0, Φ be two subsets of* $\mathrm{Fl}(\mathcal{M})$ *such that $\Phi \subset \Phi_0^\sim$ (or what amounts to the same thing, $\Phi^\sim \subset \Phi_0^\sim$). Then $v_!(\Phi) \subset v_!(\Phi_0)^\sim$, i.e. $v_!(\Phi)^\sim \subset v_!(\Phi_0)^\sim$.*

Indeed, this relation is also equivalent to $v_!(\Phi_0)_* \ \subset \ v_!(\Phi)_*$, that is (by 1.6) $(v^*)^{-1}(\Phi_{0\,*}) \ \subset \ (v^*)^{-1}(\Phi_*)$, and is implied by $\Phi_{0\,*} \ \subset \ \Phi_*$. The latter on the other hand is equivalent to the hypothesis $\Phi^\sim \subset \Phi_0^\sim$, q.e.d.

If Ψ is a subset of $\mathrm{Fl}(\mathcal{M})$, one obtains from it for every I in Cat a subset [page 10]

$$\begin{cases} \Psi(I) \subset \mathrm{Fl}(\mathcal{M}(I)), \quad \text{with} \ \ \mathcal{M}(I) = \underline{\mathrm{Hom}}(I^\circ, \mathcal{M}) \ , \\ f \in \Psi(I) \iff f(i) \in \Psi \ \ \forall\, i \in \mathrm{Ob}\, I \ . \end{cases} \qquad (1.11)$$

If one introduces the discrete category I_0 underlying $\mathrm{Ob}\, I$, and the canonical functor

$$v_I = v \ : \ I_0 \longrightarrow I \ , \qquad (1.12)$$

one finds a pair of adjoint functors

$$\begin{array}{c} \mathcal{M}(I) \underset{v^*}{\overset{v_!}{\leftrightarrows}} \mathcal{M}(I_0) \\ \big(\ \leftarrow\!\cdots\!\cdots \ \big) \\ {\scriptstyle v_*} \end{array} \quad \simeq \ \mathcal{M}^{I_0} \ , \qquad (1.13)$$

and one has

$$\Psi(I) \ = \ (v^*)^{-1}(\Psi(I_0)) \ .$$

Suppose now that Φ contains the identities and that one has

$$\begin{cases} \Psi \ = \ \Phi_* \\ \text{whence} \ \ \Psi(I_0) \ = \ \Phi(I_0)_* \end{cases} \qquad (1.14)$$

(immediate verification), one then concludes by 1.6

$$\Psi(I) \ = \ (v_!(\Phi(I_0)))_* \ = \ (\Phi\langle I\rangle)_* \ , \qquad (1.15)$$

with

$$\Phi\langle I\rangle \ \overset{\text{def}}{=} \ v_!(\Phi(I_0))^\sim \ = \ \Psi(I)^* \ . \qquad (1.16)$$

If one has a functor p. 7

$$u \ : \ I \longrightarrow I' \ , \qquad (1.17)$$ [page 11]

whence a triple of adjoint functors

$$\left\{ \mathcal{M}(I) \underset{u_*}{\overset{u_!}{\underset{\longleftarrow}{\overset{\longrightarrow}{\longrightarrow}}}} \mathcal{M}(I') \right. , \tag{1.18}$$

one will have in a $\pm$ tautological way

$$\left\{ \begin{array}{rcl} u_!(\Phi\langle I\rangle) & \subset & \Phi\langle I'\rangle \\ u^*(\Psi(I')) & \subset & \Psi(I) \ . \end{array} \right. \tag{1.19}$$

Corollary 1.8 gives:

Proposition 1.10. *Suppose that* $\Phi \longleftarrow\!\!\!\!\!\!+ \Psi$ *is a* special *Quillen couple in* $\mathcal{M}$*, i.e. of the form* (Φ, Φ_*)*, with* Φ *a special left Q-closed subset in* $\mathcal{M}$ *(*$\mathcal{M}$ *satisfying condition* (1.10. a))*. Then for every* I *in* Cat, $(\Phi\langle I\rangle, \Psi(I))$ *is a special couple in* $\mathcal{M}(I)$*, in particular it satisfies the factorization theorem.*

Let us consider the dual construction, which leads us to introduce, starting from a couple $\Phi \longleftarrow\!\!\!\!\!\!+ \Psi$ in $\mathcal{M}$, the couple

$$\Phi(I) \ \longleftarrow\!\!\!\!\!\!+ \ \Psi[I] \ , \tag{1.20}$$

where $\Phi(I)$ is defined by the general formula (1.11) (where Ψ is replaced by Φ), and where $\Psi[I]$ is defined either as $\Phi(I)_*$, or as

$$\Psi[I] \ = \ v_*(\Psi(I_0))^- \ = \ \Phi(I)_* \ . \tag{1.21}$$

One then has, for $u : I \longrightarrow I'$

[page 12]

$$\left\{ \begin{array}{rcl} u^*(\Phi(I')) & \subset & \Phi(I) \\ u_*(\Psi[I]) & \subset & \Psi[I'] \ . \end{array} \right. \tag{1.22}$$

There still arises the question whether the couple $(\Phi(I), \Psi[I])$ satisfies the factorization theorem, under suitable hypotheses on (Φ, Ψ). I can only venture the

Question-conjecture 1.11. *Suppose that* (Φ, Ψ) *is a* special *Quillen couple, is the same true of* $(\Phi(I), \Psi[I])$ *for every* I *in* (Cat)*? In other words, if* $\mathcal{M}$ *is special, condition* (1.10. a))*, and if* Φ *is a* special *left Q-closed subset of* $\mathcal{M}$*, i.e. admitting a small Q-generating subset, is it true that for every* I *in* Cat, $\Phi(I)$ *is also special? I conjecture that yes, making if need be additional hypotheses on* $\mathcal{M}$ *(existence of a cardinal filtration?). E.g. if* Φ_0 *is Q-generating in* Φ*, and small, then is* $\Phi_0(I)$ *Q-generating in* $\Phi(I)$*?*

Proposition 1.12 [5]. *Let* $\Phi \longleftarrow\!\!\!\!\!\!+ \Psi$ *in* $\mathcal{M}$*, let* $u : I \longrightarrow I'$ *be an arrow in* Cat*, whence*

$$\left\{ \mathcal{M}(I) \underset{u_*}{\overset{u_!}{\underset{\longleftarrow}{\overset{\longrightarrow}{\longrightarrow}}}} \mathcal{M}(I') \right. \ . \tag{1.12.1}$$

One then has (1.19)*, i.e.*

$$\left\{ \begin{array}{rcl} u_!(\Phi\langle I\rangle) & \subset & \Phi\langle I'\rangle \\ u^*(\Psi(I')) & \subset & \Psi(I) \ , \end{array} \right. \tag{1.13.1}$$

p. 8

and moreover the equivalence

$$u^*(\Phi\langle I'\rangle) \subset \Phi\langle I\rangle \qquad \Longleftrightarrow \qquad u_*(\Psi(I)) \subset \Psi(I') \ . \tag{1.13.2}$$

These equivalent relations are satisfied in each of the three cases [6] *:*

[4] should come before 1.8.

[5] cf. XVII 6.4 (p. 117), but where the notations for $\Phi\langle I\rangle$, $\Psi(I)$ are reversed.

[6] **N.B.** One has $1^\circ \Longrightarrow 3^\circ$, $2^\circ \Longrightarrow 3^\circ$.

1°) *u is coétale.*

2°) *I is discrete.*

3°) *$\forall\, i' \in \mathrm{Ob}\, I'$, every connected component of I/i' admits a final object.*

[page 13]

2 Reminders on closed Quillen triples and special Quillen triples

Let $(\mathcal{M}, W)$ be a *category of models,* i.e. $\mathcal{M}$ equipped with a localizer $W \subset \mathrm{Fl}(\mathcal{M})$, i.e. a subset satisfying

$$\begin{array}{l} W \text{ contains the isomorphisms, and if } f,\ g \text{ are composable arrows,} \\ \text{if two among } f,\ g,\ gf \text{ are in } W, \text{ the third also } \textit{(weak saturation).} \end{array} \tag{2.1}$$

I recall that the datum of a *closed Quillen triple* (W, C, F), or of a closed Quillen structure, on $\mathcal{M}$, having W as set of quasi-isomorphisms, amounts to the datum of two Quillen couples

$$C \;\leftarrow\!\!\!\!\!\!\rightarrow\; TF\ , \qquad TC \;\leftarrow\!\!\!\!\!\!\rightarrow\; F\ , \tag{2.2}$$

satisfying

$$TC \;\subset\; C \cap W\ , \qquad TF \;\subset\; F \cap W\ , \tag{2.3}$$

plus one of the three equivalent conditions:

$$\left\{\begin{array}{lllll} \text{a)} & TC & = & C \cap W & (\text{i.e. } C \cap W \subset TC) \\ \text{b)} & TF & = & F \cap W & (\text{i.e. } F \cap W \subset TF) \\ \text{c)} & W & = & TF \circ TC\ . & \end{array}\right. \tag{2.4}$$

Let us also recall that this implies that W *is strongly saturated,* and a fortiori that $W = W^{\natural}$, i.e. W stable under direct factors.

Let us also recall that by virtue of (1.4), one has:

p. 9
[page 14]

(2.5) Given the two couples (2.2), with $TC \subset C$ (equivalent to $TF \subset F$), for the first relation (2.3) to hold, i.e. $TC \subset W$, it is necessary, and sufficient if $W = W^{\natural}$, that every arrow f of $\mathcal{M}$ (or only every arrow of TC) factor as $f = pi$, with $i \in W$, $p \in F$. Dually, for $TF \subset W$ to hold, it is necessary and sufficient if $W = W^{\natural}$, that every arrow f of $\mathcal{M}$ (or only every arrow of TF) factor as $f = pi$, with $i \in C$, $p \in W$.

We say that the Quillen structure (W, C, F) is *special* if the two Quillen couples (2.2) are special, i.e. if the following conditions are satisfied.

(2.6) a) $\mathcal{M}$ is stable under small $\varinjlim$, and its elements are accessible ($\mathcal{M}$ "special").

b) There exist *small* subsets $C_0 \subset C$, $TC_0 \subset TC$ which are Q-generating on the left, i.e. such that we have

$$\left\{\begin{array}{lllllllllll} C & \subset & C_0^{\sim}\ , & \text{i.e.} & C & = & C_0^{\sim}\ , & \text{i.e.} & TF & = & C_{0*}\ ; \\ TC & \subset & (TC_0)^{\sim}\ , & \text{i.e.} & TC & = & (TC_0)^{\sim}\ , & \text{i.e.} & F & = & (TC_0)_*\ . \end{array}\right.$$

[page 15]

Thus, the datum of such a *special* Quillen triple in $\mathcal{M}$, when $\mathcal{M}$ satisfies (2.6. a), amounts to the datum of the two subsets

$$TC \ \subset \ C \tag{2.7}$$

of $\mathrm{Fl}(\mathcal{M})$, satisfying the conditions:

$$\left\{ \begin{array}{l} C,\ TC \text{ "special"} \left\{ \begin{array}{l} \text{a)}\quad C,\ TC \text{ are Q-closed on the left.} \\ \text{b)}\quad C,\ TC \text{ are Q-generated on the left by } \textit{small} \text{ subsets } C_0,\ TC_0 \text{ of } C,\ TC \text{ respectively.} \end{array} \right. \\ \quad \text{c)}\quad TC \subset W,\ TF \overset{\text{def}}{=} C_* \subset W. \\ \quad \text{d)}\quad \text{We have one of the three equivalent additional conditions of (2.4), i.e.:} \\ \qquad \alpha)\quad TC \ = \ C \cap W\ , \qquad \text{i.e.}\quad C \cap W \ \subset \ TC\ . \\ \qquad \beta)\quad TF \ = \ F \cap W\ , \qquad \text{i.e.}\quad F \cap W \ \subset \ TF\ . \\ \qquad \gamma)\quad W \ = \ TF \circ TC\ . \end{array} \right. \tag{2.8}$$

Proposition 2.1. *Let* p. 10

$$\mathcal{M}' \underset{v^*}{\overset{v_!}{\leftrightarrows}} \mathcal{M} \tag{2.1.1}$$

be a pair of adjoint functors, $\mathcal{M}$ being equipped with a closed Quillen structure (W, C, F). Put

$$\left\{ \begin{array}{rcl} W' & = & (v^*)^{-1}(W) \\ F' & = & (v^*)^{-1}(F) \\ TF' & = & F' \cap W' \ = \ (v^*)^{-1}(TF) \\ C' & = & v_!(C)^{\sim} \\ TC' & = & v_!(TC)^{\sim}\ . \end{array} \right. \tag{2.1.2}$$

1°) *We then have*

$$\left\{ \begin{array}{lcllcl} C' & \longleftrightarrow & TF'\ , & TC' & \longleftrightarrow & F'\ , \\ TC' & \subset & C'\ , & TF' & \subset & F'\ . \end{array} \right. \tag{2.1.3}$$

For the data (2.1.2) *in $\mathcal{M}'$ to correspond there to a Quillen triple (W', C', F'), it is necessary and sufficient that the following conditions be satisfied:* [page 16]

$$\left\{ \begin{array}{ll} \text{a)} & \textit{Factorization for } C' \longleftrightarrow TF' \textit{ and } TC' \longleftrightarrow F', \textit{ i.e. the relations } C' \longleftarrow\!\!\shortmid\!\!\longrightarrow TF' \textit{ and } TC' \longleftarrow\!\!\shortmid\!\!\longrightarrow F', \\ \text{b)} & TC' \subset W', \end{array} \right. \tag{2.1.4}$$

or again, in place of b), *the equivalent condition* (in the presence of a)):

$$\left\{ \begin{array}{ll} \text{b}') & \textit{Every arrow } f' \textit{ in } \mathcal{M}' \textit{ (not only } f' \in TC') \textit{ factors as } f' = p'i', \\ & \textit{with } i' \in W',\ p' \in F'. \end{array} \right. \tag{2.1.5}$$

2°) *Suppose* (2.1.4. a) *is satisfied (factorization). Then, for (W', C', F') to be a closed Quillen triple, it* suffices *that we have*

$$\text{b}'') \qquad \boxed{v^* v_!(TC) \ \subset \ TC}\ , \tag{2.1.6}$$

and that in addition we be in one of the following three cases:

A) v^* *admits a right adjoint* v_*.

B) v^* *commutes with small* $\varinjlim$, (W, C, F) *is special, and* $\mathcal{M}'$ *is special, i.e. satisfies condition* (2.6. a).

C) v^* *is fully faithful, and* $v^* v_!(F) \subset F$.

Let us note at once the

Corollary 2.2. *Suppose that* (W, C, F) *is a special Quillen triple, that* $\mathcal{M}'$ *is special, that* v^* *commutes with small* $\varinjlim$*, finally that we have* (2.1.6)*, i.e.* $v^* v_!(TC) \subset TC$*. Then* (W', C', F') *is a special Quillen triple.*

Indeed, by 2.1. 2°, case B, to see that (W', C', F') is a closed Quillen triple, one must check (2.1.4. a). But this follows from 1.8, applied to the two special Quillen couples (C, TF), (TC, F) in $\mathcal{M}$: the couples (C', TF') and (TC', F') are special Quillen couples, which proves at the same time that the triple (W', C', F') is special, q.e.d. p. 11 [page 17]

Let us prove 2.1. The first two relations (2.1.3) are contained in 1.6, the relations $TC' \subset C'$, $TF' \subset F'$ are tautological. The criterion (2.1.4) for (W', C', F') to be a closed Quillen triple, is contained in the reminder on page 13, taking into account that we already know that $TF' = F' \cap W'$ (relation (2.4. b)). That the condition (2.1.4. b), in the presence of (2.1.4. a), is equivalent to the condition (2.1.5. b′), is contained in the first assertion (2.5), noting that we have $W' = W'^{\natural}$ (which follows from the definition of W' and from $W = W^{\natural}$).

It remains to prove 2°), i.e. that condition b) is satisfied assuming a) and b″) (2.1.6) in each of the three cases A), B), C).

Case A. By definition of $W' = (v^*)^{-1}(W)$, to prove that $TC' \subset W'$, it suffices to prove

$$v^*(TC') \ \subset \ W \ ,$$

and a fortiori, it suffices to prove

$$v^*(TC') \ \subset \ TC \ , \qquad \text{i.e. } v^*(TC') \lhook\joinrel\longrightarrow F \ . \tag{2.1.7}$$

Now by 1.5, applied to the pair of adjoint functors (v^*, v_*), this relation is equivalent to [page 18]

$$TC' \lhook\joinrel\longrightarrow v_*(F) \ .$$

Since $TC' \overset{\text{def}}{=} v_!(TC)^{\sim}$, this is equivalent to

$$v_!(TC) \lhook\joinrel\longrightarrow v_*(F) \ ,$$

or again (applying 1.5 again to $(v_!(TC), F)$ instead of (TC', F)), to

$$v^* v_!(TC) \lhook\joinrel\longrightarrow F \ , \qquad \text{i.e. } v^* v_!(TC) \subset \underbrace{TC}_{= F^*} \ ,$$

now this is hypothesis b″).

Case B. It again suffices to prove 2.1.7. Now, since $v_!(TC)$ admits a small Q-generating set on the left, $v_!(TC)^{\sim}$ is the closure of $v_!(TC)$ under the operations of cobase change, transfinite composition, direct factors. Since v^* commutes with all these operations (as it commutes with small $\varinjlim$), and since TC in $\mathrm{Fl}(\mathcal{M})$ is stable under these operations, we thus see that to show that v^* maps $v_!(TC)^{\sim}$ into TC, it suffices to see that it maps $v_!(TC)$ into TC, which is again hypothesis b″).

Case C. We are going to check b) in the equivalent form b′). So let $f' \in \mathrm{Fl}(\mathcal{M}')$, $f' : X' \longrightarrow Y'$, put p. 12 [page 19]

$$f \ = \ v^*(f'),$$

and factor f as pi, with $i \in TC$, $p \in F$, whence

$$v_!v^*(f') = v_!(f) = v_!(p)v_!(i) \ .$$

But v^* being fully faithful, $v_!v^* \xrightarrow{\sim}$ id, so we find a factorization of f' as $v_!(p)v_!(i)$. It therefore suffices to see that

$$v_!(p) \ \in \ F' \ , \qquad v_!(i) \ \in \ W' \ ,$$

which is equivalent to

$$v^*v_!(p) \in F \ , \qquad v^*v_!(i) \in W \ ,$$

and follows from the hypotheses that $v^*v_!(F) \subset F$ and $v^*v_!(TC) \subset TC \subset W$, q.e.d.

N.B. I have noted cases A) and C) for conscience's sake, with no precise application in view. The most useful case will fall under case B), and deserves to be spelled out again in a

Theorem 2.3[7]. *Let (W, C, F) be a special Quillen triple in $\mathcal{M}$. For every I in* Cat*, let us consider the triple $(W(I), C\langle I\rangle, F(I))$ in $\mathcal{M}(I) = \underline{\mathrm{Hom}}(I^\circ, \mathcal{M})$, where the notations are those of page 10, so $W(I)$, $F(I)$ and $TF(I)$ are defined fiber by fiber, and $C\langle I\rangle = (TF(I))^* = (v_{I\,!}(C(I_0)))^{\sim}$*[8].

1°) *For every I in* Cat*, $(W(I), C\langle I\rangle, F(I))$ is a special Quillen triple in $\mathcal{M}(I)$.* [page 20]

2°) *If $u : I \longrightarrow I'$ is an arrow in* Cat*, whence*

$$\mathcal{M}(I) \underset{u^*}{\overset{u_!}{\rightleftarrows}} \mathcal{M}(I'), \tag{2.3.1}$$

then

$$\left\{ \begin{array}{rcl} u^*(W(I')) & \subset & W(I) \\ u^*(F(I')) & \subset & F(I) \\ u^*(TF(I')) & \subset & TF(I) \\ u_!(C\langle I\rangle) & \subset & C\langle I'\rangle \\ u_!(TC\langle I\rangle) & \subset & TC\langle I'\rangle \ , \end{array} \right. \begin{array}{l} \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \textit{tautology} \\ \\ \\ \end{array} \tag{2.3.2}$$

finally, $u_!$ induces a functor

$$\mathcal{M}(I)_{\mathrm{c}} \longrightarrow \mathcal{M}(I')_{\mathrm{c}} \qquad {}^{(9)} \tag{2.3.3}$$

compatible with the localizers $W(I)_{\mathrm{c}} = W(I)|\mathcal{M}(I)_{\mathrm{c}}$ *and* $W(I')_{\mathrm{c}} = W(I')|\mathcal{M}(I')_{\mathrm{c}}$*, hence induces a functor*

$$u_!^W \ : \ \mathrm{Ho}_W(I) \longrightarrow \mathrm{Ho}_W(I') \ . \tag{2.3.4}$$

This functor is a left adjoint to the functor u_W^*,

$$u_W^* \ : \ \mathrm{Ho}_W(I') \longrightarrow \mathrm{Ho}_W(I) \ , \tag{2.3.5}$$ p. 13

induced by u^* (2.3.1).

Proof of 2.3.

Let us prove 1°). We consider as in 1.10 the functor

$$v = v_I \ : \ I_0 \longrightarrow I$$

and the pair of adjoint functors

$$\mathcal{M}(I) \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}(I_0) \ , \tag{2.3.6}$$ [page 21]

to which we can apply 2.2. It remains only to check

$$v^* v_!(TC) \ \subset \ TC \ , \tag{2.3.7}$$

now we have, for $X = (X_i)_{i \in I_0} \in \mathcal{M}(I_0)$,

$$\begin{aligned} v_!(X)(i) \ &\simeq \ \coprod_{j \in I_0} v_{j\,!}(X_j)(i) \\ &\simeq \ \coprod_{j \in I_0} \ \underbrace{X_j^{\mathrm{Hom}(i,j)}}_{\substack{\text{direct sum} \\ \text{of } \mathrm{Hom}(i,j) \text{ copies}}} \end{aligned} \quad ,$$

and relation (2.3.7) follows from the fact that TC, being Q-closed on the left, is stable under small direct sums.

Let us prove 2°). Formulas (2.3.2) are essentially tautological, the last two follow from the third and the second, by 1.5 applied to the pair of adjoint functors $(u_!, u^*)$. The relation $u_!(C\langle I\rangle) \subset C\langle I'\rangle$, and the fact that $u_!$ transforms initial object into initial object, imply (2.3.3). That $u_!$ restricted to $\mathcal{M}(I)_c$ is compatible with the localizers comes from $u_!(TC\langle I\rangle) \subset TC\langle I'\rangle$, and from the fact that every arrow f in $\mathcal{M}(I)_c$ which is in $W(I)$, decomposes as $f = pi$, with $i \in TC$ and p a left inverse of a $j \in TC\langle I\rangle$. This seen, together with the first two formulas (2.3.2) shows that we are under the conditions of Quillen's adjunction theorem, whence the adjunction of $u_!^W$ and u_W^*, q.e.d. [page 22]

Corollary 2.4. *Let W be a localizer in $\mathcal{M}$, such that there exists a special Quillen triple (W, C, F) including W. Then the prederivator* Ho_W *on* Cat *satisfies the axioms Der 1 (decomposition), Der 2 (localization), Der 3 (final object), Der 4 bis (existence of the $u_!^W$), Der 5 bis (computation of the fibers of the $u_!^W$), Der 6 bis (left exactness, i.e. left exactness of W-cocartesian squares in the* $\mathrm{Ho}_W(I)$*), Der 6 (right exactness of W-cartesian squares).*

We have just proved Der 4 bis, and Der 5 bis is practically contained in the manner of obtaining of $u_!$, taking into account proposition 1.12, which implies that for X in $\mathcal{M}(I)_c$, and i' in I', the inverse image of X in $\mathcal{M}(i'\backslash I)$ is in $\mathcal{M}(i'\backslash I)_c$ …

The decomposition axiom is satisfied for every prederivator of the form $\mathbf{D}_{\mathcal{M},W}$. The left exactness axiom Der 6 bis is valid, provided that the $(\mathcal{M}(I), W(I))$ fit into a structure with cofibrations, which is the case. Likewise for the localization axiom Der 2, provided that in addition W is strongly saturated, which is also the case. Finally, the final object axiom says that if I has a final object, then p. 14 [page 23]

$$\mathrm{Ho}_W(e) \longrightarrow \mathrm{Ho}_W(I)$$

is fully faithful, or again that the adjunction morphism

$$u_! u^*(\xi) \longrightarrow \xi$$

is an isomorphism. But by virtue of 1.12, case 3°, in this case u^* maps $\mathcal{M}(e)_c = \mathcal{M}_c$ into $\mathcal{M}(I)_c$, so the computation of the pair of adjoint functors $u_!^W$, u_W^* can be carried out using the restrictions of $u_!$, u^* to $\mathcal{M}(I)_c$ and $\mathcal{M}_c$ respectively. Since the adjunction morphism $u_!^c u_c^* \longrightarrow \mathrm{id}$ is an isomorphism, the same therefore holds for $u_!^W u_W^* \longrightarrow \mathrm{id}$, which is deduced from it by localizing, q.e.d.

2.5. To finish constructing Ho_W as a derivator on all of Cat, one must therefore establish Der 4, Der 5: existence of the u_*^W, and computation of the fibers of u_*^W. As [page 24]

[7] cf. scholie 2.10, which should precede theorem 2.3.

[8] where $v_I : I_0 \longrightarrow I$ (I_0 discrete category having the same objects as I).

[9] Subcategories of cofibrant objects.

for Der 6, it is obtained automatically, from the fact that each $(\mathcal{M}(I), W(I))$ fits into a fibration structure $(W(I), F(I))$. (Omission in the proof of 2.4.) The natural idea is now to use the triple structure

$$(W(I), C(I), \underbrace{F[I]}_{\stackrel{\text{def}}{=} TC(I)_*}) , \tag{2.5.1}$$

but the difficulty is to prove the factorization for a couple

$$C(I) \longleftrightarrow TF[I] , \qquad \underbrace{TC(I)}_{\stackrel{\text{def}}{=} C(I) \cap W(I)} \longleftrightarrow F[I] , \tag{2.5.2}$$

for want of being able to show, under the general conditions of 2.3 alone (special Quillen triple) that $C(I)$ and $TC(I)$ are also special. (Cf. question 1.11.) But with suitable conditions on $\mathcal{M}$, we shall manage to get by if $C = mono$. The relevant statement, a little technical, is found in XV cor. 7.4 (p. 98-99), which I am going to recall here.

p. 15
[page 25]

Theorem 2.6. *Let $\mathcal{M}$ be a category satisfying the following conditions.*

a) *$\mathcal{M}$ is "special", i.e. stable under small $\varinjlim$, and its elements are accessible.*

b) *Filtering direct limits in $\mathcal{M}$ are left exact (and in $\mathcal{M}$ finite $\varprojlim$ are representable – follows from* a *and* c*).*

c) *$\mathcal{M}$ has a cardinal filtration $(\mathrm{Filt}^\pi)_{\pi \geq \pi_0}$.*

d) *The Filt^π are stable under $\varinjlim$ indexed by categories I such that $\operatorname{card} \mathrm{Fl}(I) \leq \pi$, and under quotients ($Z \in \mathrm{Filt}^\pi$ and $Z \longrightarrow Z'$ an epimorphism implies $Z' \in \mathrm{Filt}^\pi$).*

e) *If A_1, A_2 are two subobjects of an object B of $\mathcal{M}$, and $A_0 = A_1 \cap A_2$, then $A_1 \amalg_{A_0} A_2 \longrightarrow B$ is a monomorphism.*

f) *If we have a cocartesian square*

$$\begin{array}{ccc} A_0 & \xrightarrow{i} & A_1 \\ \downarrow & & \downarrow \\ A_2 & \xrightarrow{i'} & A_3 \end{array}$$

with i a monomorphism, then i' is a monomorphism.

(2.6.1)

[10,11,12,13]. *(We shall then say that $\mathcal{M}$ is* "very special"*). Let moreover*

$$W \subset \mathrm{Fl}(\mathcal{M}) \tag{2.6.2}$$

[10] **N.B.** a + c also means that $\mathcal{M}$ is a "$\mathfrak{U}$-paratopos category", and implies the existence of a cardinal filtration satisfying d) (except the assertion concerning the quotients Z' of Z). Thus we are dealing with a $\mathfrak{U}$-paratopos satisfying in addition b) e) f). (Is it then a topos? *No,* for e), f) are O.K. in *abelian categories!*)

[11] **N.B.** Condition d) is probably unnecessary, at least if epimorphisms are effective, for d) will be satisfied in any case for $\pi = 2^c$, with $c \geq \pi_0$.

[12] **N.B.** Condition e) is *not* satisfied for the category (Gr), and the theorem fails for this category, which does *not* have "enough injectives", cf. SGA 4 I 8.12.9. On the other hand, e) and f) are satisfied by *abelian* categories.

[13] Cf. XVIII 4.6.7 (p. 157) for a more congenial formulation of the set of conditions a) to f) for $\mathcal{M}$.

be a localizer in $\mathcal{M}$, which satisfies the conditions

$$\begin{cases} \text{a) } W \textit{ is stable under filtering } \varinjlim\ ^{(14)}. \\ \text{b) } W \textit{ is moreover } \mathrm{L}_1\textit{-accessible, i.e. there exists a small subset } W_0 \subset W \textit{ such that } W \textit{ is the } \mathrm{L}_1\textit{-envelope of } W_0\textit{, i.e. such that } W \textit{ is the smallest subset of } \mathrm{Fl}(\mathcal{M}) \textit{ containing } W_0\textit{, and stable under filtering } \varinjlim. \end{cases} \tag{2.6.3}$$

Let us then put

p. 16
[page 26]

$$C = \mathrm{Mono}(\mathcal{M}), \qquad TC = C \cap W\ , \qquad TF = C_*\ , \qquad F = (TC)_*\ . \tag{2.6.4}$$

Under these conditions:

1°) *$C \leftarrow\!\!\!\!\!\!\rightarrow TF$, more precisely $C \rightarrowtail TF$ is a special Quillen couple.*

2°) *For $TC \rightarrowtail F$ to be a Quillen couple, it is necessary and sufficient that TC be stable under cobase change, and then it is a special Quillen couple.*

3°) *For (W, C, F) to be a Quillen triple (which will necessarily be special by 1°, 2°, and in particular closed), it is necessary and sufficient that the following conditions be satisfied.*

$$\begin{cases} \text{a)} \quad TC \ (\overset{\text{def}}{=} W \cap \mathrm{Mono}) \textit{ is stable under cobase change.} \\ \text{b)} \quad TF \ (\overset{\text{def}}{=} (\mathrm{Mono})_*) \ = \ \textit{“maps of injective type”} \subset W. \end{cases} \tag{2.6.5}\ ^{(15)}$$

And condition b) *is also equivalent to:*

$$\text{b}')\ \textit{Every arrow } f \textit{ of } \mathcal{M} \textit{ (or only of } TF\textit{) factors as } f = pi\textit{, with } i \textit{ a monomorphism, } p \in W. \tag{2.6.6}$$

Corollary 2.7. *$\mathcal{M}$ and W being as in* (2.6.1), (2.6.3), *for (W, C, F) to be a Quillen triple that is* left proper *(i.e. $C \subset \mathrm{Cof}_W$), it is necessary and sufficient that W satisfy condition* (2.6.5. b) *(i.e. W contains the arrows of injective type), and moreover*

[page 27]

$C \subset \mathrm{Cof}_W$, i.e. the condition:

a′) *For every cocartesian diagram*

$$\begin{array}{ccc} A_0 & \xrightarrow{\ i\ } & A_1 \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ A_2 & \xrightarrow{\ i'\ } & A_3 \end{array} \tag{2.7.7}$$

with i a monomorphism, if $f \in W$, then $f' \in W$. (Stability of W under cobase change along monomorphisms.)

Under these conditions, this Quillen triple is special, and a fortiori closed.

This corollary follows from 2.6. 3°), since $C \subset \mathrm{Cof}_W$ implies $TC \overset{\text{def}}{=} C \cap W = C \cap W^{\mathrm{univ}}$ (for $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$), hence that TC is stable under cobase change, whence condition (2.6.5. a).

[14] **N.B.** a) already implies that $W = W^{\natural}$.

[15] **N.B.** These conditions (2.6.5) are equivalent to saying that (W, C) is a structure of a category with cofibrations on $\mathcal{M}$.

p. 17

2.8. We say that a *Quillen structure* (W, C, F) on $\mathcal{M}$ is *very special,* if it is obtained as in theorem 2.6, i.e. if we have the conditions:

$$\begin{cases} \text{a) } \mathcal{M} \text{ is very special, i.e. satisfies conditions (2.6.1).} \\ \text{b) } W \text{ is stable under small filtering } \varinjlim, \text{ and is accessible (conditions (2.6.3)).} \\ \text{c) } C = \text{Mono}. \end{cases} \tag{2.8.1}$$

Thus for a localizer W in $\mathcal{M}$ to correspond to a very special Quillen structure in $\mathcal{M}$ (which will necessarily be unique), it is necessary and sufficient that $\mathcal{M}$ and W satisfy conditions (2.8.1. a) b)), and that moreover we have: [page 28]

$$\begin{cases} \text{a) } TC \overset{\text{def}}{=} \text{Mono} \cap W \text{ is stable under cobase change.} \\ \text{b) } TF \overset{\text{def}}{=} (\text{Mono})_* \subset W, \text{ i.e. } W \text{ contains the arrows of injective type in } \mathcal{M}. \end{cases} \tag{2.8.2}$$

(16).

We now find the

Theorem 2.9. *Let* $(W, C = \text{Mono}, F = (TC)_* = (W \cap \text{Mono})_*)$ *be a very special Quillen triple in a category* $\mathcal{M}$ (2.8). *Then:*

1°) *For every* I *in* Cat, $(W(I), C(I), F[I])$ *is a very special Quillen triple in* $\mathcal{M}(I)$*, and in particular we have*

$$C(I) \;=\; \text{Mono}(\mathcal{M}(I)) \;, \tag{2.9.1}$$

and moreover

$$F[I] \cap W(I) \;=\; (TF)[I] \;. \tag{2.9.2}$$

2°) *For every arrow* $u : I \longrightarrow I'$ *in* Cat, *whence the pair of adjoint functors*

$$\mathcal{M}(I) \underset{u_*}{\overset{u^*}{\leftrightarrows}} \mathcal{M}(I') \quad , \tag{2.9.3}$$

we have

$$\begin{cases} u^*(W(I')) \;\subset\; W(I) \\ u^*(C(I')) \;\subset\; C(I) \\ u^*(TC(I')) \;\subset\; TC(I) \\ u_*(F[I]) \;\subset\; F[I'] \\ u_*(TF[I]) \;\subset\; TF[I'] \;, \end{cases} \tag{2.9.4}$$

finally u_* *induces a functor*

$$u_*^{\mathrm{f}} \;:\; \mathcal{M}(I)_{\mathrm{f}} \longrightarrow \mathcal{M}(I')_{\mathrm{f}} \tag{17} \tag{2.9.5}$$

p. 18 [page 29]

compatible with the localizers $W(I)_{\mathrm{f}}$ *and* $W(I')_{\mathrm{f}}$ *induced by* $W(I)$, $W(I')$ *respectively, hence induces a functor*

$$u_*^W \;:\; \text{Ho}_W(I) \longrightarrow \text{Ho}_W(I') \;. \tag{2.9.6}$$

This functor is a right adjoint *to the functor*

$$u_W^* \;:\; \text{Ho}_W(I') \longrightarrow \text{Ho}_W(I) \tag{2.9.7}$$

induced by u^* (2.9.3).

[16] We also say that the localizer W is very special, if it defines a very special Quillen structure.
[17] full subcategories of fibrant objects.

PROOF OF 2.9. To prove 1°), we must first check that the preliminary conditions (2.8.1. a) b)) on $\mathcal{M}$, W are stable under passage from $\mathcal{M}$, W to $\mathcal{M}(I)$, $W(I)$, which is done by AQT . Besides, one checks right away that

$$C(I) \;=\; \mathrm{Mono}(\mathcal{M}(I)) \;,$$

and theorem 2.6 (1°, 2°) then implies that

$$\begin{array}{ccccc} C(I) & \rightarrowtail & C(I)_* & \overset{\text{def}}{=} & TF[I] \\ \underbrace{TC(I)}_{\overset{\text{def}}{=} C(I)\cap W(I)} & \rightarrowtail & (TC(I))_* & \overset{\text{def}}{=} & F[I] \end{array}$$

are special Quillen couples. (Noting for the second, that $TC(I)$ is indeed stable under cobase change, TC being so.) To check that we have a very special Quillen triple, it remains by virtue of 2.8 to check that [page 30]

$$TF[I] \;\subset\; W(I) \;,$$

which is written

$$v^*(TF[I]) \;\subset\; W(I_0) \;,$$

where

$$v \;:\; I_0 \longrightarrow I$$

is defined as in (1.12) (p. 10). In fact, we shall even have

$$v^*(TF[I]) \;\subset\; \underbrace{TF(I_0)}_{=F(I_0)\cap W(I_0)} \quad ,$$

and the proof is dual to that of

$$v^* v_!(TC(I_0)) \;\subset\; TC(I_0)$$

(cf. (2.3.7), p. 21), it does not depend on questions of factorization in $\mathcal{M}(I)$, and comes here from the stability of TF under small products. This proves 1°.

The proof of 2° is now the dual of that of (2.3. 2°). Since this makes us repeat the same proof twice, this shows that we have not "set up the sorites" quite carefully enough. So by way of making up for this, I add a

Scholie 2.10. *Let (W, C, F) be a closed Quillen triple in $\mathcal{M}$. For I in* Cat*, for $(W(I), C\langle I\rangle, F(I))$ to be a closed Quillen triple for $\mathcal{M}(I)$, it is necessary and sufficient that the couples $C\langle I\rangle \longleftrightarrow TF(I)$ and $TC\langle I\rangle \longleftrightarrow F(I)$ be factorizing. Suppose that this condition is satisfied for I and for I', and let $u : I \longrightarrow I'$ be an arrow from I to I'. Then the assertions of* 2.3. 2° *hold, in particular we find a functor $u_!^W : \mathrm{Ho}_W(I) \longrightarrow \mathrm{Ho}_W(I')$, left adjoint to u_W^**[18]*. These results dualize at once (by $\mathcal{M} \longmapsto \mathcal{M}^\circ$, $I \longmapsto I^\circ$) for the triples $(W(I), C(I), F[I])$, and for the construction of u_*^W, right adjoint to u_W^*. In particular, for the preceding triple to be a Quillen triple (necessarily closed), it is necessary and sufficient that the two couples $C(I) \longleftrightarrow TF[I]$ and $TC(I) \longleftrightarrow F[I]$ be factorizing; and when this condition is satisfied for I and for I' (for a given arrow $u : I \longrightarrow I'$ in* Cat*), then u_*^W exists and is constructed dually to the construction of $u_!^W$ made explicit in* 2.3. 2°. p. 19 [page 31] [page 32]

[18] Say also that the computation of the fibers of $u_!^W(X)$, provided for in Der 5 bis, is OK.

Corollary 2.11. *The prederivator* $\mathbf{D}_{\mathcal{M},W}$ *defined for a* very special *localizer* W *in* $\mathcal{M}$ *is a derivator in the full sense of the term, i.e. it satisfies all the axioms Der 1 to Der 6 (including Der 4 bis and Der 5 bis and Der 6 bis) of derivators.*

N.B. To my own surprise, I have not been able to find a case of a localizer W which defines a derivator on all of Cat (with existence of $u_!^W$, u_*^W), *except* for W a very special localizer, hence fitting into a very special Quillen triple – or when one can reduce to this in the standard way. (For example in the case where $\mathcal{M} = \mathrm{Cat}$, W a basic localizer satisfying Loc 6 a) and Loc 8, 8 bis)[19] – the monomorphisms in W flatly refuse to be stable under cobase change!) I have absolutely no idea whether the spectacular result "proved" by ANDERSON is true or false. It looks almost too good to be true …Besides, I know of no special Quillen structure which is not very special – sorry!

[page 33]

3 Case of categories with cofibrations, of categories with fibrations

I recall the axioms of a *cofibration structure* (W, C) on a category $\mathcal{M}$.

Cof_0 : W is a localizer.

Cof_1 : C contains the isomorphisms, is stable under cobase change, under composition.

Cof_2 : $C \cap W$ is stable under cobase change (and also under composition, and contains the isomorphisms, by virtue of Cof_0, Cof_1).

Cof_3 : Every arrow f in $\mathcal{M}$ factors as $f = pi$, with $i \in C$, $p \in W$.

(Axioms of K. S. BROWN, slightly generalized by ANDERSON). There is an optional "axiom of continuity" or *"transfinite axiom"*, which is the following: p. 20

Cof_4 : $\mathcal{M}$ stable under small filtering $\varinjlim$, and C and $TC = C \cap W$ are stable under transfinite composition.

Thus, Cof_1, Cof_2, Cof_4 can be summarized by saying that C and TC are transfinitely Q-saturated on the left.

The cofibration structure is called *special* if it satisfies Cof_4, if $\mathcal{M}$ is special, and if C admits a *small* subset C_0 which is Q-generating on the left, i.e. such that

$$C \ \subset \ C_0\tilde{}\,, \qquad \text{i.e.} \quad \tilde{C} = C_0\tilde{}\,. \tag{3.1}$$

Then, by the factorization theorem 1.1, we have [page 34]

$$C \ \leftarrow\!\!\!\!\rightarrow \ TF \ \overset{\mathrm{def}}{=} \ C_* \ ,$$

and the factorization axiom Cof_3 implies (cf. 1.4) $TF \subset W^{\pitchfork}$. More generally, we have:

Proposition 3.1. *Suppose that* $\mathcal{M}$ *is special (stable under small* $\varinjlim$ *, and accessible elements), that* (W, C) *satisfies* Cof_0*,* Cof_1*, and that* C *is stable under transfinite composition (which, together with* Cof_1*, is stated by saying that* C *is transfinitely Q-saturated on the left), finally that* C *admits a small subset* C_0 *Q-generating on the left, so that* $C \leftarrow\!\!\!\!\rightarrow TF \overset{\mathrm{def}}{=} C_*$*. Then for the factorization axiom* Cof_3 *to be satisfied, it suffices that we have*

$$TF \ \subset \ W \ , \tag{3.1.1}$$

[19] axiom of cocartesian squares (direct version), axiom of direct limits, accessibility.

and it is necessary to have $TF \subset W^{\natural}$*, hence if* $W = W^{\natural}$*, the axiom Cof*$_3$ *is equivalent to* (3.1.1).

The cofibration structure is called *semi-closed* if C is Q-closed on the left, i.e. if we have $C = C^{\check{}}$ (which implies that C is stable under transfinite composition), *closed* if in addition $TC = (TC)^{\check{}}$ (which implies the transfinite axiom Cof_4).

[page 35] I do not return here to the theory of K. S. Brown on categories with cofibrations. It is, to my knowledge, the most economical set of axioms known, for ensuring on the one hand a good grip on Ho_W (through a corresponding notion of homotopy of maps, and a calculus of fractions in $\Pi\mathcal{M}_c$), and on the other hand the existence of the *exact* sequence of cofibrations, hence a theory of W-homotopic cofibers. (**N.B.** When the structure is *proper,* i.e. $C \subset \mathrm{Cof}_W$, then the existence of this structure amounts simply to saying that "W admits enough W-cofibrations", i.e. that every $f \in \mathrm{Fl}(\mathcal{M})$ factors as pi, with $i \in \mathrm{Cof}_W$, $p \in W$; and we can therefore take $C = \mathrm{Cof}_W$. In the case where W is stable under small filtering $\varinjlim$, $C = \mathrm{Cof}_W$ and $TC = W^{\mathrm{univ}}$ satisfy in addition the transfinite axiom Cof_4.) Finally, K. S. Brown shows that every arrow $i : X \longrightarrow Y$ in C, *with* X, Y *cofibrant* (i.e. $\emptyset_{\mathcal{M}} \longrightarrow X$, $\emptyset_{\mathcal{M}} \longrightarrow Y$ in C) is in Cof_W ("Brown's Lemma").

[page 36] I am interested here in the question of extending the cofibration structure to a category of the form $\mathcal{M}(I)$, and of using this extension to construct the functors $u_!^W$. Let us note already that the theory of K. S. Brown, with the supplements of Anderson[20], gives essentially this:

p. 21

Proposition 3.2. *Let* $W \subset \mathrm{Fl}(\mathcal{M})$ *be a localizer associated with a cofibration structure* (C, W) *in* $\mathcal{M}$*.*

1°) *On the category* Diag_0 *of* finite *ordered sets, this prederivator satisfies the axioms Der 1 (decomposition), Der 2 (localization), Der 3 (final object), Der 4 bis (existence of* $u_!^W$*), Der 5 bis (computation of the fibers of* $u_!^W$*), Der 6 bis (left exactness).*

2°) *This is still true on the larger subcategory* Diag_1 *in* Cat*, consisting of* noetherian *ordered sets I such that the topological space associated with* I° *is noetherian (every increasing sequence of objects of* I*, or of open sets of* I°*, is stationary).*

3°) *Finally, if the transfinite axiom Cof*$_4$ *is satisfied, i.e. if* C *and* TC *are stable under transfinite composition, then the conclusion of* 1° *remains valid on the full subcategory* Diag_2 *of* Cat *consisting of ordered sets which are merely noetherian.*

Corollary 3.3. *Suppose that we have Cof*$_3$ *in the stronger form of a factorization* $f = pi$ functorial *in* f*, which is the case in particular if the cofibration structure is special. Then for every* I *in* Cat*,* $\mathcal{M}(I)$ *equipped with* $(W(I), C(I))$ *is a category with cofibrations, which satisfies the transfinite axiom Cof*$_4$ *if* (W, C) *satisfies it. Thus we can apply to it the results of* 3.2*, which imply in particular that the axioms Der 1, 2, 3 (?) hold on all of* Cat*, and that* $u_!^W$ *exists when* u *is of the form* $\mathrm{id}_I \times v$*, with* $v : J \longrightarrow K$ *an arrow in* Diag_1 *(resp. in* Diag_2*, if the transfinite axiom is satisfied).*

[page 37]

3.4. The key to the theory is in the extension of the cofibration structure (W, C) from $\mathcal{M}$ to a category $\mathcal{M}(I)$, where I is an ordered set, as $(W(I), C_I)$. The main facts are stated in XVII, in the theorems p. 26-27, p. 50, p. 60. It is when I is *noetherian,* and we assume either that $\mathrm{Esp}(I^\circ)$ is also noetherian (e.g. I finite), or that (W, C) satisfies the transfinite axiom Cof_4, that we prove that $(W(I), C_I)$ is also a cofibration structure on $\mathcal{M}(I)$.

[page 38]

[20] revised and arranged by me.

An important fact for the construction of the functors $u_!^W$, for $u : I \longrightarrow I'$ an arrow between noetherian ordered categories, is the relation

$$C_I \cap W(I) \;=\; (TC)_I \;, \tag{3.4.1}$$

from which it follows that we have, for u as above,

$$u_!(C_I \cap W(I)) \;\subset\; C_{I'} \cap W(I') \;, \tag{3.4.2}$$

an essential ingredient of the construction of the functor $u_!^W$. To extend this construction to the case of a $\pm$ arbitrary arrow in Cat, there arises therefore the problem of extending the definition of C_I, and the formulas (3.4.1) and (3.4.2), to the case of $\pm$ arbitrary categories in Cat. Now we have proved:

Proposition 3.5. *If $\Phi = \Psi^*$, with $\Psi = \Phi_*$, is a left* Q-closed *subset of* $\mathrm{Fl}(\mathcal{M})$*, and I a noetherian ordered set, Φ_I the extension of Φ to a subset of* $\mathrm{Fl}(\mathcal{M}(I))$ *defined by* Anderson*, then we have*

$$\Phi_I \;=\; \Psi(I)^* \;=\; v_!(\Phi(I_0))^\sim \;, \tag{3.5.1}$$

where $v : I_0 \longrightarrow I$ is the canonical functor from the discrete category defined by $\mathrm{Ob}\, I$*.*

This is (with different notations) lemma 6.2.5 (p. 107) of XVII, whose proof I do not repeat here. p. 22

This gives the key to a definition of Φ_I as $\Phi\langle I\rangle$ (with the notations of paragraphs 1, 2), for arbitrary I in Cat,

[page 39]

$$\Phi_I \;\overset{\text{def}}{=}\; \Phi\langle I\rangle \;\overset{\text{def}}{=}\; \Psi(I)^* \;=\; v_!(\Phi(I_0))^\sim \;, \tag{3.5.2}$$

where $\Psi = \Phi_*$, and where we assume Φ Q-closed on the left, i.e. $\Phi = \Psi^*$. In the case where we assume only that Φ is transfinitely Q-saturated, without more, we would like to set

$$\Phi_I \;\overset{\text{def}}{=}\; \text{transfinite Q-saturation of } v_!(\Phi(I_0)) \;, \tag{3.5.3}$$

and I presume that for I ordered noetherian, this coincides with Anderson's definition. But I will restrict myself in what follows to the case where $C = \Phi$ is Q-closed on the left.

Let us then return to the case of a cofibration structure (W, C) in $\mathcal{M}$, assuming C and $TC = C \cap W$ Q-closed on the left. Let us define for every I in Cat $C_I = C\langle I\rangle$ as in (3.5.2). We then have, for $u : I \longrightarrow I'$ an arrow in Cat,

$$u_!(C\langle I\rangle) \;\subset\; C\langle I'\rangle \;, \tag{3.5.4}$$

but we have little reason here to hope to have (3.4.2) as well, without having the formula

$$C\langle I\rangle \cap W(I) \;\overset{?}{=}\; (TC)\langle I\rangle \qquad\qquad {}^{21}. \tag{3.5.5}$$

We have proved that this formula is valid if I is ordered noetherian (XVII th. 2.1, p. 26-27). This brings us to the

Question 3.6. *Let $\mathcal{M}$ be a category stable under small limits, and let (W, C) be a cofibration structure in $\mathcal{M}$ such that C and $TC = C \cap W$ are Q-closed on the left. Let I be an object of* Cat*, do we then have* [page 40]

$$C\langle I\rangle \cap W(I) \;=\; (TC)\langle I\rangle \;? \tag{3.6.1}$$

We saw in the preceding paragraph that this is so if (W, C) fits into a closed Quillen triple $(W, C, F = (TC)_*)$, and when we assume in addition that the couples

$$C\langle I\rangle \;\longleftrightarrow\; TF(I) \;, \qquad TC\langle I\rangle \;\longleftrightarrow\; F(I)$$ p. 23

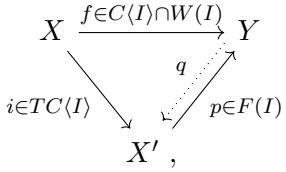
are factorizing, in which case $(W(I), C\langle I\rangle, F(I))$ is a closed Quillen structure in $\mathcal{M}(I)$. Besides, (W, C, F) is a Quillen structure (necessarily closed) if and only if the couples $C \longleftrightarrow \underbrace{TF}_{\overset{\text{def}}{=} C_*}$, $TC \longleftrightarrow \underbrace{F}_{\overset{\text{def}}{=} (TC)_*}$ themselves are factorizing, and if in addition $TF \subset W$ (which is automatic if $W = W^\natural$, for we will have in any case, by Cof_3, $TF \subset W^\natural$).

But can we get by with less, to be able to conclude (3.6.1)? Would it not suffice to know that $TC\langle I\rangle \longleftrightarrow F(I)$ is factorizing (which has little chance of being true for an I $\pm$ general, if we do not already have $TC \longleftrightarrow\!\!\!\!\!\rightarrow F$, factorization in $\mathcal{M}$ itself). This is indeed the case:

[page 41]

Proposition 3.7. *Under the general conditions of* (3.6) *on* $\mathcal{M}$, W, C, I, *suppose that the couple* $TC\langle I\rangle \longleftrightarrow F(I)$ *(where* $F \overset{\text{def}}{=} (TC)_*$*) is factorizing, i.e.* $TC\langle I\rangle \longleftrightarrow\!\!\!\!\!\rightarrow F(I)$. *Then the formula* (3.6.1) *is true.*

Indeed, if $f \in C\langle I\rangle \cap W(I)$, we factor f as pi, with $i \in TC\langle I\rangle$, $p \in F(I)$. Since $f, i \in W(I)$, we will have $p \in W(I)$, i.e. $p \in TF(I)$, and because of $\underbrace{C\langle I\rangle}_{\ni f} \longleftrightarrow \underbrace{TF(I)}_{\ni p}$, there exists $q : Y \longrightarrow X'$ such that $pq = \mathrm{id}_Y$, $qf = i$,



so f is a direct factor of $i \in TC\langle I\rangle$. Since $TC\langle I\rangle$ is stable under direct factors, we win.

Corollary 3.8. *Let* (W, C, F) *be a closed Quillen structure in* $\mathcal{M}$, *and suppose that* $TC = C \cap W$ *is a Q-special subset of* $\mathcal{M}$, *i.e. the couple* $TC \longleftrightarrow\!\!\!\!\!\rightarrow F$ *a special couple (i.e.* $\mathcal{M}$ *is* special, *and there exists a small subset* TC_0 *Q-generating on the left for* TC*). Then for every* I *in* Cat, *we have*

$$C\langle I\rangle \cap W(I) \;=\; TC\langle I\rangle \;. \tag{3.8.1}$$

[page 42]

Apparently, we have not had to assume that C too is a Q-special subset of $\mathrm{Fl}(\mathcal{M})$. But as we need the factorization for $C \longleftrightarrow TF$ (to have at least a cofibration structure on $\mathcal{M}$), we have hardly any way of establishing it without assuming C special. Not to mention that to be able to make (3.8.1) explicit for the construction of $u_!^W$, we need to know that $(W(I), C\langle I\rangle)$ does indeed define a cofibration structure on $\mathcal{M}(I)$, which can hardly be done (it seems to me) without having the factorization for $C\langle I\rangle \longleftrightarrow TF(I)$ as well; something I do not manage to establish for a general I in Cat, except by means of the hypothesis of specialness on C as well.

3.9. To get to the bottom of it, we would need a few concrete and illuminating examples, of the following type:

a) Closed Quillen structure (W, C, F) on $\mathcal{M}$, such that for the associated prederivator, $u_!^W$ does not always exist, or u_*^W does not always exist. I am thinking in particular of the case of an arrow $u : I \longrightarrow I'$ with $I' = e$ (presuming that if in this case, $u_!^W$ resp. u_*^W exists, then it exists in all cases). Such an example would be even more

instructive, of course, if $\mathcal{M}$ is stable under small $\varinjlim$, if its objects are accessible etc. and if W is L-accessible (i.e. stable under small filtering $\varinjlim$, and moreover accessible …). The first candidate for I that I think of would be a B_G (G a group, e.g. $G = \pm 1$), it is indeed the example at the antipodes of ordered sets, which, for their part, do not seem to cause too much trouble (assuming the noetherian hypothesis at least).

p. 24

[page 43]

The following examples are not all that convincing, alas!

Example 3.10. Let $\mathcal{M}$ be any category, then the couples $(\mathrm{Fl}(\mathcal{M}), \mathrm{Is}(\mathcal{M}))$ and $(\mathrm{Is}(\mathcal{M}), \mathrm{Fl}(\mathcal{M}))$ are both Quillen couples,

$$\mathrm{Fl}(\mathcal{M}) \longleftrightarrow \mathrm{Is}(\mathcal{M}) \ , \qquad \mathrm{Is}(\mathcal{M}) \longleftrightarrow \mathrm{Fl}(\mathcal{M}) \ . \tag{3.10.1}$$

It follows that if we set

$$W \ = \ \mathrm{Is}(\mathcal{M}) \ , \qquad C \ = \ F \ = \ \mathrm{Fl}(\mathcal{M}) \ , \tag{3.10.2}$$

(W, C, F) is a closed Quillen triple [(22)]. Moreover, for every I in Cat, we have $W(I) = \mathrm{Is}(\mathcal{M}(I))$, hence

$$\mathrm{Ho}_W(I) \ = \ \mathcal{M}(I) \ . \tag{3.10.3}$$

So the question of the existence of the functors $u_!^W$, u_*^W reduces to that of the usual functors $u_!$, u_*, left and right adjoints to $u^* : \mathcal{M}(I') \longrightarrow \mathcal{M}(I)$.

It is well known that the existence of these functors for a given u, calls for the existence [page 44]
of certain $\varinjlim$ resp. $\varprojlim$ in $\mathcal{M}$; e.g. if $I' = e$, it is necessary and sufficient that the $\varinjlim$ (resp. $\varprojlim$) of type I° be representable in $\mathcal{M}$. But with the hypotheses made on $\mathcal{M}$, there is no problem if $u : I \longrightarrow I'$ is a functor between categories that are *finite.* We are going to give an example where this is no longer so.

Example 3.11. Let k be a commutative ring, and $\mathcal{M} = K^{\mathrm{b}}(k\text{-Ab})$ the category of bounded cochain complexes of k-modules, W the set of quasi-isomorphisms, hence

$$\mathrm{Ho}_W \ = \ D^{\mathrm{b}}(k\text{-Ab}) \ ,$$

the bounded derived category of the abelian category of k-modules. We know, using the existence of enough projectives in k-Ab, that this structure is associated with a closed Quillen structure (W, C, F), described by Quillen in Homotopical Algebra – except that he takes $D^-(k)$ instead of $D^{\mathrm{b}}(k)$. Perhaps, to verify the axioms, we must assume that k [page 45]
has finite cohomological dimension.

Let us take $I = B_G$, where G is a group (e.g. $G = \{\pm 1\}$), and $u : I \longrightarrow e$. Then $\mathcal{M}(I)$ is the category of bounded complexes of $k[G]$-modules. We see that $u_!^W$ resp. u_*^W exists if and only if for every such bounded complex, or simply for a plain $k[G]$-module M, the complex $L\Gamma^G(M)$ resp. $R\Gamma_G(M)$ belongs to $D^{\mathrm{b}}(k)$, or what amounts to the same, if and only if we have

$$H_i(G, M) \ = \ 0 \quad \text{for } i \gg 0 \ , \qquad \text{p. 25}$$
$$\text{resp.} \quad H^i(G, M) \ = \ 0 \quad \text{for } i \gg 0 \ ,$$

for every $k[G]$-module M. This is practically never the case; when we take e.g. $k = \mathbf{Z}$ and G a finite group, this is the case only if $G = e$. The most economical example seems to me to be $G = \pm 1$, $k = \mathbf{F}_2$.

So here $u_!^W$ and u_*^W do not exist, for $I \longrightarrow e$ with I a finite category. But also $\mathcal{M}$ is [page 46]
not stable under small $\varinjlim$.

Example 3.12. Let $\mathcal{M}$ be a category equipped with a strongly saturated localizer W, such that $\mathcal{M} \longrightarrow \underbrace{\mathcal{M}W^{-1}}_{=\mathcal{H}}$ admits an adjoint – a left one, say. So we have a situation of two adjoint functors

$$\mathcal{H} \underset{v^* \text{ localization}}{\overset{v_! \text{ fully faithful}}{\rightleftarrows}} \mathcal{M} \quad , \tag{3.12.1}$$

and $W = (v^*)^{-1}(\mathrm{Iso}(\mathcal{H}))$. We find that for every I in Cat, we have

$$\mathrm{Ho}_W(I) \ \simeq \ \mathcal{H}(I) \ \overset{\mathrm{def}}{=} \ \mathrm{Hom}(I^\circ, \mathcal{H}). \tag{3.12.2}$$

[22] at least if we assume that $\mathcal{M}$ is stable under $\varinjlim$ and $\varprojlim$ that are *finite.*

So we can think of making an example where in $\mathcal{M}$ the small $\varinjlim$ and $\varprojlim$ exist, but not in $\mathcal{H}$, so that the $u_!^W$ or u_*^W do not always exist. But this seems difficult to construct, seeing that in the situation envisaged, we easily see that the small $\varprojlim$ exist in $\mathcal{H}$. So if in $\mathcal{H}$ there exists a small cogenerating family (and for this, it is enough that such a family exist in $\mathcal{M}$), there will also be small $\varinjlim$, by virtue of SGA 4 I 8.12.7. Likewise, they exist if $\mathcal{H}$ is a full subcategory of $\mathcal{M}$ *stable under small* $\varinjlim$, and it seems practically impossible to make an example (3.12.1) without $\mathcal{H}$ having this stability property. Not to mention that there is even the following perplexity: [page 47]

Question 3.13. *Is there a* $\mathfrak{U}$*-category (*$\mathfrak{U}$ *a given universe) in which the small* $\varinjlim$ *are representable, without the same being true of the small* $\varprojlim$ *?* **N.B.** Such a category does not admit a small generating family, a fortiori it is not equivalent to a small category. ! |

Question 3.14. ANDERSON claims that for the prederivator defined by a category with cofibrations satisfying the transfinite axiom Cof_4, the functors $u_!^W$ exist. His proof is totally botched, nevertheless I cannot manage to make a counterexample. Could the result he announces be true? I do not even know of an example of a localizer W, in a category $\mathcal{M}$ *with small* $\varinjlim$ *and small* $\varprojlim$ *,* such that $u_!^W$ or u_*^W do not always exist – but I am convinced that there are examples (even with $\mathcal{M}$ a small category?). I am less convinced of this if we suppose W stable under filtering $\varinjlim$, and if need be, moreover, that the elements of $\mathcal{M}$ are accessible (plus possibly other innocuous categorical properties of $\mathcal{M}$, such as the existence of a cardinal filtration etc.) [page 48] p. 26

Question 3.15. This brings me back to the particularly elusive and irritating question, of the existence of small Q-generating sets, in a subset Φ of $\mathrm{Fl}(\mathcal{M})$, when we suppose (say) that $\mathcal{M}$ is stable under small $\varprojlim$, that its elements are accessible and that it admits a cardinal filtration. Thus we have the question: does an L-accessible subset of $\mathrm{Fl}(\mathcal{M})$, which we assume if need be to be Q-saturated on the left (or what amounts to the same, containing the isomorphisms, and stable under cobase change, under composition and under direct factors) admit a *small subset* Φ_0 that is Q-generating on the left? [page 49]

On the other hand, I have not at present constructed a single example of a couple $\Phi \longleftrightarrow \Psi$, in a category $\mathcal{M}$, that is not factorizing! The case where $\mathcal{M}$ is special, was of course the most interesting. Clearly, in all this it is examples that are lacking. The key to an understanding does seem to lie in the questions of factorization between sets $\Phi \longleftrightarrow \Psi$. One ought to make a systematic study of this, but I shrink from the prospect of a digression that would perhaps take me yet more weeks! ! | [page 50]